

GMAC Subsystem RM

GMACSUBSYSRM



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Chapter 1

GMACSUBSYSRM

1.1 Introduction

This section and all its sub-sections are Synopsys Proprietary. Used with permission.

GMAC supports 10/100/1000/2500 Mbps applications in compliance with IEEE802.3-2015 specifications. GMAC has the capability of supporting advanced applications such as time sensitive networking and AVB. Also, MAC core has prominent features such as Flexible receive parser, media clock recovery and generation, and safety.

1.1.1 Features

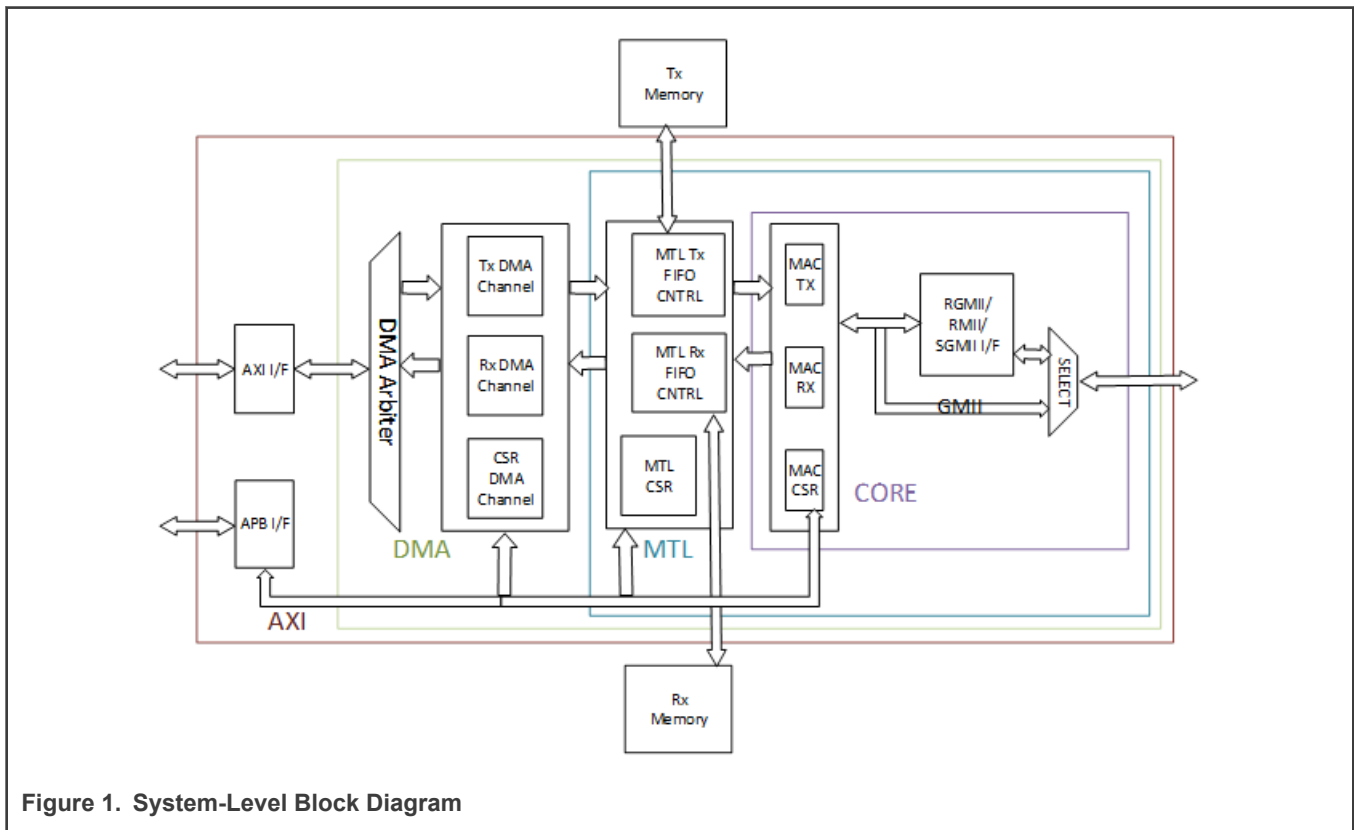
MAC core supports the Following features in addition to the default feature defined in IEEE802.3 specification

- MII 10/100Mbps, RMII (10/100Mbps), RGMII (100/1000Mbps)
- GMII to support the SGMII (1000/2500Mbps). Supported PHY interface speeds vary per device, and users should check the specifications for each chip.
- Time aware shaper (IEEE 802.1Qbv), Time synchronization (IEEE 802.1AS-rev), and Frame pre-emption (IEEE 802.1Qbu) for Time sensitive networking
- Media clock recovery & generation for AVB
- AXI4 master and APB3 slave
- RMII specification version 1.2 from RMII consortium
- RGMII specification version 2.6 from HP/Marvell
- Separate interface for Transmit, Receive and Control path
- Full duplex flow control operations (IEEE 802.3x pause packet and priority flow control)
- Network statistics with RMON counters
- Flexibility to control Pulse per second (PPS) output signals and support for auxiliary snapshot input signals
- Ethernet packet timestamping as described in IEEE 1588-2002 and IEEE 1588-2008 (64-bit timestamps given in the Tx or Rx status of PTP packet). Both one-step and two-step timestamping is supported in TX direction.
- MDIO (clause 22 and Clause 45) interface for configuration and management of PHY device
- Preamble and start of packet data (SFD) insertion/deletion
- Automatic CRC and pad generation/stripping option, option to disable CRC checking
- Programmable Inter packet gap
- Source address insertion/replacement, VLAN insertion, replacement and deletion in transmitted packets with per-packet or static-global control
- IEEE802.1Q VLAN tag detection and option to delete in the receive packets.
- Programmable safety watchdog timeout limits
- Flexible address filtering modes:
 - Up to 1 additional 48-bit perfect (DA) address filters with masks for each byte
 - Up to 1 48-bit SA address comparison check with masks for each byte
 - 256 bit Hash filter (optional) for multicast and unicast (DA) addresses
 - Option to pass all multi-cast addressed packets
 - Promiscuous mode to pass all packets without any filtering for network monitoring

- Pass all incoming packets (as per filter) with a status report
- Additional packet filtering:
 - VLAN tag-based: Perfect match and Hash-based (optional) filtering. Filtering based on either outer or inner VLAN tag is possible.
 - Layer 3 and Layer 4-based: TCP or UDP over IPv4 or IPv6
 - Extended VLAN tag based filtering 32 filter selections
 - Programmable lookup table based flexible Parser for filtering and steering the Rx packets

1.1.2 Functional Block Diagram

Figure shows the block diagram of GMAC IP. IP has four main blocks to perform all the function. AXI interface is connected to all DMA channels. DMA arbiter helps in arbitration of all paths (transmit and receive) in channels. Each channel has a separate set of control and status register (CSR) for managing the transmit and receive functions, descriptor handling, and interrupt handling.



1.2 Architecture

This section and all its sub-sections are Synopsys Proprietary. Used with permission.

This section describes about the interfaces, protocols, and functionality. Following are the main blocks of the IP.

- AXI4 master and CSR slave Interface
- DMA Controller
- MAC Transaction Layer (MTL)
- MAC
- Interrupts

1.2.1 AXI4 master and CSR slave Interface

The APB3 slave interface provides the access to control and status registers. 32-bit APB3 slave interface provides the access to DMA, MTL and MAC control and status registers. APB3 slave interface is AMBA3 APB protocol specification compliant. Each write transfer completes in two clock cycles and each read transfer completes in three clock cycles.

The AXI4 master interface is being used by DMA block to interface with the external world or system. The AXI4 master interface supports additional interface signal AxQOS and AxDOMIAN in addition to the AMBA3 AXI interface signal. AXI interface supports the simultaneous read and write transfers initiated by different DMA channels in IP. AXI interface supports multiple burst request to be queued for read and write transaction. AXI interface also support the data interleaving and burst reordering.

AXI master interface can splits DMA request in multiple bursts. Different burst length and burst type, outstanding read and write request on AXI bus can be configured through software in DMA_SysBus_Mode_register.

AXI interface supports 64-bit data width and 32-bit address width.

1.2.2 DMA Controller

The DMA has independent Transmit (Tx) and Receive (Rx) engines, and a control and status register (CSR) space. The Tx engine transfers data from the system memory to the MTL interface, whereas the Rx engine transfers data from the MTL interface to the system memory. The DMA engine uses descriptors to efficiently move data from source to destination with minimal Host intervention. The DMA is designed for packet-oriented data transfers such as packets in Ethernet. The DMA controller can be programmed to interrupt the Host for situations such as Packet Transmit and Receive Transfer completion, and other normal or error conditions.

The DMA and Host communicate through the following two data structures:

- Control and Status registers (CSR)
- Descriptor lists and data buffers

The DMA supports up to 5 Tx and 5 Rx Descriptor lists (or DMA channels). The base address of each list is written to the respective Tx Descriptor List Address register and Rx Descriptor List Address register. The descriptor list is forward linked and the next descriptor is always considered at a fixed offset to the current one. The offset is controlled by the DSL field of DMA_Ch[n]_Control register. The number of descriptors in the list is programmed in the respective Tx (or Rx) Descriptor Ring Length register. Once the DMA processes the last descriptor in the list, it automatically jumps back to the descriptor in the List Address register to create a descriptor ring.

The descriptor lists reside in the physical memory address space of the application. Each descriptor can point to a maximum of two buffers in the system memory. This enables two buffers to be used, physically addressed, rather than contiguous buffers in memory.

A data buffer resides in the application physical memory space and consists of an entire packet or part of a packet but cannot exceed a single packet. Buffers contain only data. Buffer status is maintained in the descriptor. Data chaining refers to packets that span multiple data buffers. However, a single descriptor cannot span multiple packets. The DMA skips to the data buffer of next packet when EOP is detected.

GMAC supports the ring structure for the DMA descriptor. For more information on Descriptor, see the section [Descriptors](#).

Following section discuss about the operation of DMA Controller.

1.2.2.1 DMA Controller Bus Burst Access

The DMA engines attempts to transfer data in a burst of maximum size as programmed in the PBL fields of Transmit Control and Receive Control registers of the respective DMA. The Rx and Tx descriptors are always accessed in the maximum possible (limited by PBL or $16 * 8 / \text{bus width}$) burst length for 16 bytes to be read. The burst transfers initiated by the DMA can be split into multiple burst transfers as per the AXI interface requirements and the settings of DMA_SysBus_Mode register.

The Tx DMA initiates a data transfer only when sufficient space is available in the MTL Tx Queue to accommodate either of the following:

- Bytes corresponding to the configured burst ($\text{PBL} * \text{bus_width} / 8$)
- Remaining bytes in Tx Buffer without EOP

- Number of bytes till EOP

The Rx DMA initiates a data transfer in the following conditions:

- Sufficient data is available in MTL Rx Queue to accommodate the configured burst
- EOP (when it is less than the configured burst length) is detected in the Rx Queue

The DMA indicates the start address and the number of transfers required to the AXI master interface. When the AXI Interface is configured for fixed-length burst, it transfers the data by using the best combination of INCR4, INCR8, or INCR16 and SINGLE transactions. If EOP is reached before the fixed-burst ends on the AXI interface, dummy transfers are performed in-order to complete the fixed-burst. Otherwise [Bit 0 of DMA_SysBus_Mode register is reset], the DMA transfers the data using INCR (undefined length) and SINGLE transactions.

When the AXI interface is configured for address-aligned beats, both DMA engines ensure that the first burst transfer initiated by the AXI is less than or equal to the size of the configured PBL. Therefore, all subsequent beats start at an address that is aligned to the configured PBL. The DMA can only align the address for beats up to size 256 (for PBL > 256) for AXI interface because it does not support more than BLEN256.

1.2.2.2 DMA Application Data Buffer Alignment

The Tx and Rx data buffers do not have any restrictions on start address alignment. For example, in systems with 32-bit memory, the start address for buffers can be aligned to any of the four bytes. However, the DMA always initiates write transfers with address aligned to the bus width and dummy data (old data) in the invalid byte lanes. This typically happens during the transfer of the beginning or end of an Ethernet packet. The software driver should discard the dummy bytes based on the start address of the buffer and size of the packet.

Table 1. Data Buffer Alignment Examples

Examples	
Buffer Read	If the Tx buffer address is 32'h00000FF2 (for 32-bit data bus), and 15 bytes need to be transferred, the DMA reads five full words from address 32'h00000FF0, but when transferring data to the MTL Tx queue, the extra bytes (the first two bytes) are dropped or ignored. Similarly, the last 3 bytes of the last transfer are also ignored. The DMA always ensures that it transfers a full 32-bit data to the MTL Tx queue, unless it is the end of packet.
Buffer Write	If the Rx buffer address is 32'h00000FF2 (for 64-bit data bus) and 16 bytes of a received packet need to be transferred, the DMA writes 3 full words from address 32'h00000FF0. However, the first 2 bytes of the first transfer and the last 6 bytes of the third transfer have dummy data. The DMA considers the offset address only if it is the first Rx buffer of the packet. The DMA ignores the offset address and performs full word writes for the middle and the last Rx buffer of the packet.

1.2.2.3 DMA Buffer Size Calculations

The DMA does not update the size fields in the Tx and Rx descriptors. The DMA updates only the status fields (RDES and TDES) of the descriptors. The driver has to perform the size calculations.

The Tx DMA transfers the exact number of bytes (indicated by buffer size field of TDES2) towards the MAC. If a descriptor is marked as first (FD bit of TDES3 is set), the DMA marks the first transfer from the buffer as SOP. If a descriptor is marked as last (LD bit of TDES3), the DMA marks the last transfer from that data buffer as EOP to the MTL.

The Rx DMA transfers data to a buffer until the buffer is full or the end of packet is received from the MTL. When the FD bit of a descriptor is set, the amount of valid data in a buffer is accurately indicated by the buffer size field (programmed in DMA Channel Receive Control register) minus the data buffer pointer offset. The offset is zero when the data buffer pointer is aligned to the data bus width. If a descriptor is marked as last, the buffer may not be full (as indicated by the buffer size in Bits [14:1] of Receive Control register). To compute the amount of valid data in this final buffer, the driver must read the packet length (PL bits of RDES3[14:0]) and subtract the sum of the buffer sizes of the preceding buffers in this packet. The Rx DMA always transfers the start of next packet with a new descriptor.

1.2.2.4 MAC Transaction Layer

The MAC Transaction Layer (MTL) provides the FIFO memory Interface to buffer and regulate the packets between system memory and the MAC. It also enables the data to be transferred between the system clock and MAC clock domains. The MTL layer has two data paths: Transmit path and Receive Path. The MTL communicates with the host through Application Transmit Interface (ATI) on the Transmit path and Application Receive Interface (ARI) on the Receive path.

1.2.2.4.1 Transmit Path

Internal DMA handles all transactions for the Transmit path through ATI. Internal DMA pushes the Ethernet packets read from system memory into the corresponding queue. The packet is then popped out and transferred to the MAC when the queue threshold is reached (threshold mode) or complete packet is in the queue (store-and-forward mode). When EOP is transferred, the status of the transmission is taken from the MAC and transferred back to the internal DMA.

1.2.2.4.1.1 Transmit Control Word

The following control information related to packet transmission is provided as a part of the Control Word through ATI interface:

- Packet Length (if DCB is enabled with WFQ scheduling algorithm)
- CRC Pad Control
- Source Address Insertion Control
- VLAN Insertion and Replacement Control and VLAN Tag for Outer and Inner VLAN Tags
- TCP/IP Checksum Insertion Control
- One-step Timestamping Control Correction
- Transmit Timestamp Enable

1.2.2.4.1.2 Transmit Operation

The following two modes of operation triggers reading of the data towards the MAC:

- Threshold mode: In Threshold (or cut-through) mode, as soon as the number of bytes in the Queue cross the configured threshold level (or when the end of packet is written before the threshold is crossed), the data is ready to be popped out and forwarded to the MAC. The threshold level is configured by using the TTC bits of MTL_TxQ0_Operation_Mode register corresponding to an MTL queue.
- Store-and-forward mode: In store-and-forward mode, the MTL pops out the packet towards the MAC only when one or more of the following conditions are true:
 - A complete packet is stored in the Queue
 - The Tx FIFO becomes almost full
 - The ATI watermark becomes low: The watermark becomes low when the requested Queue does not have space to accommodate the requested burst length on the ATI. Therefore, the MTL when operating in the store-and-forward mode allows packet transmission even if the packet length is bigger than the Tx Queue size.

The application can flush complete content of the Tx Queue by setting Bit 0 (FTQ) of Transmit Operation Mode register corresponding to an MTL queue. This bit is self-clearing and initializes the Queue pointers to the default state. If the FTQ bit is set during a packet transfer from the MTL to the MAC, the MTL stops further transfer because the queue is considered to be empty. Therefore, an underflow event occurs at the MAC transmitter.

1.2.2.4.1.3 Initialization Flow

Upon reset, the MTL is ready to manage the flow of data to and from the DMA, and MAC. For configurations with a single Transmit Queue, there are no initialization requirements for enabling the MTL. For configurations with multiple Transmit Queues, initialize the Queue size for each of the queues by programming the TQS bits of MTL_TxQ0_Operation_Mode register corresponding to a Transmit queue. Also, initialize the MAC block. Internal DMA controllers must be individually enabled through their respective CSRs.

Masters generating high traffic and GMAC getting inadequate bandwidth can lead to Tx FIFO underflow. To avoid this, it is recommended to use Store and Forward mode of operation selected by [MTL_TxQn_Operation_Mode\[TSF\]](#).

1.2.2.4.2 Receive Path

The MTL Rx module receives the packets from the MAC and pushes them into the Rx Queue. The status (fill level) of the queue is indicated to DMA when it crosses the configured Receive threshold (RTC bits[1:0] of MTL_RxQ0_Operation_Mode register of corresponding MTL queue), or the complete packet is received. The MTL also indicates the fill level of the queue so that the DMA can initiate preconfigured burst transfers towards the AXI interface.

1.2.2.4.2.1 Receive Operation

During a Receive operation, the MTL is a slave for the MAC. The general sequence of events is as follows:

1. When the MAC receives a packet, it indicates the availability of receive data.
2. The MAC indicates the SOP and EOP delimiters.
3. The MTL accepts the data and pushes it into corresponding Rx queue.
4. After the EOP is transferred, the MAC drives the status word which is also pushed into the corresponding Rx queue by the MTL.
5. If IEEE 1588 timestamp feature is enabled and the 64-bit timestamp is available along with the packet status, it is pushed into the Rx queue as a part of the status word. Therefore, in 32-bit data bus mode, two additional locations are taken per packet to store the timestamp in the Rx queue. In 64-bit, 128-bit, or 256-bit mode, one additional location is taken.
6. The MTL takes the data out of the queue and sends it to the DMA depending on the following modes:

1.2.2.4.2.1.1 Threshold Mode:

In the (default) Threshold mode, the MTL reads the data and indicates its availability to the application or DMA when one of the following occurs:

- data bytes equal to the threshold amount are written to the Rx queue (RTC bits[1:0] of MTL_RxQ0_Operation_Mode register of corresponding MTL queue)
- a full packet of data is received into the queue

1.2.2.4.2.1.2 Store-and-Forward Mode

In the store-and-forward mode (when Bit 5 of MTL_RxQ0_Operation_Mode register of a queue is set to 1), the initial Rx queue locations are reserved for the status words before writing the SOP. A packet is read out only after it is completely written into the Rx queue. In this mode, all error packets are dropped (if configured through Bit 4 of MTL_RxQ0_Operation_Mode register of a queue) such that only valid packets are read and forwarded to the application.

1.2.2.4.2.2 Multi-Packet Receive Operation

In Threshold mode, the packet status is available immediately after the packet data. In store-and-forward mode, the packet data is available after the packet status. The MTL is capable of storing any number of packets into the queue as long as it is not full.

If the MAC receives a packet when the corresponding Rx queue is full, the MTL ignores that packet and overflow pulse is generated on the corresponding Rx queue. In addition, the MTL increments the overflow counter in the MTL_RxQ0_Missed_Packet_Overflow_Cnt of corresponding queue.

1.2.2.4.2.3 Error Handling in Receive Operation

Following would happen if the MTL Rx queue is full before it receives the EOP data from the MAC.

1. An overflow is declared
2. The whole packet (including the status word) is dropped
3. The overflow counter in the DMA (Overflow Counter register of corresponding MTL queue) is incremented.

This is true even if Bit 4 (FEP) of MTL_RxQ0_Operation_Mode register of corresponding MTL queue is set.

If the start address of such a packet has already been transferred to the Read Controller, the rest of the packet is dropped and a dummy EOP is written to the queue along with the status word with overflow status. The status indicates a partial packet because of overflow. In such packets, the Packet Length field is invalid. If the MTL Receive Queue is configured to operate in the store-and-forward mode and the length of the received packet is more than the queue size, overflow occurs and all such packets are dropped.

The MTL Rx Control logic can filter error and undersized packets, if enabled by using the FEP and FUP bits of MTL_RxQ0_Operation_Mode register of corresponding MTL queue. If the start address of such a packet has already been transferred to the Rx Queue Read Controller, that packet is not filtered. The start address of the packet is transferred to the Read Controller after the packet crosses the receive threshold set by Bits[1:0] of MTL_RxQ0_Operation_Mode register of corresponding MTL queue.

If the MTL Receive Queue is configured to operate in the store-and-forward mode, all error packets can be filtered and dropped. For the application or DMA to flush the error packet being read from the queue, it must assert the flush signal. The MTL then stops transferring data to the application (DMA). It internally reads the rest of the packet and drops it. The MTL then starts the transfer of next packet (if available).

1.2.2.5 MAC

MAC can support MII and RMII Phy interface. MAC consist of Mac Transmit interface (MTI), MAC Receive Interface (MRI) and MAC Control Interface (MCI).

1.2.2.5.1 MAC Transmission

The MAC transmission process is as follows:

1. Transmission is initiated when the MAC Transaction Layer (MTL) pushes in data with the SOP signal asserted.
2. When the SOP signal is detected, the MAC accepts the data and begins transmitting to the RMII or MII.
3. After the EOP is transferred to the MAC, the MAC does one of the following:
 - The MAC completes the normal transmission and gives the transmission status to the MTL.
 - If a normal collision (in half-duplex mode) occurs during transmission, the MAC gives the Transmit Status with retry bitset to the MTL. The MAC gives the Retry request till one of the following is true:
 - Packet is successfully transmitted
 - Maximum retry requests expire
 - When maximum retry requests expire, the MAC aborts the packet transmission with Excessive Collision Transmit Status. The MAC accepts and drops all further data until the next SOP is received. The MTL block should retransmit the same packet from SOP on observing a Retry request (in the Status) from the MAC.
 - If any one of the following happens, the MAC aborts the packet transmission:
 - No carrier (half-duplex mode)
 - Loss of carrier (half-duplex mode)
 - Excessive deferral (half-duplex mode)
 - Late collisions (half-duplex mode)
 - Jabber
 - The MAC accepts and drops all further data until the next SOP is received.
4. The MAC issues an underflow status if the MTL is not able to provide the data continuously during the transmission. The MAC accepts and drops all further data until the next SOP is received.
5. During the normal transfer of a packet from MTL, if the MAC receives a SOP without getting an EOP for the previous packet, it ignores the SOP and considers the new packet as continuation of the previous packet.

1.2.2.5.2 MAC Reception

A receive operation is initiated when the MAC detects an SFD on RMII or MII. The MAC strips the preamble and SFD before proceeding to process the packet. The header fields are checked for filtering and the FCS field used to verify the CRC for the packet. The received packet is stored in a shallow buffer until the address filtering is performed. The packet is dropped in the MAC if it fails the address filter.

1.2.2.6 Interrupts

GMAC IP supports interrupt coalescing which reduces the number of interrupts generated by IP and reduces the CPU offload. Various interrupt event gets captured in status register. If interrupt enable bit of a interrupt is set then corresponding interrupt will be generated based on event status in status registers. INTM (interrupt mode) bit decides that interrupt signal is a level signal or pulse signal.

1.3 Signal Descriptions

This section and all its sub-sections are Synopsys Proprietary. Used with permission.

This section details all possible I/O signals in the core. For chip-specific clocking details, see Clocking chapter.

Table 2. Module signals

Port Name	I/O	Description
clk_tx_i	I	MAC Transmitter Clock. The external PHY or PLL provides this transmission clock. This is 312.5 MHz in 2.5 Gbps mode, 125 MHz in 1 Gbps mode, 25 MHz in 100 Mbps mode, 2.5 MHz in 10 Mbps mode. All transmission signals generated by the MAC are synchronous to this clock. This clock is required for all PHY interfaces.
clk_rx_i	I	MAC Receiver Clock. The external PHY provides this receive clock for RGMII, GMII, MII, and RMII interfaces. This clock is 312.5 MHz in 2.5 Gbps mode, 125 MHz in 1 Gbps mode, 25 MHz in 100 Mbps mode, 2.5 MHz in 10 Mbps mode. All RGMII, GMII, and MII receive signals received by the MAC are synchronous to this clock. This clock input is required for all PHY interfaces.
clk_rmii_i	I	RMII Clock. This is the 50-MHz clock used by the RMII interface. When the RMII interface is selected, the clk_rx_i clock should be 25 or 2.5 MHz and it should be derived from this clock.
clk_tx_180_i	I	Inverted RGMII Transmit Clock. The RGMII transmit interface uses the rising edge of clk_tx_180_i to represent the falling edge of clk_tx_i.
clk_rx_180_i	I	Inverted RGMII Receive Clock. The RGMII receive interface uses the positive edge of clk_rx_180_i to represent the falling edge of clk_rx_i.
clk_ptp_ref_i	I	PTP Reference Clock.

Table continues on the next page...

Table 2. Module signals (continued)

Port Name	I/O	Description
		This reference clock, provided by an external oscillator or a timing source, is used for the timestamp logic.
ptp_pps_o[3:0]	O	Pulse Per Second. This signal is high based on the PPS mode selected in the MAC_PPS_Control register. When PPS is programmed in Media Clock recovery Mode, this port indicates the recovered clock. See the device IOMUX sheet for details on the number of these signals available.
mcg_pst_trig_i[3:0]	I	Media Clock Generation Trigger Input Trigger Input to the DUT to capture presentation time. Based on presentation control value of MAC_PPS_Control register, this signal can be pulse or level signal. Width of this IO changes with the configured number of PPS instances. See the device IOMUX sheet for details on the number of these signals available.
phy_intf_sel_i[2:0]	I	PHY Interface Select. This signal selects the PHY interface of the MAC: • 000: GMII or MII • 001: RGMII • 100: RMII This signal is sampled only during reset assertion, and it is ignored after that. The reset is generated internally in the IP, and it gets activated when either the input reset (pwr_on_rst_n) or the software reset is asserted.
phy_txen_o	O	PHY Transmit Data Enable. The MAC drives this signal. This signal has multiple functions depending on the selected PHY interface as described in the following list: • GMII/MII: When high, indicates that valid data is being transmitted on the phy_txd_o bus. Synchronous to: clk_tx_i • RMII: When high, indicates that valid data is being transmitted on the phy_txd_o bus. Synchronous to: clk_rmii_i • RGMII: This signal is the control signal (rgmii_tctl) for the Transmit data, and it is driven on both edges of the clock. Synchronous to: clk_tx_i, clk_tx_180_i
phy_txer_o	O	PHY Transmit Error. The MAC drives this signal. It has multiple functions depending on the selected PHY interface as described in the following list: • GMII: When this signal is high, it indicates a Transmit error or carrier extension on the phy_txd_o bus. Synchronous to: clk_tx_i

Table continues on the next page...

Table 2. Module signals (continued)

Port Name	I/O	Description
		MII, RMII, RGMII: Not used and tied to low. According to the IEEE standard, the TX_ER signal (phy_txer_o signal in the GMAC) is optional in the MII mode. However, if you have enabled the Energy Efficient Ethernet feature, the phy_txer_o signal is required.
phy_txd_o[7:0]	O	PHY Transmit Data. This is a group of transmit data signals driven by the MAC. It has multiple functions depending on the selected PHY interface as described in the following list: - GMII: All eight bits provide the GMII transmit data byte. The data is valid only when the phy_txen_o and phy_txer_o signals are high. Synchronous to: clk_tx_i - MII: Bits[3:0] provide the MII transmit data nibble. The data is valid only when the phy_txen_o and phy_txer_o signals are high. Synchronous to: clk_tx_i - RGMII: Bits[3:0] provide the RGMII transmit data. The data bus changes with both rising and falling edges of the transmit clock (clk_tx_i). The data is valid only when the phy_txen_o signal is high. Synchronous to: clk_tx_i and clk_tx_180_i - RMII: Bits[1:0] provide the RMII transmit data. The data is valid only when the phy_txen_o signal is high. Synchronous to: clk_rmii_i
phy_crs_i	I	PHY CRS. This signal is valid only in the GMII or MII mode. The PHY drives this signal high when the Transmit or Receive medium is not idle. The PHY drives this signal low when both Transmit and Receive mediums are idle. This signal is not synchronous to any clock.
phy_col_i	I	PHY Collision. This signal is valid only in the GMII or MII mode. The PHY drives this signal high when a collision is detected on the medium. This signal is not synchronous to any clock.
phy_rxdv_i	I	PHY Receive Data Valid. The PHY drives this signal. It has multiple functions depending on the selected PHY interface as described in the following list: - GMII or MII: Indicates that the data on the phy_rxd_i bus is valid. It remains high continuously from the first recovered byte or nibble of the packet through the final recovered byte or nibble of the packet. Synchronous to: clk_rx_i - RGMII: This is the Receive control signal that is used to qualify the data received phy_rxd_i. This signal is sampled both edges of the clock. Synchronous to: clk_rx_i, clk_rx_180_i

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Table 2. Module signals (continued)

Port Name	I/O	Description
		RMII: Contains the crs and data valid information of the Receive interface. Synchronous to: clk_rmii_i
phy_rxer_i	I	PHY Receive Error. The PHY drives this signal. It has multiple functions depending on the selected PHY interface as described in the following list: - GMII or MII: Indicates an error or carrier extension (in GMII) in the received packet the phy_rxd_i bus. Synchronous to: clk_rx_i - RGMII, RMII: This signal is not used.
phy_rxd_i[7:0]	I	PHY Receive Data. This is a group of data signals received from the PHY. It has multiple functions depending on the selected PHY interface as described in the following list: - GMII: All eight bits provide the GMII receive data byte. The data is valid only when the phy_rxdv_i and phy_rxer_i signals are high. Synchronous to: clk_rx_i - MI: Bits [3:0] provide the MI receive data nibble. The data is valid only when the phy_rxdv_i and phy_rxer_i signals are high. Synchronous to: clk_rx_i - RGMII: Bits [3:0] provide the RGMII receive data. The data bus is sampled with both rising and falling edges of the receive clock (clk_rx_i). The data is valid only when the phy_rxdv_i signal is high. Synchronous to: clk_rx_i, clk_rx_180_i - RMII: Bits [1:0] provide the RMII receive data. The data is valid only when the phy_rxdv_i signal is high. Synchronous to: clk_rmii_i
gmii_mdc_o	O	Management Data Clock. The MAC provides timing reference for the gmii_mdi_i and gmii_mdo_o signals on GMII or MII through this aperiodic clock. This clock is generated from the application clock through a clock divider controlled by CR field of MAC_MDIO_Address register.
gmii_mdi_i	I	Management Data In. The PHY generates this signal to transfer register data during a read operation. This signal is driven synchronously with the gmii_mdc_o clock.
gmii_mdo_o	O	Management Data Out. The MAC uses this signal to transfer control and data information to the PHY.

1.4 GMAC register descriptions

This section and all its sub-sections are Synopsys Proprietary. Used with permission.

NOTE

This device has 16 KB allocated per Ethernet controller for configuration space. Only 8 KB of this memory is used in this module, and the remaining 8 KB is reserved for future use. Writing to the reserved 8 KB may lead to unintended behavior.

1.4.1 GMAC memory map

GMAC base address: 0h

Offset	Register	Width (In bits)	Access	Reset value
0h	(MAC_Configuration)	32	RW	0000_0000h
4h	(MAC_Ext_Configuration)	32	RW	0000_0000h
8h	(MAC_Packet_Filter)	32	RW	0000_0000h
Ch	(MAC_Watchdog_Timeout)	32	RW	0000_0000h
10h	(MAC_Hash_Table_Reg0)	32	RW	0000_0000h
14h	(MAC_Hash_Table_Reg1)	32	RW	0000_0000h
18h	(MAC_Hash_Table_Reg2)	32	RW	0000_0000h
1Ch	(MAC_Hash_Table_Reg3)	32	RW	0000_0000h
20h	(MAC_Hash_Table_Reg4)	32	RW	0000_0000h
24h	(MAC_Hash_Table_Reg5)	32	RW	0000_0000h
28h	(MAC_Hash_Table_Reg6)	32	RW	0000_0000h
2Ch	(MAC_Hash_Table_Reg7)	32	RW	0000_0000h
50h	(MAC_VLAN_Tag_Ctrl)	32	RW	0000_0000h
54h	(MAC_VLAN_Tag_Data)	32	RW	0000_0000h
58h	(MAC_VLAN_Hash_Table)	32	RW	0000_0000h
60h	(MAC_VLAN_Incl)	32	RW	0000_0000h
64h	(MAC_Inner_VLAN_Incl)	32	RW	0000_0000h
70h	(MAC_Q0_Tx_Flow_Ctrl)	32	RW	0000_0000h
74h	(MAC_Q1_Tx_Flow_Ctrl)	32	RW	0000_0000h
78h	(MAC_Q2_Tx_Flow_Ctrl)	32	RW	0000_0000h
7Ch	(MAC_Q3_Tx_Flow_Ctrl)	32	RW	0000_0000h
80h	(MAC_Q4_Tx_Flow_Ctrl)	32	RW	0000_0000h
90h	(MAC_Rx_Flow_Ctrl)	32	RW	0000_0000h
94h	(MAC_RxQ_Ctrl4)	32	RW	0000_0000h
98h	(MAC_TxQ_Prtty_Map0)	32	RW	0000_0000h
9Ch	(MAC_TxQ_Prtty_Map1)	32	RW	0000_0000h

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Table continued from the previous page...

Offset	Register	Width (In bits)	Access	Reset value
A0h	(MAC_RxQ_Ctrl0)	32	RW	0000_0000h
A4h	(MAC_RxQ_Ctrl1)	32	RW	0000_0000h
A8h	(MAC_RxQ_Ctrl2)	32	RW	0000_0000h
ACH	(MAC_RxQ_Ctrl3)	32	RW	0000_0000h
B0h	(MAC_Interrupt_Status)	32	RO	0000_0000h
B4h	(MAC_Interrupt_Enable)	32	RW	0000_0000h
B8h	(MAC_Rx_Tx_Status)	32	RO	0000_0000h
C0h	(MAC_PMT_Control_Status)	32	RW	0000_0000h
C4h	(MAC_RWK_Packet_Filter)	32	RW	0000_0000h
F8h	(MAC_PHYIF_Control_Status)	32	RW	0000_0000h
110h	(MAC_Version)	32	RO	0000_1051h
114h	(MAC_Debug)	32	RO	0000_0000h
11Ch	(MAC_HW_Feature0)	32	RO	0E05_53F7h
120h	(MAC_HW_Feature1)	32	RO	439B_3A28h
124h	(MAC_HW_Feature2)	32	RO	4410_4104h
128h	(MAC_HW_Feature3)	32	RO	2C37_5635h
140h	(MAC_DPP_FSM_Interrupt_Status)	32	RW	0000_0000h
148h	(MAC_FSM_Control)	32	RW	0000_0000h
14Ch	(MAC_FSM_ACT_Timer)	32	RW	0000_0000h
150h	(SCS_REG1)	32	RW	0000_0000h
200h	(MAC_MDIO_Address)	32	RW	0000_0000h
204h	(MAC_MDIO_Data)	32	RW	0000_0000h
210h	(MAC_ARP_Address)	32	RW	0000_0000h
230h	(MAC_CSR_SW_Ctrl)	32	RW	0000_0000h
234h	(MAC_FPE_CTRL_STS)	32	RW	0000_0000h
238h	(MAC_Ext_Cfg1)	32	RW	0000_0002h
240h	(MAC_Presn_Time_ns)	32	RO	0000_0000h
244h	(MAC_Presn_Time_Updt)	32	RW	0000_0000h
300h	(MAC_Address0_High)	32	RW	8000_FFFFh
304h	(MAC_Address0_Low)	32	RW	FFFF_FFFFh
308h	(MAC_Address1_High)	32	RW	0000_FFFFh

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Offset	Register	Width (In bits)	Access	Reset value
30Ch	(MAC_Address1_Low)	32	RW	FFFF_FFFFh
700h	(MMC_Control)	32	RW	0000_0000h
704h	(MMC_Rx_Interrupt)	32	RO	0000_0000h
708h	(MMC_Tx_Interrupt)	32	RO	0000_0000h
70Ch	(MMC_Rx_Interrupt_Mask)	32	RW	0000_0000h
710h	(MMC_Tx_Interrupt_Mask)	32	RW	0000_0000h
714h	(Tx_Octet_Count_Good_Bad)	32	RO	0000_0000h
718h	(Tx_Packet_Count_Good_Bad)	32	RO	0000_0000h
71Ch	(Tx_Broadcast_Packets_Good)	32	RO	0000_0000h
720h	(Tx_Multicast_Packets_Good)	32	RO	0000_0000h
724h	(Tx_64Octets_Packets_Good_Bad)	32	RO	0000_0000h
728h	(Tx_65To127Octets_Packets_Good_Bad)	32	RO	0000_0000h
72Ch	(Tx_128To255Octets_Packets_Good_Bad)	32	RO	0000_0000h
730h	(Tx_256To511Octets_Packets_Good_Bad)	32	RO	0000_0000h
734h	(Tx_512To1023Octets_Packets_Good_Bad)	32	RO	0000_0000h
738h	(Tx_1024ToMaxOctets_Packets_Good_Bad)	32	RO	0000_0000h
73Ch	(Tx_Unicast_Packets_Good_Bad)	32	RO	0000_0000h
740h	(Tx_Multicast_Packets_Good_Bad)	32	RO	0000_0000h
744h	(Tx_Broadcast_Packets_Good_Bad)	32	RO	0000_0000h
748h	(Tx_Underflow_Error_Packets)	32	RO	0000_0000h
74Ch	(Tx_Single_Collision_Good_Packets)	32	RO	0000_0000h
750h	(Tx_Multiple_Collision_Good_Packets)	32	RO	0000_0000h
754h	(Tx_Deferred_Packets)	32	RO	0000_0000h
758h	(Tx_Late_Collision_Packets)	32	RO	0000_0000h
75Ch	(Tx_Excessive_Collision_Packets)	32	RO	0000_0000h
760h	(Tx_Carrier_Error_Packets)	32	RO	0000_0000h
764h	(Tx_Octet_Count_Good)	32	RO	0000_0000h
768h	(Tx_Packet_Count_Good)	32	RO	0000_0000h
76Ch	(Tx_Excessive_Deferral_Error)	32	RO	0000_0000h
770h	(Tx_Pause_Packets)	32	RO	0000_0000h
774h	(Tx_VLAN_Packets_Good)	32	RO	0000_0000h

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Offset	Register	Width (In bits)	Access	Reset value
778h	(Tx_OSize_Packets_Good)	32	RO	0000_0000h
780h	(Rx_Packets_Count_Good_Bad)	32	RO	0000_0000h
784h	(Rx_Octet_Count_Good_Bad)	32	RO	0000_0000h
788h	(Rx_Octet_Count_Good)	32	RO	0000_0000h
78Ch	(Rx_Broadcast_Packets_Good)	32	RO	0000_0000h
790h	(Rx_Multicast_Packets_Good)	32	RO	0000_0000h
794h	(Rx_CRC_Error_Packets)	32	RO	0000_0000h
798h	(Rx_Alignment_Error_Packets)	32	RO	0000_0000h
79Ch	(Rx_Runt_Error_Packets)	32	RO	0000_0000h
7A0h	(Rx_Jabber_Error_Packets)	32	RO	0000_0000h
7A4h	(Rx_Undersize_Packets_Good)	32	RO	0000_0000h
7A8h	(Rx_Oversize_Packets_Good)	32	RO	0000_0000h
7ACh	(Rx_64Octets_Packets_Good_Bad)	32	RO	0000_0000h
7B0h	(Rx_65To127Octets_Packets_Good_Bad)	32	RO	0000_0000h
7B4h	(Rx_128To255Octets_Packets_Good_Bad)	32	RO	0000_0000h
7B8h	(Rx_256To511Octets_Packets_Good_Bad)	32	RO	0000_0000h
7BCh	(Rx_512To1023Octets_Packets_Good_Bad)	32	RO	0000_0000h
7C0h	(Rx_1024ToMaxOctets_Packets_Good_Bad)	32	RO	0000_0000h
7C4h	(Rx_Unicast_Packets_Good)	32	RO	0000_0000h
7C8h	(Rx_Length_Error_Packets)	32	RO	0000_0000h
7CCh	(Rx_Out_Of_Range_Type_Packets)	32	RO	0000_0000h
7D0h	(Rx_Pause_Packets)	32	RO	0000_0000h
7D4h	(Rx_FIFO_Overflow_Packets)	32	RO	0000_0000h
7D8h	(Rx_VLAN_Packets_Good_Bad)	32	RO	0000_0000h
7DCh	(Rx_Watchdog_Error_Packets)	32	RO	0000_0000h
7E0h	(Rx_Receive_Error_Packets)	32	RO	0000_0000h
7E4h	(Rx_Control_Packets_Good)	32	RO	0000_0000h
8A0h	(MMC_FPE_Tx_Interrupt)	32	RO	0000_0000h
8A4h	(MMC_FPE_Tx_Interrupt_Mask)	32	RW	0000_0000h
8A8h	(MMC_Tx_FPE_Fragment_Cntr)	32	RO	0000_0000h
8ACh	(MMC_Tx_Hold_Req_Cntr)	32	RO	0000_0000h

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Offset	Register	Width (In bits)	Access	Reset value
8C0h	(MMC_FPE_Rx_Interrupt)	32	RO	0000_0000h
8C4h	(MMC_FPE_Rx_Interrupt_Mask)	32	RW	0000_0000h
8C8h	(MMC_Rx_Packet_Assembly_Err_Cntr)	32	RO	0000_0000h
8CCh	(MMC_Rx_Packet_SMD_Err_Cntr)	32	RO	0000_0000h
8D0h	(MMC_Rx_Packet_Assembly_OK_Cntr)	32	RO	0000_0000h
8D4h	(MMC_Rx_FPE_Fragment_Cntr)	32	RO	0000_0000h
900h	(MAC_L3_L4_Control0)	32	RW	0000_0000h
904h	(MAC_Layer4_Address0)	32	RW	0000_0000h
910h	(MAC_Layer3_Addr0_Reg0)	32	RW	0000_0000h
914h	(MAC_Layer3_Addr1_Reg0)	32	RW	0000_0000h
918h	(MAC_Layer3_Addr2_Reg0)	32	RW	0000_0000h
91Ch	(MAC_Layer3_Addr3_Reg0)	32	RW	0000_0000h
930h	(MAC_L3_L4_Control1)	32	RW	0000_0000h
934h	(MAC_Layer4_Address1)	32	RW	0000_0000h
940h	(MAC_Layer3_Addr0_Reg1)	32	RW	0000_0000h
944h	(MAC_Layer3_Addr1_Reg1)	32	RW	0000_0000h
948h	(MAC_Layer3_Addr2_Reg1)	32	RW	0000_0000h
94Ch	(MAC_Layer3_Addr3_Reg1)	32	RW	0000_0000h
960h	(MAC_L3_L4_Control2)	32	RW	0000_0000h
964h	(MAC_Layer4_Address2)	32	RW	0000_0000h
970h	(MAC_Layer3_Addr0_Reg2)	32	RW	0000_0000h
974h	(MAC_Layer3_Addr1_Reg2)	32	RW	0000_0000h
978h	(MAC_Layer3_Addr2_Reg2)	32	RW	0000_0000h
97Ch	(MAC_Layer3_Addr3_Reg2)	32	RW	0000_0000h
990h	(MAC_L3_L4_Control3)	32	RW	0000_0000h
994h	(MAC_Layer4_Address3)	32	RW	0000_0000h
9A0h	(MAC_Layer3_Addr0_Reg3)	32	RW	0000_0000h
9A4h	(MAC_Layer3_Addr1_Reg3)	32	RW	0000_0000h
9A8h	(MAC_Layer3_Addr2_Reg3)	32	RW	0000_0000h
9ACh	(MAC_Layer3_Addr3_Reg3)	32	RW	0000_0000h
9C0h	(MAC_L3_L4_Control4)	32	RW	0000_0000h

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Offset	Register	Width (In bits)	Access	Reset value
9C4h	(MAC_Layer4_Address4)	32	RW	0000_0000h
9D0h	(MAC_Layer3_Addr0_Reg4)	32	RW	0000_0000h
9D4h	(MAC_Layer3_Addr1_Reg4)	32	RW	0000_0000h
9D8h	(MAC_Layer3_Addr2_Reg4)	32	RW	0000_0000h
9DCh	(MAC_Layer3_Addr3_Reg4)	32	RW	0000_0000h
9F0h	(MAC_L3_L4_Control5)	32	RW	0000_0000h
9F4h	(MAC_Layer4_Address5)	32	RW	0000_0000h
A00h	(MAC_Layer3_Addr0_Reg5)	32	RW	0000_0000h
A04h	(MAC_Layer3_Addr1_Reg5)	32	RW	0000_0000h
A08h	(MAC_Layer3_Addr2_Reg5)	32	RW	0000_0000h
A0Ch	(MAC_Layer3_Addr3_Reg5)	32	RW	0000_0000h
A20h	(MAC_L3_L4_Control6)	32	RW	0000_0000h
A24h	(MAC_Layer4_Address6)	32	RW	0000_0000h
A30h	(MAC_Layer3_Addr0_Reg6)	32	RW	0000_0000h
A34h	(MAC_Layer3_Addr1_Reg6)	32	RW	0000_0000h
A38h	(MAC_Layer3_Addr2_Reg6)	32	RW	0000_0000h
A3Ch	(MAC_Layer3_Addr3_Reg6)	32	RW	0000_0000h
A50h	(MAC_L3_L4_Control7)	32	RW	0000_0000h
A54h	(MAC_Layer4_Address7)	32	RW	0000_0000h
A60h	(MAC_Layer3_Addr0_Reg7)	32	RW	0000_0000h
A64h	(MAC_Layer3_Addr1_Reg7)	32	RW	0000_0000h
A68h	(MAC_Layer3_Addr2_Reg7)	32	RW	0000_0000h
A6Ch	(MAC_Layer3_Addr3_Reg7)	32	RW	0000_0000h
B00h	(MAC_Timestamp_Control)	32	RW	0000_2000h
B04h	(MAC_Sub_Second_Increment)	32	RW	0000_0000h
B08h	(MAC_System_Time_Seconds)	32	RO	0000_0000h
B0Ch	(MAC_System_Time_Nanoseconds)	32	RO	0000_0000h
B10h	(MAC_System_Time_Seconds_Update)	32	RW	0000_0000h
B14h	(MAC_System_Time_Nanoseconds_Update)	32	RW	0000_0000h
B18h	(MAC_Timestamp_Addend)	32	RW	0000_0000h
B1Ch	(MAC_System_Time_Higher_Word_Seconds)	32	RW	0000_0000h

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Offset	Register	Width (In bits)	Access	Reset value
B20h	(MAC_Timestamp_Status)	32	RO	0000_0000h
B30h	(MAC_Tx_Timestamp_Status_Nanoseconds)	32	RO	0000_0000h
B34h	(MAC_Tx_Timestamp_Status_Seconds)	32	RO	0000_0000h
B40h	(MAC_Auxiliary_Control)	32	RW	0000_0000h
B48h	(MAC_Auxiliary_Timestamp_Nanoseconds)	32	RO	0000_0000h
B4Ch	(MAC_Auxiliary_Timestamp_Seconds)	32	RO	0000_0000h
B50h	(MAC_Timestamp_Ingress_Asym_Corr)	32	RW	0000_0000h
B54h	(MAC_Timestamp_Egress_Asym_Corr)	32	RW	0000_0000h
B58h	(MAC_Timestamp_Ingress_Corr_Nanosecond)	32	RW	0000_0000h
B5Ch	(MAC_Timestamp_Egress_Corr_Nanosecond)	32	RW	0000_0000h
B60h	(MAC_Timestamp_Ingress_Corr_Subnanosec)	32	RW	0000_0000h
B64h	(MAC_Timestamp_Egress_Corr_Subnanosec)	32	RW	0000_0000h
B68h	(MAC_Timestamp_Ingress_Latency)	32	RO	0000_0000h
B6Ch	(MAC_Timestamp_Egress_Latency)	32	RO	0000_0000h
B70h	(MAC_PPS_Control)	32	RW	0000_0000h
B80h	(MAC_PPS0_Target_Time_Seconds)	32	RW	0000_0000h
B84h	(MAC_PPS0_Target_Time_Nanoseconds)	32	RW	0000_0000h
B88h	(MAC_PPS0_Interval)	32	RW	0000_0000h
B8Ch	(MAC_PPS0_Width)	32	RW	0000_0000h
B90h	(MAC_PPS1_Target_Time_Seconds)	32	RW	0000_0000h
B94h	(MAC_PPS1_Target_Time_Nanoseconds)	32	RW	0000_0000h
B98h	(MAC_PPS1_Interval)	32	RW	0000_0000h
B9Ch	(MAC_PPS1_Width)	32	RW	0000_0000h
BA0h	(MAC_PPS2_Target_Time_Seconds)	32	RW	0000_0000h
BA4h	(MAC_PPS2_Target_Time_Nanoseconds)	32	RW	0000_0000h
BA8h	(MAC_PPS2_Interval)	32	RW	0000_0000h
BACh	(MAC_PPS2_Width)	32	RW	0000_0000h
BB0h	(MAC_PPS3_Target_Time_Seconds)	32	RW	0000_0000h
BB4h	(MAC_PPS3_Target_Time_Nanoseconds)	32	RW	0000_0000h
BB8h	(MAC_PPS3_Interval)	32	RW	0000_0000h
BBCh	(MAC_PPS3_Width)	32	RW	0000_0000h

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Offset	Register	Width (In bits)	Access	Reset value
BC0h	(MAC_PTO_Control)	32	RW	0000_0000h
BC4h	(MAC_Source_Port_Identity0)	32	RW	0000_0000h
BC8h	(MAC_Source_Port_Identity1)	32	RW	0000_0000h
BCCh	(MAC_Source_Port_Identity2)	32	RW	0000_0000h
BD0h	(MAC_Log_Message_Interval)	32	RW	0000_0000h
C00h	(MTL_Operation_Mode)	32	RW	0000_0000h
C08h	(MTL_DBG_CTL)	32	RW	0000_0000h
C0Ch	(MTL_DBG_STS)	32	RW	0000_0018h
C10h	(MTL_FIFO_Debug_Data)	32	RW	0000_0000h
C20h	(MTL_Interrupt_Status)	32	RO	0000_0000h
C30h	(MTL_RxQ_DMA_Map0)	32	RW	0000_0000h
C34h	(MTL_RxQ_DMA_Map1)	32	RW	0000_0000h
C40h	(MTL_TBS_CTRL)	32	RW	0000_0000h
C50h	(MTL_EST_Control)	32	RW	0000_0000h
C58h	(MTL_EST_Status)	32	RW	0000_0000h
C60h	(MTL_EST_Sch_Error)	32	RW	0000_0000h
C64h	(MTL_EST_Frm_Size_Error)	32	RW	0000_0000h
C68h	(MTL_EST_Frm_Size_Capture)	32	RO	0000_0000h
C70h	(MTL_EST_Intr_Enable)	32	RW	0000_0000h
C80h	(MTL_EST_GCL_Control)	32	RW	0000_0000h
C84h	(MTL_EST_GCL_Data)	32	RW	0000_0000h
C90h	(MTL_FPE_CTRL_STS)	32	RW	0000_0000h
C94h	(MTL_FPE_Advance)	32	RW	0000_0000h
CA0h	(MTL_RXP_Control_Status)	32	RW	80FF_00FFh
CA4h	(MTL_RXP_Interrupt_Control_Status)	32	RW	0000_0000h
CA8h	(MTL_RXP_Drop_Cnt)	32	RO	0000_0000h
CACh	(MTL_RXP_Error_Cnt)	32	RO	0000_0000h
CB0h	(MTL_RXP_Indirect_Acc_Control_Status)	32	RW	0000_0000h
CB4h	(MTL_RXP_Indirect_Acc_Data)	32	RO	0000_0000h
CC0h	(MTL_ECC_Control)	32	RW	0000_000Fh
CC4h	(MTL_Safety_Interrupt_Status)	32	RO	0000_0000h

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Offset	Register	Width (In bits)	Access	Reset value
CC8h	(MTL_ECC_Interrupt_Enable)	32	RW	0000_1111h
CCCh	(MTL_ECC_Interrupt_Status)	32	RW	0000_0000h
CD0h	(MTL_ECC_Err_Sts_Rctl)	32	RW	0000_0000h
CD4h	(MTL_ECC_Err_Addr_Status)	32	RO	0000_0000h
CD8h	(MTL_ECC_Err_Cntr_Status)	32	RO	0000_0000h
CE0h	(MTL_DPP_Control)	32	RW	0000_0003h
D00h	(MTL_TxQ0_Operation_Mode)	32	RW	0000_0000h
D04h	(MTL_TxQ0_Underflow)	32	RO	0000_0000h
D08h	(MTL_TxQ0_Debug)	32	RO	0000_0000h
D14h	(MTL_TxQ0_ETS_Status)	32	RO	0000_0000h
D18h	(MTL_TxQ0_Quantum_Weight)	32	RW	0000_0000h
D2Ch	(MTL_Q0_Interrupt_Control_Status)	32	RW	0000_0000h
D30h	(MTL_RxQ0_Operation_Mode)	32	RW	0000_0000h
D34h	(MTL_RxQ0_Missed_Packet_Overflow_Cnt)	32	RO	0000_0000h
D38h	(MTL_RxQ0_Debug)	32	RO	0000_0000h
D3Ch	(MTL_RxQ0_Control)	32	RW	0000_0000h
D40h	(MTL_TxQ1_Operation_Mode)	32	RW	0000_0000h
D44h	(MTL_TxQ1_Underflow)	32	RO	0000_0000h
D48h	(MTL_TxQ1_Debug)	32	RO	0000_0000h
D50h	(MTL_TxQ1_ETS_Control)	32	RW	0000_0000h
D54h	(MTL_TxQ1_ETS_Status)	32	RO	0000_0000h
D58h	(MTL_TxQ1_Quantum_Weight)	32	RW	0000_0000h
D5Ch	(MTL_TxQ1_SendSlopeCredit)	32	RW	0000_0000h
D60h	(MTL_TxQ1_HiCredit)	32	RW	0000_0000h
D64h	(MTL_TxQ1_LoCredit)	32	RW	0000_0000h
D6Ch	(MTL_Q1_Interrupt_Control_Status)	32	RW	0000_0000h
D70h	(MTL_RxQ1_Operation_Mode)	32	RW	0000_0000h
D74h	(MTL_RxQ1_Missed_Packet_Overflow_Cnt)	32	RO	0000_0000h
D78h	(MTL_RxQ1_Debug)	32	RO	0000_0000h
D7Ch	(MTL_RxQ1_Control)	32	RW	0000_0000h
D80h	(MTL_TxQ2_Operation_Mode)	32	RW	0000_0000h

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Offset	Register	Width (In bits)	Access	Reset value
D84h	(MTL_TxQ2_Underflow)	32	RO	0000_0000h
D88h	(MTL_TxQ2_Debug)	32	RO	0000_0000h
D90h	(MTL_TxQ2_ETS_Control)	32	RW	0000_0000h
D94h	(MTL_TxQ2_ETS_Status)	32	RO	0000_0000h
D98h	(MTL_TxQ2_Quantum_Weight)	32	RW	0000_0000h
D9Ch	(MTL_TxQ2_SendSlopeCredit)	32	RW	0000_0000h
DA0h	(MTL_TxQ2_HiCredit)	32	RW	0000_0000h
DA4h	(MTL_TxQ2_LoCredit)	32	RW	0000_0000h
DACH	(MTL_Q2_Interrupt_Control_Status)	32	RW	0000_0000h
DB0h	(MTL_RxQ2_Operation_Mode)	32	RW	0000_0000h
DB4h	(MTL_RxQ2_Missed_Packet_Overflow_Cnt)	32	RO	0000_0000h
DB8h	(MTL_RxQ2_Debug)	32	RO	0000_0000h
DBCh	(MTL_RxQ2_Control)	32	RW	0000_0000h
DC0h	(MTL_TxQ3_Operation_Mode)	32	RW	0000_0000h
DC4h	(MTL_TxQ3_Underflow)	32	RO	0000_0000h
DC8h	(MTL_TxQ3_Debug)	32	RO	0000_0000h
DD0h	(MTL_TxQ3_ETS_Control)	32	RW	0000_0000h
DD4h	(MTL_TxQ3_ETS_Status)	32	RO	0000_0000h
DD8h	(MTL_TxQ3_Quantum_Weight)	32	RW	0000_0000h
DDCh	(MTL_TxQ3_SendSlopeCredit)	32	RW	0000_0000h
DE0h	(MTL_TxQ3_HiCredit)	32	RW	0000_0000h
DE4h	(MTL_TxQ3_LoCredit)	32	RW	0000_0000h
DECh	(MTL_Q3_Interrupt_Control_Status)	32	RW	0000_0000h
DF0h	(MTL_RxQ3_Operation_Mode)	32	RW	0000_0000h
DF4h	(MTL_RxQ3_Missed_Packet_Overflow_Cnt)	32	RO	0000_0000h
DF8h	(MTL_RxQ3_Debug)	32	RO	0000_0000h
DFCh	(MTL_RxQ3_Control)	32	RW	0000_0000h
E00h	(MTL_TxQ4_Operation_Mode)	32	RW	0000_0000h
E04h	(MTL_TxQ4_Underflow)	32	RO	0000_0000h
E08h	(MTL_TxQ4_Debug)	32	RO	0000_0000h
E10h	(MTL_TxQ4_ETS_Control)	32	RW	0000_0000h

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Offset	Register	Width (In bits)	Access	Reset value
E14h	(MTL_TxQ4_ETS_Status)	32	RO	0000_0000h
E18h	(MTL_TxQ4_Quantum_Weight)	32	RW	0000_0000h
E1Ch	(MTL_TxQ4_SendSlopeCredit)	32	RW	0000_0000h
E20h	(MTL_TxQ4_HiCredit)	32	RW	0000_0000h
E24h	(MTL_TxQ4_LoCredit)	32	RW	0000_0000h
E2Ch	(MTL_Q4_Interrupt_Control_Status)	32	RW	0000_0000h
E30h	(MTL_RxQ4_Operation_Mode)	32	RW	0000_0000h
E34h	(MTL_RxQ4_Missed_Packet_Overflow_Cnt)	32	RO	0000_0000h
E38h	(MTL_RxQ4_Debug)	32	RO	0000_0000h
E3Ch	(MTL_RxQ4_Control)	32	RW	0000_0000h
1000h	(DMA_Mode)	32	RW	0000_0000h
1004h	(DMA_SysBus_Mode)	32	RW	0101_0000h
1008h	(DMA_Interrupt_Status)	32	RO	0000_0000h
100Ch	(DMA_Debug_Status0)	32	RO	0000_0000h
1010h	(DMA_Debug_Status1)	32	RO	0000_0000h
1020h	(AXI4_Tx_AR_ACE_Control)	32	RW	0000_0000h
1024h	(AXI4_Rx_AW_ACE_Control)	32	RW	0000_0000h
1028h	(AXI4_TxRx_AWAR_ACE_Control)	32	RW	0000_0000h
1040h	(AXI_LPI_Entry_Interval)	32	RW	0000_0000h
1050h	(DMA_TBS_CTRL)	32	RW	0000_0000h
1080h	(DMA_Safety_Interrupt_Status)	32	RO	0000_0000h
1100h	(DMA_CH0_Control)	32	RW	0000_0000h
1104h	(DMA_CH0_Tx_Control)	32	RW	0000_0000h
1108h	(DMA_CH0_Rx_Control)	32	RW	0000_0000h
1114h	(DMA_CH0_TxDesc_List_Address)	32	RW	0000_0000h
111Ch	(DMA_CH0_RxDesc_List_Address)	32	RW	0000_0000h
1120h	(DMA_CH0_TxDesc_Tail_Pointer)	32	RW	0000_0000h
1128h	(DMA_CH0_RxDesc_Tail_Pointer)	32	RW	0000_0000h
112Ch	(DMA_CH0_TxDesc_Ring_Length)	32	RW	0000_0000h
1130h	(DMA_CH0_RxDesc_Ring_Length)	32	RW	0000_0000h
1134h	(DMA_CH0_Interrupt_Enable)	32	RW	0000_0000h

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Offset	Register	Width (In bits)	Access	Reset value
1138h	(DMA_CH0_Rx_Interrupt_Watchdog_Timer)	32	RW	0000_0000h
113Ch	(DMA_CH0_Slot_Function_Control_Status)	32	RW	0000_07C0h
1144h	(DMA_CH0_Current_App_TxDesc)	32	RO	0000_0000h
114Ch	(DMA_CH0_Current_App_RxDesc)	32	RO	0000_0000h
1154h	(DMA_CH0_Current_App_TxBuffer)	32	RO	0000_0000h
115Ch	(DMA_CH0_Current_App_RxBuffer)	32	RO	0000_0000h
1160h	(DMA_CH0_Status)	32	RW	0000_0000h
1164h	(DMA_CH0_Miss_Frame_Cnt)	32	RO	0000_0000h
1168h	(DMA_CH0_RXP_Accept_Cnt)	32	RO	0000_0000h
116Ch	(DMA_CH0_RX_ERI_Cnt)	32	RO	0000_0000h
1180h	(DMA_CH1_Control)	32	RW	0000_0000h
1184h	(DMA_CH1_Tx_Control)	32	RW	0000_0000h
1188h	(DMA_CH1_Rx_Control)	32	RW	0000_0000h
1194h	(DMA_CH1_TxDesc_List_Address)	32	RW	0000_0000h
119Ch	(DMA_CH1_RxDesc_List_Address)	32	RW	0000_0000h
11A0h	(DMA_CH1_TxDesc_Tail_Pointer)	32	RW	0000_0000h
11A8h	(DMA_CH1_RxDesc_Tail_Pointer)	32	RW	0000_0000h
11ACh	(DMA_CH1_TxDesc_Ring_Length)	32	RW	0000_0000h
11B0h	(DMA_CH1_RxDesc_Ring_Length)	32	RW	0000_0000h
11B4h	(DMA_CH1_Interrupt_Enable)	32	RW	0000_0000h
11B8h	(DMA_CH1_Rx_Interrupt_Watchdog_Timer)	32	RW	0000_0000h
11BCh	(DMA_CH1_Slot_Function_Control_Status)	32	RW	0000_07C0h
11C4h	(DMA_CH1_Current_App_TxDesc)	32	RO	0000_0000h
11CCh	(DMA_CH1_Current_App_RxDesc)	32	RO	0000_0000h
11D4h	(DMA_CH1_Current_App_TxBuffer)	32	RO	0000_0000h
11DCh	(DMA_CH1_Current_App_RxBuffer)	32	RO	0000_0000h
11E0h	(DMA_CH1_Status)	32	RW	0000_0000h
11E4h	(DMA_CH1_Miss_Frame_Cnt)	32	RO	0000_0000h
11E8h	(DMA_CH1_RXP_Accept_Cnt)	32	RO	0000_0000h
11ECh	(DMA_CH1_RX_ERI_Cnt)	32	RO	0000_0000h
1200h	(DMA_CH2_Control)	32	RW	0000_0000h

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Offset	Register	Width (In bits)	Access	Reset value
1204h	(DMA_CH2_Tx_Control)	32	RW	0000_0000h
1208h	(DMA_CH2_Rx_Control)	32	RW	0000_0000h
1214h	(DMA_CH2_TxDesc_List_Address)	32	RW	0000_0000h
121Ch	(DMA_CH2_RxDesc_List_Address)	32	RW	0000_0000h
1220h	(DMA_CH2_TxDesc_Tail_Pointer)	32	RW	0000_0000h
1228h	(DMA_CH2_RxDesc_Tail_Pointer)	32	RW	0000_0000h
122Ch	(DMA_CH2_TxDesc_Ring_Length)	32	RW	0000_0000h
1230h	(DMA_CH2_RxDesc_Ring_Length)	32	RW	0000_0000h
1234h	(DMA_CH2_Interrupt_Enable)	32	RW	0000_0000h
1238h	(DMA_CH2_Rx_Interrupt_Watchdog_Timer)	32	RW	0000_0000h
123Ch	(DMA_CH2_Slot_Function_Control_Status)	32	RW	0000_07C0h
1244h	(DMA_CH2_Current_App_TxDesc)	32	RO	0000_0000h
124Ch	(DMA_CH2_Current_App_RxDesc)	32	RO	0000_0000h
1254h	(DMA_CH2_Current_App_TxBuffer)	32	RO	0000_0000h
125Ch	(DMA_CH2_Current_App_RxBuffer)	32	RO	0000_0000h
1260h	(DMA_CH2_Status)	32	RW	0000_0000h
1264h	(DMA_CH2_Miss_Frame_Cnt)	32	RO	0000_0000h
1268h	(DMA_CH2_RXP_Accept_Cnt)	32	RO	0000_0000h
126Ch	(DMA_CH2_RX_ERI_Cnt)	32	RO	0000_0000h
1280h	(DMA_CH3_Control)	32	RW	0000_0000h
1284h	(DMA_CH3_Tx_Control)	32	RW	0000_0000h
1288h	(DMA_CH3_Rx_Control)	32	RW	0000_0000h
1294h	(DMA_CH3_TxDesc_List_Address)	32	RW	0000_0000h
129Ch	(DMA_CH3_RxDesc_List_Address)	32	RW	0000_0000h
12A0h	(DMA_CH3_TxDesc_Tail_Pointer)	32	RW	0000_0000h
12A8h	(DMA_CH3_RxDesc_Tail_Pointer)	32	RW	0000_0000h
12ACh	(DMA_CH3_TxDesc_Ring_Length)	32	RW	0000_0000h
12B0h	(DMA_CH3_RxDesc_Ring_Length)	32	RW	0000_0000h
12B4h	(DMA_CH3_Interrupt_Enable)	32	RW	0000_0000h
12B8h	(DMA_CH3_Rx_Interrupt_Watchdog_Timer)	32	RW	0000_0000h
12BCh	(DMA_CH3_Slot_Function_Control_Status)	32	RW	0000_07C0h

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Offset	Register	Width (In bits)	Access	Reset value
12C4h	(DMA_CH3_Current_App_TxDesc)	32	RO	0000_0000h
12CCh	(DMA_CH3_Current_App_RxDesc)	32	RO	0000_0000h
12D4h	(DMA_CH3_Current_App_TxBuffer)	32	RO	0000_0000h
12DCh	(DMA_CH3_Current_App_RxBuffer)	32	RO	0000_0000h
12E0h	(DMA_CH3_Status)	32	RW	0000_0000h
12E4h	(DMA_CH3_Miss_Frame_Cnt)	32	RO	0000_0000h
12E8h	(DMA_CH3_RXP_Accept_Cnt)	32	RO	0000_0000h
12ECh	(DMA_CH3_RX_ERI_Cnt)	32	RO	0000_0000h
1300h	(DMA_CH4_Control)	32	RW	0000_0000h
1304h	(DMA_CH4_Tx_Control)	32	RW	0000_0000h
1308h	(DMA_CH4_Rx_Control)	32	RW	0000_0000h
1314h	(DMA_CH4_TxDesc_List_Address)	32	RW	0000_0000h
131Ch	(DMA_CH4_RxDesc_List_Address)	32	RW	0000_0000h
1320h	(DMA_CH4_TxDesc_Tail_Pointer)	32	RW	0000_0000h
1328h	(DMA_CH4_RxDesc_Tail_Pointer)	32	RW	0000_0000h
132Ch	(DMA_CH4_TxDesc_Ring_Length)	32	RW	0000_0000h
1330h	(DMA_CH4_RxDesc_Ring_Length)	32	RW	0000_0000h
1334h	(DMA_CH4_Interrupt_Enable)	32	RW	0000_0000h
1338h	(DMA_CH4_Rx_Interrupt_Watchdog_Timer)	32	RW	0000_0000h
133Ch	(DMA_CH4_Slot_Function_Control_Status)	32	RW	0000_07C0h
1344h	(DMA_CH4_Current_App_TxDesc)	32	RO	0000_0000h
134Ch	(DMA_CH4_Current_App_RxDesc)	32	RO	0000_0000h
1354h	(DMA_CH4_Current_App_TxBuffer)	32	RO	0000_0000h
135Ch	(DMA_CH4_Current_App_RxBuffer)	32	RO	0000_0000h
1360h	(DMA_CH4_Status)	32	RW	0000_0000h
1364h	(DMA_CH4_Miss_Frame_Cnt)	32	RO	0000_0000h
1368h	(DMA_CH4_RXP_Accept_Cnt)	32	RO	0000_0000h
136Ch	(DMA_CH4_RX_ERI_Cnt)	32	RO	0000_0000h

1.4.2 (MAC_Configuration)

Offset

Register	Offset
MAC_Configuration	0h

Function

The MAC Configuration Register establishes the operating mode of the MAC.

Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16											
R	ARPE		SARC				IPC		IPG			GPSL		S2KP		CST		ACS		WD		BE		JD		JE	
W	N											CE															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0											

Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0												
R	PS		FES		DM		LM		ECRS		DO		DCRS		DR		Reserv		BL		DC		PRELEN		TE		RE	
W									FD																			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0												

Fields

Field	Function
31 ARPEN	ARP Offload Enable When this bit is set, the MAC can recognize an incoming ARP request packet and schedules the ARP packet for transmission. It forwards the ARP packet to the application and also indicate the events in the RxStatus. When this bit is reset, the MAC receiver does not recognize any ARP packet and indicates them as Type frame in the RxStatus. This bit is available only when the Enable IPv4 ARP Offload is selected. 0b - ARP Offload is disabled 1b - ARP Offload is enabled
30-28 SARC	Source Address Insertion or Replacement Control This field controls the source address insertion or replacement for all transmitted packets. Bit 30 specifies which MAC Address register (0 or 1) is used for source address insertion or replacement based on the values of Bits[29:28]: 2'b0x: - The mti_sa_ctrl_i and ati_sa_ctrl_i input signals control the SA field generation. 2'b10:

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Field	Function
	- If Bit 30 is set to 0, the MAC inserts the content of the MAC Address 0 registers in the SA field of all transmitted packets. - If Bit 30 is set to 1 and the Enable MAC Address Register 1 option is selected while configuring the core, the MAC inserts the content of the MAC Address 1 registers in the SA field of all transmitted packets. 2'b11: - If Bit 30 is set to 0, the MAC replaces the content of the MAC Address 0 registers in the SA field of all transmitted packets. - If Bit 30 is set to 1 and the MAC Address Register 1 is enabled, the MAC replaces the content of the MAC Address 1 registers in the SA field of all transmitted packets. Note: - Changes to this field take effect only on the start of a packet. If you write to this register field when a packet is being transmitted, only the subsequent packet can use the updated value, that is, the current packet does not use the updated value. 000b - mti_sa_ctrl_i and ati_sa_ctrl_i input signals control the SA field generation 010b - Contents of MAC Addr-0 inserted in SA field 011b - Contents of MAC Addr-0 replaces SA field 110b - Contents of MAC Addr-1 inserted in SA field 111b - Contents of MAC Addr-1 replaces SA field
27 IPC	Checksum Offload When set, this bit enables the IPv4 header checksum checking and IPv4 or IPv6 TCP, UDP, or ICMP payload checksum checking. When this bit is reset, the COE function in the receiver is disabled. The Layer 3 and Layer 4 Packet Filter and Enable Split Header features automatically selects the IPC Full Checksum Offload Engine on the Receive side. When any of these features are enabled, you must set the IPC bit. 0b - IP header/payload checksum checking is disabled 1b - IP header/payload checksum checking is enabled
26-24 IPG	Inter-Packet Gap These bits control the minimum IPG between packets during transmission. This range of minimum IPG is valid in full-duplex mode. In the half-duplex mode, the minimum IPG can be configured only for 64-bit times (IPG = 100). Lower values are not considered. When a JAM pattern is being transmitted because of backpressure activation, the MAC does not consider the minimum IPG. The above function (IPG less than 96 bit times) is valid only when EIPGEN bit in MAC_Ext_Configuration register is reset. When EIPGEN is set, then the minimum IPG (greater than 96 bit times) is controlled as per the description given in EIPG field in MAC_Ext_Configuration register. 000b - 96 bit times IPG

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Field	Function
	001b - 88 bit times IPG 010b - 80 bit times IPG 011b - 72 bit times IPG 100b - 64 bit times IPG 101b - 56 bit times IPG 110b - 48 bit times IPG 111b - 40 bit times IPG
23 GPSLCE	Giant Packet Size Limit Control Enable When this bit is set, the MAC considers the value in GPSL field in MAC_Ext_Configuration register to declare a received packet as Giant packet. This field must be programmed to more than 1,518 bytes. Otherwise, the MAC considers 1,518 bytes as giant packet limit. When this bit is reset, the MAC considers a received packet as Giant packet when its size is greater than 1,518 bytes (1522 bytes for tagged packet). The watchdog timeout limit, Jumbo Packet Enable and 2K Packet Enable have higher precedence over this bit, that is the MAC considers a received packet as Giant packet when its size is greater than 9,018 bytes (9,022 bytes for tagged packet) with Jumbo Packet Enabled and greater than 2,000 bytes with 2K Packet Enabled. The watchdog timeout, if enabled, terminates the received packet when watchdog limit is reached. Therefore, the programmed giant packet limit should be less than the watchdog limit to get the giant packet status. 0b - Giant Packet Size Limit Control is disabled 1b - Giant Packet Size Limit Control is enabled
22 S2KP	IEEE 802.3as Support for 2K Packets When this bit is set, the MAC considers all packets with up to 2,000 bytes length as normal packets. When the JE bit is not set, the MAC considers all received packets of size more than 2K bytes as Giant packets. When this bit is reset and the JE bit is not set, the MAC considers all received packets of size more than 1,518 bytes (1,522 bytes for tagged) as giant packets. For more information about how the setting of this bit and the JE bit impact the Giant packet status, see the Table, Giant Packet Status based on S2KP and JE Bits. Note: When the JE bit is set, setting this bit has no effect on the giant packet status. 0b - Support upto 2K packet is disabled 1b - Support upto 2K packet is Enabled
21 CST	CRC stripping for Type packets When this bit is set, the last four bytes (FCS) of all packets of Ether type (type field greater than 1,536) are stripped and dropped before forwarding the packet to the application. Note: For information about how the settings of the ACS bit and this bit impact the packet length, see the Table, Packet Length based on the CST and ACS Bits. 0b - CRC stripping for Type packets is disabled

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Field	Function
	1b - CRC stripping for Type packets is enabled
20 ACS	Automatic Pad or CRC Stripping When this bit is set, the MAC strips the Pad or FCS field on the incoming packets only if the value of the length field is less than 1,536 bytes. All received packets with length field greater than or equal to 1,536 bytes are passed to the application without stripping the Pad or FCS field. When this bit is reset, the MAC passes all incoming packets to the application, without any modification. Note: For information about how the settings of CST bit and this bit impact the packet length, see the Table, Packet Length based on the CST and ACS Bit . 0b - Automatic Pad or CRC Stripping is disabled 1b - Automatic Pad or CRC Stripping is enabled
19 WD	Watchdog Disable When this bit is set, the MAC disables the watchdog timer on the receiver. The MAC can receive packets of up to 16,383 bytes. When this bit is reset, the MAC does not allow more than 2,048 bytes (10,240 if JE is set high) of the packet being received. The MAC cuts off any bytes received after 2,048 bytes. 0b - Watchdog is enabled 1b - Watchdog is disabled
18 BE	Packet Burst Enable When this bit is set, the MAC allows packet bursting during transmission in the GMII half-duplex mode. 0b - Packet Burst is disabled 1b - Packet Burst is enabled
17 JD	Jabber Disable When this bit is set, the MAC disables the jabber timer on the transmitter. The MAC can transfer packets of up to 16,383 bytes. When this bit is reset, if the application sends more than 2,048 bytes of data (10,240 if JE is set high) during transmission, the MAC does not send rest of the bytes in that packet. 0b - Jabber is enabled 1b - Jabber is disabled
16 JE	Jumbo Packet Enable When this bit is set, the MAC allows jumbo packets of 9,018 bytes (9,022 bytes for VLAN tagged packets) without reporting a giant packet error in the Rx packet status. 0b - Jumbo packet is disabled 1b - Jumbo packet is enabled
15	Port Select

Table continues on the next page...

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Field	Function
PS	This bit selects the Ethernet line speed. This bit, along with Bit 14, selects the exact line speed. In the 10/100 Mbps-only (always 1) or 1000 Mbps-only (always 0) configurations, this bit is read-only (RO) with appropriate value. In default 10/100/1000 Mbps configurations, this bit is read-write (R/W). The mac_speed_o[1] signal reflects the value of this bit. 0b - For 1000 or 2500 Mbps operations 1b - For 10 or 100 Mbps operations
14 FES	Speed This bit selects the speed mode. The mac_speed_o[0] signal reflects the value of this bit. 0b - 10 Mbps when PS bit is 1 and 1 Gbps when PS bit is 0 1b - 100 Mbps when PS bit is 1 and 2.5 Gbps when PS bit is 0
13 DM	Duplex Mode When this bit is set, the MAC operates in the full-duplex mode in which it can transmit and receive simultaneously. This bit is RO with default value of 1'b1 in the full-duplex-only configurations. 0b - Half-duplex mode 1b - Full-duplex mode
12 LM	Loopback Mode When this bit is set, the MAC operates in the loopback mode at GMII or MII. The (G)MII Rx clock input (clk_rx_i) is required for the loopback to work properly. This is because the Tx clock is not internally looped back. 0b - Loopback is disabled 1b - Loopback is enabled
11 ECRSFD	Enable Carrier Sense Before Transmission in Full-Duplex Mode When this bit is set, the MAC transmitter checks the CRS signal before packet transmission in the full-duplex mode. The MAC starts the transmission only when the CRS signal is low. When this bit is reset, the MAC transmitter ignores the status of the CRS signal. 0b - ECRSFD is disabled 1b - ECRSFD is enabled
10 DO	Disable Receive Own When this bit is set, the MAC disables the reception of packets when the gmii_txen_o is asserted in the half-duplex mode. When this bit is reset, the MAC receives all packets given by the PHY. This bit is not applicable in the full-duplex mode. 0b - Enable Receive Own 1b - Disable Receive Own

Table continues on the next page...

Table continued from the previous page...

Field	Function
9 DCRS	Disable Carrier Sense During Transmission When this bit is set, the MAC transmitter ignores the (G)MII CRS signal during packet transmission in the half-duplex mode. As a result, no errors are generated because of Loss of Carrier or No Carrier during transmission. When this bit is reset, the MAC transmitter generates errors because of Carrier Sense. The MAC can even abort the transmission. 0b - Enable Carrier Sense During Transmission 1b - Disable Carrier Sense During Transmission
8 DR	Disable Retry When this bit is set, the MAC attempts only one transmission. When a collision occurs on the GMII or MII interface, the MAC ignores the current packet transmission and reports a Packet Abort with excessive collision error in the Tx packet status. When this bit is reset, the MAC retries based on the settings of the BL field. This bit is applicable only in the half-duplex mode. 0b - Enable Retry 1b - Disable Retry
7 —	Reserved.
6-5 BL	Back-Off Limit The back-off limit determines the random integer number (r) of slot time delays (4,096 bit times for 1000/2500 Mbps; 512 bit times for 10/100 Mbps) for which the MAC waits before rescheduling a transmission attempt during retries after a collision. n = retransmission attempt. The random integer r takes the value in the range $0 \leq r < 2^k$ This bit is applicable only in the half-duplex mode. 00b - $k = \min(n, 10)$ 01b - $k = \min(n, 8)$ 10b - $k = \min(n, 4)$ 11b - $k = \min(n, 1)$
4 DC	Deferral Check When this bit is set, the deferral check function is enabled in the MAC. The MAC issues a Packet Abort status, along with the excessive deferral error bit set in the Tx packet status, when the Tx state machine is deferred for more than 24,288 bit times in 10 or 100 Mbps mode. If the MAC is configured for 1000/2500 Mbps operation, the threshold for deferral is 155,680 bits times. Deferral begins when the transmitter is ready to transmit, but it is prevented because of an active carrier sense signal (CRS) on GMII or MII.

Table continues on the next page...

Table continued from the previous page...

Field	Function
	The defer time is not cumulative. For example, if the transmitter defers for 10,000 bit times because the CRS signal is active and the CRS signal becomes inactive, the transmitter transmits and collision happens. Because of collision, the transmitter needs to back off and then defer again after back off completion. In such a scenario, the deferral timer is reset to 0, and it is restarted. When this bit is reset, the deferral check function is disabled and the MAC defers until the CRS signal goes inactive. This bit is applicable only in the half-duplex mode. 0b - Deferral check function is disabled 1b - Deferral check function is enabled
3-2 PRELEN	Preamble Length for Transmit packets These bits control the number of preamble bytes that are added to the beginning of every Tx packet. The preamble reduction occurs only when the MAC is operating in the full-duplex mode. 00b - 7 bytes of preamble 01b - 5 bytes of preamble 10b - 3 bytes of preamble 11b - Reserved
1 TE	Transmitter Enable When this bit is set, the Tx state machine of the MAC is enabled for transmission on the GMII or MII interface. When this bit is reset, the MAC Tx state machine is disabled after it completes the transmission of the current packet. The Tx state machine does not transmit any more packets. 0b - Transmitter is disabled 1b - Transmitter is enabled
0 RE	Receiver Enable When this bit is set, the Rx state machine of the MAC is enabled for receiving packets from the GMII or MII interface. When this bit is reset, the MAC Rx state machine is disabled after it completes the reception of the current packet. The Rx state machine does not receive any more packets from the GMII or MII interface. 0b - Receiver is disabled 1b - Receiver is enabled

1.4.3 (MAC_Ext_Configuration)

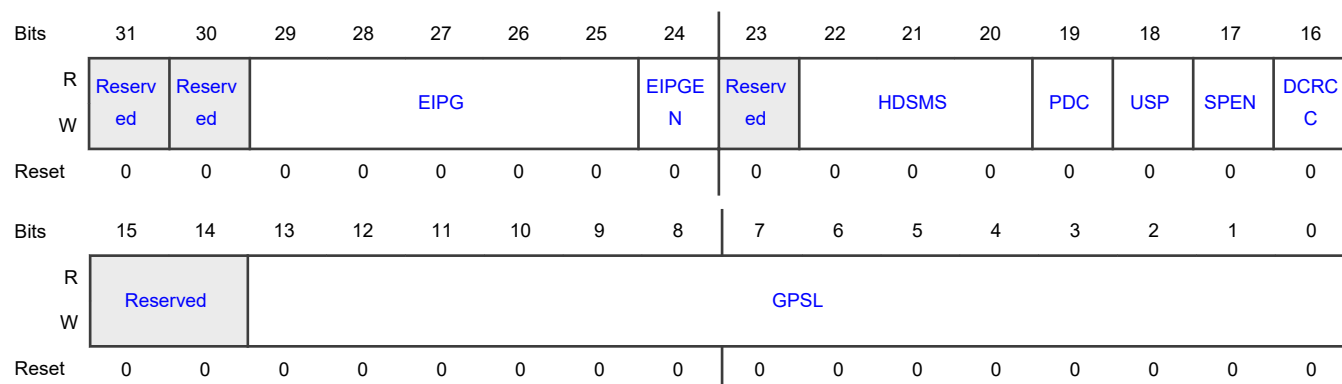
Offset

Register	Offset
MAC_Ext_Configuration	4h

Function

The MAC Extended Configuration Register establishes the operating mode of the MAC.

Diagram



Fields

Field	Function
31 —	Reserved.
30 —	Reserved.
29-25 EIPG	Extended Inter-Packet Gap The value in this field is applicable when the EIPGEN bit is set. This field (as Most Significant bits), along with IPG field in MAC_Configuration register, gives the minimum IPG greater than 96 bit times in steps of 8 bit times: {EIPG, IPG} 8'h00 - 104 bit times 8'h01 - 112 bit times 8'h02 - 120 bit times ----- 8'hFF - 2144 bit times
24 EIPGEN	Extended Inter-Packet Gap Enable When this bit is set, the MAC interprets EIPG field and IPG field in MAC_Configuration register together as minimum IPG greater than 96 bit times in steps of 8 bit times. When this bit is reset, the MAC ignores EIPG field and interprets IPG field in MAC_Configuration register as minimum IPG less than or equal to 96 bit times in steps of 8 bit times. Note: The extended Inter-Packet Gap feature must be enabled when operating in Full-Duplex mode only. There might be undesirable effects on back-pressure function and frame transmission if it is enabled in Half-Duplex mode.

Table continues on the next page...

Table continued from the previous page...

Field	Function
	0b - Extended Inter-Packet Gap is disabled 1b - Extended Inter-Packet Gap is enabled
23 —	Reserved.
22-20 HDSMS	Maximum Size for Splitting the Header Data These bits indicate the maximum header size allowed for splitting the header data in the received packet. 000b - Maximum Size for Splitting the Header Data is 64 bytes 001b - Maximum Size for Splitting the Header Data is 128 bytes 010b - Maximum Size for Splitting the Header Data is 256 bytes 011b - Maximum Size for Splitting the Header Data is 512 bytes 100b - Maximum Size for Splitting the Header Data is 1024 bytes 101b - Reserved
19 PDC	Packet Duplication Control When this bit is set, the received packet with Multicast/Broadcast Destination address is routed to multiple Receive DMA Channels. The Receive DMA Channels is identified by the DCS field of MAC_Address(#i)_High register corresponding to the MAC Address register that matches the Multicast/Broadcast Destination address in the received packet. The DCS field is interpreted to be a one-hot value, each bit corresponding to the Receive DMA Channel. When this bit is reset, the received packet is routed to single Receive DMA Channel. The Receive DMA Channel is identified by the DCS field of MAC_Address(#i)_High register corresponding to the MAC Address register that matches the Destination address in the received packet. The DCS field is interpreted as a binary value. 0b - Packet Duplication Control is disabled 1b - Packet Duplication Control is enabled
18 USP	Unicast Slow Protocol Packet Detect When this bit is set, the MAC detects the Slow Protocol packets with unicast address of the station specified in the MAC_Address0_High and MAC_Address0_Low registers. The MAC also detects the Slow Protocol packets with the Slow Protocols multicast address (01-80-C2-00-00-02). When this bit is reset, the MAC detects only Slow Protocol packets with the Slow Protocol multicast address specified in the IEEE 802.3-2015, Section 5. 0b - Unicast Slow Protocol Packet Detection is disabled 1b - Unicast Slow Protocol Packet Detection is enabled
17 SPEN	Slow Protocol Detection Enable When this bit is set, MAC processes the Slow Protocol packets (Ether Type 0x8809) and provides the Slow Protocol Sub-Type and Code fields in Rx status.

Table continues on the next page...

Table continued from the previous page...

Field	Function
	When this bit is reset, the MAC forwards all error-free Slow Protocol packets to the application. The MAC considers such packets as normal Type packets. 0b - Slow Protocol Detection is disabled 1b - Slow Protocol Detection is enabled
16 DCRCC	Disable CRC Checking for Received Packets When this bit is set, the MAC receiver does not check the CRC field in the received packets. When this bit is reset, the MAC receiver always checks the CRC field in the received packets. 0b - CRC Checking is enabled 1b - CRC Checking is disabled
15-14 —	Reserved.
13-0 GPSL	Giant Packet Size Limit If the received packet size is greater than the value programmed in this field in units of bytes, the MAC declares the received packet as Giant packet. The value programmed in this field must be greater than or equal to 1,518 bytes. Any other programmed value is considered as 1,518 bytes. For VLAN tagged packets, the MAC adds 4 bytes to the programmed value. When the Enable Double VLAN Processing option is selected, the MAC adds 8 bytes to the programmed value for double VLAN tagged packets. The value in this field is applicable when the GPSLCE bit is set in MAC_Configuration register.

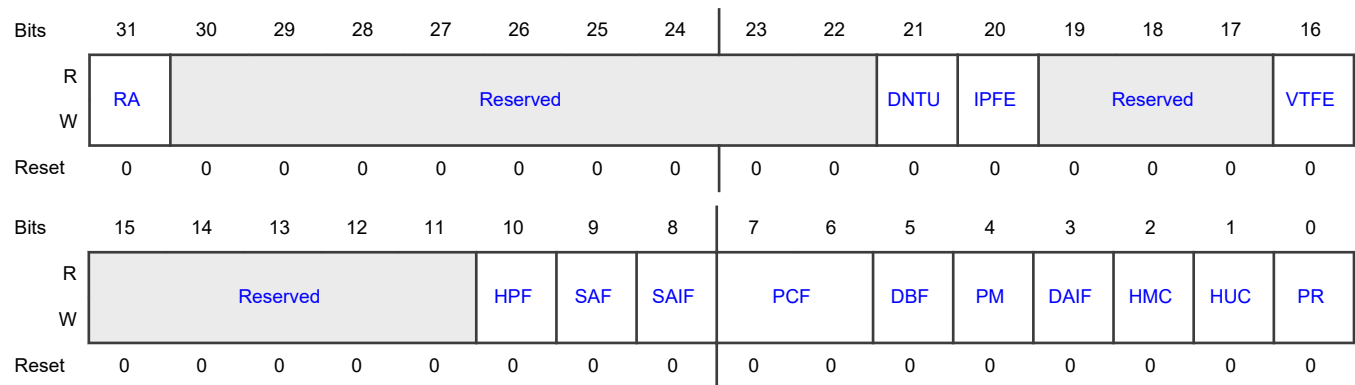
1.4.4 (MAC_Packet_Filter)

Offset

Register	Offset
MAC_Packet_Filter	8h

Function

The MAC Packet Filter register contains the filter controls for receiving packets. Some of the controls from this register go to the address check block of the MAC which performs the first level of address filtering. The second level of filtering is performed on the incoming packet based on other controls such as Pass Bad Packets and Pass Control Packets.

Diagram**Fields**

Field	Function
31 RA	Receive All When this bit is set, the MAC Receiver module passes all received packets to the application, irrespective of whether they pass the address filter or not. The result of the SA or DA filtering is updated (pass or fail) in the corresponding bit in the Rx Status Word. When this bit is reset, the Receiver module passes only those packets to the application that pass the SA or DA address filter. 0b - Receive All is disabled 1b - Receive All is enabled
30-22 —	Reserved.
21 DNTU	Drop Non-TCP/UDP over IP Packets When this bit is set, the MAC drops the non-TCP or UDP over IP packets. The MAC forward only those packets that are processed by the Layer 4 filter. When this bit is reset, the MAC forwards all non-TCP or UDP over IP packets. 0b - Forward Non-TCP/UDP over IP Packets 1b - Drop Non-TCP/UDP over IP Packets
20 IPFE	Layer 3 and Layer 4 Filter Enable When this bit is set, the MAC drops packets that do not match the enabled Layer 3 and Layer 4 filters. If Layer 3 or Layer 4 filters are not enabled for matching, this bit does not have any effect. When this bit is reset, the MAC forwards all packets irrespective of the match status of the Layer 3 and Layer 4 fields. 0b - Layer 3 and Layer 4 Filters are disabled 1b - Layer 3 and Layer 4 Filters are enabled
19-17	Reserved.

Table continues on the next page...

Table continued from the previous page...

Field	Function
—	
16 VTFE	VLAN Tag Filter Enable When this bit is set, the MAC drops the VLAN tagged packets that do not match the VLAN Tag. When this bit is reset, the MAC forwards all packets irrespective of the match status of the VLAN Tag. 0b - VLAN Tag Filter is disabled 1b - VLAN Tag Filter is enabled
15-11 —	Reserved.
10 HPF	Hash or Perfect Filter When this bit is set, the address filter passes a packet if it matches either the perfect filtering or hash filtering as set by the HMC or HUC bit. When this bit is reset and the HUC or HMC bit is set, the packet is passed only if it matches the Hash filter. 0b - Hash or Perfect Filter is disabled 1b - Hash or Perfect Filter is enabled
9 SAF	Source Address Filter Enable When this bit is set, the MAC compares the SA field of the received packets with the values programmed in the enabled SA registers. If the comparison fails, the MAC drops the packet. When this bit is reset, the MAC forwards the received packet to the application with updated SAF bit of the Rx Status depending on the SA address comparison. Note: According to the IEEE specification, Bit 47 of the SA is reserved. However, in DWC_ether_qos, the MAC compares all 48 bits. The software driver should take this into consideration while programming the MAC address registers for SA. 0b - SA Filtering is disabled 1b - SA Filtering is enabled
8 SAIF	SA Inverse Filtering When this bit is set, the Address Check block operates in the inverse filtering mode for SA address comparison. If the SA of a packet matches the values programmed in the SA registers, it is marked as failing the SA Address filter. When this bit is reset, if the SA of a packet does not match the values programmed in the SA registers, it is marked as failing the SA Address filter. 0b - SA Inverse Filtering is disabled 1b - SA Inverse Filtering is enabled
7-6 PCF	Pass Control Packets These bits control the forwarding of all control packets (including unicast and multicast Pause packets).

Table continues on the next page...

Table continued from the previous page...

Field	Function
	00b - MAC filters all control packets from reaching the application 01b - MAC forwards all control packets except Pause packets to the application even if they fail the Address filter 10b - MAC forwards all control packets to the application even if they fail the Address filter 11b - MAC forwards the control packets that pass the Address filter
5 DBF	Disable Broadcast Packets When this bit is set, the AFM module blocks all incoming broadcast packets. In addition, it overrides all other filter settings. When this bit is reset, the AFM module passes all received broadcast packets. 0b - Enable Broadcast Packets 1b - Disable Broadcast Packets
4 PM	Pass All Multicast When this bit is set, it indicates that all received packets with a multicast destination address (first bit in the destination address field is '1') are passed. When this bit is reset, filtering of multicast packet depends on HMC bit. 0b - Pass All Multicast is disabled 1b - Pass All Multicast is enabled
3 DAIF	DA Inverse Filtering When this bit is set, the Address Check block operates in inverse filtering mode for the DA address comparison for both unicast and multicast packets. When this bit is reset, normal filtering of packets is performed. 0b - DA Inverse Filtering is disabled 1b - DA Inverse Filtering is enabled
2 HMC	Hash Multicast When this bit is set, the MAC performs the destination address filtering of received multicast packets according to the hash table. When this bit is reset, the MAC performs the perfect destination address filtering for multicast packets, that is, it compares the DA field with the values programmed in DA registers. 0b - Hash Multicast is disabled 1b - Hash Multicast is enabled
1 HUC	Hash Unicast When this bit is set, the MAC performs the destination address filtering of unicast packets according to the hash table. When this bit is reset, the MAC performs a perfect destination address filtering for unicast packets, that is, it compares the DA field with the values programmed in DA registers.

Table continues on the next page...

Table continued from the previous page...

Field	Function
	0b - Hash Unicast is disabled 1b - Hash Unicast is enabled
0 PR	Promiscuous Mode When this bit is set, the Address Filtering module passes all incoming packets irrespective of the destination or source address. The SA or DA Filter Fails status bits of the Rx Status Word are always cleared when PR is set. 0b - Promiscuous Mode is disabled 1b - Promiscuous Mode is enabled

1.4.5 (MAC_Watchdog_Timeout)

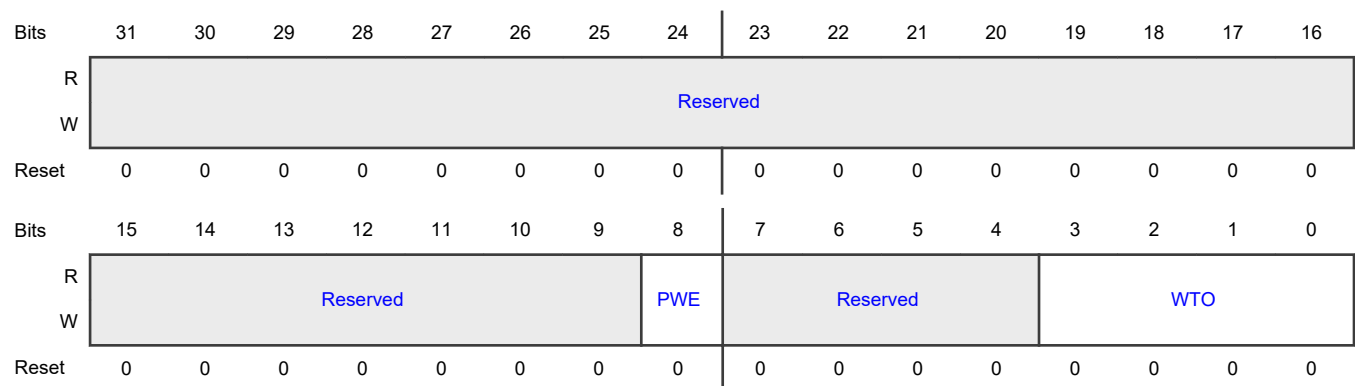
Offset

Register	Offset
MAC_Watchdog_Timeout	Ch

Function

The Watchdog Timeout register controls the watchdog timeout for received packets.

Diagram



Fields

Field	Function
31-9 —	Reserved.

Table continues on the next page...

Table continued from the previous page...

Field	Function
8 PWE	Programmable Watchdog Enable When this bit is set and the WD bit of the MAC_Configuration register is reset, the WTO field is used as watchdog timeout for a received packet. When this bit is cleared, the watchdog timeout for a received packet is controlled by setting of WD and JE bits in MAC_Configuration register. 0b - Programmable Watchdog is disabled 1b - Programmable Watchdog is enabled
7-4 —	Reserved.
3-0 WTO	Watchdog Timeout When the PWE bit is set and the WD bit of the MAC_Configuration register is reset, this field is used as watchdog timeout for a received packet. If the length of a received packet exceeds the value of this field, such packet is terminated and declared as an error packet. Note: When the PWE bit is set, the value in this field should be more than 1,522 (0x05F2). Otherwise, the IEEE 802.3-specified valid tagged packets are declared as error packets and then dropped. 0000b - 2 KB 0001b - 3 KB 0010b - 4 KB 0011b - 5 KB 0100b - 6 KB 0101b - 7 KB 0110b - 8 KB 0111b - 9 KB 1000b - 10 KB 1001b - 11 KB 1010b - 12 KB 1011b - 13 KB 1100b - 14 KB 1101b - 15 KB 1110b - 16383 Bytes 1111b - Reserved

1.4.6 (MAC_Hash_Table_Reg0)

Offset

Register	Offset
MAC_Hash_Table_Reg0	10h

Function

The Hash Table Register 0 contains the first 32 bits of the hash table, when the width of the hash table is 128 or 256 bits. You can specify the width of the hash table by using the Hash Table Size option in coreConsultant.

The Hash table is used for group address filtering. For hash filtering, the content of the destination address in the incoming packet is passed through the CRC logic and the upper six (seven in 128-bit Hash or eight in 256-bit Hash) bits of the CRC register are used to index the content of the Hash table. The most significant bits determines the register to be used (Hash Table Register X), and the least significant five bits determine the bit within the register. For example, a hash value of 6'b100000 (in 64-bit Hash) selects Bit 0 of the Hash Table Register 1, a value of 7b'1110000 (in 128-bit Hash) selects Bit 16 of the Hash Table Register 3 and a value of 8b'10111111 (in 256-bit Hash) selects Bit 31 of the Hash Table Register 5.

The hash value of the destination address is calculated in the following way:

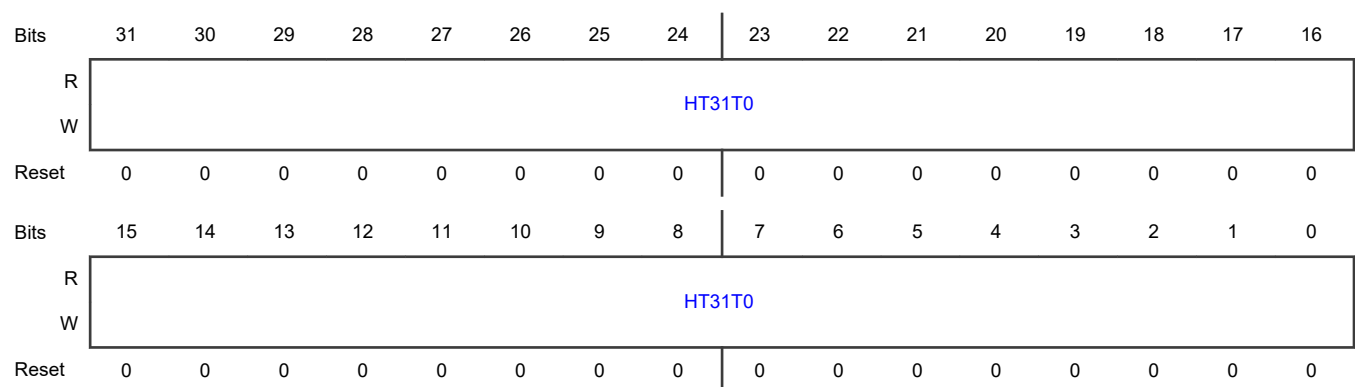
- Calculate the 32-bit CRC for the DA (See IEEE 802.3, Section 3.2.8 for the steps to calculate CRC32).
- Perform bitwise reversal for the value obtained in Step 1.
- Take the upper 6 (or 7 or 8) bits from the value obtained in Step 2.

If the corresponding bit value of the register is 1'b1, the packet is accepted. Otherwise, it is rejected. If the PM bit is set in MAC_Packet_Filter, all multicast packets are accepted regardless of the multicast hash values.

If the Hash Table register is configured to be double-synchronized to the (G)MII clock domain, the synchronization is triggered only when Bits[31:24] (in little-endian mode) or Bits[7:0] (in big-endian mode) of the Hash Table Register X registers are written.

If double-synchronization is enabled, consecutive writes to this register should be performed after at least four clock cycles in the destination clock domain.

Diagram



Fields

Field	Function
31-0	MAC Hash Table First 32 Bits
HT31T0	This field contains the first 32 Bits [31:0] of the Hash table.

1.4.7 (MAC_Hash_Table_Reg1)

Offset

Register	Offset
MAC_Hash_Table_Reg1	14h

Function

The Hash Table Register 1 contains the second 32 bits of the hash table. You can specify the width of the hash table by using the Hash Table Size option in coreConsultant.

The Hash table is used for group address filtering. For hash filtering, the content of the destination address in the incoming packet is passed through the CRC logic and the upper six (seven in 128-bit Hash or eight in 256-bit Hash) bits of the CRC register are used to index the content of the Hash table. The most significant bits determines the register to be used (Hash Table Register X), and the least significant five bits determine the bit within the register. For example, a hash value of 6'b100000 (in 64-bit Hash) selects Bit 0 of the Hash Table Register 1, a value of 7b'1110000 (in 128-bit Hash) selects Bit 16 of the Hash Table Register 3 and a value of 8b'10111111 (in 256-bit Hash) selects Bit 31 of the Hash Table Register 5.

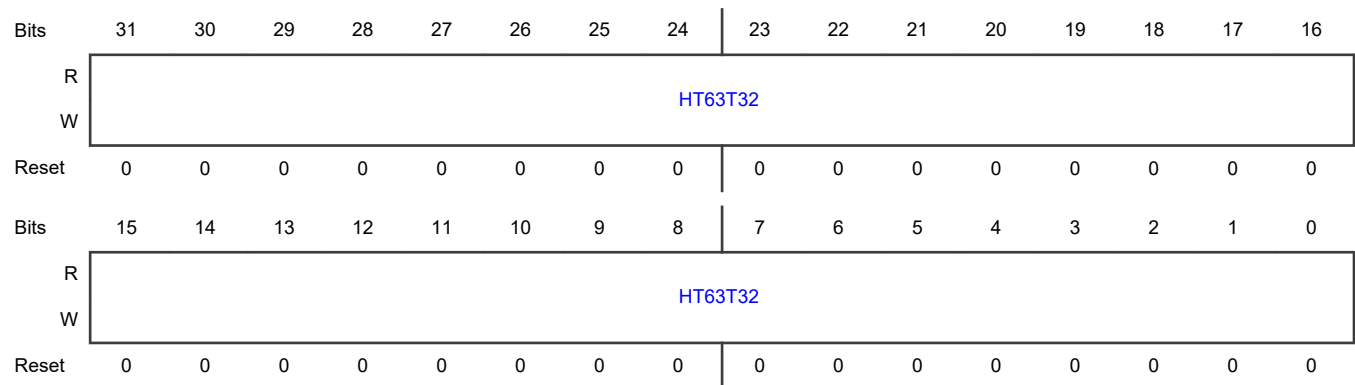
The hash value of the destination address is calculated in the following way:

- Calculate the 32-bit CRC for the DA (See IEEE 802.3, Section 3.2.8 for the steps to calculate CRC32).
- Perform bitwise reversal for the value obtained in Step 1.
- Take the upper 6 (or 7 or 8) bits from the value obtained in Step 2.

If the corresponding bit value of the register is 1'b1, the packet is accepted. Otherwise, it is rejected. If the PM bit is set in MAC_Packet_Filter, all multicast packets are accepted regardless of the multicast hash values.

If the Hash Table register is configured to be double-synchronized to the (G)MII clock domain, the synchronization is triggered only when Bits[31:24] (in little-endian mode) or Bits[7:0] (in big-endian mode) of the Hash Table Register X registers are written.

If double-synchronization is enabled, consecutive writes to this register should be performed after at least four clock cycles in the destination clock domain.

Diagram**Fields**

Field	Function
31-0 HT63T32	MAC Hash Table Second 32 Bits This field contains the second 32 Bits [63:32] of the Hash table.

1.4.8 (MAC_Hash_Table_Reg2)**Offset**

Register	Offset
MAC_Hash_Table_Reg2	18h

Function

The Hash Table Register 2 contains the third 32 bits of the hash table. You can specify the width of the hash table by using the Hash Table Size option in coreConsultant.

The Hash table is used for group address filtering. For hash filtering, the content of the destination address in the incoming packet is passed through the CRC logic and the upper six (seven in 128-bit Hash or eight in 256-bit Hash) bits of the CRC register are used to index the content of the Hash table. The most significant bits determines the register to be used (Hash Table Register X), and the least significant five bits determine the bit within the register. For example, a hash value of 6'b100000 (in 64-bit Hash) selects Bit 0 of the Hash Table Register 1, a value of 7b'1110000 (in 128-bit Hash) selects Bit 16 of the Hash Table Register 3 and a value of 8b'10111111 (in 256-bit Hash) selects Bit 31 of the Hash Table Register 5.

The hash value of the destination address is calculated in the following way:

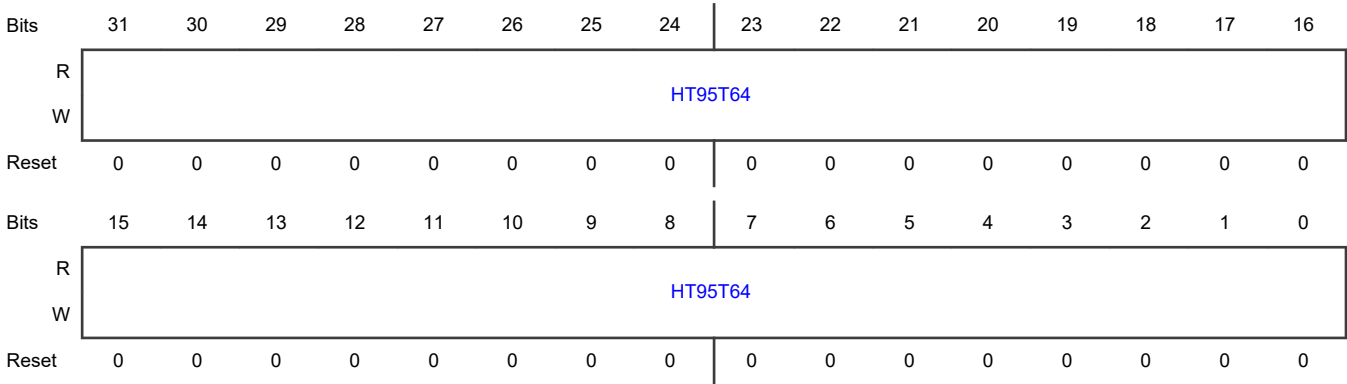
- Calculate the 32-bit CRC for the DA (See IEEE 802.3, Section 3.2.8 for the steps to calculate CRC32).
- Perform bitwise reversal for the value obtained in Step 1.
- Take the upper 6 (or 7 or 8) bits from the value obtained in Step 2.

If the corresponding bit value of the register is 1'b1, the packet is accepted. Otherwise, it is rejected. If the PM bit is set in MAC_Packet_Filter, all multicast packets are accepted regardless of the multicast hash values.

If the Hash Table register is configured to be double-synchronized to the (G)MII clock domain, the synchronization is triggered only when Bits[31:24] (in little-endian mode) or Bits[7:0] (in big-endian mode) of the Hash Table Register X registers are written.

If double-synchronization is enabled, consecutive writes to this register should be performed after at least four clock cycles in the destination clock domain.

Diagram



Fields

Field	Function
31-0	MAC Hash Table Third 32 Bits
HT95T64	This field contains the third 32 Bits [95:64] of the Hash table.

1.4.9 (MAC_Hash_Table_Reg3)

Offset

Register	Offset
MAC_Hash_Table_Reg3	1Ch

Function

The Hash Table Register 3 contains the fourth 32 bits of the hash table. You can specify the width of the hash table by using the Hash Table Size option in coreConsultant.

The Hash table is used for group address filtering. For hash filtering, the content of the destination address in the incoming packet is passed through the CRC logic and the upper six (seven in 128-bit Hash or eight in 256-bit Hash) bits of the CRC register are used to index the content of the Hash table. The most significant bits determines the register to be used (Hash Table Register X), and the least significant five bits determine the bit within the register. For example, a hash value of 6'b100000 (in 64-bit Hash) selects Bit 0 of the Hash Table Register 1, a value of 7b'1110000 (in 128-bit Hash) selects Bit 16 of the Hash Table Register 3 and a value of 8b'10111111 (in 256-bit Hash) selects Bit 31 of the Hash Table Register 5.

The hash value of the destination address is calculated in the following way:

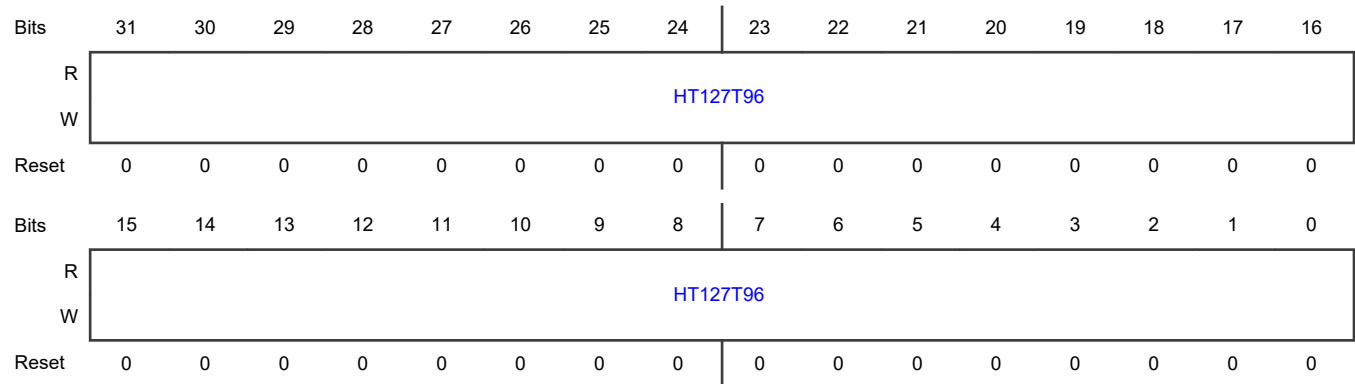
- Calculate the 32-bit CRC for the DA (See IEEE 802.3, Section 3.2.8 for the steps to calculate CRC32).
- Perform bitwise reversal for the value obtained in Step 1.
- Take the upper 6 (or 7 or 8) bits from the value obtained in Step 2.

If the corresponding bit value of the register is 1'b1, the packet is accepted. Otherwise, it is rejected. If the PM bit is set in MAC_Packet_Filter, all multicast packets are accepted regardless of the multicast hash values.

If the Hash Table register is configured to be double-synchronized to the (G)MII clock domain, the synchronization is triggered only when Bits[31:24] (in little-endian mode) or Bits[7:0] (in big-endian mode) of the Hash Table Register X registers are written.

If double-synchronization is enabled, consecutive writes to this register should be performed after at least four clock cycles in the destination clock domain.

Diagram



Fields

Field	Function
31-0	MAC Hash Table Fourth 32 Bits
HT127T96	This field contains the fourth 32 Bits [127:96] of the Hash table.

1.4.10 (MAC_Hash_Table_Reg4)

Offset

Register	Offset
MAC_Hash_Table_Reg4	20h

Function

The Hash Table Register 4 contains the fifth 32 bits of the hash table. You can specify the width of the hash table by using the Hash Table Size option in coreConsultant.

The Hash table is used for group address filtering. For hash filtering, the content of the destination address in the incoming packet is passed through the CRC logic and the upper six (seven in 128-bit Hash or eight in 256-bit Hash) bits of the CRC register are used to index the content of the Hash table. The most significant bits determines the register to be used (Hash Table Register X), and the least significant five bits determine the bit within the register. For example, a hash value of 6'b100000 (in 64-bit Hash) selects Bit 0 of the Hash Table Register 1, a value of 7b'1110000 (in 128-bit Hash) selects Bit 16 of the Hash Table Register 3 and a value of 8b'10111111 (in 256-bit Hash) selects Bit 31 of the Hash Table Register 5.

The hash value of the destination address is calculated in the following way:

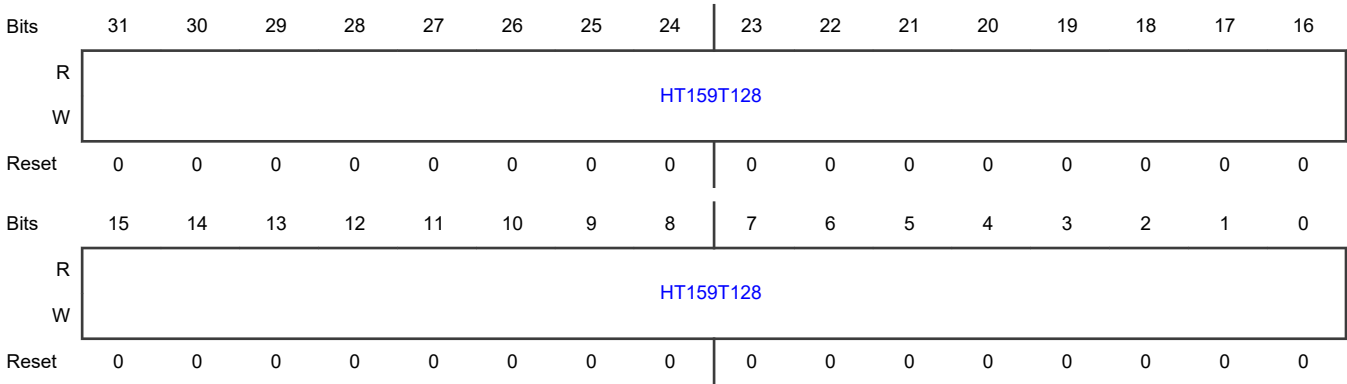
- Calculate the 32-bit CRC for the DA (See IEEE 802.3, Section 3.2.8 for the steps to calculate CRC32).
- Perform bitwise reversal for the value obtained in Step 1.
- Take the upper 6 (or 7 or 8) bits from the value obtained in Step 2.

If the corresponding bit value of the register is 1'b1, the packet is accepted. Otherwise, it is rejected. If the PM bit is set in MAC_Packet_Filter, all multicast packets are accepted regardless of the multicast hash values.

If the Hash Table register is configured to be double-synchronized to the (G)MII clock domain, the synchronization is triggered only when Bits[31:24] (in little-endian mode) or Bits[7:0] (in big-endian mode) of the Hash Table Register X registers are written.

If double-synchronization is enabled, consecutive writes to this register should be performed after at least four clock cycles in the destination clock domain.

Diagram



Fields

Field	Function
31-0	MAC Hash Table Fifth 32 Bits
HT159T128	This field contains the fifth 32 Bits [159:128] of the Hash table.

1.4.11 (MAC_Hash_Table_Reg5)

Offset

Register	Offset
MAC_Hash_Table_Reg5	24h

Function

The Hash Table Register 5 contains the sixth 32 bits of the hash table. You can specify the width of the hash table by using the Hash Table Size option in coreConsultant.

The Hash table is used for group address filtering. For hash filtering, the content of the destination address in the incoming packet is passed through the CRC logic and the upper six (seven in 128-bit Hash or eight in 256-bit Hash) bits of the CRC register are used to index the content of the Hash table. The most significant bits determines the register to be used (Hash Table Register X), and the least significant five bits determine the bit within the register. For example, a hash value of 6'b100000 (in 64-bit Hash) selects Bit 0 of the Hash Table Register 1, a value of 7b'1110000 (in 128-bit Hash) selects Bit 16 of the Hash Table Register 3 and a value of 8b'10111111 (in 256-bit Hash) selects Bit 31 of the Hash Table Register 5.

The hash value of the destination address is calculated in the following way:

- Calculate the 32-bit CRC for the DA (See IEEE 802.3, Section 3.2.8 for the steps to calculate CRC32).
- Perform bitwise reversal for the value obtained in Step 1.

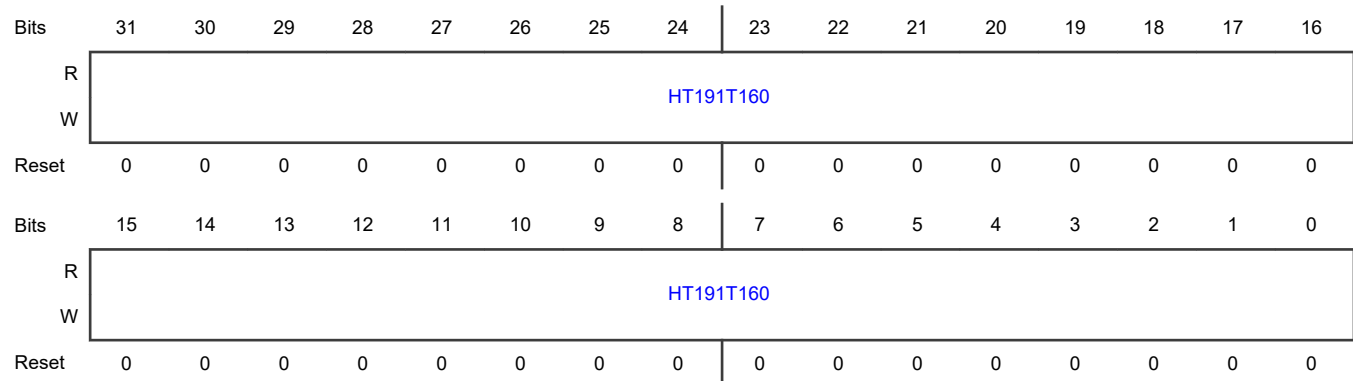
- Take the upper 6 (or 7 or 8) bits from the value obtained in Step 2.

If the corresponding bit value of the register is 1'b1, the packet is accepted. Otherwise, it is rejected. If the PM bit is set in MAC_Packet_Filter, all multicast packets are accepted regardless of the multicast hash values.

If the Hash Table register is configured to be double-synchronized to the (G)MII clock domain, the synchronization is triggered only when Bits[31:24] (in little-endian mode) or Bits[7:0] (in big-endian mode) of the Hash Table Register X registers are written.

If double-synchronization is enabled, consecutive writes to this register should be performed after at least four clock cycles in the destination clock domain.

Diagram



Fields

Field	Function
31-0	MAC Hash Table Sixth 32 Bits
HT191T160	This field contains the sixth 32 Bits [191:160] of the Hash table.

1.4.12 (MAC_Hash_Table_Reg6)

Offset

Register	Offset
MAC_Hash_Table_Reg6	28h

Function

The Hash Table Register 6 contains the seventh 32 bits of the hash table. You can specify the width of the hash table by using the Hash Table Size option in coreConsultant.

The Hash table is used for group address filtering. For hash filtering, the content of the destination address in the incoming packet is passed through the CRC logic and the upper six (seven in 128-bit Hash or eight in 256-bit Hash) bits of the CRC register are used to index the content of the Hash table. The most significant bits determines the register to be used (Hash Table Register X), and the least significant five bits determine the bit within the register. For example, a hash value of 6'b100000 (in 64-bit Hash) selects Bit 0 of the Hash Table Register 1, a value of 7b'1110000 (in 128-bit Hash) selects Bit 16 of the Hash Table Register 3 and a value of 8b'10111111 (in 256-bit Hash) selects Bit 31 of the Hash Table Register 5.

The hash value of the destination address is calculated in the following way:

- Calculate the 32-bit CRC for the DA (See IEEE 802.3, Section 3.2.8 for the steps to calculate CRC32).

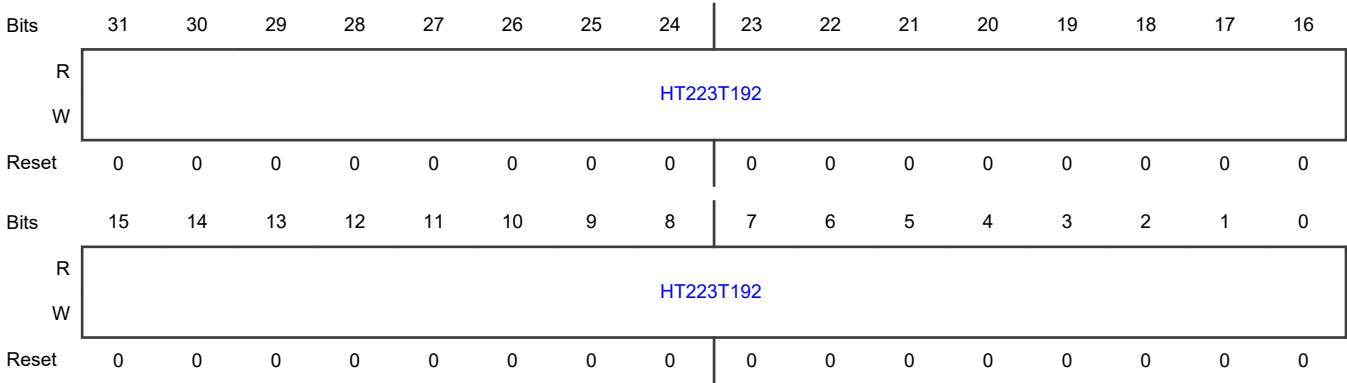
- Perform bitwise reversal for the value obtained in Step 1.
- Take the upper 6 (or 7 or 8) bits from the value obtained in Step 2.

If the corresponding bit value of the register is 1'b1, the packet is accepted. Otherwise, it is rejected. If the PM bit is set in MAC_Packet_Filter, all multicast packets are accepted regardless of the multicast hash values.

If the Hash Table register is configured to be double-synchronized to the (G)MII clock domain, the synchronization is triggered only when Bits[31:24] (in little-endian mode) or Bits[7:0] (in big-endian mode) of the Hash Table Register X registers are written.

If double-synchronization is enabled, consecutive writes to this register should be performed after at least four clock cycles in the destination clock domain.

Diagram



Fields

Field	Function
31-0	MAC Hash Table Seventh 32 Bits
HT223T192	This field contains the seventh 32 Bits [223:192] of the Hash table.

1.4.13 (MAC_Hash_Table_Reg7)

Offset

Register	Offset
MAC_Hash_Table_Reg7	2Ch

Function

The Hash Table Register 7 contains the eighth 32 bits of the hash table. You can specify the width of the hash table by using the Hash Table Size option in coreConsultant.

The Hash table is used for group address filtering. For hash filtering, the content of the destination address in the incoming packet is passed through the CRC logic and the upper six (seven in 128-bit Hash or eight in 256-bit Hash) bits of the CRC register are used to index the content of the Hash table. The most significant bits determines the register to be used (Hash Table Register X), and the least significant five bits determine the bit within the register. For example, a hash value of 6'b100000 (in 64-bit Hash) selects Bit 0 of the Hash Table Register 1, a value of 7b'1110000 (in 128-bit Hash) selects Bit 16 of the Hash Table Register 3 and a value of 8b'10111111 (in 256-bit Hash) selects Bit 31 of the Hash Table Register 5.

The hash value of the destination address is calculated in the following way:

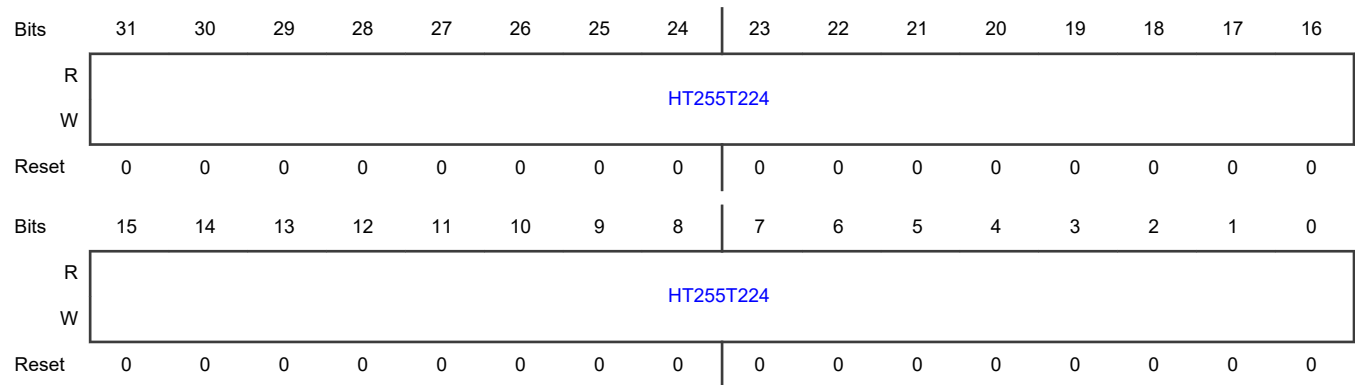
- Calculate the 32-bit CRC for the DA (See IEEE 802.3, Section 3.2.8 for the steps to calculate CRC32).
- Perform bitwise reversal for the value obtained in Step 1.
- Take the upper 6 (or 7 or 8) bits from the value obtained in Step 2.

If the corresponding bit value of the register is 1'b1, the packet is accepted. Otherwise, it is rejected. If the PM bit is set in MAC_Packet_Filter, all multicast packets are accepted regardless of the multicast hash values.

If the Hash Table register is configured to be double-synchronized to the (G)MII clock domain, the synchronization is triggered only when Bits[31:24] (in little-endian mode) or Bits[7:0] (in big-endian mode) of the Hash Table Register X registers are written.

If double-synchronization is enabled, consecutive writes to this register should be performed after at least four clock cycles in the destination clock domain.

Diagram



Fields

Field	Function
31-0	MAC Hash Table Eighth 32 Bits
HT255T224	This field contains the eighth 32 Bits [255:224] of the Hash table.

1.4.14 (MAC_VLAN_Tag_Ctrl)

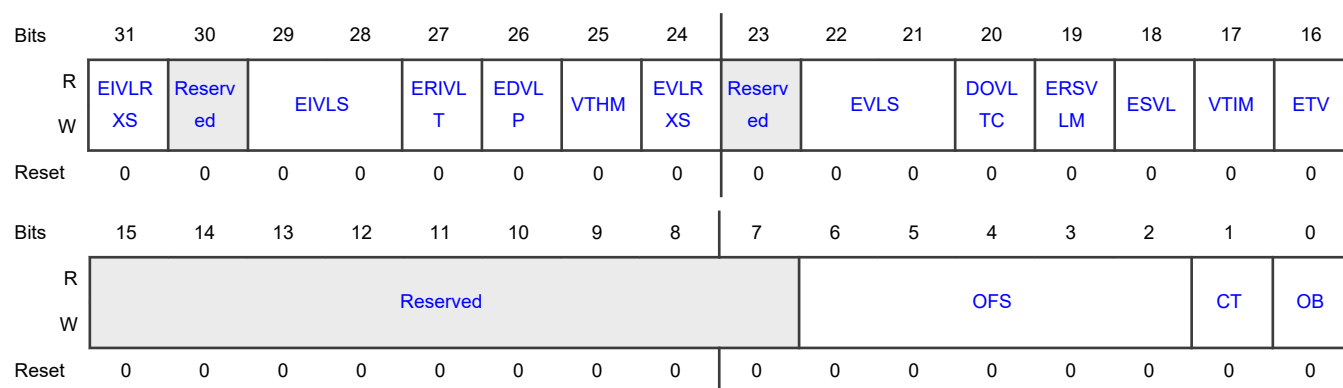
Offset

Register	Offset
MAC_VLAN_Tag_Ctrl	50h

Function

This register is the redefined format of the MAC VLAN Tag Register. It is used for indirect addressing. It contains the address offset, command type and Busy Bit for CSR access of the Per VLAN Tag registers.

Diagram



Fields

Field	Function
31 EIVLRXS	Enable Inner VLAN Tag in Rx Status When this bit is set, the MAC provides the inner VLAN Tag in the Rx status. When this bit is reset, the MAC does not provide the inner VLAN Tag in Rx status. 0b - Inner VLAN Tag in Rx status is disabled 1b - Inner VLAN Tag in Rx status is enabled
30 —	Reserved.
29-28 EIVLS	Enable Inner VLAN Tag Stripping on Receive This field indicates the stripping operation on inner VLAN Tag in received packet. 00b - Do not strip 01b - Strip if VLAN filter passes 10b - Strip if VLAN filter fails 11b - Always strip
27 ERIVLT	Enable Inner VLAN Tag Comparison Enables or disables the inner VLAN tag. When ERIVLT=VTHM=EDVLP=1, the MAC receiver enables the VLAN Hash filtering operation on the inner VLAN tag (if present). When ERIVLT=0 and VTHM=1, the MAC receiver enables the VLAN Hash filtering operation on the outer VLAN tag (if present). ERSVLM and DOVLTC determine which VLAN type is enabled for filtering. 0b - Inner VLAN tag is disabled 1b - Inner VLAN tag is enabled
26 EDVLP	Enable Double VLAN Processing

Table continues on the next page...

Table continued from the previous page...

Field	Function
	When this bit is set, the MAC enables processing of up to two VLAN Tags on Tx and Rx (if present). When this bit is reset, the MAC enables processing of up to one VLAN Tag on Tx and Rx (if present). 0b - Double VLAN Processing is disabled 1b - Double VLAN Processing is enabled
25 VTHM	VLAN Tag Hash Table Match Enable When this bit is set, the most significant four bits of CRC of VLAN Tag are used to index the content of the MAC_VLAN_Hash_Table register. A value of 1 in the VLAN Hash Table register, corresponding to the index, indicates that the packet matched the VLAN hash table. When the ETV bit is set, the CRC of the 12-bit VLAN Identifier (VID) is used for comparison. When the ETV bit is reset, the CRC of the 16-bit VLAN tag is used for comparison. When this bit is reset, the VLAN Hash Match operation is not performed. 0b - VLAN Tag Hash Table Match is disabled 1b - VLAN Tag Hash Table Match is enabled
24 EVLRXS	Enable VLAN Tag in Rx status When this bit is set, MAC provides the outer VLAN Tag in the Rx status. When this bit is reset, the MAC does not provide the outer VLAN Tag in Rx status. 0b - VLAN Tag in Rx status is disabled 1b - VLAN Tag in Rx status is enabled
23 —	Reserved.
22-21 EVLS	Enable VLAN Tag Stripping on Receive This field indicates the stripping operation on the outer VLAN Tag in received packet. 00b - Do not strip 01b - Strip if VLAN filter passes 10b - Strip if VLAN filter fails 11b - Always strip
20 DOVLTC	Disable VLAN Type Check for VLAN Hash Filtering When this bit is set, the MAC VLAN Hash Filter does not check whether the VLAN Tag specified by the ERIVLT bit is of type S-VLAN or C-VLAN. When this bit is reset, the MAC VLAN Hash Filter filters or matches the VLAN Tag specified by the ERIVLT bit only when VLAN Tag type is similar to the one specified by the ERSVLM bit. 0b - VLAN Type Check is enabled 1b - VLAN Type Check is disabled
19	Enable Receive S-VLAN Match for VLAN Hash Filtering

Table continues on the next page...

Table continued from the previous page...

Field	Function
ERSVLM	When this bit is set, the MAC receiver enables VLAN Hash filtering or matching for S-VLAN (Type = 0x88A8) packets. When this bit is reset, the MAC receiver enables VLAN Hash filtering or matching for C-VLAN (Type = 0x8100) packets. The ERIVLT bit determines the VLAN tag position considered for VLAN Hash filtering or matching. 0b - Receive S-VLAN Match is disabled 1b - Receive S-VLAN Match is enabled
18 ESVL	Enable S-VLAN When this bit is set, the MAC transmitter and receiver consider the S-VLAN packets (Type = 0x88A8) as valid VLAN tagged packets. 0b - S-VLAN is disabled 1b - S-VLAN is enabled
17 VTIM	VLAN Tag Inverse Match Enable When this bit is set, this bit enables the VLAN Tag inverse matching. The packets without matching VLAN Tag are marked as matched. When reset, this bit enables the VLAN Tag perfect matching. The packets with matched VLAN Tag are marked as matched. 0b - VLAN Tag Inverse Match is disabled 1b - VLAN Tag Inverse Match is enabled
16 ETV	Enable 12-Bit VLAN Tag Comparison for VLAN Hash Filtering When this bit is set, a 12-bit VLAN identifier is used for VLAN Hash filtering instead of the complete 16-bit VLAN tag. Bits[11:0] of VLAN tag in the received VLAN-tagged packet are used for hash-based VLAN filtering. When this bit is reset, all 16 bits of the 15th and 16th bytes of the received VLAN packet are used for VLAN hash filtering. 0b - 12-Bit VLAN Tag Comparison is disabled 1b - 12-Bit VLAN Tag Comparison is enabled
15-7 —	Reserved.
6-2 OFS	Offset This field holds the address offset of the MAC VLAN Tag Filter Register which the application is trying to access. The width of the field depends on the number of MAC VLAN Tag Registers enabled.
1 CT	Command Type This bit indicates if the current register access is a read or a write. When set, it indicate a read operation. When reset, it indicates a write operation.

Table continues on the next page...

Table continued from the previous page...

Field	Function
	0b - Write operation 1b - Read operation
0 OB	Operation Busy This bit is set along with a read or write command for initiating the indirect access to per VLAN Tag Filter register. This bit is reset when the read or write command to per VLAN Tag Filter indirect access register is complete. The next indirect register access can be initiated only after this bit is reset. During a write operation, the bit is reset only after the data has been written into the Per VLAN Tag register. During a read operation, the data should be read from the MAC_VLAN_Tag_Data register only after this bit is reset. 0b - Operation Busy is disabled 1b - Operation Busy is enabled

1.4.15 (MAC_VLAN_Tag_Data)

Offset

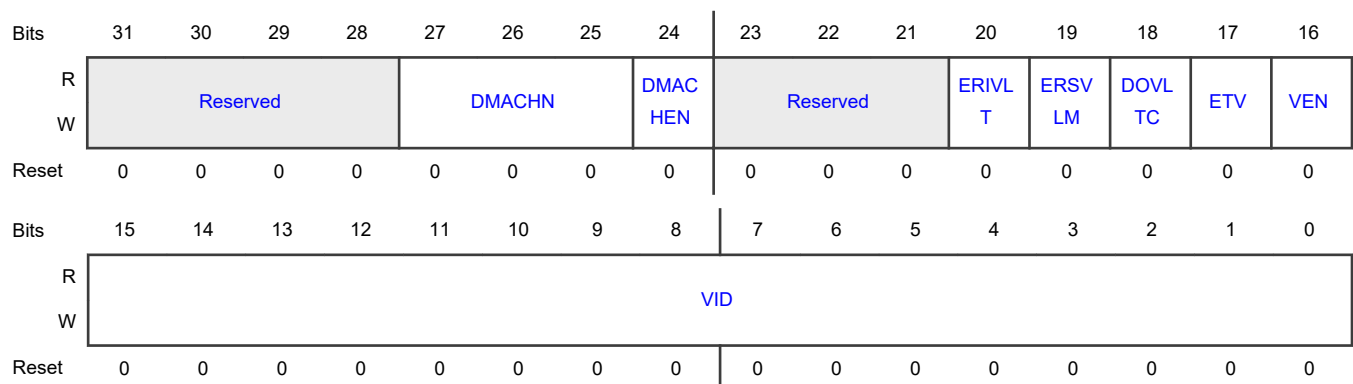
Register	Offset
MAC_VLAN_Tag_Data	54h

Function

This register holds the read/write data for Indirect Access of the Per VLAN Tag registers. During the read access, this field contains valid read data only after the OB bit is reset.

During the write access, this field should be valid prior to setting the OB bit in the MAC_VLAN_Tag_Ctrl Register.

Diagram



Fields

Field	Function
31-28 —	Reserved.
27-25 DMACHN	DMA Channel Number The DMA Channel number to which the VLAN Tagged Frame is to be routed if it passes this VLAN Tag Filter is programmed in this field. If the Routing based on VLAN Tag Filter is not necessary, this field need not be programmed.
24 DMACHEN	DMA Channel Number Enable This bit is the Enable for the DMA Channel Number value programmed in the field DMACH. When this bit is reset, the Routing does not occur based on VLAN Filter result. The frame is routed based on DA Based DMA Channel Routing. 0b - DMA Channel Number is disabled 1b - DMA Channel Number is enabled
23-21 —	Reserved.
20 ERIVLT	Enable Inner VLAN Tag Comparison This bit is valid only when Double VLAN Tag Enable of the Filter is set. When this bit and the EDVLP field are set, the MAC receiver enables operation on the inner VLAN Tag (if present). When this bit is reset, the MAC receiver enables operation on the outer VLAN Tag (if present). 0b - Inner VLAN tag comparison is disabled 1b - Inner VLAN tag comparison is enabled
19 ERSVLM	Enable S-VLAN Match for received Frames This bit is valid only when VLAN Tag Enable of the Filter is set. When this bit is set, the MAC receiver enables filtering or matching for S-VLAN (Type = 0x88A8) packets. When this bit is reset, the MAC receiver enables filtering or matching for C-VLAN (Type = 0x8100) packets. 0b - Receive S-VLAN Match is disabled 1b - Receive S-VLAN Match is enabled
18 DOVLTC	Disable VLAN Type Comparison This bit is valid only when VLAN Tag Enable of the Filter is set. When this bit is set, the MAC does not check whether the VLAN Tag specified by the Enable Inner VLAN Tag Comparison bit is of type S-VLAN or C-VLAN.

Table continues on the next page...

Table continued from the previous page...

Field	Function
	When this bit is reset, the MAC filters or matches the VLAN Tag specified by the Enable Inner VLAN Tag Comparison bit only when VLAN Tag type is similar to the one specified by the Enable S-VLAN Match for received Frames bit. 0b - VLAN type comparison is enabled 1b - VLAN type comparison is disabled
17 ETV	12bits or 16bits VLAN comparison This bit is valid only when VEN of the Filter is set. When this bit is set, a 12-bit VLAN identifier is used for comparing and filtering instead of the complete 16-bit VLAN tag. Bits [11:0] of VLAN tag are compared with the corresponding field in the received VLAN-tagged packet. 0b - 16 bit VLAN comparison 1b - 12 bit VLAN comparison
16 VEN	VLAN Tag Enable This bit is used to enable or disable the VLAN Tag. When this bit is set, the MAC compares the VLAN Tag of received packet with the VLAN Tag ID. When this bit is reset, no comparison is performed irrespective of the programming of the other fields. 0b - VLAN Tag is disabled 1b - VLAN Tag is enabled
15-0 VID	VLAN Tag ID This field holds the VLAN Tag value which is used by the MAC for perfect comparison. It is valid when VLAN Tag Enable is set.

1.4.16 (MAC_VLAN_Hash_Table)

Offset

Register	Offset
MAC_VLAN_Hash_Table	58h

Function

When VTHM bit of the MAC_VLAN_Tag register is set, the 16-bit VLAN Hash Table register is used for group address filtering based on the VLAN tag. For hash filtering, the content of the 16-bit VLAN tag or 12-bit VLAN ID (based on the ETV bit of MAC_VLAN_Tag Register) in the incoming packet is passed through the CRC logic. The upper four bits of the calculated hash value are used to index the contents of the VLAN Hash table. For example, hash value of 4b'1000 selects Bit 8 of the VLAN Hash table.

The hash value of the destination address is calculated in the following way:

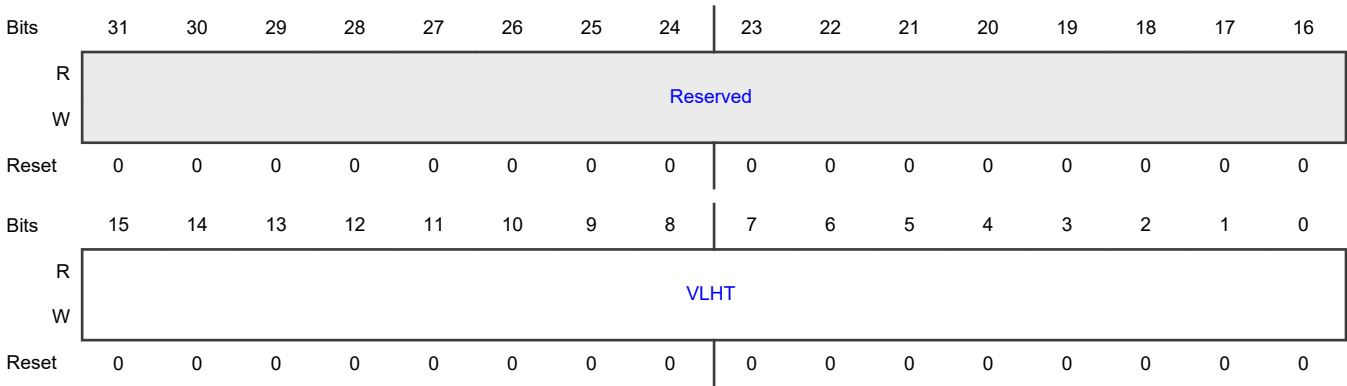
- Calculate the 32-bit CRC for the VLAN tag or ID (For steps to calculate CRC32, see Section 3.2.8 of IEEE 802.3).

- Perform bitwise reversal for the value obtained in step 1.
- Take the upper four bits from the value obtained in step 2.

If the VLAN hash Table register is configured to be double-synchronized to the (G)MII clock domain, the synchronization is triggered only when Bits[15:8] (in little-endian mode) or Bits[7:0] (in big-endian mode) of this register are written.

- If double-synchronization is enabled, consecutive writes to this register should be performed after at least four clock cycles in the destination clock domain.

Diagram



Fields

Field	Function
31-16 —	Reserved.
15-0 VLHT	VLAN Hash Table This field contains the 16-bit VLAN Hash Table.

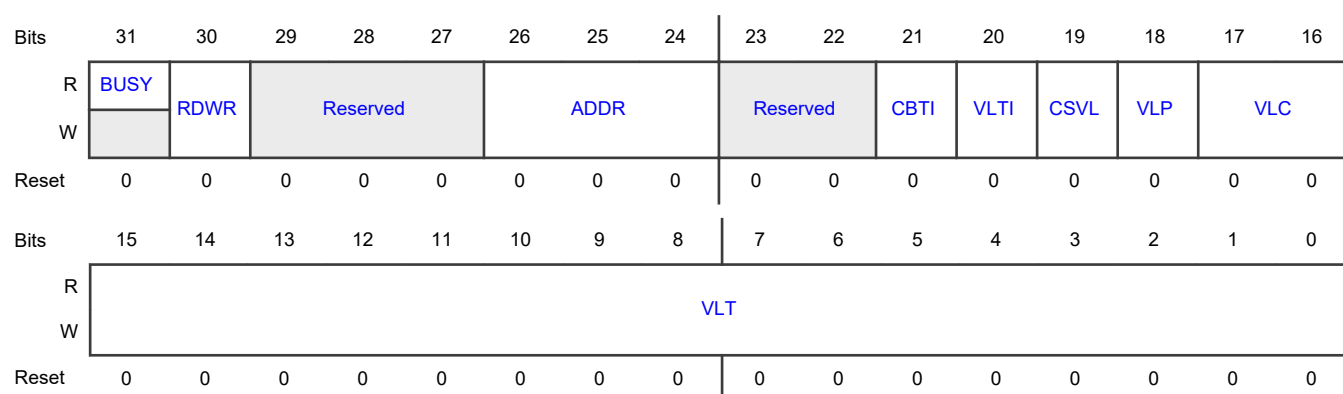
1.4.17 (MAC_VLAN_Incl)

Offset

Register	Offset
MAC_VLAN_Incl	60h

Function

The VLAN Tag Inclusion or Replacement register contains the VLAN tag for insertion or replacement in the Transmit packets. It also contains the VLAN tag insertion controls.

Diagram**Fields**

Field	Function
31 BUSY	Busy This bit indicates the status of the read/write operation of indirect access to the queue/channel specific VLAN inclusion register. For write operation write to a register is complete when this bit is reset. For read operation the read data is valid when the bit is reset. The application must make sure that this bit is reset before attempting subsequent access to this register. 0b - Busy status not detected 1b - Busy status detected
30 RDWR	Read write control This bit controls the read or write operation for indirectly accessing the queue/channel specific VLAN Inclusion register. When set indicates write operation and when reset indicates read operation. This does not have any effect when CBTI is reset. 0b - Read operation of indirect access 1b - Write operation of indirect access
29-27 —	Reserved.
26-24 ADDR	Address This field selects one of the queue/channel specific VLAN Inclusion register for read/write access. This does not have any effect when CBTI is reset.
23-22 —	Reserved.
21	Channel based tag insertion

Table continues on the next page...

Table continued from the previous page...

Field	Function
CBTI	When this bit is set, outer VLAN tag is inserted for every packets transmitted by the MAC. The tag value is taken from the queue/channel specific VLAN tag register. The VLTi, VLP, VLC, and VLT fields of this register are ignored when this bit is set. When this bit is set, a write operation to byte 3 of this register initiates the read/write access to the indirect register. When reset, outer VLAN operation is based on the setting of VLTi, VLP, VLC and VLT fields of this register. 0b - Channel based tag insertion is disabled 1b - Channel based tag insertion is enabled
20 VLTi	VLAN Tag Input When this bit is set, it indicates that the VLAN tag to be inserted or replaced in Tx packet should be taken from: - The Tx descriptor 0b - VLAN Tag Input is disabled 1b - VLAN Tag Input is enabled
19 CSVl	C-VLAN or S-VLAN When this bit is set, S-VLAN type (0x88A8) is inserted or replaced in the 13th and 14th bytes of transmitted packets. When this bit is reset, C-VLAN type (0x8100) is inserted or replaced in the 13th and 14th bytes of transmitted packets. 0b - C-VLAN type (0x8100) is inserted or replaced 1b - S-VLAN type (0x88A8) is inserted or replaced
18 VLP	VLAN Priority Control When this bit is set, the control bits[17:16] are used for VLAN deletion, insertion, or replacement. When this bit is reset, the mti_vlan_ctrl_i control input is used and bits[17:16] are ignored. 0b - VLAN Priority Control is disabled 1b - VLAN Priority Control is enabled
17-16 VLC	VLAN Tag Control in Transmit Packets - 2'b00: No VLAN tag deletion, insertion, or replacement - 2'b01: VLAN tag deletion The MAC removes the VLAN type (bytes 13 and 14) and VLAN tag (bytes 15 and 16) of all transmitted packets with VLAN tags. - 2'b10: VLAN tag insertion The MAC inserts VLT in bytes 15 and 16 of the packet after inserting the Type value (0x8100 or 0x88a8) in bytes 13 and 14. This operation is performed on all transmitted packets, irrespective of whether they already have a VLAN tag. - 2'b11: VLAN tag replacement

Table continues on the next page...

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Field	Function
	The MAC replaces VLT in bytes 15 and 16 of all VLAN-type transmitted packets (Bytes 13 and 14 are 0x8100 or 0x88a8). Note: Changes to this field take effect only on the start of a packet. If you write this register field when a packet is being transmitted, only the subsequent packet can use the updated value, that is, the current packet does not use the updated value. 00b - No VLAN tag deletion, insertion, or replacement 01b - VLAN tag deletion 10b - VLAN tag insertion 11b - VLAN tag replacement
15-0 VLT	VLAN Tag for Transmit Packets This field contains the value of the VLAN tag to be inserted or replaced. The value must only be changed when the transmit lines are inactive or during the initialization phase. Bits[15:13] are the User Priority field, Bit 12 is the CFI/DEI field, and Bits[11:0] are the VID field in the VLAN tag. The following list describes the bits of this field: - Bits[15:13]: User Priority - Bit 12: Canonical Format Indicator (CFI) or Drop Eligible Indicator (DEI) - Bits[11:0]: VLAN Identifier (VID) field of VLAN tag

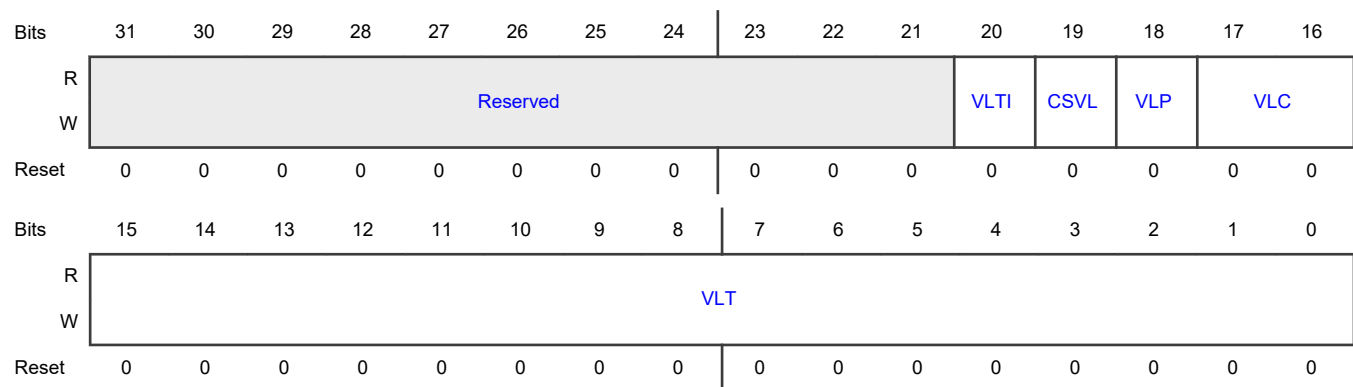
1.4.18 (MAC_Inner_VLAN_Incl)

Offset

Register	Offset
MAC_Inner_VLAN_Incl	64h

Function

The Inner VLAN Tag Inclusion or Replacement register contains the inner VLAN tag to be inserted or replaced in the Transmit packet. It also contains the inner VLAN tag insertion controls.

Diagram**Fields**

Field	Function
31-21 —	Reserved.
20 VLTi	VLAN Tag Input When this bit is set, it indicates that the VLAN tag to be inserted or replaced in Tx packet should be taken from: - The Tx descriptor 0b - VLAN Tag Input is disabled 1b - VLAN Tag Input is enabled
19 CSVL	C-VLAN or S-VLAN Controls the type insertion or replacement in the 17th and 18th bytes of transmitted packets 0b - C-VLAN type (8100h) 1b - S-VLAN type (88A8h)
18 VLP	VLAN Priority Control When this bit is set, the VLC field is used for VLAN deletion, insertion, or replacement. When this bit is reset, the mti_vlan_ctrl_i control input is used and the VLC field is ignored. 0b - VLAN Priority Control is disabled 1b - VLAN Priority Control is enabled
17-16 VLC	VLAN Tag Control in Transmit Packets - 2'b00: No VLAN tag deletion, insertion, or replacement - 2'b01: VLAN tag deletion The MAC removes the VLAN type (bytes 17 and 18) and VLAN tag (bytes 19 and 20) of all transmitted packets with VLAN tags. - 2'b10: VLAN tag insertion

Table continues on the next page...

Table continued from the previous page...

Field	Function
	The MAC inserts VLT in bytes 19 and 20 of the packet after inserting the Type value (0x8100 or 0x88a8) in bytes 17 and 18. This operation is performed on all transmitted packets, irrespective of whether they already have a VLAN tag. - 2'b11: VLAN tag replacement The MAC replaces VLT in bytes 19 and 20 of all VLAN-type transmitted packets (Bytes 17 and 18 are 0x8100 or 0x88a8). Note: Changes to this field take effect only on the start of a packet. If you write this register field when a packet is being transmitted, only the subsequent packet can use the updated value, that is, the current packet does not use the updated value. 00b - No VLAN tag deletion, insertion, or replacement 01b - VLAN tag deletion 10b - VLAN tag insertion 11b - VLAN tag replacement
15-0 VLT	VLAN Tag for Transmit Packets This field contains the value of the VLAN tag to be inserted or replaced. The value must only be changed when the transmit lines are inactive or during the initialization phase. Bits[15:13] are the User Priority field, Bit 12 is the CFI/DEI field, and Bits[11:0] are the VID field in the VLAN tag. The following list describes the bits of this field: - Bits[15:13]: User Priority - Bit 12: Canonical Format Indicator (CFI) or Drop Eligible Indicator (DEI) - Bits[11:0]: VLAN Identifier (VID) field of VLAN tag

1.4.19 (MAC_Q0_Tx_Flow_Ctrl)

Offset

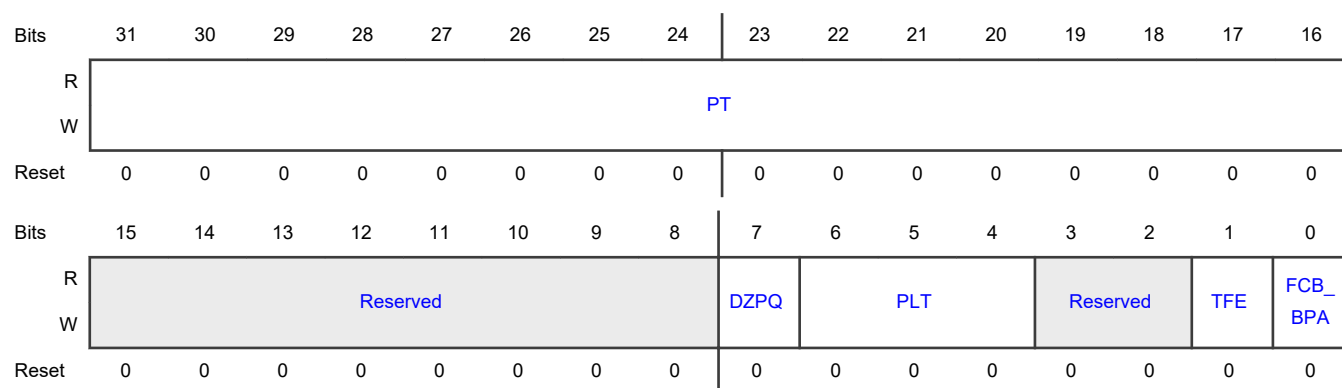
Register	Offset
MAC_Q0_Tx_Flow_Ctrl	70h

Function

The Flow Control register controls the generation and reception of the Control (Pause Command) packets by the Flow control module of the MAC. A Write to a register with the Busy bit set to 1 triggers the Flow Control block to generate a Pause packet. The fields of the control packet are selected as specified in the 802.3x specification, and the Pause Time value from this register is used in the Pause Time field of the control packet. The Busy bit remains set until the control packet is transferred onto the cable. The application must make sure that the Busy bit is cleared before writing to the register.

When the PFCE bit in the MAC_Rx_Flow_Ctrl register is enabled, this register controls the generation of Priority Flow Control (PFC) frames with priorities mapped according to PSRQ0 in the MAC_RxQ_Ctrl2 register.

Diagram



Fields

Field	Function
31-16 PT	Pause Time This field holds the value to be used in the Pause Time field in the Tx control packet. If the Pause Time bits are configured to be double-synchronized to the (G)MII clock domain, consecutive writes to this register should be performed only after at least four clock cycles in the destination clock domain.
15-8 —	Reserved.
7 DZPQ	Disable Zero-Quanta Pause When this bit is set, it disables the automatic generation of the zero-quanta Pause packets on de-assertion of the flow-control signal from the FIFO layer (MTL or external sideband flow control signal <code>sbdf_flowctrl_i</code> or <code>mti_flowctrl_i</code>). When this bit is reset, normal operation with automatic zero-quanta Pause packet generation is enabled. 0b - Zero-Quanta Pause packet generation is enabled 1b - Zero-Quanta Pause packet generation is disabled
6-4 PLT	Pause Low Threshold This field configures the threshold of the Pause timer at which the input flow control signal <code>mti_flowctrl_i</code> (or <code>sbdf_flowctrl_i</code>) is checked for automatic retransmission of the Pause packet. The threshold values should be always less than the Pause Time configured in Bits[31:16]. For example, if PT = 100H (256 slot times), and PLT = 001, a second Pause packet is automatically transmitted if the <code>mti_flowctrl_i</code> signal is asserted at 228 (256-28) slot times after the first Pause packet is transmitted. The following list provides the threshold values for different values. The slot time is defined as the time taken to transmit 512 bits (64 bytes) on the GMII or MII interface. This (approximate) computation is based on the packet size (64, 1518, 2000, 9018, 16384, or 32768) + 2 Pause Packet Size + IPG in Slot Times. 000b - Pause Time minus 4 Slot Times (PT -4 slot times) 001b - Pause Time minus 28 Slot Times (PT -28 slot times)

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Field	Function
	010b - Pause Time minus 36 Slot Times (PT -36 slot times) 011b - Pause Time minus 144 Slot Times (PT -144 slot times) 100b - Pause Time minus 256 Slot Times (PT -256 slot times) 101b - Pause Time minus 512 Slot Times (PT -512 slot times) 110b - Reserved
3-2 —	Reserved.
1 TFE	Transmit Flow Control Enable Full-Duplex Mode: In the full-duplex mode, when this bit is set, the MAC enables the flow control operation to Tx Pause packets. When this bit is reset, the flow control operation in the MAC is disabled, and the MAC does not transmit any Pause packets. Half-Duplex Mode: In the half-duplex mode, when this bit is set, the MAC enables the backpressure operation. When this bit is reset, the backpressure feature is disabled. 0b - Transmit Flow Control is disabled 1b - Transmit Flow Control is enabled
0 FCB_BPA	Flow Control Busy or Backpressure Activate This bit initiates a Pause packet in the full-duplex mode and activates the backpressure function in the half-duplex mode if the TFE bit is set. Full-Duplex Mode: In the full-duplex mode, this bit should be read as 1'b0 before writing to this register. To initiate a Pause packet, the application must set this bit to 1'b1. During Control packet transfer, this bit continues to be set to indicate that a packet transmission is in progress. When Pause packet transmission is complete, the MAC resets this bit to 1'b0. You should not write to this register until this bit is cleared. Half-Duplex Mode: When this bit is set (and TFE bit is set) in the half-duplex mode, the MAC asserts the backpressure. During backpressure, when the MAC receives a new packet, the transmitter starts sending a JAM pattern resulting in a collision. This control register bit is logically ORed with the mti_flowctrl_i input signal for the backpressure function. When the MAC is configured for the full-duplex mode, the BPA is automatically disabled. Access restriction applies. Setting 1 sets. Self-cleared. Setting 0 has no effect. 0b - Flow Control Busy or Backpressure Activate is disabled 1b - Flow Control Busy or Backpressure Activate is enabled

1.4.20 (MAC_Q1_Tx_Flow_Ctrl)

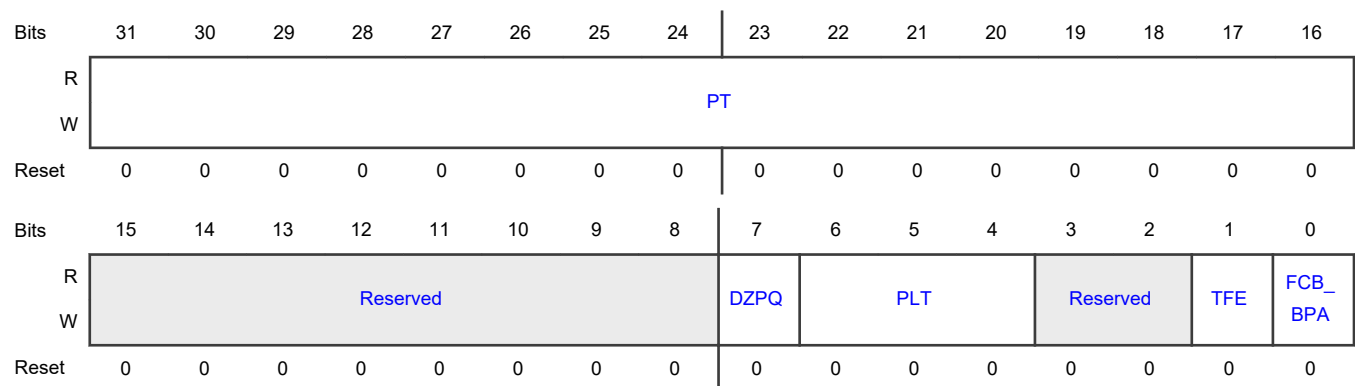
Offset

Register	Offset
MAC_Q1_Tx_Flow_Ctrl	74h

Function

This register controls the generation of PFC Control packets of priorities mapped as per the PSRQi field in the MAC_RxQ_Ctrl2/ MAC_RxQ_Ctrl3 registers.

Diagram



Fields

Field	Function
31-16 PT	Pause Time This field holds the value to be used in the Pause Time field in the Tx control packet. If the Pause Time bits are configured to be double-synchronized to the (G)MII clock domain, consecutive writes to this register should be performed only after at least four clock cycles in the destination clock domain.
15-8 —	Reserved.
7 DZPQ	Disable Zero-Quanta Pause When this bit is set, it disables the automatic generation of the zero-quanta Pause packets on de-assertion of the flow-control signal from the FIFO layer (MTL or external sideband flow control signal sbd_flowctrl_i or mti_flowctrl_i). When this bit is reset, normal operation with automatic zero-quanta Pause packet generation is enabled. 0b - Zero-Quanta Pause packet generation is enabled 1b - Zero-Quanta Pause packet generation is disabled
6-4	Pause Low Threshold

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Field	Function
PLT	This field configures the threshold of the Pause timer at which the input flow control signal mti_flowctrl_i (or sbd_flowctrl_i) is checked for automatic retransmission of the Pause packet. The threshold values should be always less than the Pause Time configured in Bits[31:16]. For example, if PT = 100H (256 slot times), and PLT = 001, a second Pause packet is automatically transmitted if the mti_flowctrl_i signal is asserted at 228 (256-28) slot times after the first Pause packet is transmitted. The following list provides the threshold values for different values. The slot time is defined as the time taken to transmit 512 bits (64 bytes) on the GMII or MII interface. This (approximate) computation is based on the packet size (64, 1518, 2000, 9018, 16384, or 32768) + 2 Pause Packet Size + IPG in Slot Times. 000b - Pause Time minus 4 Slot Times (PT -4 slot times) 001b - Pause Time minus 28 Slot Times (PT -28 slot times) 010b - Pause Time minus 36 Slot Times (PT -36 slot times) 011b - Pause Time minus 144 Slot Times (PT -144 slot times) 100b - Pause Time minus 256 Slot Times (PT -256 slot times) 101b - Pause Time minus 512 Slot Times (PT -512 slot times) 110b - Reserved
3-2 —	Reserved.
1 TFE	Transmit Flow Control Enable When this bit is set in full-duplex mode, the MAC enables the flow control operation to Tx Pause packets. When this bit is reset, the flow control operation in the MAC is disabled, and the MAC does not transmit any Pause packets. 0b - Transmit Flow Control is disabled 1b - Transmit Flow Control is enabled
0 FCB_BPA	Flow Control Busy This bit initiates a PFC packet if the TFE bit is set. To initiate a PFC packet, the application must set this bit to 1'b1. During Control packet transfer, this bit continues to be set to indicate that a packet transmission is in progress. When PFC packet transmission is complete, the MAC resets this bit to 1'b0. You should not write to this register until this bit is cleared. Access restriction applies. Setting 1 sets. Self-cleared. Setting 0 has no effect. 0b - Flow Control Busy or Backpressure Activate is disabled 1b - Flow Control Busy or Backpressure Activate is enabled

1.4.21 (MAC_Q2_Tx_Flow_Ctrl)

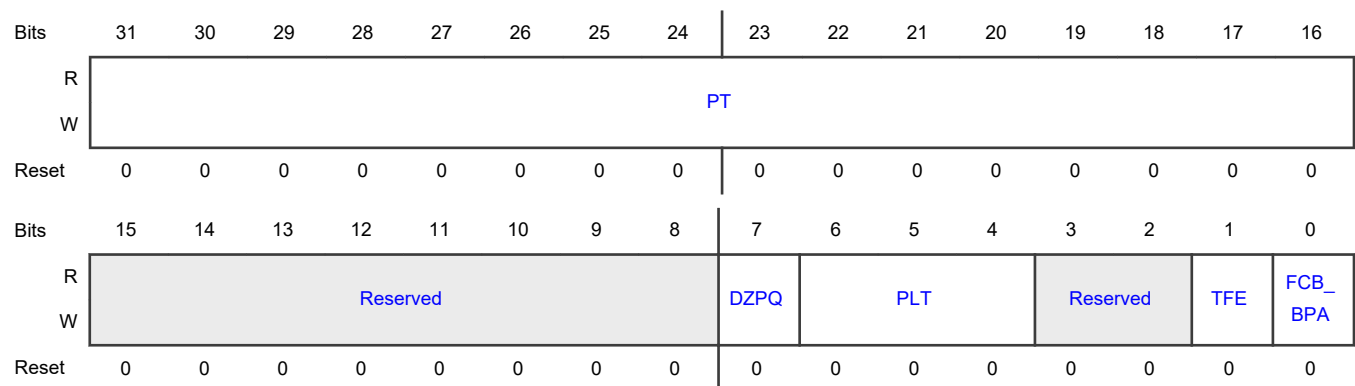
Offset

Register	Offset
MAC_Q2_Tx_Flow_Ctrl	78h

Function

This register controls the generation of PFC Control packets of priorities mapped as per the PSRQi field in the MAC_RxQ_Ctrl2/ MAC_RxQ_Ctrl3 registers.

Diagram



Fields

Field	Function
31-16 PT	Pause Time This field holds the value to be used in the Pause Time field in the Tx control packet. If the Pause Time bits are configured to be double-synchronized to the (G)MII clock domain, consecutive writes to this register should be performed only after at least four clock cycles in the destination clock domain.
15-8 —	Reserved.
7 DZPQ	Disable Zero-Quanta Pause When this bit is set, it disables the automatic generation of the zero-quanta Pause packets on de-assertion of the flow-control signal from the FIFO layer (MTL or external sideband flow control signal sbd_flowctrl_i or mti_flowctrl_i). When this bit is reset, normal operation with automatic zero-quanta Pause packet generation is enabled. 0b - Zero-Quanta Pause packet generation is enabled 1b - Zero-Quanta Pause packet generation is disabled
6-4	Pause Low Threshold

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Field	Function
PLT	This field configures the threshold of the Pause timer at which the input flow control signal mti_flowctrl_i (or sbd_flowctrl_i) is checked for automatic retransmission of the Pause packet. The threshold values should be always less than the Pause Time configured in Bits[31:16]. For example, if PT = 100H (256 slot times), and PLT = 001, a second Pause packet is automatically transmitted if the mti_flowctrl_i signal is asserted at 228 (256-28) slot times after the first Pause packet is transmitted. The following list provides the threshold values for different values. The slot time is defined as the time taken to transmit 512 bits (64 bytes) on the GMII or MII interface. This (approximate) computation is based on the packet size (64, 1518, 2000, 9018, 16384, or 32768) + 2 Pause Packet Size + IPG in Slot Times. 000b - Pause Time minus 4 Slot Times (PT -4 slot times) 001b - Pause Time minus 28 Slot Times (PT -28 slot times) 010b - Pause Time minus 36 Slot Times (PT -36 slot times) 011b - Pause Time minus 144 Slot Times (PT -144 slot times) 100b - Pause Time minus 256 Slot Times (PT -256 slot times) 101b - Pause Time minus 512 Slot Times (PT -512 slot times) 110b - Reserved
3-2 —	Reserved.
1 TFE	Transmit Flow Control Enable When this bit is set in full-duplex mode, the MAC enables the flow control operation to Tx Pause packets. When this bit is reset, the flow control operation in the MAC is disabled, and the MAC does not transmit any Pause packets. 0b - Transmit Flow Control is disabled 1b - Transmit Flow Control is enabled
0 FCB_BPA	Flow Control Busy This bit initiates a PFC packet if the TFE bit is set. To initiate a PFC packet, the application must set this bit to 1'b1. During Control packet transfer, this bit continues to be set to indicate that a packet transmission is in progress. When PFC packet transmission is complete, the MAC resets this bit to 1'b0. You should not write to this register until this bit is cleared. Access restriction applies. Setting 1 sets. Self-cleared. Setting 0 has no effect. 0b - Flow Control Busy or Backpressure Activate is disabled 1b - Flow Control Busy or Backpressure Activate is enabled

1.4.22 (MAC_Q3_Tx_Flow_Ctrl)

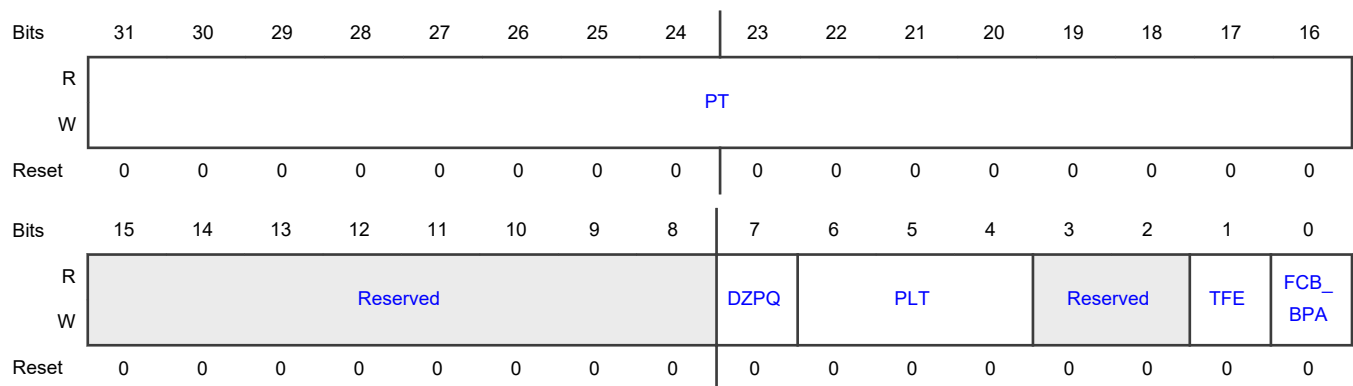
Offset

Register	Offset
MAC_Q3_Tx_Flow_Ctrl	7Ch

Function

This register controls the generation of PFC Control packets of priorities mapped as per the PSRQi field in the MAC_RxQ_Ctrl2/ MAC_RxQ_Ctrl3 registers.

Diagram



Fields

Field	Function
31-16 PT	Pause Time This field holds the value to be used in the Pause Time field in the Tx control packet. If the Pause Time bits are configured to be double-synchronized to the (G)MII clock domain, consecutive writes to this register should be performed only after at least four clock cycles in the destination clock domain.
15-8 —	Reserved.
7 DZPQ	Disable Zero-Quanta Pause When this bit is set, it disables the automatic generation of the zero-quanta Pause packets on de-assertion of the flow-control signal from the FIFO layer (MTL or external sideband flow control signal sbd_flowctrl_i or mti_flowctrl_i). When this bit is reset, normal operation with automatic zero-quanta Pause packet generation is enabled. 0b - Zero-Quanta Pause packet generation is enabled 1b - Zero-Quanta Pause packet generation is disabled
6-4	Pause Low Threshold

Table continues on the next page...

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Field	Function
PLT	This field configures the threshold of the Pause timer at which the input flow control signal mti_flowctrl_i (or sbd_flowctrl_i) is checked for automatic retransmission of the Pause packet. The threshold values should be always less than the Pause Time configured in Bits[31:16]. For example, if PT = 100H (256 slot times), and PLT = 001, a second Pause packet is automatically transmitted if the mti_flowctrl_i signal is asserted at 228 (256-28) slot times after the first Pause packet is transmitted. The following list provides the threshold values for different values. The slot time is defined as the time taken to transmit 512 bits (64 bytes) on the GMII or MII interface. This (approximate) computation is based on the packet size (64, 1518, 2000, 9018, 16384, or 32768) + 2 Pause Packet Size + IPG in Slot Times. 000b - Pause Time minus 4 Slot Times (PT -4 slot times) 001b - Pause Time minus 28 Slot Times (PT -28 slot times) 010b - Pause Time minus 36 Slot Times (PT -36 slot times) 011b - Pause Time minus 144 Slot Times (PT -144 slot times) 100b - Pause Time minus 256 Slot Times (PT -256 slot times) 101b - Pause Time minus 512 Slot Times (PT -512 slot times) 110b - Reserved
3-2 —	Reserved.
1 TFE	Transmit Flow Control Enable When this bit is set in full-duplex mode, the MAC enables the flow control operation to Tx Pause packets. When this bit is reset, the flow control operation in the MAC is disabled, and the MAC does not transmit any Pause packets. 0b - Transmit Flow Control is disabled 1b - Transmit Flow Control is enabled
0 FCB_BPA	Flow Control Busy This bit initiates a PFC packet if the TFE bit is set. To initiate a PFC packet, the application must set this bit to 1'b1. During Control packet transfer, this bit continues to be set to indicate that a packet transmission is in progress. When PFC packet transmission is complete, the MAC resets this bit to 1'b0. You should not write to this register until this bit is cleared. Access restriction applies. Setting 1 sets. Self-cleared. Setting 0 has no effect. 0b - Flow Control Busy or Backpressure Activate is disabled 1b - Flow Control Busy or Backpressure Activate is enabled

1.4.23 (MAC_Q4_Tx_Flow_Ctrl)

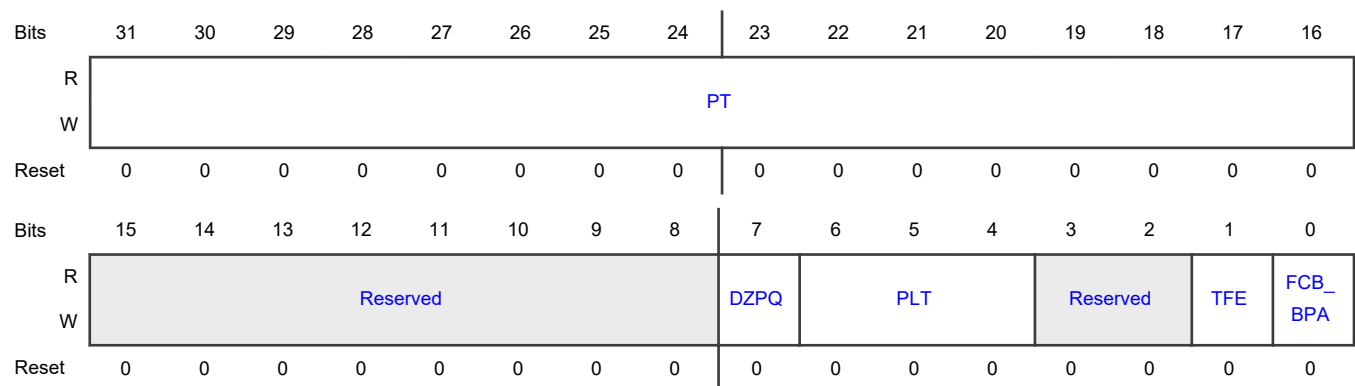
Offset

Register	Offset
MAC_Q4_Tx_Flow_Ctrl	80h

Function

This register controls the generation of PFC Control packets of priorities mapped as per the PSRQi field in the MAC_RxQ_Ctrl2/ MAC_RxQ_Ctrl3 registers.

Diagram



Fields

Field	Function
31-16 PT	Pause Time This field holds the value to be used in the Pause Time field in the Tx control packet. If the Pause Time bits are configured to be double-synchronized to the (G)MII clock domain, consecutive writes to this register should be performed only after at least four clock cycles in the destination clock domain.
15-8 —	Reserved.
7 DZPQ	Disable Zero-Quanta Pause When this bit is set, it disables the automatic generation of the zero-quanta Pause packets on de-assertion of the flow-control signal from the FIFO layer (MTL or external sideband flow control signal sbd_flowctrl_i or mti_flowctrl_i). When this bit is reset, normal operation with automatic zero-quanta Pause packet generation is enabled. 0b - Zero-Quanta Pause packet generation is enabled 1b - Zero-Quanta Pause packet generation is disabled
6-4	Pause Low Threshold

Table continues on the next page...

Table continued from the previous page...

Field	Function
PLT	This field configures the threshold of the Pause timer at which the input flow control signal mti_flowctrl_i (or sbd_flowctrl_i) is checked for automatic retransmission of the Pause packet. The threshold values should be always less than the Pause Time configured in Bits[31:16]. For example, if PT = 100H (256 slot times), and PLT = 001, a second Pause packet is automatically transmitted if the mti_flowctrl_i signal is asserted at 228 (256-28) slot times after the first Pause packet is transmitted. The following list provides the threshold values for different values. The slot time is defined as the time taken to transmit 512 bits (64 bytes) on the GMII or MII interface. This (approximate) computation is based on the packet size (64, 1518, 2000, 9018, 16384, or 32768) + 2 Pause Packet Size + IPG in Slot Times. 000b - Pause Time minus 4 Slot Times (PT -4 slot times) 001b - Pause Time minus 28 Slot Times (PT -28 slot times) 010b - Pause Time minus 36 Slot Times (PT -36 slot times) 011b - Pause Time minus 144 Slot Times (PT -144 slot times) 100b - Pause Time minus 256 Slot Times (PT -256 slot times) 101b - Pause Time minus 512 Slot Times (PT -512 slot times) 110b - Reserved
3-2 —	Reserved.
1 TFE	Transmit Flow Control Enable When this bit is set in full-duplex mode, the MAC enables the flow control operation to Tx Pause packets. When this bit is reset, the flow control operation in the MAC is disabled, and the MAC does not transmit any Pause packets. 0b - Transmit Flow Control is disabled 1b - Transmit Flow Control is enabled
0 FCB_BPA	Flow Control Busy This bit initiates a PFC packet if the TFE bit is set. To initiate a PFC packet, the application must set this bit to 1'b1. During Control packet transfer, this bit continues to be set to indicate that a packet transmission is in progress. When PFC packet transmission is complete, the MAC resets this bit to 1'b0. You should not write to this register until this bit is cleared. Access restriction applies. Setting 1 sets. Self-cleared. Setting 0 has no effect. 0b - Flow Control Busy or Backpressure Activate is disabled 1b - Flow Control Busy or Backpressure Activate is enabled

1.4.24 (MAC_Rx_Flow_Ctrl)

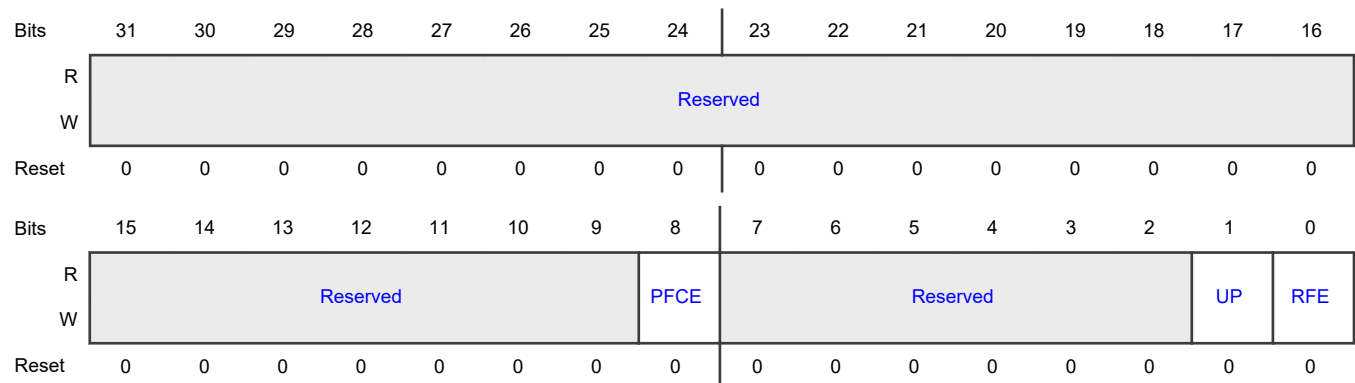
Offset

Register	Offset
MAC_Rx_Flow_Ctrl	90h

Function

The Receive Flow Control register controls the pausing of MAC Transmit based on the received Pause packet.

Diagram



Fields

Field	Function
31-9 —	Reserved.
8 PFCE	Priority Based Flow Control Enable When this bit is set, it enables generation and reception of priority-based flow control (PFC) packets. When this bit is reset, it enables generation and reception of 802.3x Pause control packets. 0b - Priority Based Flow Control is disabled 1b - Priority Based Flow Control is enabled
7-2 —	Reserved.
1 UP	Unicast Pause Packet Detect A pause packet is processed when it has the unique multicast address specified in the IEEE 802.3. When this bit is set, the MAC can also detect Pause packets with unicast address of the station. This unicast address should be as specified in MAC_Address0_High and MAC_Address0_Low. When this bit is reset, the MAC only detects Pause packets with unique multicast address.

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Field	Function
	Note: The MAC does not process a Pause packet if the multicast address is different from the unique multicast address. This is also applicable to the received PFC packet when the Priority Flow Control (PFC) is enabled. The unique multicast address (0x01_80_C2_00_00_01) is as specified in IEEE 802.1 Qbb-2011. 0b - Unicast Pause Packet Detect disabled 1b - Unicast Pause Packet Detect enabled
0 RFE	Receive Flow Control Enable When this bit is set and the MAC is operating in full-duplex mode, the MAC decodes the received Pause packet and disables its transmitter for a specified (Pause) time. When this bit is reset or the MAC is operating in half-duplex mode, the decode function of the Pause packet is disabled. When PFC is enabled, flow control is enabled for PFC packets. The MAC decodes the received PFC packet and disables the Transmit queue, with matching priorities, for a duration of received Pause time. 0b - Receive Flow Control is disabled 1b - Receive Flow Control is enabled

1.4.25 (MAC_RxQ_Ctrl4)

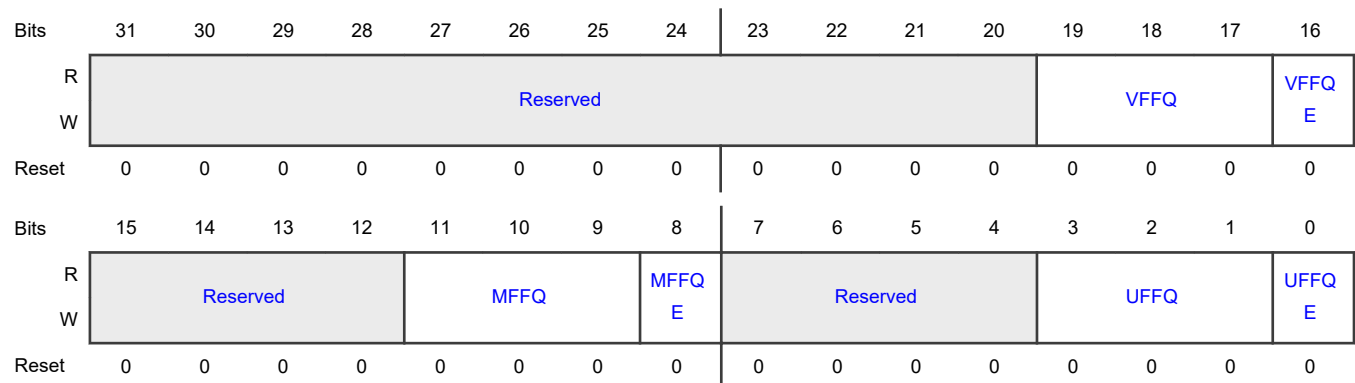
Offset

Register	Offset
MAC_RxQ_Ctrl4	94h

Function

The Receive Queue Control 4 register controls the routing of unicast and multicast packets that fail the Destination or Source address filter to the Rx queues.

Diagram



Fields

Field	Function
31-20 —	Reserved.
19-17 VFFQ	VLAN Tag Filter Fail Packets Queue This field holds the Rx queue number to which the tagged packets failing the Destination or Source Address filter (and UFFQE/MFFQE not enabled) or failing the VLAN tag filter must be routed to. This field is valid only when the VFFQE bit is set.
16 VFFQE	VLAN Tag Filter Fail Packets Queuing Enable When this bit is set, the tagged packets which fail the Destination or Source address filter or fail the VLAN tag filter, are routed to the Rx Queue Number programmed in the VFFQ. When this bit is reset, the tagged packets which fail the Destination or Source address filter or fail the VLAN tag filter are routed based on other routing options. This bit is valid only when the RA bit of the MAC_Packet_Filter register is set. 0b - VLAN tag Filter Fail Packets Queuing is disabled 1b - VLAN tag Filter Fail Packets Queuing is enabled
15-12 —	Reserved.
11-9 MFFQ	Multicast Address Filter Fail Packets Queue. This field holds the Rx queue number to which the Multicast packets failing the Destination or Source Address filter are routed to. This field is valid only when the MFFQE bit is set.
8 MFFQE	Multicast Address Filter Fail Packets Queuing Enable. When this bit is set, the Multicast packets which fail the Destination or Source address filter is routed to the Rx Queue Number programmed in the MFFQ. When this bit is reset, the Multicast packets which fail the Destination or Source address filter is routed based on other routing options. This bit is valid only when the RA bit of the MAC_Packet_Filter register is set. 0b - Multicast Address Filter Fail Packets Queuing is disabled 1b - Multicast Address Filter Fail Packets Queuing is enabled
7-4 —	Reserved.
3-1 UFFQ	Unicast Address Filter Fail Packets Queue. This field holds the Rx queue number to which the Unicast packets failing the Destination or Source Address filter are routed to. This field is valid only when the UFFQE bit is set.
0 UFFQE	Unicast Address Filter Fail Packets Queuing Enable. When this bit is set, the Unicast packets which fail the Destination or Source address filter is routed to the Rx Queue Number programmed in the UFFQ.

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Field	Function
	When this bit is reset, the Unicast packets which fail the Destination or Source address filter is routed based on other routing options. This bit is valid only when the RA bit of the MAC_Packet_Filter register is set. 0b - Unicast Address Filter Fail Packets Queuing is disabled 1b - Unicast Address Filter Fail Packets Queuing is enabled

1.4.26 (MAC_TxQ_PrtY_Map0)

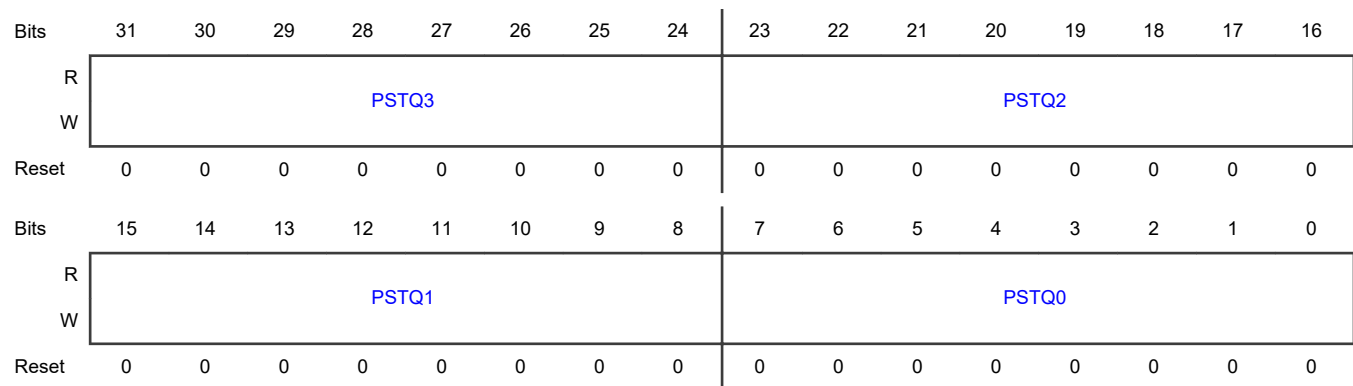
Offset

Register	Offset
MAC_TxQ_PrtY_Map0	98h

Function

The Transmit Queue Priority Mapping 0 register contains the priority values assigned to Tx Queue 0 through Tx Queue 3.

Diagram



Fields

Field	Function
31-24 PSTQ3	Priorities Selected in Transmit Queue 3 This bit is similar to the PSTQ0 bit.
23-16 PSTQ2	Priorities Selected in Transmit Queue 2 This bit is similar to the PSTQ0 bit.
15-8	Priorities Selected in Transmit Queue 1

Table continues on the next page...

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Field	Function
PSTQ1	This bit is similar to the PSTQ0 bit.
7-0 PSTQ0	Priorities Selected in Transmit Queue 0 This field holds the priorities assigned to Tx Queue 0 by the software. This field determines if Tx Queue 0 should be blocked from transmitting specified pause time when a PFC packet is received with priorities matching the priorities programmed in this field. If the content of this field is not mutually exclusive to corresponding fields of other Transmit queues, that is, same priority is mapped to multiple Tx queues, the MAC blocks all queues with matching priority for specified time.

1.4.27 (MAC_TxQ_PrtY_Map1)

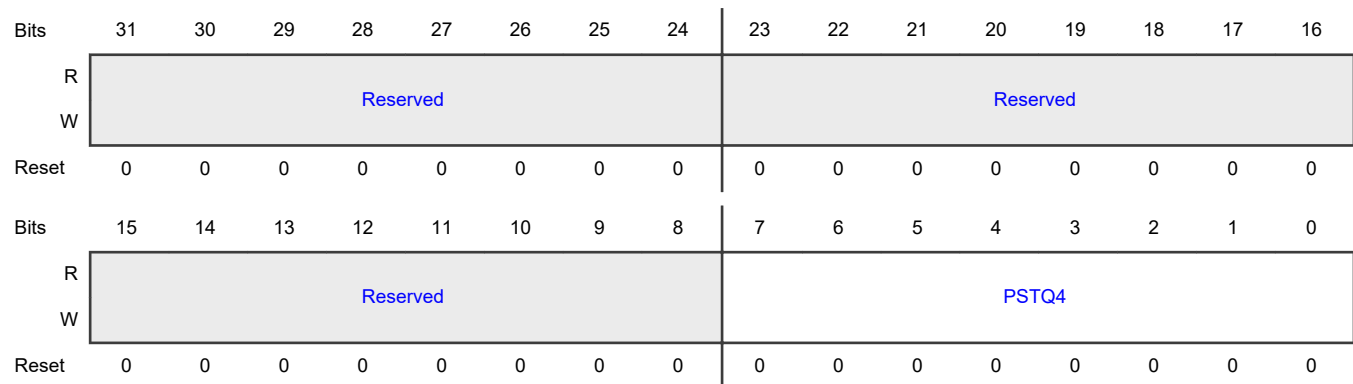
Offset

Register	Offset
MAC_TxQ_PrtY_Map1	9Ch

Function

The Transmit Queue Priority Mapping 1 register contains the priority values assigned to Tx Queue 4 through Tx Queue 7.

Diagram



Fields

Field	Function
31-24 —	Reserved.
23-16	Reserved.

Table continues on the next page...

Table continued from the previous page...

Field	Function
—	
15-8 —	Reserved.
7-0 PSTQ4	Priorities Selected in Transmit Queue 4 This field holds the priorities assigned to Tx Queue 4 by the software. This field determines if Tx Queue 4 should be blocked from transmitting specified pause time when a PFC packet is received with priorities matching the priorities programmed in this field. If the content of this field is not mutually exclusive to corresponding fields of other Transmit queues, that is, same priority is mapped to multiple Tx queues, the MAC blocks all queues with matching priority for specified time.

1.4.28 (MAC_RxQ_Ctrl0)

Offset

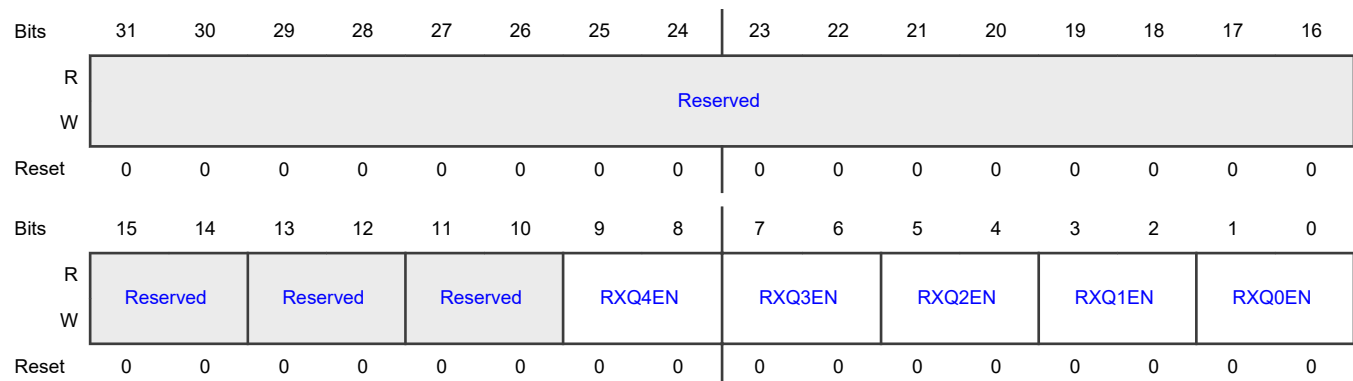
Register	Offset
MAC_RxQ_Ctrl0	A0h

Function

The Receive Queue Control 0 register controls the queue management in the MAC Receiver.

Note: In multiple Rx queues configuration, all the queues are disabled by default. Enable the Rx queue by programming the corresponding field in this register.

Diagram



Fields

Field	Function
31-16 —	Reserved.
15-14 —	Reserved.
13-12 —	Reserved.
11-10 —	Reserved.
9-8 RXQ4EN	Receive Queue 4 Enable This field is similar to the RXQ0EN field. 00b - Queue not enabled 01b - Queue enabled for AV 10b - Queue enabled for DCB/Generic 11b - Reserved
7-6 RXQ3EN	Receive Queue 3 Enable This field is similar to the RXQ0EN field. 00b - Queue not enabled 01b - Queue enabled for AV 10b - Queue enabled for DCB/Generic 11b - Reserved
5-4 RXQ2EN	Receive Queue 2 Enable This field is similar to the RXQ0EN field. 00b - Queue not enabled 01b - Queue enabled for AV 10b - Queue enabled for DCB/Generic 11b - Reserved
3-2 RXQ1EN	Receive Queue 1 Enable This field is similar to the RXQ0EN field. 00b - Queue not enabled 01b - Queue enabled for AV 10b - Queue enabled for DCB/Generic

Table continues on the next page...

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Field	Function
	11b - Reserved
1-0 RXQ0EN	Receive Queue 0 Enable This field indicates whether Rx Queue 0 is enabled for AV or DCB. 00b - Queue not enabled 01b - Queue enabled for AV 10b - Queue enabled for DCB/Generic 11b - Reserved

1.4.29 (MAC_RxQ_Ctrl1)

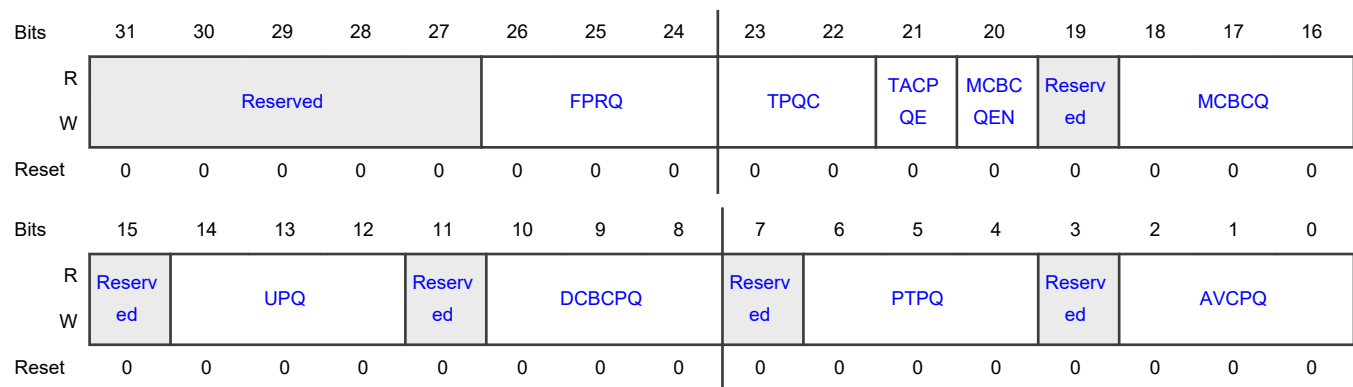
Offset

Register	Offset
MAC_RxQ_Ctrl1	A4h

Function

The Receive Queue Control 1 register controls the routing of multicast, broadcast, AV, DCB, and untagged packets to the Rx queues.

Diagram



Fields

Field	Function
31-27 —	Reserved.

Table continues on the next page...

Table continued from the previous page...

Field	Function
26-24 FPRQ	Frame Preemption Residue Queue This field holds the Rx queue number to which the residual preemption frames must be forwarded. Preemption frames that are tagged and pass the SA/DA/VLAN filtering are routed based on PSRQ and all other frames are treated as residual frames and is routed to the queue number mentioned in this field. The Queue-0 is used as a default queue for express frames, so this field cannot be programmed to a value 0.
23-22 TPQC	Tagged PTP over Ethernet Packets Queuing Control. This field controls the routing of the VLAN Tagged PTPoE packets. The following programmable options are allowed. - 2'b00: VLAN Tagged PTPoE packets are routed as generic VLAN Tagged packet (based on PSRQ for only non-AV enabled Rx Queues). - 2'b01: VLAN Tagged PTPoE packets are routed to Rx Queue specified by PTPQ field (That Rx Queue can be enabled for AV or non-AV traffic). - 2'b10: VLAN Tagged PTPoE packets are routed to only AV enabled Rx Queues based on PSRQ.
21 TACPQE	Tagged AV Control Packets Queuing Enable. When set, the MAC routes the received Tagged AV Control packets to the Rx queue specified by AVCPQ field. When reset, the MAC routes the received Tagged AV Control packets based on the tag priority matching the PSRQ fields in MAC_RxQ_Ctrl2 and MAC_RxQ_Ctrl3 registers. 0b - Tagged AV Control Packets Queuing is disabled 1b - Tagged AV Control Packets Queuing is enabled
20 MCBCQEN	Multicast and Broadcast Queue Enable This bit specifies that Multicast or Broadcast packets routing to the Rx Queue is enabled and the Multicast or Broadcast packets must be routed to Rx Queue specified in MCBCQ field. 0b - Multicast and Broadcast Queue is disabled 1b - Multicast and Broadcast Queue is enabled
19 —	Reserved.
18-16 MCBCQ	Multicast and Broadcast Queue This field specifies the Rx Queue onto which Multicast or Broadcast Packets are routed. Any Rx Queue enabled for Generic/DCB/AV traffic can be used to route the Multicast or Broadcast Packets. 000b - Receive Queue 0 001b - Receive Queue 1 010b - Receive Queue 2 011b - Receive Queue 3

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Field	Function
	100b - Receive Queue 4 101b - Receive Queue 5 110b - Receive Queue 6 111b - Receive Queue 7
15 —	Reserved.
14-12 UPQ	Untagged Packet Queue This field indicates the Rx Queue to which Untagged Packets are to be routed. Any Rx Queue enabled for Generic/DCB/AV traffic can be used to route the Untagged Packets. 000b - Receive Queue 0 001b - Receive Queue 1 010b - Receive Queue 2 011b - Receive Queue 3 100b - Receive Queue 4 101b - Receive Queue 5 110b - Receive Queue 6 111b - Receive Queue 7
11 —	Reserved.
10-8 DCBCPQ	DCB Control Packets Queue This field specifies the Rx queue on which the received DCB control packets are routed. The DCB data packets are routed based on the PSRQ field of the Transmit Flow Control Register of corresponding queue. 000b - Receive Queue 0 001b - Receive Queue 1 010b - Receive Queue 2 011b - Receive Queue 3 100b - Receive Queue 4 101b - Receive Queue 5 110b - Receive Queue 6 111b - Receive Queue 7
7	Reserved.

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Field	Function
—	
6-4 PTPQ	PTP Packets Queue This field specifies the Rx queue on which the PTP packets sent over the Ethernet payload (not over IPv4 or IPv6) are routed. When the AV8021ASMEN bit of MAC_Timestamp_Control register is set, only untagged PTP over Ethernet packets are routed on an Rx Queue. If the bit is not set, then based on programming of TPQC field, both tagged and untagged PTPoE packets can be routed to this Rx Queue. 000b - Receive Queue 0 001b - Receive Queue 1 010b - Receive Queue 2 011b - Receive Queue 3 100b - Receive Queue 4 101b - Receive Queue 5 110b - Receive Queue 6 111b - Receive Queue 7
3 —	Reserved.
2-0 AVCPQ	AV Untagged Control Packets Queue This field specifies the Receive queue on which the received AV tagged and untagged control packets are routed. The AV tagged (when TACPQE bit is set) and untagged control packets are routed to Receive queue specified by this field. 000b - Receive Queue 0 001b - Receive Queue 1 010b - Receive Queue 2 011b - Receive Queue 3 100b - Receive Queue 4 101b - Receive Queue 5 110b - Receive Queue 6 111b - Receive Queue 7

1.4.30 (MAC_RxQ_Ctrl2)

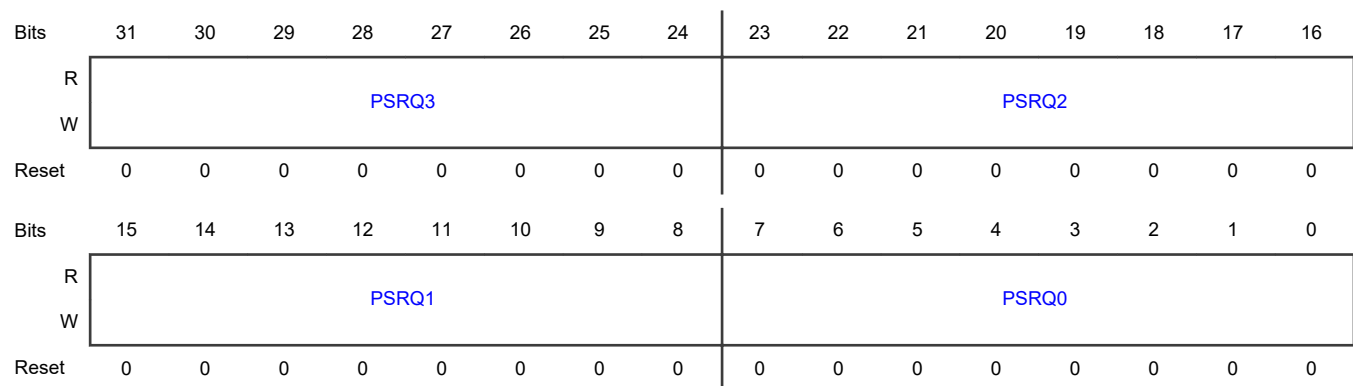
Offset

Register	Offset
MAC_RxQ_Ctrl2	A8h

Function

This register controls the routing of tagged packets based on the USP (user Priority) field of the received packets to the RxQueues 0 to 3.

Diagram



Fields

Field	Function
31-24 PSRQ3	Priorities Selected in the Receive Queue 3 This field decides the priorities assigned to Rx Queue 3. All packets with priorities that match the values set in this field are routed to Rx Queue 3. For example, if PSRQ3[6, 3] are set, packets with USP field equal to 3 or 6 are routed to Rx Queue 3. The software must ensure that the content of this field is mutually exclusive to the PSRQ fields for other queues, that is, the same priority is not mapped to multiple Rx queues. this field also determines the priorities to be included in the PFC packet sent to remote station when Rx Queue 3 crosses the flow control threshold settings.
23-16 PSRQ2	Priorities Selected in the Receive Queue 2 This field decides the priorities assigned to Rx Queue 2. All packets with priorities that match the values set in this field are routed to Rx Queue 2. For example, if PSRQ2[1, 0] are set, packets with USP field equal to 1 or 0 are routed to Rx Queue 2. The software must ensure that the content of this field is mutually exclusive to the PSRQ fields for other queues, that is, the same priority is not mapped to multiple Rx queues.

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Field	Function
	this field also determines the priorities to be included in the PFC packet sent to remote station when Rx Queue 2 crosses the flow control threshold settings.
15-8 PSRQ1	Priorities Selected in the Receive Queue 1 This field decides the priorities assigned to Rx Queue 1. All packets with priorities that match the values set in this field are routed to Rx Queue 1. For example, if PSRQ1[4] is set, packets with USP field equal to 4 are routed to Rx Queue 1. The software must ensure that the content of this field is mutually exclusive to the PSRQ fields for other queues, that is, the same priority is not mapped to multiple Rx queues. this field also determines the priorities to be included in the PFC packet sent to remote station when Rx Queue 1 crosses the flow control threshold settings.
7-0 PSRQ0	Priorities Selected in the Receive Queue 0 This field decides the priorities assigned to Rx Queue 0. All packets with priorities that match the values set in this field are routed to Rx Queue 0. For example, if PSRQ0[5] is set, packets with USP field equal to 5 are routed to Rx Queue 0. The software must ensure that the content of this field is mutually exclusive to the PSRQ fields for other queues, that is, the same priority is not mapped to multiple Rx queues. this field also determines the priorities to be included in the PFC packet sent to remote station when Rx Queue 0 crosses the flow control threshold settings.

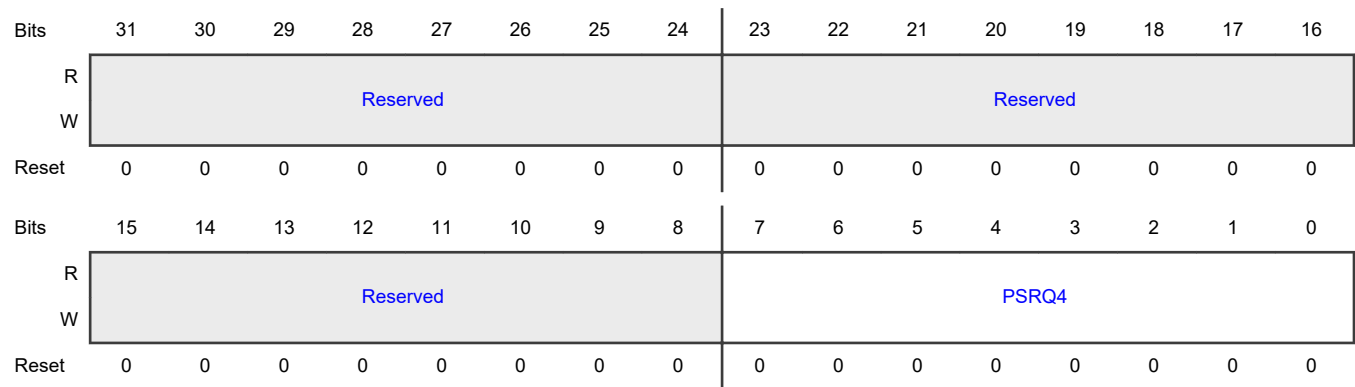
1.4.31 (MAC_RxQ_Ctrl3)

Offset

Register	Offset
MAC_RxQ_Ctrl3	ACh

Function

This register controls the routing of tagged packets based on the USP (user Priority) field of the received packets to the RxQueues 4 to 7.

Diagram**Fields**

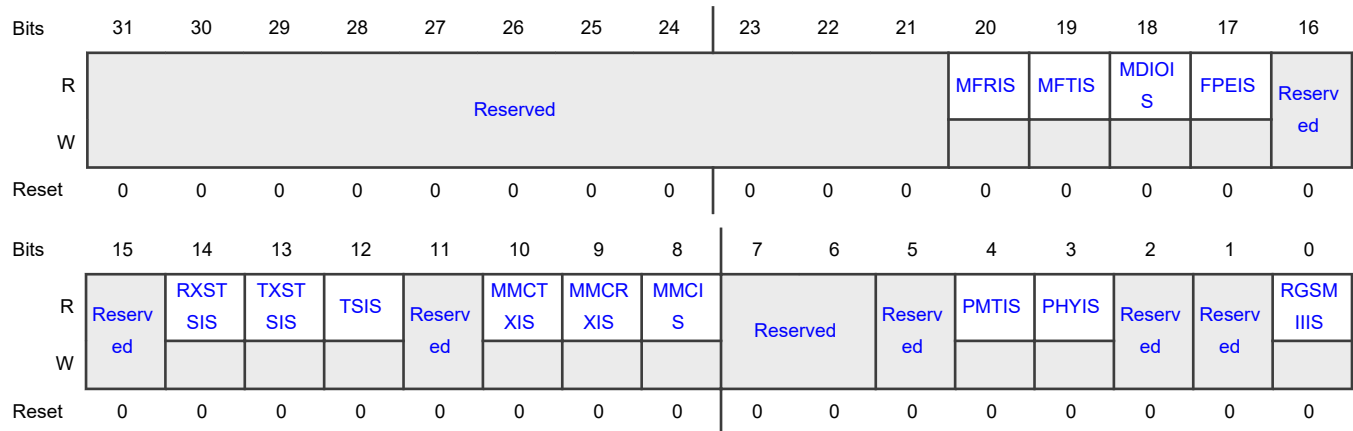
Field	Function
31-24 —	Reserved.
23-16 —	Reserved.
15-8 —	Reserved.
7-0 PSRQ4	Priorities Selected in the Receive Queue 4 This field decides the priorities assigned to Rx Queue 4. All packets with priorities that match the values set in this field are routed to Rx Queue 4. For example, if PSRQ4[7:4] is set, packets with USP field equal to 7, 6, 5, or 4 are routed to Rx Queue 4. The software must ensure that the content of this field is mutually exclusive to the PSRQ fields for other queues, that is, the same priority is not mapped to multiple Rx queues. this field also determines the priorities to be included in the PFC packet sent to remote station when Rx Queue 4 crosses the flow control threshold settings.

1.4.32 (MAC_Interrupt_Status)**Offset**

Register	Offset
MAC_Interrupt_Status	B0h

Function

The Interrupt Status register contains the status of interrupts.

Diagram**Fields**

Field	Function
31-21 —	Reserved.
20 MFRIS	MMC FPE Receive Interrupt Status This bit is set high when an interrupt is generated in the MMC FPE Receive Interrupt Register. This bit is cleared when all bits in this interrupt register are cleared. This bit is valid only when you select the Enable MAC Management Counters (MMC) option along with FPE support. 0b - MMC FPE Receive Interrupt status not active 1b - MMC FPE Receive Interrupt status active
19 MFTIS	MMC FPE Transmit Interrupt Status This bit is set high when an interrupt is generated in the MMC FPE Transmit Interrupt Register. This bit is cleared when all bits in this interrupt register are cleared. This bit is valid only when you select the Enable MAC Management Counters (MMC) option along with FPE support. 0b - MMC FPE Transmit Interrupt status not active 1b - MMC FPE Transmit Interrupt status active
18 MDIOIS	MDIO Interrupt Status This bit indicates an interrupt event after the completion of MDIO operation. To reset this bit, the application has to read this bit/Write 1 to this bit when RCWE bit of MAC_CSR_SW_Ctrl register is set. Access restriction applies. Clears on read (or write of 1 when RCWE bit in MAC_CSR_SW_Ctrl register is set). Self-set to 1 on internal event. 0b - MDIO Interrupt status not active 1b - MDIO Interrupt status active
17 FPEIS	Frame Preemption Interrupt Status

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Field	Function
	This bit indicates an interrupt event during the operation of Frame Preemption (Bits[19:16] of MAC_FPE_CTRL_STS register is set). To reset this bit, the application must clear the event in MAC_FPE_CTRL_STS that has caused the Interrupt. 0b - Frame Preemption Interrupt status not active 1b - Frame Preemption Interrupt status active
16 —	Reserved.
15 —	Reserved.
14 RXSTIS	Receive Status Interrupt This bit indicates the status of received packets. This bit is set when the RWT bit is set in the MAC_Rx_Tx_Status register. This bit is cleared when the corresponding interrupt source bit is read (or corresponding interrupt source bit is written to 1 when RCWE bit of MAC_CSR_SW_Ctrl register is set) in the MAC_Rx_Tx_Status register. 0b - Receive Interrupt status not active 1b - Receive Interrupt status active
13 TXSTIS	Transmit Status Interrupt This bit indicates the status of transmitted packets. This bit is set when any of the following bits is set in the MAC_Rx_Tx_Status register: - Excessive Collision (EXCOL) - Late Collision (LCOL) - Excessive Deferral (EXDEF) - Loss of Carrier (LCARR) - No Carrier (NCARR) - Jabber Timeout (TJT) This bit is cleared when the corresponding interrupt source bit is read (or corresponding interrupt source bit is written to 1 when RCWE bit of MAC_CSR_SW_Ctrl register is set) in the MAC_Rx_Tx_Status register. 0b - Transmit Interrupt status not active 1b - Transmit Interrupt status active
12 TSIS	Timestamp Interrupt Status If the Timestamp feature is enabled, this bit is set when any of the following conditions is true: - The system time value is equal to or exceeds the value specified in the Target Time High and Low registers. - There is an overflow in the Seconds register. - The Target Time Error occurred, that is, programmed target time already elapsed.

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Field	Function
	If the Auxiliary Snapshot feature is enabled, this bit is set when the auxiliary snapshot trigger is asserted. In configurations other than EQOS_CORE, when drop transmit status is enabled in MTL, this bit is set when the captured transmit timestamp is updated in the MAC_Tx_Timestamp_Status_Nanoseconds and Mac_TxTimestamp_Status_Seconds registers. When PTP offload feature is enabled, this bit is set when the captured transmit timestamp is updated in the MAC_Tx_Timestamp_Status_Nanoseconds and MAC_Tx_Timestamp_Status_Seconds registers, for PTO generated Delay Request and Pdelay request packets. This bit is cleared when the corresponding interrupt source bit is read (or corresponding interrupt source bit is written to 1 when RCWE bit of MAC_CSR_SW_Ctrl register is set) in the MAC_Timestamp_Status register. 0b - Timestamp Interrupt status not active 1b - Timestamp Interrupt status active
11 —	Reserved.
10 MMCTXIS	MMC Transmit Interrupt Status This bit is set high when an interrupt is generated in the MMC Transmit Interrupt Register. This bit is cleared when all bits in this interrupt register are cleared. This bit is valid only when you select the Enable MAC Management Counters (MMC) option. 0b - MMC Transmit Interrupt status not active 1b - MMC Transmit Interrupt status active
9 MMCRXIS	MMC Receive Interrupt Status This bit is set high when an interrupt is generated in the MMC Receive Interrupt Register. This bit is cleared when all bits in this interrupt register are cleared. This bit is valid only when you select the Enable MAC Management Counters (MMC) option. 0b - MMC Receive Interrupt status not active 1b - MMC Receive Interrupt status active
8 MMCIS	MMC Interrupt Status This bit is set high when Bit 11, Bit 10, or Bit 9 is set high. This bit is cleared only when all these bits are low. This bit is valid only when you select the Enable MAC Management Counters (MMC) option. 0b - MMC Interrupt status not active 1b - MMC Interrupt status active
7-6 —	Reserved.
5	Reserved.

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Field	Function
—	
4 PMTIS	PMT Interrupt Status This bit is set when a Magic packet or Wake-on-LAN packet is received in the power-down mode (RWKPRCVD and MGKPRCVD bits in MAC_PMT_Control_Status register). This bit is cleared when corresponding interrupt source bit are cleared because of a Read operation to the MAC_PMT_Control_Status register (or corresponding interrupt source bit of MAC_PMT_Control_Status register is written to 1 when RCWE bit of MAC_CSR_SW_Ctrl register is set). This bit is valid only when you select the Enable Power Management option. 0b - PMT Interrupt status not active 1b - PMT Interrupt status active
3 PHYIS	PHY Interrupt This bit is set when rising edge is detected on the phy_intr_i input. This bit is cleared when this register is read (or this bit is written to 1 when RCWE bit of MAC_CSR_SW_Ctrl register is set). 0b - PHY Interrupt not detected 1b - PHY Interrupt detected
2 —	Reserved.
1 —	Reserved.
0 RGSMIIS	RGMII or SMII Interrupt Status This bit is set because of any change in value of the Link Status of RGMII or SMII interface (LNKSTS bit in MAC_PHYIF_Control_Status register). This bit is cleared when the MAC_PHYIF_Control_Status register is read (or LNKSTS bit of MAC_PHYIF_Control_Status register is written to 1 when RCWE bit of MAC_CSR_SW_Ctrl register is set). This bit is valid only when you select the optional RGMII or SMII PHY interface. 0b - RGMII or SMII Interrupt Status is not active 1b - RGMII or SMII Interrupt Status is active

1.4.33 (MAC_Interrupt_Enable)

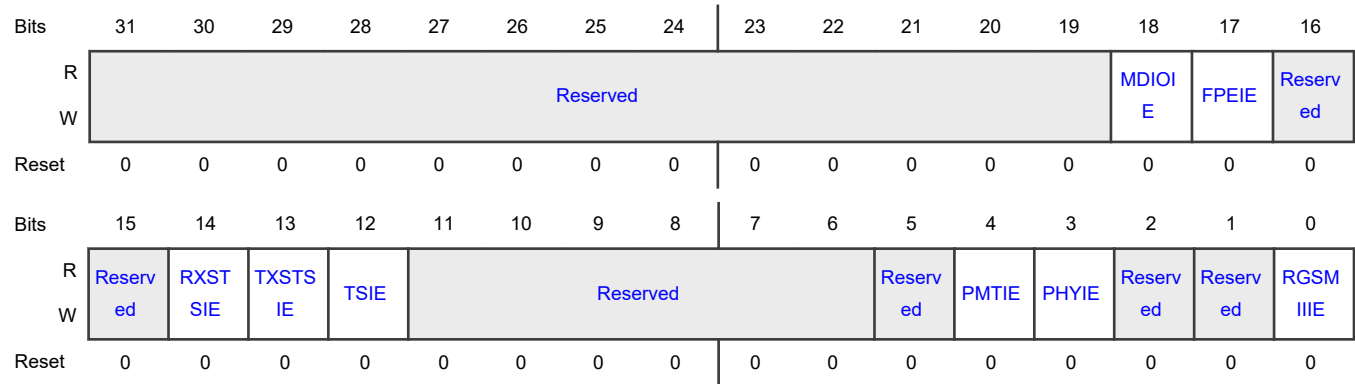
Offset

Register	Offset
MAC_Interrupt_Enable	B4h

Function

The Interrupt Enable register contains the masks for generating the interrupts.

Diagram



Fields

Field	Function
31-19 —	Reserved.
18 MDIOIE	MDIO Interrupt Enable When this bit is set, it enables the assertion of the interrupt when MDIOIS field is set in the MAC_Interrupt_Status register. 0b - MDIO Interrupt is disabled 1b - MDIO Interrupt is enabled
17 FPEIE	Frame Preemption Interrupt Enable When this bit is set, it enables the assertion of the interrupt when FPEIS field is set in the MAC_Interrupt_Status register. 0b - Frame Preemption Interrupt is disabled 1b - Frame Preemption Interrupt is enabled
16 —	Reserved.
15 —	Reserved.
14 RXSTSIE	Receive Status Interrupt Enable When this bit is set, it enables the assertion of the interrupt signal because of the setting of RXSTSIS bit in the MAC_Interrupt_Status register. 0b - Receive Status Interrupt is disabled

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Field	Function
	1b - Receive Status Interrupt is enabled
13 TXSTSIE	Transmit Status Interrupt Enable When this bit is set, it enables the assertion of the interrupt signal because of the setting of TXSTSIS bit in the MAC_Interrupt_Status register. 0b - Timestamp Status Interrupt is disabled 1b - Timestamp Status Interrupt is enabled
12 TSIE	Timestamp Interrupt Enable When this bit is set, it enables the assertion of the interrupt signal because of the setting of TSIS bit in MAC_Interrupt_Status register. 0b - Timestamp Interrupt is disabled 1b - Timestamp Interrupt is enabled
11-6 —	Reserved.
5 —	Reserved.
4 PMTIE	PMT Interrupt Enable When this bit is set, it enables the assertion of the interrupt signal because of the setting of PMTIS bit in MAC_Interrupt_Status register. 0b - PMT Interrupt is disabled 1b - PMT Interrupt is enabled
3 PHYIE	PHY Interrupt Enable When this bit is set, it enables the assertion of the interrupt signal because of the setting of PHYIS bit in MAC_Interrupt_Status register. 0b - PHY Interrupt is disabled 1b - PHY Interrupt is enabled
2 —	Reserved.
1 —	Reserved.
0 RGSMIIIE	RGMII or SMII Interrupt Enable When this bit is set, it enables the assertion of the interrupt signal because of the setting of RGSMIIIS bit in MAC_Interrupt_Status register.

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Field	Function
	0b - RGMII or SMI Interrupt is disabled 1b - RGMII or SMI Interrupt is enabled

1.4.34 (MAC_Rx_Tx_Status)

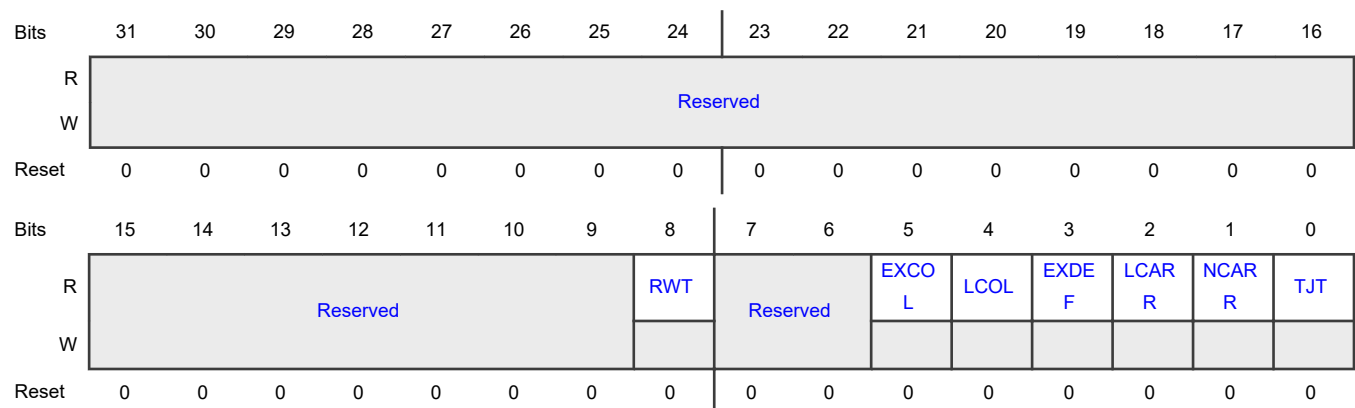
Offset

Register	Offset
MAC_Rx_Tx_Status	B8h

Function

The Receive Transmit Status register contains the Receive and Transmit Error status.

Diagram



Fields

Field	Function
31-9 —	Reserved.
8 RWT	Receive Watchdog Timeout This bit is set when a packet with length greater than 2,048 bytes is received (10, 240 bytes when Jumbo Packet mode is enabled) and the WD bit is reset in the MAC_Configuration register. This bit is set when a packet with length greater than 16,383 bytes is received and the WD bit is set in the MAC_Configuration register. Access restriction applies. Clears on read (or write of 1 when RCWE bit in MAC_CSR_SW_Ctrl register is set). Self-set to 1 on internal event.

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Field	Function
	0b - No receive watchdog timeout 1b - Receive watchdog timed out
7-6 —	Reserved.
5 EXCOL	Excessive Collisions When the DTXSTS bit is set in the MTL_Operation_Mode register, this bit indicates that the transmission aborted after 16 successive collisions while attempting to transmit the current packet. If the DR bit is set in the MAC_Configuration register, this bit is set after the first collision and the packet transmission is aborted. Access restriction applies. Clears on read (or write of 1 when RCWE bit in MAC_CSR_SW_Ctrl register is set). Self-set to 1 on internal event. 0b - No collision 1b - Excessive collision is sensed
4 LCOL	Late Collision When the DTXSTS bit is set in the MTL_Operation_Mode register, this bit indicates that the packet transmission aborted because a collision occurred after the collision window (64 bytes including Preamble in MII mode; 512 bytes including Preamble and Carrier Extension in GMII mode). This bit is not valid if the Underflow error occurs. Access restriction applies. Clears on read (or write of 1 when RCWE bit in MAC_CSR_SW_Ctrl register is set). Self-set to 1 on internal event. 0b - No collision 1b - Late collision is sensed
3 EXDEF	Excessive Deferral When the DTXSTS bit is set in the MTL_Operation_Mode register and the DC bit is set in the MAC_Configuration register, this bit indicates that the transmission ended because of excessive deferral of over 24,288 bit times (155,680 in 1000/2500 Mbps mode or when Jumbo packet is enabled). Access restriction applies. Clears on read (or write of 1 when RCWE bit in MAC_CSR_SW_Ctrl register is set). Self-set to 1 on internal event. 0b - No Excessive deferral 1b - Excessive deferral
2 LCARR	Loss of Carrier When the DTXSTS bit is set in the MTL_Operation_Mode register, this bit indicates that the loss of carrier occurred during packet transmission, that is, the phy_crs_i signal was inactive for one or more transmission clock periods during packet transmission. This bit is valid only for packets transmitted without collision. Access restriction applies. Clears on read (or write of 1 when RCWE bit in MAC_CSR_SW_Ctrl register is set). Self-set to 1 on internal event.

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Field	Function
	0b - Carrier is present 1b - Loss of carrier
1 NCARR	No Carrier When the DTXSTS bit is set in the MTL_Operation_Mode register, this bit indicates that the carrier signal from the PHY is not present at the end of preamble transmission. Access restriction applies. Clears on read (or write of 1 when RCWE bit in MAC_CSR_SW_Ctrl register is set). Self-set to 1 on internal event. 0b - Carrier is present 1b - No carrier
0 TJT	Transmit Jabber Timeout This bit indicates that the Transmit Jabber Timer expired which happens when the packet size exceeds 2,048 bytes (10,240 bytes when the Jumbo packet is enabled) and JD bit is reset in the MAC_Configuration register. This bit is set when the packet size exceeds 16,383 bytes and the JD bit is set in the MAC_Configuration register. Access restriction applies. Clears on read (or write of 1 when RCWE bit in MAC_CSR_SW_Ctrl register is set). Self-set to 1 on internal event. 0b - No Transmit Jabber Timeout 1b - Transmit Jabber Timeout occurred

1.4.35 (MAC_PMT_Control_Status)

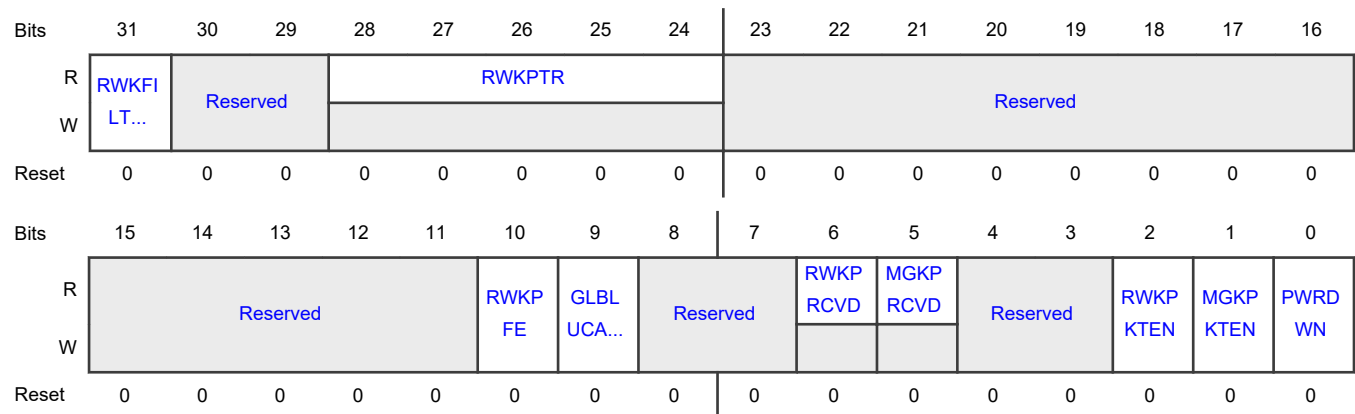
Offset

Register	Offset
MAC_PMT_Control_Status	C0h

Function

The PMT Control and Status Register.

Diagram



Fields

Field	Function
31 RWKFILTRST	Remote Wake-Up Packet Filter Register Pointer Reset When this bit is set, the remote wake-up packet filter register pointer is reset to 3'b000. It is automatically cleared after 1 clock cycle. Access restriction applies. Setting 1 sets. Self-cleared. Setting 0 has no effect. 0b - Remote Wake-Up Packet Filter Register Pointer is not Reset 1b - Remote Wake-Up Packet Filter Register Pointer is Reset
30-29 —	Reserved.
28-24 RWKPTR	Remote Wake-up FIFO Pointer This field gives the current value (0 to 7, 15, or 31 when 4, 8, or 16 Remote Wake-up Packet Filters are selected) of the Remote Wake-up Packet Filter register pointer. When the value of this pointer is equal to maximum for the selected number of Remote Wake-up Packet Filters, the contents of the Remote Wake-up Packet Filter Register are transferred to the clk_rx_i domain when a Write occurs to that register.
23-11 —	Reserved.
10 RWKPFE	Remote Wake-up Packet Forwarding Enable When this bit is set along with RWKPKTEN, the MAC receiver drops all received frames until it receives the expected Wake-up frame. All frames after that event including the received wake-up frame are forwarded to application. This bit is then self-cleared on receiving the wake-up packet. The application can also clear this bit before the expected wake-up frame is received. In such cases, the MAC reverts to the default behavior where packets received are forwarded to the application. This bit must only be set when RWKPKTEN is set high and PWRDWN is set low. The setting of this bit has no effect when PWRDWN is set high.

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Field	Function
	Note: If Magic Packet Enable and Wake-Up Frame Enable are both set along with setting of this bit and Magic Packet is received prior to wake-up frame, this bit is self-cleared on receiving Magic Packet, the received Magic packet is dropped, and all frames after received Magic Packet are forwarded to application. Access restriction applies. Setting 1 sets. Self-cleared. Setting 0 has no effect. 0b - Remote Wake-up Packet Forwarding is disabled 1b - Remote Wake-up Packet Forwarding is enabled
9 GLBLUCAST	Global Unicast When this bit set, any unicast packet filtered by the MAC (DAF) address recognition is detected as a remote wake-up packet. 0b - Global unicast is disabled 1b - Global unicast is enabled
8-7 —	Reserved.
6 RWKPRCVD	Remote Wake-Up Packet Received When this bit is set, it indicates that the power management event is generated because of the reception of a remote wake-up packet. This bit is cleared when this register is read. Access restriction applies. Clears on read (or write of 1 when RCWE bit in MAC_CSR_SW_Ctrl register is set). Self-set to 1 on internal event. 0b - Remote wake-up packet is received 1b - Remote wake-up packet is received
5 MGKPRCVD	Magic Packet Received When this bit is set, it indicates that the power management event is generated because of the reception of a magic packet. This bit is cleared when this register is read. Access restriction applies. Clears on read (or write of 1 when RCWE bit in MAC_CSR_SW_Ctrl register is set). Self-set to 1 on internal event. 0b - No Magic packet is received 1b - Magic packet is received
4-3 —	Reserved.
2 RWKPKTEN	Remote Wake-Up Packet Enable When this bit is set, a power management event is generated when the MAC receives a remote wake-up packet. 0b - Remote wake-up packet is disabled 1b - Remote wake-up packet is enabled

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Field	Function
1 MGKPKTEN	Magic Packet Enable When this bit is set, a power management event is generated when the MAC receives a magic packet. 0b - Magic Packet is disabled 1b - Magic Packet is enabled
0 PWRDWN	Power Down When this bit is set, the MAC receiver drops all received packets until it receives the expected magic packet or remote wake-up packet. This bit is then self-cleared and the power-down mode is disabled. The software can clear this bit before the expected magic packet or remote wake-up packet is received. The packets received by the MAC after this bit is cleared are forwarded to the application. This bit must only be set when the Magic Packet Enable, Global Unicast, or Remote Wake-Up Packet Enable bit is set high. Note: You can gate-off the CSR clock during the power-down mode. However, when the CSR clock is gated-off, you cannot perform any read or write operations on this register. Therefore, the Software cannot clear this bit. Access restriction applies. Setting 1 sets. Self-cleared. Setting 0 has no effect. 0b - Power down is disabled 1b - Power down is enabled

1.4.36 (MAC_RWK_Packet_Filter)

Offset

Register	Offset
MAC_RWK_Packet_Filter	C4h

Function

The Remote Wakeup Filter registers are implemented as 8, 16, or 32 indirect access registers (wkuppktfilter_reg#i) based on whether 4, 8, or 16 Remote Wakeup Filters are selected in the configuration and accessed by application through MAC_RWK_Packet_Filter register. When the Remote Wakeup Filters are to be programmed, the entire set of wkuppktfilter_reg registers must be written. The wkuppktfilter_reg register is programmed by sequentially writing the eight, sixteen or thirty-two register values in MAC_RWK_Packet_Filter register for wkuppktfilter_reg0, wkuppktfilter_reg1, ..., wkuppktfilter_reg31 respectively. The wkuppktfilter_reg register is read in a similar way. The MAC updates the wkuppktfilter_reg register current pointer value in RWKPTR field of MAC_PMT_Control_Status register.

The Remote Wakeup Filters are arranged in blocks of 4 filters each and each such block have eight 32-bit wide registers, viz. wkuppktfilter_reg0-7, wkuppktfilter_reg8-15, wkuppktfilter_reg16-23 and wkuppktfilter_reg24-31. The fields of Remote Wakeup Filter are described as follows:

Filter i Byte Mask: The filter i byte mask register defines the bytes of the packet that are examined by filter i (0, 1, 2, 3, ..., 15) to determine whether or not a packet is a wake-up packet.

- The MSB (31st bit) must be zero.
- Bit j[30:0] is the byte mask.

- If Bit j (byte number) of the byte mask is set, the CRC block processes the Filter i Offset + j of the incoming packet; otherwise Filter i Offset + j is ignored.

Filter i Command: The 4-bit filter i command controls the filter i operation.

- Bit 3 specifies the address type, defining the destination address type of the pattern. When the bit is set, the pattern applies to only multicast packets; when the bit is reset, the pattern applies only to unicast packet.

- Bit 2 (Inverse Mode), when set, reverses the logic of the CRC16 hash function signal, to reject a packet with matching CRC-16 value. Bit 2, along with Bit 1, allows a MAC to reject a subset of remote wake-up packets by creating filter logic such as "Pattern 1 AND NOT Pattern 2".

- Bit 1 (And_Previous) implements the Boolean logic. When set, the result of the current entry is logically ANDed with the result of the previous filter. This AND logic allows a filter pattern longer than 32 bytes by splitting the mask among two, three, or four filters. This depends on the number of filters that have the And_Previous bit set. The details are provided below:

-- The And_Previous bit setting is applicable within a set of 4 filters.

-- Setting of And_Previous bit of filter that is not enabled has no effect, that is setting And_Previous bit of lowest number filter in the set of 4 filters has no effect. For example, setting of And_Previous bit of Filter 0 has no effect.

-- If And_Previous bit is set for filter to form AND chained filter, the AND chain breaks at the point any filter is not enabled. For example: If Filter 2 And_Previous bit is set (bit 1 in Filter 2 command is set) but Filter 1 is not enabled (bit 0 in Filter 1 command is reset), then only Filter 2 result is considered. If Filter 2 And_Previous bit is set (bit 1 in Filter 2 command is set), Filter 3 And_Previous bit is set (bit 1 in Filter 3 command is set), but Filter 1 is not enabled (bit 0 in Filter 1 command is reset), then only Filter 2 result ANDed with Filter 3 result is considered. If Filter 2 And_Previous bit is set (bit 1 in Filter 2 command is set), Filter 3 And_Previous bit is set (bit 1 in Filter 3 command is set), but Filter 2 is not enabled (bit 0 in Filter 2 command is reset), then since setting of Filter 2 And_Previous bit has no effect only Filter 1 result ORed with Filter 3 result is considered.

-- If filters chained by And_Previous bit setting have complementary programming, then a frame might never pass the AND chained filter. For example, if Filter 2 And_Previous bit is set (bit 1 in Filter 2 command is set), Filter 1 Address_Type bit is set (bit 3 in Filter 1 command is set) indicating multicast detection and Filter 2 Address_Type bit is reset (bit 3 in Filter 2 command is reset) indicating unicast detection or vice versa, a remote wakeup frame does not pass the AND chained filter as a remote wakeup frame cannot be of both unicast and multicast address type.

- Bit 0 is the enable for filter i . If Bit 0 is not set, filter i is disabled.

Filter i Offset: The filter i offset register defines the offset (within the packet) from which the filter i examines the packets.

- This 8-bit pattern-offset is the offset for the filter i first byte to be examined.

- The minimum allowed offset is 12, which refers to the 13th byte of the packet.

- The offset value 0 refers to the first byte of the packet.

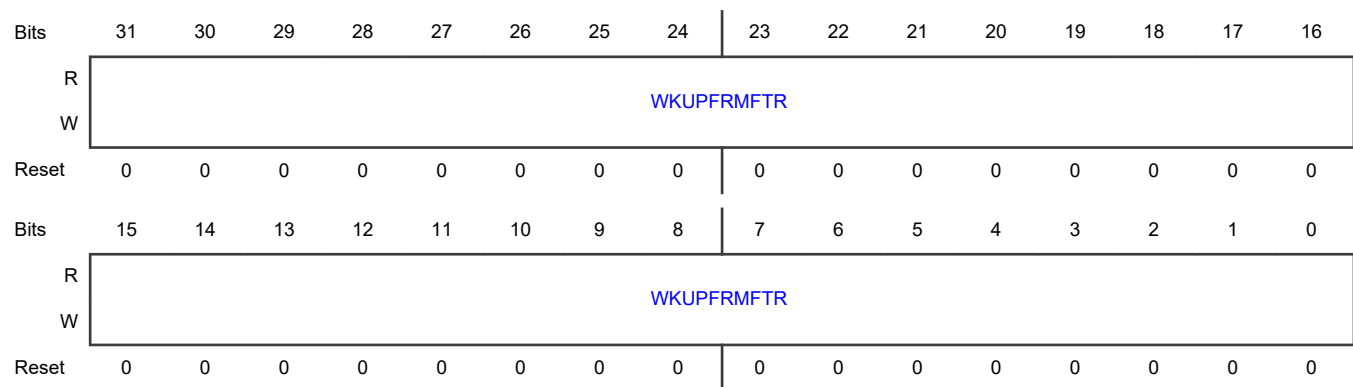
Filter i CRC-16: The filter i CRC-16 register contains the CRC-16 value calculated from the pattern and the byte mask programmed in the Remote Wakeup filter register.

- The 16-bit CRC calculation uses the following polynomial: $G(x) = x^{16} + x^{15} + x^2 + 1$

- Each mask, used in the hash function calculation, is compared with a 16-bit value associated with that mask. Each filter has the following:

-- 32-bit Mask: Each bit in this mask corresponds to one byte in the detected packet. If the bit is 1, the corresponding byte is taken into the CRC-16 calculation.

-- 8-bit Offset Pointer: Specifies the byte to start the CRC-16 computation. The pointer and the mask are used together to locate the bytes to be used in the CRC-16 calculations.

Diagram**Fields**

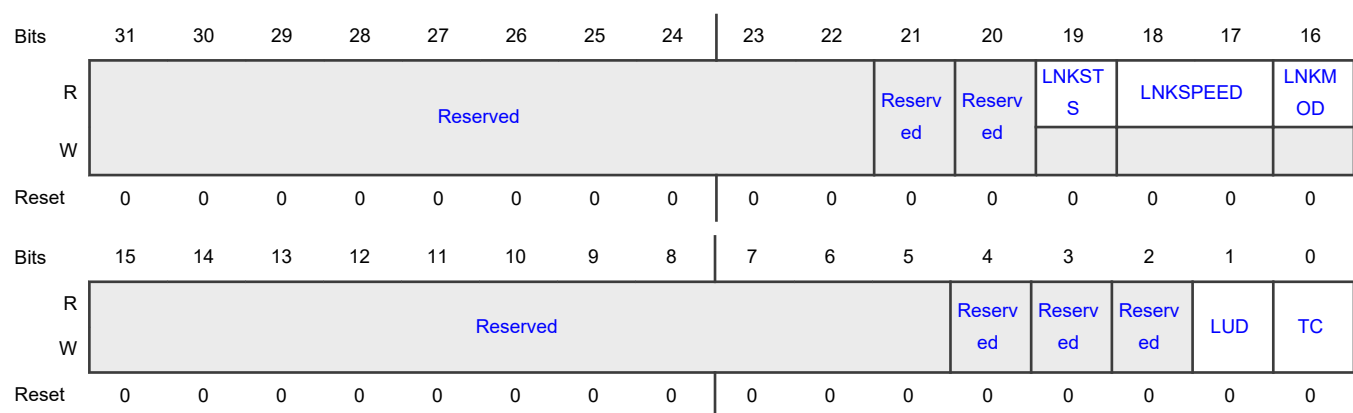
Field	Function
31-0	RWK Packet Filter
WKUPFRMFTR	This field contains the various controls of RWK Packet filter.

1.4.37 (MAC_PHYIF_Control_Status)**Offset**

Register	Offset
MAC_PHYIF_Control_Status	F8h

Function

The PHY Interface Control and Status register indicates the status signals received by the SGMII, RGMII, or SMII interface (selected at reset) from the PHY. This register is optional.

Diagram

Fields

Field	Function
31-22 —	Reserved.
21 —	Reserved.
20 —	Reserved.
19 LNKSTS	Link Status This bit indicates whether the link is up (1'b1) or down (1'b0). 0b - Link down 1b - Link up
18-17 LNKSPEED	Link Speed This bit indicates the current speed of the link. 00b - 2.5 MHz 01b - 25 MHz 10b - 125 MHz 11b - Reserved
16 LNKMOD	Link Mode This bit indicates the current mode of operation of the link. 0b - Half-duplex mode 1b - Full-duplex mode
15-5 —	Reserved.
4 —	Reserved.
3 —	Reserved.
2 —	Reserved.
1 LUD	Link Up or Down This bit indicates whether the link is up or down during transmission of configuration in the RGMII, SGMII, or SMII interface.

Table continues on the next page...

Table continued from the previous page...

Field	Function
	0b - Link down 1b - Link up
0 TC	Transmit Configuration in RGMII, SGMII, or SMII When set, this bit enables the transmission of duplex mode, link speed, and link up or down information to the PHY in the RGMII, SMII, or SGMII port. When this bit is reset, no such information is driven to the PHY. The details of this feature are provided in the following sections: - "Reduced Gigabit Media Independent Interface" - "Serial Media Independent Interface" - "Serial Gigabit Media Independent Interface" 0b - Disable Transmit Configuration in RGMII, SGMII, or SMII 1b - Enable Transmit Configuration in RGMII, SGMII, or SMII

1.4.38 (MAC_Version)

Offset

Register	Offset
MAC_Version	110h

Function

The version register identifies the version of the module.

Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	Reserved															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	CFGVER								IPVER							
W																
Reset	0	0	0	1	0	0	0	0	0	1	0	1	0	0	0	1

Fields

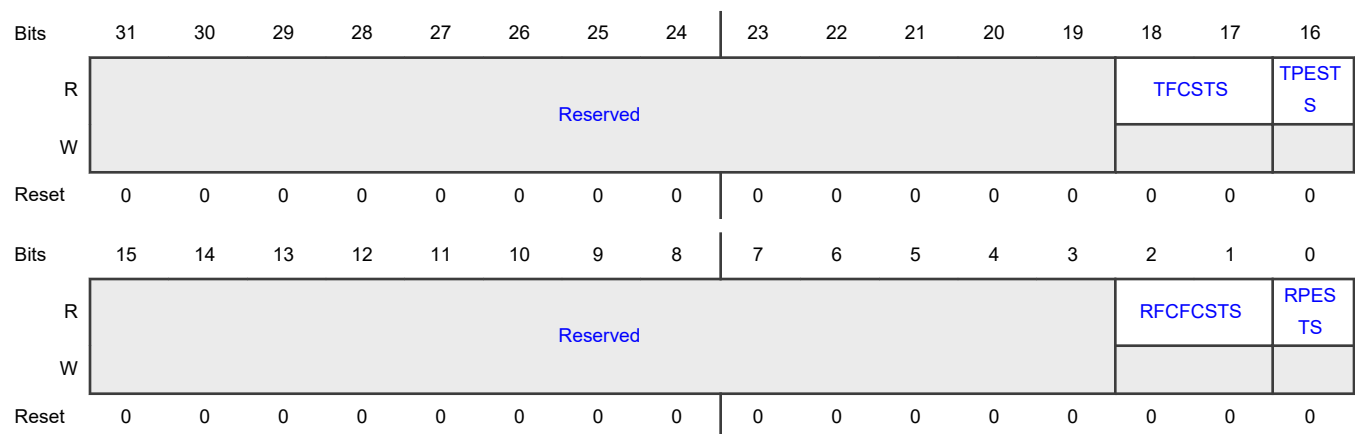
Field	Function
31-16 —	Reserved.
15-8 CFGVER	IP configuration version
7-0 IPVER	IP version

1.4.39 (MAC_Debug)**Offset**

Register	Offset
MAC_Debug	114h

Function

The Debug register provides the debug status of various MAC blocks.

Diagram**Fields**

Field	Function
31-19 —	Reserved.
18-17	MAC Transmit Packet Controller Status

Table continues on the next page...

Table continued from the previous page...

Field	Function
TFCSTS	This field indicates the state of the MAC Transmit Packet Controller module. 00b - Idle state 01b - Waiting for one of the following: Status of the previous packet OR IPG or back off period to be over 10b - Generating and transmitting a Pause control packet (in full-duplex mode) 11b - Transferring input packet for transmission
16 TPESTS	MAC GMII or MII Transmit Protocol Engine Status When this bit is set, it indicates that the MAC GMII or MII transmit protocol engine is actively transmitting data, and it is not in the Idle state. 0b - MAC GMII or MII Transmit Protocol Engine Status not detected 1b - MAC GMII or MII Transmit Protocol Engine Status detected
15-3 —	Reserved.
2-1 RFCFCSTS	MAC Receive Packet Controller FIFO Status When this bit is set, this field indicates the active state of the small FIFO Read and Write controllers of the MAC Receive Packet Controller module.
0 RPESTS	MAC GMII or MII Receive Protocol Engine Status When this bit is set, it indicates that the MAC GMII or MII receive protocol engine is actively receiving data, and it is not in the Idle state. 0b - MAC GMII or MII Receive Protocol Engine Status not detected 1b - MAC GMII or MII Receive Protocol Engine Status detected

1.4.40 (MAC_HW_Feature0)

Offset

Register	Offset
MAC_HW_Feature0	11Ch

Function

This register indicates the presence of first set of the optional features or functions of the DWC_ether_qos. The software driver can use this register to dynamically enable or disable the programs related to the optional blocks.

Note: All bits are set or reset according to the features selected while configuring the core in coreConsultant.

Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	Reserv ed	ACTPHYSEL			SAVLA NI...	TSSTSSEL		MACA DR6...	MACA DR3...	ADDMACADRSEL					Reserv ed	RXCO ESEL
W																
Reset	0	0	0	0	1	1	1	0	0	0	0	0	0	1	0	1
Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	Reserv ed	TXCO ESEL	EEES EL	TSSEL	Reserved		ARPO FFS...	MMCS EL	MGKS EL	RWKS EL	SMAS EL	VLHA SH	PCSS EL	HDSE L	GMIIS EL	MIISE L
W																
Reset	0	1	0	1	0	0	1	1	1	1	1	1	0	1	1	1

Fields

Field	Function
31 —	Reserved.
30-28 ACTPHYSEL	Active PHY Selected When you have multiple PHY interfaces in your configuration, this field indicates the sampled value of phy_intf_sel_i during reset de-assertion. 000b - GMII or MII 001b - RGMII 010b - SGMII 011b - TBI 100b - RMII 101b - RTBI 110b - SMII 111b - RevMII
27 SAVLANS	Source Address or VLAN Insertion Enable This bit is set to 1 when the Enable SA and VLAN Insertion on Tx option is selected 0b - Source Address or VLAN Insertion Enable option is not selected 1b - Source Address or VLAN Insertion Enable option is selected
26-25 TSSTSSEL	Timestamp System Time Source This bit indicates the source of the Timestamp system time: This bit is set to 1 when the Enable IEEE 1588 Timestamp Support option is selected 00b - Internal 01b - External

Table continues on the next page...

Table continued from the previous page...

Field	Function
	10b - Both 11b - Reserved
24 MACADR64SEL	MAC Addresses 64-127 Selected This bit is set to 1 when the Enable Additional 64 MAC Address Registers (64-127) option is selected 0b - MAC Addresses 64-127 Select option is not selected 1b - MAC Addresses 64-127 Select option is selected
23 MACADR32SEL	MAC Addresses 32-63 Selected This bit is set to 1 when the Enable Additional 32 MAC Address Registers (32-63) option is selected 0b - MAC Addresses 32-63 Select option is not selected 1b - MAC Addresses 32-63 Select option is selected
22-18 ADDMACADRS EL	MAC Addresses 1-31 Selected This bit is set to 1 when the non-zero value is selected for Enable Additional 1-31 MAC Address Registers option
17 —	Reserved.
16 RXCOESEL	Receive Checksum Offload Enabled This bit is set to 1 when the Enable Receive TCP/IP Checksum Check option is selected 0b - Receive Checksum Offload Enable option is not selected 1b - Receive Checksum Offload Enable option is selected
15 —	Reserved.
14 TXCOESEL	Transmit Checksum Offload Enabled This bit is set to 1 when the Enable Transmit TCP/IP Checksum Insertion option is selected 0b - Transmit Checksum Offload Enable option is not selected 1b - Transmit Checksum Offload Enable option is selected
13 EEEESEL	Energy Efficient Ethernet Enabled This bit is set to 1 when the Enable Energy Efficient Ethernet (EEE) option is selected 0b - Energy Efficient Ethernet Enable option is not selected 1b - Energy Efficient Ethernet Enable option is selected
12 TSSEL	IEEE 1588-2008 Timestamp Enabled This bit is set to 1 when the Enable IEEE 1588 Timestamp Support option is selected

Table continues on the next page...

Table continued from the previous page...

Field	Function
	0b - IEEE 1588-2008 Timestamp Enable option is not selected 1b - IEEE 1588-2008 Timestamp Enable option is selected
11-10 —	Reserved.
9 ARPOFFSEL	ARP Offload Enabled This bit is set to 1 when the Enable IPv4 ARP Offload option is selected 0b - ARP Offload Enable option is not selected 1b - ARP Offload Enable option is selected
8 MMCSEL	RMON Module Enable This bit is set to 1 when the Enable MAC Management Counters (MMC) option is selected 0b - RMON Module Enable option is not selected 1b - RMON Module Enable option is selected
7 MGKSEL	PMT Magic Packet Enable This bit is set to 1 when the Enable Magic Packet Detection option is selected 0b - PMT Magic Packet Enable option is not selected 1b - PMT Magic Packet Enable option is selected
6 RWKSEL	PMT Remote Wake-up Packet Enable This bit is set to 1 when the Enable Remote Wake-Up Packet Detection option is selected 0b - PMT Remote Wake-up Packet Enable option is not selected 1b - PMT Remote Wake-up Packet Enable option is selected
5 SMASEL	SMA (MDIO) Interface This bit is set to 1 when the Enable Station Management (MDIO Interface) option is selected 0b - SMA (MDIO) Interface not selected 1b - SMA (MDIO) Interface selected
4 VLHASH	VLAN Hash Filter Selected This bit is set to 1 when the Enable VLAN Hash Table Based Filtering option is selected 0b - VLAN Hash Filter not selected 1b - VLAN Hash Filter selected
3 PCSSEL	PCS Registers (TBI, SGMII, or RTBI PHY interface) This bit is set to 1 when the TBI, SGMII, or RTBI PHY interface option is selected 0b - No PCS Registers (TBI, SGMII, or RTBI PHY interface)

Table continues on the next page...

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Field	Function
	1b - PCS Registers (TBI, SGMII, or RTBI PHY interface)
2 HDSEL	Half-duplex Support This bit is set to 1 when the half-duplex mode is selected 0b - No Half-duplex support 1b - Half-duplex support
1 GMIISEL	1000 Mbps Support This bit is set to 1 when 1000 Mbps is selected as the Mode of Operation 0b - No 1000 Mbps support 1b - 1000 Mbps support
0 MIISEL	10 or 100 Mbps Support This bit is set to 1 when 10/100 Mbps is selected as the Mode of Operation 0b - No 10 or 100 Mbps support 1b - 10 or 100 Mbps support

1.4.41 (MAC_HW_Feature1)

Offset

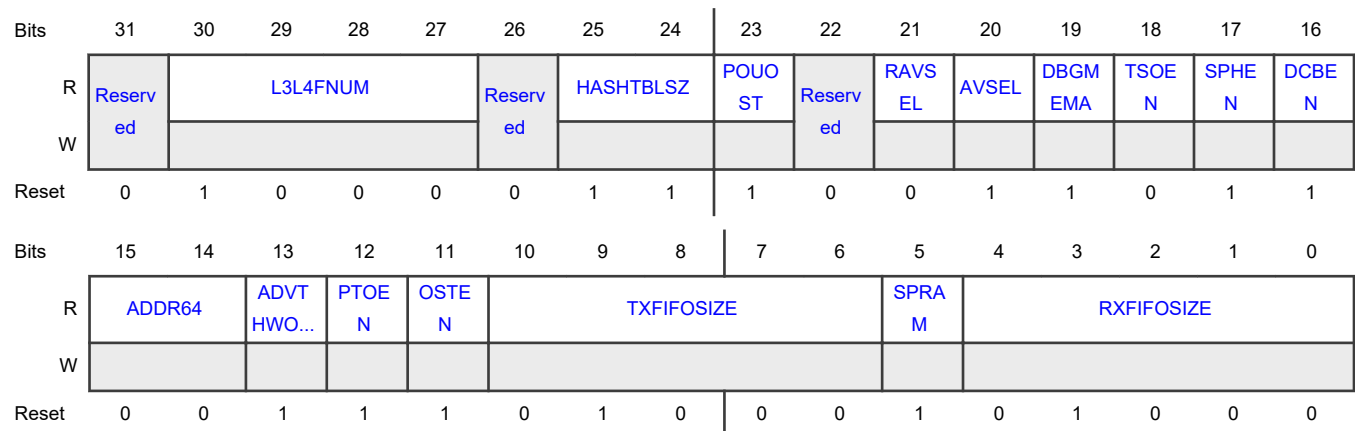
Register	Offset
MAC_HW_Feature1	120h

Function

This register indicates the presence of second set of the optional features or functions of the DWC_ether_qos. The software driver can use this register to dynamically enable or disable the programs related to the optional blocks.

Note: All bits are set or reset according to the features selected while configuring the core in coreConsultant.

Diagram



Fields

Field	Function
31 —	Reserved.
30-27 L3L4FNUM	Total number of L3 or L4 Filters This field indicates the total number of L3 or L4 filters: 0000b - No L3 or L4 Filter 0001b - 1 L3 or L4 Filter 0010b - 2 L3 or L4 Filters 0011b - 3 L3 or L4 Filters 0100b - 4 L3 or L4 Filters 0101b - 5 L3 or L4 Filters 0110b - 6 L3 or L4 Filters 0111b - 7 L3 or L4 Filters 1000b - 8 L3 or L4 Filters
26 —	Reserved.
25-24 HASHTBLSZ	Hash Table Size This field indicates the size of the hash table: 00b - No hash table 01b - 64 10b - 128 11b - 256

Table continues on the next page...

Table continued from the previous page...

Field	Function
23 POUOST	One Step for PTP over UDP/IP Feature Enable This bit is set to 1 when the Enable One step timestamp for PTP over UDP/IP feature is selected. 0b - One Step for PTP over UDP/IP Feature is not selected 1b - One Step for PTP over UDP/IP Feature is selected
22 —	Reserved.
21 RAVSEL	Rx Side Only AV Feature Enable This bit is set to 1 when the Enable Audio Video Bridging option on Rx Side Only is selected. 0b - Rx Side Only AV Feature is not selected 1b - Rx Side Only AV Feature is selected
20 AVSEL	AV Feature Enable This bit is set to 1 when the Enable Audio Video Bridging option is selected. 0b - AV Feature is not selected 1b - AV Feature is selected
19 DBGMEMA	DMA Debug Registers Enable This bit is set to 1 when the Debug Mode Enable option is selected 0b - DMA Debug Registers option is not selected 1b - DMA Debug Registers option is selected
18 TSOEN	TCP Segmentation Offload Enable This bit is set to 1 when the Enable TCP Segmentation Offloading for TCP/IP Packets option is selected 0b - TCP Segmentation Offload Feature is not selected 1b - TCP Segmentation Offload Feature is selected
17 SPHEN	Split Header Feature Enable This bit is set to 1 when the Enable Split Header Structure option is selected 0b - Split Header Feature is not selected 1b - Split Header Feature is selected
16 DCBEN	DCB Feature Enable This bit is set to 1 when the Enable Data Center Bridging option is selected 0b - DCB Feature is not selected 1b - DCB Feature is selected
15-14	Address Width.

Table continues on the next page...

Table continued from the previous page...

Field	Function
ADDR64	This field indicates the configured address width: 00b - 32 01b - 40 10b - 48 11b - Reserved
13 ADVTHWORD	IEEE 1588 High Word Register Enable This bit is set to 1 when the Add IEEE 1588 Higher Word Register option is selected 0b - IEEE 1588 High Word Register option is not selected 1b - IEEE 1588 High Word Register option is selected
12 PTOEN	PTP Offload Enable This bit is set to 1 when the Enable PTP Timestamp Offload Feature is selected. 0b - PTP Offload feature is not selected 1b - PTP Offload feature is selected
11 OSTEN	One-Step Timestamping Enable This bit is set to 1 when the Enable One-Step Timestamp Feature is selected. 0b - One-Step Timestamping feature is not selected 1b - One-Step Timestamping feature is selected
10-6 TXFIFOSIZE	MTL Transmit FIFO Size This field contains the configured value of MTL Tx FIFO in bytes expressed as Log to base 2 minus 7, that is, $\text{Log}_2(\text{TXFIFO_SIZE}) - 7$: 00000b - 128 bytes 00001b - 256 bytes 00010b - 512 bytes 00011b - 1024 bytes 00100b - 2048 bytes 00101b - 4096 bytes 00110b - 8192 bytes 00111b - 16384 bytes 01000b - 32 KB 01001b - 64 KB 01010b - 128 KB 01011b - Reserved

Table continues on the next page...

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Field	Function
5 SPRAM	Single Port RAM Enable This bit is set to 1 when the Use single port RAM Feature is selected. 0b - Single Port RAM feature is not selected 1b - Single Port RAM feature is selected
4-0 RXFIFOSIZE	MTL Receive FIFO Size This field contains the configured value of MTL Rx FIFO in bytes expressed as Log to base 2 minus 7, that is, $\text{Log}_2(\text{RXFIFO_SIZE}) - 7$: 00000b - 128 bytes 00001b - 256 bytes 00010b - 512 bytes 00011b - 1024 bytes 00100b - 2048 bytes 00101b - 4096 bytes 00110b - 8192 bytes 00111b - 16384 bytes 01000b - 32 KB 01001b - 64 KB 01010b - 128 KB 01011b - 256 KB 01100b - Reserved

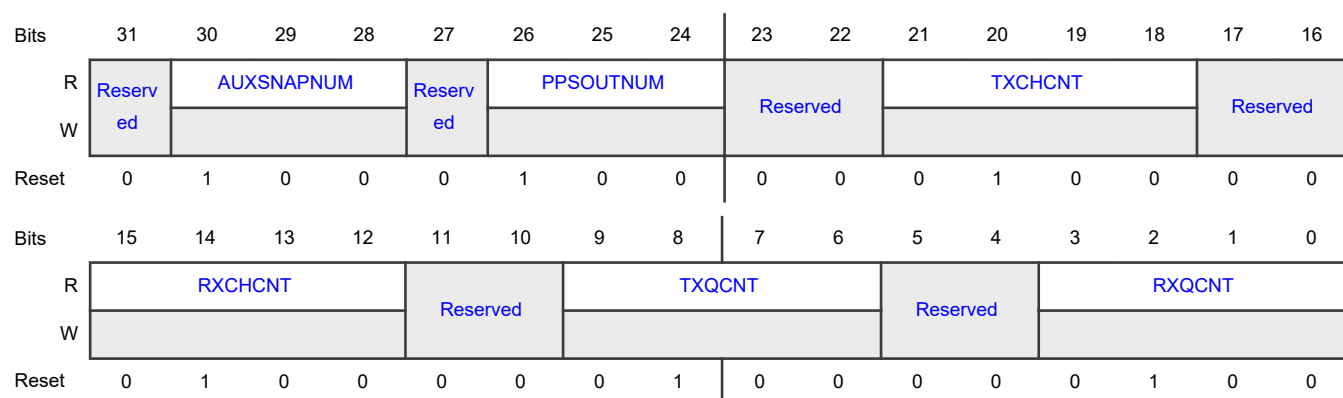
1.4.42 (MAC_HW_Feature2)

Offset

Register	Offset
MAC_HW_Feature2	124h

Function

This register indicates the presence of third set of the optional features or functions of the DWC_ether_qos. The software driver can use this register to dynamically enable or disable the programs related to the optional blocks.

Diagram**Fields**

Field	Function
31 —	Reserved.
30-28 AUXSNAPNUM	Number of Auxiliary Snapshot Inputs This field indicates the number of auxiliary snapshot inputs: 000b - No auxiliary input 001b - 1 auxiliary input 010b - 2 auxiliary input 011b - 3 auxiliary input 100b - 4 auxiliary input 101b - Reserved
27 —	Reserved.
26-24 PPSOUTNUM	Number of PPS Outputs This field indicates the number of PPS outputs: 000b - No PPS output 001b - 1 PPS output 010b - 2 PPS output 011b - 3 PPS output 100b - 4 PPS output 101b - Reserved
23-22 —	Reserved.

Table continues on the next page...

Table continued from the previous page...

Field	Function
21-18 TXHCNT	Number of DMA Transmit Channels This field indicates the number of DMA Transmit channels: 0000b - 1 MTL Tx Channel 0001b - 2 MTL Tx Channels 0010b - 3 MTL Tx Channels 0011b - 4 MTL Tx Channels 0100b - 5 MTL Tx Channels 0101b - 6 MTL Tx Channels 0110b - 7 MTL Tx Channels 0111b - 8 MTL Tx Channels
17-16 —	Reserved.
15-12 RXHCNT	Number of DMA Receive Channels This field indicates the number of DMA Receive channels: 0000b - 1 MTL Rx Channel 0001b - 2 MTL Rx Channels 0010b - 3 MTL Rx Channels 0011b - 4 MTL Rx Channels 0100b - 5 MTL Rx Channels 0101b - 6 MTL Rx Channels 0110b - 7 MTL Rx Channels 0111b - 8 MTL Rx Channels
11-10 —	Reserved.
9-6 TXQCNT	Number of MTL Transmit Queues This field indicates the number of MTL Transmit queues: 0000b - 1 MTL Tx Queue 0001b - 2 MTL Tx Queues 0010b - 3 MTL Tx Queues 0011b - 4 MTL Tx Queues 0100b - 5 MTL Tx Queues 0101b - 6 MTL Tx Queues

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Field	Function
	0110b - 7 MTL Tx Queues 0111b - 8 MTL Tx Queues
5-4 —	Reserved.
3-0 RXQCNT	Number of MTL Receive Queues This field indicates the number of MTL Receive queues: 0000b - 1 MTL Rx Queue 0001b - 2 MTL Rx Queues 0010b - 3 MTL Rx Queues 0011b - 4 MTL Rx Queues 0100b - 5 MTL Rx Queues 0101b - 6 MTL Rx Queues 0110b - 7 MTL Rx Queues 0111b - 8 MTL Rx Queues

1.4.43 (MAC_HW_Feature3)

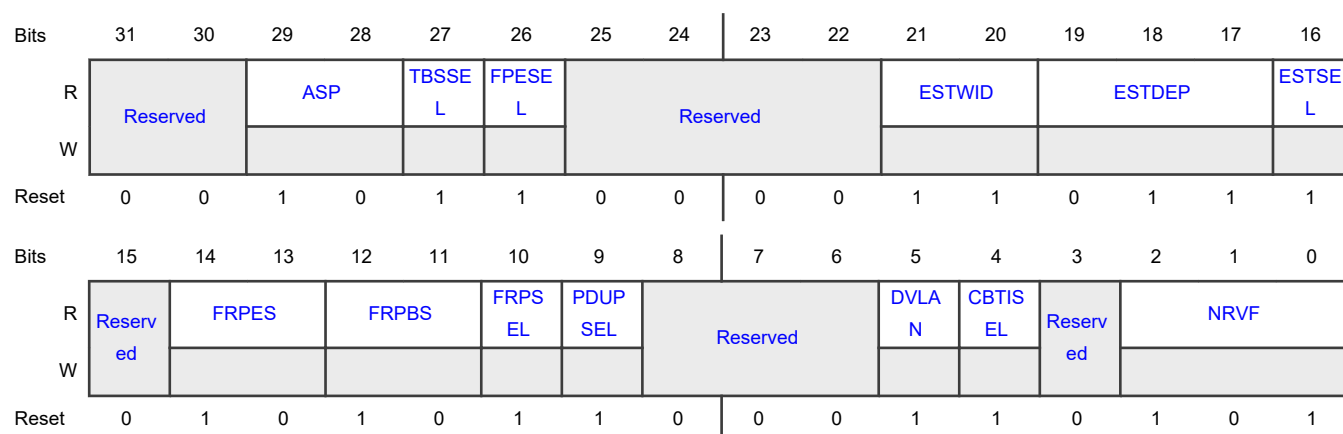
Offset

Register	Offset
MAC_HW_Feature3	128h

Function

This register indicates the presence of fourth set the optional features or functions of the DWC_ether_qos. The software driver can use this register to dynamically enable or disable the programs related to the optional blocks.

Diagram



Fields

Field	Function
31-30 —	Reserved.
29-28 ASP	Automotive Safety Package Following are the encoding for the different Safety features 00b - No Safety features selected 01b - Only "ECC protection for external memory" feature is selected 10b - All the Automotive Safety features are selected without the "Parity Port Enable for external interface" feature 11b - All the Automotive Safety features are selected with the "Parity Port Enable for external interface" feature
27 TBSSEL	Time Based Scheduling Enable This bit is set to 1 when the Time Based Scheduling feature is selected. 0b - Time Based Scheduling Enable feature is not selected 1b - Time Based Scheduling Enable feature is selected
26 FPESEL	Frame Preemption Enable This bit is set to 1 when the Enable Frame preemption feature is selected. 0b - Frame Preemption Enable feature is not selected 1b - Frame Preemption Enable feature is selected
25-22 —	Reserved.
21-20	Width of the Time Interval field in the Gate Control List

Table continues on the next page...

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Field	Function
ESTWID	This field indicates the width of the Configured Time Interval Field 00b - Width not configured 01b - 16 10b - 20 11b - 24
19-17 ESTDEP	Depth of the Gate Control List This field indicates the depth of Gate Control list expressed as $\text{Log}_2(\text{DWC_EQOS_EST_DEP})-5$ 000b - No Depth configured 001b - 64 010b - 128 011b - 256 100b - 512 101b - 1024 110b - Reserved
16 ESTSEL	Enhancements to Scheduling Traffic Enable This bit is set to 1 when the Enable Enhancements to Scheduling Traffic feature is selected. 0b - Enable Enhancements to Scheduling Traffic feature is not selected 1b - Enable Enhancements to Scheduling Traffic feature is selected
15 —	Reserved.
14-13 FRPES	Flexible Receive Parser Table Entries size This field indicates the Max Number of Parser Entries supported by Flexible Receive Parser. 00b - 64 Entries 01b - 128 Entries 10b - 256 Entries 11b - Reserved
12-11 FRPBS	Flexible Receive Parser Buffer size This field indicates the supported Max Number of bytes of the packet data to be Parsed by Flexible Receive Parser. 00b - 64 Bytes 01b - 128 Bytes 10b - 256 Bytes

Table continues on the next page...

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Field	Function
	11b - Reserved
10 FRPSEL	Flexible Receive Parser Selected This bit is set to 1 when the Enable Flexible Programmable Receive Parser option is selected. 0b - Flexible Receive Parser feature is not selected 1b - Flexible Receive Parser feature is selected
9 PDUPSEL	Broadcast/Multicast Packet Duplication This bit is set to 1 when the Broadcast/Multicast Packet Duplication feature is selected. 0b - Broadcast/Multicast Packet Duplication feature is not selected 1b - Broadcast/Multicast Packet Duplication feature is selected
8-6 —	Reserved.
5 DVLAN	Double VLAN Tag Processing Selected This bit is set to 1 when the Enable Double VLAN Processing Feature is selected. 0b - Double VLAN option is not selected 1b - Double VLAN option is selected
4 CBTISEL	Queue/Channel based VLAN tag insertion on Tx Enable This bit is set to 1 when the Enable Queue/Channel based VLAN tag insertion on Tx Feature is selected. 0b - Enable Queue/Channel based VLAN tag insertion on Tx feature is not selected 1b - Enable Queue/Channel based VLAN tag insertion on Tx feature is selected
3 —	Reserved.
2-0 NRVF	Number of Extended VLAN Tag Filters Enabled This field indicates the Number of Extended VLAN Tag Filters selected: 000b - No Extended Rx VLAN Filters 001b - 4 Extended Rx VLAN Filters 010b - 8 Extended Rx VLAN Filters 011b - 16 Extended Rx VLAN Filters 100b - 24 Extended Rx VLAN Filters 101b - 32 Extended Rx VLAN Filters 110b - Reserved

1.4.44 (MAC_DPP_FSM_Interrupt_Status)

Offset

Register	Offset
MAC_DPP_FSM_Interrupt_Status	140h

Function

This register contains the status of Automotive Safety related Data Path Parity Errors, Interface Timeout Errors, FSM State Parity Errors and FSM State Timeout Errors. All the non-Reserved bits are cleared on read.

Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16			
R	Reserved								FSMP	Reserved								Reserv	MSTT
W	Reserved								ES	Reserved								ed	ES
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0			

Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	Reserv	Reserv	Reserv	PTES	ATES	Reserv	RTES	TTES	Reserv	Reserv	ARPE	MTSP	MPES	RDPE	Reserv	Reserv
W	ed	ed	ed			ed			ed	ed	S	ES		S	ed	ed
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Fields

Field	Function
31-25 —	Reserved.
24 FSMPES	FSM State Parity Error Status This field when set indicates one of the FSMs State registers has a parity error detected. 0b - FSM State Parity Error Status not detected 1b - FSM State Parity Error Status detected
23-18 —	Reserved.
17 —	Reserved.
16 MSTTES	Master Read/Write Timeout Error Status

Table continues on the next page...

Table continued from the previous page...

Field	Function
	This field when set indicates that an Application/CSR Timeout has occurred on the master (AXI/AHB/ARI/ATI) interface. 0b - Master Read/Write Timeout Error Status not detected 1b - Master Read/Write Timeout Error Status detected
15 —	Reserved.
14 —	Reserved.
13 —	Reserved.
12 PTES	PTP FSM Timeout Error Status This field when set indicates that one of the PTP FSM Timeout has occurred. 0b - PTP FSM Timeout Error Status not detected 1b - PTP FSM Timeout Error Status detected
11 ATES	APP FSM Timeout Error Status This field when set indicates that one of the APP FSM Timeout has occurred. 0b - APP FSM Timeout Error Status not detected 1b - APP FSM Timeout Error Status detected
10 —	Reserved.
9 RTES	Rx FSM Timeout Error Status This field when set indicates that one of the Rx FSM Timeout has occurred. 0b - Rx FSM Timeout Error Status not detected 1b - Rx FSM Timeout Error Status detected
8 TTES	Tx FSM Timeout Error Status This field when set indicates that one of the Tx FSM Timeout has occurred. 0b - Tx FSM Timeout Error Status not detected 1b - Tx FSM Timeout Error Status detected
7 —	Reserved.

Table continues on the next page...

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Field	Function
6 —	Reserved.
5 ARPES	This bit when set indicates that a parity error is detected at checker. In AHB configuration, parity checker (PC6 as shown in Fig.Receive Data path Parity protection) at AHB master interface. 0b - Application Receive interface data path Parity Error Status not detected 1b - Application Receive interface data path Parity Error Status detected
4 MTSPES	MTL TX Status data path Parity checker Error Status This field when set indicates that, parity error is detected on the MTL TX Status data on ati interface (or at PC5 as shown in Transmit data path parity protection diagram). 0b - MTL TX Status data path Parity checker Error Status not detected 1b - MTL TX Status data path Parity checker Error Status detected
3 MPES	MTL data path Parity checker Error Status This bit when set indicates that a parity error is detected at the MTL transmit write controller parity checker (or at PC4 as shown in Transmit data path parity protection diagram). 0b - MTL data path Parity checker Error Status not detected 1b - MTL data path Parity checker Error Status detected
2 RDPES	Read Descriptor Parity checker Error Status This bit when set indicates that a parity error is detected at the DMA Read descriptor parity checker (or at PC3 as shown in Transmit data path parity protection diagram). 0b - Read Descriptor Parity checker Error Status not detected 1b - Read Descriptor Parity checker Error Status detected
1 —	Reserved.
0 —	Reserved.

1.4.45 (MAC_FSM_Control)

Offset

Register	Offset
MAC_FSM_Control	148h

Function

This register is used to control the FSM State parity and timeout error injection in Debug mode.

Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	Reserv ed	Reserv ed	Reserv ed	PLGR NML	ALGR NML	Reserv ed	RLGR NML	TLGR NML	Reserv ed	Reserv ed	Reserv ed	PPEIN	APEIN	Reserv ed	RPEIN	TPEIN
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	Reserv ed	Reserv ed	Reserv ed	PTEIN	ATEIN	Reserv ed	RTEIN	TTEIN	Reserved						PRTY EN	TMOU TEN
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Fields

Field	Function
31 —	Reserved.
30 —	Reserved.
29 —	Reserved.
28 PLGRNML	PTP Large/Normal Mode Select This field when set indicates that large mode tic generation is used for PTP domain, else normal mode tic generation is used. 0b - normal mode tic generation is used for PTP domain 1b - large mode tic generation is used for PTP domain
27 ALGRNML	APP Large/Normal Mode Select This field when set indicates that large mode tic generation is used for APP domain, else normal mode tic generation is used. 0b - normal mode tic generation is used for APP domain 1b - large mode tic generation is used for APP domain
26 —	Reserved.
25 RLGRNML	Rx Large/Normal Mode Select This field when set indicates that large mode tic generation is used for Rx domain, else normal mode tic generation is used. 0b - normal mode tic generation is used for Rx domain

Table continues on the next page...

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Field	Function
	1b - large mode tic generation is used for Rx domain
24 TLGRNML	Tx Large/Normal Mode Select This field when set indicates that large mode tic generation is used for Tx domain, else normal mode tic generation is used. 0b - normal mode tic generation is used for Tx domain 1b - large mode tic generation is used for Tx domain
23 —	Reserved.
22 —	Reserved.
21 —	Reserved.
20 PPEIN	PTP FSM Parity Error Injection This field when set indicates that Error Injection for PTP FSM Parity is enabled. 0b - PTP FSM Parity Error Injection is disabled 1b - PTP FSM Parity Error Injection is enabled
19 APEIN	APP FSM Parity Error Injection This field when set indicates that Error Injection for APP FSM Parity is enabled. 0b - APP FSM Parity Error Injection is disabled 1b - APP FSM Parity Error Injection is enabled
18 —	Reserved.
17 RPEIN	Rx FSM Parity Error Injection This field when set indicates that Error Injection for RX FSM Parity is enabled. 0b - Rx FSM Parity Error Injection is disabled 1b - Rx FSM Parity Error Injection is enabled
16 TPEIN	Tx FSM Parity Error Injection This field when set indicates that Error Injection for TX FSM Parity is enabled. 0b - Tx FSM Parity Error Injection is disabled 1b - Tx FSM Parity Error Injection is enabled

Table continues on the next page...

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Field	Function
15 —	Reserved.
14 —	Reserved.
13 —	Reserved.
12 PTEIN	PTP FSM Timeout Error Injection This field when set indicates that Error Injection for PTP FSM timeout is enabled. 0b - PTP FSM Timeout Error Injection is disabled 1b - PTP FSM Timeout Error Injection is enabled
11 ATEIN	APP FSM Timeout Error Injection This field when set indicates that Error Injection for APP FSM timeout is enabled. 0b - APP FSM Timeout Error Injection is disabled 1b - APP FSM Timeout Error Injection is enabled
10 —	Reserved.
9 RTEIN	Rx FSM Timeout Error Injection This field when set indicates that Error Injection for RX FSM timeout is enabled. 0b - Rx FSM Timeout Error Injection is disabled 1b - Rx FSM Timeout Error Injection is enabled
8 TTEIN	Tx FSM Timeout Error Injection This field when set indicates that Error Injection for TX FSM timeout is enabled. 0b - Tx FSM Timeout Error Injection is disabled 1b - Tx FSM Timeout Error Injection is enabled
7-2 —	Reserved.
1 PRTYEN	. This bit when set indicates that the FSM parity feature is enabled. 0b - FSM Parity feature is disabled 1b - FSM Parity feature is enabled

Table continues on the next page...

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Field	Function
0 TMOUTEN	This bit when set indicates that the FSM timeout feature is enabled. 0b - FSM timeout feature is disabled 1b - FSM timeout feature is enabled

1.4.46 (MAC_FSM_ACT_Timer)

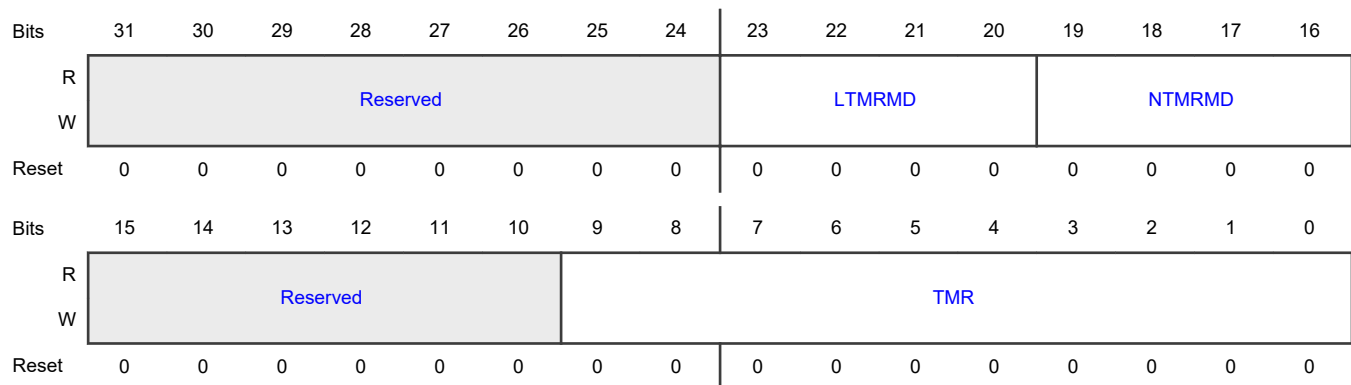
Offset

Register	Offset
MAC_FSM_ACT_Timer	14Ch

Function

This register is used to select the FSM and Interface Timeout values.

Diagram



Fields

Field	Function
31-24 —	Reserved.
23-20 LTMRMD	This field provides the mode value to be used for large mode FSM and other interface time outs. The timeout duration based on the mode value is given below 0000b - Timer disabled

Table continues on the next page...

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Field	Function
	0001b - 1us 0010b - 1.024ms (~4ms) 0011b - 16.384ms (~16ms) 0100b - 65.536ms (~64ms) 0101b - 262.144ms (~256ms) 0110b - 1.048sec (~1sec) 0111b - 4.194sec (~4sec) 1000b - 16.777sec (~16sec) 1001b - 33.554sec (~32sec) 1010b - 67.108sec (~64sec) 1011b - Reserved
19-16 NTMRMD	. This field provides the value to be used for normal mode FSM and other interface time outs. The timeout duration based on the mode value is given below 0000b - Timer disabled 0001b - 1us 0010b - 1.024ms (~4ms) 0011b - 16.384ms (~16ms) 0100b - 65.536ms (~64ms) 0101b - 262.144ms (~256ms) 0110b - 1.048sec (~1sec) 0111b - 4.194sec (~4sec) 1000b - 16.777sec (~16sec) 1001b - 33.554sec (~32sec) 1010b - 67.108sec (~64sec) 1011b - Reserved
15-10 —	Reserved.
9-0 TMR	. This field indicates the number of CSR clocks required to generate 1us tic.

1.4.47 (SCS_REG1)

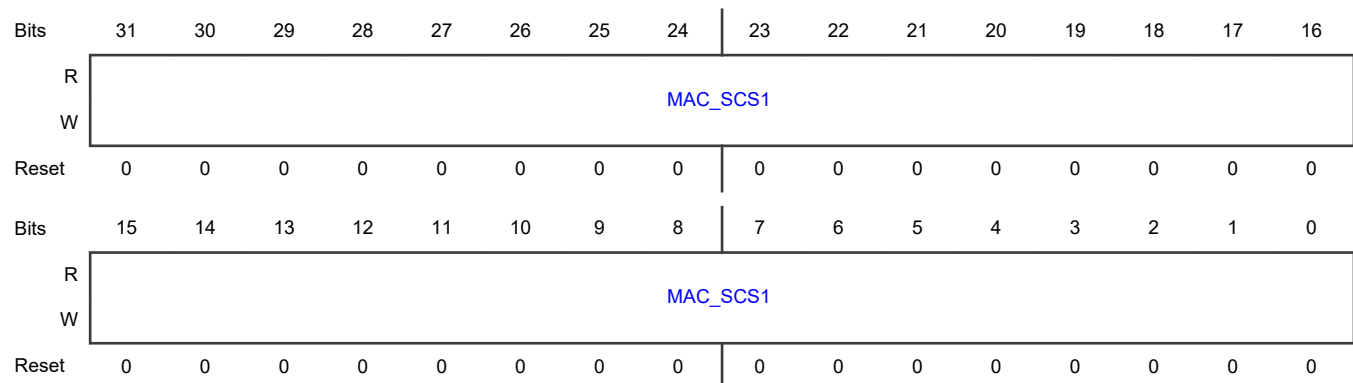
Offset

Register	Offset
SCS_REG1	150h

Function

NXP Reserved Register

Diagram



Fields

Field	Function
31-0	NXP Reserved, All the bits must be set to "0".
MAC_SCS1	This field is reserved for NXP Internal use, and must always be set to "0" unless instructed by NXP. Setting any bit to "1" might cause unexpected behavior in the IP.

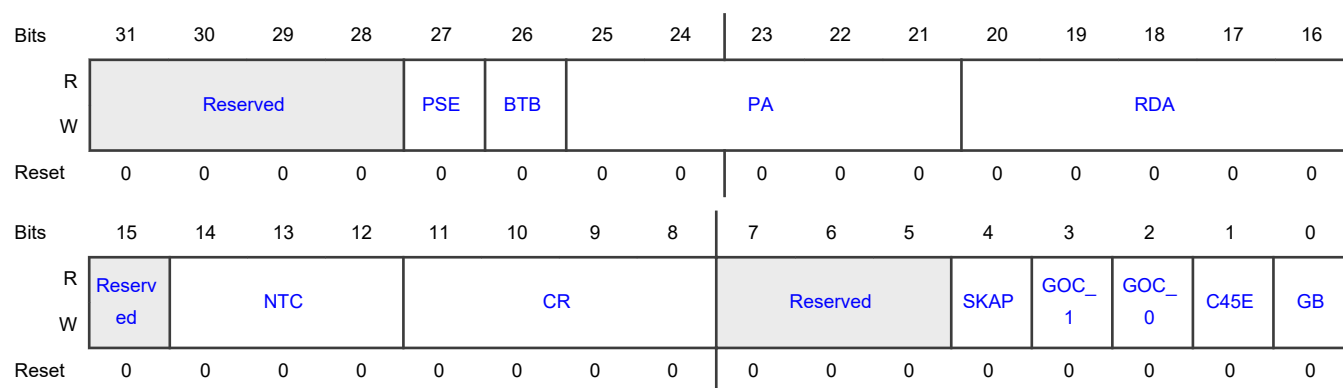
1.4.48 (MAC_MDIO_Address)

Offset

Register	Offset
MAC_MDIO_Address	200h

Function

The MDIO Address register controls the management cycles to external PHY through a management interface.

Diagram**Fields**

Field	Function
31-28 —	Reserved.
27 PSE	Preamble Suppression Enable When this bit is set, the SMA suppresses the 32-bit preamble and transmits MDIO frames with only 1 preamble bit. When this bit is 0, the MDIO frame always has 32 bits of preamble as defined in the IEEE specifications. 0b - Preamble Suppression disabled 1b - Preamble Suppression enabled
26 BTB	Back to Back transactions When this bit is set and the NTC has value greater than 0, then the MAC informs the completion of a read or write command at the end of frame transfer (before the trailing clocks are transmitted). The software can thus initiate the next command which is executed immediately irrespective of the number trailing clocks generated for the previous frame. When this bit is reset, then the read/write command completion (GB is cleared) only after the trailing clocks are generated. In this mode, it is ensured that the NTC is always generated after each frame. This bit must not be set when NTC=0. 0b - Back to Back transactions disabled 1b - Back to Back transactions enabled
25-21	Physical Layer Address

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Field	Function
PA	This field indicates which Clause 22 PHY devices (out of 32 devices) the MAC is accessing. This field indicates which Clause 45 capable PHYs (out of 32 PHYs) the MAC is accessing.
20-16 RDA	Register/Device Address These bits select the PHY register in selected Clause 22 PHY device. These bits select the Device (MMD) in selected Clause 45 capable PHY.
15 —	Reserved.
14-12 NTC	Number of Trailing Clocks This field controls the number of trailing clock cycles generated on gmii_mdc_o (MDC) after the end of transmission of MDIO frame. The valid values can be from 0 to 7. Programming the value to 3'h3 indicates that there are additional three clock cycles on the MDC line after the end of MDIO frame transfer.
11-8 CR	CSR Clock Range The CSR Clock Range selection determines the frequency of the MDC clock according to the CSR clock frequency used in your design: - 0000: CSR clock = 60-100 MHz; MDC clock = CSR clock/42 - 0001: CSR clock = 100-150 MHz; MDC clock = CSR clock/62 - 0010: CSR clock = 20-35 MHz; MDC clock = CSR clock/16 - 0011: CSR clock = 35-60 MHz; MDC clock = CSR clock/26 - 0100: CSR clock = 150-250 MHz; MDC clock = CSR clock/102 - 0101: CSR clock = 250-300 MHz; MDC clock = CSR clock/124 - 0110: CSR clock = 300-500 MHz; MDC clock = CSR clock/204 - 0111: CSR clock = 500-800 MHz; MDC clock = CSR clock/324 The suggested range of CSR clock frequency applicable for each value (when Bit 11 = 0) ensures that the MDC clock is approximately between 1.0 MHz to 2.5 MHz frequency range. When Bit 11 is set, you can achieve a higher frequency of the MDC clock than the frequency limit of 2.5 MHz (specified in the IEEE 802.3) and program a clock divider of lower value. For example, when CSR clock is of 100 MHz frequency and you program these bits as 1010, the resultant MDC clock is of 12.5 MHz which is above the range specified in IEEE 802.3. Program the following values only if the interfacing chips support faster MDC clocks: - 1000: CSR clock/4 - 1001: CSR clock/6 - 1010: CSR clock/8 - 1011: CSR clock/10 - 1100: CSR clock/12 - 1101: CSR clock/14

Table continues on the next page...

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Field	Function
	- 1110: CSR clock/16 - 1111: CSR clock/18 These bits are not used for accessing RevMII. These bits are read-only if the RevMII interface is selected as single PHY interface.
7-5 —	Reserved.
4 SKAP	Skip Address Packet When this bit is set, the SMA does not send the address packets before read, write, or post-read increment address packets. This bit is valid only when C45E is set. 0b - Skip Address Packet is disabled 1b - Skip Address Packet is enabled
3 GOC_1	GMII Operation Command 1 This bit is higher bit of the operation command to the PHY or RevMII, GOC_1 and GOC_O is encoded as follows: - 00: Reserved - 01: Write - 10: Post Read Increment Address for Clause 45 PHY - 11: Read When Clause 22 PHY or RevMII is enabled, only Write and Read commands are valid. 0b - GMII Operation Command 1 is disabled 1b - GMII Operation Command 1 is enabled
2 GOC_0	GMII Operation Command 0 This is the lower bit of the operation command to the PHY or RevMII. When in SMA mode (MDIO master) this bit along with GOC_1 determines the operation to be performed to the PHY. When only RevMII is selected in configuration this bit is read-only and tied to 1. 0b - GMII Operation Command 0 is disabled 1b - GMII Operation Command 0 is enabled
1 C45E	Clause 45 PHY Enable When this bit is set, Clause 45 capable PHY is connected to MDIO. When this bit is reset, Clause 22 capable PHY is connected to MDIO. 0b - Clause 45 PHY is disabled 1b - Clause 45 PHY is enabled
0	GMII Busy

Table continues on the next page...

Table continued from the previous page...

Field	Function
GB	The application sets this bit to instruct the SMA to initiate a Read or Write access to the MDIO slave. The MAC clears this bit after the MDIO frame transfer is completed. Hence the software must not write or change any of the fields in MAC_MDIO_Address and MAC_MDIO_Data registers as long as this bit is set. For write transfers, the application must first write 16-bit data in the GDI field (and also RA field when C45E is set) in MAC_MDIO_Data register before setting this bit. When C45E is set, it should also write into the RA field of MAC_MDIO_Data register before initiating a read transfer. When a read transfer is completed (GB=0), the data read from the PHY register is valid in the GD field of the MAC_MDIO_Data register. Note: Even if the addressed PHY is not present, there is no change in the functionality of this bit. Access restriction applies. Setting 1 sets. Self-cleared. Setting 0 has no effect. 0b - GMII Busy is disabled 1b - GMII Busy is enabled

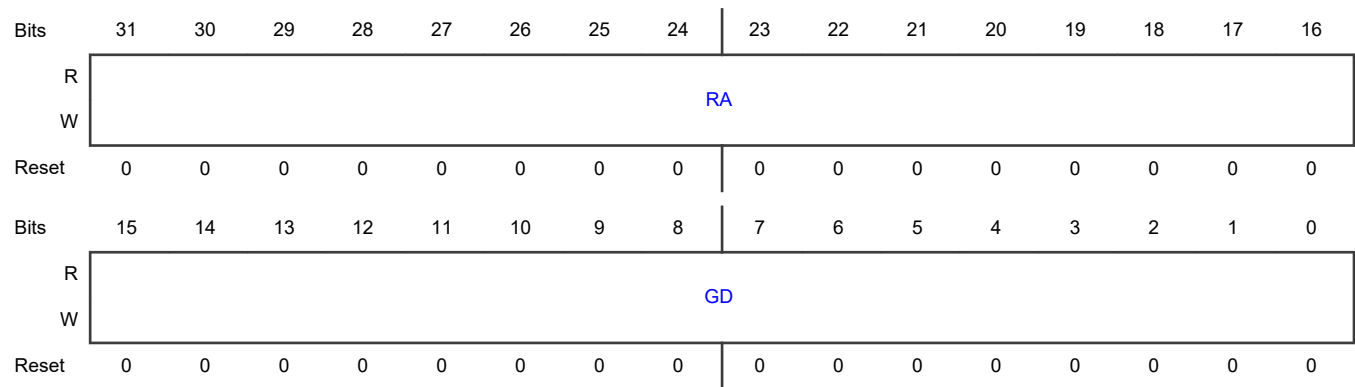
1.4.49 (MAC_MDIO_Data)

Offset

Register	Offset
MAC_MDIO_Data	204h

Function

The MDIO Data register stores the Write data to be written to the PHY register located at the address specified in MAC_MDIO_Address. This register also stores the Read data from the PHY register located at the address specified by MDIO Address register.

Diagram**Fields**

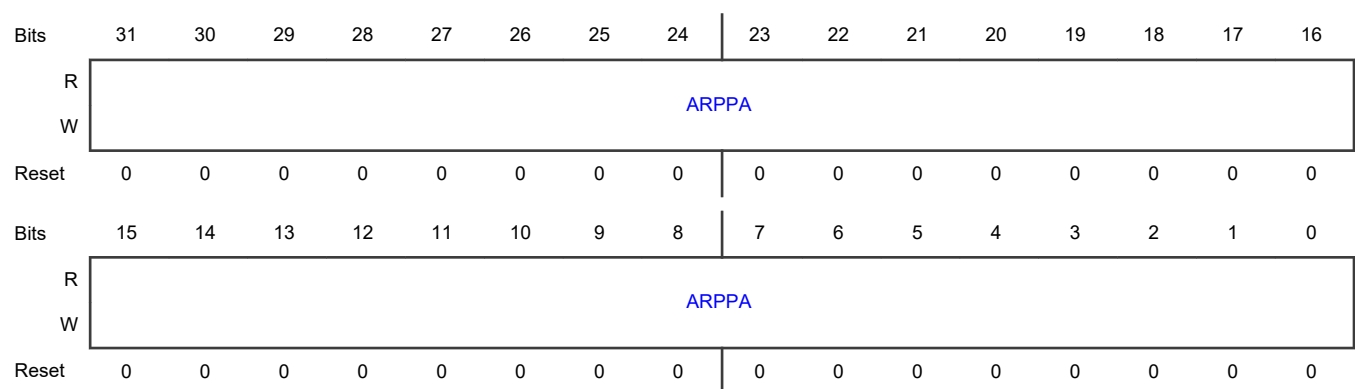
Field	Function
31-16 RA	Register Address This field is valid only when C45E is set. It contains the Register Address in the PHY to which the MDIO frame is intended for.
15-0 GD	GMII Data This field contains the 16-bit data value read from the PHY or RevMII after a Management Read operation or the 16-bit data value to be written to the PHY or RevMII before a Management Write operation.

1.4.50 (MAC_ARP_Address)**Offset**

Register	Offset
MAC_ARP_Address	210h

Function

The ARP Address register contains the IPv4 Destination Address of the MAC.

Diagram

Fields

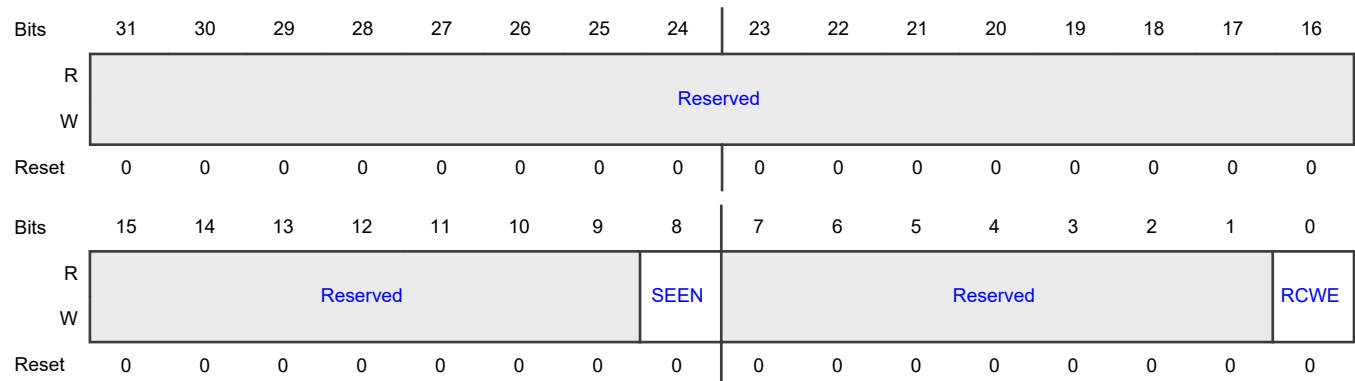
Field	Function
31-0	ARP Protocol Address
ARPPA	This field contains the IPv4 Destination Address of the MAC. This address is used for perfect match with the Protocol Address of Target field in the received ARP packet. This field is available only when the Enable IPv4 ARP Offload option is selected.

1.4.51 (MAC_CSR_SW_Ctrl)**Offset**

Register	Offset
MAC_CSR_SW_Ctrl	230h

Function

This register contains SW programmable controls for changing the CSR access response and status bits clearing.

Diagram**Fields**

Field	Function
31-9 —	Reserved.
8 SEEN	Slave Error Response Enable When this bit is set, the MAC responds with Slave Error for accesses to reserved registers in CSR space. When this bit is reset, the MAC responds with Okay response to any register accessed from CSR space. 0b - Slave Error Response is disabled 1b - Slave Error Response is enabled

Table continues on the next page...

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Field	Function
7-1 —	Reserved.
0 RCWE	Register Clear on Write 1 Enable When this bit is set, the access mode of some register fields changes to Clear on Write 1, the application needs to set that respective bit to 1 to clear it. When this bit is reset, the access mode of these register fields remain as Clear on Read. 0b - Register Clear on Write 1 is disabled 1b - Register Clear on Write 1 is enabled

1.4.52 (MAC_FPE_CTRL_STS)

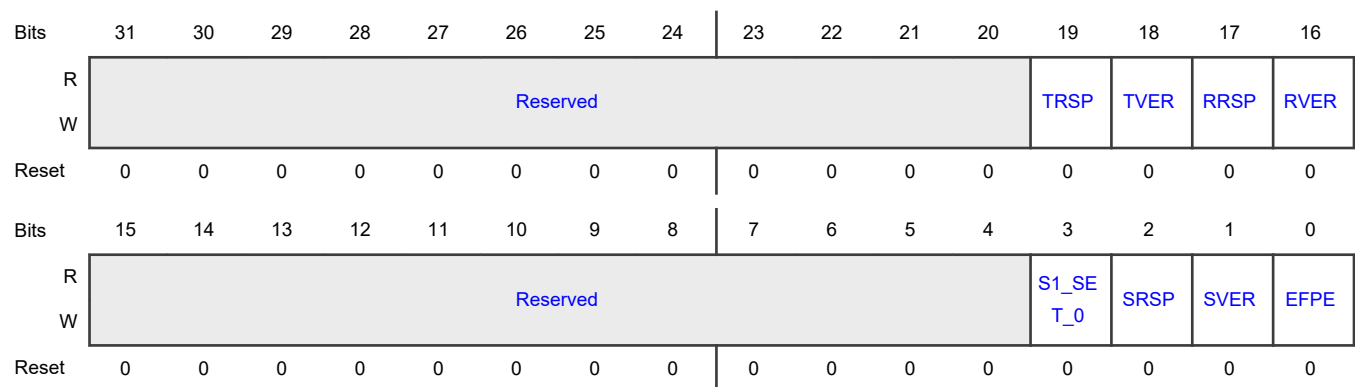
Offset

Register	Offset
MAC_FPE_CTRL_STS	234h

Function

This register controls the operation of Frame Preemption.

Diagram



Fields

Field	Function
31-20 —	Reserved.

Table continues on the next page...

Table continued from the previous page...

Field	Function
19 TRSP	Transmitted Respond Frame Set when a Respond mPacket is transmitted (triggered by setting SRSP field). An interrupt can be generated for this event if FPEIE bit of MAC_Interrupt_Enable is set. Access restriction applies. Clears on read (or write of 1 when RCWE bit in MAC_CSR_SW_Ctrl register is set). Self-set to 1 on internal event. 0b - Not transmitted Respond Frame 1b - transmitted Respond Frame
18 TVER	Transmitted Verify Frame Set when a Verify mPacket is transmitted (triggered by setting SVER field). An interrupt can be generated for this event if FPEIE bit of MAC_Interrupt_Enable is set. Access restriction applies. Clears on read (or write of 1 when RCWE bit in MAC_CSR_SW_Ctrl register is set). Self-set to 1 on internal event. 0b - Not transmitted Verify Frame 1b - transmitted Verify Frame
17 RRSP	Received Respond Frame Set when a Respond mPacket is received. An interrupt can be generated for this event if FPEIE bit of MAC_Interrupt_Enable is set. Access restriction applies. Clears on read (or write of 1 when RCWE bit in MAC_CSR_SW_Ctrl register is set). Self-set to 1 on internal event. 0b - Not received Respond Frame 1b - Received Respond Frame
16 RVER	Received Verify Frame Set when a Verify mPacket is received. An interrupt can be generated for this event if FPEIE bit of MAC_Interrupt_Enable is set. Access restriction applies. Clears on read (or write of 1 when RCWE bit in MAC_CSR_SW_Ctrl register is set). Self-set to 1 on internal event. 0b - Not received Verify Frame 1b - Received Verify Frame
15-4 —	Reserved.
3 S1_SET_0	NXP Reserved, Must be set to "0". This field is reserved for NXP Internal use, and must always be set to "0" unless instructed by NXP. Setting to "1" might cause unexpected behavior in the IP.
2	Send Respond mPacket

Table continues on the next page...

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Field	Function
SRSP	When set indicates hardware to send a Respond mPacket. Reset by hardware after sending the Respond mPacket. Access restriction applies. Setting 1 sets. Self-cleared. Setting 0 has no effect. 0b - Send Respond mPacket is disabled 1b - Send Respond mPacket is enabled
1 SVER	Send Verify mPacket When set indicates hardware to send a verify mPacket. Reset by hardware after sending the Verify mPacket. Access restriction applies. Setting 1 sets. Self-cleared. Setting 0 has no effect. 0b - Send Verify mPacket is disabled 1b - Send Verify mPacket is enabled
0 EFPE	Enable Tx Frame Preemption When set Frame Preemption Tx functionality is enabled. 0b - Tx Frame Preemption is disabled 1b - Tx Frame Preemption is enabled

1.4.53 (MAC_Ext_Cfg1)

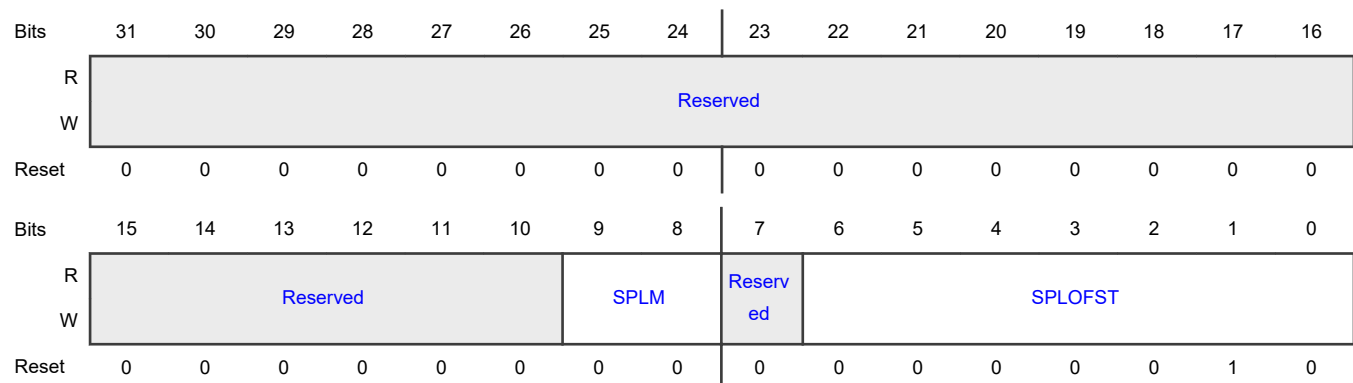
Offset

Register	Offset
MAC_Ext_Cfg1	238h

Function

This register contains Split mode control field and offset field for Split Header feature.

Diagram



Fields

Field	Function
31-10 —	Reserved.
9-8 SPLM	Split Mode These bits indicate the mode of splitting the incoming Rx packets. They are 00b - Split at L3/L4 header 01b - Split at L2 header with an offset. Always Split at SPLOFST bytes from the beginning of Length/Type field of the Frame 10b - Combination mode: Split similar to SPLM=00 for IP packets that are untagged or tagged and VLAN stripped 11b - Reserved
7 —	Reserved.
6-0 SPLOFST	Split Offset These bits indicate the value of offset from the beginning of Length/Type field at which header split should take place when the appropriate SPLM is selected. The reset value of this field is 2 bytes indicating a split at L2 header. Value is in terms of bytes.

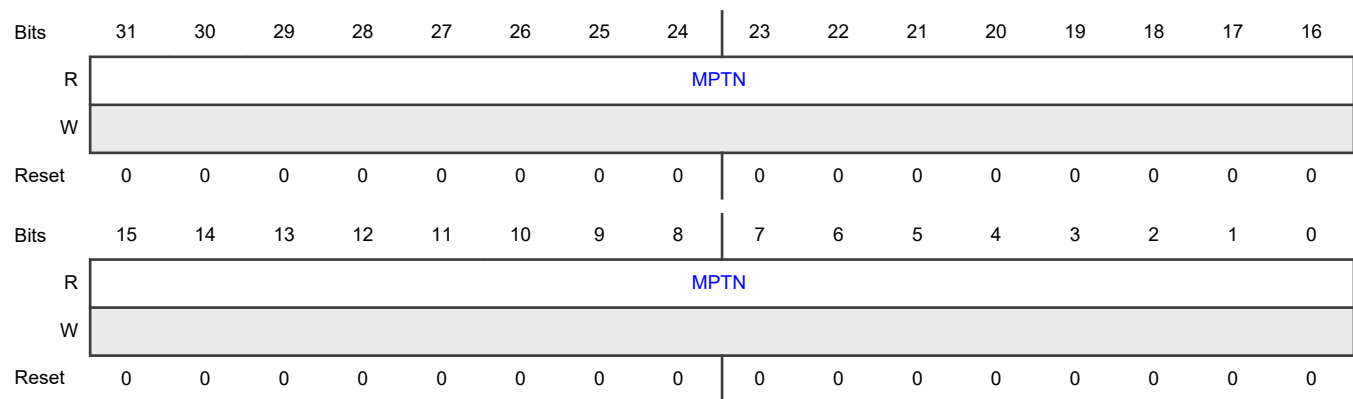
1.4.54 (MAC_Presn_Time_ns)**Offset**

Register	Offset
MAC_Presn_Time_ns	240h

Function

This register contains the 32-bit binary rollover equivalent time of the PTP System Time in ns

Exists when DWC_EQOS_FLEXI_PPS_OUT_EN is configured

Diagram**Fields**

Field	Function
31-0	MAC 1722 Presentation Time in ns
MPTN	These bits indicate the value of the 32-bit binary rollover equivalent time of the PTP System Time in ns

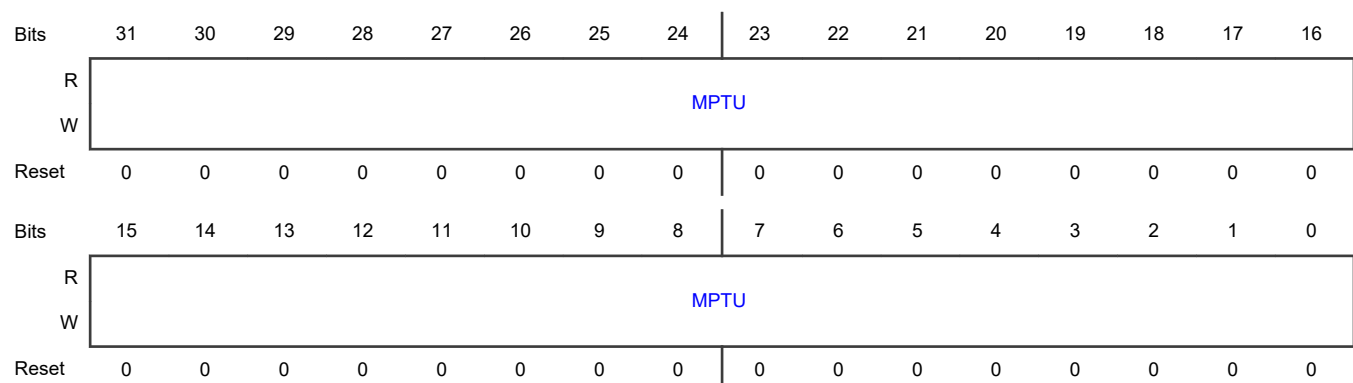
1.4.55 (MAC_Presn_Time_Updt)**Offset**

Register	Offset
MAC_Presn_Time_Updt	244h

Function

This field holds the 32-bit value of MAC 1722 Presentation Time in ns, that should be added to the Current Presentation Time Counter value. Init happens when TSINIT is set, and update happens when the TSUPDT bit is set (TSINIT and TSUPDT defined in MAC_Timestamp_Control register).

Exists when DWC_EQOS_FLEXI_PPS_OUT_EN is configured

Diagram

Fields

Field	Function
31-0 MPTU	MAC 1722 Presentation Time Update This field holds the init value or the update value for the presentation time. When used for update, this field holds the 32-bit value in ns, that should be added to the Current Presentation Time Counter value. Init happens when TSINIT is set, and update happens when the TSUPDT bit is set (TSINIT and TSUPDT defined in MAC_Timestamp_Control register). When ADDSUB field of MAC_System_Time_Nanoseconds_Update is set, this value is directly used for subtraction

1.4.56 (MAC_Address0_High)

Offset

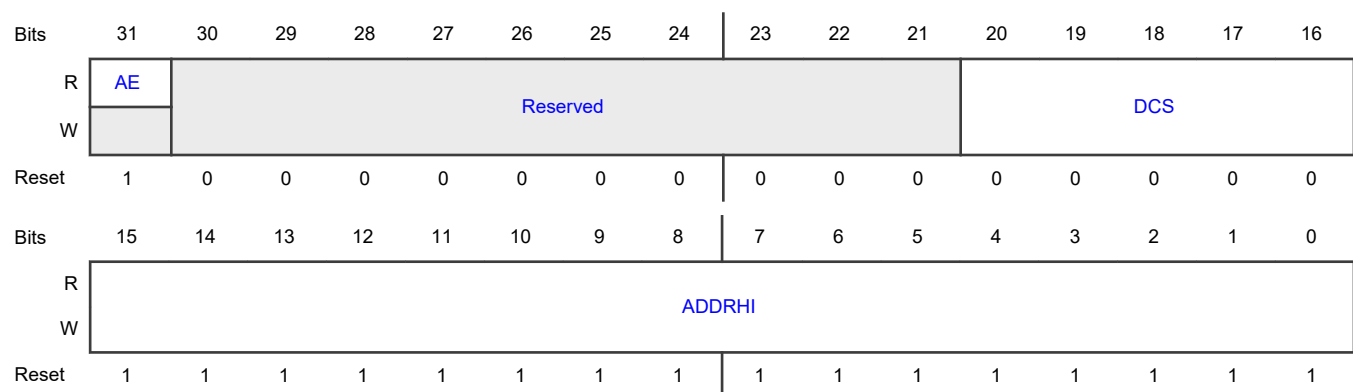
Register	Offset
MAC_Address0_High	300h

Function

The MAC Address0 High register holds the upper 16 bits of the first 6-byte MAC address of the station. The first DA byte that is received on the (G)MII interface corresponds to the LS byte (Bits [7:0]) of the MAC Address Low register. For example, if 0x112233445566 is received (0x11 in lane 0 of the first column) on the (G)MII as the destination address, then the MacAddress0 Register [47:0] is compared with 0x665544332211.

If the MAC address registers are configured to be double-synchronized to the (G)MII clock domains, then the synchronization is triggered only when Bits[31:24] (in little-endian mode) or Bits[7:0] (in big-endian mode) of the MAC Address0 Low Register are written. For proper synchronization updates, the consecutive writes to this Address Low Register should be performed after at least four clock cycles in the destination clock domain.

Diagram



Fields

Field	Function
31	Address Enable

Table continues on the next page...

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Field	Function
AE	This bit is always set to 1. 0b - INVALID : This bit must be always set to 1 1b - This bit is always set to 1
30-21 —	Reserved.
20-16 DCS	DMA Channel Select If the PDC bit of MAC_Ext_Configuration register is not set: This field contains the binary representation of the DMA Channel number to which an Rx packet whose DA matches the MAC Address0 content is routed. If the PDC bit of MAC_Ext_Configuration register is set: This field contains the one-hot representation of one or more DMA Channel numbers to which an Rx packet whose DA matches the MAC Address0 content is routed.
15-0 ADDRHI	MAC Address0[47:32] This field contains the upper 16 bits [47:32] of the first 6-byte MAC address. The MAC uses this field for filtering the received packets and inserting the MAC address in the Transmit Flow Control (Pause) Packets.

1.4.57 (MAC_Address0_Low)

Offset

Register	Offset
MAC_Address0_Low	304h

Function

The MAC Address0 Low register holds the lower 32 bits of the 6-byte first MAC address of the station.

Diagram



Fields

Field	Function
31-0 ADDRLO	MAC Address0[31:0] This field contains the lower 32 bits of the first 6-byte MAC address. The MAC uses this field for filtering the received packets and inserting the MAC address in the Transmit Flow Control (Pause) Packets.

1.4.58 (MAC_Address1_High)

Offset

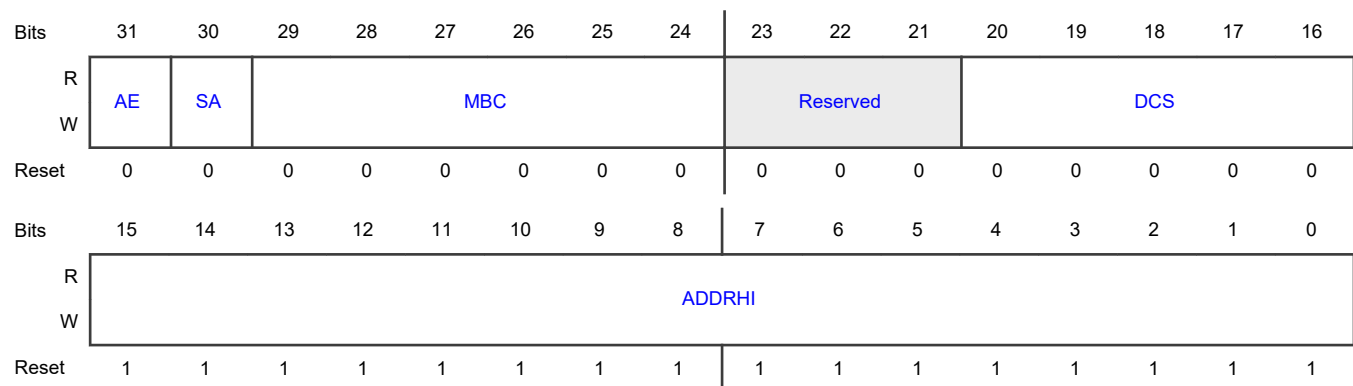
Register	Offset
MAC_Address1_High	308h

Function

The MAC Address1 High register holds the upper 16 bits of the second 6-byte MAC address of the station.

If the MAC address registers are configured to be double-synchronized to the (G)MII clock domains, the synchronization is triggered only when Bits[31:24] (in little-endian mode) or Bits[7:0] (in big-endian mode) of the MAC Address1 Low Register are written. For proper synchronization updates, the consecutive writes to this Address Low Register should be performed after at least four clock cycles in the destination clock domain.

Diagram



Fields

Field	Function
31 AE	Address Enable When this bit is set, the address filter module uses the second MAC address for perfect filtering. When this bit is reset, the address filter module ignores the address for filtering. 0b - Address is ignored 1b - Address is enabled

Table continues on the next page...

Table continued from the previous page...

Field	Function
30 SA	Source Address When this bit is set, the MAC Address1[47:0] is used to compare with the SA fields of the received packet. When this bit is reset, the MAC Address1[47:0] is used to compare with the DA fields of the received packet. 0b - Compare with Destination Address 1b - Compare with Source Address
29-24 MBC	Mask Byte Control These bits are mask control bits for comparing each of the MAC Address bytes. When set high, the MAC does not compare the corresponding byte of received DA or SA with the contents of MAC Address1 registers. Each bit controls the masking of the bytes as follows: - Bit 29: MAC_Address\${i}_High[15:8] - Bit 28: MAC_Address\${i}_High[7:0] - Bit 27: MAC_Address\${i}_Low[31:24] - .. - Bit 24: MAC_Address\${i}_Low[7:0] You can filter a group of addresses (known as group address filtering) by masking one or more bytes of the address.
23-21 —	Reserved.
20-16 DCS	DMA Channel Select If the PDC bit of MAC_Ext_Configuration register is not set: This field contains the binary representation of the DMA Channel number to which an Rx packet whose DA matches the MAC Address(#i) content is routed. If the PDC bit of MAC_Ext_Configuration register is set: This field contains the one-hot representation of one or more DMA Channel numbers to which an Rx packet whose DA matches the MAC Address(#i) content is routed.
15-0 ADDRHI	MAC Address1 [47:32] This field contains the upper 16 bits[47:32] of the second 6-byte MAC address.

1.4.59 (MAC_Address1_Low)

Offset

Register	Offset
MAC_Address1_Low	30Ch

Function

The MAC Address1 Low register holds the lower 32 bits of the second 6-byte MAC address of the station.

Diagram**Fields**

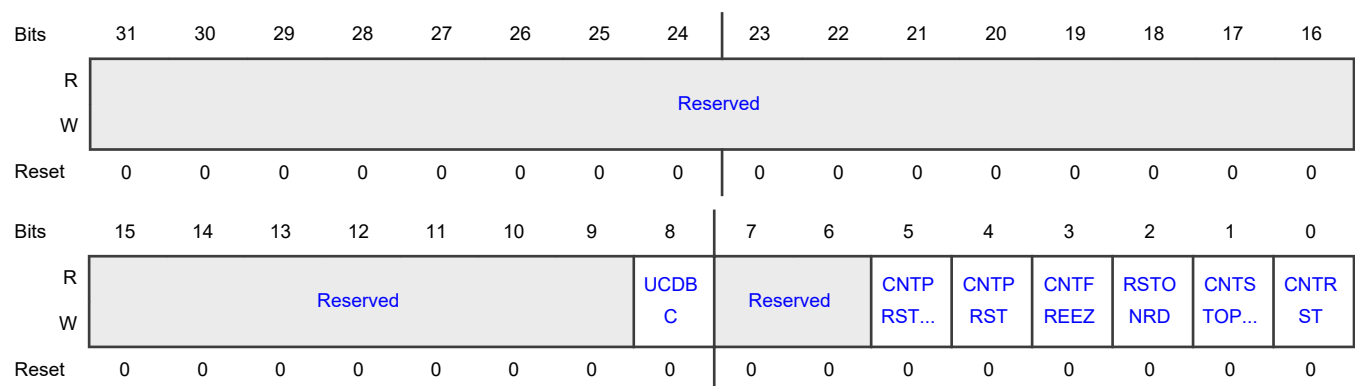
Field	Function
31-0	MAC Address1 [31:0]
ADDRLO	This field contains the lower 32 bits of second 6-byte MAC address. The content of this field is undefined until loaded by the application after the initialization process.

1.4.60 (MMC_Control)**Offset**

Register	Offset
MMC_Control	700h

Function

This register establishes the operating mode of MMC.

Diagram

Fields

Field	Function
31-9 —	Reserved.
8 UCDBC	Update MMC Counters for Dropped Broadcast Packets Note: The CNTRST bit has a higher priority than the CNTPRST bit. Therefore, when the software tries to set both bits in the same write cycle, all counters are cleared and the CNTPRST bit is not set. When set, the MAC updates all related MMC Counters for Broadcast packets that are dropped because of the setting of the DBF bit of MAC_Packet_Filter register. When reset, the MMC Counters are not updated for dropped Broadcast packets. 0b - Update MMC Counters for Dropped Broadcast Packets is disabled 1b - Update MMC Counters for Dropped Broadcast Packets is enabled
7-6 —	Reserved.
5 CNTPRSTLVL	Full-Half Preset When this bit is low and the CNTPRST bit is set, all MMC counters get preset to almost-half value. All octet counters get preset to 0x7FFF_F800 (Half 2KBytes) and all packet-counters gets preset to 0x7FFF_FFF0 (Half 16). When this bit is high and the CNTPRST bit is set, all MMC counters get preset to almost-full value. All octet counters get preset to 0xFFFF_F800 (Full 2KBytes) and all packet-counters gets preset to 0xFFFF_FFF0 (Full 16). For 16-bit counters, the almost-half preset values are 0x7800 and 0x7FF0 for the respective octet and packet counters. Similarly, the almost-full preset values for the 16-bit counters are 0xF800 and 0xFFFF. 0b - Full-Half Preset is disabled 1b - Full-Half Preset is enabled
4 CNTPRST	Counters Preset When this bit is set, all counters are initialized or preset to almost full or almost half according to the CNTPRSTLVL bit. This bit is cleared automatically after 1 clock cycle. This bit, along with the CNTPRSTLVL bit, is useful for debugging and testing the assertion of interrupts because of MMC counter becoming half-full or full. Access restriction applies. Self-cleared. Setting 0 clears. Setting 1 sets. 0b - Counters Preset is disabled 1b - Counters Preset is enabled
3 CNTFREEZ	MMC Counter Freeze When this bit is set, it freezes all MMC counters to their current value. Until this bit is reset to 0, no MMC counter is updated because of any transmitted or received packet. If any MMC counter is read with the Reset on Read bit set, then that counter is also cleared in this mode.

Table continues on the next page...

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Field	Function
	0b - MMC Counter Freeze is disabled 1b - MMC Counter Freeze is enabled
2 RSTONRD	Reset on Read When this bit is set, the MMC counters are reset to zero after Read (self-clearing after reset). The counters are cleared when the least significant byte lane (Bits[7:0]) is read. 0b - Reset on Read is disabled 1b - Reset on Read is enabled
1 CNTSTOPRO	Counter Stop Rollover When this bit is set, the counter does not roll over to zero after reaching the maximum value. 0b - Counter Stop Rollover is disabled 1b - Counter Stop Rollover is enabled
0 CNTRST	Counters Reset When this bit is set, all counters are reset. This bit is cleared automatically after 1 clock cycle. Access restriction applies. Self-cleared. Setting 0 clears. Setting 1 sets. 0b - Counters are not reset 1b - All counters are reset

1.4.61 (MMC_Rx_Interrupt)

Offset

Register	Offset
MMC_Rx_Interrupt	704h

Function

This register maintains the interrupts generated from all Receive statistics counters.

The MMC Receive Interrupt register maintains the interrupts that are generated when the following occur:

- Receive statistic counters reach half of their maximum values (0x8000_0000 for 32 bit counter and 0x8000 for 16 bit counter).
- Receive statistic counters cross their maximum values (0xFFFF_FFFF for 32 bit counter and 0xFFFF for 16 bit counter).

When the Counter Stop Rollover is set, interrupts are set but the counter remains at all-ones. The MMC Receive Interrupt register is a 32 bit register. An interrupt bit is cleared when the respective MMC counter that caused the interrupt is read. The least significant byte lane (Bits[7:0]) of the respective counter must be read to clear the interrupt bit.

Note: R_SS_RC means that this register bit is set internally, and it is cleared when the Counter register is read.

Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	Reserved				Reserved	Reserved	RXCT RLP...	RXRC VER...	RXWD OGP...	RXVL ANG...	RXFO VPIS	RXPA USP...	RXOR ANG...	RXLE NER...	RXUC GPIS	RX102 4T...
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	RX512 T1...	RX256 T5...	RX128 T2...	RX65T 12...	RX64 OCT...	RXOSI ZE...	RXUSI ZE...	RXJA BER...	RXRU NTP...	RXAL GNE...	RXCR CER...	RXMC GPIS	RXBC GPIS	RXGO CTIS	RXGB OCT...	RXGB PKT...
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Fields

Field	Function
31-28 —	Reserved.
27 —	Reserved.
26 —	Reserved.
25 RXCTRLPIS	MMC Receive Control Packet Counter Interrupt Status This bit is set when the rxctrlpackets_g counter reaches half of the maximum value or the maximum value. Access restriction applies. Clears on read. Self-set to 1 on internal event. 0b - MMC Receive Control Packet Counter Interrupt Status not detected 1b - MMC Receive Control Packet Counter Interrupt Status detected
24 RXRCVERRPIS	MMC Receive Error Packet Counter Interrupt Status This bit is set when the rxrcverror counter reaches half of the maximum value or the maximum value. Access restriction applies. Clears on read. Self-set to 1 on internal event. 0b - MMC Receive Error Packet Counter Interrupt Status not detected 1b - MMC Receive Error Packet Counter Interrupt Status detected
23 RXWDOGPIS	MMC Receive Watchdog Error Packet Counter Interrupt Status This bit is set when the rxwatchdog error counter reaches half of the maximum value or the maximum value. Access restriction applies. Clears on read. Self-set to 1 on internal event. 0b - MMC Receive Watchdog Error Packet Counter Interrupt Status not detected 1b - MMC Receive Watchdog Error Packet Counter Interrupt Status detected

Table continues on the next page...

Table continued from the previous page...

Field	Function
22 RXVLANGBPIS	MMC Receive VLAN Good Bad Packet Counter Interrupt Status This bit is set when the rxvlanpackets_gb counter reaches half of the maximum value or the maximum value. Access restriction applies. Clears on read. Self-set to 1 on internal event. 0b - MMC Receive VLAN Good Bad Packet Counter Interrupt Status not detected 1b - MMC Receive VLAN Good Bad Packet Counter Interrupt Status detected
21 RXFOVPIS	MMC Receive FIFO Overflow Packet Counter Interrupt Status This bit is set when the rxfifooverflow counter reaches half of the maximum value or the maximum value. Access restriction applies. Clears on read. Self-set to 1 on internal event. 0b - MMC Receive FIFO Overflow Packet Counter Interrupt Status not detected 1b - MMC Receive FIFO Overflow Packet Counter Interrupt Status detected
20 RXPAUSPIS	MMC Receive Pause Packet Counter Interrupt Status This bit is set when the rxpausepackets counter reaches half of the maximum value or the maximum value. Access restriction applies. Clears on read. Self-set to 1 on internal event. 0b - MMC Receive Pause Packet Counter Interrupt Status not detected 1b - MMC Receive Pause Packet Counter Interrupt Status detected
19 RXORANGEPI S	MMC Receive Out Of Range Error Packet Counter Interrupt Status. This bit is set when the rxoutofrangetype counter reaches half of the maximum value or the maximum value. Access restriction applies. Clears on read. Self-set to 1 on internal event. 0b - MMC Receive Out Of Range Error Packet Counter Interrupt Status not detected 1b - MMC Receive Out Of Range Error Packet Counter Interrupt Status detected
18 RXLENERPIS	MMC Receive Length Error Packet Counter Interrupt Status This bit is set when the rxlengtherror counter reaches half of the maximum value or the maximum value. Access restriction applies. Clears on read. Self-set to 1 on internal event. 0b - MMC Receive Length Error Packet Counter Interrupt Status not detected 1b - MMC Receive Length Error Packet Counter Interrupt Status detected
17 RXUCGPIS	MMC Receive Unicast Good Packet Counter Interrupt Status This bit is set when the rxunicastpackets_g counter reaches half of the maximum value or the maximum value. Access restriction applies. Clears on read. Self-set to 1 on internal event. 0b - MMC Receive Unicast Good Packet Counter Interrupt Status not detected 1b - MMC Receive Unicast Good Packet Counter Interrupt Status detected

Table continues on the next page...

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Field	Function
16 RX1024TMAXO CTGBPIS	MMC Receive 1024 to Maximum Octet Good Bad Packet Counter Interrupt Status This bit is set when the rx1024tomaxoctets_gb counter reaches half of the maximum value or the maximum value. Access restriction applies. Clears on read. Self-set to 1 on internal event. 0b - MMC Receive 1024 to Maximum Octet Good Bad Packet Counter Interrupt Status not detected 1b - MMC Receive 1024 to Maximum Octet Good Bad Packet Counter Interrupt Status detected
15 RX512T1023O CTGBPIS	MMC Receive 512 to 1023 Octet Good Bad Packet Counter Interrupt Status This bit is set when the rx512to1023octets_gb counter reaches half of the maximum value or the maximum value. Access restriction applies. Clears on read. Self-set to 1 on internal event. 0b - MMC Receive 512 to 1023 Octet Good Bad Packet Counter Interrupt Status not detected 1b - MMC Receive 512 to 1023 Octet Good Bad Packet Counter Interrupt Status detected
14 RX256T511OC TGBPIS	MMC Receive 256 to 511 Octet Good Bad Packet Counter Interrupt Status This bit is set when the rx256to511octets_gb counter reaches half of the maximum value or the maximum value. Access restriction applies. Clears on read. Self-set to 1 on internal event. 0b - MMC Receive 256 to 511 Octet Good Bad Packet Counter Interrupt Status not detected 1b - MMC Receive 256 to 511 Octet Good Bad Packet Counter Interrupt Status detected
13 RX128T255OC TGBPIS	MMC Receive 128 to 255 Octet Good Bad Packet Counter Interrupt Status This bit is set when the rx128to255octets_gb counter reaches half of the maximum value or the maximum value. Access restriction applies. Clears on read. Self-set to 1 on internal event. 0b - MMC Receive 128 to 255 Octet Good Bad Packet Counter Interrupt Status not detected 1b - MMC Receive 128 to 255 Octet Good Bad Packet Counter Interrupt Status detected
12 RX65T127OCT GBPIS	MMC Receive 65 to 127 Octet Good Bad Packet Counter Interrupt Status This bit is set when the rx65to127octets_gb counter reaches half of the maximum value or the maximum value. Access restriction applies. Clears on read. Self-set to 1 on internal event. 0b - MMC Receive 65 to 127 Octet Good Bad Packet Counter Interrupt Status not detected 1b - MMC Receive 65 to 127 Octet Good Bad Packet Counter Interrupt Status detected
11 RX64OCTGBPIS	MMC Receive 64 Octet Good Bad Packet Counter Interrupt Status This bit is set when the rx64octets_gb counter reaches half of the maximum value or the maximum value. Access restriction applies. Clears on read. Self-set to 1 on internal event.

Table continues on the next page...

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Field	Function
	0b - MMC Receive 64 Octet Good Bad Packet Counter Interrupt Status not detected 1b - MMC Receive 64 Octet Good Bad Packet Counter Interrupt Status detected
10 RXOSIZEGPIS	MMC Receive Oversize Good Packet Counter Interrupt Status This bit is set when the rxoversize_g counter reaches half of the maximum value or the maximum value. Access restriction applies. Clears on read. Self-set to 1 on internal event. 0b - MMC Receive Oversize Good Packet Counter Interrupt Status not detected 1b - MMC Receive Oversize Good Packet Counter Interrupt Status detected
9 RXUSIZEGPIS	MMC Receive Undersize Good Packet Counter Interrupt Status This bit is set when the rxundersize_g counter reaches half of the maximum value or the maximum value. Access restriction applies. Clears on read. Self-set to 1 on internal event. 0b - MMC Receive Undersize Good Packet Counter Interrupt Status not detected 1b - MMC Receive Undersize Good Packet Counter Interrupt Status detected
8 RXJABBERPIS	MMC Receive Jabber Error Packet Counter Interrupt Status This bit is set when the rxjabbererror counter reaches half of the maximum value or the maximum value. Access restriction applies. Clears on read. Self-set to 1 on internal event. 0b - MMC Receive Jabber Error Packet Counter Interrupt Status not detected 1b - MMC Receive Jabber Error Packet Counter Interrupt Status detected
7 RXRUNTPIS	MMC Receive Runt Packet Counter Interrupt Status This bit is set when the rxrunterror counter reaches half of the maximum value or the maximum value. Access restriction applies. Clears on read. Self-set to 1 on internal event. 0b - MMC Receive Runt Packet Counter Interrupt Status not detected 1b - MMC Receive Runt Packet Counter Interrupt Status detected
6 RXALGNERPIS	MMC Receive Alignment Error Packet Counter Interrupt Status This bit is set when the rxalignmenterror counter reaches half of the maximum value or the maximum value. Access restriction applies. Clears on read. Self-set to 1 on internal event. 0b - MMC Receive Alignment Error Packet Counter Interrupt Status not detected 1b - MMC Receive Alignment Error Packet Counter Interrupt Status detected
5 RXCRCERPIS	MMC Receive CRC Error Packet Counter Interrupt Status This bit is set when the rxrcrcerror counter reaches half of the maximum value or the maximum value. Access restriction applies. Clears on read. Self-set to 1 on internal event. 0b - MMC Receive CRC Error Packet Counter Interrupt Status not detected

Table continues on the next page...

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Field	Function
	1b - MMC Receive CRC Error Packet Counter Interrupt Status detected
4 RXMCGPIS	MMC Receive Multicast Good Packet Counter Interrupt Status This bit is set when the rxmulticastpackets_g counter reaches half of the maximum value or the maximum value. Access restriction applies. Clears on read. Self-set to 1 on internal event. 0b - MMC Receive Multicast Good Packet Counter Interrupt Status not detected 1b - MMC Receive Multicast Good Packet Counter Interrupt Status detected
3 RXBCGPIS	MMC Receive Broadcast Good Packet Counter Interrupt Status This bit is set when the rxbroadcastpackets_g counter reaches half of the maximum value or the maximum value. Access restriction applies. Clears on read. Self-set to 1 on internal event. 0b - MMC Receive Broadcast Good Packet Counter Interrupt Status not detected 1b - MMC Receive Broadcast Good Packet Counter Interrupt Status detected
2 RXGOCTIS	MMC Receive Good Octet Counter Interrupt Status This bit is set when the rxoctetcount_g counter reaches half of the maximum value or the maximum value. Access restriction applies. Clears on read. Self-set to 1 on internal event. 0b - MMC Receive Good Octet Counter Interrupt Status not detected 1b - MMC Receive Good Octet Counter Interrupt Status detected
1 RXGBOCTIS	MMC Receive Good Bad Octet Counter Interrupt Status This bit is set when the rxoctetcount_gb counter reaches half of the maximum value or the maximum value. Access restriction applies. Clears on read. Self-set to 1 on internal event. 0b - MMC Receive Good Bad Octet Counter Interrupt Status not detected 1b - MMC Receive Good Bad Octet Counter Interrupt Status detected
0 RXGBPKTIS	MMC Receive Good Bad Packet Counter Interrupt Status This bit is set when the rxpacketcount_gb counter reaches half of the maximum value or the maximum value. Access restriction applies. Clears on read. Self-set to 1 on internal event. 0b - MMC Receive Good Bad Packet Counter Interrupt Status not detected 1b - MMC Receive Good Bad Packet Counter Interrupt Status detected

1.4.62 (MMC_Tx_Interrupt)

Offset

Register	Offset
MMC_Tx_Interrupt	708h

Function

This register maintains the interrupts generated from all Transmit statistics counters.

The MMC Transmit Interrupt register maintains the interrupts generated when transmit statistic counters reach half their maximum values

(0x8000_0000 for 32 bit counter and 0x8000 for 16 bit counter), and when they cross their maximum values (0xFFFF_FFFF for 32-bit counter and 0xFFFF for 16-bit counter).

When Counter Stop Rollover is set, the interrupts are set but the counter remains at all-ones.

The MMC Transmit Interrupt register is a 32 bit register. An interrupt bit is cleared when the respective MMC counter that caused the interrupt is read.

The least significant byte lane (Bits[7:0]) of the respective counter must be read to clear the interrupt bit.

Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	Reserved				Reserv ed	Reserv ed	TXOSI ZE...	TXVLA NG...	TXPA USP...	TXEX DEF...	TXGP KTIS	TXGO CTIS	TXCA RER...	TXEX COL...	TXLAT CO...	TXDE FPIS
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	TXMC OLG...	TXSC OLG...	TXUFL OW...	TXBC GBP...	TXMC GBP...	TXUC GBP...	TX102 4T...	TX512 T1...	TX256 T5...	TX128 T2...	TX65T 12...	TX64O CT...	TXMC GPIS	TXBC GPIS	TXGB PKT...	TXGB OCT...
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Fields

Field	Function
31-28 —	Reserved.
27 —	Reserved.
26 —	Reserved.

Table continues on the next page...

Table continued from the previous page...

Field	Function
25 TXOSIZEGPIS	MMC Transmit Oversize Good Packet Counter Interrupt Status This bit is set when the txoversize_g counter reaches half of the maximum value or the maximum value. Access restriction applies. Clears on read. Self-set to 1 on internal event. 0b - MMC Transmit Oversize Good Packet Counter Interrupt Status not detected 1b - MMC Transmit Oversize Good Packet Counter Interrupt Status detected
24 TXVLANGPIS	MMC Transmit VLAN Good Packet Counter Interrupt Status This bit is set when the txvlanpackets_g counter reaches half of the maximum value or the maximum value. Access restriction applies. Clears on read. Self-set to 1 on internal event. 0b - MMC Transmit VLAN Good Packet Counter Interrupt Status not detected 1b - MMC Transmit VLAN Good Packet Counter Interrupt Status detected
23 TXPAUSPIS	MMC Transmit Pause Packet Counter Interrupt Status This bit is set when the txpausepacketerror counter reaches half of the maximum value or the maximum value. Access restriction applies. Clears on read. Self-set to 1 on internal event. 0b - MMC Transmit Pause Packet Counter Interrupt Status not detected 1b - MMC Transmit Pause Packet Counter Interrupt Status detected
22 TXEXDEFPIS	MMC Transmit Excessive Deferral Packet Counter Interrupt Status This bit is set when the txexcessdef counter reaches half of the maximum value or the maximum value. Access restriction applies. Clears on read. Self-set to 1 on internal event. 0b - MMC Transmit Excessive Deferral Packet Counter Interrupt Status not detected 1b - MMC Transmit Excessive Deferral Packet Counter Interrupt Status detected
21 TXGPKTIS	MMC Transmit Good Packet Counter Interrupt Status This bit is set when the txpacketcount_g counter reaches half of the maximum value or the maximum value. Access restriction applies. Clears on read. Self-set to 1 on internal event. 0b - MMC Transmit Good Packet Counter Interrupt Status not detected 1b - MMC Transmit Good Packet Counter Interrupt Status detected
20 TXGOCTIS	MMC Transmit Good Octet Counter Interrupt Status This bit is set when the txoctetcount_g counter reaches half of the maximum value or the maximum value. Access restriction applies. Clears on read. Self-set to 1 on internal event. 0b - MMC Transmit Good Octet Counter Interrupt Status not detected 1b - MMC Transmit Good Octet Counter Interrupt Status detected
19	MMC Transmit Carrier Error Packet Counter Interrupt Status

Table continues on the next page...

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Field	Function
TXCARERPIS	This bit is set when the txcarriererror counter reaches half of the maximum value or the maximum value. Access restriction applies. Clears on read. Self-set to 1 on internal event. 0b - MMC Transmit Carrier Error Packet Counter Interrupt Status not detected 1b - MMC Transmit Carrier Error Packet Counter Interrupt Status detected
18 TXEXCOLPIS	MMC Transmit Excessive Collision Packet Counter Interrupt Status This bit is set when the txexesscol counter reaches half of the maximum value or the maximum value. Access restriction applies. Clears on read. Self-set to 1 on internal event. 0b - MMC Transmit Excessive Collision Packet Counter Interrupt Status not detected 1b - MMC Transmit Excessive Collision Packet Counter Interrupt Status detected
17 TXLATCOLPIS	MMC Transmit Late Collision Packet Counter Interrupt Status This bit is set when the txlatecol counter reaches half of the maximum value or the maximum value. Access restriction applies. Clears on read. Self-set to 1 on internal event. 0b - MMC Transmit Late Collision Packet Counter Interrupt Status not detected 1b - MMC Transmit Late Collision Packet Counter Interrupt Status detected
16 TXDEFPIIS	MMC Transmit Deferred Packet Counter Interrupt Status This bit is set when the txdeferred counter reaches half of the maximum value or the maximum value. Access restriction applies. Clears on read. Self-set to 1 on internal event. 0b - MMC Transmit Deferred Packet Counter Interrupt Status not detected 1b - MMC Transmit Deferred Packet Counter Interrupt Status detected
15 TXMCOLGPIS	MMC Transmit Multiple Collision Good Packet Counter Interrupt Status This bit is set when the txmulticol_g counter reaches half of the maximum value or the maximum value. Access restriction applies. Clears on read. Self-set to 1 on internal event. 0b - MMC Transmit Multiple Collision Good Packet Counter Interrupt Status not detected 1b - MMC Transmit Multiple Collision Good Packet Counter Interrupt Status detected
14 TXSCOLGPIS	MMC Transmit Single Collision Good Packet Counter Interrupt Status This bit is set when the txsinglecol_g counter reaches half of the maximum value or the maximum value. Access restriction applies. Clears on read. Self-set to 1 on internal event. 0b - MMC Transmit Single Collision Good Packet Counter Interrupt Status not detected 1b - MMC Transmit Single Collision Good Packet Counter Interrupt Status detected
13 TXUFLOWERPIS	MMC Transmit Underflow Error Packet Counter Interrupt Status This bit is set when the txunderflowerror counter reaches half of the maximum value or the maximum value. Access restriction applies. Clears on read. Self-set to 1 on internal event.

Table continues on the next page...

Table continued from the previous page...

Field	Function
	0b - MMC Transmit Underflow Error Packet Counter Interrupt Status not detected 1b - MMC Transmit Underflow Error Packet Counter Interrupt Status detected
12 TXBCGBPIS	MMC Transmit Broadcast Good Bad Packet Counter Interrupt Status This bit is set when the txbroadcastpackets_gb counter reaches half of the maximum value or the maximum value. Access restriction applies. Clears on read. Self-set to 1 on internal event. 0b - MMC Transmit Broadcast Good Bad Packet Counter Interrupt Status not detected 1b - MMC Transmit Broadcast Good Bad Packet Counter Interrupt Status detected
11 TXMCGBPIS	MMC Transmit Multicast Good Bad Packet Counter Interrupt Status The bit is set when the txmulticastpackets_gb counter reaches half of the maximum value or the maximum value. Access restriction applies. Clears on read. Self-set to 1 on internal event. 0b - MMC Transmit Multicast Good Bad Packet Counter Interrupt Status not detected 1b - MMC Transmit Multicast Good Bad Packet Counter Interrupt Status detected
10 TXUCGBPIS	MMC Transmit Unicast Good Bad Packet Counter Interrupt Status This bit is set when the txunicastpackets_gb counter reaches half of the maximum value or the maximum value. Access restriction applies. Clears on read. Self-set to 1 on internal event. 0b - MMC Transmit Unicast Good Bad Packet Counter Interrupt Status not detected 1b - MMC Transmit Unicast Good Bad Packet Counter Interrupt Status detected
9 TX1024TMAXO CTGBPIS	MMC Transmit 1024 to Maximum Octet Good Bad Packet Counter Interrupt Status This bit is set when the tx1024tomaxoctets_gb counter reaches half of the maximum value or the maximum value. Access restriction applies. Clears on read. Self-set to 1 on internal event. 0b - MMC Transmit 1024 to Maximum Octet Good Bad Packet Counter Interrupt Status not detected 1b - MMC Transmit 1024 to Maximum Octet Good Bad Packet Counter Interrupt Status detected
8 TX512T1023O CTGBPIS	MMC Transmit 512 to 1023 Octet Good Bad Packet Counter Interrupt Status This bit is set when the tx512to1023octets_gb counter reaches half of the maximum value or the maximum value. Access restriction applies. Clears on read. Self-set to 1 on internal event. 0b - MMC Transmit 512 to 1023 Octet Good Bad Packet Counter Interrupt Status not detected 1b - MMC Transmit 512 to 1023 Octet Good Bad Packet Counter Interrupt Status detected

Table continues on the next page...

Table continued from the previous page...

Field	Function
7 TX256T511OC TGBPIS	MMC Transmit 256 to 511 Octet Good Bad Packet Counter Interrupt Status This bit is set when the tx256to511octets_gb counter reaches half of the maximum value or the maximum value. Access restriction applies. Clears on read. Self-set to 1 on internal event. 0b - MMC Transmit 256 to 511 Octet Good Bad Packet Counter Interrupt Status not detected 1b - MMC Transmit 256 to 511 Octet Good Bad Packet Counter Interrupt Status detected
6 TX128T255OC TGBPIS	MMC Transmit 128 to 255 Octet Good Bad Packet Counter Interrupt Status This bit is set when the tx128to255octets_gb counter reaches half of the maximum value or the maximum value. Access restriction applies. Clears on read. Self-set to 1 on internal event. 0b - MMC Transmit 128 to 255 Octet Good Bad Packet Counter Interrupt Status not detected 1b - MMC Transmit 128 to 255 Octet Good Bad Packet Counter Interrupt Status detected
5 TX65T127OCT GBPIS	MMC Transmit 65 to 127 Octet Good Bad Packet Counter Interrupt Status This bit is set when the tx65to127octets_gb counter reaches half the maximum value, and also when it reaches the maximum value. Access restriction applies. Clears on read. Self-set to 1 on internal event. 0b - MMC Transmit 65 to 127 Octet Good Bad Packet Counter Interrupt Status not detected 1b - MMC Transmit 65 to 127 Octet Good Bad Packet Counter Interrupt Status detected
4 TX64OCTGBPIS	MMC Transmit 64 Octet Good Bad Packet Counter Interrupt Status This bit is set when the tx64octets_gb counter reaches half of the maximum value or the maximum value. Access restriction applies. Clears on read. Self-set to 1 on internal event. 0b - MMC Transmit 64 Octet Good Bad Packet Counter Interrupt Status not detected 1b - MMC Transmit 64 Octet Good Bad Packet Counter Interrupt Status detected
3 TXMCGPIS	MMC Transmit Multicast Good Packet Counter Interrupt Status This bit is set when the txmulticastpackets_g counter reaches half of the maximum value or the maximum value. Access restriction applies. Clears on read. Self-set to 1 on internal event. 0b - MMC Transmit Multicast Good Packet Counter Interrupt Status not detected 1b - MMC Transmit Multicast Good Packet Counter Interrupt Status detected
2 TXBCGPIS	MMC Transmit Broadcast Good Packet Counter Interrupt Status This bit is set when the txbroadcastpackets_g counter reaches half of the maximum value or the maximum value. Access restriction applies. Clears on read. Self-set to 1 on internal event.

Table continues on the next page...

Table continued from the previous page...

Field	Function
	0b - MMC Transmit Broadcast Good Packet Counter Interrupt Status not detected 1b - MMC Transmit Broadcast Good Packet Counter Interrupt Status detected
1 TXGBPKTIS	MMC Transmit Good Bad Packet Counter Interrupt Status This bit is set when the txpacketcount_gb counter reaches half of the maximum value or the maximum value. Access restriction applies. Clears on read. Self-set to 1 on internal event. 0b - MMC Transmit Good Bad Packet Counter Interrupt Status not detected 1b - MMC Transmit Good Bad Packet Counter Interrupt Status detected
0 TXGBOCTIS	MMC Transmit Good Bad Octet Counter Interrupt Status This bit is set when the txoctetcount_gb counter reaches half of the maximum value or the maximum value. Access restriction applies. Clears on read. Self-set to 1 on internal event. 0b - MMC Transmit Good Bad Octet Counter Interrupt Status not detected 1b - MMC Transmit Good Bad Octet Counter Interrupt Status detected

1.4.63 (MMC_Rx_Interrupt_Mask)

Offset

Register	Offset
MMC_Rx_Interrupt_Mask	70Ch

Function

This register maintains the masks for interrupts generated from all Receive statistics counters.

The MMC Receive Interrupt Mask register maintains the masks for the interrupts generated when receive statistic counters reach half of their maximum value or the maximum values.

This register is 32 bit wide.

Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	Reserved				Reserv	Reserv	RXCT	RXRC	RXWD	RXVL	RXFO	RXPA	RXOR	RXLE	RXUC	RX102
W					ed	ed	RLP...	VER...	OGP...	ANG...	VPIM	USP...	ANG...	NER...	GPIM	4T...
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	RX512	RX256	RX128	RX65T	RX64	RXOSI	RXUSI	RXJA	RXRU	RXAL	RXCR	RXMC	RXBC	RXGO	RXGB	RXGB
W	T1...	T5...	T2...	12...	OCT...	ZE...	ZE...	BER...	NTP...	GNE...	CER...	GPIM	GPIM	CTIM	OCT...	PKT...
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Fields

Field	Function
31-28 —	Reserved.
27 —	Reserved.
26 —	Reserved.
25 RXCTRLPIM	MMC Receive Control Packet Counter Interrupt Mask Setting this bit masks the interrupt when the rxctrlpackets_g counter reaches half of the maximum value or the maximum value. 0b - MMC Receive Control Packet Counter Interrupt Mask is disabled 1b - MMC Receive Control Packet Counter Interrupt Mask is enabled
24 RXRCVERRPIM	MMC Receive Error Packet Counter Interrupt Mask Setting this bit masks the interrupt when the rxrcverror counter reaches half of the maximum value or the maximum value. 0b - MMC Receive Error Packet Counter Interrupt Mask is disabled 1b - MMC Receive Error Packet Counter Interrupt Mask is enabled
23 RXWDOGPIM	MMC Receive Watchdog Error Packet Counter Interrupt Mask Setting this bit masks the interrupt when the rxwatchdog counter reaches half of the maximum value or the maximum value. 0b - MMC Receive Watchdog Error Packet Counter Interrupt Mask is disabled 1b - MMC Receive Watchdog Error Packet Counter Interrupt Mask is enabled
22	MMC Receive VLAN Good Bad Packet Counter Interrupt Mask

Table continues on the next page...

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Field	Function
RXVLANGBPIM	Setting this bit masks the interrupt when the rxvlanpackets_gb counter reaches half of the maximum value or the maximum value. 0b - MMC Receive VLAN Good Bad Packet Counter Interrupt Mask is disabled 1b - MMC Receive VLAN Good Bad Packet Counter Interrupt Mask is enabled
21 RXFOVPIM	MMC Receive FIFO Overflow Packet Counter Interrupt Mask Setting this bit masks the interrupt when the rxfifooverflow counter reaches half of the maximum value or the maximum value. 0b - MMC Receive FIFO Overflow Packet Counter Interrupt Mask is disabled 1b - MMC Receive FIFO Overflow Packet Counter Interrupt Mask is enabled
20 RXPAUSPIM	MMC Receive Pause Packet Counter Interrupt Mask Setting this bit masks the interrupt when the rxpausepackets counter reaches half of the maximum value or the maximum value. 0b - MMC Receive Pause Packet Counter Interrupt Mask is disabled 1b - MMC Receive Pause Packet Counter Interrupt Mask is enabled
19 RXORANGEPI M	MMC Receive Out Of Range Error Packet Counter Interrupt Mask Setting this bit masks the interrupt when the rxoutofrangetype counter reaches half of the maximum value or the maximum value. 0b - MMC Receive Out Of Range Error Packet Counter Interrupt Mask is disabled 1b - MMC Receive Out Of Range Error Packet Counter Interrupt Mask is enabled
18 RXLENERPIM	MMC Receive Length Error Packet Counter Interrupt Mask Setting this bit masks the interrupt when the rxlengtherror counter reaches half of the maximum value or the maximum value. 0b - MMC Receive Length Error Packet Counter Interrupt Mask is disabled 1b - MMC Receive Length Error Packet Counter Interrupt Mask is enabled
17 RXUCGPIM	MMC Receive Unicast Good Packet Counter Interrupt Mask Setting this bit masks the interrupt when the rxunicastpackets_g counter reaches half of the maximum value or the maximum value. 0b - MMC Receive Unicast Good Packet Counter Interrupt Mask is disabled 1b - MMC Receive Unicast Good Packet Counter Interrupt Mask is enabled
16 RX1024TMAXO CTGBPIM	MMC Receive 1024 to Maximum Octet Good Bad Packet Counter Interrupt Mask. Setting this bit masks the interrupt when the rx1024tomaxoctets_gb counter reaches half of the maximum value or the maximum value. 0b - MMC Receive 1024 to Maximum Octet Good Bad Packet Counter Interrupt Mask is disabled 1b - MMC Receive 1024 to Maximum Octet Good Bad Packet Counter Interrupt Mask is enabled

Table continues on the next page...

Table continued from the previous page...

Field	Function
15 RX512T1023O CTGBPIM	MMC Receive 512 to 1023 Octet Good Bad Packet Counter Interrupt Mask Setting this bit masks the interrupt when the rx512to1023octets_gb counter reaches half of the maximum value or the maximum value. 0b - MMC Receive 512 to 1023 Octet Good Bad Packet Counter Interrupt Mask is disabled 1b - MMC Receive 512 to 1023 Octet Good Bad Packet Counter Interrupt Mask is enabled
14 RX256T511OC TGBPIM	MMC Receive 256 to 511 Octet Good Bad Packet Counter Interrupt Mask Setting this bit masks the interrupt when the rx256to511octets_gb counter reaches half of the maximum value or the maximum value. 0b - MMC Receive 256 to 511 Octet Good Bad Packet Counter Interrupt Mask is disabled 1b - MMC Receive 256 to 511 Octet Good Bad Packet Counter Interrupt Mask is enabled
13 RX128T255OC TGBPIM	MMC Receive 128 to 255 Octet Good Bad Packet Counter Interrupt Mask Setting this bit masks the interrupt when the rx128to255octets_gb counter reaches half of the maximum value or the maximum value. 0b - MMC Receive 128 to 255 Octet Good Bad Packet Counter Interrupt Mask is disabled 1b - MMC Receive 128 to 255 Octet Good Bad Packet Counter Interrupt Mask is enabled
12 RX65T127OCT GBPIM	MMC Receive 65 to 127 Octet Good Bad Packet Counter Interrupt Mask Setting this bit masks the interrupt when the rx65to127octets_gb counter reaches half of the maximum value or the maximum value. 0b - MMC Receive 65 to 127 Octet Good Bad Packet Counter Interrupt Mask is disabled 1b - MMC Receive 65 to 127 Octet Good Bad Packet Counter Interrupt Mask is enabled
11 RX64OCTGBPIM	MMC Receive 64 Octet Good Bad Packet Counter Interrupt Mask Setting this bit masks the interrupt when the rx64octets_gb counter reaches half of the maximum value or the maximum value. 0b - MMC Receive 64 Octet Good Bad Packet Counter Interrupt Mask is disabled 1b - MMC Receive 64 Octet Good Bad Packet Counter Interrupt Mask is enabled
10 RXOSIZEGPIM	MMC Receive Oversize Good Packet Counter Interrupt Mask Setting this bit masks the interrupt when the rxoversize_g counter reaches half of the maximum value or the maximum value. 0b - MMC Receive Oversize Good Packet Counter Interrupt Mask is disabled 1b - MMC Receive Oversize Good Packet Counter Interrupt Mask is enabled
9 RXUSIZEGPIM	MMC Receive Undersize Good Packet Counter Interrupt Mask Setting this bit masks the interrupt when the rxundersize_g counter reaches half of the maximum value or the maximum value.

Table continues on the next page...

Table continued from the previous page...

Field	Function
	0b - MMC Receive Undersize Good Packet Counter Interrupt Mask is disabled 1b - MMC Receive Undersize Good Packet Counter Interrupt Mask is enabled
8 RXJABERPI	MMC Receive Jabber Error Packet Counter Interrupt Mask Setting this bit masks the interrupt when the rxjabbererror counter reaches half of the maximum value or the maximum value. 0b - MMC Receive Jabber Error Packet Counter Interrupt Mask is disabled 1b - MMC Receive Jabber Error Packet Counter Interrupt Mask is enabled
7 RXRUNTPIM	MMC Receive Runt Packet Counter Interrupt Mask Setting this bit masks the interrupt when the rxrunterror counter reaches half of the maximum value or the maximum value. 0b - MMC Receive Runt Packet Counter Interrupt Mask is disabled 1b - MMC Receive Runt Packet Counter Interrupt Mask is enabled
6 RXALGNERPIM	MMC Receive Alignment Error Packet Counter Interrupt Mask Setting this bit masks the interrupt when the rxalignmenterror counter reaches half of the maximum value or the maximum value. 0b - MMC Receive Alignment Error Packet Counter Interrupt Mask is disabled 1b - MMC Receive Alignment Error Packet Counter Interrupt Mask is enabled
5 RXCRCERPIM	MMC Receive CRC Error Packet Counter Interrupt Mask Setting this bit masks the interrupt when the rxrcerror counter reaches half of the maximum value or the maximum value. 0b - MMC Receive CRC Error Packet Counter Interrupt Mask is disabled 1b - MMC Receive CRC Error Packet Counter Interrupt Mask is enabled
4 RXMCGPIM	MMC Receive Multicast Good Packet Counter Interrupt Mask Setting this bit masks the interrupt when the rxmulticastpackets_g counter reaches half of the maximum value or the maximum value. 0b - MMC Receive Multicast Good Packet Counter Interrupt Mask is disabled 1b - MMC Receive Multicast Good Packet Counter Interrupt Mask is enabled
3 RXBCGPIM	MMC Receive Broadcast Good Packet Counter Interrupt Mask Setting this bit masks the interrupt when the rxbroadcastpackets_g counter reaches half of the maximum value or the maximum value. 0b - MMC Receive Broadcast Good Packet Counter Interrupt Mask is disabled 1b - MMC Receive Broadcast Good Packet Counter Interrupt Mask is enabled
2	MMC Receive Good Octet Counter Interrupt Mask

Table continues on the next page...

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Field	Function
RXGOCTIM	Setting this bit masks the interrupt when the rxoctetcount_g counter reaches half of the maximum value or the maximum value. 0b - MMC Receive Good Octet Counter Interrupt Mask is disabled 1b - MMC Receive Good Octet Counter Interrupt Mask is enabled
1 RXGBOCTIM	MMC Receive Good Bad Octet Counter Interrupt Mask Setting this bit masks the interrupt when the rxoctetcount_gb counter reaches half of the maximum value or the maximum value. 0b - MMC Receive Good Bad Octet Counter Interrupt Mask is disabled 1b - MMC Receive Good Bad Octet Counter Interrupt Mask is enabled
0 RXGBPCTIM	MMC Receive Good Bad Packet Counter Interrupt Mask Setting this bit masks the interrupt when the rxpacketcount_gb counter reaches half of the maximum value or the maximum value. 0b - MMC Receive Good Bad Packet Counter Interrupt Mask is disabled 1b - MMC Receive Good Bad Packet Counter Interrupt Mask is enabled

1.4.64 (MMC_Tx_Interrupt_Mask)

Offset

Register	Offset
MMC_Tx_Interrupt_Mask	710h

Function

This register maintains the masks for interrupts generated from all Transmit statistics counters.

The MMC Transmit Interrupt Mask register maintains the masks for the interrupts generated when the transmit statistic counters reach half of their maximum value or the maximum values. This register is 32 bit wide.

Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	Reserved				Reserv	Reserv	TXOSI	TXVLA	TXPA	TXEX	TXGP	TXGO	TXCA	TXEX	TXLAT	TXDE
W					ed	ed	ZE...	NG...	USP...	DEF...	KTIM	CTIM	RER...	COL...	CO...	FPIM
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	TXMC	TXSC	TXUFL	TXBC	TXMC	TXUC	TX102	TX512	TX256	TX128	TX65T	TX64O	TXMC	TXBC	TXGB	TXGB
W	OLG...	OLG...	OW...	GBP...	GBP...	GBP...	4T...	T1...	T5...	T2...	12...	CT...	GPIM	GPIM	PKT...	OCT...
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Fields

Field	Function
31-28 —	Reserved.
27 —	Reserved.
26 —	Reserved.
25 TXOSIZEGPIM	MMC Transmit Oversize Good Packet Counter Interrupt Mask Setting this bit masks the interrupt when the txoversize_g counter reaches half of the maximum value or the maximum value. 0b - MMC Transmit Oversize Good Packet Counter Interrupt Mask is disabled 1b - MMC Transmit Oversize Good Packet Counter Interrupt Mask is enabled
24 TXVLANGPIM	MMC Transmit VLAN Good Packet Counter Interrupt Mask Setting this bit masks the interrupt when the txvlanpackets_g counter reaches half of the maximum value or the maximum value. 0b - MMC Transmit VLAN Good Packet Counter Interrupt Mask is disabled 1b - MMC Transmit VLAN Good Packet Counter Interrupt Mask is enabled
23 TXPAUSPIM	MMC Transmit Pause Packet Counter Interrupt Mask Setting this bit masks the interrupt when the txpausepackets counter reaches half of the maximum value or the maximum value. 0b - MMC Transmit Pause Packet Counter Interrupt Mask is disabled 1b - MMC Transmit Pause Packet Counter Interrupt Mask is enabled
22	MMC Transmit Excessive Deferral Packet Counter Interrupt Mask

Table continues on the next page...

Table continued from the previous page...

Field	Function
TXEXDEFPIM	Setting this bit masks the interrupt when the txexcessdef counter reaches half of the maximum value or the maximum value. 0b - MMC Transmit Excessive Deferral Packet Counter Interrupt Mask is disabled 1b - MMC Transmit Excessive Deferral Packet Counter Interrupt Mask is enabled
21 TXGPKTIM	MMC Transmit Good Packet Counter Interrupt Mask Setting this bit masks the interrupt when the txpacketcount_g counter reaches half of the maximum value or the maximum value. 0b - MMC Transmit Good Packet Counter Interrupt Mask is disabled 1b - MMC Transmit Good Packet Counter Interrupt Mask is enabled
20 TXGOCTIM	MMC Transmit Good Octet Counter Interrupt Mask Setting this bit masks the interrupt when the txoctetcount_g counter reaches half of the maximum value or the maximum value. 0b - MMC Transmit Good Octet Counter Interrupt Mask is disabled 1b - MMC Transmit Good Octet Counter Interrupt Mask is enabled
19 TXCARERPIM	MMC Transmit Carrier Error Packet Counter Interrupt Mask Setting this bit masks the interrupt when the txcarriererror counter reaches half of the maximum value or the maximum value. 0b - MMC Transmit Carrier Error Packet Counter Interrupt Mask is disabled 1b - MMC Transmit Carrier Error Packet Counter Interrupt Mask is enabled
18 TXEXCOLPIM	MMC Transmit Excessive Collision Packet Counter Interrupt Mask Setting this bit masks the interrupt when the txexcesscol counter reaches half of the maximum value or the maximum value. 0b - MMC Transmit Excessive Collision Packet Counter Interrupt Mask is disabled 1b - MMC Transmit Excessive Collision Packet Counter Interrupt Mask is enabled
17 TXLATCOLPIM	MMC Transmit Late Collision Packet Counter Interrupt Mask Setting this bit masks the interrupt when the txlatecol counter reaches half of the maximum value or the maximum value. 0b - MMC Transmit Late Collision Packet Counter Interrupt Mask is disabled 1b - MMC Transmit Late Collision Packet Counter Interrupt Mask is enabled
16 TXDEFPIM	MMC Transmit Deferred Packet Counter Interrupt Mask Setting this bit masks the interrupt when the txdeferred counter reaches half of the maximum value or the maximum value. 0b - MMC Transmit Deferred Packet Counter Interrupt Mask is disabled 1b - MMC Transmit Deferred Packet Counter Interrupt Mask is enabled

Table continues on the next page...

Table continued from the previous page...

Field	Function
15 TXMCOLGPIM	MMC Transmit Multiple Collision Good Packet Counter Interrupt Mask Setting this bit masks the interrupt when the txmulticol_g counter reaches half of the maximum value or the maximum value. 0b - MMC Transmit Multiple Collision Good Packet Counter Interrupt Mask is disabled 1b - MMC Transmit Multiple Collision Good Packet Counter Interrupt Mask is enabled
14 TXSCOLGPIM	MMC Transmit Single Collision Good Packet Counter Interrupt Mask Setting this bit masks the interrupt when the txsinglecol_g counter reaches half of the maximum value or the maximum value. 0b - MMC Transmit Single Collision Good Packet Counter Interrupt Mask is disabled 1b - MMC Transmit Single Collision Good Packet Counter Interrupt Mask is enabled
13 TXUFLOWERPIM	MMC Transmit Underflow Error Packet Counter Interrupt Mask Setting this bit masks the interrupt when the txunderflowerror counter reaches half of the maximum value or the maximum value. 0b - MMC Transmit Underflow Error Packet Counter Interrupt Mask is disabled 1b - MMC Transmit Underflow Error Packet Counter Interrupt Mask is enabled
12 TXBCGBPIM	MMC Transmit Broadcast Good Bad Packet Counter Interrupt Mask Setting this bit masks the interrupt when the txbroadcastpackets_gb counter reaches half of the maximum value or the maximum value. 0b - MMC Transmit Broadcast Good Bad Packet Counter Interrupt Mask is disabled 1b - MMC Transmit Broadcast Good Bad Packet Counter Interrupt Mask is enabled
11 TXMCGBPIM	MMC Transmit Multicast Good Bad Packet Counter Interrupt Mask Setting this bit masks the interrupt when the txmulticastpackets_gb counter reaches half of the maximum value or the maximum value. 0b - MMC Transmit Multicast Good Bad Packet Counter Interrupt Mask is disabled 1b - MMC Transmit Multicast Good Bad Packet Counter Interrupt Mask is enabled
10 TXUCGBPIM	MMC Transmit Unicast Good Bad Packet Counter Interrupt Mask Setting this bit masks the interrupt when the txunicastpackets_gb counter reaches half of the maximum value or the maximum value. 0b - MMC Transmit Unicast Good Bad Packet Counter Interrupt Mask is disabled 1b - MMC Transmit Unicast Good Bad Packet Counter Interrupt Mask is enabled
9 TX1024TMAXOCTGBPIM	MMC Transmit 1024 to Maximum Octet Good Bad Packet Counter Interrupt Mask Setting this bit masks the interrupt when the tx1024tomaxoctets_gb counter reaches half of the maximum value or the maximum value.

Table continues on the next page...

Table continued from the previous page...

Field	Function
	0b - MMC Transmit 1024 to Maximum Octet Good Bad Packet Counter Interrupt Mask is disabled 1b - MMC Transmit 1024 to Maximum Octet Good Bad Packet Counter Interrupt Mask is enabled
8 TX512T1023O CTGBPIM	MMC Transmit 512 to 1023 Octet Good Bad Packet Counter Interrupt Mask Setting this bit masks the interrupt when the tx512to1023octets_gb counter reaches half of the maximum value or the maximum value. 0b - MMC Transmit 512 to 1023 Octet Good Bad Packet Counter Interrupt Mask is disabled 1b - MMC Transmit 512 to 1023 Octet Good Bad Packet Counter Interrupt Mask is enabled
7 TX256T511OC TGBPIM	MMC Transmit 256 to 511 Octet Good Bad Packet Counter Interrupt Mask Setting this bit masks the interrupt when the tx256to511octets_gb counter reaches half of the maximum value or the maximum value. 0b - MMC Transmit 256 to 511 Octet Good Bad Packet Counter Interrupt Mask is disabled 1b - MMC Transmit 256 to 511 Octet Good Bad Packet Counter Interrupt Mask is enabled
6 TX128T255OC TGBPIM	MMC Transmit 128 to 255 Octet Good Bad Packet Counter Interrupt Mask Setting this bit masks the interrupt when the tx128to255octets_gb counter reaches half of the maximum value or the maximum value. 0b - MMC Transmit 128 to 255 Octet Good Bad Packet Counter Interrupt Mask is disabled 1b - MMC Transmit 128 to 255 Octet Good Bad Packet Counter Interrupt Mask is enabled
5 TX65T127OCT GBPIM	MMC Transmit 65 to 127 Octet Good Bad Packet Counter Interrupt Mask Setting this bit masks the interrupt when the tx65to127octets_gb counter reaches half of the maximum value or the maximum value. 0b - MMC Transmit 65 to 127 Octet Good Bad Packet Counter Interrupt Mask is disabled 1b - MMC Transmit 65 to 127 Octet Good Bad Packet Counter Interrupt Mask is enabled
4 TX64OCTGBPIM	MMC Transmit 64 Octet Good Bad Packet Counter Interrupt Mask Setting this bit masks the interrupt when the tx64octets_gb counter reaches half of the maximum value or the maximum value. 0b - MMC Transmit 64 Octet Good Bad Packet Counter Interrupt Mask is disabled 1b - MMC Transmit 64 Octet Good Bad Packet Counter Interrupt Mask is enabled
3 TXMCGPIM	MMC Transmit Multicast Good Packet Counter Interrupt Mask Setting this bit masks the interrupt when the txmulticastpackets_g counter reaches half of the maximum value or the maximum value. 0b - MMC Transmit Multicast Good Packet Counter Interrupt Mask is disabled 1b - MMC Transmit Multicast Good Packet Counter Interrupt Mask is enabled
2	MMC Transmit Broadcast Good Packet Counter Interrupt Mask

Table continues on the next page...

Table continued from the previous page...

Field	Function
TXBCGPIM	Setting this bit masks the interrupt when the txbroadcastpackets_g counter reaches half of the maximum value or the maximum value. 0b - MMC Transmit Broadcast Good Packet Counter Interrupt Mask is disabled 1b - MMC Transmit Broadcast Good Packet Counter Interrupt Mask is enabled
1 TXGBPKTIM	MMC Transmit Good Bad Packet Counter Interrupt Mask Setting this bit masks the interrupt when the txpacketcount_gb counter reaches half of the maximum value or the maximum value. 0b - MMC Transmit Good Bad Packet Counter Interrupt Mask is disabled 1b - MMC Transmit Good Bad Packet Counter Interrupt Mask is enabled
0 TXGBOCTIM	MMC Transmit Good Bad Octet Counter Interrupt Mask Setting this bit masks the interrupt when the txoctetcount_gb counter reaches half of the maximum value or the maximum value. 0b - MMC Transmit Good Bad Octet Counter Interrupt Mask is disabled 1b - MMC Transmit Good Bad Octet Counter Interrupt Mask is enabled

1.4.65 (Tx_Octet_Count_Good_Bad)

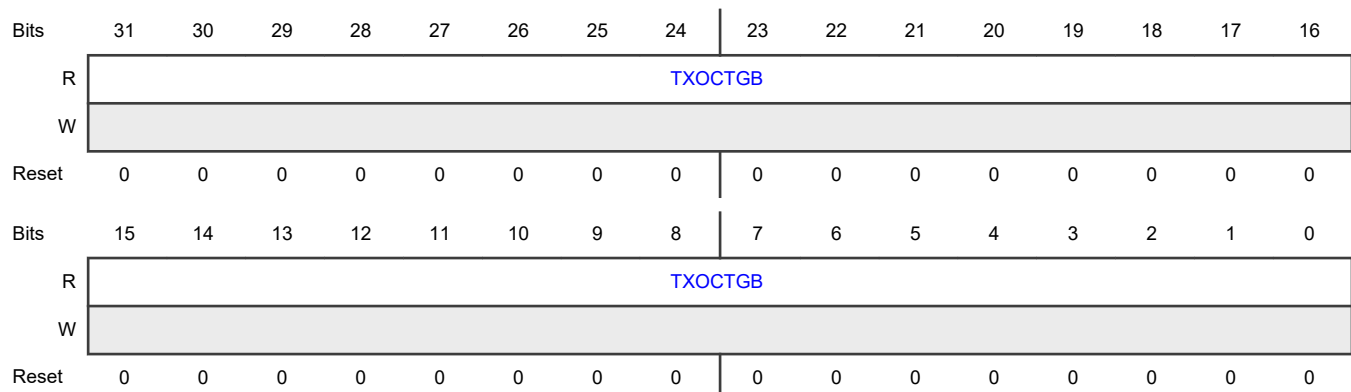
Offset

Register	Offset
Tx_Octet_Count_Good_Bad	714h

Function

This register provides the number of bytes transmitted by the DWC_ether_qos, exclusive of preamble and retried bytes, in good and bad packets.

Diagram



Fields

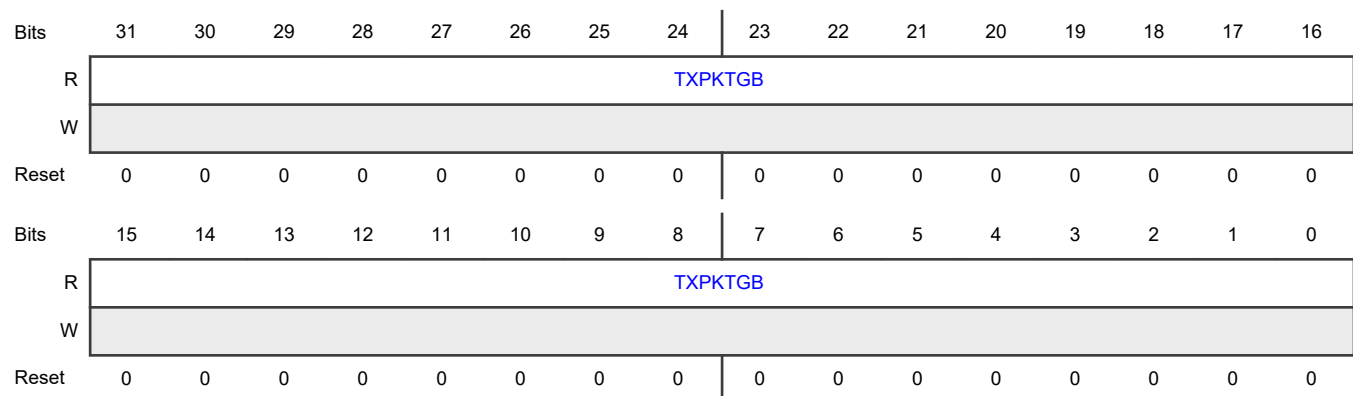
Field	Function
31-0 TXOCTGB	Tx Octet Count Good Bad This field indicates the number of bytes transmitted, exclusive of preamble and retried bytes, in good and bad packets.

1.4.66 (Tx_Packet_Count_Good_Bad)**Offset**

Register	Offset
Tx_Packet_Count_Good_Bad	718h

Function

This register provides the number of good and bad packets transmitted by DWC_ether_qos, exclusive of retried packets.

Diagram**Fields**

Field	Function
31-0 TXPKTGB	Tx Packet Count Good Bad This field indicates the number of good and bad packets transmitted, exclusive of retried packets.

1.4.67 (Tx_Broadcast_Packets_Good)

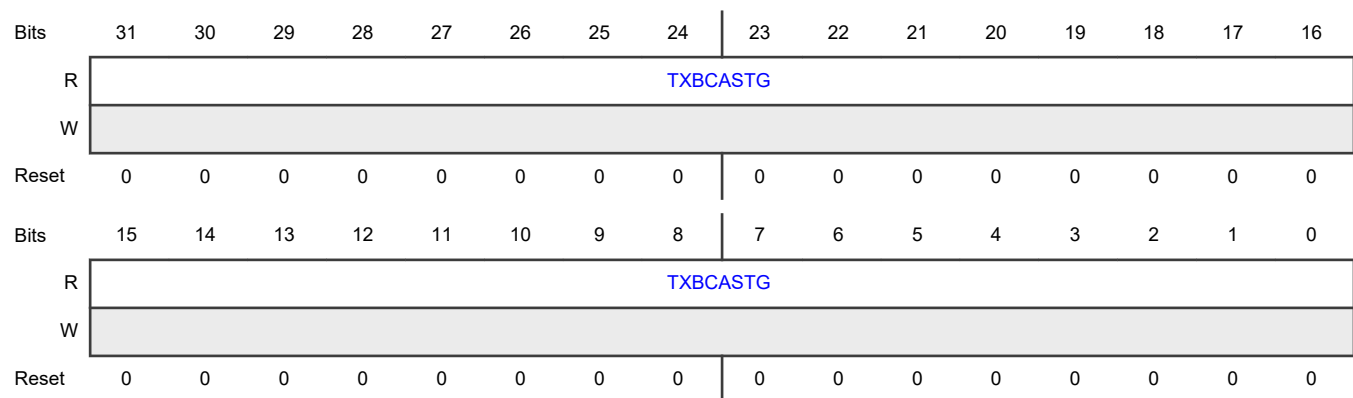
Offset

Register	Offset
Tx_Broadcast_Packets_Good	71Ch

Function

This register provides the number of good broadcast packets transmitted by DWC_ether_qos.

Diagram



Fields

Field	Function
31-0	Tx Broadcast Packets Good
TXBCASTG	This field indicates the number of good broadcast packets transmitted.

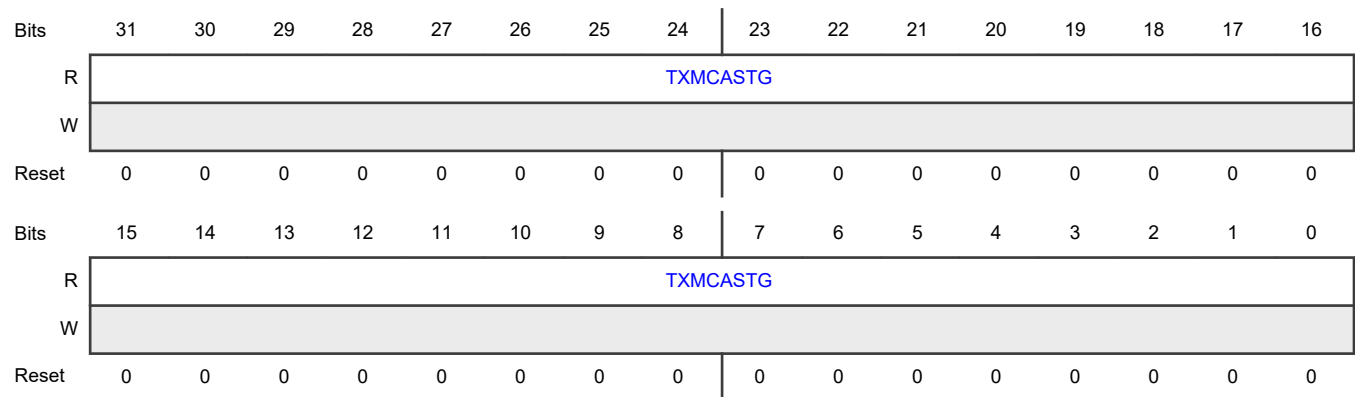
1.4.68 (Tx_Multicast_Packets_Good)

Offset

Register	Offset
Tx_Multicast_Packets_Good	720h

Function

This register provides the number of good multicast packets transmitted by DWC_ether_qos.

Diagram**Fields**

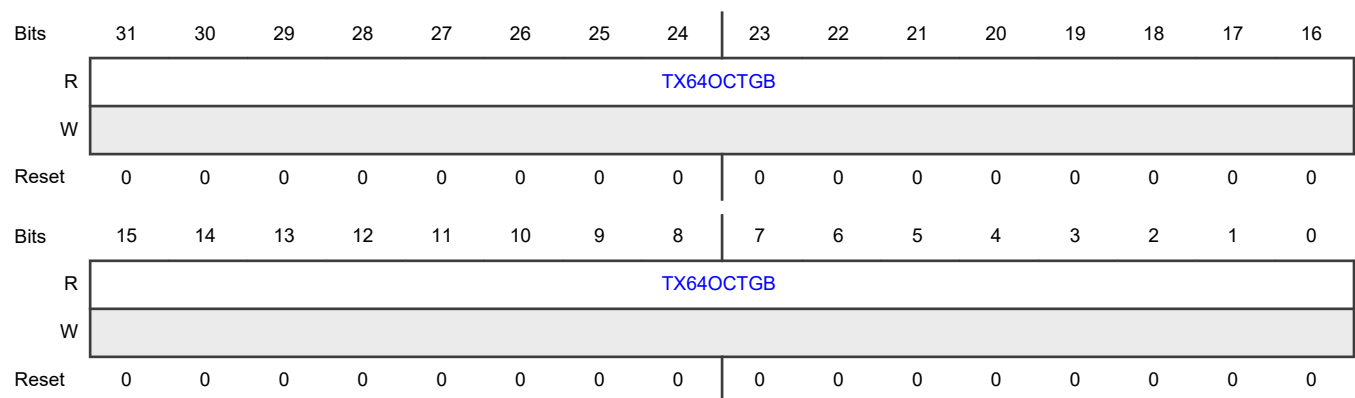
Field	Function
31-0	Tx Multicast Packets Good
TXMCASTG	This field indicates the number of good multicast packets transmitted.

1.4.69 (Tx_64Octets_Packets_Good_Bad)**Offset**

Register	Offset
Tx_64Octets_Packets_Good_Bad	724h

Function

This register provides the number of good and bad packets transmitted by DWC_ether_qos with length 64 bytes, exclusive of preamble and retried packets.

Diagram

Fields

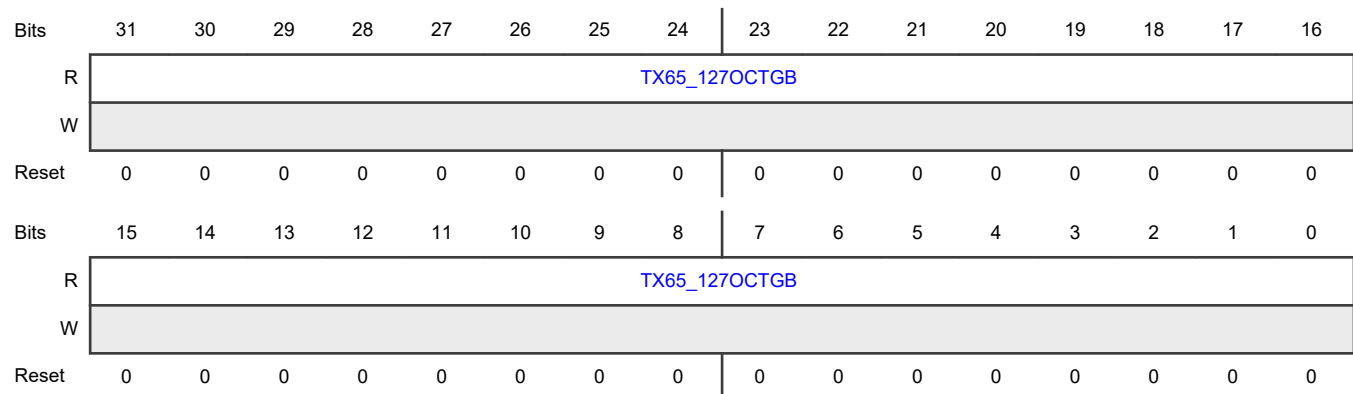
Field	Function
31-0 TX64OCTGB	Tx 64Octets Packets Good_Bad This field indicates the number of good and bad packets transmitted with length 64 bytes, exclusive of preamble and retried packets.

1.4.70 (Tx_65To127Octets_Packets_Good_Bad)**Offset**

Register	Offset
Tx_65To127Octets_Packets_Good_Bad	728h

Function

This register provides the number of good and bad packets transmitted by DWC_ether_qos with length between 65 and 127 (inclusive) bytes, exclusive of preamble and retried packets.

Diagram**Fields**

Field	Function
31-0 TX65_127OCTGB	Tx 65To127Octets Packets Good Bad This field indicates the number of good and bad packets transmitted with length between 65 and 127 (inclusive) bytes, exclusive of preamble and retried packets.

1.4.71 (Tx_128To255Octets_Packets_Good_Bad)

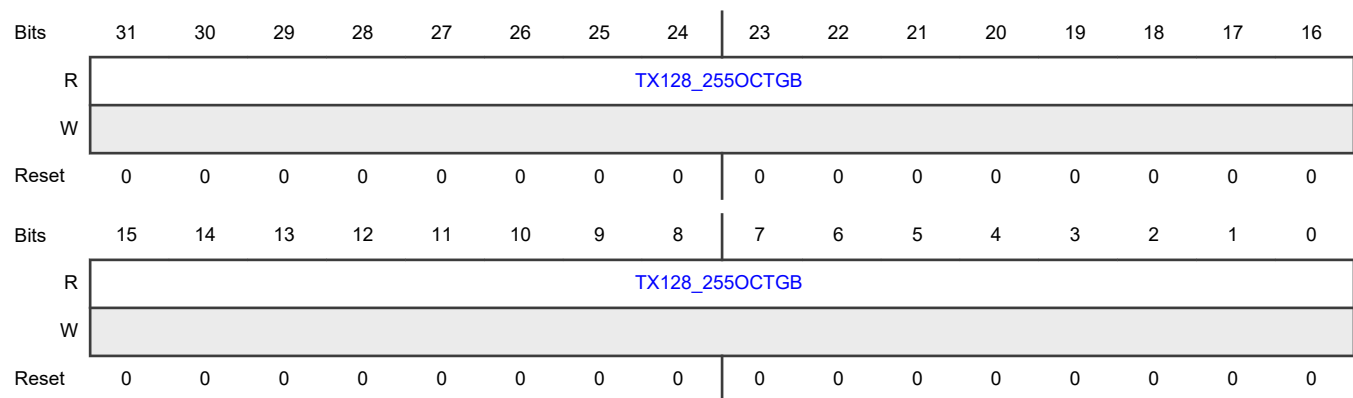
Offset

Register	Offset
Tx_128To255Octets_Packets_Good_Bad	72Ch

Function

This register provides the number of good and bad packets transmitted by DWC_ether_qos with length between 128 to 255 (inclusive) bytes, exclusive of preamble and retried packets.

Diagram



Fields

Field	Function
31-0	Tx 128To255Octets Packets Good Bad
TX128_255OCTGB	This field indicates the number of good and bad packets transmitted with length between 128 and 255 (inclusive) bytes, exclusive of preamble and retried packets.

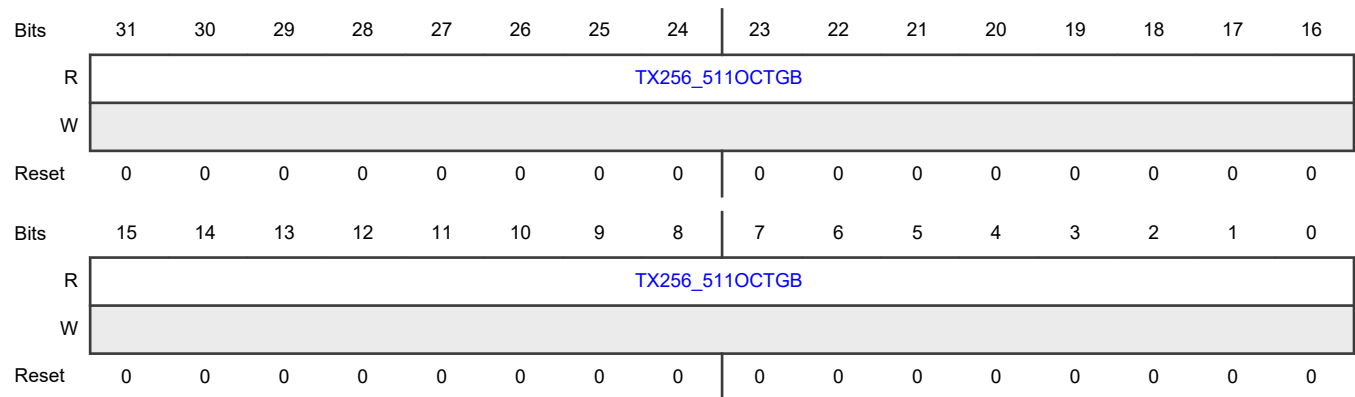
1.4.72 (Tx_256To511Octets_Packets_Good_Bad)

Offset

Register	Offset
Tx_256To511Octets_Packets_Good_Bad	730h

Function

This register provides the number of good and bad packets transmitted by DWC_ether_qos with length between 256 to 511 (inclusive) bytes, exclusive of preamble and retried packets.

Diagram**Fields**

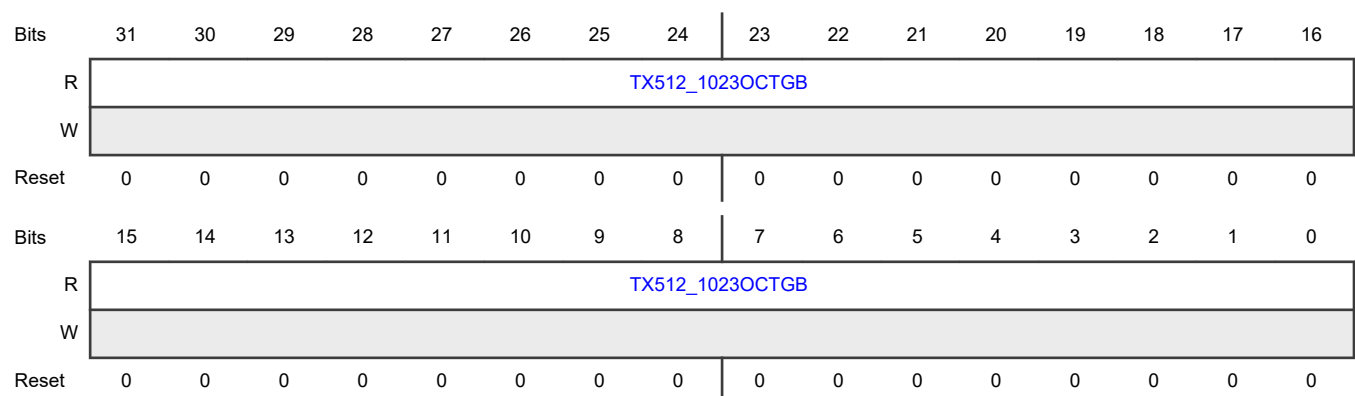
Field	Function
31-0	Tx 256To511Octets Packets Good Bad
TX256_511OCTGB	This field indicates the number of good and bad packets transmitted with length between 256 and 511 (inclusive) bytes, exclusive of preamble and retried packets.

1.4.73 (Tx_512To1023Octets_Packets_Good_Bad)**Offset**

Register	Offset
Tx_512To1023Octets_Packets_Good_Bad	734h

Function

This register provides the number of good and bad packets transmitted by DWC_ether_qos with length 512 to 1023 (inclusive) bytes, exclusive of preamble and retried packets.

Diagram

Fields

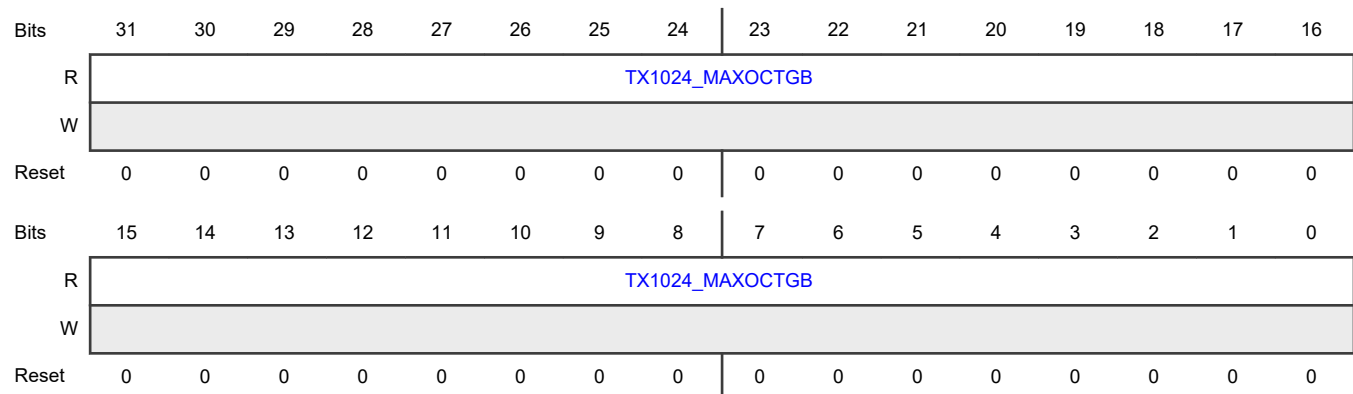
Field	Function
31-0 TX512_1023OC TGB	Tx 512To1023Octets Packets Good Bad This field indicates the number of good and bad packets transmitted with length between 512 and 1023 (inclusive) bytes, exclusive of preamble and retried packets.

1.4.74 (Tx_1024ToMaxOctets_Packets_Good_Bad)**Offset**

Register	Offset
Tx_1024ToMaxOctets_P ackets_Good_Bad	738h

Function

This register provides the number of good and bad packets transmitted by DWC_ether_qos with length 1024 to maxsize (inclusive) bytes, exclusive of preamble and retried packets.

Diagram**Fields**

Field	Function
31-0 TX1024_MAXO CTGB	Tx 1024ToMaxOctets Packets Good Bad This field indicates the number of good and bad packets transmitted with length between 1024 and maxsize (inclusive) bytes, exclusive of preamble and retried packets.

1.4.75 (Tx_Unicast_Packets_Good_Bad)

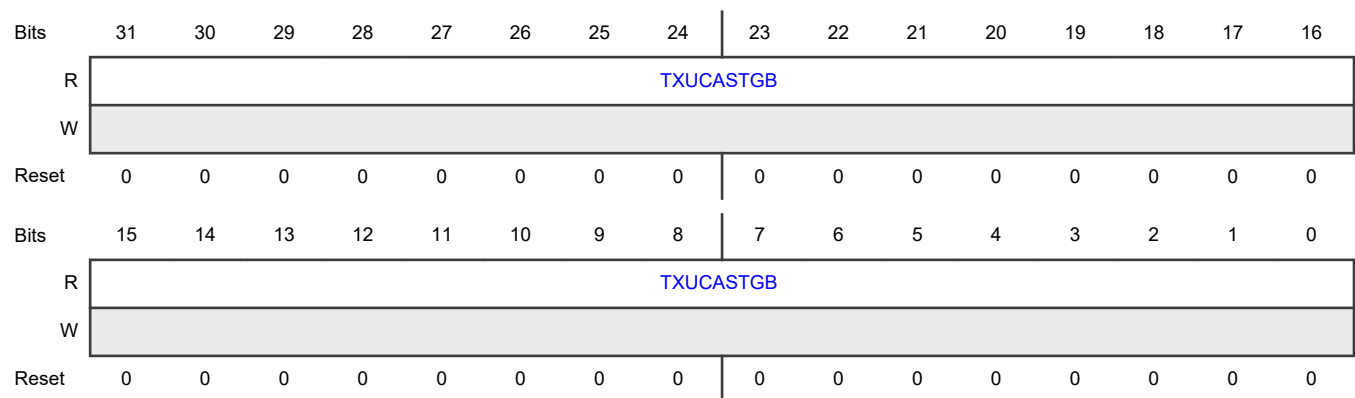
Offset

Register	Offset
Tx_Unicast_Packets_Good_Bad	73Ch

Function

This register provides the number of good and bad unicast packets transmitted by DWC_ether_qos.

Diagram



Fields

Field	Function
31-0	Tx Unicast Packets Good Bad
TXUCASTGB	This field indicates the number of good and bad unicast packets transmitted.

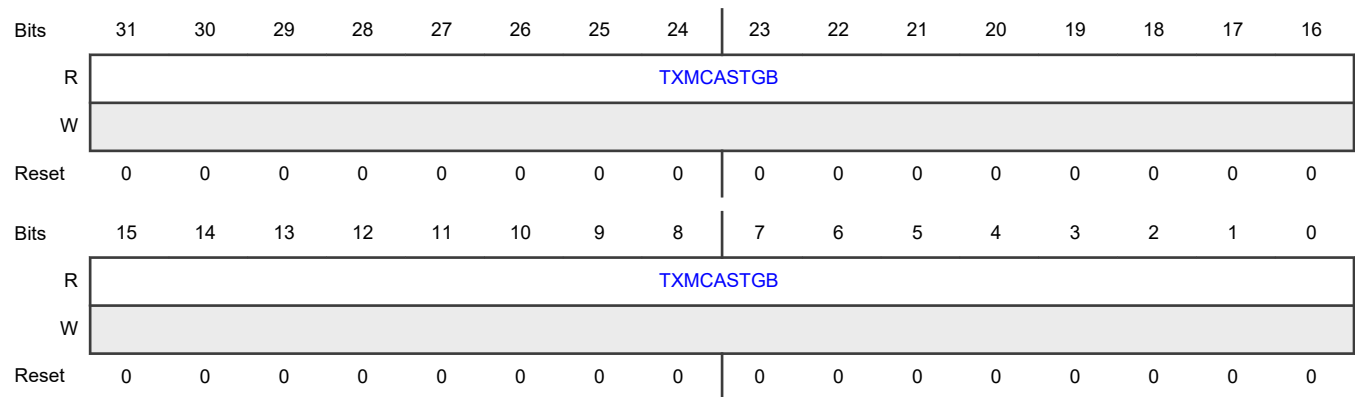
1.4.76 (Tx_Multicast_Packets_Good_Bad)

Offset

Register	Offset
Tx_Multicast_Packets_Good_Bad	740h

Function

This register provides the number of good and bad multicast packets transmitted by DWC_ether_qos.

Diagram**Fields**

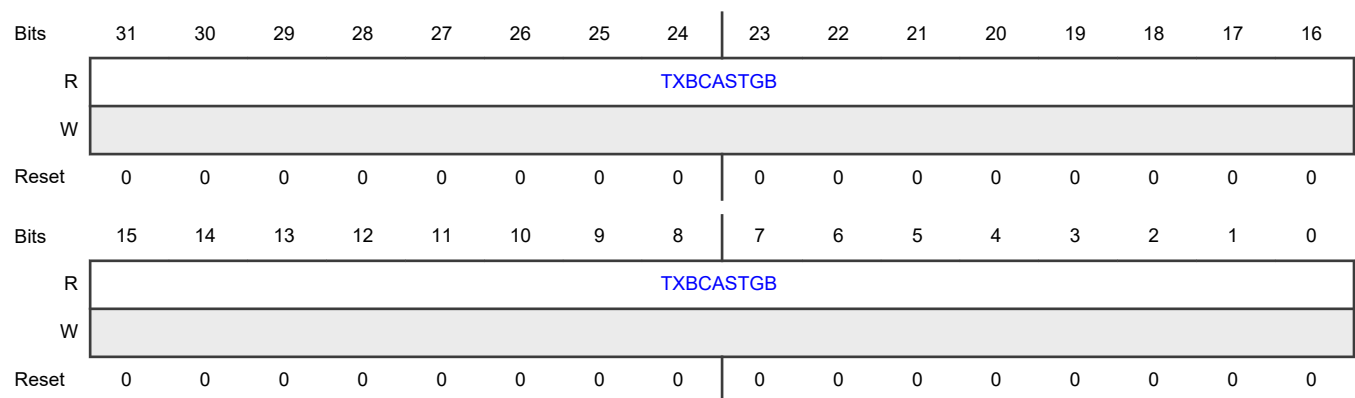
Field	Function
31-0	Tx Multicast Packets Good Bad
TXMCASTGB	This field indicates the number of good and bad multicast packets transmitted.

1.4.77 (Tx_Broadcast_Packets_Good_Bad)**Offset**

Register	Offset
Tx_Broadcast_Packets_Good_Bad	744h

Function

This register provides the number of good and bad broadcast packets transmitted by DWC_ether_qos.

Diagram

Fields

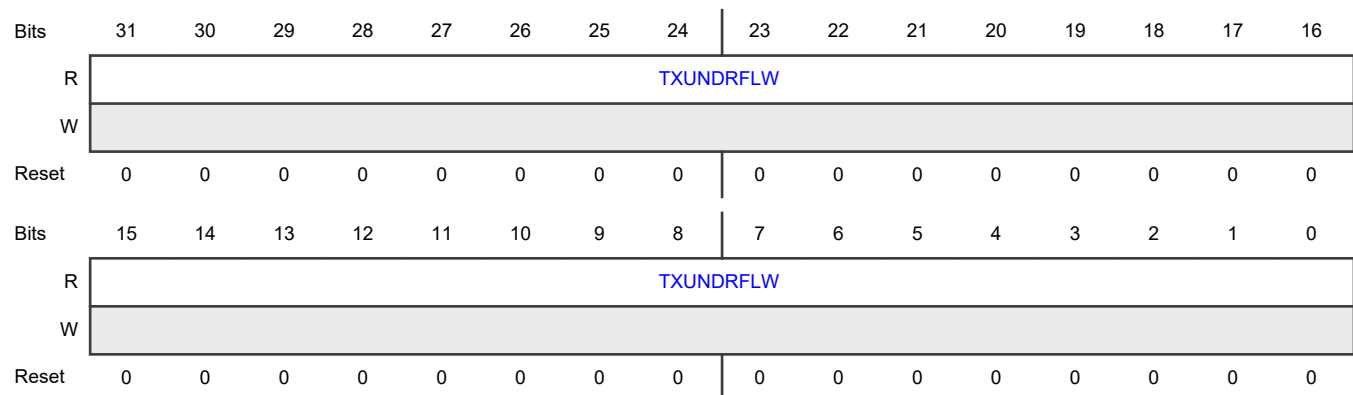
Field	Function
31-0 TXBCASTGB	Tx Broadcast Packets Good Bad This field indicates the number of good and bad broadcast packets transmitted.

1.4.78 (Tx_Underflow_Error_Packets)**Offset**

Register	Offset
Tx_Underflow_Error_Packets	748h

Function

This register provides the number of packets aborted by DWC_ether_qos because of packets underflow error.

Diagram**Fields**

Field	Function
31-0 TXUNDRFLW	Tx Underflow Error Packets This field indicates the number of packets aborted because of packets underflow error.

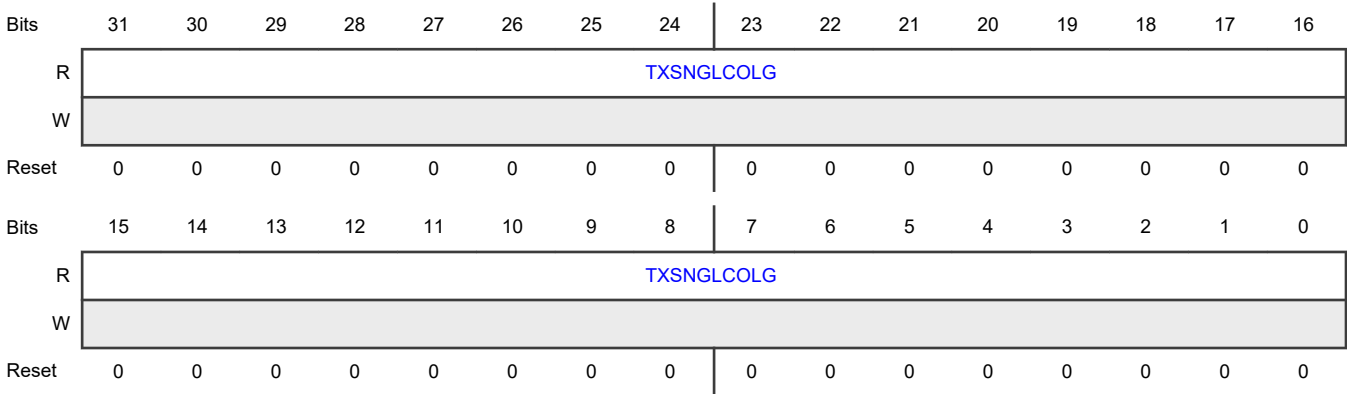
1.4.79 (Tx_Single_Collision_Good_Packets)**Offset**

Register	Offset
Tx_Single_Collision_Good_Packets	74Ch

Function

This register provides the number of successfully transmitted packets by DWC_ether_qos after a single collision in the half-duplex mode.

Diagram



Fields

Field	Function
31-0 TXSNGLCOLG	Tx Single Collision Good Packets This field indicates the number of successfully transmitted packets after a single collision in the half-duplex mode.

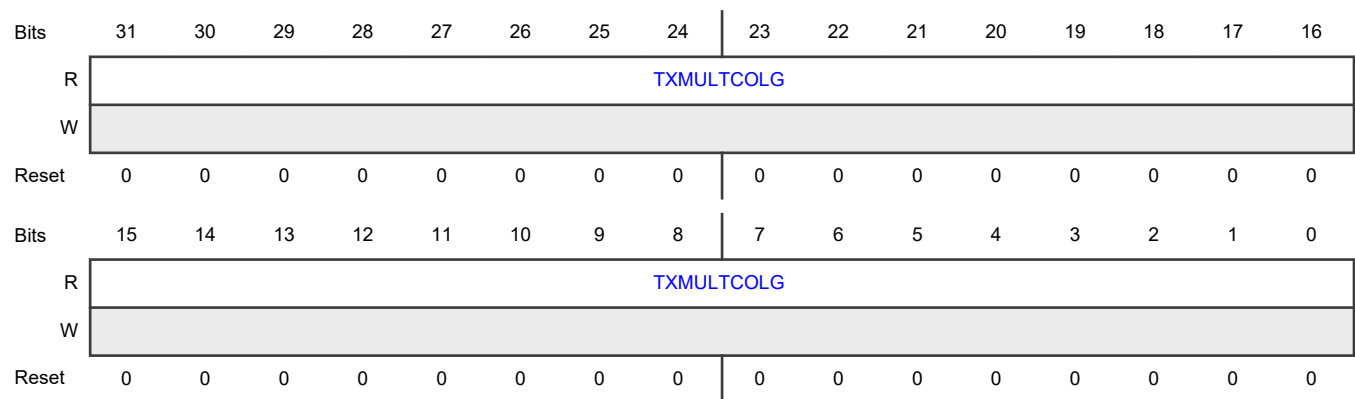
1.4.80 (Tx_Multiple_Collision_Good_Packets)

Offset

Register	Offset
Tx_Multiple_Collision_Good_Packets	750h

Function

This register provides the number of successfully transmitted packets by DWC_ether_qos after multiple collisions in the half-duplex mode.

Diagram**Fields**

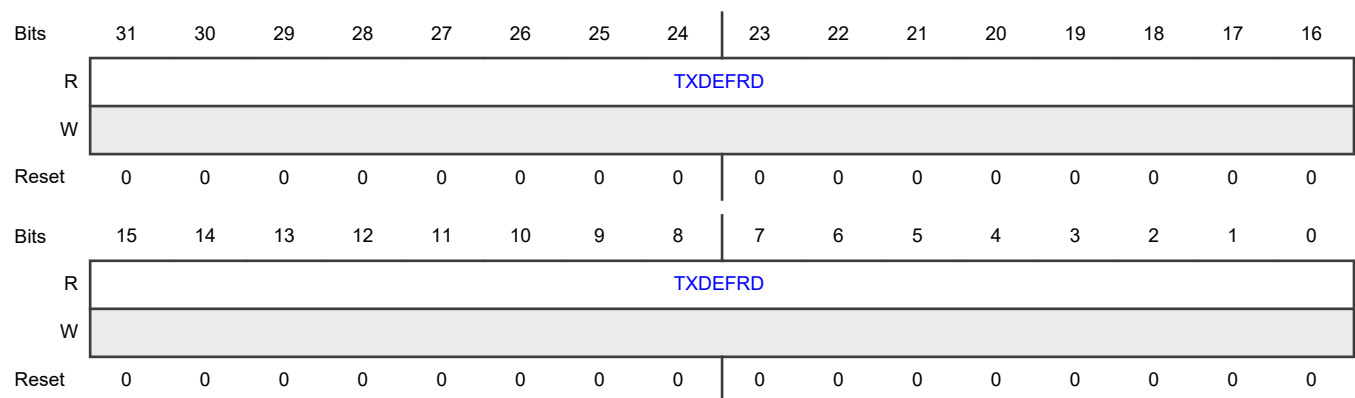
Field	Function
31-0	Tx Multiple Collision Good Packets
TXMULTCOLG	This field indicates the number of successfully transmitted packets after multiple collisions in the half-duplex mode.

1.4.81 (Tx_Deferred_Packets)**Offset**

Register	Offset
Tx_Deferred_Packets	754h

Function

This register provides the number of successfully transmitted by DWC_ether_qos after a deferral in the half-duplex mode.

Diagram

Fields

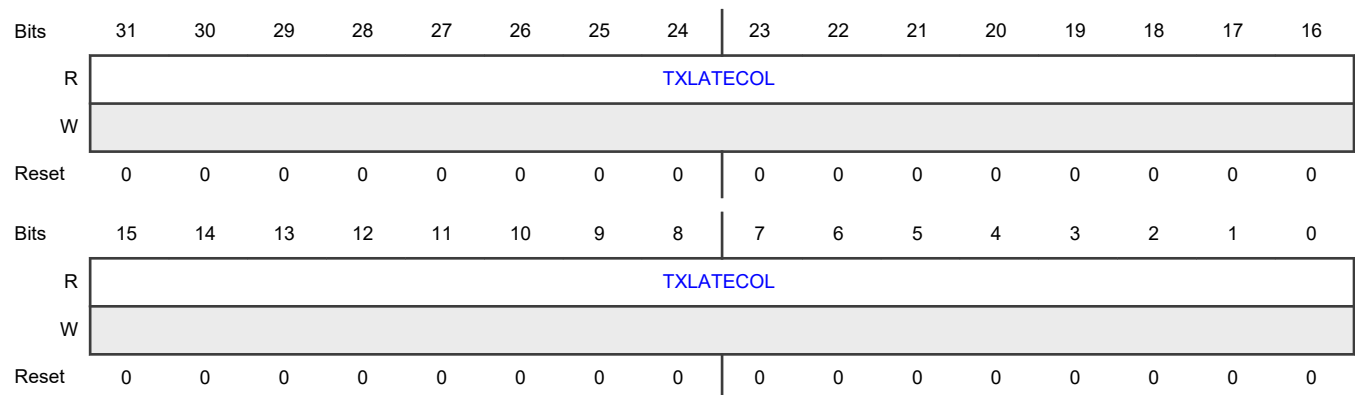
Field	Function
31-0 TXDEFRD	Tx Deferred Packets This field indicates the number of successfully transmitted after a deferral in the half-duplex mode.

1.4.82 (Tx_Late_Collision_Packets)**Offset**

Register	Offset
Tx_Late_Collision_Packets	758h

Function

This register provides the number of packets aborted by DWC_ether_qos because of late collision error.

Diagram**Fields**

Field	Function
31-0 TXLATECOL	Tx Late Collision Packets This field indicates the number of packets aborted because of late collision error.

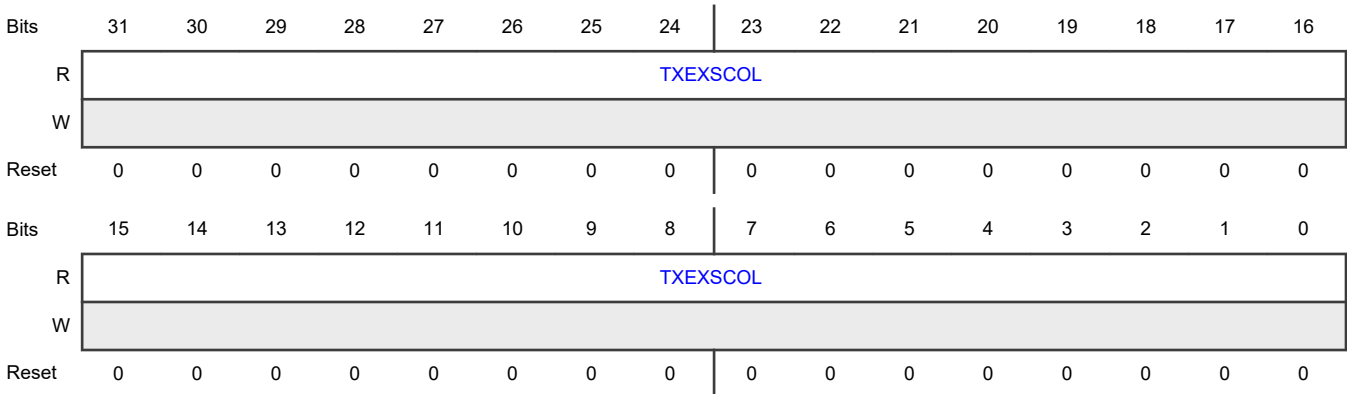
1.4.83 (Tx_Excessive_Collision_Packets)**Offset**

Register	Offset
Tx_Excessive_Collision_Packets	75Ch

Function

This register provides the number of packets aborted by DWC_ether_qos because of excessive (16) collision errors.

Diagram



Fields

Field	Function
31-0	Tx Excessive Collision Packets
TXEXSCOL	This field indicates the number of packets aborted because of excessive (16) collision errors.

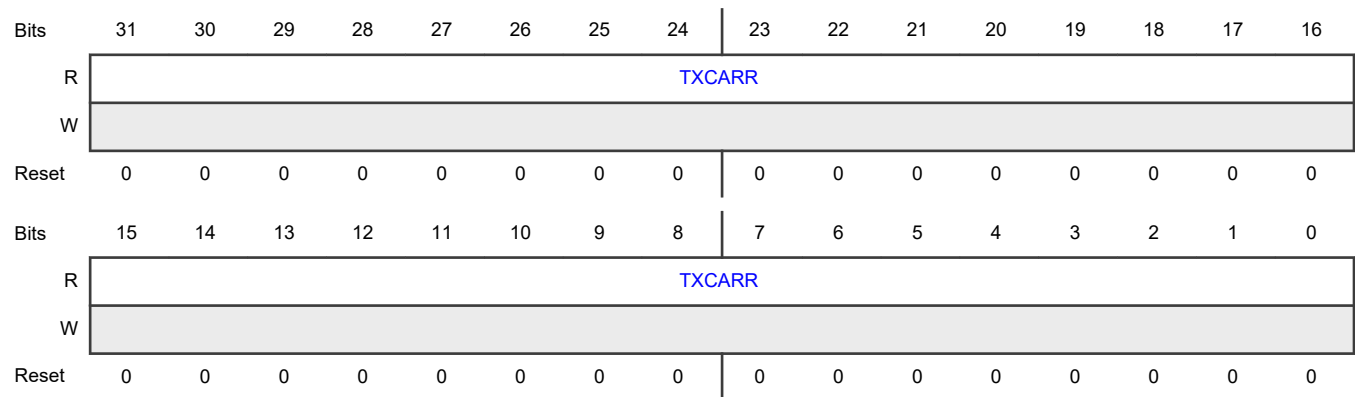
1.4.84 (Tx_Carrier_Error_Packets)

Offset

Register	Offset
Tx_Carrier_Error_Packets	760h

Function

This register provides the number of packets aborted by DWC_ether_qos because of carrier sense error (no carrier or loss of carrier).

Diagram**Fields**

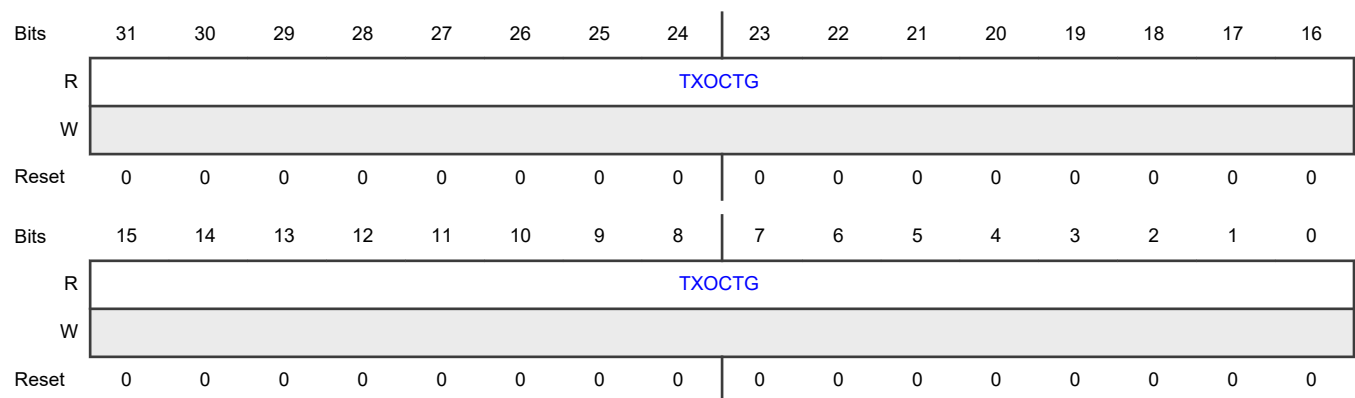
Field	Function
31-0 TXCARR	Tx Carrier Error Packets This field indicates the number of packets aborted because of carrier sense error (no carrier or loss of carrier).

1.4.85 (Tx_Octet_Count_Good)**Offset**

Register	Offset
Tx_Octet_Count_Good	764h

Function

This register provides the number of bytes transmitted by DWC_ether_qos, exclusive of preamble, only in good packets.

Diagram

Fields

Field	Function
31-0 TXOCTG	Tx Octet Count Good This field indicates the number of bytes transmitted, exclusive of preamble, only in good packets.

1.4.86 (Tx_Packet_Count_Good)

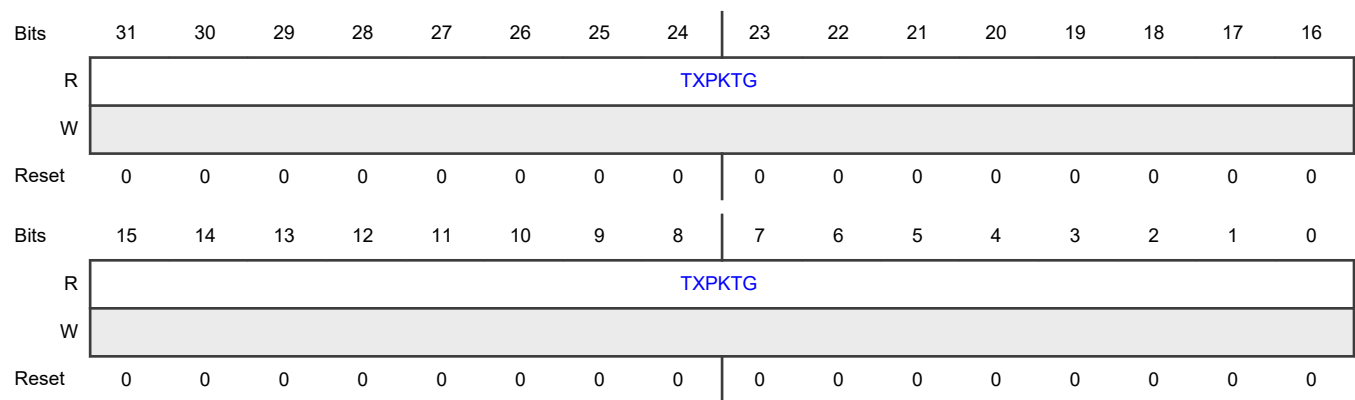
Offset

Register	Offset
Tx_Packet_Count_Good	768h

Function

This register provides the number of good packets transmitted by DWC_ether_qos.

Diagram



Fields

Field	Function
31-0 TXPKTG	Tx Packet Count Good This field indicates the number of good packets transmitted.

1.4.87 (Tx_Excessive_Deferral_Error)

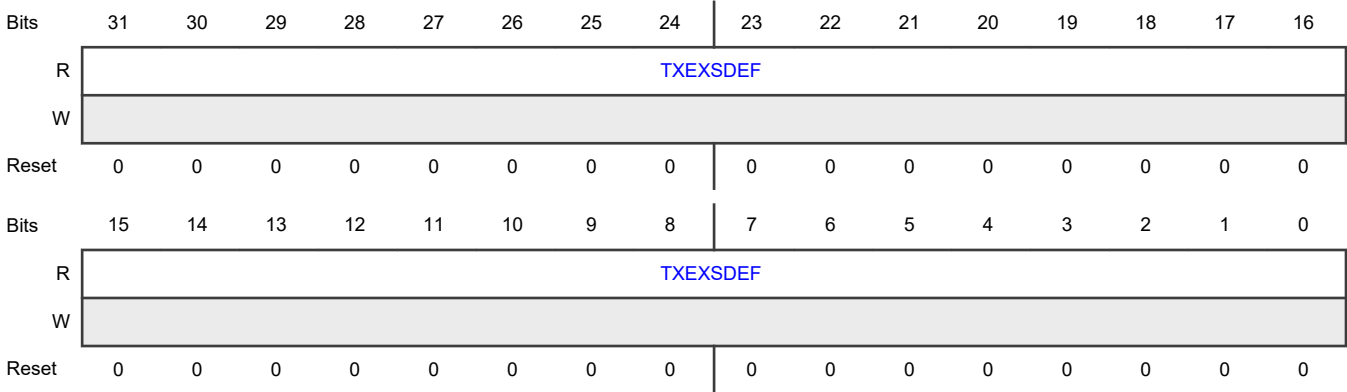
Offset

Register	Offset
Tx_Excessive_Deferral_Error	76Ch

Function

This register provides the number of packets aborted by DWC_ether_qos because of excessive deferral error (deferred for more than two max-sized packet times).

Diagram



Fields

Field	Function
31-0 TXEXSDEF	Tx Excessive Deferral Error This field indicates the number of packets aborted because of excessive deferral error (deferred for more than two max-sized packet times).

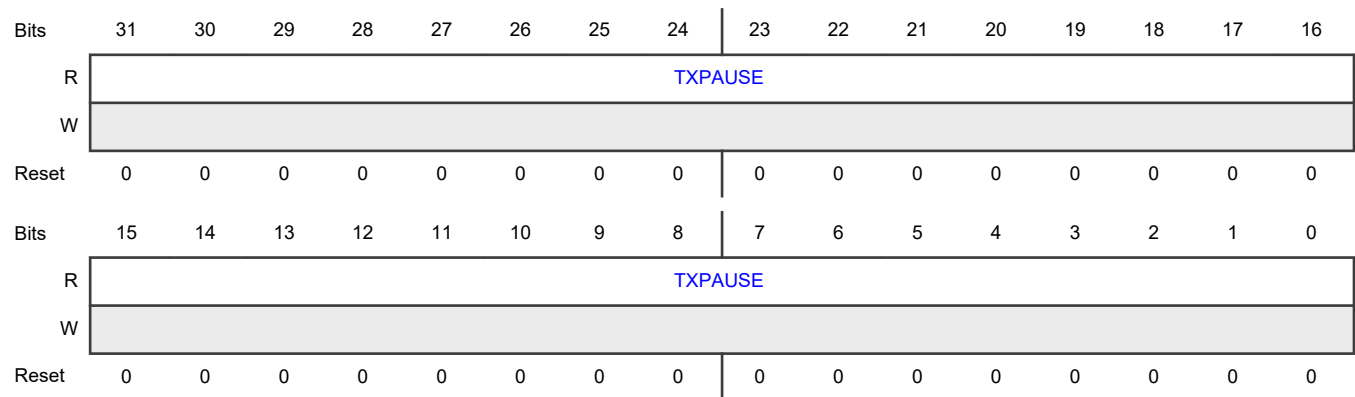
1.4.88 (Tx_Pause_Packets)

Offset

Register	Offset
Tx_Pause_Packets	770h

Function

This register provides the number of good Pause packets transmitted by DWC_ether_qos.

Diagram**Fields**

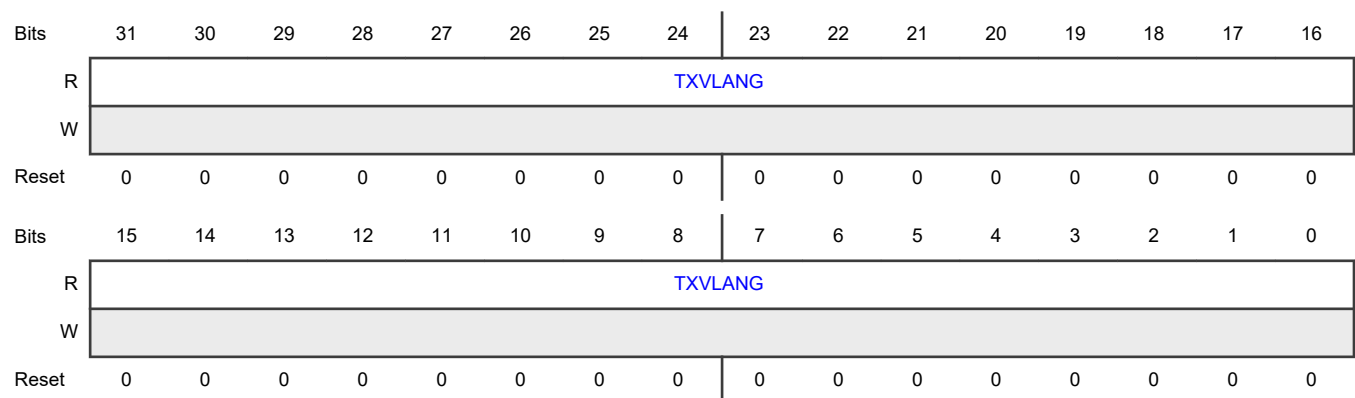
Field	Function
31-0	Tx Pause Packets
TXPAUSE	This field indicates the number of good Pause packets transmitted.

1.4.89 (Tx_VLAN_Packets_Good)**Offset**

Register	Offset
Tx_VLAN_Packets_Good	774h

Function

This register provides the number of good VLAN packets transmitted by DWC_ether_qos.

Diagram

Fields

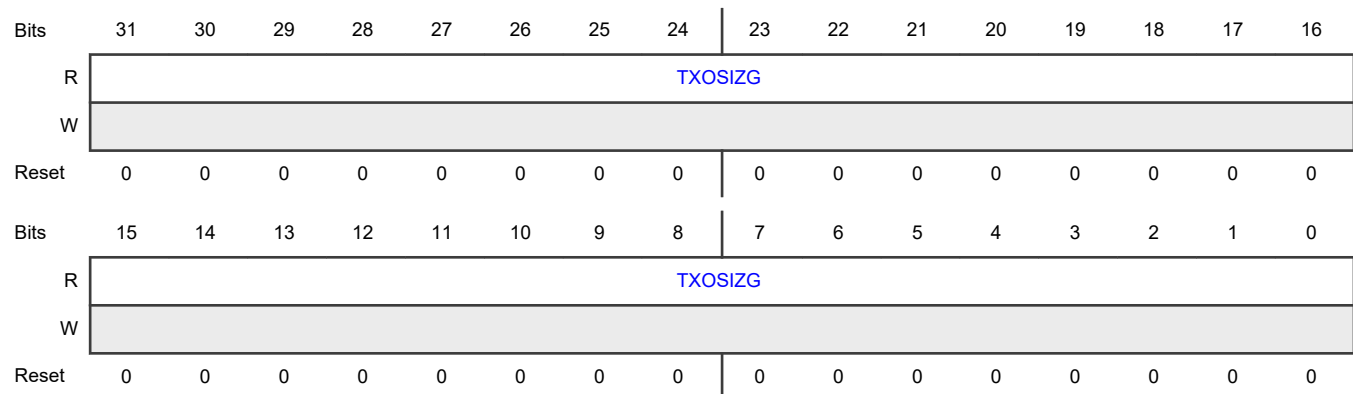
Field	Function
31-0 TXVLANG	Tx VLAN Packets Good This field provides the number of good VLAN packets transmitted.

1.4.90 (Tx_OSize_Packets_Good)**Offset**

Register	Offset
Tx_OSize_Packets_Good	778h

Function

This register provides the number of packets transmitted by DWC_ether_qos without errors and with length greater than the maxsize (1,518 or 1,522 bytes for VLAN tagged packets; 2000 bytes if enabled in S2KP bit of the MAC_Configuration register).

Diagram**Fields**

Field	Function
31-0 TXOSIZG	Tx OSize Packets Good This field indicates the number of packets transmitted without errors and with length greater than the maxsize (1,518 or 1,522 bytes for VLAN tagged packets; 2000 bytes if enabled in S2KP bit of the MAC_Configuration register).

1.4.91 (Rx_Packets_Count_Good_Bad)

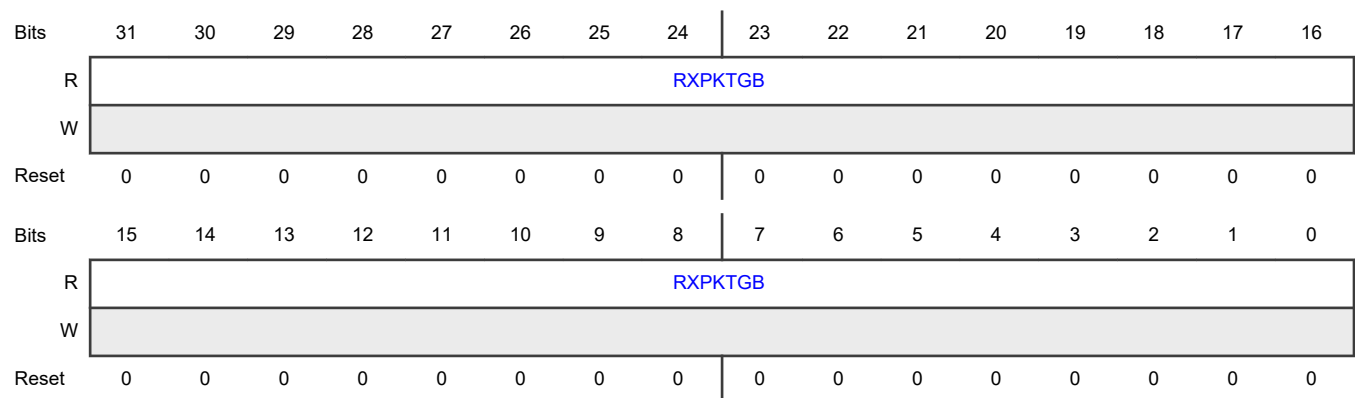
Offset

Register	Offset
Rx_Packets_Count_Good_Bad	780h

Function

This register provides the number of good and bad packets received by DWC_ether_qos.

Diagram



Fields

Field	Function
31-0	Rx Packets Count Good Bad
RXPKTGB	This field indicates the number of good and bad packets received.

1.4.92 (Rx_Octet_Count_Good_Bad)

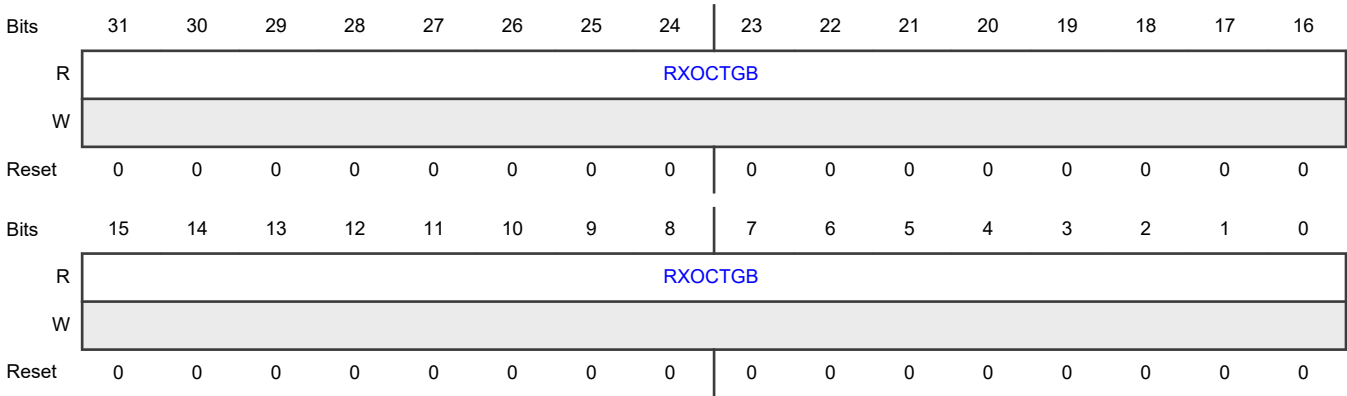
Offset

Register	Offset
Rx_Octet_Count_Good_Bad	784h

Function

This register provides the number of bytes received by DWC_ther_qos, exclusive of preamble, in good and bad packets.

Diagram



Fields

Field	Function
31-0	Rx Octet Count Good Bad
RXOCTGB	This field indicates the number of bytes received, exclusive of preamble, in good and bad packets.

1.4.93 (Rx_Octet_Count_Good)

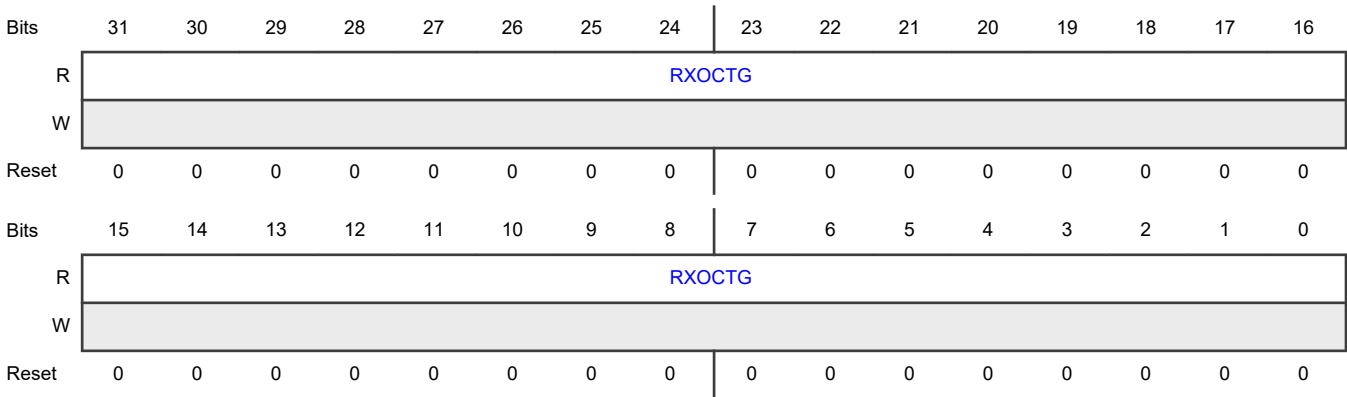
Offset

Register	Offset
Rx_Octet_Count_Good	788h

Function

This register provides the number of bytes received by DWC_ether_qos, exclusive of preamble, only in good packets.

Diagram



Fields

Field	Function
31-0 RXOCTG	Rx Octet Count Good This field indicates the number of bytes received, exclusive of preamble, only in good packets.

1.4.94 (Rx_Broadcast_Packets_Good)

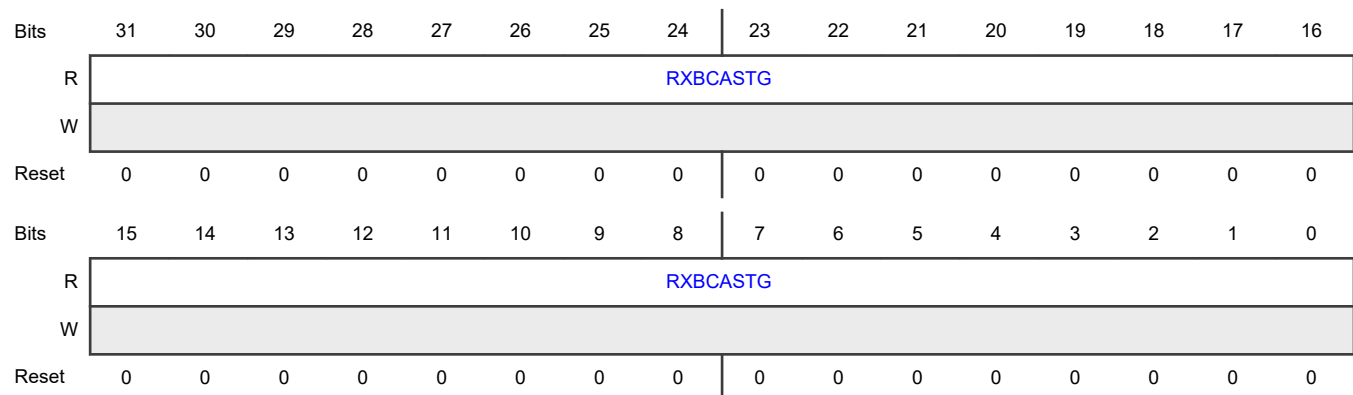
Offset

Register	Offset
Rx_Broadcast_Packets_Good	78Ch

Function

This register provides the number of good broadcast packets received by DWC_ether_qos.

Diagram



Fields

Field	Function
31-0 RXBCASTG	Rx Broadcast Packets Good This field indicates the number of good broadcast packets received.

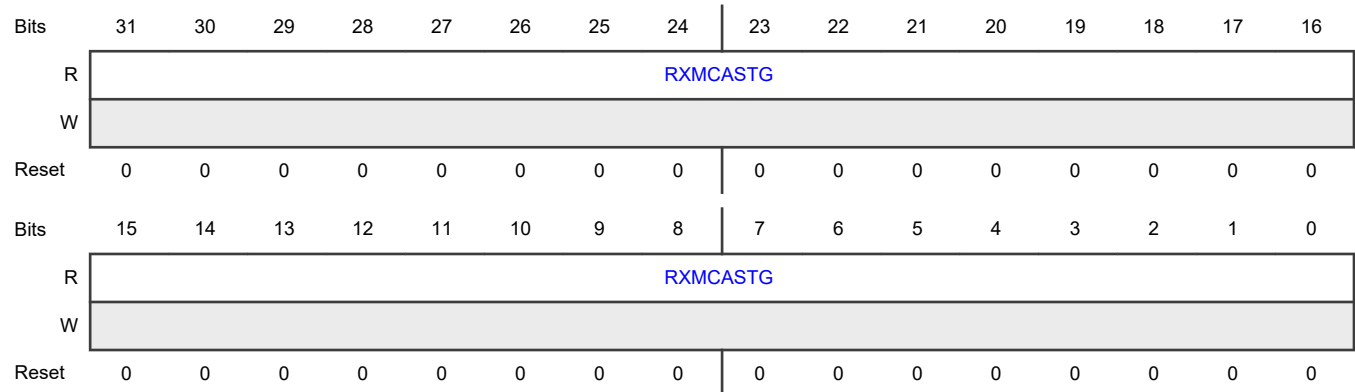
1.4.95 (Rx_Multicast_Packets_Good)

Offset

Register	Offset
Rx_Multicast_Packets_Good	790h

Function

This register provides the number of good multicast packets received by DWC_ether_qos.

Diagram**Fields**

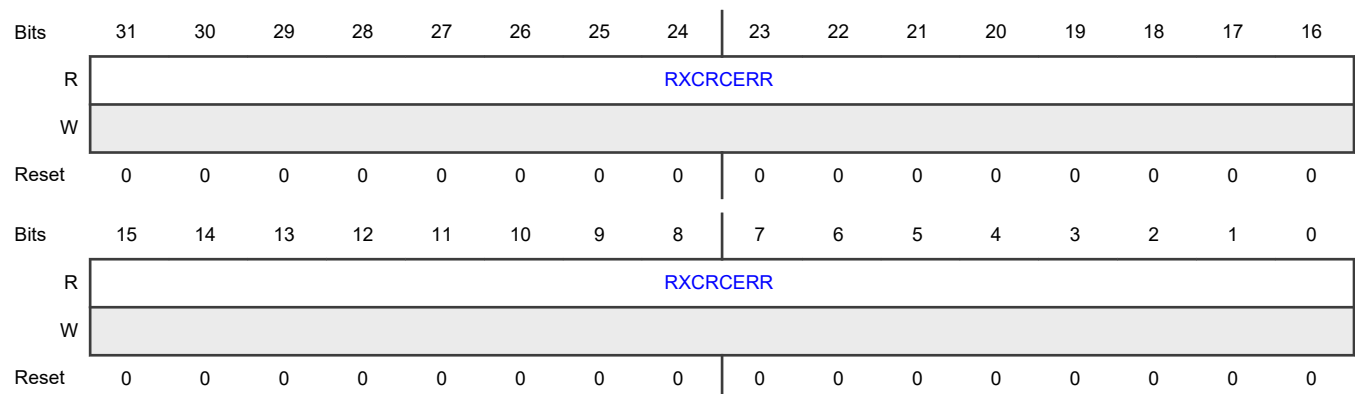
Field	Function
31-0 RXMCASTG	Rx Multicast Packets Good This field indicates the number of good multicast packets received.

1.4.96 (Rx_CRC_Error_Packets)**Offset**

Register	Offset
Rx_CRC_Error_Packets	794h

Function

This register provides the number of packets received by DWC_ether_qos with CRC error.

Diagram

Fields

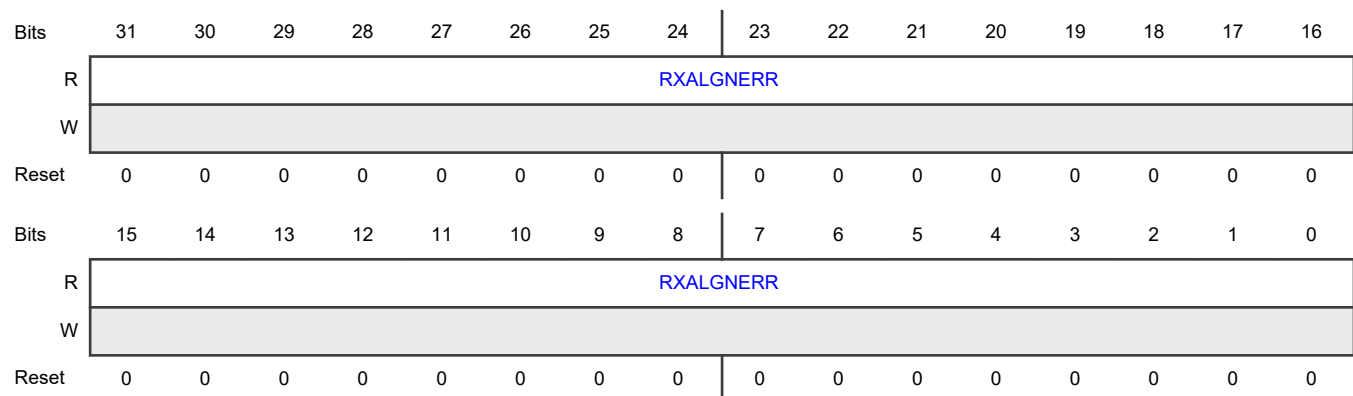
Field	Function
31-0 RXCRCERR	Rx CRC Error Packets This field indicates the number of packets received with CRC error.

1.4.97 (Rx_Alignment_Error_Packets)**Offset**

Register	Offset
Rx_Alignment_Error_Packets	798h

Function

This register provides the number of packets received by DWC_ether_qos with alignment (dribble) error. It is valid only in 10/100 mode.

Diagram**Fields**

Field	Function
31-0 RXALGNERR	Rx Alignment Error Packets This field indicates the number of packets received with alignment (dribble) error. It is valid only in 10/100 mode.

1.4.98 (Rx_Runt_Error_Packets)

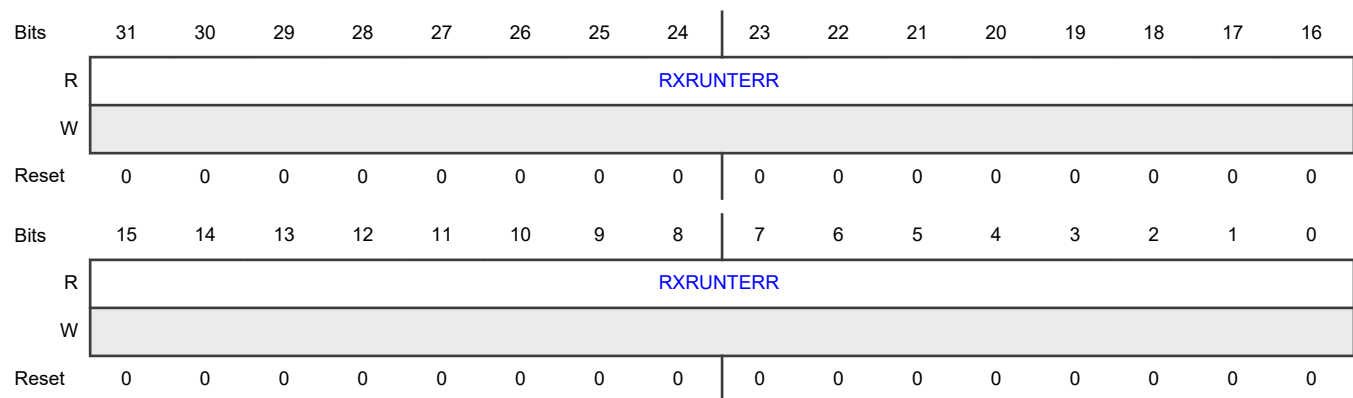
Offset

Register	Offset
Rx_Runt_Error_Packets	79Ch

Function

This register provides the number of packets received by DWC_ether_qos with runt (length less than 64 bytes and CRC error) error.

Diagram



Fields

Field	Function
31-0	Rx Runt Error Packets
RXRUNTERR	This field indicates the number of packets received with runt (length less than 64 bytes and CRC error) error.

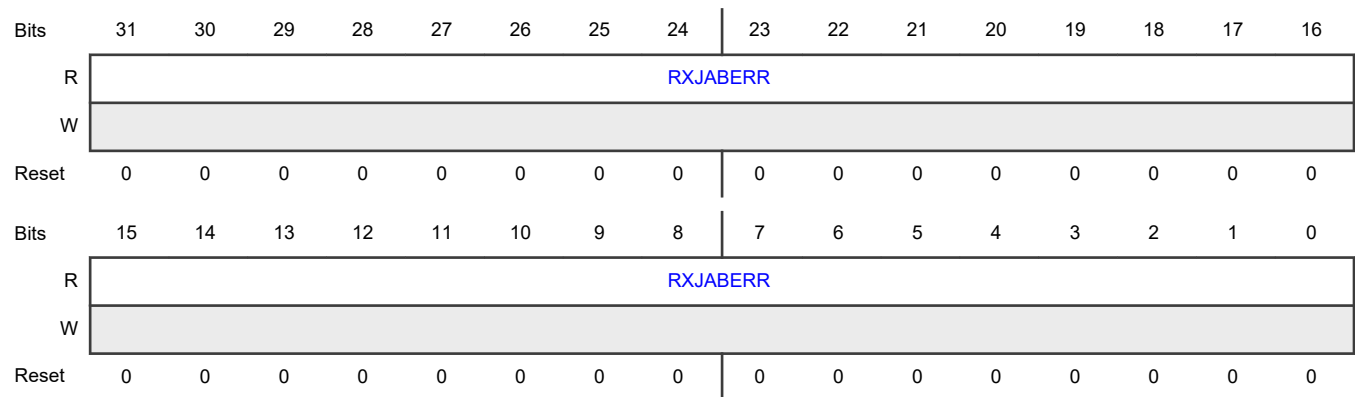
1.4.99 (Rx_Jabber_Error_Packets)

Offset

Register	Offset
Rx_Jabber_Error_Packets	7A0h

Function

This register provides the number of giant packets received by DWC_ether_qos with length (including CRC) greater than 1,518 bytes (1,522 bytes for VLAN tagged) and with CRC error. If Jumbo Packet mode is enabled, packets of length greater than 9,018 bytes (9,022 bytes for VLAN tagged) are considered as giant packets.

Diagram**Fields**

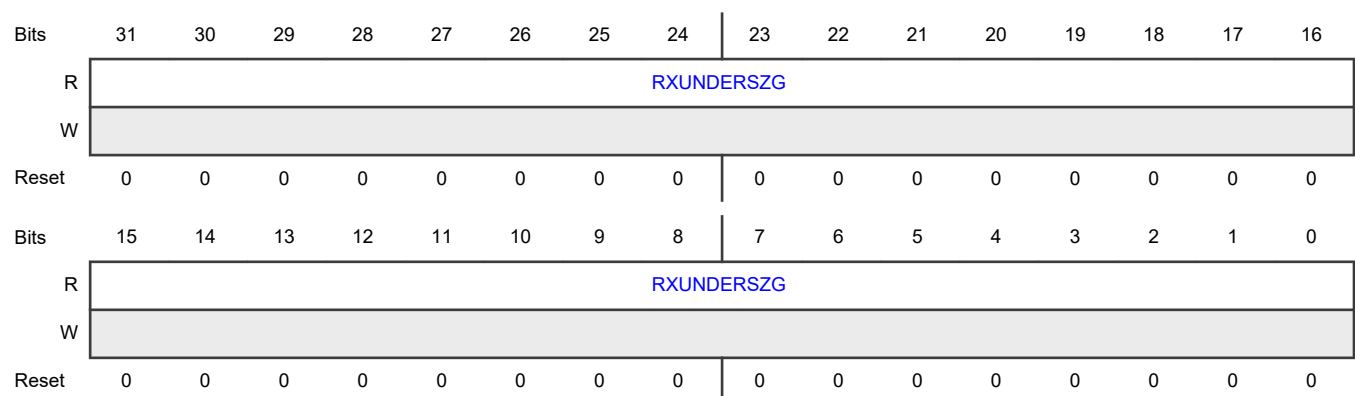
Field	Function
31-0 RXJABERR	Rx Jabber Error Packets This field indicates the number of giant packets received with length (including CRC) greater than 1,518 bytes (1,522 bytes for VLAN tagged) and with CRC error. If Jumbo Packet mode is enabled, packets of length greater than 9,018 bytes (9,022 bytes for VLAN tagged) are considered as giant packets.

1.4.100 (Rx_Undersize_Packets_Good)**Offset**

Register	Offset
Rx_Undersize_Packets_Good	7A4h

Function

This register provides the number of packets received by DWC_ether_qos with length less than 64 bytes, without any errors.

Diagram

Fields

Field	Function
31-0 RXUNDERSZG	Rx Undersize Packets Good This field indicates the number of packets received with length less than 64 bytes, without any errors.

1.4.101 (Rx_Oversize_Packets_Good)

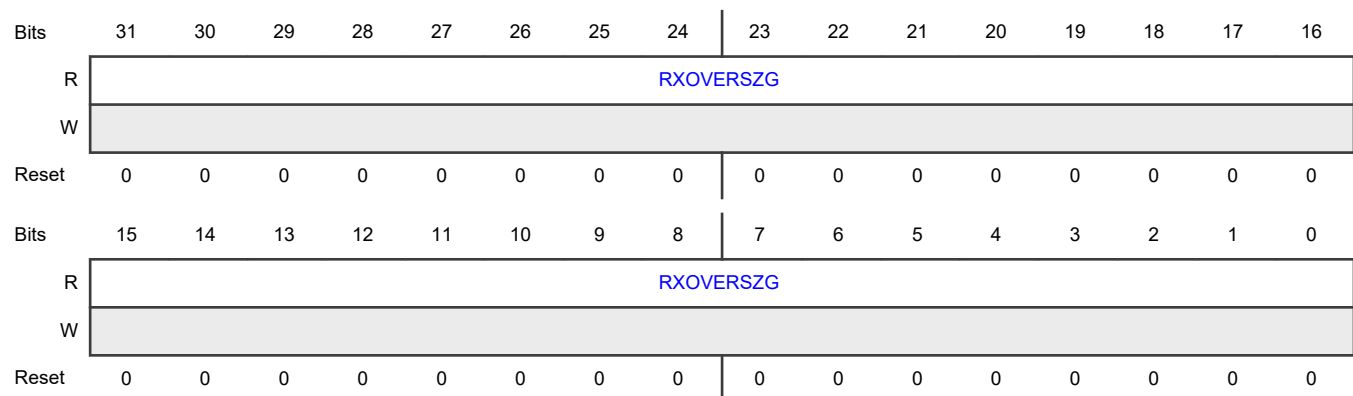
Offset

Register	Offset
Rx_Oversize_Packets_Good	7A8h

Function

This register provides the number of packets received by DWC_ether_qos without errors, with length greater than the maxsize (1,518 bytes or 1,522 bytes for VLAN tagged packets; 2000 bytes if enabled in the S2KP bit of the MAC_Configuration register).

Diagram



Fields

Field	Function
31-0 RXOVERSZG	Rx Oversize Packets Good This field indicates the number of packets received without errors, with length greater than the maxsize (1,518 bytes or 1,522 bytes for VLAN tagged packets; 2000 bytes if enabled in the S2KP bit of the MAC_Configuration register).

1.4.102 (Rx_64Octets_Packets_Good_Bad)

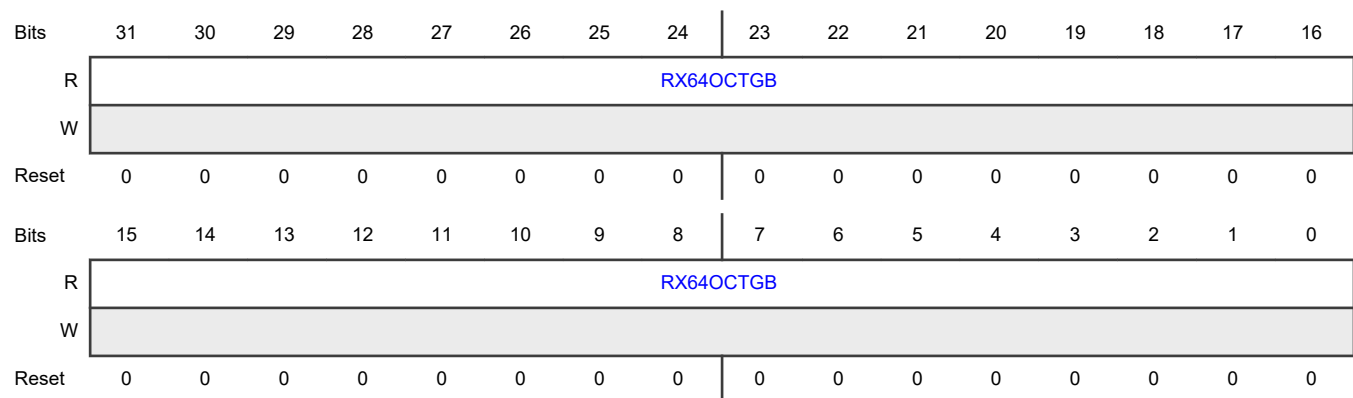
Offset

Register	Offset
Rx_64Octets_Packets_Good_Bad	7ACh

Function

This register provides the number of good and bad packets received by DWC_ether_qos with length 64 bytes, exclusive of the preamble.

Diagram



Fields

Field	Function
31-0 RX64OCTGB	Rx 64 Octets Packets Good Bad This field indicates the number of good and bad packets received with length 64 bytes, exclusive of the preamble.

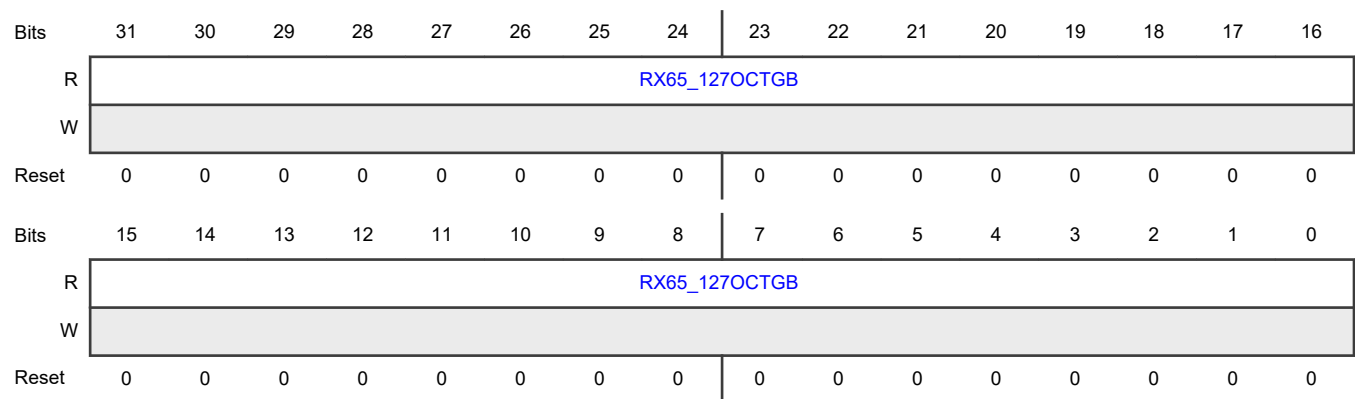
1.4.103 (Rx_65To127Octets_Packets_Good_Bad)

Offset

Register	Offset
Rx_65To127Octets_Packets_Good_Bad	7B0h

Function

This register provides the number of good and bad packets received by DWC_ether_qos with length between 65 and 127 (inclusive) bytes, exclusive of the preamble.

Diagram**Fields**

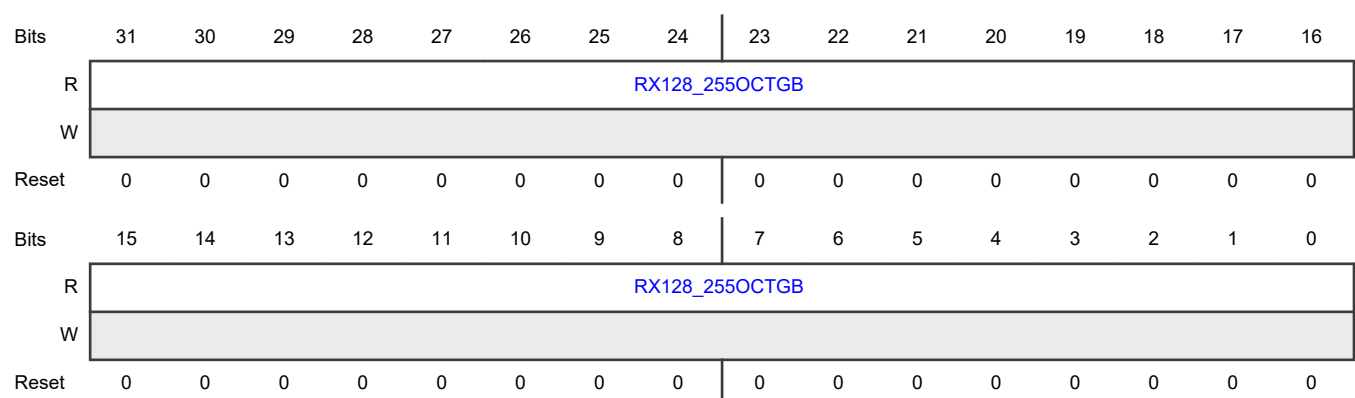
Field	Function
31-0	Rx 65-127 Octets Packets Good Bad
RX65_127OCTGB	This field indicates the number of good and bad packets received with length between 65 and 127 (inclusive) bytes, exclusive of the preamble.

1.4.104 (Rx_128To255Octets_Packets_Good_Bad)**Offset**

Register	Offset
Rx_128To255Octets_Packets_Good_Bad	7B4h

Function

This register provides the number of good and bad packets received by DWC_ether_qos with length between 128 and 255 (inclusive) bytes, exclusive of the preamble.

Diagram

Fields

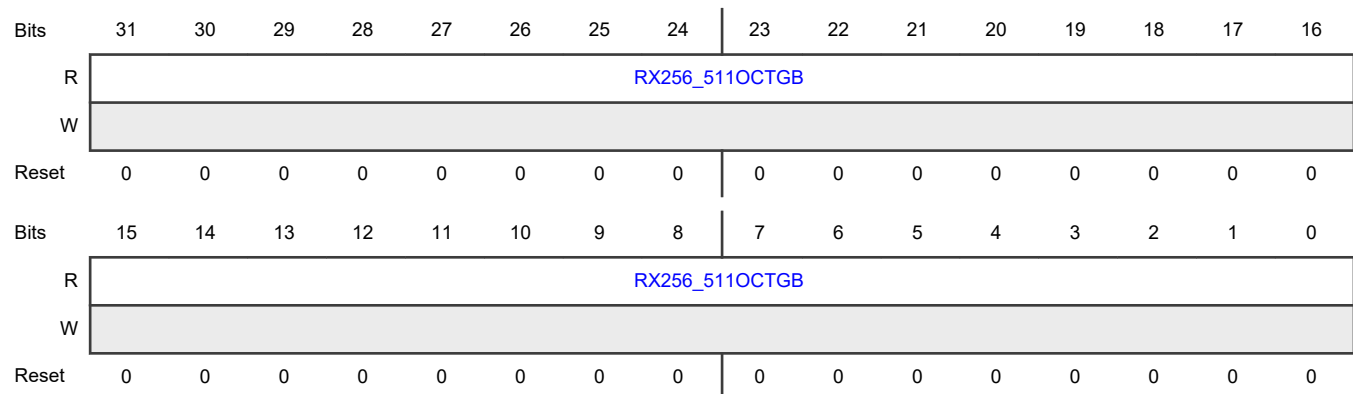
Field	Function
31-0 RX128_255OC TGB	Rx 128-255 Octets Packets Good Bad This field indicates the number of good and bad packets received with length between 128 and 255 (inclusive) bytes, exclusive of the preamble.

1.4.105 (Rx_256To511Octets_Packets_Good_Bad)**Offset**

Register	Offset
Rx_256To511Octets_Pa ckets_Good_Bad	7B8h

Function

This register provides the number of good and bad packets received by DWC_ether_qos with length between 256 and 511 (inclusive) bytes, exclusive of the preamble.

Diagram**Fields**

Field	Function
31-0 RX256_511OC TGB	Rx 256-511 Octets Packets Good Bad This field indicates the number of good and bad packets received with length between 256 and 511 (inclusive) bytes, exclusive of the preamble.

1.4.106 (Rx_512To1023Octets_Packets_Good_Bad)

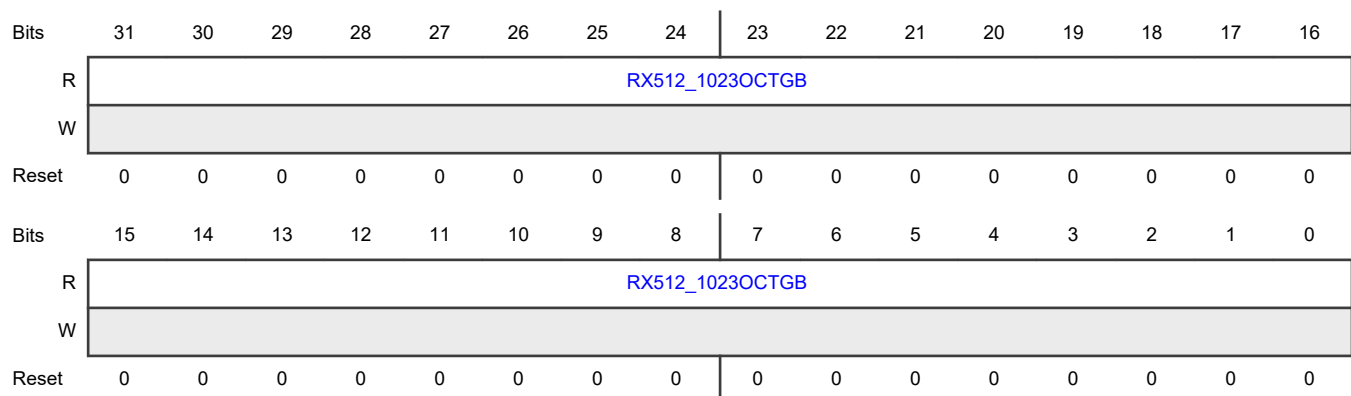
Offset

Register	Offset
Rx_512To1023Octets_Packets_Good_Bad	7BCh

Function

This register provides the number of good and bad packets received by DWC_ether_qos with length between 512 and 1023 (inclusive) bytes, exclusive of the preamble.

Diagram



Fields

Field	Function
31-0	RX 512-1023 Octets Packets Good Bad
RX512_1023OCTGB	This field indicates the number of good and bad packets received with length between 512 and 1023 (inclusive) bytes, exclusive of the preamble.

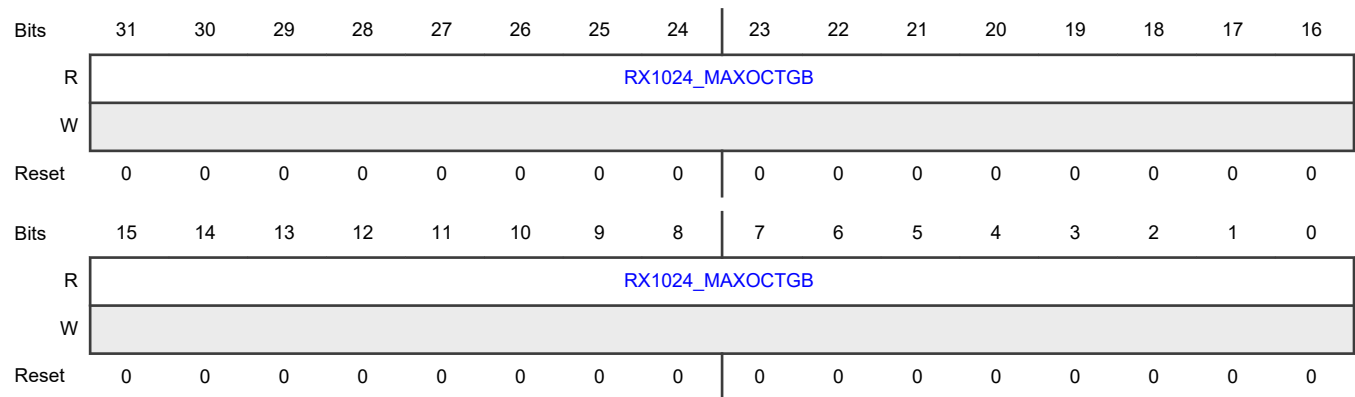
1.4.107 (Rx_1024ToMaxOctets_Packets_Good_Bad)

Offset

Register	Offset
Rx_1024ToMaxOctets_Packets_Good_Bad	7C0h

Function

This register provides the number of good and bad packets received by DWC_ether_qos with length between 1024 and maxsize (inclusive) bytes, exclusive of the preamble.

Diagram**Fields**

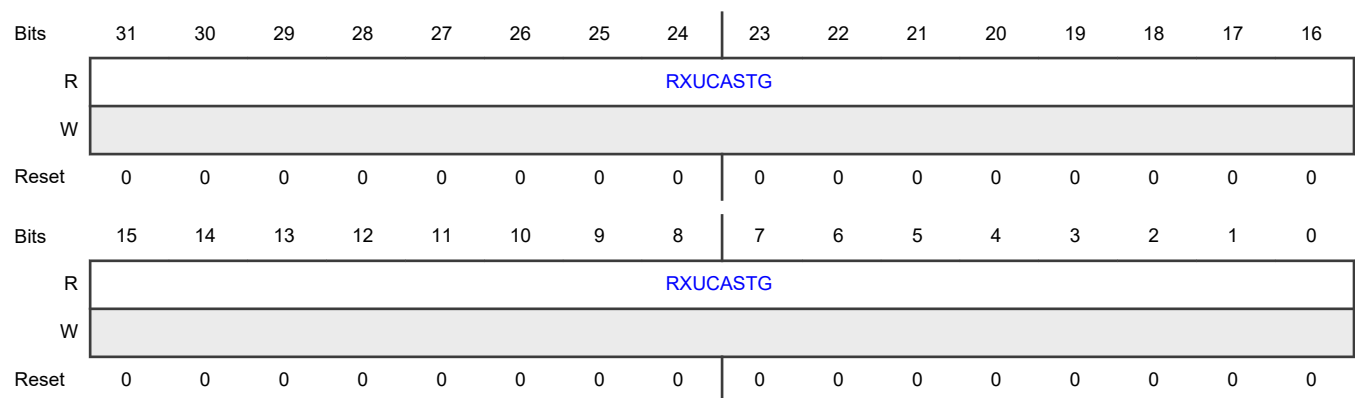
Field	Function
31-0	Rx 1024-Max Octets Good Bad
RX1024_MAXOCTGB	This field indicates the number of good and bad packets received with length between 1024 and maxsize (inclusive) bytes, exclusive of the preamble.

1.4.108 (Rx_Unicast_Packets_Good)**Offset**

Register	Offset
Rx_Unicast_Packets_Good	7C4h

Function

This register provides the number of good unicast packets received by DWC_ether_qos.

Diagram

Fields

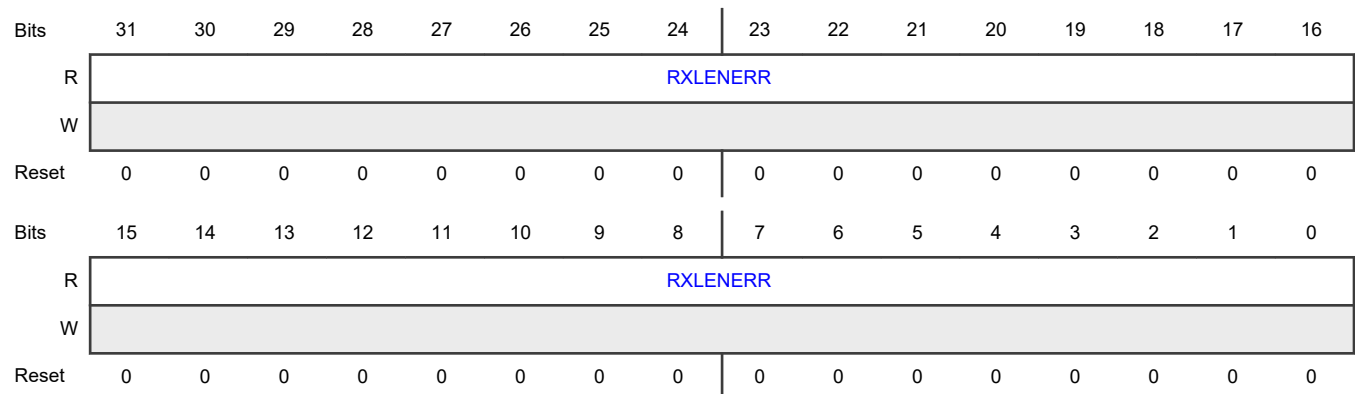
Field	Function
31-0 RXUCASTG	Rx Unicast Packets Good This field indicates the number of good unicast packets received.

1.4.109 (Rx_Length_Error_Packets)**Offset**

Register	Offset
Rx_Length_Error_Packets	7C8h

Function

This register provides the number of packets received by DWC_ether_qos with length error (Length Type field not equal to packet size), for all packets with valid length field.

Diagram**Fields**

Field	Function
31-0 RXLENERR	Rx Length Error Packets This field indicates the number of packets received with length error (Length Type field not equal to packet size), for all packets with valid length field.

1.4.110 (Rx_Out_Of_Range_Type_Packets)

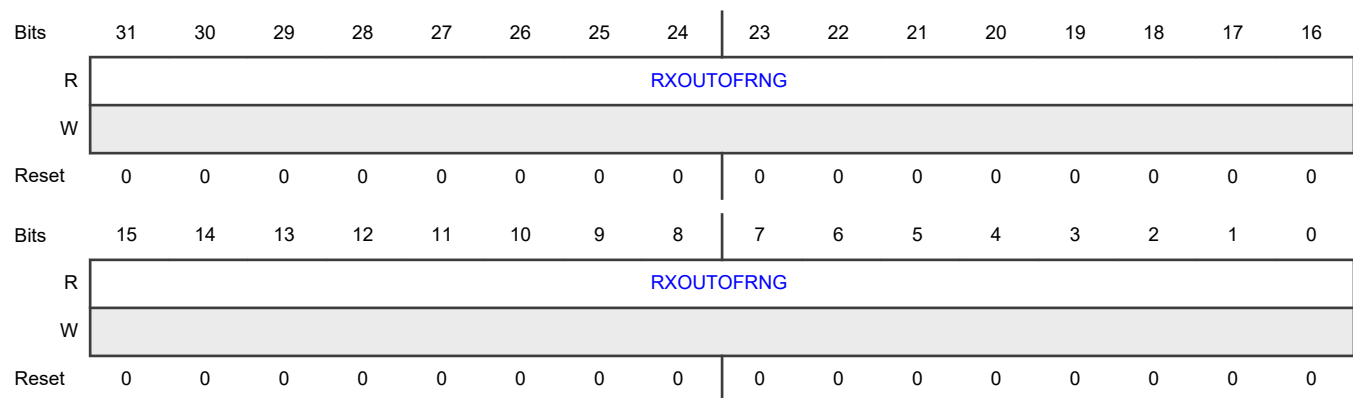
Offset

Register	Offset
Rx_Out_Of_Range_Type_Packets	7CCh

Function

This register provides the number of packets received by DWC_ether_qos with length field not equal to the valid packet size (greater than 1,500 but less than 1,536).

Diagram



Fields

Field	Function
31-0 RXOUTOFRNG	Rx Out of Range Type Packet This field indicates the number of packets received with length field not equal to the valid packet size (greater than 1,500 but less than 1,536).

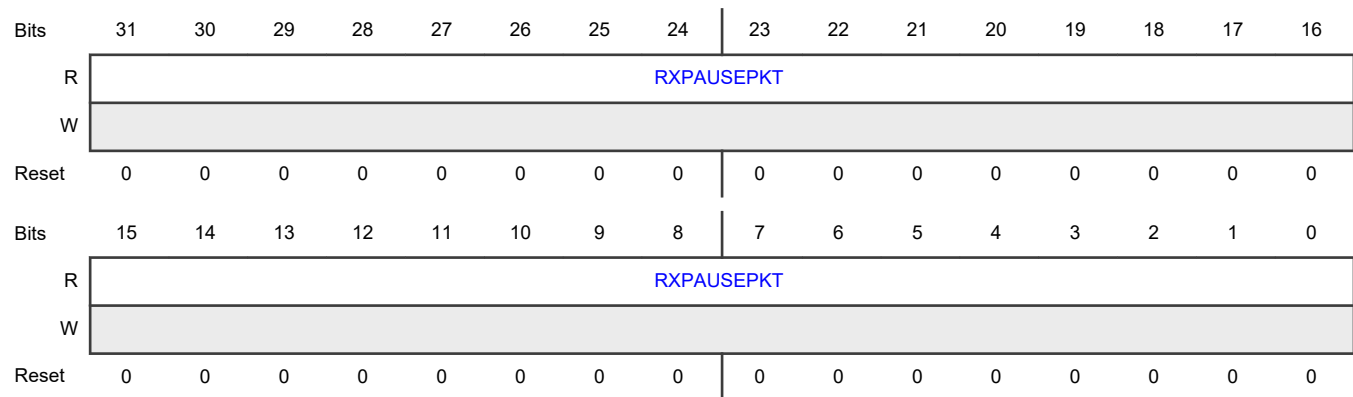
1.4.111 (Rx_Pause_Packets)

Offset

Register	Offset
Rx_Pause_Packets	7D0h

Function

This register provides the number of good and valid Pause packets received by DWC_ether_qos.

Diagram**Fields**

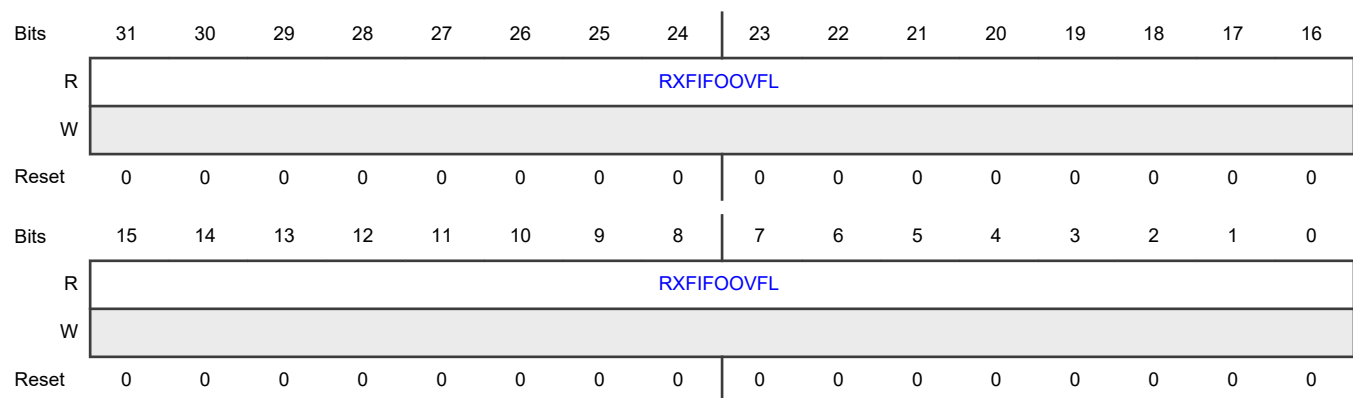
Field	Function
31-0	Rx Pause Packets
RXPAUSEPKT	This field indicates the number of good and valid Pause packets received.

1.4.112 (Rx_FIFO_Overflow_Packets)**Offset**

Register	Offset
Rx_FIFO_Overflow_Packets	7D4h

Function

This register provides the number of missed received packets because of FIFO overflow in DWC_ether_qos.

Diagram

Fields

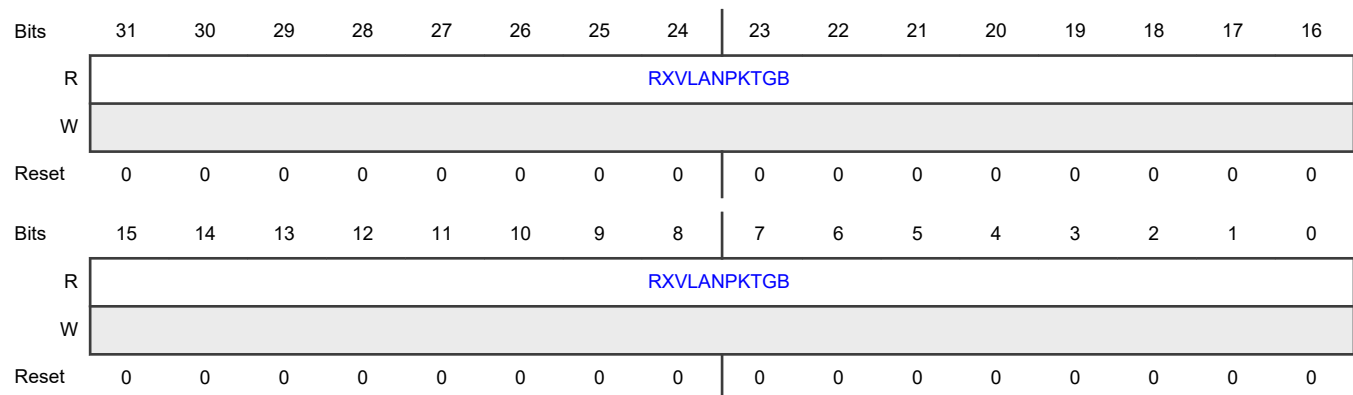
Field	Function
31-0 RXFIFOVFL	Rx FIFO Overflow Packets This field indicates the number of missed received packets because of FIFO overflow.

1.4.113 (Rx_VLAN_Packets_Good_Bad)**Offset**

Register	Offset
Rx_VLAN_Packets_Good_Bad	7D8h

Function

This register provides the number of good and bad VLAN packets received by DWC_ether_qos.

Diagram**Fields**

Field	Function
31-0 RXVLANPKTGB	Rx VLAN Packets Good Bad This field indicates the number of good and bad VLAN packets received.

1.4.114 (Rx_Watchdog_Error_Packets)

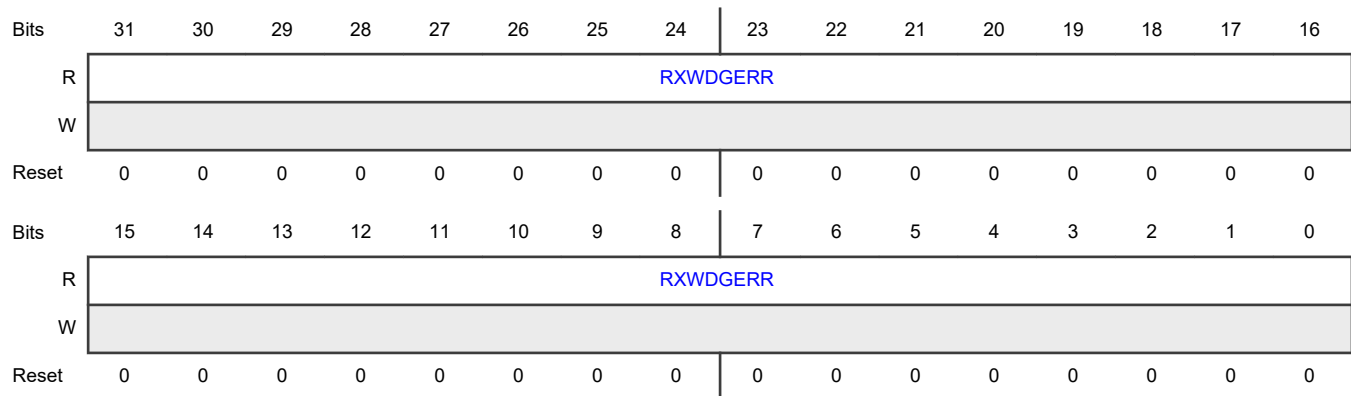
Offset

Register	Offset
Rx_Watchdog_Error_Packets	7DCh

Function

This register provides the number of packets received by DWC_ether_qos with error because of watchdog timeout error (packets with a data load larger than 2,048 bytes (when JE and WD bits are reset in MAC_Configuration register), 10,240 bytes (when JE bit is set and WD bit is reset in MAC_Configuration register), 16,384 bytes (when WD bit is set in MAC_Configuration register) or the value programmed in the MAC_Watchdog_Timeout register).

Diagram



Fields

Field	Function
31-0 RXWDGERR	Rx Watchdog Error Packets This field indicates the number of packets received with error because of watchdog timeout error (packets with a data load larger than 2,048 bytes (when JE and WD bits are reset in MAC_Configuration register), 10,240 bytes (when JE bit is set and WD bit is reset in MAC_Configuration register), 16,384 bytes (when WD bit is set in MAC_Configuration register) or the value programmed in the MAC_Watchdog_Timeout register).

1.4.115 (Rx_Receive_Error_Packets)

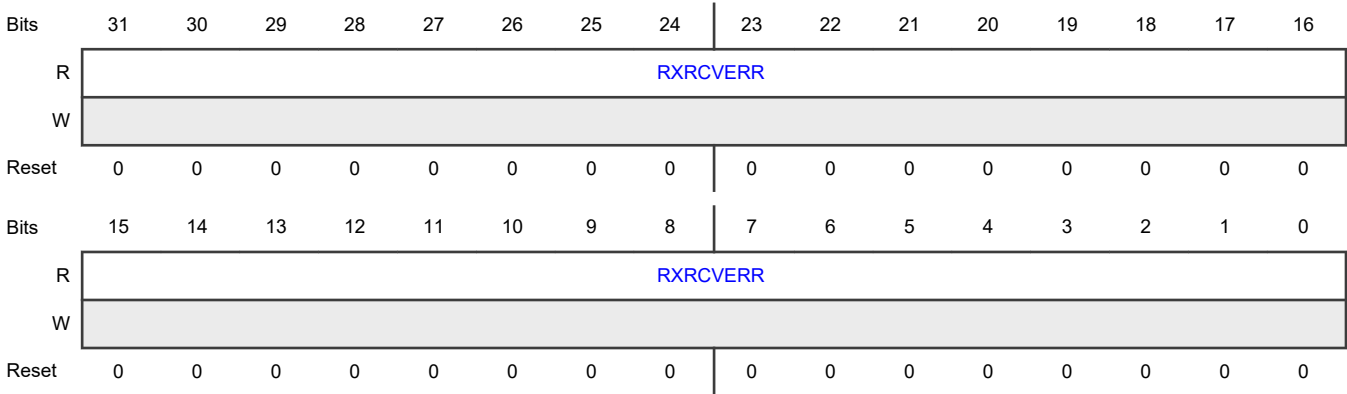
Offset

Register	Offset
Rx_Receive_Error_Packets	7E0h

Function

This register provides the number of packets received by DWC_ether_qos with Receive error or Packet Extension error on the GMII or MII interface.

Diagram



Fields

Field	Function
31-0 RXRCVERR	Rx Receive Error Packets This field indicates the number of packets received with Receive error or Packet Extension error on the GMII or MII interface.

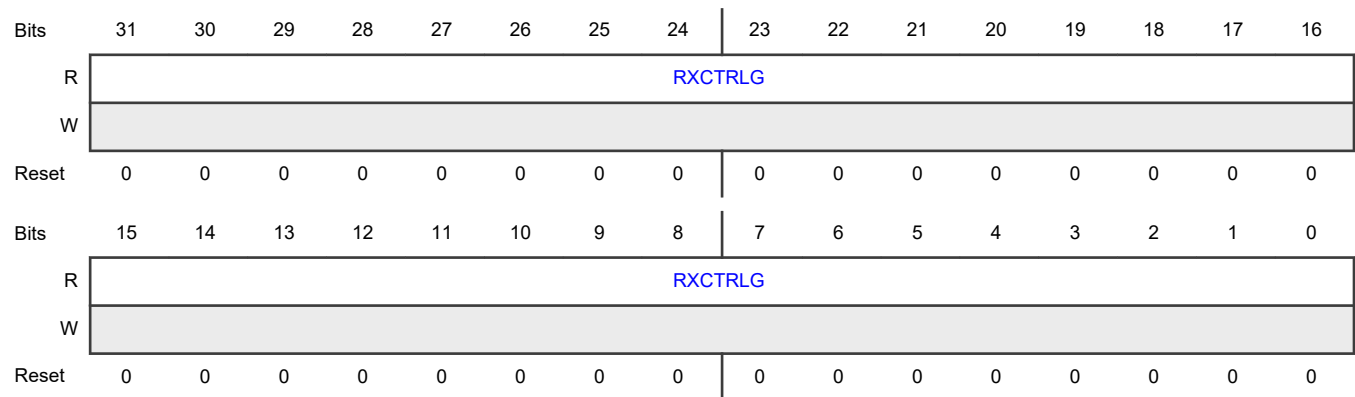
1.4.116 (Rx_Control_Packets_Good)

Offset

Register	Offset
Rx_Control_Packets_Good	7E4h

Function

This register provides the number of good control packets received by DWC_ether_qos.

Diagram**Fields**

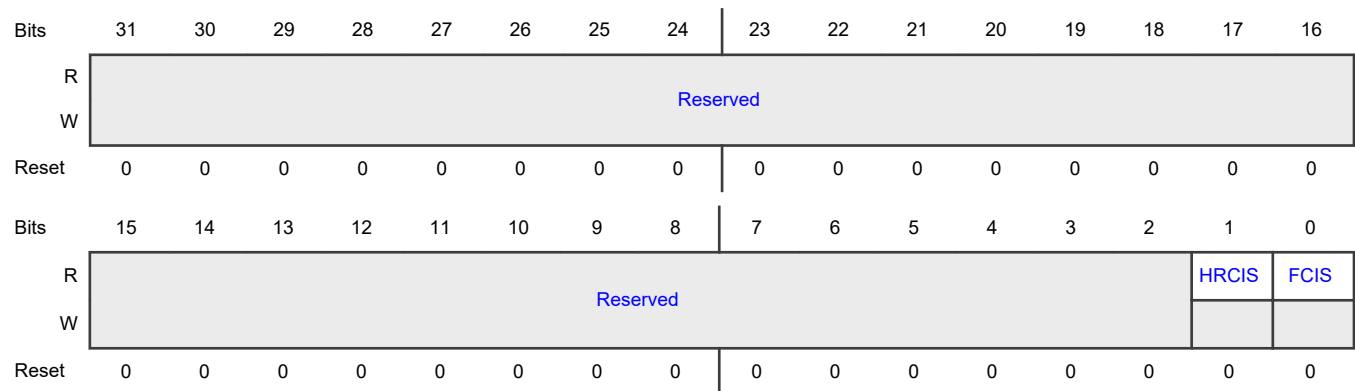
Field	Function
31-0	Rx Control Packets Good
RXCTRLG	This field indicates the number of good control packets received.

1.4.117 (MMC_FPE_Tx_Interrupt)**Offset**

Register	Offset
MMC_FPE_Tx_Interrupt	8A0h

Function

This register maintains the interrupts generated from all FPE related Transmit statistics counters. The MMC FPE Transmit Interrupt register maintains the interrupts generated when transmit statistic counters reach half their maximum values (0x8000_0000 for 32 bit counter and 0x8000 for 16 bit counter), and when they cross their maximum values (0xFFFF_FFFF for 32-bit counter and 0xFFFF for 16-bit counter). When Counter Stop Rollover is set, the interrupts are set but the counter remains at all-ones. The MMC FPE Transmit Interrupt register is a 32 bit register. An interrupt bit is cleared when the respective MMC counter that caused the interrupt is read. The least significant byte lane (Bits[7:0]) of the respective counter must be read to clear the interrupt bit.

Diagram**Fields**

Field	Function
31-2 —	Reserved.
1 HRCIS	MMC Tx Hold Request Counter Interrupt Status This bit is set when the Tx_Hold_Req_Cntr counter reaches half of the maximum value or the maximum value. Access restriction applies. Clears on read. Self-set to 1 on internal event. Exists when any one of the RX/TX MMC counters are enabled during FPE with AV_EST Enabled configuration. 0b - MMC Tx Hold Request Counter Interrupt Status not detected 1b - MMC Tx Hold Request Counter Interrupt Status detected
0 FCIS	MMC Tx FPE Fragment Counter Interrupt status This bit is set when the Tx_FPE_Fragment_Cntr counter reaches half of the maximum value or the maximum value. Access restriction applies. Clears on read. Self-set to 1 on internal event. Exists when any one of the RX/TX MMC counters are enabled during FPE Enabled configuration. 0b - MMC Tx FPE Fragment Counter Interrupt status not detected 1b - MMC Tx FPE Fragment Counter Interrupt status detected

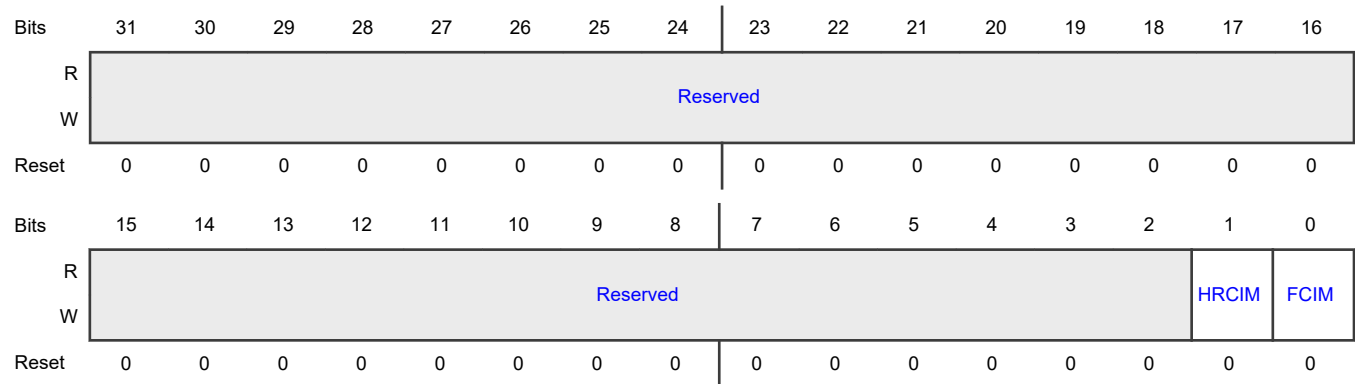
1.4.118 (MMC_FPE_Tx_Interrupt_Mask)**Offset**

Register	Offset
MMC_FPE_Tx_Interrupt_Mask	8A4h

Function

This register maintains the masks for interrupts generated from all FPE related Transmit statistics counters. The MMC Receive Interrupt Mask register maintains the masks for the interrupts generated when FPE related receive statistic counters reach half of their maximum value or the maximum values. This register is 32 bit wide.

Diagram



Fields

Field	Function
31-2 —	Reserved.
1 HRCIM	MMC Transmit Hold Request Counter Interrupt Mask Setting this bit masks the interrupt when the Tx_Hold_Req_Cntr counter reaches half of the maximum value or the maximum value. Exists when any one of the RX/TX MMC counters are enabled during FPE with AV_EST Enabled configuration. 0b - MMC Transmit Hold Request Counter Interrupt Mask is disabled 1b - MMC Transmit Hold Request Counter Interrupt Mask is enabled
0 FCIM	MMC Transmit Fragment Counter Interrupt Mask Setting this bit masks the interrupt when the Tx_FPE_Fragment_Cntr counter reaches half of the maximum value or the maximum value. Exists when any one of the RX/TX MMC counters are enabled during FPE Enabled configuration. 0b - MMC Transmit Fragment Counter Interrupt Mask is disabled 1b - MMC Transmit Fragment Counter Interrupt Mask is enabled

1.4.119 (MMC_Tx_FPE_Fragment_Cntr)

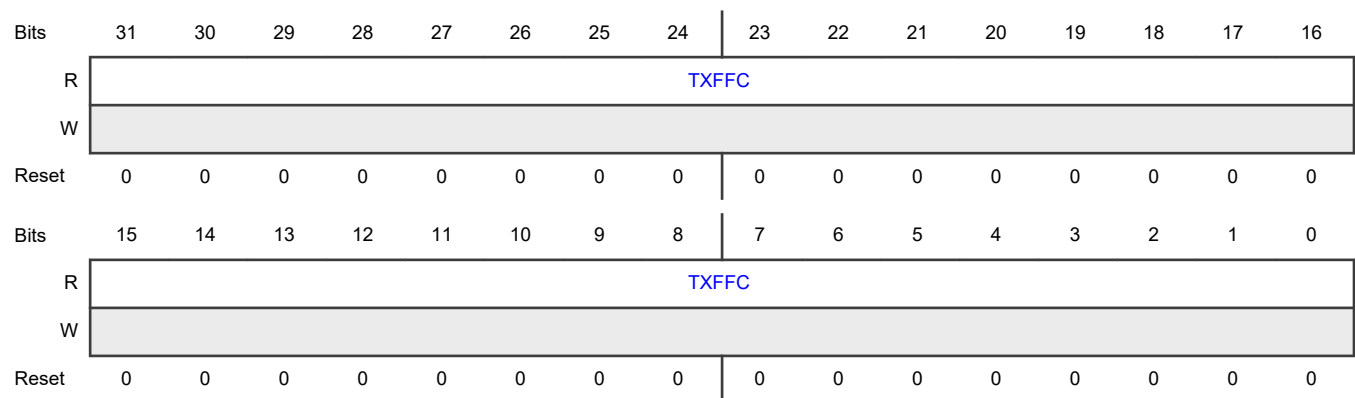
Offset

Register	Offset
MMC_Tx_FPE_Fragment_Cntr	8A8h

Function

This register provides the number of additional mPackets transmitted due to preemption.

Diagram



Fields

Field	Function
31-0	Tx FPE Fragment counter
TXFFC	This field indicates the number of additional mPackets that has been transmitted due to preemption Exists when any one of the RX/TX MMC counters are enabled during FPE Enabled configuration.

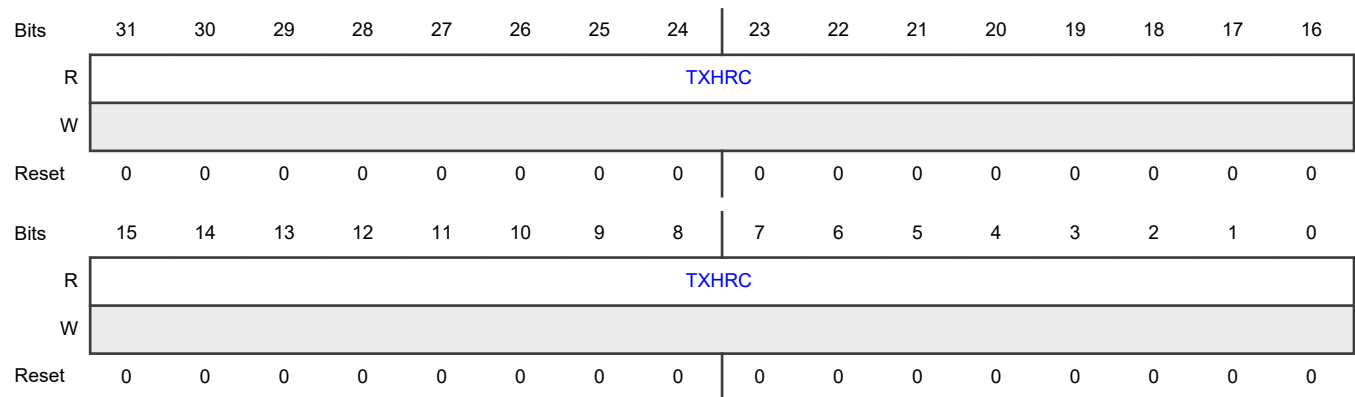
1.4.120 (MMC_Tx_Hold_Req_Cntr)

Offset

Register	Offset
MMC_Tx_Hold_Req_Cntr	8ACh

Function

This register provides the count of number of times a hold request is given to MAC

Diagram**Fields**

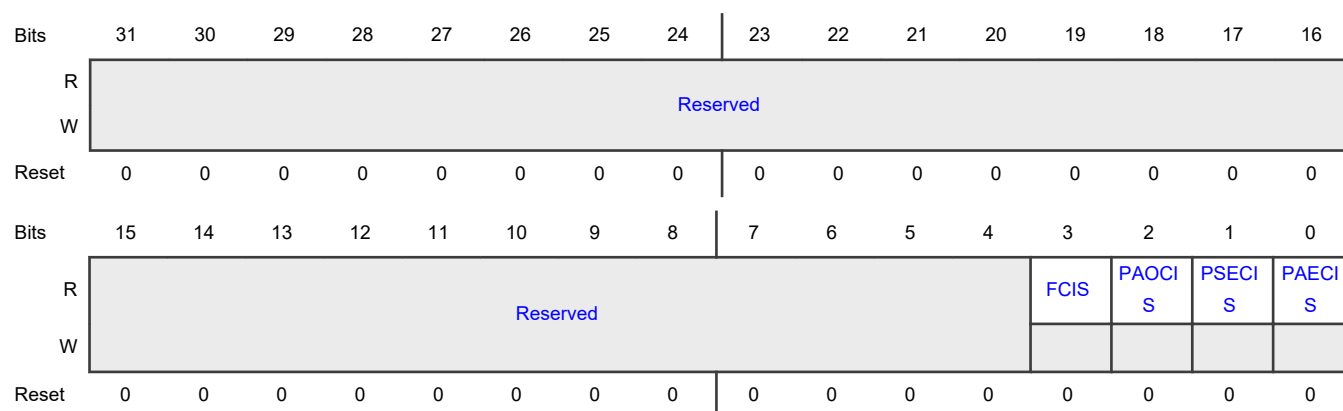
Field	Function
31-0	Tx Hold Request Counter
TXHRC	This field indicates count of number of a hold request is given to MAC. Exists when any one of the RX/TX MMC counters are enabled during FPE with AV_EST Enabled configuration.

1.4.121 (MMC_FPE_Rx_Interrupt)**Offset**

Register	Offset
MMC_FPE_Rx_Interrupt	8C0h

Function

This register maintains the interrupts generated from all FPE related Receive statistics counters. The MMC FPE Receive Interrupt register maintains the interrupts generated when transmit statistic counters reach half their maximum values (0x8000_0000 for 32 bit counter and 0x8000 for 16 bit counter), and when they cross their maximum values (0xFFFF_FFFF for 32-bit counter and 0xFFFF for 16-bit counter). When Counter Stop Rollover is set, the interrupts are set but the counter remains at all-ones. The MMC FPE Receive Interrupt register is a 32 bit register. An interrupt bit is cleared when the respective MMC counter that caused the interrupt is read. The least significant byte lane (Bits[7:0]) of the respective counter must be read to clear the interrupt bit.

Diagram**Fields**

Field	Function
31-4 —	Reserved.
3 FCIS	MMC Rx FPE Fragment Counter Interrupt Status This bit is set when the Rx_FPE_Fragment_Cntr counter reaches half of the maximum value or the maximum value. Access restriction applies. Clears on read. Self-set to 1 on internal event. Exists when any one of the RX/TX MMC counters are enabled during FPE Enabled configuration. 0b - MMC Rx FPE Fragment Counter Interrupt Status not detected 1b - MMC Rx FPE Fragment Counter Interrupt Status detected
2 PAOCIS	MMC Rx Packet Assembly OK Counter Interrupt Status This bit is set when the Rx_Packet_Assemble_Ok_Cntr counter reaches half of the maximum value or the maximum value. Access restriction applies. Clears on read. Self-set to 1 on internal event. Exists when any one of the RX/TX MMC counters are enabled during FPE Enabled configuration. 0b - MMC Rx Packet Assembly OK Counter Interrupt Status not detected 1b - MMC Rx Packet Assembly OK Counter Interrupt Status detected
1 PSECI	MMC Rx Packet SMD Error Counter Interrupt Status This bit is set when the Rx_Packet_SMD_Err_Cntr counter reaches half of the maximum value or the maximum value. Access restriction applies. Clears on read. Self-set to 1 on internal event. Exists when any one of the RX/TX MMC counters are enabled during FPE Enabled configuration. 0b - MMC Rx Packet SMD Error Counter Interrupt Status not detected 1b - MMC Rx Packet SMD Error Counter Interrupt Status detected

Table continues on the next page...

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Field	Function
0 PAECIS	MMC Rx Packet Assembly Error Counter Interrupt Status This bit is set when the Rx_Packet_Assemble_Err_Cntr counter reaches half of the maximum value or the maximum value. Access restriction applies. Clears on read. Self-set to 1 on internal event. Exists when any one of the RX/TX MMC counters are enabled during FPE Enabled configuration. 0b - MMC Rx Packet Assembly Error Counter Interrupt Status not detected 1b - MMC Rx Packet Assembly Error Counter Interrupt Status detected

1.4.122 (MMC_FPE_Rx_Interrupt_Mask)

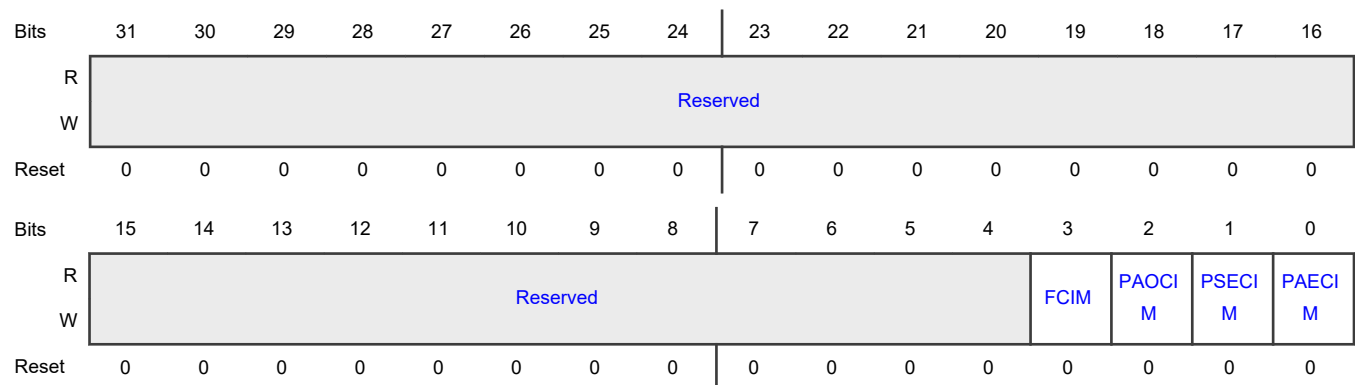
Offset

Register	Offset
MMC_FPE_Rx_Interrupt_Mask	8C4h

Function

This register maintains the masks for interrupts generated from all FPE related Receive statistics counters. The MMC Receive Interrupt Mask register maintains the masks for the interrupts generated when FPE related receive statistic counters reach half of their maximum value or the maximum values. This register is 32 bit wide.

Diagram



Fields

Field	Function
31-4 —	Reserved.

Table continues on the next page...

Table continued from the previous page...

Field	Function
3 FCIM	MMC Rx FPE Fragment Counter Interrupt Mask Setting this bit masks the interrupt when the Tx_FPE_Fragment_Cntr counter reaches half of the maximum value or the maximum value. Exists when any one of the RX/TX MMC counters are enabled during FPE Enabled configuration. 0b - MMC Rx FPE Fragment Counter Interrupt Mask is disabled 1b - MMC Rx FPE Fragment Counter Interrupt Mask is enabled
2 PAOCIM	MMC Rx Packet Assembly OK Counter Interrupt Mask Setting this bit masks the interrupt when the Rx_Packet_Assemble_Ok_Cntr counter reaches half of the maximum value or the maximum value. Exists when any one of the RX/TX MMC counters are enabled during FPE Enabled configuration. 0b - MMC Rx Packet Assembly OK Counter Interrupt Mask is disabled 1b - MMC Rx Packet Assembly OK Counter Interrupt Mask is enabled
1 PSECIM	MMC Rx Packet SMD Error Counter Interrupt Mask Setting this bit masks the interrupt when the R Rx_Packet_SMD_Err_Cntr counter reaches half of the maximum value or the maximum value. Exists when any one of the RX/TX MMC counters are enabled during FPE Enabled configuration. 0b - MMC Rx Packet SMD Error Counter Interrupt Mask is disabled 1b - MMC Rx Packet SMD Error Counter Interrupt Mask is enabled
0 PAECIM	MMC Rx Packet Assembly Error Counter Interrupt Mask Setting this bit masks the interrupt when the R Rx_Packet_Assemble_Err_Cntr counter reaches half of the maximum value or the maximum value. Exists when any one of the RX/TX MMC counters are enabled during FPE Enabled configuration. 0b - MMC Rx Packet Assembly Error Counter Interrupt Mask is disabled 1b - MMC Rx Packet Assembly Error Counter Interrupt Mask is enabled

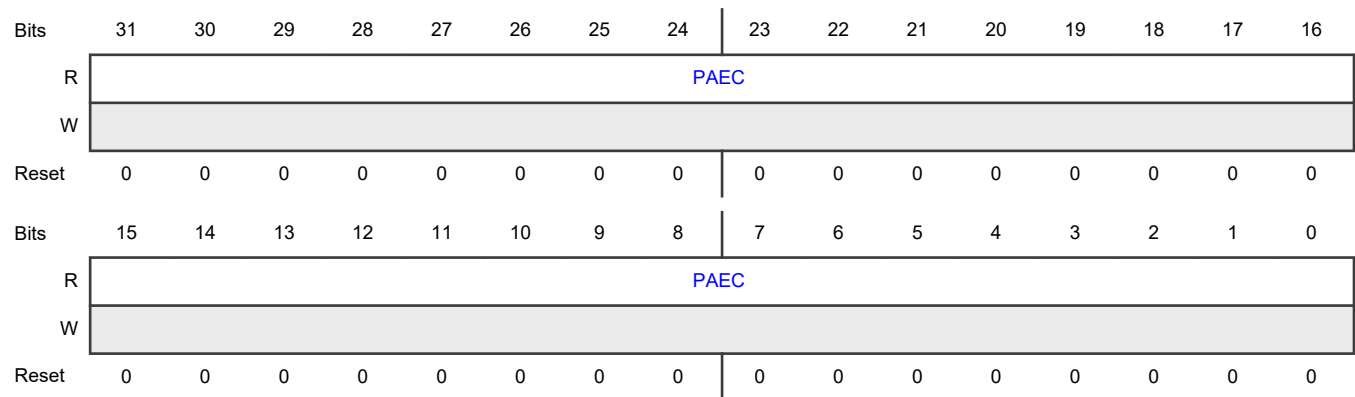
1.4.123 (MMC_Rx_Packet_Assembly_Err_Cntr)

Offset

Register	Offset
MMC_Rx_Packet_Assembly_Err_Cntr	8C8h

Function

This register provides the number of MAC frames with reassembly errors on the Receiver, due to mismatch in the Fragment Count value.

Diagram**Fields**

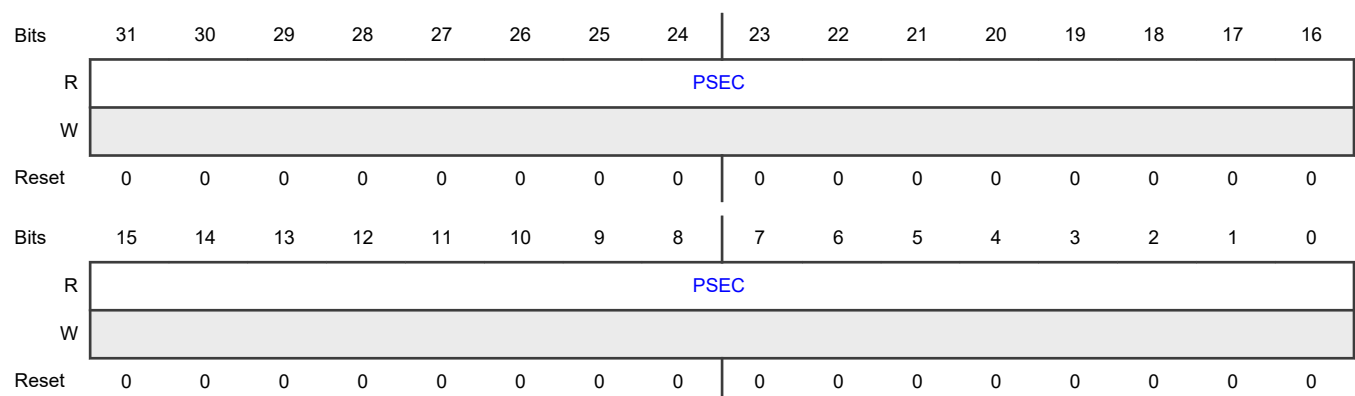
Field	Function
31-0	Rx Packet Assembly Error Counter
PAEC	This field indicates the number of MAC frames with reassembly errors on the Receiver, due to mismatch in the Fragment Count value. Exists when any one of the RX/TX MMC counters are enabled during FPE Enabled configuration.

1.4.124 (MMC_Rx_Packet_SMD_Err_Cntr)**Offset**

Register	Offset
MMC_Rx_Packet_SMD_Err_Cntr	8CCh

Function

This register provides the number of received MAC frames rejected due to unknown SMD value and MAC frame fragments rejected due to arriving with an SMD-C when there was no preceding preempted frame.

Diagram

Fields

Field	Function
31-0 PSEC	Rx Packet SMD Error Counter This field indicates the number of MAC frames rejected due to unknown SMD value and MAC frame fragments rejected due to arriving with an SMD-C when there was no preceding preempted frame. Exists when at least one of the RX/TX MMC counters are enabled during FPE Enabled configuration.

1.4.125 (MMC_Rx_Packet_Assembly_OK_Cntr)

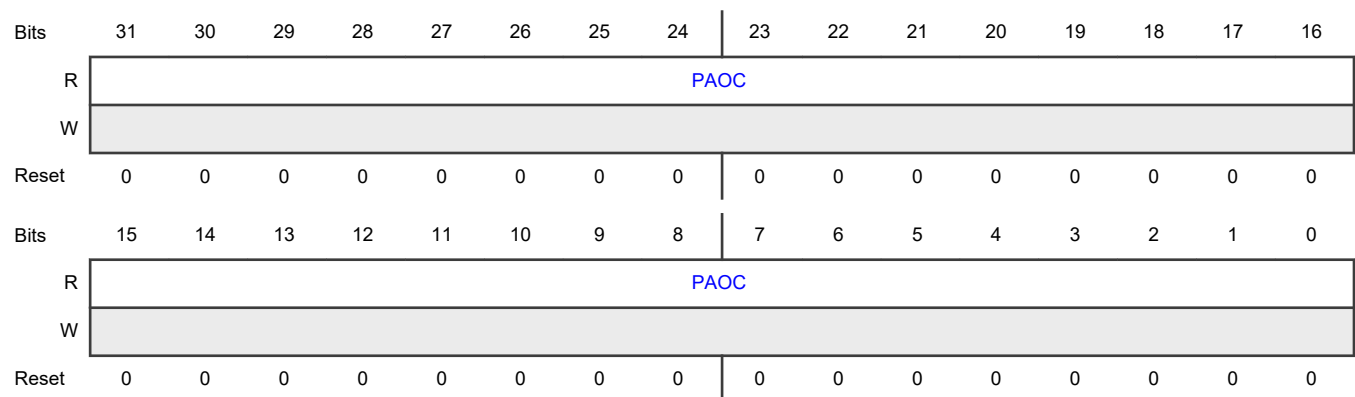
Offset

Register	Offset
MMC_Rx_Packet_Assembly_OK_Cntr	8D0h

Function

This register provides the number of MAC frames that were successfully reassembled and delivered to MAC.

Diagram



Fields

Field	Function
31-0 PAOC	Rx Packet Assembly OK Counter This field indicates the number of MAC frames that were successfully reassembled and delivered to MAC. Exists when at least one of the RX/TX MMC counters are enabled during FPE Enabled configuration.

1.4.126 (MMC_Rx_FPE_Fragment_Cntr)

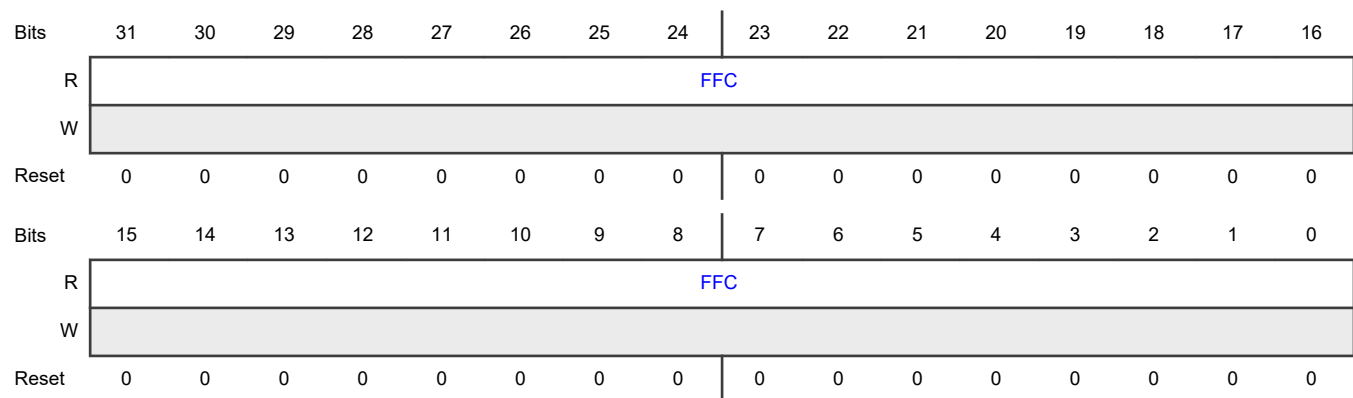
Offset

Register	Offset
MMC_Rx_FPE_Fragment_Cntr	8D4h

Function

This register provides the number of additional mPackets received due to preemption.

Diagram



Fields

Field	Function
31-0	Rx FPE Fragment Counter
FFC	This field indicates the number of additional mPackets received due to preemption Exists when at least one of the RX/TX MMC counters are enabled during FPE Enabled configuration.

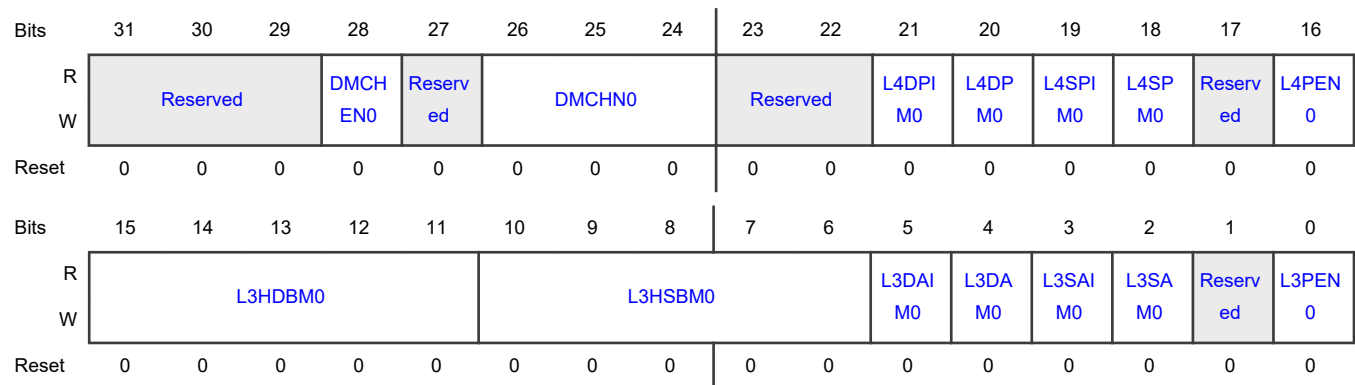
1.4.127 (MAC_L3_L4_Control0)

Offset

Register	Offset
MAC_L3_L4_Control0	900h

Function

The Layer 3 and Layer 4 Control register controls the operations of filter 0 of Layer 3 and Layer 4.

Diagram**Fields**

Field	Function
31-29 —	Reserved.
28 DMCHEN0	DMA Channel Select Enable When set, this bit enables the selection of the DMA channel number for the packet that is passed by this L3_L4 filter. The DMA channel is indicated by the DMCHN bits. When this bit is reset, the DMA channel is not decided by this filter. 0b - DMA Channel Select is disabled 1b - DMA Channel Select is enabled
27 —	Reserved.
26-24 DMCHN0	DMA Channel Number When DMCHEN is set high, this field selects the DMA Channel number to which the packet passed by this filter is routed. The width of this field depends on the number of the DMA channels present in your configuration.
23-22 —	Reserved.
21 L4DPIM0	Layer 4 Destination Port Inverse Match Enable When this bit is set, the Layer 4 Destination Port number field is enabled for inverse matching. When this bit is reset, the Layer 4 Destination Port number field is enabled for perfect matching. This bit is valid and applicable only when the L4DPM0 bit is set high. 0b - Layer 4 Destination Port Inverse Match is disabled 1b - Layer 4 Destination Port Inverse Match is enabled
20	Layer 4 Destination Port Match Enable

Table continues on the next page...

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Field	Function
L4DPM0	When this bit is set, the Layer 4 Destination Port number field is enabled for matching. When this bit is reset, the MAC ignores the Layer 4 Destination Port number field for matching. 0b - Layer 4 Destination Port Match is disabled 1b - Layer 4 Destination Port Match is enabled
19 L4SPIM0	Layer 4 Source Port Inverse Match Enable When this bit is set, the Layer 4 Source Port number field is enabled for inverse matching. When this bit is reset, the Layer 4 Source Port number field is enabled for perfect matching. This bit is valid and applicable only when the L4SPM0 bit is set high. 0b - Layer 4 Source Port Inverse Match is disabled 1b - Layer 4 Source Port Inverse Match is enabled
18 L4SPM0	Layer 4 Source Port Match Enable When this bit is set, the Layer 4 Source Port number field is enabled for matching. When this bit is reset, the MAC ignores the Layer 4 Source Port number field for matching. 0b - Layer 4 Source Port Match is disabled 1b - Layer 4 Source Port Match is enabled
17 —	Reserved.
16 L4PEN0	Layer 4 Protocol Enable When this bit is set, the Source and Destination Port number fields of UDP packets are used for matching. When this bit is reset, the Source and Destination Port number fields of TCP packets are used for matching. The Layer 4 matching is done only when the L4SPM0 or L4DPM0 bit is set. 0b - Layer 4 Protocol is disabled 1b - Layer 4 Protocol is enabled
15-11 L3HDBM0	Layer 3 IP DA Higher Bits Match IPv4 Packets: This field contains the number of higher bits of IP Destination Address that are matched in the IPv4 packets. The following list describes the values of this field: - 0: No bits are masked. - 1: LSb[0] is masked - 2: Two LSbs [1:0] are masked - .. - 31: All bits except MSb are masked. IPv6 Packets:

Table continues on the next page...

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Field	Function
	Bits[12:11] of this field correspond to Bits[6:5] of L3HSBM0 which indicate the number of lower bits of IP Source or Destination Address that are masked in the IPv6 packets. The following list describes the concatenated values of the L3HDBM0[1:0] and L3HSBM0 bits: - 0: No bits are masked. - 1: LSb[0] is masked. - 2: Two LSbs [1:0] are masked - .. - 127: All bits except MSb are masked. This field is valid and applicable only when the L3DAM0 or L3SAM0 bit is set.
10-6 L3HSBM0	Layer 3 IP SA Higher Bits Match IPv4 Packets: This field contains the number of lower bits of IP Source Address that are masked for matching in the IPv4 packets. The following list describes the values of this field: - 0: No bits are masked. - 1: LSb[0] is masked - 2: Two LSbs [1:0] are masked - .. - 31: All bits except MSb are masked. IPv6 Packets: This field contains Bits[4:0] of L3HSBM0. These bits indicate the number of higher bits of IP Source or Destination Address matched in the IPv6 packets. This field is valid and applicable only when the L3DAM0 or L3SAM0 bit is set high.
5 L3DAIM0	Layer 3 IP DA Inverse Match Enable When this bit is set, the Layer 3 IP Destination Address field is enabled for inverse matching. When this bit is reset, the Layer 3 IP Destination Address field is enabled for perfect matching. This bit is valid and applicable only when the L3DAM0 bit is set high. 0b - Layer 3 IP DA Inverse Match is disabled 1b - Layer 3 IP DA Inverse Match is enabled
4 L3DAM0	Layer 3 IP DA Match Enable When this bit is set, the Layer 3 IP Destination Address field is enabled for matching. When this bit is reset, the MAC ignores the Layer 3 IP Destination Address field for matching. Note: When the L3PEN0 bit is set, you should set either this bit or the L3SAM0 bit because either IPv6 DA or SA can be checked for filtering. 0b - Layer 3 IP DA Match is disabled 1b - Layer 3 IP DA Match is enabled

Table continues on the next page...

Table continued from the previous page...

Field	Function
3 L3SAIM0	Layer 3 IP SA Inverse Match Enable When this bit is set, the Layer 3 IP Source Address field is enabled for inverse matching. When this bit is reset, the Layer 3 IP Source Address field is enabled for perfect matching. This bit is valid and applicable only when the L3SAM0 bit is set. 0b - Layer 3 IP SA Inverse Match is disabled 1b - Layer 3 IP SA Inverse Match is enabled
2 L3SAM0	Layer 3 IP SA Match Enable When this bit is set, the Layer 3 IP Source Address field is enabled for matching. When this bit is reset, the MAC ignores the Layer 3 IP Source Address field for matching. Note: When the L3PEN0 bit is set, you should set either this bit or the L3DAM0 bit because either IPv6 SA or DA can be checked for filtering. 0b - Layer 3 IP SA Match is disabled 1b - Layer 3 IP SA Match is enabled
1 —	Reserved.
0 L3PEN0	Layer 3 Protocol Enable When this bit is set, the Layer 3 IP Source or Destination Address matching is enabled for IPv6 packets. When this bit is reset, the Layer 3 IP Source or Destination Address matching is enabled for IPv4 packets. The Layer 3 matching is done only when the L3SAM0 or L3DAM0 bit is set. 0b - Layer 3 Protocol is disabled 1b - Layer 3 Protocol is enabled

1.4.128 (MAC_Layer4_Address0)

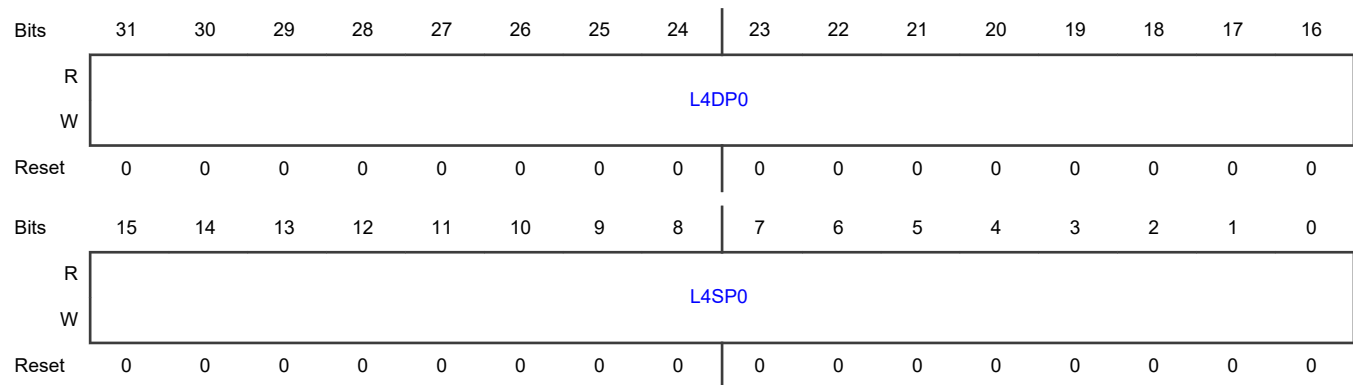
Offset

Register	Offset
MAC_Layer4_Address0	904h

Function

The MAC_Layer4_Address(#i), MAC_L3_L4_Control(#i), MAC_Layer3_Addr0_Reg(#i), MAC_Layer3_Addr1_Reg(#i), MAC_Layer3_Addr2_Reg(#i) and MAC_Layer3_Addr3_Reg(#i) registers are reserved (RO with default value) if Enable Layer 3 and Layer 4 Packet Filter option is not selected while configuring the core.

You can configure the Layer 3 and Layer 4 Address Registers to be double-synchronized by selecting the Synchronize Layer 3 and Layer 4 Address Registers to Rx Clock Domain option while configuring the core. When you select this option, the synchronization is triggered only when Bits[31:24] (in little-endian mode) or Bits[7:0] (in big-endian mode) of the Layer 3 and Layer 4 Address Registers are written. For proper synchronization updates, you should perform consecutive writes to same Layer 3 and Layer 4 Address Registers after at least four clock cycles delay of the destination clock.

Diagram**Fields**

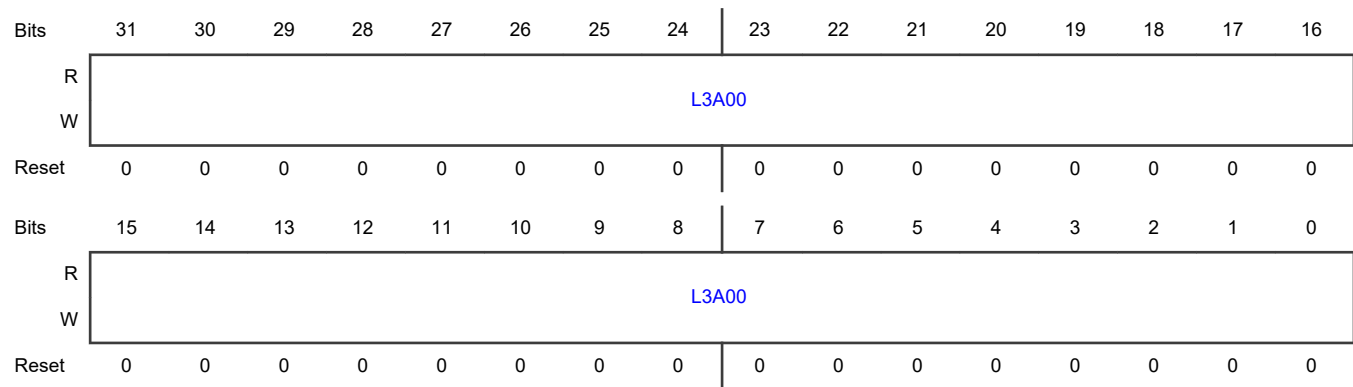
Field	Function
31-16 L4DP0	Layer 4 Destination Port Number Field When the L4PEN0 bit is reset and the L4DPM0 bit is set in the MAC_L3_L4_Control0 register, this field contains the value to be matched with the TCP Destination Port Number field in the IPv4 or IPv6 packets. When the L4PEN0 and L4DPM0 bits are set in MAC_L3_L4_Control0 register, this field contains the value to be matched with the UDP Destination Port Number field in the IPv4 or IPv6 packets.
15-0 L4SP0	Layer 4 Source Port Number Field When the L4PEN0 bit is reset and the L4SPM0 bit is set in the MAC_L3_L4_Control0 register, this field contains the value to be matched with the TCP Source Port Number field in the IPv4 or IPv6 packets. When the L4PEN0 and L4SPM0 bits are set in MAC_L3_L4_Control0 register, this field contains the value to be matched with the UDP Source Port Number field in the IPv4 or IPv6 packets.

1.4.129 (MAC_Layer3_Addr0_Reg0)**Offset**

Register	Offset
MAC_Layer3_Addr0_Reg0	910h

Function

For IPv4 packets, the Layer 3 Address 0 Register 0 register contains the 32-bit IP Source Address field. For IPv6 packets, it contains Bits[31:0] of the 128-bit IP Source Address or Destination Address field.

Diagram**Fields**

Field	Function
31-0 L3A00	Layer 3 Address 0 Field When the L3PEN0 and L3SAM0 bits are set in the MAC_L3_L4_Control0 register, this field contains the value to be matched with Bits[31:0] of the IP Source Address field in the IPv6 packets. When the L3PEN0 and L3DAM0 bits are set in the MAC_L3_L4_Control0 register, this field contains the value to be matched with Bits[31:0] of the IP Destination Address field in the IPv6 packets. When the L3PEN0 bit is reset and the L3SAM0 bit is set in the MAC_L3_L4_Control0 register, this field contains the value to be matched with the IP Source Address field in the IPv4 packets.

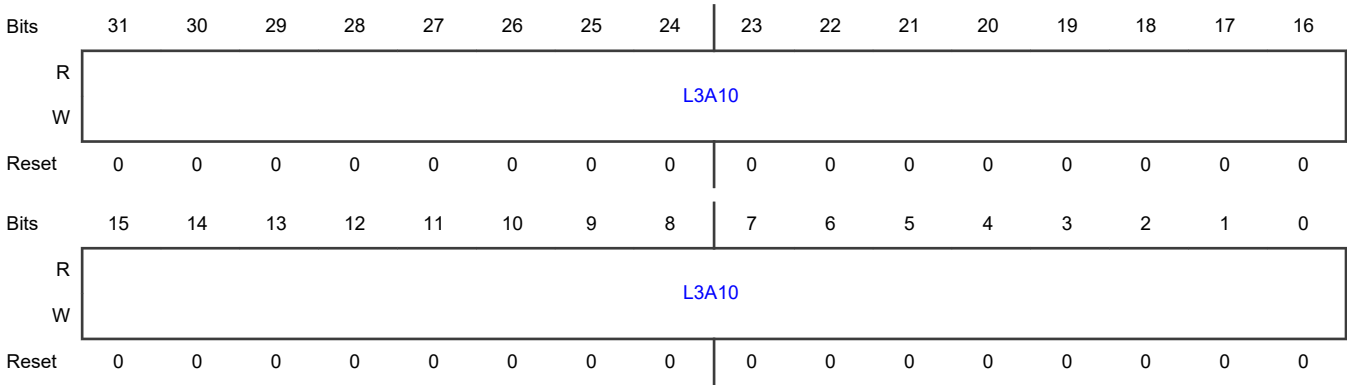
1.4.130 (MAC_Layer3_Addr1_Reg0)**Offset**

Register	Offset
MAC_Layer3_Addr1_Reg0	914h

Function

For IPv4 packets, the Layer 3 Address 1 Register 0 register contains the 32-bit IP Destination Address field. For IPv6 packets, it contains Bits[63:32] of the 128-bit IP Source Address or Destination Address field.

Diagram



Fields

Field	Function
31-0 L3A10	Layer 3 Address 1 Field When the L3PEN0 and L3SAM0 bits are set in the MAC_L3_L4_Control0 register, this field contains the value to be matched with Bits[63:32] of the IP Source Address field in the IPv6 packets. When the L3PEN0 and L3DAM0 bits are set in the MAC_L3_L4_Control0 register, this field contains the value to be matched with Bits[63:32] of the IP Destination Address field in the IPv6 packets. When the L3PEN0 bit is reset and the L3SAM0 bit is set in the MAC_L3_L4_Control0 register, this field contains the value to be matched with the IP Destination Address field in the IPv4 packets.

1.4.131 (MAC_Layer3_Addr2_Reg0)

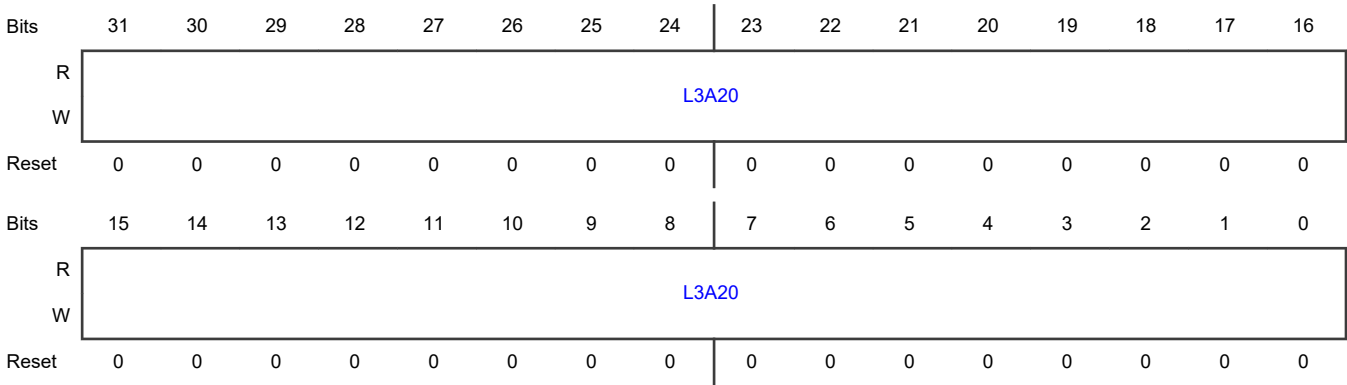
Offset

Register	Offset
MAC_Layer3_Addr2_Reg0	918h

Function

The Layer 3 Address 2 Register 0 register is reserved for IPv4 packets. For IPv6 packets, it contains Bits[95:64] of 128-bit IP Source Address or Destination Address field.

Diagram



Fields

Field	Function
31-0 L3A20	Layer 3 Address 2 Field When the L3PEN0 and L3SAM0 bits are set in the MAC_L3_L4_Control0 register, this field contains the value to be matched with Bits[95:64] of the IP Source Address field in the IPv6 packets. When the L3PEN0 and L3DAM0 bits are set in the MAC_L3_L4_Control0 register, this field contains the value to be matched with Bits[95:64] of the IP Destination Address field in the IPv6 packets. When the L3PEN0 bit is reset in the MAC_L3_L4_Control0 register, this field is not used.

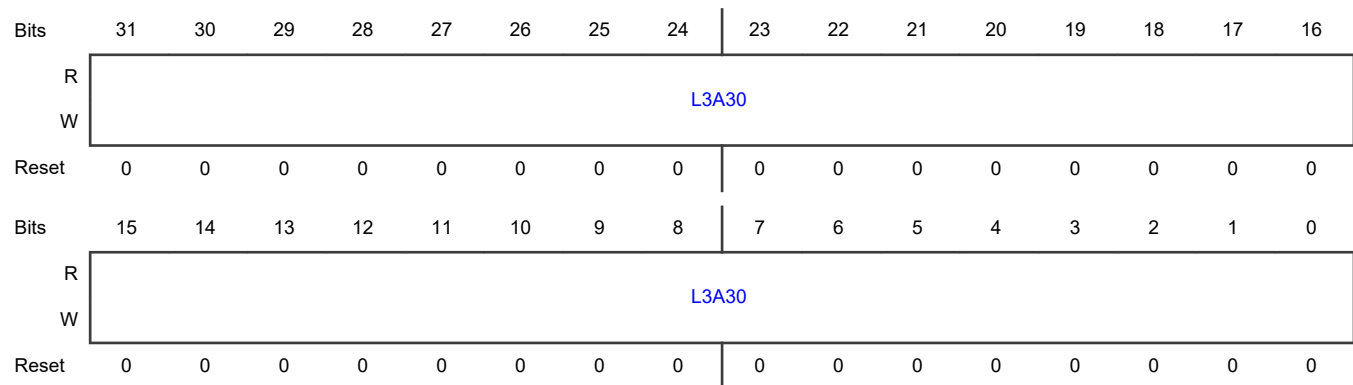
1.4.132 (MAC_Layer3_Addr3_Reg0)

Offset

Register	Offset
MAC_Layer3_Addr3_Reg0	91Ch

Function

The Layer 3 Address 3 Register 0 register is reserved for IPv4 packets. For IPv6 packets, it contains Bits[127:96] of 128-bit IP Source Address or Destination Address field.

Diagram**Fields**

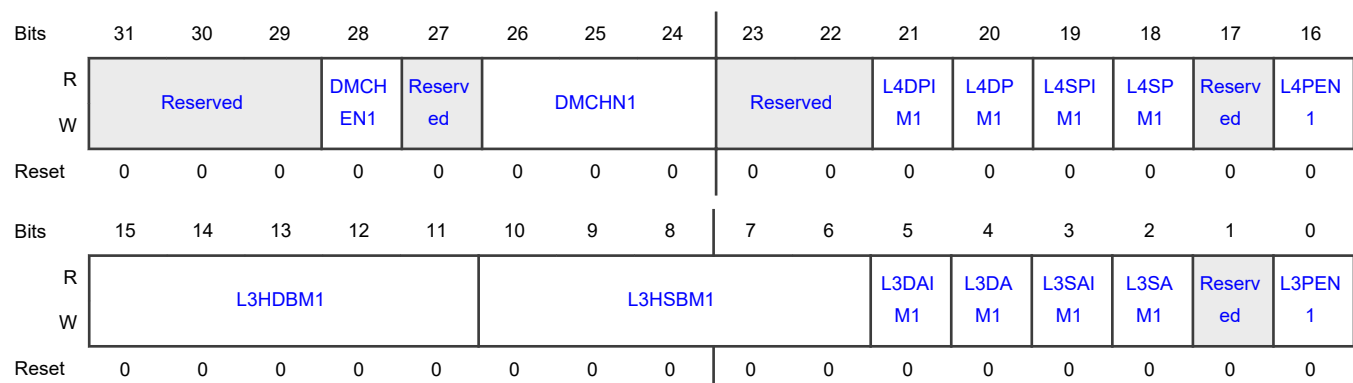
Field	Function
31-0 L3A30	Layer 3 Address 3 Field When the L3PEN0 and L3SAM0 bits are set in the MAC_L3_L4_Control0 register, this field contains the value to be matched with Bits[127:96] of the IP Source Address field in the IPv6 packets. When the L3PEN0 and L3DAM0 bits are set in the MAC_L3_L4_Control0 register, this field contains the value to be matched with Bits[127:96] of the IP Destination Address field in the IPv6 packets. When the L3PEN0 bit is reset in the MAC_L3_L4_Control0 register, this field is not used.

1.4.133 (MAC_L3_L4_Control1)**Offset**

Register	Offset
MAC_L3_L4_Control1	930h

Function

The Layer 3 and Layer 4 Control register controls the operations of filter 0 of Layer 3 and Layer 4.

Diagram

Fields

Field	Function
31-29 —	Reserved.
28 DMCHEN1	DMA Channel Select Enable When set, this bit enables the selection of the DMA channel number for the packet that is passed by this L3_L4 filter. The DMA channel is indicated by the DMCHN bits. When this bit is reset, the DMA channel is not decided by this filter. 0b - DMA Channel Select is disabled 1b - DMA Channel Select is enabled
27 —	Reserved.
26-24 DMCHN1	DMA Channel Number When DMCHEN is set high, this field selects the DMA Channel number to which the packet passed by this filter is routed. The width of this field depends on the number of the DMA channels present in your configuration.
23-22 —	Reserved.
21 L4DPIM1	Layer 4 Destination Port Inverse Match Enable When this bit is set, the Layer 4 Destination Port number field is enabled for inverse matching. When this bit is reset, the Layer 4 Destination Port number field is enabled for perfect matching. This bit is valid and applicable only when the L4DPM0 bit is set high. 0b - Layer 4 Destination Port Inverse Match is disabled 1b - Layer 4 Destination Port Inverse Match is enabled
20 L4DPM1	Layer 4 Destination Port Match Enable When this bit is set, the Layer 4 Destination Port number field is enabled for matching. When this bit is reset, the MAC ignores the Layer 4 Destination Port number field for matching. 0b - Layer 4 Destination Port Match is disabled 1b - Layer 4 Destination Port Match is enabled
19 L4SPIM1	Layer 4 Source Port Inverse Match Enable When this bit is set, the Layer 4 Source Port number field is enabled for inverse matching. When this bit is reset, the Layer 4 Source Port number field is enabled for perfect matching. This bit is valid and applicable only when the L4SPM0 bit is set high. 0b - Layer 4 Source Port Inverse Match is disabled 1b - Layer 4 Source Port Inverse Match is enabled

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Field	Function
18 L4SPM1	Layer 4 Source Port Match Enable When this bit is set, the Layer 4 Source Port number field is enabled for matching. When this bit is reset, the MAC ignores the Layer 4 Source Port number field for matching. 0b - Layer 4 Source Port Match is disabled 1b - Layer 4 Source Port Match is enabled
17 —	Reserved.
16 L4PEN1	Layer 4 Protocol Enable When this bit is set, the Source and Destination Port number fields of UDP packets are used for matching. When this bit is reset, the Source and Destination Port number fields of TCP packets are used for matching. The Layer 4 matching is done only when the L4SPM0 or L4DPM0 bit is set. 0b - Layer 4 Protocol is disabled 1b - Layer 4 Protocol is enabled
15-11 L3HDBM1	Layer 3 IP DA Higher Bits Match IPv4 Packets: This field contains the number of higher bits of IP Destination Address that are matched in the IPv4 packets. The following list describes the values of this field: - 0: No bits are masked. - 1: LSb[0] is masked - 2: Two LSbs [1:0] are masked - .. - 31: All bits except MSb are masked. IPv6 Packets: Bits[12:11] of this field correspond to Bits[6:5] of L3HSBM0 which indicate the number of lower bits of IP Source or Destination Address that are masked in the IPv6 packets. The following list describes the concatenated values of the L3HDBM0[1:0] and L3HSBM0 bits: - 0: No bits are masked. - 1: LSb[0] is masked. - 2: Two LSbs [1:0] are masked - .. - 127: All bits except MSb are masked. This field is valid and applicable only when the L3DAM0 or L3SAM0 bit is set.
10-6 L3HSBM1	Layer 3 IP SA Higher Bits Match IPv4 Packets:

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Field	Function
	This field contains the number of lower bits of IP Source Address that are masked for matching in the IPv4 packets. The following list describes the values of this field: - 0: No bits are masked. - 1: LSb[0] is masked - 2: Two LSbs [1:0] are masked - .. - 31: All bits except MSb are masked. IPv6 Packets: This field contains Bits[4:0] of L3HSBM0. These bits indicate the number of higher bits of IP Source or Destination Address matched in the IPv6 packets. This field is valid and applicable only when the L3DAM0 or L3SAM0 bit is set high.
5 L3DAIM1	Layer 3 IP DA Inverse Match Enable When this bit is set, the Layer 3 IP Destination Address field is enabled for inverse matching. When this bit is reset, the Layer 3 IP Destination Address field is enabled for perfect matching. This bit is valid and applicable only when the L3DAM0 bit is set high. 0b - Layer 3 IP DA Inverse Match is disabled 1b - Layer 3 IP DA Inverse Match is enabled
4 L3DAM1	Layer 3 IP DA Match Enable When this bit is set, the Layer 3 IP Destination Address field is enabled for matching. When this bit is reset, the MAC ignores the Layer 3 IP Destination Address field for matching. Note: When the L3PEN0 bit is set, you should set either this bit or the L3SAM0 bit because either IPv6 DA or SA can be checked for filtering. 0b - Layer 3 IP DA Match is disabled 1b - Layer 3 IP DA Match is enabled
3 L3SAIM1	Layer 3 IP SA Inverse Match Enable When this bit is set, the Layer 3 IP Source Address field is enabled for inverse matching. When this bit is reset, the Layer 3 IP Source Address field is enabled for perfect matching. This bit is valid and applicable only when the L3SAM0 bit is set. 0b - Layer 3 IP SA Inverse Match is disabled 1b - Layer 3 IP SA Inverse Match is enabled
2 L3SAM1	Layer 3 IP SA Match Enable When this bit is set, the Layer 3 IP Source Address field is enabled for matching. When this bit is reset, the MAC ignores the Layer 3 IP Source Address field for matching. Note: When the L3PEN0 bit is set, you should set either this bit or the L3DAM0 bit because either IPv6 SA or DA can be checked for filtering.

Table continues on the next page...

Table continued from the previous page...

Field	Function
	0b - Layer 3 IP SA Match is disabled 1b - Layer 3 IP SA Match is enabled
1 —	Reserved.
0 L3PEN1	Layer 3 Protocol Enable When this bit is set, the Layer 3 IP Source or Destination Address matching is enabled for IPv6 packets. When this bit is reset, the Layer 3 IP Source or Destination Address matching is enabled for IPv4 packets. The Layer 3 matching is done only when the L3SAM0 or L3DAM0 bit is set. 0b - Layer 3 Protocol is disabled 1b - Layer 3 Protocol is enabled

1.4.134 (MAC_Layer4_Address1)

Offset

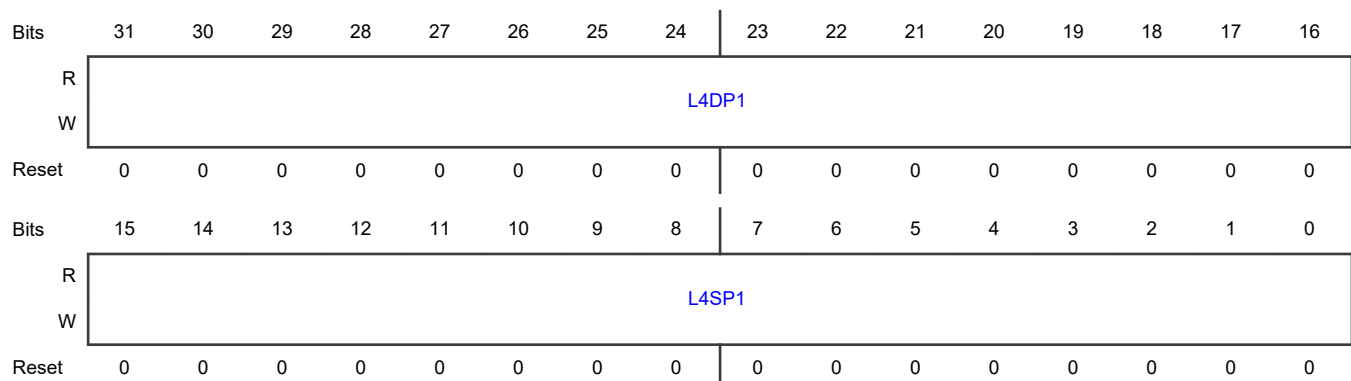
Register	Offset
MAC_Layer4_Address1	934h

Function

The MAC_Layer4_Address(#i), MAC_L3_L4_Control(#i), MAC_Layer3_Addr0_Reg(#i), MAC_Layer3_Addr1_Reg(#i), MAC_Layer3_Addr2_Reg(#i) and MAC_Layer3_Addr3_Reg(#i) registers are reserved (RO with default value) if Enable Layer 3 and Layer 4 Packet Filter option is not selected while configuring the core.

You can configure the Layer 3 and Layer 4 Address Registers to be double-synchronized by selecting the Synchronize Layer 3 and Layer 4 Address Registers to Rx Clock Domain option while configuring the core. When you select this option, the synchronization is triggered only when Bits[31:24] (in little-endian mode) or Bits[7:0] (in big-endian mode) of the Layer 3 and Layer 4 Address Registers are written. For proper synchronization updates, you should perform consecutive writes to same Layer 3 and Layer 4 Address Registers after at least four clock cycles delay of the destination clock.

Diagram



Fields

Field	Function
31-16 L4DP1	Layer 4 Destination Port Number Field When the L4PEN0 bit is reset and the L4DPM0 bit is set in the MAC_L3_L4_Control0 register, this field contains the value to be matched with the TCP Destination Port Number field in the IPv4 or IPv6 packets. When the L4PEN0 and L4DPM0 bits are set in MAC_L3_L4_Control0 register, this field contains the value to be matched with the UDP Destination Port Number field in the IPv4 or IPv6 packets.
15-0 L4SP1	Layer 4 Source Port Number Field When the L4PEN0 bit is reset and the L4SPM0 bit is set in the MAC_L3_L4_Control0 register, this field contains the value to be matched with the TCP Source Port Number field in the IPv4 or IPv6 packets. When the L4PEN0 and L4SPM0 bits are set in MAC_L3_L4_Control0 register, this field contains the value to be matched with the UDP Source Port Number field in the IPv4 or IPv6 packets.

1.4.135 (MAC_Layer3_Addr0_Reg1)

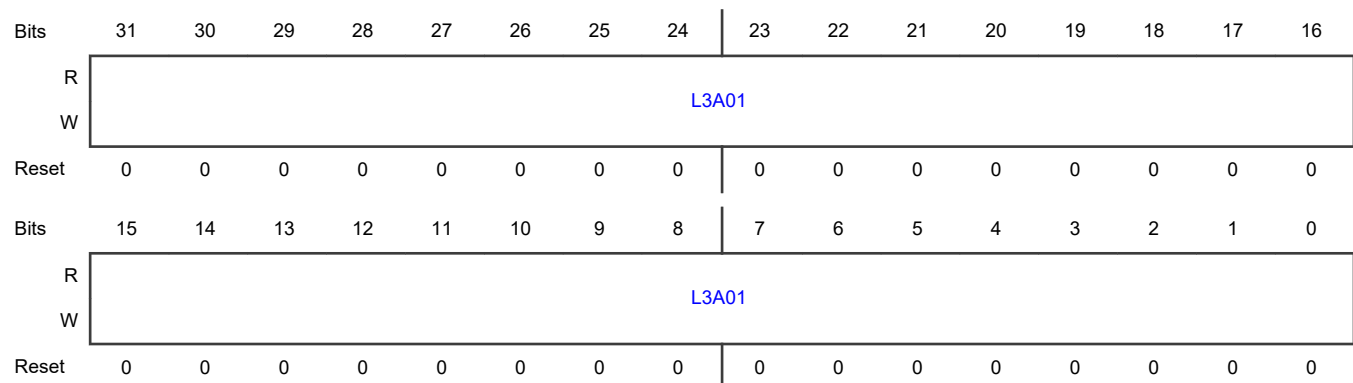
Offset

Register	Offset
MAC_Layer3_Addr0_Reg1	940h

Function

For IPv4 packets, the Layer 3 Address 0 Register 0 register contains the 32-bit IP Source Address field. For IPv6 packets, it contains Bits[31:0] of the 128-bit IP Source Address or Destination Address field.

Diagram



Fields

Field	Function
31-0 L3A01	Layer 3 Address 0 Field When the L3PEN0 and L3SAM0 bits are set in the MAC_L3_L4_Control0 register, this field contains the value to be matched with Bits[31:0] of the IP Source Address field in the IPv6 packets. When the L3PEN0 and L3DAM0 bits are set in the MAC_L3_L4_Control0 register, this field contains the value to be matched with Bits[31:0] of the IP Destination Address field in the IPv6 packets. When the L3PEN0 bit is reset and the L3SAM0 bit is set in the MAC_L3_L4_Control0 register, this field contains the value to be matched with the IP Source Address field in the IPv4 packets.

1.4.136 (MAC_Layer3_Addr1_Reg1)

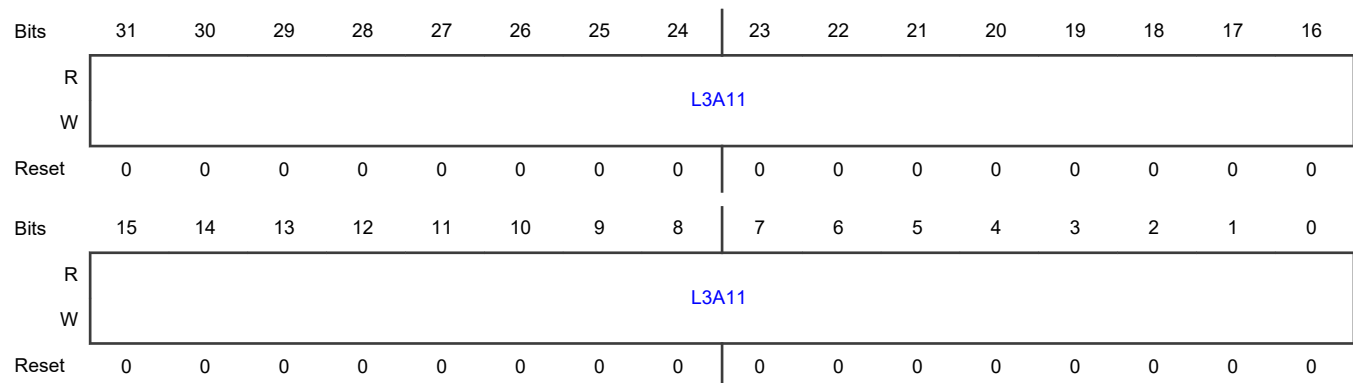
Offset

Register	Offset
MAC_Layer3_Addr1_Reg1	944h

Function

For IPv4 packets, the Layer 3 Address 1 Register 0 register contains the 32-bit IP Destination Address field. For IPv6 packets, it contains Bits[63:32] of the 128-bit IP Source Address or Destination Address field.

Diagram



Fields

Field	Function
31-0 L3A11	Layer 3 Address 1 Field When the L3PEN0 and L3SAM0 bits are set in the MAC_L3_L4_Control0 register, this field contains the value to be matched with Bits[63:32] of the IP Source Address field in the IPv6 packets.

Table continues on the next page...

Field	Function
	When the L3PEN0 and L3DAM0 bits are set in the MAC_L3_L4_Control0 register, this field contains the value to be matched with Bits[63:32] of the IP Destination Address field in the IPv6 packets.
	When the L3PEN0 bit is reset and the L3SAM0 bit is set in the MAC_L3_L4_Control0 register, this field contains the value to be matched with the IP Destination Address field in the IPv4 packets.

1.4.137 (MAC_Layer3_Addr2_Reg1)

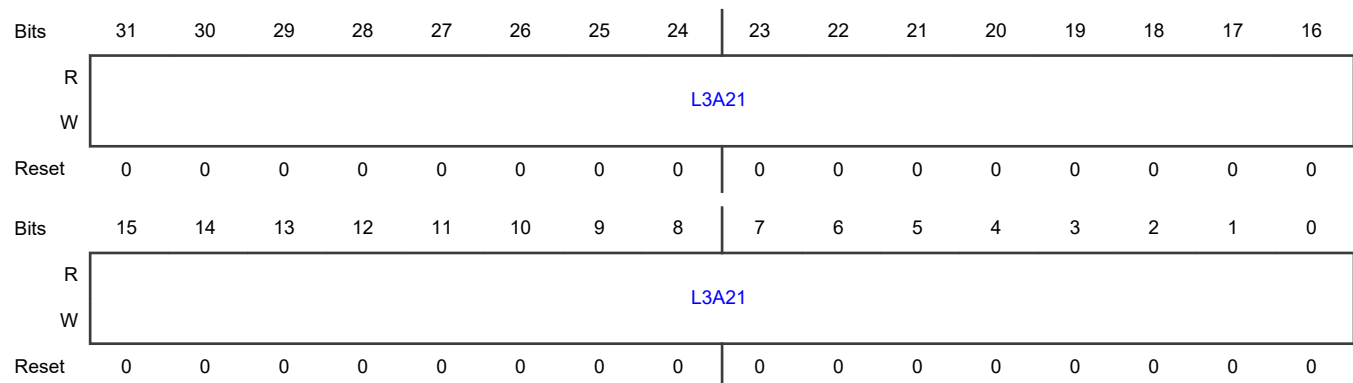
Offset

Register	Offset
MAC_Layer3_Addr2_Reg1	948h

Function

The Layer 3 Address 2 Register 0 register is reserved for IPv4 packets. For IPv6 packets, it contains Bits[95:64] of 128-bit IP Source Address or Destination Address field.

Diagram



Fields

Field	Function
31-0	Layer 3 Address 2 Field
L3A21	When the L3PEN0 and L3SAM0 bits are set in the MAC_L3_L4_Control0 register, this field contains the value to be matched with Bits[95:64] of the IP Source Address field in the IPv6 packets.
	When the L3PEN0 and L3DAM0 bits are set in the MAC_L3_L4_Control0 register, this field contains the value to be matched with Bits[95:64] of the IP Destination Address field in the IPv6 packets.
	When the L3PEN0 bit is reset in the MAC_L3_L4_Control0 register, this field is not used.

1.4.138 (MAC_Layer3_Addr3_Reg1)

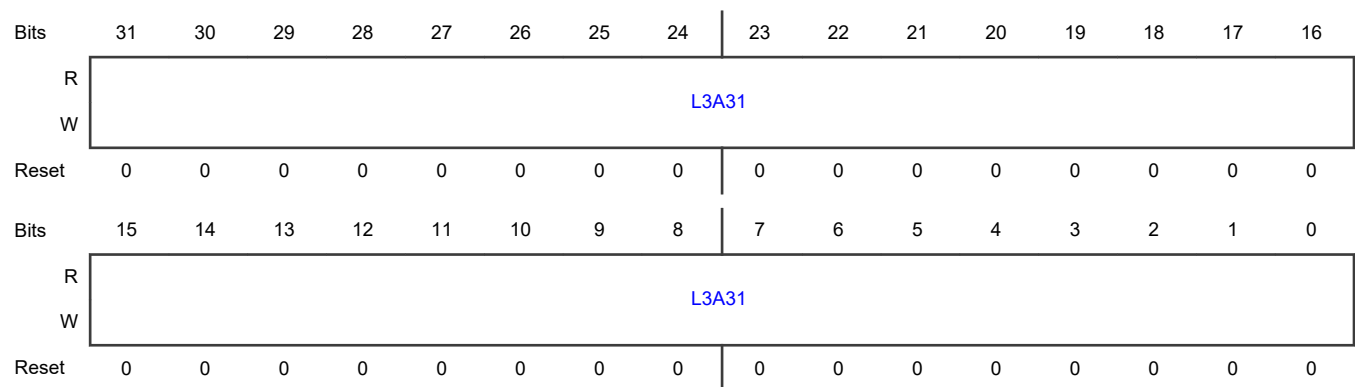
Offset

Register	Offset
MAC_Layer3_Addr3_Reg1	94Ch

Function

The Layer 3 Address 3 Register 0 register is reserved for IPv4 packets. For IPv6 packets, it contains Bits[127:96] of 128-bit IP Source Address or Destination Address field.

Diagram



Fields

Field	Function
31-0	Layer 3 Address 3 Field
L3A31	When the L3PEN0 and L3SAM0 bits are set in the MAC_L3_L4_Control0 register, this field contains the value to be matched with Bits[127:96] of the IP Source Address field in the IPv6 packets. When the L3PEN0 and L3DAM0 bits are set in the MAC_L3_L4_Control0 register, this field contains the value to be matched with Bits[127:96] of the IP Destination Address field in the IPv6 packets. When the L3PEN0 bit is reset in the MAC_L3_L4_Control0 register, this field is not used.

1.4.139 (MAC_L3_L4_Control2)

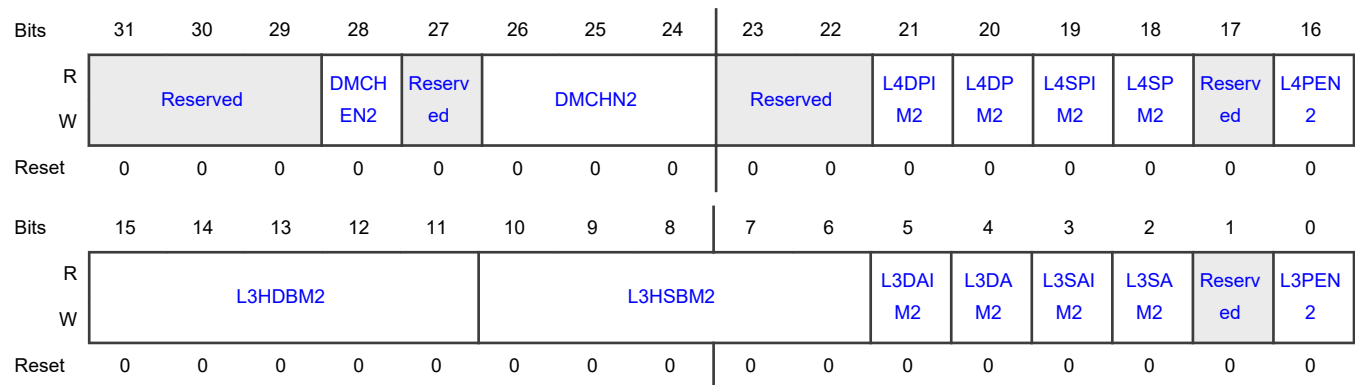
Offset

Register	Offset
MAC_L3_L4_Control2	960h

Function

The Layer 3 and Layer 4 Control register controls the operations of filter 0 of Layer 3 and Layer 4.

Diagram



Fields

Field	Function
31-29 —	Reserved.
28 DMCHEN2	DMA Channel Select Enable When set, this bit enables the selection of the DMA channel number for the packet that is passed by this L3_L4 filter. The DMA channel is indicated by the DMCHN bits. When this bit is reset, the DMA channel is not decided by this filter. 0b - DMA Channel Select is disabled 1b - DMA Channel Select is enabled
27 —	Reserved.
26-24 DMCHN2	DMA Channel Number When DMCHEN is set high, this field selects the DMA Channel number to which the packet passed by this filter is routed. The width of this field depends on the number of the DMA channels present in your configuration.
23-22 —	Reserved.
21 L4DPIM2	Layer 4 Destination Port Inverse Match Enable When this bit is set, the Layer 4 Destination Port number field is enabled for inverse matching. When this bit is reset, the Layer 4 Destination Port number field is enabled for perfect matching. This bit is valid and applicable only when the L4DPM0 bit is set high. 0b - Layer 4 Destination Port Inverse Match is disabled 1b - Layer 4 Destination Port Inverse Match is enabled
20	Layer 4 Destination Port Match Enable

Table continues on the next page...

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Field	Function
L4DPM2	When this bit is set, the Layer 4 Destination Port number field is enabled for matching. When this bit is reset, the MAC ignores the Layer 4 Destination Port number field for matching. 0b - Layer 4 Destination Port Match is disabled 1b - Layer 4 Destination Port Match is enabled
19 L4SPIM2	Layer 4 Source Port Inverse Match Enable When this bit is set, the Layer 4 Source Port number field is enabled for inverse matching. When this bit is reset, the Layer 4 Source Port number field is enabled for perfect matching. This bit is valid and applicable only when the L4SPM0 bit is set high. 0b - Layer 4 Source Port Inverse Match is disabled 1b - Layer 4 Source Port Inverse Match is enabled
18 L4SPM2	Layer 4 Source Port Match Enable When this bit is set, the Layer 4 Source Port number field is enabled for matching. When this bit is reset, the MAC ignores the Layer 4 Source Port number field for matching. 0b - Layer 4 Source Port Match is disabled 1b - Layer 4 Source Port Match is enabled
17 —	Reserved.
16 L4PEN2	Layer 4 Protocol Enable When this bit is set, the Source and Destination Port number fields of UDP packets are used for matching. When this bit is reset, the Source and Destination Port number fields of TCP packets are used for matching. The Layer 4 matching is done only when the L4SPM0 or L4DPM0 bit is set. 0b - Layer 4 Protocol is disabled 1b - Layer 4 Protocol is enabled
15-11 L3HDBM2	Layer 3 IP DA Higher Bits Match IPv4 Packets: This field contains the number of higher bits of IP Destination Address that are matched in the IPv4 packets. The following list describes the values of this field: - 0: No bits are masked. - 1: LSb[0] is masked - 2: Two LSbs [1:0] are masked - .. - 31: All bits except MSb are masked. IPv6 Packets:

Table continues on the next page...

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Field	Function
	Bits[12:11] of this field correspond to Bits[6:5] of L3HSBM0 which indicate the number of lower bits of IP Source or Destination Address that are masked in the IPv6 packets. The following list describes the concatenated values of the L3HDBM0[1:0] and L3HSBM0 bits: - 0: No bits are masked. - 1: LSb[0] is masked. - 2: Two LSbs [1:0] are masked - .. - 127: All bits except MSb are masked. This field is valid and applicable only when the L3DAM0 or L3SAM0 bit is set.
10-6 L3HSBM2	Layer 3 IP SA Higher Bits Match IPv4 Packets: This field contains the number of lower bits of IP Source Address that are masked for matching in the IPv4 packets. The following list describes the values of this field: - 0: No bits are masked. - 1: LSb[0] is masked - 2: Two LSbs [1:0] are masked - .. - 31: All bits except MSb are masked. IPv6 Packets: This field contains Bits[4:0] of L3HSBM0. These bits indicate the number of higher bits of IP Source or Destination Address matched in the IPv6 packets. This field is valid and applicable only when the L3DAM0 or L3SAM0 bit is set high.
5 L3DAIM2	Layer 3 IP DA Inverse Match Enable When this bit is set, the Layer 3 IP Destination Address field is enabled for inverse matching. When this bit is reset, the Layer 3 IP Destination Address field is enabled for perfect matching. This bit is valid and applicable only when the L3DAM0 bit is set high. 0b - Layer 3 IP DA Inverse Match is disabled 1b - Layer 3 IP DA Inverse Match is enabled
4 L3DAM2	Layer 3 IP DA Match Enable When this bit is set, the Layer 3 IP Destination Address field is enabled for matching. When this bit is reset, the MAC ignores the Layer 3 IP Destination Address field for matching. Note: When the L3PEN0 bit is set, you should set either this bit or the L3SAM0 bit because either IPv6 DA or SA can be checked for filtering. 0b - Layer 3 IP DA Match is disabled 1b - Layer 3 IP DA Match is enabled

Table continues on the next page...

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Field	Function
3 L3SAIM2	Layer 3 IP SA Inverse Match Enable When this bit is set, the Layer 3 IP Source Address field is enabled for inverse matching. When this bit is reset, the Layer 3 IP Source Address field is enabled for perfect matching. This bit is valid and applicable only when the L3SAM0 bit is set. 0b - Layer 3 IP SA Inverse Match is disabled 1b - Layer 3 IP SA Inverse Match is enabled
2 L3SAM2	Layer 3 IP SA Match Enable When this bit is set, the Layer 3 IP Source Address field is enabled for matching. When this bit is reset, the MAC ignores the Layer 3 IP Source Address field for matching. Note: When the L3PEN0 bit is set, you should set either this bit or the L3DAM0 bit because either IPv6 SA or DA can be checked for filtering. 0b - Layer 3 IP SA Match is disabled 1b - Layer 3 IP SA Match is enabled
1 —	Reserved.
0 L3PEN2	Layer 3 Protocol Enable When this bit is set, the Layer 3 IP Source or Destination Address matching is enabled for IPv6 packets. When this bit is reset, the Layer 3 IP Source or Destination Address matching is enabled for IPv4 packets. The Layer 3 matching is done only when the L3SAM0 or L3DAM0 bit is set. 0b - Layer 3 Protocol is disabled 1b - Layer 3 Protocol is enabled

1.4.140 (MAC_Layer4_Address2)

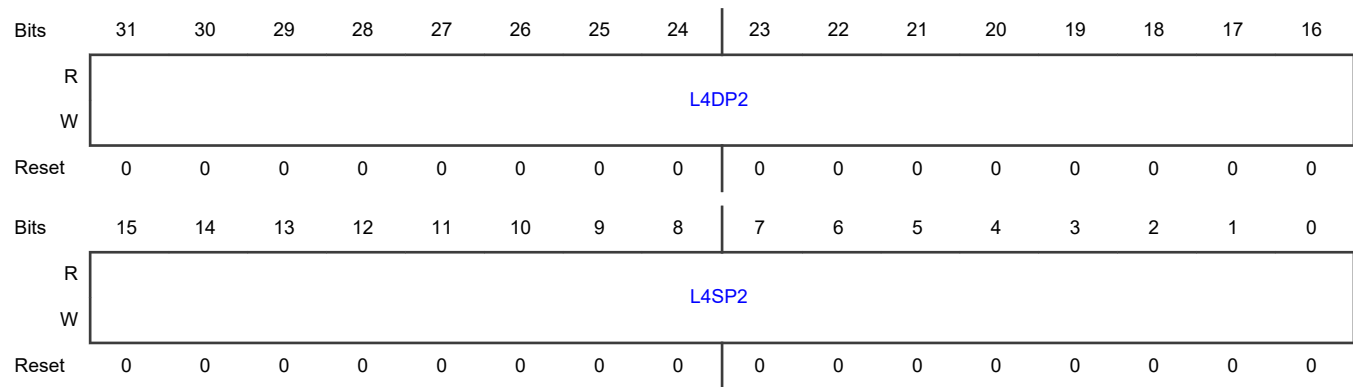
Offset

Register	Offset
MAC_Layer4_Address2	964h

Function

The MAC_Layer4_Address(#i), MAC_L3_L4_Control(#i), MAC_Layer3_Addr0_Reg(#i), MAC_Layer3_Addr1_Reg(#i), MAC_Layer3_Addr2_Reg(#i) and MAC_Layer3_Addr3_Reg(#i) registers are reserved (RO with default value) if Enable Layer 3 and Layer 4 Packet Filter option is not selected while configuring the core.

You can configure the Layer 3 and Layer 4 Address Registers to be double-synchronized by selecting the Synchronize Layer 3 and Layer 4 Address Registers to Rx Clock Domain option while configuring the core. When you select this option, the synchronization is triggered only when Bits[31:24] (in little-endian mode) or Bits[7:0] (in big-endian mode) of the Layer 3 and Layer 4 Address Registers are written. For proper synchronization updates, you should perform consecutive writes to same Layer 3 and Layer 4 Address Registers after at least four clock cycles delay of the destination clock.

Diagram**Fields**

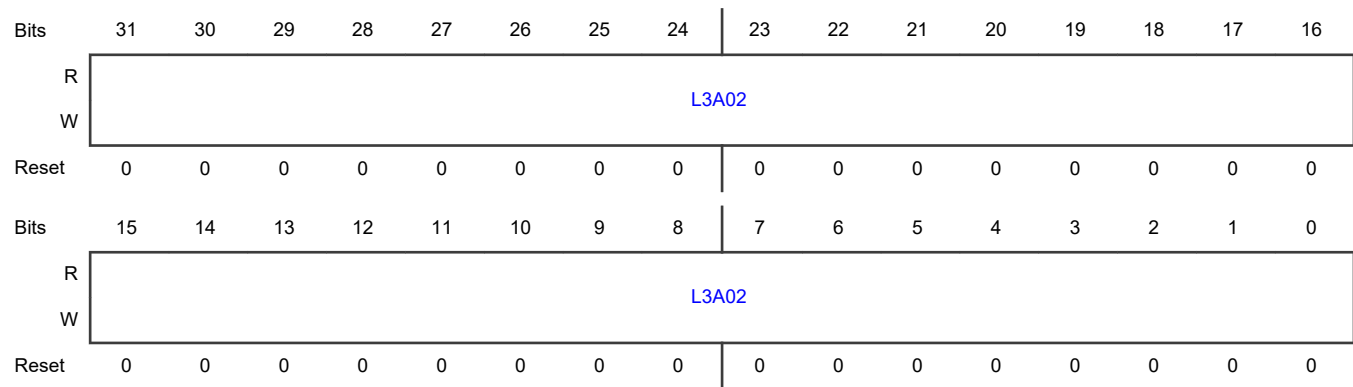
Field	Function
31-16 L4DP2	Layer 4 Destination Port Number Field When the L4PEN0 bit is reset and the L4DPM0 bit is set in the MAC_L3_L4_Control0 register, this field contains the value to be matched with the TCP Destination Port Number field in the IPv4 or IPv6 packets. When the L4PEN0 and L4DPM0 bits are set in MAC_L3_L4_Control0 register, this field contains the value to be matched with the UDP Destination Port Number field in the IPv4 or IPv6 packets.
15-0 L4SP2	Layer 4 Source Port Number Field When the L4PEN0 bit is reset and the L4SPM0 bit is set in the MAC_L3_L4_Control0 register, this field contains the value to be matched with the TCP Source Port Number field in the IPv4 or IPv6 packets. When the L4PEN0 and L4SPM0 bits are set in MAC_L3_L4_Control0 register, this field contains the value to be matched with the UDP Source Port Number field in the IPv4 or IPv6 packets.

1.4.141 (MAC_Layer3_Addr0_Reg2)**Offset**

Register	Offset
MAC_Layer3_Addr0_Reg2	970h

Function

For IPv4 packets, the Layer 3 Address 0 Register 0 register contains the 32-bit IP Source Address field. For IPv6 packets, it contains Bits[31:0] of the 128-bit IP Source Address or Destination Address field.

Diagram**Fields**

Field	Function
31-0 L3A02	Layer 3 Address 0 Field When the L3PEN0 and L3SAM0 bits are set in the MAC_L3_L4_Control0 register, this field contains the value to be matched with Bits[31:0] of the IP Source Address field in the IPv6 packets. When the L3PEN0 and L3DAM0 bits are set in the MAC_L3_L4_Control0 register, this field contains the value to be matched with Bits[31:0] of the IP Destination Address field in the IPv6 packets. When the L3PEN0 bit is reset and the L3SAM0 bit is set in the MAC_L3_L4_Control0 register, this field contains the value to be matched with the IP Source Address field in the IPv4 packets.

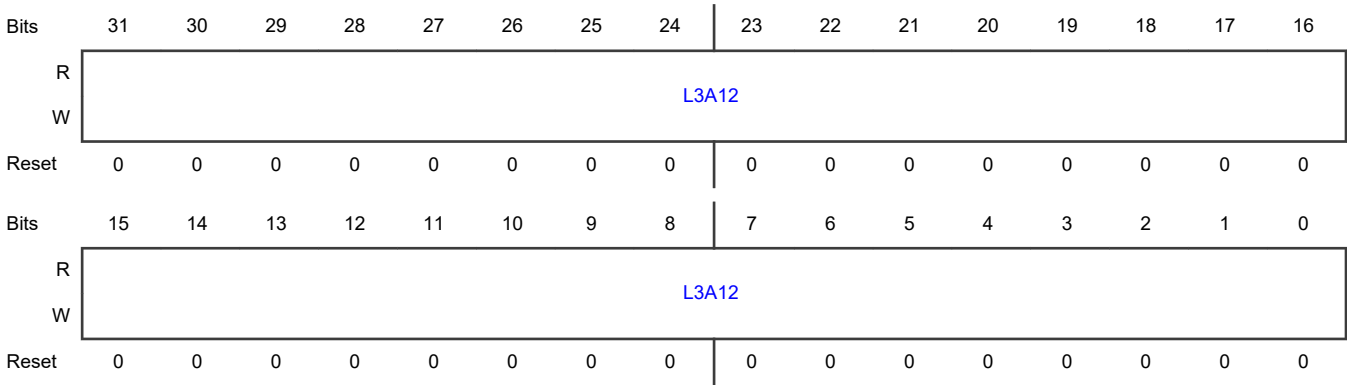
1.4.142 (MAC_Layer3_Addr1_Reg2)**Offset**

Register	Offset
MAC_Layer3_Addr1_Reg2	974h

Function

For IPv4 packets, the Layer 3 Address 1 Register 0 register contains the 32-bit IP Destination Address field. For IPv6 packets, it contains Bits[63:32] of the 128-bit IP Source Address or Destination Address field.

Diagram



Fields

Field	Function
31-0 L3A12	Layer 3 Address 1 Field When the L3PEN0 and L3SAM0 bits are set in the MAC_L3_L4_Control0 register, this field contains the value to be matched with Bits[63:32] of the IP Source Address field in the IPv6 packets. When the L3PEN0 and L3DAM0 bits are set in the MAC_L3_L4_Control0 register, this field contains the value to be matched with Bits[63:32] of the IP Destination Address field in the IPv6 packets. When the L3PEN0 bit is reset and the L3SAM0 bit is set in the MAC_L3_L4_Control0 register, this field contains the value to be matched with the IP Destination Address field in the IPv4 packets.

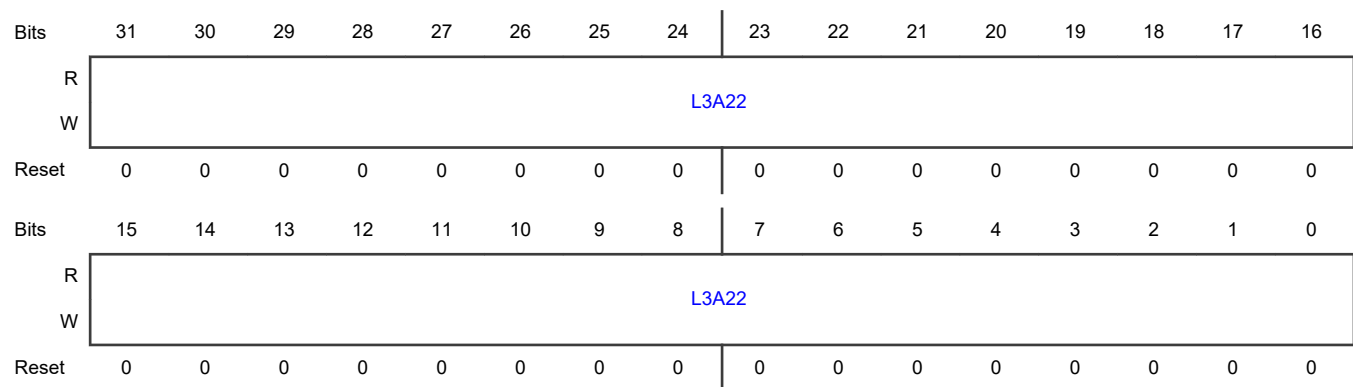
1.4.143 (MAC_Layer3_Addr2_Reg2)

Offset

Register	Offset
MAC_Layer3_Addr2_Reg2	978h

Function

The Layer 3 Address 2 Register 0 register is reserved for IPv4 packets. For IPv6 packets, it contains Bits[95:64] of 128-bit IP Source Address or Destination Address field.

Diagram**Fields**

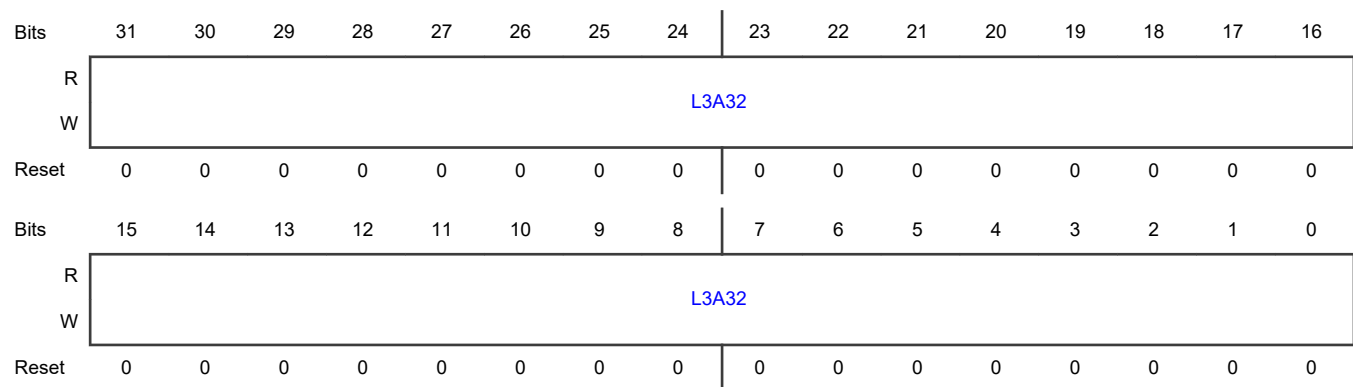
Field	Function
31-0 L3A22	Layer 3 Address 2 Field When the L3PEN0 and L3SAM0 bits are set in the MAC_L3_L4_Control0 register, this field contains the value to be matched with Bits[95:64] of the IP Source Address field in the IPv6 packets. When the L3PEN0 and L3DAM0 bits are set in the MAC_L3_L4_Control0 register, this field contains the value to be matched with Bits[95:64] of the IP Destination Address field in the IPv6 packets. When the L3PEN0 bit is reset in the MAC_L3_L4_Control0 register, this field is not used.

1.4.144 (MAC_Layer3_Addr3_Reg2)**Offset**

Register	Offset
MAC_Layer3_Addr3_Reg2	97Ch

Function

The Layer 3 Address 3 Register 0 register is reserved for IPv4 packets. For IPv6 packets, it contains Bits[127:96] of 128-bit IP Source Address or Destination Address field.

Diagram**Fields**

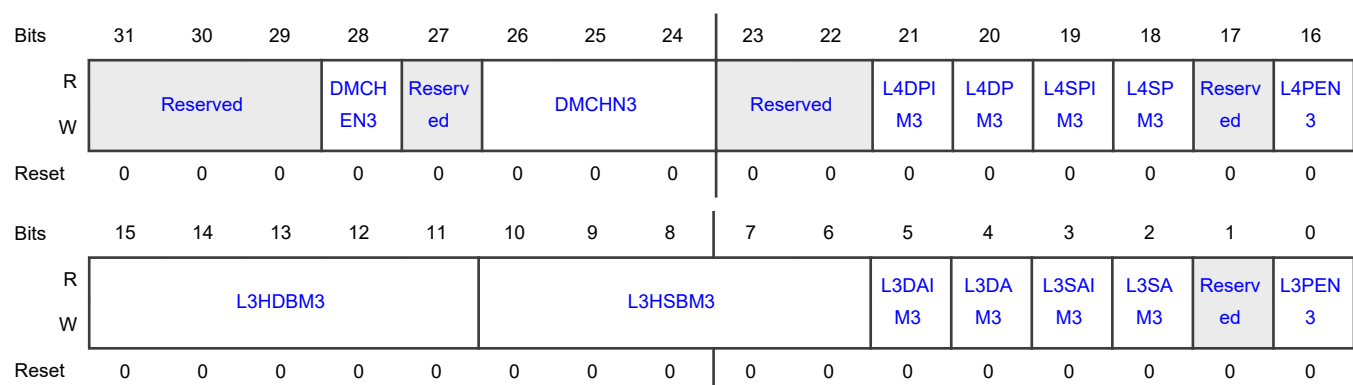
Field	Function
31-0 L3A32	Layer 3 Address 3 Field When the L3PEN0 and L3SAM0 bits are set in the MAC_L3_L4_Control0 register, this field contains the value to be matched with Bits[127:96] of the IP Source Address field in the IPv6 packets. When the L3PEN0 and L3DAM0 bits are set in the MAC_L3_L4_Control0 register, this field contains the value to be matched with Bits[127:96] of the IP Destination Address field in the IPv6 packets. When the L3PEN0 bit is reset in the MAC_L3_L4_Control0 register, this field is not used.

1.4.145 (MAC_L3_L4_Control3)**Offset**

Register	Offset
MAC_L3_L4_Control3	990h

Function

The Layer 3 and Layer 4 Control register controls the operations of filter 0 of Layer 3 and Layer 4.

Diagram

Fields

Field	Function
31-29 —	Reserved.
28 DMCHEN3	DMA Channel Select Enable When set, this bit enables the selection of the DMA channel number for the packet that is passed by this L3_L4 filter. The DMA channel is indicated by the DMCHN bits. When this bit is reset, the DMA channel is not decided by this filter. 0b - DMA Channel Select is disabled 1b - DMA Channel Select is enabled
27 —	Reserved.
26-24 DMCHN3	DMA Channel Number When DMCHEN is set high, this field selects the DMA Channel number to which the packet passed by this filter is routed. The width of this field depends on the number of the DMA channels present in your configuration.
23-22 —	Reserved.
21 L4DPIM3	Layer 4 Destination Port Inverse Match Enable When this bit is set, the Layer 4 Destination Port number field is enabled for inverse matching. When this bit is reset, the Layer 4 Destination Port number field is enabled for perfect matching. This bit is valid and applicable only when the L4DPM0 bit is set high. 0b - Layer 4 Destination Port Inverse Match is disabled 1b - Layer 4 Destination Port Inverse Match is enabled
20 L4DPM3	Layer 4 Destination Port Match Enable When this bit is set, the Layer 4 Destination Port number field is enabled for matching. When this bit is reset, the MAC ignores the Layer 4 Destination Port number field for matching. 0b - Layer 4 Destination Port Match is disabled 1b - Layer 4 Destination Port Match is enabled
19 L4SPIM3	Layer 4 Source Port Inverse Match Enable When this bit is set, the Layer 4 Source Port number field is enabled for inverse matching. When this bit is reset, the Layer 4 Source Port number field is enabled for perfect matching. This bit is valid and applicable only when the L4SPM0 bit is set high. 0b - Layer 4 Source Port Inverse Match is disabled 1b - Layer 4 Source Port Inverse Match is enabled

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Field	Function
18 L4SPM3	Layer 4 Source Port Match Enable When this bit is set, the Layer 4 Source Port number field is enabled for matching. When this bit is reset, the MAC ignores the Layer 4 Source Port number field for matching. 0b - Layer 4 Source Port Match is disabled 1b - Layer 4 Source Port Match is enabled
17 —	Reserved.
16 L4PEN3	Layer 4 Protocol Enable When this bit is set, the Source and Destination Port number fields of UDP packets are used for matching. When this bit is reset, the Source and Destination Port number fields of TCP packets are used for matching. The Layer 4 matching is done only when the L4SPM0 or L4DPM0 bit is set. 0b - Layer 4 Protocol is disabled 1b - Layer 4 Protocol is enabled
15-11 L3HDBM3	Layer 3 IP DA Higher Bits Match IPv4 Packets: This field contains the number of higher bits of IP Destination Address that are matched in the IPv4 packets. The following list describes the values of this field: - 0: No bits are masked. - 1: LSb[0] is masked - 2: Two LSbs [1:0] are masked - .. - 31: All bits except MSb are masked. IPv6 Packets: Bits[12:11] of this field correspond to Bits[6:5] of L3HSBM0 which indicate the number of lower bits of IP Source or Destination Address that are masked in the IPv6 packets. The following list describes the concatenated values of the L3HDBM0[1:0] and L3HSBM0 bits: - 0: No bits are masked. - 1: LSb[0] is masked. - 2: Two LSbs [1:0] are masked - .. - 127: All bits except MSb are masked. This field is valid and applicable only when the L3DAM0 or L3SAM0 bit is set.
10-6 L3HSBM3	Layer 3 IP SA Higher Bits Match IPv4 Packets:

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Field	Function
	This field contains the number of lower bits of IP Source Address that are masked for matching in the IPv4 packets. The following list describes the values of this field: - 0: No bits are masked. - 1: LSb[0] is masked - 2: Two LSbs [1:0] are masked - .. - 31: All bits except MSb are masked. IPv6 Packets: This field contains Bits[4:0] of L3HSBM0. These bits indicate the number of higher bits of IP Source or Destination Address matched in the IPv6 packets. This field is valid and applicable only when the L3DAM0 or L3SAM0 bit is set high.
5 L3DAIM3	Layer 3 IP DA Inverse Match Enable When this bit is set, the Layer 3 IP Destination Address field is enabled for inverse matching. When this bit is reset, the Layer 3 IP Destination Address field is enabled for perfect matching. This bit is valid and applicable only when the L3DAM0 bit is set high. 0b - Layer 3 IP DA Inverse Match is disabled 1b - Layer 3 IP DA Inverse Match is enabled
4 L3DAM3	Layer 3 IP DA Match Enable When this bit is set, the Layer 3 IP Destination Address field is enabled for matching. When this bit is reset, the MAC ignores the Layer 3 IP Destination Address field for matching. Note: When the L3PEN0 bit is set, you should set either this bit or the L3SAM0 bit because either IPv6 DA or SA can be checked for filtering. 0b - Layer 3 IP DA Match is disabled 1b - Layer 3 IP DA Match is enabled
3 L3SAIM3	Layer 3 IP SA Inverse Match Enable When this bit is set, the Layer 3 IP Source Address field is enabled for inverse matching. When this bit is reset, the Layer 3 IP Source Address field is enabled for perfect matching. This bit is valid and applicable only when the L3SAM0 bit is set. 0b - Layer 3 IP SA Inverse Match is disabled 1b - Layer 3 IP SA Inverse Match is enabled
2 L3SAM3	Layer 3 IP SA Match Enable When this bit is set, the Layer 3 IP Source Address field is enabled for matching. When this bit is reset, the MAC ignores the Layer 3 IP Source Address field for matching. Note: When the L3PEN0 bit is set, you should set either this bit or the L3DAM0 bit because either IPv6 SA or DA can be checked for filtering.

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Table continued from the previous page...

Field	Function
	0b - Layer 3 IP SA Match is disabled 1b - Layer 3 IP SA Match is enabled
1 —	Reserved.
0 L3PEN3	Layer 3 Protocol Enable When this bit is set, the Layer 3 IP Source or Destination Address matching is enabled for IPv6 packets. When this bit is reset, the Layer 3 IP Source or Destination Address matching is enabled for IPv4 packets. The Layer 3 matching is done only when the L3SAM0 or L3DAM0 bit is set. 0b - Layer 3 Protocol is disabled 1b - Layer 3 Protocol is enabled

1.4.146 (MAC_Layer4_Address3)

Offset

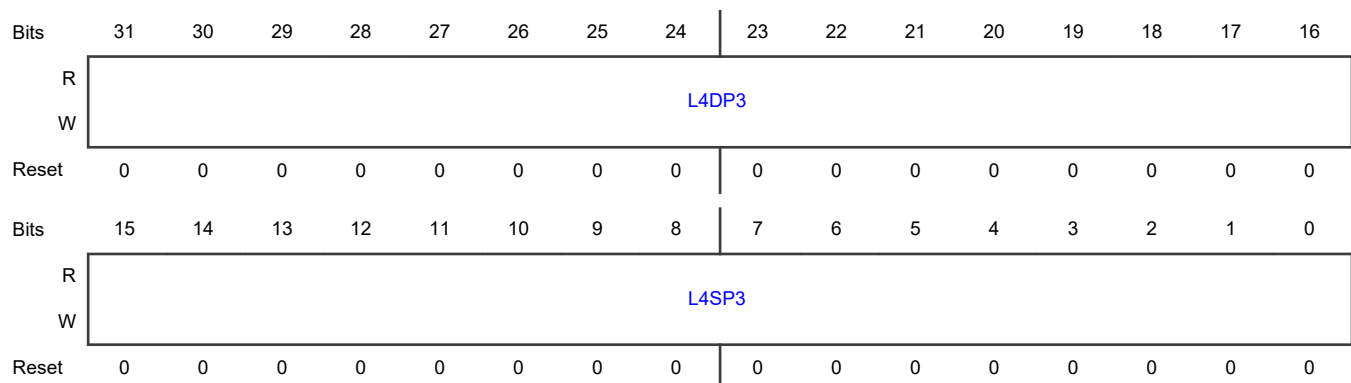
Register	Offset
MAC_Layer4_Address3	994h

Function

The MAC_Layer4_Address(#i), MAC_L3_L4_Control(#i), MAC_Layer3_Addr0_Reg(#i), MAC_Layer3_Addr1_Reg(#i), MAC_Layer3_Addr2_Reg(#i) and MAC_Layer3_Addr3_Reg(#i) registers are reserved (RO with default value) if Enable Layer 3 and Layer 4 Packet Filter option is not selected while configuring the core.

You can configure the Layer 3 and Layer 4 Address Registers to be double-synchronized by selecting the Synchronize Layer 3 and Layer 4 Address Registers to Rx Clock Domain option while configuring the core. When you select this option, the synchronization is triggered only when Bits[31:24] (in little-endian mode) or Bits[7:0] (in big-endian mode) of the Layer 3 and Layer 4 Address Registers are written. For proper synchronization updates, you should perform consecutive writes to same Layer 3 and Layer 4 Address Registers after at least four clock cycles delay of the destination clock.

Diagram



Fields

Field	Function
31-16 L4DP3	Layer 4 Destination Port Number Field When the L4PEN0 bit is reset and the L4DPM0 bit is set in the MAC_L3_L4_Control0 register, this field contains the value to be matched with the TCP Destination Port Number field in the IPv4 or IPv6 packets. When the L4PEN0 and L4DPM0 bits are set in MAC_L3_L4_Control0 register, this field contains the value to be matched with the UDP Destination Port Number field in the IPv4 or IPv6 packets.
15-0 L4SP3	Layer 4 Source Port Number Field When the L4PEN0 bit is reset and the L4SPM0 bit is set in the MAC_L3_L4_Control0 register, this field contains the value to be matched with the TCP Source Port Number field in the IPv4 or IPv6 packets. When the L4PEN0 and L4SPM0 bits are set in MAC_L3_L4_Control0 register, this field contains the value to be matched with the UDP Source Port Number field in the IPv4 or IPv6 packets.

1.4.147 (MAC_Layer3_Addr0_Reg3)

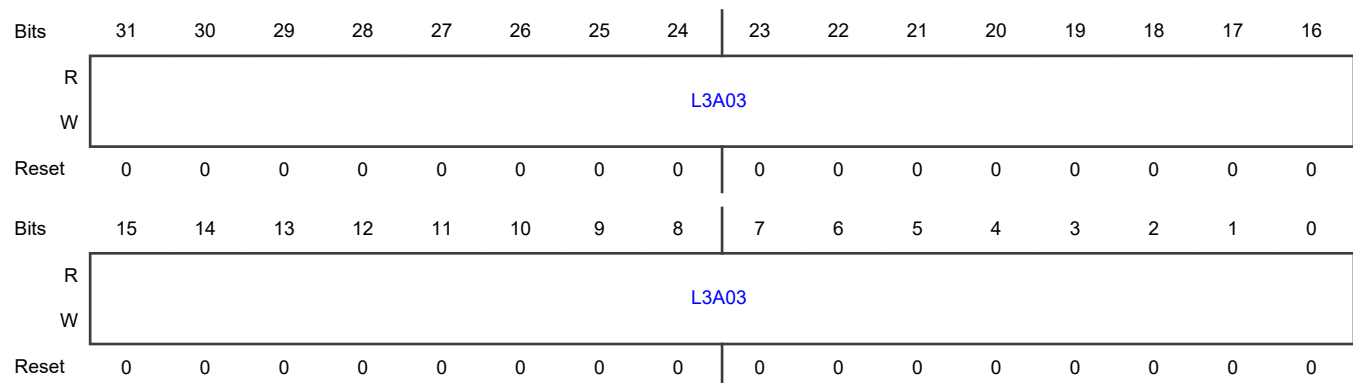
Offset

Register	Offset
MAC_Layer3_Addr0_Reg3	9A0h

Function

For IPv4 packets, the Layer 3 Address 0 Register 0 register contains the 32-bit IP Source Address field. For IPv6 packets, it contains Bits[31:0] of the 128-bit IP Source Address or Destination Address field.

Diagram



Fields

Field	Function
31-0 L3A03	Layer 3 Address 0 Field When the L3PEN0 and L3SAM0 bits are set in the MAC_L3_L4_Control0 register, this field contains the value to be matched with Bits[31:0] of the IP Source Address field in the IPv6 packets. When the L3PEN0 and L3DAM0 bits are set in the MAC_L3_L4_Control0 register, this field contains the value to be matched with Bits[31:0] of the IP Destination Address field in the IPv6 packets. When the L3PEN0 bit is reset and the L3SAM0 bit is set in the MAC_L3_L4_Control0 register, this field contains the value to be matched with the IP Source Address field in the IPv4 packets.

1.4.148 (MAC_Layer3_Addr1_Reg3)

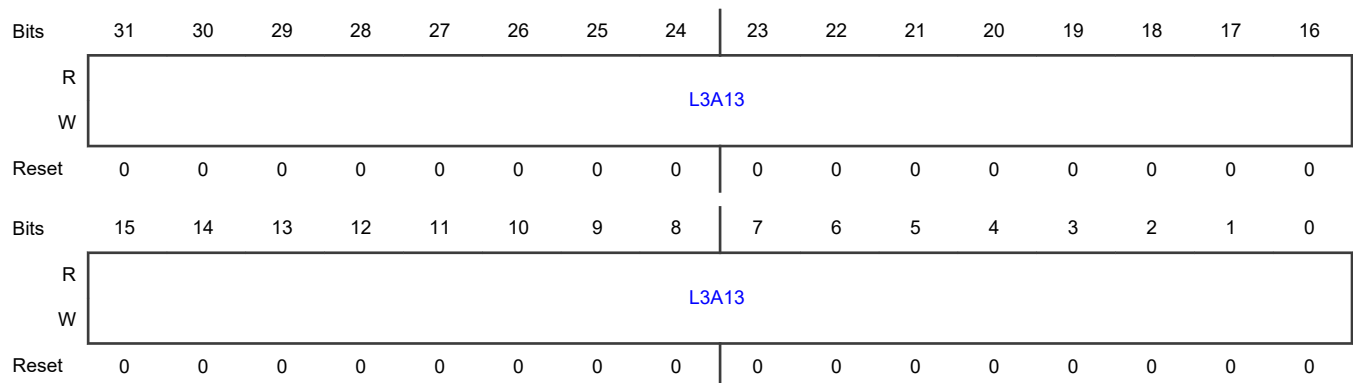
Offset

Register	Offset
MAC_Layer3_Addr1_Reg3	9A4h

Function

For IPv4 packets, the Layer 3 Address 1 Register 0 register contains the 32-bit IP Destination Address field. For IPv6 packets, it contains Bits[63:32] of the 128-bit IP Source Address or Destination Address field.

Diagram



Fields

Field	Function
31-0 L3A13	Layer 3 Address 1 Field When the L3PEN0 and L3SAM0 bits are set in the MAC_L3_L4_Control0 register, this field contains the value to be matched with Bits[63:32] of the IP Source Address field in the IPv6 packets.

Table continues on the next page...

Field	Function
	When the L3PEN0 and L3DAM0 bits are set in the MAC_L3_L4_Control0 register, this field contains the value to be matched with Bits[63:32] of the IP Destination Address field in the IPv6 packets.
	When the L3PEN0 bit is reset and the L3SAM0 bit is set in the MAC_L3_L4_Control0 register, this field contains the value to be matched with the IP Destination Address field in the IPv4 packets.

1.4.149 (MAC_Layer3_Addr2_Reg3)

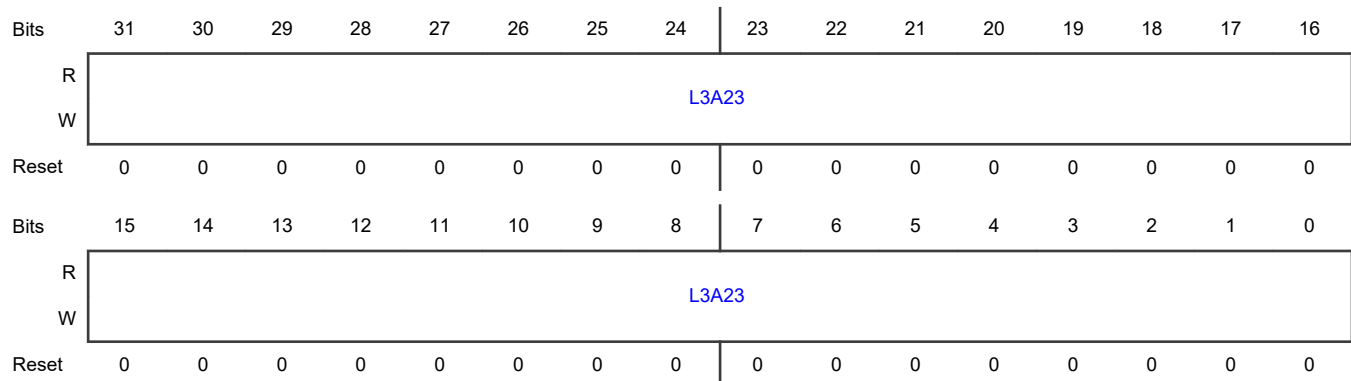
Offset

Register	Offset
MAC_Layer3_Addr2_Reg3	9A8h

Function

The Layer 3 Address 2 Register 0 register is reserved for IPv4 packets. For IPv6 packets, it contains Bits[95:64] of 128-bit IP Source Address or Destination Address field.

Diagram



Fields

Field	Function
31-0 L3A23	Layer 3 Address 2 Field When the L3PEN0 and L3SAM0 bits are set in the MAC_L3_L4_Control0 register, this field contains the value to be matched with Bits[95:64] of the IP Source Address field in the IPv6 packets. When the L3PEN0 and L3DAM0 bits are set in the MAC_L3_L4_Control0 register, this field contains the value to be matched with Bits[95:64] of the IP Destination Address field in the IPv6 packets. When the L3PEN0 bit is reset in the MAC_L3_L4_Control0 register, this field is not used.

1.4.150 (MAC_Layer3_Addr3_Reg3)

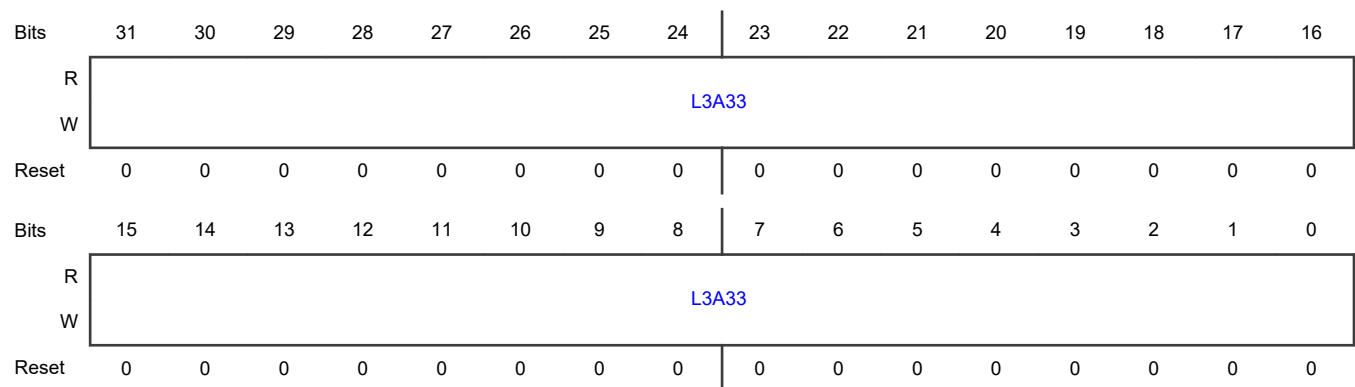
Offset

Register	Offset
MAC_Layer3_Addr3_Reg3	9ACh

Function

The Layer 3 Address 3 Register 0 register is reserved for IPv4 packets. For IPv6 packets, it contains Bits[127:96] of 128-bit IP Source Address or Destination Address field.

Diagram



Fields

Field	Function
31-0	Layer 3 Address 3 Field
L3A33	When the L3PEN0 and L3SAM0 bits are set in the MAC_L3_L4_Control0 register, this field contains the value to be matched with Bits[127:96] of the IP Source Address field in the IPv6 packets. When the L3PEN0 and L3DAM0 bits are set in the MAC_L3_L4_Control0 register, this field contains the value to be matched with Bits[127:96] of the IP Destination Address field in the IPv6 packets. When the L3PEN0 bit is reset in the MAC_L3_L4_Control0 register, this field is not used.

1.4.151 (MAC_L3_L4_Control4)

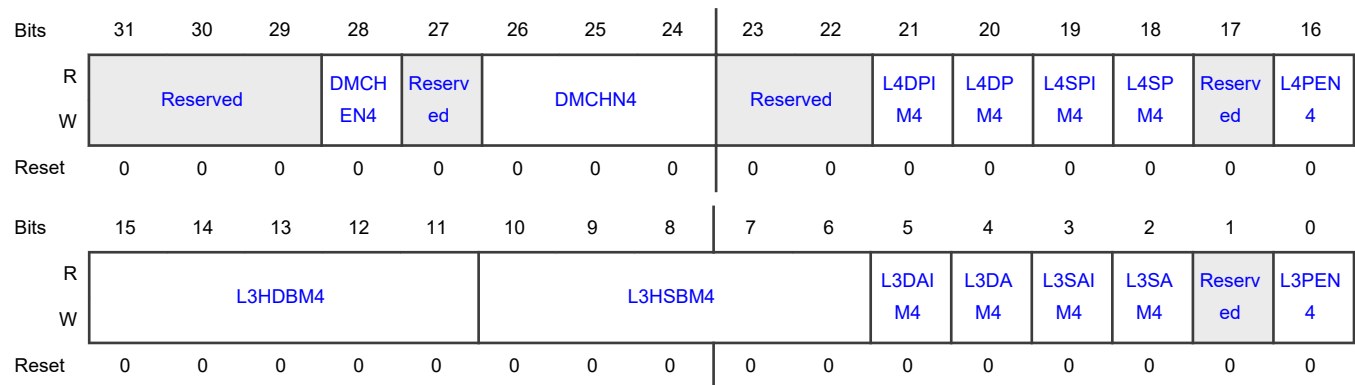
Offset

Register	Offset
MAC_L3_L4_Control4	9C0h

Function

The Layer 3 and Layer 4 Control register controls the operations of filter 0 of Layer 3 and Layer 4.

Diagram



Fields

Field	Function
31-29 —	Reserved.
28 DMCHEN4	DMA Channel Select Enable When set, this bit enables the selection of the DMA channel number for the packet that is passed by this L3_L4 filter. The DMA channel is indicated by the DMCHN bits. When this bit is reset, the DMA channel is not decided by this filter. 0b - DMA Channel Select is disabled 1b - DMA Channel Select is enabled
27 —	Reserved.
26-24 DMCHN4	DMA Channel Number When DMCHEN is set high, this field selects the DMA Channel number to which the packet passed by this filter is routed. The width of this field depends on the number of the DMA channels present in your configuration.
23-22 —	Reserved.
21 L4DPIM4	Layer 4 Destination Port Inverse Match Enable When this bit is set, the Layer 4 Destination Port number field is enabled for inverse matching. When this bit is reset, the Layer 4 Destination Port number field is enabled for perfect matching. This bit is valid and applicable only when the L4DPM0 bit is set high. 0b - Layer 4 Destination Port Inverse Match is disabled 1b - Layer 4 Destination Port Inverse Match is enabled
20	Layer 4 Destination Port Match Enable

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Field	Function
L4DPM4	When this bit is set, the Layer 4 Destination Port number field is enabled for matching. When this bit is reset, the MAC ignores the Layer 4 Destination Port number field for matching. 0b - Layer 4 Destination Port Match is disabled 1b - Layer 4 Destination Port Match is enabled
19 L4SPIM4	Layer 4 Source Port Inverse Match Enable When this bit is set, the Layer 4 Source Port number field is enabled for inverse matching. When this bit is reset, the Layer 4 Source Port number field is enabled for perfect matching. This bit is valid and applicable only when the L4SPM0 bit is set high. 0b - Layer 4 Source Port Inverse Match is disabled 1b - Layer 4 Source Port Inverse Match is enabled
18 L4SPM4	Layer 4 Source Port Match Enable When this bit is set, the Layer 4 Source Port number field is enabled for matching. When this bit is reset, the MAC ignores the Layer 4 Source Port number field for matching. 0b - Layer 4 Source Port Match is disabled 1b - Layer 4 Source Port Match is enabled
17 —	Reserved.
16 L4PEN4	Layer 4 Protocol Enable When this bit is set, the Source and Destination Port number fields of UDP packets are used for matching. When this bit is reset, the Source and Destination Port number fields of TCP packets are used for matching. The Layer 4 matching is done only when the L4SPM0 or L4DPM0 bit is set. 0b - Layer 4 Protocol is disabled 1b - Layer 4 Protocol is enabled
15-11 L3HDBM4	Layer 3 IP DA Higher Bits Match IPv4 Packets: This field contains the number of higher bits of IP Destination Address that are matched in the IPv4 packets. The following list describes the values of this field: - 0: No bits are masked. - 1: LSb[0] is masked - 2: Two LSbs [1:0] are masked - .. - 31: All bits except MSb are masked. IPv6 Packets:

Table continues on the next page...

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Field	Function
	Bits[12:11] of this field correspond to Bits[6:5] of L3HSBM0 which indicate the number of lower bits of IP Source or Destination Address that are masked in the IPv6 packets. The following list describes the concatenated values of the L3HDBM0[1:0] and L3HSBM0 bits: - 0: No bits are masked. - 1: LSb[0] is masked. - 2: Two LSbs [1:0] are masked - .. - 127: All bits except MSb are masked. This field is valid and applicable only when the L3DAM0 or L3SAM0 bit is set.
10-6 L3HSBM4	Layer 3 IP SA Higher Bits Match IPv4 Packets: This field contains the number of lower bits of IP Source Address that are masked for matching in the IPv4 packets. The following list describes the values of this field: - 0: No bits are masked. - 1: LSb[0] is masked - 2: Two LSbs [1:0] are masked - .. - 31: All bits except MSb are masked. IPv6 Packets: This field contains Bits[4:0] of L3HSBM0. These bits indicate the number of higher bits of IP Source or Destination Address matched in the IPv6 packets. This field is valid and applicable only when the L3DAM0 or L3SAM0 bit is set high.
5 L3DAIM4	Layer 3 IP DA Inverse Match Enable When this bit is set, the Layer 3 IP Destination Address field is enabled for inverse matching. When this bit is reset, the Layer 3 IP Destination Address field is enabled for perfect matching. This bit is valid and applicable only when the L3DAM0 bit is set high. 0b - Layer 3 IP DA Inverse Match is disabled 1b - Layer 3 IP DA Inverse Match is enabled
4 L3DAM4	Layer 3 IP DA Match Enable When this bit is set, the Layer 3 IP Destination Address field is enabled for matching. When this bit is reset, the MAC ignores the Layer 3 IP Destination Address field for matching. Note: When the L3PEN0 bit is set, you should set either this bit or the L3SAM0 bit because either IPv6 DA or SA can be checked for filtering. 0b - Layer 3 IP DA Match is disabled 1b - Layer 3 IP DA Match is enabled

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Field	Function
3 L3SAIM4	Layer 3 IP SA Inverse Match Enable When this bit is set, the Layer 3 IP Source Address field is enabled for inverse matching. When this bit is reset, the Layer 3 IP Source Address field is enabled for perfect matching. This bit is valid and applicable only when the L3SAM0 bit is set. 0b - Layer 3 IP SA Inverse Match is disabled 1b - Layer 3 IP SA Inverse Match is enabled
2 L3SAM4	Layer 3 IP SA Match Enable When this bit is set, the Layer 3 IP Source Address field is enabled for matching. When this bit is reset, the MAC ignores the Layer 3 IP Source Address field for matching. Note: When the L3PEN0 bit is set, you should set either this bit or the L3DAM0 bit because either IPv6 SA or DA can be checked for filtering. 0b - Layer 3 IP SA Match is disabled 1b - Layer 3 IP SA Match is enabled
1 —	Reserved.
0 L3PEN4	Layer 3 Protocol Enable When this bit is set, the Layer 3 IP Source or Destination Address matching is enabled for IPv6 packets. When this bit is reset, the Layer 3 IP Source or Destination Address matching is enabled for IPv4 packets. The Layer 3 matching is done only when the L3SAM0 or L3DAM0 bit is set. 0b - Layer 3 Protocol is disabled 1b - Layer 3 Protocol is enabled

1.4.152 (MAC_Layer4_Address4)

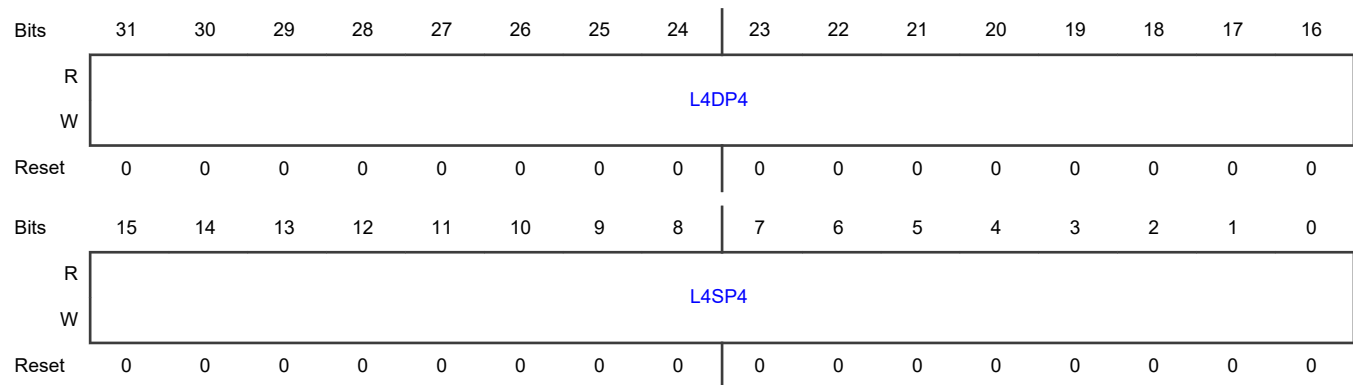
Offset

Register	Offset
MAC_Layer4_Address4	9C4h

Function

The MAC_Layer4_Address(#i), MAC_L3_L4_Control(#i), MAC_Layer3_Addr0_Reg(#i), MAC_Layer3_Addr1_Reg(#i), MAC_Layer3_Addr2_Reg(#i) and MAC_Layer3_Addr3_Reg(#i) registers are reserved (RO with default value) if Enable Layer 3 and Layer 4 Packet Filter option is not selected while configuring the core.

You can configure the Layer 3 and Layer 4 Address Registers to be double-synchronized by selecting the Synchronize Layer 3 and Layer 4 Address Registers to Rx Clock Domain option while configuring the core. When you select this option, the synchronization is triggered only when Bits[31:24] (in little-endian mode) or Bits[7:0] (in big-endian mode) of the Layer 3 and Layer 4 Address Registers are written. For proper synchronization updates, you should perform consecutive writes to same Layer 3 and Layer 4 Address Registers after at least four clock cycles delay of the destination clock.

Diagram**Fields**

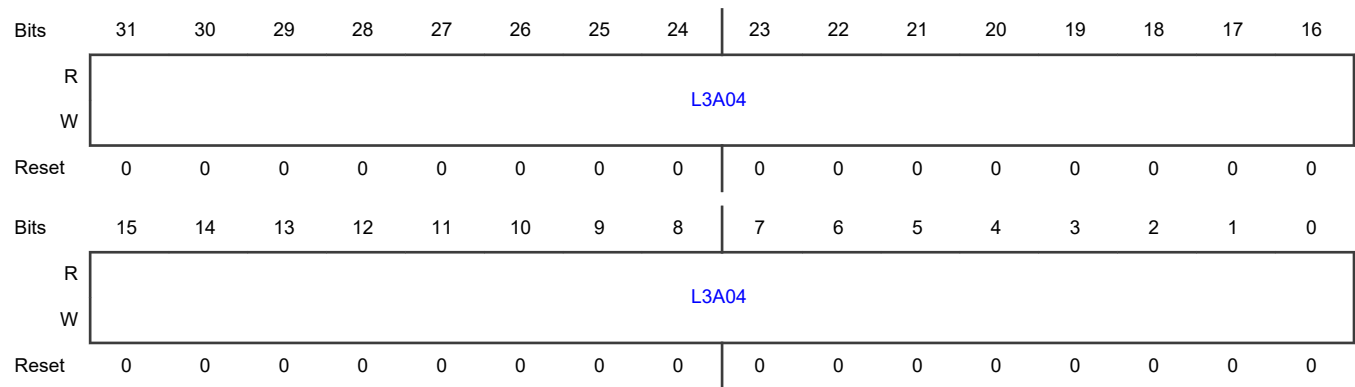
Field	Function
31-16 L4DP4	Layer 4 Destination Port Number Field When the L4PEN0 bit is reset and the L4DPM0 bit is set in the MAC_L3_L4_Control0 register, this field contains the value to be matched with the TCP Destination Port Number field in the IPv4 or IPv6 packets. When the L4PEN0 and L4DPM0 bits are set in MAC_L3_L4_Control0 register, this field contains the value to be matched with the UDP Destination Port Number field in the IPv4 or IPv6 packets.
15-0 L4SP4	Layer 4 Source Port Number Field When the L4PEN0 bit is reset and the L4SPM0 bit is set in the MAC_L3_L4_Control0 register, this field contains the value to be matched with the TCP Source Port Number field in the IPv4 or IPv6 packets. When the L4PEN0 and L4SPM0 bits are set in MAC_L3_L4_Control0 register, this field contains the value to be matched with the UDP Source Port Number field in the IPv4 or IPv6 packets.

1.4.153 (MAC_Layer3_Addr0_Reg4)**Offset**

Register	Offset
MAC_Layer3_Addr0_Reg4	9D0h

Function

For IPv4 packets, the Layer 3 Address 0 Register 0 register contains the 32-bit IP Source Address field. For IPv6 packets, it contains Bits[31:0] of the 128-bit IP Source Address or Destination Address field.

Diagram**Fields**

Field	Function
31-0 L3A04	Layer 3 Address 0 Field When the L3PEN0 and L3SAM0 bits are set in the MAC_L3_L4_Control0 register, this field contains the value to be matched with Bits[31:0] of the IP Source Address field in the IPv6 packets. When the L3PEN0 and L3DAM0 bits are set in the MAC_L3_L4_Control0 register, this field contains the value to be matched with Bits[31:0] of the IP Destination Address field in the IPv6 packets. When the L3PEN0 bit is reset and the L3SAM0 bit is set in the MAC_L3_L4_Control0 register, this field contains the value to be matched with the IP Source Address field in the IPv4 packets.

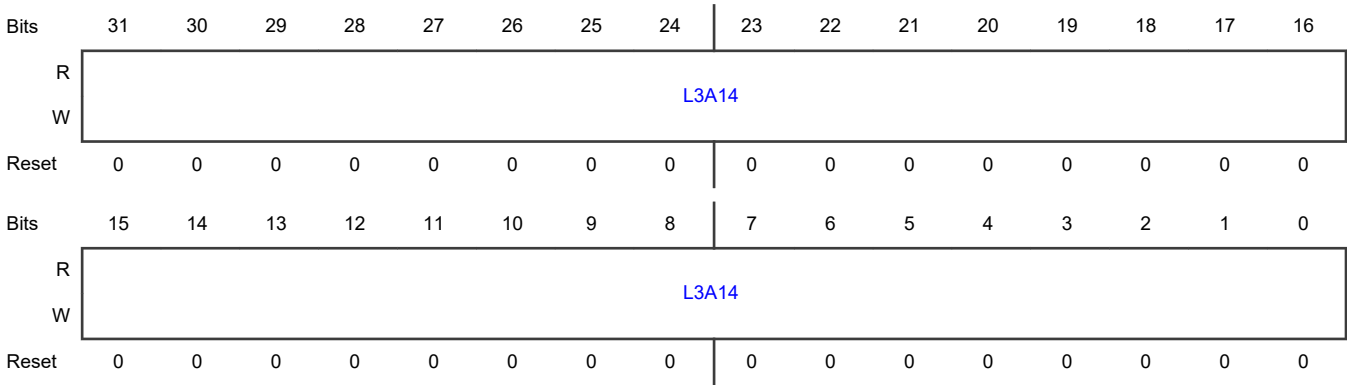
1.4.154 (MAC_Layer3_Addr1_Reg4)**Offset**

Register	Offset
MAC_Layer3_Addr1_Reg4	9D4h

Function

For IPv4 packets, the Layer 3 Address 1 Register 0 register contains the 32-bit IP Destination Address field. For IPv6 packets, it contains Bits[63:32] of the 128-bit IP Source Address or Destination Address field.

Diagram



Fields

Field	Function
31-0 L3A14	Layer 3 Address 1 Field When the L3PEN0 and L3SAM0 bits are set in the MAC_L3_L4_Control0 register, this field contains the value to be matched with Bits[63:32] of the IP Source Address field in the IPv6 packets. When the L3PEN0 and L3DAM0 bits are set in the MAC_L3_L4_Control0 register, this field contains the value to be matched with Bits[63:32] of the IP Destination Address field in the IPv6 packets. When the L3PEN0 bit is reset and the L3SAM0 bit is set in the MAC_L3_L4_Control0 register, this field contains the value to be matched with the IP Destination Address field in the IPv4 packets.

1.4.155 (MAC_Layer3_Addr2_Reg4)

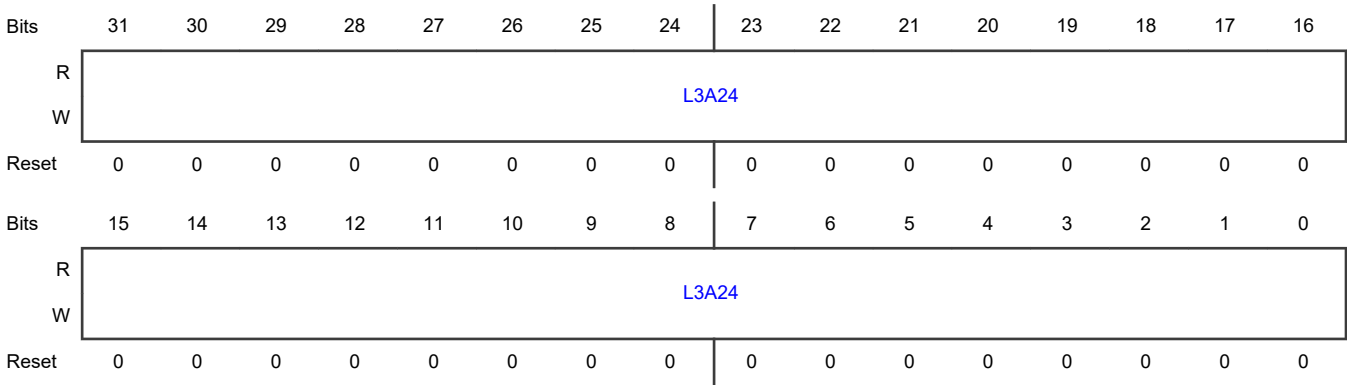
Offset

Register	Offset
MAC_Layer3_Addr2_Reg4	9D8h

Function

The Layer 3 Address 2 Register 0 register is reserved for IPv4 packets. For IPv6 packets, it contains Bits[95:64] of 128-bit IP Source Address or Destination Address field.

Diagram



Fields

Field	Function
31-0 L3A24	Layer 3 Address 2 Field When the L3PEN0 and L3SAM0 bits are set in the MAC_L3_L4_Control0 register, this field contains the value to be matched with Bits[95:64] of the IP Source Address field in the IPv6 packets. When the L3PEN0 and L3DAM0 bits are set in the MAC_L3_L4_Control0 register, this field contains the value to be matched with Bits[95:64] of the IP Destination Address field in the IPv6 packets. When the L3PEN0 bit is reset in the MAC_L3_L4_Control0 register, this field is not used.

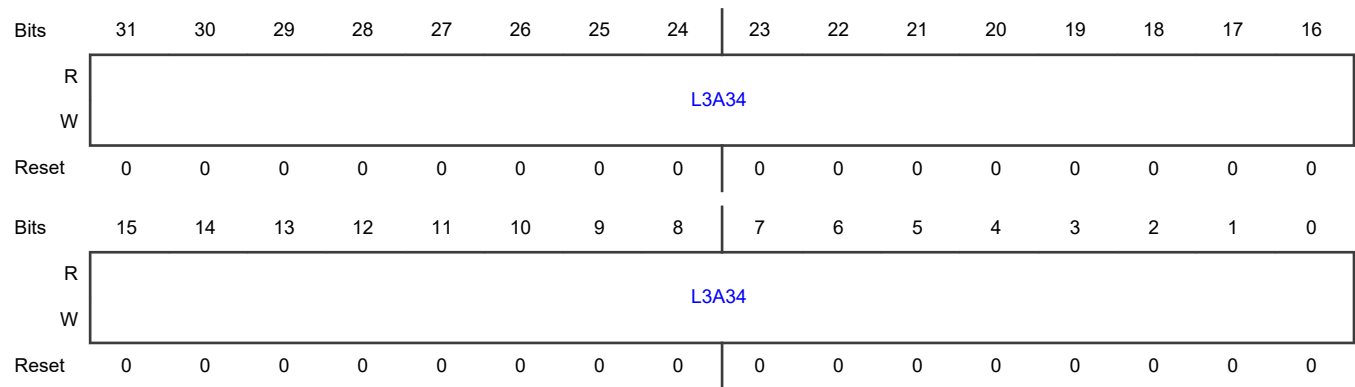
1.4.156 (MAC_Layer3_Addr3_Reg4)

Offset

Register	Offset
MAC_Layer3_Addr3_Reg4	9DCh

Function

The Layer 3 Address 3 Register 0 register is reserved for IPv4 packets. For IPv6 packets, it contains Bits[127:96] of 128-bit IP Source Address or Destination Address field.

Diagram**Fields**

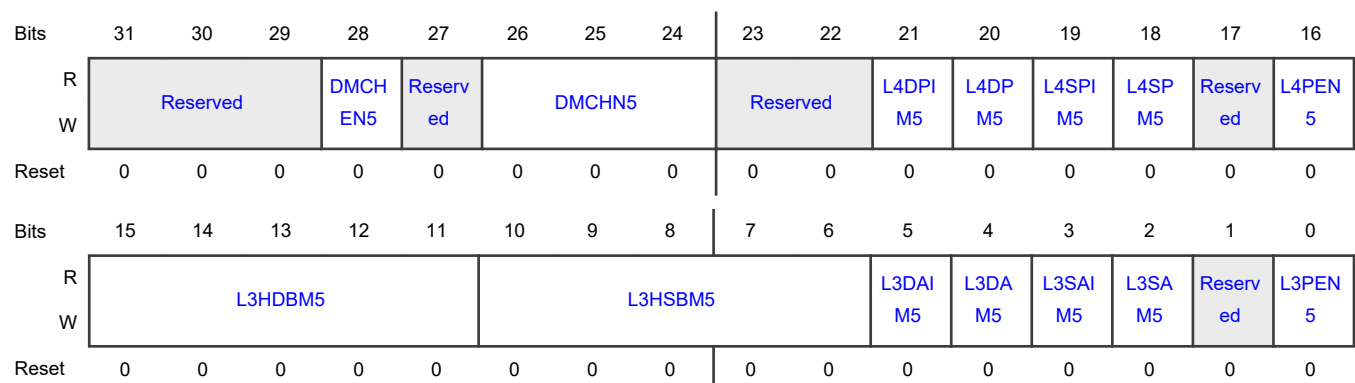
Field	Function
31-0 L3A34	Layer 3 Address 3 Field When the L3PEN0 and L3SAM0 bits are set in the MAC_L3_L4_Control0 register, this field contains the value to be matched with Bits[127:96] of the IP Source Address field in the IPv6 packets. When the L3PEN0 and L3DAM0 bits are set in the MAC_L3_L4_Control0 register, this field contains the value to be matched with Bits[127:96] of the IP Destination Address field in the IPv6 packets. When the L3PEN0 bit is reset in the MAC_L3_L4_Control0 register, this field is not used.

1.4.157 (MAC_L3_L4_Control5)**Offset**

Register	Offset
MAC_L3_L4_Control5	9F0h

Function

The Layer 3 and Layer 4 Control register controls the operations of filter 0 of Layer 3 and Layer 4.

Diagram

Fields

Field	Function
31-29 —	Reserved.
28 DMCHEN5	DMA Channel Select Enable When set, this bit enables the selection of the DMA channel number for the packet that is passed by this L3_L4 filter. The DMA channel is indicated by the DMCHN bits. When this bit is reset, the DMA channel is not decided by this filter. 0b - DMA Channel Select is disabled 1b - DMA Channel Select is enabled
27 —	Reserved.
26-24 DMCHN5	DMA Channel Number When DMCHEN is set high, this field selects the DMA Channel number to which the packet passed by this filter is routed. The width of this field depends on the number of the DMA channels present in your configuration.
23-22 —	Reserved.
21 L4DPIM5	Layer 4 Destination Port Inverse Match Enable When this bit is set, the Layer 4 Destination Port number field is enabled for inverse matching. When this bit is reset, the Layer 4 Destination Port number field is enabled for perfect matching. This bit is valid and applicable only when the L4DPM0 bit is set high. 0b - Layer 4 Destination Port Inverse Match is disabled 1b - Layer 4 Destination Port Inverse Match is enabled
20 L4DPM5	Layer 4 Destination Port Match Enable When this bit is set, the Layer 4 Destination Port number field is enabled for matching. When this bit is reset, the MAC ignores the Layer 4 Destination Port number field for matching. 0b - Layer 4 Destination Port Match is disabled 1b - Layer 4 Destination Port Match is enabled
19 L4SPIM5	Layer 4 Source Port Inverse Match Enable When this bit is set, the Layer 4 Source Port number field is enabled for inverse matching. When this bit is reset, the Layer 4 Source Port number field is enabled for perfect matching. This bit is valid and applicable only when the L4SPM0 bit is set high. 0b - Layer 4 Source Port Inverse Match is disabled 1b - Layer 4 Source Port Inverse Match is enabled

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Field	Function
18 L4SPM5	Layer 4 Source Port Match Enable When this bit is set, the Layer 4 Source Port number field is enabled for matching. When this bit is reset, the MAC ignores the Layer 4 Source Port number field for matching. 0b - Layer 4 Source Port Match is disabled 1b - Layer 4 Source Port Match is enabled
17 —	Reserved.
16 L4PEN5	Layer 4 Protocol Enable When this bit is set, the Source and Destination Port number fields of UDP packets are used for matching. When this bit is reset, the Source and Destination Port number fields of TCP packets are used for matching. The Layer 4 matching is done only when the L4SPM0 or L4DPM0 bit is set. 0b - Layer 4 Protocol is disabled 1b - Layer 4 Protocol is enabled
15-11 L3HDBM5	Layer 3 IP DA Higher Bits Match IPv4 Packets: This field contains the number of higher bits of IP Destination Address that are matched in the IPv4 packets. The following list describes the values of this field: - 0: No bits are masked. - 1: LSb[0] is masked - 2: Two LSbs [1:0] are masked - .. - 31: All bits except MSb are masked. IPv6 Packets: Bits[12:11] of this field correspond to Bits[6:5] of L3HSBM0 which indicate the number of lower bits of IP Source or Destination Address that are masked in the IPv6 packets. The following list describes the concatenated values of the L3HDBM0[1:0] and L3HSBM0 bits: - 0: No bits are masked. - 1: LSb[0] is masked. - 2: Two LSbs [1:0] are masked - .. - 127: All bits except MSb are masked. This field is valid and applicable only when the L3DAM0 or L3SAM0 bit is set.
10-6 L3HSBM5	Layer 3 IP SA Higher Bits Match IPv4 Packets:

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Field	Function
	This field contains the number of lower bits of IP Source Address that are masked for matching in the IPv4 packets. The following list describes the values of this field: - 0: No bits are masked. - 1: LSb[0] is masked - 2: Two LSbs [1:0] are masked - .. - 31: All bits except MSb are masked. IPv6 Packets: This field contains Bits[4:0] of L3HSBM0. These bits indicate the number of higher bits of IP Source or Destination Address matched in the IPv6 packets. This field is valid and applicable only when the L3DAM0 or L3SAM0 bit is set high.
5 L3DAIM5	Layer 3 IP DA Inverse Match Enable When this bit is set, the Layer 3 IP Destination Address field is enabled for inverse matching. When this bit is reset, the Layer 3 IP Destination Address field is enabled for perfect matching. This bit is valid and applicable only when the L3DAM0 bit is set high. 0b - Layer 3 IP DA Inverse Match is disabled 1b - Layer 3 IP DA Inverse Match is enabled
4 L3DAM5	Layer 3 IP DA Match Enable When this bit is set, the Layer 3 IP Destination Address field is enabled for matching. When this bit is reset, the MAC ignores the Layer 3 IP Destination Address field for matching. Note: When the L3PEN0 bit is set, you should set either this bit or the L3SAM0 bit because either IPv6 DA or SA can be checked for filtering. 0b - Layer 3 IP DA Match is disabled 1b - Layer 3 IP DA Match is enabled
3 L3SAIM5	Layer 3 IP SA Inverse Match Enable When this bit is set, the Layer 3 IP Source Address field is enabled for inverse matching. When this bit is reset, the Layer 3 IP Source Address field is enabled for perfect matching. This bit is valid and applicable only when the L3SAM0 bit is set. 0b - Layer 3 IP SA Inverse Match is disabled 1b - Layer 3 IP SA Inverse Match is enabled
2 L3SAM5	Layer 3 IP SA Match Enable When this bit is set, the Layer 3 IP Source Address field is enabled for matching. When this bit is reset, the MAC ignores the Layer 3 IP Source Address field for matching. Note: When the L3PEN0 bit is set, you should set either this bit or the L3DAM0 bit because either IPv6 SA or DA can be checked for filtering.

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Table continued from the previous page...

Field	Function
	0b - Layer 3 IP SA Match is disabled 1b - Layer 3 IP SA Match is enabled
1 —	Reserved.
0 L3PEN5	Layer 3 Protocol Enable When this bit is set, the Layer 3 IP Source or Destination Address matching is enabled for IPv6 packets. When this bit is reset, the Layer 3 IP Source or Destination Address matching is enabled for IPv4 packets. The Layer 3 matching is done only when the L3SAM0 or L3DAM0 bit is set. 0b - Layer 3 Protocol is disabled 1b - Layer 3 Protocol is enabled

1.4.158 (MAC_Layer4_Address5)

Offset

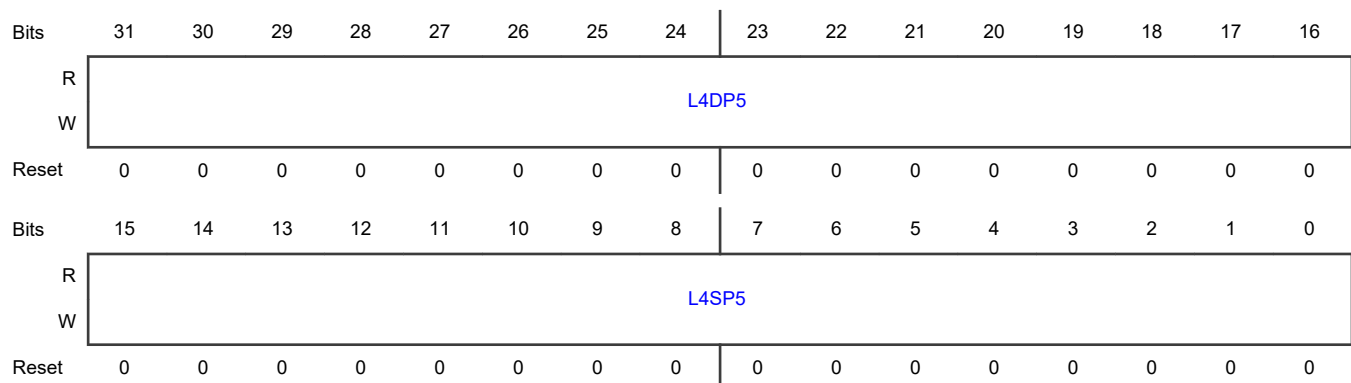
Register	Offset
MAC_Layer4_Address5	9F4h

Function

The MAC_Layer4_Address(#i), MAC_L3_L4_Control(#i), MAC_Layer3_Addr0_Reg(#i), MAC_Layer3_Addr1_Reg(#i), MAC_Layer3_Addr2_Reg(#i) and MAC_Layer3_Addr3_Reg(#i) registers are reserved (RO with default value) if Enable Layer 3 and Layer 4 Packet Filter option is not selected while configuring the core.

You can configure the Layer 3 and Layer 4 Address Registers to be double-synchronized by selecting the Synchronize Layer 3 and Layer 4 Address Registers to Rx Clock Domain option while configuring the core. When you select this option, the synchronization is triggered only when Bits[31:24] (in little-endian mode) or Bits[7:0] (in big-endian mode) of the Layer 3 and Layer 4 Address Registers are written. For proper synchronization updates, you should perform consecutive writes to same Layer 3 and Layer 4 Address Registers after at least four clock cycles delay of the destination clock.

Diagram



Fields

Field	Function
31-16 L4DP5	Layer 4 Destination Port Number Field When the L4PEN0 bit is reset and the L4DPM0 bit is set in the MAC_L3_L4_Control0 register, this field contains the value to be matched with the TCP Destination Port Number field in the IPv4 or IPv6 packets. When the L4PEN0 and L4DPM0 bits are set in MAC_L3_L4_Control0 register, this field contains the value to be matched with the UDP Destination Port Number field in the IPv4 or IPv6 packets.
15-0 L4SP5	Layer 4 Source Port Number Field When the L4PEN0 bit is reset and the L4SPM0 bit is set in the MAC_L3_L4_Control0 register, this field contains the value to be matched with the TCP Source Port Number field in the IPv4 or IPv6 packets. When the L4PEN0 and L4SPM0 bits are set in MAC_L3_L4_Control0 register, this field contains the value to be matched with the UDP Source Port Number field in the IPv4 or IPv6 packets.

1.4.159 (MAC_Layer3_Addr0_Reg5)

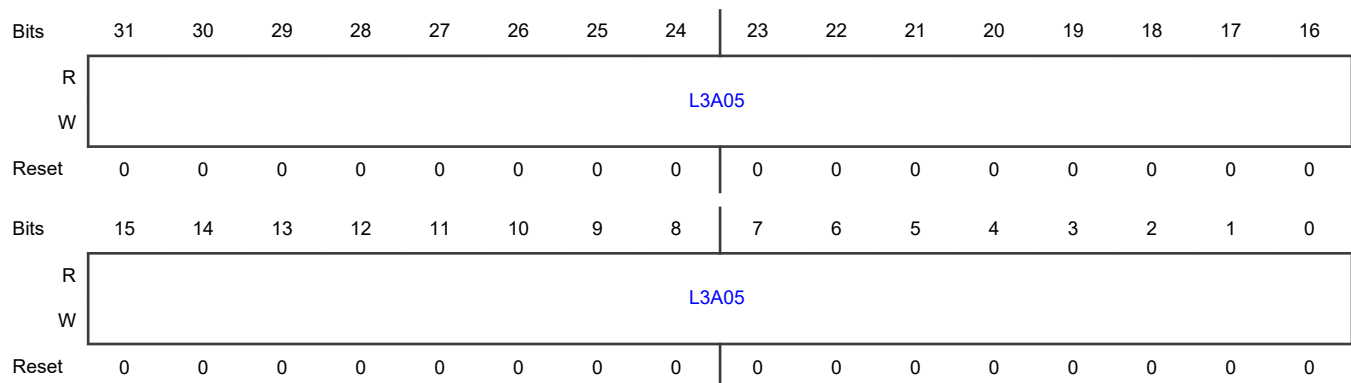
Offset

Register	Offset
MAC_Layer3_Addr0_Reg5	A00h

Function

For IPv4 packets, the Layer 3 Address 0 Register 0 register contains the 32-bit IP Source Address field. For IPv6 packets, it contains Bits[31:0] of the 128-bit IP Source Address or Destination Address field.

Diagram



Fields

Field	Function
31-0 L3A05	Layer 3 Address 0 Field When the L3PEN0 and L3SAM0 bits are set in the MAC_L3_L4_Control0 register, this field contains the value to be matched with Bits[31:0] of the IP Source Address field in the IPv6 packets. When the L3PEN0 and L3DAM0 bits are set in the MAC_L3_L4_Control0 register, this field contains the value to be matched with Bits[31:0] of the IP Destination Address field in the IPv6 packets. When the L3PEN0 bit is reset and the L3SAM0 bit is set in the MAC_L3_L4_Control0 register, this field contains the value to be matched with the IP Source Address field in the IPv4 packets.

1.4.160 (MAC_Layer3_Addr1_Reg5)

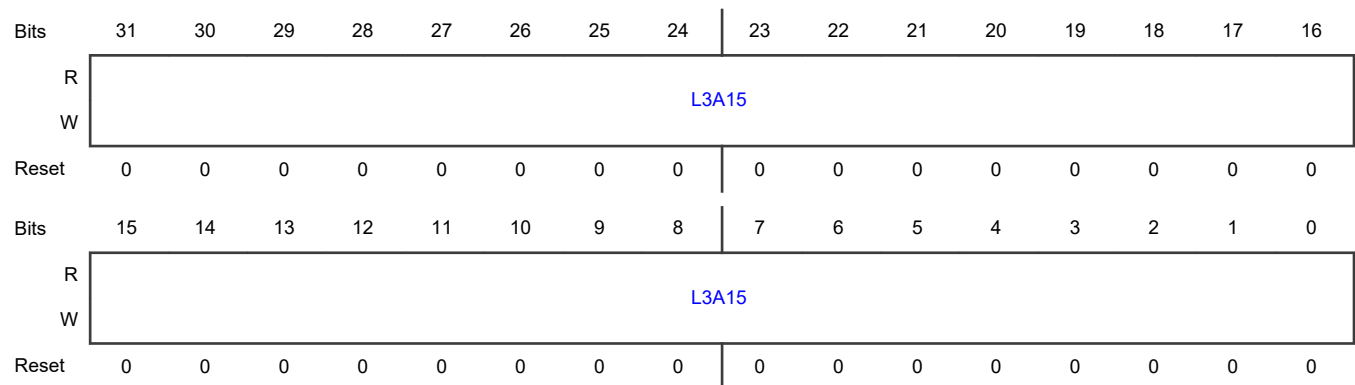
Offset

Register	Offset
MAC_Layer3_Addr1_Reg5	A04h

Function

For IPv4 packets, the Layer 3 Address 1 Register 0 register contains the 32-bit IP Destination Address field. For IPv6 packets, it contains Bits[63:32] of the 128-bit IP Source Address or Destination Address field.

Diagram



Fields

Field	Function
31-0 L3A15	Layer 3 Address 1 Field When the L3PEN0 and L3SAM0 bits are set in the MAC_L3_L4_Control0 register, this field contains the value to be matched with Bits[63:32] of the IP Source Address field in the IPv6 packets.

Table continues on the next page...

Field	Function
	When the L3PEN0 and L3DAM0 bits are set in the MAC_L3_L4_Control0 register, this field contains the value to be matched with Bits[63:32] of the IP Destination Address field in the IPv6 packets.
	When the L3PEN0 bit is reset and the L3SAM0 bit is set in the MAC_L3_L4_Control0 register, this field contains the value to be matched with the IP Destination Address field in the IPv4 packets.

1.4.161 (MAC_Layer3_Addr2_Reg5)

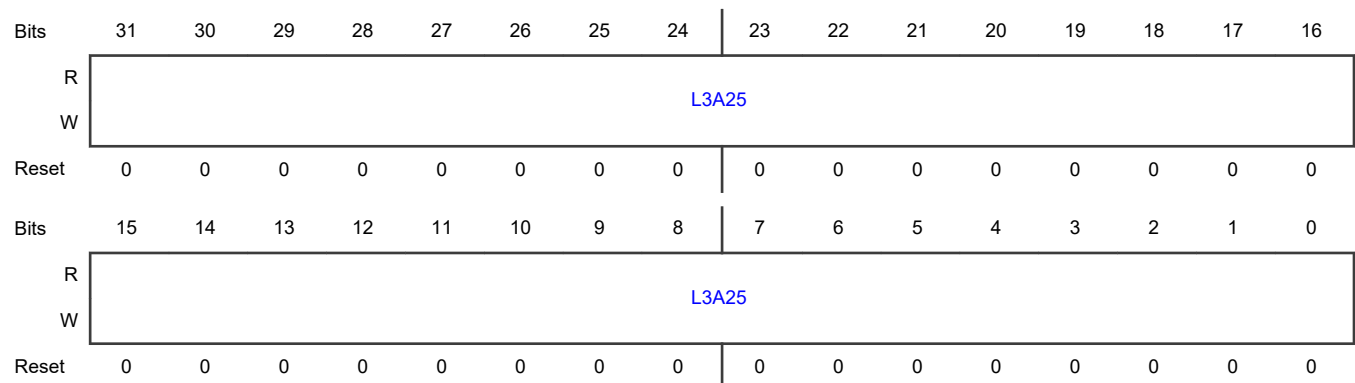
Offset

Register	Offset
MAC_Layer3_Addr2_Reg5	A08h

Function

The Layer 3 Address 2 Register 0 register is reserved for IPv4 packets. For IPv6 packets, it contains Bits[95:64] of 128-bit IP Source Address or Destination Address field.

Diagram



Fields

Field	Function
31-0 L3A25	Layer 3 Address 2 Field When the L3PEN0 and L3SAM0 bits are set in the MAC_L3_L4_Control0 register, this field contains the value to be matched with Bits[95:64] of the IP Source Address field in the IPv6 packets. When the L3PEN0 and L3DAM0 bits are set in the MAC_L3_L4_Control0 register, this field contains the value to be matched with Bits[95:64] of the IP Destination Address field in the IPv6 packets. When the L3PEN0 bit is reset in the MAC_L3_L4_Control0 register, this field is not used.

1.4.162 (MAC_Layer3_Addr3_Reg5)

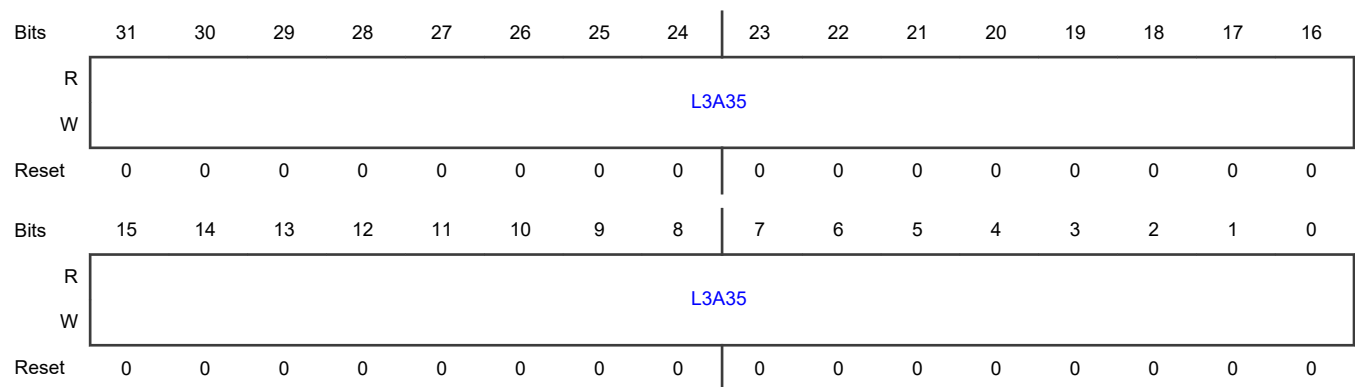
Offset

Register	Offset
MAC_Layer3_Addr3_Reg5	A0Ch

Function

The Layer 3 Address 3 Register 0 register is reserved for IPv4 packets. For IPv6 packets, it contains Bits[127:96] of 128-bit IP Source Address or Destination Address field.

Diagram



Fields

Field	Function
31-0	Layer 3 Address 3 Field
L3A35	When the L3PEN0 and L3SAM0 bits are set in the MAC_L3_L4_Control0 register, this field contains the value to be matched with Bits[127:96] of the IP Source Address field in the IPv6 packets. When the L3PEN0 and L3DAM0 bits are set in the MAC_L3_L4_Control0 register, this field contains the value to be matched with Bits[127:96] of the IP Destination Address field in the IPv6 packets. When the L3PEN0 bit is reset in the MAC_L3_L4_Control0 register, this field is not used.

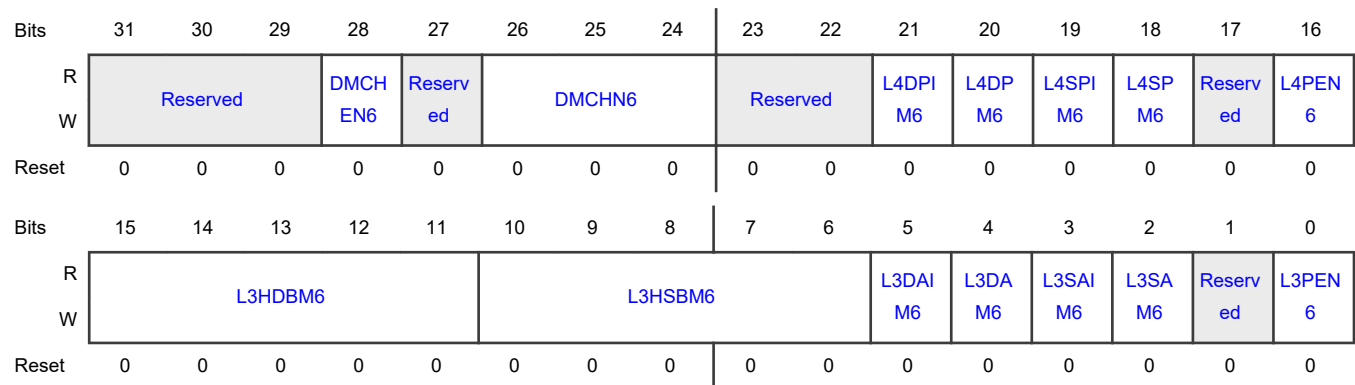
1.4.163 (MAC_L3_L4_Control6)

Offset

Register	Offset
MAC_L3_L4_Control6	A20h

Function

The Layer 3 and Layer 4 Control register controls the operations of filter 0 of Layer 3 and Layer 4.

Diagram**Fields**

Field	Function
31-29 —	Reserved.
28 DMCHEN6	DMA Channel Select Enable When set, this bit enables the selection of the DMA channel number for the packet that is passed by this L3_L4 filter. The DMA channel is indicated by the DMCHN bits. When this bit is reset, the DMA channel is not decided by this filter. 0b - DMA Channel Select is disabled 1b - DMA Channel Select is enabled
27 —	Reserved.
26-24 DMCHN6	DMA Channel Number When DMCHEN is set high, this field selects the DMA Channel number to which the packet passed by this filter is routed. The width of this field depends on the number of the DMA channels present in your configuration.
23-22 —	Reserved.
21 L4DPIM6	Layer 4 Destination Port Inverse Match Enable When this bit is set, the Layer 4 Destination Port number field is enabled for inverse matching. When this bit is reset, the Layer 4 Destination Port number field is enabled for perfect matching. This bit is valid and applicable only when the L4DPM0 bit is set high. 0b - Layer 4 Destination Port Inverse Match is disabled 1b - Layer 4 Destination Port Inverse Match is enabled
20	Layer 4 Destination Port Match Enable

Table continues on the next page...

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Field	Function
L4DPM6	When this bit is set, the Layer 4 Destination Port number field is enabled for matching. When this bit is reset, the MAC ignores the Layer 4 Destination Port number field for matching. 0b - Layer 4 Destination Port Match is disabled 1b - Layer 4 Destination Port Match is enabled
19 L4SPIM6	Layer 4 Source Port Inverse Match Enable When this bit is set, the Layer 4 Source Port number field is enabled for inverse matching. When this bit is reset, the Layer 4 Source Port number field is enabled for perfect matching. This bit is valid and applicable only when the L4SPM0 bit is set high. 0b - Layer 4 Source Port Inverse Match is disabled 1b - Layer 4 Source Port Inverse Match is enabled
18 L4SPM6	Layer 4 Source Port Match Enable When this bit is set, the Layer 4 Source Port number field is enabled for matching. When this bit is reset, the MAC ignores the Layer 4 Source Port number field for matching. 0b - Layer 4 Source Port Match is disabled 1b - Layer 4 Source Port Match is enabled
17 —	Reserved.
16 L4PEN6	Layer 4 Protocol Enable When this bit is set, the Source and Destination Port number fields of UDP packets are used for matching. When this bit is reset, the Source and Destination Port number fields of TCP packets are used for matching. The Layer 4 matching is done only when the L4SPM0 or L4DPM0 bit is set. 0b - Layer 4 Protocol is disabled 1b - Layer 4 Protocol is enabled
15-11 L3HDBM6	Layer 3 IP DA Higher Bits Match IPv4 Packets: This field contains the number of higher bits of IP Destination Address that are matched in the IPv4 packets. The following list describes the values of this field: - 0: No bits are masked. - 1: LSb[0] is masked - 2: Two LSbs [1:0] are masked - .. - 31: All bits except MSb are masked. IPv6 Packets:

Table continues on the next page...

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Field	Function
	Bits[12:11] of this field correspond to Bits[6:5] of L3HSBM0 which indicate the number of lower bits of IP Source or Destination Address that are masked in the IPv6 packets. The following list describes the concatenated values of the L3HDBM0[1:0] and L3HSBM0 bits: - 0: No bits are masked. - 1: LSb[0] is masked. - 2: Two LSbs [1:0] are masked - .. - 127: All bits except MSb are masked. This field is valid and applicable only when the L3DAM0 or L3SAM0 bit is set.
10-6 L3HSBM6	Layer 3 IP SA Higher Bits Match IPv4 Packets: This field contains the number of lower bits of IP Source Address that are masked for matching in the IPv4 packets. The following list describes the values of this field: - 0: No bits are masked. - 1: LSb[0] is masked - 2: Two LSbs [1:0] are masked - .. - 31: All bits except MSb are masked. IPv6 Packets: This field contains Bits[4:0] of L3HSBM0. These bits indicate the number of higher bits of IP Source or Destination Address matched in the IPv6 packets. This field is valid and applicable only when the L3DAM0 or L3SAM0 bit is set high.
5 L3DAIM6	Layer 3 IP DA Inverse Match Enable When this bit is set, the Layer 3 IP Destination Address field is enabled for inverse matching. When this bit is reset, the Layer 3 IP Destination Address field is enabled for perfect matching. This bit is valid and applicable only when the L3DAM0 bit is set high. 0b - Layer 3 IP DA Inverse Match is disabled 1b - Layer 3 IP DA Inverse Match is enabled
4 L3DAM6	Layer 3 IP DA Match Enable When this bit is set, the Layer 3 IP Destination Address field is enabled for matching. When this bit is reset, the MAC ignores the Layer 3 IP Destination Address field for matching. Note: When the L3PEN0 bit is set, you should set either this bit or the L3SAM0 bit because either IPv6 DA or SA can be checked for filtering. 0b - Layer 3 IP DA Match is disabled 1b - Layer 3 IP DA Match is enabled

Table continues on the next page...

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Field	Function
3 L3SAIM6	Layer 3 IP SA Inverse Match Enable When this bit is set, the Layer 3 IP Source Address field is enabled for inverse matching. When this bit is reset, the Layer 3 IP Source Address field is enabled for perfect matching. This bit is valid and applicable only when the L3SAM0 bit is set. 0b - Layer 3 IP SA Inverse Match is disabled 1b - Layer 3 IP SA Inverse Match is enabled
2 L3SAM6	Layer 3 IP SA Match Enable When this bit is set, the Layer 3 IP Source Address field is enabled for matching. When this bit is reset, the MAC ignores the Layer 3 IP Source Address field for matching. Note: When the L3PEN0 bit is set, you should set either this bit or the L3DAM0 bit because either IPv6 SA or DA can be checked for filtering. 0b - Layer 3 IP SA Match is disabled 1b - Layer 3 IP SA Match is enabled
1 —	Reserved.
0 L3PEN6	Layer 3 Protocol Enable When this bit is set, the Layer 3 IP Source or Destination Address matching is enabled for IPv6 packets. When this bit is reset, the Layer 3 IP Source or Destination Address matching is enabled for IPv4 packets. The Layer 3 matching is done only when the L3SAM0 or L3DAM0 bit is set. 0b - Layer 3 Protocol is disabled 1b - Layer 3 Protocol is enabled

1.4.164 (MAC_Layer4_Address6)

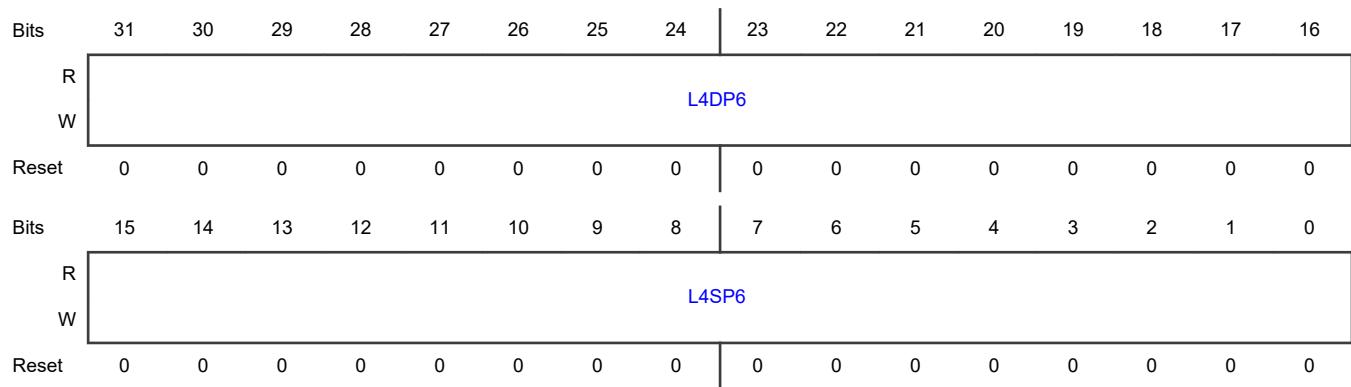
Offset

Register	Offset
MAC_Layer4_Address6	A24h

Function

The MAC_Layer4_Address(#i), MAC_L3_L4_Control(#i), MAC_Layer3_Addr0_Reg(#i), MAC_Layer3_Addr1_Reg(#i), MAC_Layer3_Addr2_Reg(#i) and MAC_Layer3_Addr3_Reg(#i) registers are reserved (RO with default value) if Enable Layer 3 and Layer 4 Packet Filter option is not selected while configuring the core.

You can configure the Layer 3 and Layer 4 Address Registers to be double-synchronized by selecting the Synchronize Layer 3 and Layer 4 Address Registers to Rx Clock Domain option while configuring the core. When you select this option, the synchronization is triggered only when Bits[31:24] (in little-endian mode) or Bits[7:0] (in big-endian mode) of the Layer 3 and Layer 4 Address Registers are written. For proper synchronization updates, you should perform consecutive writes to same Layer 3 and Layer 4 Address Registers after at least four clock cycles delay of the destination clock.

Diagram**Fields**

Field	Function
31-16 L4DP6	Layer 4 Destination Port Number Field When the L4PEN0 bit is reset and the L4DPM0 bit is set in the MAC_L3_L4_Control0 register, this field contains the value to be matched with the TCP Destination Port Number field in the IPv4 or IPv6 packets. When the L4PEN0 and L4DPM0 bits are set in MAC_L3_L4_Control0 register, this field contains the value to be matched with the UDP Destination Port Number field in the IPv4 or IPv6 packets.
15-0 L4SP6	Layer 4 Source Port Number Field When the L4PEN0 bit is reset and the L4SPM0 bit is set in the MAC_L3_L4_Control0 register, this field contains the value to be matched with the TCP Source Port Number field in the IPv4 or IPv6 packets. When the L4PEN0 and L4SPM0 bits are set in MAC_L3_L4_Control0 register, this field contains the value to be matched with the UDP Source Port Number field in the IPv4 or IPv6 packets.

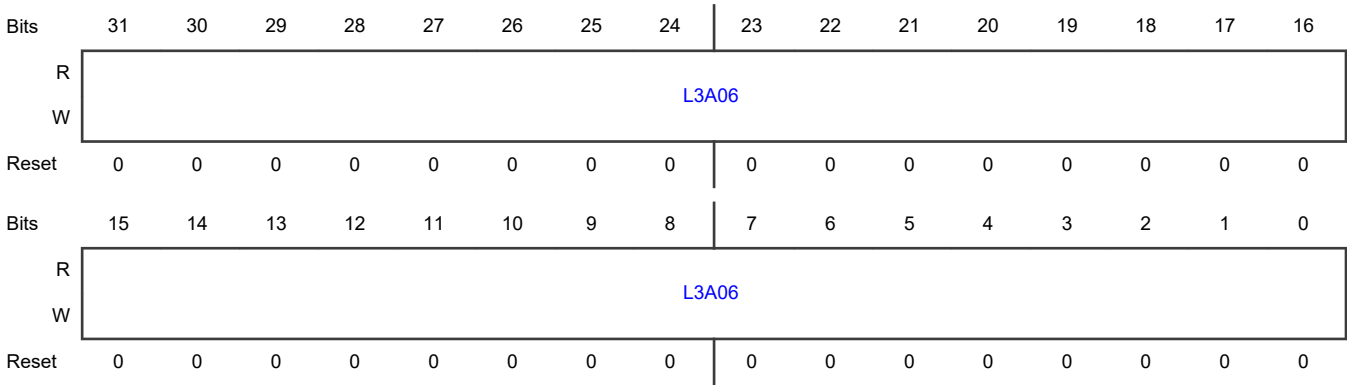
1.4.165 (MAC_Layer3_Addr0_Reg6)**Offset**

Register	Offset
MAC_Layer3_Addr0_Reg6	A30h

Function

For IPv4 packets, the Layer 3 Address 0 Register 0 register contains the 32-bit IP Source Address field. For IPv6 packets, it contains Bits[31:0] of the 128-bit IP Source Address or Destination Address field.

Diagram



Fields

Field	Function
31-0 L3A06	Layer 3 Address 0 Field When the L3PEN0 and L3SAM0 bits are set in the MAC_L3_L4_Control0 register, this field contains the value to be matched with Bits[31:0] of the IP Source Address field in the IPv6 packets. When the L3PEN0 and L3DAM0 bits are set in the MAC_L3_L4_Control0 register, this field contains the value to be matched with Bits[31:0] of the IP Destination Address field in the IPv6 packets. When the L3PEN0 bit is reset and the L3SAM0 bit is set in the MAC_L3_L4_Control0 register, this field contains the value to be matched with the IP Source Address field in the IPv4 packets.

1.4.166 (MAC_Layer3_Addr1_Reg6)

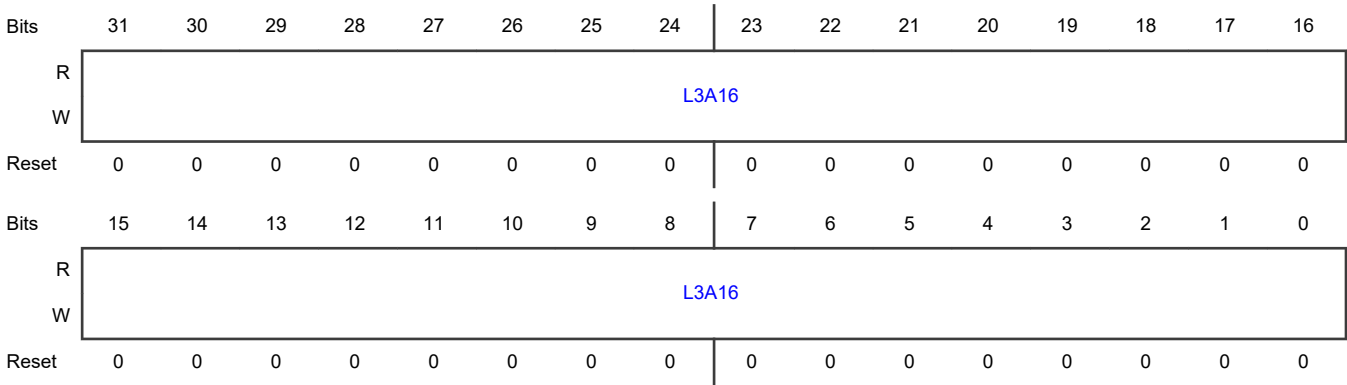
Offset

Register	Offset
MAC_Layer3_Addr1_Reg6	A34h

Function

For IPv4 packets, the Layer 3 Address 1 Register 0 register contains the 32-bit IP Destination Address field. For IPv6 packets, it contains Bits[63:32] of the 128-bit IP Source Address or Destination Address field.

Diagram



Fields

Field	Function
31-0 L3A16	Layer 3 Address 1 Field When the L3PEN0 and L3SAM0 bits are set in the MAC_L3_L4_Control0 register, this field contains the value to be matched with Bits[63:32] of the IP Source Address field in the IPv6 packets. When the L3PEN0 and L3DAM0 bits are set in the MAC_L3_L4_Control0 register, this field contains the value to be matched with Bits[63:32] of the IP Destination Address field in the IPv6 packets. When the L3PEN0 bit is reset and the L3SAM0 bit is set in the MAC_L3_L4_Control0 register, this field contains the value to be matched with the IP Destination Address field in the IPv4 packets.

1.4.167 (MAC_Layer3_Addr2_Reg6)

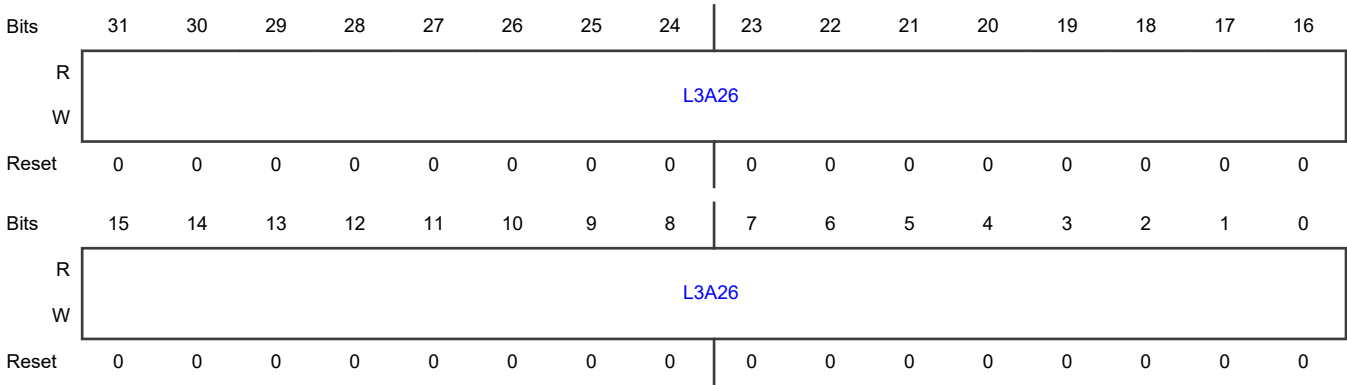
Offset

Register	Offset
MAC_Layer3_Addr2_Reg6	A38h

Function

The Layer 3 Address 2 Register 0 register is reserved for IPv4 packets. For IPv6 packets, it contains Bits[95:64] of 128-bit IP Source Address or Destination Address field.

Diagram



Fields

Field	Function
31-0 L3A26	Layer 3 Address 2 Field When the L3PEN0 and L3SAM0 bits are set in the MAC_L3_L4_Control0 register, this field contains the value to be matched with Bits[95:64] of the IP Source Address field in the IPv6 packets. When the L3PEN0 and L3DAM0 bits are set in the MAC_L3_L4_Control0 register, this field contains the value to be matched with Bits[95:64] of the IP Destination Address field in the IPv6 packets. When the L3PEN0 bit is reset in the MAC_L3_L4_Control0 register, this field is not used.

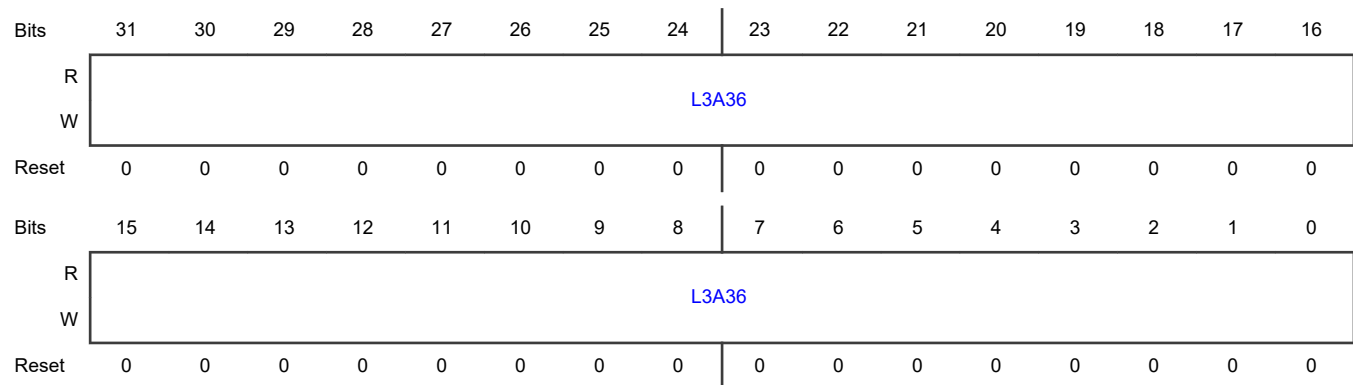
1.4.168 (MAC_Layer3_Addr3_Reg6)

Offset

Register	Offset
MAC_Layer3_Addr3_Reg6	A3Ch

Function

The Layer 3 Address 3 Register 0 register is reserved for IPv4 packets. For IPv6 packets, it contains Bits[127:96] of 128-bit IP Source Address or Destination Address field.

Diagram**Fields**

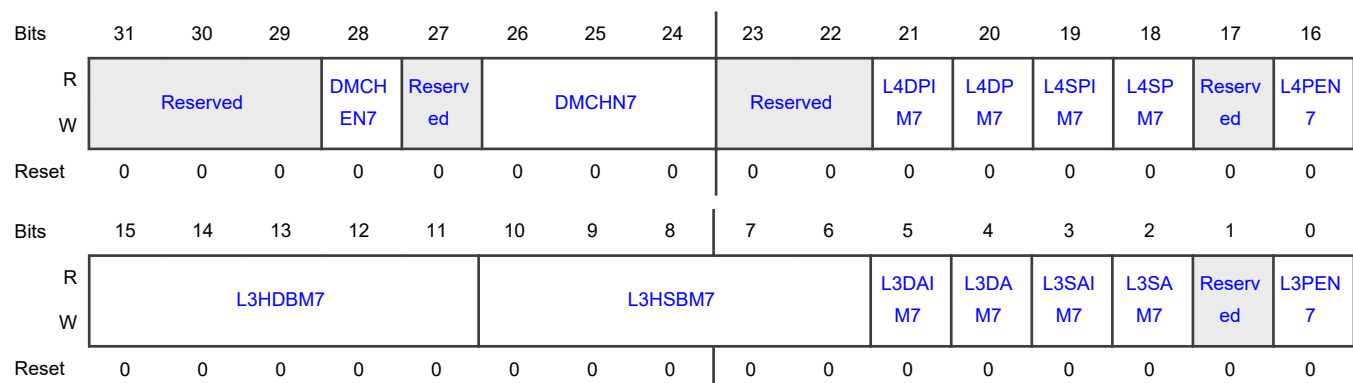
Field	Function
31-0 L3A36	Layer 3 Address 3 Field When the L3PEN0 and L3SAM0 bits are set in the MAC_L3_L4_Control0 register, this field contains the value to be matched with Bits[127:96] of the IP Source Address field in the IPv6 packets. When the L3PEN0 and L3DAM0 bits are set in the MAC_L3_L4_Control0 register, this field contains the value to be matched with Bits[127:96] of the IP Destination Address field in the IPv6 packets. When the L3PEN0 bit is reset in the MAC_L3_L4_Control0 register, this field is not used.

1.4.169 (MAC_L3_L4_Control7)**Offset**

Register	Offset
MAC_L3_L4_Control7	A50h

Function

The Layer 3 and Layer 4 Control register controls the operations of filter 0 of Layer 3 and Layer 4.

Diagram

Fields

Field	Function
31-29 —	Reserved.
28 DMCHEN7	DMA Channel Select Enable When set, this bit enables the selection of the DMA channel number for the packet that is passed by this L3_L4 filter. The DMA channel is indicated by the DMCHN bits. When this bit is reset, the DMA channel is not decided by this filter. 0b - DMA Channel Select is disabled 1b - DMA Channel Select is enabled
27 —	Reserved.
26-24 DMCHN7	DMA Channel Number When DMCHEN is set high, this field selects the DMA Channel number to which the packet passed by this filter is routed. The width of this field depends on the number of the DMA channels present in your configuration.
23-22 —	Reserved.
21 L4DPIM7	Layer 4 Destination Port Inverse Match Enable When this bit is set, the Layer 4 Destination Port number field is enabled for inverse matching. When this bit is reset, the Layer 4 Destination Port number field is enabled for perfect matching. This bit is valid and applicable only when the L4DPM0 bit is set high. 0b - Layer 4 Destination Port Inverse Match is disabled 1b - Layer 4 Destination Port Inverse Match is enabled
20 L4DPM7	Layer 4 Destination Port Match Enable When this bit is set, the Layer 4 Destination Port number field is enabled for matching. When this bit is reset, the MAC ignores the Layer 4 Destination Port number field for matching. 0b - Layer 4 Destination Port Match is disabled 1b - Layer 4 Destination Port Match is enabled
19 L4SPIM7	Layer 4 Source Port Inverse Match Enable When this bit is set, the Layer 4 Source Port number field is enabled for inverse matching. When this bit is reset, the Layer 4 Source Port number field is enabled for perfect matching. This bit is valid and applicable only when the L4SPM0 bit is set high. 0b - Layer 4 Source Port Inverse Match is disabled 1b - Layer 4 Source Port Inverse Match is enabled

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Field	Function
18 L4SPM7	Layer 4 Source Port Match Enable When this bit is set, the Layer 4 Source Port number field is enabled for matching. When this bit is reset, the MAC ignores the Layer 4 Source Port number field for matching. 0b - Layer 4 Source Port Match is disabled 1b - Layer 4 Source Port Match is enabled
17 —	Reserved.
16 L4PEN7	Layer 4 Protocol Enable When this bit is set, the Source and Destination Port number fields of UDP packets are used for matching. When this bit is reset, the Source and Destination Port number fields of TCP packets are used for matching. The Layer 4 matching is done only when the L4SPM0 or L4DPM0 bit is set. 0b - Layer 4 Protocol is disabled 1b - Layer 4 Protocol is enabled
15-11 L3HDBM7	Layer 3 IP DA Higher Bits Match IPv4 Packets: This field contains the number of higher bits of IP Destination Address that are matched in the IPv4 packets. The following list describes the values of this field: - 0: No bits are masked. - 1: LSb[0] is masked - 2: Two LSbs [1:0] are masked - .. - 31: All bits except MSb are masked. IPv6 Packets: Bits[12:11] of this field correspond to Bits[6:5] of L3HSBM0 which indicate the number of lower bits of IP Source or Destination Address that are masked in the IPv6 packets. The following list describes the concatenated values of the L3HDBM0[1:0] and L3HSBM0 bits: - 0: No bits are masked. - 1: LSb[0] is masked. - 2: Two LSbs [1:0] are masked - .. - 127: All bits except MSb are masked. This field is valid and applicable only when the L3DAM0 or L3SAM0 bit is set.
10-6 L3HSBM7	Layer 3 IP SA Higher Bits Match IPv4 Packets:

Table continues on the next page...

Table continued from the previous page...

Field	Function
	This field contains the number of lower bits of IP Source Address that are masked for matching in the IPv4 packets. The following list describes the values of this field: - 0: No bits are masked. - 1: LSb[0] is masked - 2: Two LSbs [1:0] are masked - .. - 31: All bits except MSb are masked. IPv6 Packets: This field contains Bits[4:0] of L3HSBM0. These bits indicate the number of higher bits of IP Source or Destination Address matched in the IPv6 packets. This field is valid and applicable only when the L3DAM0 or L3SAM0 bit is set high.
5 L3DAIM7	Layer 3 IP DA Inverse Match Enable When this bit is set, the Layer 3 IP Destination Address field is enabled for inverse matching. When this bit is reset, the Layer 3 IP Destination Address field is enabled for perfect matching. This bit is valid and applicable only when the L3DAM0 bit is set high. 0b - Layer 3 IP DA Inverse Match is disabled 1b - Layer 3 IP DA Inverse Match is enabled
4 L3DAM7	Layer 3 IP DA Match Enable When this bit is set, the Layer 3 IP Destination Address field is enabled for matching. When this bit is reset, the MAC ignores the Layer 3 IP Destination Address field for matching. Note: When the L3PEN0 bit is set, you should set either this bit or the L3SAM0 bit because either IPv6 DA or SA can be checked for filtering. 0b - Layer 3 IP DA Match is disabled 1b - Layer 3 IP DA Match is enabled
3 L3SAIM7	Layer 3 IP SA Inverse Match Enable When this bit is set, the Layer 3 IP Source Address field is enabled for inverse matching. When this bit is reset, the Layer 3 IP Source Address field is enabled for perfect matching. This bit is valid and applicable only when the L3SAM0 bit is set. 0b - Layer 3 IP SA Inverse Match is disabled 1b - Layer 3 IP SA Inverse Match is enabled
2 L3SAM7	Layer 3 IP SA Match Enable When this bit is set, the Layer 3 IP Source Address field is enabled for matching. When this bit is reset, the MAC ignores the Layer 3 IP Source Address field for matching. Note: When the L3PEN0 bit is set, you should set either this bit or the L3DAM0 bit because either IPv6 SA or DA can be checked for filtering.

Table continues on the next page...

Table continued from the previous page...

Field	Function
	0b - Layer 3 IP SA Match is disabled 1b - Layer 3 IP SA Match is enabled
1 —	Reserved.
0 L3PEN7	Layer 3 Protocol Enable When this bit is set, the Layer 3 IP Source or Destination Address matching is enabled for IPv6 packets. When this bit is reset, the Layer 3 IP Source or Destination Address matching is enabled for IPv4 packets. The Layer 3 matching is done only when the L3SAM0 or L3DAM0 bit is set. 0b - Layer 3 Protocol is disabled 1b - Layer 3 Protocol is enabled

1.4.170 (MAC_Layer4_Address7)

Offset

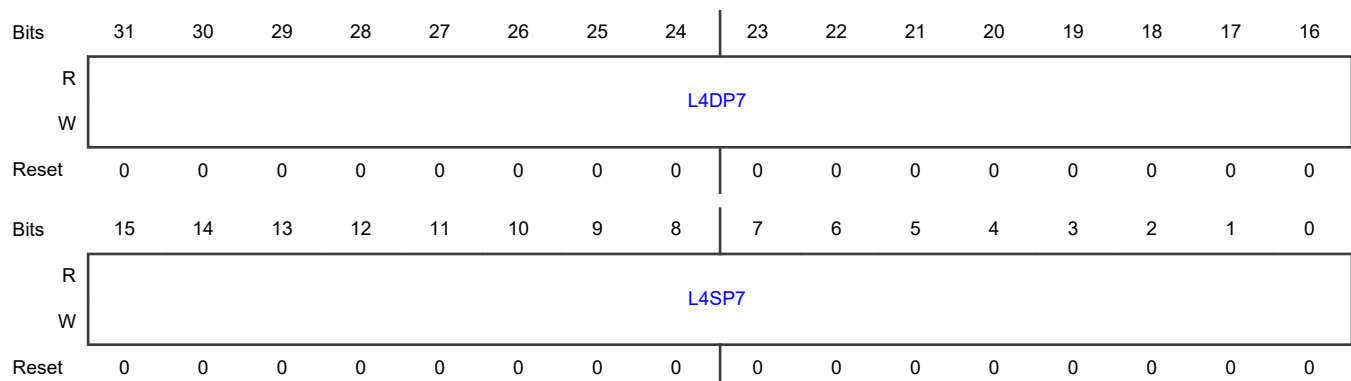
Register	Offset
MAC_Layer4_Address7	A54h

Function

The MAC_Layer4_Address(#i), MAC_L3_L4_Control(#i), MAC_Layer3_Addr0_Reg(#i), MAC_Layer3_Addr1_Reg(#i), MAC_Layer3_Addr2_Reg(#i) and MAC_Layer3_Addr3_Reg(#i) registers are reserved (RO with default value) if Enable Layer 3 and Layer 4 Packet Filter option is not selected while configuring the core.

You can configure the Layer 3 and Layer 4 Address Registers to be double-synchronized by selecting the Synchronize Layer 3 and Layer 4 Address Registers to Rx Clock Domain option while configuring the core. When you select this option, the synchronization is triggered only when Bits[31:24] (in little-endian mode) or Bits[7:0] (in big-endian mode) of the Layer 3 and Layer 4 Address Registers are written. For proper synchronization updates, you should perform consecutive writes to same Layer 3 and Layer 4 Address Registers after at least four clock cycles delay of the destination clock.

Diagram



Fields

Field	Function
31-16 L4DP7	Layer 4 Destination Port Number Field When the L4PEN0 bit is reset and the L4DPM0 bit is set in the MAC_L3_L4_Control0 register, this field contains the value to be matched with the TCP Destination Port Number field in the IPv4 or IPv6 packets. When the L4PEN0 and L4DPM0 bits are set in MAC_L3_L4_Control0 register, this field contains the value to be matched with the UDP Destination Port Number field in the IPv4 or IPv6 packets.
15-0 L4SP7	Layer 4 Source Port Number Field When the L4PEN0 bit is reset and the L4SPM0 bit is set in the MAC_L3_L4_Control0 register, this field contains the value to be matched with the TCP Source Port Number field in the IPv4 or IPv6 packets. When the L4PEN0 and L4SPM0 bits are set in MAC_L3_L4_Control0 register, this field contains the value to be matched with the UDP Source Port Number field in the IPv4 or IPv6 packets.

1.4.171 (MAC_Layer3_Addr0_Reg7)

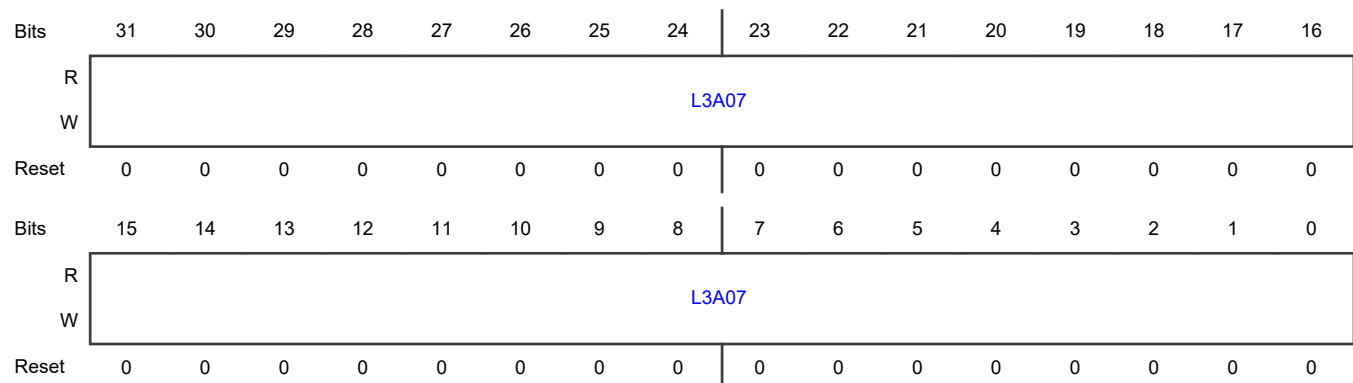
Offset

Register	Offset
MAC_Layer3_Addr0_Reg7	A60h

Function

For IPv4 packets, the Layer 3 Address 0 Register 0 register contains the 32-bit IP Source Address field. For IPv6 packets, it contains Bits[31:0] of the 128-bit IP Source Address or Destination Address field.

Diagram



Fields

Field	Function
31-0 L3A07	Layer 3 Address 0 Field When the L3PEN0 and L3SAM0 bits are set in the MAC_L3_L4_Control0 register, this field contains the value to be matched with Bits[31:0] of the IP Source Address field in the IPv6 packets. When the L3PEN0 and L3DAM0 bits are set in the MAC_L3_L4_Control0 register, this field contains the value to be matched with Bits[31:0] of the IP Destination Address field in the IPv6 packets. When the L3PEN0 bit is reset and the L3SAM0 bit is set in the MAC_L3_L4_Control0 register, this field contains the value to be matched with the IP Source Address field in the IPv4 packets.

1.4.172 (MAC_Layer3_Addr1_Reg7)

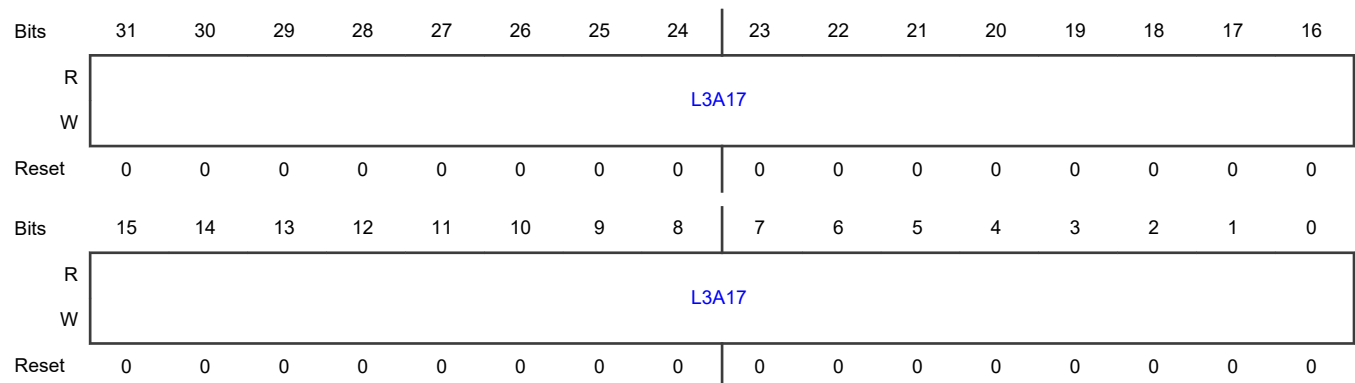
Offset

Register	Offset
MAC_Layer3_Addr1_Reg7	A64h

Function

For IPv4 packets, the Layer 3 Address 1 Register 0 register contains the 32-bit IP Destination Address field. For IPv6 packets, it contains Bits[63:32] of the 128-bit IP Source Address or Destination Address field.

Diagram



Fields

Field	Function
31-0 L3A17	Layer 3 Address 1 Field When the L3PEN0 and L3SAM0 bits are set in the MAC_L3_L4_Control0 register, this field contains the value to be matched with Bits[63:32] of the IP Source Address field in the IPv6 packets.

Table continues on the next page...

Field	Function
	When the L3PEN0 and L3DAM0 bits are set in the MAC_L3_L4_Control0 register, this field contains the value to be matched with Bits[63:32] of the IP Destination Address field in the IPv6 packets.
	When the L3PEN0 bit is reset and the L3SAM0 bit is set in the MAC_L3_L4_Control0 register, this field contains the value to be matched with the IP Destination Address field in the IPv4 packets.

1.4.173 (MAC_Layer3_Addr2_Reg7)

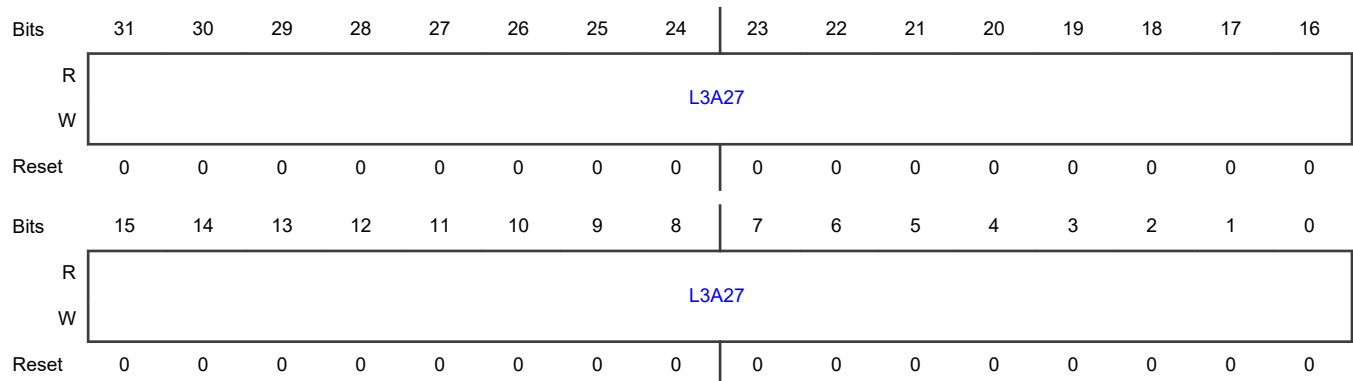
Offset

Register	Offset
MAC_Layer3_Addr2_Reg7	A68h

Function

The Layer 3 Address 2 Register 0 register is reserved for IPv4 packets. For IPv6 packets, it contains Bits[95:64] of 128-bit IP Source Address or Destination Address field.

Diagram



Fields

Field	Function
31-0 L3A27	Layer 3 Address 2 Field When the L3PEN0 and L3SAM0 bits are set in the MAC_L3_L4_Control0 register, this field contains the value to be matched with Bits[95:64] of the IP Source Address field in the IPv6 packets. When the L3PEN0 and L3DAM0 bits are set in the MAC_L3_L4_Control0 register, this field contains the value to be matched with Bits[95:64] of the IP Destination Address field in the IPv6 packets. When the L3PEN0 bit is reset in the MAC_L3_L4_Control0 register, this field is not used.

1.4.174 (MAC_Layer3_Addr3_Reg7)

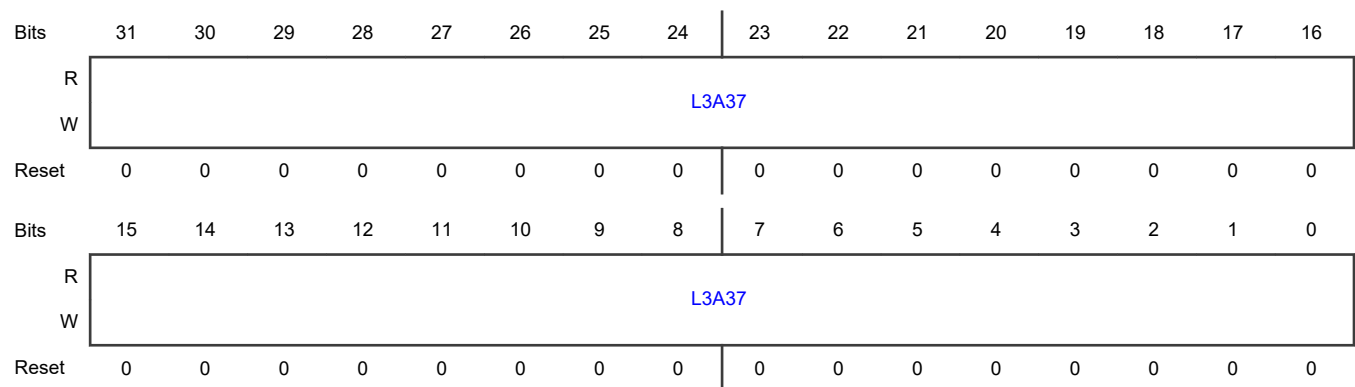
Offset

Register	Offset
MAC_Layer3_Addr3_Reg7	A6Ch

Function

The Layer 3 Address 3 Register 0 register is reserved for IPv4 packets. For IPv6 packets, it contains Bits[127:96] of 128-bit IP Source Address or Destination Address field.

Diagram



Fields

Field	Function
31-0	Layer 3 Address 3 Field
L3A37	When the L3PEN0 and L3SAM0 bits are set in the MAC_L3_L4_Control0 register, this field contains the value to be matched with Bits[127:96] of the IP Source Address field in the IPv6 packets. When the L3PEN0 and L3DAM0 bits are set in the MAC_L3_L4_Control0 register, this field contains the value to be matched with Bits[127:96] of the IP Destination Address field in the IPv6 packets. When the L3PEN0 bit is reset in the MAC_L3_L4_Control0 register, this field is not used.

1.4.175 (MAC_Timestamp_Control)

Offset

Register	Offset
MAC_Timestamp_Control	B00h

Function

This register controls the operation of the System Time generator and processing of PTP packets for timestamping in the Receiver.

Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	Reserved			AV802 1A...	Reserved			TXTSS TSM	Reserved			ESTI	CSC	TSEN MAC...	SNAPTYPSEL	
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	TSMS TRE...	TSEV NTE...	TSIPV 4E...	TSIPV 6E...	TSIPE NA	TSVE R2E...	TSCT RLS...	TSEN ALL	Reserv ed	PTGE	TSAD DREG	Reserv ed	TSUP DT	TSINIT	TSCF UPDT	TSEN A
W																
Reset	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0

Fields

Field	Function
31-29 —	Reserved.
28 AV8021ASMEN	AV 802.1AS Mode Enable When this bit is set, the MAC processes only untagged PTP over Ethernet packets for providing PTP status and capturing timestamp snapshots, that is, IEEE 802.1AS mode of operation. When PTP offload feature is enabled, for the purpose of PTP offload, the transport specific field in the PTP header is generated and checked based on the value of this bit. 0b - AV 802.1AS Mode is disabled 1b - AV 802.1AS Mode is enabled
27-25 —	Reserved.
24 TXTSSTSM	Transmit Timestamp Status Mode When this bit is set, the MAC overwrites the earlier transmit timestamp status even if it is not read by the software. The MAC indicates this by setting the TXTSSMIS bit of the MAC_Tx_Timestamp_Status_Nanoseconds register. When this bit is reset, the MAC ignores the timestamp status of current packet if the timestamp status of previous packet is not read by the software. The MAC indicates this by setting the TXTSSMIS bit of the MAC_Tx_Timestamp_Status_Nanoseconds register. 0b - Transmit Timestamp Status Mode is disabled 1b - Transmit Timestamp Status Mode is enabled
23-21	Reserved.

Table continues on the next page...

Table continued from the previous page...

Field	Function
—	
20 ESTI	External System Time Input When this bit is set, the MAC uses the external 64-bit reference System Time input for the following: - To take the timestamp provided as status - To insert the timestamp in transmit PTP packets when One-step Timestamp or Timestamp Offload feature is enabled. When this bit is reset, the MAC uses the internal reference System Time. 0b - External System Time Input is disabled 1b - External System Time Input is enabled
19 CSC	Enable checksum correction during OST for PTP over UDP/IPv4 packets When this bit is set, the last two bytes of PTP message sent over UDP/IPv4 is updated to keep the UDP checksum correct, for changes made to origin timestamp and/or correction field as part of one step timestamp operation. The application shall form the packet with these two dummy bytes. When reset, no updates are done to keep the UDP checksum correct. The application shall form the packet with UDP checksum set to 0. 0b - checksum correction during OST for PTP over UDP/IPv4 packets is disabled 1b - checksum correction during OST for PTP over UDP/IPv4 packets is enabled
18 TSENMACADDR	Enable MAC Address for PTP Packet Filtering When this bit is set, the DA MAC address (that matches any MAC Address register) is used to filter the PTP packets when PTP is directly sent over Ethernet. When this bit is set, received PTP packets with DA containing a special multicast or unicast address that matches the one programmed in MAC address registers are considered for processing as indicated below, when PTP is directly sent over Ethernet. For normal time stamping operation, MAC address registers 0 to 31 is considered for unicast destination address matching. For PTP offload, only MAC address register 0 is considered for unicast destination address matching. 0b - MAC Address for PTP Packet Filtering is disabled 1b - MAC Address for PTP Packet Filtering is enabled
17-16 SNAPTYPSEL	Select PTP packets for Taking Snapshots These bits, along with Bits 15 and 14, decide the set of PTP packet types for which snapshot needs to be taken. The encoding is given in Timestamp Snapshot Dependency on Register Bits Table.
15 TSMSTRENA	Enable Snapshot for Messages Relevant to Master When this bit is set, the snapshot is taken only for the messages that are relevant to the master node. Otherwise, the snapshot is taken for the messages relevant to the slave node. 0b - Snapshot for Messages Relevant to Master is disabled

Table continues on the next page...

Table continued from the previous page...

Field	Function
	1b - Snapshot for Messages Relevant to Master is enabled
14 TSEVNTENA	Enable Timestamp Snapshot for Event Messages When this bit is set, the timestamp snapshot is taken only for event messages (SYNC, Delay_Req, Pdelay_Req, or Pdelay_Resp). When this bit is reset, the snapshot is taken for all messages except Announce, Management, and Signaling. For more information about the timestamp snapshots, see Timestamp Snapshot Dependency on Register Bits Table. 0b - Timestamp Snapshot for Event Messages is disabled 1b - Timestamp Snapshot for Event Messages is enabled
13 TSIPV4ENA	Enable Processing of PTP Packets Sent over IPv4-UDP When this bit is set, the MAC receiver processes the PTP packets encapsulated in IPv4-UDP packets. When this bit is reset, the MAC ignores the PTP transported over IPv4-UDP packets. This bit is set by default. 0b - Processing of PTP Packets Sent over IPv4-UDP is disabled 1b - Processing of PTP Packets Sent over IPv4-UDP is enabled
12 TSIPV6ENA	Enable Processing of PTP Packets Sent over IPv6-UDP When this bit is set, the MAC receiver processes the PTP packets encapsulated in IPv6-UDP packets. When this bit is clear, the MAC ignores the PTP transported over IPv6-UDP packets. 0b - Processing of PTP Packets Sent over IPv6-UDP is disabled 1b - Processing of PTP Packets Sent over IPv6-UDP is enabled
11 TSIPENA	Enable Processing of PTP over Ethernet Packets When this bit is set, the MAC receiver processes the PTP packets encapsulated directly in the Ethernet packets. When this bit is reset, the MAC ignores the PTP over Ethernet packets. 0b - Processing of PTP over Ethernet Packets is disabled 1b - Processing of PTP over Ethernet Packets is enabled
10 TSVER2ENA	Enable PTP Packet Processing for Version 2 Format When this bit is set, the IEEE 1588 version 2 format is used to process the PTP packets. When this bit is reset, the IEEE 1588 version 1 format is used to process the PTP packets. The IEEE 1588 formats are described in 'PTP Processing and Control'. 0b - PTP Packet Processing for Version 2 Format is disabled 1b - PTP Packet Processing for Version 2 Format is enabled
9 TSCTRLSSR	Timestamp Digital or Binary Rollover Control When this bit is set, the Timestamp Low register rolls over after 0x3B9A_C9FF value (that is, 1 nanosecond accuracy) and increments the timestamp (High) seconds. When this bit is reset, the rollover value of sub-second register is 0x7FFF_FFFF. The sub-second increment must be programmed correctly depending on the PTP reference clock frequency and the value of this bit.

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Field	Function
	0b - Timestamp Digital or Binary Rollover Control is disabled 1b - Timestamp Digital or Binary Rollover Control is enabled
8 TSENALL	Enable Timestamp for All Packets When this bit is set, the timestamp snapshot is enabled for all packets received by the MAC. 0b - Timestamp for All Packets disabled 1b - Timestamp for All Packets enabled
7 —	Reserved.
6 PTGE	Presentation Time Generation Enable When this bit is set the Presentation Time generation is enabled. 0b - Presentation Time Generation is disabled 1b - Presentation Time Generation is enabled
5 TSADDREG	Update Addend Register When this bit is set, the content of the Timestamp Addend register is updated in the PTP block for fine correction. This bit is cleared when the update is complete. This bit should be zero before it is set. Access restriction applies. Setting 1 sets. Self-cleared. Setting 0 has no effect. 0b - Addend Register is not updated 1b - Addend Register is updated
4 —	Reserved.
3 TSUPDT	Update Timestamp When this bit is set, the system time is updated (added or subtracted) with the value specified in MAC_System_Time_Seconds_Update and MAC_System_Time_Nanoseconds_Update registers. This bit should be zero before updating it. This bit is reset when the update is complete in hardware. The Timestamp Higher Word register (if enabled during core configuration) is not updated. When Media Clock Generation and Recovery is configured (DWC_EQOS_FLEXI_PPS_OUT_EN) and enabled MAC_Presn_Time_Updt should also be updated before setting this field. Access restriction applies. Setting 1 sets. Self-cleared. Setting 0 has no effect. 0b - Timestamp is not updated 1b - Timestamp is updated
2 TSINIT	Initialize Timestamp When this bit is set, the system time is initialized (overwritten) with the value specified in the MAC_System_Time_Seconds_Update and MAC_System_Time_Nanoseconds_Update registers.

Table continues on the next page...

Table continued from the previous page...

Field	Function
	This bit should be zero before it is updated. This bit is reset when the initialization is complete. The Timestamp Higher Word register (if enabled during core configuration) can only be initialized. When Media Clock Generation and Recovery is configured (DWC_EQOS_FLEXI_PPS_OUT_EN) and enabled MAC_Presn_Time_Updt should also be updated before setting this field. Access restriction applies. Setting 1 sets. Self-cleared. Setting 0 has no effect. 0b - Timestamp is not initialized 1b - Timestamp is initialized
1 TSCFUPDT	Fine or Coarse Timestamp Update When this bit is set, the Fine method is used to update system timestamp. When this bit is reset, Coarse method is used to update the system timestamp. 0b - Coarse method is used to update system timestamp 1b - Fine method is used to update system timestamp
0 TSENA	Enable Timestamp When this bit is set, the timestamp is added for Transmit and Receive packets. When disabled, timestamp is not added for transmit and receive packets and the Timestamp Generator is also suspended. You need to initialize the Timestamp (system time) after enabling this mode. On the Receive side, the MAC processes the 1588 packets only if this bit is set. 0b - Timestamp is disabled 1b - Timestamp is enabled

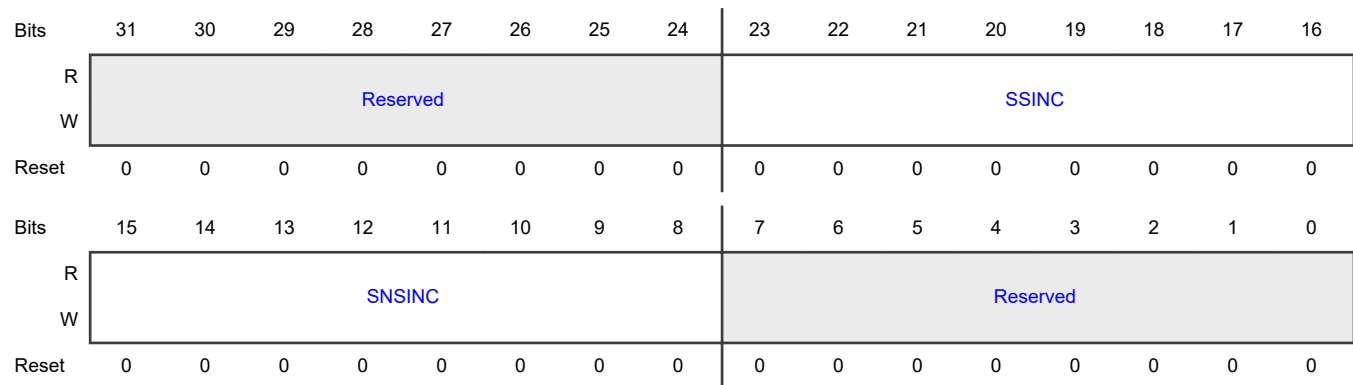
1.4.176 (MAC_Sub_Second_Increment)

Offset

Register	Offset
MAC_Sub_Second_Increment	B04h

Function

This register specifies the value to be added to the internal system time register every cycle of clk_ptp_ref_i clock.

Diagram**Fields**

Field	Function
31-24 —	Reserved.
23-16 SSINC	Sub-second Increment Value The value programmed in this field is accumulated every clock cycle (of clk_ptp_i) with the contents of the sub-second register. For example, when the PTP clock is 50 MHz (period is 20 ns), you should program 20 (0x14) when the System Time Nanoseconds register has an accuracy of 1 ns [Bit 9 (TSCTRLSSR) is set in MAC_Timestamp_Control]. When TSCTRLSSR is clear, the Nanoseconds register has a resolution of ~0.465 ns. In this case, you should program a value of 43 (0x2B) which is derived by 20 ns/0.465.
15-8 SNSINC	Sub-nanosecond Increment Value This field contains the sub-nanosecond increment value, represented in nanoseconds multiplied by 2⁸. This value is accumulated with the sub-nanoseconds field of the subsecond register. For example, when TSCTRLSSR field in the MAC_Timestamp_Control register is set. and if the required increment is 5.3ns, then SSINC should be 0x05 and SNSINC should be 0x4C.
7-0 —	Reserved.

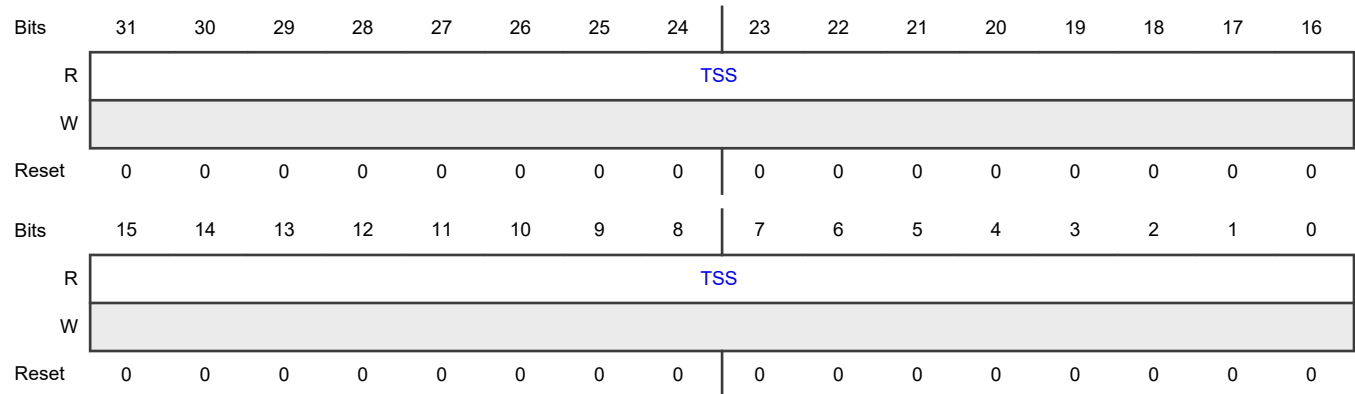
1.4.177 (MAC_System_Time_Seconds)**Offset**

Register	Offset
MAC_System_Time_Seconds	B08h

Function

The System Time Seconds register, along with System Time Nanoseconds register, indicates the current value of the system time maintained by the MAC. Though it is updated on a continuous basis, there is some delay from the actual time because of clock domain transfer latencies (from clk_ptp_ref_i to CSR clock).

Diagram



Fields

Field	Function
31-0	Timestamp Second
TSS	The value in this field indicates the current value in seconds of the System Time maintained by the MAC.

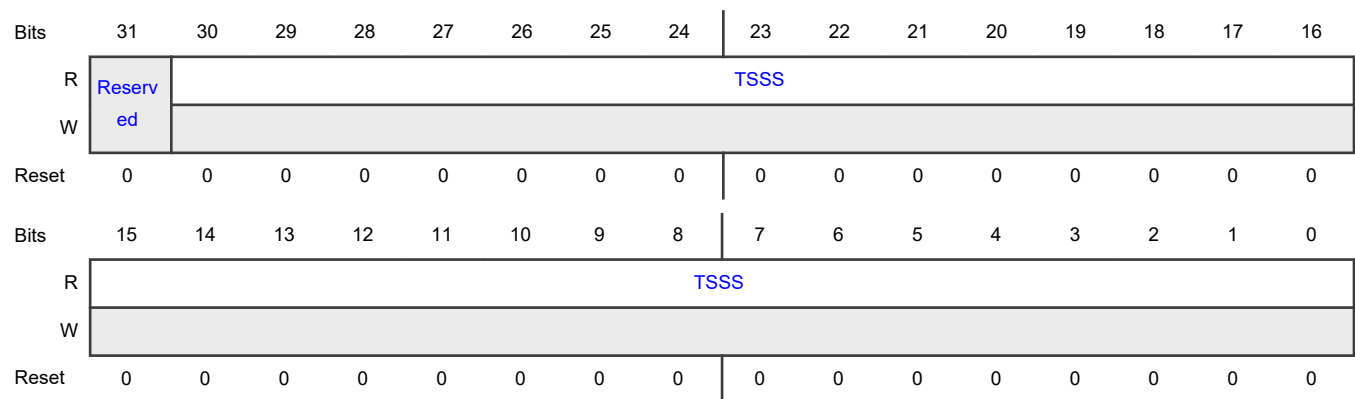
1.4.178 (MAC_System_Time_Nanoseconds)

Offset

Register	Offset
MAC_System_Time_Nanoseconds	B0Ch

Function

The System Time Nanoseconds register, along with System Time Seconds register, indicates the current value of the system time maintained by the MAC.

Diagram**Fields**

Field	Function
31 —	Reserved.
30-0 TSSS	Timestamp Sub Seconds The value in this field has the sub-second representation of time, with an accuracy of 0.46 ns. When Bit 9 is set in MAC_Timestamp_Control, each bit represents 1 ns. The maximum value is 0x3B9A_C9FF after which it rolls-over to zero.

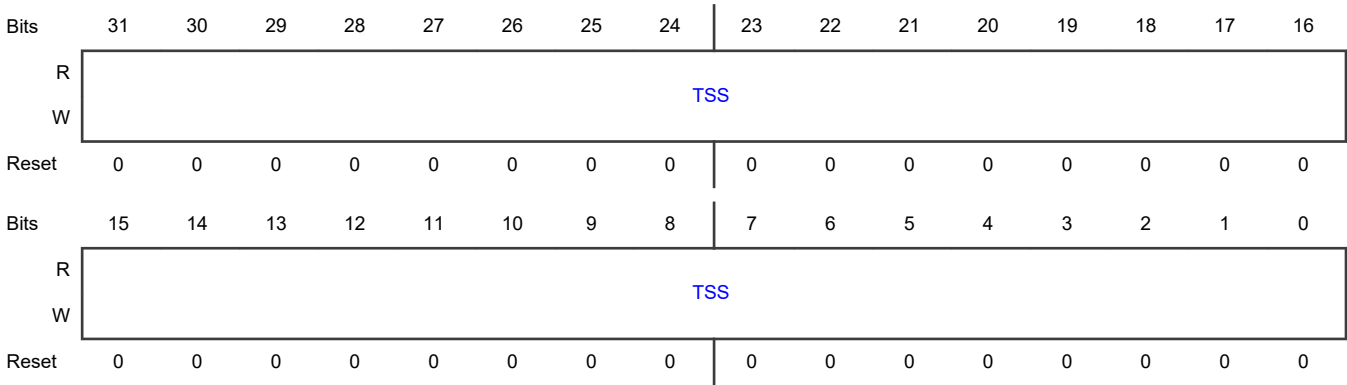
1.4.179 (MAC_System_Time_Seconds_Update)**Offset**

Register	Offset
MAC_System_Time_Seconds_Update	B10h

Function

The System Time Seconds Update register, along with the System Time Nanoseconds Update register, initializes or updates the system time maintained by the MAC. You must write both registers before setting the TSINIT or TSUPDT bits in MAC_Timestamp_Control.

Diagram



Fields

Field	Function
31-0	Timestamp Seconds
TSS	The value in this field is the seconds part of the update. When ADDSUB is reset, this field must be programmed with the seconds part of the update value. When ADDSUB is set, this field must be programmed with the complement of the seconds part of the update value. For example, if 2.000000001 seconds need to be subtracted from the system time, the TSS field in the MAC_Timestamp_Seconds_Update register must be 0xFFFF_FFFE (that is, 2^32 - 2).

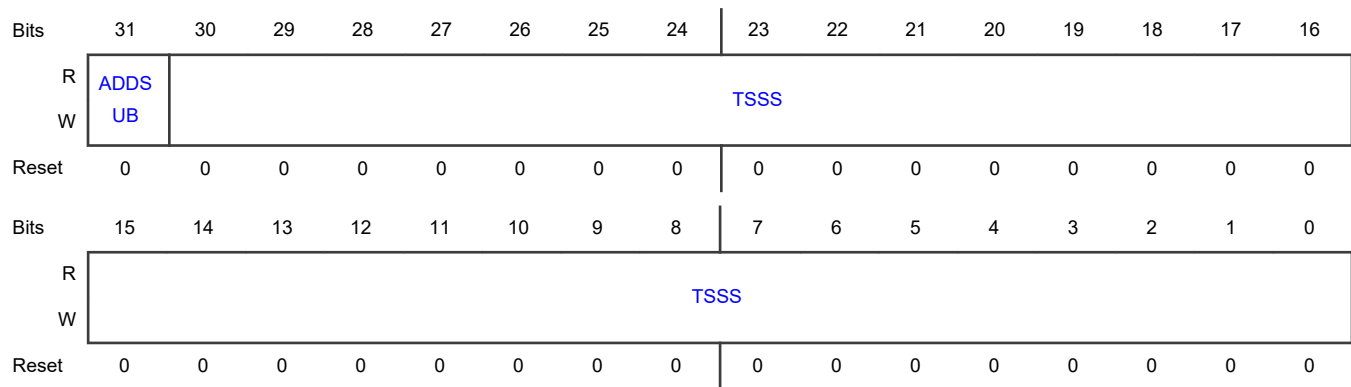
1.4.180 (MAC_System_Time_Nanoseconds_Update)

Offset

Register	Offset
MAC_System_Time_Nanoseconds_Update	B14h

Function

MAC System Time Nanoseconds Update register.

Diagram**Fields**

Field	Function
31 ADDSUB	Add or Subtract Time When this bit is set, the time value is subtracted with the contents of the update register. When this bit is reset, the time value is added with the contents of the update register. 0b - Add time 1b - Subtract time
30-0 TSSS	Timestamp Sub Seconds The value in this field is the sub-seconds part of the update. When ADDSUB is reset, this field must be programmed with the sub-seconds part of the update value, with an accuracy based on the TSCTRLSSR bit of the MAC_Timestamp_Control register. When ADDSUB is set, this field must be programmed with the complement of the sub-seconds part of the update value as described below. When TSCTRLSSR bit in MAC_Timestamp_Control is set, the programmed value must be $10^9 - \text{<sub-second value>}$. When TSCTRLSSR bit in MAC_Timestamp_Control is reset, the programmed value must be $2^{31} - \text{<sub-second value>}$. When the TSCTRLSSR bit is reset in the MAC_Timestamp_Control register, each bit represents an accuracy of 0.46 ns. When the TSCTRLSSR bit is set in the MAC_Timestamp_Control register, each bit represents 1 ns and the programmed value should not exceed 0x3B9A_C9FF. For example, if 2.000000001 seconds need to be subtracted from the system time, then the TSSS field in the MAC_Timestamp_Nanoseconds_Update register must be 0x7FFF_FFFF (that is, $2^{31} - 1$), when TSCTRLSSR bit in MAC_Timestamp_Control is reset and 0x3B9A_C9FF (that is, $10^9 - 1$), when TSCTRLSSR bit in MAC_Timestamp_Control is set.

1.4.181 (MAC_Timestamp_Addend)

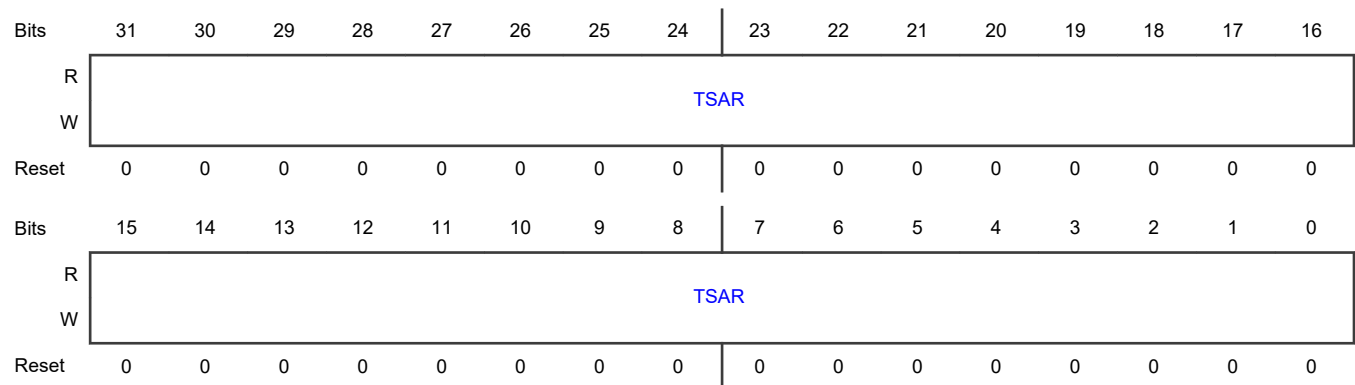
Offset

Register	Offset
MAC_Timestamp_Addend	B18h

Function

Timestamp Addend register. This register value is used only when the system time is configured for Fine Update mode (TSCFUPDT bit in the MAC_Timestamp_Control register). The content of this register is added to a 32-bit accumulator in every clock cycle (of clk_ptp_ref_i) and the system time is updated whenever the accumulator overflows.

Diagram



Fields

Field	Function
31-0	Timestamp Addend Register
TSAR	This field indicates the 32-bit time value to be added to the Accumulator register to achieve time synchronization.

1.4.182 (MAC_System_Time_Higher_Word_Seconds)

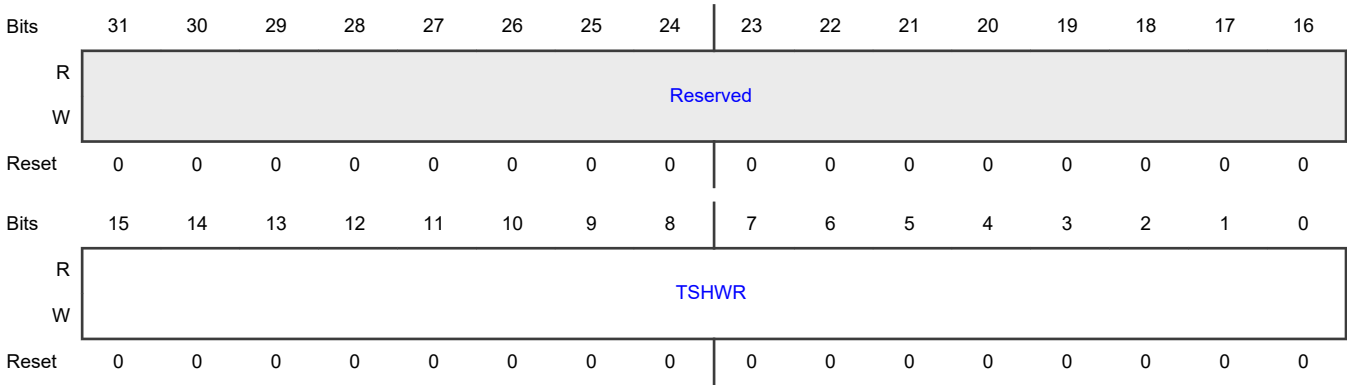
Offset

Register	Offset
MAC_System_Time_Higher_Word_Seconds	B1Ch

Function

System Time - Higher Word Seconds register.

Diagram



Fields

Field	Function
31-16 —	Reserved.
15-0 TSHWR	Timestamp Higher Word Register This field contains the most-significant 16-bits of timestamp seconds value. This register is optional. You can add this register by selecting the Add IEEE 1588 Higher Word Register option. This register is directly written to initialize the value and it is incremented when there is an overflow from 32-bits of the System Time Seconds register. Access restriction applies. Updated based on the event. Setting 1 sets. Setting 0 clears.

1.4.183 (MAC_Timestamp_Status)

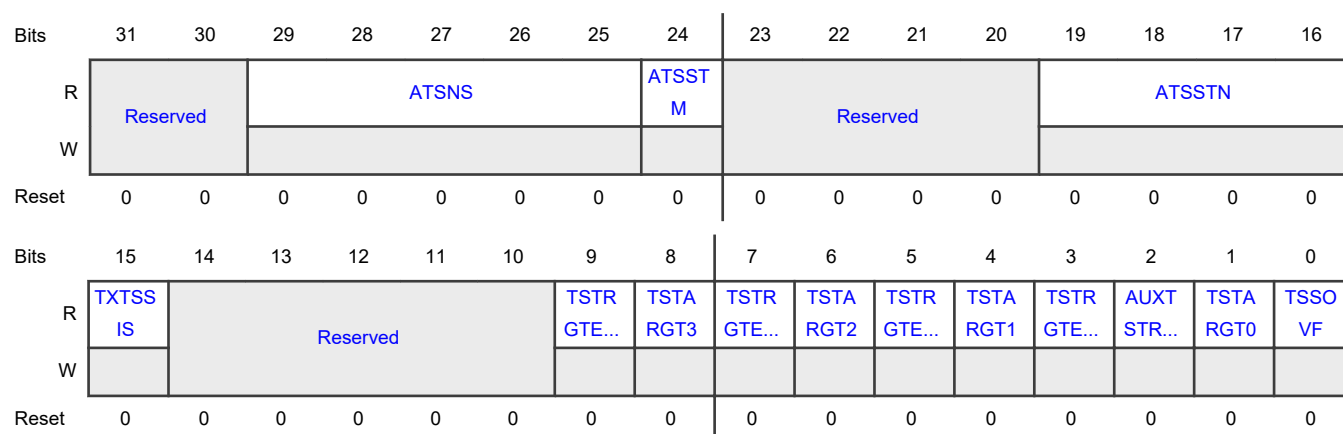
Offset

Register	Offset
MAC_Timestamp_Status	B20h

Function

Timestamp Status register. All bits except Bits[27:25] gets cleared when the application reads this register.

Diagram



Fields

Field	Function
31-30 —	Reserved.
29-25 ATSNS	Number of Auxiliary Timestamp Snapshots This field indicates the number of Snapshots available in the FIFO. A value equal to the selected depth of FIFO (4, 8, or 16) indicates that the Auxiliary Snapshot FIFO is full. These bits are cleared (to 00000) when the Auxiliary snapshot FIFO clear bit is set. This bit is valid only if the Add IEEE 1588 Auxiliary Snapshot option is selected.
24 ATSSTM	Auxiliary Timestamp Snapshot Trigger Missed This bit is set when the Auxiliary timestamp snapshot FIFO is full and external trigger was set. This indicates that the latest snapshot is not stored in the FIFO. This bit is valid only if the Add IEEE 1588 Auxiliary Snapshot option is selected. 0b - Auxiliary Timestamp Snapshot Trigger Missed status not detected 1b - Auxiliary Timestamp Snapshot Trigger Missed status detected
23-20 —	Reserved.
19-16 ATSSTN	Auxiliary Timestamp Snapshot Trigger Identifier These bits identify the Auxiliary trigger inputs for which the timestamp available in the Auxiliary Snapshot Register is applicable. When more than one bit is set at the same time, it means that corresponding auxiliary triggers were sampled at the same clock. These bits are applicable only if the number of Auxiliary snapshots is more than one. One bit is assigned for each trigger as shown in the following list: - Bit 16: Auxiliary trigger 0 - Bit 17: Auxiliary trigger 1 - Bit 18: Auxiliary trigger 2 - Bit 19: Auxiliary trigger 3

Table continues on the next page...

Table continued from the previous page...

Field	Function
	The software can read this register to find the triggers that are set when the timestamp is taken.
15 TXTSSIS	Tx Timestamp Status Interrupt Status In non-EQOS_CORE configurations when drop transmit status is enabled in MTL, this bit is set when the captured transmit timestamp is updated in the MAC_Tx_Timestamp_Status_Nanoseconds and MAC_Tx_Timestamp_Status_Seconds registers. When PTP offload feature is enabled, this bit is set when the captured transmit timestamp is updated in the MAC_Tx_Timestamp_Status_Nanoseconds and MAC_Tx_Timestamp_Status_Seconds registers, for PTO generated Delay Request and Pdelay request packets. This bit is cleared when the MAC_Tx_Timestamp_Status_Seconds register is read (or write to MAC_Tx_Timestamp_Status_Seconds register when RCWE bit of MAC_CSR_SW_Ctrl register is set). 0b - Tx Timestamp Status Interrupt status not detected 1b - Tx Timestamp Status Interrupt status detected
14-10 —	Reserved.
9 TSTRGTERR3	Timestamp Target Time Error This bit is set when the latest target time programmed in the MAC_PPS3_Target_Time_Seconds and MAC_PPS3_Target_Time_Nanoseconds registers elapses. This bit is cleared when the application reads this bit. Access restriction applies. Clears on read (or this bit is written to 1 when RCWE bit in MAC_CSR_SW_Ctrl register is set). Self-set to 1 on internal event. 0b - Timestamp Target Time Error status not detected 1b - Timestamp Target Time Error status detected
8 TSTARGET3	Timestamp Target Time Reached for Target Time PPS3 When this bit is set, it indicates that the value of system time is greater than or equal to the value specified in the MAC_PPS3_Target_Time_Seconds and MAC_PPS3_Target_Time_Nanoseconds registers. Access restriction applies. Clears on read (or this bit is written to 1 when RCWE bit in MAC_CSR_SW_Ctrl register is set). Self-set to 1 on internal event. 0b - Timestamp Target Time Reached for Target Time PPS3 status not detected 1b - Timestamp Target Time Reached for Target Time PPS3 status detected
7 TSTRGTERR2	Timestamp Target Time Error This bit is set when the latest target time programmed in the MAC_PPS2_Target_Time_Seconds and MAC_PPS2_Target_Time_Nanoseconds registers elapses. This bit is cleared when the application reads this bit. Access restriction applies. Clears on read (or this bit is written to 1 when RCWE bit in MAC_CSR_SW_Ctrl register is set). Self-set to 1 on internal event. 0b - Timestamp Target Time Error status not detected

Table continues on the next page...

Table continued from the previous page...

Field	Function
	1b - Timestamp Target Time Error status detected
6 TSTARGET2	Timestamp Target Time Reached for Target Time PPS2 When set, this bit indicates that the value of system time is greater than or equal to the value specified in the MAC_PPS2_Target_Time_Seconds and MAC_PPS2_Target_Time_Nanoseconds registers. Access restriction applies. Clears on read (or this bit is written to 1 when RCWE bit in MAC_CSR_SW_Ctrl register is set). Self-set to 1 on internal event. 0b - Timestamp Target Time Reached for Target Time PPS2 status not detected 1b - Timestamp Target Time Reached for Target Time PPS2 status detected
5 TSTRGTERR1	Timestamp Target Time Error This bit is set when the latest target time programmed in the MAC_PPS1_Target_Time_Seconds and MAC_PPS1_Target_Time_Nanoseconds registers elapses. This bit is cleared when the application reads this bit. Access restriction applies. Clears on read (or this bit is written to 1 when RCWE bit in MAC_CSR_SW_Ctrl register is set). Self-set to 1 on internal event. 0b - Timestamp Target Time Error status not detected 1b - Timestamp Target Time Error status detected
4 TSTARGET1	Timestamp Target Time Reached for Target Time PPS1 When set, this bit indicates that the value of system time is greater than or equal to the value specified in the MAC_PPS1_Target_Time_Seconds and MAC_PPS1_Target_Time_Nanoseconds registers. Access restriction applies. Clears on read (or this bit is written to 1 when RCWE bit in MAC_CSR_SW_Ctrl register is set). Self-set to 1 on internal event. 0b - Timestamp Target Time Reached for Target Time PPS1 status not detected 1b - Timestamp Target Time Reached for Target Time PPS1 status detected
3 TSTRGTERR0	Timestamp Target Time Error This bit is set when the latest target time programmed in the MAC_PPS0_Target_Time_Seconds and MAC_PPS0_Target_Time_Nanoseconds registers elapses. This bit is cleared when the application reads this bit. Access restriction applies. Clears on read (or this bit is written to 1 when RCWE bit in MAC_CSR_SW_Ctrl register is set). Self-set to 1 on internal event. 0b - Timestamp Target Time Error status not detected 1b - Timestamp Target Time Error status detected
2 AUXTSTRIG	Auxiliary Timestamp Trigger Snapshot This bit is set high when the auxiliary snapshot is written to the FIFO. This bit is valid only if the Add IEEE 1588 Auxiliary Snapshot option is selected.

Table continues on the next page...

Table continued from the previous page...

Field	Function
	Access restriction applies. Clears on read (or this bit is written to 1 when RCWE bit in MAC_CSR_SW_Ctrl register is set). Self-set to 1 on internal event. 0b - Auxiliary Timestamp Trigger Snapshot status not detected 1b - Auxiliary Timestamp Trigger Snapshot status detected
1 TSTARGET0	Timestamp Target Time Reached When set, this bit indicates that the value of system time is greater than or equal to the value specified in the MAC_PPS0_Target_Time_Seconds and MAC_PPS0_Target_Time_Nanoseconds registers. Access restriction applies. Clears on read (or this bit is written to 1 when RCWE bit in MAC_CSR_SW_Ctrl register is set). Self-set to 1 on internal event. 0b - Timestamp Target Time Reached status not detected 1b - Timestamp Target Time Reached status detected
0 TSSOVF	Timestamp Seconds Overflow When this bit is set, it indicates that the seconds value of the timestamp (when supporting version 2 format) has overflowed beyond 32'hFFFF_FFFF. Access restriction applies. Clears on read (or this bit is written to 1 when RCWE bit in MAC_CSR_SW_Ctrl register is set). Self-set to 1 on internal event. 0b - Timestamp Seconds Overflow status not detected 1b - Timestamp Seconds Overflow status detected

1.4.184 (MAC_Tx_Timestamp_Status_Nanoseconds)

Offset

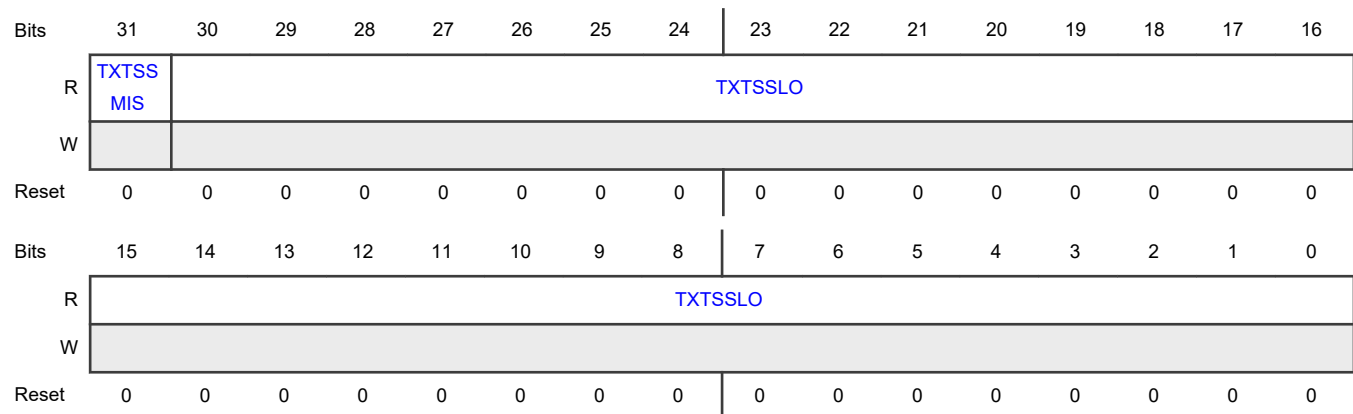
Register	Offset
MAC_Tx_Timestamp_Status_Nanoseconds	B30h

Function

This register contains the nanosecond part of timestamp captured for Transmit packets when Tx status is disabled.

The MAC_Tx_Timestamp_Status_Nanoseconds register, along with MAC_Tx_Timestamp_Status_Seconds, gives the 64-bit timestamp captured for the PTP packet successfully transmitted by the MAC. This value is considered to be read by the application when the last byte of MAC_Tx_Timestamp_Status_Nanoseconds is read. In the little-endian mode, this means when bits[31:24] are read; in big-endian mode, bits[7:0] are read.

If the application does not read these registers and timestamp of another packet is captured, then either the current timestamp is lost (overwritten) or the new timestamp is lost (dropped), depending on the setting of the TXTSSTSM bit of the MAC_Timestamp_Control register. The status bit TXTSC bit [15] in MAC_Timestamp_Status register is set whenever the MAC transmitter captures the timestamp.

Diagram**Fields**

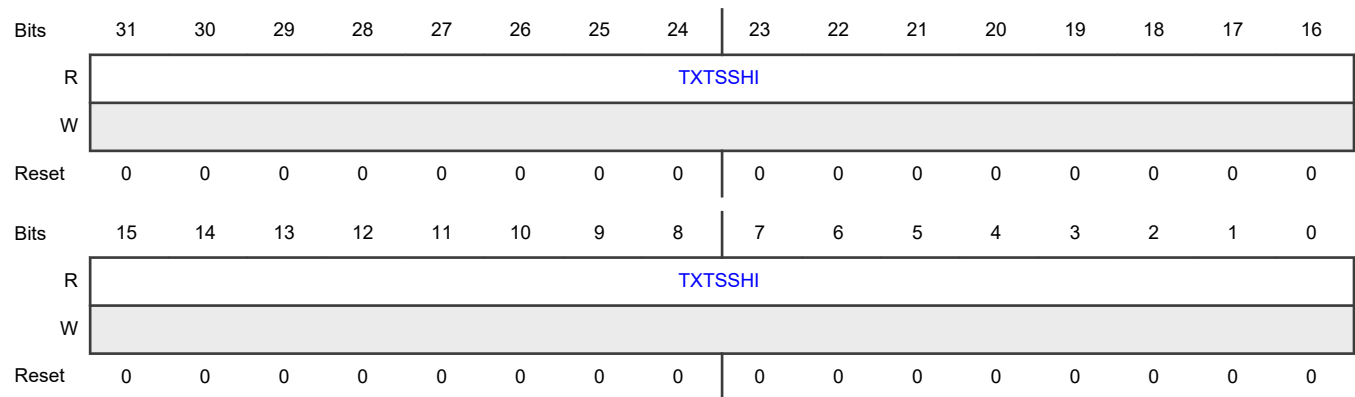
Field	Function
31 TXTSSMIS	Transmit Timestamp Status Missed When this bit is set, it indicates one of the following: - The timestamp of the current packet is ignored if TXTSSTSM bit of the MAC_Timestamp_Control register is reset - The timestamp of the previous packet is overwritten with timestamp of the current packet if TXTSSTSM bit of the MAC_Timestamp_Control register is set. Access restriction applies. Clears on read. Self-set to 1 on internal event. 0b - Transmit Timestamp Status Missed status not detected 1b - Transmit Timestamp Status Missed status detected
30-0 TXTSSLO	Transmit Timestamp Status Low This field contains the 31 bits of the Nanoseconds field of the Transmit packet's captured timestamp.

1.4.185 (MAC_Tx_Timestamp_Status_Seconds)**Offset**

Register	Offset
MAC_Tx_Timestamp_Status_Seconds	B34h

Function

The register contains the higher 32 bits of the timestamp (in seconds) captured when a PTP packet is transmitted.

Diagram**Fields**

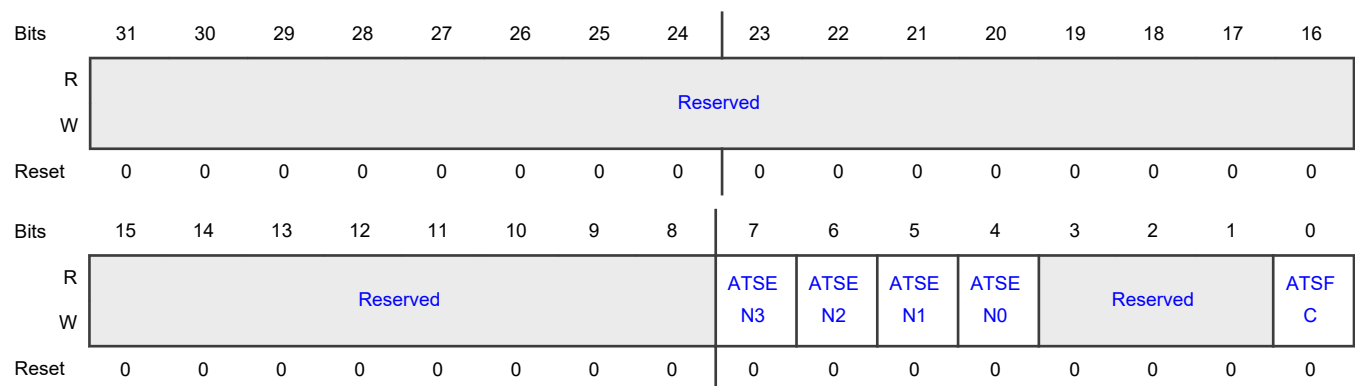
Field	Function
31-0	Transmit Timestamp Status High
TXTSSHI	This field contains the lower 32 bits of the Seconds field of Transmit packet's captured timestamp.

1.4.186 (MAC_Auxiliary_Control)**Offset**

Register	Offset
MAC_Auxiliary_Control	B40h

Function

The Auxiliary Timestamp Control register controls the Auxiliary Timestamp snapshot.

Diagram

Fields

Field	Function
31-8 —	Reserved.
7 ATSEN3	Auxiliary Snapshot 3 Enable This bit controls the capturing of Auxiliary Snapshot Trigger 3. When this bit is set, the auxiliary snapshot of the event on ptp_aux_trig_i[3] input is enabled. When this bit is reset, the events on this input are ignored. 0b - Auxiliary Snapshot \$i is disabled 1b - Auxiliary Snapshot \$i is enabled
6 ATSEN2	Auxiliary Snapshot 2 Enable This bit controls the capturing of Auxiliary Snapshot Trigger 2. When this bit is set, the auxiliary snapshot of the event on ptp_aux_trig_i[2] input is enabled. When this bit is reset, the events on this input are ignored. 0b - Auxiliary Snapshot \$i is disabled 1b - Auxiliary Snapshot \$i is enabled
5 ATSEN1	Auxiliary Snapshot 1 Enable This bit controls the capturing of Auxiliary Snapshot Trigger 1. When this bit is set, the auxiliary snapshot of the event on ptp_aux_trig_i[1] input is enabled. When this bit is reset, the events on this input are ignored. 0b - Auxiliary Snapshot \$i is disabled 1b - Auxiliary Snapshot \$i is enabled
4 ATSEN0	Auxiliary Snapshot 0 Enable This bit controls the capturing of Auxiliary Snapshot Trigger 0. When this bit is set, the auxiliary snapshot of the event on ptp_aux_trig_i[0] input is enabled. When this bit is reset, the events on this input are ignored. 0b - Auxiliary Snapshot \$i is disabled 1b - Auxiliary Snapshot \$i is enabled
3-1 —	Reserved.
0 ATSFC	Auxiliary Snapshot FIFO Clear When set, this bit resets the pointers of the Auxiliary Snapshot FIFO. This bit is cleared when the pointers are reset and the FIFO is empty. When this bit is high, the auxiliary snapshots are stored in the FIFO. Access restriction applies. Setting 1 sets. Self-cleared. Setting 0 has no effect. 0b - Auxiliary Snapshot FIFO Clear is disabled 1b - Auxiliary Snapshot FIFO Clear is enabled

1.4.187 (MAC_Auxiliary_Timestamp_Nanoseconds)

Offset

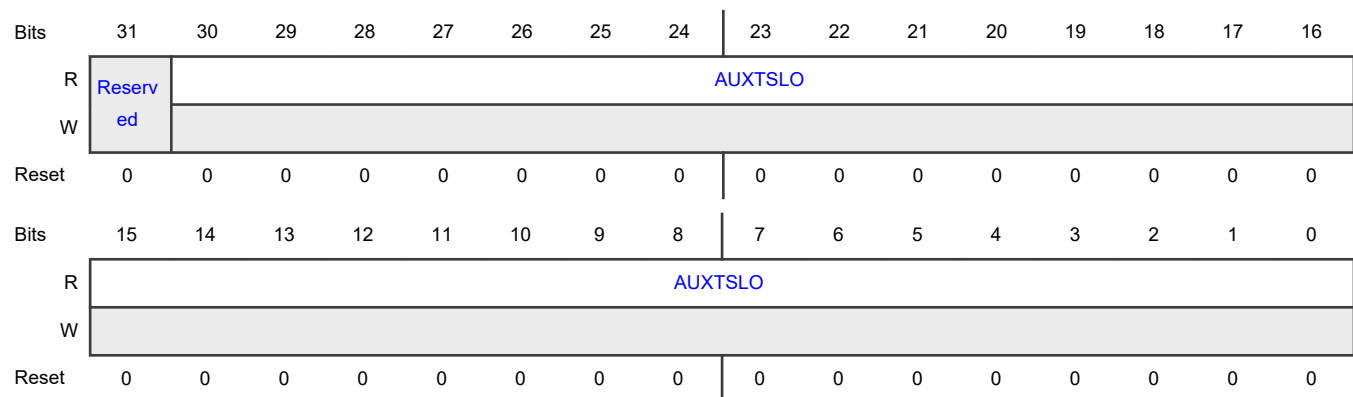
Register	Offset
MAC_Auxiliary_Timestamp_Nanoseconds	B48h

Function

The Auxiliary Timestamp Nanoseconds register, along with MAC_Auxiliary_Timestamp_Seconds, gives the 64-bit timestamp stored as auxiliary snapshot. These two registers form the read port of a 64-bit wide FIFO with a depth of 4, 8, or 16 as selected while configuring the core.

You can store multiple snapshots in this FIFO. Bits[29:25] in MAC_Timestamp_Status indicate the fill-level of the FIFO. The top of the FIFO is removed only when the last byte of MAC_Auxiliary_Timestamp_Seconds register is read. In the little-endian mode, this means when Bits[31:24] are read and in big-endian mode, Bits[7:0] are read.

Diagram



Fields

Field	Function
31 —	Reserved.
30-0 AUXTSLO	Auxiliary Timestamp Contains the lower 31 bits (nanoseconds field) of the auxiliary timestamp.

1.4.188 (MAC_Auxiliary_Timestamp_Seconds)

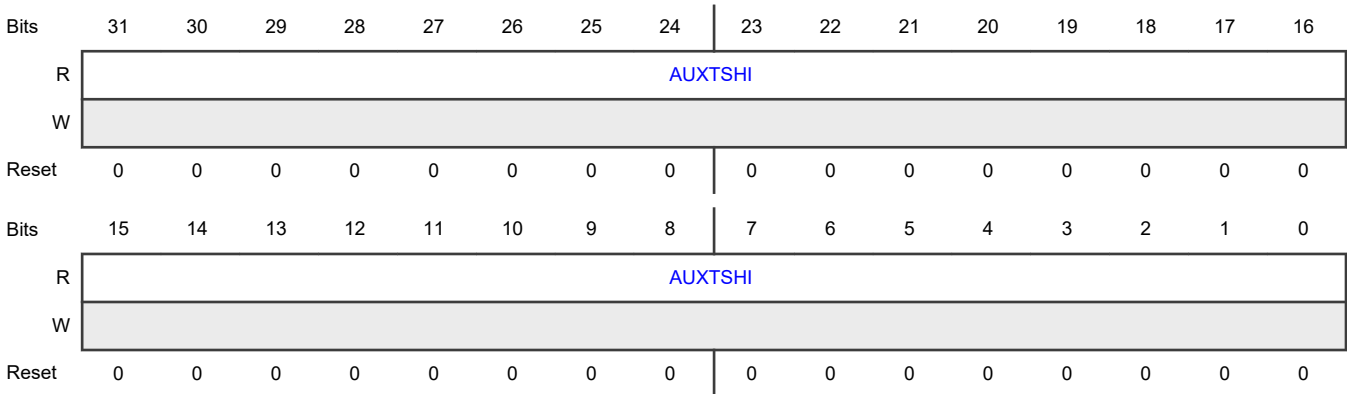
Offset

Register	Offset
MAC_Auxiliary_Timestamp_Seconds	B4Ch

Function

The Auxiliary Timestamp - Seconds register contains the lower 32 bits of the Seconds field of the auxiliary timestamp register.

Diagram



Fields

Field	Function
31-0 AUXTSHI	Auxiliary Timestamp Contains the lower 32 bits of the Seconds field of the auxiliary timestamp.

1.4.189 (MAC_Timestamp_Ingress_Asym_Corr)

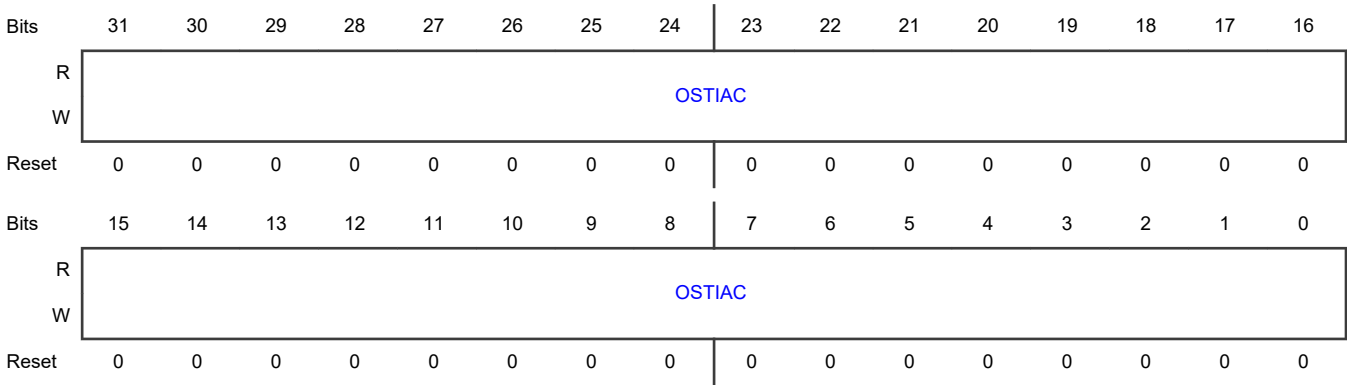
Offset

Register	Offset
MAC_Timestamp_Ingress_Asym_Corr	B50h

Function

The MAC Timestamp Ingress Asymmetry Correction register contains the Ingress Asymmetry Correction value to be used while updating correction field in PDelay_Resp PTP messages.

Diagram



Fields

Field	Function
31-0 OSTIAC	One-Step Timestamp Ingress Asymmetry Correction This field contains the ingress path asymmetry value to be added to correctionField of Pdelay_Resp PTP packet. The programmed value should be in units of nanoseconds and multiplied by 2¹⁶. For example, 2.5 ns is represented as 0x00028000. The value can also be negative, which is represented in 2's complement form with bit 31 representing the sign bit.

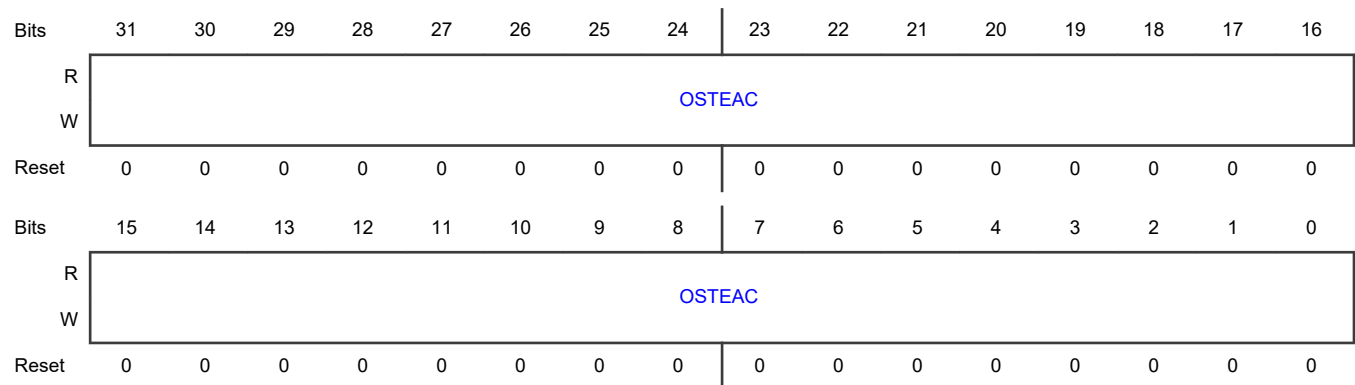
1.4.190 (MAC_Timestamp_Egress_Asym_Corr)

Offset

Register	Offset
MAC_Timestamp_Egress_Asym_Corr	B54h

Function

The MAC Timestamp Egress Asymmetry Correction register contains the Egress Asymmetry Correction value to be used while updating the correction field in PDelay_Req PTP messages.

Diagram**Fields**

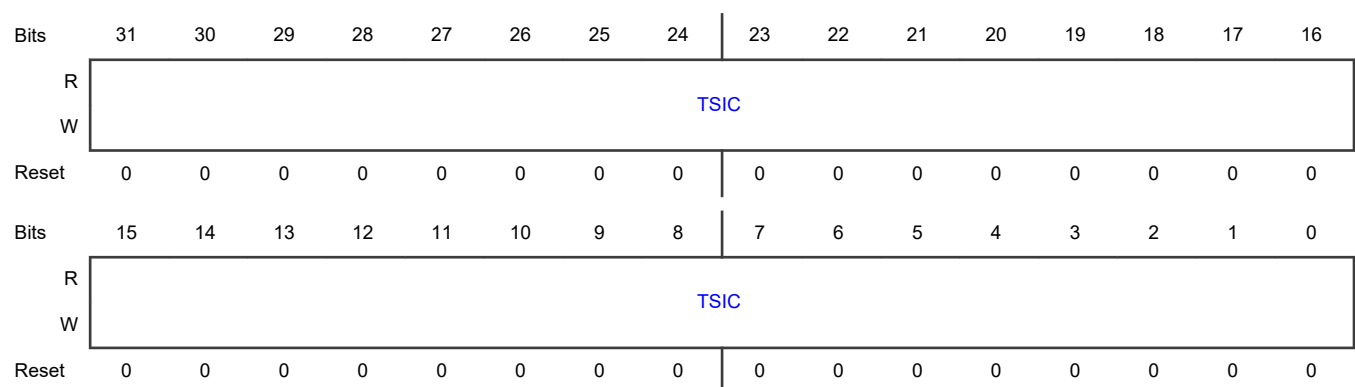
Field	Function
31-0 OSTEAC	One-Step Timestamp Egress Asymmetry Correction This field contains the egress path asymmetry value to be subtracted from correctionField of Pdelay_Resp PTP packet. The programmed value must be the negated value in units of nanoseconds multiplied by 2^{16}. For example, if the required correction is +2.5 ns, the programmed value must be 0xFFFD_8000, which is the 2's complement of 0x0002_8000 ($2.5 * 2^{16}$). Similarly, if the required correction is -3.3 ns, the programmed value is 0x0003_4CCC ($3.3 * 2^{16}$).

1.4.191 (MAC_Timestamp_Ingress_Corr_Nanosecond)**Offset**

Register	Offset
MAC_Timestamp_Ingress_Corr_Nanosecond	B58h

Function

This register contains the correction value in nanoseconds to be used with the captured timestamp value in the ingress path.

Diagram

Fields

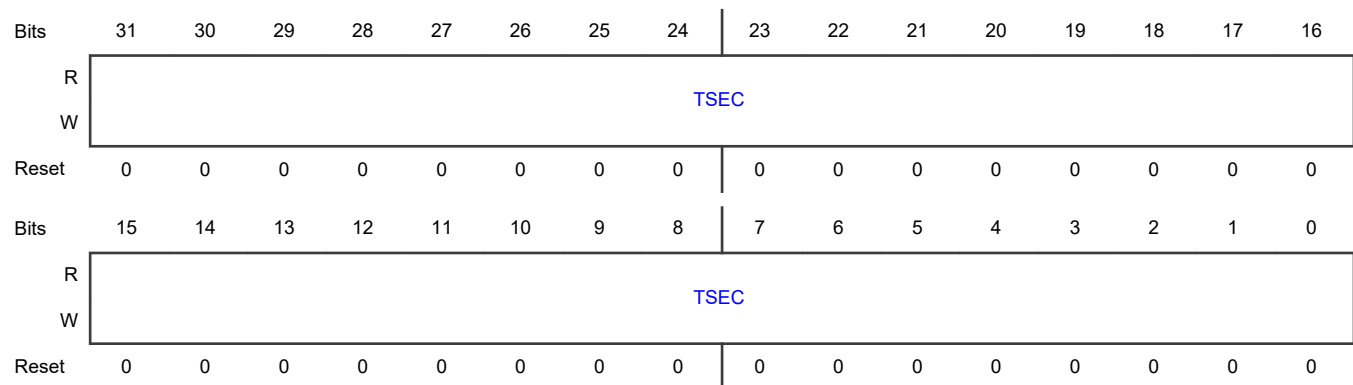
Field	Function
31-0	Timestamp Ingress Correction
TSIC	This field contains the ingress path correction value as defined by the Ingress Correction expression.

1.4.192 (MAC_Timestamp_Egress_Corr_Nanosecond)**Offset**

Register	Offset
MAC_Timestamp_Egress_Corr_Nanosecond	B5Ch

Function

This register contains the correction value in nanoseconds to be used with the captured timestamp value in the egress path.

Diagram**Fields**

Field	Function
31-0	Timestamp Egress Correction
TSEC	This field contains the nanoseconds part of the egress path correction value as defined by the Egress Correction expression.

1.4.193 (MAC_Timestamp_Ingress_Corr_Subnanosec)

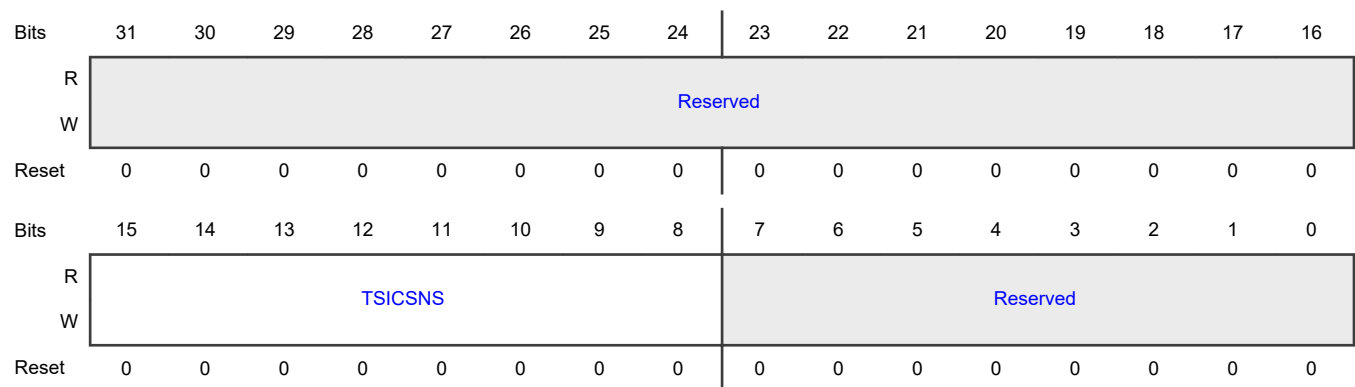
Offset

Register	Offset
MAC_Timestamp_Ingress_Corr_Subnanosec	B60h

Function

This register contains the sub-nanosecond part of the correction value to be used with the captured timestamp value, for ingress direction.

Diagram



Fields

Field	Function
31-16 —	Reserved.
15-8 TSICSNS	Timestamp Ingress Correction, sub-nanoseconds This field contains the sub-nanoseconds part of the ingress path correction value as defined by the "Ingress Correction" expression.
7-0 —	Reserved.

1.4.194 (MAC_Timestamp_Egress_Corr_Subnanosec)

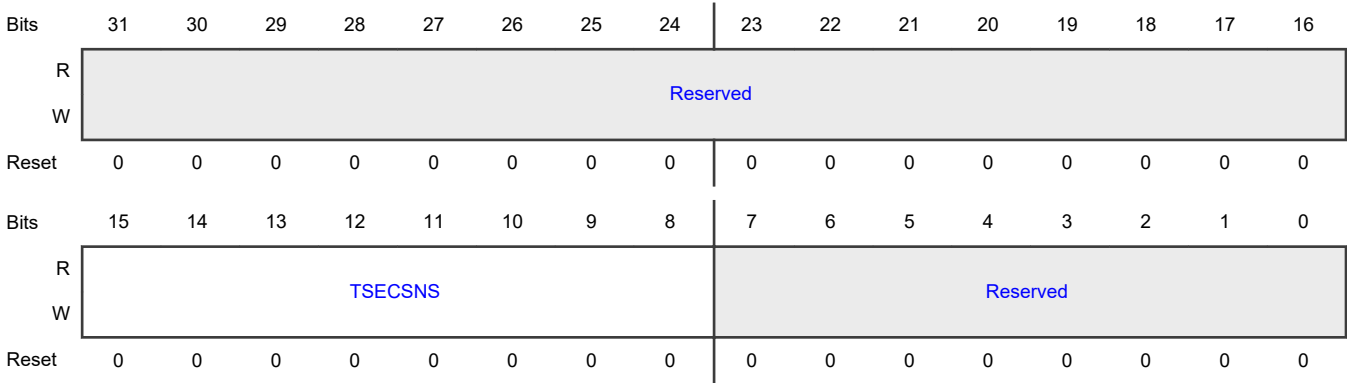
Offset

Register	Offset
MAC_Timestamp_Egress_Corr_Subnanosec	B64h

Function

This register contains the sub-nanosecond part of the correction value to be used with the captured timestamp value, for egress direction.

Diagram



Fields

Field	Function
31-16 —	Reserved.
15-8 TSECSNS	Timestamp Egress Correction, sub-nanoseconds This field contains the sub-nanoseconds part of the egress path correction value as defined by the "Egress Correction" expression.
7-0 —	Reserved.

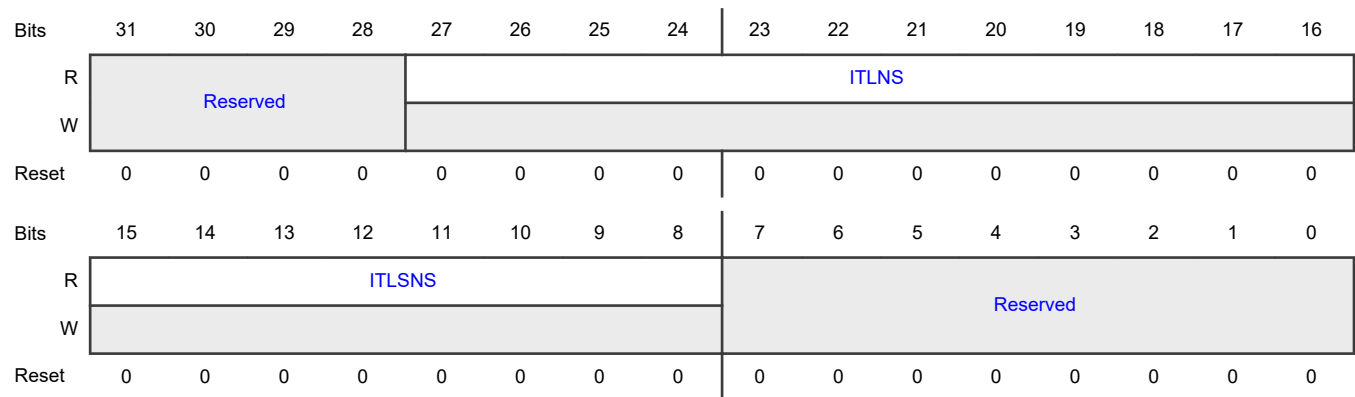
1.4.195 (MAC_Timestamp_Ingress_Latency)

Offset

Register	Offset
MAC_Timestamp_Ingress_Latency	B68h

Function

This register holds the Ingress MAC latency.

Diagram**Fields**

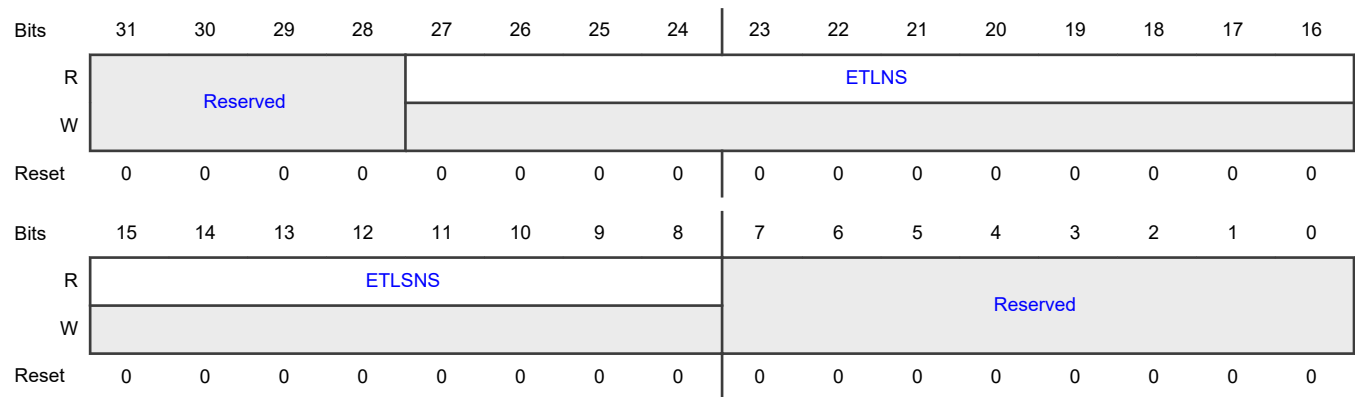
Field	Function
31-28 —	Reserved.
27-16 ITLNS	Ingress Timestamp Latency, in sub-nanoseconds This register holds the average latency in sub-nanoseconds between the input ports (phy_rxd_i) of MAC and the actual point (GMII/MII) where the ingress timestamp is taken. Ingress correction value is computed as described in the section 7.1.2.4.1 of QoS Databook.
15-8 ITLSNS	Ingress Timestamp Latency, in nanoseconds This register holds the average latency in nanoseconds between the input ports (phy_rxd_i) of MAC and the actual point (GMII/MII) where the ingress timestamp is taken. Ingress correction value is computed as described in the section 7.1.2.4.1 of QoS Databook.
7-0 —	Reserved.

1.4.196 (MAC_Timestamp_Egress_Latency)**Offset**

Register	Offset
MAC_Timestamp_Egress_Latency	B6Ch

Function

This register holds the Egress MAC latency.

Diagram**Fields**

Field	Function
31-28 —	Reserved.
27-16 ETLNS	Egress Timestamp Latency, in nanoseconds This register holds the average latency in nanoseconds between the actual point (GMII/MII) where the egress timestamp is taken and the output ports (phy_txd_o) of the MAC. Ingress correction value is computed as described in the section 7.1.2.4.2 of QoS Databook.
15-8 ETLSNS	Egress Timestamp Latency, in sub-nanoseconds This register holds the average latency in sub-nanoseconds between the actual point (GMII/MII) where the egress timestamp is taken and the output ports (phy_txd_o) of the MAC. Ingress correction value is computed as described in the section 7.1.2.4.2 of QoS Databook.
7-0 —	Reserved.

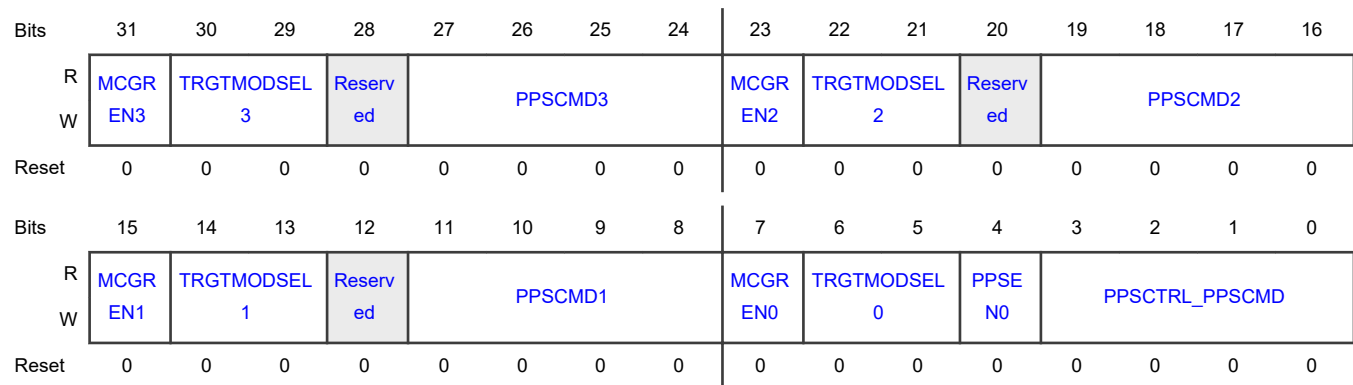
1.4.197 (MAC_PPS_Control)**Offset**

Register	Offset
MAC_PPS_Control	B70h

Function

PPS Control register.

Bits[30:24] of this register are valid only when four Flexible PPS outputs are selected. Bits[22:16] are valid only when three or more Flexible PPS outputs are selected. Bits[14:8] are valid only when two or more Flexible PPS outputs are selected. Bits[6:4] are valid only when Flexible PPS feature is selected.

Diagram**Fields**

Field	Function
31 MCGREN3	MCGR Mode Enable for PPS3 Output This field enables the 3rd PPS instance to operate in PPS or MCGR mode. When set it operates in MCGR mode and on reset it operates in PPS mode.
30-29 TRGTMODSEL 3	Target Time Register Mode for PPS3 Output This field indicates the Target Time registers (MAC_PPS3_Target_Time_Seconds and MAC_PPS3_Target_Time_Nanoseconds) mode for PPS3 output signal. 00b - Target Time registers are programmed only for generating the interrupt event. The Flexible PPS function must not be enabled in this mode, otherwise spurious transitions may be observed on the corresponding ptp_pps_o output port 01b - Reserved 10b - Target Time registers are programmed for generating the interrupt event and starting or stopping the PPS0 output signal generation 11b - Target Time registers are programmed only for starting or stopping the PPS0 output signal generation. No interrupt is asserted
28 —	Reserved.
27-24 PPSCMD3	Flexible PPS3 Output Control This field controls the flexible PPS3 output (ptp_pps_o[3]) signal. This field is similar to the PPSCMD0[2:0] field. If MCGREN3 is set, then PPSCMD3 indicated by these 4 bits [27:24] are taken as Presentation Time Control bits for media clock generation and recovery for comparator instance 3. This field is similar to the PPSCMD0 Presentation Time Control bits. If MCGREN3 is not set then only 3 bits from [26:24] is used as PPSCMD3 and the 4th bit is to be set as 0. Access restriction applies. Setting 1 sets. Self-cleared. Setting 0 has no effect.
23	MCGR Mode Enable for PPS2 Output

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Field	Function
MCGREN2	This field enables the 2nd PPS instance to operate in PPS or MCGR mode. When set it operates in MCGR mode and on reset it operates in PPS mode. 0b - 2nd PPS instance is disabled to operate in PPS or MCGR mode 1b - 2nd PPS instance is enabled to operate in PPS or MCGR mode
22-21 TRGTMODSEL 2	Target Time Register Mode for PPS2 Output This field indicates the Target Time registers (MAC_PPS2_Target_Time_Seconds and MAC_PPS2_Target_Time_Nanoseconds) mode for PPS2 output signal. 00b - Target Time registers are programmed only for generating the interrupt event. The Flexible PPS function must not be enabled in this mode, otherwise spurious transitions may be observed on the corresponding ptp_pps_o output port 01b - Reserved 10b - Target Time registers are programmed for generating the interrupt event and starting or stopping the PPS0 output signal generation 11b - Target Time registers are programmed only for starting or stopping the PPS0 output signal generation. No interrupt is asserted
20 —	Reserved.
19-16 PPSCMD2	Flexible PPS2 Output Control This field controls the flexible PPS2 output (ptp_pps_o[2]) signal. This field is similar to the PPSCMD0 field. If MCGREN2 is set, then PPSCMD2 indicated by these 4 bits [19:16] are taken as Presentation Time Control bits for media clock generation and recovery for comparator instance 2. This field is similar to the PPSCMD0 Presentation Time Control bits. If MCGREN2 is not set then only 3 bits from [18:16] is used as PPSCMD2 and the 4th bit is to be set as 0. Access restriction applies. Setting 1 sets. Self-cleared. Setting 0 has no effect.
15 MCGREN1	MCGR Mode Enable for PPS1 Output This field enables the 1st PPS instance to operate in PPS or MCGR mode. When set it operates in MCGR mode and on reset it operates in PPS mode. 0b - 1st PPS instance is disabled to operate in PPS or MCGR mode 1b - 1st PPS instance is enabled to operate in PPS or MCGR mode
14-13 TRGTMODSEL 1	Target Time Register Mode for PPS1 Output This field indicates the Target Time registers (MAC_PPS1_Target_Time_Seconds and MAC_PPS1_Target_Time_Nanoseconds) mode for PPS1 output signal. 00b - Target Time registers are programmed only for generating the interrupt event. The Flexible PPS function must not be enabled in this mode, otherwise spurious transitions may be observed on the corresponding ptp_pps_o output port 01b - Reserved

Table continues on the next page...

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Field	Function
	10b - Target Time registers are programmed for generating the interrupt event and starting or stopping the PPS0 output signal generation 11b - Target Time registers are programmed only for starting or stopping the PPS0 output signal generation. No interrupt is asserted
12 —	Reserved.
11-8 PPSCMD1	Flexible PPS1 Output Control This field controls the flexible PPS1 output (ptp_pps_o[1]) signal. This field is similar to the PPSCMD0 field. If MCGREN1 is set, then PPSCMD1 indicated by these 4 bits [11:8] are taken as Presentation Time Control bits for media clock generation and recovery for comparator instance 1. This field is similar to the PPSCMD0 Presentation Time Control bits. If MCGREN1 is not set then only 3 bits from [10:8] is used as PPSCMD1 and the 4th bit is to be set as 0. Access restriction applies. Setting 1 sets. Self-cleared. Setting 0 has no effect.
7 MCGREN0	MCGR Mode Enable for PPS0 Output This field enables the 0th PPS instance to operate in PPS or MCGR mode. When set it operates in MCGR mode and on reset it operates in PPS mode. 0b - 0th PPS instance is enabled to operate in PPS mode 1b - 0th PPS instance is enabled to operate in MCGR mode
6-5 TRGTMODSEL 0	Target Time Register Mode for PPS0 Output This field indicates the Target Time registers (MAC_PPS0_Target_Time_Seconds and MAC_PPS0_Target_Time_Nanoseconds) mode for PPS0 output signal: 00b - Target Time registers are programmed only for generating the interrupt event. The Flexible PPS function must not be enabled in this mode, otherwise spurious transitions may be observed on the corresponding ptp_pps_o output port 01b - Reserved 10b - Target Time registers are programmed for generating the interrupt event and starting or stopping the PPS0 output signal generation 11b - Target Time registers are programmed only for starting or stopping the PPS0 output signal generation. No interrupt is asserted
4 PPSEN0	Flexible PPS Output Mode Enable When this bit is set, Bits[3:0] function as PPSCMD. When this bit is reset, Bits[3:0] function as PPSCTRL (Fixed PPS mode). 0b - Flexible PPS Output Mode is disabled 1b - Flexible PPS Output Mode is enabled
3-0	PPS Output Frequency Control

Table continues on the next page...

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Field	Function
PPSCTRL_PPS CMD	This field controls the frequency of the PPS0 output (ptp_pps_o[0]) signal. The default value of PPSCTRL is 0000, and the PPS output is 1 pulse (of width clk_ptp_i) every second. For other values of PPSCTRL, the PPS output becomes a generated clock of following frequencies: - 0001: The binary rollover is 2 Hz, and the digital rollover is 1 Hz. - 0010: The binary rollover is 4 Hz, and the digital rollover is 2 Hz. - 0011: The binary rollover is 8 Hz, and the digital rollover is 4 Hz. - 0100: The binary rollover is 16 Hz, and the digital rollover is 8 Hz. - .. - 1111: The binary rollover is 32.768 KHz and the digital rollover is 16.384 KHz. Note: In the binary rollover mode, the PPS output (ptp_pps_o) has a duty cycle of 50 percent with these frequencies. In the digital rollover mode, the PPS output frequency is an average number. The actual clock is of different frequency that gets synchronized every second. For example: - When PPSCTRL = 0001, the PPS (1 Hz) has a low period of 537 ms and a high period of 463 ms - When PPSCTRL = 0010, the PPS (2 Hz) is a sequence of - One clock of 50 percent duty cycle and 537 ms period - Second clock of 463 ms period (268 ms low and 195 ms high) - When PPSCTRL = 0011, the PPS (4 Hz) is a sequence of - Three clocks of 50 percent duty cycle and 268 ms period - Fourth clock of 195 ms period (134 ms low and 61 ms high) This behavior is because of the non-linear toggling of bits in the digital rollover mode in the MAC_System_Time_Nanoseconds register. or Flexible PPS Output (ptp_pps_o[0]) Control Programming these bits with a non-zero value instructs the MAC to initiate an event. When the command is transferred or synchronized to the PTP clock domain, these bits get cleared automatically. The software should ensure that these bits are programmed only when they are 'all-zero'. The following list describes the values of PPSCMD0: - 0000: No Command - 0001: START Single Pulse This command generates single pulse rising at the start point defined in MAC_PPS0_Target_Time_Seconds and MAC_PPS0_Target_Time_Nanoseconds register and of a duration defined in the PPS0 Width Register. - 0010: START Pulse Train This command generates the train of pulses rising at the start point defined in the Target Time Registers and of a duration defined in the PPS0 Width Register and repeated at interval defined in the PPS Interval

Table continues on the next page...

Table continued from the previous page...

Field	Function
	Register. By default, the PPS pulse train is free-running unless stopped by the 'Stop Pulse train at time' or 'Stop Pulse Train immediately' commands. - 0011: Cancel START This command cancels the START Single Pulse and START Pulse Train commands if the system time has not crossed the programmed start time. - 0100: STOP Pulse train at time This command stops the train of pulses initiated by the START Pulse Train command (PPSCMD = 0010) after the time programmed in the Target Time registers elapses. - 0101: STOP Pulse Train immediately This command immediately stops the train of pulses initiated by the START Pulse Train command (PPSCMD = 0010). - 110: Cancel STOP Pulse train This command cancels the STOP pulse train at time command if the programmed stop time has not elapsed. The PPS pulse train becomes free-running on the successful execution of this command. - 0111-1111: Reserved or Presentation Time Control If MCGREN0 is set then these bits are treated as Presentation time control bits. The following list describes the values of PPSCMD0: - 0000: MCGR operation is not carried out. If set to this value in the mid of clock recovery or generation, all the processing inputs are flushed - 0001: Capture the Presentation time at the rising edge of mcg_pst_trig_i[0] into the MAC_PPS0_Target_Time_Seconds register - 0010: Capture the Presentation time at the falling edge of mcg_pst_trig_i[0] into the MAC_PPS0_Target_Time_Seconds register - 0011: Capture the Presentation time at both edges of mcg_pst_trig_i[0] into the MAC_PPS0_Target_Time_Seconds register - 0100-1000: Reserved - 1001: Toggle output on compare - 1010: Pulse output low on compare for one PTP-clock cycle - 1011: Pulse output high on compare for one PTP-clock cycle - 1100-1111: Reserved

1.4.198 (MAC_PPS0_Target_Time_Seconds)

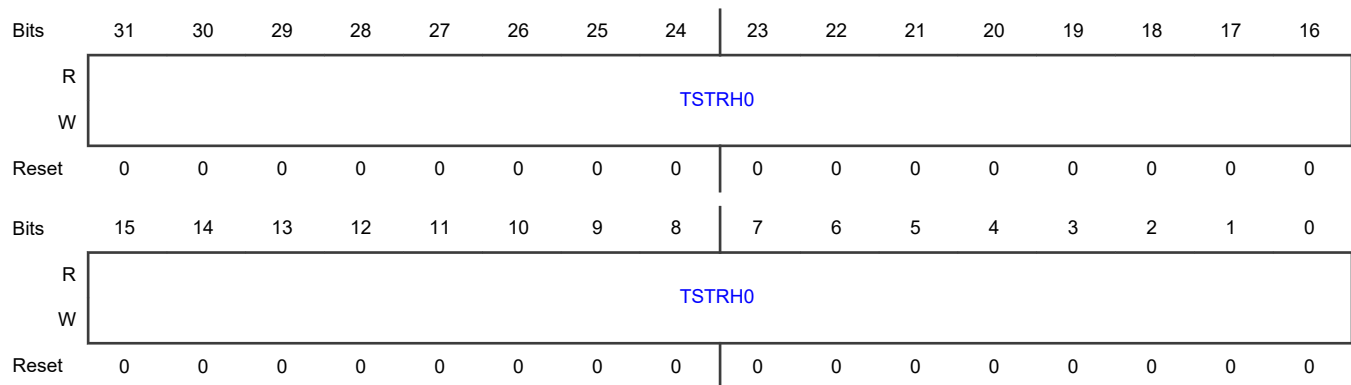
Offset

Register	Offset
MAC_PPS0_Target_Time_Seconds	B80h

Function

The PPS Target Time Seconds register, along with PPS Target Time Nanoseconds register, is used to schedule an interrupt event [Bit 1 of MAC_Timestamp_Status] when the system time exceeds the value programmed in these registers.

Diagram



Fields

Field	Function
31-0 TSTRH0	PPS Target Time Seconds Register This field stores the time in seconds. When the timestamp value matches or exceeds both Target Timestamp registers, the MAC starts or stops the PPS signal output and generates an interrupt (if enabled) based on Target Time mode selected for the corresponding PPS output in the MAC_PPS_Control register. If DWC_EQOS_FLEXI_PPS_OUT_EN is enabled in the configuration and PTGE field of MAC_Timestamp_Control Register is set with Presentation time control set in recovery mode, then these bits indicate the TPT being programmed by the application and in generation mode it indicates the CPT generated at the sampled trigger.

1.4.199 (MAC_PPS0_Target_Time_Nanoseconds)

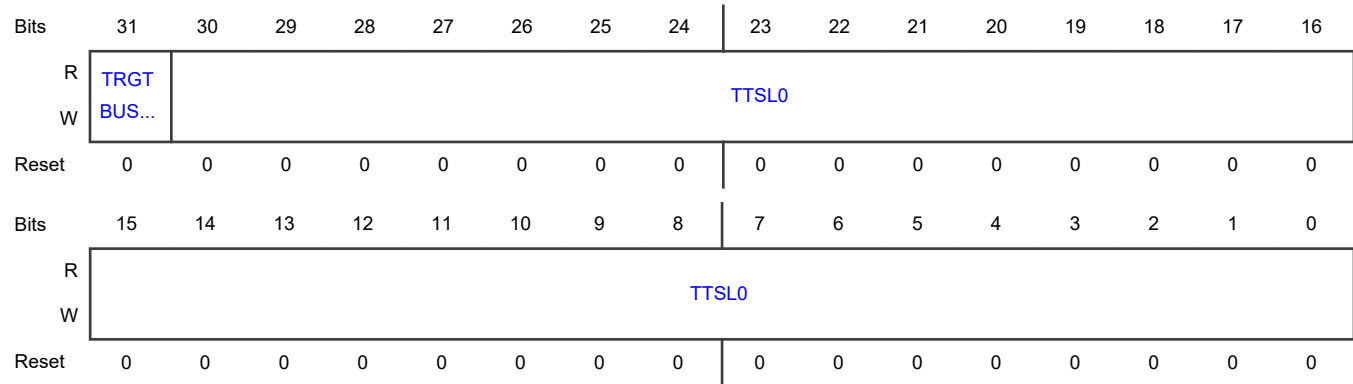
Offset

Register	Offset
MAC_PPS0_Target_Time_Nanoseconds	B84h

Function

PPS0 Target Time Nanoseconds register.

Diagram



Fields

Field	Function
31 TRGTBUSY0	PPS Target Time Register Busy The MAC sets this bit when the PPSCMD0 field in the MAC_PPS_Control register is programmed to 010 or 011. Programming the PPSCMD0 field to 010 or 011 instructs the MAC to synchronize the Target Time Registers to the PTP clock domain. The MAC clears this bit after synchronizing the Target Time Registers to the PTP clock domain. The application must not update the Target Time Registers when this bit is read as 1. Otherwise, the synchronization of the previous programmed time gets corrupted. 0b - PPS Target Time Register Busy status is not detected 1b - PPS Target Time Register Busy is detected
30-0 TTSL0	Target Time Low for PPS Register This register stores the time in (signed) nanoseconds. When the value of the timestamp matches the value in both Target Timestamp registers, the MAC starts or stops the PPS signal output and generates an interrupt (if enabled) based on the TRGTMODSEL0 field (Bits [6:5]) in MAC_PPS_Control. When the TSCTRLSSR bit is reset in the MAC_Timestamp_Control register, this value should be (time in ns / 0.465). The actual start or stop time of the PPS signal output might have an error margin up to one unit of sub-second increment value. When the TSCTRLSSR bit is set in the MAC_Timestamp_Control register, this value should not exceed 0x3B9A_C9FF. The actual start or stop time of the PPS signal output might have an error margin up to one unit of sub-second increment value. Access restriction applies. Setting 1 sets. Self-cleared. Setting 0 has no effect.

1.4.200 (MAC_PPS0_Interval)

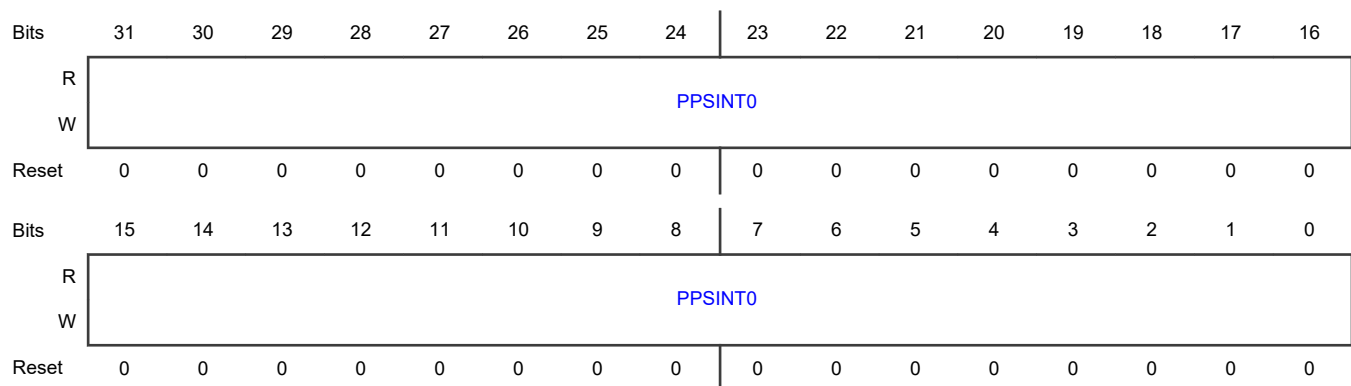
Offset

Register	Offset
MAC_PPS0_Interval	B88h

Function

The PPS0 Interval register contains the number of units of sub-second increment value between the rising edges of PPS0 signal output (ptp_pps_o[0]).

Diagram



Fields

Field	Function
31-0 PPSINT0	PPS Output Signal Interval These bits store the interval between the rising edges of PPS0 signal output. The interval is stored in terms of number of units of sub-second increment value. You need to program one value less than the required interval. For example, if the PTP reference clock is 50 MHz (period of 20 ns), and desired interval between the rising edges of PPS0 signal output is 100 ns (that is, 5 units of sub-second increment value), you should program value 4 (5-1) in this register.

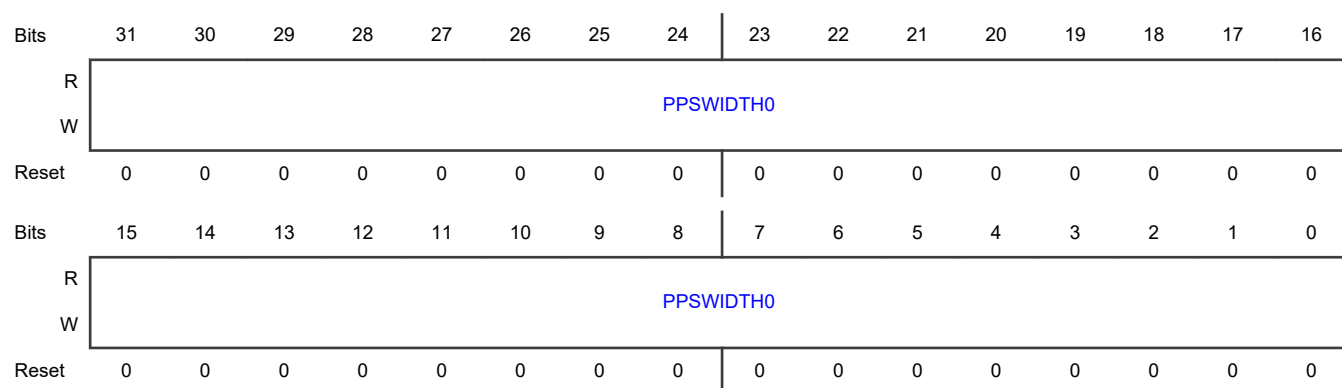
1.4.201 (MAC_PPS0_Width)

Offset

Register	Offset
MAC_PPS0_Width	B8Ch

Function

The PPS0 Width register contains the number of units of sub-second increment value between the rising and corresponding falling edges of PPS0 signal output (ptp_pps_o[0]).

Diagram**Fields**

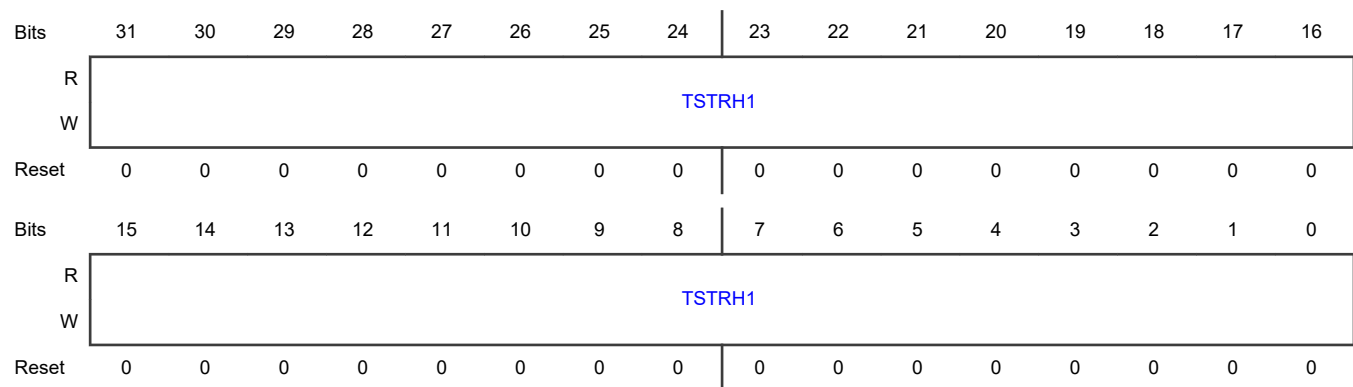
Field	Function
31-0 PPSWIDTH0	PPS Output Signal Width These bits store the width between the rising edge and corresponding falling edge of PPS0 signal output. The width is stored in terms of number of units of sub-second increment value. You need to program one value less than the required interval. For example, if PTP reference clock is 50 MHz (period of 20 ns), and width between the rising and corresponding falling edges of PPS0 signal output is 80 ns (that is, four units of sub-second increment value), you should program value 3 (4-1) in this register. Note: The value programmed in this register must be lesser than the value programmed in MAC_PPS0_Interval.

1.4.202 (MAC_PPS1_Target_Time_Seconds)**Offset**

Register	Offset
MAC_PPS1_Target_Time_Seconds	B90h

Function

The PPS Target Time Seconds register, along with PPS Target Time Nanoseconds register, is used to schedule an interrupt event [Bit 1 of MAC_Timestamp_Status] when the system time exceeds the value programmed in these registers.

Diagram**Fields**

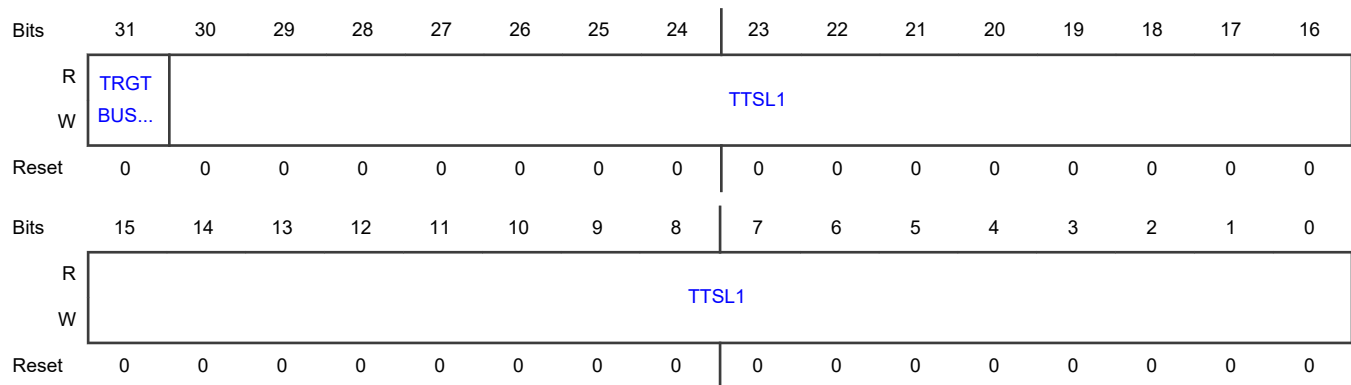
Field	Function
31-0 TSTRH1	PPS Target Time Seconds Register This field stores the time in seconds. When the timestamp value matches or exceeds both Target Timestamp registers, the MAC starts or stops the PPS signal output and generates an interrupt (if enabled) based on Target Time mode selected for the corresponding PPS output in the MAC_PPS_Control register. If DWC_EQOS_FLEXI_PPS_OUT_EN is enabled in the configuration and PTGE field of MAC_Timestamp_Control Register is set with Presentation time control set in recovery mode, then these bits indicate the TPT being programmed by the application and in generation mode it indicates the CPT generated at the sampled trigger.

1.4.203 (MAC_PPS1_Target_Time_Nanoseconds)**Offset**

Register	Offset
MAC_PPS1_Target_Time_Nanoseconds	B94h

Function

PPS0 Target Time Nanoseconds register.

Diagram**Fields**

Field	Function
31 TRGTBUSY1	PPS Target Time Register Busy The MAC sets this bit when the PPSCMD0 field in the MAC_PPS_Control register is programmed to 010 or 011. Programming the PPSCMD0 field to 010 or 011 instructs the MAC to synchronize the Target Time Registers to the PTP clock domain. The MAC clears this bit after synchronizing the Target Time Registers to the PTP clock domain. The application must not update the Target Time Registers when this bit is read as 1. Otherwise, the synchronization of the previous programmed time gets corrupted. 0b - PPS Target Time Register Busy status is not detected 1b - PPS Target Time Register Busy is detected
30-0 TTSL1	Target Time Low for PPS Register This register stores the time in (signed) nanoseconds. When the value of the timestamp matches the value in both Target Timestamp registers, the MAC starts or stops the PPS signal output and generates an interrupt (if enabled) based on the TRGTMODSEL0 field (Bits [6:5]) in MAC_PPS_Control. When the TSCTRLSSR bit is reset in the MAC_Timestamp_Control register, this value should be (time in ns / 0.465). The actual start or stop time of the PPS signal output might have an error margin up to one unit of sub-second increment value. When the TSCTRLSSR bit is set in the MAC_Timestamp_Control register, this value should not exceed 0x3B9A_C9FF. The actual start or stop time of the PPS signal output might have an error margin up to one unit of sub-second increment value. Access restriction applies. Setting 1 sets. Self-cleared. Setting 0 has no effect.

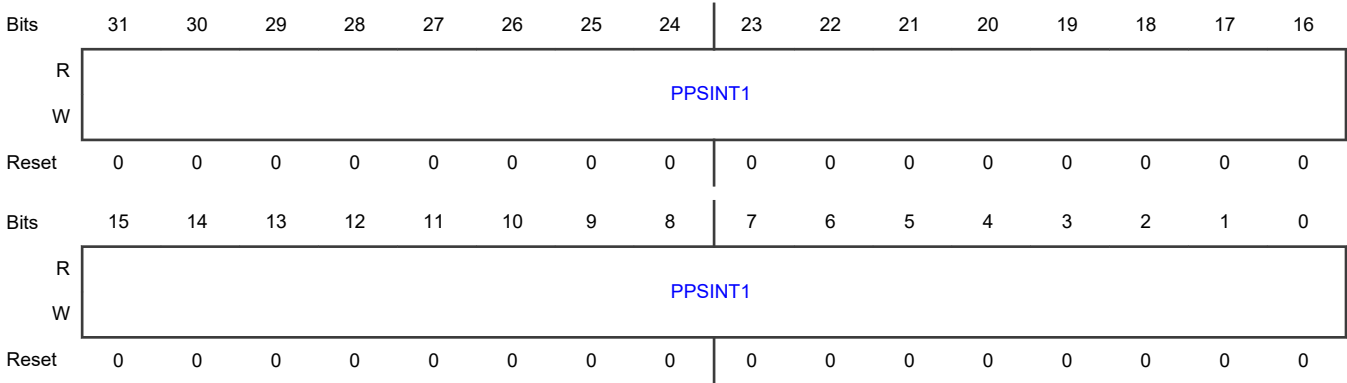
1.4.204 (MAC_PPS1_Interval)**Offset**

Register	Offset
MAC_PPS1_Interval	B98h

Function

The PPS0 Interval register contains the number of units of sub-second increment value between the rising edges of PPS0 signal output (ptp_pps_o[0]).

Diagram



Fields

Field	Function
31-0 PPSINT1	PPS Output Signal Interval These bits store the interval between the rising edges of PPS0 signal output. The interval is stored in terms of number of units of sub-second increment value. You need to program one value less than the required interval. For example, if the PTP reference clock is 50 MHz (period of 20 ns), and desired interval between the rising edges of PPS0 signal output is 100 ns (that is, 5 units of sub-second increment value), you should program value 4 (5-1) in this register.

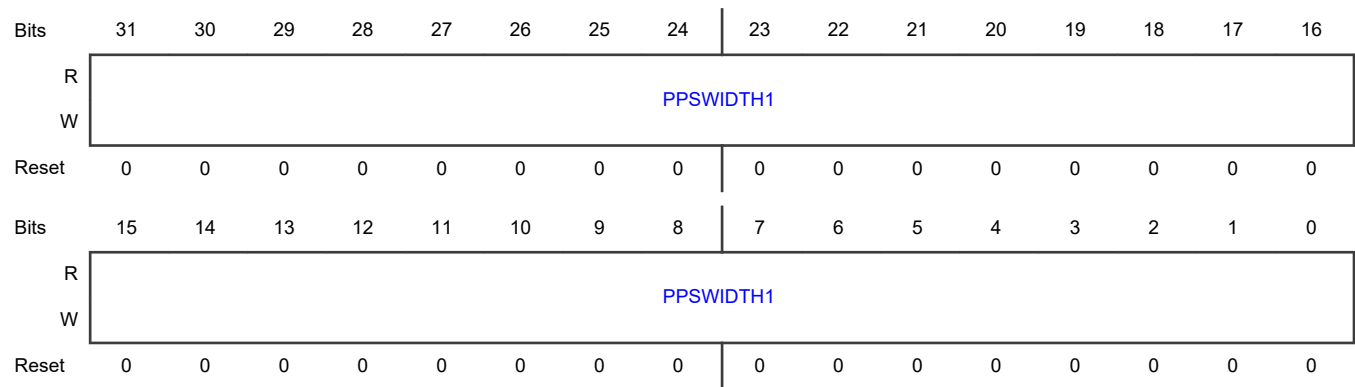
1.4.205 (MAC_PPS1_Width)

Offset

Register	Offset
MAC_PPS1_Width	B9Ch

Function

The PPS0 Width register contains the number of units of sub-second increment value between the rising and corresponding falling edges of PPS0 signal output (ptp_pps_o[0]).

Diagram**Fields**

Field	Function
31-0 PPSWIDTH1	PPS Output Signal Width These bits store the width between the rising edge and corresponding falling edge of PPS0 signal output. The width is stored in terms of number of units of sub-second increment value. You need to program one value less than the required interval. For example, if PTP reference clock is 50 MHz (period of 20 ns), and width between the rising and corresponding falling edges of PPS0 signal output is 80 ns (that is, four units of sub-second increment value), you should program value 3 (4-1) in this register. Note: The value programmed in this register must be lesser than the value programmed in MAC_PPS0_Interval.

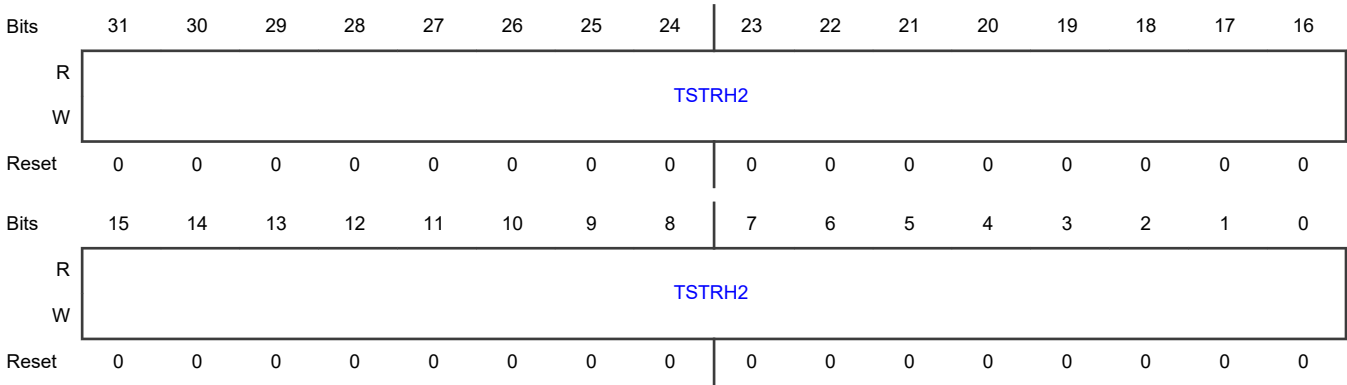
1.4.206 (MAC_PPS2_Target_Time_Seconds)**Offset**

Register	Offset
MAC_PPS2_Target_Time_Seconds	BA0h

Function

The PPS Target Time Seconds register, along with PPS Target Time Nanoseconds register, is used to schedule an interrupt event [Bit 1 of MAC_Timestamp_Status] when the system time exceeds the value programmed in these registers.

Diagram



Fields

Field	Function
31-0 TSTRH2	PPS Target Time Seconds Register This field stores the time in seconds. When the timestamp value matches or exceeds both Target Timestamp registers, the MAC starts or stops the PPS signal output and generates an interrupt (if enabled) based on Target Time mode selected for the corresponding PPS output in the MAC_PPS_Control register. If DWC_EQOS_FLEXI_PPS_OUT_EN is enabled in the configuration and PTGE field of MAC_Timestamp_Control Register is set with Presentation time control set in recovery mode, then these bits indicate the TPT being programmed by the application and in generation mode it indicates the CPT generated at the sampled trigger.

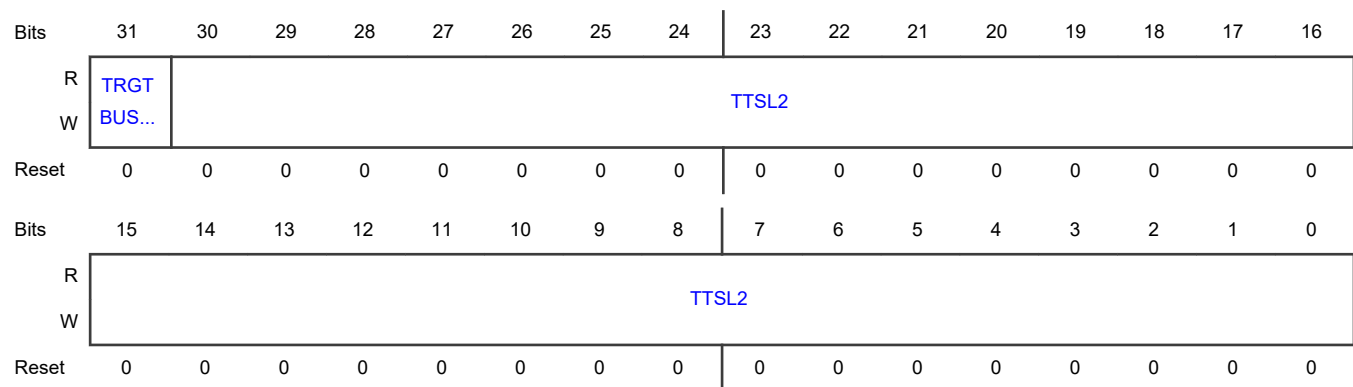
1.4.207 (MAC_PPS2_Target_Time_Nanoseconds)

Offset

Register	Offset
MAC_PPS2_Target_Time_Nanoseconds	BA4h

Function

PPS0 Target Time Nanoseconds register.

Diagram**Fields**

Field	Function
31 TRGTBUSY2	PPS Target Time Register Busy The MAC sets this bit when the PPSCMD0 field in the MAC_PPS_Control register is programmed to 010 or 011. Programming the PPSCMD0 field to 010 or 011 instructs the MAC to synchronize the Target Time Registers to the PTP clock domain. The MAC clears this bit after synchronizing the Target Time Registers to the PTP clock domain. The application must not update the Target Time Registers when this bit is read as 1. Otherwise, the synchronization of the previous programmed time gets corrupted. 0b - PPS Target Time Register Busy status is not detected 1b - PPS Target Time Register Busy is detected
30-0 TTSL2	Target Time Low for PPS Register This register stores the time in (signed) nanoseconds. When the value of the timestamp matches the value in both Target Timestamp registers, the MAC starts or stops the PPS signal output and generates an interrupt (if enabled) based on the TRGTMODSEL0 field (Bits [6:5]) in MAC_PPS_Control. When the TSCTRLSSR bit is reset in the MAC_Timestamp_Control register, this value should be (time in ns / 0.465). The actual start or stop time of the PPS signal output might have an error margin up to one unit of sub-second increment value. When the TSCTRLSSR bit is set in the MAC_Timestamp_Control register, this value should not exceed 0x3B9A_C9FF. The actual start or stop time of the PPS signal output might have an error margin up to one unit of sub-second increment value. Access restriction applies. Setting 1 sets. Self-cleared. Setting 0 has no effect.

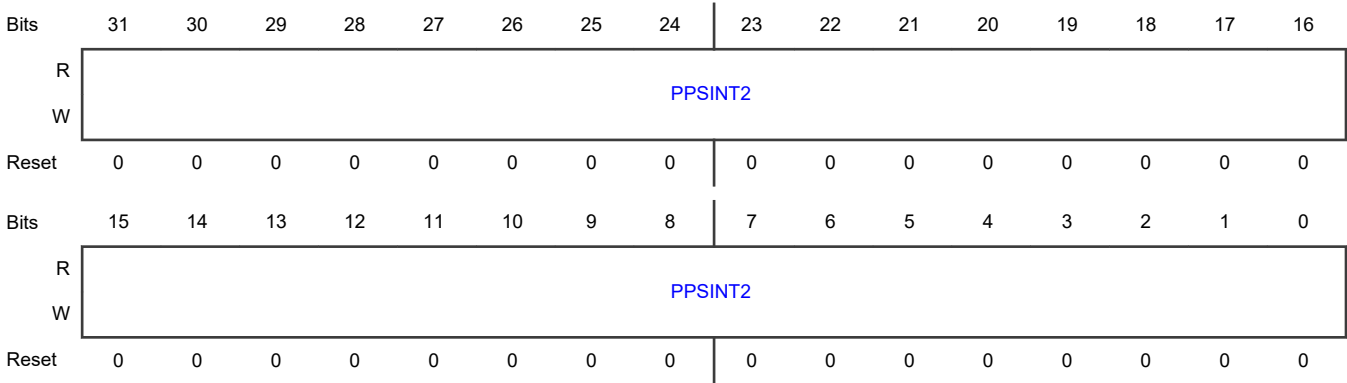
1.4.208 (MAC_PPS2_Interval)**Offset**

Register	Offset
MAC_PPS2_Interval	BA8h

Function

The PPS0 Interval register contains the number of units of sub-second increment value between the rising edges of PPS0 signal output (ptp_pps_o[0]).

Diagram



Fields

Field	Function
31-0 PPSINT2	PPS Output Signal Interval These bits store the interval between the rising edges of PPS0 signal output. The interval is stored in terms of number of units of sub-second increment value. You need to program one value less than the required interval. For example, if the PTP reference clock is 50 MHz (period of 20 ns), and desired interval between the rising edges of PPS0 signal output is 100 ns (that is, 5 units of sub-second increment value), you should program value 4 (5-1) in this register.

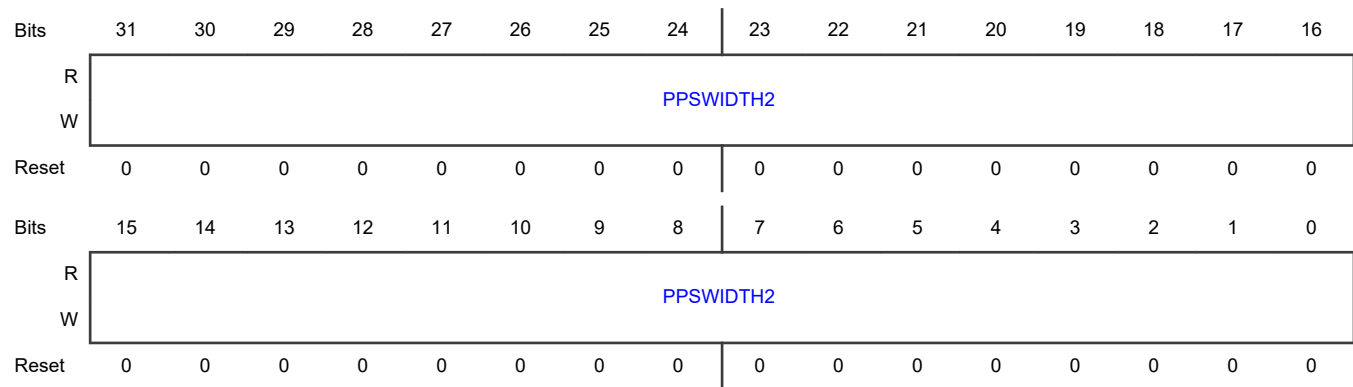
1.4.209 (MAC_PPS2_Width)

Offset

Register	Offset
MAC_PPS2_Width	BACH

Function

The PPS0 Width register contains the number of units of sub-second increment value between the rising and corresponding falling edges of PPS0 signal output (ptp_pps_o[0]).

Diagram**Fields**

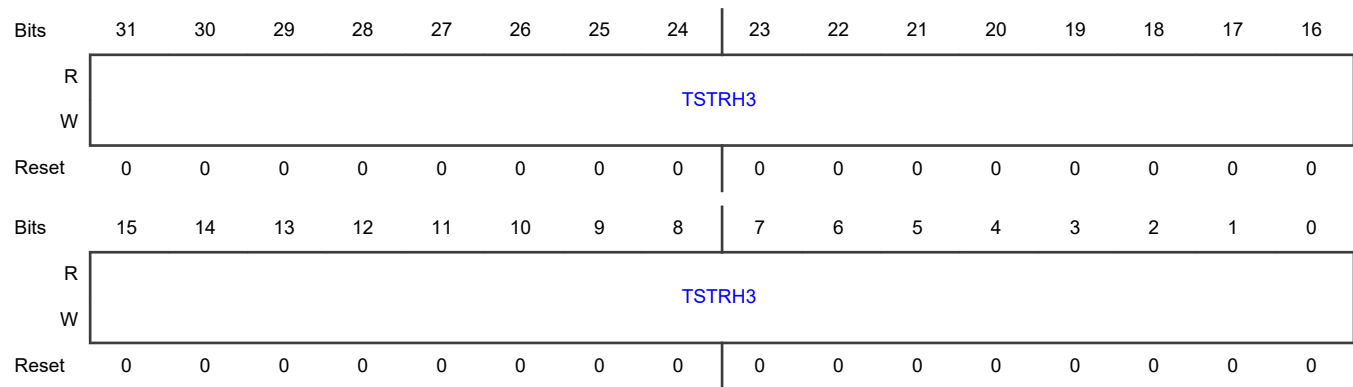
Field	Function
31-0 PPSWIDTH2	PPS Output Signal Width These bits store the width between the rising edge and corresponding falling edge of PPS0 signal output. The width is stored in terms of number of units of sub-second increment value. You need to program one value less than the required interval. For example, if PTP reference clock is 50 MHz (period of 20 ns), and width between the rising and corresponding falling edges of PPS0 signal output is 80 ns (that is, four units of sub-second increment value), you should program value 3 (4-1) in this register. Note: The value programmed in this register must be lesser than the value programmed in MAC_PPS0_Interval.

1.4.210 (MAC_PPS3_Target_Time_Seconds)**Offset**

Register	Offset
MAC_PPS3_Target_Time_Seconds	BB0h

Function

The PPS Target Time Seconds register, along with PPS Target Time Nanoseconds register, is used to schedule an interrupt event [Bit 1 of MAC_Timestamp_Status] when the system time exceeds the value programmed in these registers.

Diagram**Fields**

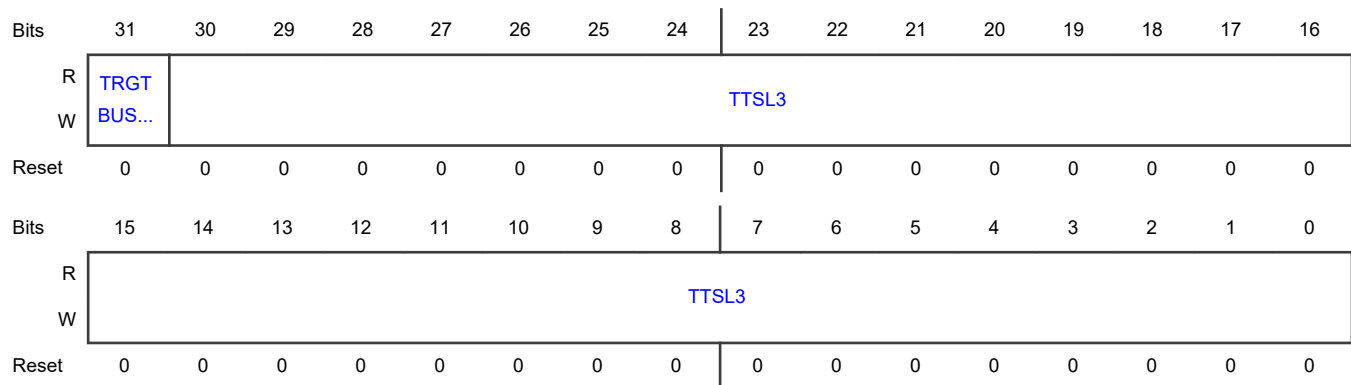
Field	Function
31-0 TSTRH3	PPS Target Time Seconds Register This field stores the time in seconds. When the timestamp value matches or exceeds both Target Timestamp registers, the MAC starts or stops the PPS signal output and generates an interrupt (if enabled) based on Target Time mode selected for the corresponding PPS output in the MAC_PPS_Control register. If DWC_EQOS_FLEXI_PPS_OUT_EN is enabled in the configuration and PTGE field of MAC_Timestamp_Control Register is set with Presentation time control set in recovery mode, then these bits indicate the TPT being programmed by the application and in generation mode it indicates the CPT generated at the sampled trigger.

1.4.211 (MAC_PPS3_Target_Time_Nanoseconds)**Offset**

Register	Offset
MAC_PPS3_Target_Time_Nanoseconds	BB4h

Function

PPS0 Target Time Nanoseconds register.

Diagram**Fields**

Field	Function
31 TRGTBUSY3	PPS Target Time Register Busy The MAC sets this bit when the PPSCMD0 field in the MAC_PPS_Control register is programmed to 010 or 011. Programming the PPSCMD0 field to 010 or 011 instructs the MAC to synchronize the Target Time Registers to the PTP clock domain. The MAC clears this bit after synchronizing the Target Time Registers to the PTP clock domain. The application must not update the Target Time Registers when this bit is read as 1. Otherwise, the synchronization of the previous programmed time gets corrupted. 0b - PPS Target Time Register Busy status is not detected 1b - PPS Target Time Register Busy is detected
30-0 TTSL3	Target Time Low for PPS Register This register stores the time in (signed) nanoseconds. When the value of the timestamp matches the value in both Target Timestamp registers, the MAC starts or stops the PPS signal output and generates an interrupt (if enabled) based on the TRGTMODSEL0 field (Bits [6:5]) in MAC_PPS_Control. When the TSCTRLSSR bit is reset in the MAC_Timestamp_Control register, this value should be (time in ns / 0.465). The actual start or stop time of the PPS signal output might have an error margin up to one unit of sub-second increment value. When the TSCTRLSSR bit is set in the MAC_Timestamp_Control register, this value should not exceed 0x3B9A_C9FF. The actual start or stop time of the PPS signal output might have an error margin up to one unit of sub-second increment value. Access restriction applies. Setting 1 sets. Self-cleared. Setting 0 has no effect.

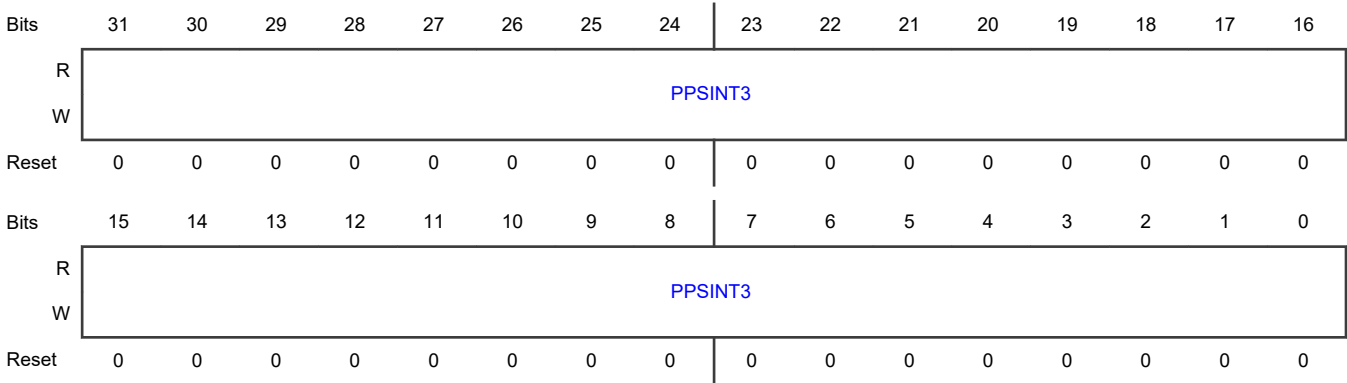
1.4.212 (MAC_PPS3_Interval)**Offset**

Register	Offset
MAC_PPS3_Interval	BB8h

Function

The PPS0 Interval register contains the number of units of sub-second increment value between the rising edges of PPS0 signal output (ptp_pps_o[0]).

Diagram



Fields

Field	Function
31-0 PPSINT3	PPS Output Signal Interval These bits store the interval between the rising edges of PPS0 signal output. The interval is stored in terms of number of units of sub-second increment value. You need to program one value less than the required interval. For example, if the PTP reference clock is 50 MHz (period of 20 ns), and desired interval between the rising edges of PPS0 signal output is 100 ns (that is, 5 units of sub-second increment value), you should program value 4 (5-1) in this register.

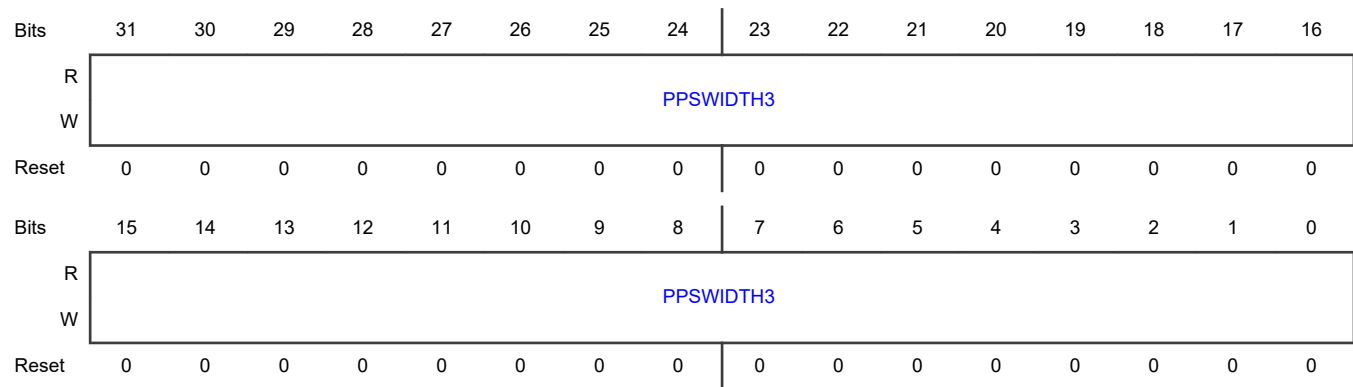
1.4.213 (MAC_PPS3_Width)

Offset

Register	Offset
MAC_PPS3_Width	BBCh

Function

The PPS0 Width register contains the number of units of sub-second increment value between the rising and corresponding falling edges of PPS0 signal output (ptp_pps_o[0]).

Diagram**Fields**

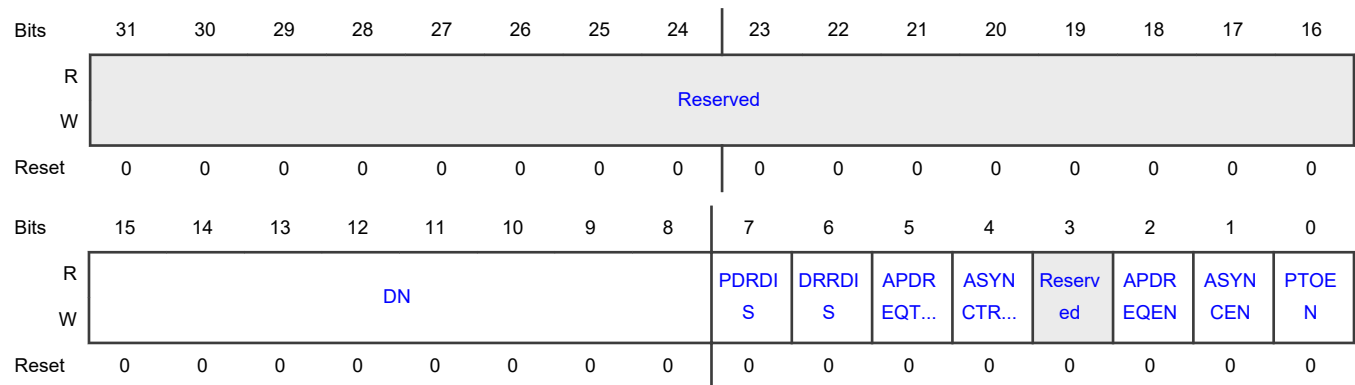
Field	Function
31-0 PPSWIDTH3	PPS Output Signal Width These bits store the width between the rising edge and corresponding falling edge of PPS0 signal output. The width is stored in terms of number of units of sub-second increment value. You need to program one value less than the required interval. For example, if PTP reference clock is 50 MHz (period of 20 ns), and width between the rising and corresponding falling edges of PPS0 signal output is 80 ns (that is, four units of sub-second increment value), you should program value 3 (4-1) in this register. Note: The value programmed in this register must be lesser than the value programmed in MAC_PPS0_Interval.

1.4.214 (MAC_PTO_Control)**Offset**

Register	Offset
MAC_PTO_Control	BC0h

Function

This register controls the PTP Offload Engine operation. This register is available only when the Enable PTP Timestamp Offload feature is selected.

Diagram**Fields**

Field	Function
31-16 —	Reserved.
15-8 DN	Domain Number This field indicates the domain Number in which the PTP node is operating.
7 PDRDIS	Disable Peer Delay Response response generation When this bit is set, the Peer Delay Response (Pdelay_Resp) response is not be generated for received Peer Delay Request (Pdelay_Req) request packet, as required by the programmed mode. Note: Setting this bit to 1 affects the normal PTP Offload operation and the time synchronization. So, this bit must be set only if there is problem with Pdelay_Resp generation in Hardware and/or Pdelay_Resp generation is handled by Software. 0b - Peer Delay Response response generation is enabled 1b - Peer Delay Response response generation is disabled
6 DRRDIS	Disable PTO Delay Request/Response response generation When this bit is set, the Delay Request and Delay response is not generated for received SYNC and Delay request packet respectively, as required by the programmed mode. 0b - PTO Delay Request/Response response generation is enabled 1b - PTO Delay Request/Response response generation is disabled
5 APDREQTRIG	Automatic PTP Pdelay_Req message Trigger When this bit is set, one PTP Pdelay_Req message is transmitted. This bit is automatically cleared after the PTP Pdelay_Req message is transmitted. The application should set the APDREQEN bit for this operation. Access restriction applies. Setting 1 sets. Self-cleared. Setting 0 has no effect. 0b - Automatic PTP Pdelay_Req message Trigger is disabled 1b - Automatic PTP Pdelay_Req message Trigger is enabled

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Field	Function
4 ASYNCTRIG	Automatic PTP SYNC message Trigger When this bit is set, one PTP SYNC message is transmitted. This bit is automatically cleared after the PTP SYNC message is transmitted. The application should set the ASYNCEN bit for this operation. Access restriction applies. Setting 1 sets. Self-cleared. Setting 0 has no effect. 0b - Automatic PTP SYNC message Trigger is disabled 1b - Automatic PTP SYNC message Trigger is enabled
3 —	Reserved.
2 APDREQEN	Automatic PTP Pdelay_Req message Enable When this bit is set, PTP Pdelay_Req message is generated periodically based on interval programmed or trigger from application, when the MAC is programmed to be in Peer-to-Peer Transparent mode. 0b - Automatic PTP Pdelay_Req message is disabled 1b - Automatic PTP Pdelay_Req message is enabled
1 ASYNCEN	Automatic PTP SYNC message Enable When this bit is set, PTP SYNC message is generated periodically based on interval programmed or trigger from application, when the MAC is programmed to be in Clock Master mode. 0b - Automatic PTP SYNC message is disabled 1b - Automatic PTP SYNC message is enabled
0 PTOEN	PTP Offload Enable When this bit is set, the PTP Offload feature is enabled. 0b - PTP Offload feature is disabled 1b - PTP Offload feature is enabled

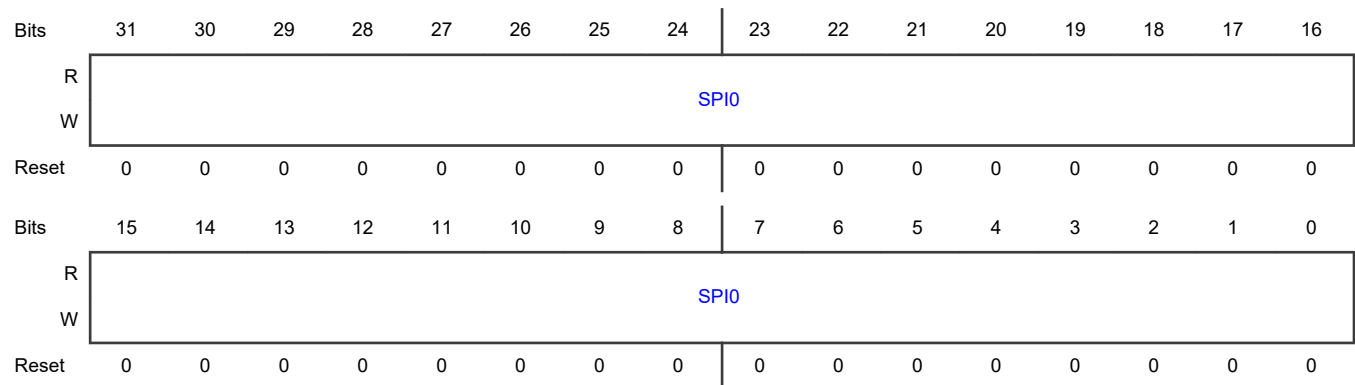
1.4.215 (MAC_Source_Port_Identity0)

Offset

Register	Offset
MAC_Source_Port_Identity0	BC4h

Function

This register contains Bits[31:0] of the 80-bit Source Port Identity of the PTP node. This register is available only when the Enable PTP Timestamp Offload feature is selected.

Diagram**Fields**

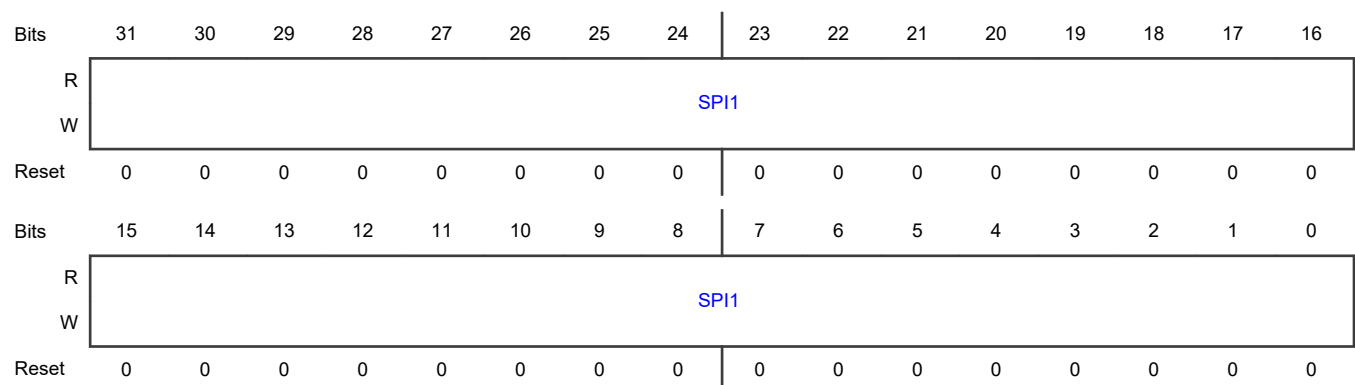
Field	Function
31-0	Source Port Identity 0
SPI0	This field indicates bits [31:0] of sourcePortIdentity of PTP node.

1.4.216 (MAC_Source_Port_Identity1)**Offset**

Register	Offset
MAC_Source_Port_Identity1	BC8h

Function

This register contains Bits[63:32] of the 80-bit Source Port Identity of the PTP node. This register is available only when the Enable PTP Timestamp Offload feature is selected.

Diagram

Fields

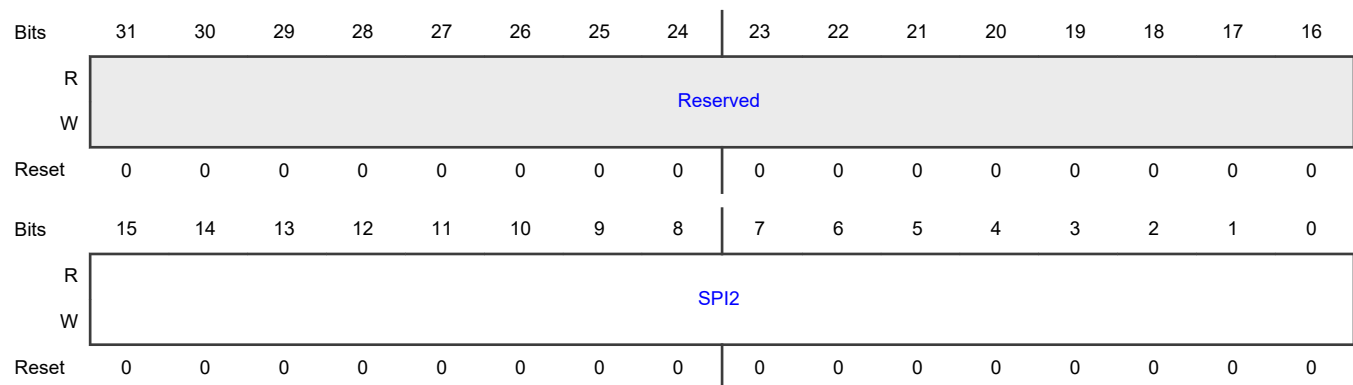
Field	Function
31-0	Source Port Identity 1
SPI1	This field indicates bits [63:32] of sourcePortIdentity of PTP node.

1.4.217 (MAC_Source_Port_Identity2)**Offset**

Register	Offset
MAC_Source_Port_Identity2	BCCh

Function

This register contains Bits[79:64] of the 80-bit Source Port Identity of the PTP node. This register is available only when the Enable PTP Timestamp Offload feature is selected.

Diagram**Fields**

Field	Function
31-16	Reserved.
—	
15-0	Source Port Identity 2
SPI2	This field indicates bits [79:64] of sourcePortIdentity of PTP node.

1.4.218 (MAC_Log_Message_Interval)

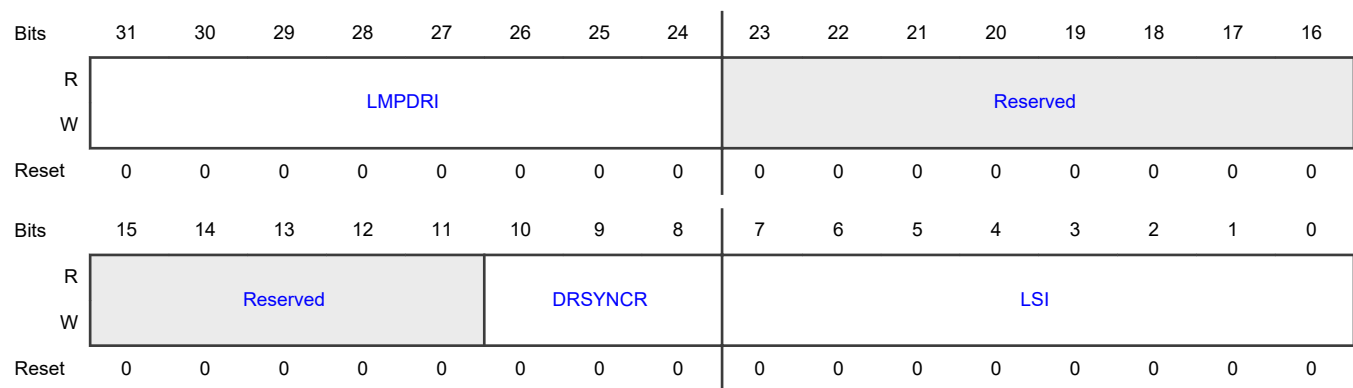
Offset

Register	Offset
MAC_Log_Message_Interval	BD0h

Function

This register contains the periodic intervals for automatic PTP packet generation. This register is available only when the Enable PTP Timestamp Offload feature is selected.

Diagram



Fields

Field	Function
31-24 LMPDRI	Log Min Pdelay_Req Interval This field indicates logMinPdelayReqInterval of PTP node. This is used to schedule the periodic Pdelay request packet transmission. Allowed values are -15 to 15. Negative value must be represented in 2's-complement form. For example, if the required value is -1, the value programmed must be 0xFF.
23-11 —	Reserved.
10-8 DRSYNCR	. Delay_Req to SYNC Ratio In Slave mode, it is used for controlling frequency of Delay_Req messages transmitted. - 0: DelayReq generated for every received SYNC - 1: DelayReq generated every alternate reception of SYNC - 2: for every 4 SYNC messages

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Field	Function
	- 3: for every 8 SYNC messages - 4: for every 16 SYNC messages - 5: for every 32 SYNC messages - 6-7: Reserved The master sends this information (logMinDelayReqInterval) in the DelayResp PTP messages to the slave. The DWC_ether_qos Receiver processes this value from the received DelayResp messages and updates this field accordingly. In the Slave mode, the host must not write/update this register unless it has to override the received value. In Master mode, the sum of this field and logSyncInterval (LSI) field is provided in the logMinDelayReqInterval field of the generated multicast Delay_Resp PTP message. Access restriction applies. Updated based on the event. Setting 1 sets. Setting 0 clears. 000b - DelayReq generated for every received SYNC 001b - DelayReq generated every alternate reception of SYNC 010b - for every 4 SYNC messages 011b - for every 8 SYNC messages 100b - for every 16 SYNC messages 101b - for every 32 SYNC messages 110b - Reserved
7-0 LSI	. Log Sync Interval This field indicates the periodicity of the automatically generated SYNC message when the PTP node is Master. Allowed values are -15 to 15. Negative value must be represented in 2's-complement form. For example, if the required value is -1, the value programmed must be 0xFF.

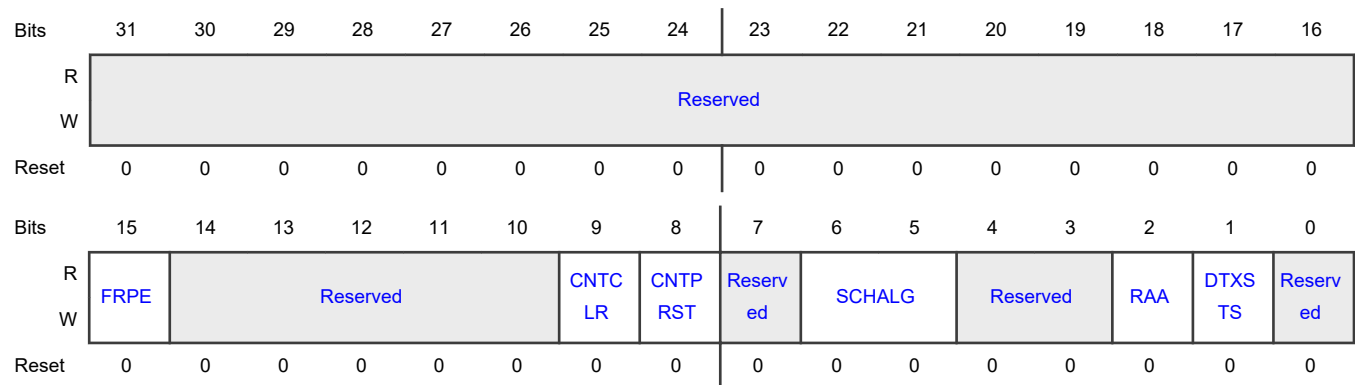
1.4.219 (MTL_Operation_Mode)

Offset

Register	Offset
MTL_Operation_Mode	C00h

Function

The Operation Mode register establishes the Transmit and Receive operating modes and commands.

Diagram**Fields**

Field	Function
31-16 —	Reserved.
15 FRPE	Flexible Rx parser Enable When this bit is set to 1, the Programmable Rx Parser functionality is enabled. When the Rx parser is disabled and if the Rx parser is in the middle of the parsing then it gets disabled only after completing the current packet parsing. When the Rx parser is enabled from disabled state then the Rx parser gets activated for the next immediate packet. 0b - Flexible Rx parser is disabled 1b - Flexible Rx parser is enabled
14-10 —	Reserved.
9 CNTCLR	Counters Reset When this bit is set, all counters are reset. This bit is cleared automatically after 1 clock cycle. If this bit is set along with CNT_PRESET bit, CNT_PRESET has precedence. Access restriction applies. Setting 1 sets. Self-cleared. Setting 0 has no effect. 0b - Counters are not reset 1b - All counters are reset
8 CNTPRST	Counters Preset When this bit is set, - MTL_TxQ[0-7]_Underflow register is initialized/preset to 12'h7F0. - Missed Packet and Overflow Packet counters in MTL_RxQ[0-7]_Missed_Packet_Overflow_Cnt register is initialized/preset to 12'h7F0.

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Field	Function
	Access restriction applies. Setting 1 sets. Self-cleared. Setting 0 has no effect. 0b - Counters Preset is disabled 1b - Counters Preset is enabled
7 —	Reserved.
6-5 SCHALG	Tx Scheduling Algorithm This field indicates the algorithm for Tx scheduling: 00b - WRR algorithm 01b - WFQ algorithm when DCB feature is selected. Otherwise, Reserved 10b - DWRR algorithm when DCB feature is selected. Otherwise, Reserved 11b - Strict priority algorithm
4-3 —	Reserved.
2 RAA	Receive Arbitration Algorithm This field is used to select the arbitration algorithm for the Rx side. - 0: Strict priority (SP) Queue 0 has the lowest priority and the last queue has the highest priority. - 1: Weighted Strict Priority (WSP) 0b - Strict priority (SP) 1b - Weighted Strict Priority (WSP)
1 DTXSTS	Drop Transmit Status When this bit is set, the Tx packet status received from the MAC is dropped in the MTL. When this bit is reset, the Tx packet status received from the MAC is forwarded to the application. 0b - Drop Transmit Status is disabled 1b - Drop Transmit Status is enabled
0 —	Reserved.

1.4.220 (MTL_DBG_CTL)

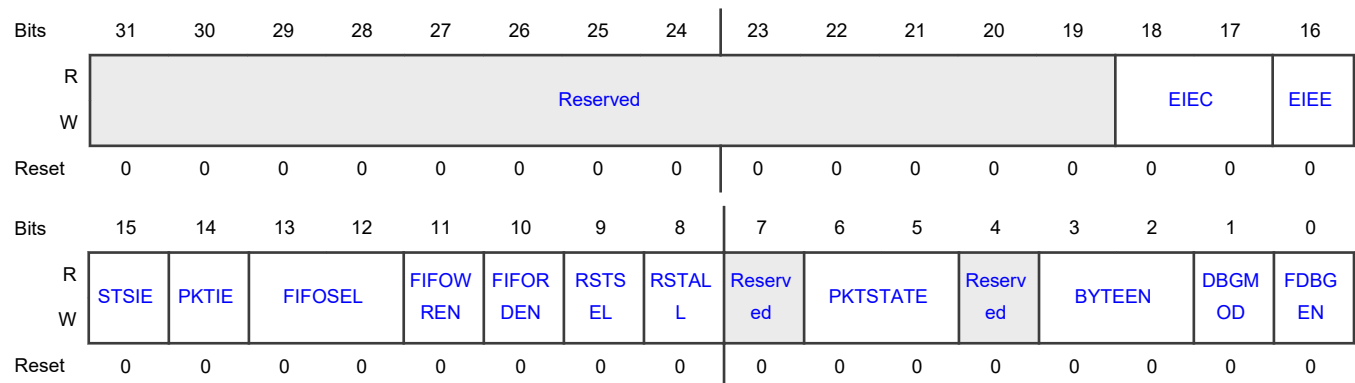
Offset

Register	Offset
MTL_DBG_CTL	C08h

Function

The FIFO Debug Access Control and Status register controls the operation mode of FIFO debug access.

Diagram



Fields

Field	Function
31-19 —	Reserved.
18-17 EIEC	ECC Inject Error Control for Tx, Rx and TSO memories When EIEE bit of this register is set, following are the errors inserted based on the value encoded in this field. 00b - Insert 1 bit error 01b - Insert 2 bit errors 10b - Insert 3 bit errors 11b - Insert 1 bit error in address field
16 EIEE	ECC Inject Error Enable for Tx, Rx and TSO memories When set, enables the ECC error injection feature. When reset, disables the ECC error injection feature. 0b - ECC Inject Error for Tx, Rx and TSO memories is disabled 1b - ECC Inject Error for Tx, Rx and TSO memories is enabled

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Field	Function
15 STSIE	Transmit Status Available Interrupt Status Enable When this bit is set, an interrupt is generated when Transmit status is available in slave mode. 0b - Transmit Packet Available Interrupt Status is disabled 1b - Transmit Packet Available Interrupt Status is enabled
14 PKTIE	Receive Packet Available Interrupt Status Enable When this bit is set, an interrupt is generated when EOP of received packet is written to the Rx FIFO. 0b - Receive Packet Available Interrupt Status is disabled 1b - Receive Packet Available Interrupt Status is enabled
13-12 FIFOSEL	FIFO Selected for Access This field indicates the FIFO selected for debug access: 00b - Tx FIFO 01b - Tx Status FIFO (only read access when SLVMOD is set) 10b - TSO FIFO (cannot be accessed when SLVMOD is set) 11b - Rx FIFO
11 FIFOWREN	FIFO Write Enable When this bit is set, it enables the Write operation on selected FIFO when FIFO Debug Access is enabled. This bit must not be written to 1 when FIFO Debug Access is not enabled, that is FDBGEN bit is 0. Access restriction applies. Self-cleared. Setting 0 clears. Setting 1 sets. 0b - FIFO Write is disabled 1b - FIFO Write is enabled
10 FIFORDEN	FIFO Read Enable When this bit is set, it enables the Read operation on selected FIFO when FIFO Debug Access is enabled. This bit must not be written to 1 when FIFO Debug Access is not enabled, that is FDBGEN bit is 0. Access restriction applies. Self-cleared. Setting 0 clears. Setting 1 sets. 0b - FIFO Read is disabled 1b - FIFO Read is enabled
9 RSTSEL	Reset Pointers of Selected FIFO When this bit is set, the pointers of the currently-selected FIFO are reset when FIFO Debug Access is enabled. This bit must not be written to 1 when FIFO Debug Access is not enabled, that is FDBGEN bit is 0. Access restriction applies. Self-cleared. Setting 0 clears. Setting 1 sets. 0b - Reset Pointers of Selected FIFO is disabled

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Field	Function
	1b - Reset Pointers of Selected FIFO is enabled
8 RSTALL	Reset All Pointers When this bit is set, the pointers of all FIFOs are reset when FIFO Debug Access is enabled. This bit must not be written to 1 when FIFO Debug Access is not enabled, that is FDBGEN bit is 0. Access restriction applies. Self-cleared. Setting 0 clears. Setting 1 sets. 0b - Reset All Pointers is disabled 1b - Reset All Pointers is enabled
7 —	Reserved.
6-5 PKTSTATE	Encoded Packet State This field is used to write the control information to the Tx FIFO or Rx FIFO. Tx FIFO: - 00: Packet Data - 01: Control Word - 10: SOP Data - 11: EOP Data Rx FIFO: - 00: Packet Data - 01: Normal Status - 10: Last Status - 11: EOP 00b - Packet Data 01b - Control Word/Normal Status 10b - SOP Data/Last Status 11b - EOP Data/EOP
4 —	Reserved.
3-2 BYTEEN	Byte Enables This field indicates the number of data bytes valid in the data register during Write operation. This is valid only when PKTSTATE is 2'b10 (EOP) and Tx FIFO or Rx FIFO is selected. 00b - Byte 0 valid 01b - Byte 0 and Byte 1 are valid

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Field	Function
	10b - Byte 0, Byte 1, and Byte 2 are valid 11b - All four bytes are valid
1 DBGMOD	Debug Mode Access to FIFO When this bit is set, it indicates that the current access to the FIFO is read, write, and debug access. In this mode, the following access types are allowed: - Read and Write access to Tx FIFO, TSO FIFO, and Rx FIFO - Read access is allowed to Tx Status FIFO. When this bit is reset, it indicates that the current access to the FIFO is slave access bypassing the DMA. In this mode, the following access are allowed: - Write access to the Tx FIFO - Read access to the Rx FIFO and Tx Status FIFO 0b - Debug Mode Access to FIFO is disabled 1b - Debug Mode Access to FIFO is enabled
0 FDBGEN	FIFO Debug Access Enable When this bit is set, it indicates that the debug mode access to the FIFO is enabled. When this bit is reset, it indicates that the FIFO can be accessed only through a master interface. 0b - FIFO Debug Access is disabled 1b - FIFO Debug Access is enabled

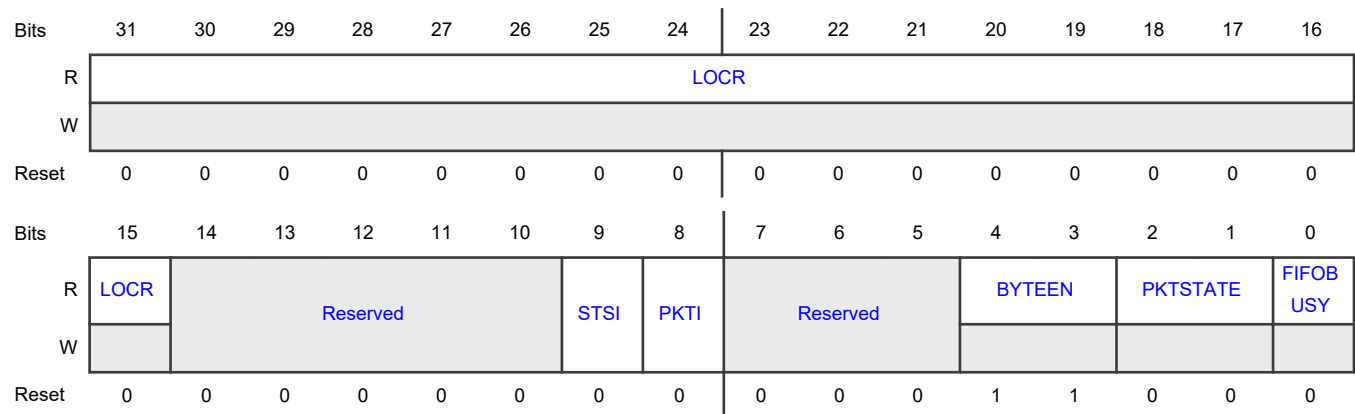
1.4.221 (MTL_DBG_STS)

Offset

Register	Offset
MTL_DBG_STS	C0Ch

Function

The FIFO Debug Status register contains the status of FIFO debug access.

Diagram**Fields**

Field	Function
31-15 LOCR	Remaining Locations in the FIFO Slave Access Mode: This field indicates the space available in selected FIFO. Debug Access Mode: This field contains the Write or Read pointer value of the selected FIFO during Write or Read operation, respectively. Reset: In single Tx Queue configurations, (DWC_EQOS_TXFIFO_SIZE/(DWC_EQOS_DATAWIDTH/8)), Otherwise 0000H
14-10 —	Reserved.
9 STSI	Transmit Status Available Interrupt Status When set, this bit indicates that the Slave mode Tx packet is transmitted, and the status is available in Tx Status FIFO. This bit is reset when 1 is written to this bit. 0b - Transmit Status Available Interrupt Status not detected 1b - Transmit Status Available Interrupt Status detected
8 PKTI	Receive Packet Available Interrupt Status When set, this bit indicates that MAC layer has written the EOP of received packet to the Rx FIFO. This bit is reset when 1 is written to this bit. 0b - Receive Packet Available Interrupt Status not detected 1b - Receive Packet Available Interrupt Status detected
7-5 —	Reserved.
4-3	Byte Enables

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Field	Function
BYTEEN	This field indicates the number of data bytes valid in the data register during Read operation. This is valid only when PKTSTATE is 2'b10 (EOP) and Tx FIFO or Rx FIFO is selected. 00b - Byte 0 valid 01b - Byte 0 and Byte 1 are valid 10b - Byte 0, Byte 1, and Byte 2 are valid 11b - All four bytes are valid
2-1 PKTSTATE	Encoded Packet State This field is used to get the control or status information of the selected FIFO. Tx FIFO: - 00: Packet Data - 01: Control Word - 10: SOP Data - 11: EOP Data Rx FIFO: - 00: Packet Data - 01: Normal Status - 10: Last Status - 11: EOP This field is applicable only for Tx FIFO and Rx FIFO during Read operation. 00b - Packet Data 01b - Control Word/Normal Status 10b - SOP Data/Last Status 11b - EOP Data/EOP
0 FIFOBUSY	FIFO Busy When set, this bit indicates that a FIFO operation is in progress in the MAC and content of the following fields is not valid: - All other fields of this register - All fields of the MTL_FIFO_Debug_Data register 0b - FIFO Busy not detected 1b - FIFO Busy detected

1.4.222 (MTL_FIFO_Debug_Data)

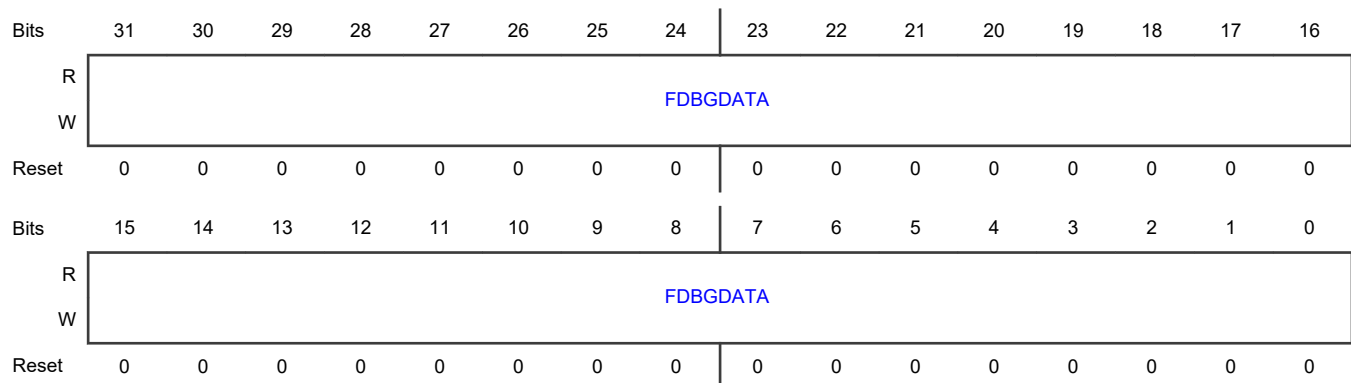
Offset

Register	Offset
MTL_FIFO_Debug_Data	C10h

Function

The FIFO Debug Data register contains the data to be written to or read from the FIFOs.

Diagram



Fields

Field	Function
31-0 FDBGDATA	FIFO Debug Data During debug or slave access write operation, this field contains the data to be written to the Tx FIFO, Rx FIFO, or TSO FIFO. During debug or slave access read operation, this field contains the data read from the Tx FIFO, Rx FIFO, TSO FIFO, or Tx Status FIFO.

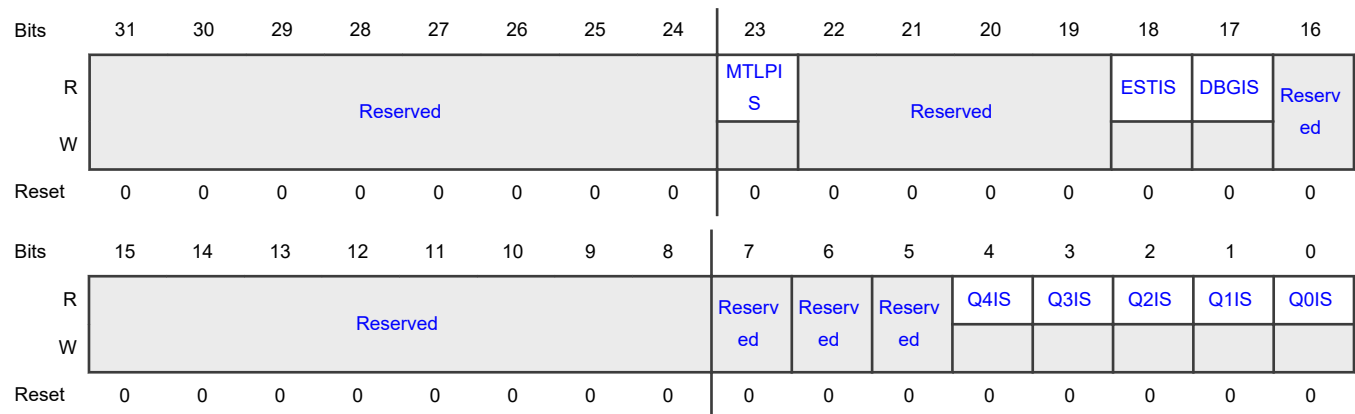
1.4.223 (MTL_Interrupt_Status)

Offset

Register	Offset
MTL_Interrupt_Status	C20h

Function

The software driver (application) reads this register during interrupt service routine or polling to determine the interrupt status of MTL queues and the MAC.

Diagram**Fields**

Field	Function
31-24 —	Reserved.
23 MTLPIS	MTL Rx Parser Interrupt Status This bit indicates that there is an interrupt from Rx Parser Block. To reset this bit, the application must read the MTL_Rxp_Interrupt_Status register to get the exact cause of the interrupt and clear its source. 0b - MTL Rx Parser Interrupt status not detected 1b - MTL Rx Parser Interrupt status detected
22-19 —	Reserved.
18 ESTIS	EST (TAS- 802.1Qbv) Interrupt Status This bit indicates an interrupt event during the operation of 802.1Qbv. To reset this bit, the application must clear the error/event that has caused the Interrupt. 0b - EST (TAS- 802.1Qbv) Interrupt status not detected 1b - EST (TAS- 802.1Qbv) Interrupt status detected
17 DBGIS	Debug Interrupt status This bit indicates an interrupt event during the slave access. To reset this bit, the application must read the FIFO Debug Access Status register to get the exact cause of the interrupt and clear its source. 0b - Debug Interrupt status not detected 1b - Debug Interrupt status detected
16 —	Reserved.
15-8	Reserved.

Table continues on the next page...

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Field	Function
—	
7 —	Reserved.
6 —	Reserved.
5 —	Reserved.
4 Q4IS	Queue 4 Interrupt status This bit indicates that there is an interrupt from Queue 4. To reset this bit, the application must read the MTL_Q4_Interrupt_Control_Status register to get the exact cause of the interrupt and clear its source. 0b - Queue 4 Interrupt status not detected 1b - Queue 4 Interrupt status detected
3 Q3IS	Queue 3 Interrupt status This bit indicates that there is an interrupt from Queue 3. To reset this bit, the application must read the MTL_Q3_Interrupt_Control_Status register to get the exact cause of the interrupt and clear its source. 0b - Queue 3 Interrupt status not detected 1b - Queue 3 Interrupt status detected
2 Q2IS	Queue 2 Interrupt status This bit indicates that there is an interrupt from Queue 2. To reset this bit, the application must read the MTL_Q2_Interrupt_Control_Status register to get the exact cause of the interrupt and clear its source. 0b - Queue 2 Interrupt status not detected 1b - Queue 2 Interrupt status detected
1 Q1IS	Queue 1 Interrupt status This bit indicates that there is an interrupt from Queue 1. To reset this bit, the application must read the MTL_Q1_Interrupt_Control_Status register to get the exact cause of the interrupt and clear its source. 0b - Queue 1 Interrupt status not detected 1b - Queue 1 Interrupt status detected
0 Q0IS	Queue 0 Interrupt status This bit indicates that there is an interrupt from Queue 0. To reset this bit, the application must read Queue 0 Interrupt Control and Status register to get the exact cause of the interrupt and clear its source. 0b - Queue 0 Interrupt status not detected 1b - Queue 0 Interrupt status detected

1.4.224 (MTL_RxQ_DMA_Map0)

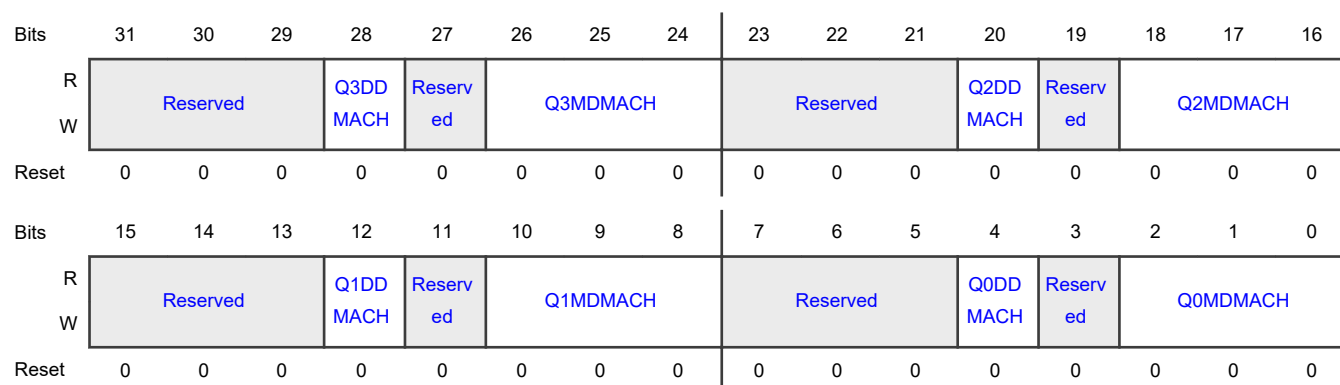
Offset

Register	Offset
MTL_RxQ_DMA_Map0	C30h

Function

The Receive Queue and DMA Channel Mapping 0 register is reserved in EQOS-CORE and EQOS-MTL configurations.

Diagram



Fields

Field	Function
31-29 —	Reserved.
28 Q3DDMACH	Queue 3 Enabled for Dynamic (per packet) DMA Channel Selection When set, this bit indicates that the packets received in Queue 3 are routed to a particular DMA channel as decided in the MAC Receiver based on the DMA channel number programmed in the L3-L4 filter registers, or the Ethernet DA address. When reset, this bit indicates that the packets received in Queue 3 are routed to the DMA Channel programmed in the Q3MDMACH field (Bits[26:24]). 0b - Queue 3 disabled for DA-based DMA Channel Selection 1b - Queue 3 enabled for DA-based DMA Channel Selection
27 —	Reserved.
26-24 Q3MDMACH	Queue 3 Mapped to DMA Channel This field controls the routing of the received packet in Queue 3 to the DMA channel: - 000: DMA Channel 0

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Field	Function
	- 001: DMA Channel 1 - 010: DMA Channel 2 - 011: DMA Channel 3 - 100: DMA Channel 4 - 101: DMA Channel 5 - 110: DMA Channel 6 - 111: DMA Channel 7 This field is valid when the Q3DDMACH field is reset. Note: The width of this field depends on the number of RX DMA channels and not all the values might be valid in some configurations. For example, if the number of RX DMA channels selected is 2, only 000 and 001 are valid, the others are reserved
23-21 —	Reserved.
20 Q2DDMACH	Queue 2 Enabled for DA-based DMA Channel Selection When set, this bit indicates that the packets received in Queue 2 are routed to a particular DMA channel as decided in the MAC Receiver based on the DMA channel number programmed in the L3-L4 filter registers, or the Ethernet DA address. When reset, this bit indicates that the packets received in Queue 2 are routed to the DMA Channel programmed in the Q2MDMACH field (Bits[18:16]). 0b - Queue 2 disabled for DA-based DMA Channel Selection 1b - Queue 2 enabled for DA-based DMA Channel Selection
19 —	Reserved.
18-16 Q2MDMACH	Queue 2 Mapped to DMA Channel This field controls the routing of the received packet in Queue 2 to the DMA channel: - 000: DMA Channel 0 - 001: DMA Channel 1 - 010: DMA Channel 2 - 011: DMA Channel 3 - 100: DMA Channel 4 - 101: DMA Channel 5 - 110: DMA Channel 6 - 111: DMA Channel 7 This field is valid when the Q2DDMACH field is reset.

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Field	Function
15-13 —	Reserved.
12 Q1DDMACH	Queue 1 Enabled for DA-based DMA Channel Selection When set, this bit indicates that the packets received in Queue 1 are routed to a particular DMA channel as decided in the MAC Receiver based on the DMA channel number programmed in the L3-L4 filter registers, or the Ethernet DA address. When reset, this bit indicates that the packets received in Queue 1 are routed to the DMA Channel programmed in the Q1MDMACH field (Bits[10:8]). 0b - Queue 1 disabled for DA-based DMA Channel Selection 1b - Queue 1 enabled for DA-based DMA Channel Selection
11 —	Reserved.
10-8 Q1MDMACH	Queue 1 Mapped to DMA Channel This field controls the routing of the received packet in Queue 1 to the DMA channel: - 000: DMA Channel 0 - 001: DMA Channel 1 - 010: DMA Channel 2 - 011: DMA Channel 3 - 100: DMA Channel 4 - 101: DMA Channel 5 - 110: DMA Channel 6 - 111: DMA Channel 7 This field is valid when the Q1DDMACH field is reset. The width of this field depends on the number of RX DMA channels and not all the values might be valid in some configurations. For example, if the number of RX DMA channels selected is 2, only 000 and 001 are valid, the other bits are reserved.
7-5 —	Reserved.
4 Q0DDMACH	Queue 0 Enabled for DA-based DMA Channel Selection When set, this bit indicates that the packets received in Queue 0 are routed to a particular DMA channel as decided in the MAC Receiver based on the DMA channel number programmed in the L3-L4 filter registers, or the Ethernet DA address. When reset, this bit indicates that the packets received in Queue 0 are routed to the DMA Channel programmed in the Q0MDMACH field.

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Field	Function
	0b - Queue 0 disabled for DA-based DMA Channel Selection 1b - Queue 0 enabled for DA-based DMA Channel Selection
3 —	Reserved.
2-0 Q0MDMACH	Queue 0 Mapped to DMA Channel This field controls the routing of the packet received in Queue 0 to the DMA channel: - 000: DMA Channel 0 - 001: DMA Channel 1 - 010: DMA Channel 2 - 011: DMA Channel 3 - 100: DMA Channel 4 - 101: DMA Channel 5 - 110: DMA Channel 6 - 111: DMA Channel 7 This field is valid when the Q0DDMACH field is reset. The width of this field depends on the number of RX DMA channels and not all the values might be valid in some configurations. For example, if the number of RX DMA channels selected is 2, only 000 and 001 are valid, the other bits are reserved.

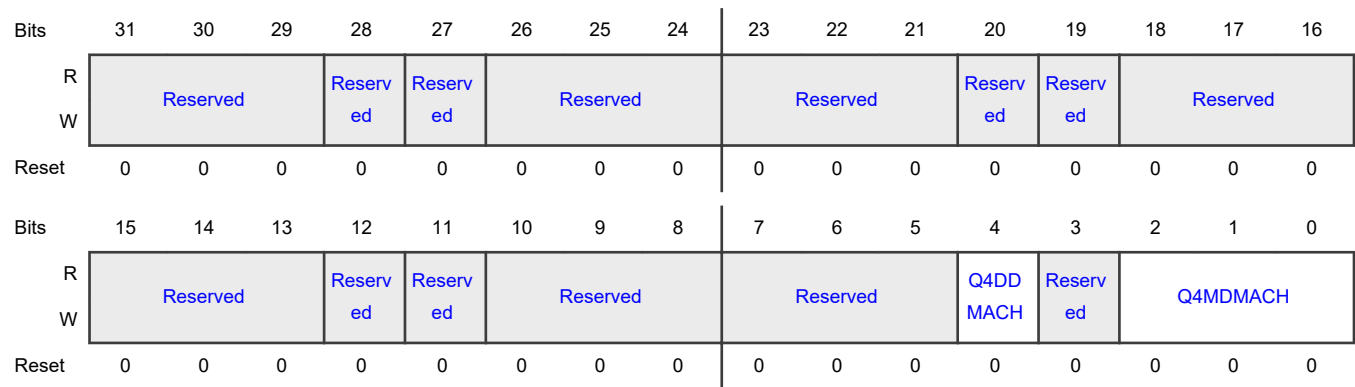
1.4.225 (MTL_RxQ_DMA_Map1)

Offset

Register	Offset
MTL_RxQ_DMA_Map1	C34h

Function

The Receive Queue and DMA Channel Mapping 1 register is reserved in EQOS-CORE and EQOS-MTL configurations.

Diagram**Fields**

Field	Function
31-29 —	Reserved.
28 —	Reserved.
27 —	Reserved.
26-24 —	Reserved.
23-21 —	Reserved.
20 —	Reserved.
19 —	Reserved.
18-16 —	Reserved.
15-13 —	Reserved.
12 —	Reserved.

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Field	Function
11 —	Reserved.
10-8 —	Reserved.
7-5 —	Reserved.
4 Q4DDMACH	Queue 4 Enabled for DA-based DMA Channel Selection When set, this bit indicates that the packets received in Queue 4 are routed to a particular DMA channel as decided in the MAC Receiver based on the DMA channel number programmed in the L3-L4 filter registers, or the Ethernet DA address. When reset, this bit indicates that the packets received in Queue 4 are routed to the DMA Channel programmed in the Q4MDMACH field. 0b - Queue 4 disabled for DA-based DMA Channel Selection 1b - Queue 4 enabled for DA-based DMA Channel Selection
3 —	Reserved.
2-0 Q4MDMACH	Queue 4 Mapped to DMA Channel This field controls the routing of the packet received in Queue 4 to the DMA channel: - 000: DMA Channel 0 - 001: DMA Channel 1 - 010: DMA Channel 2 - 011: DMA Channel 3 - 100: DMA Channel 4 - 101: DMA Channel 5 - 110: DMA Channel 6 - 111: DMA Channel 7 This field is valid when the Q4DDMACH field is reset. The width of this field depends on the number of RX DMA channels and not all the values might be valid in some configurations. For example, if the number of RX DMA channels selected is 2, only 000 and 001 are valid, the other bits are reserved.

1.4.226 (MTL_TBS_CTRL)

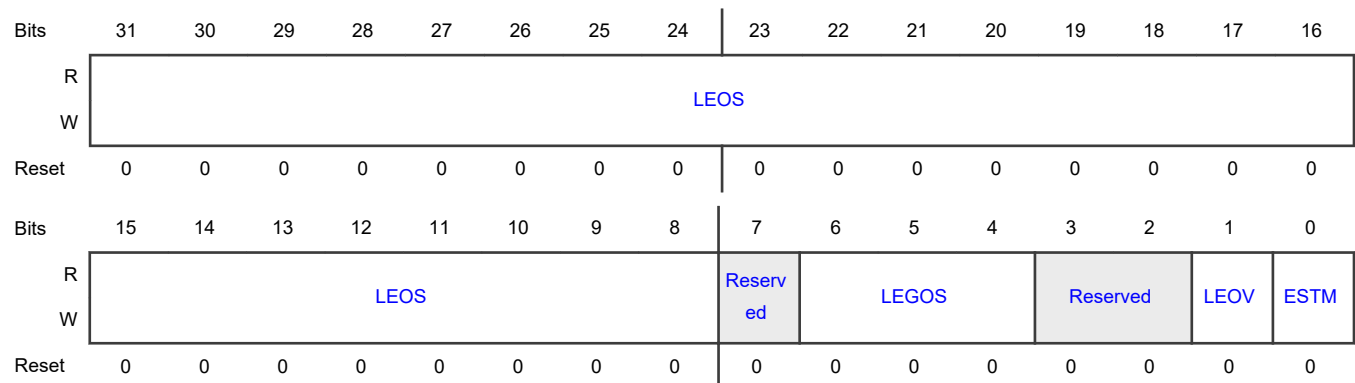
Offset

Register	Offset
MTL_TBS_CTRL	C40h

Function

This register controls the operation of Time Based Scheduling.

Diagram



Fields

Field	Function
31-8 LEOS	Launch Expiry Offset The value in units of 256 nanoseconds that has to be added to the Launch time to compute the Launch Expiry time. Value valid only when LEOV is set. Max value: 999,999,999 ns, additionally should be smaller than CTR-1 value when ESTM mode is set since this value is a modulo CTR value.
7 —	Reserved.
6-4 LEGOS	Launch Expiry GSN Offset The number GSN slots that has to be added to the Launch GSN to compute the Launch Expiry time. Value valid only when LEOV is set.
3-2 —	Reserved.
1 LEOV	Launch Expiry Offset Valid When set indicates the LEOS field is valid. When not set, indicates the Launch Expiry Offset is not valid and the MTL must not check for Launch expiry time.

Table continues on the next page...

Table continued from the previous page...

Field	Function
	0b - LEOS field is invalid 1b - LEOS field is valid
0 ESTM	EST offset Mode When this bit is set, the Launch Time value used in Time Based Scheduling is interpreted as an EST offset value and is added to the Base Time Register (BTR) of the current list. When reset, the Launch Time value is used as an absolute value that should be compared with the System time [39:8]. 0b - EST offset Mode is disabled 1b - EST offset Mode is enabled

1.4.227 (MTL_EST_Control)

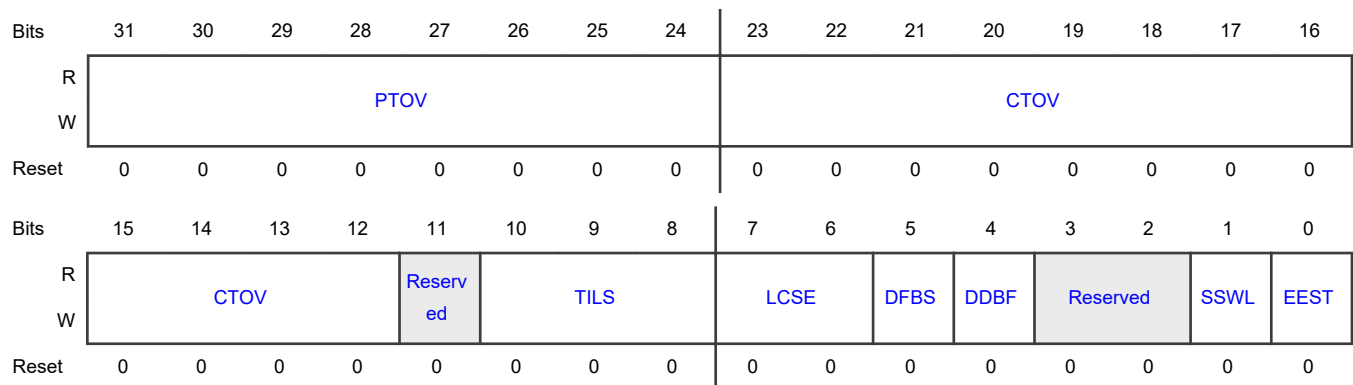
Offset

Register	Offset
MTL_EST_Control	C50h

Function

This register controls the operation of Enhancements to Scheduled Transmission (IEEE802.1Qbv).

Diagram



Fields

Field	Function
31-24 PTOV	PTP Time Offset Value

Table continues on the next page...

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Field	Function
	The value of PTP Clock period multiplied by 6 in nanoseconds. This value is needed to avoid transmission overruns at the beginning of the installation of a new GCL.
23-12 CTOV	Current Time Offset Value Provides a 12 bit time offset value in nano second that is added to the current time to compensate for all the implementation pipeline delays such as the CDC sync delay, buffering delays, data path delays etc. This offset helps to ensure that the impact of gate controls is visible on the line exactly at the pre-determined schedule (or as close to the schedule as possible).
11 —	Reserved.
10-8 TILS	Time Interval Left Shift Amount This field provides the left shift amount for the programmed Time Interval values used in the Gate Control Lists. - 000: No left shift needed (equal to x1ns) - 001: Left shift TI by 1 bit (equal to x2ns) - 010: Left shift TI by 2 bits (equal to x4ns) - . - . - 100: Left shift TI by 7 bits (equal to x128ns) Based on the configuration one or more bits of this field should be treated as Reserved/Read-Only.
7-6 LCSE	Loop Count to report Scheduling Error Programmable number of GCL list iterations before reporting an HLBS error defined in EST_Status register. 00b - 4 iterations 01b - 8 iterations 10b - 16 iterations 11b - 32 iterations
5 DFBS	Drop Frames causing Scheduling Error When set frames reported to cause HOL Blocking due to not getting scheduled (HLBS field of EST_Status register) after 4,8,16,32 (based on LCSE field of this register) GCL iterations are dropped. 0b - Do not Drop Frames causing Scheduling Error 1b - Drop Frames causing Scheduling Error
4 DDBF	Do not Drop frames during Frame Size Error When set, frames are not be dropped during Head-of-Line blocking due to Frame Size Error (HLBF field of EST_Status register). 0b - Drop frames during Frame Size Error

Table continues on the next page...

Table continued from the previous page...

Field	Function
	1b - Do not Drop frames during Frame Size Error
3-2 —	Reserved.
1 SSWL	Switch to S/W owned list When set indicates that the software has programmed that list that it currently owns (SWOL) and the hardware should switch to the new list based on the new BTR. Hardware clears this bit when the switch to the SWOL happens to indicate the completion of the switch or when an BTR error (BTRE in Status register) is set. When BTRE is set this bit is cleared but SWOL is not updated as the switch was not successful. Access restriction applies. Setting 1 sets. Self-cleared. Setting 0 has no effect. 0b - Switch to S/W owned list is disabled 1b - Switch to S/W owned list is enabled
0 EEST	Enable EST When reset, the gate control list processing is halted and all gates are assumed to be in Open state. Should be set for the hardware to start processing the gate control lists. During the toggle from 0 to 1, the gate control list processing starts only after the SSWL bit is set. When DWC_EQOS_ASP_ECC is selected during the configuration, if any uncorrectable error is detected in the EST memory the hardware resets this bit and disables the EST function. 0b - EST is disabled 1b - EST is enabled

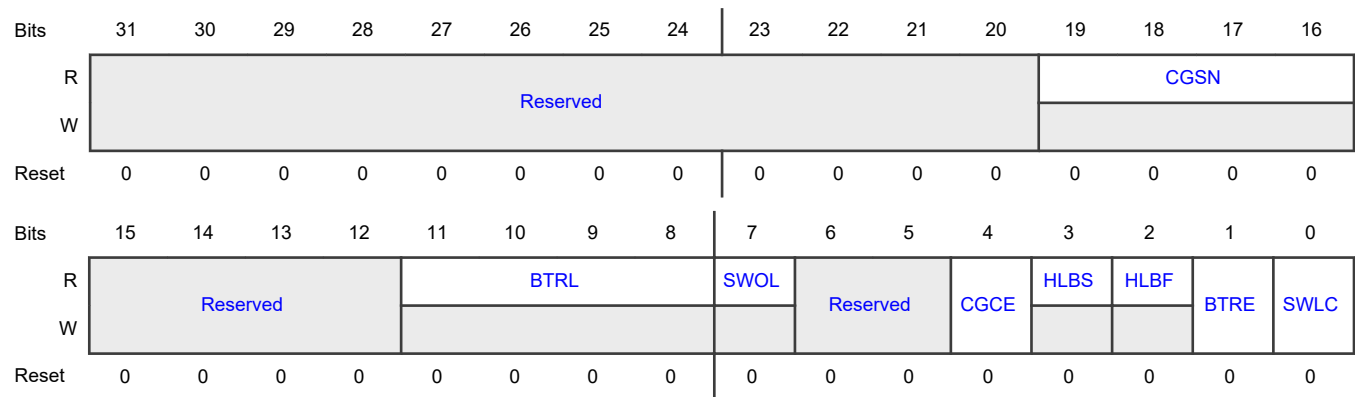
1.4.228 (MTL_EST_Status)

Offset

Register	Offset
MTL_EST_Status	C58h

Function

This register provides Status related to Enhancements to Scheduled Transmission (IEEE802.1Qbv).

Diagram**Fields**

Field	Function
31-20 —	Reserved.
19-16 CGSN	Current GCL Slot Number Indicates the slot number of the GCL list. Slot number is a modulo 16 count of the GCL List loops executed so far. Even if a new GCL list is installed, the count is incremental.
15-12 —	Reserved.
11-8 BTRL	BTR Error Loop Count Provides the minimum count (N) for which the equation $\text{Current Time} \leq \text{New BTR} + (N * \text{New Cycle Time})$ becomes true. N = "1111" indicates the iterations exceeded the value of 8 and the hardware was not able to update New BTR to be equal to or greater than Current Time. Software intervention is needed to update the New BTR. Value cleared when BTRE field of this register is cleared.
7 SWOL	S/W owned list When '0' indicates Gate control list number "0" is owned by software and when "1" indicates the Gate Control list "1" is owned by the software. Any reads/writes by the software (using indirect access via GCL_Control) is directed to the list indicated by this value by default. The inverse of this value is treated as HWOL. R/W operations performed by hardware are directed to the list pointed by HWOL by default. 0b - Gate control list number "0" is owned by software 1b - Gate control list number "1" is owned by software
6-5 —	Reserved.
4	Constant Gate Control Error

Table continues on the next page...

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Field	Function
CGCE	This error occurs when the list length (LLR) is 1 and the Cycle Time (CTR) is less than or equal to the programmed Time Interval (TI) value after the optional Left Shifting. The above programming implies Gates are either always Closed or always Open based on the Gate Control values; the same effect can be achieved by other simpler (non TSN) programming mechanisms. Since the implementation does not support such a programming an error is reported. Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect. 0b - Constant Gate Control Error not detected 1b - Constant Gate Control Error detected
3 HLBS	Head-Of-Line Blocking due to Scheduling Set when the frame is not able to win arbitration and get scheduled even after 4 iterations of the GCL. Indicates to software a potential programming error. The one hot encoded values of the Queue Numbers that are not able to make progress are indicated in the MTL_EST_Sch_Error register. Bit cleared when MTL_EST_Sch_Error register is all zeros. 0b - Head-Of-Line Blocking due to Scheduling not detected 1b - Head-Of-Line Blocking due to Scheduling detected
2 HLBF	Head-Of-Line Blocking due to Frame Size Set when HOL Blocking is noticed on one or more Queues as a result of none of the Time Intervals of gate open in the GCL being greater than or equal to the duration needed for frame size (or frame fragment size when preemption is enabled) transmission. The one hot encoded Queue numbers that are experiencing HLBF are indicated in the MTL_EST_Frm_Size_Error register. Additionally, the first Queue number that experienced HLBF along with the frame size is captured in MTL_EST_Frm_Size_Capture register. Bit cleared when MTL_EST_Frame_Size_Error register is all zeros. 0b - Head-Of-Line Blocking due to Frame Size not detected 1b - Head-Of-Line Blocking due to Frame Size detected
1 BTRE	BTR Error When "1" indicates a programming error in the BTR of SWOL where the programmed value is less than current time. If the BTRL = "1111", SWOL is not updated and Software should reprogram the BTR to a value greater than current time and then set SSWL to reinitiate the switch to SWOL. Else if the value of BTRL < "1111", SWOL is updated and this field indicates the number of iterations (of + CycleTime) taken by hardware to update the BTR to a value greater than Current Time. Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect. 0b - BTR Error not detected 1b - BTR Error detected
0 SWLC	Switch to S/W owned list Complete When "1" indicates the hardware has successfully switched to the SWOL, and the SWOL bit has been updated to that effect. Cleared when the SSWL of EST_Control register transitions from 0 to 1, or on a software write. Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect.

Table continues on the next page...

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Field	Function
	0b - Switch to S/W owned list Complete not detected 1b - Switch to S/W owned list Complete detected

1.4.229 (MTL_EST_Sch_Error)

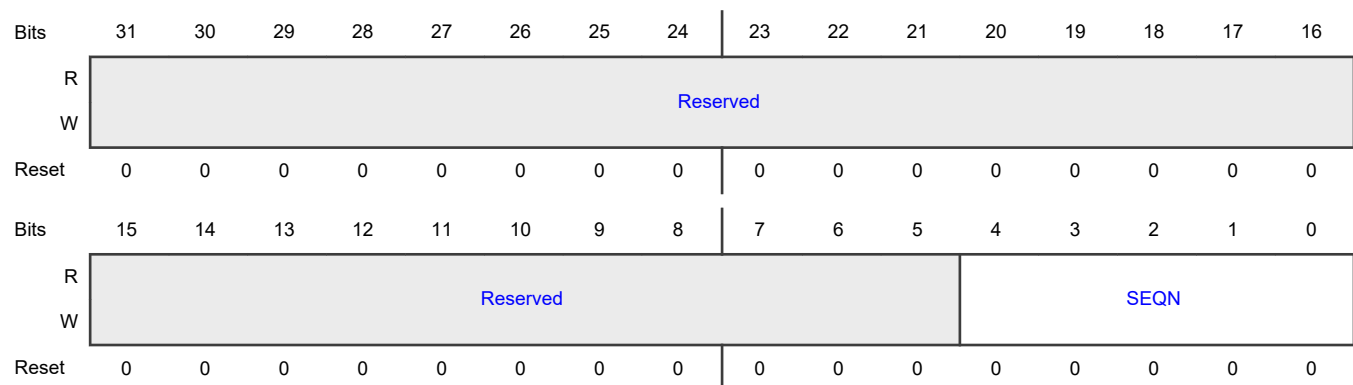
Offset

Register	Offset
MTL_EST_Sch_Error	C60h

Function

This register provides the One Hot encoded Queue Numbers that are having the Scheduling related error (timeout).

Diagram



Fields

Field	Function
31-5 —	Reserved.
4-0 SEQN	Schedule Error Queue Number The One Hot Encoded Queue Numbers that have experienced error/timeout described in HLBS field of status register. Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect.

1.4.230 (MTL_EST_Frm_Size_Error)

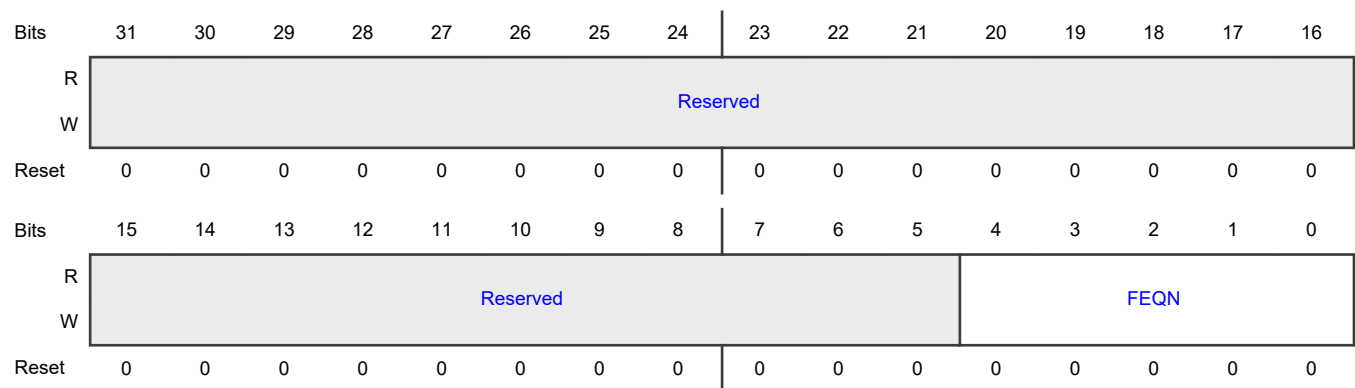
Offset

Register	Offset
MTL_EST_Frm_Size_Err or	C64h

Function

This register provides the One Hot encoded Queue Numbers that are having the Frame Size related error.

Diagram



Fields

Field	Function
31-5 —	Reserved.
4-0 FEQN	Frame Size Error Queue Number The One Hot Encoded Queue Numbers that have experienced error described in HLBFF field of status register. Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect.

1.4.231 (MTL_EST_Frm_Size_Capture)

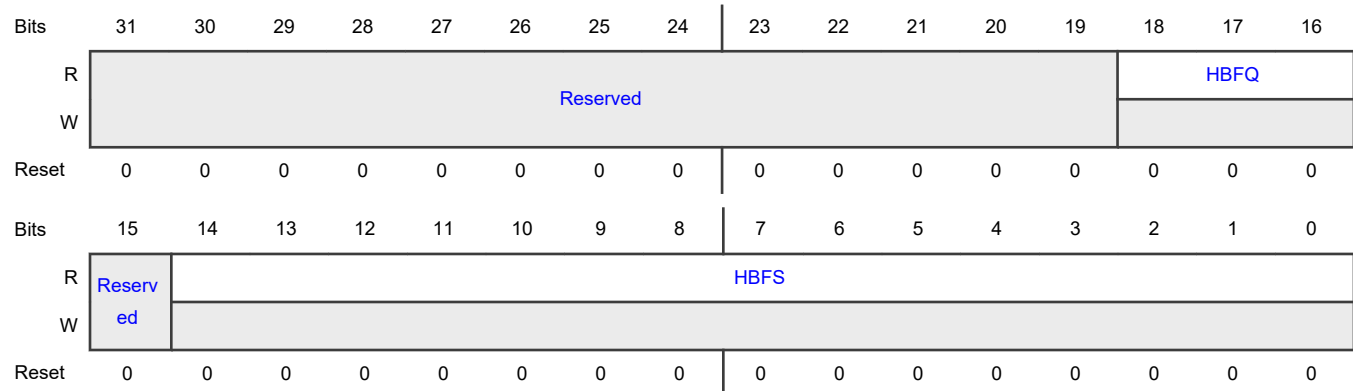
Offset

Register	Offset
MTL_EST_Frm_Size_Ca pture	C68h

Function

This register captures the Frame Size and Queue Number of the first occurrence of the Frame Size related error. Up on clearing it captures the data of immediate next occurrence of a similar error.

Diagram



Fields

Field	Function
31-19 —	Reserved.
18-16 HBFQ	Queue Number of HLBF Captures the binary value of the of the first Queue (number) experiencing HLBF error (see HLBF field of status register). Value once written is not altered by any subsequent queue errors of similar nature. Once cleared the queue number of the next occurring HLBF error is captured. Width is based on the number of Tx Queues configured; remaining bits are Read-Only. Cleared when MTL_EST_Frm_Size_Error register is all zeros.
15 —	Reserved.
14-0 HBFS	Frame Size of HLBF Captures the Frame Size of the dropped frame related to queue number indicated in HBFQ field of this register. Contents of this register should be considered invalid, if this field is zero. Cleared when MTL_EST_Frm_Size_Error register is all zeros.

1.4.232 (MTL_EST_Intr_Enable)

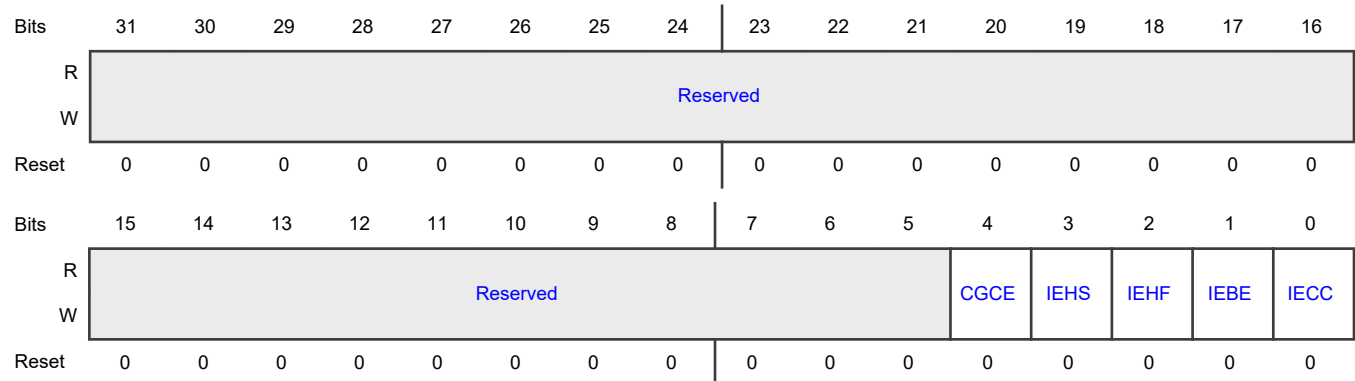
Offset

Register	Offset
MTL_EST_Intr_Enable	C70h

Function

This register implements the Interrupt Enable bits for the various events that generate an interrupt. Bit positions have a 1 to 1 correlation with the status bit positions in MTL_ETS_Status register.

Diagram



Fields

Field	Function
31-5 —	Reserved.
4 CGCE	Interrupt Enable for CGCE When set, generates interrupt when the Constant Gate Control Error occurs and is indicated in the status. When reset this event does not generate an interrupt 0b - Interrupt for CGCE is disabled 1b - Interrupt for CGCE is enabled
3 IEHS	Interrupt Enable for HLBS When set, generates interrupt when the Head-of-Line Blocking due to Scheduling issue and is indicated in the status. When reset this event does not generate an interrupt. 0b - Interrupt for HLBS is disabled 1b - Interrupt for HLBS is enabled
2 IEHF	Interrupt Enable for HLBF When set, generates interrupt when the Head-of-Line Blocking due to Frame Size error occurs and is indicated in the status. When reset this event does not generate an interrupt. 0b - Interrupt for HLBF is disabled 1b - Interrupt for HLBF is enabled
1 IEBE	Interrupt Enable for BTR Error When set, generates interrupt when the BTR Error occurs and is indicated in the status. When reset this event does not generate an interrupt.

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Field	Function
	0b - Interrupt for BTR Error is disabled 1b - Interrupt for BTR Error is enabled
0 IECC	Interrupt Enable for Switch List When set, generates interrupt when the configuration change is successful and the hardware has switched to the new list. When reset this event does not generate an interrupt. 0b - Interrupt for Switch List is disabled 1b - Interrupt for Switch List is enabled

1.4.233 (MTL_EST_GCL_Control)

Offset

Register	Offset
MTL_EST_GCL_Control	C80h

Function

This register provides the control information for reading/writing to the Gate Control lists.

Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	Reserved								ESTEIEC		ESTEIEE	ERR0	Reserved			
W	Reserved												Reserved			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	ADDR								Reserved		DBGB	DBGM	Reserved	GCRR	R1W0	SRWO
W	ADDR								Reserved		DBGB	DBGM	Reserved	GCRR	R1W0	SRWO
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Fields

Field	Function
31-24 —	Reserved.
23-22	ECC Inject Error Control for EST Memory

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Field	Function
ESTEIEC	When EIEE bit of this register is set, following are the errors inserted based on the value encoded in this field. This field is valid only if DWC_EQOS_ASP_ECC feature is selected during the configuration, else it is reserved. 00b - Insert 1 bit error 01b - Insert 2 bit errors 10b - Insert 3 bit errors 11b - Insert 1 bit error in address field
21 ESTEIEE	EST ECC Inject Error Enable When set along with EEST bit of MTL_EST_Control register, enables the ECC error injection feature. When reset, disables the ECC error injection feature. 0b - EST ECC Inject Error is disabled 1b - EST ECC Inject Error is enabled
20 ERR0	When set indicates the last write operation was aborted as software writes to GCL and GCL registers is prohibited when SSWL bit of MTL_EST_Control Register is set. Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect. 0b - ERR0 is disabled 1b - ERR1 is enabled
19-16 —	Reserved.
15-8 ADDR	Gate Control List Address: (GCLA when GCRR is "0"). Provides the address (row number) of the Gate Control List at which the R/W operation has to be performed. By default the Gate Control List pointed by SWOL of MTL_EST_Status is selected for R/W, however if the DBGm bit of this register is set, a debug mode access is given to R/W from DBGb. The width of this field is dependent on DWC_EQOS_EST_DEP; unused bits should be treated as read only. Gate Control list Related Registers Address: (GCRA when GCRR is "1"). By default the GCL related register set pointed by SWOL of MTL_EST_Status is selected for R/W, however if the DBGm bit of this register is set, a debug mode access is given to R/W from DBGb. Lower 3 bits are only used in this mode, higher order bits are treated as dont cares. - 000: BTR Low (31:0) - 001: BTR High (63:31) - 010: CTR Low (31:0) - 011: CTR High (39:32) - 100: TER (31:0) - 101: LLR (n:0) (where n is $(\log\{DWC_EQOS_EST_DEP\} / \log 2)$)

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Field	Function
	- Others: Reserved
7-6 —	Reserved.
5 DBGB	Debug Mode Bank Select When set to "0" indicates R/W in debug mode should be directed to Bank 0 (GCL0 and corresponding Time related registers). When set to "1" indicates R/W in debug mode should be directed to Bank 1 (GCL1 and corresponding Time related registers). This value is used when DBGGM is set and overrides by value of SWOL which is normally used. 0b - R/W in debug mode should be directed to Bank 0 1b - R/W in debug mode should be directed to Bank 1
4 DBGGM	Debug Mode When set to "1" indicates R/W in debug mode where the memory bank (for GCL and Time related registers) is explicitly provided by DBGGB value, when set to "0" SWOL bit is used to determine which bank to use. 0b - Debug Mode is disabled 1b - Debug Mode is enabled
3 —	Reserved.
2 GCRR	Gate Control Related Registers When set to "1" indicates the R/W access is for the GCL related registers (BTR, CTR, TER, LLR) whose address is provided by GCRA. When "0" indicates R/W should be directed to GCL from the address provided by GCLA. 0b - Gate Control Related Registers are disabled 1b - Gate Control Related Registers are enabled
1 R1W0	Read '1', Write '0': When set to '1': Read Operation When set to '0': Write Operation. 0b - Write Operation 1b - Read Operation
0 SRWO	Start Read/Write Op When set indicates a Read/Write Op has started and is in progress. When reset by hardware indicates the R/W Op has completed or an error has occurred (when bit 20 is set) Reads: Data can be read from MTL_EST_GCL_Data register after this bit is reset Writes: MTL_EST_GCL_Data should be programmed with write data before setting SRWO.

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Field	Function
	Access restriction applies. Setting 1 sets. Self-cleared. Setting 0 has no effect. 0b - Start Read/Write Op disabled 1b - Start Read/Write Op enabled

1.4.234 (MTL_EST_GCL_Data)

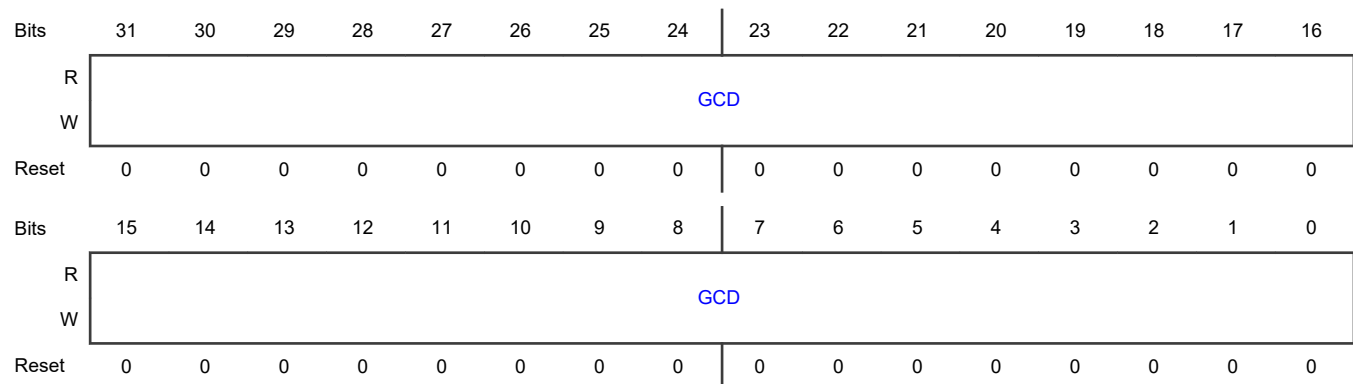
Offset

Register	Offset
MTL_EST_GCL_Data	C84h

Function

This register holds the read data or write data in case of reads and writes respectively.

Diagram



Fields

Field	Function
31-0	Gate Control Data
GCD	The data corresponding to the address selected in the GCL_Control register. Used for both Read and Write operations.

1.4.235 (MTL_FPE_CTRL_STS)

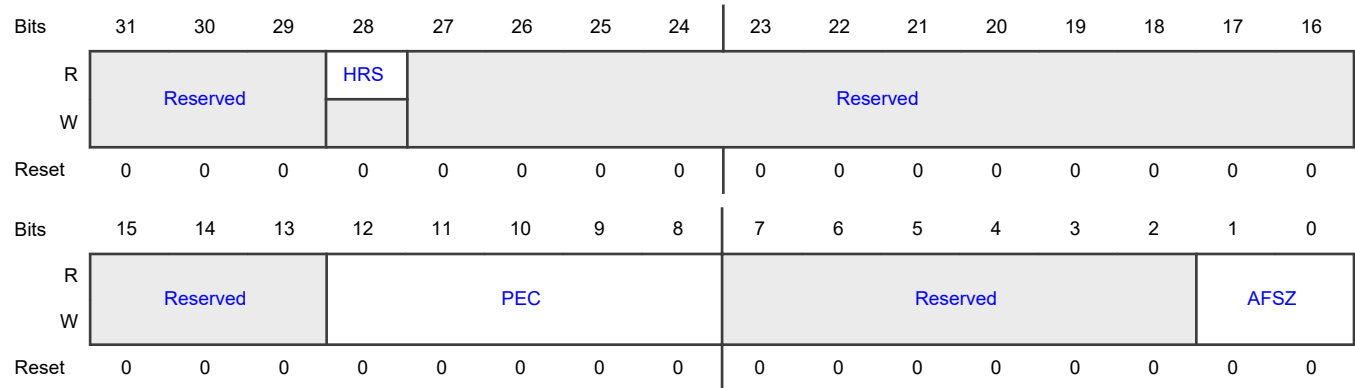
Offset

Register	Offset
MTL_FPE_CTRL_STS	C90h

Function

This register controls the operation of, and provides status for Frame Preemption (IEEE802.1Qbu/802.3br).

Diagram



Fields

Field	Function
31-29 —	Reserved.
28 HRS	Hold/Release Status - 1: Indicates a Set-and-Hold-MAC operation was last executed and the pMAC is in Hold State. - 0: Indicates a Set-and-Release-MAC operation was last executed and the pMAC is in Release State. 0b - Indicates a Set-and-Release-MAC operation was last executed and the pMAC is in Release State 1b - Indicates a Set-and-Hold-MAC operation was last executed and the pMAC is in Hold State
27-16 —	Reserved.
15-13 —	Reserved.
12-8 PEC	Preemption Classification When set indicates the corresponding Queue must be classified as preemptable, when '0' Queue is classified as express. When both EST (Qbv) and Preemption are enabled, Queue-0 is always assumed to be preemptable. When EST (Qbv) is enabled Queues categorized as preemptable here are always assumed to be in "Open" state in the Gate Control List.
7-2 —	Reserved.
1-0	Additional Fragment Size

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Field	Function
AFSZ	used to indicate, in units of 64 bytes, the minimum number of bytes over 64 bytes required in non-final fragments of preempted frames. The minimum non-final fragment size is (AFSZ +1) * 64 bytes

1.4.236 (MTL_FPE_Advance)

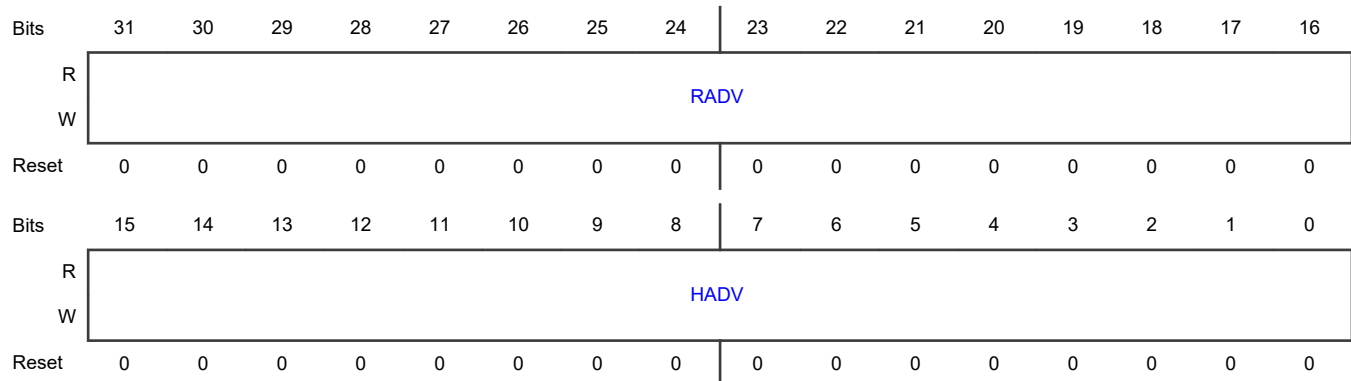
Offset

Register	Offset
MTL_FPE_Advance	C94h

Function

This register holds the Hold and Release Advance time.

Diagram



Fields

Field	Function
31-16 RADV	Release Advance The maximum time in nanoseconds that can elapse between issuing a RELEASE to the MAC and the MAC being ready to resume transmission of preemptable frames, in the absence of there being any express frames available for transmission.
15-0 HADV	Hold Advance The maximum time in nanoseconds that can elapse between issuing a HOLD to the MAC and the MAC ceasing to transmit any preemptable frame that is in the process of transmission or any preemptable frames that are queued for transmission.

1.4.237 (MTL_RXP_Control_Status)

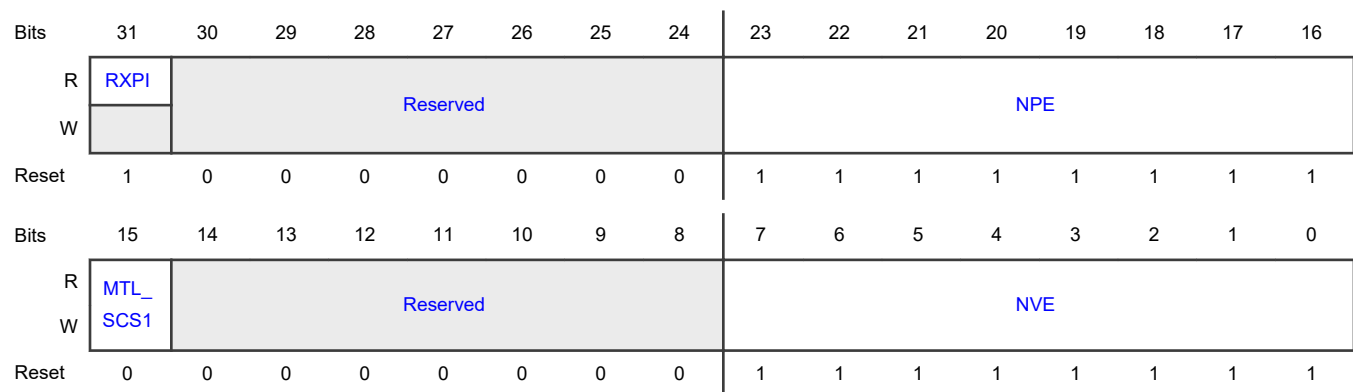
Offset

Register	Offset
MTL_RXP_Control_Status	CA0h

Function

The MTL_RXP_Control_Status register establishes the operating mode of Rx Parser and provides some status.

Diagram



Fields

Field	Function
31 RXPI	RX Parser in Idle state This status bit is set to 1 when the Rx parser is in Idle State and waiting for a new packet for processing. This bit is used as a handshake with software when parser gets disabled. After disabling, when bit is set then software can update the Rx parser instruction table. 0b - RX Parser not in Idle state 1b - RX Parser in Idle state
30-24 —	Reserved.
23-16 NPE	Number of parsable entries in the Instruction table This control indicates the number of parsable entries in the Instruction Memory. This is used in Rx parser logic to detect programming Error. In case number of parsed entries for a packet is more than this entry then NPEOVIS bit in the MTL_RXP_Interrupt_Control_Status register is set.
15	NXP Reserved, All the bits must be set to "0".

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Field	Function
MTL_SCS1	This field is reserved for NXP Internal use, and must always be set to "0" unless instructed by NXP. Setting any bit to "1" might cause unexpected behavior in the IP.
14-8 —	Reserved.
7-0 NVE	Number of valid entry address/index in the Instruction table This control indicates the number of valid entries address/index in the Instruction Memory (i.e. when NVE field in register=31, the maximum valid entry address is NVE+1 i.e. addresses/indices=0 to 32, or 33 entries). This is used in Rx parser logic to detect any programming Error. In case while parsing Table address (memory address) found to be more than this maximum valid entry address then NVEOVIS bit in the MTL_RXP_Interrupt_Control_Status register is set. Note: The minimum value of this should be 2.

1.4.238 (MTL_RXP_Interrupt_Control_Status)

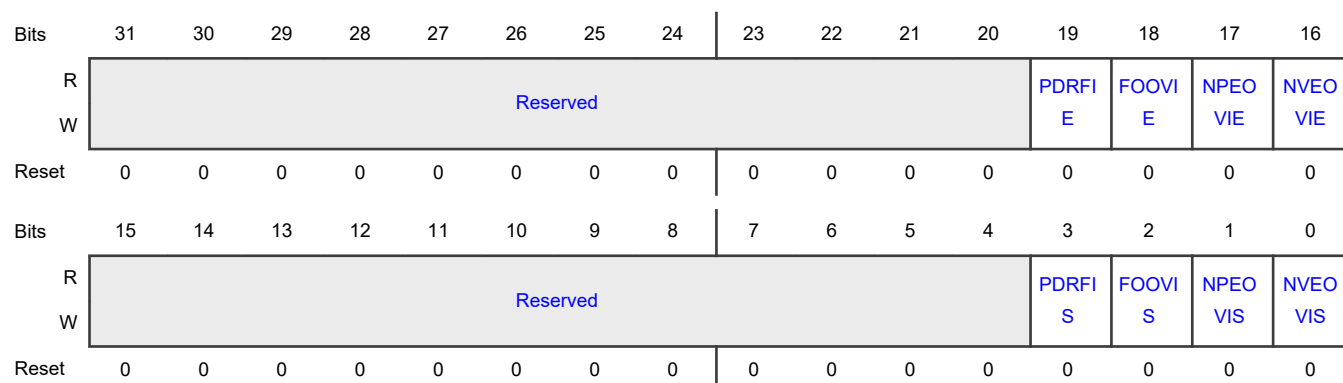
Offset

Register	Offset
MTL_RXP_Interrupt_Control_Status	CA4h

Function

The MTL_RXP_Interrupt_Control_Status registers provides enable control for the interrupts and provides interrupt status.

Diagram



Fields

Field	Function
31-20 —	Reserved.
19 PDRFIE	Packet Drop due to RF Interrupt Enable When this bit is set, the PDRFIS interrupt is enabled. When this bit is reset, the PDRFIS interrupt is disabled. 0b - Packet Drop due to RF Interrupt is disabled 1b - Packet Drop due to RF Interrupt is enabled
18 FOOVIE	Frame Offset Overflow Interrupt Enable When this bit is set, the FOOVIS interrupt is enabled. When this bit is reset, the FOOVIS interrupt is disabled. 0b - Frame Offset Overflow Interrupt is disabled 1b - Frame Offset Overflow Interrupt is enabled
17 NPEOVIE	Number of Parsable Entries Overflow Interrupt Enable When this bit is set, the NPEOVIS interrupt is enabled. When this bit is reset, the NPEOVIS interrupt is disabled. 0b - Number of Parsable Entries Overflow Interrupt is disabled 1b - Number of Parsable Entries Overflow Interrupt is enabled
16 NVEOVIE	Number of Valid Entries Overflow Interrupt Enable When this bit is set, the NVEOVIS interrupt is enabled. When this bit is reset, the NVEOVIS interrupt is disabled. 0b - Number of Valid Entries Overflow Interrupt is disabled 1b - Number of Valid Entries Overflow Interrupt is enabled
15-4 —	Reserved.
3 PDRFIS	Packet Dropped due to RF Interrupt Status If the Rx Parser result says to drop the packet by setting RF=1 in the instruction memory, then this bit is set to 1. This bit is cleared when the application writes 1 to this bit. Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect. 0b - Packet Dropped due to RF Interrupt Status not detected 1b - Packet Dropped due to RF Interrupt Status detected
2 FOOVIS	Frame Offset Overflow Interrupt Status While parsing if the Instruction table entry's 'Frame Offset' found to be more than EOF offset, then then this bit is set. This bit is cleared when the application writes 1 to this bit.

Table continues on the next page...

Table continued from the previous page...

Field	Function
	Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect. 0b - Frame Offset Overflow Interrupt Status not detected 1b - Frame Offset Overflow Interrupt Status detected
1 NPEOVIS	Number of Parsable Entries Overflow Interrupt Status While parsing a packet if the number of parsed entries found to be more than NPE[] (Number of Parseable Entries in MTL_RXP_Control register), then this bit is set to 1. This bit is cleared when the application writes 1 to this bit. Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect. 0b - Number of Parsable Entries Overflow Interrupt Status not detected 1b - Number of Parsable Entries Overflow Interrupt Status detected
0 NVEOVIS	Number of Valid Entry Address/Index Overflow Interrupt Status While parsing if the Instruction address found to be more than NVE (Number of Valid Entry Address/index in MTL_RXP_Control register), then this bit is set to 1. For example, when NVE field in register=31, the maximum valid entry address/index is NVE+1 i.e. 32 (addresses/indices=0 to 32, or 33 entries), so NVEOVIS is set when currently processed entry indicates next address is 33 or more i.e. 34th or later entries. This bit is cleared when the application writes 1 to this bit. Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect. 0b - Number of Valid Entries Overflow Interrupt Status not detected 1b - Number of Valid Entries Overflow Interrupt Status detected

1.4.239 (MTL_RXP_Drop_Cnt)

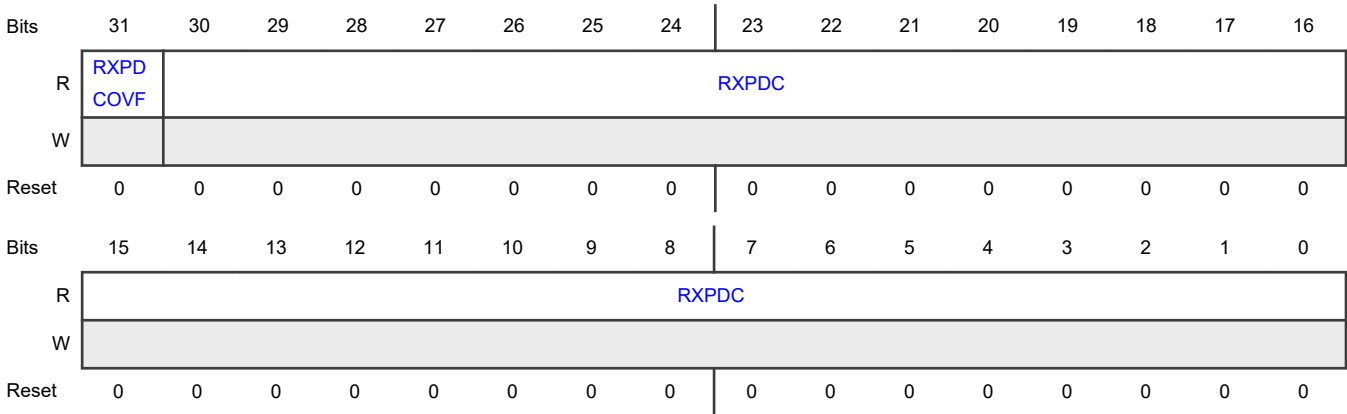
Offset

Register	Offset
MTL_RXP_Drop_Cnt	CA8h

Function

The MTL_RXP_Drop_Cnt register provides the drop count of Rx Parser initiated drops.

Diagram



Fields

Field	Function
31 RXPDCOVF	Rx Parser Drop Counter Overflow Bit When set, this bit indicates that the MTL_RXP_Drop_cnt (RXPDC) Counter field crossed the maximum limit. Access restriction applies. Clears on read. Self-set to 1 on internal event. 0b - Rx Parser Drop count overflow not occurred 1b - Rx Parser Drop count overflow occurred
30-0 RXPDC	Rx Parser Drop count This 31-bit counter is implemented whenever a Rx Parser Drops a packet due to RF =1. The counter is cleared when the register is read.

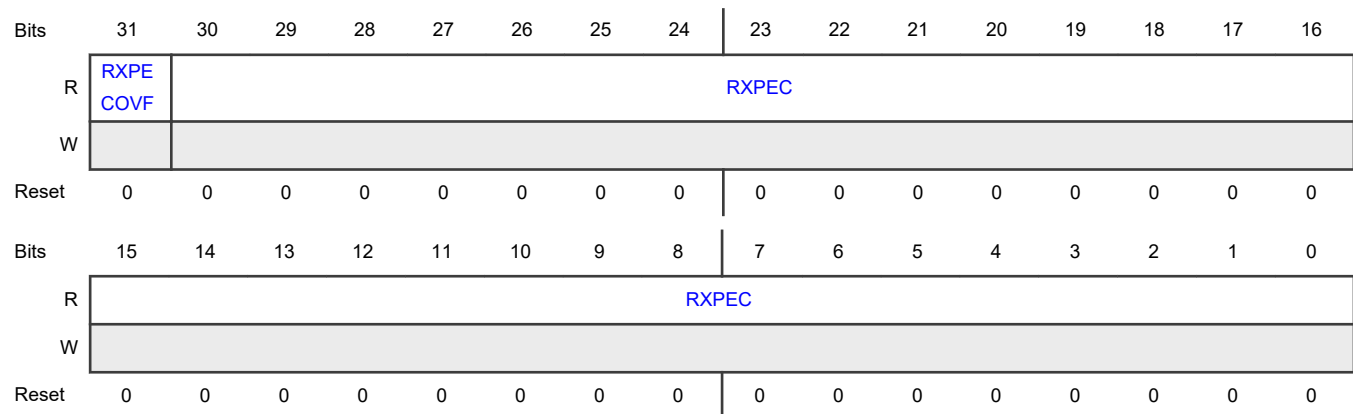
1.4.240 (MTL_RXP_Error_Cnt)

Offset

Register	Offset
MTL_RXP_Error_Cnt	CACH

Function

The MTL_RXP_Error_Cnt register provides the Rx Parser related error occurrence count.

Diagram**Fields**

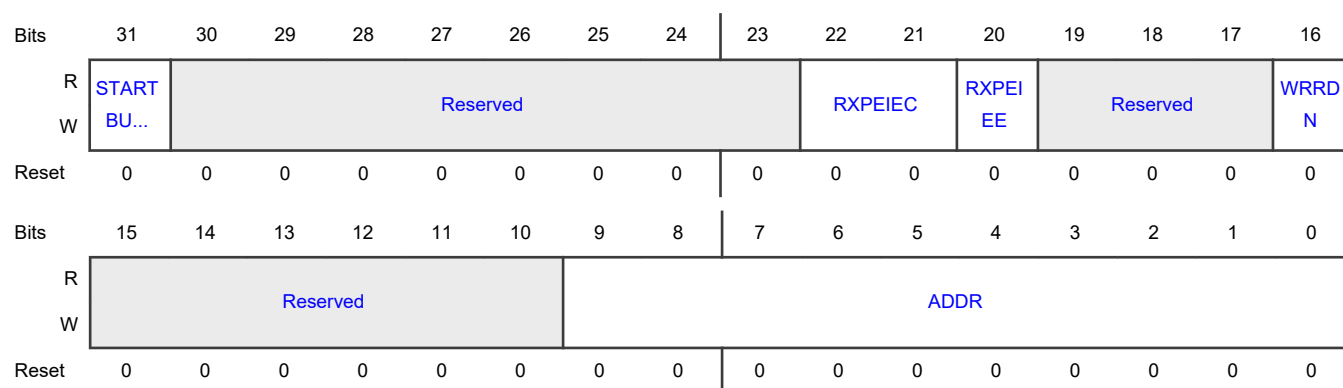
Field	Function
31 RXPECOVF	Rx Parser Error Counter Overflow Bit When set, this bit indicates that the MTL_RXP_Error_cnt (RXPEC) Counter field crossed the maximum limit. Access restriction applies. Clears on read. Self-set to 1 on internal event. 0b - Rx Parser Error count overflow not occurred 1b - Rx Parser Error count overflow occurred
30-0 RXPEC	Rx Parser Error count This 31-bit counter is implemented whenever a Rx Parser encounters following Error scenarios - Entry address >= NVE[] - Number Parsed Entries >= NPE[] - Entry address > EOF data entry address The counter is cleared when the register is read.

1.4.241 (MTL_RXP_Indirect_Acc_Control_Status)**Offset**

Register	Offset
MTL_RXP_Indirect_Acc_Control_Status	CB0h

Function

The MTL_RXP_Indirect_Acc_Control_Status register provides the Indirect Access control and status for Rx Parser memory.

Diagram**Fields**

Field	Function
31 STARTBUSY	FRP Instruction Table Access Busy When this bit is set to 1 by the software then it indicates to start the Read/Write operation from/to the Rx Parser Memory. Software should read this bit as 0 before issuing read or write request to access the Parser Memory Instructions. This bit when set to 1 indicates that hardware is busy until its gets cleared by hardware and software should not issue any read or write operation. 0b - hardware not busy 1b - hardware is busy (Read/Write operation from/to the Rx Parser Memory)
30-23 —	Reserved.
22-21 RXPEIEC	ECC Inject Error Control for Rx Parser Memory When EIEE bit of this register is set, following are the errors inserted based on the value encoded in this field. 00b - Insert 1 bit error 01b - Insert 2 bit errors 10b - Insert 3 bit errors 11b - Insert 1 bit error in address field
20 RXPEIEE	ECC Inject Error Enable for Rx Parser Memory When set, enables the ECC error injection feature. When reset, disables the ECC error injection feature. 0b - ECC Inject Error for Rx Parser Memory is disabled 1b - ECC Inject Error for Rx Parser Memory is enabled

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Field	Function
19-17 —	Reserved.
16 WRRDN	Read Write Control When this bit is set to 1 indicates the write operation to the Rx Parser Memory. When this bit is set to 0 indicates the read operation to the Rx Parser Memory. 0b - Read operation to the Rx Parser Memory 1b - Write operation to the Rx Parser Memory
15-10 —	Reserved.
9-0 ADDR	FRP Instruction Table Offset Address This field indicates the ADDR of the 32-bit entry in Rx parser instruction table. Each entry has 128-bit (4x32-bit words). The X depends on the configurable DWC_EQOS_FRP_ENTRIES If DWC_EQOS_FRP_ENTRIES == 256 , then X = 9 If DWC_EQOS_FRP_ENTRIES = 128 , then X = 8 If DWC_EQOS_FRP_ENTRIES = 64, then X = 7

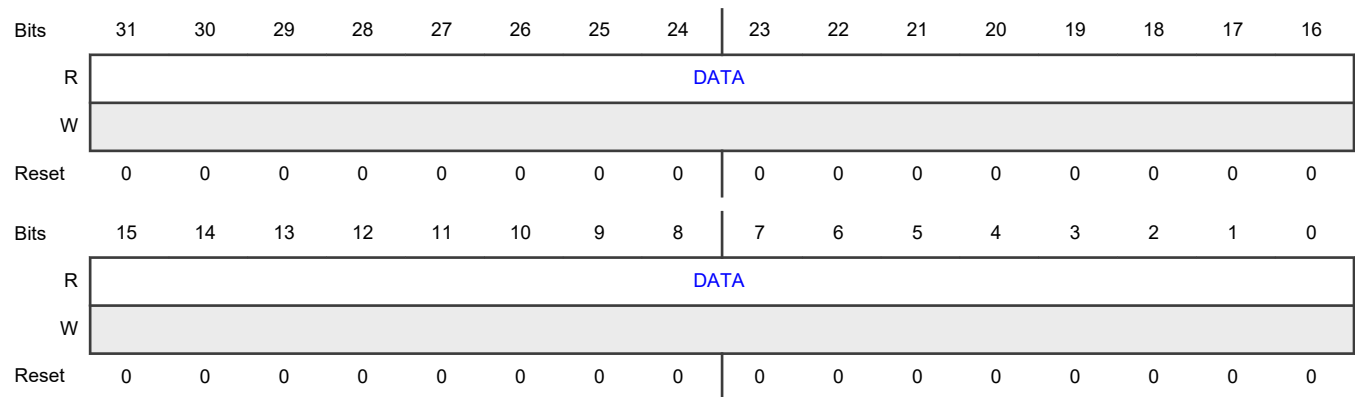
1.4.242 (MTL_RXP_Indirect_Acc_Data)

Offset

Register	Offset
MTL_RXP_Indirect_Acc_Data	CB4h

Function

The MTL_RXP_Indirect_Acc_Data registers holds the data associated to Indirect Access to Rx Parser memory.

Diagram**Fields**

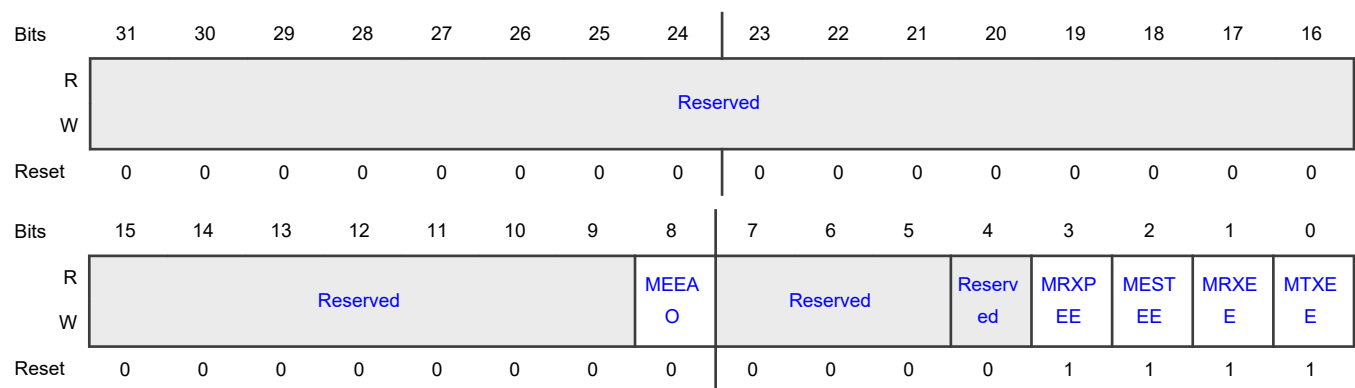
Field	Function
31-0 DATA	FRP Instruction Table Write/Read Data Software should write this register before issuing any write command. The hardware provides the read data from the Rx Parser Memory for read operation when STARTBUSY =0 after read command.

1.4.243 (MTL_ECC_Control)**Offset**

Register	Offset
MTL_ECC_Control	CC0h

Function

The MTL_ECC_Control register establishes the operating mode of ECC related to MTL memories.

Diagram

Fields

Field	Function
31-9 —	Reserved.
8 MEEAO	MTL ECC Error Address Status Over-ride When set, the following error address fields hold the last valid address where the error is detected. When reset, the following error address fields hold the first address where the error is detected. EUEAS/ECEAS of MTL_ECC_Err_Addr_Status register. 0b - MTL ECC Error Address Status Over-ride is disabled 1b - MTL ECC Error Address Status Over-ride is enabled
7-5 —	Reserved.
4 —	Reserved.
3 MRXPEE	MTL Rx Parser ECC Enable When set to 1, enables the ECC feature for Rx Parser memory. When set to zero, disables the ECC feature for Rx Parser memory. 0b - MTL Rx Parser ECC is disabled 1b - MTL Rx Parser ECC is enabled
2 MESTEE	MTL EST ECC Enable When set to 1, enables the ECC feature for EST memory. When set to zero, disables the ECC feature for EST memory. 0b - MTL EST ECC is disabled 1b - MTL EST ECC is enabled
1 MRXEE	MTL Rx FIFO ECC Enable When set to 1, enables the ECC feature for MTL Rx FIFO memory. When set to zero, disables the ECC feature for MTL Rx FIFO memory. 0b - MTL Rx FIFO ECC is disabled 1b - MTL Rx FIFO ECC is enabled
0 MTXEE	MTL Tx FIFO ECC Enable When set to 1, enables the ECC feature for MTL Tx FIFO memory. When set to zero, disables the ECC feature for MTL Tx FIFO memory. 0b - MTL Tx FIFO ECC is disabled 1b - MTL Tx FIFO ECC is enabled

1.4.244 (MTL_Safety_Interrupt_Status)

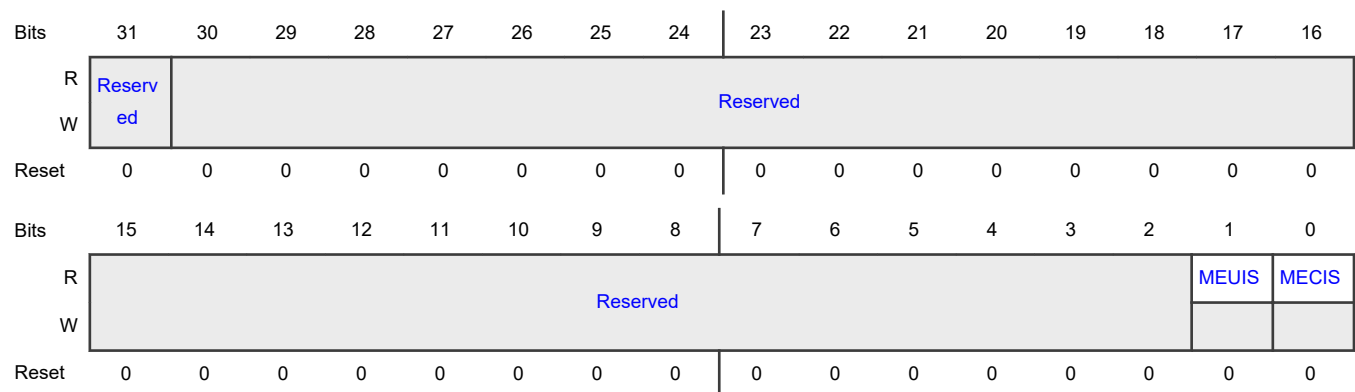
Offset

Register	Offset
MTL_Safety_Interrupt_Status	CC4h

Function

The MTL_Safety_Interrupt_Status registers provides Safety interrupt status.

Diagram



Fields

Field	Function
31 —	Reserved. 0b - MAC Safety Uncorrectable Interrupt Status not detected 1b - MAC Safety Uncorrectable Interrupt Status detected
30-2 —	Reserved.
1 MEUIS	MTL ECC Uncorrectable error Interrupt Status This bit indicates that an uncorrectable error interrupt event in the MTL ECC safety feature. To get the exact cause of the interrupt the application should read the MTL_ECC_Interrupt_Status register. 0b - MTL ECC Uncorrectable error Interrupt Status not detected 1b - MTL ECC Uncorrectable error Interrupt Status detected
0 MECIS	MTL ECC Correctable error Interrupt Status This bit indicates that a correctable error interrupt event in the MTL ECC safety feature. To get the exact cause of the interrupt the application should read the MTL_ECC_Interrupt_Status register.

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Field	Function
	0b - MTL ECC Correctable error Interrupt Status not detected 1b - MTL ECC Correctable error Interrupt Status detected

1.4.245 (MTL_ECC_Interrupt_Enable)

Offset

Register	Offset
MTL_ECC_Interrupt_Enable	CC8h

Function

The MTL_ECC_Interrupt_Enable register provides enable bits for the ECC interrupts.

Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	Reserved															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	Reserved			RPCEIE	Reserved				ECEIE	Reserved			RXCEIE	Reserved		TXCEIE
W																
Reset	0	0	0	1	0	0	0	1	0	0	0	1	0	0	0	1

Fields

Field	Function
31-13 —	Reserved.
12 RPCEIE	Rx Parser memory Correctable Error Interrupt Enable When set, generates an interrupt when an uncorrectable error is detected at the Rx Parser memory interface. It is indicated in RPCES status bit of MTL_ECC_Interrupt_Status register. When reset this event does not generates an interrupt. 0b - Rx Parser memory Correctable Error Interrupt is disabled 1b - Rx Parser memory Correctable Error Interrupt is enabled

Table continues on the next page...

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Field	Function
11-9 —	Reserved.
8 ECEIE	EST memory Correctable Error Interrupt Enable When set, generates an interrupt when a correctable error is detected at the MTL EST memory interface. It is indicated in the ECES bit of MTL_ECC_Interrupt_Status register. When reset this event does not generates an interrupt. 0b - EST memory Correctable Error Interrupt is disabled 1b - EST memory Correctable Error Interrupt is enabled
7-5 —	Reserved.
4 RXCEIE	Rx memory Correctable Error Interrupt Enable When set, generates an interrupt when a correctable error is detected at the MTL Rx memory interface. It is indicated in the RXCES bit of MTL_ECC_Interrupt_Status register. When reset this event does not generates an interrupt. 0b - Rx memory Correctable Error Interrupt is disabled 1b - Rx memory Correctable Error Interrupt is enabled
3-1 —	Reserved.
0 TXCEIE	Tx memory Correctable Error Interrupt Enable When set, generates an interrupt when a correctable error is detected at the MTL Tx memory interface. It is indicated in the TXCES bit of MTL_ECC_Interrupt_Status register. When reset this event does not generates an interrupt. 0b - Tx memory Correctable Error Interrupt is disabled 1b - Tx memory Correctable Error Interrupt is enabled

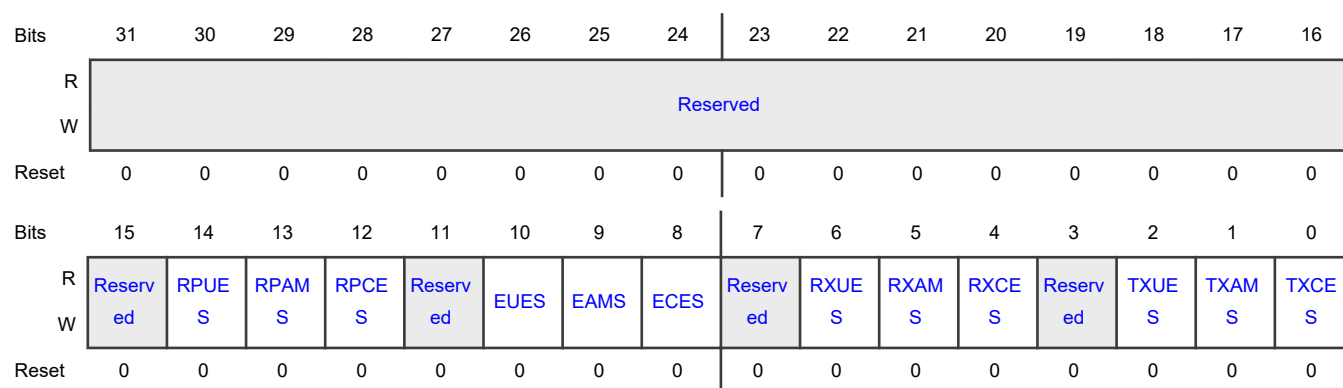
1.4.246 (MTL_ECC_Interrupt_Status)

Offset

Register	Offset
MTL_ECC_Interrupt_Status	CCCh

Function

The MTL_ECC_Interrupt_Status register provides MTL ECC Interrupt Status.

Diagram**Fields**

Field	Function
31-15 —	Reserved.
14 RPUES	Rx Parser memory Uncorrectable Error Status When set, indicates that an uncorrectable error is detected at Rx Parser memory interface. 0b - Rx Parser memory Uncorrectable Error Status not detected 1b - Rx Parser memory Uncorrectable Error Status detected
13 RPAMS	MTL Rx Parser memory Address Mismatch Status This bit when set indicates that address mismatch is found for address bus of Rx Parser memory. 0b - MTL Rx Parser memory Address Mismatch Status not detected 1b - MTL Rx Parser memory Address Mismatch Status detected
12 RPCES	MTL Rx Parser memory Correctable Error Status This bit when set indicates that correctable error is detected at RX Parser memory interface. 0b - MTL Rx Parser memory Correctable Error Status not detected 1b - MTL Rx Parser memory Correctable Error Status detected
11 —	Reserved.
10 EUES	MTL EST memory Uncorrectable Error Status When set, indicates that an uncorrectable error is detected at MTL EST memory interface. 0b - MTL EST memory Uncorrectable Error Status not detected 1b - MTL EST memory Uncorrectable Error Status detected
9	MTL EST memory Address Mismatch Status This bit when set indicates that address mismatch is found for address bus of MTL EST memory.

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Field	Function
EAMS	0b - MTL EST memory Address Mismatch Status not detected 1b - MTL EST memory Address Mismatch Status detected
8 ECES	MTL EST memory Correctable Error Status This bit when set indicates that correctable error is detected at the MTL EST memory. 0b - MTL EST memory Correctable Error Status not detected 1b - MTL EST memory Correctable Error Status detected
7 —	Reserved.
6 RXUES	MTL Rx memory Uncorrectable Error Status When set, indicates that an uncorrectable error is detected at the MTL Rx memory interface. 0b - MTL Rx memory Uncorrectable Error Status not detected 1b - MTL Rx memory Uncorrectable Error Status detected
5 RXAMS	MTL Rx memory Address Mismatch Status This bit when set indicates that address mismatch is found for address bus of the MTL Rx memory. 0b - MTL Rx memory Address Mismatch Status not detected 1b - MTL Rx memory Address Mismatch Status detected
4 RXCES	MTL Rx memory Correctable Error Status This bit when set indicates that correctable error is detected at the MTL Rx memory. 0b - MTL Rx memory correctable Error Status not detected 1b - MTL Rx memory correctable Error Status detected
3 —	Reserved.
2 TXUES	MTL Tx memory Uncorrectable Error Status When set, indicates that an uncorrectable error is detected at the MTL TX memory interface. 0b - MTL Tx memory Uncorrectable Error Status not detected 1b - MTL Tx memory Uncorrectable Error Status detected
1 TXAMS	MTL Tx memory Address Mismatch Status This bit when set indicates that address mismatch is found for address bus of the MTL Tx memory. 0b - MTL Tx memory Address Mismatch Status not detected 1b - MTL Tx memory Address Mismatch Status detected
0	MTL Tx memory Correctable Error Status

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Field	Function
TXCES	This bit when set indicates that a correctable error is detected at the MTL Tx memory. 0b - MTL Tx memory Correctable Error Status not detected 1b - MTL Tx memory Correctable Error Status detected

1.4.247 (MTL_ECC_Err_Sts_Rctl)

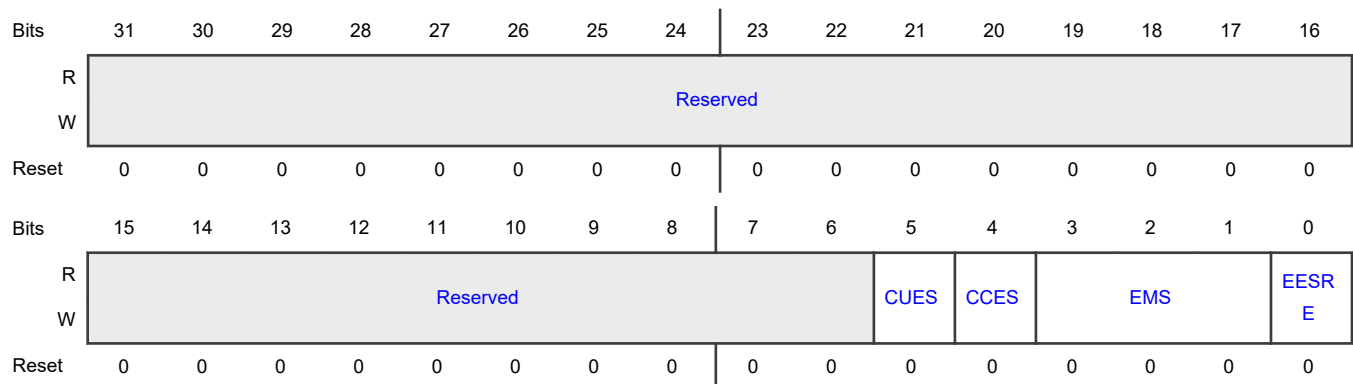
Offset

Register	Offset
MTL_ECC_Err_Sts_Rctl	CD0h

Function

The MTL_ECC_Err_Sts_Rctl register establishes the control for ECC Error status capture.

Diagram



Fields

Field	Function
31-6 —	Reserved.
5 CUES	Clear Uncorrectable Error Status When this bit is set along with EESRE bit of this register, based on the EMS field of this register, the respective memory's uncorrectable error address and uncorrectable error count values are cleared upon reading. Hardware resets this bit when all the error status values are cleared. 0b - Clear Uncorrectable Error Status not detected

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Field	Function
	1b - Clear Uncorrectable Error Status detected
4 CCES	Clear Correctable Error Status When this bit is set along with EESRE bit of this register, based on the EMS field of this register, the respective memory's correctable error address and correctable error count values are cleared upon reading. Hardware resets this bit when all the error status values are cleared. 0b - Clear Correctable Error Status not detected 1b - Clear Correctable Error Status detected
3-1 EMS	MTL ECC Memory Selection When EESRE bit of this register is set, this field indicates which memory's error status value to be read. The memory selection encoding is as described below. 000b - MTL Tx memory 001b - MTL Rx memory 010b - MTL EST memory 011b - MTL Rx Parser memory 100b - DMA TSO memory
0 EESRE	MTL ECC Error Status Read Enable When this bit is set, based on the EMS field of this register, the respective memory's error status values are captured as described: - The correctable and uncorrectable error count values are captured into MTL_ECC_Err_Cnt_Status register - The address location's of correctable and uncorrectable errors are captured into MTL_ECC_Err_Addr_Status register. Hardware resets this bit when all the status values are captured into the MTL_ECC_Err_Cnt_Status and MTL_ECC_Err_Addr_Status registers. 0b - MTL ECC Error Status Read is disabled 1b - MTL ECC Error Status Read is enabled

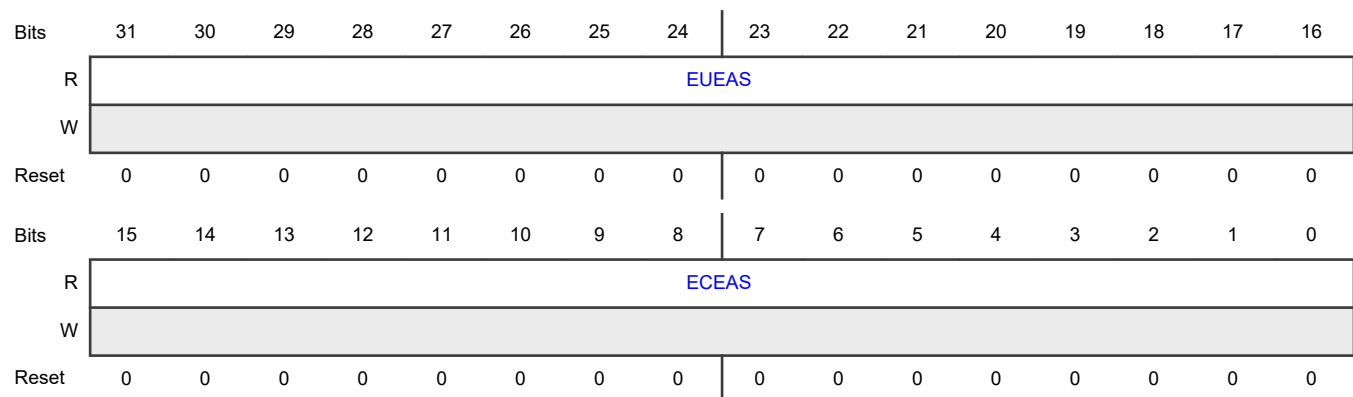
1.4.248 (MTL_ECC_Err_Addr_Status)

Offset

Register	Offset
MTL_ECC_Err_Addr_Status	CD4h

Function

The MTL_ECC_Err_Addr_Status register provides the memory addresses for the correctable and uncorrectable errors.

Diagram**Fields**

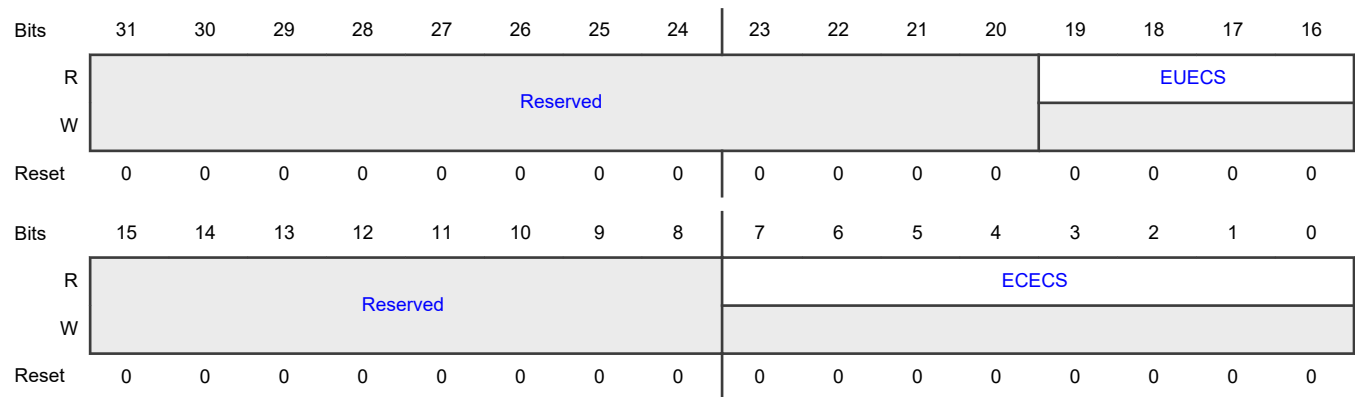
Field	Function
31-16 EUEAS	MTL ECC Uncorrectable Error Address Status Based on the EMS field of MTL_ECC_Err_Sts_Rctl register, this field holds the respective memory's address locations for which an uncorrectable error or address mismatch is detected. When MEEAO bit of MTL_ECC_Control register is set, this field holds the last valid address of memory for which either an uncorrectable error or an address mismatch is detected. When MEEAO bit of MTL_ECC_Control register is reset, this field holds the first address of the memory for which either an uncorrectable error or address mismatch is detected.
15-0 ECEAS	MTL ECC Correctable Error Address Status Based on the EMS field of MTL_ECC_Err_Sts_Rctl register, this field holds the respective memory's address locations for which a correctable error is detected. When MEEAO bit of MTL_ECC_Control register is set, this field holds the last valid address of memory for which correctable error or address mismatch is detected. When MEEAO bit of MTL_ECC_Control register is reset, this field holds the first address of the memory for which correctable error is detected.

1.4.249 (MTL_ECC_Err_Cntr_Status)**Offset**

Register	Offset
MTL_ECC_Err_Cntr_Status	CD8h

Function

The MTL_ECC_Err_Cntr_Status register provides ECC Error count for Correctable and uncorrectable errors.

Diagram**Fields**

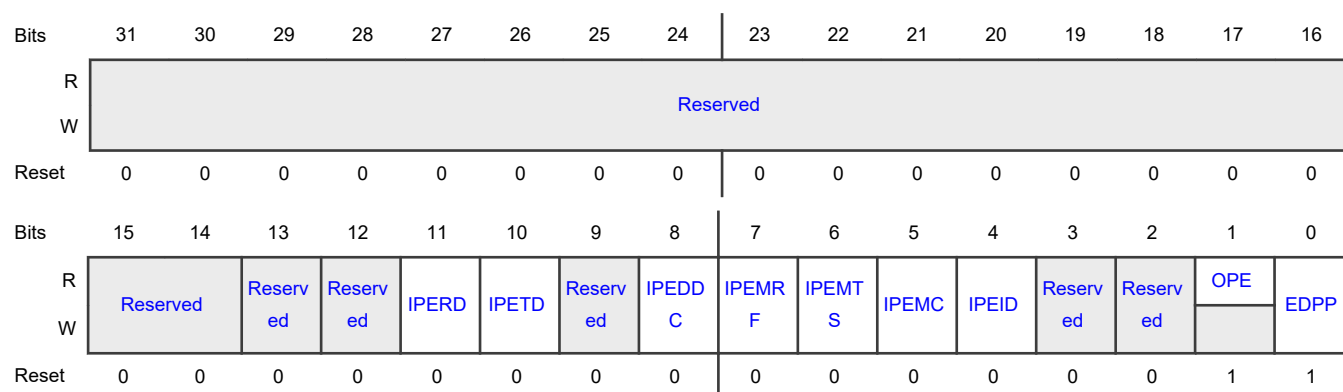
Field	Function
31-20 —	Reserved.
19-16 EUECS	MTL ECC Uncorrectable Error Counter Status Based on the EMS field of MTL_ECC_Err_Cntr_Rctl register, this field holds the respective memory's uncorrectable error count value.
15-8 —	Reserved.
7-0 ECECS	MTL ECC Correctable Error Counter Status Based on the EMS field of MTL_ECC_Err_Cntr_Rctl register, this field holds the respective memory's correctable error count value.

1.4.250 (MTL_DPP_Control)**Offset**

Register	Offset
MTL_DPP_Control	CE0h

Function

The MTL_DPP_Control establishes the operating mode of Data Parity protection and error injection.

Diagram**Fields**

Field	Function
31-14 —	Reserved.
13 —	Reserved.
12 —	Reserved.
11 IPERD	Insert Parity error in Rx write-back Descriptor parity generator When set to 1, parity bit of first valid data generated by the DMA Rx write-back descriptor parity generator (or at PG8 as shown in Receive data path parity protection diagram) is flipped. 0b - Insert Parity error in Rx write-back Descriptor parity generator is disabled 1b - Insert Parity error in Rx write-back Descriptor parity generator is enabled
10 IPETD	Insert Parity error in Tx write-back Descriptor parity generator When set to 1, parity bit of first valid data generated by the DMA Tx write-back descriptor parity generator (or at PG4 as shown in Transmit data path parity protection diagram) is flipped. Hardware clears this bit once respective parity bit is flipped. 0b - Insert Parity error in Tx write-back Descriptor parity generator is disabled 1b - Insert Parity error in Tx write-back Descriptor parity generator is enabled
9 —	Reserved.
8 IPEDDC	Insert Parity Error in DMA DTX Control word parity generator When set to 1, parity bit of first valid data generated by the DMA DTX Control word parity generator (or at PG2 as shown in Transmit data path parity protection diagram) is flipped.

Table continues on the next page...

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Field	Function
	Hardware clears this bit once respective parity bit is flipped. 0b - Insert Parity Error in DMA DTX Control word parity generator is disabled 1b - Insert Parity Error in DMA DTX Control word parity generator is enabled
7 IPEMRF	Insert Parity Error in MTL Rx FIFO read control parity generator When set to 1, parity bit of first valid data generated by the MTL Rx FIFO read control parity generator (or at PG7 as shown in Receive data path parity protection diagram) is flipped. Hardware clears this bit once respective parity bit is flipped. 0b - Insert Parity Error in MTL Rx FIFO read control parity generator is disabled 1b - Insert Parity Error in MTL Rx FIFO read control parity generator is enabled
6 IPEMTS	Insert Parity Error in MTL Tx Status parity generator When set to 1, parity bit of first valid data generated by the MTL Tx Status parity generator (or at PG6 as shown in Transmit data path parity protection diagram) is flipped. Hardware clears this bit once respective parity bit is flipped. 0b - Insert Parity Error in MTL Tx Status parity generator is disabled 1b - Insert Parity Error in MTL Tx Status parity generator is enabled
5 IPEMC	Insert Parity Error in MTL checksum parity generator When set to 1, parity bit of first valid data generated by the MTL checksum parity generator (or at PG5 as shown in Transmit data path parity protection diagram) is flipped. Hardware clears this bit once the respective parity bit is flipped. 0b - Insert Parity Error in MTL checksum parity generator is disabled 1b - Insert Parity Error in MTL checksum parity generator is enabled
4 IPEID	Insert Parity Error in Interface Data parity generator When set to 1, parity bit of first valid input data generated by the Interface data parity generator (or at PG1 as shown in Transmit data path parity protection diagram) is flipped. Following are the input data bus on which parity bits are generated based on configuration selected In AHB Config, hrdata_i In AXI config, rdata_m_i In DMA Config, mdc_rdata_i In MTL Config, ati_data_i Hardware clears this bit once the respective parity bit is flipped. 0b - Insert Parity Error in Interface Data parity generator is disabled 1b - Insert Parity Error in Interface Data parity generator is enabled
3	Reserved.

Table continues on the next page...

Table continued from the previous page...

Field	Function
—	
2 —	Reserved.
1 OPE	Odd Parity Enable When set to 1, enables odd parity protection on all the external interfaces and when set to 0, enables even parity protection on all the external interfaces. 0b - Odd Parity is disabled 1b - Odd Parity is enabled
0 EDPP	Enable Data path Parity Protection When set to 1, enables the parity protection for EQOS datapath by generating and checking the parity on EQOS datapath. When set to 0, disables the parity protection for EQOS datapath. 0b - Data path Parity Protection is disabled 1b - Data path Parity Protection is enabled

1.4.251 (MTL_TxQ0_Operation_Mode)

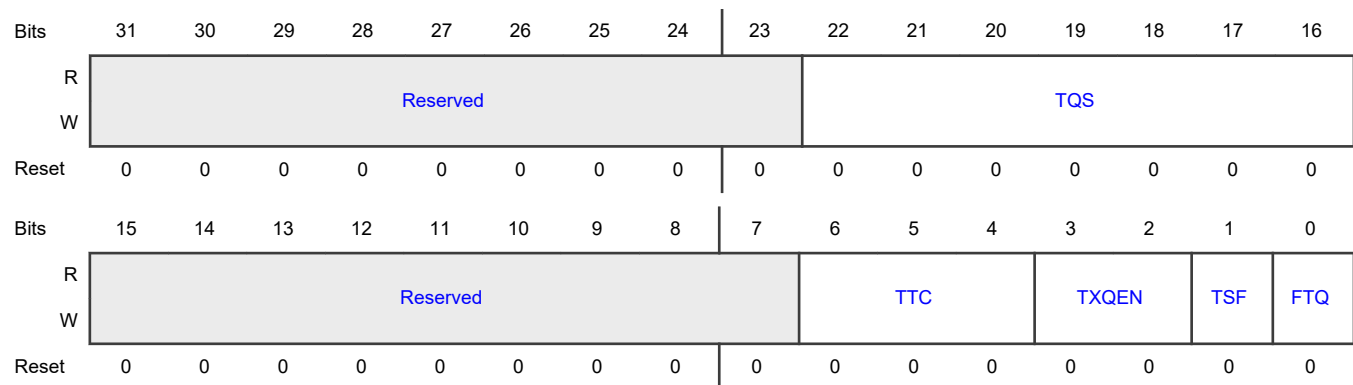
Offset

Register	Offset
MTL_TxQ0_Operation_Mode	D00h

Function

The Queue 0 Transmit Operation Mode register establishes the Transmit queue operating modes and commands.

Diagram



Fields

Field	Function
31-23 —	Reserved.
22-16 TQS	Transmit Queue Size This field indicates the size of the allocated Transmit queues in blocks of 256 bytes. The TQS field is read-write only if the number of Tx Queues more than one, the reset value is 0x0 and indicates size of 256 bytes. This means that value of 0x0 = 256 bytes, 0x1 = 512 bytes and so on, so user should be programming TQS [5:0] = 6'b001111 to allocate queue size of 4096 (4K) bytes. In general, the size of the Queue = (TQS + 1)*256 bytes. When the number of Tx Queues is one, the field is read-only and the configured TX FIFO size in blocks of 256 bytes is reflected in the reset value. The width of this field depends on the Tx memory size selected in your configuration. For example, if the memory size is 2048, the width of this field is 3 bits: $\text{LOG2}(2048/256) = \text{LOG2}(8) = 3 \text{ bits}$
15-7 —	Reserved.
6-4 TTC	Transmit Threshold Control These bits control the threshold level of the MTL Tx Queue. The transmission starts when the packet size within the MTL Tx Queue is larger than the threshold. In addition, full packets with length less than the threshold are also transmitted. These bits are used only when the TSF bit is reset. 000b - 32 001b - 64 010b - 96 011b - 128 100b - 192 101b - 256 110b - 384 111b - 512
3-2 TXQEN	Transmit Queue Enable This field is used to enable/disable the transmit queue 0. - 2'b00: Not enabled - 2'b01: Reserved - 2'b10: Enabled - 2'b11: Reserved This field is Read Only in Single Queue configurations and Read Write in Multiple Queue configurations.

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Field	Function
	Note: In multiple Tx queues configuration, all the queues are disabled by default. Enable the Tx queue by programming this field. 00b - Not enabled 01b - Enable in AV mode (Reserved in non-AV) 10b - Enabled 11b - Reserved
1 TSF	Transmit Store and Forward When this bit is set, the transmission starts when a full packet resides in the MTL Tx queue. When this bit is set, the TTC values specified in Bits[6:4] of this register are ignored. This bit should be changed only when the transmission is stopped. 0b - Transmit Store and Forward is disabled 1b - Transmit Store and Forward is enabled
0 FTQ	Flush Transmit Queue When this bit is set, the Tx queue controller logic is reset to its default values. Therefore, all the data in the Tx queue is lost or flushed. This bit is internally reset when the flushing operation is complete. Until this bit is reset, you should not write to the MTL_TxQ1_Operation_Mode register. The data which is already accepted by the MAC transmitter is not flushed. It is scheduled for transmission and results in underflow and runt packet transmission. Note: The flush operation is complete only when the Tx queue is empty and the application has accepted the pending Tx Status of all transmitted packets. To complete this flush operation, the PHY Tx clock (clk_tx_i) should be active. Access restriction applies. Setting 1 sets. Self-cleared. Setting 0 has no effect. 0b - Flush Transmit Queue is disabled 1b - Flush Transmit Queue is enabled

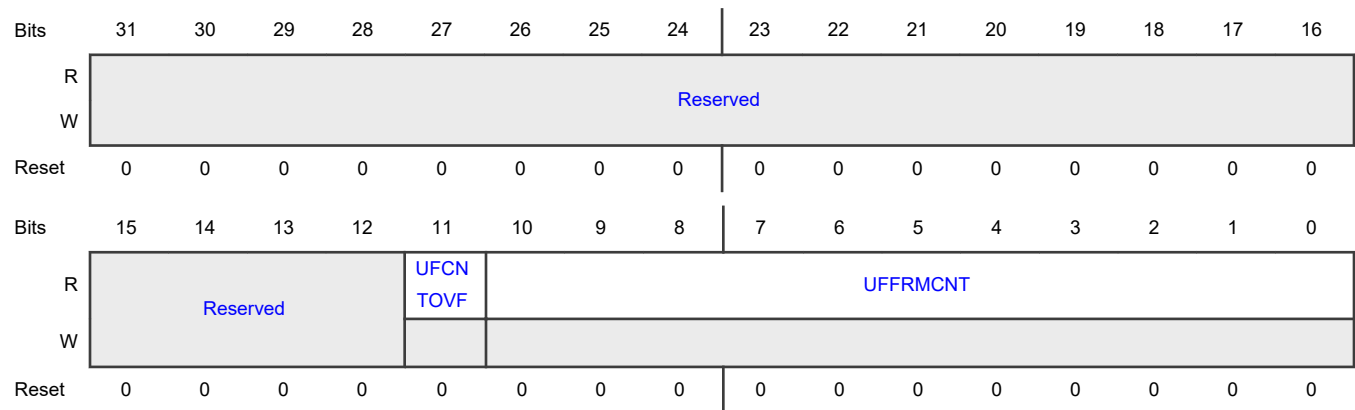
1.4.252 (MTL_TxQ0_Underflow)

Offset

Register	Offset
MTL_TxQ0_Underflow	D04h

Function

The Queue 0 Underflow Counter register contains the counter for packets aborted because of Transmit queue underflow and packets missed because of Receive queue packet flush

Diagram**Fields**

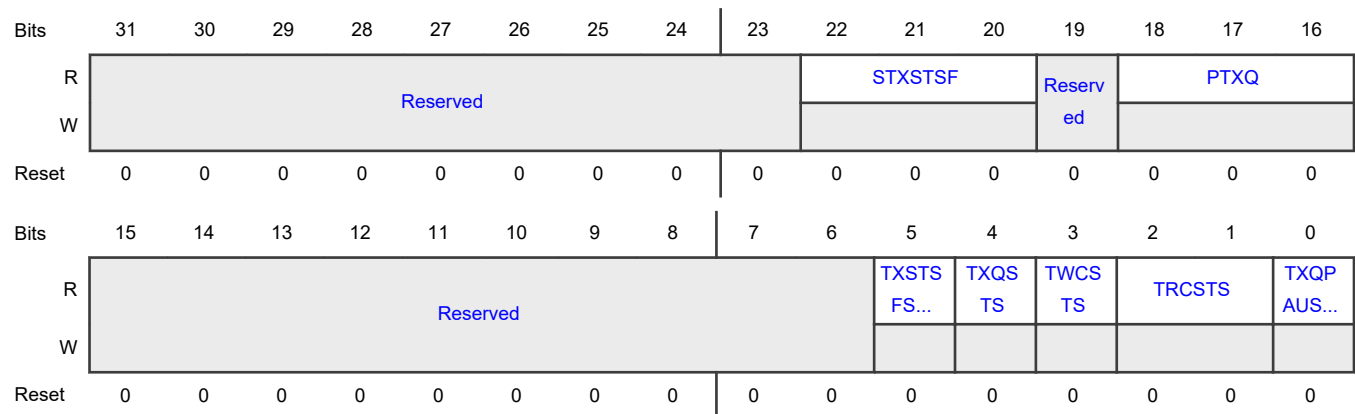
Field	Function
31-12 —	Reserved.
11 UFCNTOVF	Overflow Bit for Underflow Packet Counter This bit is set every time the Tx queue Underflow Packet Counter field overflows, that is, it has crossed the maximum count. In such a scenario, the overflow packet counter is reset to all-zeros and this bit indicates that the rollover happened. Access restriction applies. Clears on read. Self-set to 1 on internal event. 0b - Overflow not detected for Underflow Packet Counter 1b - Overflow detected for Underflow Packet Counter
10-0 UFFRMCNT	Underflow Packet Counter This field indicates the number of packets aborted by the controller because of Tx Queue Underflow. This counter is incremented each time the MAC aborts outgoing packet because of underflow. The counter is cleared when this register is read with mci_be_[0] at 1'b1. Access restriction applies. Clears on read. Self-set to 1 on internal event.

1.4.253 (MTL_TxQ0_Debug)**Offset**

Register	Offset
MTL_TxQ0_Debug	D08h

Function

The Queue 0 Transmit Debug register gives the debug status of various blocks related to the Transmit queue.

Diagram**Fields**

Field	Function
31-23 —	Reserved.
22-20 STXSTS	Number of Status Words in Tx Status FIFO of Queue This field indicates the current number of status in the Tx Status FIFO of this queue. When the DTXSTS bit of MTL_Operation_Mode register is set to 1, this field does not reflect the number of status words in Tx Status FIFO.
19 —	Reserved.
18-16 PTXQ	Number of Packets in the Transmit Queue This field indicates the current number of packets in the Tx Queue. When the DTXSTS bit of MTL_Operation_Mode register is set to 1, this field does not reflect the number of packets in the Transmit queue.
15-6 —	Reserved.
5 TXSTS	MTL Tx Status FIFO Full Status When high, this bit indicates that the MTL Tx Status FIFO is full. Therefore, the MTL cannot accept any more packets for transmission. 0b - MTL Tx Status FIFO Full status is not detected 1b - MTL Tx Status FIFO Full status is detected
4 TXQSTS	MTL Tx Queue Not Empty Status When this bit is high, it indicates that the MTL Tx Queue is not empty and some data is left for transmission. 0b - MTL Tx Queue Not Empty status is not detected

Table continues on the next page...

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Field	Function
	1b - MTL Tx Queue Not Empty status is detected
3 TWCSTS	MTL Tx Queue Write Controller Status When high, this bit indicates that the MTL Tx Queue Write Controller is active, and it is transferring the data to the Tx Queue. 0b - MTL Tx Queue Write Controller status is not detected 1b - MTL Tx Queue Write Controller status is detected
2-1 TRCSTS	MTL Tx Queue Read Controller Status This field indicates the state of the Tx Queue Read Controller: 00b - Idle state 01b - Read state (transferring data to the MAC transmitter) 10b - Waiting for pending Tx Status from the MAC transmitter 11b - Flushing the Tx queue because of the Packet Abort request from the MAC
0 TXQPAUSED	Transmit Queue in Pause When this bit is high and the Rx flow control is enabled, it indicates that the Tx Queue is in the Pause condition (in the full-duplex only mode) because of the following: - Reception of the PFC packet for the priorities assigned to the Tx Queue when PFC is enabled - Reception of 802.3x Pause packet when PFC is disabled 0b - Transmit Queue in Pause status is not detected 1b - Transmit Queue in Pause status is detected

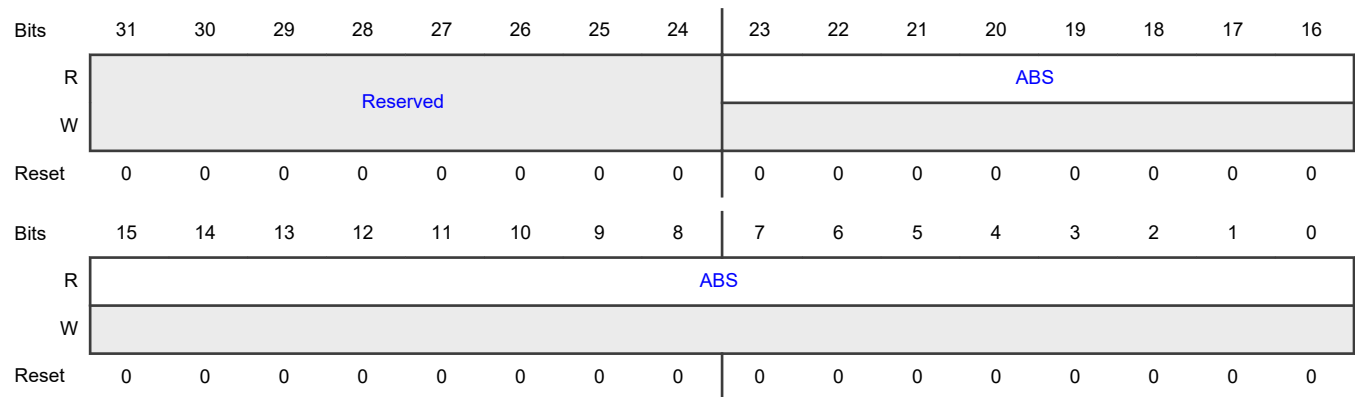
1.4.254 (MTL_TxQ0_ETS_Status)

Offset

Register	Offset
MTL_TxQ0_ETS_Status	D14h

Function

The Queue 0 ETS Status register provides the average traffic transmitted in Queue 0.

Diagram**Fields**

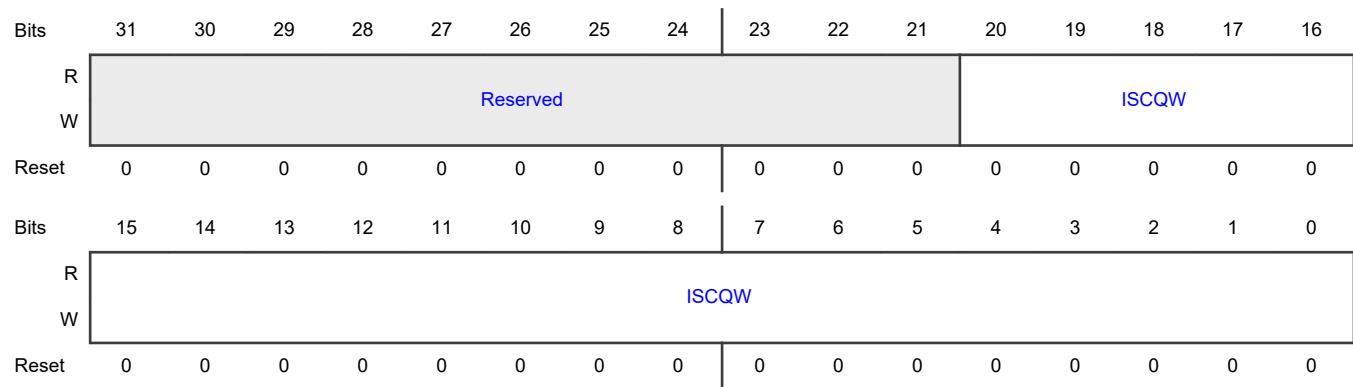
Field	Function
31-24 —	Reserved.
23-0 ABS	Average Bits per Slot This field contains the average transmitted bits per slot. When the DCB operation is enabled for Queue 0, this field is computed over every 10 million bit times slot (4 ms in 2500 Mbps; 10 ms in 1000 Mbps; 100 ms in 100 Mbps). The maximum value is 0x989680.

1.4.255 (MTL_TxQ0_Quantum_Weight)**Offset**

Register	Offset
MTL_TxQ0_Quantum_Weight	D18h

Function

The Queue 0 Quantum or Weights register contains the quantum value for Deficit Weighted Round Robin (DWRR), weights for the Weighted Round Robin (WRR), and Weighted Fair Queuing (WFQ) for Queue 0.

Diagram**Fields**

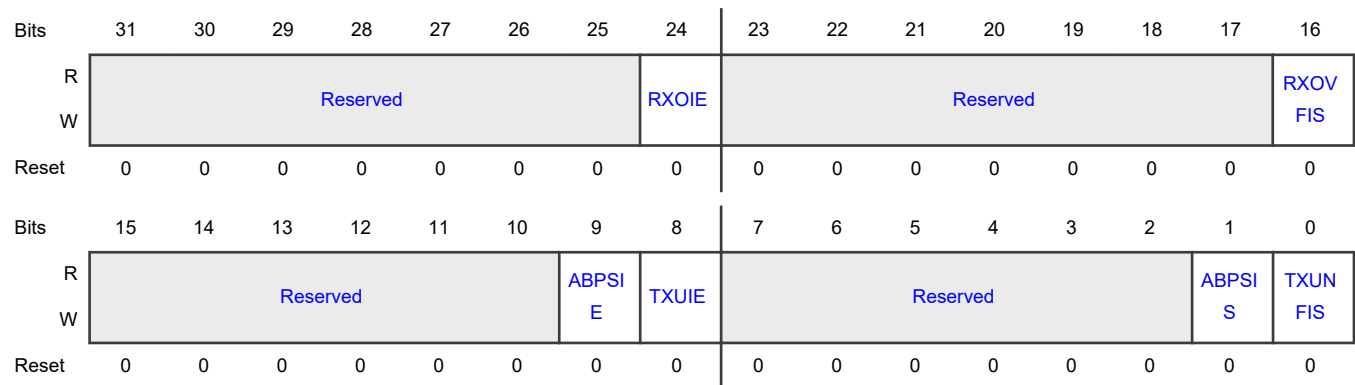
Field	Function
31-21 —	Reserved.
20-0 ISCQW	Quantum or Weights When the DCB operation is enabled with DWRR algorithm for Queue 0 traffic, this field contains the quantum value in bytes to be added to credit during every queue scanning cycle. The maximum value is 0x1312D0 bytes. When DCB operation is enabled with WFQ algorithm for Queue 0 traffic, this field contains the weight for this queue. The maximum value is 0x3FFF where weight of 0 indicates 100% bandwidth. Bits[20:14] must be written to zero. The higher the programmed weights lesser the bandwidth allocated for the particular Transmit Queue. This is because the weights are used to compute the packet finish time (weights*packet_size). Lesser the finish time, higher the probability of the packet getting scheduled first and using more bandwidth. When DCB operation or generic queuing operation is enabled with WRR algorithm for Queue 0 traffic, this field contains the weight for this queue. The maximum value is 0x64. Bits [20:7] must be written to zero.

1.4.256 (MTL_Q0_Interrupt_Control_Status)**Offset**

Register	Offset
MTL_Q0_Interrupt_Control_Status	D2Ch

Function

This register contains the interrupt enable and status bits for the queue 0 interrupts.

Diagram**Fields**

Field	Function
31-25 —	Reserved.
24 RXOIE	Receive Queue Overflow Interrupt Enable When this bit is set, the Receive Queue Overflow interrupt is enabled. When this bit is reset, the Receive Queue Overflow interrupt is disabled. 0b - Receive Queue Overflow Interrupt is disabled 1b - Receive Queue Overflow Interrupt is enabled
23-17 —	Reserved.
16 RXOVFIS	Receive Queue Overflow Interrupt Status This bit indicates that the Receive Queue had an overflow while receiving the packet. If a partial packet is transferred to the application, the overflow status is set in RDES3[21]. This bit is cleared when the application writes 1 to this bit. Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect. 0b - Receive Queue Overflow Interrupt Status not detected 1b - Receive Queue Overflow Interrupt Status detected
15-10 —	Reserved.
9 ABPSIE	Average Bits Per Slot Interrupt Enable When this bit is set, the MAC asserts the sbd_intr_o or mci_intr_o interrupt when the average bits per slot status is updated. When this bit is cleared, the interrupt is not asserted for such an event. 0b - Average Bits Per Slot Interrupt is disabled

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Field	Function
	1b - Average Bits Per Slot Interrupt is enabled
8 TXUIE	Transmit Queue Underflow Interrupt Enable When this bit is set, the Transmit Queue Underflow interrupt is enabled. When this bit is reset, the Transmit Queue Underflow interrupt is disabled. 0b - Transmit Queue Underflow Interrupt Status is disabled 1b - Transmit Queue Underflow Interrupt Status is enabled
7-2 —	Reserved.
1 ABPSIS	Average Bits Per Slot Interrupt Status When set, this bit indicates that the MAC has updated the ABS value. This bit is cleared when the application writes 1 to this bit. Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect. 0b - Average Bits Per Slot Interrupt Status not detected 1b - Average Bits Per Slot Interrupt Status detected
0 TXUNFIS	Transmit Queue Underflow Interrupt Status This bit indicates that the Transmit Queue had an underflow while transmitting the packet. Transmission is suspended and an Underflow Error TDES3[2] is set. This bit is cleared when the application writes 1 to this bit. Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect. 0b - Transmit Queue Underflow Interrupt Status not detected 1b - Transmit Queue Underflow Interrupt Status detected

1.4.257 (MTL_RxQ0_Operation_Mode)

Offset

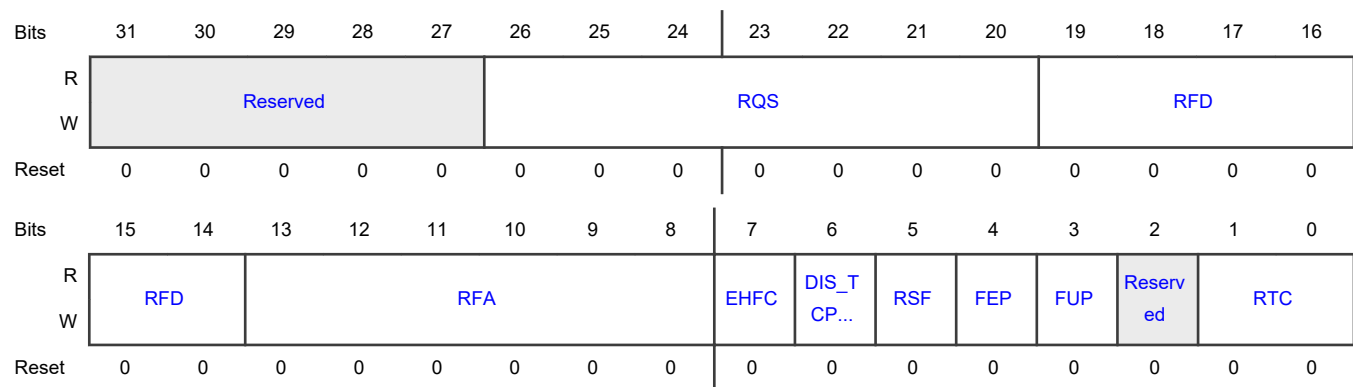
Register	Offset
MTL_RxQ0_Operation_Mode	D30h

Function

The Queue 0 Receive Operation Mode register establishes the Receive queue operating modes and command.

The RFA and RFD fields are not backward compatible with the RFA and RFD fields of 4.00a release

Diagram



Fields

Field	Function
31-27 —	Reserved.
26-20 RQS	Receive Queue Size This field indicates the size of the allocated Receive queues in blocks of 256 bytes. The RQS field is read-write only if the number of Rx Queues more than one, the reset value is 0x0 and indicates size of 256 bytes. This means that value of 0x0 = 256 bytes, 0x1 = 512 bytes and so on, so user should be programming RQS [5:0] = 6'b001111 to allocate queue size of 4096 (4K) bytes. In general, the size of the Queue = (RQS + 1)*256 bytes. When the number of Rx Queues is one, the field is read-only and the configured RX FIFO size in blocks of 256 bytes is reflected in the reset value. The width of this field depends on the Rx memory size selected in your configuration. For example, if the memory size is 2048, the width of this field is 3 bits: $\text{LOG2}(2048/256) = \text{LOG2}(8) = 3 \text{ bits}$
19-14 RFD	Threshold for Deactivating Flow Control (in half-duplex and full-duplex modes) These bits control the threshold (fill-level of Rx queue) at which the flow control is de-asserted after activation: - 0: Full minus 1 KB, that is, FULL 1 KB - 1: Full minus 1.5 KB, that is, FULL 1.5 KB - 2: Full minus 2 KB, that is, FULL 2 KB - 3: Full minus 2.5 KB, that is, FULL 2.5 KB - ... - 62: Full minus 32 KB, that is, FULL 32 KB - 63: Full minus 32.5 KB, that is, FULL 32.5 KB The de-assertion is effective only after flow control is asserted. Note: The value must be programmed in such a way to make sure that the threshold is a positive number.

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Field	Function
	When the EHFC is set high, these values are applicable only when the Rx queue size determined by the RQS field of this register, is equal to or greater than 4 KB. For a given queue size, the values ranges between 0 and the encoding for FULL minus (QSIZE - 0.5 KB) and all other values are illegal. Here the term FULL and QSIZE refers to the queue size determined by the RQS field of this register. The width of this field depends on RX FIFO size selected during the configuration. Remaining bits are reserved and read only.
13-8 RFA	Threshold for Activating Flow Control (in half-duplex and full-duplex) These bits control the threshold (fill-level of Rx queue) at which the flow control is activated: For more information on encoding for this field, see RFD.
7 EHFC	Enable Hardware Flow Control When this bit is set, the flow control signal operation, based on the fill-level of Rx queue, is enabled. When reset, the flow control operation is disabled. 0b - Hardware Flow Control is disabled 1b - Hardware Flow Control is enabled
6 DIS_TCP_EF	Disable Dropping of TCP/IP Checksum Error Packets When this bit is set, the MAC does not drop the packets which only have the errors detected by the Receive Checksum Offload engine. Such packets have errors only in the encapsulated payload. There are no errors (including FCS error) in the Ethernet packet received by the MAC. When this bit is reset, all error packets are dropped if the FEP bit is reset. 0b - Dropping of TCP/IP Checksum Error Packets is enabled 1b - Dropping of TCP/IP Checksum Error Packets is disabled
5 RSF	Receive Queue Store and Forward When this bit is set, the DWC_ether_qos reads a packet from the Rx queue only after the complete packet has been written to it, ignoring the RTC field of this register. When this bit is reset, the Rx queue operates in the Threshold (cut-through) mode, subject to the threshold specified by the RTC field of this register. 0b - Receive Queue Store and Forward is disabled 1b - Receive Queue Store and Forward is enabled
4 FEP	Forward Error Packets When this bit is reset, the Rx queue drops packets with error status (CRC error, GMII_ER, watchdog timeout, or overflow). However, if the start byte (write) pointer of a packet is already transferred to the read controller side (in Threshold mode), the packet is not dropped. When this bit is set, all packets except the runt error packets are forwarded to the application or DMA. If the RSF bit is set and the Rx queue overflows when a partial packet is written, the packet is dropped irrespective of the setting of this bit. However, if the RSF bit is reset and the Rx queue overflows when a partial packet is written, a partial packet might be forwarded to the application or DMA.

Table continues on the next page...

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Field	Function
	0b - Forward Error Packets is disabled 1b - Forward Error Packets is enabled
3 FUP	Forward Undersized Good Packets When this bit is set, the Rx queue forwards the undersized good packets (packets with no error and length less than 64 bytes), including pad-bytes and CRC. When this bit is reset, the Rx queue drops all packets of less than 64 bytes, unless a packet is already transferred because of the lower value of Rx Threshold, for example, RTC = 01. 0b - Forward Undersized Good Packets is disabled 1b - Forward Undersized Good Packets is enabled
2 —	Reserved.
1-0 RTC	Receive Queue Threshold Control These bits control the threshold level of the MTL Rx queue (in bytes): The received packet is transferred to the application or DMA when the packet size within the MTL Rx queue is larger than the threshold. In addition, full packets with length less than the threshold are automatically transferred. This field is valid only when the RSF bit is zero. This field is ignored when the RSF bit is set to 1. 00b - 64 01b - 32 10b - 96 11b - 128

1.4.258 (MTL_RxQ0_Missed_Packet_Overflow_Cnt)

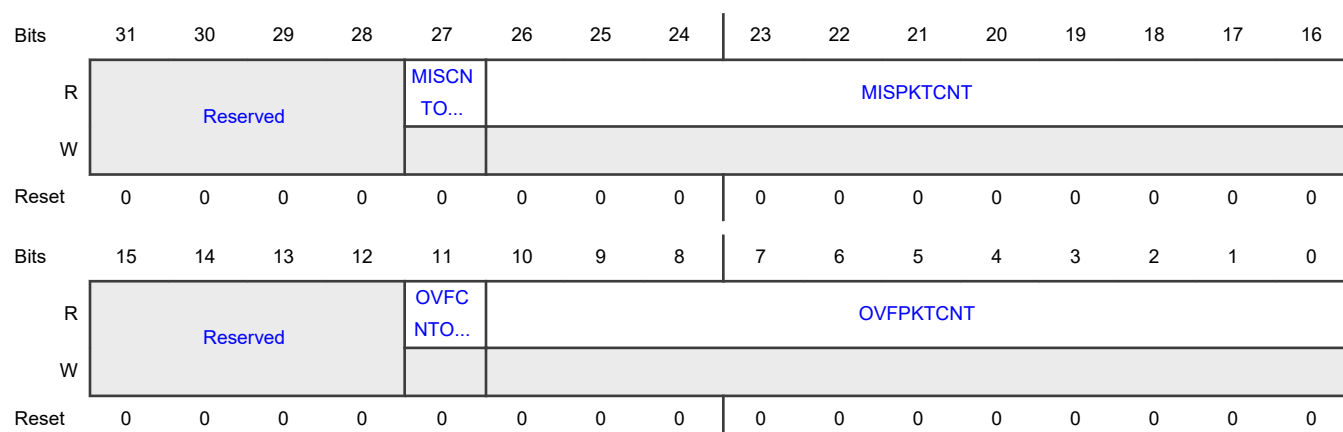
Offset

Register	Offset
MTL_RxQ0_Missed_Packet_Overflow_Cnt	D34h

Function

The Queue 0 Missed Packet and Overflow Counter register contains the counter for packets missed because of Receive queue packet flush and packets discarded because of Receive queue overflow.

Diagram



Fields

Field	Function
31-28 —	Reserved.
27 MISCNTOVF	Missed Packet Counter Overflow Bit When set, this bit indicates that the Rx Queue Missed Packet Counter crossed the maximum limit. Access restriction applies. Clears on read. Self-set to 1 on internal event. 0b - Missed Packet Counter overflow not detected 1b - Missed Packet Counter overflow detected
26-16 MISPKTCNT	Missed Packet Counter This field indicates the number of packets missed by the DWC_ether_qos because the application asserted ari_pkt_flush_i[] for this queue. This counter is reset when this register is read with mci_be_i[0] at 1b1. This counter is incremented by 1 when the DMA discards the packet because of buffer unavailability. Access restriction applies. Clears on read. Self-set to 1 on internal event.
15-12 —	Reserved.
11 OVFCNTOVF	Overflow Counter Overflow Bit When set, this bit indicates that the Rx Queue Overflow Packet Counter field crossed the maximum limit. Access restriction applies. Clears on read. Self-set to 1 on internal event. 0b - Overflow Counter overflow not detected 1b - Overflow Counter overflow detected
10-0 OVFPKTCNT	Overflow Packet Counter

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Field	Function
	This field indicates the number of packets discarded by the DWC_ether_qos because of Receive queue overflow. This counter is incremented each time the DWC_ether_qos discards an incoming packet because of overflow. This counter is reset when this register is read with mci_be_i[0] at 1'b1. Access restriction applies. Clears on read. Self-set to 1 on internal event.

1.4.259 (MTL_RxQ0_Debug)

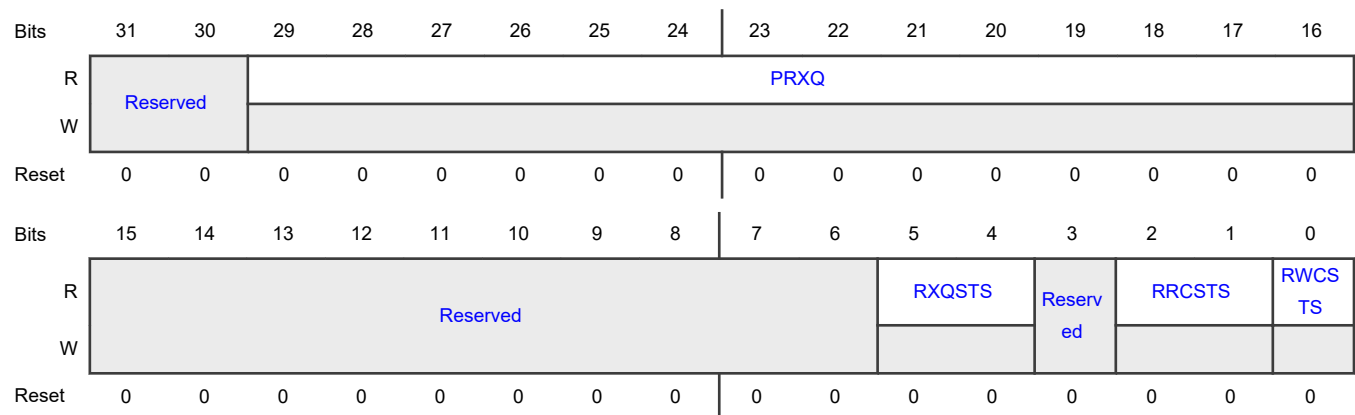
Offset

Register	Offset
MTL_RxQ0_Debug	D38h

Function

The Queue 0 Receive Debug register gives the debug status of various blocks related to the Receive queue.

Diagram



Fields

Field	Function
31-30 —	Reserved.
29-16 PRXQ	Number of Packets in Receive Queue This field indicates the current number of packets in the Rx Queue. The theoretical maximum value for this field is 256KB/16B = 16K Packets, that is, Max_Queue_Size/Min_Packet_Size.
15-6	Reserved.

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Field	Function
—	
5-4 RXQSTS	MTL Rx Queue Fill-Level Status This field gives the status of the fill-level of the Rx Queue: 00b - Rx Queue empty 01b - Rx Queue fill-level below flow-control deactivate threshold 10b - Rx Queue fill-level above flow-control activate threshold 11b - Rx Queue full
3 —	Reserved.
2-1 RRCSTS	MTL Rx Queue Read Controller State This field gives the state of the Rx queue Read controller: 00b - Idle state 01b - Reading packet data 10b - Reading packet status (or timestamp) 11b - Flushing the packet data and status
0 RWCSTS	MTL Rx Queue Write Controller Active Status When high, this bit indicates that the MTL Rx queue Write controller is active, and it is transferring a received packet to the Rx Queue. 0b - MTL Rx Queue Write Controller Active Status not detected 1b - MTL Rx Queue Write Controller Active Status detected

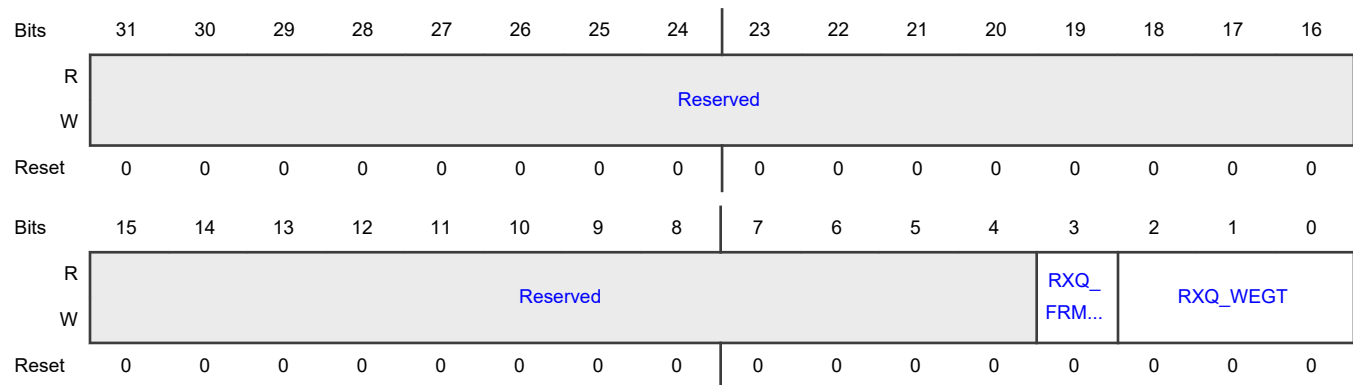
1.4.260 (MTL_RxQ0_Control)

Offset

Register	Offset
MTL_RxQ0_Control	D3Ch

Function

The Queue Receive Control register controls the receive arbitration and passing of received packets to the application.

Diagram**Fields**

Field	Function
31-4 —	Reserved.
3 RXQ_FRM_AR BIT	Receive Queue Packet Arbitration When this bit is set, the DWC_ether_qos drives the packet data to the ARI interface such that the entire packet data of currently-selected queue is transmitted before switching to other queue. When this bit is reset, the DWC_ether_qos drives the packet data to the ARI interface such that the following amount of data of currently-selected queue is transmitted before switching to other queue: - PBL amount of data (indicated by ari_qN_pbl_i[]) or - Complete data of a packet The status and the timestamp are not a part of the PBL data. Therefore, the DWC_ether_qos drives the complete status (including timestamp status) during first PBL request for the packet (in store-and-forward mode) or the last PBL request for the packet (in Threshold mode). 0b - Receive Queue Packet Arbitration is disabled 1b - Receive Queue Packet Arbitration is enabled
2-0 RXQ_WEGT	Receive Queue Weight This field indicates the weight assigned to the Rx Queue 0. This field needs to be programmed with one value less than the required weight, i.e. reset value of 0 indicates weight of 1, value of 1 indicates weight of 2, and so on. The weight is used as the number of continuous PBL or packets requests (depending on the RXQ_FRM_ARBIT) allocated to the queue in one arbitration cycle. Note: The change in value of RXQ_WEGT takes effect only after the completion of current service round or when there is change from RAA=SP to RAA=WSP algorithm. This approach is taken so that there is smooth transition. For the RXQ_WEGT value to take effect at the start, the MTL_RxQ(##)_Control registers must be programmed before the MTL_Operation_Mode register.

1.4.261 (MTL_TxQ1_Operation_Mode)

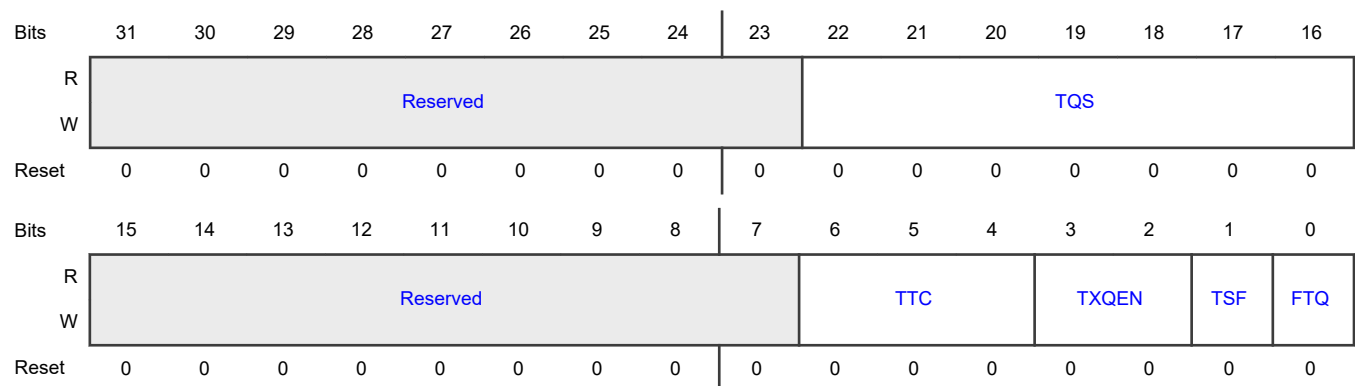
Offset

Register	Offset
MTL_TxQ1_Operation_Mode	D40h

Function

The Queue 1 Transmit Operation Mode register establishes the Transmit queue operating modes and commands.

Diagram



Fields

Field	Function
31-23 —	Reserved.
22-16 TQS	Transmit Queue Size This field indicates the size of the allocated Transmit queues in blocks of 256 bytes. The TQS field is read-write only if the number of Tx Queues more than one, the reset value is 0x0 and indicates size of 256 bytes. This means that value of 0x0 = 256 bytes, 0x1 = 512 bytes and so on, so user should be programming TQS [5:0] = 6'b001111 to allocate queue size of 4096 (4K) bytes. In general, the size of the Queue = (TQS + 1)*256 bytes. When the number of Tx Queues is one, the field is read-only and the configured TX FIFO size in blocks of 256 bytes is reflected in the reset value. The width of this field depends on the Tx memory size selected in your configuration. For example, if the memory size is 2048, the width of this field is 3 bits: $\text{LOG2}(2048/256) = \text{LOG2}(8) = 3 \text{ bits}$
15-7 —	Reserved.

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Field	Function
6-4 TTC	Transmit Threshold Control These bits control the threshold level of the MTL Tx Queue. The transmission starts when the packet size within the MTL Tx Queue is larger than the threshold. In addition, full packets with length less than the threshold are also transmitted. These bits are used only when the TSF bit is reset. 000b - 32 001b - 64 010b - 96 011b - 128 100b - 192 101b - 256 110b - 384 111b - 512
3-2 TXQEN	Transmit Queue Enable This field is used to enable/disable the transmit queue 0. - 2'b00: Not enabled - 2'b01: Enable in AV mode - 2'b10: Enabled - 2'b11: Reserved Note: In multiple Tx queues configuration, all the queues are disabled by default. Enable the Tx queue by programming this field. 00b - Not enabled 01b - Enable in AV mode (Reserved in non-AV) 10b - Enabled 11b - Reserved
1 TSF	Transmit Store and Forward When this bit is set, the transmission starts when a full packet resides in the MTL Tx queue. When this bit is set, the TTC values specified in Bits[6:4] of this register are ignored. This bit should be changed only when the transmission is stopped. 0b - Transmit Store and Forward is disabled 1b - Transmit Store and Forward is enabled
0 FTQ	Flush Transmit Queue When this bit is set, the Tx queue controller logic is reset to its default values. Therefore, all the data in the Tx queue is lost or flushed. This bit is internally reset when the flushing operation is complete. Until this bit is reset, you should not write to the MTL_TxQ1_Operation_Mode register. The data which is already

Table continues on the next page...

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Field	Function
	accepted by the MAC transmitter is not flushed. It is scheduled for transmission and results in underflow and runt packet transmission.
	Note: The flush operation is complete only when the Tx queue is empty and the application has accepted the pending Tx Status of all transmitted packets. To complete this flush operation, the PHY Tx clock (clk_tx_i) should be active.
	Access restriction applies. Setting 1 sets. Self-cleared. Setting 0 has no effect.
	0b - Flush Transmit Queue is disabled
	1b - Flush Transmit Queue is enabled

1.4.262 (MTL_TxQ1_Underflow)

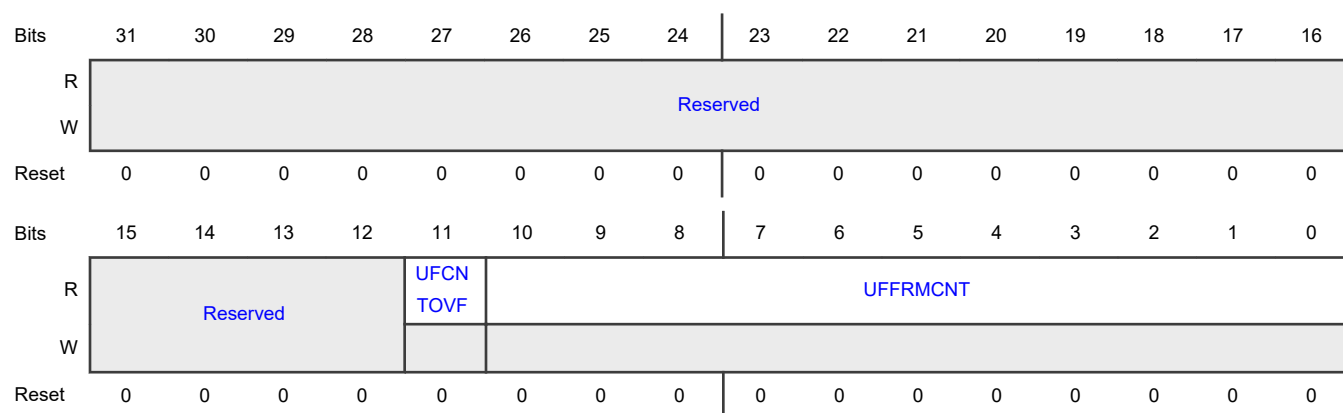
Offset

Register	Offset
MTL_TxQ1_Underflow	D44h

Function

The Queue 1 Underflow Counter register contains the counter for packets aborted because of Transmit queue underflow and packets missed because of Receive queue packet flush

Diagram



Fields

Field	Function
31-12	Reserved.
—	

Table continues on the next page...

Table continued from the previous page...

Field	Function
11 UFCNTOVF	Overflow Bit for Underflow Packet Counter This bit is set every time the Tx queue Underflow Packet Counter field overflows, that is, it has crossed the maximum count. In such a scenario, the overflow packet counter is reset to all-zeros and this bit indicates that the rollover happened. Access restriction applies. Clears on read. Self-set to 1 on internal event. 0b - Overflow not detected for Underflow Packet Counter 1b - Overflow detected for Underflow Packet Counter
10-0 UFFRMCNT	Underflow Packet Counter This field indicates the number of packets aborted by the controller because of Tx Queue Underflow. This counter is incremented each time the MAC aborts outgoing packet because of underflow. The counter is cleared when this register is read with mci_be_i[0] at 1'b1. Access restriction applies. Clears on read. Self-set to 1 on internal event.

1.4.263 (MTL_TxQ1_Debug)

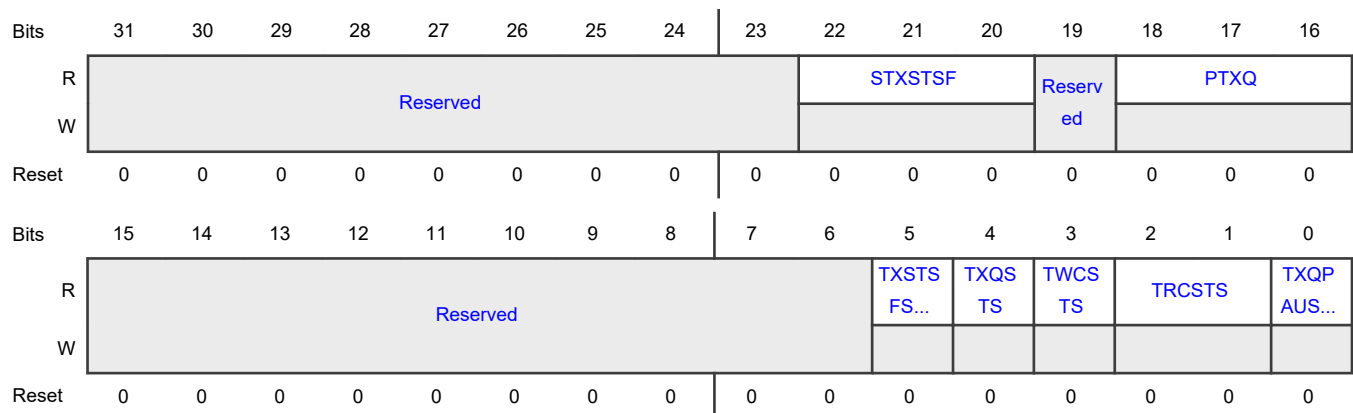
Offset

Register	Offset
MTL_TxQ1_Debug	D48h

Function

The Queue 1 Transmit Debug register gives the debug status of various blocks related to the Transmit queue.

Diagram



Fields

Field	Function
31-23 —	Reserved.
22-20 STXSTS	Number of Status Words in Tx Status FIFO of Queue This field indicates the current number of status in the Tx Status FIFO of this queue. When the DTXSTS bit of MTL_Operation_Mode register is set to 1, this field does not reflect the number of status words in Tx Status FIFO.
19 —	Reserved.
18-16 PTXQ	Number of Packets in the Transmit Queue This field indicates the current number of packets in the Tx Queue. When the DTXSTS bit of MTL_Operation_Mode register is set to 1, this field does not reflect the number of packets in the Transmit queue.
15-6 —	Reserved.
5 TXSTS	MTL Tx Status FIFO Full Status When high, this bit indicates that the MTL Tx Status FIFO is full. Therefore, the MTL cannot accept any more packets for transmission. 0b - MTL Tx Status FIFO Full status is not detected 1b - MTL Tx Status FIFO Full status is detected
4 TXQSTS	MTL Tx Queue Not Empty Status When this bit is high, it indicates that the MTL Tx Queue is not empty and some data is left for transmission. 0b - MTL Tx Queue Not Empty status is not detected 1b - MTL Tx Queue Not Empty status is detected
3 TWCSTS	MTL Tx Queue Write Controller Status When high, this bit indicates that the MTL Tx Queue Write Controller is active, and it is transferring the data to the Tx Queue. 0b - MTL Tx Queue Write Controller status is not detected 1b - MTL Tx Queue Write Controller status is detected
2-1 TRCSTS	MTL Tx Queue Read Controller Status This field indicates the state of the Tx Queue Read Controller: 00b - Idle state 01b - Read state (transferring data to the MAC transmitter)

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Field	Function
	10b - Waiting for pending Tx Status from the MAC transmitter 11b - Flushing the Tx queue because of the Packet Abort request from the MAC
0 TXQPAUSED	Transmit Queue in Pause When this bit is high and the Rx flow control is enabled, it indicates that the Tx Queue is in the Pause condition (in the full-duplex only mode) because of the following: - Reception of the PFC packet for the priorities assigned to the Tx Queue when PFC is enabled - Reception of 802.3x Pause packet when PFC is disabled 0b - Transmit Queue in Pause status is not detected 1b - Transmit Queue in Pause status is detected

1.4.264 (MTL_TxQ1_ETS_Control)

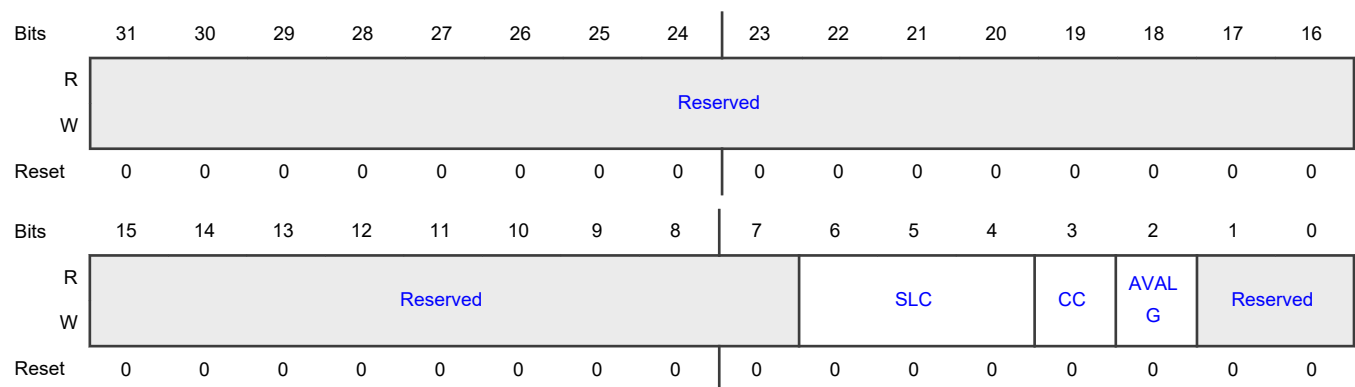
Offset

Register	Offset
MTL_TxQ1_ETS_Control	D50h

Function

The Queue ETS Control register controls the enhanced transmission selection operation.

Diagram



Fields

Field	Function
31-7	Reserved.

Table continues on the next page...

Table continued from the previous page...

Field	Function
—	
6-4 SLC	Slot Count If the credit-based shaper algorithm is enabled, the software can program the number of slots (of duration programmed in DMA_CH(#i)_Slot_Interval register) over which the average transmitted bits per slot, provided in the MTL_TxQ(#i)_ETS_Status register, need to be computed for Queue. The encoding is as follows: 000b - 1 slot 001b - 2 slots 010b - 4 slots 011b - 8 slots 100b - 16 slots 101b - Reserved
3 CC	Credit Control When this bit is set, the accumulated credit parameter in the credit-based shaper algorithm logic is not reset to zero when there is positive credit and no packet to transmit in Channel 1. The credit accumulates even when there is no packet waiting in Channel 1 and another channel is transmitting. When this bit is reset, the accumulated credit parameter in the credit-based shaper algorithm logic is set to zero when there is positive credit and no packet to transmit in Channel 1. When there is no packet waiting in Channel 1 and other channel is transmitting, no credit is accumulated. 0b - Credit Control is disabled 1b - Credit Control is enabled
2 AVALG	AV Algorithm When Queue 1 is programmed for AV, this field configures the scheduling algorithm for this queue: This bit when set, indicates credit based shaper algorithm (CBS) is selected for Queue 1 traffic. When reset, strict priority is selected. 0b - CBS Algorithm is disabled 1b - CBS Algorithm is enabled
1-0 —	Reserved.

1.4.265 (MTL_TxQ1_ETS_Status)

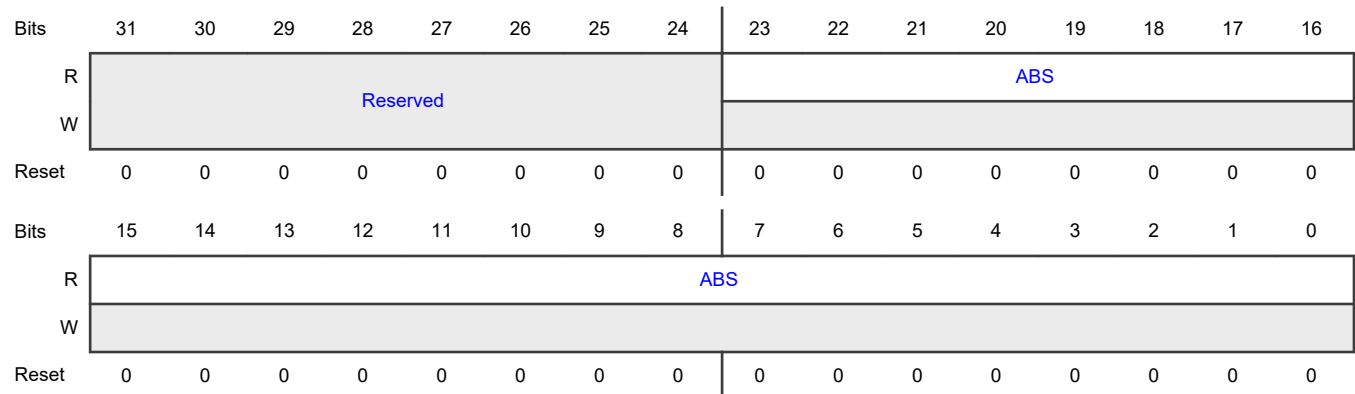
Offset

Register	Offset
MTL_TxQ1_ETS_Status	D54h

Function

The Queue 1 ETS Status register provides the average traffic transmitted in Queue 1.

Diagram



Fields

Field	Function
31-24 —	Reserved.
23-0 ABS	Average Bits per Slot This field contains the average transmitted bits per slot. If AV operation is enabled, this field is computed over number of slots, programmed in the SLC field of MTL_TxQ(#i)_ETS_CONTROL register. The maximum value of this field is 0x6_4000 in 100 Mbps, 0x3E_8000 in 1000 Mbps and 9C_4000 in 2500 Mbps mode respectively. When the DCB operation is enabled for Queue, this field is computed over every 10 million bit times slot (4 ms in 2500 Mbps; 10 ms in 1000 Mbps; 100 ms in 100 Mbps). The maximum value is 0x989680.

1.4.266 (MTL_TxQ1_Quantum_Weight)

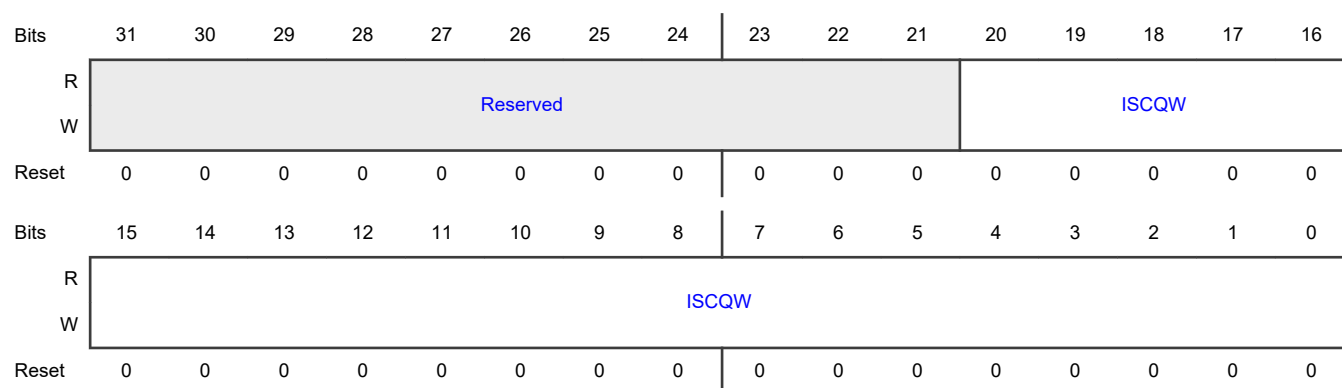
Offset

Register	Offset
MTL_TxQ1_Quantum_Weight	D58h

Function

The Queue 1 idleSlopeCredit, Quantum or Weights register provides the average traffic transmitted in Queue 1.

Diagram



Fields

Field	Function
31-21 —	Reserved.
20-0 ISCQW	idleSlopeCredit, Quantum or Weights - idleSlopeCredit When AV feature is enabled, this field contains the idleSlopeCredit value required for the credit-based shaper algorithm for Queue 1. This is the rate of change of credit in bits per cycle (40 ns for 100 Mbps; 8 ns for 1000 Mbps; 3.2 ns for 2500 Mbps) when the credit is increasing. The software should program this field with computed credit in bits per cycle scaled by 1,024. The maximum value is portTransmitRate, that is, 0x2000 in 1000/2500 Mbps mode and 0x1000 in 100 Mbps mode. Bits[20:14] must be written to zero. - Quantum When the DCB operation is enabled with DWRR algorithm for Queue 1 traffic, this field contains the quantum value in bytes to be added to credit during every queue scanning cycle. The maximum value is 0x1312D0 bytes. - Weights When DCB operation is enabled with WFQ algorithm for Queue 1 traffic, this field contains the weight for this queue. The maximum value is 0x3FFF where weight of 0 indicates 100% bandwidth. Bits[20:14] must be written to zero. When DCB operation or generic queuing operation is enabled with WRR algorithm for Queue 1 traffic, this field contains the weight for this queue. The maximum value is 0x64. Bits [20:7] must be written to zero. - Note 1: In multiple Queue configuration this field in respective per queue register must be programmed to some non-zero value when multiple queues are enabled or single queue other than Q0 is enabled. This field need not be programmed when only Q0 is enabled. In general, when WRR algorithm is selected a non-zero value must be programmed on both Receive and Transmit. In Receive, the register is MTL_Operation_Mode register. - Note 2: For WFQ algorithm, higher the programmed weights lesser the bandwidth allocated for that Transmit Queue. The finish time is not a function of particular packet alone but it is as per the formula: (previous_finish_time of particular Transmit Queue + (weights*packet_size))

Table continues on the next page...

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Field	Function
	- Note 3: The weights programmed do not correspond to the number of packets but the fraction of bandwidth or time allocated for particular queue w.r.t. total BW or time.

1.4.267 (MTL_TxQ1_SendSlopeCredit)

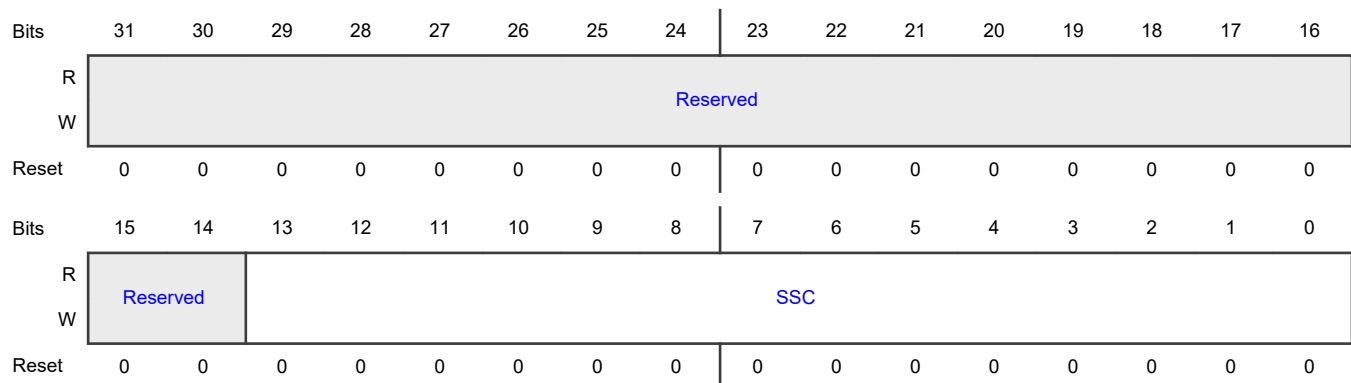
Offset

Register	Offset
MTL_TxQ1_SendSlopeCredit	D5Ch

Function

The sendSlopeCredit register contains the sendSlope credit value required for the credit-based shaper algorithm for the Queue.

Diagram



Fields

Field	Function
31-14 —	Reserved.
13-0 SSC	sendSlopeCredit Value When AV operation is enabled, this field contains the sendSlopeCredit value required for credit-based shaper algorithm for Queue 1. This is the rate of change of credit in bits per cycle (40 ns, 8 ns and 3.2 ns for 100 Mbps, 1000 Mbps and 2500 Mbps respectively) when the credit is decreasing. The software should program this field with computed credit in bits per cycle scaled by 1,024. The maximum value is portTransmitRate, that is, 0x2000 in 1000/2500 Mbps mode and 0x1000 in 100 Mbps mode. This field should be programmed with absolute sendSlopeCredit value. The credit-based shaper logic subtracts it from the accumulated credit when Channel 1 is selected for transmission.

1.4.268 (MTL_TxQ1_HiCredit)

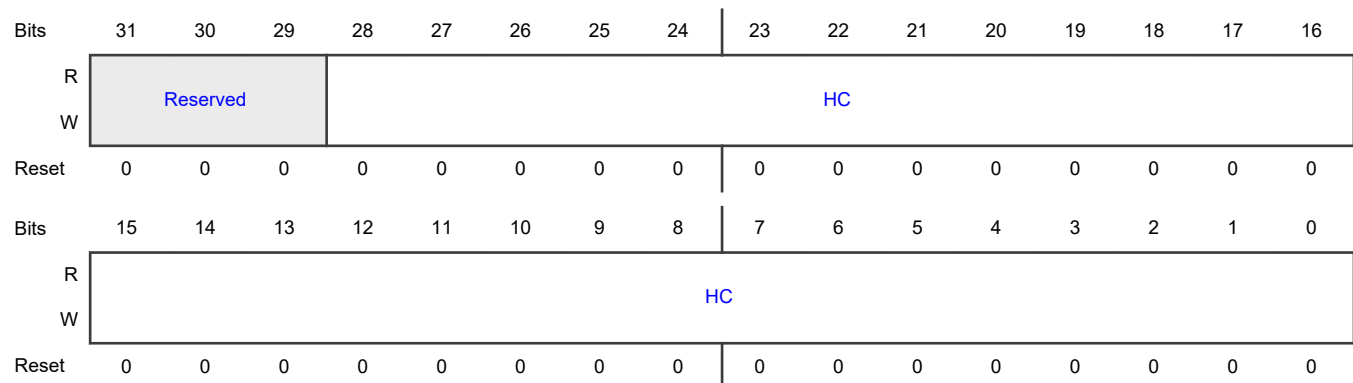
Offset

Register	Offset
MTL_TxQ1_HiCredit	D60h

Function

The hiCredit register contains the hiCredit value required for the credit-based shaper algorithm for the Queue.

Diagram



Fields

Field	Function
31-29 —	Reserved.
28-0 HC	hiCredit Value When the AV feature is enabled, this field contains the hiCredit value required for the credit-based shaper algorithm. This is the maximum value that can be accumulated in the credit parameter. This is specified in bits scaled by 1,024. The maximum value is maxInterferenceSize, that is, best-effort maximum packet size (16,384 bytes or 131,072 bits). The value to be specified is $131,072 * 1,024 = 134,217,728$ or 0x0800_0000.

1.4.269 (MTL_TxQ1_LoCredit)

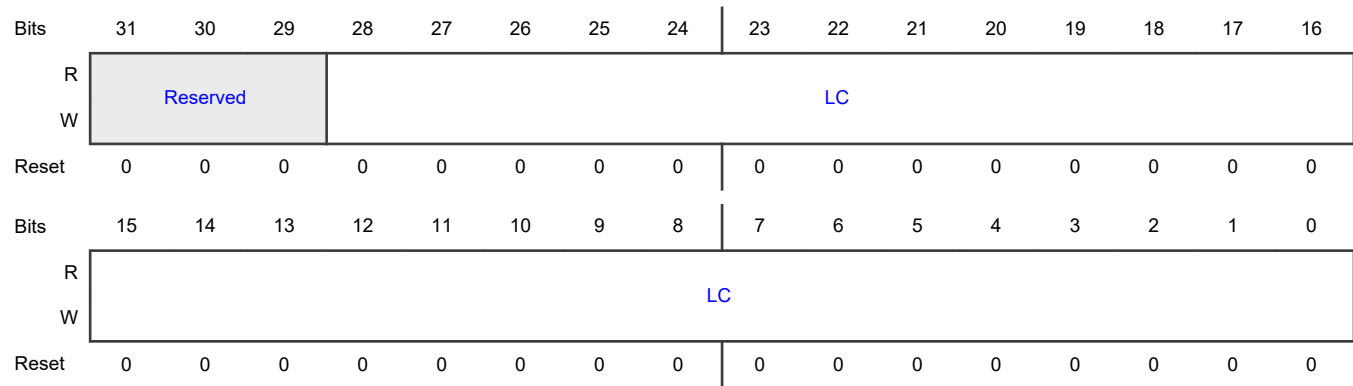
Offset

Register	Offset
MTL_TxQ1_LoCredit	D64h

Function

The loCredit register contains the loCredit value required for the credit-based shaper algorithm for the Queue.

Diagram



Fields

Field	Function
31-29 —	Reserved.
28-0 LC	loCredit Value When AV operation is enabled, this field contains the loCredit value required for the credit-based shaper algorithm. This is the minimum value that can be accumulated in the credit parameter. This is specified in bits scaled by 1,024. The maximum value to be programmed corresponds to twice the maxFrameSize transmitted from this queue. If the maxFrameSize is 8192 bytes, then $(8192*2) * 8 * 1024 = 134,217,728$ or 0x0800_0000. Because it is a negative value, the programmed value is 2's complement of the value, that is, 0x1800_0000.

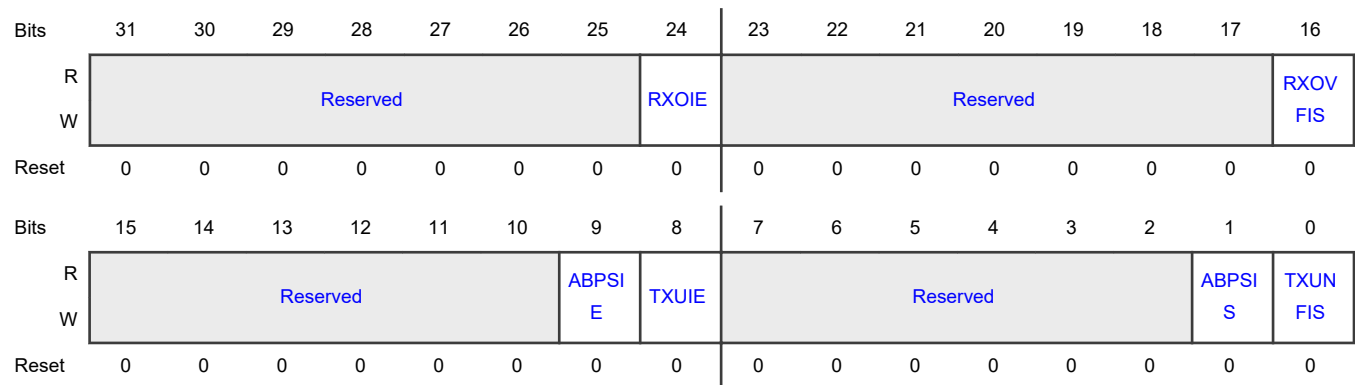
1.4.270 (MTL_Q1_Interrupt_Control_Status)

Offset

Register	Offset
MTL_Q1_Interrupt_Contr ol_Status	D6Ch

Function

This register contains the interrupt enable and status bits for the queue 1 interrupts.

Diagram**Fields**

Field	Function
31-25 —	Reserved.
24 RXOIE	Receive Queue Overflow Interrupt Enable When this bit is set, the Receive Queue Overflow interrupt is enabled. When this bit is reset, the Receive Queue Overflow interrupt is disabled. 0b - Receive Queue Overflow Interrupt is disabled 1b - Receive Queue Overflow Interrupt is enabled
23-17 —	Reserved.
16 RXOVFIS	Receive Queue Overflow Interrupt Status This bit indicates that the Receive Queue had an overflow while receiving the packet. If a partial packet is transferred to the application, the overflow status is set in RDES3[21]. This bit is cleared when the application writes 1 to this bit. Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect. 0b - Receive Queue Overflow Interrupt Status not detected 1b - Receive Queue Overflow Interrupt Status detected
15-10 —	Reserved.
9 ABPSIE	Average Bits Per Slot Interrupt Enable When this bit is set, the MAC asserts the sbd_intr_o or mci_intr_o interrupt when the average bits per slot status is updated. When this bit is cleared, the interrupt is not asserted for such an event. 0b - Average Bits Per Slot Interrupt is disabled

Table continues on the next page...

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Field	Function
	1b - Average Bits Per Slot Interrupt is enabled
8 TXUIE	Transmit Queue Underflow Interrupt Enable When this bit is set, the Transmit Queue Underflow interrupt is enabled. When this bit is reset, the Transmit Queue Underflow interrupt is disabled. 0b - Transmit Queue Underflow Interrupt Status is disabled 1b - Transmit Queue Underflow Interrupt Status is enabled
7-2 —	Reserved.
1 ABPSIS	Average Bits Per Slot Interrupt Status When set, this bit indicates that the MAC has updated the ABS value. This bit is cleared when the application writes 1 to this bit. Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect. 0b - Average Bits Per Slot Interrupt Status not detected 1b - Average Bits Per Slot Interrupt Status detected
0 TXUNFIS	Transmit Queue Underflow Interrupt Status This bit indicates that the Transmit Queue had an underflow while transmitting the packet. Transmission is suspended and an Underflow Error TDES3[2] is set. This bit is cleared when the application writes 1 to this bit. Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect. 0b - Transmit Queue Underflow Interrupt Status not detected 1b - Transmit Queue Underflow Interrupt Status detected

1.4.271 (MTL_RxQ1_Operation_Mode)

Offset

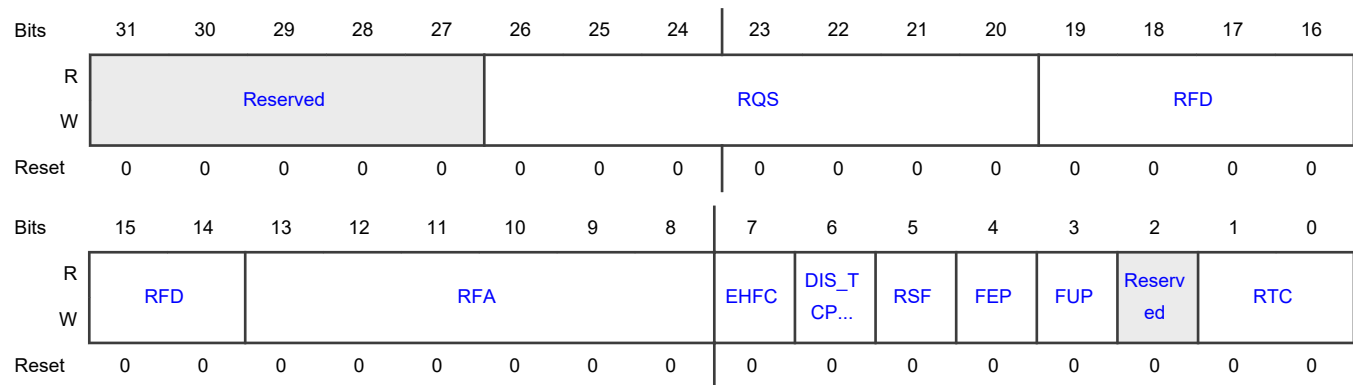
Register	Offset
MTL_RxQ1_Operation_Mode	D70h

Function

The Queue 1 Receive Operation Mode register establishes the Receive queue operating modes and command.

The RFA and RFD fields are not backward compatible with the RFA and RFD fields of 4.00a release

Diagram



Fields

Field	Function
31-27 —	Reserved.
26-20 RQS	Receive Queue Size This field indicates the size of the allocated Receive queues in blocks of 256 bytes. The RQS field is read-write only if the number of Rx Queues more than one, the reset value is 0x0 and indicates size of 256 bytes. This means that value of 0x0 = 256 bytes, 0x1 = 512 bytes and so on, so user should be programming RQS [5:0] = 6'b001111 to allocate queue size of 4096 (4K) bytes. In general, the size of the Queue = (RQS + 1)*256 bytes. When the number of Rx Queues is one, the field is read-only and the configured RX FIFO size in blocks of 256 bytes is reflected in the reset value. The width of this field depends on the Rx memory size selected in your configuration. For example, if the memory size is 2048, the width of this field is 3 bits: $\text{LOG2}(2048/256) = \text{LOG2}(8) = 3 \text{ bits}$
19-14 RFD	Threshold for Deactivating Flow Control (in half-duplex and full-duplex modes) These bits control the threshold (fill-level of Rx queue) at which the flow control is de-asserted after activation: - 0: Full minus 1 KB, that is, FULL 1 KB - 1: Full minus 1.5 KB, that is, FULL 1.5 KB - 2: Full minus 2 KB, that is, FULL 2 KB - 3: Full minus 2.5 KB, that is, FULL 2.5 KB - ... - 62: Full minus 32 KB, that is, FULL 32 KB - 63: Full minus 32.5 KB, that is, FULL 32.5 KB The de-assertion is effective only after flow control is asserted. Note: The value must be programmed in such a way to make sure that the threshold is a positive number.

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Field	Function
	When the EHFC is set high, these values are applicable only when the Rx queue size determined by the RQS field of this register, is equal to or greater than 4 KB. For a given queue size, the values ranges between 0 and the encoding for FULL minus (QSIZE - 0.5 KB) and all other values are illegal. Here the term FULL and QSIZE refers to the queue size determined by the RQS field of this register. The width of this field depends on RX FIFO size selected during the configuration. Remaining bits are reserved and read only.
13-8 RFA	Threshold for Activating Flow Control (in half-duplex and full-duplex) These bits control the threshold (fill-level of Rx queue) at which the flow control is activated: For more information on encoding for this field, see RFD.
7 EHFC	Enable Hardware Flow Control When this bit is set, the flow control signal operation, based on the fill-level of Rx queue, is enabled. When reset, the flow control operation is disabled. 0b - Hardware Flow Control is disabled 1b - Hardware Flow Control is enabled
6 DIS_TCP_EF	Disable Dropping of TCP/IP Checksum Error Packets When this bit is set, the MAC does not drop the packets which only have the errors detected by the Receive Checksum Offload engine. Such packets have errors only in the encapsulated payload. There are no errors (including FCS error) in the Ethernet packet received by the MAC. When this bit is reset, all error packets are dropped if the FEP bit is reset. 0b - Dropping of TCP/IP Checksum Error Packets is enabled 1b - Dropping of TCP/IP Checksum Error Packets is disabled
5 RSF	Receive Queue Store and Forward When this bit is set, the DWC_ether_qos reads a packet from the Rx queue only after the complete packet has been written to it, ignoring the RTC field of this register. When this bit is reset, the Rx queue operates in the Threshold (cut-through) mode, subject to the threshold specified by the RTC field of this register. 0b - Receive Queue Store and Forward is disabled 1b - Receive Queue Store and Forward is enabled
4 FEP	Forward Error Packets When this bit is reset, the Rx queue drops packets with error status (CRC error, GMII_ER, watchdog timeout, or overflow). However, if the start byte (write) pointer of a packet is already transferred to the read controller side (in Threshold mode), the packet is not dropped. When this bit is set, all packets except the runt error packets are forwarded to the application or DMA. If the RSF bit is set and the Rx queue overflows when a partial packet is written, the packet is dropped irrespective of the setting of this bit. However, if the RSF bit is reset and the Rx queue overflows when a partial packet is written, a partial packet might be forwarded to the application or DMA.

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Field	Function
	0b - Forward Error Packets is disabled 1b - Forward Error Packets is enabled
3 FUP	Forward Undersized Good Packets When this bit is set, the Rx queue forwards the undersized good packets (packets with no error and length less than 64 bytes), including pad-bytes and CRC. When this bit is reset, the Rx queue drops all packets of less than 64 bytes, unless a packet is already transferred because of the lower value of Rx Threshold, for example, RTC = 01. 0b - Forward Undersized Good Packets is disabled 1b - Forward Undersized Good Packets is enabled
2 —	Reserved.
1-0 RTC	Receive Queue Threshold Control These bits control the threshold level of the MTL Rx queue (in bytes): The received packet is transferred to the application or DMA when the packet size within the MTL Rx queue is larger than the threshold. In addition, full packets with length less than the threshold are automatically transferred. This field is valid only when the RSF bit is zero. This field is ignored when the RSF bit is set to 1. 00b - 64 01b - 32 10b - 96 11b - 128

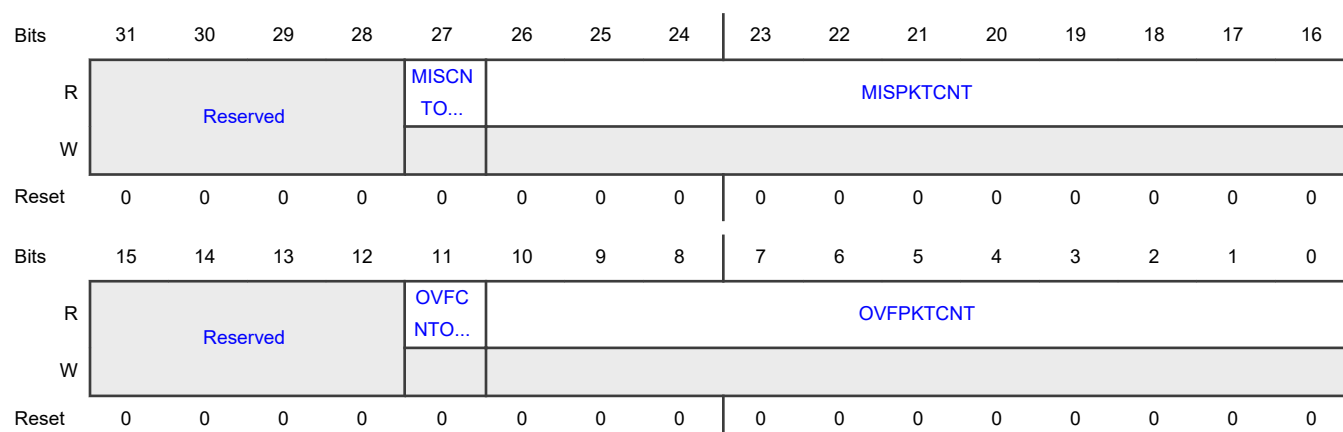
1.4.272 (MTL_RxQ1_Missed_Packet_Overflow_Cnt)

Offset

Register	Offset
MTL_RxQ1_Missed_Packet_Overflow_Cnt	D74h

Function

The Queue 1 Missed Packet and Overflow Counter register contains the counter for packets missed because of Receive queue packet flush and packets discarded because of Receive queue overflow.

Diagram**Fields**

Field	Function
31-28 —	Reserved.
27 MISCNTOVF	Missed Packet Counter Overflow Bit When set, this bit indicates that the Rx Queue Missed Packet Counter crossed the maximum limit. Access restriction applies. Clears on read. Self-set to 1 on internal event. 0b - Missed Packet Counter overflow not detected 1b - Missed Packet Counter overflow detected
26-16 MISPKTCNT	Missed Packet Counter This field indicates the number of packets missed by the DWC_ether_qos because the application asserted ari_pkt_flush_i[] for this queue. This counter is reset when this register is read with mci_be_i[0] at 1b1. This counter is incremented by 1 when the DMA discards the packet because of buffer unavailability. Access restriction applies. Clears on read. Self-set to 1 on internal event.
15-12 —	Reserved.
11 OVFCNTOVF	Overflow Counter Overflow Bit When set, this bit indicates that the Rx Queue Overflow Packet Counter field crossed the maximum limit. Access restriction applies. Clears on read. Self-set to 1 on internal event. 0b - Overflow Counter overflow not detected 1b - Overflow Counter overflow detected
10-0 OVFPKTCNT	Overflow Packet Counter

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Field	Function
	This field indicates the number of packets discarded by the DWC_ether_qos because of Receive queue overflow. This counter is incremented each time the DWC_ether_qos discards an incoming packet because of overflow. This counter is reset when this register is read with mci_be_i[0] at 1'b1. Access restriction applies. Clears on read. Self-set to 1 on internal event.

1.4.273 (MTL_RxQ1_Debug)

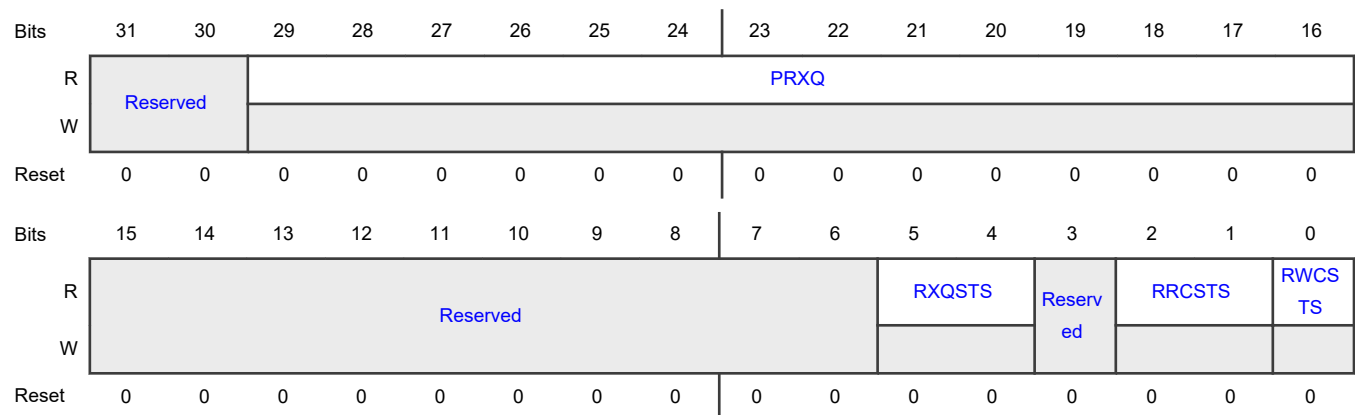
Offset

Register	Offset
MTL_RxQ1_Debug	D78h

Function

The Queue 1 Receive Debug register gives the debug status of various blocks related to the Receive queue.

Diagram



Fields

Field	Function
31-30 —	Reserved.
29-16 PRXQ	Number of Packets in Receive Queue This field indicates the current number of packets in the Rx Queue. The theoretical maximum value for this field is 256KB/16B = 16K Packets, that is, Max_Queue_Size/Min_Packet_Size.
15-6	Reserved.

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Field	Function
—	
5-4 RXQSTS	MTL Rx Queue Fill-Level Status This field gives the status of the fill-level of the Rx Queue: 00b - Rx Queue empty 01b - Rx Queue fill-level below flow-control deactivate threshold 10b - Rx Queue fill-level above flow-control activate threshold 11b - Rx Queue full
3 —	Reserved.
2-1 RRCSTS	MTL Rx Queue Read Controller State This field gives the state of the Rx queue Read controller: 00b - Idle state 01b - Reading packet data 10b - Reading packet status (or timestamp) 11b - Flushing the packet data and status
0 RWCSTS	MTL Rx Queue Write Controller Active Status When high, this bit indicates that the MTL Rx queue Write controller is active, and it is transferring a received packet to the Rx Queue. 0b - MTL Rx Queue Write Controller Active Status not detected 1b - MTL Rx Queue Write Controller Active Status detected

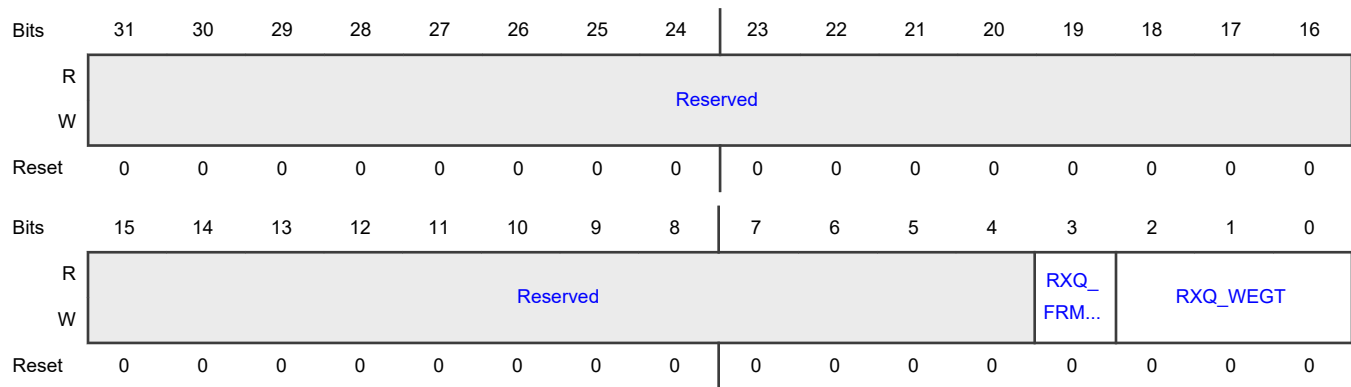
1.4.274 (MTL_RxQ1_Control)

Offset

Register	Offset
MTL_RxQ1_Control	D7Ch

Function

The Queue Receive Control register controls the receive arbitration and passing of received packets to the application.

Diagram**Fields**

Field	Function
31-4 —	Reserved.
3 RXQ_FRM_AR BIT	Receive Queue Packet Arbitration When this bit is set, the DWC_ether_qos drives the packet data to the ARI interface such that the entire packet data of currently-selected queue is transmitted before switching to other queue. When this bit is reset, the DWC_ether_qos drives the packet data to the ARI interface such that the following amount of data of currently-selected queue is transmitted before switching to other queue: - PBL amount of data (indicated by ari_qN_pbl_i[]) or - Complete data of a packet The status and the timestamp are not a part of the PBL data. Therefore, the DWC_ether_qos drives the complete status (including timestamp status) during first PBL request for the packet (in store-and-forward mode) or the last PBL request for the packet (in Threshold mode). 0b - Receive Queue Packet Arbitration is disabled 1b - Receive Queue Packet Arbitration is enabled
2-0 RXQ_WEGT	Receive Queue Weight This field indicates the weight assigned to the Rx Queue 0. This field needs to be programmed with one value less than the required weight, i.e. reset value of 0 indicates weight of 1, value of 1 indicates weight of 2, and so on. The weight is used as the number of continuous PBL or packets requests (depending on the RXQ_FRM_ARBIT) allocated to the queue in one arbitration cycle. Note: The change in value of RXQ_WEGT takes effect only after the completion of current service round or when there is change from RAA=SP to RAA=WSP algorithm. This approach is taken so that there is smooth transition. For the RXQ_WEGT value to take effect at the start, the MTL_RxQ(##)_Control registers must be programmed before the MTL_Operation_Mode register.

1.4.275 (MTL_TxQ2_Operation_Mode)

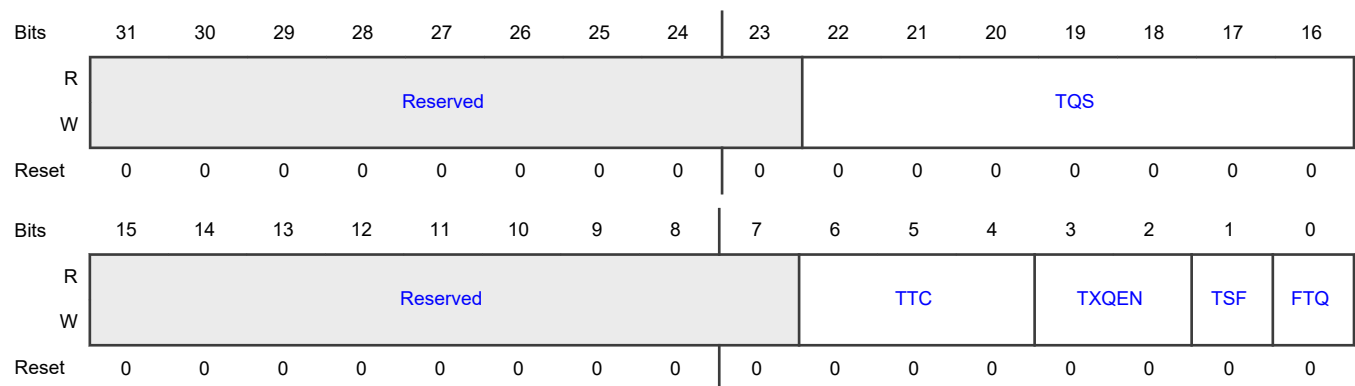
Offset

Register	Offset
MTL_TxQ2_Operation_Mode	D80h

Function

The Queue 2 Transmit Operation Mode register establishes the Transmit queue operating modes and commands.

Diagram



Fields

Field	Function
31-23 —	Reserved.
22-16 TQS	Transmit Queue Size This field indicates the size of the allocated Transmit queues in blocks of 256 bytes. The TQS field is read-write only if the number of Tx Queues more than one, the reset value is 0x0 and indicates size of 256 bytes. This means that value of 0x0 = 256 bytes, 0x1 = 512 bytes and so on, so user should be programming TQS [5:0] = 6'b001111 to allocate queue size of 4096 (4K) bytes. In general, the size of the Queue = (TQS + 1)*256 bytes. When the number of Tx Queues is one, the field is read-only and the configured TX FIFO size in blocks of 256 bytes is reflected in the reset value. The width of this field depends on the Tx memory size selected in your configuration. For example, if the memory size is 2048, the width of this field is 3 bits: $\text{LOG}_2(2048/256) = \text{LOG}_2(8) = 3 \text{ bits}$
15-7 —	Reserved.

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Field	Function
6-4 TTC	Transmit Threshold Control These bits control the threshold level of the MTL Tx Queue. The transmission starts when the packet size within the MTL Tx Queue is larger than the threshold. In addition, full packets with length less than the threshold are also transmitted. These bits are used only when the TSF bit is reset. 000b - 32 001b - 64 010b - 96 011b - 128 100b - 192 101b - 256 110b - 384 111b - 512
3-2 TXQEN	Transmit Queue Enable This field is used to enable/disable the transmit queue 0. - 2'b00: Not enabled - 2'b01: Enable in AV mode - 2'b10: Enabled - 2'b11: Reserved Note: In multiple Tx queues configuration, all the queues are disabled by default. Enable the Tx queue by programming this field. 00b - Not enabled 01b - Enable in AV mode (Reserved in non-AV) 10b - Enabled 11b - Reserved
1 TSF	Transmit Store and Forward When this bit is set, the transmission starts when a full packet resides in the MTL Tx queue. When this bit is set, the TTC values specified in Bits[6:4] of this register are ignored. This bit should be changed only when the transmission is stopped. 0b - Transmit Store and Forward is disabled 1b - Transmit Store and Forward is enabled
0 FTQ	Flush Transmit Queue When this bit is set, the Tx queue controller logic is reset to its default values. Therefore, all the data in the Tx queue is lost or flushed. This bit is internally reset when the flushing operation is complete. Until this bit is reset, you should not write to the MTL_TxQ1_Operation_Mode register. The data which is already

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Field	Function
	accepted by the MAC transmitter is not flushed. It is scheduled for transmission and results in underflow and runt packet transmission. Note: The flush operation is complete only when the Tx queue is empty and the application has accepted the pending Tx Status of all transmitted packets. To complete this flush operation, the PHY Tx clock (clk_tx_i) should be active. Access restriction applies. Setting 1 sets. Self-cleared. Setting 0 has no effect. 0b - Flush Transmit Queue is disabled 1b - Flush Transmit Queue is enabled

1.4.276 (MTL_TxQ2_Underflow)

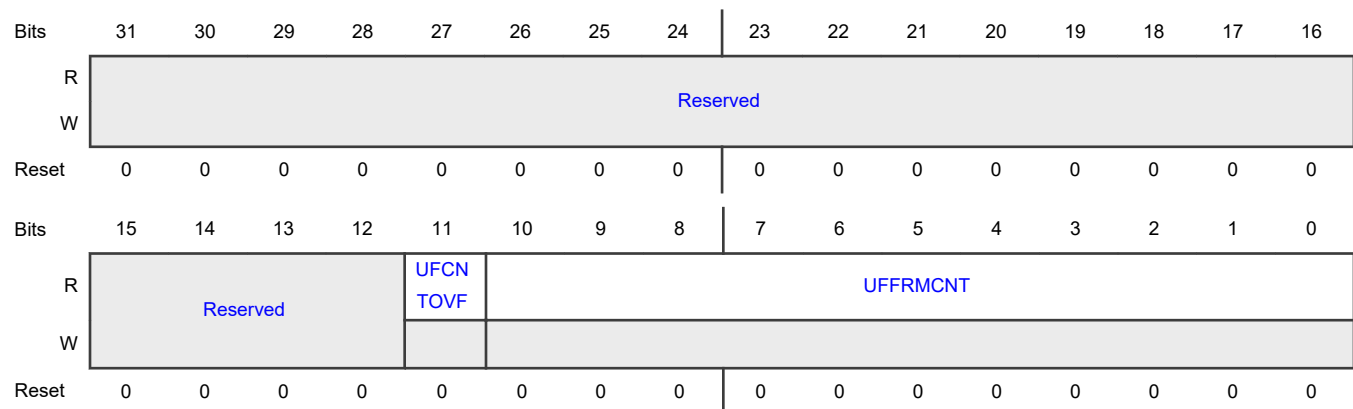
Offset

Register	Offset
MTL_TxQ2_Underflow	D84h

Function

The Queue 2 Underflow Counter register contains the counter for packets aborted because of Transmit queue underflow and packets missed because of Receive queue packet flush

Diagram



Fields

Field	Function
31-12	Reserved.
—	

Table continues on the next page...

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Field	Function
11 UFCNTOVF	Overflow Bit for Underflow Packet Counter This bit is set every time the Tx queue Underflow Packet Counter field overflows, that is, it has crossed the maximum count. In such a scenario, the overflow packet counter is reset to all-zeros and this bit indicates that the rollover happened. Access restriction applies. Clears on read. Self-set to 1 on internal event. 0b - Overflow not detected for Underflow Packet Counter 1b - Overflow detected for Underflow Packet Counter
10-0 UFFRMCNT	Underflow Packet Counter This field indicates the number of packets aborted by the controller because of Tx Queue Underflow. This counter is incremented each time the MAC aborts outgoing packet because of underflow. The counter is cleared when this register is read with mci_be_i[0] at 1'b1. Access restriction applies. Clears on read. Self-set to 1 on internal event.

1.4.277 (MTL_TxQ2_Debug)

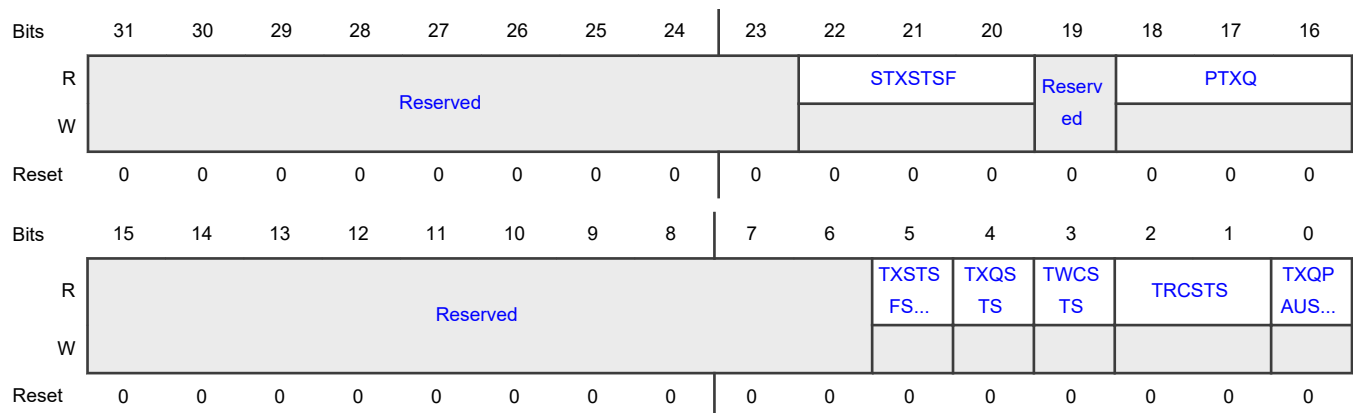
Offset

Register	Offset
MTL_TxQ2_Debug	D88h

Function

The Queue 2 Transmit Debug register gives the debug status of various blocks related to the Transmit queue.

Diagram



Fields

Field	Function
31-23 —	Reserved.
22-20 STXSTS	Number of Status Words in Tx Status FIFO of Queue This field indicates the current number of status in the Tx Status FIFO of this queue. When the DTXSTS bit of MTL_Operation_Mode register is set to 1, this field does not reflect the number of status words in Tx Status FIFO.
19 —	Reserved.
18-16 PTXQ	Number of Packets in the Transmit Queue This field indicates the current number of packets in the Tx Queue. When the DTXSTS bit of MTL_Operation_Mode register is set to 1, this field does not reflect the number of packets in the Transmit queue.
15-6 —	Reserved.
5 TXSTS	MTL Tx Status FIFO Full Status When high, this bit indicates that the MTL Tx Status FIFO is full. Therefore, the MTL cannot accept any more packets for transmission. 0b - MTL Tx Status FIFO Full status is not detected 1b - MTL Tx Status FIFO Full status is detected
4 TXQSTS	MTL Tx Queue Not Empty Status When this bit is high, it indicates that the MTL Tx Queue is not empty and some data is left for transmission. 0b - MTL Tx Queue Not Empty status is not detected 1b - MTL Tx Queue Not Empty status is detected
3 TWCSTS	MTL Tx Queue Write Controller Status When high, this bit indicates that the MTL Tx Queue Write Controller is active, and it is transferring the data to the Tx Queue. 0b - MTL Tx Queue Write Controller status is not detected 1b - MTL Tx Queue Write Controller status is detected
2-1 TRCSTS	MTL Tx Queue Read Controller Status This field indicates the state of the Tx Queue Read Controller: 00b - Idle state 01b - Read state (transferring data to the MAC transmitter)

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Field	Function
	10b - Waiting for pending Tx Status from the MAC transmitter 11b - Flushing the Tx queue because of the Packet Abort request from the MAC
0 TXQPAUSED	Transmit Queue in Pause When this bit is high and the Rx flow control is enabled, it indicates that the Tx Queue is in the Pause condition (in the full-duplex only mode) because of the following: - Reception of the PFC packet for the priorities assigned to the Tx Queue when PFC is enabled - Reception of 802.3x Pause packet when PFC is disabled 0b - Transmit Queue in Pause status is not detected 1b - Transmit Queue in Pause status is detected

1.4.278 (MTL_TxQ2_ETS_Control)

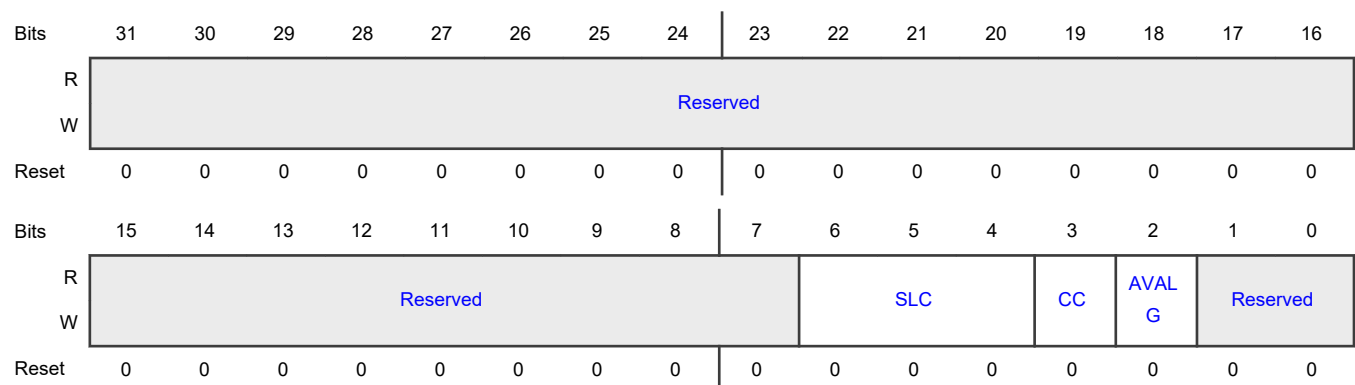
Offset

Register	Offset
MTL_TxQ2_ETS_Control	D90h

Function

The Queue ETS Control register controls the enhanced transmission selection operation.

Diagram



Fields

Field	Function
31-7	Reserved.

Table continues on the next page...

Table continued from the previous page...

Field	Function
—	
6-4 SLC	Slot Count If the credit-based shaper algorithm is enabled, the software can program the number of slots (of duration programmed in DMA_CH(#i)_Slot_Interval register) over which the average transmitted bits per slot, provided in the MTL_TxQ(#i)_ETS_Status register, need to be computed for Queue. The encoding is as follows: 000b - 1 slot 001b - 2 slots 010b - 4 slots 011b - 8 slots 100b - 16 slots 101b - Reserved
3 CC	Credit Control When this bit is set, the accumulated credit parameter in the credit-based shaper algorithm logic is not reset to zero when there is positive credit and no packet to transmit in Channel 1. The credit accumulates even when there is no packet waiting in Channel 1 and another channel is transmitting. When this bit is reset, the accumulated credit parameter in the credit-based shaper algorithm logic is set to zero when there is positive credit and no packet to transmit in Channel 1. When there is no packet waiting in Channel 1 and other channel is transmitting, no credit is accumulated. 0b - Credit Control is disabled 1b - Credit Control is enabled
2 AVALG	AV Algorithm When Queue 1 is programmed for AV, this field configures the scheduling algorithm for this queue: This bit when set, indicates credit based shaper algorithm (CBS) is selected for Queue 1 traffic. When reset, strict priority is selected. 0b - CBS Algorithm is disabled 1b - CBS Algorithm is enabled
1-0 —	Reserved.

1.4.279 (MTL_TxQ2_ETS_Status)

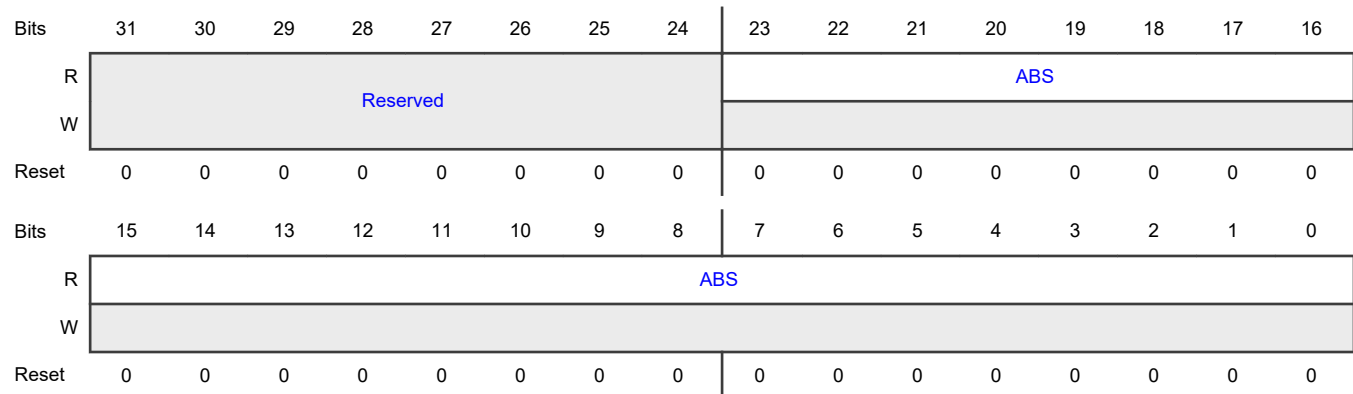
Offset

Register	Offset
MTL_TxQ2_ETS_Status	D94h

Function

The Queue 2 ETS Status register provides the average traffic transmitted in Queue 2.

Diagram



Fields

Field	Function
31-24 —	Reserved.
23-0 ABS	Average Bits per Slot This field contains the average transmitted bits per slot. If AV operation is enabled, this field is computed over number of slots, programmed in the SLC field of MTL_TxQ(#i)_ETS_CONTROL register. The maximum value of this field is 0x6_4000 in 100 Mbps, 0x3E_8000 in 1000 Mbps and 9C_4000 in 2500 Mbps mode respectively. When the DCB operation is enabled for Queue, this field is computed over every 10 million bit times slot (4 ms in 2500 Mbps; 10 ms in 1000 Mbps; 100 ms in 100 Mbps). The maximum value is 0x989680.

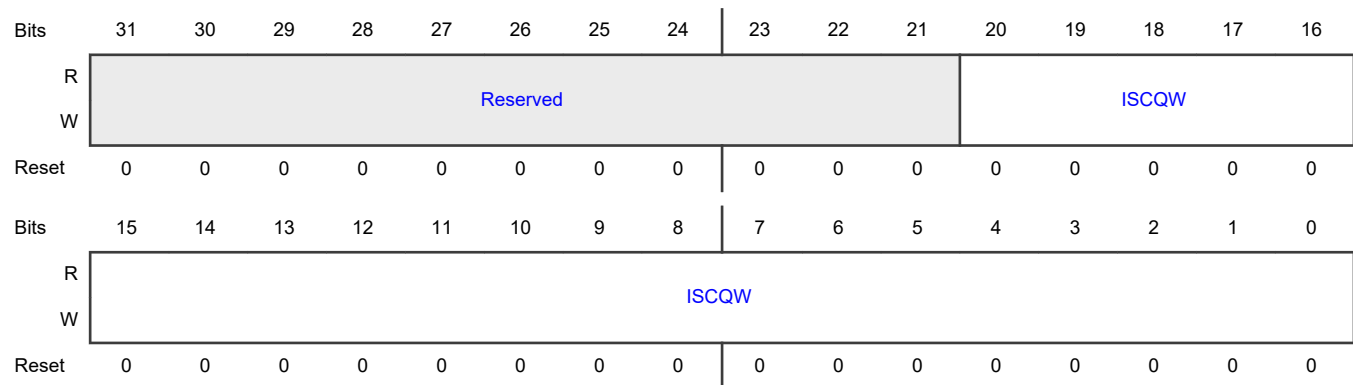
1.4.280 (MTL_TxQ2_Quantum_Weight)

Offset

Register	Offset
MTL_TxQ2_Quantum_Weight	D98h

Function

The Queue 2 idleSlopeCredit, Quantum or Weights register provides the average traffic transmitted in Queue 2.

Diagram**Fields**

Field	Function
31-21 —	Reserved.
20-0 ISCQW	idleSlopeCredit, Quantum or Weights - idleSlopeCredit When AV feature is enabled, this field contains the idleSlopeCredit value required for the credit-based shaper algorithm for Queue 1. This is the rate of change of credit in bits per cycle (40 ns for 100 Mbps; 8 ns for 1000 Mbps; 3.2 ns for 2500 Mbps) when the credit is increasing. The software should program this field with computed credit in bits per cycle scaled by 1,024. The maximum value is portTransmitRate, that is, 0x2000 in 1000/2500 Mbps mode and 0x1000 in 100 Mbps mode. Bits[20:14] must be written to zero. - Quantum When the DCB operation is enabled with DWRR algorithm for Queue 1 traffic, this field contains the quantum value in bytes to be added to credit during every queue scanning cycle. The maximum value is 0x1312D0 bytes. - Weights When DCB operation is enabled with WFQ algorithm for Queue 1 traffic, this field contains the weight for this queue. The maximum value is 0x3FFF where weight of 0 indicates 100% bandwidth. Bits[20:14] must be written to zero. When DCB operation or generic queuing operation is enabled with WRR algorithm for Queue 1 traffic, this field contains the weight for this queue. The maximum value is 0x64. Bits [20:7] must be written to zero. - Note 1: In multiple Queue configuration this field in respective per queue register must be programmed to some non-zero value when multiple queues are enabled or single queue other than Q0 is enabled. This field need not be programmed when only Q0 is enabled. In general, when WRR algorithm is selected a non-zero value must be programmed on both Receive and Transmit. In Receive, the register is MTL_Operation_Mode register. - Note 2: For WFQ algorithm, higher the programmed weights lesser the bandwidth allocated for that Transmit Queue. The finish time is not a function of particular packet alone but it is as per the formula: (previous_finish_time of particular Transmit Queue + (weights*packet_size))

Table continues on the next page...

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Field	Function
	- Note 3: The weights programmed do not correspond to the number of packets but the fraction of bandwidth or time allocated for particular queue w.r.t. total BW or time.

1.4.281 (MTL_TxQ2_SendSlopeCredit)

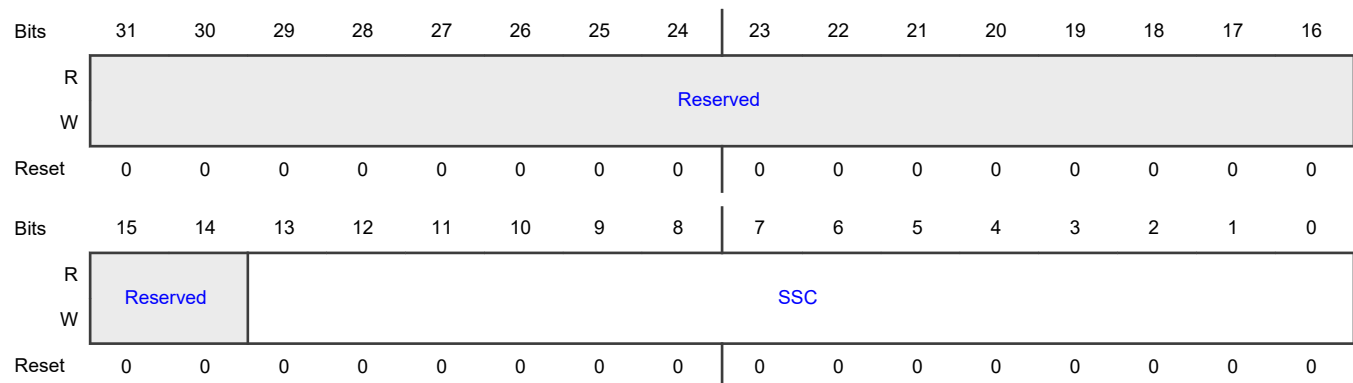
Offset

Register	Offset
MTL_TxQ2_SendSlopeCredit	D9Ch

Function

The sendSlopeCredit register contains the sendSlope credit value required for the credit-based shaper algorithm for the Queue.

Diagram



Fields

Field	Function
31-14 —	Reserved.
13-0 SSC	sendSlopeCredit Value When AV operation is enabled, this field contains the sendSlopeCredit value required for credit-based shaper algorithm for Queue 1. This is the rate of change of credit in bits per cycle (40 ns, 8 ns and 3.2 ns for 100 Mbps, 1000 Mbps and 2500 Mbps respectively) when the credit is decreasing. The software should program this field with computed credit in bits per cycle scaled by 1,024. The maximum value is portTransmitRate, that is, 0x2000 in 1000/2500 Mbps mode and 0x1000 in 100 Mbps mode. This field should be programmed with absolute sendSlopeCredit value. The credit-based shaper logic subtracts it from the accumulated credit when Channel 1 is selected for transmission.

1.4.282 (MTL_TxQ2_HiCredit)

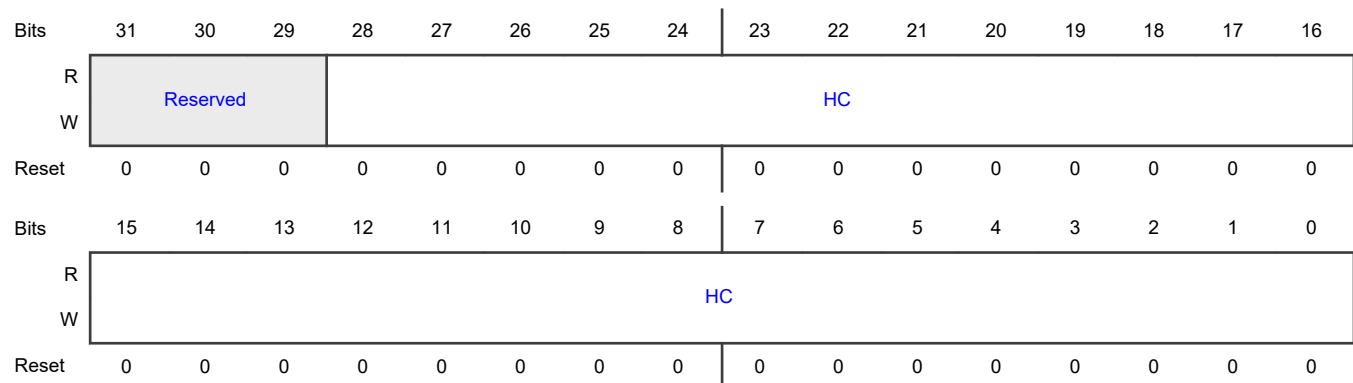
Offset

Register	Offset
MTL_TxQ2_HiCredit	DA0h

Function

The hiCredit register contains the hiCredit value required for the credit-based shaper algorithm for the Queue.

Diagram



Fields

Field	Function
31-29 —	Reserved.
28-0 HC	hiCredit Value When the AV feature is enabled, this field contains the hiCredit value required for the credit-based shaper algorithm. This is the maximum value that can be accumulated in the credit parameter. This is specified in bits scaled by 1,024. The maximum value is maxInterferenceSize, that is, best-effort maximum packet size (16,384 bytes or 131,072 bits). The value to be specified is $131,072 * 1,024 = 134,217,728$ or 0x0800_0000.

1.4.283 (MTL_TxQ2_LoCredit)

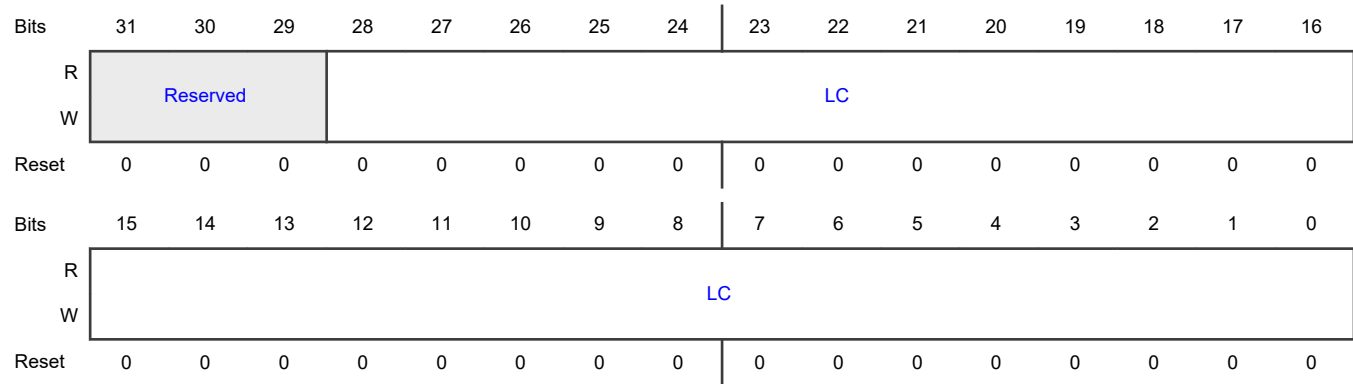
Offset

Register	Offset
MTL_TxQ2_LoCredit	DA4h

Function

The loCredit register contains the loCredit value required for the credit-based shaper algorithm for the Queue.

Diagram



Fields

Field	Function
31-29 —	Reserved.
28-0 LC	loCredit Value When AV operation is enabled, this field contains the loCredit value required for the credit-based shaper algorithm. This is the minimum value that can be accumulated in the credit parameter. This is specified in bits scaled by 1,024. The maximum value to be programmed is corresponds to twice the maxFrameSize transmitted from this queue. If the maxFrameSize is 8192 bytes, then $(8192*2) * 8 * 1024 = 134,217,728$ or 0x0800_0000. Because it is a negative value, the programmed value is 2's complement of the value, that is, 0x1800_0000.

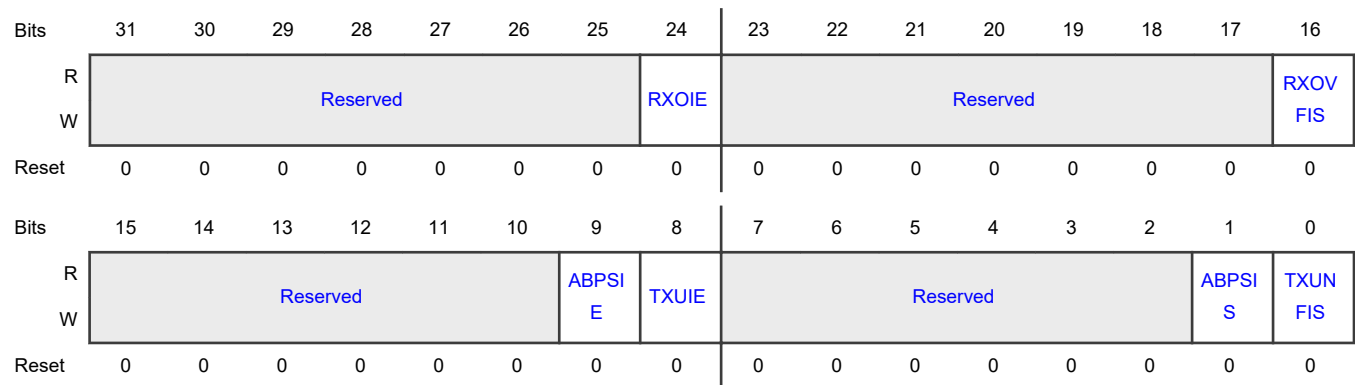
1.4.284 (MTL_Q2_Interrupt_Control_Status)

Offset

Register	Offset
MTL_Q2_Interrupt_Contr ol_Status	DACH

Function

This register contains the interrupt enable and status bits for the queue 2 interrupts.

Diagram**Fields**

Field	Function
31-25 —	Reserved.
24 RXOIE	Receive Queue Overflow Interrupt Enable When this bit is set, the Receive Queue Overflow interrupt is enabled. When this bit is reset, the Receive Queue Overflow interrupt is disabled. 0b - Receive Queue Overflow Interrupt is disabled 1b - Receive Queue Overflow Interrupt is enabled
23-17 —	Reserved.
16 RXOVFIS	Receive Queue Overflow Interrupt Status This bit indicates that the Receive Queue had an overflow while receiving the packet. If a partial packet is transferred to the application, the overflow status is set in RDES3[21]. This bit is cleared when the application writes 1 to this bit. Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect. 0b - Receive Queue Overflow Interrupt Status not detected 1b - Receive Queue Overflow Interrupt Status detected
15-10 —	Reserved.
9 ABPSIE	Average Bits Per Slot Interrupt Enable When this bit is set, the MAC asserts the sbd_intr_o or mci_intr_o interrupt when the average bits per slot status is updated. When this bit is cleared, the interrupt is not asserted for such an event. 0b - Average Bits Per Slot Interrupt is disabled

Table continues on the next page...

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Field	Function
	1b - Average Bits Per Slot Interrupt is enabled
8 TXUIE	Transmit Queue Underflow Interrupt Enable When this bit is set, the Transmit Queue Underflow interrupt is enabled. When this bit is reset, the Transmit Queue Underflow interrupt is disabled. 0b - Transmit Queue Underflow Interrupt Status is disabled 1b - Transmit Queue Underflow Interrupt Status is enabled
7-2 —	Reserved.
1 ABPSIS	Average Bits Per Slot Interrupt Status When set, this bit indicates that the MAC has updated the ABS value. This bit is cleared when the application writes 1 to this bit. Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect. 0b - Average Bits Per Slot Interrupt Status not detected 1b - Average Bits Per Slot Interrupt Status detected
0 TXUNFIS	Transmit Queue Underflow Interrupt Status This bit indicates that the Transmit Queue had an underflow while transmitting the packet. Transmission is suspended and an Underflow Error TDES3[2] is set. This bit is cleared when the application writes 1 to this bit. Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect. 0b - Transmit Queue Underflow Interrupt Status not detected 1b - Transmit Queue Underflow Interrupt Status detected

1.4.285 (MTL_RxQ2_Operation_Mode)

Offset

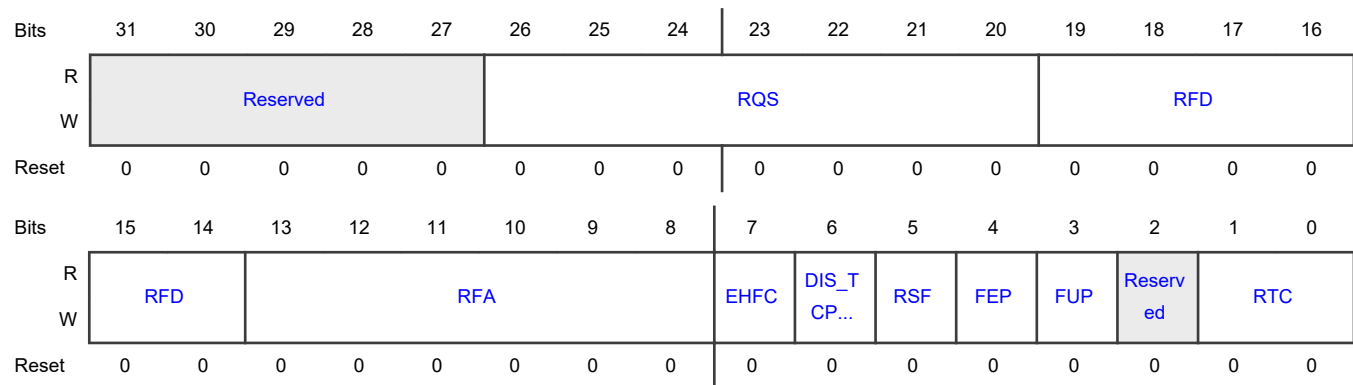
Register	Offset
MTL_RxQ2_Operation_Mode	DB0h

Function

The Queue 2 Receive Operation Mode register establishes the Receive queue operating modes and command.

The RFA and RFD fields are not backward compatible with the RFA and RFD fields of 4.00a release

Diagram



Fields

Field	Function
31-27 —	Reserved.
26-20 RQS	Receive Queue Size This field indicates the size of the allocated Receive queues in blocks of 256 bytes. The RQS field is read-write only if the number of Rx Queues more than one, the reset value is 0x0 and indicates size of 256 bytes. This means that value of 0x0 = 256 bytes, 0x1 = 512 bytes and so on, so user should be programming RQS [5:0] = 6'b001111 to allocate queue size of 4096 (4K) bytes. In general, the size of the Queue = (RQS + 1)*256 bytes. When the number of Rx Queues is one, the field is read-only and the configured RX FIFO size in blocks of 256 bytes is reflected in the reset value. The width of this field depends on the Rx memory size selected in your configuration. For example, if the memory size is 2048, the width of this field is 3 bits: $\text{LOG2}(2048/256) = \text{LOG2}(8) = 3 \text{ bits}$
19-14 RFD	Threshold for Deactivating Flow Control (in half-duplex and full-duplex modes) These bits control the threshold (fill-level of Rx queue) at which the flow control is de-asserted after activation: - 0: Full minus 1 KB, that is, FULL 1 KB - 1: Full minus 1.5 KB, that is, FULL 1.5 KB - 2: Full minus 2 KB, that is, FULL 2 KB - 3: Full minus 2.5 KB, that is, FULL 2.5 KB - ... - 62: Full minus 32 KB, that is, FULL 32 KB - 63: Full minus 32.5 KB, that is, FULL 32.5 KB The de-assertion is effective only after flow control is asserted. Note: The value must be programmed in such a way to make sure that the threshold is a positive number.

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Field	Function
	When the EHFC is set high, these values are applicable only when the Rx queue size determined by the RQS field of this register, is equal to or greater than 4 KB. For a given queue size, the values ranges between 0 and the encoding for FULL minus (QSIZE - 0.5 KB) and all other values are illegal. Here the term FULL and QSIZE refers to the queue size determined by the RQS field of this register. The width of this field depends on RX FIFO size selected during the configuration. Remaining bits are reserved and read only.
13-8 RFA	Threshold for Activating Flow Control (in half-duplex and full-duplex) These bits control the threshold (fill-level of Rx queue) at which the flow control is activated: For more information on encoding for this field, see RFD.
7 EHFC	Enable Hardware Flow Control When this bit is set, the flow control signal operation, based on the fill-level of Rx queue, is enabled. When reset, the flow control operation is disabled. 0b - Hardware Flow Control is disabled 1b - Hardware Flow Control is enabled
6 DIS_TCP_EF	Disable Dropping of TCP/IP Checksum Error Packets When this bit is set, the MAC does not drop the packets which only have the errors detected by the Receive Checksum Offload engine. Such packets have errors only in the encapsulated payload. There are no errors (including FCS error) in the Ethernet packet received by the MAC. When this bit is reset, all error packets are dropped if the FEP bit is reset. 0b - Dropping of TCP/IP Checksum Error Packets is enabled 1b - Dropping of TCP/IP Checksum Error Packets is disabled
5 RSF	Receive Queue Store and Forward When this bit is set, the DWC_ether_qos reads a packet from the Rx queue only after the complete packet has been written to it, ignoring the RTC field of this register. When this bit is reset, the Rx queue operates in the Threshold (cut-through) mode, subject to the threshold specified by the RTC field of this register. 0b - Receive Queue Store and Forward is disabled 1b - Receive Queue Store and Forward is enabled
4 FEP	Forward Error Packets When this bit is reset, the Rx queue drops packets with error status (CRC error, GMII_ER, watchdog timeout, or overflow). However, if the start byte (write) pointer of a packet is already transferred to the read controller side (in Threshold mode), the packet is not dropped. When this bit is set, all packets except the runt error packets are forwarded to the application or DMA. If the RSF bit is set and the Rx queue overflows when a partial packet is written, the packet is dropped irrespective of the setting of this bit. However, if the RSF bit is reset and the Rx queue overflows when a partial packet is written, a partial packet might be forwarded to the application or DMA.

Table continues on the next page...

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Field	Function
	0b - Forward Error Packets is disabled 1b - Forward Error Packets is enabled
3 FUP	Forward Undersized Good Packets When this bit is set, the Rx queue forwards the undersized good packets (packets with no error and length less than 64 bytes), including pad-bytes and CRC. When this bit is reset, the Rx queue drops all packets of less than 64 bytes, unless a packet is already transferred because of the lower value of Rx Threshold, for example, RTC = 01. 0b - Forward Undersized Good Packets is disabled 1b - Forward Undersized Good Packets is enabled
2 —	Reserved.
1-0 RTC	Receive Queue Threshold Control These bits control the threshold level of the MTL Rx queue (in bytes): The received packet is transferred to the application or DMA when the packet size within the MTL Rx queue is larger than the threshold. In addition, full packets with length less than the threshold are automatically transferred. This field is valid only when the RSF bit is zero. This field is ignored when the RSF bit is set to 1. 00b - 64 01b - 32 10b - 96 11b - 128

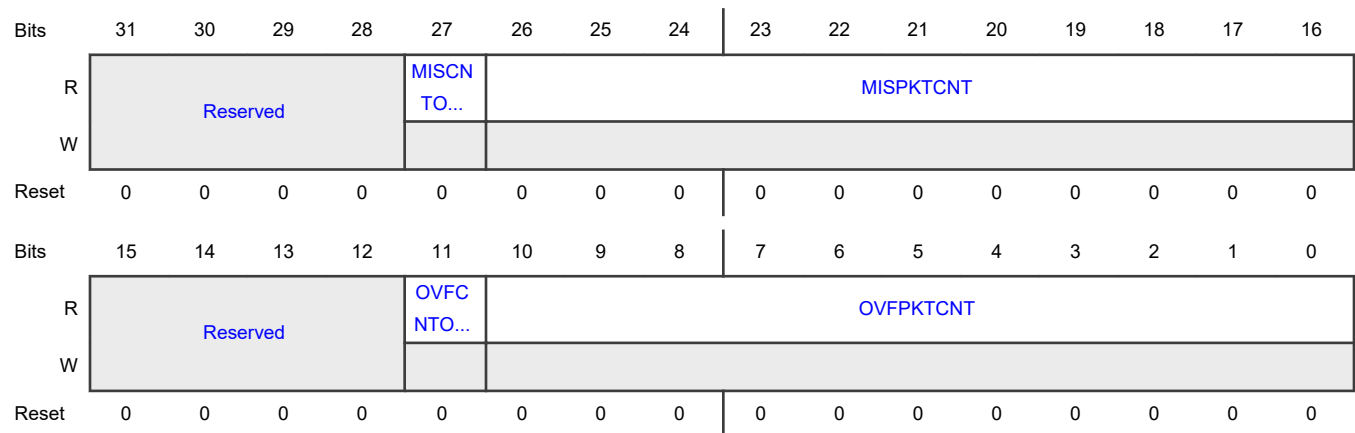
1.4.286 (MTL_RxQ2_Missed_Packet_Overflow_Cnt)

Offset

Register	Offset
MTL_RxQ2_Missed_Packet_Overflow_Cnt	DB4h

Function

The Queue 2 Missed Packet and Overflow Counter register contains the counter for packets missed because of Receive queue packet flush and packets discarded because of Receive queue overflow.

Diagram**Fields**

Field	Function
31-28 —	Reserved.
27 MISCNTOVF	Missed Packet Counter Overflow Bit When set, this bit indicates that the Rx Queue Missed Packet Counter crossed the maximum limit. Access restriction applies. Clears on read. Self-set to 1 on internal event. 0b - Missed Packet Counter overflow not detected 1b - Missed Packet Counter overflow detected
26-16 MISPKTCNT	Missed Packet Counter This field indicates the number of packets missed by the DWC_ether_qos because the application asserted ari_pkt_flush_i[] for this queue. This counter is reset when this register is read with mci_be_i[0] at 1b1. This counter is incremented by 1 when the DMA discards the packet because of buffer unavailability. Access restriction applies. Clears on read. Self-set to 1 on internal event.
15-12 —	Reserved.
11 OVFCNTOVF	Overflow Counter Overflow Bit When set, this bit indicates that the Rx Queue Overflow Packet Counter field crossed the maximum limit. Access restriction applies. Clears on read. Self-set to 1 on internal event. 0b - Overflow Counter overflow not detected 1b - Overflow Counter overflow detected
10-0 OVFPKTCNT	Overflow Packet Counter

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Field	Function
	This field indicates the number of packets discarded by the DWC_ether_qos because of Receive queue overflow. This counter is incremented each time the DWC_ether_qos discards an incoming packet because of overflow. This counter is reset when this register is read with mci_be_i[0] at 1'b1. Access restriction applies. Clears on read. Self-set to 1 on internal event.

1.4.287 (MTL_RxQ2_Debug)

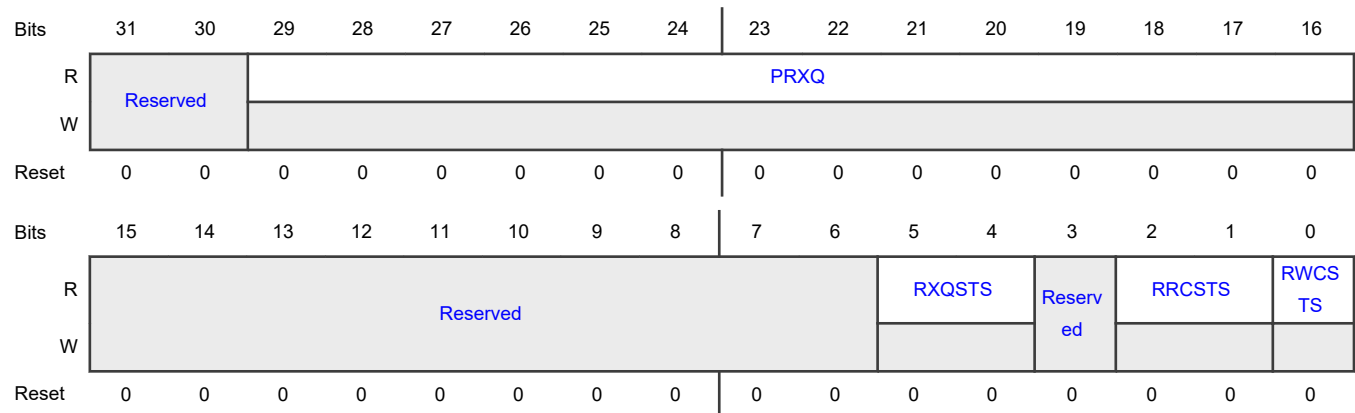
Offset

Register	Offset
MTL_RxQ2_Debug	DB8h

Function

The Queue 2 Receive Debug register gives the debug status of various blocks related to the Receive queue.

Diagram



Fields

Field	Function
31-30 —	Reserved.
29-16 PRXQ	Number of Packets in Receive Queue This field indicates the current number of packets in the Rx Queue. The theoretical maximum value for this field is 256KB/16B = 16K Packets, that is, Max_Queue_Size/Min_Packet_Size.
15-6	Reserved.

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Field	Function
—	
5-4 RXQSTS	MTL Rx Queue Fill-Level Status This field gives the status of the fill-level of the Rx Queue: 00b - Rx Queue empty 01b - Rx Queue fill-level below flow-control deactivate threshold 10b - Rx Queue fill-level above flow-control activate threshold 11b - Rx Queue full
3 —	Reserved.
2-1 RRCSTS	MTL Rx Queue Read Controller State This field gives the state of the Rx queue Read controller: 00b - Idle state 01b - Reading packet data 10b - Reading packet status (or timestamp) 11b - Flushing the packet data and status
0 RWCSTS	MTL Rx Queue Write Controller Active Status When high, this bit indicates that the MTL Rx queue Write controller is active, and it is transferring a received packet to the Rx Queue. 0b - MTL Rx Queue Write Controller Active Status not detected 1b - MTL Rx Queue Write Controller Active Status detected

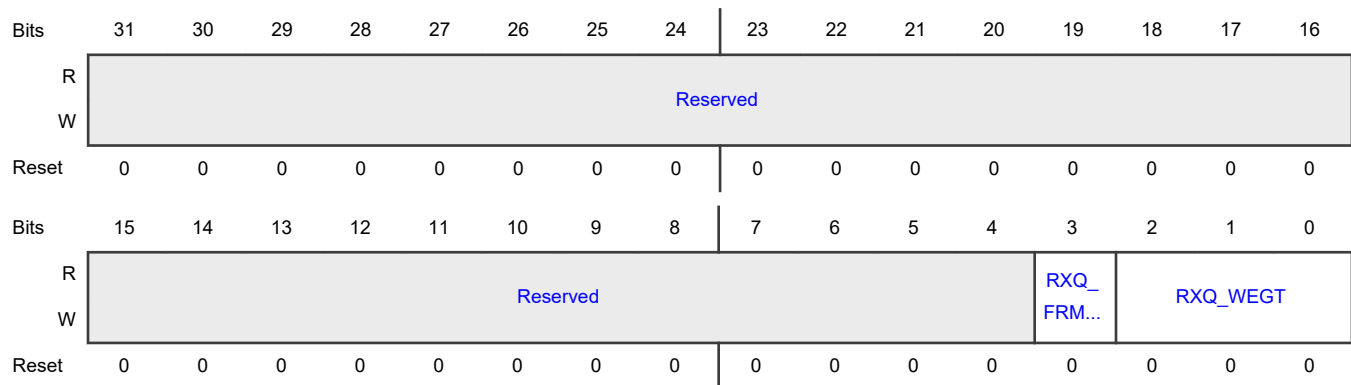
1.4.288 (MTL_RxQ2_Control)

Offset

Register	Offset
MTL_RxQ2_Control	DBCh

Function

The Queue Receive Control register controls the receive arbitration and passing of received packets to the application.

Diagram**Fields**

Field	Function
31-4 —	Reserved.
3 RXQ_FRM_AR BIT	Receive Queue Packet Arbitration When this bit is set, the DWC_ether_qos drives the packet data to the ARI interface such that the entire packet data of currently-selected queue is transmitted before switching to other queue. When this bit is reset, the DWC_ether_qos drives the packet data to the ARI interface such that the following amount of data of currently-selected queue is transmitted before switching to other queue: - PBL amount of data (indicated by ari_qN_pbl_i[]) or - Complete data of a packet The status and the timestamp are not a part of the PBL data. Therefore, the DWC_ether_qos drives the complete status (including timestamp status) during first PBL request for the packet (in store-and-forward mode) or the last PBL request for the packet (in Threshold mode). 0b - Receive Queue Packet Arbitration is disabled 1b - Receive Queue Packet Arbitration is enabled
2-0 RXQ_WEGT	Receive Queue Weight This field indicates the weight assigned to the Rx Queue 0. This field needs to be programmed with one value less than the required weight, i.e. reset value of 0 indicates weight of 1, value of 1 indicates weight of 2, and so on. The weight is used as the number of continuous PBL or packets requests (depending on the RXQ_FRM_ARBIT) allocated to the queue in one arbitration cycle. Note: The change in value of RXQ_WEGT takes effect only after the completion of current service round or when there is change from RAA=SP to RAA=WSP algorithm. This approach is taken so that there is smooth transition. For the RXQ_WEGT value to take effect at the start, the MTL_RxQ(#i)_Control registers must be programmed before the MTL_Operation_Mode register.

1.4.289 (MTL_TxQ3_Operation_Mode)

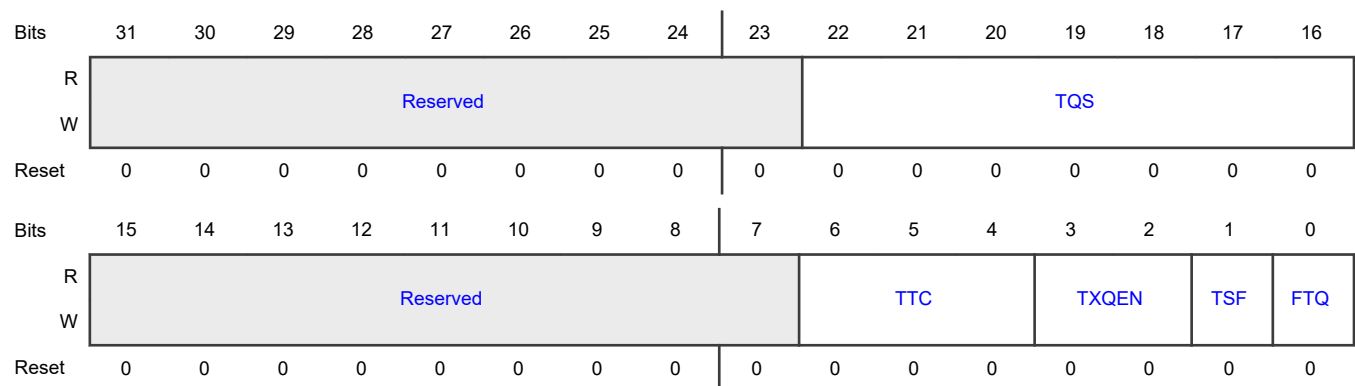
Offset

Register	Offset
MTL_TxQ3_Operation_Mode	DC0h

Function

The Queue 3 Transmit Operation Mode register establishes the Transmit queue operating modes and commands.

Diagram



Fields

Field	Function
31-23 —	Reserved.
22-16 TQS	Transmit Queue Size This field indicates the size of the allocated Transmit queues in blocks of 256 bytes. The TQS field is read-write only if the number of Tx Queues more than one, the reset value is 0x0 and indicates size of 256 bytes. This means that value of 0x0 = 256 bytes, 0x1 = 512 bytes and so on, so user should be programming TQS [5:0] = 6'b001111 to allocate queue size of 4096 (4K) bytes. In general, the size of the Queue = (TQS + 1)*256 bytes. When the number of Tx Queues is one, the field is read-only and the configured TX FIFO size in blocks of 256 bytes is reflected in the reset value. The width of this field depends on the Tx memory size selected in your configuration. For example, if the memory size is 2048, the width of this field is 3 bits: $\text{LOG}_2(2048/256) = \text{LOG}_2(8) = 3 \text{ bits}$
15-7 —	Reserved.

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Field	Function
6-4 TTC	Transmit Threshold Control These bits control the threshold level of the MTL Tx Queue. The transmission starts when the packet size within the MTL Tx Queue is larger than the threshold. In addition, full packets with length less than the threshold are also transmitted. These bits are used only when the TSF bit is reset. 000b - 32 001b - 64 010b - 96 011b - 128 100b - 192 101b - 256 110b - 384 111b - 512
3-2 TXQEN	Transmit Queue Enable This field is used to enable/disable the transmit queue 0. - 2'b00: Not enabled - 2'b01: Enable in AV mode - 2'b10: Enabled - 2'b11: Reserved Note: In multiple Tx queues configuration, all the queues are disabled by default. Enable the Tx queue by programming this field. 00b - Not enabled 01b - Enable in AV mode (Reserved in non-AV) 10b - Enabled 11b - Reserved
1 TSF	Transmit Store and Forward When this bit is set, the transmission starts when a full packet resides in the MTL Tx queue. When this bit is set, the TTC values specified in Bits[6:4] of this register are ignored. This bit should be changed only when the transmission is stopped. 0b - Transmit Store and Forward is disabled 1b - Transmit Store and Forward is enabled
0 FTQ	Flush Transmit Queue When this bit is set, the Tx queue controller logic is reset to its default values. Therefore, all the data in the Tx queue is lost or flushed. This bit is internally reset when the flushing operation is complete. Until this bit is reset, you should not write to the MTL_TxQ1_Operation_Mode register. The data which is already

Table continues on the next page...

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Field	Function
	accepted by the MAC transmitter is not flushed. It is scheduled for transmission and results in underflow and runt packet transmission. Note: The flush operation is complete only when the Tx queue is empty and the application has accepted the pending Tx Status of all transmitted packets. To complete this flush operation, the PHY Tx clock (clk_tx_i) should be active. Access restriction applies. Setting 1 sets. Self-cleared. Setting 0 has no effect. 0b - Flush Transmit Queue is disabled 1b - Flush Transmit Queue is enabled

1.4.290 (MTL_TxQ3_Underflow)

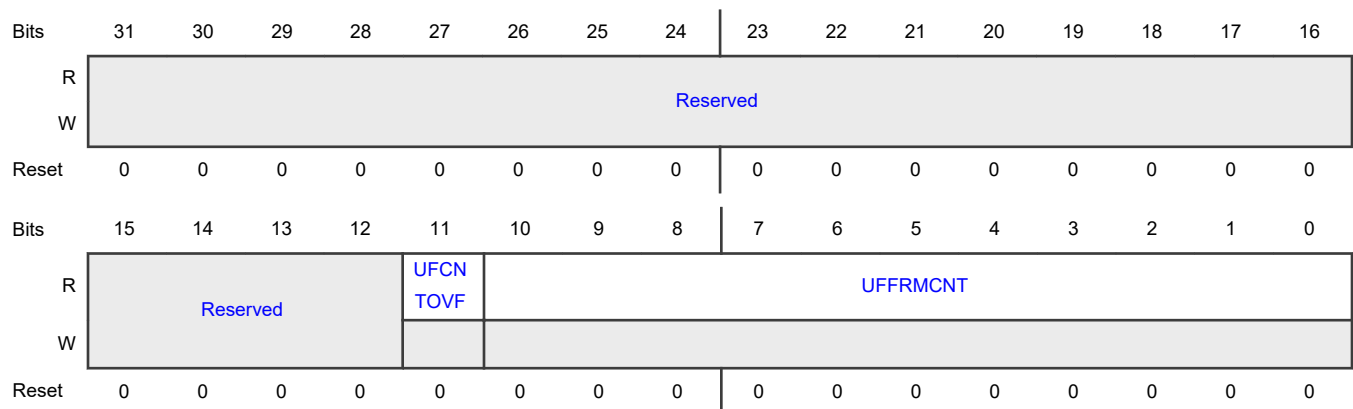
Offset

Register	Offset
MTL_TxQ3_Underflow	DC4h

Function

The Queue 3 Underflow Counter register contains the counter for packets aborted because of Transmit queue underflow and packets missed because of Receive queue packet flush

Diagram



Fields

Field	Function
31-12	Reserved.
—	

Table continues on the next page...

Table continued from the previous page...

Field	Function
11 UFCNTOVF	Overflow Bit for Underflow Packet Counter This bit is set every time the Tx queue Underflow Packet Counter field overflows, that is, it has crossed the maximum count. In such a scenario, the overflow packet counter is reset to all-zeros and this bit indicates that the rollover happened. Access restriction applies. Clears on read. Self-set to 1 on internal event. 0b - Overflow not detected for Underflow Packet Counter 1b - Overflow detected for Underflow Packet Counter
10-0 UFFRMCNT	Underflow Packet Counter This field indicates the number of packets aborted by the controller because of Tx Queue Underflow. This counter is incremented each time the MAC aborts outgoing packet because of underflow. The counter is cleared when this register is read with mci_be_i[0] at 1'b1. Access restriction applies. Clears on read. Self-set to 1 on internal event.

1.4.291 (MTL_TxQ3_Debug)

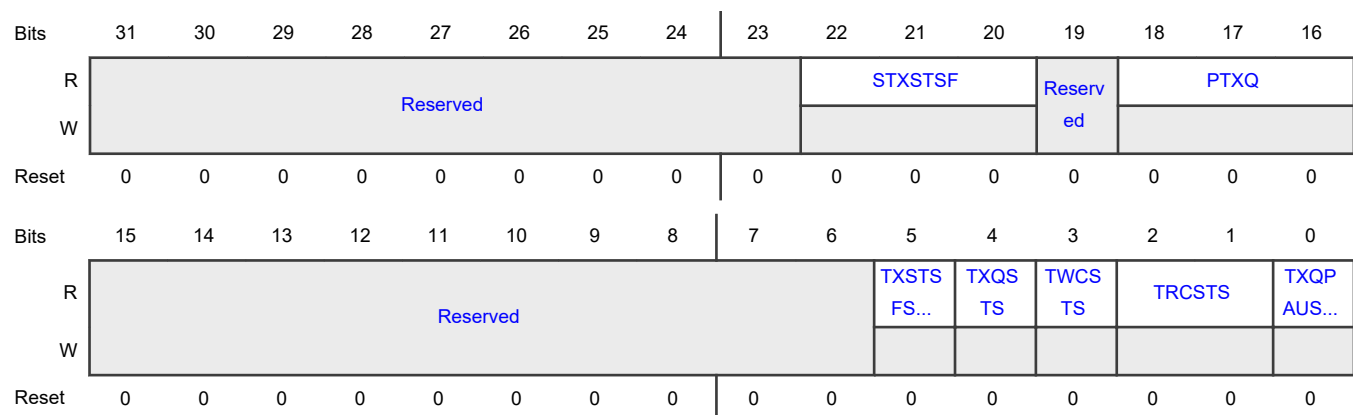
Offset

Register	Offset
MTL_TxQ3_Debug	DC8h

Function

The Queue 3 Transmit Debug register gives the debug status of various blocks related to the Transmit queue.

Diagram



Fields

Field	Function
31-23 —	Reserved.
22-20 STXSTS	Number of Status Words in Tx Status FIFO of Queue This field indicates the current number of status in the Tx Status FIFO of this queue. When the DTXSTS bit of MTL_Operation_Mode register is set to 1, this field does not reflect the number of status words in Tx Status FIFO.
19 —	Reserved.
18-16 PTXQ	Number of Packets in the Transmit Queue This field indicates the current number of packets in the Tx Queue. When the DTXSTS bit of MTL_Operation_Mode register is set to 1, this field does not reflect the number of packets in the Transmit queue.
15-6 —	Reserved.
5 TXSTS	MTL Tx Status FIFO Full Status When high, this bit indicates that the MTL Tx Status FIFO is full. Therefore, the MTL cannot accept any more packets for transmission. 0b - MTL Tx Status FIFO Full status is not detected 1b - MTL Tx Status FIFO Full status is detected
4 TXQSTS	MTL Tx Queue Not Empty Status When this bit is high, it indicates that the MTL Tx Queue is not empty and some data is left for transmission. 0b - MTL Tx Queue Not Empty status is not detected 1b - MTL Tx Queue Not Empty status is detected
3 TWCSTS	MTL Tx Queue Write Controller Status When high, this bit indicates that the MTL Tx Queue Write Controller is active, and it is transferring the data to the Tx Queue. 0b - MTL Tx Queue Write Controller status is not detected 1b - MTL Tx Queue Write Controller status is detected
2-1 TRCSTS	MTL Tx Queue Read Controller Status This field indicates the state of the Tx Queue Read Controller: 00b - Idle state 01b - Read state (transferring data to the MAC transmitter)

Table continues on the next page...

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Field	Function
	10b - Waiting for pending Tx Status from the MAC transmitter 11b - Flushing the Tx queue because of the Packet Abort request from the MAC
0 TXQPAUSED	Transmit Queue in Pause When this bit is high and the Rx flow control is enabled, it indicates that the Tx Queue is in the Pause condition (in the full-duplex only mode) because of the following: - Reception of the PFC packet for the priorities assigned to the Tx Queue when PFC is enabled - Reception of 802.3x Pause packet when PFC is disabled 0b - Transmit Queue in Pause status is not detected 1b - Transmit Queue in Pause status is detected

1.4.292 (MTL_TxQ3_ETS_Control)

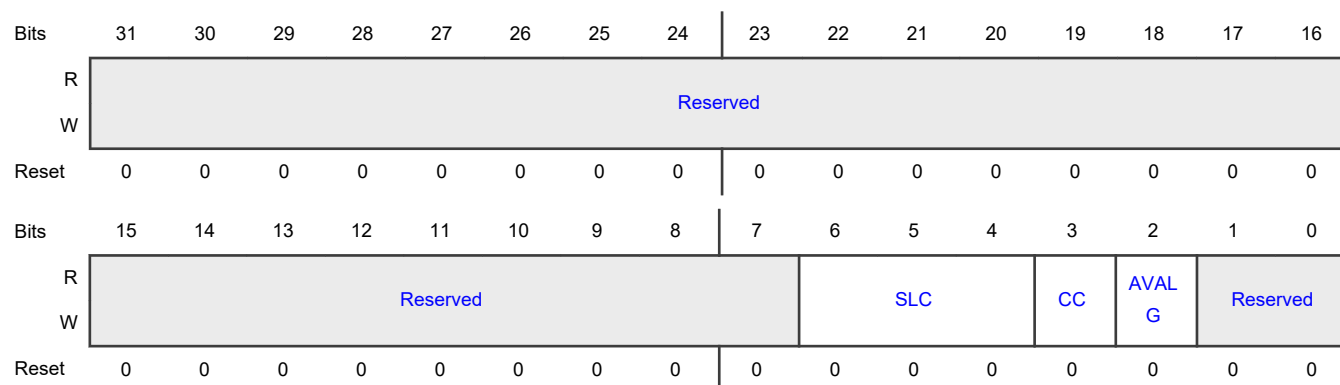
Offset

Register	Offset
MTL_TxQ3_ETS_Control	DD0h

Function

The Queue ETS Control register controls the enhanced transmission selection operation.

Diagram



Fields

Field	Function
31-7	Reserved.

Table continues on the next page...

Table continued from the previous page...

Field	Function
—	
6-4 SLC	Slot Count If the credit-based shaper algorithm is enabled, the software can program the number of slots (of duration programmed in DMA_CH(#i)_Slot_Interval register) over which the average transmitted bits per slot, provided in the MTL_TxQ(#i)_ETS_Status register, need to be computed for Queue. The encoding is as follows: 000b - 1 slot 001b - 2 slots 010b - 4 slots 011b - 8 slots 100b - 16 slots 101b - Reserved
3 CC	Credit Control When this bit is set, the accumulated credit parameter in the credit-based shaper algorithm logic is not reset to zero when there is positive credit and no packet to transmit in Channel 1. The credit accumulates even when there is no packet waiting in Channel 1 and another channel is transmitting. When this bit is reset, the accumulated credit parameter in the credit-based shaper algorithm logic is set to zero when there is positive credit and no packet to transmit in Channel 1. When there is no packet waiting in Channel 1 and other channel is transmitting, no credit is accumulated. 0b - Credit Control is disabled 1b - Credit Control is enabled
2 AVALG	AV Algorithm When Queue 1 is programmed for AV, this field configures the scheduling algorithm for this queue: This bit when set, indicates credit based shaper algorithm (CBS) is selected for Queue 1 traffic. When reset, strict priority is selected. 0b - CBS Algorithm is disabled 1b - CBS Algorithm is enabled
1-0 —	Reserved.

1.4.293 (MTL_TxQ3_ETS_Status)

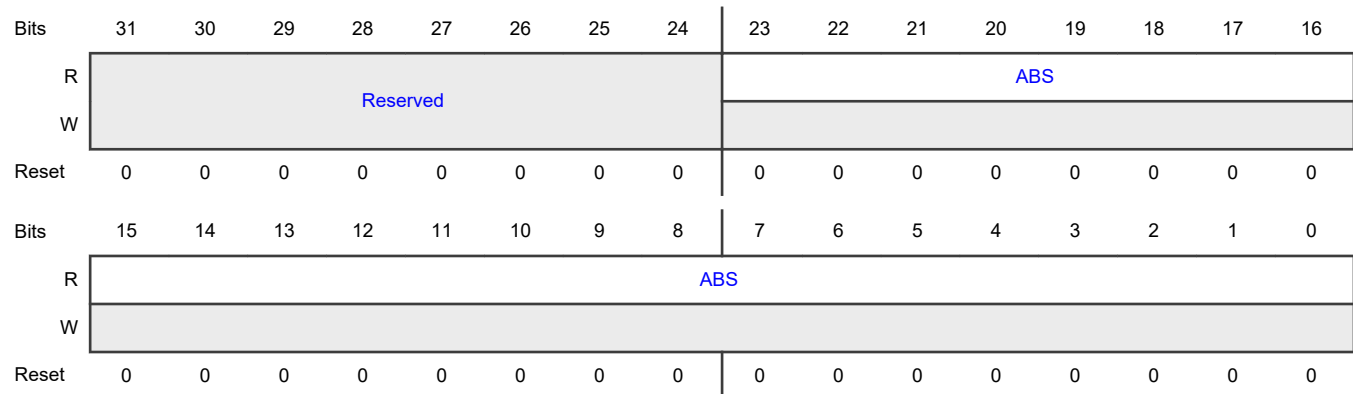
Offset

Register	Offset
MTL_TxQ3_ETS_Status	DD4h

Function

The Queue 3 ETS Status register provides the average traffic transmitted in Queue 3.

Diagram



Fields

Field	Function
31-24 —	Reserved.
23-0 ABS	Average Bits per Slot This field contains the average transmitted bits per slot. If AV operation is enabled, this field is computed over number of slots, programmed in the SLC field of MTL_TxQ(#i)_ETS_CONTROL register. The maximum value of this field is 0x6_4000 in 100 Mbps, 0x3E_8000 in 1000 Mbps and 9C_4000 in 2500 Mbps mode respectively. When the DCB operation is enabled for Queue, this field is computed over every 10 million bit times slot (4 ms in 2500 Mbps; 10 ms in 1000 Mbps; 100 ms in 100 Mbps). The maximum value is 0x989680.

1.4.294 (MTL_TxQ3_Quantum_Weight)

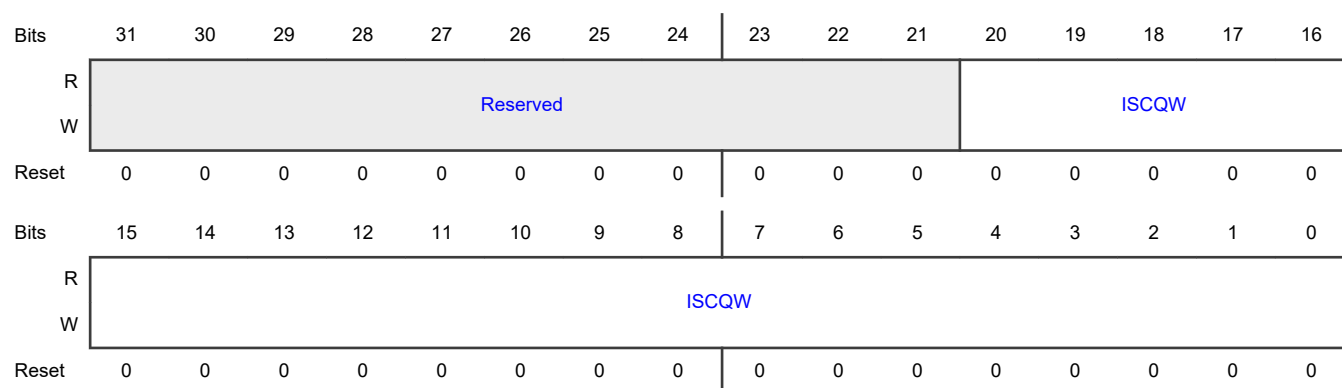
Offset

Register	Offset
MTL_TxQ3_Quantum_Weight	DD8h

Function

The Queue 3 idleSlopeCredit, Quantum or Weights register provides the average traffic transmitted in Queue 3.

Diagram



Fields

Field	Function
31-21 —	Reserved.
20-0 ISCQW	idleSlopeCredit, Quantum or Weights - idleSlopeCredit When AV feature is enabled, this field contains the idleSlopeCredit value required for the credit-based shaper algorithm for Queue 1. This is the rate of change of credit in bits per cycle (40 ns for 100 Mbps; 8 ns for 1000 Mbps; 3.2 ns for 2500 Mbps) when the credit is increasing. The software should program this field with computed credit in bits per cycle scaled by 1,024. The maximum value is portTransmitRate, that is, 0x2000 in 1000/2500 Mbps mode and 0x1000 in 100 Mbps mode. Bits[20:14] must be written to zero. - Quantum When the DCB operation is enabled with DWRR algorithm for Queue 1 traffic, this field contains the quantum value in bytes to be added to credit during every queue scanning cycle. The maximum value is 0x1312D0 bytes. - Weights When DCB operation is enabled with WFQ algorithm for Queue 1 traffic, this field contains the weight for this queue. The maximum value is 0x3FFF where weight of 0 indicates 100% bandwidth. Bits[20:14] must be written to zero. When DCB operation or generic queuing operation is enabled with WRR algorithm for Queue 1 traffic, this field contains the weight for this queue. The maximum value is 0x64. Bits [20:7] must be written to zero. - Note 1: In multiple Queue configuration this field in respective per queue register must be programmed to some non-zero value when multiple queues are enabled or single queue other than Q0 is enabled. This field need not be programmed when only Q0 is enabled. In general, when WRR algorithm is selected a non-zero value must be programmed on both Receive and Transmit. In Receive, the register is MTL_Operation_Mode register. - Note 2: For WFQ algorithm, higher the programmed weights lesser the bandwidth allocated for that Transmit Queue. The finish time is not a function of particular packet alone but it is as per the formula: (previous_finish_time of particular Transmit Queue + (weights*packet_size))

Table continues on the next page...

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Field	Function
	- Note 3: The weights programmed do not correspond to the number of packets but the fraction of bandwidth or time allocated for particular queue w.r.t. total BW or time.

1.4.295 (MTL_TxQ3_SendSlopeCredit)

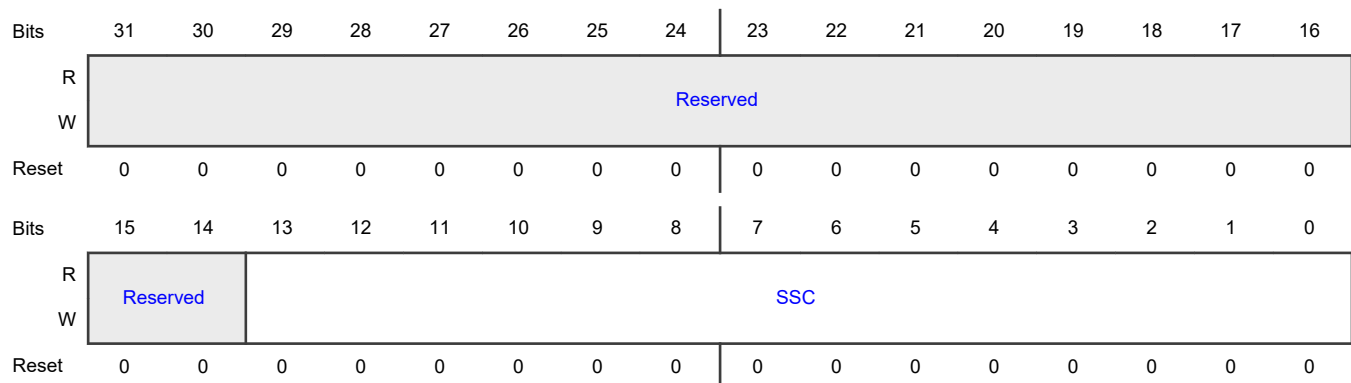
Offset

Register	Offset
MTL_TxQ3_SendSlopeCredit	DDCh

Function

The sendSlopeCredit register contains the sendSlope credit value required for the credit-based shaper algorithm for the Queue.

Diagram



Fields

Field	Function
31-14 —	Reserved.
13-0 SSC	sendSlopeCredit Value When AV operation is enabled, this field contains the sendSlopeCredit value required for credit-based shaper algorithm for Queue 1. This is the rate of change of credit in bits per cycle (40 ns, 8 ns and 3.2 ns for 100 Mbps, 1000 Mbps and 2500 Mbps respectively) when the credit is decreasing. The software should program this field with computed credit in bits per cycle scaled by 1,024. The maximum value is portTransmitRate, that is, 0x2000 in 1000/2500 Mbps mode and 0x1000 in 100 Mbps mode. This field should be programmed with absolute sendSlopeCredit value. The credit-based shaper logic subtracts it from the accumulated credit when Channel 1 is selected for transmission.

1.4.296 (MTL_TxQ3_HiCredit)

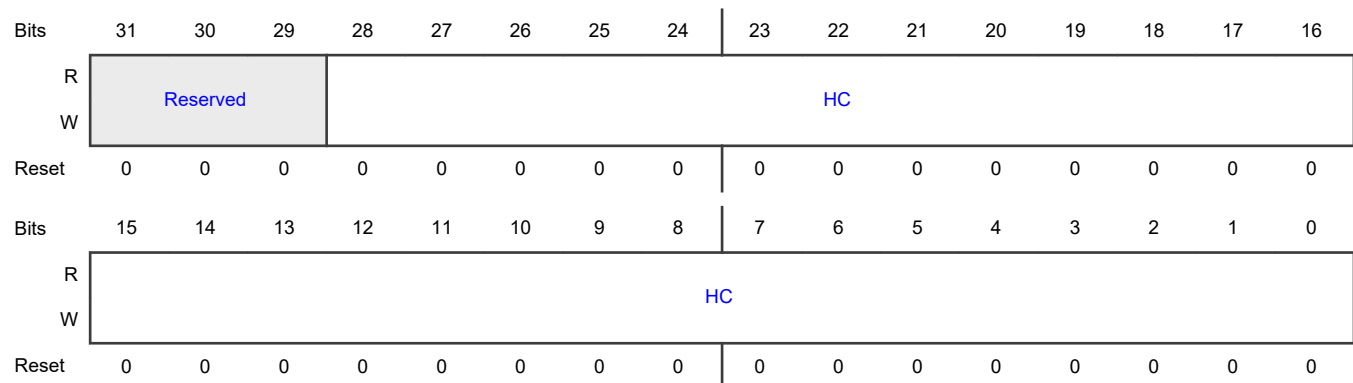
Offset

Register	Offset
MTL_TxQ3_HiCredit	DE0h

Function

The hiCredit register contains the hiCredit value required for the credit-based shaper algorithm for the Queue.

Diagram



Fields

Field	Function
31-29 —	Reserved.
28-0 HC	hiCredit Value When the AV feature is enabled, this field contains the hiCredit value required for the credit-based shaper algorithm. This is the maximum value that can be accumulated in the credit parameter. This is specified in bits scaled by 1,024. The maximum value is maxInterferenceSize, that is, best-effort maximum packet size (16,384 bytes or 131,072 bits). The value to be specified is $131,072 * 1,024 = 134,217,728$ or 0x0800_0000.

1.4.297 (MTL_TxQ3_LoCredit)

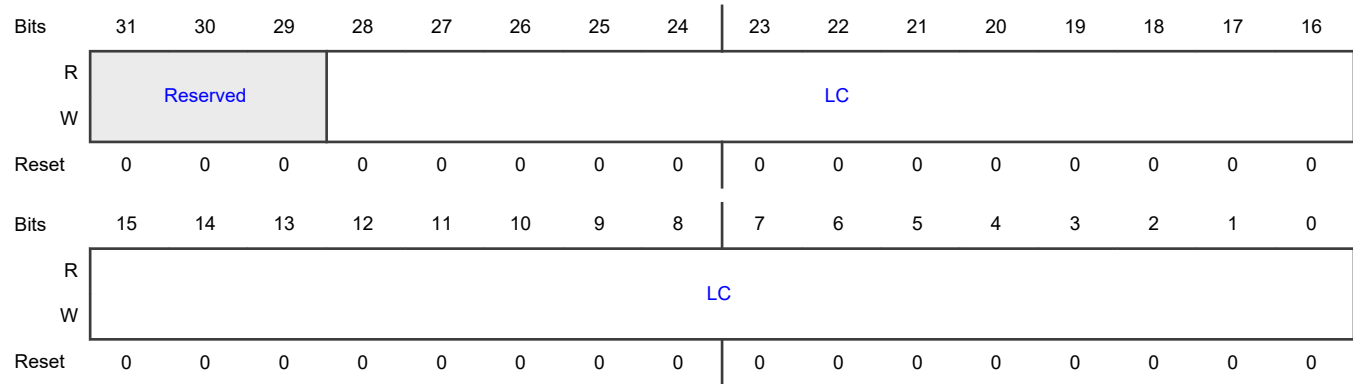
Offset

Register	Offset
MTL_TxQ3_LoCredit	DE4h

Function

The loCredit register contains the loCredit value required for the credit-based shaper algorithm for the Queue.

Diagram



Fields

Field	Function
31-29 —	Reserved.
28-0 LC	loCredit Value When AV operation is enabled, this field contains the loCredit value required for the credit-based shaper algorithm. This is the minimum value that can be accumulated in the credit parameter. This is specified in bits scaled by 1,024. The maximum value to be programmed is corresponds to twice the maxFrameSize transmitted from this queue. If the maxFrameSize is 8192 bytes, then $(8192*2) * 8 * 1024 = 134,217,728$ or 0x0800_0000. Because it is a negative value, the programmed value is 2's complement of the value, that is, 0x1800_0000.

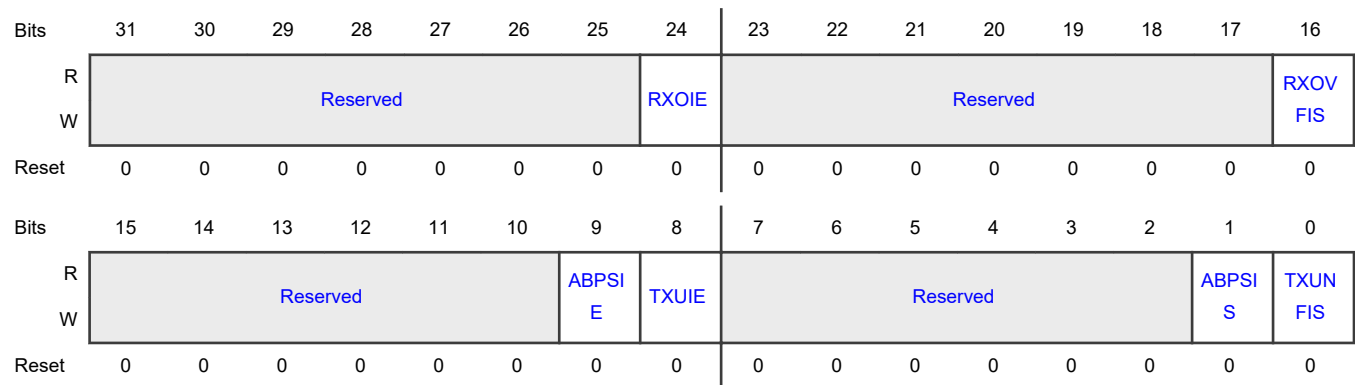
1.4.298 (MTL_Q3_Interrupt_Control_Status)

Offset

Register	Offset
MTL_Q3_Interrupt_Contr ol_Status	DECh

Function

This register contains the interrupt enable and status bits for the queue 3 interrupts.

Diagram**Fields**

Field	Function
31-25 —	Reserved.
24 RXOIE	Receive Queue Overflow Interrupt Enable When this bit is set, the Receive Queue Overflow interrupt is enabled. When this bit is reset, the Receive Queue Overflow interrupt is disabled. 0b - Receive Queue Overflow Interrupt is disabled 1b - Receive Queue Overflow Interrupt is enabled
23-17 —	Reserved.
16 RXOVFIS	Receive Queue Overflow Interrupt Status This bit indicates that the Receive Queue had an overflow while receiving the packet. If a partial packet is transferred to the application, the overflow status is set in RDES3[21]. This bit is cleared when the application writes 1 to this bit. Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect. 0b - Receive Queue Overflow Interrupt Status not detected 1b - Receive Queue Overflow Interrupt Status detected
15-10 —	Reserved.
9 ABPSIE	Average Bits Per Slot Interrupt Enable When this bit is set, the MAC asserts the sbd_intr_o or mci_intr_o interrupt when the average bits per slot status is updated. When this bit is cleared, the interrupt is not asserted for such an event. 0b - Average Bits Per Slot Interrupt is disabled

Table continues on the next page...

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Field	Function
	1b - Average Bits Per Slot Interrupt is enabled
8 TXUIE	Transmit Queue Underflow Interrupt Enable When this bit is set, the Transmit Queue Underflow interrupt is enabled. When this bit is reset, the Transmit Queue Underflow interrupt is disabled. 0b - Transmit Queue Underflow Interrupt Status is disabled 1b - Transmit Queue Underflow Interrupt Status is enabled
7-2 —	Reserved.
1 ABPSIS	Average Bits Per Slot Interrupt Status When set, this bit indicates that the MAC has updated the ABS value. This bit is cleared when the application writes 1 to this bit. Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect. 0b - Average Bits Per Slot Interrupt Status not detected 1b - Average Bits Per Slot Interrupt Status detected
0 TXUNFIS	Transmit Queue Underflow Interrupt Status This bit indicates that the Transmit Queue had an underflow while transmitting the packet. Transmission is suspended and an Underflow Error TDES3[2] is set. This bit is cleared when the application writes 1 to this bit. Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect. 0b - Transmit Queue Underflow Interrupt Status not detected 1b - Transmit Queue Underflow Interrupt Status detected

1.4.299 (MTL_RxQ3_Operation_Mode)

Offset

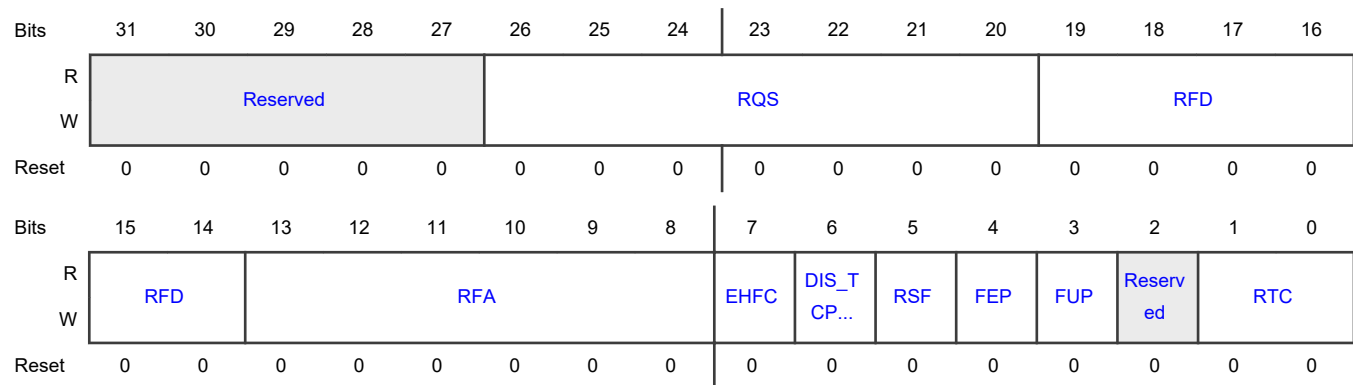
Register	Offset
MTL_RxQ3_Operation_Mode	DF0h

Function

The Queue 3 Receive Operation Mode register establishes the Receive queue operating modes and command.

The RFA and RFD fields are not backward compatible with the RFA and RFD fields of 4.00a release

Diagram



Fields

Field	Function
31-27 —	Reserved.
26-20 RQS	Receive Queue Size This field indicates the size of the allocated Receive queues in blocks of 256 bytes. The RQS field is read-write only if the number of Rx Queues more than one, the reset value is 0x0 and indicates size of 256 bytes. This means that value of 0x0 = 256 bytes, 0x1 = 512 bytes and so on, so user should be programming RQS [5:0] = 6'b001111 to allocate queue size of 4096 (4K) bytes. In general, the size of the Queue = (RQS + 1)*256 bytes. When the number of Rx Queues is one, the field is read-only and the configured RX FIFO size in blocks of 256 bytes is reflected in the reset value. The width of this field depends on the Rx memory size selected in your configuration. For example, if the memory size is 2048, the width of this field is 3 bits: $\text{LOG2}(2048/256) = \text{LOG2}(8) = 3 \text{ bits}$
19-14 RFD	Threshold for Deactivating Flow Control (in half-duplex and full-duplex modes) These bits control the threshold (fill-level of Rx queue) at which the flow control is de-asserted after activation: - 0: Full minus 1 KB, that is, FULL 1 KB - 1: Full minus 1.5 KB, that is, FULL 1.5 KB - 2: Full minus 2 KB, that is, FULL 2 KB - 3: Full minus 2.5 KB, that is, FULL 2.5 KB - ... - 62: Full minus 32 KB, that is, FULL 32 KB - 63: Full minus 32.5 KB, that is, FULL 32.5 KB The de-assertion is effective only after flow control is asserted. Note: The value must be programmed in such a way to make sure that the threshold is a positive number.

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Field	Function
	When the EHFC is set high, these values are applicable only when the Rx queue size determined by the RQS field of this register, is equal to or greater than 4 KB. For a given queue size, the values ranges between 0 and the encoding for FULL minus (QSIZE - 0.5 KB) and all other values are illegal. Here the term FULL and QSIZE refers to the queue size determined by the RQS field of this register. The width of this field depends on RX FIFO size selected during the configuration. Remaining bits are reserved and read only.
13-8 RFA	Threshold for Activating Flow Control (in half-duplex and full-duplex) These bits control the threshold (fill-level of Rx queue) at which the flow control is activated: For more information on encoding for this field, see RFD.
7 EHFC	Enable Hardware Flow Control When this bit is set, the flow control signal operation, based on the fill-level of Rx queue, is enabled. When reset, the flow control operation is disabled. 0b - Hardware Flow Control is disabled 1b - Hardware Flow Control is enabled
6 DIS_TCP_EF	Disable Dropping of TCP/IP Checksum Error Packets When this bit is set, the MAC does not drop the packets which only have the errors detected by the Receive Checksum Offload engine. Such packets have errors only in the encapsulated payload. There are no errors (including FCS error) in the Ethernet packet received by the MAC. When this bit is reset, all error packets are dropped if the FEP bit is reset. 0b - Dropping of TCP/IP Checksum Error Packets is enabled 1b - Dropping of TCP/IP Checksum Error Packets is disabled
5 RSF	Receive Queue Store and Forward When this bit is set, the DWC_ether_qos reads a packet from the Rx queue only after the complete packet has been written to it, ignoring the RTC field of this register. When this bit is reset, the Rx queue operates in the Threshold (cut-through) mode, subject to the threshold specified by the RTC field of this register. 0b - Receive Queue Store and Forward is disabled 1b - Receive Queue Store and Forward is enabled
4 FEP	Forward Error Packets When this bit is reset, the Rx queue drops packets with error status (CRC error, GMII_ER, watchdog timeout, or overflow). However, if the start byte (write) pointer of a packet is already transferred to the read controller side (in Threshold mode), the packet is not dropped. When this bit is set, all packets except the runt error packets are forwarded to the application or DMA. If the RSF bit is set and the Rx queue overflows when a partial packet is written, the packet is dropped irrespective of the setting of this bit. However, if the RSF bit is reset and the Rx queue overflows when a partial packet is written, a partial packet might be forwarded to the application or DMA.

Table continues on the next page...

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Field	Function
	0b - Forward Error Packets is disabled 1b - Forward Error Packets is enabled
3 FUP	Forward Undersized Good Packets When this bit is set, the Rx queue forwards the undersized good packets (packets with no error and length less than 64 bytes), including pad-bytes and CRC. When this bit is reset, the Rx queue drops all packets of less than 64 bytes, unless a packet is already transferred because of the lower value of Rx Threshold, for example, RTC = 01. 0b - Forward Undersized Good Packets is disabled 1b - Forward Undersized Good Packets is enabled
2 —	Reserved.
1-0 RTC	Receive Queue Threshold Control These bits control the threshold level of the MTL Rx queue (in bytes): The received packet is transferred to the application or DMA when the packet size within the MTL Rx queue is larger than the threshold. In addition, full packets with length less than the threshold are automatically transferred. This field is valid only when the RSF bit is zero. This field is ignored when the RSF bit is set to 1. 00b - 64 01b - 32 10b - 96 11b - 128

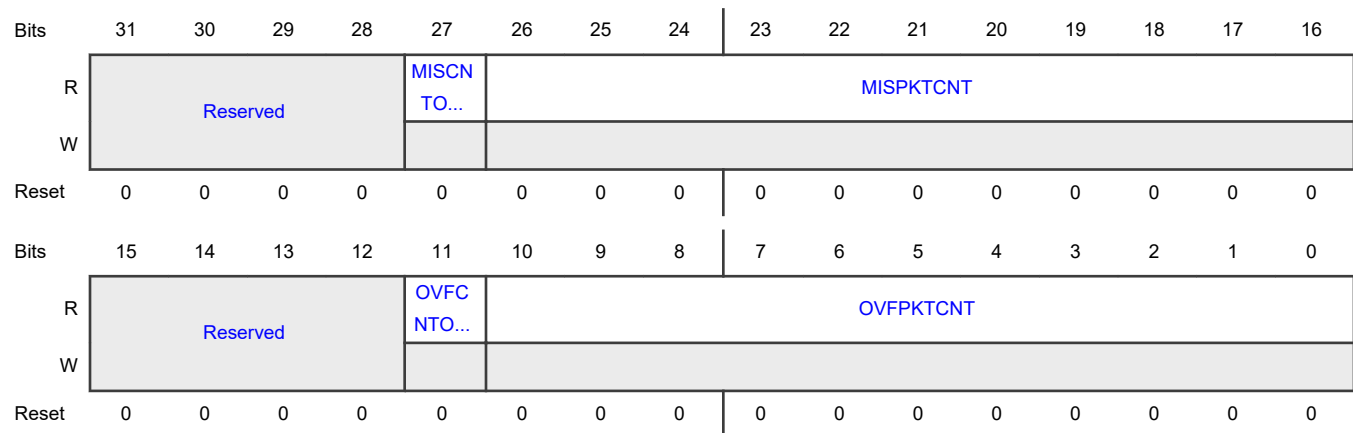
1.4.300 (MTL_RxQ3_Missed_Packet_Overflow_Cnt)

Offset

Register	Offset
MTL_RxQ3_Missed_Packet_Overflow_Cnt	DF4h

Function

The Queue 3 Missed Packet and Overflow Counter register contains the counter for packets missed because of Receive queue packet flush and packets discarded because of Receive queue overflow.

Diagram**Fields**

Field	Function
31-28 —	Reserved.
27 MISCNTOVF	Missed Packet Counter Overflow Bit When set, this bit indicates that the Rx Queue Missed Packet Counter crossed the maximum limit. Access restriction applies. Clears on read. Self-set to 1 on internal event. 0b - Missed Packet Counter overflow not detected 1b - Missed Packet Counter overflow detected
26-16 MISPKTCNT	Missed Packet Counter This field indicates the number of packets missed by the DWC_ether_qos because the application asserted ari_pkt_flush_i[] for this queue. This counter is reset when this register is read with mci_be_i[0] at 1b1. This counter is incremented by 1 when the DMA discards the packet because of buffer unavailability. Access restriction applies. Clears on read. Self-set to 1 on internal event.
15-12 —	Reserved.
11 OVFCNTOVF	Overflow Counter Overflow Bit When set, this bit indicates that the Rx Queue Overflow Packet Counter field crossed the maximum limit. Access restriction applies. Clears on read. Self-set to 1 on internal event. 0b - Overflow Counter overflow not detected 1b - Overflow Counter overflow detected
10-0 OVFPKTCNT	Overflow Packet Counter

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Field	Function
	This field indicates the number of packets discarded by the DWC_ether_qos because of Receive queue overflow. This counter is incremented each time the DWC_ether_qos discards an incoming packet because of overflow. This counter is reset when this register is read with mci_be_i[0] at 1'b1. Access restriction applies. Clears on read. Self-set to 1 on internal event.

1.4.301 (MTL_RxQ3_Debug)

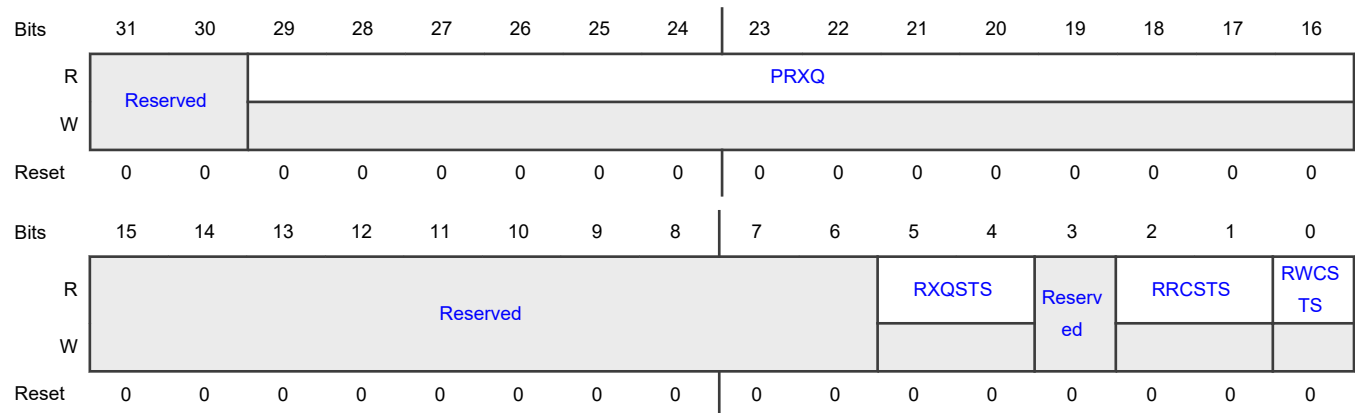
Offset

Register	Offset
MTL_RxQ3_Debug	DF8h

Function

The Queue 3 Receive Debug register gives the debug status of various blocks related to the Receive queue.

Diagram



Fields

Field	Function
31-30 —	Reserved.
29-16 PRXQ	Number of Packets in Receive Queue This field indicates the current number of packets in the Rx Queue. The theoretical maximum value for this field is 256KB/16B = 16K Packets, that is, Max_Queue_Size/Min_Packet_Size.
15-6	Reserved.

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Field	Function
—	
5-4 RXQSTS	MTL Rx Queue Fill-Level Status This field gives the status of the fill-level of the Rx Queue: 00b - Rx Queue empty 01b - Rx Queue fill-level below flow-control deactivate threshold 10b - Rx Queue fill-level above flow-control activate threshold 11b - Rx Queue full
3 —	Reserved.
2-1 RRCSTS	MTL Rx Queue Read Controller State This field gives the state of the Rx queue Read controller: 00b - Idle state 01b - Reading packet data 10b - Reading packet status (or timestamp) 11b - Flushing the packet data and status
0 RWCSTS	MTL Rx Queue Write Controller Active Status When high, this bit indicates that the MTL Rx queue Write controller is active, and it is transferring a received packet to the Rx Queue. 0b - MTL Rx Queue Write Controller Active Status not detected 1b - MTL Rx Queue Write Controller Active Status detected

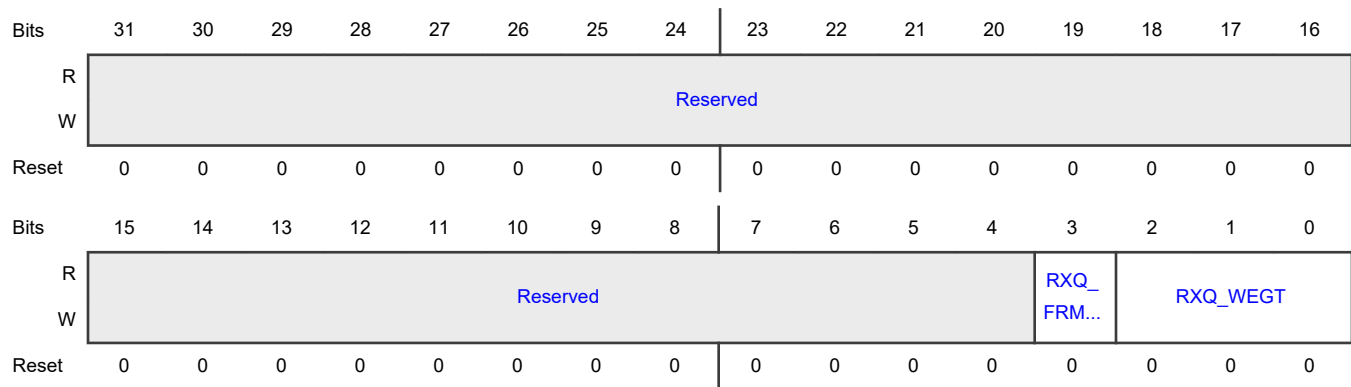
1.4.302 (MTL_RxQ3_Control)

Offset

Register	Offset
MTL_RxQ3_Control	DFCh

Function

The Queue Receive Control register controls the receive arbitration and passing of received packets to the application.

Diagram**Fields**

Field	Function
31-4 —	Reserved.
3 RXQ_FRM_AR BIT	Receive Queue Packet Arbitration When this bit is set, the DWC_ether_qos drives the packet data to the ARI interface such that the entire packet data of currently-selected queue is transmitted before switching to other queue. When this bit is reset, the DWC_ether_qos drives the packet data to the ARI interface such that the following amount of data of currently-selected queue is transmitted before switching to other queue: - PBL amount of data (indicated by ari_qN_pbl_i[]) or - Complete data of a packet The status and the timestamp are not a part of the PBL data. Therefore, the DWC_ether_qos drives the complete status (including timestamp status) during first PBL request for the packet (in store-and-forward mode) or the last PBL request for the packet (in Threshold mode). 0b - Receive Queue Packet Arbitration is disabled 1b - Receive Queue Packet Arbitration is enabled
2-0 RXQ_WEGT	Receive Queue Weight This field indicates the weight assigned to the Rx Queue 0. This field needs to be programmed with one value less than the required weight, i.e. reset value of 0 indicates weight of 1, value of 1 indicates weight of 2, and so on. The weight is used as the number of continuous PBL or packets requests (depending on the RXQ_FRM_ARBIT) allocated to the queue in one arbitration cycle. Note: The change in value of RXQ_WEGT takes effect only after the completion of current service round or when there is change from RAA=SP to RAA=WSP algorithm. This approach is taken so that there is smooth transition. For the RXQ_WEGT value to take effect at the start, the MTL_RxQ(##)_Control registers must be programmed before the MTL_Operation_Mode register.

1.4.303 (MTL_TxQ4_Operation_Mode)

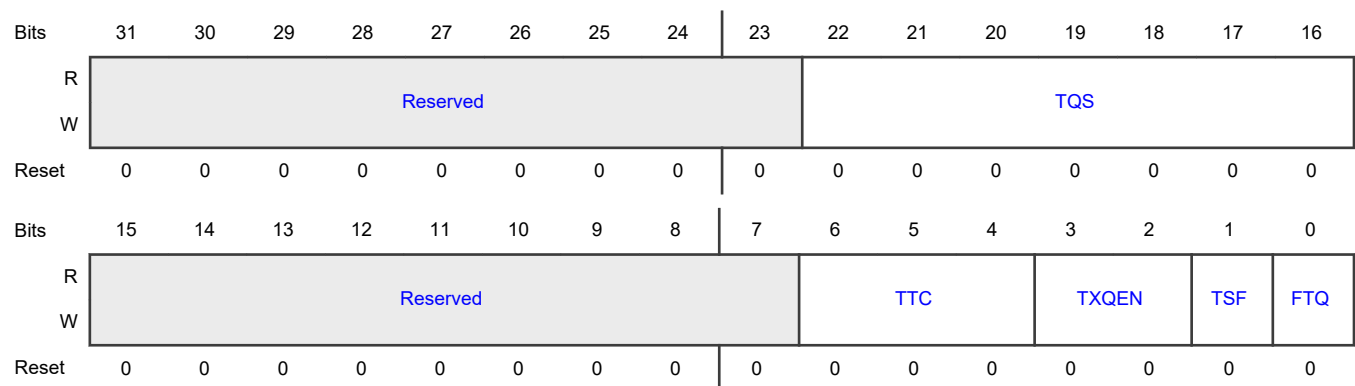
Offset

Register	Offset
MTL_TxQ4_Operation_Mode	E00h

Function

The Queue 4 Transmit Operation Mode register establishes the Transmit queue operating modes and commands.

Diagram



Fields

Field	Function
31-23 —	Reserved.
22-16 TQS	Transmit Queue Size This field indicates the size of the allocated Transmit queues in blocks of 256 bytes. The TQS field is read-write only if the number of Tx Queues more than one, the reset value is 0x0 and indicates size of 256 bytes. This means that value of 0x0 = 256 bytes, 0x1 = 512 bytes and so on, so user should be programming TQS [5:0] = 6'b001111 to allocate queue size of 4096 (4K) bytes. In general, the size of the Queue = (TQS + 1)*256 bytes. When the number of Tx Queues is one, the field is read-only and the configured TX FIFO size in blocks of 256 bytes is reflected in the reset value. The width of this field depends on the Tx memory size selected in your configuration. For example, if the memory size is 2048, the width of this field is 3 bits: $\text{LOG}_2(2048/256) = \text{LOG}_2(8) = 3 \text{ bits}$
15-7 —	Reserved.

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Field	Function
6-4 TTC	Transmit Threshold Control These bits control the threshold level of the MTL Tx Queue. The transmission starts when the packet size within the MTL Tx Queue is larger than the threshold. In addition, full packets with length less than the threshold are also transmitted. These bits are used only when the TSF bit is reset. 000b - 32 001b - 64 010b - 96 011b - 128 100b - 192 101b - 256 110b - 384 111b - 512
3-2 TXQEN	Transmit Queue Enable This field is used to enable/disable the transmit queue 0. - 2'b00: Not enabled - 2'b01: Enable in AV mode - 2'b10: Enabled - 2'b11: Reserved Note: In multiple Tx queues configuration, all the queues are disabled by default. Enable the Tx queue by programming this field. 00b - Not enabled 01b - Enable in AV mode (Reserved in non-AV) 10b - Enabled 11b - Reserved
1 TSF	Transmit Store and Forward When this bit is set, the transmission starts when a full packet resides in the MTL Tx queue. When this bit is set, the TTC values specified in Bits[6:4] of this register are ignored. This bit should be changed only when the transmission is stopped. 0b - Transmit Store and Forward is disabled 1b - Transmit Store and Forward is enabled
0 FTQ	Flush Transmit Queue When this bit is set, the Tx queue controller logic is reset to its default values. Therefore, all the data in the Tx queue is lost or flushed. This bit is internally reset when the flushing operation is complete. Until this bit is reset, you should not write to the MTL_TxQ1_Operation_Mode register. The data which is already

Table continues on the next page...

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Field	Function
	accepted by the MAC transmitter is not flushed. It is scheduled for transmission and results in underflow and runt packet transmission.
	Note: The flush operation is complete only when the Tx queue is empty and the application has accepted the pending Tx Status of all transmitted packets. To complete this flush operation, the PHY Tx clock (clk_tx_i) should be active.
	Access restriction applies. Setting 1 sets. Self-cleared. Setting 0 has no effect.
	0b - Flush Transmit Queue is disabled
	1b - Flush Transmit Queue is enabled

1.4.304 (MTL_TxQ4_Underflow)

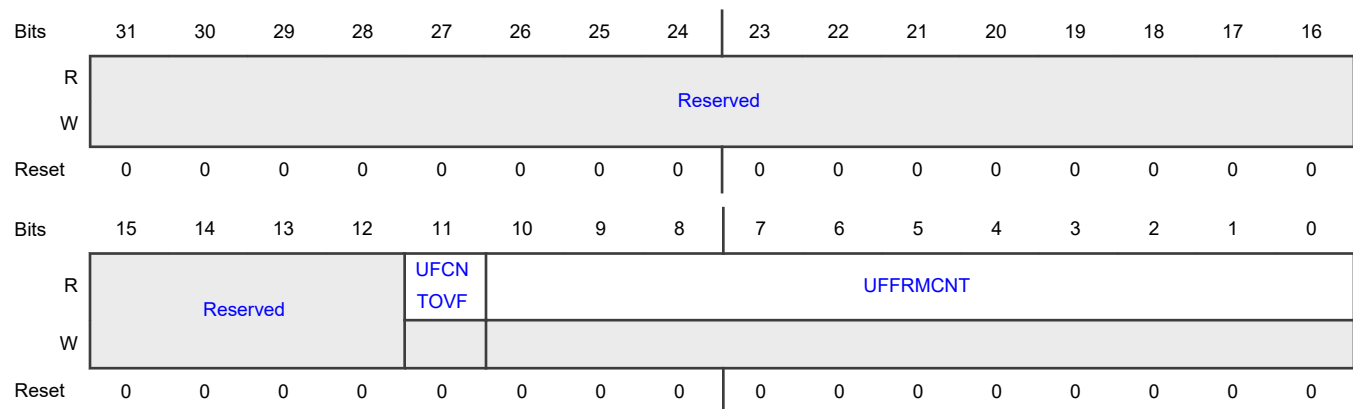
Offset

Register	Offset
MTL_TxQ4_Underflow	E04h

Function

The Queue 4 Underflow Counter register contains the counter for packets aborted because of Transmit queue underflow and packets missed because of Receive queue packet flush

Diagram



Fields

Field	Function
31-12	Reserved.
—	

Table continues on the next page...

Table continued from the previous page...

Field	Function
11 UFCNTOVF	Overflow Bit for Underflow Packet Counter This bit is set every time the Tx queue Underflow Packet Counter field overflows, that is, it has crossed the maximum count. In such a scenario, the overflow packet counter is reset to all-zeros and this bit indicates that the rollover happened. Access restriction applies. Clears on read. Self-set to 1 on internal event. 0b - Overflow not detected for Underflow Packet Counter 1b - Overflow detected for Underflow Packet Counter
10-0 UFFRMCNT	Underflow Packet Counter This field indicates the number of packets aborted by the controller because of Tx Queue Underflow. This counter is incremented each time the MAC aborts outgoing packet because of underflow. The counter is cleared when this register is read with mci_be_i[0] at 1'b1. Access restriction applies. Clears on read. Self-set to 1 on internal event.

1.4.305 (MTL_TxQ4_Debug)

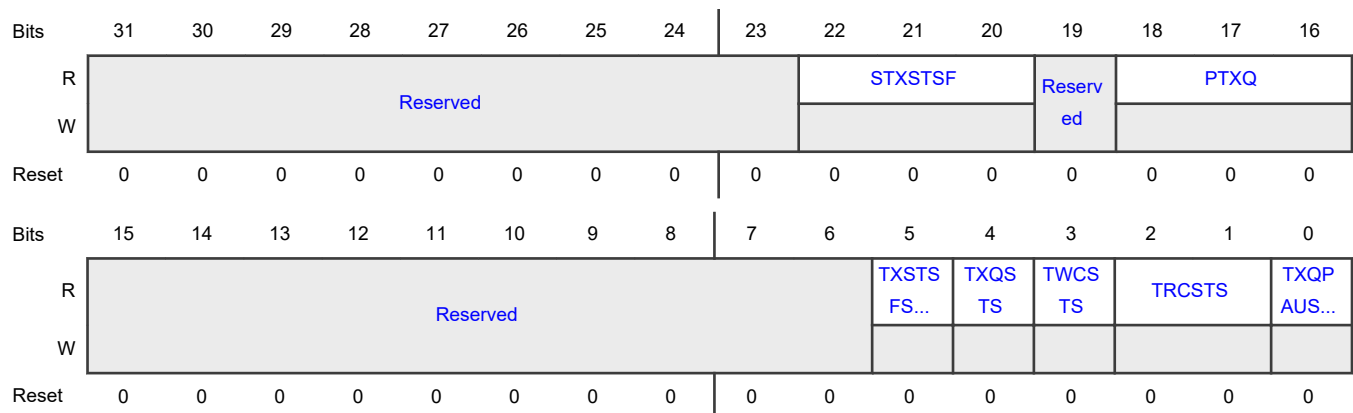
Offset

Register	Offset
MTL_TxQ4_Debug	E08h

Function

The Queue 4 Transmit Debug register gives the debug status of various blocks related to the Transmit queue.

Diagram



Fields

Field	Function
31-23 —	Reserved.
22-20 STXSTS	Number of Status Words in Tx Status FIFO of Queue This field indicates the current number of status in the Tx Status FIFO of this queue. When the DTXSTS bit of MTL_Operation_Mode register is set to 1, this field does not reflect the number of status words in Tx Status FIFO.
19 —	Reserved.
18-16 PTXQ	Number of Packets in the Transmit Queue This field indicates the current number of packets in the Tx Queue. When the DTXSTS bit of MTL_Operation_Mode register is set to 1, this field does not reflect the number of packets in the Transmit queue.
15-6 —	Reserved.
5 TXSTS	MTL Tx Status FIFO Full Status When high, this bit indicates that the MTL Tx Status FIFO is full. Therefore, the MTL cannot accept any more packets for transmission. 0b - MTL Tx Status FIFO Full status is not detected 1b - MTL Tx Status FIFO Full status is detected
4 TXQSTS	MTL Tx Queue Not Empty Status When this bit is high, it indicates that the MTL Tx Queue is not empty and some data is left for transmission. 0b - MTL Tx Queue Not Empty status is not detected 1b - MTL Tx Queue Not Empty status is detected
3 TWCSTS	MTL Tx Queue Write Controller Status When high, this bit indicates that the MTL Tx Queue Write Controller is active, and it is transferring the data to the Tx Queue. 0b - MTL Tx Queue Write Controller status is not detected 1b - MTL Tx Queue Write Controller status is detected
2-1 TRCSTS	MTL Tx Queue Read Controller Status This field indicates the state of the Tx Queue Read Controller: 00b - Idle state 01b - Read state (transferring data to the MAC transmitter)

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Field	Function
	10b - Waiting for pending Tx Status from the MAC transmitter 11b - Flushing the Tx queue because of the Packet Abort request from the MAC
0 TXQPAUSED	Transmit Queue in Pause When this bit is high and the Rx flow control is enabled, it indicates that the Tx Queue is in the Pause condition (in the full-duplex only mode) because of the following: - Reception of the PFC packet for the priorities assigned to the Tx Queue when PFC is enabled - Reception of 802.3x Pause packet when PFC is disabled 0b - Transmit Queue in Pause status is not detected 1b - Transmit Queue in Pause status is detected

1.4.306 (MTL_TxQ4_ETS_Control)

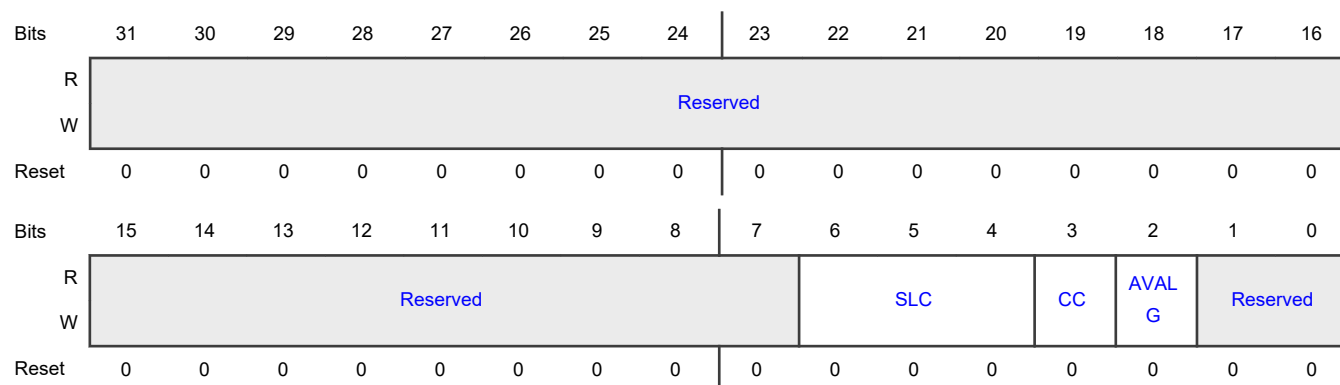
Offset

Register	Offset
MTL_TxQ4_ETS_Control	E10h

Function

The Queue ETS Control register controls the enhanced transmission selection operation.

Diagram



Fields

Field	Function
31-7	Reserved.

Table continues on the next page...

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Field	Function
—	
6-4 SLC	Slot Count If the credit-based shaper algorithm is enabled, the software can program the number of slots (of duration programmed in DMA_CH(#i)_Slot_Interval register) over which the average transmitted bits per slot, provided in the MTL_TxQ(#i)_ETS_Status register, need to be computed for Queue. The encoding is as follows: 000b - 1 slot 001b - 2 slots 010b - 4 slots 011b - 8 slots 100b - 16 slots 101b - Reserved
3 CC	Credit Control When this bit is set, the accumulated credit parameter in the credit-based shaper algorithm logic is not reset to zero when there is positive credit and no packet to transmit in Channel 1. The credit accumulates even when there is no packet waiting in Channel 1 and another channel is transmitting. When this bit is reset, the accumulated credit parameter in the credit-based shaper algorithm logic is set to zero when there is positive credit and no packet to transmit in Channel 1. When there is no packet waiting in Channel 1 and other channel is transmitting, no credit is accumulated. 0b - Credit Control is disabled 1b - Credit Control is enabled
2 AVALG	AV Algorithm When Queue 1 is programmed for AV, this field configures the scheduling algorithm for this queue: This bit when set, indicates credit based shaper algorithm (CBS) is selected for Queue 1 traffic. When reset, strict priority is selected. 0b - CBS Algorithm is disabled 1b - CBS Algorithm is enabled
1-0 —	Reserved.

1.4.307 (MTL_TxQ4_ETS_Status)

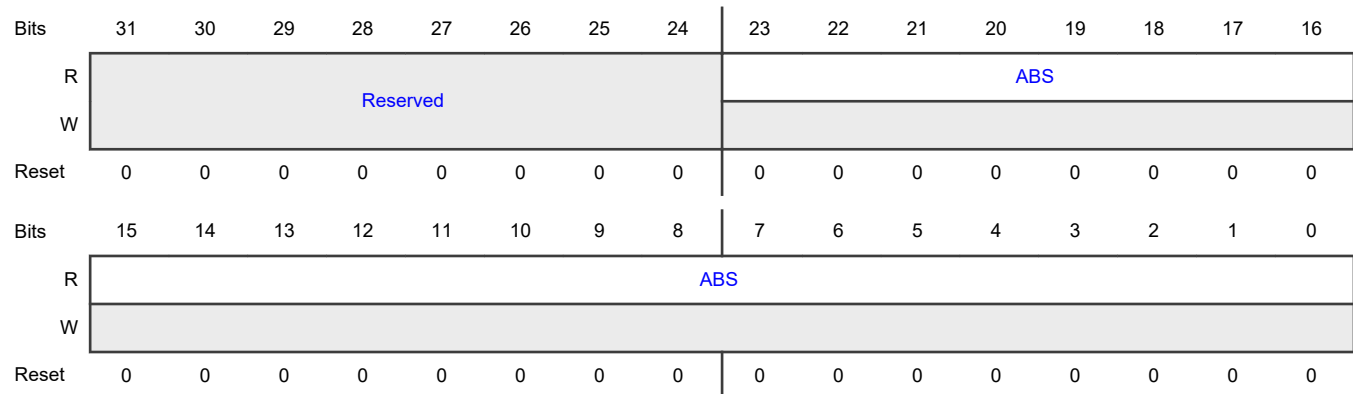
Offset

Register	Offset
MTL_TxQ4_ETS_Status	E14h

Function

The Queue 4 ETS Status register provides the average traffic transmitted in Queue 4.

Diagram



Fields

Field	Function
31-24 —	Reserved.
23-0 ABS	Average Bits per Slot This field contains the average transmitted bits per slot. If AV operation is enabled, this field is computed over number of slots, programmed in the SLC field of MTL_TxQ(#i)_ETS_CONTROL register. The maximum value of this field is 0x6_4000 in 100 Mbps, 0x3E_8000 in 1000 Mbps and 9C_4000 in 2500 Mbps mode respectively. When the DCB operation is enabled for Queue, this field is computed over every 10 million bit times slot (4 ms in 2500 Mbps; 10 ms in 1000 Mbps; 100 ms in 100 Mbps). The maximum value is 0x989680.

1.4.308 (MTL_TxQ4_Quantum_Weight)

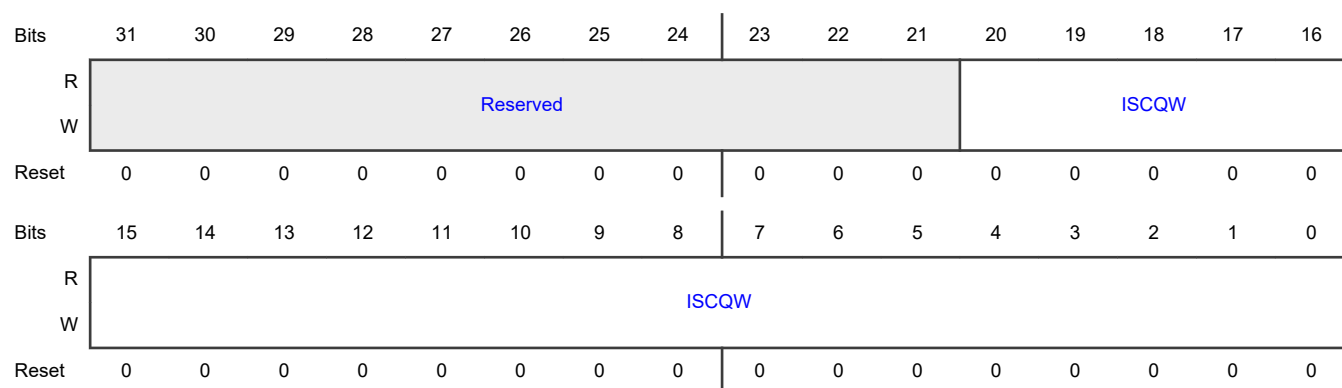
Offset

Register	Offset
MTL_TxQ4_Quantum_Weight	E18h

Function

The Queue 4 idleSlopeCredit, Quantum or Weights register provides the average traffic transmitted in Queue 4.

Diagram



Fields

Field	Function
31-21 —	Reserved.
20-0 ISCQW	idleSlopeCredit, Quantum or Weights - idleSlopeCredit When AV feature is enabled, this field contains the idleSlopeCredit value required for the credit-based shaper algorithm for Queue 1. This is the rate of change of credit in bits per cycle (40 ns for 100 Mbps; 8 ns for 1000 Mbps; 3.2 ns for 2500 Mbps) when the credit is increasing. The software should program this field with computed credit in bits per cycle scaled by 1,024. The maximum value is portTransmitRate, that is, 0x2000 in 1000/2500 Mbps mode and 0x1000 in 100 Mbps mode. Bits[20:14] must be written to zero. - Quantum When the DCB operation is enabled with DWRR algorithm for Queue 1 traffic, this field contains the quantum value in bytes to be added to credit during every queue scanning cycle. The maximum value is 0x1312D0 bytes. - Weights When DCB operation is enabled with WFQ algorithm for Queue 1 traffic, this field contains the weight for this queue. The maximum value is 0x3FFF where weight of 0 indicates 100% bandwidth. Bits[20:14] must be written to zero. When DCB operation or generic queuing operation is enabled with WRR algorithm for Queue 1 traffic, this field contains the weight for this queue. The maximum value is 0x64. Bits [20:7] must be written to zero. - Note 1: In multiple Queue configuration this field in respective per queue register must be programmed to some non-zero value when multiple queues are enabled or single queue other than Q0 is enabled. This field need not be programmed when only Q0 is enabled. In general, when WRR algorithm is selected a non-zero value must be programmed on both Receive and Transmit. In Receive, the register is MTL_Operation_Mode register. - Note 2: For WFQ algorithm, higher the programmed weights lesser the bandwidth allocated for that Transmit Queue. The finish time is not a function of particular packet alone but it is as per the formula: (previous_finish_time of particular Transmit Queue + (weights*packet_size))

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Field	Function
	- Note 3: The weights programmed do not correspond to the number of packets but the fraction of bandwidth or time allocated for particular queue w.r.t. total BW or time.

1.4.309 (MTL_TxQ4_SendSlopeCredit)

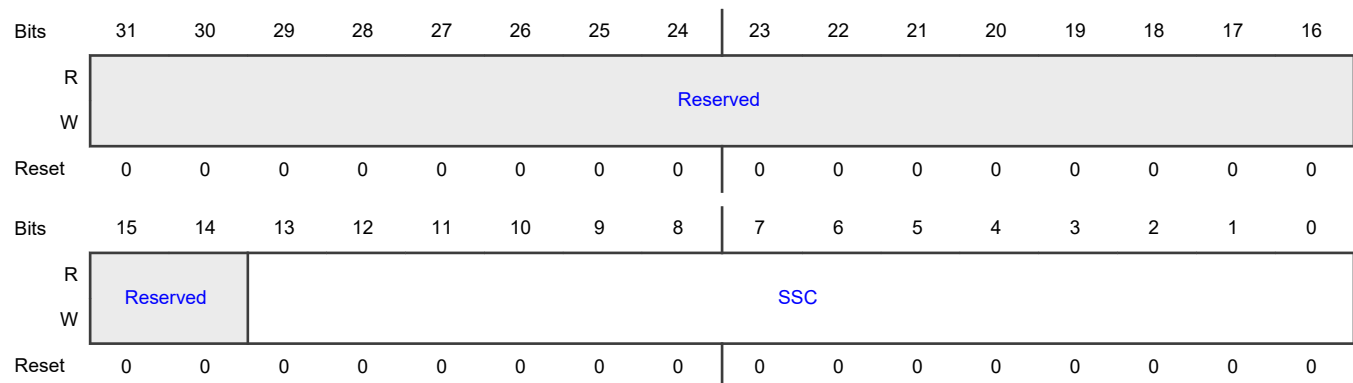
Offset

Register	Offset
MTL_TxQ4_SendSlopeCredit	E1Ch

Function

The sendSlopeCredit register contains the sendSlope credit value required for the credit-based shaper algorithm for the Queue.

Diagram



Fields

Field	Function
31-14 —	Reserved.
13-0 SSC	sendSlopeCredit Value When AV operation is enabled, this field contains the sendSlopeCredit value required for credit-based shaper algorithm for Queue 1. This is the rate of change of credit in bits per cycle (40 ns, 8 ns and 3.2 ns for 100 Mbps, 1000 Mbps and 2500 Mbps respectively) when the credit is decreasing. The software should program this field with computed credit in bits per cycle scaled by 1,024. The maximum value is portTransmitRate, that is, 0x2000 in 1000/2500 Mbps mode and 0x1000 in 100 Mbps mode. This field should be programmed with absolute sendSlopeCredit value. The credit-based shaper logic subtracts it from the accumulated credit when Channel 1 is selected for transmission.

1.4.310 (MTL_TxQ4_HiCredit)

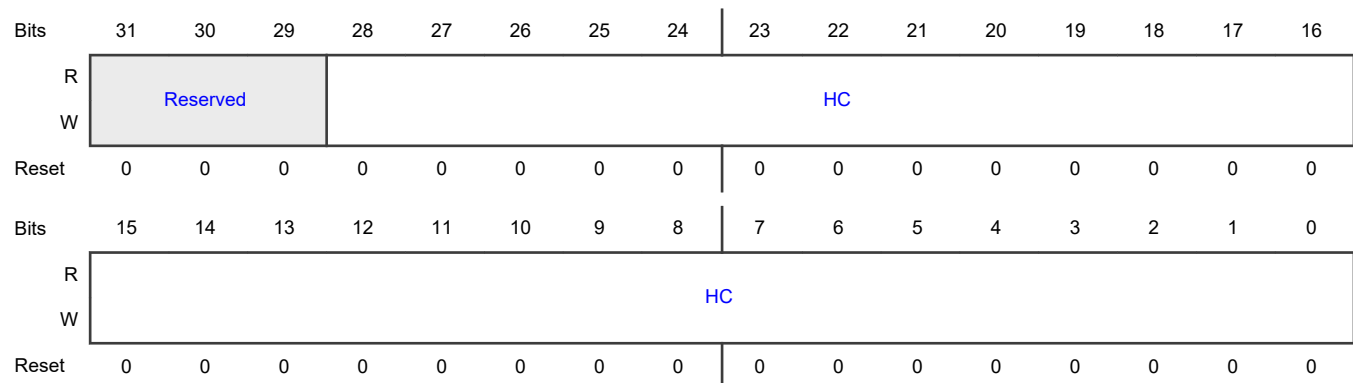
Offset

Register	Offset
MTL_TxQ4_HiCredit	E20h

Function

The hiCredit register contains the hiCredit value required for the credit-based shaper algorithm for the Queue.

Diagram



Fields

Field	Function
31-29 —	Reserved.
28-0 HC	hiCredit Value When the AV feature is enabled, this field contains the hiCredit value required for the credit-based shaper algorithm. This is the maximum value that can be accumulated in the credit parameter. This is specified in bits scaled by 1,024. The maximum value is maxInterferenceSize, that is, best-effort maximum packet size (16,384 bytes or 131,072 bits). The value to be specified is $131,072 * 1,024 = 134,217,728$ or 0x0800_0000.

1.4.311 (MTL_TxQ4_LoCredit)

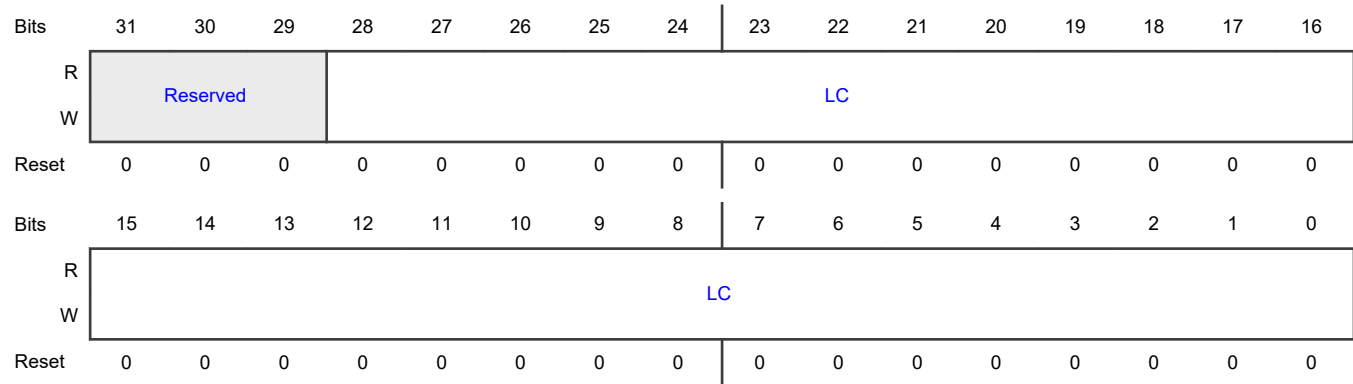
Offset

Register	Offset
MTL_TxQ4_LoCredit	E24h

Function

The loCredit register contains the loCredit value required for the credit-based shaper algorithm for the Queue.

Diagram



Fields

Field	Function
31-29 —	Reserved.
28-0 LC	loCredit Value When AV operation is enabled, this field contains the loCredit value required for the credit-based shaper algorithm. This is the minimum value that can be accumulated in the credit parameter. This is specified in bits scaled by 1,024. The maximum value to be programmed corresponds to twice the maxFrameSize transmitted from this queue. If the maxFrameSize is 8192 bytes, then $(8192 \times 2) \times 8 \times 1024 = 134,217,728$ or 0x0800_0000. Because it is a negative value, the programmed value is 2's complement of the value, that is, 0x1800_0000.

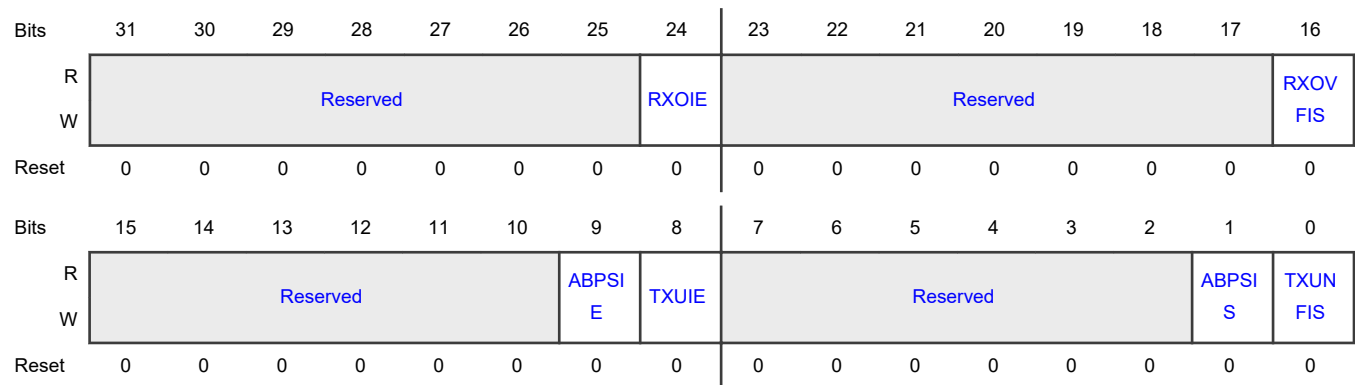
1.4.312 (MTL_Q4_Interrupt_Control_Status)

Offset

Register	Offset
MTL_Q4_Interrupt_Contr ol_Status	E2Ch

Function

This register contains the interrupt enable and status bits for the queue 4 interrupts.

Diagram**Fields**

Field	Function
31-25 —	Reserved.
24 RXOIE	Receive Queue Overflow Interrupt Enable When this bit is set, the Receive Queue Overflow interrupt is enabled. When this bit is reset, the Receive Queue Overflow interrupt is disabled. 0b - Receive Queue Overflow Interrupt is disabled 1b - Receive Queue Overflow Interrupt is enabled
23-17 —	Reserved.
16 RXOVFIS	Receive Queue Overflow Interrupt Status This bit indicates that the Receive Queue had an overflow while receiving the packet. If a partial packet is transferred to the application, the overflow status is set in RDES3[21]. This bit is cleared when the application writes 1 to this bit. Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect. 0b - Receive Queue Overflow Interrupt Status not detected 1b - Receive Queue Overflow Interrupt Status detected
15-10 —	Reserved.
9 ABPSIE	Average Bits Per Slot Interrupt Enable When this bit is set, the MAC asserts the sbd_intr_o or mci_intr_o interrupt when the average bits per slot status is updated. When this bit is cleared, the interrupt is not asserted for such an event. 0b - Average Bits Per Slot Interrupt is disabled

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Field	Function
	1b - Average Bits Per Slot Interrupt is enabled
8 TXUIE	Transmit Queue Underflow Interrupt Enable When this bit is set, the Transmit Queue Underflow interrupt is enabled. When this bit is reset, the Transmit Queue Underflow interrupt is disabled. 0b - Transmit Queue Underflow Interrupt Status is disabled 1b - Transmit Queue Underflow Interrupt Status is enabled
7-2 —	Reserved.
1 ABPSIS	Average Bits Per Slot Interrupt Status When set, this bit indicates that the MAC has updated the ABS value. This bit is cleared when the application writes 1 to this bit. Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect. 0b - Average Bits Per Slot Interrupt Status not detected 1b - Average Bits Per Slot Interrupt Status detected
0 TXUNFIS	Transmit Queue Underflow Interrupt Status This bit indicates that the Transmit Queue had an underflow while transmitting the packet. Transmission is suspended and an Underflow Error TDES3[2] is set. This bit is cleared when the application writes 1 to this bit. Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect. 0b - Transmit Queue Underflow Interrupt Status not detected 1b - Transmit Queue Underflow Interrupt Status detected

1.4.313 (MTL_RxQ4_Operation_Mode)

Offset

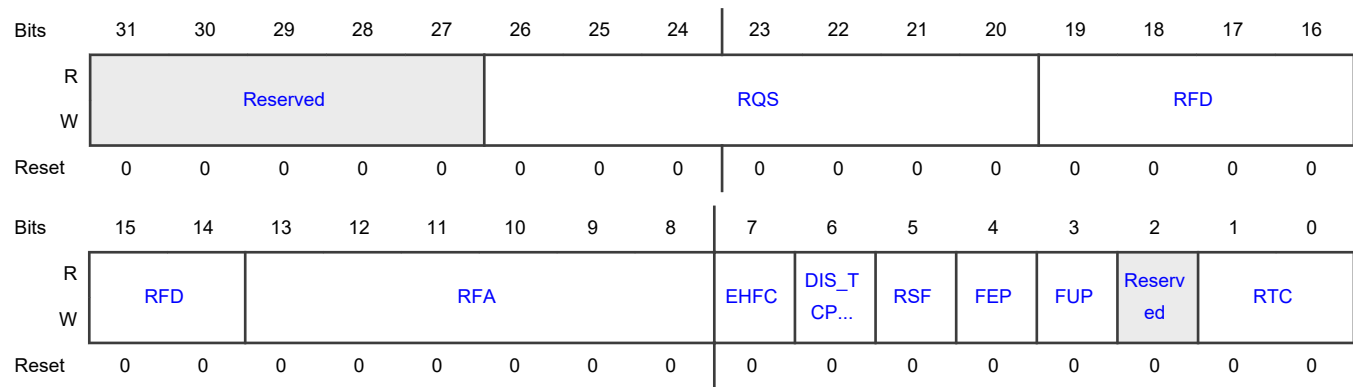
Register	Offset
MTL_RxQ4_Operation_Mode	E30h

Function

The Queue 4 Receive Operation Mode register establishes the Receive queue operating modes and command.

The RFA and RFD fields are not backward compatible with the RFA and RFD fields of 4.00a release

Diagram



Fields

Field	Function
31-27 —	Reserved.
26-20 RQS	Receive Queue Size This field indicates the size of the allocated Receive queues in blocks of 256 bytes. The RQS field is read-write only if the number of Rx Queues more than one, the reset value is 0x0 and indicates size of 256 bytes. This means that value of 0x0 = 256 bytes, 0x1 = 512 bytes and so on, so user should be programming RQS [5:0] = 6'b001111 to allocate queue size of 4096 (4K) bytes. In general, the size of the Queue = (RQS + 1)*256 bytes. When the number of Rx Queues is one, the field is read-only and the configured RX FIFO size in blocks of 256 bytes is reflected in the reset value. The width of this field depends on the Rx memory size selected in your configuration. For example, if the memory size is 2048, the width of this field is 3 bits: $\text{LOG2}(2048/256) = \text{LOG2}(8) = 3 \text{ bits}$
19-14 RFD	Threshold for Deactivating Flow Control (in half-duplex and full-duplex modes) These bits control the threshold (fill-level of Rx queue) at which the flow control is de-asserted after activation: - 0: Full minus 1 KB, that is, FULL 1 KB - 1: Full minus 1.5 KB, that is, FULL 1.5 KB - 2: Full minus 2 KB, that is, FULL 2 KB - 3: Full minus 2.5 KB, that is, FULL 2.5 KB - ... - 62: Full minus 32 KB, that is, FULL 32 KB - 63: Full minus 32.5 KB, that is, FULL 32.5 KB The de-assertion is effective only after flow control is asserted. Note: The value must be programmed in such a way to make sure that the threshold is a positive number.

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Field	Function
	When the EHFC is set high, these values are applicable only when the Rx queue size determined by the RQS field of this register, is equal to or greater than 4 KB. For a given queue size, the values ranges between 0 and the encoding for FULL minus (QSIZE - 0.5 KB) and all other values are illegal. Here the term FULL and QSIZE refers to the queue size determined by the RQS field of this register. The width of this field depends on RX FIFO size selected during the configuration. Remaining bits are reserved and read only.
13-8 RFA	Threshold for Activating Flow Control (in half-duplex and full-duplex) These bits control the threshold (fill-level of Rx queue) at which the flow control is activated: For more information on encoding for this field, see RFD.
7 EHFC	Enable Hardware Flow Control When this bit is set, the flow control signal operation, based on the fill-level of Rx queue, is enabled. When reset, the flow control operation is disabled. 0b - Hardware Flow Control is disabled 1b - Hardware Flow Control is enabled
6 DIS_TCP_EF	Disable Dropping of TCP/IP Checksum Error Packets When this bit is set, the MAC does not drop the packets which only have the errors detected by the Receive Checksum Offload engine. Such packets have errors only in the encapsulated payload. There are no errors (including FCS error) in the Ethernet packet received by the MAC. When this bit is reset, all error packets are dropped if the FEP bit is reset. 0b - Dropping of TCP/IP Checksum Error Packets is enabled 1b - Dropping of TCP/IP Checksum Error Packets is disabled
5 RSF	Receive Queue Store and Forward When this bit is set, the DWC_ether_qos reads a packet from the Rx queue only after the complete packet has been written to it, ignoring the RTC field of this register. When this bit is reset, the Rx queue operates in the Threshold (cut-through) mode, subject to the threshold specified by the RTC field of this register. 0b - Receive Queue Store and Forward is disabled 1b - Receive Queue Store and Forward is enabled
4 FEP	Forward Error Packets When this bit is reset, the Rx queue drops packets with error status (CRC error, GMII_ER, watchdog timeout, or overflow). However, if the start byte (write) pointer of a packet is already transferred to the read controller side (in Threshold mode), the packet is not dropped. When this bit is set, all packets except the runt error packets are forwarded to the application or DMA. If the RSF bit is set and the Rx queue overflows when a partial packet is written, the packet is dropped irrespective of the setting of this bit. However, if the RSF bit is reset and the Rx queue overflows when a partial packet is written, a partial packet might be forwarded to the application or DMA.

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Field	Function
	0b - Forward Error Packets is disabled 1b - Forward Error Packets is enabled
3 FUP	Forward Undersized Good Packets When this bit is set, the Rx queue forwards the undersized good packets (packets with no error and length less than 64 bytes), including pad-bytes and CRC. When this bit is reset, the Rx queue drops all packets of less than 64 bytes, unless a packet is already transferred because of the lower value of Rx Threshold, for example, RTC = 01. 0b - Forward Undersized Good Packets is disabled 1b - Forward Undersized Good Packets is enabled
2 —	Reserved.
1-0 RTC	Receive Queue Threshold Control These bits control the threshold level of the MTL Rx queue (in bytes): The received packet is transferred to the application or DMA when the packet size within the MTL Rx queue is larger than the threshold. In addition, full packets with length less than the threshold are automatically transferred. This field is valid only when the RSF bit is zero. This field is ignored when the RSF bit is set to 1. 00b - 64 01b - 32 10b - 96 11b - 128

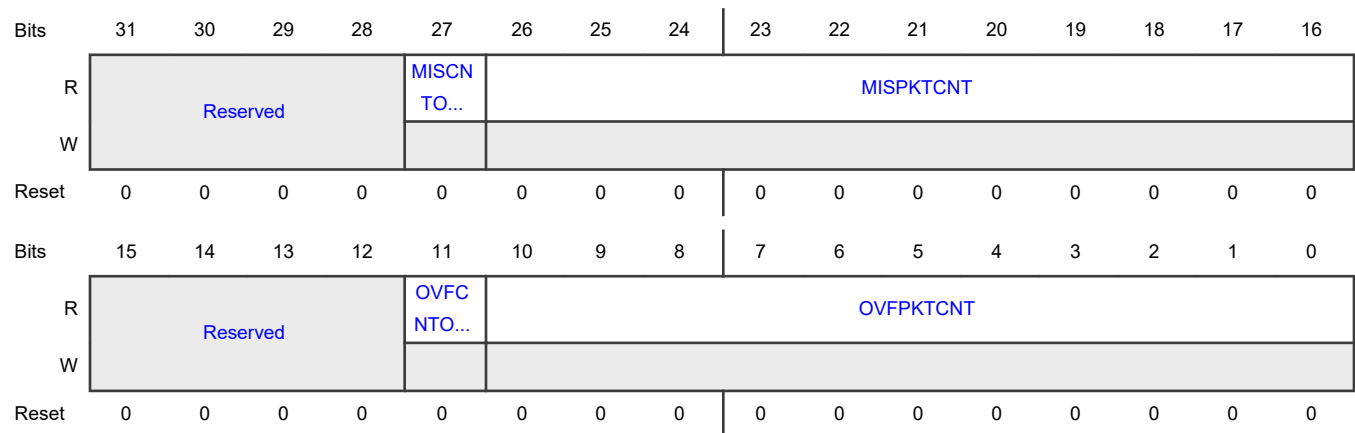
1.4.314 (MTL_RxQ4_Missed_Packet_Overflow_Cnt)

Offset

Register	Offset
MTL_RxQ4_Missed_Packet_Overflow_Cnt	E34h

Function

The Queue 4 Missed Packet and Overflow Counter register contains the counter for packets missed because of Receive queue packet flush and packets discarded because of Receive queue overflow.

Diagram**Fields**

Field	Function
31-28 —	Reserved.
27 MISCNTOVF	Missed Packet Counter Overflow Bit When set, this bit indicates that the Rx Queue Missed Packet Counter crossed the maximum limit. Access restriction applies. Clears on read. Self-set to 1 on internal event. 0b - Missed Packet Counter overflow not detected 1b - Missed Packet Counter overflow detected
26-16 MISPKTCNT	Missed Packet Counter This field indicates the number of packets missed by the DWC_ether_qos because the application asserted ari_pkt_flush_i[] for this queue. This counter is reset when this register is read with mci_be_i[0] at 1b1. This counter is incremented by 1 when the DMA discards the packet because of buffer unavailability. Access restriction applies. Clears on read. Self-set to 1 on internal event.
15-12 —	Reserved.
11 OVFCNTOVF	Overflow Counter Overflow Bit When set, this bit indicates that the Rx Queue Overflow Packet Counter field crossed the maximum limit. Access restriction applies. Clears on read. Self-set to 1 on internal event. 0b - Overflow Counter overflow not detected 1b - Overflow Counter overflow detected
10-0 OVFPKTCNT	Overflow Packet Counter

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Field	Function
	This field indicates the number of packets discarded by the DWC_ether_qos because of Receive queue overflow. This counter is incremented each time the DWC_ether_qos discards an incoming packet because of overflow. This counter is reset when this register is read with mci_be_i[0] at 1'b1. Access restriction applies. Clears on read. Self-set to 1 on internal event.

1.4.315 (MTL_RxQ4_Debug)

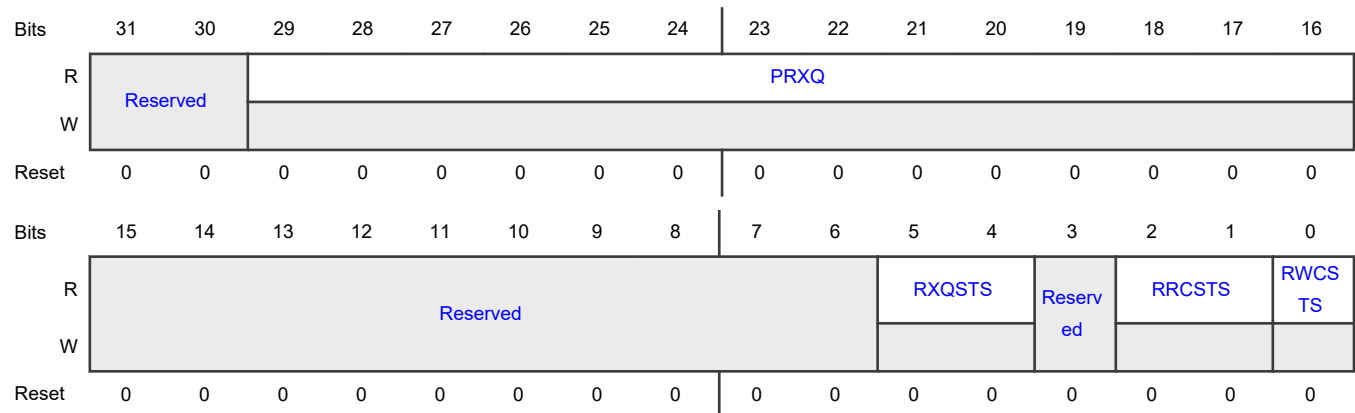
Offset

Register	Offset
MTL_RxQ4_Debug	E38h

Function

The Queue 4 Receive Debug register gives the debug status of various blocks related to the Receive queue.

Diagram



Fields

Field	Function
31-30 —	Reserved.
29-16 PRXQ	Number of Packets in Receive Queue This field indicates the current number of packets in the Rx Queue. The theoretical maximum value for this field is 256KB/16B = 16K Packets, that is, Max_Queue_Size/Min_Packet_Size.
15-6	Reserved.

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Field	Function
—	
5-4 RXQSTS	MTL Rx Queue Fill-Level Status This field gives the status of the fill-level of the Rx Queue: 00b - Rx Queue empty 01b - Rx Queue fill-level below flow-control deactivate threshold 10b - Rx Queue fill-level above flow-control activate threshold 11b - Rx Queue full
3 —	Reserved.
2-1 RRCSTS	MTL Rx Queue Read Controller State This field gives the state of the Rx queue Read controller: 00b - Idle state 01b - Reading packet data 10b - Reading packet status (or timestamp) 11b - Flushing the packet data and status
0 RWCSTS	MTL Rx Queue Write Controller Active Status When high, this bit indicates that the MTL Rx queue Write controller is active, and it is transferring a received packet to the Rx Queue. 0b - MTL Rx Queue Write Controller Active Status not detected 1b - MTL Rx Queue Write Controller Active Status detected

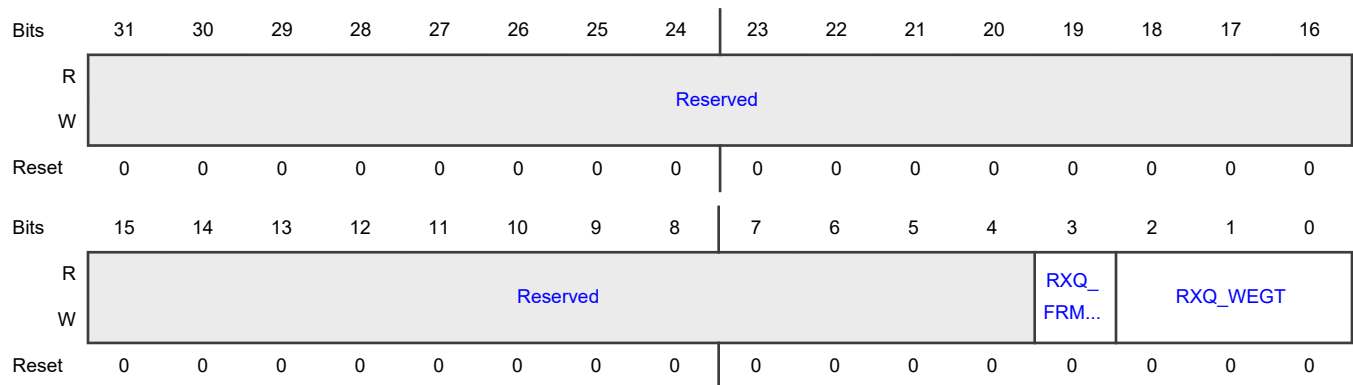
1.4.316 (MTL_RxQ4_Control)

Offset

Register	Offset
MTL_RxQ4_Control	E3Ch

Function

The Queue Receive Control register controls the receive arbitration and passing of received packets to the application.

Diagram**Fields**

Field	Function
31-4 —	Reserved.
3 RXQ_FRM_AR BIT	Receive Queue Packet Arbitration When this bit is set, the DWC_ether_qos drives the packet data to the ARI interface such that the entire packet data of currently-selected queue is transmitted before switching to other queue. When this bit is reset, the DWC_ether_qos drives the packet data to the ARI interface such that the following amount of data of currently-selected queue is transmitted before switching to other queue: - PBL amount of data (indicated by ari_qN_pbl_i[]) or - Complete data of a packet The status and the timestamp are not a part of the PBL data. Therefore, the DWC_ether_qos drives the complete status (including timestamp status) during first PBL request for the packet (in store-and-forward mode) or the last PBL request for the packet (in Threshold mode). 0b - Receive Queue Packet Arbitration is disabled 1b - Receive Queue Packet Arbitration is enabled
2-0 RXQ_WEGT	Receive Queue Weight This field indicates the weight assigned to the Rx Queue 0. This field needs to be programmed with one value less than the required weight, i.e. reset value of 0 indicates weight of 1, value of 1 indicates weight of 2, and so on. The weight is used as the number of continuous PBL or packets requests (depending on the RXQ_FRM_ARBIT) allocated to the queue in one arbitration cycle. Note: The change in value of RXQ_WEGT takes effect only after the completion of current service round or when there is change from RAA=SP to RAA=WSP algorithm. This approach is taken so that there is smooth transition. For the RXQ_WEGT value to take effect at the start, the MTL_RxQ(##)_Control registers must be programmed before the MTL_Operation_Mode register.

1.4.317 (DMA_Mode)

Offset

Register	Offset
DMA_Mode	1000h

Function

The Bus Mode register establishes the bus operating modes for the DMA.

Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	Reserved														INTM	
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	Reserved	Reserved				Reserved	Reserved	Reserved	Reserved				Reserved		Reserved	SWR
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Fields

Field	Function
31-18 —	Reserved.
17-16 INTM	Interrupt Mode This field defines the interrupt mode of DWC_ether_qos. The behavior of the following outputs changes depending on the following settings: - sbd_perch_tx_intr_o[] (Transmit Per Channel Interrupt) - sbd_perch_rx_intr_o[] (Receive Per Channel Interrupt) - sbd_intr_o (Common Interrupt) It also changes the behavior of the RI/TI bits in the DMA_CH0_Status. 00: Reserved 01: sbd_perch_* are level signals asserted on TX/RX packet transfer completion event when corresponding interrupts are enabled and de-asserted when the software clears the corresponding RI/TI status bits. The sbd_intr_o is not asserted for these TX/RX packet transfer completion events. 10: sbd_perch_* are level signals asserted on TX/RX packet transfer completion event when corresponding interrupts are enabled and de-asserted when the software clears the corresponding RI/TI status bits.

Table continues on the next page...

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Field	Function
	However, the signal is asserted again if the same event occurred again before it was cleared. The <code>sbd_intr_o</code> is not asserted for these TX/RX packet transfer completion events. 11: Reserved For more details please refer Table "DWC_ether_qos Transfer Complete Interrupt Behavior". 00b - See above description 01b - See above description 10b - See above description 11b - Reserved
15 —	Reserved.
14-12 —	Reserved.
11 —	Reserved.
10 —	Reserved.
9 —	Reserved.
8 DSPW	Descriptor Posted Write When this bit is set to 0, the descriptor writes are always non-posted. When this bit is set to 1, the descriptor writes are non-posted only when IOC (Interrupt on completion) is set in last descriptor, otherwise the descriptor writes are always posted. 0b - Descriptor Posted Write is disabled 1b - Descriptor Posted Write is enabled
7-5 —	Reserved.
4-2 —	Reserved.
1 —	Reserved.
0	Software Reset

Table continues on the next page...

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Field	Function
SWR	When this bit is set, the MAC and the DMA controller reset the logic and all internal registers of the DMA, MTL, and MAC. This bit is automatically cleared after the reset operation is complete in all DWC_ether_qos clock domains. Before reprogramming any DWC_ether_qos register, a value of zero should be read in this bit. This bit must be read at least 4 CSR clock cycles after it is written to 1. Note: The reset operation is complete only when all resets in all active clock domains are de-asserted. Therefore, it is essential that all PHY inputs clocks (applicable for the selected PHY interface) are present for software reset completion. The time to complete the software reset operation depends on the frequency of the slowest active clock. Access restriction applies. Setting 1 sets. Self-cleared. Setting 0 has no effect. 0b - Software Reset is disabled 1b - Software Reset is enabled

1.4.318 (DMA_SysBus_Mode)

Offset

Register	Offset
DMA_SysBus_Mode	1004h

Function

The System Bus mode register controls the behavior of the AHB or AXI master. It mainly controls burst splitting and number of outstanding requests.

Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	EN_LPI	LPI_XI T...	Reserved		WR_OSR_LMT				Reserved				RD_OSR_LMT			
W																
Reset	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	1
Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	Reserv ed	Reserv ed	ONEK BBE	AAL	Reserv ed	AAL	Reserved		BLN2 56	BLN1 28	BLN6 4	BLN3 2	BLN1 6	BLN8	BLN4	FB
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Fields

Field	Function
31 EN_LPI	Enable Low Power Interface (LPI) When set to 1, this bit enables the LPI mode supported by the EQOS-AXI configuration and accepts the LPI request from the AXI System Clock controller. When set to 0, this bit disables the LPI mode and always denies the LPI request from the AXI System Clock controller. 0b - Low Power Interface (LPI) is disabled 1b - Low Power Interface (LPI) is enabled
30 LPI_XIT_PKT	Unlock on Magic Packet or Remote Wake-Up Packet When set to 1, this bit enables the AXI master to come out of the LPI mode only when the magic packet or remote wake-up packet is received. When set to 0, this bit enables the AXI master to come out of the LPI mode when any packet is received. 0b - Unlock on Magic Packet or Remote Wake-Up Packet is disabled 1b - Unlock on Magic Packet or Remote Wake-Up Packet is enabled
29-28 —	Reserved.
27-24 WR_OSR_LMT	AXI Maximum Write Outstanding Request Limit This value limits the maximum outstanding request on the AXI write interface. Maximum outstanding requests = WR_OSR_LMT + 1 Note: - Bit 26 is reserved if DWC_ETHER_QOS_AXI_MAX_WR_REQ = 4 - Bit 27 is reserved if DWC_ETHER_QOS_AXI_MAX_WR_REQ != 16
23-20 —	Reserved.
19-16 RD_OSR_LMT	AXI Maximum Read Outstanding Request Limit This value limits the maximum outstanding request on the AXI read interface. Maximum outstanding requests = RD_OSR_LMT + 1 Note: - Bit 18 is reserved if parameter DWC_ETHER_QOS_AXI_MAX_RD_REQ = 4 - Bit 19 is reserved if parameter DWC_ETHER_QOS_AXI_MAX_RD_REQ != 16
15 —	Reserved.
14 —	Reserved.

Table continues on the next page...

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Field	Function
13 ONEKBBE	1 KB Boundary Crossing Enable for the EQOS-AXI Master When set, the burst transfers performed by the EQOS-AXI master do not cross 1 KB boundary. When reset, the burst transfers performed by the EQOS-AXI master do not cross 4 KB boundary. 0b - 1 KB Boundary Crossing for the EQOS-AXI Master Beats is disabled 1b - 1 KB Boundary Crossing for the EQOS-AXI Master Beats is enabled
12 AAL	Address-Aligned Beats When this bit is set to 1, the EQOS-AXI or EQOS-AHB master performs address-aligned burst transfers on Read and Write channels. 0b - Address-Aligned Beats is disabled 1b - Address-Aligned Beats is enabled
11 —	Reserved.
10 AALE	Automatic AXI LPI enable When set to 1, enables the AXI master to enter into LPI state when there is no activity in the DWC_ether_qos for number of system clock cycles programmed in the LPIEI field of AXI_LPI_Entry_Interval register. 0b - Automatic AXI LPI is disabled 1b - Automatic AXI LPI is enabled
9-8 —	Reserved.
7 BLEN256	AXI Burst Length 256 When this bit is set to 1, the EQOS-AXI master can select a burst length of 256 on the AXI interface. 0b - No effect 1b - AXI Burst Length 256
6 BLEN128	AXI Burst Length 128 When this bit is set to 1, the EQOS-AXI master can select a burst length of 128 on the AXI interface. 0b - No effect 1b - AXI Burst Length 128
5 BLEN64	AXI Burst Length 64 When this bit is set to 1, the EQOS-AXI master can select a burst length of 64 on the AXI interface. 0b - No effect 1b - AXI Burst Length 64
4	AXI Burst Length 32

Table continues on the next page...

Table continued from the previous page...

Field	Function
BLEN32	When this bit is set to 1, the EQOS-AXI master can select a burst length of 32 on the AXI interface. 0b - No effect 1b - AXI Burst Length 32
3 BLEN16	AXI Burst Length 16 When this bit is set to 1 or the FB bit is set to 0, the EQOS-AXI master can select a burst length of 16 on the AXI interface. When the FB bit is set to 0, setting this bit has no effect. 0b - No effect 1b - AXI Burst Length 16
2 BLEN8	AXI Burst Length 8 When this bit is set to 1 or the FB bit is set to 0, the EQOS-AXI master can select a burst length of 8 on the AXI interface. When the FB bit is set to 0, setting this bit has no effect. 0b - No effect 1b - AXI Burst Length 8
1 BLEN4	AXI Burst Length 4 When this bit is set to 1 or the FB bit is set to 0, the EQOS-AXI master can select a burst length of 4 on the AXI interface. When the FB bit is set to 0, setting this bit has no effect. 0b - No effect 1b - AXI Burst Length 4
0 FB	Fixed Burst Length When this bit is set to 1, the EQOS-AXI master initiates burst transfers of specified lengths as given below. - Burst transfers of fixed burst lengths as indicated by the BLEN256, BLEN128, BLEN64, BLEN32, BLEN16, BLEN8, or BLEN4 field - Burst transfers of length 1 When this bit is set to 0, the EQOS-AXI master initiates burst transfers that are equal to or less than the maximum allowed burst length programmed in Bits[7:1]. 0b - Fixed Burst Length is disabled 1b - Fixed Burst Length is enabled

1.4.319 (DMA_Interrupt_Status)

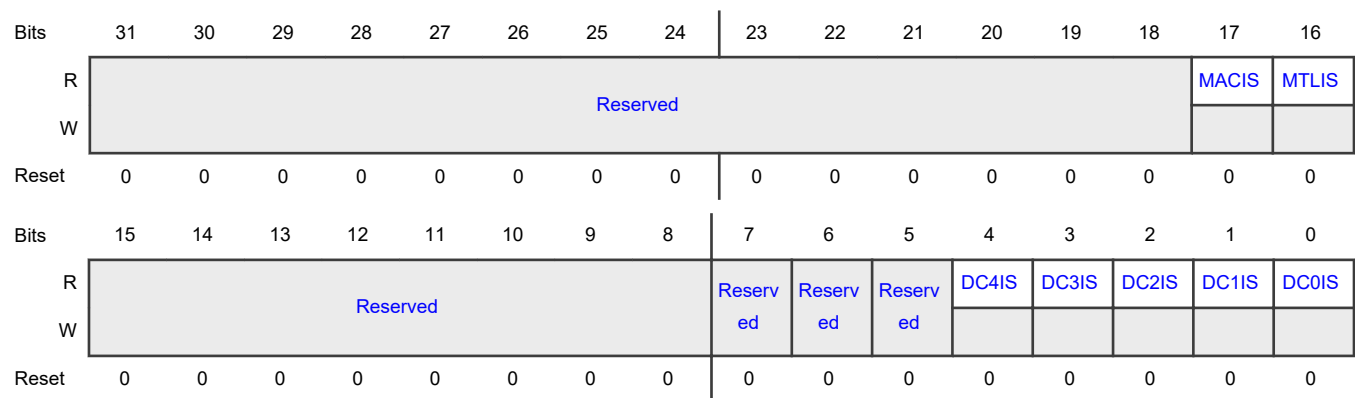
Offset

Register	Offset
DMA_Interrupt_Status	1008h

Function

The application reads this Interrupt Status register during interrupt service routine or polling to determine the interrupt status of DMA channels, MTL queues, and the MAC.

Diagram



Fields

Field	Function
31-18 —	Reserved.
17 MACIS	MAC Interrupt Status This bit indicates an interrupt event in the MAC. To reset this bit to 1'b0, the software must read the corresponding register in the MAC to get the exact cause of the interrupt and clear its source. 0b - MAC Interrupt Status not detected 1b - MAC Interrupt Status detected
16 MTLIS	MTL Interrupt Status This bit indicates an interrupt event in the MTL. To reset this bit to 1'b0, the software must read the corresponding register in the MTL to get the exact cause of the interrupt and clear its source. 0b - MTL Interrupt Status not detected 1b - MTL Interrupt Status detected
15-8	Reserved.

Table continues on the next page...

Table continued from the previous page...

Field	Function
—	
7 —	Reserved.
6 —	Reserved.
5 —	Reserved.
4 DC4IS	DMA Channel 4 Interrupt Status This bit indicates an interrupt event in DMA Channel 4. To reset this bit to 1'b0, the software must read the corresponding register in DMA Channel 4 to get the exact cause of the interrupt and clear its source. 0b - DMA Channel 4 Interrupt Status not detected 1b - DMA Channel 4 Interrupt Status detected
3 DC3IS	DMA Channel 3 Interrupt Status This bit indicates an interrupt event in DMA Channel 3. To reset this bit to 1'b0, the software must read the corresponding register in DMA Channel 3 to get the exact cause of the interrupt and clear its source. 0b - DMA Channel 3 Interrupt Status not detected 1b - DMA Channel 3 Interrupt Status detected
2 DC2IS	DMA Channel 2 Interrupt Status This bit indicates an interrupt event in DMA Channel 2. To reset this bit to 1'b0, the software must read the corresponding register in DMA Channel 2 to get the exact cause of the interrupt and clear its source. 0b - DMA Channel 2 Interrupt Status not detected 1b - DMA Channel 2 Interrupt Status detected
1 DC1IS	DMA Channel 1 Interrupt Status This bit indicates an interrupt event in DMA Channel 1. To reset this bit to 1'b0, the software must read the corresponding register in DMA Channel 1 to get the exact cause of the interrupt and clear its source. 0b - DMA Channel 1 Interrupt Status not detected 1b - DMA Channel 1 Interrupt Status detected
0 DC0IS	DMA Channel 0 Interrupt Status This bit indicates an interrupt event in DMA Channel 0. To reset this bit to 1'b0, the software must read the corresponding register in DMA Channel 0 to get the exact cause of the interrupt and clear its source. 0b - DMA Channel 0 Interrupt Status not detected 1b - DMA Channel 0 Interrupt Status detected

1.4.320 (DMA_Debug_Status0)

Offset

Register	Offset
DMA_Debug_Status0	100Ch

Function

The Debug Status 0 register gives the Receive and Transmit process status for DMA Channel 0-Channel 2 for debugging purpose.

Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	TPS2				RPS2				TPS1				RPS1			
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	TPS0				RPS0				Reserved						AXRH STS	AXWH STS
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Fields

Field	Function
31-28 TPS2	DMA Channel 2 Transmit Process State This field indicates the Tx DMA FSM state for Channel 2. The MSB of this field always returns 0. This field does not generate an interrupt. 0000b - Stopped (Reset or Stop Transmit Command issued) 0001b - Running (Fetching Tx Transfer Descriptor) 0010b - Running (Waiting for status) 0011b - Running (Reading Data from system memory buffer and queuing it to the Tx buffer (Tx FIFO)) 0100b - Timestamp write state 0101b - Reserved for future use 0110b - Suspended (Tx Descriptor Unavailable or Tx Buffer Underflow) 0111b - Running (Closing Tx Descriptor)
27-24 RPS2	DMA Channel 2 Receive Process State This field indicates the Rx DMA FSM state for Channel 2.

Table continues on the next page...

Table continued from the previous page...

Field	Function
	The MSB of this field always returns 0. This field does not generate an interrupt. 0000b - Stopped (Reset or Stop Receive Command issued) 0001b - Running (Fetching Rx Transfer Descriptor) 0010b - Reserved for future use 0011b - Running (Waiting for Rx packet) 0100b - Suspended (Rx Descriptor Unavailable) 0101b - Running (Closing the Rx Descriptor) 0110b - Timestamp write state 0111b - Running (Transferring the received packet data from the Rx buffer to the system memory)
23-20 TPS1	DMA Channel 1 Transmit Process State This field indicates the Tx DMA FSM state for Channel 1. The MSB of this field always returns 0. This field does not generate an interrupt. 0000b - Stopped (Reset or Stop Transmit Command issued) 0001b - Running (Fetching Tx Transfer Descriptor) 0010b - Running (Waiting for status) 0011b - Running (Reading Data from system memory buffer and queuing it to the Tx buffer (Tx FIFO)) 0100b - Timestamp write state 0101b - Reserved for future use 0110b - Suspended (Tx Descriptor Unavailable or Tx Buffer Underflow) 0111b - Running (Closing Tx Descriptor)
19-16 RPS1	DMA Channel 1 Receive Process State This field indicates the Rx DMA FSM state for Channel 1. The MSB of this field always returns 0. This field does not generate an interrupt. 0000b - Stopped (Reset or Stop Receive Command issued) 0001b - Running (Fetching Rx Transfer Descriptor) 0010b - Reserved for future use 0011b - Running (Waiting for Rx packet) 0100b - Suspended (Rx Descriptor Unavailable) 0101b - Running (Closing the Rx Descriptor) 0110b - Timestamp write state 0111b - Running (Transferring the received packet data from the Rx buffer to the system memory)
15-12	DMA Channel 0 Transmit Process State

Table continues on the next page...

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Field	Function
TPS0	This field indicates the Tx DMA FSM state for Channel 0. The MSB of this field always returns 0. This field does not generate an interrupt. 0000b - Stopped (Reset or Stop Transmit Command issued) 0001b - Running (Fetching Tx Transfer Descriptor) 0010b - Running (Waiting for status) 0011b - Running (Reading Data from system memory buffer and queuing it to the Tx buffer (Tx FIFO)) 0100b - Timestamp write state 0101b - Reserved for future use 0110b - Suspended (Tx Descriptor Unavailable or Tx Buffer Underflow) 0111b - Running (Closing Tx Descriptor)
11-8 RPS0	DMA Channel 0 Receive Process State This field indicates the Rx DMA FSM state for Channel 0. The MSB of this field always returns 0. This field does not generate an interrupt. 0000b - Stopped (Reset or Stop Receive Command issued) 0001b - Running (Fetching Rx Transfer Descriptor) 0010b - Reserved for future use 0011b - Running (Waiting for Rx packet) 0100b - Suspended (Rx Descriptor Unavailable) 0101b - Running (Closing the Rx Descriptor) 0110b - Timestamp write state 0111b - Running (Transferring the received packet data from the Rx buffer to the system memory)
7-2 —	Reserved.
1 AXRHSTS	AXI Master Read Channel Status When high, this bit indicates that the read channel of the AXI master is active, and it is transferring the data. 0b - AXI Master Read Channel Status not detected 1b - AXI Master Read Channel Status detected
0 AXWHSTS	AXI Master Write Channel When high, this bit indicates that the write channel of the AXI master is active, and it is transferring data. 0b - AXI Master Write Channel or AHB Master Status not detected 1b - AXI Master Write Channel or AHB Master Status detected

1.4.321 (DMA_Debug_Status1)

Offset

Register	Offset
DMA_Debug_Status1	1010h

Function

The Debug Status1 register gives the Receive and Transmit process status for DMA Channel 3-Channel 6.

Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	Reserved				Reserved				Reserved				Reserved			
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	TPS4				RPS4				TPS3				RPS3			
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Fields

Field	Function
31-28 —	Reserved.
27-24 —	Reserved.
23-20 —	Reserved.
19-16 —	Reserved.
15-12 TPS4	DMA Channel 4 Transmit Process State This field indicates the Tx DMA FSM state for Channel 4. The MSB of this field always returns 0. This field does not generate an interrupt. 0000b - Stopped (Reset or Stop Transmit Command issued) 0001b - Running (Fetching Tx Transfer Descriptor)

Table continues on the next page...

Table continued from the previous page...

Field	Function
	0010b - Running (Waiting for status) 0011b - Running (Reading Data from system memory buffer and queuing it to the Tx buffer (Tx FIFO)) 0100b - Timestamp write state 0101b - Reserved for future use 0110b - Suspended (Tx Descriptor Unavailable or Tx Buffer Underflow) 0111b - Running (Closing Tx Descriptor)
11-8 RPS4	DMA Channel 4 Receive Process State This field indicates the Rx DMA FSM state for Channel 4. The MSB of this field always returns 0. This field does not generate an interrupt. 0000b - Stopped (Reset or Stop Receive Command issued) 0001b - Running (Fetching Rx Transfer Descriptor) 0010b - Reserved for future use 0011b - Running (Waiting for Rx packet) 0100b - Suspended (Rx Descriptor Unavailable) 0101b - Running (Closing the Rx Descriptor) 0110b - Timestamp write state 0111b - Running (Transferring the received packet data from the Rx buffer to the system memory)
7-4 TPS3	DMA Channel 3 Transmit Process State This field indicates the Tx DMA FSM state for Channel 3. The MSB of this field always returns 0. This field does not generate an interrupt. 0000b - Stopped (Reset or Stop Transmit Command issued) 0001b - Running (Fetching Tx Transfer Descriptor) 0010b - Running (Waiting for status) 0011b - Running (Reading Data from system memory buffer and queuing it to the Tx buffer (Tx FIFO)) 0100b - Timestamp write state 0101b - Reserved for future use 0110b - Suspended (Tx Descriptor Unavailable or Tx Buffer Underflow) 0111b - Running (Closing Tx Descriptor)
3-0 RPS3	DMA Channel 3 Receive Process State This field indicates the Rx DMA FSM state for Channel 3. The MSB of this field always returns 0. This field does not generate an interrupt.

Table continues on the next page...

Table continued from the previous page...

Field	Function
	0000b - Stopped (Reset or Stop Receive Command issued)
	0001b - Running (Fetching Rx Transfer Descriptor)
	0010b - Reserved for future use
	0011b - Running (Waiting for Rx packet)
	0100b - Suspended (Rx Descriptor Unavailable)
	0101b - Running (Closing the Rx Descriptor)
	0110b - Timestamp write state
	0111b - Running (Transferring the received packet data from the Rx buffer to the system memory)

1.4.322 (AXI4_Tx_AR_ACE_Control)

Offset

Register	Offset
AXI4_Tx_AR_ACE_Control	1020h

Function

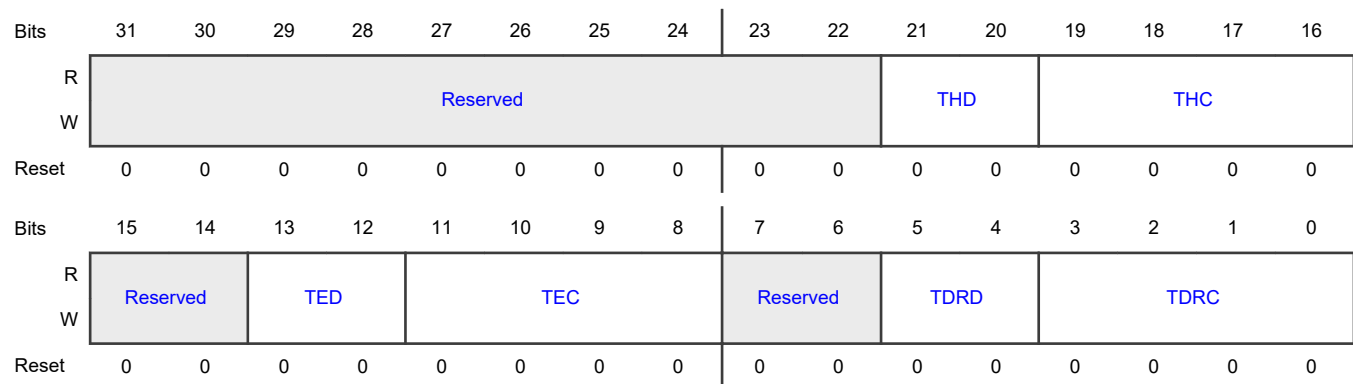
This register is used to control the AXI4 Cache Coherency Signals for read transactions by all the Transmit DMA channels. The following signals of the AXI4 interface are driven with different values as programmed

for corresponding type (descriptor, buffer1, buffer2) of access.

- arcache_m_o[3:0]

- ardomain_m_o[1:0]

Diagram



Fields

Field	Function
31-22 —	Reserved.
21-20 THD	Transmit DMA First Packet Buffer This field is used to drive ardomain_o[1:0] signal when Transmit DMA is accessing First Buffer of the Packet (First valid buffer with FD being set in the TDES3 of the Descriptor).
19-16 THC	Transmit DMA First Packet Buffer This field is used to drive arcache_o[3:0] signal when Transmit DMA is accessing First Buffer of the Packet (First valid buffer with FD being set in the TDES3 of the Descriptor)..
15-14 —	Reserved.
13-12 TED	Transmit DMA Extended Packet Buffer This field is used to drive ardomain_o[1:0] signal when Transmit DMA is accessing the extended buffers (when packet is distributed across multiple buffers).
11-8 TEC	Transmit DMA Extended Packet Buffer This field is used to drive arcache_o[3:0] signal when Transmit DMA is accessing the extended buffers (when packet is distributed across multiple buffers).
7-6 —	Reserved.
5-4 TDRD	Transmit DMA Read Descriptor Domain Control This field is used to drive ardomain_o[1:0] signal when Transmit DMA engines access the Descriptor.
3-0 TDRC	Transmit DMA Read Descriptor Cache Control This field is used to drive arcache_o[3:0] signal when Transmit DMA engines access the Descriptor.

1.4.323 (AXI4_Rx_AW_ACE_Control)**Offset**

Register	Offset
AXI4_Rx_AW_ACE_Control	1024h

Function

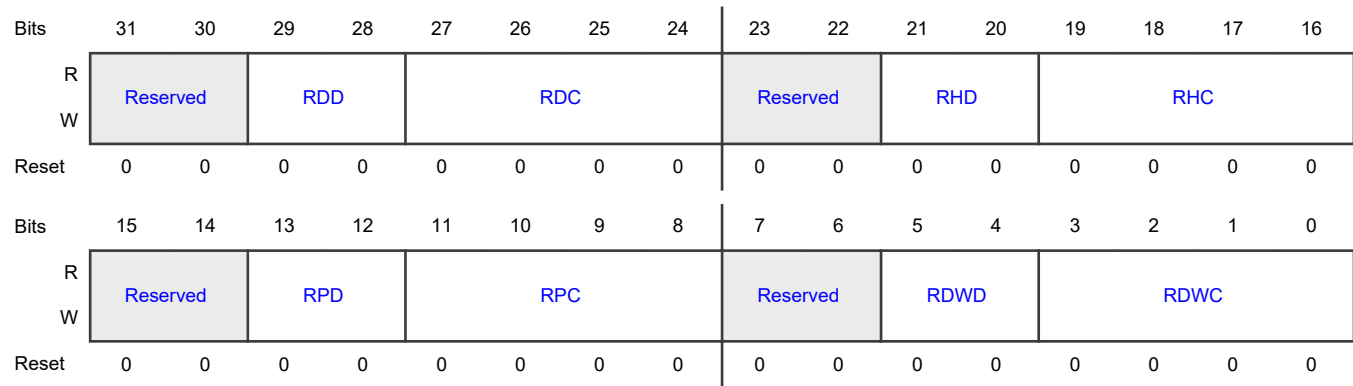
This register is used to control the AXI4 Cache Coherency Signals for write transactions by all the Receive DMA channels. The following signals of the AXI4 interface are driven with different values as programmed

for corresponding type (descriptor, buffer1, buffer2) of access.

- awcache_m_o[3:0]

- awdomain_m_o[1:0]

Diagram



Fields

Field	Function
31-30 —	Reserved.
29-28 RDD	Receive DMA Buffer Domain Control This field is used to drive the awdomain_o[1:0] signal when Receive DMA is accessing the Buffer when Header and payload are NOT separated.
27-24 RDC	Receive DMA Buffer Cache Control This field is used to drive awcache_o[3:0] signal when Receive DMA is accessing the Buffer when Header and payload are NOT separated.
23-22 —	Reserved.
21-20 RHD	Receive DMA Header Domain Control This field is used to drive awdomain_o[1:0] and signal when Receive DMA is accessing the header Buffer when Header and payload are separated.
19-16 RHC	Receive DMA Header Cache Control This field is used to drive awcache_o[3:0] and signal when Receive DMA is accessing the header Buffer when Header and payload are separated.
15-14 —	Reserved.
13-12 RPD	Receive DMA Payload Domain Control

Table continues on the next page...

Table continued from the previous page...

Field	Function
	This field is used to drive awdomain_o[1:0] signal when Receive DMA is accessing the Payload Buffer when Header and payload are separated.
11-8 RPC	Receive DMA Payload Cache Control This field is used to drive awcache_o[3:0] signal when Receive DMA is accessing the Payload Buffer when Header and payload are separated.
7-6 —	Reserved.
5-4 RDWD	Receive DMA Write Descriptor Domain Control This field is used to drive awdomain_o[1:0] signal when Receive DMA accesses the Descriptor.
3-0 RDWC	Receive DMA Write Descriptor Cache Control This field is used to drive awcache_o[3:0] signal when Receive DMA accesses the Descriptor.

1.4.324 (AXI4_TxRx_AWAR_ACE_Control)

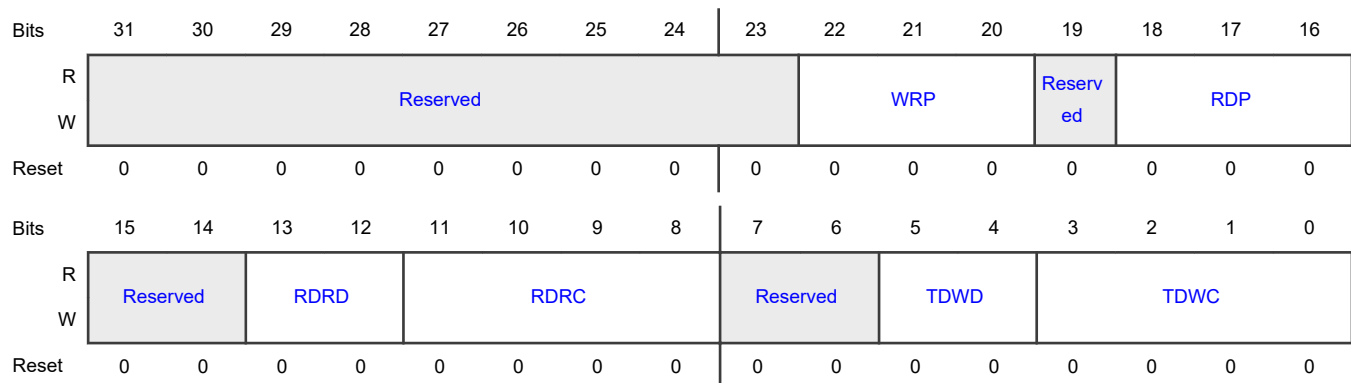
Offset

Register	Offset
AXI4_TxRx_AWAR_ACE _Control	1028h

Function

This register is used to control the AXI4 Cache Coherency Signals for Descriptor write transactions by all the TxDMA channels and Descriptor read transactions by all the RxDMA channels. It also controls the values to be driven on awprot_m_o and aprot_m_o.

Diagram



Fields

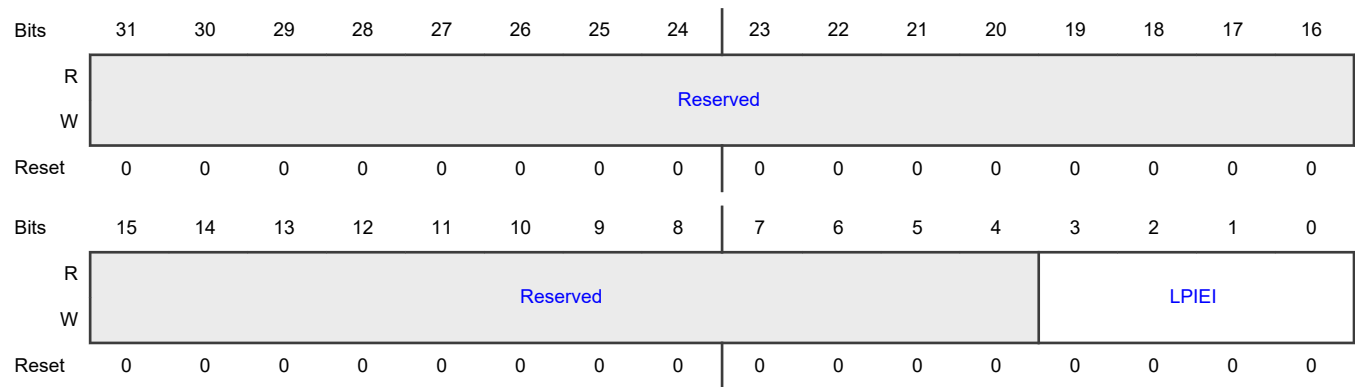
Field	Function
31-23 —	Reserved.
22-20 WRP	DMA Write Protection control This field is used to drive awprot_m_o[2:0] signal on the AXI Write Channel.
19 —	Reserved.
18-16 RDP	DMA Read Protection control This field is used to drive arprot_m_o[2:0] signal during all read requests.
15-14 —	Reserved.
13-12 RDRD	Receive DMA Read Descriptor Domain control This field is used to drive ardomain_o[1:0] signal when Receive DMA engines read the Descriptor.
11-8 RDRC	Receive DMA Read Descriptor Cache control This field is used to drive arcache_o[3:0] signal when Receive DMA engines read the Descriptor.
7-6 —	Reserved.
5-4 TDWD	Transmit DMA Write Descriptor Domain control This field is used to drive awdomain_o[1:0] signal when Transmit DMA write to the Descriptor.
3-0 TDWC	Transmit DMA Write Descriptor Cache control This field is used to drive awcache_o[3:0] signal when Transmit DMA writes to the Descriptor.

1.4.325 (AXI_LPI_Entry_Interval)**Offset**

Register	Offset
AXI_LPI_Entry_Interval	1040h

Function

This register is used to control the AXI LPI entry interval.

Diagram**Fields**

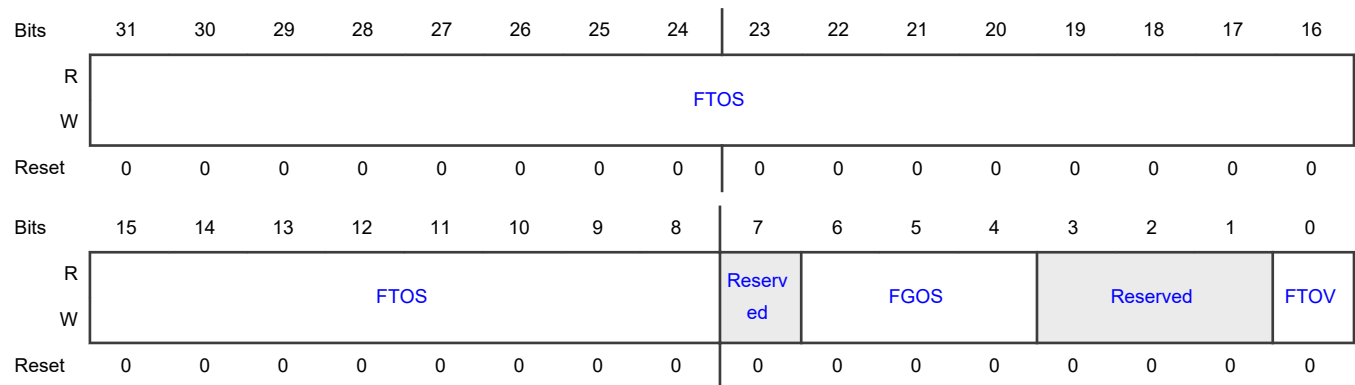
Field	Function
31-4 —	Reserved.
3-0 LPIEI	LPI Entry Interval Contains the number of system clock cycles, multiplied by 64, to wait for an activity in the DWC_ether_qos to enter into the AXI low power state 0 indicates 64 clock cycles

1.4.326 (DMA_TBS_CTRL)**Offset**

Register	Offset
DMA_TBS_CTRL	1050h

Function

This register is used to control the TBS attributes.

Diagram**Fields**

Field	Function
31-8 FTOS	Fetch Time Offset The value in units of 256 nanoseconds, that has to be deducted from the Launch time to compute the Fetch Time. Max value: 999,999,999 ns, additionally should be smaller than CTR-1 value when ESTM mode is set since this value is a modulo CTR value.
7 —	Reserved.
6-4 FGOS	Fetch GSN Offset The number GSN slots that must be deducted from the Launch GSN to compute the Fetch GSN. Value valid only when FTOV is set.
3-1 —	Reserved.
0 FTOV	Fetch Time Offset Valid When set indicates the FTOS field is valid. When not set, indicates the Fetch Offset is not valid and the DMA engine can fetch the frames from host memory without any time restrictions. 0b - Fetch Time Offset is invalid 1b - Fetch Time Offset is valid

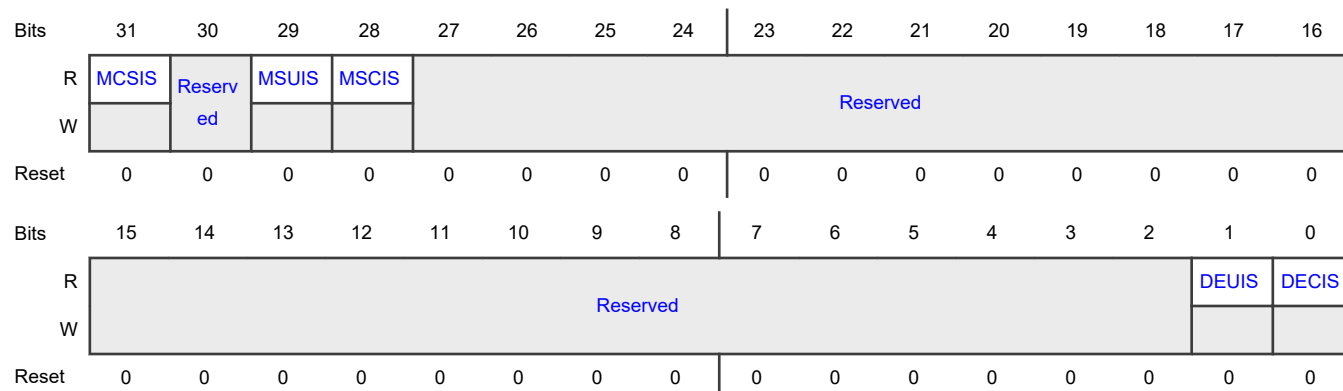
1.4.327 (DMA_Safety_Interrupt_Status)**Offset**

Register	Offset
DMA_Safety_Interrupt_Status	1080h

Function

This register indicates summary (whether error occurred in DMA/MTL/MAC and correctable/uncorrectable) of the Automotive Safety related error interrupts.

Diagram



Fields

Field	Function
31 MCSIS	MAC Safety Uncorrectable Interrupt Status Indicates a uncorrectable Safety related Interrupt is set in the MAC module. MAC_DPP_FSM_Interrupt_Status register should be read when this bit is set, to get the cause of the Safety Interrupt in MAC. 0b - MAC Safety Uncorrectable Interrupt Status not detected 1b - MAC Safety Uncorrectable Interrupt Status detected
30 —	Reserved.
29 MSUIS	MTL Safety Uncorrectable error Interrupt Status This bit indicates an uncorrectable error interrupt event in MTL. To get exact cause of the interrupt the software should read the MTL_Safety_Interrupt_Status register. 0b - MTL Safety Uncorrectable error Interrupt Status not detected 1b - MTL Safety Uncorrectable error Interrupt Status detected
28 MSCIS	MTL Safety Correctable error Interrupt Status This bit indicates a correctable error interrupt event in MTL. To get exact cause of the interrupt the software should read the MTL_Safety_Interrupt_Status register. 0b - MTL Safety Correctable error Interrupt Status not detected 1b - MTL Safety Correctable error Interrupt Status detected
27-2 —	Reserved.

Table continues on the next page...

Table continued from the previous page...

Field	Function
1 DEUIS	DMA ECC Uncorrectable error Interrupt Status This bit indicates an interrupt event in the DMA ECC safety feature. To get the exact cause of the interrupt the application should read the DMA_ECC_Interrupt_Status register. 0b - DMA ECC Uncorrectable error Interrupt Status not detected 1b - DMA ECC Uncorrectable error Interrupt Status detected
0 DECIS	DMA ECC Correctable error Interrupt Status This bit indicates an interrupt event in the DMA ECC safety feature. To get the exact cause of the interrupt the application should read the DMA_ECC_Interrupt_Status register. 0b - DMA ECC Correctable error Interrupt Status not detected 1b - DMA ECC Correctable error Interrupt Status detected

1.4.328 (DMA_CH0_Control)

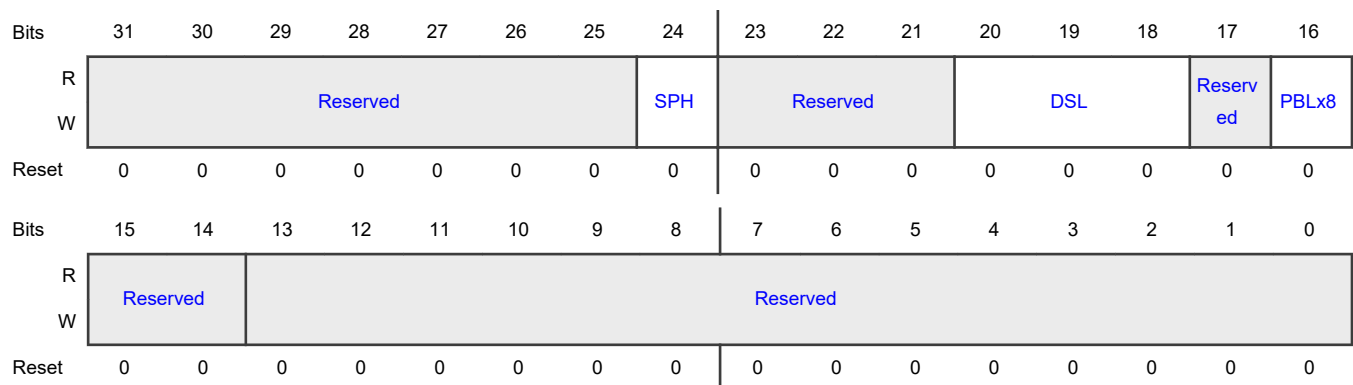
Offset

Register	Offset
DMA_CH0_Control	1100h

Function

The DMA Channel Control register specifies the MSS value for segmentation, length to skip between two descriptors, and also the features such as header splitting and 8xPBL mode.

Diagram



Fields

Field	Function
31-25 —	Reserved.
24 SPH	Split Headers When this bit is set, the DMA splits the header and payload in the Receive path. The DMA writes the header to the Buffer Address1 of RDES0. The DMA writes the payload to the buffer to which the Buffer Address2 is pointing. The software must ensure that the header fits into the Receive buffers. If the header length exceeds the receive buffer size, the DMA does not split the header and payload. This bit is available only if Enable Split Header Structure option is selected. 0b - Split Headers feature is disabled 1b - Split Headers feature is enabled
23-21 —	Reserved.
20-18 DSL	Descriptor Skip Length This bit specifies the Word, Dword, or Lword number (depending on the 32-bit, 64-bit, or 128-bit bus) to skip between two unchained descriptors. The address skipping starts from the end of the current descriptor to the start of the next descriptor. When the DSL value is equal to zero, the DMA takes the descriptor table as contiguous.
17 —	Reserved.
16 PBLx8	8xPBL mode When this bit is set, the PBL value programmed in Bits[21:16] in DMA_CH(#i)_Tx_Control and Bits[21:16] in DMA_CH(#i)_Rx_Control is multiplied by eight times. Therefore, the DMA transfers the data in 8, 16, 32, 64, 128, and 256 beats depending on the PBL value. 0b - 8xPBL mode is disabled 1b - 8xPBL mode is enabled
15-14 —	Reserved.
13-0 —	Reserved.

1.4.329 (DMA_CH0_Tx_Control)

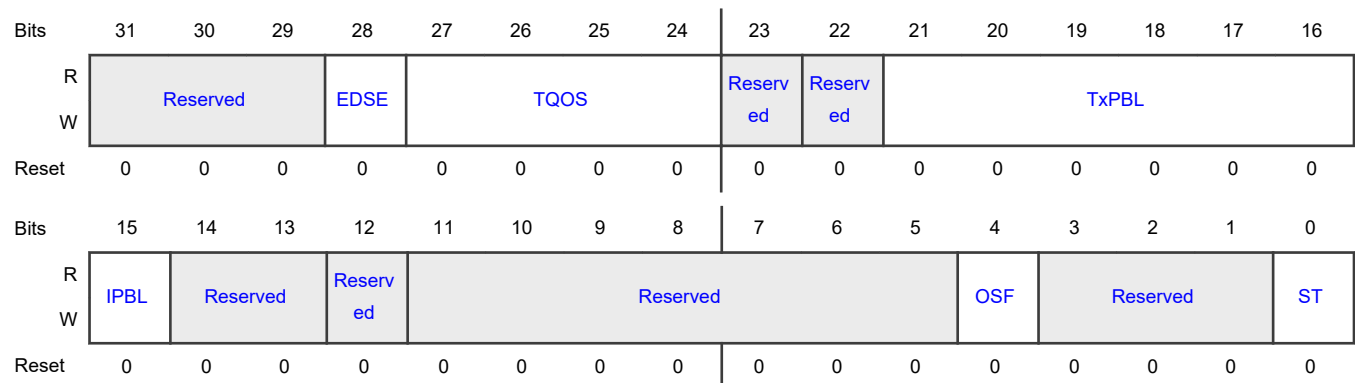
Offset

Register	Offset
DMA_CH0_Tx_Control	1104h

Function

The DMA Channel Transmit Control register controls the Tx features such as PBL, TCP segmentation, and Tx Channel weights.

Diagram



Fields

Field	Function
31-29 —	Reserved.
28 EDSE	Enhanced Descriptor Enable When this bit is set, the corresponding channel uses Enhanced Descriptors that are 32 Bytes for both Normal and Context Descriptors. When reset, the corresponding channel uses the descriptors that are 16 Bytes. 0b - Enhanced Descriptor is disabled 1b - Enhanced Descriptor is enabled
27-24 TQOS	Transmit QOS. This field is used to drive arqos_m_o[3:0] or awqos_m_o[3:0] output signals for all transactions of DMA Tx Channel0.
23 —	Reserved.
22	Reserved.

Table continues on the next page...

Table continued from the previous page...

Field	Function
—	
21-16 TxPBL	Transmit Programmable Burst Length These bits indicate the maximum number of beats to be transferred in one DMA block data transfer. The DMA always attempts max burst as specified in PBL each time it starts a burst transfer on the application bus. You can program PBL with any of the following values: 1, 2, 4, 8, 16, or 32. Any other value results in undefined behavior. To transfer more than 32 beats, perform the following steps: 1. Set the 8xPBL mode in DMA_CH0_Control register. 2. Set the TxPBL. Note: The maximum value of TxPBL must be less than or equal to half the Tx Queue size (TQS field of MTL_TxQ(#!)_Operation_Mode register) in terms of beats. This is required so that the Tx Queue has space to store at least another Tx PBL worth of data while the MTL Tx Queue Controller is transferring data to MAC. For example, in 64-bit data width configurations the total locations in Tx Queue of size 512 bytes is 64, TxPBL and 8xPBL needs to be programmed to less than or equal to 32.
15 IPBL	Ignore PBL Requirement When this bit is set, the DMA does not check for PBL number of locations in the MTL before initiating a transfer. If space is not available, the MTL might use handshaking to slow the DMA. Note: This bit/mode must not be used when multiple Transmit DMA Channels are enabled as it might block other Transmit and Receive DMA Channels from accessing the Read Data Channel of AXI bus until space is available in Transmit Queue for current transfer. 0b - Ignore PBL Requirement is disabled 1b - Ignore PBL Requirement is enabled
14-13 —	Reserved.
12 —	Reserved.
11-5 —	Reserved.
4 OSF	Operate on Second Packet When this bit is set, it instructs the DMA to process the second packet of the Transmit data even before the status for the first packet is obtained. 0b - Operate on Second Packet disabled 1b - Operate on Second Packet enabled
3-1 —	Reserved.

Table continues on the next page...

Table continued from the previous page...

Field	Function
0	Start or Stop Transmission Command
ST	When this bit is set, transmission is placed in the Running state. The DMA checks the Transmit list at the current position for a packet to be transmitted. The DMA tries to acquire descriptor from either of the following positions: - The current position in the list This is the base address of the Transmit list set by the DMA_CH0_TxDesc_List_Address register. - The position at which the transmission was previously stopped If the DMA does not own the current descriptor, the transmission enters the Suspended state and the TBU bit of the DMA_CH0_Status register is set. The Start Transmission command is effective only when the transmission is stopped. If the command is issued before setting the DMA_CH0_TxDesc_List_Address register, the DMA behavior is unpredictable. When this bit is reset, the transmission process is placed in the Stopped state after completing the transmission of the current packet. The Next Descriptor position in the Transmit list is saved, and it becomes the current position when the transmission is restarted. To change the list address, you need to program DMA_CH0_TxDesc_List_Address register with a new value when this bit is reset. The new value is considered when this bit is set again. The stop transmission command is effective only when the transmission of the current packet is complete or the transmission is in the Suspended state. 0b - Stop Transmission Command 1b - Start Transmission Command

1.4.330 (DMA_CH0_Rx_Control)

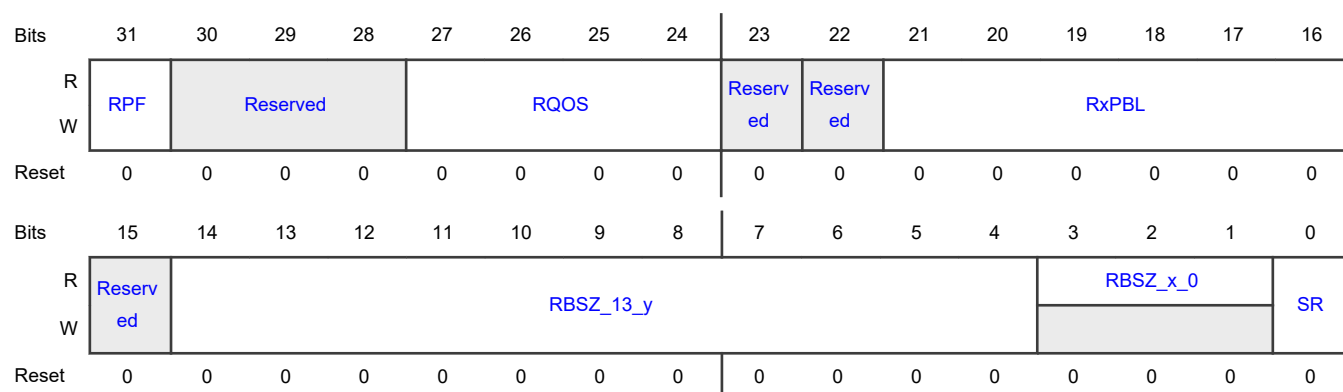
Offset

Register	Offset
DMA_CH0_Rx_Control	1108h

Function

The DMA Channel0 Receive Control register controls the Rx features such as PBL, buffer size, and extended status.

Diagram



Fields

Field	Function
31 RPF	Rx Packet Flush. When this bit is set to 1, then <code>DWC_ether_qos</code> automatically flushes the packet from the Rx Queues destined to this DMA Rx Channel, when it is stopped. When this bit remains set and the DMA is re-started by the software driver, the packets residing in the Rx Queues that were received when this RxDMA was stopped, get flushed out. The packets that are received by the MAC after the RxDMA is re-started are routed to the RxDMA. The flushing happens on the Read side of the Rx Queue. When this bit is set to 0, the <code>DWC_ether_qos</code> not flush the packet in the Rx Queue destined to this RxDMA Channel when it is STOP state. This might in turn cause head-of-line blocking in the corresponding RxQueue. Note: The stopping of packet flow from a Rx DMA Channel to the application by setting RPF works only when there is one-to-one mapping of Rx Queue to Rx DMA channels. In Dynamic mapping mode, setting RPF bit in any <code>DMA_CH(#i)_Rx_Control</code> register might flush packets from unintended Rx Queues which are destined to the stopped Rx DMA Channel. 0b - Rx Packet Flush is disabled 1b - Rx Packet Flush is enabled
30-28 —	Reserved.
27-24 RQOS	Rx AXI4 QOS. This field is used to drive <code>arqos_m_o[3:0]</code> or <code>awqos_m_o[3:0]</code> output signals for all transactions of DMA Rx Channel0.
23 —	Reserved.
22 —	Reserved.
21-16	Receive Programmable Burst Length

Table continues on the next page...

Table continued from the previous page...

Field	Function
RxPBL	These bits indicate the maximum number of beats to be transferred in one DMA block data transfer. The DMA always attempts max burst as specified in PBL each time it starts a burst transfer on the application bus. You can program PBL with any of the following values: 1, 2, 4, 8, 16, or 32. Any other value results in undefined behavior. To transfer more than 32 beats, perform the following steps: 1. Set the 8xPBL mode in the DMA_CH0_Control register. 2. Set the RxPBL. Note: The maximum value of RxPBL must be less than or equal to half the Rx Queue size (RQS field of MTL_RxQ(##)_Operation_Mode register) in terms of beats. This is required so that the Rx Queue has space to store at least another Rx PBL worth of data while the Rx DMA is transferring a block of data. For example, in 64-bit data width configurations the total locations in Rx Queue of size 512 bytes is 64, so RxPBL and 8xPBL needs to be programmed to less than or equal to 32.
15 —	Reserved.
14-4 RBSZ_13_y	Receive Buffer size High RBSZ[13:0] is split into two fields higher RBSZ_13_y and lower RBSZ_x_0. The RBSZ[13:0] field indicates the size of the Rx buffers specified in bytes. The maximum buffer size is limited to 16K bytes. The buffer size is applicable to payload buffers when split headers are enabled. Note: The buffer size must be a multiple of 4, 8, or 16 depending on the data bus widths (32-bit, 64-bit, or 128-bit respectively). This is required even if the value of buffer address pointer is not aligned to data bus width. Hence the lower RBSZ_x_0 bits are read-only and the value is considered as all-zero. Thus the RBSZ_13_y indicates the buffer size in terms of locations (with the width same as bus-width).
3-1 RBSZ_x_0	Receive Buffer size Low RBSZ[13:0] is split into two fields RBSZ_13_y and RBSZ_x_0. The RBSZ_x_0 is the lower field whose width is based on data bus width of the configuration. This field is of width 2, 3, or 4 bits for 32-bit, 64-bit, or 128-bit data bus width respectively. This field is read-only (RO).
0 SR	Start or Stop Receive When this bit is set, the DMA tries to acquire the descriptor from the Receive list and processes the incoming packets. The DMA tries to acquire descriptor from either of the following positions: - The current position in the list This is the address set by the DMA_CH0_RxDesc_List_Address register. - The position at which the Rx process was previously stopped If the DMA does not own the current descriptor, the reception is suspended and the RBU bit of the DMA_CH0_Status register is set. The Start Receive command is effective only when the reception is stopped. If the command is issued before setting the DMA_CH0_RxDesc_List_Address register, the DMA behavior is unpredictable.

Table continues on the next page...

Table continued from the previous page...

Field	Function
	When this bit is reset, the Rx DMA operation is stopped after the transfer of the current packet. The next descriptor position in the Receive list is saved, and it becomes the current position after the Rx process is restarted. The Stop Receive command is effective only when the Rx process is in the Running (waiting for Rx packet) or Suspended state.
	0b - Stop Receive
	1b - Start Receive

1.4.331 (DMA_CH0_TxDesc_List_Address)

Offset

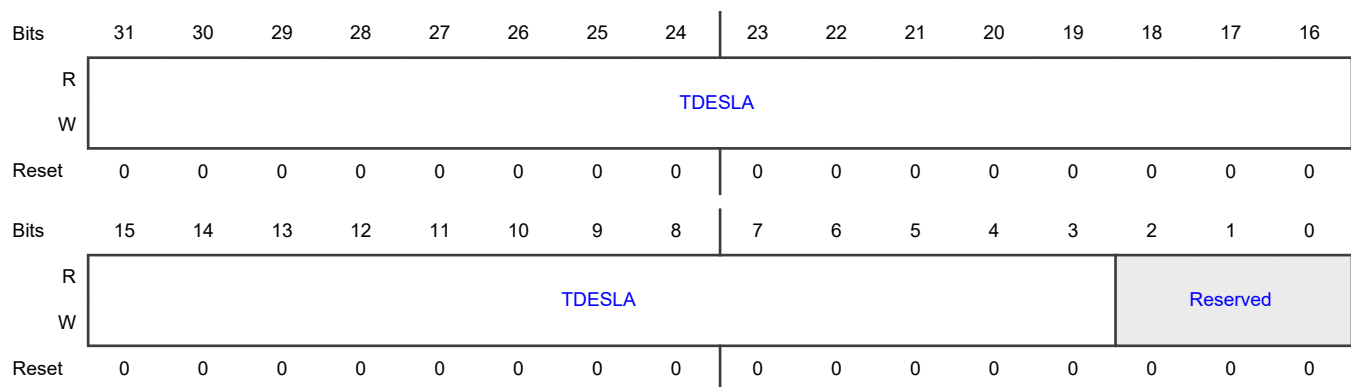
Register	Offset
DMA_CH0_TxDesc_List_Address	1114h

Function

The Channeli Tx Descriptor List Address register points the DMA to the start of Transmit descriptor list. The descriptor lists reside in the physical memory space of the application and must be Word, Dword, or Lword-aligned (for 32-bit, 64-bit, or 128-bit data bus). The DMA internally converts it to bus width aligned address by making the corresponding LSB to low.

You can write to this register only when the Tx DMA has stopped, that is, the ST bit is set to zero in DMA_CH0_Tx_Control register. When stopped, this register can be written with a new descriptor list address. When you set the ST bit to 1, the DMA takes the newly-programmed descriptor base address. If this register is not changed when the ST bit is set to 0, the DMA takes the descriptor address where it was stopped earlier.

Diagram



Fields

Field	Function
31-3	Start of Transmit List

Table continues on the next page...

Table continued from the previous page...

Field	Function
TDESLA	This field contains the base address of the first descriptor in the Transmit descriptor list. The DMA ignores the LSB bits (1:0, 2:0, or 3:0) for 32-bit, 64-bit, or 128-bit bus width and internally takes these bits as all-zero. Therefore, these LSB bits are read-only (RO). The width of this field depends on the configuration: - 31:2 for 32-bit configuration - 31:3 for 64-bit configuration - 31:4 for 128-bit configuration
2-0 —	Reserved.

1.4.332 (DMA_CH0_RxDesc_List_Address)

Offset

Register	Offset
DMA_CH0_RxDesc_List_Address	111Ch

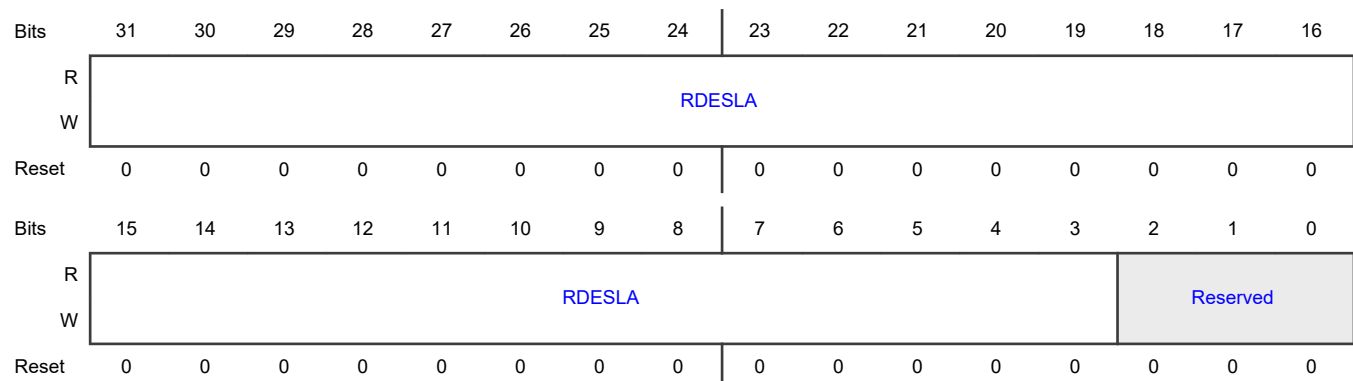
Function

The Channel Rx Descriptor List Address register points the DMA to the start of Receive descriptor list.

This register points to the start of the Receive Descriptor List. The descriptor lists reside in the physical memory space of the application and must be Word, Dword, or Lword-aligned (for 32-bit, 64-bit, or 128-bit data bus). The DMA internally converts it to bus width aligned address by making the corresponding LS bits low. Writing to this register is permitted only when reception is stopped. When stopped, this register must be written to before the receive Start command is given. You can write to this register only when Rx DMA has stopped, that is, SR bit is set to zero in DMA_CH0_Rx_Control register. When stopped, this register can be written with a new descriptor list address.

When you set the SR bit to 1, the DMA takes the newly programmed descriptor base address.

Diagram



Fields

Field	Function
31-3 RDESLA	Start of Receive List This field contains the base address of the first descriptor in the Rx Descriptor list. The DMA ignores the LSB bits (1:0, 2:0, or 3:0) for 32-bit, 64-bit, or 128-bit bus width and internally takes these bits as all-zero. Therefore, these LSB bits are read-only (RO). The width of this field depends on the configuration: - 31:2 for 32-bit configuration - 31:3 for 64-bit configuration - 31:4 for 128-bit configuration
2-0 —	Reserved.

1.4.333 (DMA_CH0_TxDesc_Tail_Pointer)

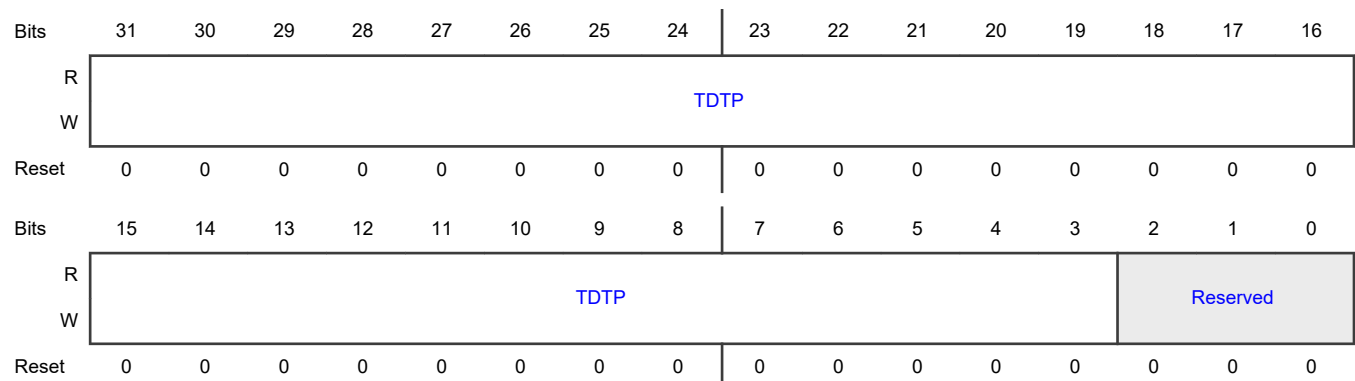
Offset

Register	Offset
DMA_CH0_TxDesc_Tail_Pointer	1120h

Function

The Channel_i Tx Descriptor Tail Pointer register points to an offset from the base and indicates the location of the last valid descriptor.

Diagram



Fields

Field	Function
31-3 TDTP	Transmit Descriptor Tail Pointer This field contains the tail pointer for the Tx descriptor ring. The software writes the tail pointer to add more descriptors to the Tx channel. The hardware tries to transmit all packets referenced by the descriptors between the head and the tail pointer registers. The width of this field depends on the configuration: - 31:2 for 32-bit configuration - 31:3 for 64-bit configuration - 31:4 for 128-bit configuration
2-0 —	Reserved.

1.4.334 (DMA_CH0_RxDesc_Tail_Pointer)

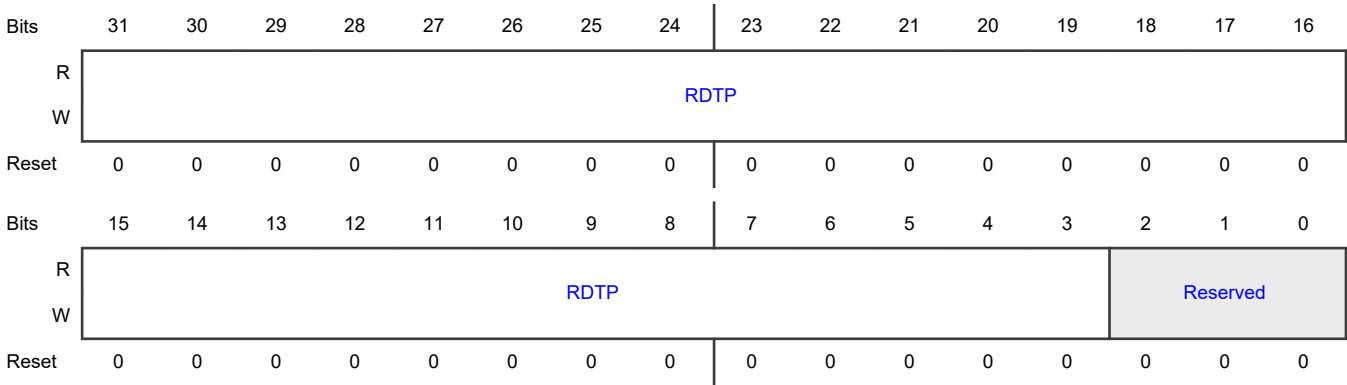
Offset

Register	Offset
DMA_CH0_RxDesc_Tail_Pointer	1128h

Function

The Channeli Rx Descriptor Tail Pointer Points to an offset from the base and indicates the location of the last valid descriptor.

Diagram



Fields

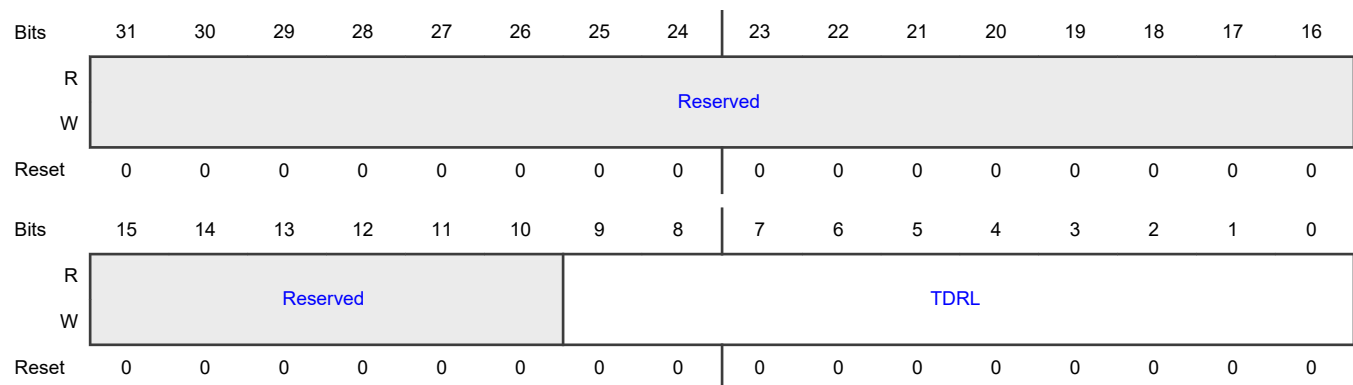
Field	Function
31-3 RDTP	Receive Descriptor Tail Pointer This field contains the tail pointer for the Rx descriptor ring. The software writes the tail pointer to add more descriptors to the Rx channel. The hardware tries to write all received packets to the descriptors referenced between the head and the tail pointer registers. The width of this field depends on the configuration: - 31:2 for 32-bit configuration - 31:3 for 64-bit configuration - 31:4 for 128-bit configuration
2-0 —	Reserved.

1.4.335 (DMA_CH0_TxDesc_Ring_Length)**Offset**

Register	Offset
DMA_CH0_TxDesc_Ring_Length	112Ch

Function

The Tx Descriptor Ring Length register contains the length of the Transmit descriptor ring.

Diagram

Fields

Field	Function
31-10 —	Reserved.
9-0 TDRL	Transmit Descriptor Ring Length This field sets the maximum number of Tx descriptors in the circular descriptor ring. The maximum number of descriptors is limited to 1K descriptors. NXP recommends a minimum ring descriptor length of 4. For example, You can program any value up to 0x3FF in this field. This field is 10 bits wide, if you program 0x3FF, you can have 1024 descriptors. If you want to have 10 descriptors, program it to a value of 0x9.

1.4.336 (DMA_CH0_RxDesc_Ring_Length)

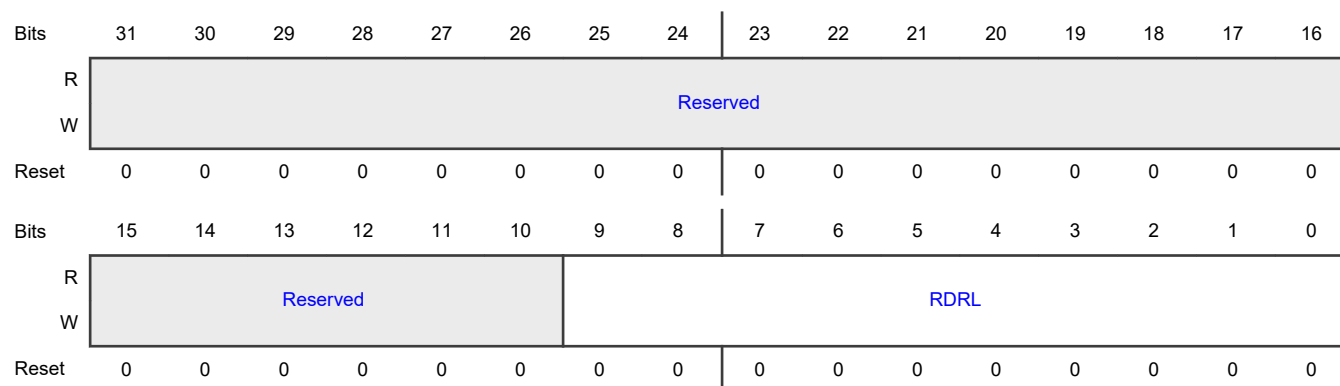
Offset

Register	Offset
DMA_CH0_RxDesc_Ring_Length	1130h

Function

The Channel0 Rx Descriptor Ring Length register contains the length of the Receive descriptor circular ring.

Diagram



Fields

Field	Function
31-10 —	Reserved.
9-0	Receive Descriptor Ring Length

Table continues on the next page...

Table continued from the previous page...

Field	Function
RDRL	This register sets the maximum number of Rx descriptors in the circular descriptor ring. The maximum number of descriptors is limited to 1K descriptors. For example, You can program any value up to 0x3FF in this field. This field is 10 bits wide, if you program 0x3FF, you can have 1024 descriptors. If you want to have 10 descriptors, program it to a value of 0x9.

1.4.337 (DMA_CH0_Interrupt_Enable)

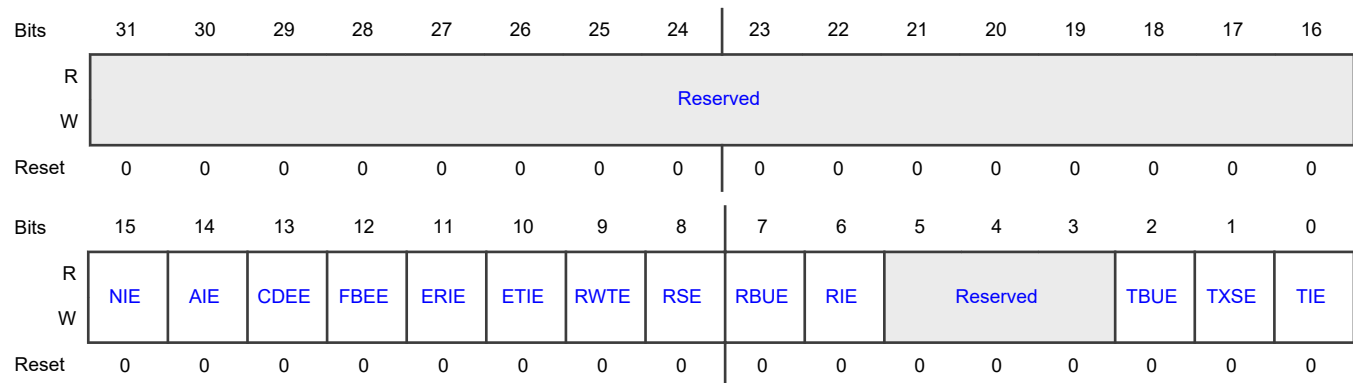
Offset

Register	Offset
DMA_CH0_Interrupt_Enable	1134h

Function

The Channeli Interrupt Enable register enables the interrupts reported by the Status register.

Diagram



Fields

Field	Function
31-16 —	Reserved.
15 NIE	Normal Interrupt Summary Enable When this bit is set, the normal interrupt summary is enabled. This bit enables the following interrupts in the DMA_CH0_Status register: - Bit 0: Transmit Interrupt - Bit 2: Transmit Buffer Unavailable

Table continues on the next page...

Table continued from the previous page...

Field	Function
	- Bit 6: Receive Interrupt - Bit 11: Early Receive Interrupt When this bit is reset, the normal interrupt summary is disabled. 0b - Normal Interrupt Summary is disabled 1b - Normal Interrupt Summary is enabled
14 AIE	Abnormal Interrupt Summary Enable When this bit is set, the abnormal interrupt summary is enabled. This bit enables the following interrupts in the DMA_CH0_Status register: - Bit 1: Transmit Process Stopped - Bit 7: Rx Buffer Unavailable - Bit 8: Receive Process Stopped - Bit 9: Receive Watchdog Timeout - Bit 10: Early Transmit Interrupt - Bit 12: Fatal Bus Error - Bit 13: Context Descriptor Error When this bit is reset, the abnormal interrupt summary is disabled. 0b - Abnormal Interrupt Summary is disabled 1b - Abnormal Interrupt Summary is enabled
13 CDEE	Context Descriptor Error Enable When this bit is set along with the AIE bit, the Descriptor error interrupt is enabled. When this bit is reset, the Descriptor error interrupt is disabled. 0b - Context Descriptor Error is disabled 1b - Context Descriptor Error is enabled
12 FBEE	Fatal Bus Error Enable When this bit is set along with the AIE bit, the Fatal Bus error interrupt is enabled. When this bit is reset, the Fatal Bus Error error interrupt is disabled. 0b - Fatal Bus Error is disabled 1b - Fatal Bus Error is enabled
11 ERIE	Early Receive Interrupt Enable When this bit is set along with the NIE bit, the Early Receive interrupt is enabled. When this bit is reset, the Early Receive interrupt is disabled. 0b - Early Receive Interrupt is disabled 1b - Early Receive Interrupt is enabled

Table continues on the next page...

Table continued from the previous page...

Field	Function
10 ETIE	Early Transmit Interrupt Enable When this bit is set along with the AIE bit, the Early Transmit interrupt is enabled. When this bit is reset, the Early Transmit interrupt is disabled. 0b - Early Transmit Interrupt is disabled 1b - Early Transmit Interrupt is enabled
9 RWTE	Receive Watchdog Timeout Enable When this bit is set along with the AIE bit, the Receive Watchdog Timeout interrupt is enabled. When this bit is reset, the Receive Watchdog Timeout interrupt is disabled. 0b - Receive Watchdog Timeout is disabled 1b - Receive Watchdog Timeout is enabled
8 RSE	Receive Stopped Enable When this bit is set along with the AIE bit, the Receive Stopped Interrupt is enabled. When this bit is reset, the Receive Stopped interrupt is disabled. 0b - Receive Stopped is disabled 1b - Receive Stopped is enabled
7 RBUE	Receive Buffer Unavailable Enable When this bit is set along with the AIE bit, the Receive Buffer Unavailable interrupt is enabled. When this bit is reset, the Receive Buffer Unavailable interrupt is disabled. 0b - Receive Buffer Unavailable is disabled 1b - Receive Buffer Unavailable is enabled
6 RIE	Receive Interrupt Enable When this bit is set along with the NIE bit, the Receive Interrupt is enabled. When this bit is reset, the Receive Interrupt is disabled. 0b - Receive Interrupt is disabled 1b - Receive Interrupt is enabled
5-3 —	Reserved.
2 TBUE	Transmit Buffer Unavailable Enable When this bit is set along with the NIE bit, the Transmit Buffer Unavailable interrupt is enabled. When this bit is reset, the Transmit Buffer Unavailable interrupt is disabled. 0b - Transmit Buffer Unavailable is disabled 1b - Transmit Buffer Unavailable is enabled
1	Transmit Stopped Enable

Table continues on the next page...

Table continued from the previous page...

Field	Function
TXSE	When this bit is set along with the AIE bit, the Transmission Stopped interrupt is enabled. When this bit is reset, the Transmission Stopped interrupt is disabled. 0b - Transmit Stopped is disabled 1b - Transmit Stopped is enabled
0 TIE	Transmit Interrupt Enable When this bit is set along with the NIE bit, the Transmit Interrupt is enabled. When this bit is reset, the Transmit Interrupt is disabled. 0b - Transmit Interrupt is disabled 1b - Transmit Interrupt is enabled

1.4.338 (DMA_CH0_Rx_Interrupt_Watchdog_Timer)

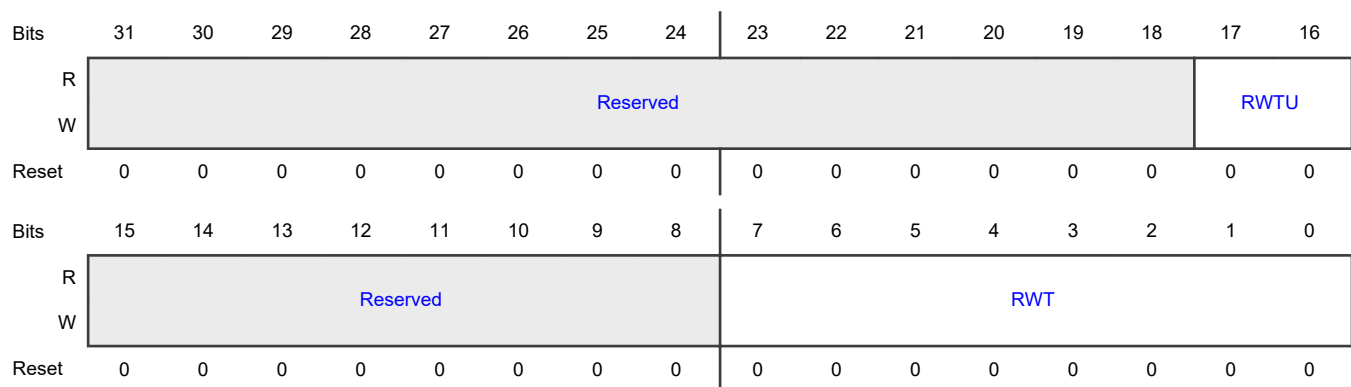
Offset

Register	Offset
DMA_CH0_Rx_Interrupt_Watchdog_Timer	1138h

Function

The Receive Interrupt Watchdog Timer register indicates the watchdog timeout for Receive Interrupt (RI) from the DMA. When this register is written with a non-zero value, it enables the watchdog timer for the RI bit of the DMA_CHi_Status register.

Diagram



Fields

Field	Function
31-18	Reserved.

Table continues on the next page...

Table continued from the previous page...

Field	Function
—	
17-16 RWTU	Receive Interrupt Watchdog Timer Count Units This fields indicates the number of system clock cycles corresponding to one unit in RWT field. - 2'b00: 256 - 2'b01: 512 - 2'b10: 1024 - 2'b11: 2048 For example, when RWT=2 and RWTU=1, the watchdog timer is set for 2*512=1024 system clock cycles.
15-8 —	Reserved.
7-0 RWT	Receive Interrupt Watchdog Timer Count This field indicates the number of system clock cycles, multiplied by factor indicated in RWTU field, for which the watchdog timer is set. The watchdog timer is triggered with the programmed value after the Rx DMA completes the transfer of a packet for which the RI bit is not set in the DMA_CH(#i)_Status register, because of the setting of Interrupt Enable bit in the corresponding descriptor RDES3[30]. When the watchdog timer runs out, the RI bit is set and the timer is stopped. The watchdog timer is reset when the RI bit is set high because of automatic setting of RI as per the Interrupt Enable bit RDES3[30] of any received packet.

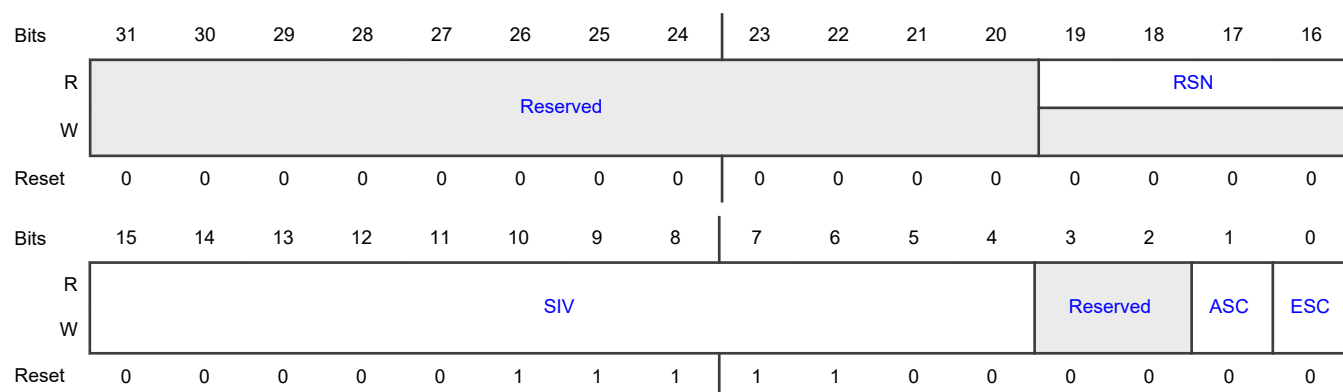
1.4.339 (DMA_CH0_Slot_Function_Control_Status)

Offset

Register	Offset
DMA_CH0_Slot_Function_Control_Status	113Ch

Function

The Slot Function Control and Status register contains the control bits for slot function and the status for Transmit path.

Diagram**Fields**

Field	Function
31-20 —	Reserved.
19-16 RSN	Reference Slot Number This field gives the current value of the reference slot number in the DMA. It is used for slot comparison.
15-4 SIV	Slot Interval Value This field controls the period of the slot interval in which the TxDMA fetches the scheduled packets. A value of 0 specifies the slot interval of 1 us while the maximum value 4095 specifies the slot interval of 4096us. The default/reset value is 0x07C which corresponds to slot interval of 125us
3-2 —	Reserved.
1 ASC	Advance Slot Check When set, this bit enables the DMA to fetch the data from the buffer when the slot number (SLOTNUM) programmed in the Tx descriptor is - equal to the reference slot number given in the RSN field or - ahead of the reference slot number by up to two slots This bit is applicable only when the ESC bit is set. 0b - Advance Slot Check is disabled 1b - Advance Slot Check is enabled
0 ESC	Enable Slot Comparison When set, this bit enables the checking of the slot numbers programmed in the Tx descriptor with the current reference given in the RSN field. The DMA fetches the data from the corresponding buffer only when the slot number is

Table continues on the next page...

Table continued from the previous page...

Field	Function
	- equal to the reference slot number or - ahead of the reference slot number by one slot When reset, this bit disables the checking of the slot numbers. The DMA fetches the data immediately after the descriptor is processed. Note: The UFO (UDP Fragmentation over IPv4)/TSO/USO should not be enabled along with TBS/AVB Slot number check. The UFO/TSO/USO involves multiple packets/segments/fragments transmission for single packet received from application and the slot number check are applicable for fetching of only first segment/fragment. As a result it might be difficult for software to specify slot number for subsequent packets. 0b - Slot Comparison is disabled 1b - Slot Comparison is enabled

1.4.340 (DMA_CH0_Current_App_TxDesc)

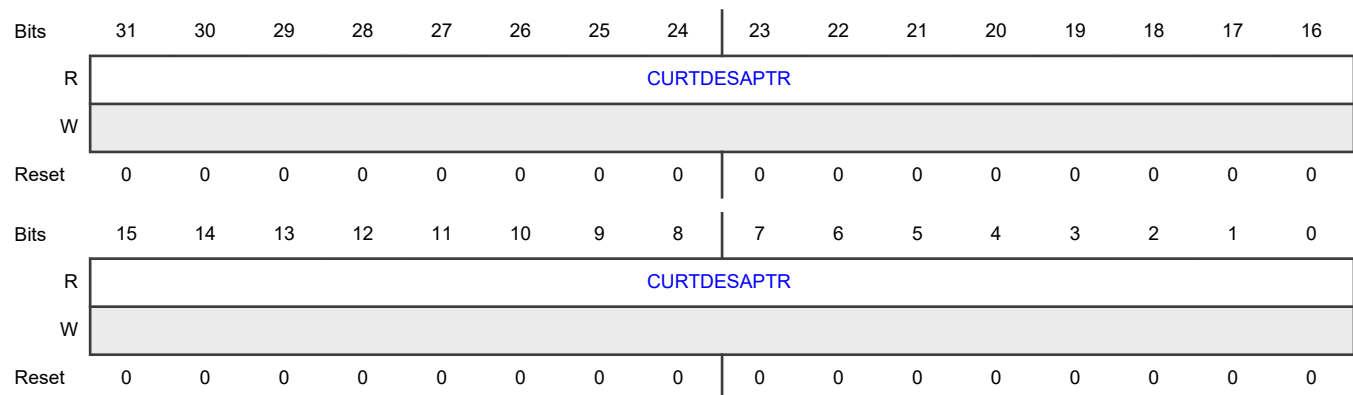
Offset

Register	Offset
DMA_CH0_Current_App_TxDesc	1144h

Function

The Channeli Current Application Transmit Descriptor register points to the current Transmit descriptor read by the DMA.

Diagram



Fields

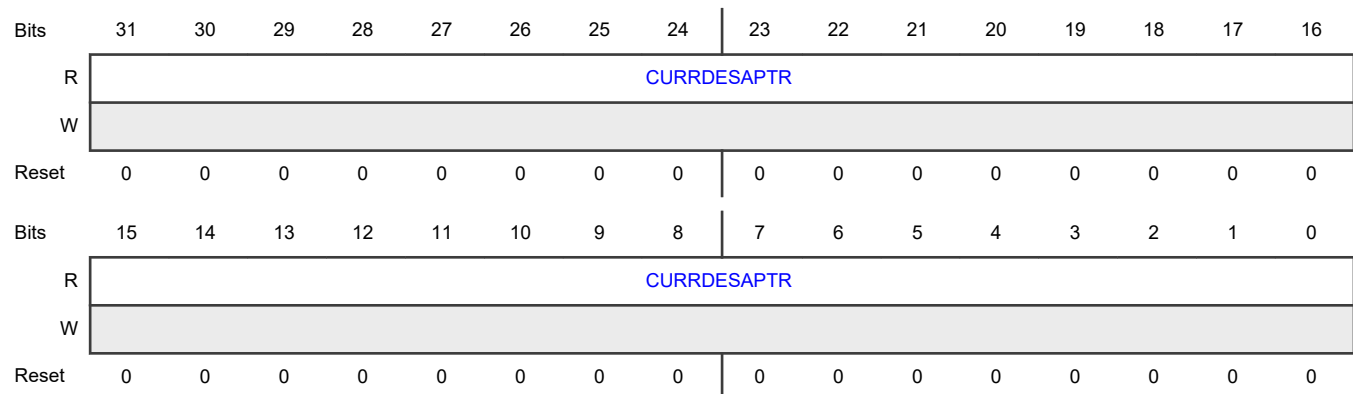
Field	Function
31-0 CURDESAPTR	Application Transmit Descriptor Address Pointer The DMA updates this pointer during Tx operation. This pointer is cleared on reset.

1.4.341 (DMA_CH0_Current_App_RxDesc)**Offset**

Register	Offset
DMA_CH0_Current_App_RxDesc	114Ch

Function

The Channeli Current Application Receive Descriptor register points to the current Receive descriptor read by the DMA.

Diagram**Fields**

Field	Function
31-0 CURRDESAPTR	Application Receive Descriptor Address Pointer The DMA updates this pointer during Rx operation. This pointer is cleared on reset.

1.4.342 (DMA_CH0_Current_App_TxBuffer)

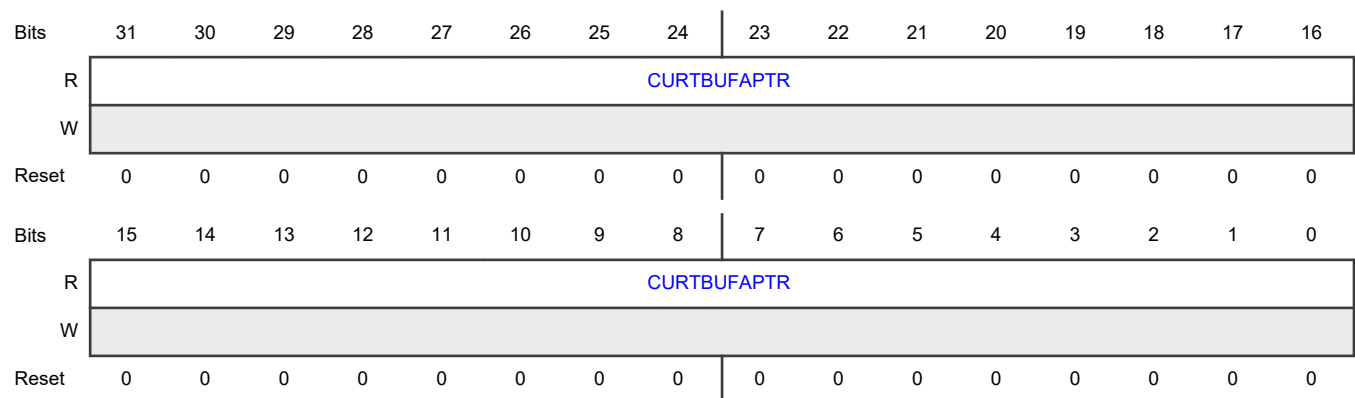
Offset

Register	Offset
DMA_CH0_Current_App_TxBuffer	1154h

Function

The Channel0 Current Application Transmit Buffer Address register points to the current Tx buffer address read by the DMA.

Diagram



Fields

Field	Function
31-0	Application Transmit Buffer Address Pointer
CURTBUFAPT R	The DMA updates this pointer during Tx operation. This pointer is cleared on reset.

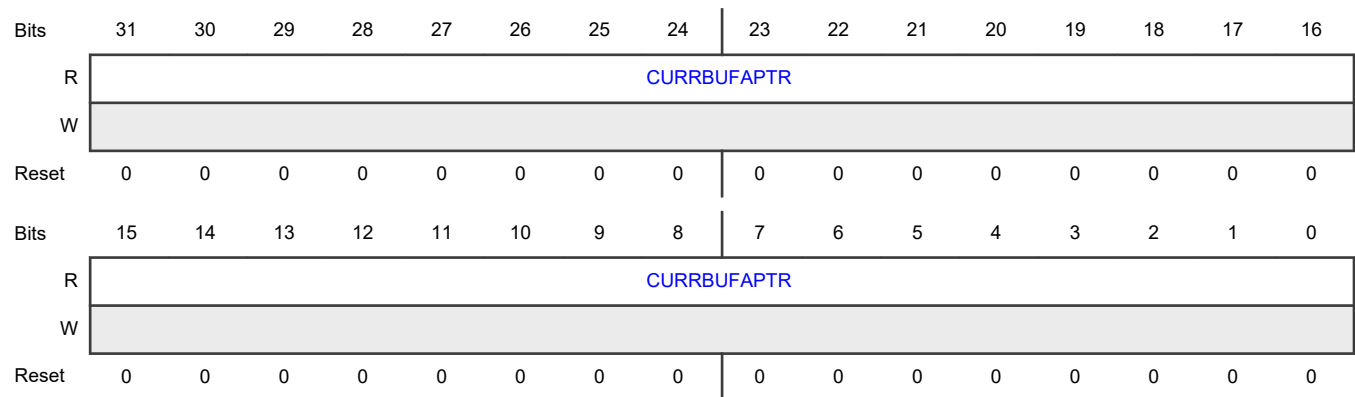
1.4.343 (DMA_CH0_Current_App_RxBuffer)

Offset

Register	Offset
DMA_CH0_Current_App_RxBuffer	115Ch

Function

The Channel 0 Current Application Receive Buffer Address register points to the current Rx buffer address read by the DMA.

Diagram**Fields**

Field	Function
31-0 CURRBUFAPT R	Application Receive Buffer Address Pointer The DMA updates this pointer during Rx operation. This pointer is cleared on reset.

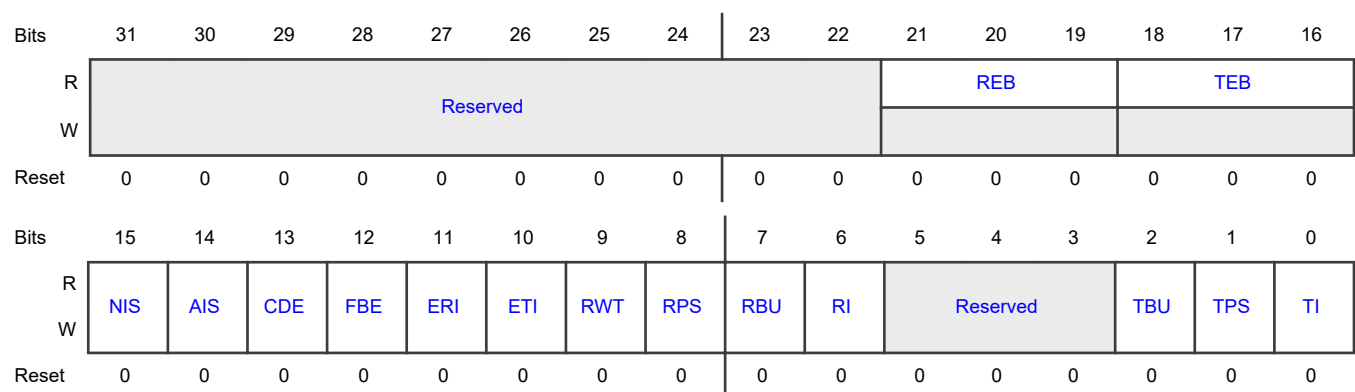
1.4.344 (DMA_CH0_Status)**Offset**

Register	Offset
DMA_CH0_Status	1160h

Function

The software driver (application) reads the Status register during interrupt service routine or polling to determine the status of the DMA.

Note: The number of DMA_CH(##)_Status register in the configuration is the higher of number of Rx DMA Channels and Tx DMA Channels.

Diagram

Fields

Field	Function
31-22 —	Reserved.
21-19 REB	Rx DMA Error Bits This field indicates the type of error that caused a Bus Error. For example, error response on the AHB or AXI interface. - Bit 21 -- 1'b1: Error during data transfer by Rx DMA -- 1'b0: No Error during data transfer by Rx DMA - Bit 20 -- 1'b1: Error during descriptor access -- 1'b0: Error during data buffer access - Bit 19 -- 1'b1: Error during read transfer -- 1'b0: Error during write transfer This field is valid only when the FBE bit is set. This field does not generate an interrupt.
18-16 TEB	Tx DMA Error Bits This field indicates the type of error that caused a Bus Error. For example, error response on the AHB or AXI interface. - Bit 18 -- 1'b1: Error during data transfer by Tx DMA -- 1'b0: No Error during data transfer by Tx DMA - Bit 17 -- 1'b1: Error during descriptor access -- 1'b0: Error during data buffer access - Bit 16 -- 1'b1: Error during read transfer -- 1'b0: Error during write transfer This field is valid only when the FBE bit is set. This field does not generate an interrupt.
15 NIS	Normal Interrupt Summary Normal Interrupt Summary bit value is the logical OR of the following bits when the corresponding interrupt bits are enabled in the DMA_CH0_Interrupt_Enable register: - Bit 0: Transmit Interrupt - Bit 2: Transmit Buffer Unavailable - Bit 6: Receive Interrupt

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Field	Function
	- Bit 11: Early Receive Interrupt Only unmasked bits (interrupts for which interrupt enable is set in DMA_CH0_Interrupt_Enable register) affect the Normal Interrupt Summary bit. This is a sticky bit. You must clear this bit (by writing 1 to this bit) each time a corresponding bit which causes NIS to be set is cleared. Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect. 0b - Normal Interrupt Summary status not detected 1b - Normal Interrupt Summary status detected
14 AIS	Abnormal Interrupt Summary Abnormal Interrupt Summary bit value is the logical OR of the following when the corresponding interrupt bits are enabled in the DMA_CH0_Interrupt_Enable register: - Bit 1: Transmit Process Stopped - Bit 7: Receive Buffer Unavailable - Bit 8: Receive Process Stopped - Bit 10: Early Transmit Interrupt - Bit 12: Fatal Bus Error - Bit 13: Context Descriptor Error Only unmasked bits affect the Abnormal Interrupt Summary bit. This is a sticky bit. You must clear this bit (by writing 1 to this bit) each time a corresponding bit, which causes AIS to be set, is cleared. Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect. 0b - Abnormal Interrupt Summary status not detected 1b - Abnormal Interrupt Summary status detected
13 CDE	Context Descriptor Error This bit indicates that the DMA Tx/Rx engine received a descriptor error, which indicates invalid context in the middle of packet flow (intermediate descriptor) or all one's descriptor in Tx case and on Rx side it indicates DMA has read a descriptor with either of the buffer address as ones which is considered to be invalid. Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect. 0b - Context Descriptor Error status not detected 1b - Context Descriptor Error status detected
12 FBE	Fatal Bus Error This bit indicates that a bus error occurred (as described in the EB field). When this bit is set, the corresponding DMA channel engine disables all bus accesses. Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect.

Table continues on the next page...

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Field	Function
	0b - Fatal Bus Error status not detected 1b - Fatal Bus Error status detected
11 ERI	Early Receive Interrupt This bit when set indicates that the RxDMA has completed the transfer of packet data to the memory. In configs supporting ERIC, When ERIC=0, this bit is set only after the Rx DMA has filled up a complete receive buffer with packet data. When ERIC=1, this bit is set after every burst transfer of data from the Rx DMA to the buffer. The setting of RI bit automatically clears this bit. Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect. 0b - Early Receive Interrupt status not detected 1b - Early Receive Interrupt status detected
10 ETI	Early Transmit Interrupt This bit when set indicates that the TxDMA has completed the transfer of packet data to the MTL TXFIFO memory. In configs supporting ERIC: When ETIC=0, this bit is set only after the Tx DMA has transferred a complete packet to MTL. When ETIC=1, this bit is set after completion of (partial) packet data transfer from buffers in the Transmit descriptor in which IOC=1. Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect. 0b - Early Transmit Interrupt status not detected 1b - Early Transmit Interrupt status detected
9 RWT	Receive Watchdog Timeout This bit is asserted when a packet with length greater than 2,048 bytes (10,240 bytes when Jumbo Packet mode is enabled) is received. 0b - Receive Watchdog Timeout status not detected 1b - Receive Watchdog Timeout status detected
8 RPS	Receive Process Stopped This bit is asserted when the Rx process enters the Stopped state. Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect. 0b - Receive Process Stopped status not detected 1b - Receive Process Stopped status detected
7 RBU	Receive Buffer Unavailable This bit indicates that the application owns the next descriptor in the Receive list, and the DMA cannot acquire it. The Rx process is suspended. To resume processing Rx descriptors, the application should change the ownership of the descriptor and issue a Receive Poll Demand command. If this command is

Table continues on the next page...

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Field	Function
	not issued, the Rx process resumes when the next recognized incoming packet is received. In ring mode, the application should advance the Receive Descriptor Tail Pointer register of a channel. This bit is set only when the DMA owns the previous Rx descriptor. Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect. 0b - Receive Buffer Unavailable status not detected 1b - Receive Buffer Unavailable status detected
6 RI	Receive Interrupt This bit indicates that the packet reception is complete. When packet reception is complete, Bit 31 of RDES3 is reset in the last descriptor, and the specific packet status information is updated in the descriptor. The reception remains in the Running state. Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect. 0b - Receive Interrupt status not detected 1b - Receive Interrupt status detected
5-3 —	Reserved.
2 TBU	Transmit Buffer Unavailable This bit indicates that the application owns the next descriptor in the Transmit list, and the DMA cannot acquire it. Transmission is suspended. The TPS0 field of the DMA_Debug_Status0 register explains the Transmit Process state transitions. To resume processing the Transmit descriptors, the application should do the following: 1. Change the ownership of the descriptor by setting Bit 31 of TDES3. 2. Issue a Transmit Poll Demand command. For ring mode, the application should advance the Transmit Descriptor Tail Pointer register of a channel. Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect. 0b - Transmit Buffer Unavailable status not detected 1b - Transmit Buffer Unavailable status detected
1 TPS	Transmit Process Stopped This bit is set when the transmission is stopped. Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect. 0b - Transmit Process Stopped status not detected 1b - Transmit Process Stopped status detected
0 TI	Transmit Interrupt This bit indicates that the packet transmission is complete. When transmission is complete, Bit 31 of TDES3 is reset in the last descriptor, and the specific packet status information is updated in the descriptor.

Table continues on the next page...

Table continued from the previous page...

Field	Function
	Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect. 0b - Transmit Interrupt status not detected 1b - Transmit Interrupt status detected

1.4.345 (DMA_CH0_Miss_Frame_Cnt)

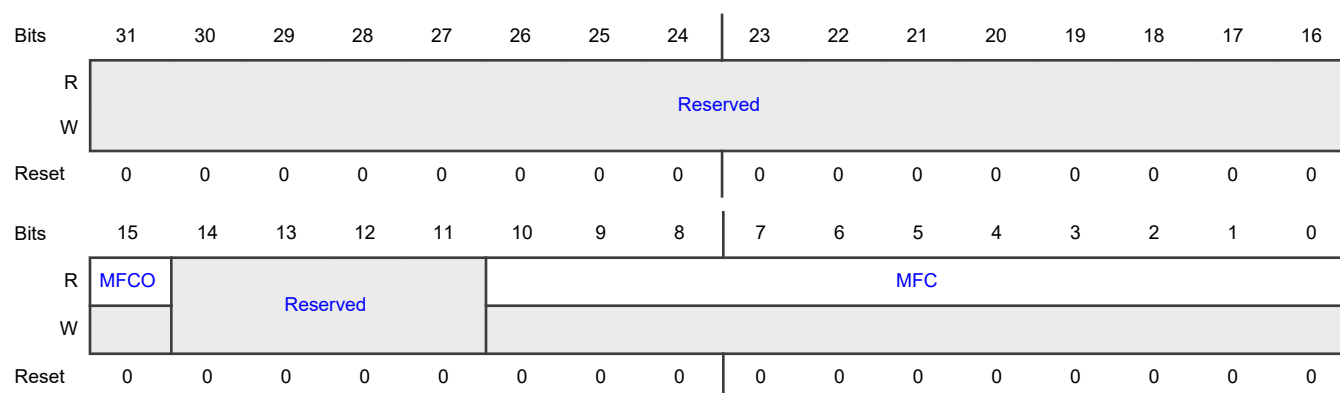
Offset

Register	Offset
DMA_CH0_Miss_Frame_Cnt	1164h

Function

This register has the number of packet counter that got dropped by the DMA either due to Bus Error or due to programming RPF field in DMA_CH*{i}*_Rx_Control register.

Diagram



Fields

Field	Function
31-16 —	Reserved.
15 MFCO	Overflow status of the MFC Counter When this bit is set then the MFC counter does not get incremented further. The bit gets cleared when this register is read. Access restriction applies. Clears on read. Self-set to 1 on internal event.

Table continues on the next page...

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Field	Function
	0b - Miss Frame Counter overflow not occurred 1b - Miss Frame Counter overflow occurred
14-11 —	Reserved.
10-0 MFC	Dropped Packet Counters This counter indicates the number of packet counters that are dropped by the DMA either because of bus error or because of programing RPF field in DMA_CH\${i}_Rx_Control register. The counter gets cleared when this register is read. Access restriction applies. Clears on read. Self-set to 1 on internal event.

1.4.346 (DMA_CH0_RXP_Accept_Cnt)

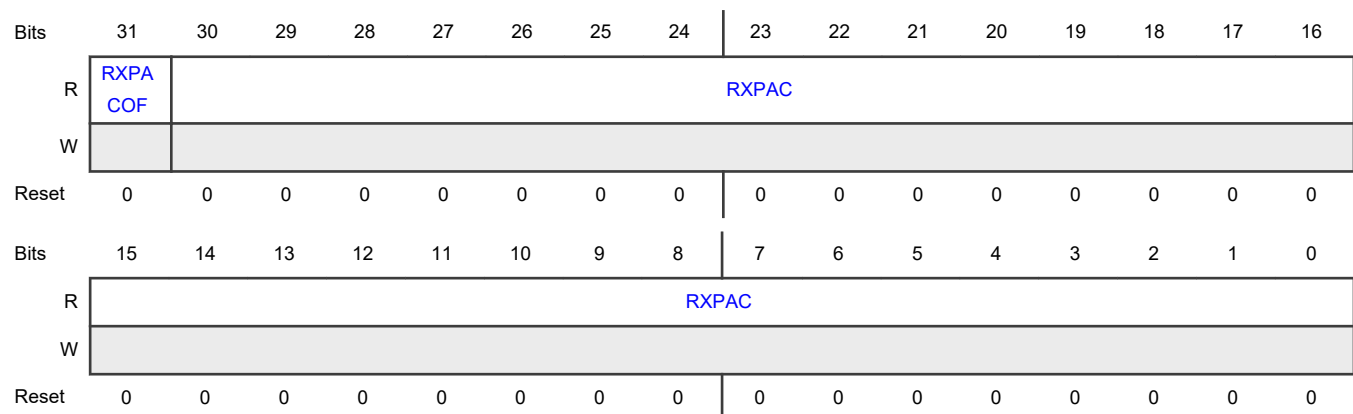
Offset

Register	Offset
DMA_CH0_RXP_Accept_Cnt	1168h

Function

The DMA_CH(#i)_RXP_Accept_Cnt registers provides the count of the number of frames accepted by Rx Parser.

Diagram



Fields

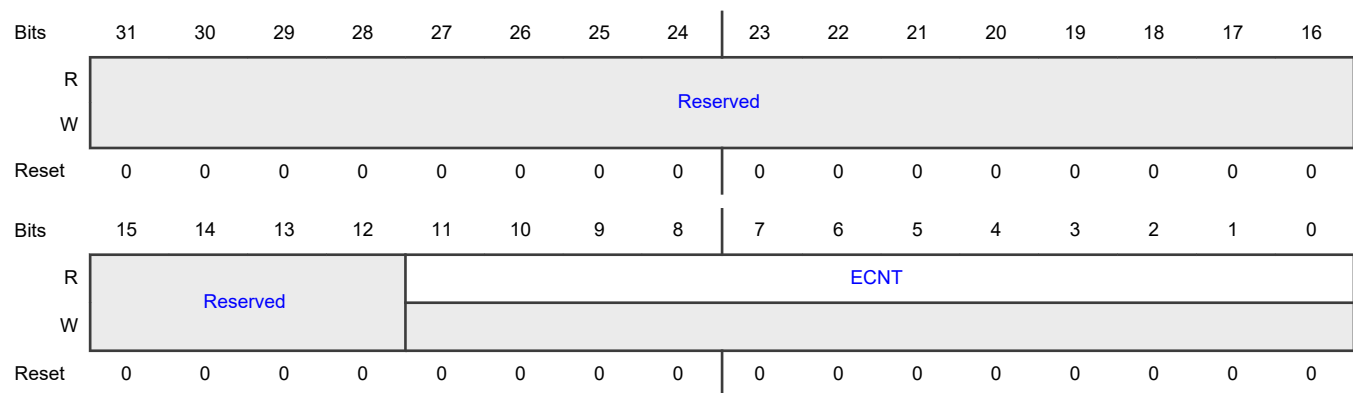
Field	Function
31 RXPACOF	Rx Parser Accept Counter Overflow Bit When set, this bit indicates that the RXPAC Counter field crossed the maximum limit. Access restriction applies. Clears on read. Self-set to 1 on internal event. 0b - Rx Parser Accept Counter overflow not occurred 1b - Rx Parser Accept Counter overflow occurred
30-0 RXPAC	Rx Parser Accept Counter This 31-bit counter is implemented whenever a Rx Parser Accept a packet due to AF =1. The counter is cleared when the register is read.

1.4.347 (DMA_CH0_RX_ERI_Cnt)**Offset**

Register	Offset
DMA_CH0_RX_ERI_Cnt	116Ch

Function

The DMA_CH(#i)_RX_ERI_Cnt registers provides the count of the number of times ERI was asserted.

Diagram**Fields**

Field	Function
31-12 —	Reserved.

Table continues on the next page...

Table continued from the previous page...

Field	Function
11-0 ECNT	ERI Counter When ERIC bit of DMA_CH(#i)_RX_Control register is set, this counter increments for burst transfer completed by the Rx DMA from the start of packet transfer. This counter is reset at the start of new packet.

1.4.348 (DMA_CH1_Control)

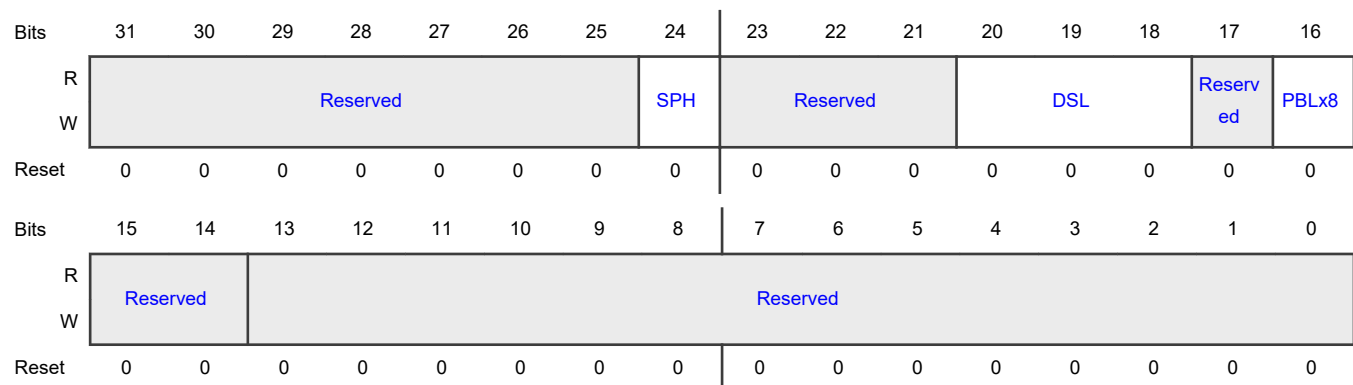
Offset

Register	Offset
DMA_CH1_Control	1180h

Function

The DMA Channeli Control register specifies the MSS value for segmentation, length to skip between two descriptors, and also the features such as header splitting and 8xPBL mode.

Diagram



Fields

Field	Function
31-25 —	Reserved.
24 SPH	Split Headers When this bit is set, the DMA splits the header and payload in the Receive path. The DMA writes the header to the Buffer Address1 of RDES0. The DMA writes the payload to the buffer to which the Buffer Address2 is pointing. The software must ensure that the header fits into the Receive buffers. If the header length exceeds the receive buffer size, the DMA does not split the header and payload.

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Table continued from the previous page...

Field	Function
	This bit is available only if Enable Split Header Structure option is selected. 0b - Split Headers feature is disabled 1b - Split Headers feature is enabled
23-21 —	Reserved.
20-18 DSL	Descriptor Skip Length This bit specifies the Word, Dword, or Lword number (depending on the 32-bit, 64-bit, or 128-bit bus) to skip between two unchained descriptors. The address skipping starts from the end of the current descriptor to the start of the next descriptor. When the DSL value is equal to zero, the DMA takes the descriptor table as contiguous.
17 —	Reserved.
16 PBLx8	8xPBL mode When this bit is set, the PBL value programmed in Bits[21:16] in DMA_CH(#i)_Tx_Control and Bits[21:16] in DMA_CH(#i)_Rx_Control is multiplied by eight times. Therefore, the DMA transfers the data in 8, 16, 32, 64, 128, and 256 beats depending on the PBL value. 0b - 8xPBL mode is disabled 1b - 8xPBL mode is enabled
15-14 —	Reserved.
13-0 —	Reserved.

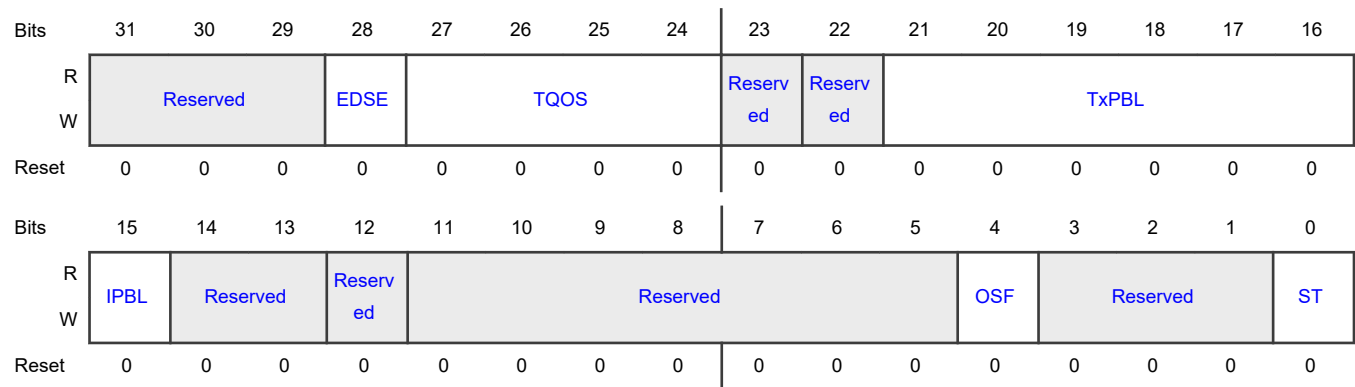
1.4.349 (DMA_CH1_Tx_Control)

Offset

Register	Offset
DMA_CH1_Tx_Control	1184h

Function

The DMA Channeli Transmit Control register controls the Tx features such as PBL, TCP segmentation, and Tx Channel weights.

Diagram**Fields**

Field	Function
31-29 —	Reserved.
28 EDSE	Enhanced Descriptor Enable When this bit is set, the corresponding channel uses Enhanced Descriptors that are 32 Bytes for both Normal and Context Descriptors. When reset, the corresponding channel uses the descriptors that are 16 Bytes. 0b - Enhanced Descriptor is disabled 1b - Enhanced Descriptor is enabled
27-24 TQOS	Transmit QOS. This field is used to drive arqos_m_o[3:0] or awqos_m_o[3:0] output signals for all transactions of DMA Tx Channel0.
23 —	Reserved.
22 —	Reserved.
21-16 TxPBL	Transmit Programmable Burst Length These bits indicate the maximum number of beats to be transferred in one DMA block data transfer. The DMA always attempts max burst as specified in PBL each time it starts a burst transfer on the application bus. You can program PBL with any of the following values: 1, 2, 4, 8, 16, or 32. Any other value results in undefined behavior. To transfer more than 32 beats, perform the following steps: 1. Set the 8xPBL mode in DMA_CH0_Control register. 2. Set the TxPBL.

Table continues on the next page...

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Field	Function
	Note: The maximum value of TxPBL must be less than or equal to half the Tx Queue size (TQS field of MTL_TxQ(#i)_Operation_Mode register) in terms of beats. This is required so that the Tx Queue has space to store at least another Tx PBL worth of data while the MTL Tx Queue Controller is transferring data to MAC. For example, in 64-bit data width configurations the total locations in Tx Queue of size 512 bytes is 64, TxPBL and 8xPBL needs to be programmed to less than or equal to 32.
15 IPBL	Ignore PBL Requirement When this bit is set, the DMA does not check for PBL number of locations in the MTL before initiating a transfer. If space is not available, the MTL might use handshaking to slow the DMA. Note: This bit/mode must not be used when multiple Transmit DMA Channels are enabled as it might block other Transmit and Receive DMA Channels from accessing the Read Data Channel of AXI bus until space is available in Transmit Queue for current transfer. 0b - Ignore PBL Requirement is disabled 1b - Ignore PBL Requirement is enabled
14-13 —	Reserved.
12 —	Reserved.
11-5 —	Reserved.
4 OSF	Operate on Second Packet When this bit is set, it instructs the DMA to process the second packet of the Transmit data even before the status for the first packet is obtained. 0b - Operate on Second Packet disabled 1b - Operate on Second Packet enabled
3-1 —	Reserved.
0 ST	Start or Stop Transmission Command When this bit is set, transmission is placed in the Running state. The DMA checks the Transmit list at the current position for a packet to be transmitted. The DMA tries to acquire descriptor from either of the following positions: - The current position in the list This is the base address of the Transmit list set by the DMA_CH0_TxDesc_List_Address register. - The position at which the transmission was previously stopped

Table continues on the next page...

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Field	Function
	If the DMA does not own the current descriptor, the transmission enters the Suspended state and the TBU bit of the DMA_CH0_Status register is set. The Start Transmission command is effective only when the transmission is stopped. If the command is issued before setting the DMA_CH0_TxDesc_List_Address register, the DMA behavior is unpredictable. When this bit is reset, the transmission process is placed in the Stopped state after completing the transmission of the current packet. The Next Descriptor position in the Transmit list is saved, and it becomes the current position when the transmission is restarted. To change the list address, you need to program DMA_CH0_TxDesc_List_Address register with a new value when this bit is reset. The new value is considered when this bit is set again. The stop transmission command is effective only when the transmission of the current packet is complete or the transmission is in the Suspended state. 0b - Stop Transmission Command 1b - Start Transmission Command

1.4.350 (DMA_CH1_Rx_Control)

Offset

Register	Offset
DMA_CH1_Rx_Control	1188h

Function

The DMA Channel1 Receive Control register controls the Rx features such as PBL, buffer size, and extended status.

Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R																
W	RPF	Reserved				RQOS				Reserv ed	Reserv ed	RxPBL				
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	Reserv ed	RBSZ_13_y											RBSZ_x_0		SR	
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Fields

Field	Function
31	Rx Packet Flush.

Table continues on the next page...

Table continued from the previous page...

Field	Function
RPF	When this bit is set to 1, then DWC_ether_qos automatically flushes the packet from the Rx Queues destined to this DMA Rx Channel, when it is stopped. When this bit remains set and the DMA is re-started by the software driver, the packets residing in the Rx Queues that were received when this RxDMA was stopped, get flushed out. The packets that are received by the MAC after the RxDMA is re-started are routed to the RxDMA. The flushing happens on the Read side of the Rx Queue. When this bit is set to 0, the DWC_ether_qos not flush the packet in the Rx Queue destined to this RxDMA Channel when it is STOP state. This might in turn cause head-of-line blocking in the corresponding RxQueue. Note: The stopping of packet flow from a Rx DMA Channel to the application by setting RPF works only when there is one-to-one mapping of Rx Queue to Rx DMA channels. In Dynamic mapping mode, setting RPF bit in any DMA_CH(#i)_Rx_Control register might flush packets from unintended Rx Queues which are destined to the stopped Rx DMA Channel. 0b - Rx Packet Flush is disabled 1b - Rx Packet Flush is enabled
30-28 —	Reserved.
27-24 RQOS	Rx AXI4 QOS. This field is used to drive arqos_m_o[3:0] or awqos_m_o[3:0] output signals for all transactions of DMA Rx Channel0.
23 —	Reserved.
22 —	Reserved.
21-16 RxPBL	Receive Programmable Burst Length These bits indicate the maximum number of beats to be transferred in one DMA block data transfer. The DMA always attempts max burst as specified in PBL each time it starts a burst transfer on the application bus. You can program PBL with any of the following values: 1, 2, 4, 8, 16, or 32. Any other value results in undefined behavior. To transfer more than 32 beats, perform the following steps: 1. Set the 8xPBL mode in the DMA_CH0_Control register. 2. Set the RxPBL. Note: The maximum value of RxPBL must be less than or equal to half the Rx Queue size (RQS field of MTL_RxQ(#i)_Operation_Mode register) in terms of beats. This is required so that the Rx Queue has space to store at least another Rx PBL worth of data while the Rx DMA is transferring a block of data. For example, in 64-bit data width configurations the total locations in Rx Queue of size 512 bytes is 64, so RxPBL and 8xPBL needs to be programmed to less than or equal to 32.
15	Reserved.

Table continues on the next page...

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Field	Function
—	
14-4 RBSZ_13_y	Receive Buffer size High RBSZ[13:0] is split into two fields higher RBSZ_13_y and lower RBSZ_x_0. The RBSZ[13:0] field indicates the size of the Rx buffers specified in bytes. The maximum buffer size is limited to 16K bytes. The buffer size is applicable to payload buffers when split headers are enabled. Note: The buffer size must be a multiple of 4, 8, or 16 depending on the data bus widths (32-bit, 64-bit, or 128-bit respectively). This is required even if the value of buffer address pointer is not aligned to data bus width. Hence the lower RBSZ_x_0 bits are read-only and the value is considered as all-zero. Thus the RBSZ_13_y indicates the buffer size in terms of locations (with the width same as bus-width).
3-1 RBSZ_x_0	Receive Buffer size Low RBSZ[13:0] is split into two fields RBSZ_13_y and RBSZ_x_0. The RBSZ_x_0 is the lower field whose width is based on data bus width of the configuration. This field is of width 2, 3, or 4 bits for 32-bit, 64-bit, or 128-bit data bus width respectively. This field is read-only (RO).
0 SR	Start or Stop Receive When this bit is set, the DMA tries to acquire the descriptor from the Receive list and processes the incoming packets. The DMA tries to acquire descriptor from either of the following positions: - The current position in the list This is the address set by the DMA_CH0_RxDesc_List_Address register. - The position at which the Rx process was previously stopped If the DMA does not own the current descriptor, the reception is suspended and the RBU bit of the DMA_CH0_Status register is set. The Start Receive command is effective only when the reception is stopped. If the command is issued before setting the DMA_CH0_RxDesc_List_Address register, the DMA behavior is unpredictable. When this bit is reset, the Rx DMA operation is stopped after the transfer of the current packet. The next descriptor position in the Receive list is saved, and it becomes the current position after the Rx process is restarted. The Stop Receive command is effective only when the Rx process is in the Running (waiting for Rx packet) or Suspended state. 0b - Stop Receive 1b - Start Receive

1.4.351 (DMA_CH1_TxDesc_List_Address)

Offset

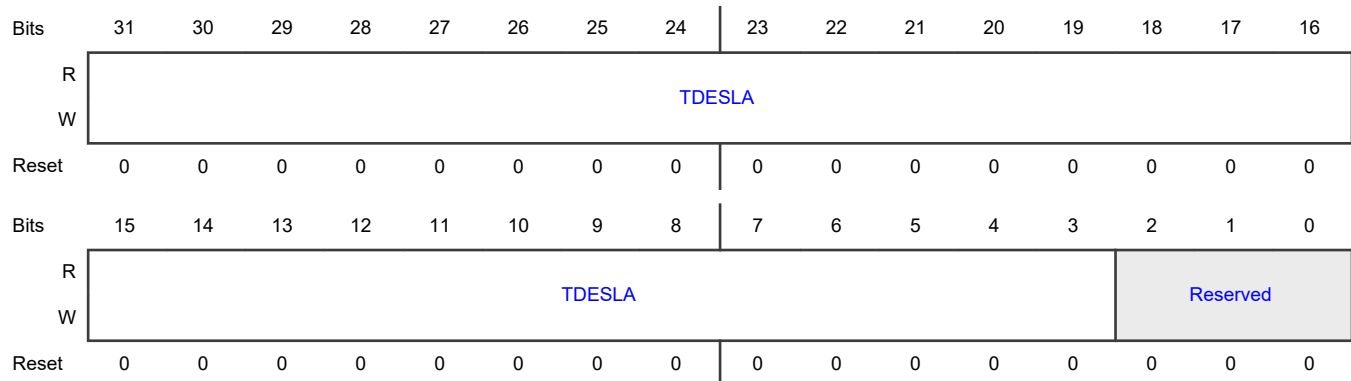
Register	Offset
DMA_CH1_TxDesc_List_Address	1194h

Function

The Channel Tx Descriptor List Address register points the DMA to the start of Transmit descriptor list. The descriptor lists reside in the physical memory space of the application and must be Word, Dword, or Lword-aligned (for 32-bit, 64-bit, or 128-bit data bus). The DMA internally converts it to bus width aligned address by making the corresponding LSB to low.

You can write to this register only when the Tx DMA has stopped, that is, the ST bit is set to zero in DMA_CH0_Tx_Control register. When stopped, this register can be written with a new descriptor list address. When you set the ST bit to 1, the DMA takes the newly-programmed descriptor base address. If this register is not changed when the ST bit is set to 0, the DMA takes the descriptor address where it was stopped earlier.

Diagram



Fields

Field	Function
31-3 TDES LA	Start of Transmit List This field contains the base address of the first descriptor in the Transmit descriptor list. The DMA ignores the LSB bits (1:0, 2:0, or 3:0) for 32-bit, 64-bit, or 128-bit bus width and internally takes these bits as all-zero. Therefore, these LSB bits are read-only (RO). The width of this field depends on the configuration: - 31:2 for 32-bit configuration - 31:3 for 64-bit configuration - 31:4 for 128-bit configuration
2-0 —	Reserved.

1.4.352 (DMA_CH1_RxDesc_List_Address)

Offset

Register	Offset
DMA_CH1_RxDesc_List_Address	119Ch

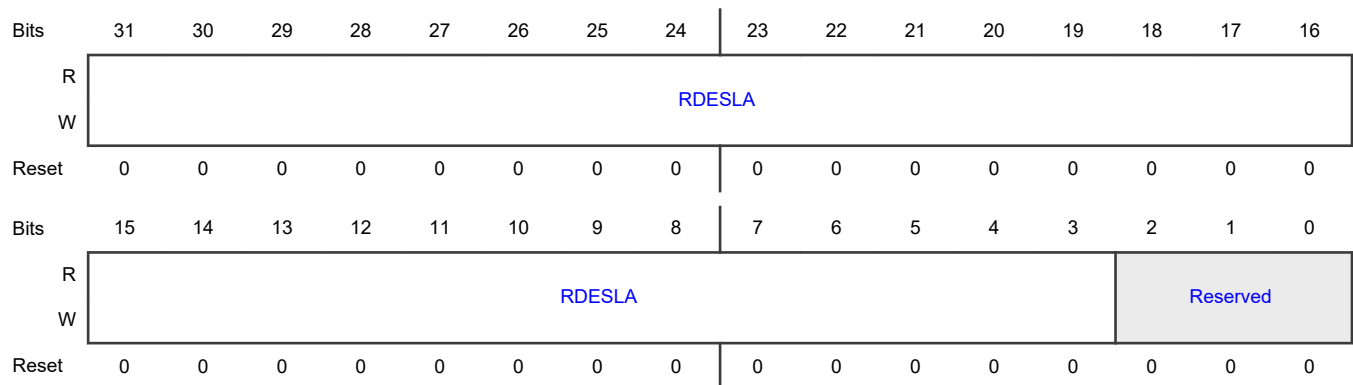
Function

The Channel Rx Descriptor List Address register points the DMA to the start of Receive descriptor list.

This register points to the start of the Receive Descriptor List. The descriptor lists reside in the physical memory space of the application and must be Word, Dword, or Lword-aligned (for 32-bit, 64-bit, or 128-bit data bus). The DMA internally converts it to bus width aligned address by making the corresponding LS bits low. Writing to this register is permitted only when reception is stopped. When stopped, this register must be written to before the receive Start command is given. You can write to this register only when Rx DMA has stopped, that is, SR bit is set to zero in DMA_CH0_Rx_Control register. When stopped, this register can be written with a new descriptor list address.

When you set the SR bit to 1, the DMA takes the newly programmed descriptor base address.

Diagram



Fields

Field	Function
31-3 RDESLA	Start of Receive List This field contains the base address of the first descriptor in the Rx Descriptor list. The DMA ignores the LSB bits (1:0, 2:0, or 3:0) for 32-bit, 64-bit, or 128-bit bus width and internally takes these bits as all-zero. Therefore, these LSB bits are read-only (RO). The width of this field depends on the configuration: - 31:2 for 32-bit configuration - 31:3 for 64-bit configuration - 31:4 for 128-bit configuration
2-0 —	Reserved.

1.4.353 (DMA_CH1_TxDesc_Tail_Pointer)

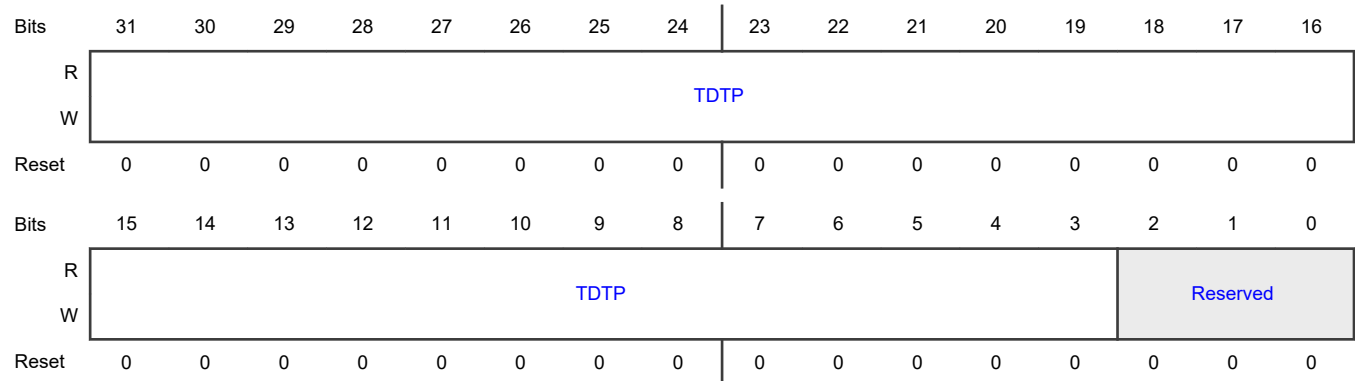
Offset

Register	Offset
DMA_CH1_TxDesc_Tail_Pointer	11A0h

Function

The Channel1 Tx Descriptor Tail Pointer register points to an offset from the base and indicates the location of the last valid descriptor.

Diagram



Fields

Field	Function
31-3 TDTP	Transmit Descriptor Tail Pointer This field contains the tail pointer for the Tx descriptor ring. The software writes the tail pointer to add more descriptors to the Tx channel. The hardware tries to transmit all packets referenced by the descriptors between the head and the tail pointer registers. The width of this field depends on the configuration: - 31:2 for 32-bit configuration - 31:3 for 64-bit configuration - 31:4 for 128-bit configuration
2-0 —	Reserved.

1.4.354 (DMA_CH1_RxDesc_Tail_Pointer)

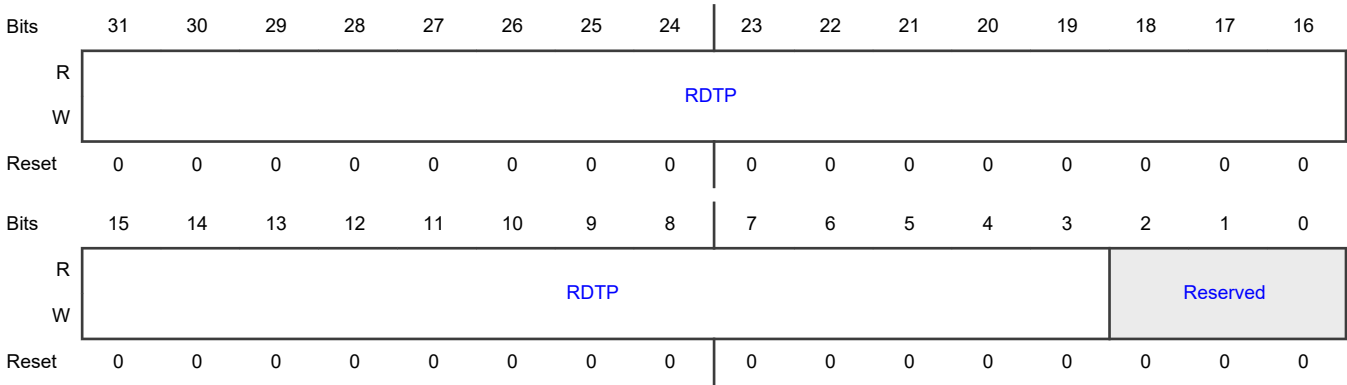
Offset

Register	Offset
DMA_CH1_RxDesc_Tail_Pointer	11A8h

Function

The Channel1 Rx Descriptor Tail Pointer Points to an offset from the base and indicates the location of the last valid descriptor.

Diagram



Fields

Field	Function
31-3 RDTP	Receive Descriptor Tail Pointer This field contains the tail pointer for the Rx descriptor ring. The software writes the tail pointer to add more descriptors to the Rx channel. The hardware tries to write all received packets to the descriptors referenced between the head and the tail pointer registers. The width of this field depends on the configuration: - 31:2 for 32-bit configuration - 31:3 for 64-bit configuration - 31:4 for 128-bit configuration
2-0 —	Reserved.

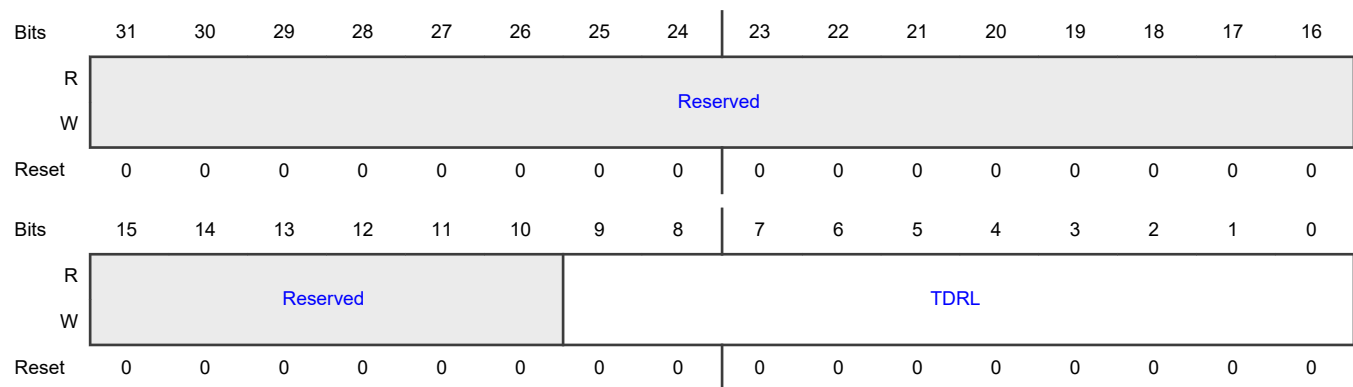
1.4.355 (DMA_CH1_TxDesc_Ring_Length)

Offset

Register	Offset
DMA_CH1_TxDesc_Ring_Length	11ACh

Function

The Tx Descriptor Ring Length register contains the length of the Transmit descriptor ring.

Diagram**Fields**

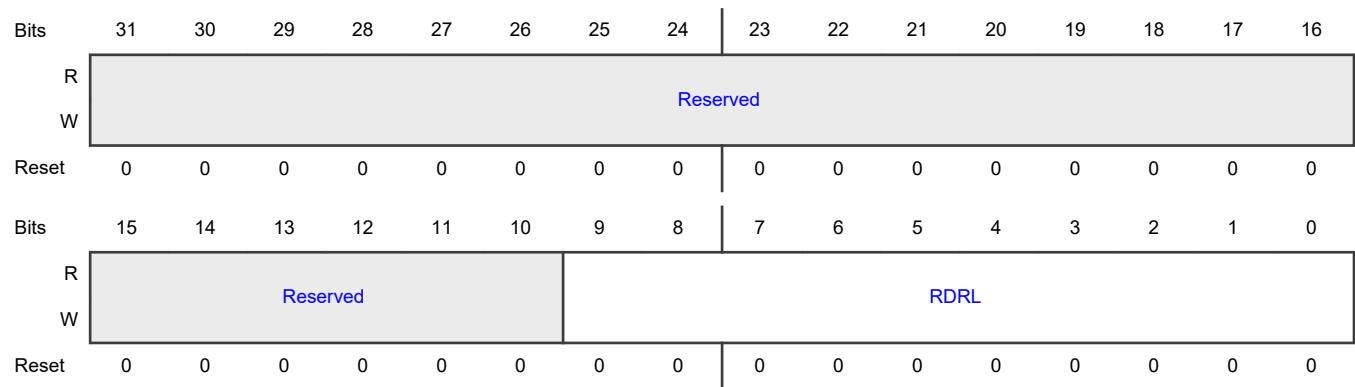
Field	Function
31-10 —	Reserved.
9-0 TDRL	Transmit Descriptor Ring Length This field sets the maximum number of Tx descriptors in the circular descriptor ring. The maximum number of descriptors is limited to 1K descriptors. NXP recommends a minimum ring descriptor length of 4. For example, You can program any value up to 0x3FF in this field. This field is 10 bits wide, if you program 0x3FF, you can have 1024 descriptors. If you want to have 10 descriptors, program it to a value of 0x9.

1.4.356 (DMA_CH1_RxDesc_Ring_Length)**Offset**

Register	Offset
DMA_CH1_RxDesc_Ring_Length	11B0h

Function

The Channel1 Rx Descriptor Ring Length register contains the length of the Receive descriptor circular ring.

Diagram**Fields**

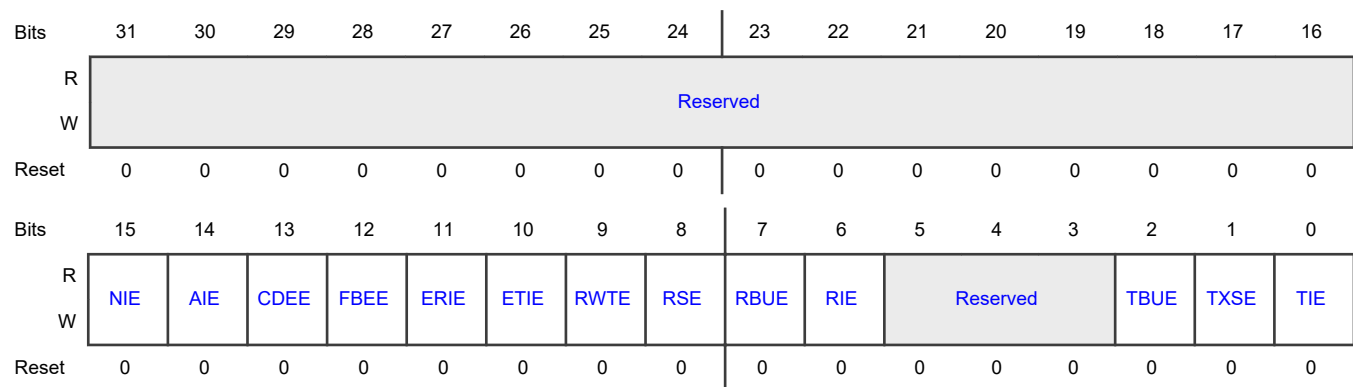
Field	Function
31-10 —	Reserved.
9-0 RDRL	Receive Descriptor Ring Length This register sets the maximum number of Rx descriptors in the circular descriptor ring. The maximum number of descriptors is limited to 1K descriptors. For example, You can program any value up to 0x3FF in this field. This field is 10 bits wide, if you program 0x3FF, you can have 1024 descriptors. If you want to have 10 descriptors, program it to a value of 0x9.

1.4.357 (DMA_CH1_Interrupt_Enable)**Offset**

Register	Offset
DMA_CH1_Interrupt_Enable	11B4h

Function

The Channel1 Interrupt Enable register enables the interrupts reported by the Status register.

Diagram**Fields**

Field	Function
31-16 —	Reserved.
15 NIE	Normal Interrupt Summary Enable When this bit is set, the normal interrupt summary is enabled. This bit enables the following interrupts in the DMA_CH0_Status register: - Bit 0: Transmit Interrupt - Bit 2: Transmit Buffer Unavailable - Bit 6: Receive Interrupt - Bit 11: Early Receive Interrupt When this bit is reset, the normal interrupt summary is disabled. 0b - Normal Interrupt Summary is disabled 1b - Normal Interrupt Summary is enabled
14 AIE	Abnormal Interrupt Summary Enable When this bit is set, the abnormal interrupt summary is enabled. This bit enables the following interrupts in the DMA_CH0_Status register: - Bit 1: Transmit Process Stopped - Bit 7: Rx Buffer Unavailable - Bit 8: Receive Process Stopped - Bit 9: Receive Watchdog Timeout - Bit 10: Early Transmit Interrupt - Bit 12: Fatal Bus Error - Bit 13: Context Descriptor Error When this bit is reset, the abnormal interrupt summary is disabled.

Table continues on the next page...

Table continued from the previous page...

Field	Function
	0b - Abnormal Interrupt Summary is disabled 1b - Abnormal Interrupt Summary is enabled
13 CDEE	Context Descriptor Error Enable When this bit is set along with the AIE bit, the Descriptor error interrupt is enabled. When this bit is reset, the Descriptor error interrupt is disabled. 0b - Context Descriptor Error is disabled 1b - Context Descriptor Error is enabled
12 FBEE	Fatal Bus Error Enable When this bit is set along with the AIE bit, the Fatal Bus error interrupt is enabled. When this bit is reset, the Fatal Bus Error error interrupt is disabled. 0b - Fatal Bus Error is disabled 1b - Fatal Bus Error is enabled
11 ERIE	Early Receive Interrupt Enable When this bit is set along with the NIE bit, the Early Receive interrupt is enabled. When this bit is reset, the Early Receive interrupt is disabled. 0b - Early Receive Interrupt is disabled 1b - Early Receive Interrupt is enabled
10 ETIE	Early Transmit Interrupt Enable When this bit is set along with the AIE bit, the Early Transmit interrupt is enabled. When this bit is reset, the Early Transmit interrupt is disabled. 0b - Early Transmit Interrupt is disabled 1b - Early Transmit Interrupt is enabled
9 RWTE	Receive Watchdog Timeout Enable When this bit is set along with the AIE bit, the Receive Watchdog Timeout interrupt is enabled. When this bit is reset, the Receive Watchdog Timeout interrupt is disabled. 0b - Receive Watchdog Timeout is disabled 1b - Receive Watchdog Timeout is enabled
8 RSE	Receive Stopped Enable When this bit is set along with the AIE bit, the Receive Stopped Interrupt is enabled. When this bit is reset, the Receive Stopped interrupt is disabled. 0b - Receive Stopped is disabled 1b - Receive Stopped is enabled
7	Receive Buffer Unavailable Enable

Table continues on the next page...

Table continued from the previous page...

Field	Function
RBUE	When this bit is set along with the AIE bit, the Receive Buffer Unavailable interrupt is enabled. When this bit is reset, the Receive Buffer Unavailable interrupt is disabled. 0b - Receive Buffer Unavailable is disabled 1b - Receive Buffer Unavailable is enabled
6 RIE	Receive Interrupt Enable When this bit is set along with the NIE bit, the Receive Interrupt is enabled. When this bit is reset, the Receive Interrupt is disabled. 0b - Receive Interrupt is disabled 1b - Receive Interrupt is enabled
5-3 —	Reserved.
2 TBUE	Transmit Buffer Unavailable Enable When this bit is set along with the NIE bit, the Transmit Buffer Unavailable interrupt is enabled. When this bit is reset, the Transmit Buffer Unavailable interrupt is disabled. 0b - Transmit Buffer Unavailable is disabled 1b - Transmit Buffer Unavailable is enabled
1 TXSE	Transmit Stopped Enable When this bit is set along with the AIE bit, the Transmission Stopped interrupt is enabled. When this bit is reset, the Transmission Stopped interrupt is disabled. 0b - Transmit Stopped is disabled 1b - Transmit Stopped is enabled
0 TIE	Transmit Interrupt Enable When this bit is set along with the NIE bit, the Transmit Interrupt is enabled. When this bit is reset, the Transmit Interrupt is disabled. 0b - Transmit Interrupt is disabled 1b - Transmit Interrupt is enabled

1.4.358 (DMA_CH1_Rx_Interrupt_Watchdog_Timer)

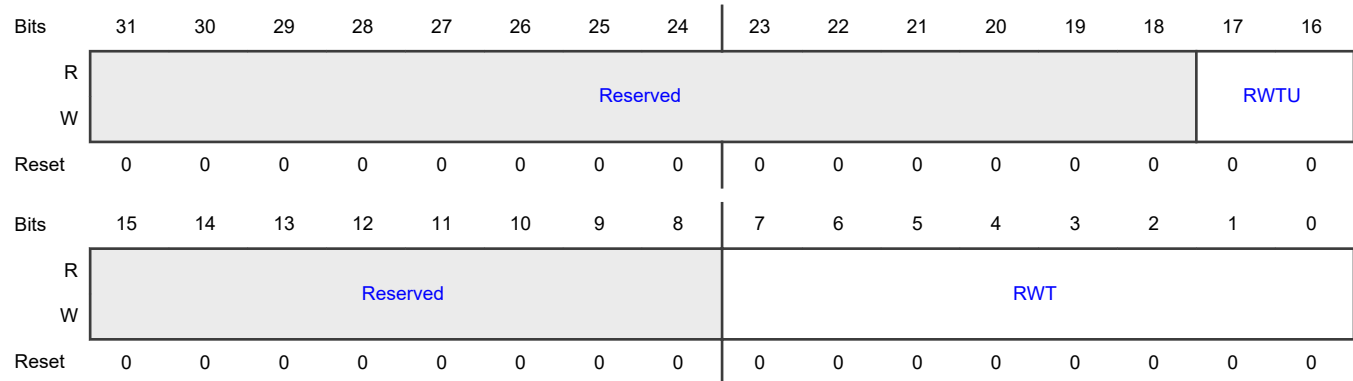
Offset

Register	Offset
DMA_CH1_Rx_Interrupt_Watchdog_Timer	11B8h

Function

The Receive Interrupt Watchdog Timer register indicates the watchdog timeout for Receive Interrupt (RI) from the DMA. When this register is written with a non-zero value, it enables the watchdog timer for the RI bit of the DMA_CHi_Status register.

Diagram



Fields

Field	Function
31-18 —	Reserved.
17-16 RWTU	Receive Interrupt Watchdog Timer Count Units This fields indicates the number of system clock cycles corresponding to one unit in RWT field. - 2'b00: 256 - 2'b01: 512 - 2'b10: 1024 - 2'b11: 2048 For example, when RWT=2 and RWTU=1, the watchdog timer is set for 2*512=1024 system clock cycles.
15-8 —	Reserved.
7-0 RWT	Receive Interrupt Watchdog Timer Count This field indicates the number of system clock cycles, multiplied by factor indicated in RWTU field, for which the watchdog timer is set. The watchdog timer is triggered with the programmed value after the Rx DMA completes the transfer of a packet for which the RI bit is not set in the DMA_CH(i)_Status register, because of the setting of Interrupt Enable bit in the corresponding descriptor RDES3[30]. When the watchdog timer runs out, the RI bit is set and the timer is stopped. The watchdog timer is reset when the RI bit is set high because of automatic setting of RI as per the Interrupt Enable bit RDES3[30] of any received packet.

1.4.359 (DMA_CH1_Slot_Function_Control_Status)

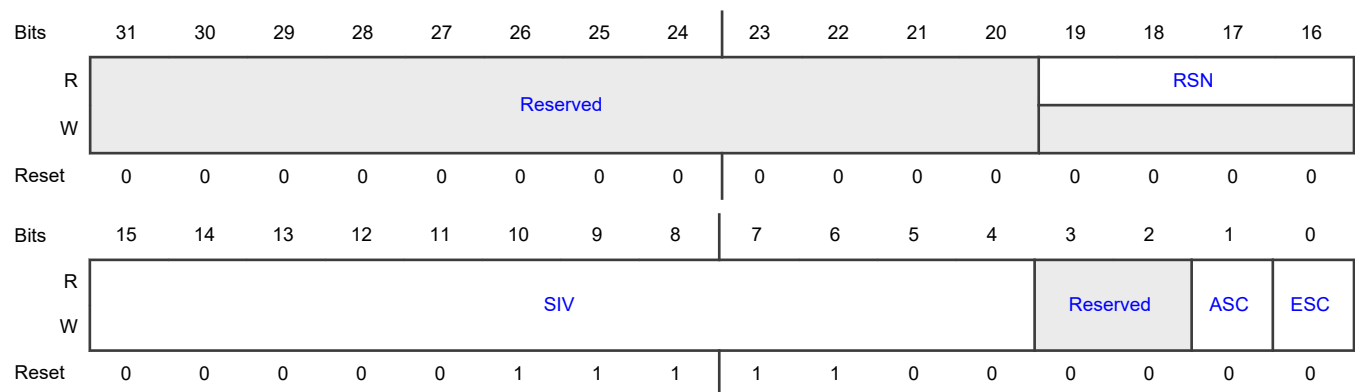
Offset

Register	Offset
DMA_CH1_Slot_Function_Control_Status	11BCh

Function

The Slot Function Control and Status register contains the control bits for slot function and the status for Transmit path.

Diagram



Fields

Field	Function
31-20 —	Reserved.
19-16 RSN	Reference Slot Number This field gives the current value of the reference slot number in the DMA. It is used for slot comparison.
15-4 SIV	Slot Interval Value This field controls the period of the slot interval in which the TxDMA fetches the scheduled packets. A value of 0 specifies the slot interval of 1 us while the maximum value 4095 specifies the slot interval of 4096us. The default/reset value is 0x07C which corresponds to slot interval of 125us
3-2 —	Reserved.
1 ASC	Advance Slot Check When set, this bit enables the DMA to fetch the data from the buffer when the slot number (SLOTNUM) programmed in the Tx descriptor is - equal to the reference slot number given in the RSN field

Table continues on the next page...

Table continued from the previous page...

Field	Function
	or - ahead of the reference slot number by up to two slots This bit is applicable only when the ESC bit is set. 0b - Advance Slot Check is disabled 1b - Advance Slot Check is enabled
0 ESC	Enable Slot Comparison When set, this bit enables the checking of the slot numbers programmed in the Tx descriptor with the current reference given in the RSN field. The DMA fetches the data from the corresponding buffer only when the slot number is - equal to the reference slot number or - ahead of the reference slot number by one slot When reset, this bit disables the checking of the slot numbers. The DMA fetches the data immediately after the descriptor is processed. Note: The UFO (UDP Fragmentation over IPv4)/TSO/USO should not be enabled along with TBS/AVB Slot number check. The UFO/TSO/USO involves multiple packets/segments/fragments transmission for single packet received from application and the slot number check are applicable for fetching of only first segment/fragment. As a result it might be difficult for software to specify slot number for subsequent packets. 0b - Slot Comparison is disabled 1b - Slot Comparison is enabled

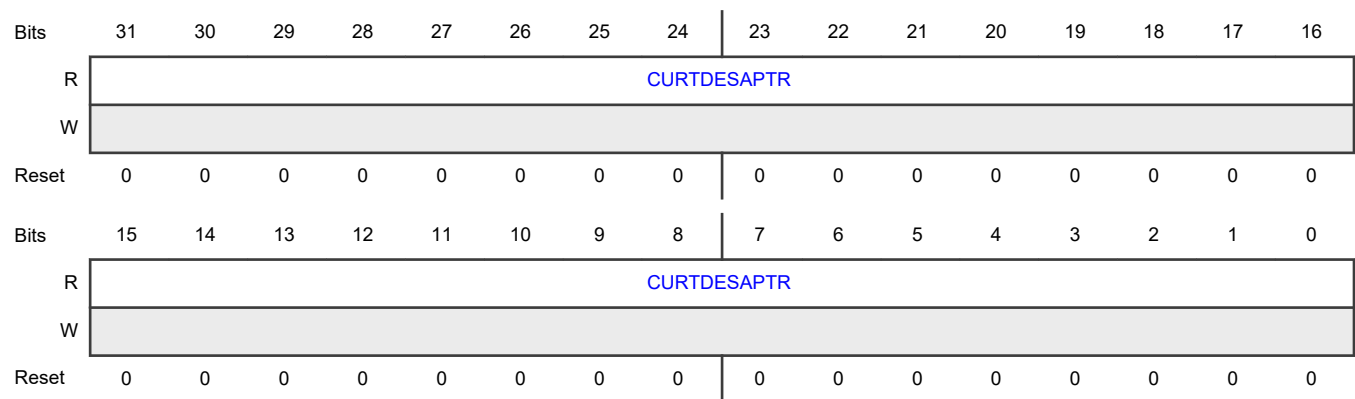
1.4.360 (DMA_CH1_Current_App_TxDesc)

Offset

Register	Offset
DMA_CH1_Current_App_TxDesc	11C4h

Function

The Channel1 Current Application Transmit Descriptor register points to the current Transmit descriptor read by the DMA.

Diagram**Fields**

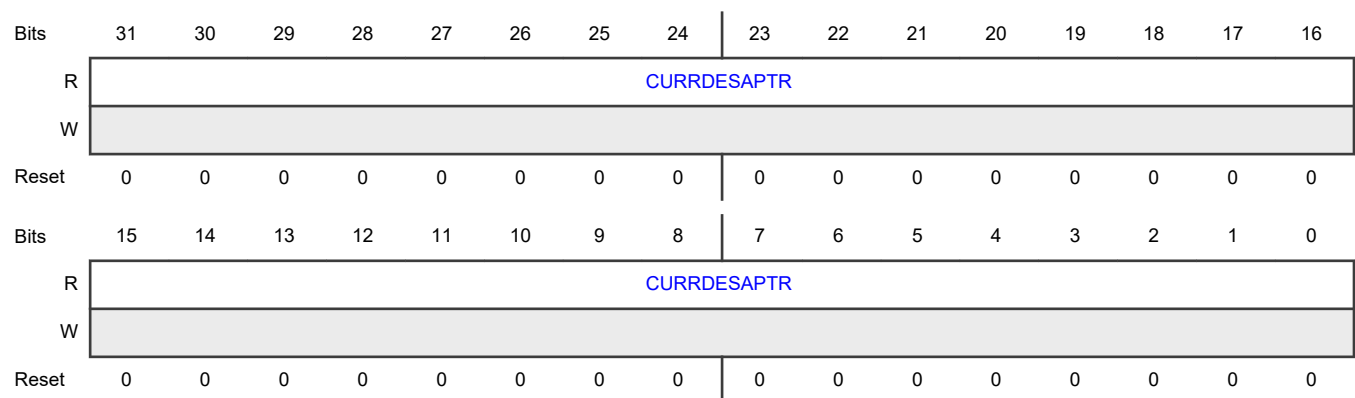
Field	Function
31-0	Application Transmit Descriptor Address Pointer
CURTDESAPTR	The DMA updates this pointer during Tx operation. This pointer is cleared on reset.
R	

1.4.361 (DMA_CH1_Current_App_RxDesc)**Offset**

Register	Offset
DMA_CH1_Current_App_RxDesc	11CCh

Function

The Channel1 Current Application Receive Descriptor register points to the current Receive descriptor read by the DMA.

Diagram

Fields

Field	Function
31-0 CURDESAPTR	Application Receive Descriptor Address Pointer The DMA updates this pointer during Rx operation. This pointer is cleared on reset.

1.4.362 (DMA_CH1_Current_App_TxBuffer)

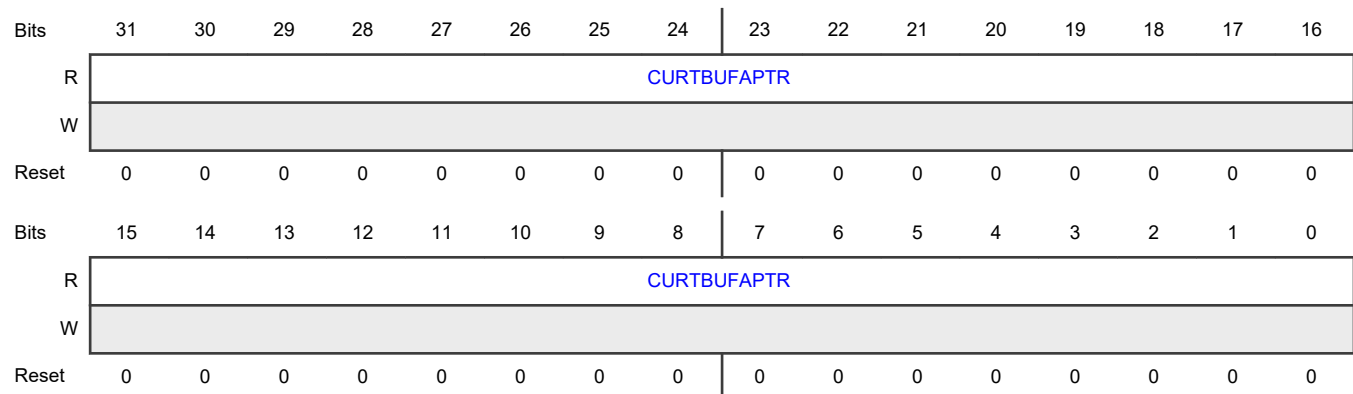
Offset

Register	Offset
DMA_CH1_Current_App_TxBuffer	11D4h

Function

The Channel1 Current Application Transmit Buffer Address register points to the current Tx buffer address read by the DMA.

Diagram



Fields

Field	Function
31-0 CURTBUFAPTR	Application Transmit Buffer Address Pointer The DMA updates this pointer during Tx operation. This pointer is cleared on reset.

1.4.363 (DMA_CH1_Current_App_RxBuffer)

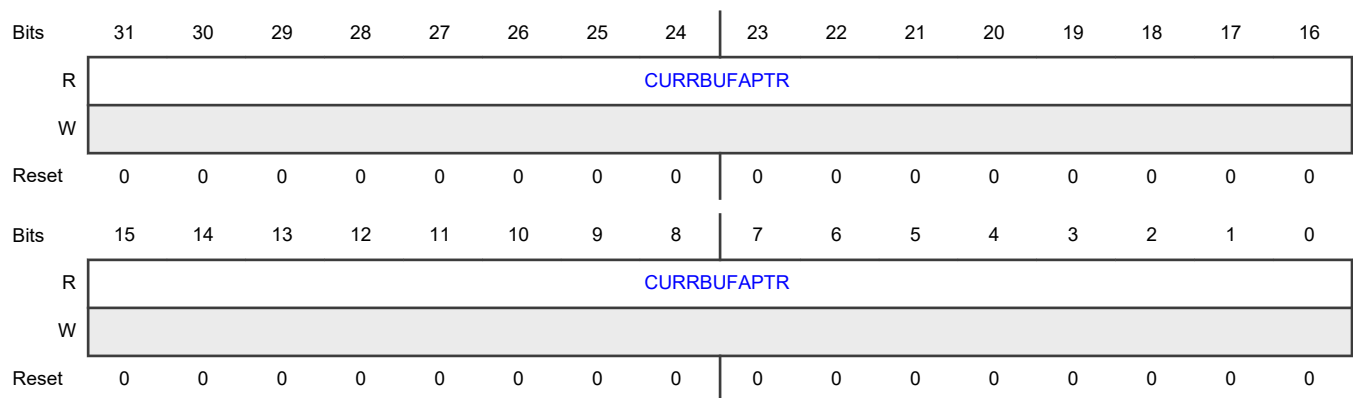
Offset

Register	Offset
DMA_CH1_Current_App_RxBuffer	11DCh

Function

The Channel 0 Current Application Receive Buffer Address register points to the current Rx buffer address read by the DMA.

Diagram



Fields

Field	Function
31-0 CURRBUFAPTR	Application Receive Buffer Address Pointer
R	The DMA updates this pointer during Rx operation. This pointer is cleared on reset.

1.4.364 (DMA_CH1_Status)

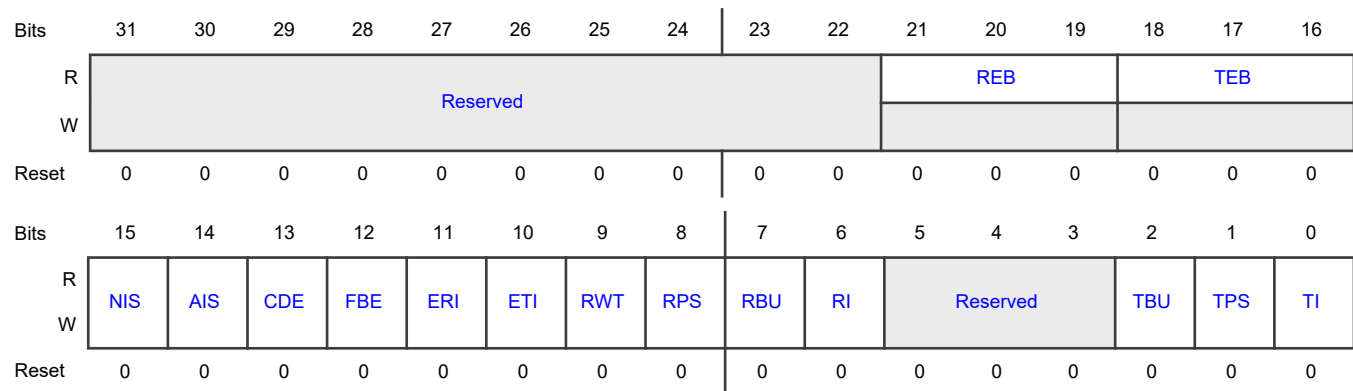
Offset

Register	Offset
DMA_CH1_Status	11E0h

Function

The software driver (application) reads the Status register during interrupt service routine or polling to determine the status of the DMA.

Note: The number of DMA_CH(##)_Status register in the configuration is the higher of number of Rx DMA Channels and Tx DMA Channels.

Diagram**Fields**

Field	Function
31-22 —	Reserved.
21-19 REB	Rx DMA Error Bits This field indicates the type of error that caused a Bus Error. For example, error response on the AHB or AXI interface. - Bit 21 -- 1'b1: Error during data transfer by Rx DMA -- 1'b0: No Error during data transfer by Rx DMA - Bit 20 -- 1'b1: Error during descriptor access -- 1'b0: Error during data buffer access - Bit 19 -- 1'b1: Error during read transfer -- 1'b0: Error during write transfer This field is valid only when the FBE bit is set. This field does not generate an interrupt.
18-16 TEB	Tx DMA Error Bits This field indicates the type of error that caused a Bus Error. For example, error response on the AHB or AXI interface. - Bit 18 -- 1'b1: Error during data transfer by Tx DMA -- 1'b0: No Error during data transfer by Tx DMA - Bit 17 -- 1'b1: Error during descriptor access

Table continues on the next page...

Table continued from the previous page...

Field	Function
	-- 1'b0: Error during data buffer access - Bit 16 -- 1'b1: Error during read transfer -- 1'b0: Error during write transfer This field is valid only when the FBE bit is set. This field does not generate an interrupt.
15 NIS	Normal Interrupt Summary Normal Interrupt Summary bit value is the logical OR of the following bits when the corresponding interrupt bits are enabled in the DMA_CH0_Interrupt_Enable register: - Bit 0: Transmit Interrupt - Bit 2: Transmit Buffer Unavailable - Bit 6: Receive Interrupt - Bit 11: Early Receive Interrupt Only unmasked bits (interrupts for which interrupt enable is set in DMA_CH0_Interrupt_Enable register) affect the Normal Interrupt Summary bit. This is a sticky bit. You must clear this bit (by writing 1 to this bit) each time a corresponding bit which causes NIS to be set is cleared. Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect. 0b - Normal Interrupt Summary status not detected 1b - Normal Interrupt Summary status detected
14 AIS	Abnormal Interrupt Summary Abnormal Interrupt Summary bit value is the logical OR of the following when the corresponding interrupt bits are enabled in the DMA_CH0_Interrupt_Enable register: - Bit 1: Transmit Process Stopped - Bit 7: Receive Buffer Unavailable - Bit 8: Receive Process Stopped - Bit 10: Early Transmit Interrupt - Bit 12: Fatal Bus Error - Bit 13: Context Descriptor Error Only unmasked bits affect the Abnormal Interrupt Summary bit. This is a sticky bit. You must clear this bit (by writing 1 to this bit) each time a corresponding bit, which causes AIS to be set, is cleared. Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect. 0b - Abnormal Interrupt Summary status not detected 1b - Abnormal Interrupt Summary status detected

Table continues on the next page...

Table continued from the previous page...

Field	Function
13 CDE	Context Descriptor Error This bit indicates that the DMA Tx/Rx engine received a descriptor error, which indicates invalid context in the middle of packet flow (intermediate descriptor) or all one's descriptor in Tx case and on Rx side it indicates DMA has read a descriptor with either of the buffer address as ones which is considered to be invalid. Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect. 0b - Context Descriptor Error status not detected 1b - Context Descriptor Error status detected
12 FBE	Fatal Bus Error This bit indicates that a bus error occurred (as described in the EB field). When this bit is set, the corresponding DMA channel engine disables all bus accesses. Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect. 0b - Fatal Bus Error status not detected 1b - Fatal Bus Error status detected
11 ERI	Early Receive Interrupt This bit when set indicates that the RxDMA has completed the transfer of packet data to the memory. In configs supporting ERIC, When ERIC=0, this bit is set only after the Rx DMA has filled up a complete receive buffer with packet data. When ERIC=1, this bit is set after every burst transfer of data from the Rx DMA to the buffer. The setting of RI bit automatically clears this bit. Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect. 0b - Early Receive Interrupt status not detected 1b - Early Receive Interrupt status detected
10 ETI	Early Transmit Interrupt This bit when set indicates that the TxDMA has completed the transfer of packet data to the MTL TXFIFO memory. In configs supporting ERIC: When ETIC=0, this bit is set only after the Tx DMA has transferred a complete packet to MTL. When ETIC=1, this bit is set after completion of (partial) packet data transfer from buffers in the Transmit descriptor in which IOC=1. Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect. 0b - Early Transmit Interrupt status not detected 1b - Early Transmit Interrupt status detected
9 RWT	Receive Watchdog Timeout This bit is asserted when a packet with length greater than 2,048 bytes (10,240 bytes when Jumbo Packet mode is enabled) is received.

Table continues on the next page...

Table continued from the previous page...

Field	Function
	0b - Receive Watchdog Timeout status not detected 1b - Receive Watchdog Timeout status detected
8 RPS	Receive Process Stopped This bit is asserted when the Rx process enters the Stopped state. Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect. 0b - Receive Process Stopped status not detected 1b - Receive Process Stopped status detected
7 RBU	Receive Buffer Unavailable This bit indicates that the application owns the next descriptor in the Receive list, and the DMA cannot acquire it. The Rx process is suspended. To resume processing Rx descriptors, the application should change the ownership of the descriptor and issue a Receive Poll Demand command. If this command is not issued, the Rx process resumes when the next recognized incoming packet is received. In ring mode, the application should advance the Receive Descriptor Tail Pointer register of a channel. This bit is set only when the DMA owns the previous Rx descriptor. Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect. 0b - Receive Buffer Unavailable status not detected 1b - Receive Buffer Unavailable status detected
6 RI	Receive Interrupt This bit indicates that the packet reception is complete. When packet reception is complete, Bit 31 of RDES3 is reset in the last descriptor, and the specific packet status information is updated in the descriptor. The reception remains in the Running state. Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect. 0b - Receive Interrupt status not detected 1b - Receive Interrupt status detected
5-3 —	Reserved.
2 TBU	Transmit Buffer Unavailable This bit indicates that the application owns the next descriptor in the Transmit list, and the DMA cannot acquire it. Transmission is suspended. The TPS0 field of the DMA_Debug_Status0 register explains the Transmit Process state transitions. To resume processing the Transmit descriptors, the application should do the following: 1. Change the ownership of the descriptor by setting Bit 31 of TDES3. 2. Issue a Transmit Poll Demand command. For ring mode, the application should advance the Transmit Descriptor Tail Pointer register of a channel.

Table continues on the next page...

Table continued from the previous page...

Field	Function
	Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect. 0b - Transmit Buffer Unavailable status not detected 1b - Transmit Buffer Unavailable status detected
1 TPS	Transmit Process Stopped This bit is set when the transmission is stopped. Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect. 0b - Transmit Process Stopped status not detected 1b - Transmit Process Stopped status detected
0 TI	Transmit Interrupt This bit indicates that the packet transmission is complete. When transmission is complete, Bit 31 of TDES3 is reset in the last descriptor, and the specific packet status information is updated in the descriptor. Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect. 0b - Transmit Interrupt status not detected 1b - Transmit Interrupt status detected

1.4.365 (DMA_CH1_Miss_Frame_Cnt)

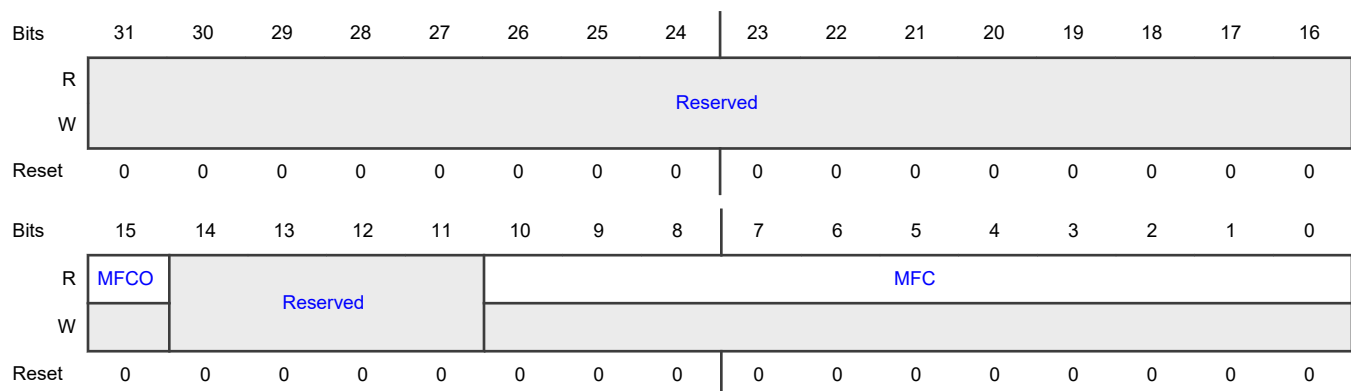
Offset

Register	Offset
DMA_CH1_Miss_Frame_Cnt	11E4h

Function

This register has the number of packet counter that got dropped by the DMA either due to Bus Error or due to programming RPF field in DMA_CH\${i}_Rx_Control register.

Diagram



Fields

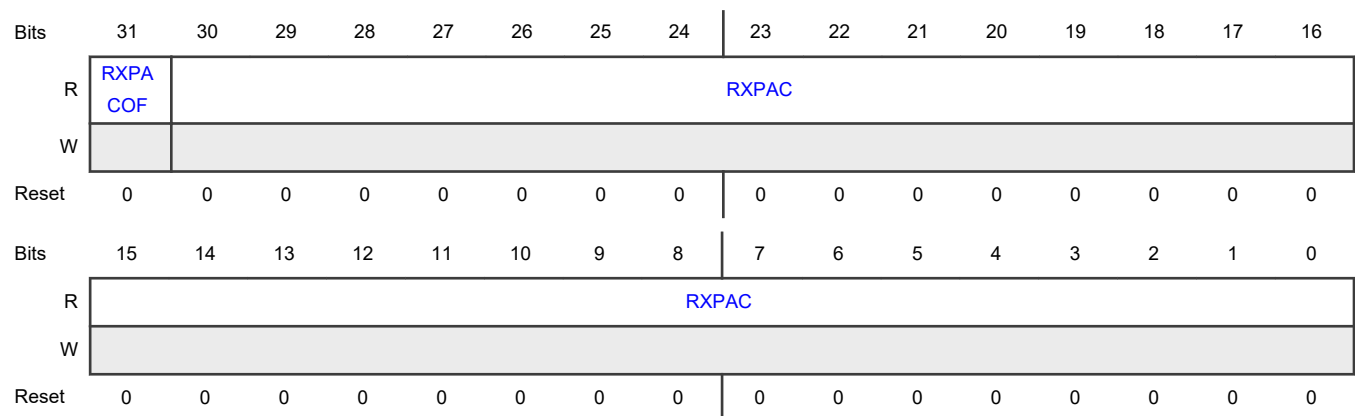
Field	Function
31-16 —	Reserved.
15 MFCO	Overflow status of the MFC Counter When this bit is set then the MFC counter does not get incremented further. The bit gets cleared when this register is read. Access restriction applies. Clears on read. Self-set to 1 on internal event. 0b - Miss Frame Counter overflow not occurred 1b - Miss Frame Counter overflow occurred
14-11 —	Reserved.
10-0 MFC	Dropped Packet Counters This counter indicates the number of packet counters that are dropped by the DMA either because of bus error or because of programming RPF field in DMA_CH\${i}_Rx_Control register. The counter gets cleared when this register is read. Access restriction applies. Clears on read. Self-set to 1 on internal event.

1.4.366 (DMA_CH1_RXP_Accept_Cnt)**Offset**

Register	Offset
DMA_CH1_RXP_Accept_Cnt	11E8h

Function

The DMA_CH(#i)_RXP_Accept_Cnt registers provides the count of the number of frames accepted by Rx Parser.

Diagram**Fields**

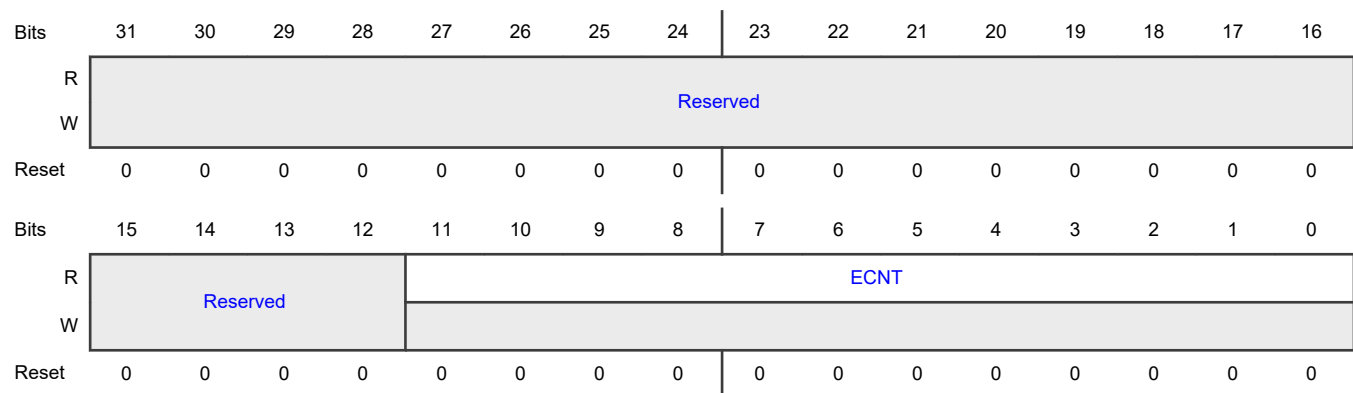
Field	Function
31 RXPACOF	Rx Parser Accept Counter Overflow Bit When set, this bit indicates that the RXPAC Counter field crossed the maximum limit. Access restriction applies. Clears on read. Self-set to 1 on internal event. 0b - Rx Parser Accept Counter overflow not occurred 1b - Rx Parser Accept Counter overflow occurred
30-0 RXPAC	Rx Parser Accept Counter This 31-bit counter is implemented whenever a Rx Parser Accept a packet due to AF =1. The counter is cleared when the register is read.

1.4.367 (DMA_CH1_RX_ERI_Cnt)**Offset**

Register	Offset
DMA_CH1_RX_ERI_Cnt	11ECh

Function

The DMA_CH(#i)_RX_ERI_Cnt registers provides the count of the number of times ERI was asserted.

Diagram**Fields**

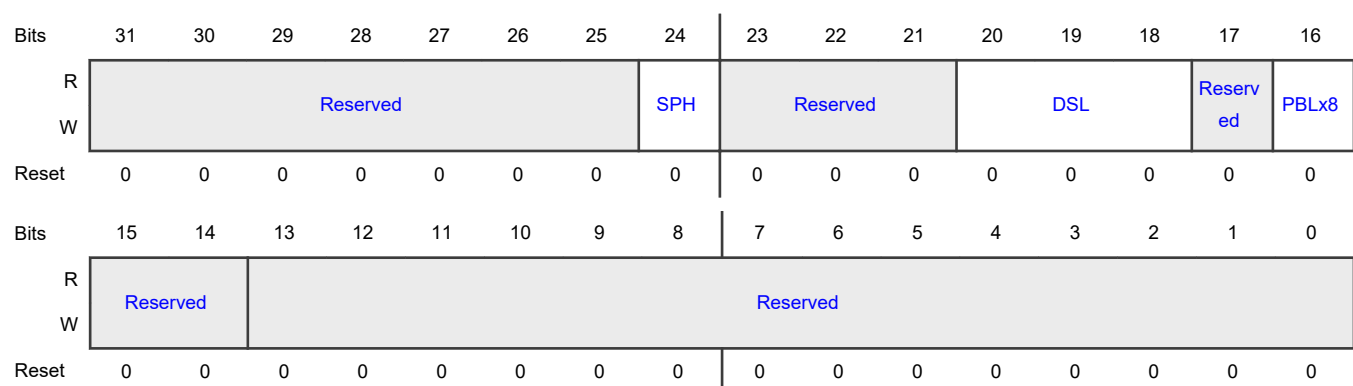
Field	Function
31-12 —	Reserved.
11-0 ECNT	ERI Counter When ERIC bit of DMA_CH(#)_RX_Control register is set, this counter increments for burst transfer completed by the Rx DMA from the start of packet transfer. This counter is reset at the start of new packet.

1.4.368 (DMA_CH2_Control)**Offset**

Register	Offset
DMA_CH2_Control	1200h

Function

The DMA Channel Control register specifies the MSS value for segmentation, length to skip between two descriptors, and also the features such as header splitting and 8xPBL mode.

Diagram

Fields

Field	Function
31-25 —	Reserved.
24 SPH	Split Headers When this bit is set, the DMA splits the header and payload in the Receive path. The DMA writes the header to the Buffer Address1 of RDES0. The DMA writes the payload to the buffer to which the Buffer Address2 is pointing. The software must ensure that the header fits into the Receive buffers. If the header length exceeds the receive buffer size, the DMA does not split the header and payload. This bit is available only if Enable Split Header Structure option is selected. 0b - Split Headers feature is disabled 1b - Split Headers feature is enabled
23-21 —	Reserved.
20-18 DSL	Descriptor Skip Length This bit specifies the Word, Dword, or Lword number (depending on the 32-bit, 64-bit, or 128-bit bus) to skip between two unchained descriptors. The address skipping starts from the end of the current descriptor to the start of the next descriptor. When the DSL value is equal to zero, the DMA takes the descriptor table as contiguous.
17 —	Reserved.
16 PBLx8	8xPBL mode When this bit is set, the PBL value programmed in Bits[21:16] in DMA_CH(#i)_Tx_Control and Bits[21:16] in DMA_CH(#i)_Rx_Control is multiplied by eight times. Therefore, the DMA transfers the data in 8, 16, 32, 64, 128, and 256 beats depending on the PBL value. 0b - 8xPBL mode is disabled 1b - 8xPBL mode is enabled
15-14 —	Reserved.
13-0 —	Reserved.

1.4.369 (DMA_CH2_Tx_Control)

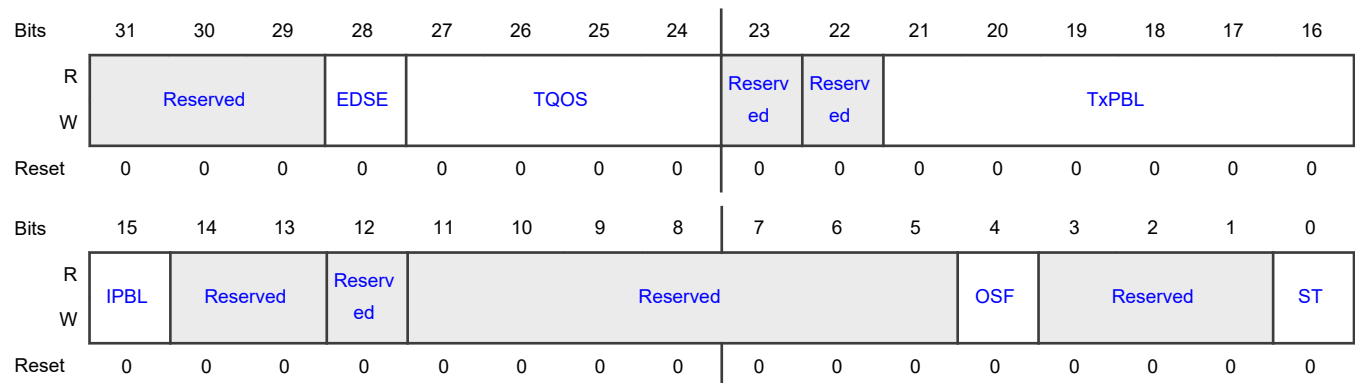
Offset

Register	Offset
DMA_CH2_Tx_Control	1204h

Function

The DMA Channel Transmit Control register controls the Tx features such as PBL, TCP segmentation, and Tx Channel weights.

Diagram



Fields

Field	Function
31-29 —	Reserved.
28 EDSE	Enhanced Descriptor Enable When this bit is set, the corresponding channel uses Enhanced Descriptors that are 32 Bytes for both Normal and Context Descriptors. When reset, the corresponding channel uses the descriptors that are 16 Bytes. 0b - Enhanced Descriptor is disabled 1b - Enhanced Descriptor is enabled
27-24 TQOS	Transmit QOS. This field is used to drive arqos_m_o[3:0] or awqos_m_o[3:0] output signals for all transactions of DMA Tx Channel0.
23 —	Reserved.
22	Reserved.

Table continues on the next page...

Table continued from the previous page...

Field	Function
—	
21-16 TxPBL	Transmit Programmable Burst Length These bits indicate the maximum number of beats to be transferred in one DMA block data transfer. The DMA always attempts max burst as specified in PBL each time it starts a burst transfer on the application bus. You can program PBL with any of the following values: 1, 2, 4, 8, 16, or 32. Any other value results in undefined behavior. To transfer more than 32 beats, perform the following steps: 1. Set the 8xPBL mode in DMA_CH0_Control register. 2. Set the TxPBL. Note: The maximum value of TxPBL must be less than or equal to half the Tx Queue size (TQS field of MTL_TxQ(##)_Operation_Mode register) in terms of beats. This is required so that the Tx Queue has space to store at least another Tx PBL worth of data while the MTL Tx Queue Controller is transferring data to MAC. For example, in 64-bit data width configurations the total locations in Tx Queue of size 512 bytes is 64, TxPBL and 8xPBL needs to be programmed to less than or equal to 32.
15 IPBL	Ignore PBL Requirement When this bit is set, the DMA does not check for PBL number of locations in the MTL before initiating a transfer. If space is not available, the MTL might use handshaking to slow the DMA. Note: This bit/mode must not be used when multiple Transmit DMA Channels are enabled as it might block other Transmit and Receive DMA Channels from accessing the Read Data Channel of AXI bus until space is available in Transmit Queue for current transfer. 0b - Ignore PBL Requirement is disabled 1b - Ignore PBL Requirement is enabled
14-13 —	Reserved.
12 —	Reserved.
11-5 —	Reserved.
4 OSF	Operate on Second Packet When this bit is set, it instructs the DMA to process the second packet of the Transmit data even before the status for the first packet is obtained. 0b - Operate on Second Packet disabled 1b - Operate on Second Packet enabled
3-1 —	Reserved.

Table continues on the next page...

Table continued from the previous page...

Field	Function
0	Start or Stop Transmission Command
ST	When this bit is set, transmission is placed in the Running state. The DMA checks the Transmit list at the current position for a packet to be transmitted. The DMA tries to acquire descriptor from either of the following positions: - The current position in the list This is the base address of the Transmit list set by the DMA_CH0_TxDesc_List_Address register. - The position at which the transmission was previously stopped If the DMA does not own the current descriptor, the transmission enters the Suspended state and the TBU bit of the DMA_CH0_Status register is set. The Start Transmission command is effective only when the transmission is stopped. If the command is issued before setting the DMA_CH0_TxDesc_List_Address register, the DMA behavior is unpredictable. When this bit is reset, the transmission process is placed in the Stopped state after completing the transmission of the current packet. The Next Descriptor position in the Transmit list is saved, and it becomes the current position when the transmission is restarted. To change the list address, you need to program DMA_CH0_TxDesc_List_Address register with a new value when this bit is reset. The new value is considered when this bit is set again. The stop transmission command is effective only when the transmission of the current packet is complete or the transmission is in the Suspended state. 0b - Stop Transmission Command 1b - Start Transmission Command

1.4.370 (DMA_CH2_Rx_Control)

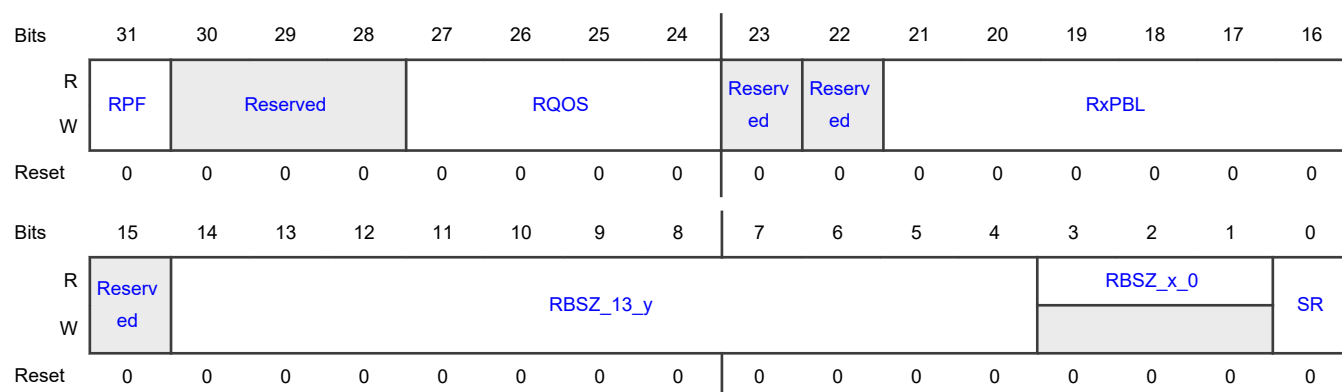
Offset

Register	Offset
DMA_CH2_Rx_Control	1208h

Function

The DMA Channel1 Receive Control register controls the Rx features such as PBL, buffer size, and extended status.

Diagram



Fields

Field	Function
31 RPF	Rx Packet Flush. When this bit is set to 1, then <code>DWC_ether_qos</code> automatically flushes the packet from the Rx Queues destined to this DMA Rx Channel, when it is stopped. When this bit remains set and the DMA is re-started by the software driver, the packets residing in the Rx Queues that were received when this RxDMA was stopped, get flushed out. The packets that are received by the MAC after the RxDMA is re-started are routed to the RxDMA. The flushing happens on the Read side of the Rx Queue. When this bit is set to 0, the <code>DWC_ether_qos</code> not flush the packet in the Rx Queue destined to this RxDMA Channel when it is STOP state. This might in turn cause head-of-line blocking in the corresponding RxQueue. Note: The stopping of packet flow from a Rx DMA Channel to the application by setting RPF works only when there is one-to-one mapping of Rx Queue to Rx DMA channels. In Dynamic mapping mode, setting RPF bit in any <code>DMA_CH(#i)_Rx_Control</code> register might flush packets from unintended Rx Queues which are destined to the stopped Rx DMA Channel. 0b - Rx Packet Flush is disabled 1b - Rx Packet Flush is enabled
30-28 —	Reserved.
27-24 RQOS	Rx AXI4 QOS. This field is used to drive <code>arqos_m_o[3:0]</code> or <code>awqos_m_o[3:0]</code> output signals for all transactions of DMA Rx Channel0.
23 —	Reserved.
22 —	Reserved.
21-16	Receive Programmable Burst Length

Table continues on the next page...

Table continued from the previous page...

Field	Function
RxPBL	These bits indicate the maximum number of beats to be transferred in one DMA block data transfer. The DMA always attempts max burst as specified in PBL each time it starts a burst transfer on the application bus. You can program PBL with any of the following values: 1, 2, 4, 8, 16, or 32. Any other value results in undefined behavior. To transfer more than 32 beats, perform the following steps: 1. Set the 8xPBL mode in the DMA_CH0_Control register. 2. Set the RxPBL. Note: The maximum value of RxPBL must be less than or equal to half the Rx Queue size (RQS field of MTL_RxQ(##)_Operation_Mode register) in terms of beats. This is required so that the Rx Queue has space to store at least another Rx PBL worth of data while the Rx DMA is transferring a block of data. For example, in 64-bit data width configurations the total locations in Rx Queue of size 512 bytes is 64, so RxPBL and 8xPBL needs to be programmed to less than or equal to 32.
15 —	Reserved.
14-4 RBSZ_13_y	Receive Buffer size High RBSZ[13:0] is split into two fields higher RBSZ_13_y and lower RBSZ_x_0. The RBSZ[13:0] field indicates the size of the Rx buffers specified in bytes. The maximum buffer size is limited to 16K bytes. The buffer size is applicable to payload buffers when split headers are enabled. Note: The buffer size must be a multiple of 4, 8, or 16 depending on the data bus widths (32-bit, 64-bit, or 128-bit respectively). This is required even if the value of buffer address pointer is not aligned to data bus width. Hence the lower RBSZ_x_0 bits are read-only and the value is considered as all-zero. Thus the RBSZ_13_y indicates the buffer size in terms of locations (with the width same as bus-width).
3-1 RBSZ_x_0	Receive Buffer size Low RBSZ[13:0] is split into two fields RBSZ_13_y and RBSZ_x_0. The RBSZ_x_0 is the lower field whose width is based on data bus width of the configuration. This field is of width 2, 3, or 4 bits for 32-bit, 64-bit, or 128-bit data bus width respectively. This field is read-only (RO).
0 SR	Start or Stop Receive When this bit is set, the DMA tries to acquire the descriptor from the Receive list and processes the incoming packets. The DMA tries to acquire descriptor from either of the following positions: - The current position in the list This is the address set by the DMA_CH0_RxDesc_List_Address register. - The position at which the Rx process was previously stopped If the DMA does not own the current descriptor, the reception is suspended and the RBU bit of the DMA_CH0_Status register is set. The Start Receive command is effective only when the reception is stopped. If the command is issued before setting the DMA_CH0_RxDesc_List_Address register, the DMA behavior is unpredictable.

Table continues on the next page...

Table continued from the previous page...

Field	Function
	When this bit is reset, the Rx DMA operation is stopped after the transfer of the current packet. The next descriptor position in the Receive list is saved, and it becomes the current position after the Rx process is restarted. The Stop Receive command is effective only when the Rx process is in the Running (waiting for Rx packet) or Suspended state.
	0b - Stop Receive
	1b - Start Receive

1.4.371 (DMA_CH2_TxDesc_List_Address)

Offset

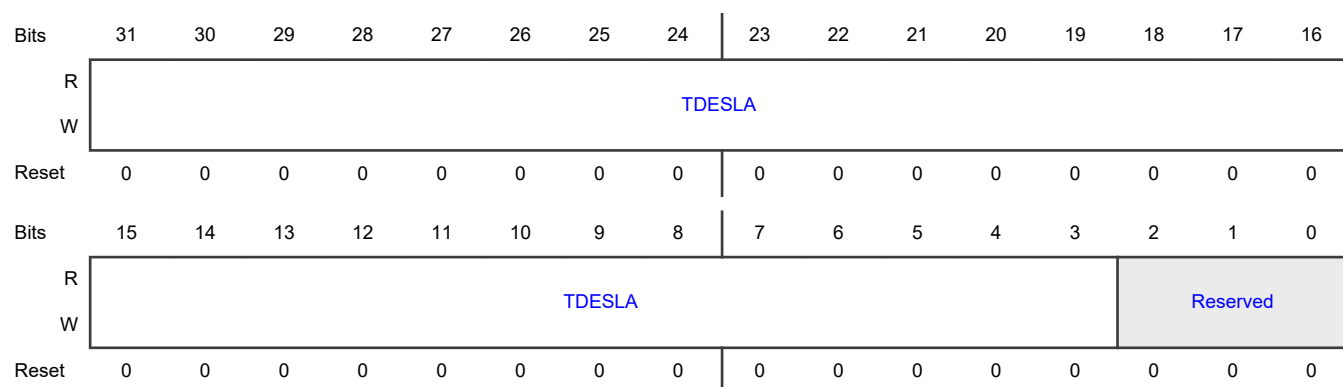
Register	Offset
DMA_CH2_TxDesc_List_Address	1214h

Function

The Channeli Tx Descriptor List Address register points the DMA to the start of Transmit descriptor list. The descriptor lists reside in the physical memory space of the application and must be Word, Dword, or Lword-aligned (for 32-bit, 64-bit, or 128-bit data bus). The DMA internally converts it to bus width aligned address by making the corresponding LSB to low.

You can write to this register only when the Tx DMA has stopped, that is, the ST bit is set to zero in DMA_CH0_Tx_Control register. When stopped, this register can be written with a new descriptor list address. When you set the ST bit to 1, the DMA takes the newly-programmed descriptor base address. If this register is not changed when the ST bit is set to 0, the DMA takes the descriptor address where it was stopped earlier.

Diagram



Fields

Field	Function
31-3	Start of Transmit List

Table continues on the next page...

Table continued from the previous page...

Field	Function
TDESLA	This field contains the base address of the first descriptor in the Transmit descriptor list. The DMA ignores the LSB bits (1:0, 2:0, or 3:0) for 32-bit, 64-bit, or 128-bit bus width and internally takes these bits as all-zero. Therefore, these LSB bits are read-only (RO). The width of this field depends on the configuration: - 31:2 for 32-bit configuration - 31:3 for 64-bit configuration - 31:4 for 128-bit configuration
2-0 —	Reserved.

1.4.372 (DMA_CH2_RxDesc_List_Address)

Offset

Register	Offset
DMA_CH2_RxDesc_List_Address	121Ch

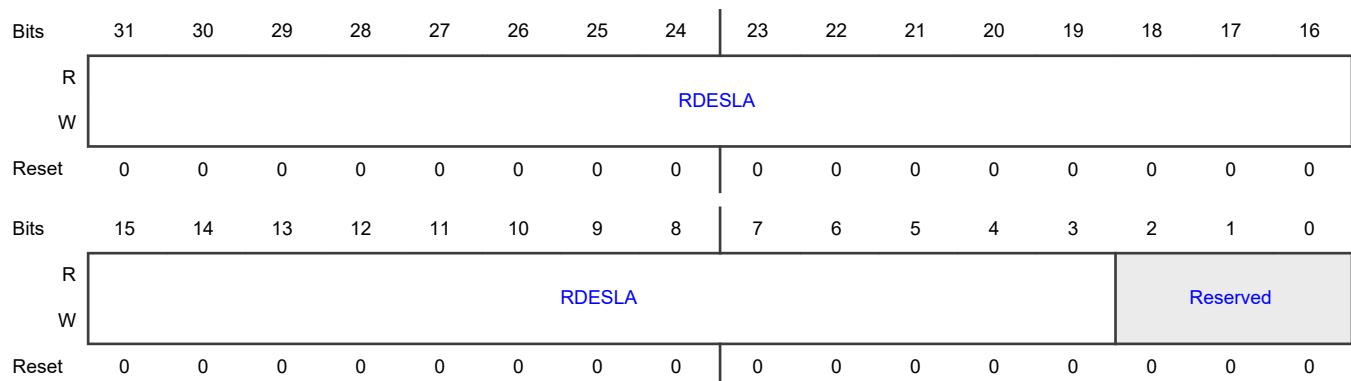
Function

The Channel1 Rx Descriptor List Address register points the DMA to the start of Receive descriptor list.

This register points to the start of the Receive Descriptor List. The descriptor lists reside in the physical memory space of the application and must be Word, Dword, or Lword-aligned (for 32-bit, 64-bit, or 128-bit data bus). The DMA internally converts it to bus width aligned address by making the corresponding LS bits low. Writing to this register is permitted only when reception is stopped. When stopped, this register must be written to before the receive Start command is given. You can write to this register only when Rx DMA has stopped, that is, SR bit is set to zero in DMA_CH0_Rx_Control register. When stopped, this register can be written with a new descriptor list address.

When you set the SR bit to 1, the DMA takes the newly programmed descriptor base address.

Diagram



Fields

Field	Function
31-3 RDESLA	Start of Receive List This field contains the base address of the first descriptor in the Rx Descriptor list. The DMA ignores the LSB bits (1:0, 2:0, or 3:0) for 32-bit, 64-bit, or 128-bit bus width and internally takes these bits as all-zero. Therefore, these LSB bits are read-only (RO). The width of this field depends on the configuration: - 31:2 for 32-bit configuration - 31:3 for 64-bit configuration - 31:4 for 128-bit configuration
2-0 —	Reserved.

1.4.373 (DMA_CH2_TxDesc_Tail_Pointer)

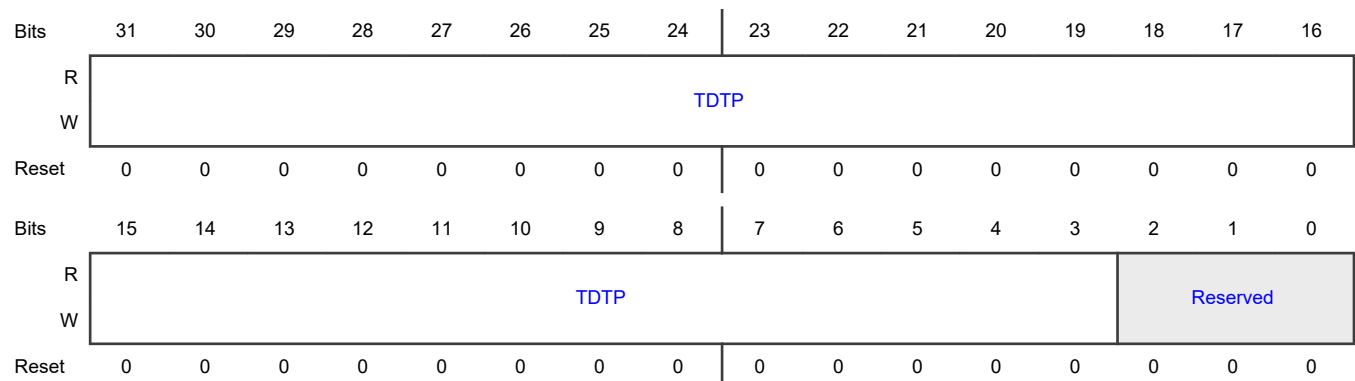
Offset

Register	Offset
DMA_CH2_TxDesc_Tail_Pointer	1220h

Function

The Channel1 Tx Descriptor Tail Pointer register points to an offset from the base and indicates the location of the last valid descriptor.

Diagram



Fields

Field	Function
31-3 TDTP	Transmit Descriptor Tail Pointer This field contains the tail pointer for the Tx descriptor ring. The software writes the tail pointer to add more descriptors to the Tx channel. The hardware tries to transmit all packets referenced by the descriptors between the head and the tail pointer registers. The width of this field depends on the configuration: - 31:2 for 32-bit configuration - 31:3 for 64-bit configuration - 31:4 for 128-bit configuration
2-0 —	Reserved.

1.4.374 (DMA_CH2_RxDesc_Tail_Pointer)

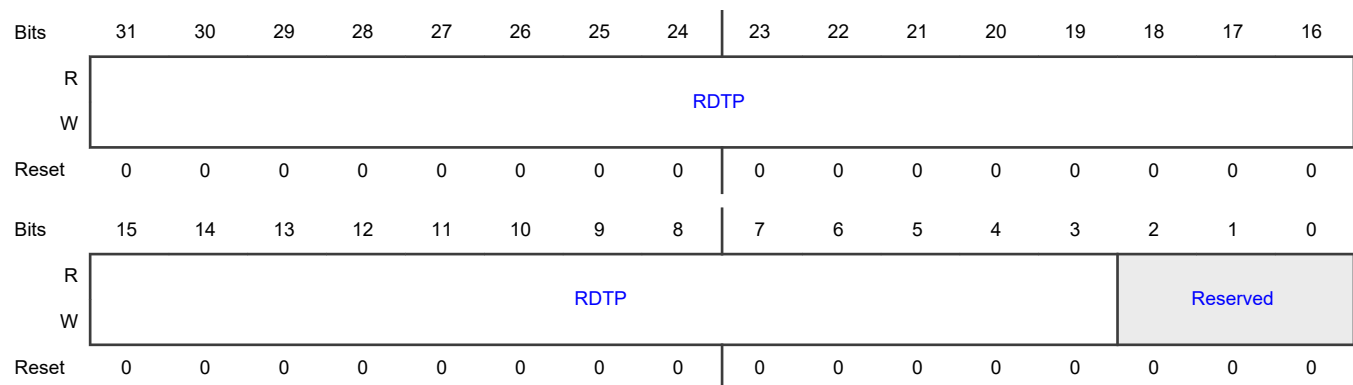
Offset

Register	Offset
DMA_CH2_RxDesc_Tail_Pointer	1228h

Function

The Channel1 Rx Descriptor Tail Pointer Points to an offset from the base and indicates the location of the last valid descriptor.

Diagram



Fields

Field	Function
31-3 RDTP	Receive Descriptor Tail Pointer This field contains the tail pointer for the Rx descriptor ring. The software writes the tail pointer to add more descriptors to the Rx channel. The hardware tries to write all received packets to the descriptors referenced between the head and the tail pointer registers. The width of this field depends on the configuration: - 31:2 for 32-bit configuration - 31:3 for 64-bit configuration - 31:4 for 128-bit configuration
2-0 —	Reserved.

1.4.375 (DMA_CH2_TxDesc_Ring_Length)

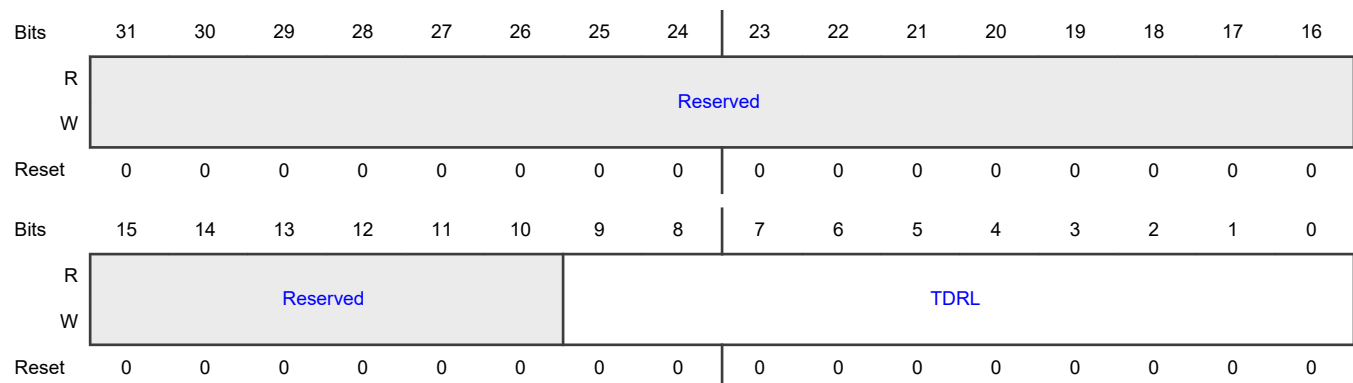
Offset

Register	Offset
DMA_CH2_TxDesc_Ring_Length	122Ch

Function

The Tx Descriptor Ring Length register contains the length of the Transmit descriptor ring.

Diagram



Fields

Field	Function
31-10 —	Reserved.
9-0 TDRL	Transmit Descriptor Ring Length This field sets the maximum number of Tx descriptors in the circular descriptor ring. The maximum number of descriptors is limited to 1K descriptors. Synopsys recommends a minimum ring descriptor length of 4. For example, You can program any value up to 0x3FF in this field. This field is 10 bits wide, if you program 0x3FF, you can have 1024 descriptors. If you want to have 10 descriptors, program it to a value of 0x9.

1.4.376 (DMA_CH2_RxDesc_Ring_Length)

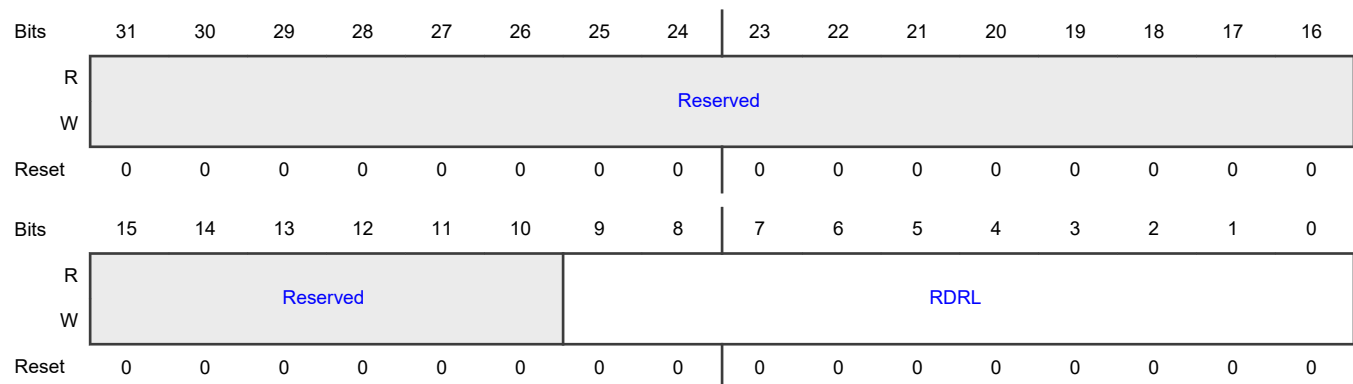
Offset

Register	Offset
DMA_CH2_RxDesc_Ring_Length	1230h

Function

The Channel1 Rx Descriptor Ring Length register contains the length of the Receive descriptor circular ring.

Diagram



Fields

Field	Function
31-10 —	Reserved.
9-0	Receive Descriptor Ring Length

Table continues on the next page...

Table continued from the previous page...

Field	Function
RDRL	This register sets the maximum number of Rx descriptors in the circular descriptor ring. The maximum number of descriptors is limited to 1K descriptors. For example, You can program any value up to 0x3FF in this field. This field is 10 bits wide, if you program 0x3FF, you can have 1024 descriptors. If you want to have 10 descriptors, program it to a value of 0x9.

1.4.377 (DMA_CH2_Interrupt_Enable)

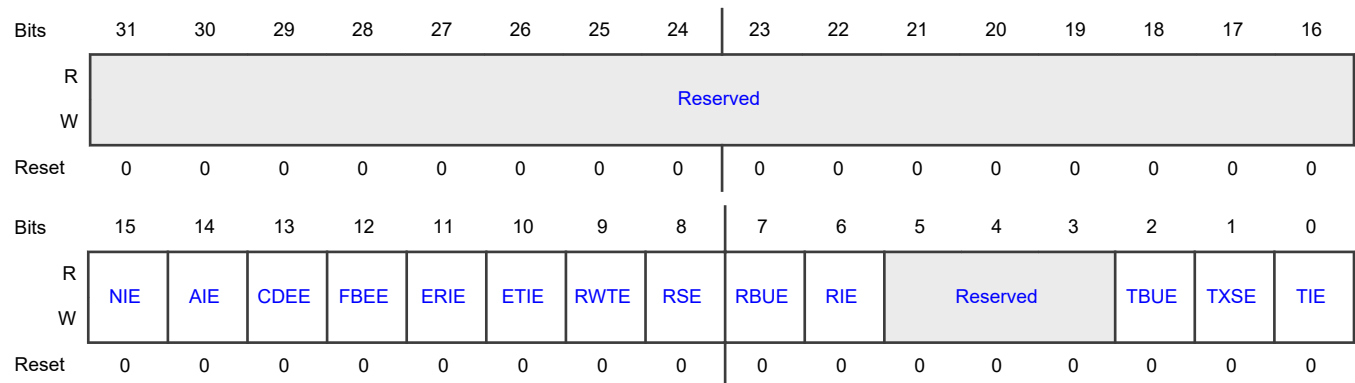
Offset

Register	Offset
DMA_CH2_Interrupt_Enable	1234h

Function

The Channeli Interrupt Enable register enables the interrupts reported by the Status register.

Diagram



Fields

Field	Function
31-16 —	Reserved.
15 NIE	Normal Interrupt Summary Enable When this bit is set, the normal interrupt summary is enabled. This bit enables the following interrupts in the DMA_CH0_Status register: - Bit 0: Transmit Interrupt - Bit 2: Transmit Buffer Unavailable

Table continues on the next page...

Table continued from the previous page...

Field	Function
	- Bit 6: Receive Interrupt - Bit 11: Early Receive Interrupt When this bit is reset, the normal interrupt summary is disabled. 0b - Normal Interrupt Summary is disabled 1b - Normal Interrupt Summary is enabled
14 AIE	Abnormal Interrupt Summary Enable When this bit is set, the abnormal interrupt summary is enabled. This bit enables the following interrupts in the DMA_CH0_Status register: - Bit 1: Transmit Process Stopped - Bit 7: Rx Buffer Unavailable - Bit 8: Receive Process Stopped - Bit 9: Receive Watchdog Timeout - Bit 10: Early Transmit Interrupt - Bit 12: Fatal Bus Error - Bit 13: Context Descriptor Error When this bit is reset, the abnormal interrupt summary is disabled. 0b - Abnormal Interrupt Summary is disabled 1b - Abnormal Interrupt Summary is enabled
13 CDEE	Context Descriptor Error Enable When this bit is set along with the AIE bit, the Descriptor error interrupt is enabled. When this bit is reset, the Descriptor error interrupt is disabled. 0b - Context Descriptor Error is disabled 1b - Context Descriptor Error is enabled
12 FBEE	Fatal Bus Error Enable When this bit is set along with the AIE bit, the Fatal Bus error interrupt is enabled. When this bit is reset, the Fatal Bus Error error interrupt is disabled. 0b - Fatal Bus Error is disabled 1b - Fatal Bus Error is enabled
11 ERIE	Early Receive Interrupt Enable When this bit is set along with the NIE bit, the Early Receive interrupt is enabled. When this bit is reset, the Early Receive interrupt is disabled. 0b - Early Receive Interrupt is disabled 1b - Early Receive Interrupt is enabled

Table continues on the next page...

Table continued from the previous page...

Field	Function
10 ETIE	Early Transmit Interrupt Enable When this bit is set along with the AIE bit, the Early Transmit interrupt is enabled. When this bit is reset, the Early Transmit interrupt is disabled. 0b - Early Transmit Interrupt is disabled 1b - Early Transmit Interrupt is enabled
9 RWTE	Receive Watchdog Timeout Enable When this bit is set along with the AIE bit, the Receive Watchdog Timeout interrupt is enabled. When this bit is reset, the Receive Watchdog Timeout interrupt is disabled. 0b - Receive Watchdog Timeout is disabled 1b - Receive Watchdog Timeout is enabled
8 RSE	Receive Stopped Enable When this bit is set along with the AIE bit, the Receive Stopped Interrupt is enabled. When this bit is reset, the Receive Stopped interrupt is disabled. 0b - Receive Stopped is disabled 1b - Receive Stopped is enabled
7 RBUE	Receive Buffer Unavailable Enable When this bit is set along with the AIE bit, the Receive Buffer Unavailable interrupt is enabled. When this bit is reset, the Receive Buffer Unavailable interrupt is disabled. 0b - Receive Buffer Unavailable is disabled 1b - Receive Buffer Unavailable is enabled
6 RIE	Receive Interrupt Enable When this bit is set along with the NIE bit, the Receive Interrupt is enabled. When this bit is reset, the Receive Interrupt is disabled. 0b - Receive Interrupt is disabled 1b - Receive Interrupt is enabled
5-3 —	Reserved.
2 TBUE	Transmit Buffer Unavailable Enable When this bit is set along with the NIE bit, the Transmit Buffer Unavailable interrupt is enabled. When this bit is reset, the Transmit Buffer Unavailable interrupt is disabled. 0b - Transmit Buffer Unavailable is disabled 1b - Transmit Buffer Unavailable is enabled
1	Transmit Stopped Enable

Table continues on the next page...

Table continued from the previous page...

Field	Function
TXSE	When this bit is set along with the AIE bit, the Transmission Stopped interrupt is enabled. When this bit is reset, the Transmission Stopped interrupt is disabled. 0b - Transmit Stopped is disabled 1b - Transmit Stopped is enabled
0 TIE	Transmit Interrupt Enable When this bit is set along with the NIE bit, the Transmit Interrupt is enabled. When this bit is reset, the Transmit Interrupt is disabled. 0b - Transmit Interrupt is disabled 1b - Transmit Interrupt is enabled

1.4.378 (DMA_CH2_Rx_Interrupt_Watchdog_Timer)

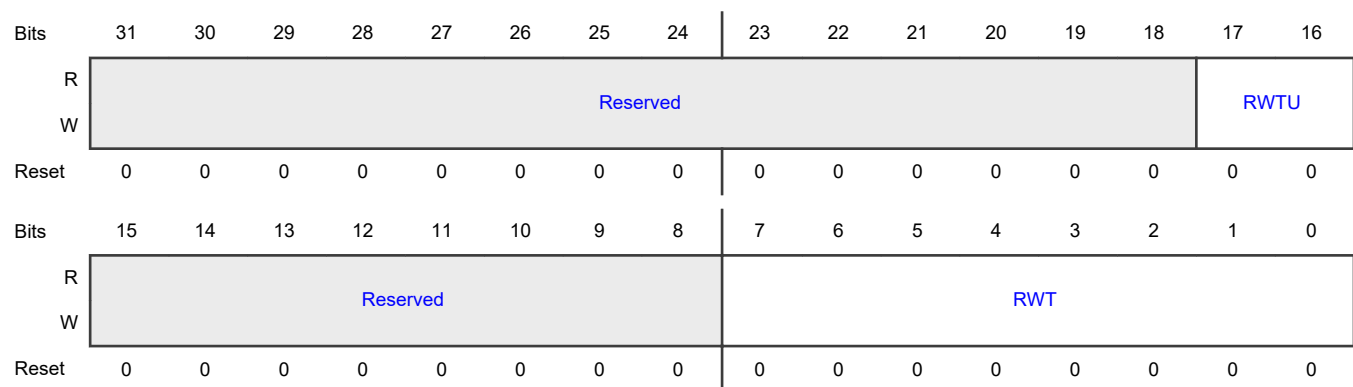
Offset

Register	Offset
DMA_CH2_Rx_Interrupt_Watchdog_Timer	1238h

Function

The Receive Interrupt Watchdog Timer register indicates the watchdog timeout for Receive Interrupt (RI) from the DMA. When this register is written with a non-zero value, it enables the watchdog timer for the RI bit of the DMA_CHi_Status register.

Diagram



Fields

Field	Function
31-18	Reserved.

Table continues on the next page...

Table continued from the previous page...

Field	Function
—	
17-16 RWTU	Receive Interrupt Watchdog Timer Count Units This field indicates the number of system clock cycles corresponding to one unit in RWT field. - 2'b00: 256 - 2'b01: 512 - 2'b10: 1024 - 2'b11: 2048 For example, when RWT=2 and RWTU=1, the watchdog timer is set for 2*512=1024 system clock cycles.
15-8 —	Reserved.
7-0 RWT	Receive Interrupt Watchdog Timer Count This field indicates the number of system clock cycles, multiplied by factor indicated in RWTU field, for which the watchdog timer is set. The watchdog timer is triggered with the programmed value after the Rx DMA completes the transfer of a packet for which the RI bit is not set in the DMA_CH(#i)_Status register, because of the setting of Interrupt Enable bit in the corresponding descriptor RDES3[30]. When the watchdog timer runs out, the RI bit is set and the timer is stopped. The watchdog timer is reset when the RI bit is set high because of automatic setting of RI as per the Interrupt Enable bit RDES3[30] of any received packet.

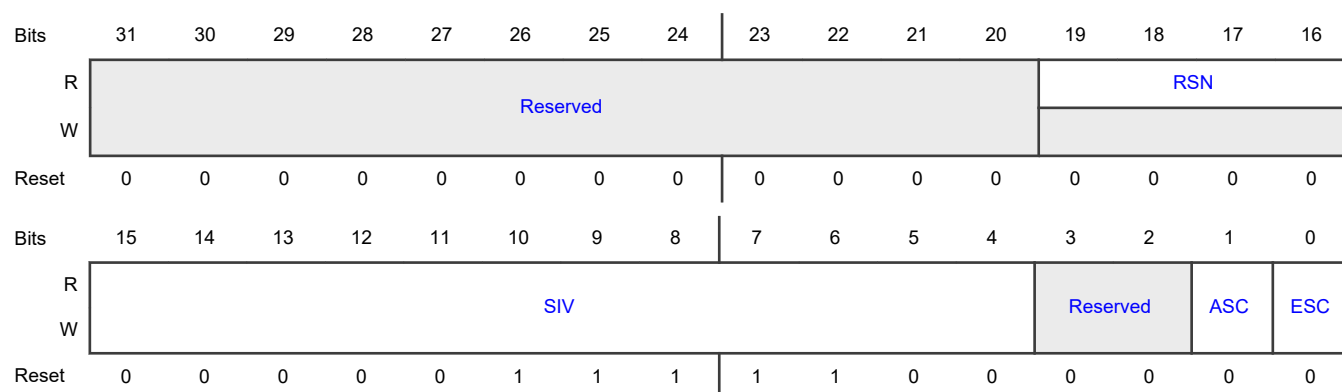
1.4.379 (DMA_CH2_Slot_Function_Control_Status)

Offset

Register	Offset
DMA_CH2_Slot_Function_Control_Status	123Ch

Function

The Slot Function Control and Status register contains the control bits for slot function and the status for Transmit path.

Diagram**Fields**

Field	Function
31-20 —	Reserved.
19-16 RSN	Reference Slot Number This field gives the current value of the reference slot number in the DMA. It is used for slot comparison.
15-4 SIV	Slot Interval Value This field controls the period of the slot interval in which the TxDMA fetches the scheduled packets. A value of 0 specifies the slot interval of 1 us while the maximum value 4095 specifies the slot interval of 4096us. The default/reset value is 0x07C which corresponds to slot interval of 125us
3-2 —	Reserved.
1 ASC	Advance Slot Check When set, this bit enables the DMA to fetch the data from the buffer when the slot number (SLOTNUM) programmed in the Tx descriptor is - equal to the reference slot number given in the RSN field or - ahead of the reference slot number by up to two slots This bit is applicable only when the ESC bit is set. 0b - Advance Slot Check is disabled 1b - Advance Slot Check is enabled
0 ESC	Enable Slot Comparison When set, this bit enables the checking of the slot numbers programmed in the Tx descriptor with the current reference given in the RSN field. The DMA fetches the data from the corresponding buffer only when the slot number is

Table continues on the next page...

Table continued from the previous page...

Field	Function
	- equal to the reference slot number or - ahead of the reference slot number by one slot When reset, this bit disables the checking of the slot numbers. The DMA fetches the data immediately after the descriptor is processed. Note: The UFO (UDP Fragmentation over IPv4)/TSO/USO should not be enabled along with TBS/AVB Slot number check. The UFO/TSO/USO involves multiple packets/segments/fragments transmission for single packet received from application and the slot number check are applicable for fetching of only first segment/fragment. As a result it might be difficult for software to specify slot number for subsequent packets. 0b - Slot Comparison is disabled 1b - Slot Comparison is enabled

1.4.380 (DMA_CH2_Current_App_TxDesc)

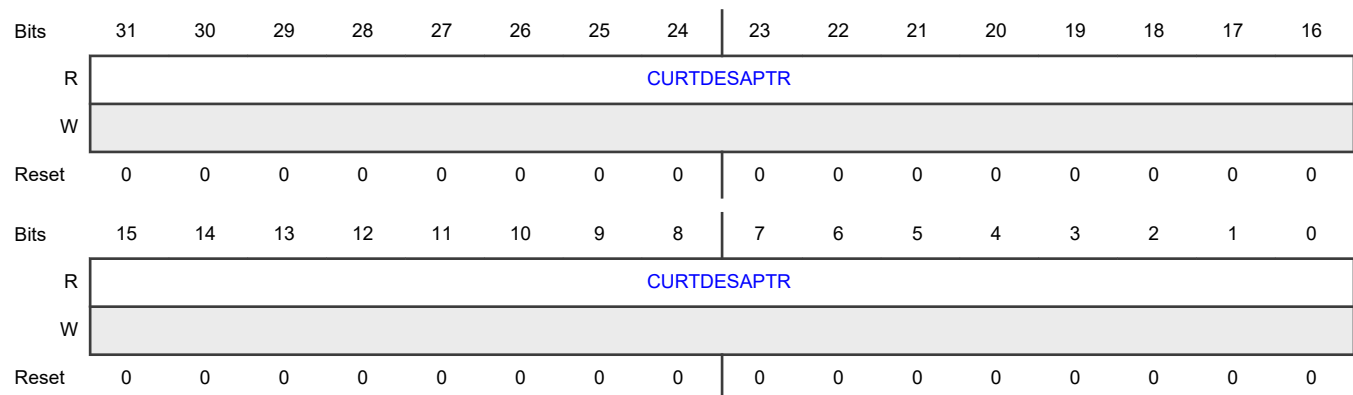
Offset

Register	Offset
DMA_CH2_Current_App_TxDesc	1244h

Function

The Channeli Current Application Transmit Descriptor register points to the current Transmit descriptor read by the DMA.

Diagram



Fields

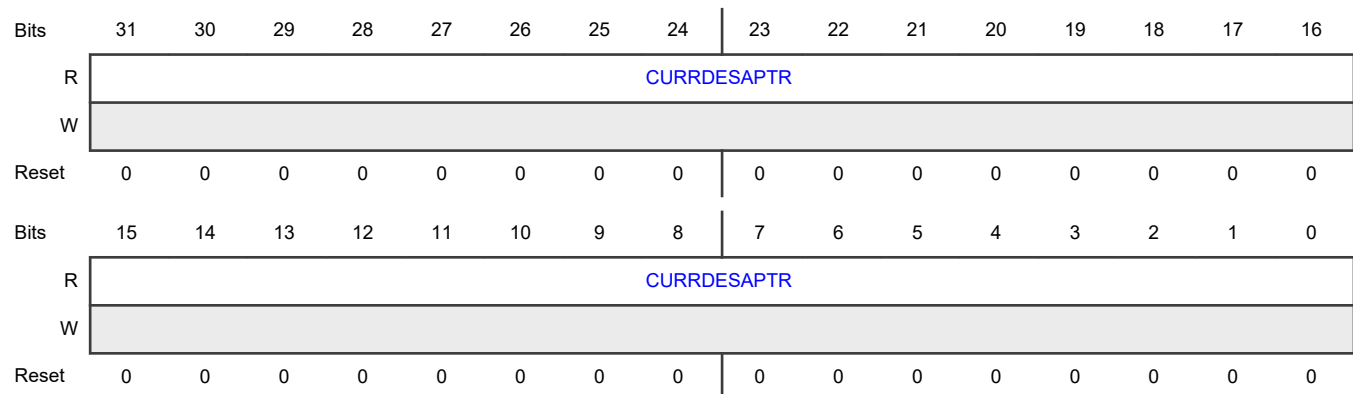
Field	Function
31-0 CURTDESAPTR	Application Transmit Descriptor Address Pointer The DMA updates this pointer during Tx operation. This pointer is cleared on reset.

1.4.381 (DMA_CH2_Current_App_RxDesc)**Offset**

Register	Offset
DMA_CH2_Current_App_RxDesc	124Ch

Function

The Channeli Current Application Receive Descriptor register points to the current Receive descriptor read by the DMA.

Diagram**Fields**

Field	Function
31-0 CURRDESAPTR	Application Receive Descriptor Address Pointer The DMA updates this pointer during Rx operation. This pointer is cleared on reset.

1.4.382 (DMA_CH2_Current_App_TxBuffer)

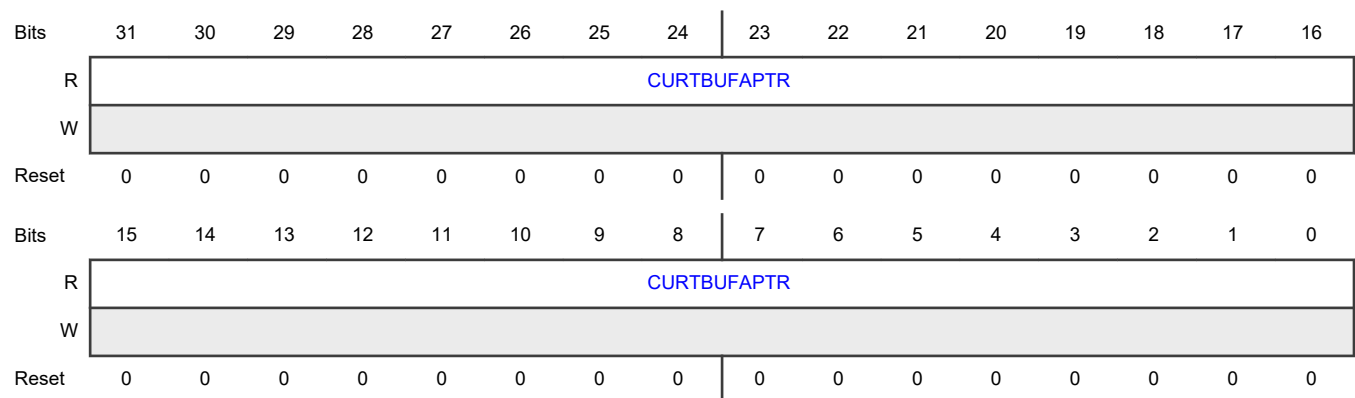
Offset

Register	Offset
DMA_CH2_Current_App_TxBuffer	1254h

Function

The Channel1 Current Application Transmit Buffer Address register points to the current Tx buffer address read by the DMA.

Diagram



Fields

Field	Function
31-0	Application Transmit Buffer Address Pointer
CURTBUFAPTR	The DMA updates this pointer during Tx operation. This pointer is cleared on reset.
R	

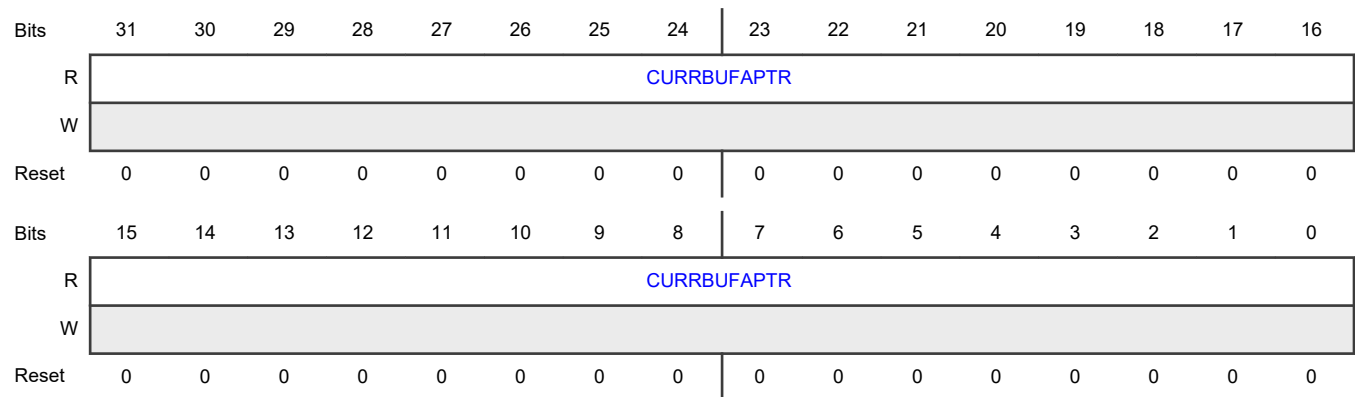
1.4.383 (DMA_CH2_Current_App_RxBuffer)

Offset

Register	Offset
DMA_CH2_Current_App_RxBuffer	125Ch

Function

The Channel 0 Current Application Receive Buffer Address register points to the current Rx buffer address read by the DMA.

Diagram**Fields**

Field	Function
31-0 CURRBUFAPT R	Application Receive Buffer Address Pointer The DMA updates this pointer during Rx operation. This pointer is cleared on reset.

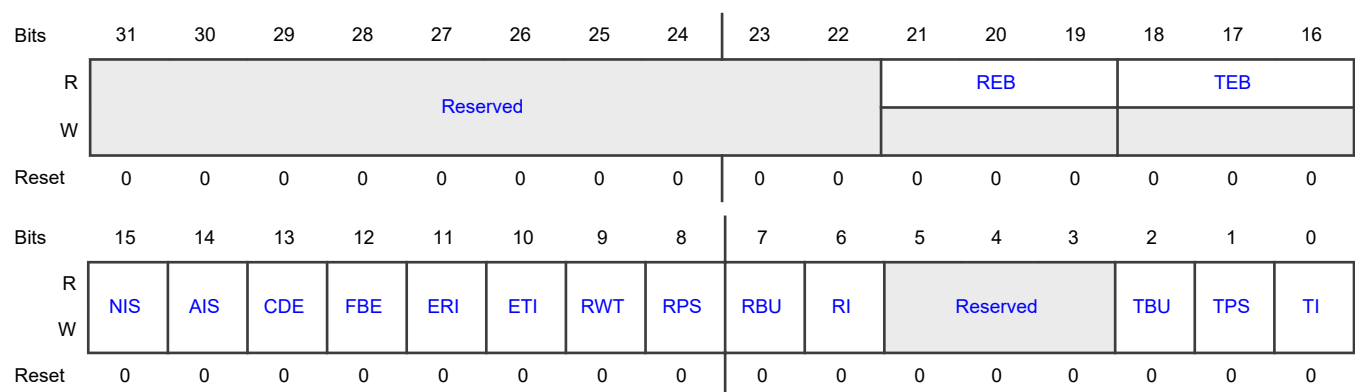
1.4.384 (DMA_CH2_Status)**Offset**

Register	Offset
DMA_CH2_Status	1260h

Function

The software driver (application) reads the Status register during interrupt service routine or polling to determine the status of the DMA.

Note: The number of DMA_CH(#i)_Status register in the configuration is the higher of number of Rx DMA Channels and Tx DMA Channels.

Diagram

Fields

Field	Function
31-22 —	Reserved.
21-19 REB	Rx DMA Error Bits This field indicates the type of error that caused a Bus Error. For example, error response on the AHB or AXI interface. - Bit 21 -- 1'b1: Error during data transfer by Rx DMA -- 1'b0: No Error during data transfer by Rx DMA - Bit 20 -- 1'b1: Error during descriptor access -- 1'b0: Error during data buffer access - Bit 19 -- 1'b1: Error during read transfer -- 1'b0: Error during write transfer This field is valid only when the FBE bit is set. This field does not generate an interrupt.
18-16 TEB	Tx DMA Error Bits This field indicates the type of error that caused a Bus Error. For example, error response on the AHB or AXI interface. - Bit 18 -- 1'b1: Error during data transfer by Tx DMA -- 1'b0: No Error during data transfer by Tx DMA - Bit 17 -- 1'b1: Error during descriptor access -- 1'b0: Error during data buffer access - Bit 16 -- 1'b1: Error during read transfer -- 1'b0: Error during write transfer This field is valid only when the FBE bit is set. This field does not generate an interrupt.
15 NIS	Normal Interrupt Summary Normal Interrupt Summary bit value is the logical OR of the following bits when the corresponding interrupt bits are enabled in the DMA_CH0_Interrupt_Enable register: - Bit 0: Transmit Interrupt - Bit 2: Transmit Buffer Unavailable - Bit 6: Receive Interrupt

Table continues on the next page...

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Field	Function
	- Bit 11: Early Receive Interrupt Only unmasked bits (interrupts for which interrupt enable is set in DMA_CH0_Interrupt_Enable register) affect the Normal Interrupt Summary bit. This is a sticky bit. You must clear this bit (by writing 1 to this bit) each time a corresponding bit which causes NIS to be set is cleared. Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect. 0b - Normal Interrupt Summary status not detected 1b - Normal Interrupt Summary status detected
14 AIS	Abnormal Interrupt Summary Abnormal Interrupt Summary bit value is the logical OR of the following when the corresponding interrupt bits are enabled in the DMA_CH0_Interrupt_Enable register: - Bit 1: Transmit Process Stopped - Bit 7: Receive Buffer Unavailable - Bit 8: Receive Process Stopped - Bit 10: Early Transmit Interrupt - Bit 12: Fatal Bus Error - Bit 13: Context Descriptor Error Only unmasked bits affect the Abnormal Interrupt Summary bit. This is a sticky bit. You must clear this bit (by writing 1 to this bit) each time a corresponding bit, which causes AIS to be set, is cleared. Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect. 0b - Abnormal Interrupt Summary status not detected 1b - Abnormal Interrupt Summary status detected
13 CDE	Context Descriptor Error This bit indicates that the DMA Tx/Rx engine received a descriptor error, which indicates invalid context in the middle of packet flow (intermediate descriptor) or all one's descriptor in Tx case and on Rx side it indicates DMA has read a descriptor with either of the buffer address as ones which is considered to be invalid. Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect. 0b - Context Descriptor Error status not detected 1b - Context Descriptor Error status detected
12 FBE	Fatal Bus Error This bit indicates that a bus error occurred (as described in the EB field). When this bit is set, the corresponding DMA channel engine disables all bus accesses. Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect.

Table continues on the next page...

Table continued from the previous page...

Field	Function
	0b - Fatal Bus Error status not detected 1b - Fatal Bus Error status detected
11 ERI	Early Receive Interrupt This bit when set indicates that the RxDMA has completed the transfer of packet data to the memory. In configs supporting ERIC, When ERIC=0, this bit is set only after the Rx DMA has filled up a complete receive buffer with packet data. When ERIC=1, this bit is set after every burst transfer of data from the Rx DMA to the buffer. The setting of RI bit automatically clears this bit. Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect. 0b - Early Receive Interrupt status not detected 1b - Early Receive Interrupt status detected
10 ETI	Early Transmit Interrupt This bit when set indicates that the TxDMA has completed the transfer of packet data to the MTL TXFIFO memory. In configs supporting ERIC: When ETIC=0, this bit is set only after the Tx DMA has transferred a complete packet to MTL. When ETIC=1, this bit is set after completion of (partial) packet data transfer from buffers in the Transmit descriptor in which IOC=1. Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect. 0b - Early Transmit Interrupt status not detected 1b - Early Transmit Interrupt status detected
9 RWT	Receive Watchdog Timeout This bit is asserted when a packet with length greater than 2,048 bytes (10,240 bytes when Jumbo Packet mode is enabled) is received. 0b - Receive Watchdog Timeout status not detected 1b - Receive Watchdog Timeout status detected
8 RPS	Receive Process Stopped This bit is asserted when the Rx process enters the Stopped state. Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect. 0b - Receive Process Stopped status not detected 1b - Receive Process Stopped status detected
7 RBU	Receive Buffer Unavailable This bit indicates that the application owns the next descriptor in the Receive list, and the DMA cannot acquire it. The Rx process is suspended. To resume processing Rx descriptors, the application should change the ownership of the descriptor and issue a Receive Poll Demand command. If this command is

Table continues on the next page...

Table continued from the previous page...

Field	Function
	not issued, the Rx process resumes when the next recognized incoming packet is received. In ring mode, the application should advance the Receive Descriptor Tail Pointer register of a channel. This bit is set only when the DMA owns the previous Rx descriptor. Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect. 0b - Receive Buffer Unavailable status not detected 1b - Receive Buffer Unavailable status detected
6 RI	Receive Interrupt This bit indicates that the packet reception is complete. When packet reception is complete, Bit 31 of RDES3 is reset in the last descriptor, and the specific packet status information is updated in the descriptor. The reception remains in the Running state. Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect. 0b - Receive Interrupt status not detected 1b - Receive Interrupt status detected
5-3 —	Reserved.
2 TBU	Transmit Buffer Unavailable This bit indicates that the application owns the next descriptor in the Transmit list, and the DMA cannot acquire it. Transmission is suspended. The TPS0 field of the DMA_Debug_Status0 register explains the Transmit Process state transitions. To resume processing the Transmit descriptors, the application should do the following: 1. Change the ownership of the descriptor by setting Bit 31 of TDES3. 2. Issue a Transmit Poll Demand command. For ring mode, the application should advance the Transmit Descriptor Tail Pointer register of a channel. Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect. 0b - Transmit Buffer Unavailable status not detected 1b - Transmit Buffer Unavailable status detected
1 TPS	Transmit Process Stopped This bit is set when the transmission is stopped. Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect. 0b - Transmit Process Stopped status not detected 1b - Transmit Process Stopped status detected
0 TI	Transmit Interrupt This bit indicates that the packet transmission is complete. When transmission is complete, Bit 31 of TDES3 is reset in the last descriptor, and the specific packet status information is updated in the descriptor.

Table continues on the next page...

Table continued from the previous page...

Field	Function
	Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect. 0b - Transmit Interrupt status not detected 1b - Transmit Interrupt status detected

1.4.385 (DMA_CH2_Miss_Frame_Cnt)

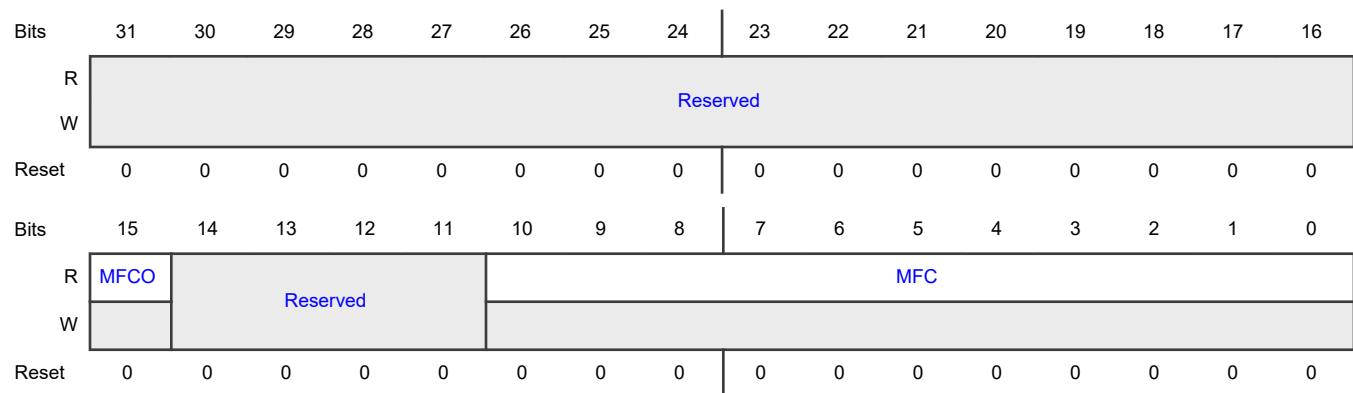
Offset

Register	Offset
DMA_CH2_Miss_Frame_Cnt	1264h

Function

This register has the number of packet counter that got dropped by the DMA either due to Bus Error or due to programming RPF field in DMA_CH*{i}*_Rx_Control register.

Diagram



Fields

Field	Function
31-16 —	Reserved.
15 MFCO	Overflow status of the MFC Counter When this bit is set then the MFC counter does not get incremented further. The bit gets cleared when this register is read. Access restriction applies. Clears on read. Self-set to 1 on internal event.

Table continues on the next page...

Table continued from the previous page...

Field	Function
	0b - Miss Frame Counter overflow not occurred 1b - Miss Frame Counter overflow occurred
14-11 —	Reserved.
10-0 MFC	Dropped Packet Counters This counter indicates the number of packet counters that are dropped by the DMA either because of bus error or because of programing RPF field in DMA_CH\${i}_Rx_Control register. The counter gets cleared when this register is read. Access restriction applies. Clears on read. Self-set to 1 on internal event.

1.4.386 (DMA_CH2_RXP_Accept_Cnt)

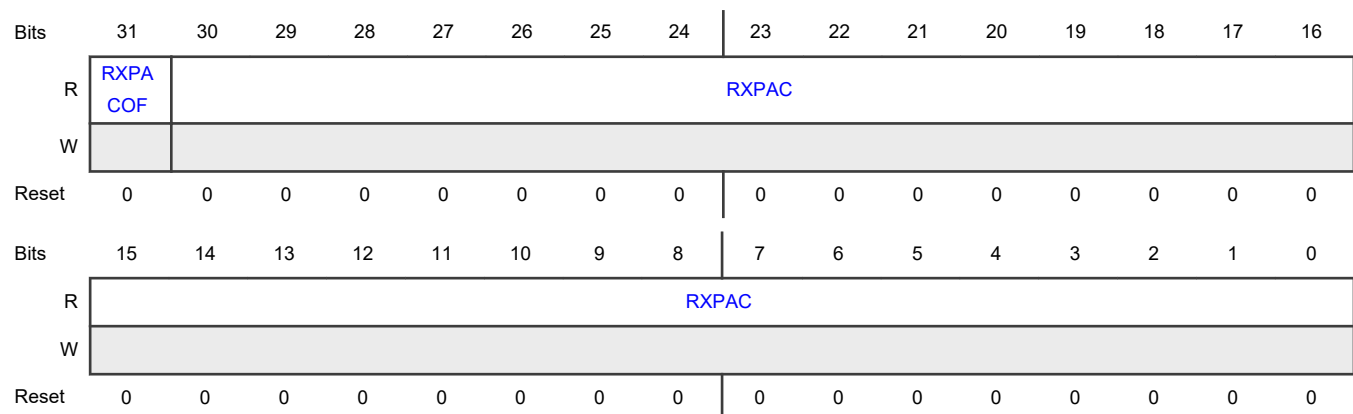
Offset

Register	Offset
DMA_CH2_RXP_Accept_Cnt	1268h

Function

The DMA_CH(#i)_RXP_Accept_Cnt registers provides the count of the number of frames accepted by Rx Parser.

Diagram



Fields

Field	Function
31 RXPACOF	Rx Parser Accept Counter Overflow Bit When set, this bit indicates that the RXPAC Counter field crossed the maximum limit. Access restriction applies. Clears on read. Self-set to 1 on internal event. 0b - Rx Parser Accept Counter overflow not occurred 1b - Rx Parser Accept Counter overflow occurred
30-0 RXPAC	Rx Parser Accept Counter This 31-bit counter is implemented whenever a Rx Parser Accept a packet due to AF =1. The counter is cleared when the register is read.

1.4.387 (DMA_CH2_RX_ERI_Cnt)

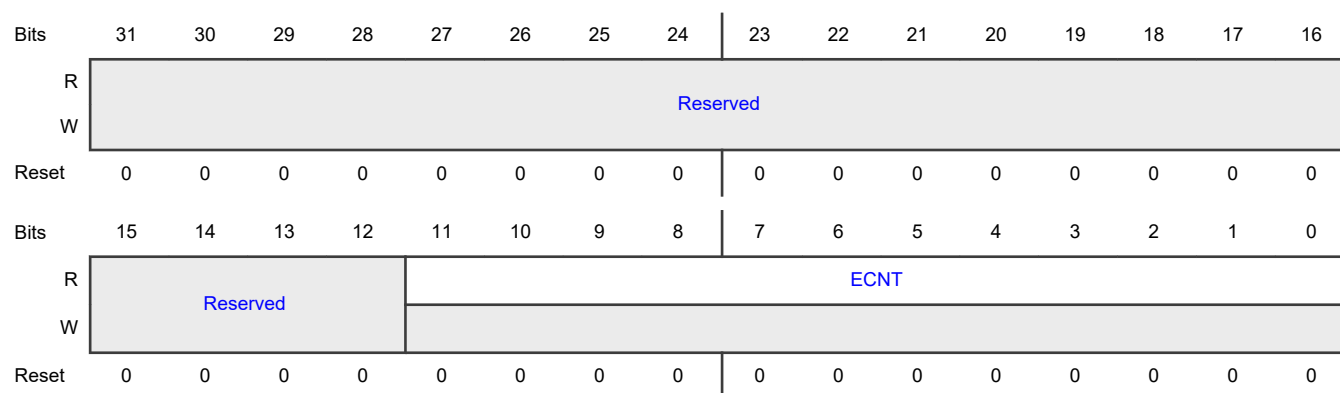
Offset

Register	Offset
DMA_CH2_RX_ERI_Cnt	126Ch

Function

The DMA_CH(#i)_RX_ERI_Cnt registers provides the count of the number of times ERI was asserted.

Diagram



Fields

Field	Function
31-12 —	Reserved.

Table continues on the next page...

Table continued from the previous page...

Field	Function
11-0 ECNT	ERI Counter When ERIC bit of DMA_CH(#i)_RX_Control register is set, this counter increments for burst transfer completed by the Rx DMA from the start of packet transfer. This counter is reset at the start of new packet.

1.4.388 (DMA_CH3_Control)

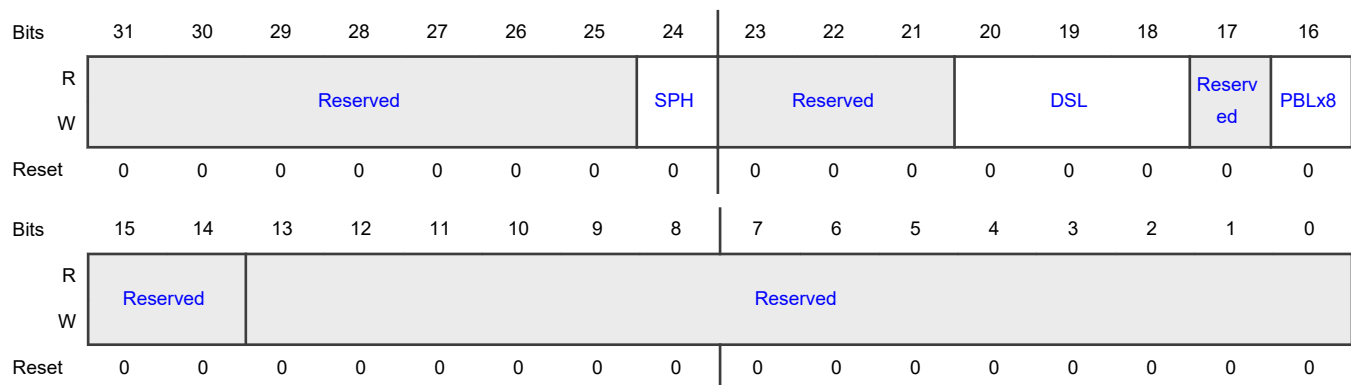
Offset

Register	Offset
DMA_CH3_Control	1280h

Function

The DMA Channel Control register specifies the MSS value for segmentation, length to skip between two descriptors, and also the features such as header splitting and 8xPBL mode.

Diagram



Fields

Field	Function
31-25 —	Reserved.
24 SPH	Split Headers When this bit is set, the DMA splits the header and payload in the Receive path. The DMA writes the header to the Buffer Address1 of RDES0. The DMA writes the payload to the buffer to which the Buffer Address2 is pointing. The software must ensure that the header fits into the Receive buffers. If the header length exceeds the receive buffer size, the DMA does not split the header and payload.

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Table continued from the previous page...

Field	Function
	This bit is available only if Enable Split Header Structure option is selected. 0b - Split Headers feature is disabled 1b - Split Headers feature is enabled
23-21 —	Reserved.
20-18 DSL	Descriptor Skip Length This bit specifies the Word, Dword, or Lword number (depending on the 32-bit, 64-bit, or 128-bit bus) to skip between two unchained descriptors. The address skipping starts from the end of the current descriptor to the start of the next descriptor. When the DSL value is equal to zero, the DMA takes the descriptor table as contiguous.
17 —	Reserved.
16 PBLx8	8xPBL mode When this bit is set, the PBL value programmed in Bits[21:16] in DMA_CH(#i)_Tx_Control and Bits[21:16] in DMA_CH(#i)_Rx_Control is multiplied by eight times. Therefore, the DMA transfers the data in 8, 16, 32, 64, 128, and 256 beats depending on the PBL value. 0b - 8xPBL mode is disabled 1b - 8xPBL mode is enabled
15-14 —	Reserved.
13-0 —	Reserved.

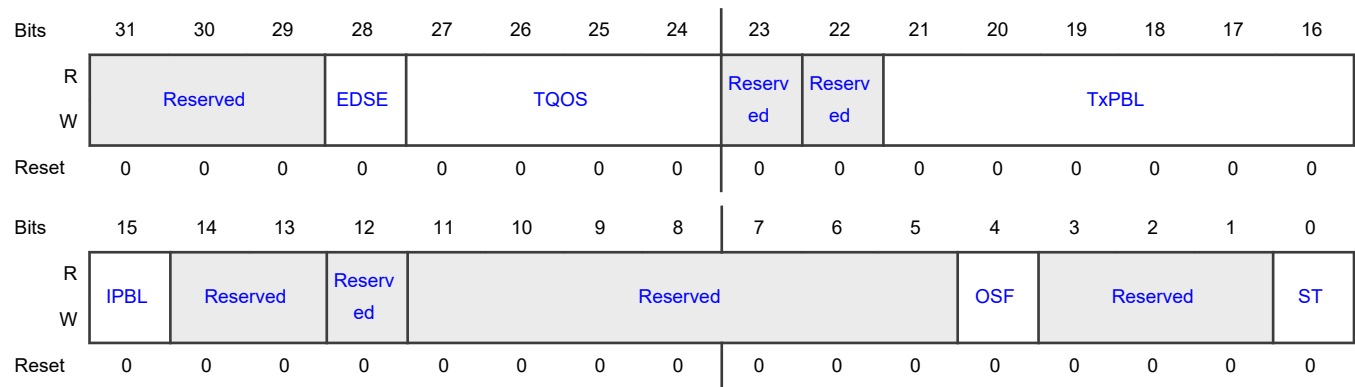
1.4.389 (DMA_CH3_Tx_Control)

Offset

Register	Offset
DMA_CH3_Tx_Control	1284h

Function

The DMA Channeli Transmit Control register controls the Tx features such as PBL, TCP segmentation, and Tx Channel weights.

Diagram**Fields**

Field	Function
31-29 —	Reserved.
28 EDSE	Enhanced Descriptor Enable When this bit is set, the corresponding channel uses Enhanced Descriptors that are 32 Bytes for both Normal and Context Descriptors. When reset, the corresponding channel uses the descriptors that are 16 Bytes. 0b - Enhanced Descriptor is disabled 1b - Enhanced Descriptor is enabled
27-24 TQOS	Transmit QOS. This field is used to drive arqos_m_o[3:0] or awqos_m_o[3:0] output signals for all transactions of DMA Tx Channel0.
23 —	Reserved.
22 —	Reserved.
21-16 TxPBL	Transmit Programmable Burst Length These bits indicate the maximum number of beats to be transferred in one DMA block data transfer. The DMA always attempts max burst as specified in PBL each time it starts a burst transfer on the application bus. You can program PBL with any of the following values: 1, 2, 4, 8, 16, or 32. Any other value results in undefined behavior. To transfer more than 32 beats, perform the following steps: 1. Set the 8xPBL mode in DMA_CH0_Control register. 2. Set the TxPBL.

Table continues on the next page...

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Field	Function
	Note: The maximum value of TxPBL must be less than or equal to half the Tx Queue size (TQS field of MTL_TxQ(#i)_Operation_Mode register) in terms of beats. This is required so that the Tx Queue has space to store at least another Tx PBL worth of data while the MTL Tx Queue Controller is transferring data to MAC. For example, in 64-bit data width configurations the total locations in Tx Queue of size 512 bytes is 64, TxPBL and 8xPBL needs to be programmed to less than or equal to 32.
15 IPBL	Ignore PBL Requirement When this bit is set, the DMA does not check for PBL number of locations in the MTL before initiating a transfer. If space is not available, the MTL might use handshaking to slow the DMA. Note: This bit/mode must not be used when multiple Transmit DMA Channels are enabled as it might block other Transmit and Receive DMA Channels from accessing the Read Data Channel of AXI bus until space is available in Transmit Queue for current transfer. 0b - Ignore PBL Requirement is disabled 1b - Ignore PBL Requirement is enabled
14-13 —	Reserved.
12 —	Reserved.
11-5 —	Reserved.
4 OSF	Operate on Second Packet When this bit is set, it instructs the DMA to process the second packet of the Transmit data even before the status for the first packet is obtained. 0b - Operate on Second Packet disabled 1b - Operate on Second Packet enabled
3-1 —	Reserved.
0 ST	Start or Stop Transmission Command When this bit is set, transmission is placed in the Running state. The DMA checks the Transmit list at the current position for a packet to be transmitted. The DMA tries to acquire descriptor from either of the following positions: - The current position in the list This is the base address of the Transmit list set by the DMA_CH0_TxDesc_List_Address register. - The position at which the transmission was previously stopped

Table continues on the next page...

Table continued from the previous page...

Field	Function
	If the DMA does not own the current descriptor, the transmission enters the Suspended state and the TBU bit of the DMA_CH0_Status register is set. The Start Transmission command is effective only when the transmission is stopped. If the command is issued before setting the DMA_CH0_TxDesc_List_Address register, the DMA behavior is unpredictable. When this bit is reset, the transmission process is placed in the Stopped state after completing the transmission of the current packet. The Next Descriptor position in the Transmit list is saved, and it becomes the current position when the transmission is restarted. To change the list address, you need to program DMA_CH0_TxDesc_List_Address register with a new value when this bit is reset. The new value is considered when this bit is set again. The stop transmission command is effective only when the transmission of the current packet is complete or the transmission is in the Suspended state. 0b - Stop Transmission Command 1b - Start Transmission Command

1.4.390 (DMA_CH3_Rx_Control)

Offset

Register	Offset
DMA_CH3_Rx_Control	1288h

Function

The DMA Channel1 Receive Control register controls the Rx features such as PBL, buffer size, and extended status.

Diagram

Bits	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	RPF	Reserved				RQOS				Reserv ed	Reserv ed	RxPBL				
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bits	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	Reserv ed	RBSZ_13_y										RBSZ_x_0			SR	
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Fields

Field	Function
31	Rx Packet Flush.

Table continues on the next page...

Table continued from the previous page...

Field	Function
RPF	When this bit is set to 1, then <code>DWC_ether_qos</code> automatically flushes the packet from the Rx Queues destined to this DMA Rx Channel, when it is stopped. When this bit remains set and the DMA is re-started by the software driver, the packets residing in the Rx Queues that were received when this RxDMA was stopped, get flushed out. The packets that are received by the MAC after the RxDMA is re-started are routed to the RxDMA. The flushing happens on the Read side of the Rx Queue. When this bit is set to 0, the <code>DWC_ether_qos</code> not flush the packet in the Rx Queue destined to this RxDMA Channel when it is STOP state. This might in turn cause head-of-line blocking in the corresponding RxQueue. Note: The stopping of packet flow from a Rx DMA Channel to the application by setting RPF works only when there is one-to-one mapping of Rx Queue to Rx DMA channels. In Dynamic mapping mode, setting RPF bit in any <code>DMA_CH(#i)_Rx_Control</code> register might flush packets from unintended Rx Queues which are destined to the stopped Rx DMA Channel. 0b - Rx Packet Flush is disabled 1b - Rx Packet Flush is enabled
30-28 —	Reserved.
27-24 RQOS	Rx AXI4 QOS. This field is used to drive <code>arqos_m_o[3:0]</code> or <code>awqos_m_o[3:0]</code> output signals for all transactions of DMA Rx Channel0.
23 —	Reserved.
22 —	Reserved.
21-16 RxPBL	Receive Programmable Burst Length These bits indicate the maximum number of beats to be transferred in one DMA block data transfer. The DMA always attempts max burst as specified in PBL each time it starts a burst transfer on the application bus. You can program PBL with any of the following values: 1, 2, 4, 8, 16, or 32. Any other value results in undefined behavior. To transfer more than 32 beats, perform the following steps: 1. Set the 8xPBL mode in the <code>DMA_CH0_Control</code> register. 2. Set the RxPBL. Note: The maximum value of RxPBL must be less than or equal to half the Rx Queue size (RQS field of <code>MTL_RxQ(#i)_Operation_Mode</code> register) in terms of beats. This is required so that the Rx Queue has space to store at least another Rx PBL worth of data while the Rx DMA is transferring a block of data. For example, in 64-bit data width configurations the total locations in Rx Queue of size 512 bytes is 64, so RxPBL and 8xPBL needs to be programmed to less than or equal to 32.
15	Reserved.

Table continues on the next page...

Table continued from the previous page...

Field	Function
—	
14-4 RBSZ_13_y	Receive Buffer size High RBSZ[13:0] is split into two fields higher RBSZ_13_y and lower RBSZ_x_0. The RBSZ[13:0] field indicates the size of the Rx buffers specified in bytes. The maximum buffer size is limited to 16K bytes. The buffer size is applicable to payload buffers when split headers are enabled. Note: The buffer size must be a multiple of 4, 8, or 16 depending on the data bus widths (32-bit, 64-bit, or 128-bit respectively). This is required even if the value of buffer address pointer is not aligned to data bus width. Hence the lower RBSZ_x_0 bits are read-only and the value is considered as all-zero. Thus the RBSZ_13_y indicates the buffer size in terms of locations (with the width same as bus-width).
3-1 RBSZ_x_0	Receive Buffer size Low RBSZ[13:0] is split into two fields RBSZ_13_y and RBSZ_x_0. The RBSZ_x_0 is the lower field whose width is based on data bus width of the configuration. This field is of width 2, 3, or 4 bits for 32-bit, 64-bit, or 128-bit data bus width respectively. This field is read-only (RO).
0 SR	Start or Stop Receive When this bit is set, the DMA tries to acquire the descriptor from the Receive list and processes the incoming packets. The DMA tries to acquire descriptor from either of the following positions: - The current position in the list This is the address set by the DMA_CH0_RxDesc_List_Address register. - The position at which the Rx process was previously stopped If the DMA does not own the current descriptor, the reception is suspended and the RBU bit of the DMA_CH0_Status register is set. The Start Receive command is effective only when the reception is stopped. If the command is issued before setting the DMA_CH0_RxDesc_List_Address register, the DMA behavior is unpredictable. When this bit is reset, the Rx DMA operation is stopped after the transfer of the current packet. The next descriptor position in the Receive list is saved, and it becomes the current position after the Rx process is restarted. The Stop Receive command is effective only when the Rx process is in the Running (waiting for Rx packet) or Suspended state. 0b - Stop Receive 1b - Start Receive

1.4.391 (DMA_CH3_TxDesc_List_Address)

Offset

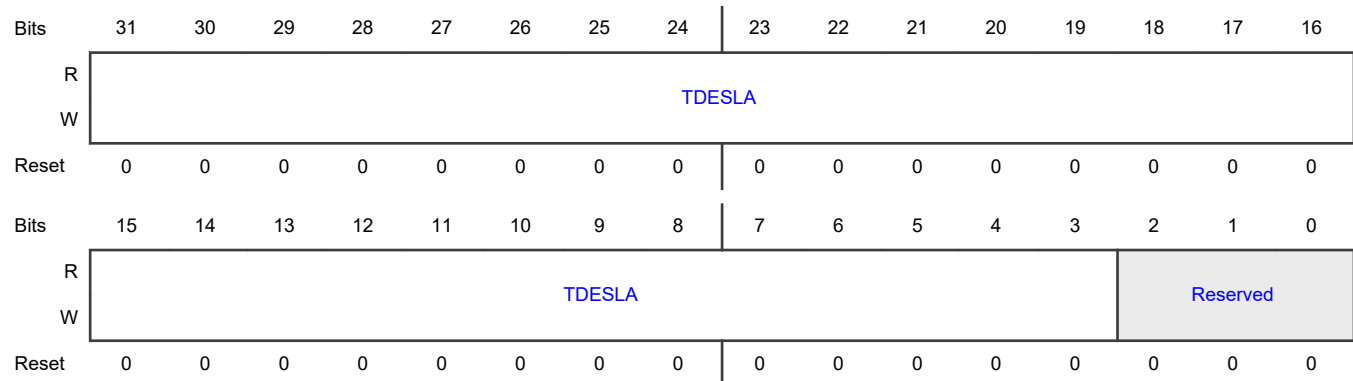
Register	Offset
DMA_CH3_TxDesc_List_Address	1294h

Function

The Channel Tx Descriptor List Address register points the DMA to the start of Transmit descriptor list. The descriptor lists reside in the physical memory space of the application and must be Word, Dword, or Lword-aligned (for 32-bit, 64-bit, or 128-bit data bus). The DMA internally converts it to bus width aligned address by making the corresponding LSB to low.

You can write to this register only when the Tx DMA has stopped, that is, the ST bit is set to zero in DMA_CH0_Tx_Control register. When stopped, this register can be written with a new descriptor list address. When you set the ST bit to 1, the DMA takes the newly-programmed descriptor base address. If this register is not changed when the ST bit is set to 0, the DMA takes the descriptor address where it was stopped earlier.

Diagram



Fields

Field	Function
31-3 TDESLA	Start of Transmit List This field contains the base address of the first descriptor in the Transmit descriptor list. The DMA ignores the LSB bits (1:0, 2:0, or 3:0) for 32-bit, 64-bit, or 128-bit bus width and internally takes these bits as all-zero. Therefore, these LSB bits are read-only (RO). The width of this field depends on the configuration: - 31:2 for 32-bit configuration - 31:3 for 64-bit configuration - 31:4 for 128-bit configuration
2-0 —	Reserved.

1.4.392 (DMA_CH3_RxDesc_List_Address)

Offset

Register	Offset
DMA_CH3_RxDesc_List_Address	129Ch

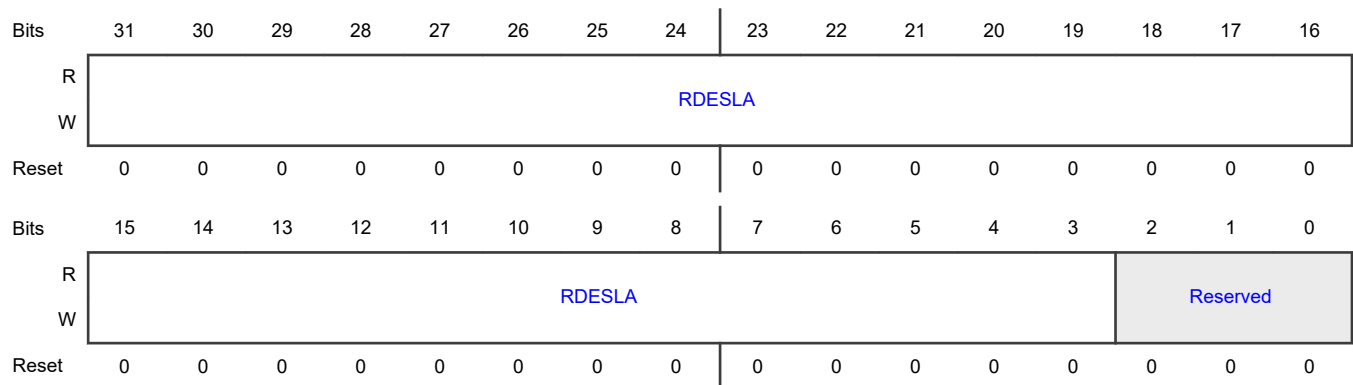
Function

The Channel Rx Descriptor List Address register points the DMA to the start of Receive descriptor list.

This register points to the start of the Receive Descriptor List. The descriptor lists reside in the physical memory space of the application and must be Word, Dword, or Lword-aligned (for 32-bit, 64-bit, or 128-bit data bus). The DMA internally converts it to bus width aligned address by making the corresponding LS bits low. Writing to this register is permitted only when reception is stopped. When stopped, this register must be written to before the receive Start command is given. You can write to this register only when Rx DMA has stopped, that is, SR bit is set to zero in DMA_CH0_Rx_Control register. When stopped, this register can be written with a new descriptor list address.

When you set the SR bit to 1, the DMA takes the newly programmed descriptor base address.

Diagram



Fields

Field	Function
31-3 RDESLA	Start of Receive List This field contains the base address of the first descriptor in the Rx Descriptor list. The DMA ignores the LSB bits (1:0, 2:0, or 3:0) for 32-bit, 64-bit, or 128-bit bus width and internally takes these bits as all-zero. Therefore, these LSB bits are read-only (RO). The width of this field depends on the configuration: - 31:2 for 32-bit configuration - 31:3 for 64-bit configuration - 31:4 for 128-bit configuration
2-0 —	Reserved.

1.4.393 (DMA_CH3_TxDesc_Tail_Pointer)

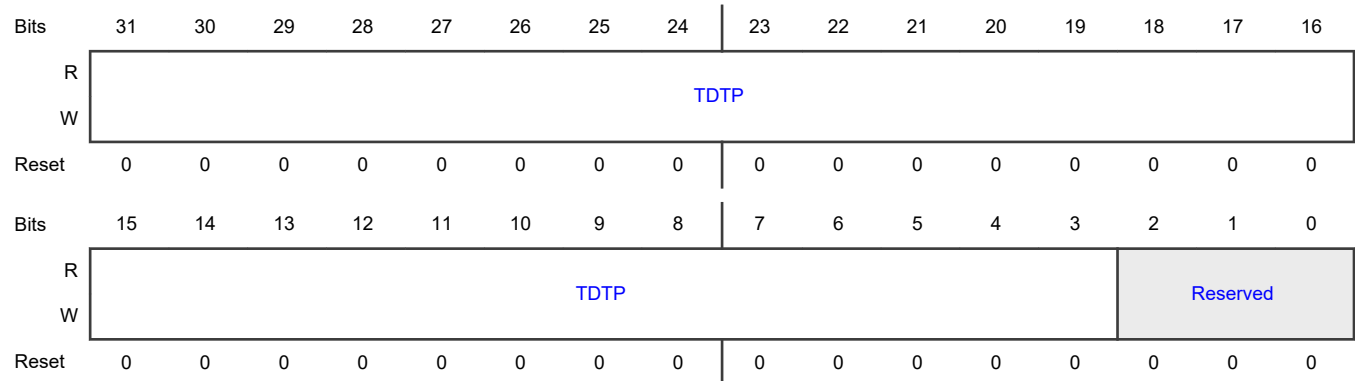
Offset

Register	Offset
DMA_CH3_TxDesc_Tail_Pointer	12A0h

Function

The Channeli Tx Descriptor Tail Pointer register points to an offset from the base and indicates the location of the last valid descriptor.

Diagram



Fields

Field	Function
31-3 TDTP	Transmit Descriptor Tail Pointer This field contains the tail pointer for the Tx descriptor ring. The software writes the tail pointer to add more descriptors to the Tx channel. The hardware tries to transmit all packets referenced by the descriptors between the head and the tail pointer registers. The width of this field depends on the configuration: - 31:2 for 32-bit configuration - 31:3 for 64-bit configuration - 31:4 for 128-bit configuration
2-0 —	Reserved.

1.4.394 (DMA_CH3_RxDesc_Tail_Pointer)

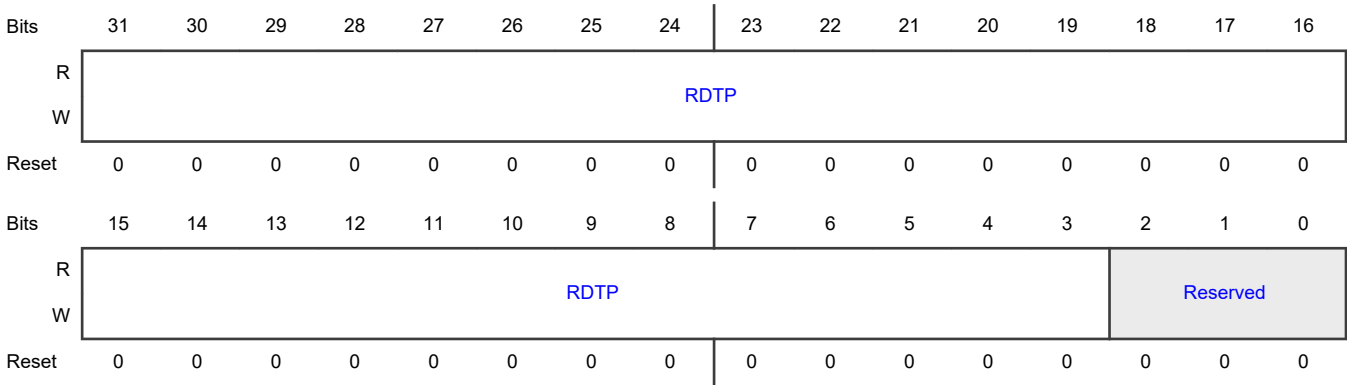
Offset

Register	Offset
DMA_CH3_RxDesc_Tail_Pointer	12A8h

Function

The Channeli Rx Descriptor Tail Pointer Points to an offset from the base and indicates the location of the last valid descriptor.

Diagram



Fields

Field	Function
31-3 RDTP	Receive Descriptor Tail Pointer This field contains the tail pointer for the Rx descriptor ring. The software writes the tail pointer to add more descriptors to the Rx channel. The hardware tries to write all received packets to the descriptors referenced between the head and the tail pointer registers. The width of this field depends on the configuration: - 31:2 for 32-bit configuration - 31:3 for 64-bit configuration - 31:4 for 128-bit configuration
2-0 —	Reserved.

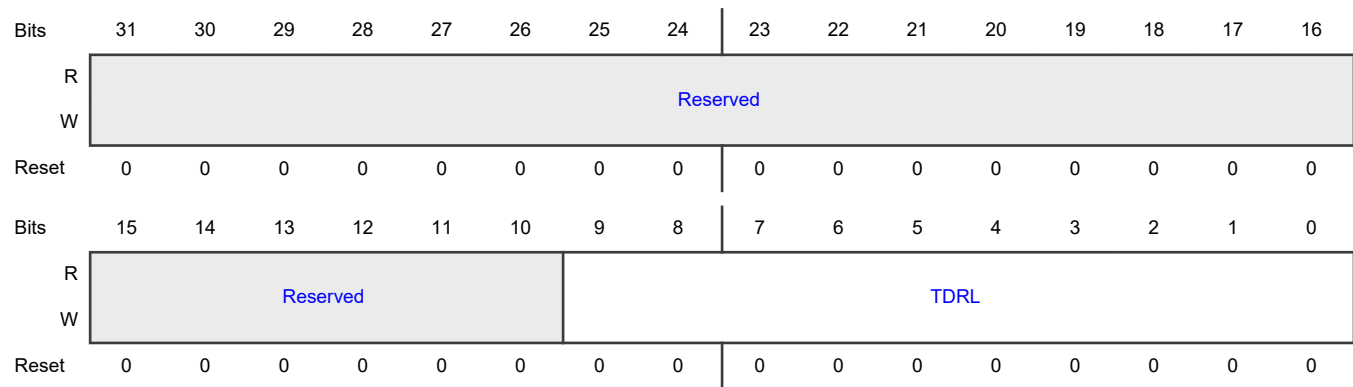
1.4.395 (DMA_CH3_TxDesc_Ring_Length)

Offset

Register	Offset
DMA_CH3_TxDesc_Ring_Length	12ACh

Function

The Tx Descriptor Ring Length register contains the length of the Transmit descriptor ring.

Diagram**Fields**

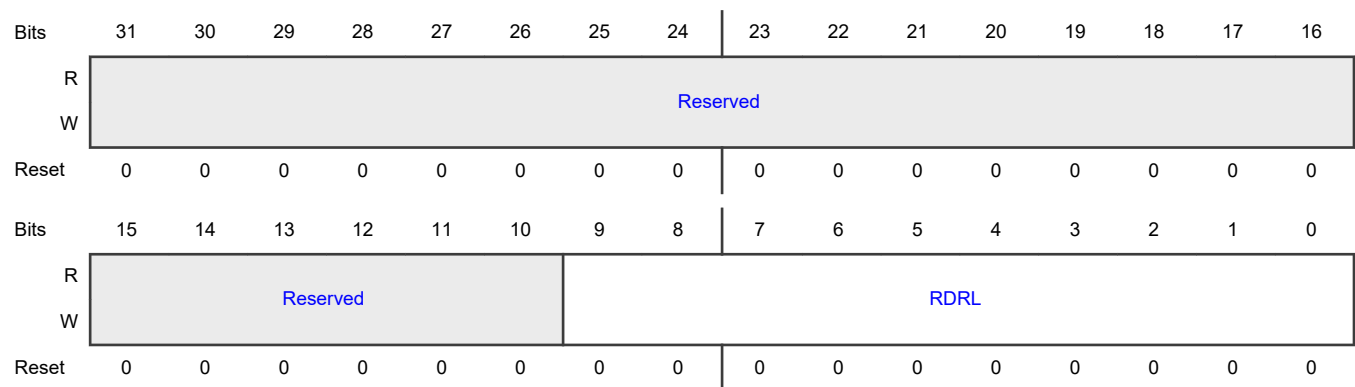
Field	Function
31-10 —	Reserved.
9-0 TDRL	Transmit Descriptor Ring Length This field sets the maximum number of Tx descriptors in the circular descriptor ring. The maximum number of descriptors is limited to 1K descriptors. Synopsys recommends a minimum ring descriptor length of 4. For example, You can program any value up to 0x3FF in this field. This field is 10 bits wide, if you program 0x3FF, you can have 1024 descriptors. If you want to have 10 descriptors, program it to a value of 0x9.

1.4.396 (DMA_CH3_RxDesc_Ring_Length)**Offset**

Register	Offset
DMA_CH3_RxDesc_Ring_Length	12B0h

Function

The Channel Rx Descriptor Ring Length register contains the length of the Receive descriptor circular ring.

Diagram**Fields**

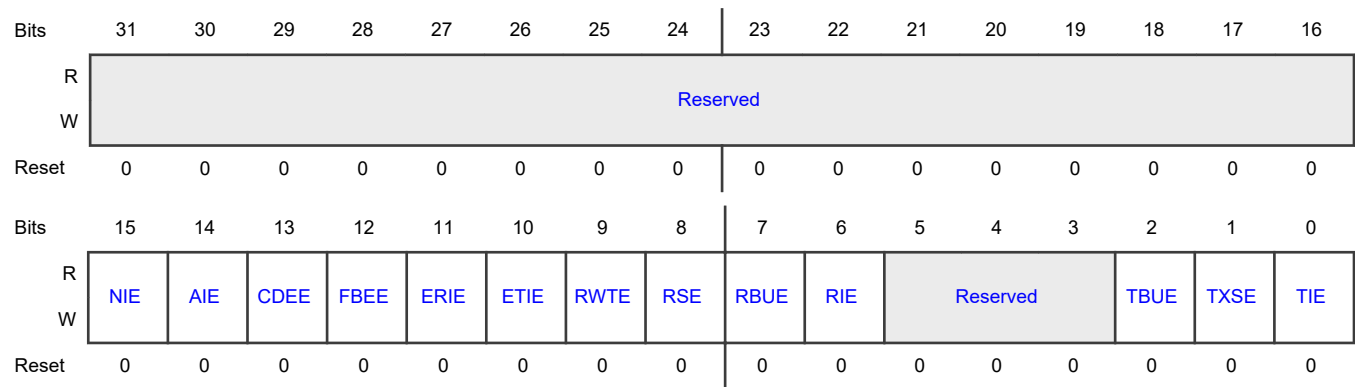
Field	Function
31-10 —	Reserved.
9-0 RDRL	Receive Descriptor Ring Length This register sets the maximum number of Rx descriptors in the circular descriptor ring. The maximum number of descriptors is limited to 1K descriptors. For example, You can program any value up to 0x3FF in this field. This field is 10 bits wide, if you program 0x3FF, you can have 1024 descriptors. If you want to have 10 descriptors, program it to a value of 0x9.

1.4.397 (DMA_CH3_Interrupt_Enable)**Offset**

Register	Offset
DMA_CH3_Interrupt_Enable	12B4h

Function

The Channeli Interrupt Enable register enables the interrupts reported by the Status register.

Diagram**Fields**

Field	Function
31-16 —	Reserved.
15 NIE	Normal Interrupt Summary Enable When this bit is set, the normal interrupt summary is enabled. This bit enables the following interrupts in the DMA_CH0_Status register: - Bit 0: Transmit Interrupt - Bit 2: Transmit Buffer Unavailable - Bit 6: Receive Interrupt - Bit 11: Early Receive Interrupt When this bit is reset, the normal interrupt summary is disabled. 0b - Normal Interrupt Summary is disabled 1b - Normal Interrupt Summary is enabled
14 AIE	Abnormal Interrupt Summary Enable When this bit is set, the abnormal interrupt summary is enabled. This bit enables the following interrupts in the DMA_CH0_Status register: - Bit 1: Transmit Process Stopped - Bit 7: Rx Buffer Unavailable - Bit 8: Receive Process Stopped - Bit 9: Receive Watchdog Timeout - Bit 10: Early Transmit Interrupt - Bit 12: Fatal Bus Error - Bit 13: Context Descriptor Error When this bit is reset, the abnormal interrupt summary is disabled.

Table continues on the next page...

Table continued from the previous page...

Field	Function
	0b - Abnormal Interrupt Summary is disabled 1b - Abnormal Interrupt Summary is enabled
13 CDEE	Context Descriptor Error Enable When this bit is set along with the AIE bit, the Descriptor error interrupt is enabled. When this bit is reset, the Descriptor error interrupt is disabled. 0b - Context Descriptor Error is disabled 1b - Context Descriptor Error is enabled
12 FBEE	Fatal Bus Error Enable When this bit is set along with the AIE bit, the Fatal Bus error interrupt is enabled. When this bit is reset, the Fatal Bus Error error interrupt is disabled. 0b - Fatal Bus Error is disabled 1b - Fatal Bus Error is enabled
11 ERIE	Early Receive Interrupt Enable When this bit is set along with the NIE bit, the Early Receive interrupt is enabled. When this bit is reset, the Early Receive interrupt is disabled. 0b - Early Receive Interrupt is disabled 1b - Early Receive Interrupt is enabled
10 ETIE	Early Transmit Interrupt Enable When this bit is set along with the AIE bit, the Early Transmit interrupt is enabled. When this bit is reset, the Early Transmit interrupt is disabled. 0b - Early Transmit Interrupt is disabled 1b - Early Transmit Interrupt is enabled
9 RWTE	Receive Watchdog Timeout Enable When this bit is set along with the AIE bit, the Receive Watchdog Timeout interrupt is enabled. When this bit is reset, the Receive Watchdog Timeout interrupt is disabled. 0b - Receive Watchdog Timeout is disabled 1b - Receive Watchdog Timeout is enabled
8 RSE	Receive Stopped Enable When this bit is set along with the AIE bit, the Receive Stopped Interrupt is enabled. When this bit is reset, the Receive Stopped interrupt is disabled. 0b - Receive Stopped is disabled 1b - Receive Stopped is enabled
7	Receive Buffer Unavailable Enable

Table continues on the next page...

Table continued from the previous page...

Field	Function
RBUE	When this bit is set along with the AIE bit, the Receive Buffer Unavailable interrupt is enabled. When this bit is reset, the Receive Buffer Unavailable interrupt is disabled. 0b - Receive Buffer Unavailable is disabled 1b - Receive Buffer Unavailable is enabled
6 RIE	Receive Interrupt Enable When this bit is set along with the NIE bit, the Receive Interrupt is enabled. When this bit is reset, the Receive Interrupt is disabled. 0b - Receive Interrupt is disabled 1b - Receive Interrupt is enabled
5-3 —	Reserved.
2 TBUE	Transmit Buffer Unavailable Enable When this bit is set along with the NIE bit, the Transmit Buffer Unavailable interrupt is enabled. When this bit is reset, the Transmit Buffer Unavailable interrupt is disabled. 0b - Transmit Buffer Unavailable is disabled 1b - Transmit Buffer Unavailable is enabled
1 TXSE	Transmit Stopped Enable When this bit is set along with the AIE bit, the Transmission Stopped interrupt is enabled. When this bit is reset, the Transmission Stopped interrupt is disabled. 0b - Transmit Stopped is disabled 1b - Transmit Stopped is enabled
0 TIE	Transmit Interrupt Enable When this bit is set along with the NIE bit, the Transmit Interrupt is enabled. When this bit is reset, the Transmit Interrupt is disabled. 0b - Transmit Interrupt is disabled 1b - Transmit Interrupt is enabled

1.4.398 (DMA_CH3_Rx_Interrupt_Watchdog_Timer)

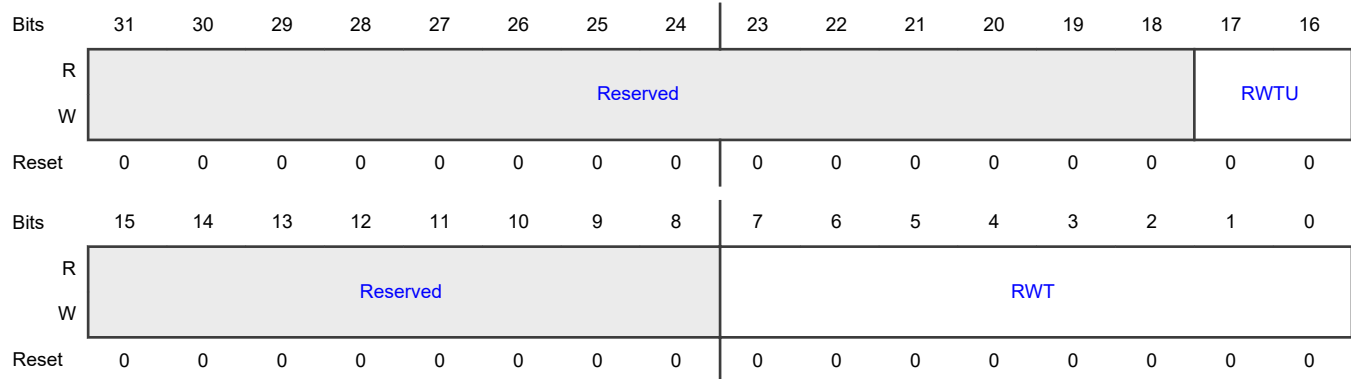
Offset

Register	Offset
DMA_CH3_Rx_Interrupt_Watchdog_Timer	12B8h

Function

The Receive Interrupt Watchdog Timer register indicates the watchdog timeout for Receive Interrupt (RI) from the DMA. When this register is written with a non-zero value, it enables the watchdog timer for the RI bit of the DMA_CHi_Status register.

Diagram



Fields

Field	Function
31-18 —	Reserved.
17-16 RWTU	Receive Interrupt Watchdog Timer Count Units This fields indicates the number of system clock cycles corresponding to one unit in RWT field. - 2'b00: 256 - 2'b01: 512 - 2'b10: 1024 - 2'b11: 2048 For example, when RWT=2 and RWTU=1, the watchdog timer is set for 2*512=1024 system clock cycles.
15-8 —	Reserved.
7-0 RWT	Receive Interrupt Watchdog Timer Count This field indicates the number of system clock cycles, multiplied by factor indicated in RWTU field, for which the watchdog timer is set. The watchdog timer is triggered with the programmed value after the Rx DMA completes the transfer of a packet for which the RI bit is not set in the DMA_CH(i)_Status register, because of the setting of Interrupt Enable bit in the corresponding descriptor RDES3[30]. When the watchdog timer runs out, the RI bit is set and the timer is stopped. The watchdog timer is reset when the RI bit is set high because of automatic setting of RI as per the Interrupt Enable bit RDES3[30] of any received packet.

1.4.399 (DMA_CH3_Slot_Function_Control_Status)

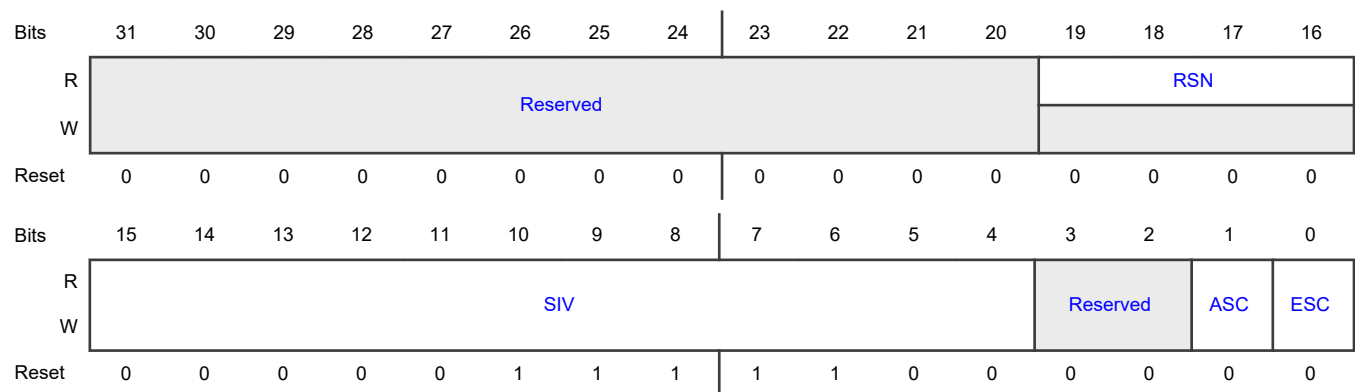
Offset

Register	Offset
DMA_CH3_Slot_Function_Control_Status	12BCh

Function

The Slot Function Control and Status register contains the control bits for slot function and the status for Transmit path.

Diagram



Fields

Field	Function
31-20 —	Reserved.
19-16 RSN	Reference Slot Number This field gives the current value of the reference slot number in the DMA. It is used for slot comparison.
15-4 SIV	Slot Interval Value This field controls the period of the slot interval in which the TxDMA fetches the scheduled packets. A value of 0 specifies the slot interval of 1 us while the maximum value 4095 specifies the slot interval of 4096us. The default/reset value is 0x07C which corresponds to slot interval of 125us
3-2 —	Reserved.
1 ASC	Advance Slot Check When set, this bit enables the DMA to fetch the data from the buffer when the slot number (SLOTNUM) programmed in the Tx descriptor is - equal to the reference slot number given in the RSN field

Table continues on the next page...

Table continued from the previous page...

Field	Function
	or - ahead of the reference slot number by up to two slots This bit is applicable only when the ESC bit is set. 0b - Advance Slot Check is disabled 1b - Advance Slot Check is enabled
0 ESC	Enable Slot Comparison When set, this bit enables the checking of the slot numbers programmed in the Tx descriptor with the current reference given in the RSN field. The DMA fetches the data from the corresponding buffer only when the slot number is - equal to the reference slot number or - ahead of the reference slot number by one slot When reset, this bit disables the checking of the slot numbers. The DMA fetches the data immediately after the descriptor is processed. Note: The UFO (UDP Fragmentation over IPv4)/TSO/USO should not be enabled along with TBS/AVB Slot number check. The UFO/TSO/USO involves multiple packets/segments/fragments transmission for single packet received from application and the slot number check are applicable for fetching of only first segment/fragment. As a result it might be difficult for software to specify slot number for subsequent packets. 0b - Slot Comparison is disabled 1b - Slot Comparison is enabled

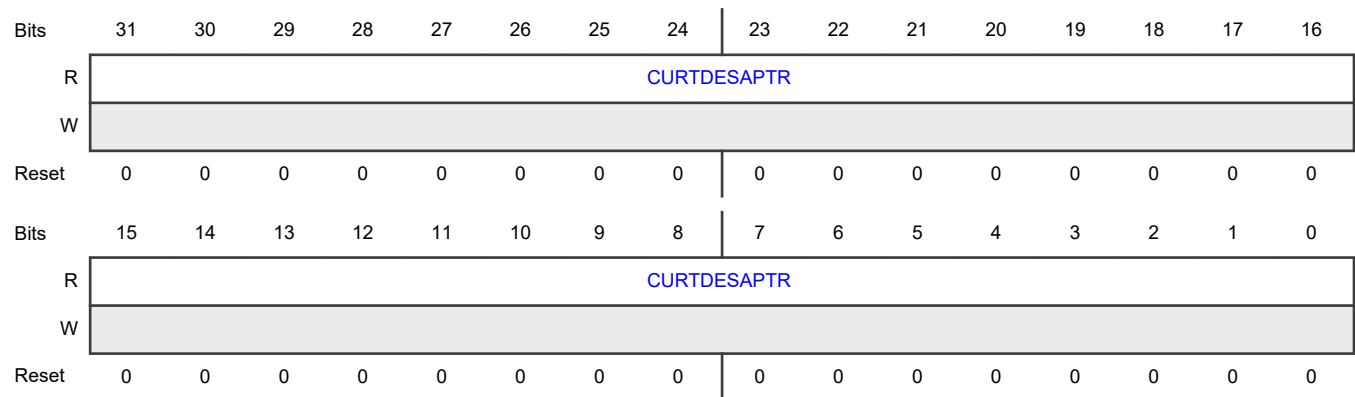
1.4.400 (DMA_CH3_Current_App_TxDesc)

Offset

Register	Offset
DMA_CH3_Current_App_TxDesc	12C4h

Function

The Channel1 Current Application Transmit Descriptor register points to the current Transmit descriptor read by the DMA.

Diagram**Fields**

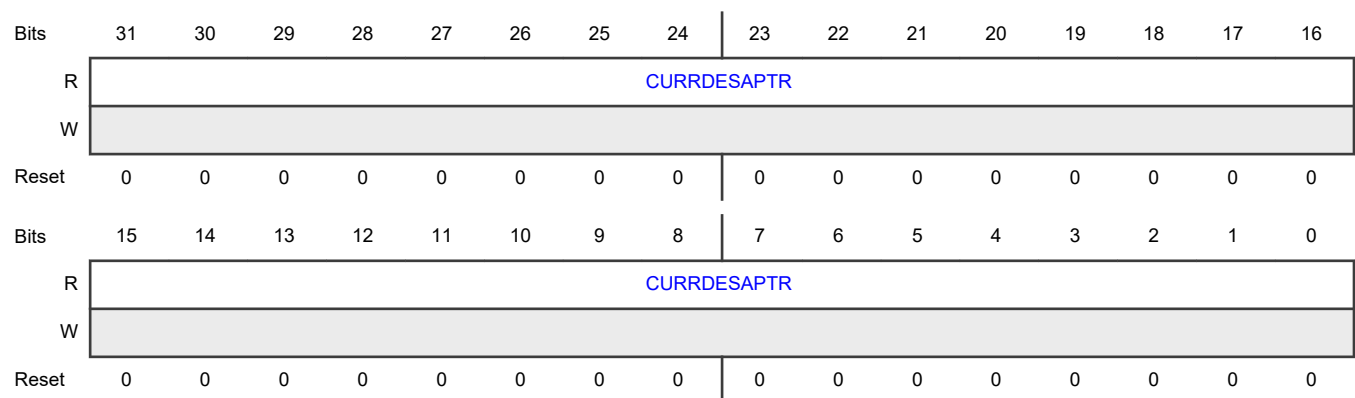
Field	Function
31-0	Application Transmit Descriptor Address Pointer
CURTDESAPT R	The DMA updates this pointer during Tx operation. This pointer is cleared on reset.

1.4.401 (DMA_CH3_Current_App_RxDesc)**Offset**

Register	Offset
DMA_CH3_Current_App_RxDesc	12CCh

Function

The Channel1 Current Application Receive Descriptor register points to the current Receive descriptor read by the DMA.

Diagram

Fields

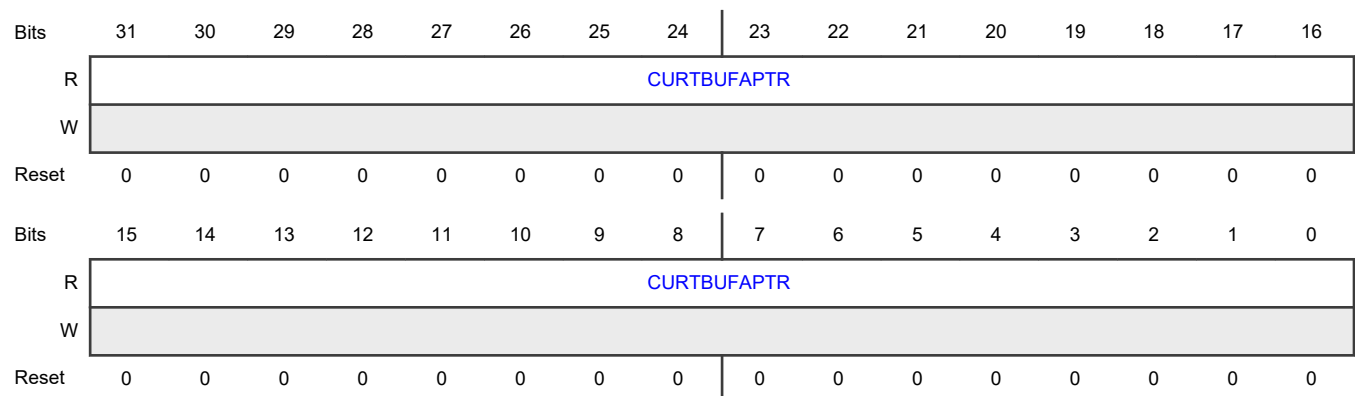
Field	Function
31-0 CURDESAPTR	Application Receive Descriptor Address Pointer The DMA updates this pointer during Rx operation. This pointer is cleared on reset.

1.4.402 (DMA_CH3_Current_App_TxBuffer)**Offset**

Register	Offset
DMA_CH3_Current_App_TxBuffer	12D4h

Function

The Channeli Current Application Transmit Buffer Address register points to the current Tx buffer address read by the DMA.

Diagram**Fields**

Field	Function
31-0 CURTBUFAPTR	Application Transmit Buffer Address Pointer The DMA updates this pointer during Tx operation. This pointer is cleared on reset.

1.4.403 (DMA_CH3_Current_App_RxBuffer)

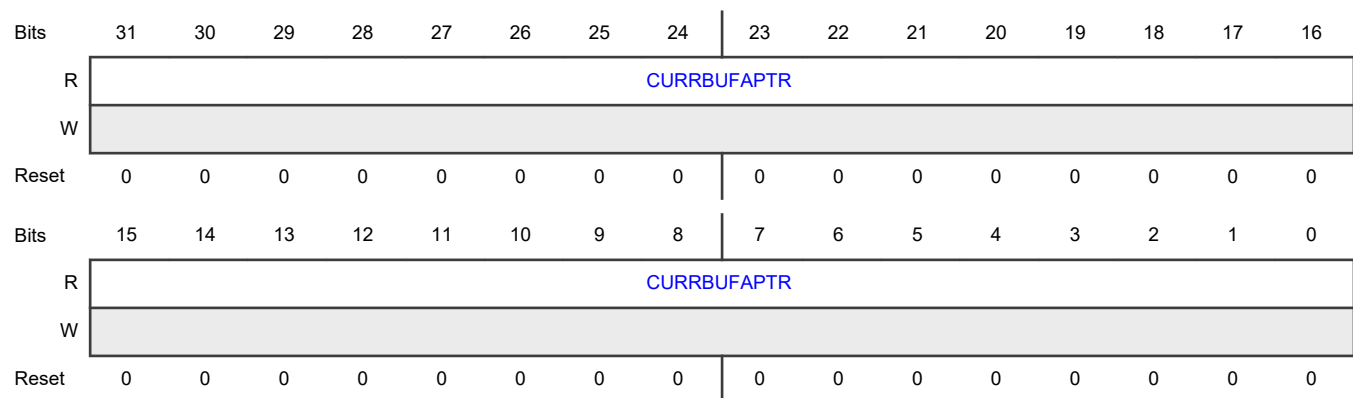
Offset

Register	Offset
DMA_CH3_Current_App_RxBuffer	12DCh

Function

The Channel 0 Current Application Receive Buffer Address register points to the current Rx buffer address read by the DMA.

Diagram



Fields

Field	Function
31-0	Application Receive Buffer Address Pointer
CURRBUFAPTR	The DMA updates this pointer during Rx operation. This pointer is cleared on reset.
R	

1.4.404 (DMA_CH3_Status)

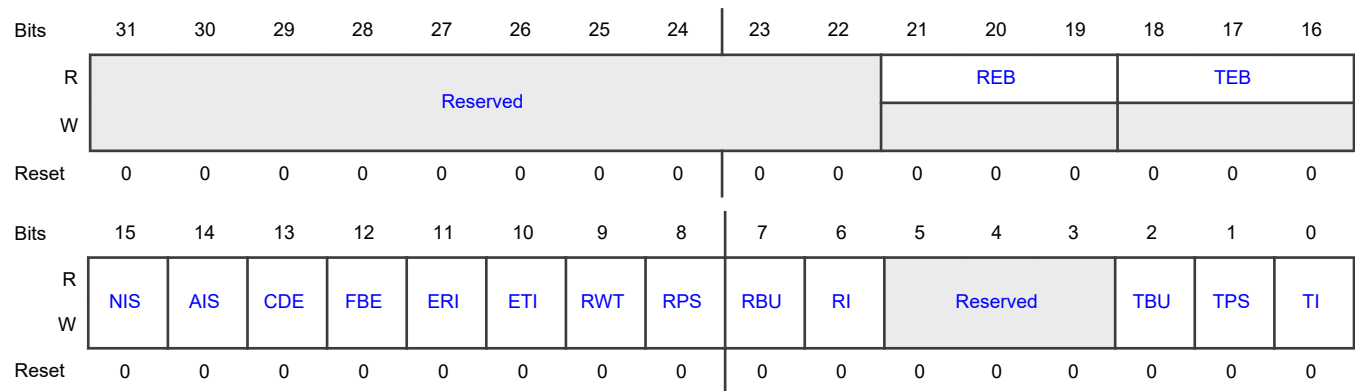
Offset

Register	Offset
DMA_CH3_Status	12E0h

Function

The software driver (application) reads the Status register during interrupt service routine or polling to determine the status of the DMA.

Note: The number of DMA_CH(#i)_Status register in the configuration is the higher of number of Rx DMA Channels and Tx DMA Channels.

Diagram**Fields**

Field	Function
31-22 —	Reserved.
21-19 REB	Rx DMA Error Bits This field indicates the type of error that caused a Bus Error. For example, error response on the AHB or AXI interface. - Bit 21 -- 1'b1: Error during data transfer by Rx DMA -- 1'b0: No Error during data transfer by Rx DMA - Bit 20 -- 1'b1: Error during descriptor access -- 1'b0: Error during data buffer access - Bit 19 -- 1'b1: Error during read transfer -- 1'b0: Error during write transfer This field is valid only when the FBE bit is set. This field does not generate an interrupt.
18-16 TEB	Tx DMA Error Bits This field indicates the type of error that caused a Bus Error. For example, error response on the AHB or AXI interface. - Bit 18 -- 1'b1: Error during data transfer by Tx DMA -- 1'b0: No Error during data transfer by Tx DMA - Bit 17 -- 1'b1: Error during descriptor access

Table continues on the next page...

Table continued from the previous page...

Field	Function
	-- 1'b0: Error during data buffer access - Bit 16 -- 1'b1: Error during read transfer -- 1'b0: Error during write transfer This field is valid only when the FBE bit is set. This field does not generate an interrupt.
15 NIS	Normal Interrupt Summary Normal Interrupt Summary bit value is the logical OR of the following bits when the corresponding interrupt bits are enabled in the DMA_CH0_Interrupt_Enable register: - Bit 0: Transmit Interrupt - Bit 2: Transmit Buffer Unavailable - Bit 6: Receive Interrupt - Bit 11: Early Receive Interrupt Only unmasked bits (interrupts for which interrupt enable is set in DMA_CH0_Interrupt_Enable register) affect the Normal Interrupt Summary bit. This is a sticky bit. You must clear this bit (by writing 1 to this bit) each time a corresponding bit which causes NIS to be set is cleared. Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect. 0b - Normal Interrupt Summary status not detected 1b - Normal Interrupt Summary status detected
14 AIS	Abnormal Interrupt Summary Abnormal Interrupt Summary bit value is the logical OR of the following when the corresponding interrupt bits are enabled in the DMA_CH0_Interrupt_Enable register: - Bit 1: Transmit Process Stopped - Bit 7: Receive Buffer Unavailable - Bit 8: Receive Process Stopped - Bit 10: Early Transmit Interrupt - Bit 12: Fatal Bus Error - Bit 13: Context Descriptor Error Only unmasked bits affect the Abnormal Interrupt Summary bit. This is a sticky bit. You must clear this bit (by writing 1 to this bit) each time a corresponding bit, which causes AIS to be set, is cleared. Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect. 0b - Abnormal Interrupt Summary status not detected 1b - Abnormal Interrupt Summary status detected

Table continues on the next page...

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Field	Function
13 CDE	Context Descriptor Error This bit indicates that the DMA Tx/Rx engine received a descriptor error, which indicates invalid context in the middle of packet flow (intermediate descriptor) or all one's descriptor in Tx case and on Rx side it indicates DMA has read a descriptor with either of the buffer address as ones which is considered to be invalid. Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect. 0b - Context Descriptor Error status not detected 1b - Context Descriptor Error status detected
12 FBE	Fatal Bus Error This bit indicates that a bus error occurred (as described in the EB field). When this bit is set, the corresponding DMA channel engine disables all bus accesses. Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect. 0b - Fatal Bus Error status not detected 1b - Fatal Bus Error status detected
11 ERI	Early Receive Interrupt This bit when set indicates that the RxDMA has completed the transfer of packet data to the memory. In configs supporting ERIC, When ERIC=0, this bit is set only after the Rx DMA has filled up a complete receive buffer with packet data. When ERIC=1, this bit is set after every burst transfer of data from the Rx DMA to the buffer. The setting of RI bit automatically clears this bit. Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect. 0b - Early Receive Interrupt status not detected 1b - Early Receive Interrupt status detected
10 ETI	Early Transmit Interrupt This bit when set indicates that the TxDMA has completed the transfer of packet data to the MTL TXFIFO memory. In configs supporting ERIC: When ETIC=0, this bit is set only after the Tx DMA has transferred a complete packet to MTL. When ETIC=1, this bit is set after completion of (partial) packet data transfer from buffers in the Transmit descriptor in which IOC=1. Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect. 0b - Early Transmit Interrupt status not detected 1b - Early Transmit Interrupt status detected
9 RWT	Receive Watchdog Timeout This bit is asserted when a packet with length greater than 2,048 bytes (10,240 bytes when Jumbo Packet mode is enabled) is received.

Table continues on the next page...

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Field	Function
	0b - Receive Watchdog Timeout status not detected 1b - Receive Watchdog Timeout status detected
8 RPS	Receive Process Stopped This bit is asserted when the Rx process enters the Stopped state. Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect. 0b - Receive Process Stopped status not detected 1b - Receive Process Stopped status detected
7 RBU	Receive Buffer Unavailable This bit indicates that the application owns the next descriptor in the Receive list, and the DMA cannot acquire it. The Rx process is suspended. To resume processing Rx descriptors, the application should change the ownership of the descriptor and issue a Receive Poll Demand command. If this command is not issued, the Rx process resumes when the next recognized incoming packet is received. In ring mode, the application should advance the Receive Descriptor Tail Pointer register of a channel. This bit is set only when the DMA owns the previous Rx descriptor. Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect. 0b - Receive Buffer Unavailable status not detected 1b - Receive Buffer Unavailable status detected
6 RI	Receive Interrupt This bit indicates that the packet reception is complete. When packet reception is complete, Bit 31 of RDES3 is reset in the last descriptor, and the specific packet status information is updated in the descriptor. The reception remains in the Running state. Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect. 0b - Receive Interrupt status not detected 1b - Receive Interrupt status detected
5-3 —	Reserved.
2 TBU	Transmit Buffer Unavailable This bit indicates that the application owns the next descriptor in the Transmit list, and the DMA cannot acquire it. Transmission is suspended. The TPS0 field of the DMA_Debug_Status0 register explains the Transmit Process state transitions. To resume processing the Transmit descriptors, the application should do the following: 1. Change the ownership of the descriptor by setting Bit 31 of TDES3. 2. Issue a Transmit Poll Demand command. For ring mode, the application should advance the Transmit Descriptor Tail Pointer register of a channel.

Table continues on the next page...

Table continued from the previous page...

Field	Function
	Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect. 0b - Transmit Buffer Unavailable status not detected 1b - Transmit Buffer Unavailable status detected
1 TPS	Transmit Process Stopped This bit is set when the transmission is stopped. Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect. 0b - Transmit Process Stopped status not detected 1b - Transmit Process Stopped status detected
0 TI	Transmit Interrupt This bit indicates that the packet transmission is complete. When transmission is complete, Bit 31 of TDES3 is reset in the last descriptor, and the specific packet status information is updated in the descriptor. Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect. 0b - Transmit Interrupt status not detected 1b - Transmit Interrupt status detected

1.4.405 (DMA_CH3_Miss_Frame_Cnt)

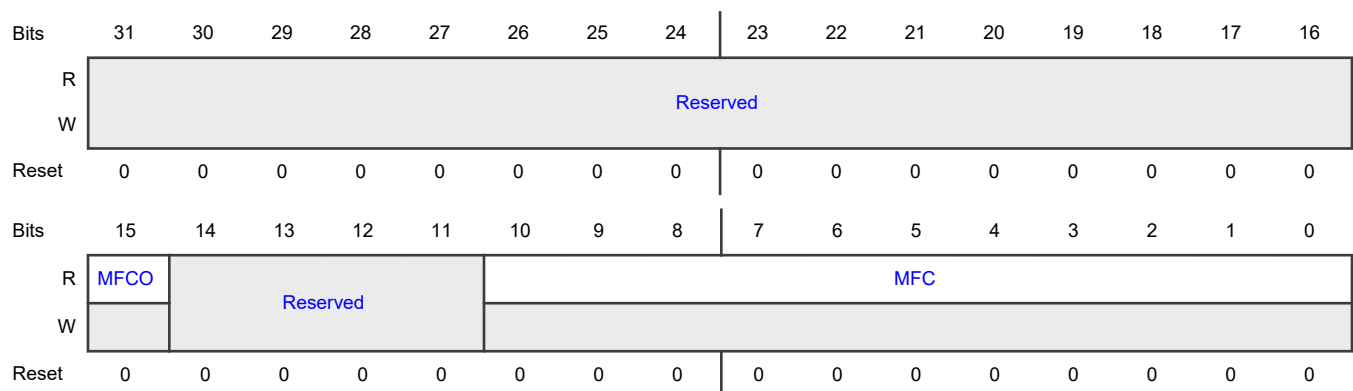
Offset

Register	Offset
DMA_CH3_Miss_Frame_Cnt	12E4h

Function

This register has the number of packet counter that got dropped by the DMA either due to Bus Error or due to programming RPF field in DMA_CH3{i}_Rx_Control register.

Diagram



Fields

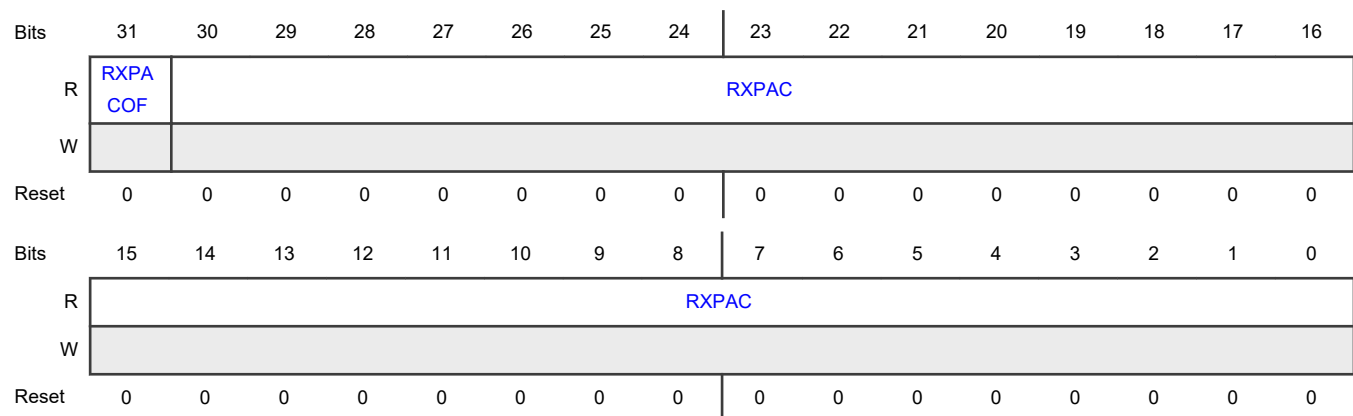
Field	Function
31-16 —	Reserved.
15 MFCO	Overflow status of the MFC Counter When this bit is set then the MFC counter does not get incremented further. The bit gets cleared when this register is read. Access restriction applies. Clears on read. Self-set to 1 on internal event. 0b - Miss Frame Counter overflow not occurred 1b - Miss Frame Counter overflow occurred
14-11 —	Reserved.
10-0 MFC	Dropped Packet Counters This counter indicates the number of packet counters that are dropped by the DMA either because of bus error or because of programming RPF field in DMA_CH\${i}_Rx_Control register. The counter gets cleared when this register is read. Access restriction applies. Clears on read. Self-set to 1 on internal event.

1.4.406 (DMA_CH3_RXP_Accept_Cnt)**Offset**

Register	Offset
DMA_CH3_RXP_Accept_Cnt	12E8h

Function

The DMA_CH(#i)_RXP_Accept_Cnt registers provides the count of the number of frames accepted by Rx Parser.

Diagram**Fields**

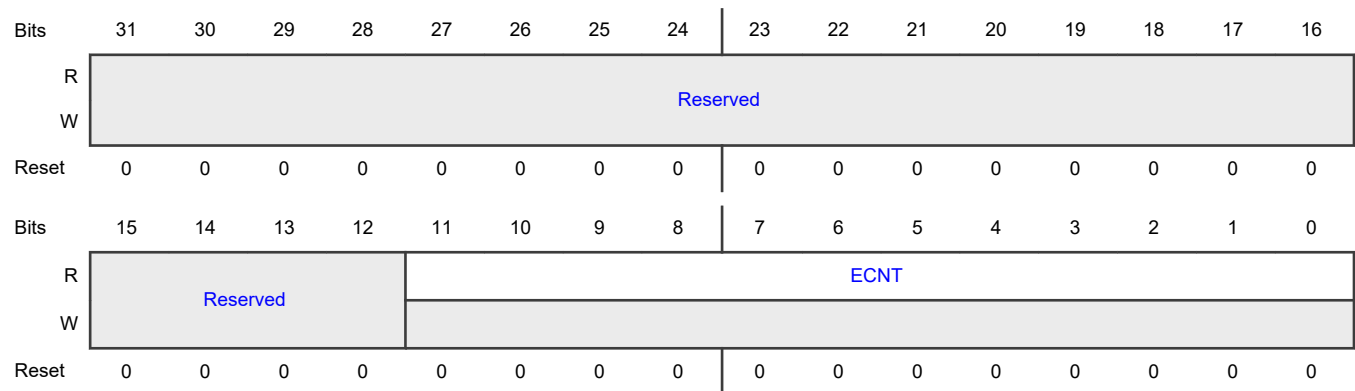
Field	Function
31 RXPACOF	Rx Parser Accept Counter Overflow Bit When set, this bit indicates that the RXPAC Counter field crossed the maximum limit. Access restriction applies. Clears on read. Self-set to 1 on internal event. 0b - Rx Parser Accept Counter overflow not occurred 1b - Rx Parser Accept Counter overflow occurred
30-0 RXPAC	Rx Parser Accept Counter This 31-bit counter is implemented whenever a Rx Parser Accept a packet due to AF =1. The counter is cleared when the register is read.

1.4.407 (DMA_CH3_RX_ERI_Cnt)**Offset**

Register	Offset
DMA_CH3_RX_ERI_Cnt	12ECh

Function

The DMA_CH(#i)_RX_ERI_Cnt registers provides the count of the number of times ERI was asserted.

Diagram**Fields**

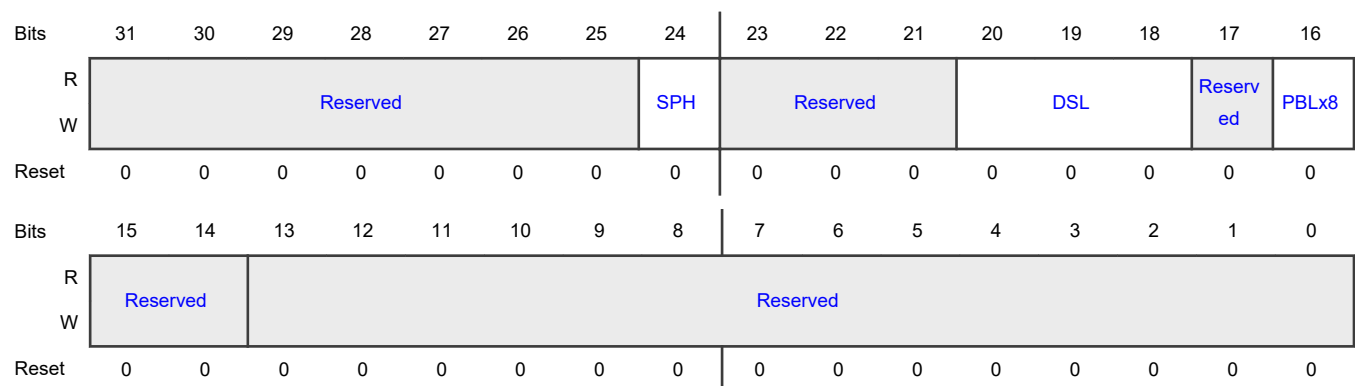
Field	Function
31-12 —	Reserved.
11-0 ECNT	ERI Counter When ERIC bit of DMA_CH(#)_RX_Control register is set, this counter increments for burst transfer completed by the Rx DMA from the start of packet transfer. This counter is reset at the start of new packet.

1.4.408 (DMA_CH4_Control)**Offset**

Register	Offset
DMA_CH4_Control	1300h

Function

The DMA Channel Control register specifies the MSS value for segmentation, length to skip between two descriptors, and also the features such as header splitting and 8xPBL mode.

Diagram

Fields

Field	Function
31-25 —	Reserved.
24 SPH	Split Headers When this bit is set, the DMA splits the header and payload in the Receive path. The DMA writes the header to the Buffer Address1 of RDES0. The DMA writes the payload to the buffer to which the Buffer Address2 is pointing. The software must ensure that the header fits into the Receive buffers. If the header length exceeds the receive buffer size, the DMA does not split the header and payload. This bit is available only if Enable Split Header Structure option is selected. 0b - Split Headers feature is disabled 1b - Split Headers feature is enabled
23-21 —	Reserved.
20-18 DSL	Descriptor Skip Length This bit specifies the Word, Dword, or Lword number (depending on the 32-bit, 64-bit, or 128-bit bus) to skip between two unchained descriptors. The address skipping starts from the end of the current descriptor to the start of the next descriptor. When the DSL value is equal to zero, the DMA takes the descriptor table as contiguous.
17 —	Reserved.
16 PBLx8	8xPBL mode When this bit is set, the PBL value programmed in Bits[21:16] in DMA_CH(#i)_Tx_Control and Bits[21:16] in DMA_CH(#i)_Rx_Control is multiplied by eight times. Therefore, the DMA transfers the data in 8, 16, 32, 64, 128, and 256 beats depending on the PBL value. 0b - 8xPBL mode is disabled 1b - 8xPBL mode is enabled
15-14 —	Reserved.
13-0 —	Reserved.

1.4.409 (DMA_CH4_Tx_Control)

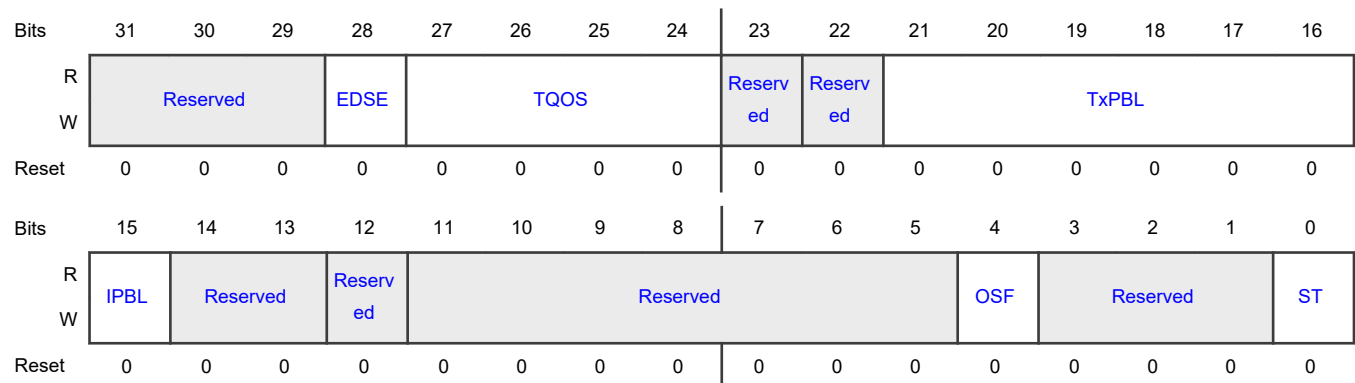
Offset

Register	Offset
DMA_CH4_Tx_Control	1304h

Function

The DMA Channel Transmit Control register controls the Tx features such as PBL, TCP segmentation, and Tx Channel weights.

Diagram



Fields

Field	Function
31-29 —	Reserved.
28 EDSE	Enhanced Descriptor Enable When this bit is set, the corresponding channel uses Enhanced Descriptors that are 32 Bytes for both Normal and Context Descriptors. When reset, the corresponding channel uses the descriptors that are 16 Bytes. 0b - Enhanced Descriptor is disabled 1b - Enhanced Descriptor is enabled
27-24 TQOS	Transmit QOS. This field is used to drive arqos_m_o[3:0] or awqos_m_o[3:0] output signals for all transactions of DMA Tx Channel0.
23 —	Reserved.
22	Reserved.

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Field	Function
—	
21-16 TxPBL	Transmit Programmable Burst Length These bits indicate the maximum number of beats to be transferred in one DMA block data transfer. The DMA always attempts max burst as specified in PBL each time it starts a burst transfer on the application bus. You can program PBL with any of the following values: 1, 2, 4, 8, 16, or 32. Any other value results in undefined behavior. To transfer more than 32 beats, perform the following steps: 1. Set the 8xPBL mode in DMA_CH0_Control register. 2. Set the TxPBL. Note: The maximum value of TxPBL must be less than or equal to half the Tx Queue size (TQS field of MTL_TxQ(#!)_Operation_Mode register) in terms of beats. This is required so that the Tx Queue has space to store at least another Tx PBL worth of data while the MTL Tx Queue Controller is transferring data to MAC. For example, in 64-bit data width configurations the total locations in Tx Queue of size 512 bytes is 64, TxPBL and 8xPBL needs to be programmed to less than or equal to 32.
15 IPBL	Ignore PBL Requirement When this bit is set, the DMA does not check for PBL number of locations in the MTL before initiating a transfer. If space is not available, the MTL might use handshaking to slow the DMA. Note: This bit/mode must not be used when multiple Transmit DMA Channels are enabled as it might block other Transmit and Receive DMA Channels from accessing the Read Data Channel of AXI bus until space is available in Transmit Queue for current transfer. 0b - Ignore PBL Requirement is disabled 1b - Ignore PBL Requirement is enabled
14-13 —	Reserved.
12 —	Reserved.
11-5 —	Reserved.
4 OSF	Operate on Second Packet When this bit is set, it instructs the DMA to process the second packet of the Transmit data even before the status for the first packet is obtained. 0b - Operate on Second Packet disabled 1b - Operate on Second Packet enabled
3-1 —	Reserved.

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Field	Function
0	Start or Stop Transmission Command
ST	When this bit is set, transmission is placed in the Running state. The DMA checks the Transmit list at the current position for a packet to be transmitted. The DMA tries to acquire descriptor from either of the following positions: - The current position in the list This is the base address of the Transmit list set by the DMA_CH0_TxDesc_List_Address register. - The position at which the transmission was previously stopped If the DMA does not own the current descriptor, the transmission enters the Suspended state and the TBU bit of the DMA_CH0_Status register is set. The Start Transmission command is effective only when the transmission is stopped. If the command is issued before setting the DMA_CH0_TxDesc_List_Address register, the DMA behavior is unpredictable. When this bit is reset, the transmission process is placed in the Stopped state after completing the transmission of the current packet. The Next Descriptor position in the Transmit list is saved, and it becomes the current position when the transmission is restarted. To change the list address, you need to program DMA_CH0_TxDesc_List_Address register with a new value when this bit is reset. The new value is considered when this bit is set again. The stop transmission command is effective only when the transmission of the current packet is complete or the transmission is in the Suspended state. 0b - Stop Transmission Command 1b - Start Transmission Command

1.4.410 (DMA_CH4_Rx_Control)

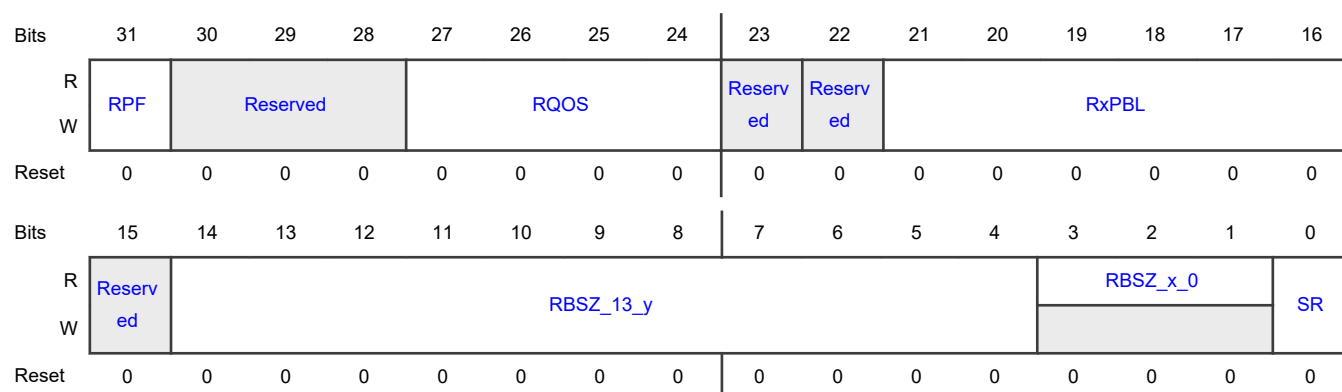
Offset

Register	Offset
DMA_CH4_Rx_Control	1308h

Function

The DMA Channel4 Receive Control register controls the Rx features such as PBL, buffer size, and extended status.

Diagram



Fields

Field	Function
31 RPF	Rx Packet Flush. When this bit is set to 1, then <code>DWC_ether_qos</code> automatically flushes the packet from the Rx Queues destined to this DMA Rx Channel, when it is stopped. When this bit remains set and the DMA is re-started by the software driver, the packets residing in the Rx Queues that were received when this RxDMA was stopped, get flushed out. The packets that are received by the MAC after the RxDMA is re-started are routed to the RxDMA. The flushing happens on the Read side of the Rx Queue. When this bit is set to 0, the <code>DWC_ether_qos</code> not flush the packet in the Rx Queue destined to this RxDMA Channel when it is STOP state. This might in turn cause head-of-line blocking in the corresponding RxQueue. Note: The stopping of packet flow from a Rx DMA Channel to the application by setting RPF works only when there is one-to-one mapping of Rx Queue to Rx DMA channels. In Dynamic mapping mode, setting RPF bit in any <code>DMA_CH(#i)_Rx_Control</code> register might flush packets from unintended Rx Queues which are destined to the stopped Rx DMA Channel. 0b - Rx Packet Flush is disabled 1b - Rx Packet Flush is enabled
30-28 —	Reserved.
27-24 RQOS	Rx AXI4 QOS. This field is used to drive <code>arqos_m_o[3:0]</code> or <code>awqos_m_o[3:0]</code> output signals for all transactions of DMA Rx Channel0.
23 —	Reserved.
22 —	Reserved.
21-16	Receive Programmable Burst Length

Table continues on the next page...

Table continued from the previous page...

Field	Function
RxPBL	These bits indicate the maximum number of beats to be transferred in one DMA block data transfer. The DMA always attempts max burst as specified in PBL each time it starts a burst transfer on the application bus. You can program PBL with any of the following values: 1, 2, 4, 8, 16, or 32. Any other value results in undefined behavior. To transfer more than 32 beats, perform the following steps: 1. Set the 8xPBL mode in the DMA_CH0_Control register. 2. Set the RxPBL. Note: The maximum value of RxPBL must be less than or equal to half the Rx Queue size (RQS field of MTL_RxQ(#)_Operation_Mode register) in terms of beats. This is required so that the Rx Queue has space to store at least another Rx PBL worth of data while the Rx DMA is transferring a block of data. For example, in 64-bit data width configurations the total locations in Rx Queue of size 512 bytes is 64, so RxPBL and 8xPBL needs to be programmed to less than or equal to 32.
15 —	Reserved.
14-4 RBSZ_13_y	Receive Buffer size High RBSZ[13:0] is split into two fields higher RBSZ_13_y and lower RBSZ_x_0. The RBSZ[13:0] field indicates the size of the Rx buffers specified in bytes. The maximum buffer size is limited to 16K bytes. The buffer size is applicable to payload buffers when split headers are enabled. Note: The buffer size must be a multiple of 4, 8, or 16 depending on the data bus widths (32-bit, 64-bit, or 128-bit respectively). This is required even if the value of buffer address pointer is not aligned to data bus width. Hence the lower RBSZ_x_0 bits are read-only and the value is considered as all-zero. Thus the RBSZ_13_y indicates the buffer size in terms of locations (with the width same as bus-width).
3-1 RBSZ_x_0	Receive Buffer size Low RBSZ[13:0] is split into two fields RBSZ_13_y and RBSZ_x_0. The RBSZ_x_0 is the lower field whose width is based on data bus width of the configuration. This field is of width 2, 3, or 4 bits for 32-bit, 64-bit, or 128-bit data bus width respectively. This field is read-only (RO).
0 SR	Start or Stop Receive When this bit is set, the DMA tries to acquire the descriptor from the Receive list and processes the incoming packets. The DMA tries to acquire descriptor from either of the following positions: - The current position in the list This is the address set by the DMA_CH0_RxDesc_List_Address register. - The position at which the Rx process was previously stopped If the DMA does not own the current descriptor, the reception is suspended and the RBU bit of the DMA_CH0_Status register is set. The Start Receive command is effective only when the reception is stopped. If the command is issued before setting the DMA_CH0_RxDesc_List_Address register, the DMA behavior is unpredictable.

Table continues on the next page...

Table continued from the previous page...

Field	Function
	When this bit is reset, the Rx DMA operation is stopped after the transfer of the current packet. The next descriptor position in the Receive list is saved, and it becomes the current position after the Rx process is restarted. The Stop Receive command is effective only when the Rx process is in the Running (waiting for Rx packet) or Suspended state.
	0b - Stop Receive
	1b - Start Receive

1.4.411 (DMA_CH4_TxDesc_List_Address)

Offset

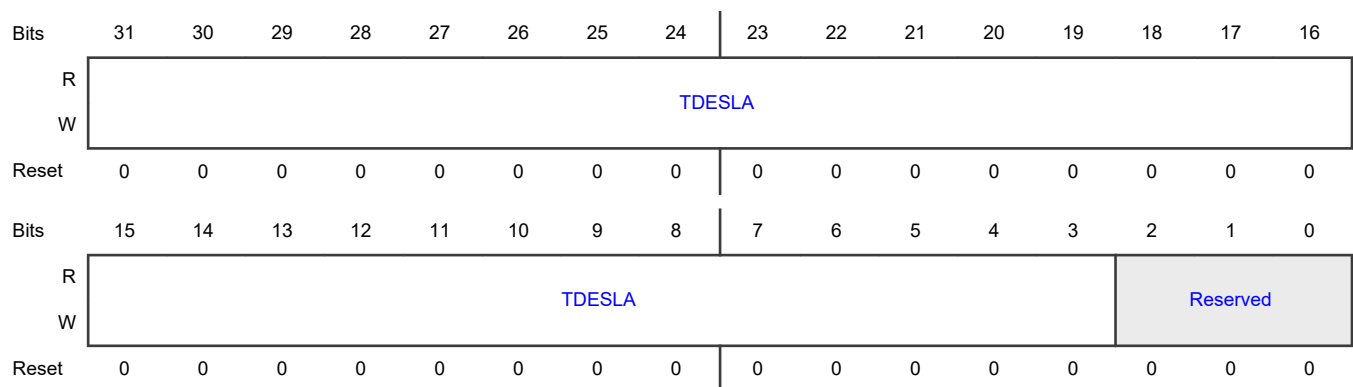
Register	Offset
DMA_CH4_TxDesc_List_Address	1314h

Function

The Channeli Tx Descriptor List Address register points the DMA to the start of Transmit descriptor list. The descriptor lists reside in the physical memory space of the application and must be Word, Dword, or Lword-aligned (for 32-bit, 64-bit, or 128-bit data bus). The DMA internally converts it to bus width aligned address by making the corresponding LSB to low.

You can write to this register only when the Tx DMA has stopped, that is, the ST bit is set to zero in DMA_CH0_Tx_Control register. When stopped, this register can be written with a new descriptor list address. When you set the ST bit to 1, the DMA takes the newly-programmed descriptor base address. If this register is not changed when the ST bit is set to 0, the DMA takes the descriptor address where it was stopped earlier.

Diagram



Fields

Field	Function
31-3	Start of Transmit List

Table continues on the next page...

Table continued from the previous page...

Field	Function
TDESLA	This field contains the base address of the first descriptor in the Transmit descriptor list. The DMA ignores the LSB bits (1:0, 2:0, or 3:0) for 32-bit, 64-bit, or 128-bit bus width and internally takes these bits as all-zero. Therefore, these LSB bits are read-only (RO). The width of this field depends on the configuration: - 31:2 for 32-bit configuration - 31:3 for 64-bit configuration - 31:4 for 128-bit configuration
2-0 —	Reserved.

1.4.412 (DMA_CH4_RxDesc_List_Address)

Offset

Register	Offset
DMA_CH4_RxDesc_List_Address	131Ch

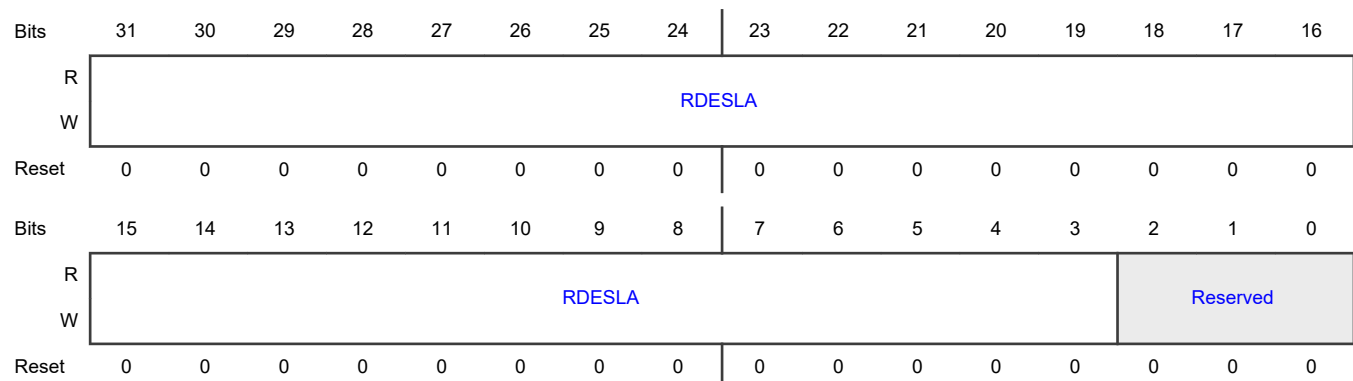
Function

The Channel Rx Descriptor List Address register points the DMA to the start of Receive descriptor list.

This register points to the start of the Receive Descriptor List. The descriptor lists reside in the physical memory space of the application and must be Word, Dword, or Lword-aligned (for 32-bit, 64-bit, or 128-bit data bus). The DMA internally converts it to bus width aligned address by making the corresponding LS bits low. Writing to this register is permitted only when reception is stopped. When stopped, this register must be written to before the receive Start command is given. You can write to this register only when Rx DMA has stopped, that is, SR bit is set to zero in DMA_CH0_Rx_Control register. When stopped, this register can be written with a new descriptor list address.

When you set the SR bit to 1, the DMA takes the newly programmed descriptor base address.

Diagram



Fields

Field	Function
31-3 RDESLA	Start of Receive List This field contains the base address of the first descriptor in the Rx Descriptor list. The DMA ignores the LSB bits (1:0, 2:0, or 3:0) for 32-bit, 64-bit, or 128-bit bus width and internally takes these bits as all-zero. Therefore, these LSB bits are read-only (RO). The width of this field depends on the configuration: - 31:2 for 32-bit configuration - 31:3 for 64-bit configuration - 31:4 for 128-bit configuration
2-0 —	Reserved.

1.4.413 (DMA_CH4_TxDesc_Tail_Pointer)

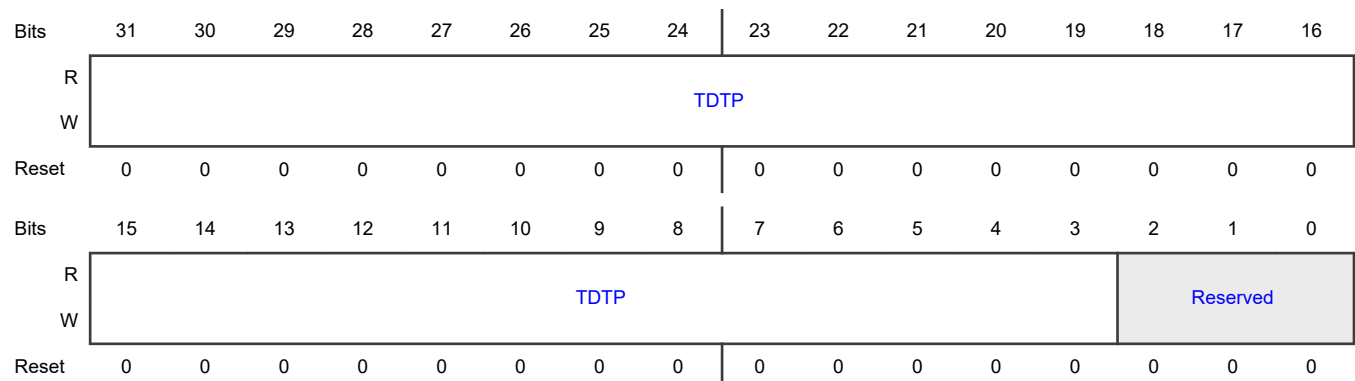
Offset

Register	Offset
DMA_CH4_TxDesc_Tail_Pointer	1320h

Function

The Channel_i Tx Descriptor Tail Pointer register points to an offset from the base and indicates the location of the last valid descriptor.

Diagram



Fields

Field	Function
31-3 TDTP	Transmit Descriptor Tail Pointer This field contains the tail pointer for the Tx descriptor ring. The software writes the tail pointer to add more descriptors to the Tx channel. The hardware tries to transmit all packets referenced by the descriptors between the head and the tail pointer registers. The width of this field depends on the configuration: - 31:2 for 32-bit configuration - 31:3 for 64-bit configuration - 31:4 for 128-bit configuration
2-0 —	Reserved.

1.4.414 (DMA_CH4_RxDesc_Tail_Pointer)

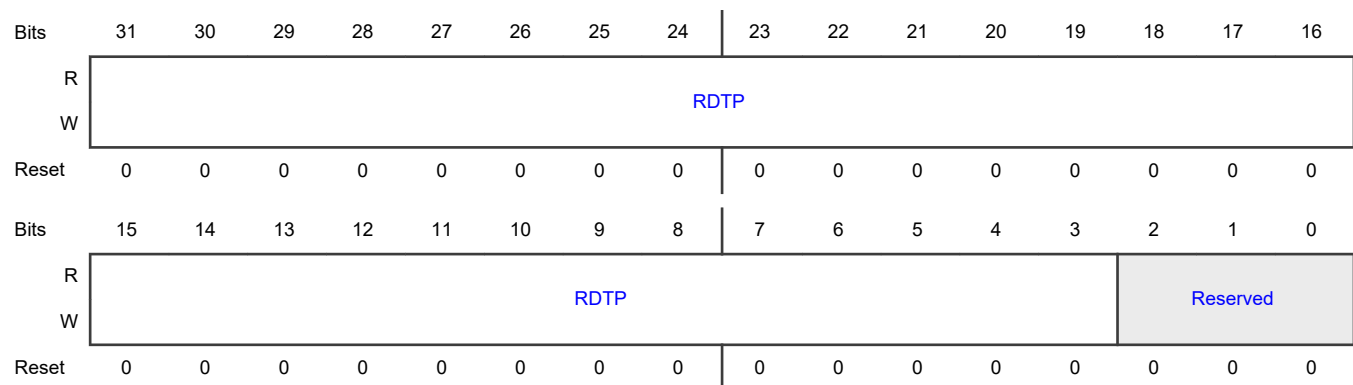
Offset

Register	Offset
DMA_CH4_RxDesc_Tail_Pointer	1328h

Function

The Channel Rx Descriptor Tail Pointer Points to an offset from the base and indicates the location of the last valid descriptor.

Diagram



Fields

Field	Function
31-3 RDTP	Receive Descriptor Tail Pointer This field contains the tail pointer for the Rx descriptor ring. The software writes the tail pointer to add more descriptors to the Rx channel. The hardware tries to write all received packets to the descriptors referenced between the head and the tail pointer registers. The width of this field depends on the configuration: - 31:2 for 32-bit configuration - 31:3 for 64-bit configuration - 31:4 for 128-bit configuration
2-0 —	Reserved.

1.4.415 (DMA_CH4_TxDesc_Ring_Length)

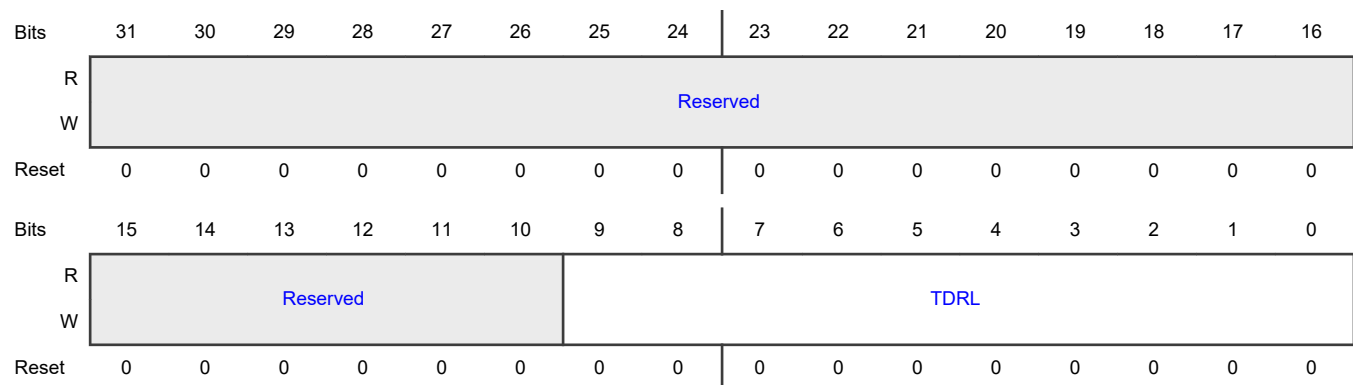
Offset

Register	Offset
DMA_CH4_TxDesc_Ring_Length	132Ch

Function

The Tx Descriptor Ring Length register contains the length of the Transmit descriptor ring.

Diagram



Fields

Field	Function
31-10 —	Reserved.
9-0 TDRL	Transmit Descriptor Ring Length This field sets the maximum number of Tx descriptors in the circular descriptor ring. The maximum number of descriptors is limited to 1K descriptors. NXP recommends a minimum ring descriptor length of 4. For example, You can program any value up to 0x3FF in this field. This field is 10 bits wide, if you program 0x3FF, you can have 1024 descriptors. If you want to have 10 descriptors, program it to a value of 0x9.

1.4.416 (DMA_CH4_RxDesc_Ring_Length)

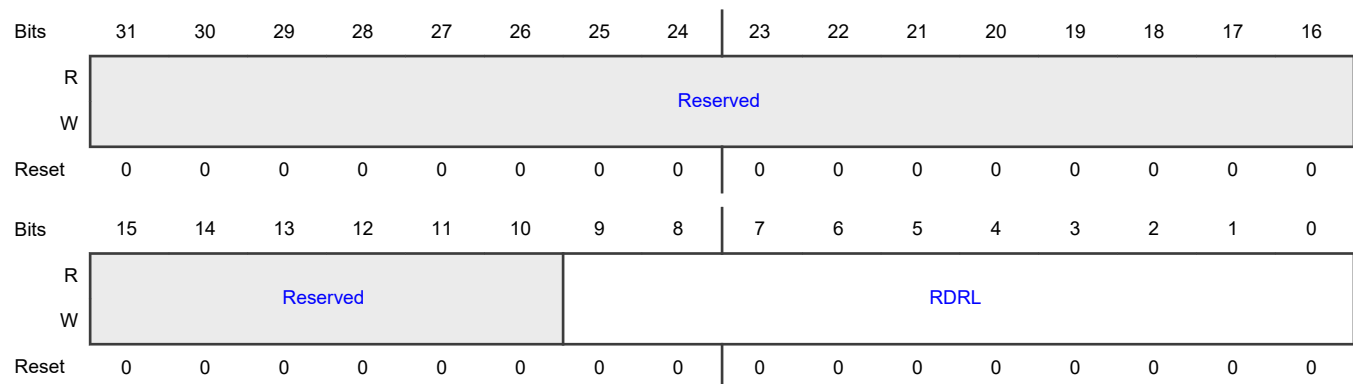
Offset

Register	Offset
DMA_CH4_RxDesc_Ring_Length	1330h

Function

The Channeli Rx Descriptor Ring Length register contains the length of the Receive descriptor circular ring.

Diagram



Fields

Field	Function
31-10 —	Reserved.
9-0	Receive Descriptor Ring Length

Table continues on the next page...

Table continued from the previous page...

Field	Function
RDRL	This register sets the maximum number of Rx descriptors in the circular descriptor ring. The maximum number of descriptors is limited to 1K descriptors. For example, You can program any value up to 0x3FF in this field. This field is 10 bits wide, if you program 0x3FF, you can have 1024 descriptors. If you want to have 10 descriptors, program it to a value of 0x9.

1.4.417 (DMA_CH4_Interrupt_Enable)

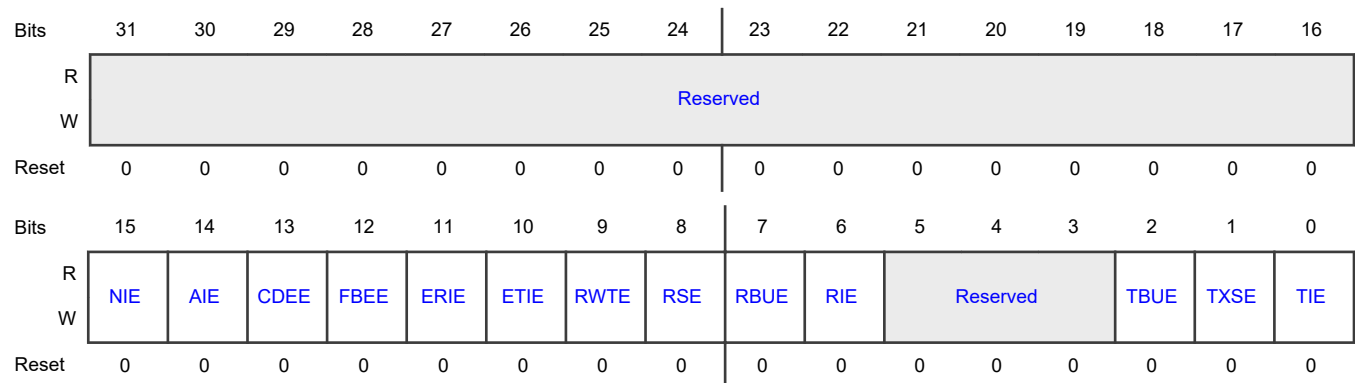
Offset

Register	Offset
DMA_CH4_Interrupt_Enable	1334h

Function

The Channeli Interrupt Enable register enables the interrupts reported by the Status register.

Diagram



Fields

Field	Function
31-16 —	Reserved.
15 NIE	Normal Interrupt Summary Enable When this bit is set, the normal interrupt summary is enabled. This bit enables the following interrupts in the DMA_CH0_Status register: - Bit 0: Transmit Interrupt - Bit 2: Transmit Buffer Unavailable

Table continues on the next page...

Table continued from the previous page...

Field	Function
	- Bit 6: Receive Interrupt - Bit 11: Early Receive Interrupt When this bit is reset, the normal interrupt summary is disabled. 0b - Normal Interrupt Summary is disabled 1b - Normal Interrupt Summary is enabled
14 AIE	Abnormal Interrupt Summary Enable When this bit is set, the abnormal interrupt summary is enabled. This bit enables the following interrupts in the DMA_CH0_Status register: - Bit 1: Transmit Process Stopped - Bit 7: Rx Buffer Unavailable - Bit 8: Receive Process Stopped - Bit 9: Receive Watchdog Timeout - Bit 10: Early Transmit Interrupt - Bit 12: Fatal Bus Error - Bit 13: Context Descriptor Error When this bit is reset, the abnormal interrupt summary is disabled. 0b - Abnormal Interrupt Summary is disabled 1b - Abnormal Interrupt Summary is enabled
13 CDEE	Context Descriptor Error Enable When this bit is set along with the AIE bit, the Descriptor error interrupt is enabled. When this bit is reset, the Descriptor error interrupt is disabled. 0b - Context Descriptor Error is disabled 1b - Context Descriptor Error is enabled
12 FBEE	Fatal Bus Error Enable When this bit is set along with the AIE bit, the Fatal Bus error interrupt is enabled. When this bit is reset, the Fatal Bus Error error interrupt is disabled. 0b - Fatal Bus Error is disabled 1b - Fatal Bus Error is enabled
11 ERIE	Early Receive Interrupt Enable When this bit is set along with the NIE bit, the Early Receive interrupt is enabled. When this bit is reset, the Early Receive interrupt is disabled. 0b - Early Receive Interrupt is disabled 1b - Early Receive Interrupt is enabled

Table continues on the next page...

Table continued from the previous page...

Field	Function
10 ETIE	Early Transmit Interrupt Enable When this bit is set along with the AIE bit, the Early Transmit interrupt is enabled. When this bit is reset, the Early Transmit interrupt is disabled. 0b - Early Transmit Interrupt is disabled 1b - Early Transmit Interrupt is enabled
9 RWTE	Receive Watchdog Timeout Enable When this bit is set along with the AIE bit, the Receive Watchdog Timeout interrupt is enabled. When this bit is reset, the Receive Watchdog Timeout interrupt is disabled. 0b - Receive Watchdog Timeout is disabled 1b - Receive Watchdog Timeout is enabled
8 RSE	Receive Stopped Enable When this bit is set along with the AIE bit, the Receive Stopped Interrupt is enabled. When this bit is reset, the Receive Stopped interrupt is disabled. 0b - Receive Stopped is disabled 1b - Receive Stopped is enabled
7 RBUE	Receive Buffer Unavailable Enable When this bit is set along with the AIE bit, the Receive Buffer Unavailable interrupt is enabled. When this bit is reset, the Receive Buffer Unavailable interrupt is disabled. 0b - Receive Buffer Unavailable is disabled 1b - Receive Buffer Unavailable is enabled
6 RIE	Receive Interrupt Enable When this bit is set along with the NIE bit, the Receive Interrupt is enabled. When this bit is reset, the Receive Interrupt is disabled. 0b - Receive Interrupt is disabled 1b - Receive Interrupt is enabled
5-3 —	Reserved.
2 TBUE	Transmit Buffer Unavailable Enable When this bit is set along with the NIE bit, the Transmit Buffer Unavailable interrupt is enabled. When this bit is reset, the Transmit Buffer Unavailable interrupt is disabled. 0b - Transmit Buffer Unavailable is disabled 1b - Transmit Buffer Unavailable is enabled
1	Transmit Stopped Enable

Table continues on the next page...

Table continued from the previous page...

Field	Function
TXSE	When this bit is set along with the AIE bit, the Transmission Stopped interrupt is enabled. When this bit is reset, the Transmission Stopped interrupt is disabled. 0b - Transmit Stopped is disabled 1b - Transmit Stopped is enabled
0 TIE	Transmit Interrupt Enable When this bit is set along with the NIE bit, the Transmit Interrupt is enabled. When this bit is reset, the Transmit Interrupt is disabled. 0b - Transmit Interrupt is disabled 1b - Transmit Interrupt is enabled

1.4.418 (DMA_CH4_Rx_Interrupt_Watchdog_Timer)

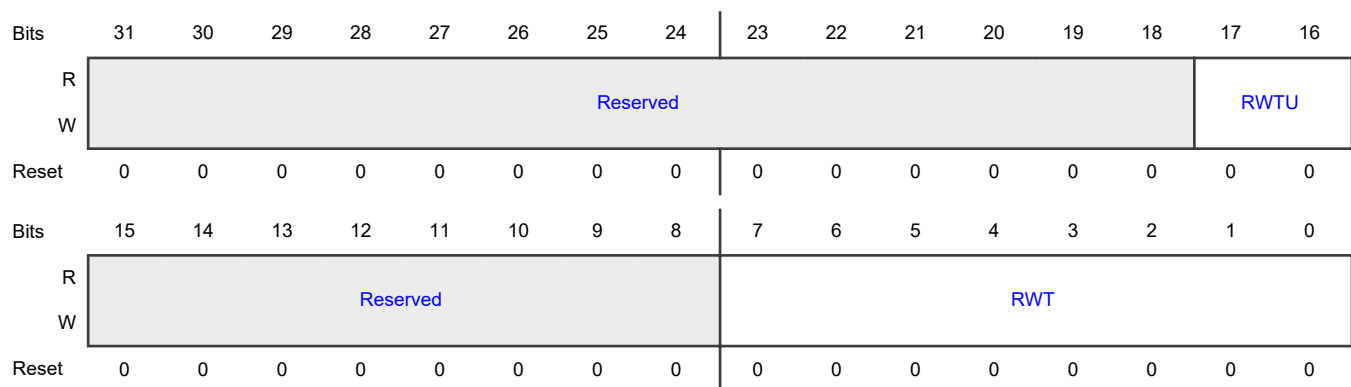
Offset

Register	Offset
DMA_CH4_Rx_Interrupt_Watchdog_Timer	1338h

Function

The Receive Interrupt Watchdog Timer register indicates the watchdog timeout for Receive Interrupt (RI) from the DMA. When this register is written with a non-zero value, it enables the watchdog timer for the RI bit of the DMA_CHi_Status register.

Diagram



Fields

Field	Function
31-18	Reserved.

Table continues on the next page...

Table continued from the previous page...

Field	Function
—	
17-16 RWTU	Receive Interrupt Watchdog Timer Count Units This field indicates the number of system clock cycles corresponding to one unit in RWT field. - 2'b00: 256 - 2'b01: 512 - 2'b10: 1024 - 2'b11: 2048 For example, when RWT=2 and RWTU=1, the watchdog timer is set for 2*512=1024 system clock cycles.
15-8 —	Reserved.
7-0 RWT	Receive Interrupt Watchdog Timer Count This field indicates the number of system clock cycles, multiplied by factor indicated in RWTU field, for which the watchdog timer is set. The watchdog timer is triggered with the programmed value after the Rx DMA completes the transfer of a packet for which the RI bit is not set in the DMA_CH(#i)_Status register, because of the setting of Interrupt Enable bit in the corresponding descriptor RDES3[30]. When the watchdog timer runs out, the RI bit is set and the timer is stopped. The watchdog timer is reset when the RI bit is set high because of automatic setting of RI as per the Interrupt Enable bit RDES3[30] of any received packet.

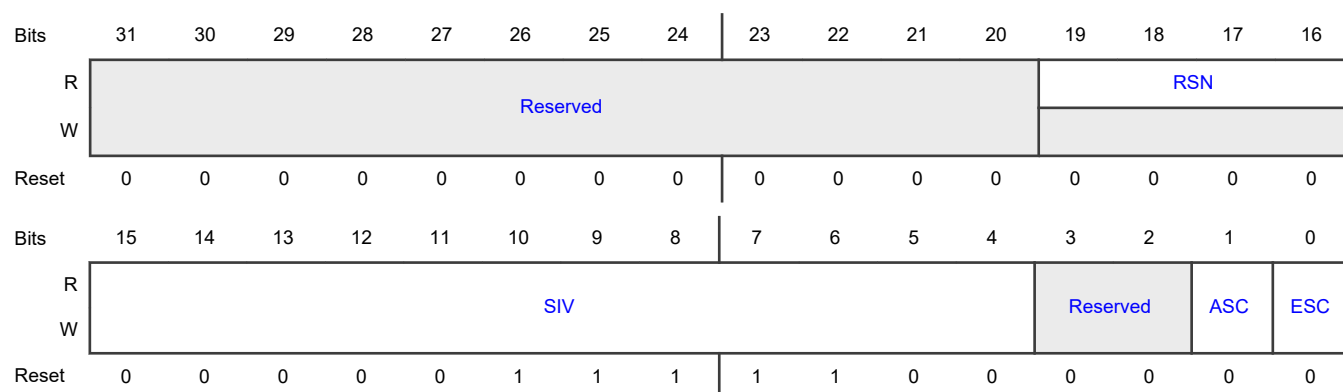
1.4.419 (DMA_CH4_Slot_Function_Control_Status)

Offset

Register	Offset
DMA_CH4_Slot_Function_Control_Status	133Ch

Function

The Slot Function Control and Status register contains the control bits for slot function and the status for Transmit path.

Diagram**Fields**

Field	Function
31-20 —	Reserved.
19-16 RSN	Reference Slot Number This field gives the current value of the reference slot number in the DMA. It is used for slot comparison.
15-4 SIV	Slot Interval Value This field controls the period of the slot interval in which the TxDMA fetches the scheduled packets. A value of 0 specifies the slot interval of 1 us while the maximum value 4095 specifies the slot interval of 4096us. The default/reset value is 0x07C which corresponds to slot interval of 125us
3-2 —	Reserved.
1 ASC	Advance Slot Check When set, this bit enables the DMA to fetch the data from the buffer when the slot number (SLOTNUM) programmed in the Tx descriptor is - equal to the reference slot number given in the RSN field or - ahead of the reference slot number by up to two slots This bit is applicable only when the ESC bit is set. 0b - Advance Slot Check is disabled 1b - Advance Slot Check is enabled
0 ESC	Enable Slot Comparison When set, this bit enables the checking of the slot numbers programmed in the Tx descriptor with the current reference given in the RSN field. The DMA fetches the data from the corresponding buffer only when the slot number is

Table continues on the next page...

Table continued from the previous page...

Field	Function
	- equal to the reference slot number or - ahead of the reference slot number by one slot When reset, this bit disables the checking of the slot numbers. The DMA fetches the data immediately after the descriptor is processed. Note: The UFO (UDP Fragmentation over IPv4)/TSO/USO should not be enabled along with TBS/AVB Slot number check. The UFO/TSO/USO involves multiple packets/segments/fragments transmission for single packet received from application and the slot number check are applicable for fetching of only first segment/fragment. As a result it might be difficult for software to specify slot number for subsequent packets. 0b - Slot Comparison is disabled 1b - Slot Comparison is enabled

1.4.420 (DMA_CH4_Current_App_TxDesc)

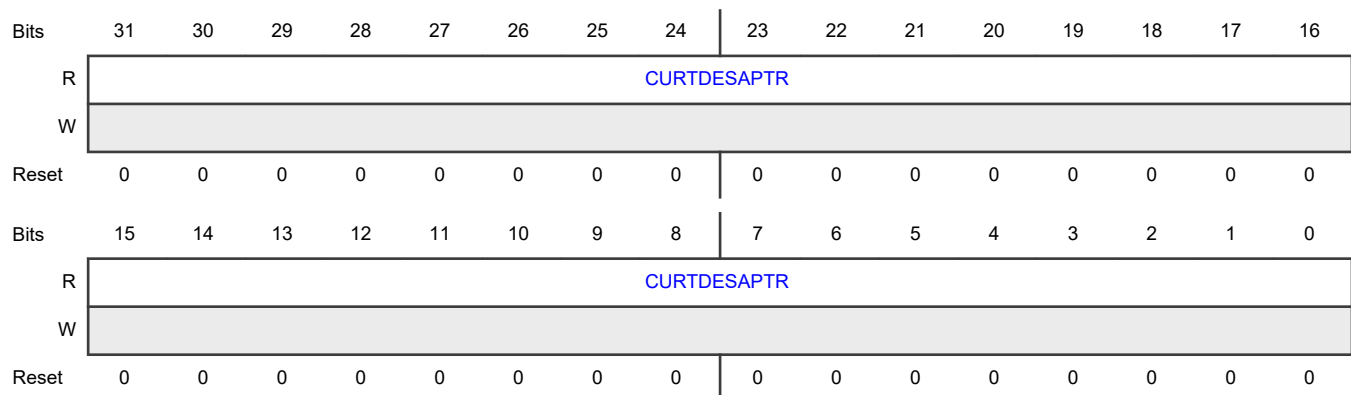
Offset

Register	Offset
DMA_CH4_Current_App_TxDesc	1344h

Function

The Channeli Current Application Transmit Descriptor register points to the current Transmit descriptor read by the DMA.

Diagram



Fields

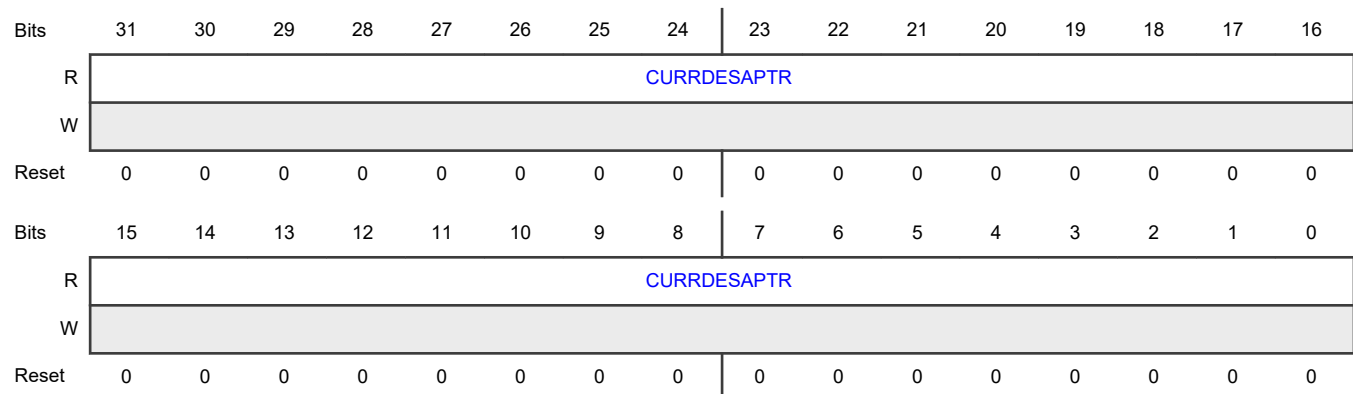
Field	Function
31-0 CURDESAPTR	Application Transmit Descriptor Address Pointer The DMA updates this pointer during Tx operation. This pointer is cleared on reset.

1.4.421 (DMA_CH4_Current_App_RxDesc)**Offset**

Register	Offset
DMA_CH4_Current_App_RxDesc	134Ch

Function

The Channeli Current Application Receive Descriptor register points to the current Receive descriptor read by the DMA.

Diagram**Fields**

Field	Function
31-0 CURRDESAPTR	Application Receive Descriptor Address Pointer The DMA updates this pointer during Rx operation. This pointer is cleared on reset.

1.4.422 (DMA_CH4_Current_App_TxBuffer)

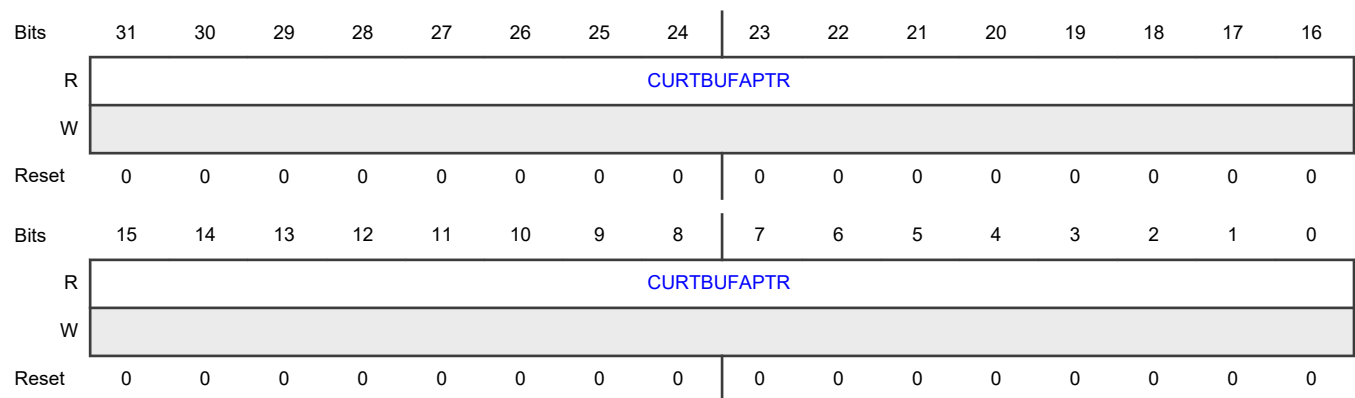
Offset

Register	Offset
DMA_CH4_Current_App_TxBuffer	1354h

Function

The Channel1 Current Application Transmit Buffer Address register points to the current Tx buffer address read by the DMA.

Diagram



Fields

Field	Function
31-0	Application Transmit Buffer Address Pointer
CURTBUFAPTR	The DMA updates this pointer during Tx operation. This pointer is cleared on reset.
R	

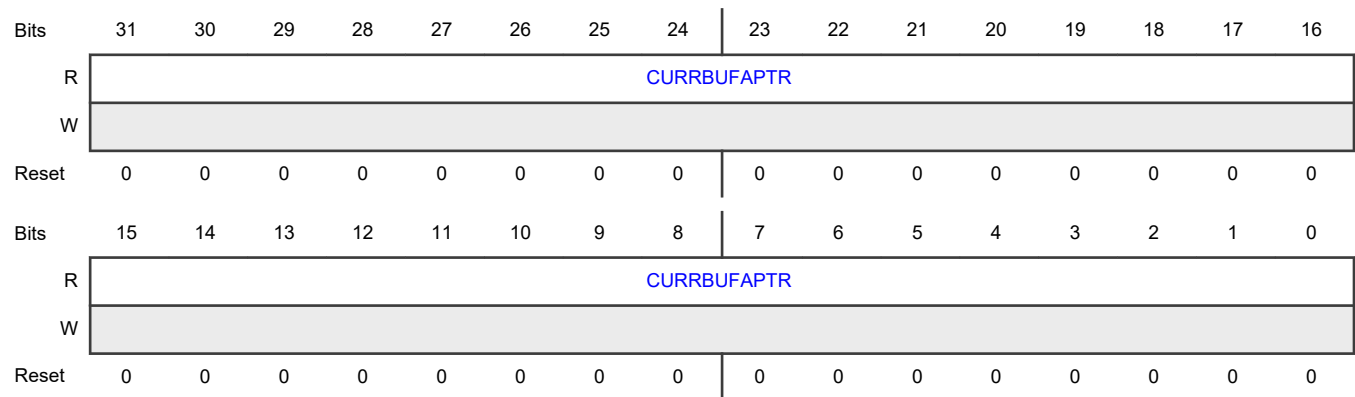
1.4.423 (DMA_CH4_Current_App_RxBuffer)

Offset

Register	Offset
DMA_CH4_Current_App_RxBuffer	135Ch

Function

The Channel 0 Current Application Receive Buffer Address register points to the current Rx buffer address read by the DMA.

Diagram**Fields**

Field	Function
31-0 CURRBUFAPT R	Application Receive Buffer Address Pointer The DMA updates this pointer during Rx operation. This pointer is cleared on reset.

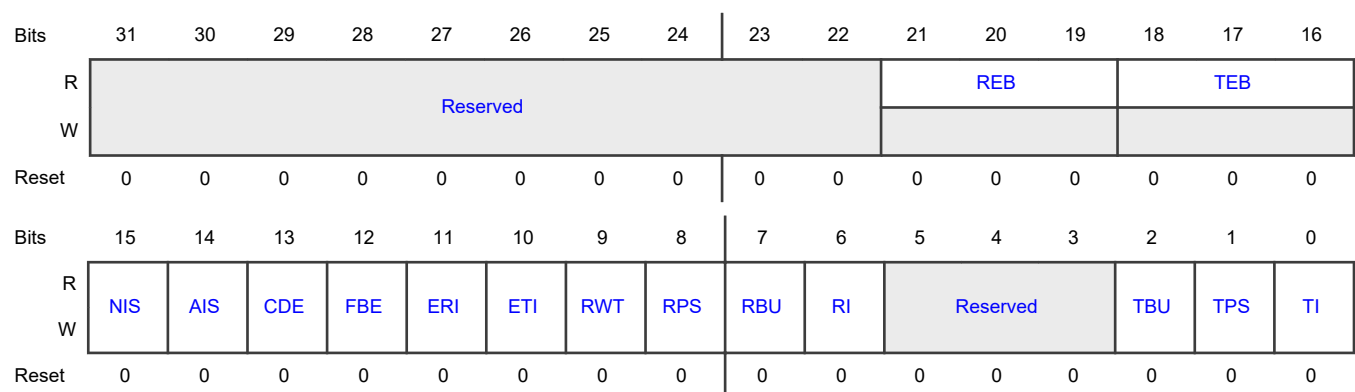
1.4.424 (DMA_CH4_Status)**Offset**

Register	Offset
DMA_CH4_Status	1360h

Function

The software driver (application) reads the Status register during interrupt service routine or polling to determine the status of the DMA.

Note: The number of DMA_CH(##)_Status register in the configuration is the higher of number of Rx DMA Channels and Tx DMA Channels.

Diagram

Fields

Field	Function
31-22 —	Reserved.
21-19 REB	Rx DMA Error Bits This field indicates the type of error that caused a Bus Error. For example, error response on the AHB or AXI interface. - Bit 21 -- 1'b1: Error during data transfer by Rx DMA -- 1'b0: No Error during data transfer by Rx DMA - Bit 20 -- 1'b1: Error during descriptor access -- 1'b0: Error during data buffer access - Bit 19 -- 1'b1: Error during read transfer -- 1'b0: Error during write transfer This field is valid only when the FBE bit is set. This field does not generate an interrupt.
18-16 TEB	Tx DMA Error Bits This field indicates the type of error that caused a Bus Error. For example, error response on the AHB or AXI interface. - Bit 18 -- 1'b1: Error during data transfer by Tx DMA -- 1'b0: No Error during data transfer by Tx DMA - Bit 17 -- 1'b1: Error during descriptor access -- 1'b0: Error during data buffer access - Bit 16 -- 1'b1: Error during read transfer -- 1'b0: Error during write transfer This field is valid only when the FBE bit is set. This field does not generate an interrupt.
15 NIS	Normal Interrupt Summary Normal Interrupt Summary bit value is the logical OR of the following bits when the corresponding interrupt bits are enabled in the DMA_CH0_Interrupt_Enable register: - Bit 0: Transmit Interrupt - Bit 2: Transmit Buffer Unavailable - Bit 6: Receive Interrupt

Table continues on the next page...

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Field	Function
	- Bit 11: Early Receive Interrupt Only unmasked bits (interrupts for which interrupt enable is set in DMA_CH0_Interrupt_Enable register) affect the Normal Interrupt Summary bit. This is a sticky bit. You must clear this bit (by writing 1 to this bit) each time a corresponding bit which causes NIS to be set is cleared. Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect. 0b - Normal Interrupt Summary status not detected 1b - Normal Interrupt Summary status detected
14 AIS	Abnormal Interrupt Summary Abnormal Interrupt Summary bit value is the logical OR of the following when the corresponding interrupt bits are enabled in the DMA_CH0_Interrupt_Enable register: - Bit 1: Transmit Process Stopped - Bit 7: Receive Buffer Unavailable - Bit 8: Receive Process Stopped - Bit 10: Early Transmit Interrupt - Bit 12: Fatal Bus Error - Bit 13: Context Descriptor Error Only unmasked bits affect the Abnormal Interrupt Summary bit. This is a sticky bit. You must clear this bit (by writing 1 to this bit) each time a corresponding bit, which causes AIS to be set, is cleared. Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect. 0b - Abnormal Interrupt Summary status not detected 1b - Abnormal Interrupt Summary status detected
13 CDE	Context Descriptor Error This bit indicates that the DMA Tx/Rx engine received a descriptor error, which indicates invalid context in the middle of packet flow (intermediate descriptor) or all one's descriptor in Tx case and on Rx side it indicates DMA has read a descriptor with either of the buffer address as ones which is considered to be invalid. Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect. 0b - Context Descriptor Error status not detected 1b - Context Descriptor Error status detected
12 FBE	Fatal Bus Error This bit indicates that a bus error occurred (as described in the EB field). When this bit is set, the corresponding DMA channel engine disables all bus accesses. Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect.

Table continues on the next page...

Table continued from the previous page...

Field	Function
	0b - Fatal Bus Error status not detected 1b - Fatal Bus Error status detected
11 ERI	Early Receive Interrupt This bit when set indicates that the RxDMA has completed the transfer of packet data to the memory. In configs supporting ERIC, When ERIC=0, this bit is set only after the Rx DMA has filled up a complete receive buffer with packet data. When ERIC=1, this bit is set after every burst transfer of data from the Rx DMA to the buffer. The setting of RI bit automatically clears this bit. Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect. 0b - Early Receive Interrupt status not detected 1b - Early Receive Interrupt status detected
10 ETI	Early Transmit Interrupt This bit when set indicates that the TxDMA has completed the transfer of packet data to the MTL TXFIFO memory. In configs supporting ERIC: When ETIC=0, this bit is set only after the Tx DMA has transferred a complete packet to MTL. When ETIC=1, this bit is set after completion of (partial) packet data transfer from buffers in the Transmit descriptor in which IOC=1. Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect. 0b - Early Transmit Interrupt status not detected 1b - Early Transmit Interrupt status detected
9 RWT	Receive Watchdog Timeout This bit is asserted when a packet with length greater than 2,048 bytes (10,240 bytes when Jumbo Packet mode is enabled) is received. 0b - Receive Watchdog Timeout status not detected 1b - Receive Watchdog Timeout status detected
8 RPS	Receive Process Stopped This bit is asserted when the Rx process enters the Stopped state. Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect. 0b - Receive Process Stopped status not detected 1b - Receive Process Stopped status detected
7 RBU	Receive Buffer Unavailable This bit indicates that the application owns the next descriptor in the Receive list, and the DMA cannot acquire it. The Rx process is suspended. To resume processing Rx descriptors, the application should change the ownership of the descriptor and issue a Receive Poll Demand command. If this command is

Table continues on the next page...

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Field	Function
	not issued, the Rx process resumes when the next recognized incoming packet is received. In ring mode, the application should advance the Receive Descriptor Tail Pointer register of a channel. This bit is set only when the DMA owns the previous Rx descriptor. Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect. 0b - Receive Buffer Unavailable status not detected 1b - Receive Buffer Unavailable status detected
6 RI	Receive Interrupt This bit indicates that the packet reception is complete. When packet reception is complete, Bit 31 of RDES3 is reset in the last descriptor, and the specific packet status information is updated in the descriptor. The reception remains in the Running state. Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect. 0b - Receive Interrupt status not detected 1b - Receive Interrupt status detected
5-3 —	Reserved.
2 TBU	Transmit Buffer Unavailable This bit indicates that the application owns the next descriptor in the Transmit list, and the DMA cannot acquire it. Transmission is suspended. The TPS0 field of the DMA_Debug_Status0 register explains the Transmit Process state transitions. To resume processing the Transmit descriptors, the application should do the following: 1. Change the ownership of the descriptor by setting Bit 31 of TDES3. 2. Issue a Transmit Poll Demand command. For ring mode, the application should advance the Transmit Descriptor Tail Pointer register of a channel. Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect. 0b - Transmit Buffer Unavailable status not detected 1b - Transmit Buffer Unavailable status detected
1 TPS	Transmit Process Stopped This bit is set when the transmission is stopped. Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect. 0b - Transmit Process Stopped status not detected 1b - Transmit Process Stopped status detected
0 TI	Transmit Interrupt This bit indicates that the packet transmission is complete. When transmission is complete, Bit 31 of TDES3 is reset in the last descriptor, and the specific packet status information is updated in the descriptor.

Table continues on the next page...

Table continued from the previous page...

Field	Function
	Access restriction applies. Self-set to 1 on internal event. Setting 1 clears. Setting 0 has no effect. 0b - Transmit Interrupt status not detected 1b - Transmit Interrupt status detected

1.4.425 (DMA_CH4_Miss_Frame_Cnt)

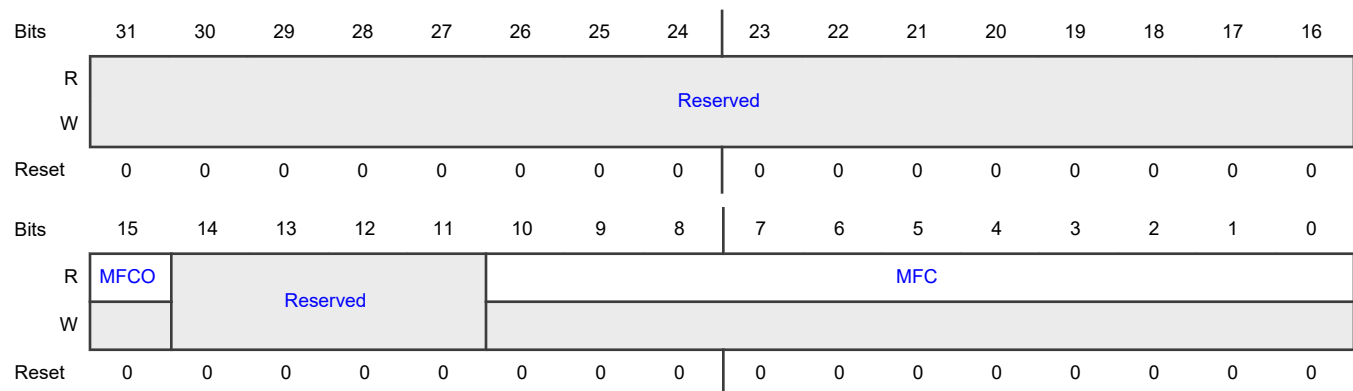
Offset

Register	Offset
DMA_CH4_Miss_Frame_Cnt	1364h

Function

This register has the number of packet counter that got dropped by the DMA either due to Bus Error or due to programming RPF field in DMA_CH*{i}*_Rx_Control register.

Diagram



Fields

Field	Function
31-16 —	Reserved.
15 MFCO	Overflow status of the MFC Counter When this bit is set then the MFC counter does not get incremented further. The bit gets cleared when this register is read. Access restriction applies. Clears on read. Self-set to 1 on internal event.

Table continues on the next page...

Table continued from the previous page...

Field	Function
	0b - Miss Frame Counter overflow not occurred 1b - Miss Frame Counter overflow occurred
14-11 —	Reserved.
10-0 MFC	Dropped Packet Counters This counter indicates the number of packet counters that are dropped by the DMA either because of bus error or because of programing RPF field in DMA_CH\${i}_Rx_Control register. The counter gets cleared when this register is read. Access restriction applies. Clears on read. Self-set to 1 on internal event.

1.4.426 (DMA_CH4_RXP_Accept_Cnt)

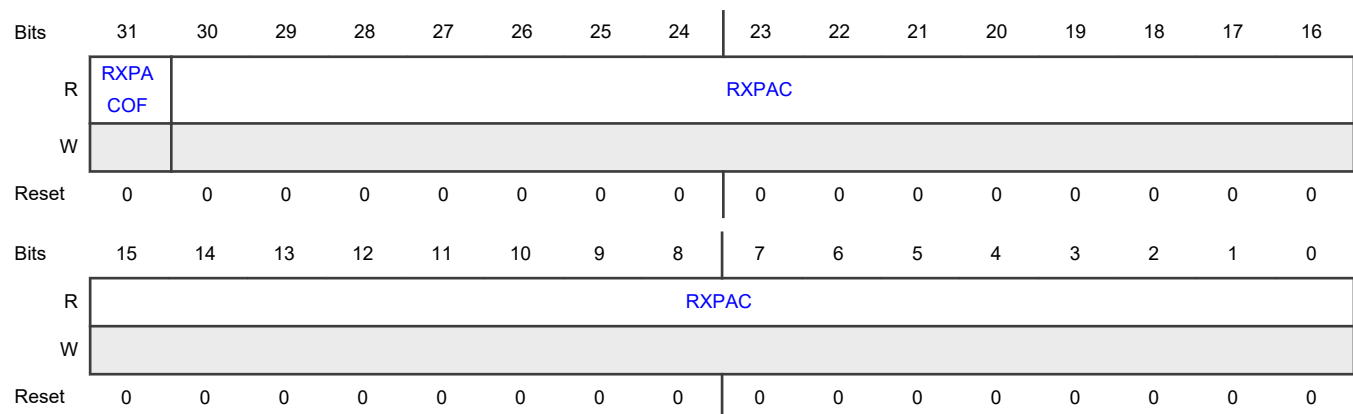
Offset

Register	Offset
DMA_CH4_RXP_Accept_Cnt	1368h

Function

The DMA_CH(#i)_RXP_Accept_Cnt registers provides the count of the number of frames accepted by Rx Parser.

Diagram



Fields

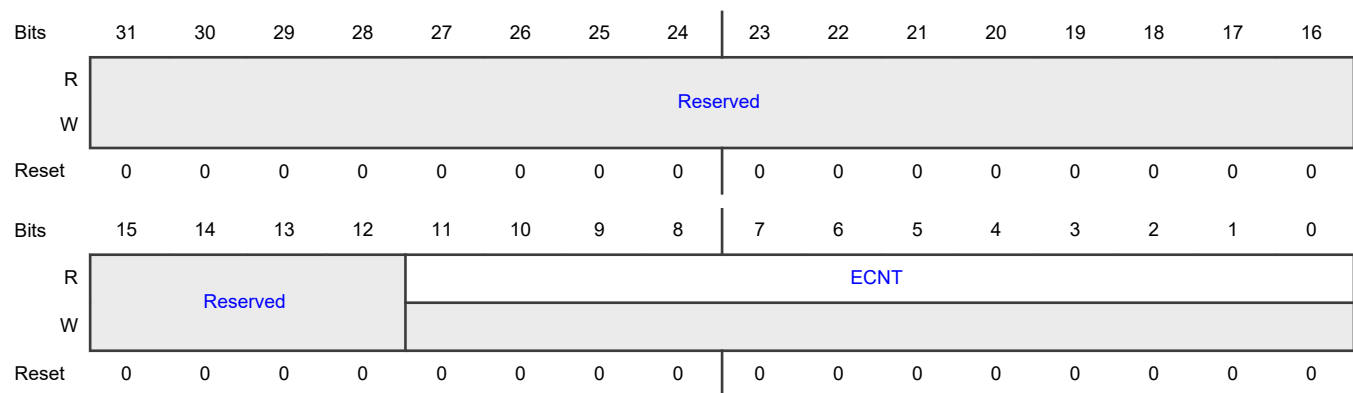
Field	Function
31 RXPACOF	Rx Parser Accept Counter Overflow Bit When set, this bit indicates that the RXPAC Counter field crossed the maximum limit. Access restriction applies. Clears on read. Self-set to 1 on internal event. 0b - Rx Parser Accept Counter overflow not occurred 1b - Rx Parser Accept Counter overflow occurred
30-0 RXPAC	Rx Parser Accept Counter This 31-bit counter is implemented whenever a Rx Parser Accept a packet due to AF =1. The counter is cleared when the register is read.

1.4.427 (DMA_CH4_RX_ERI_Cnt)**Offset**

Register	Offset
DMA_CH4_RX_ERI_Cnt	136Ch

Function

The DMA_CH(#i)_RX_ERI_Cnt registers provides the count of the number of times ERI was asserted.

Diagram**Fields**

Field	Function
31-12 —	Reserved.

Table continues on the next page...

Table continued from the previous page...

Field	Function
11-0 ECNT	ERI Counter When ERIC bit of DMA_CH(#i)_RX_Control register is set, this counter increments for burst transfer completed by the Rx DMA from the start of packet transfer. This counter is reset at the start of new packet.

1.5 Ethernet Modes and PHY Management Interface

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GMAC IP support the following modes.

- MII 10/100 Mbps interface
- RMII 10/100 Mbps interface
- RGMII 100/1000 Mbps interface
- SGMII 1000/2500 Mbps through GMII interface (2500 Mbps support varies per device)

Phy_intf_sel input signal decides which mode has been selected. Phy_intf_sel signal gets sampled at reset. Please see the signal description. PS and FES bit of MAC configuration register selects the speed of mode.

GMAC IP supports the IEEE802.3-2015 specification for MII/GMII, RMII specification version 1.2 from RMII consortium, and RGMII specification version 2.6 from HP/Marvell.

NOTE

- RMII reference clock (50 MHz) can be fed to IP from an external source or internally from PLL on SoC.
- All phy interface modes are being supported on same set of I/O pin at IP depending upon mode configuration.

GMAC IP can access the phy registers through Station Management Agent (SMA) which is a two wire Station management interface (MIM). One wire is management data clock (MDC) and other wire is management data input/output (MDIO). As per IEEE 802.3 specification, maximum operating frequency of MDC is 2.5 Mhz which can be derived from system clock using a divider. The CR bit of MAC_MDIO_Address register can be programmed to generate different MDC clock frequency.

SMA supports both MDIO clause 45 and clause 22 frame structure as per IEEE802.3 specification. The C45E bit of MAC_MDIO_Address register can be set to enable the clause 45 frame structure.

If the MII interface is overclocked to run at 200 Mbps, it reduces the accuracy of the 1588 timestamp. This is because the MAC logic uses the MII interface speed, as specified in [MAC_Configuration\[FES\]](#), to adjust or compensate for the timestamps taken at MII, as compared to the time generated in the PTP clock domain.

1.6 VLAN and Double VLAN Insertion, Deletion, Replacement and Tagging

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Following section describes the VLAN and Double VLAN features.

1.6.1 Double VLAN Processing

IP can support processing of two VLAN tags in double VLAN tagging processing. These two VLAN tags named as inner (C-VLAN) and outer (S-VLAN) VLAN tag. If packet is having only one tag then it would be considered as outer tag.

IP supports the following function in Double VLAN processing feature using the MAC_VLAN_Incl, MAC_Inner_VLAN_Incl and MAC_VLAN_Tag_ctrl registers.

- Insertion, replacement, or deletion of up to two VLAN tags in the Transmit path.

- Packet filtering and stripping based on any one of the two VLAN Tags in the Receive path. Stripping and providing up to two VLAN Tags in the Receive path as a part of the Receive status.

1.6.1.1 Transmit Path

Following table describes the features supported by the MAC on the Transmit side.

Table 3. Double VLAN Processing Features in Transmit Path

Feature	Description
Support for C-VLAN and S-VLAN Tag types	The inner or outer VLAN tag can be of C-VLAN and S-VLAN type. The VLAN type is specified through the CSVL bit of MAC_VLAN_Incl and MAC_Inner_VLAN_Incl registers. IP supports processing of any sequence of outer and inner VLAN tags. Note: IP does not support the C-VLAN S-VLAN sequence. The MAC does not check whether the packet provided by the application has a valid sequence of the VLAN Tag types or the insertion or replacement operation results in invalid sequence of VLAN Tag type. Therefore, the application must provide correct sequence of VLAN Tag types and program the MAC in such a way that it results in correct sequence of VLAN Tag types in the transmitted packet. The application must ensure the following: • The inner tag should not be S-VLAN when outer C-VLAN Tag insertion is enabled. • The outer tag should not be C-VLAN when inner S-VLAN Tag insertion is enabled. • The inner tag should not be S-VLAN when outer tag should be replaced with C-VLAN. • The outer tag should not be C-VLAN when inner tag should be replaced with S-VLAN.
VLAN Tag deletion	VLAN tag deletion for outer or inner tag can be enabled through VLC field in the MAC_VLAN_Incl or MAC_Inner_VLAN_Incl register, respectively. When VLAN deletion is enabled, the MAC deletes the tag present at the corresponding position. When a packet has only one tag, it is considered as the outer tag. If inner tag deletion is enabled and the packet has only one tag, the MAC does not delete the tag.
VLAN Tag Insertion or Replacement	VLAN tag insertion or replacement for outer or inner tag can be enabled through VLC field in the MAC_VLAN_Incl or MAC_Inner_VLAN_Incl register, respectively. When VLAN tag insertion or replacement is enabled, the VLTi bit in the MAC_VLAN_Incl or MAC_Inner_VLAN_Incl register is used to determine whether the VLAN tag should be taken from the register or the Control Word.

1.6.1.2 Receive Path

Following table describes the features supported by the MAC on the Receive side and the corresponding bits in the MAC_VLAN_Tag_ctrl register.

Table 4. Double VLAN Processing in Receive Path

Feature	Description
Outer or inner VLAN tag-based filtering	The MAC can filter packets based on the outer or inner VLAN tag through the ERIVLT bit.
C-VLAN or S-VLAN tag-based filtering	The MAC can filter packets based on the C-VLAN or S-VLAN type based on the ERSVLM bit.

Table continues on the next page...

Table 4. Double VLAN Processing in Receive Path (continued)

Outer and Inner VLAN Tag stripping	The MAC can strip the outer and inner VLAN Tags from received frame based on the EVLS and EIVLS bits.
16-bit outer and inner VLAN Tag and Type in Rx status	The MAC can provide the 16-bit outer and inner VLAN Tag and Type in the Rx status based on the EVLRXS and EIVLRXS bits, respectively.
Disabling or skipping checking of outer VLAN Tag type	The MAC can disable or skip checking of outer VLAN Tag type to match C-VLAN or SVLAN based on the DOVLTC bit.

1.6.1.3 Source Address and VLAN Insertion, Replacement, or Deletion:

IP supports the SA and VLAN insertion, replacement and deletion. The source address (SA) and VLAN fields are Tx packet-related control information that are provided as part of the control word through ATI interfaces. IP supports the feature to insert or replace the source address based on the information in the MAC Address Registers, and also the feature to insert, replace or delete the VLAN fields (VLAN Type and VLAN Tag) based on the setting of the VLTi bit in the MAC_VLAN_Incl register. SA insertion or replacement feature can be enabled for all Transmit packets or selective packets. Similarly, VLAN insertion, replacement, or deletion feature can be enabled for all Tx packets or selective packets.

1.6.1.4 Programming Source Address Insertion or Replacement:

The software can use the SA insertion or replacement feature to instruct the MAC to do the following for Tx packets:

- Insert the content of the MAC Address Registers in the SA field
- Replace the content of the SA field with the content of the MAC Address Registers

When SA insertion is enabled, the application must ensure that the packets sent to the MAC do not have the SA field. The MAC does not check whether the SA field is present in the Transmit packet and it inserts the content of MAC Address Registers in the SA field. Similarly, when SA replacement is enabled, the application must ensure that the SA field is present in the packets sent to the MAC.

The MAC replaces the six bytes following the Destination Address field in the Transmit packet with the content of the MAC Address Registers.

SA insertion/replacement can be enabled for all the transmit packet or selective transmit packets.

- SARC bit of MAC_Configuration register can be program to enable the SA insertion or replacement for all Tx packets.
- SA insertion for selective packet can be enabled by programming the SA Insertion Control field (bits [25:23] of TDES3) in the first Transmit descriptor of the packet. When Bit 25 of TDES3 is set, the SA Insertion Control field indicates insertion or replacement by MAC Address1 registers. When Bit 25 of TDES3 is reset, it indicates insertion or replacement by MAC Address 0 registers.

If MAC Address1 Registers are not enabled, the MAC Address0 registers are used for insertion or replacement irrespective of the value of the most-significant bit of the SA Insertion Control field.

1.6.1.5 Programming VLAN Insertion, Replacement, or Deletion

The software can use the VLAN insertion, replacement, or deletion feature to instruct the MAC to do the following for Tx packets:

- Delete the VLAN Type and VLAN Tag fields
- Insert or replace the VLAN Type and VLAN Tag fields

Insertion or replacement is done based on the setting of VLTi bit in the MAC_VLAN_Incl register as described in the following table.

Table 5. VLAN Insertion or Replacement Based on VLTi Bit

Condition	Description
VLTi bit is set	The MAC inserts or replaces the following: - VLAN Type field (C-VLAN or S-VLAN as indicated by the CSVL bit of MAC_VLAN_Incl register) - VLAN Tag field depending on the Content of the VT field of Transmit context descriptor of the packet
VLTi bit is reset	The MAC inserts or replaces the following: - VLAN Type field (C-VLAN or S-VLAN as indicated by the CSVL bit of MAC_VLAN_Incl register) - VLAN Tag field with the VLT field of MAC_VLAN_Incl register

When VLAN replacement or deletion is enabled, the MAC checks if the VLAN Type field (0x8100 or 0x88a8) is present after the DA and SA fields in the Transmit packet. The replace or delete operation does not occur if the VLAN Type field is not detected in two bytes following the DA and SA fields. However, when VLAN insertion is enabled, the MAC does not check the presence of VLAN Type field in the Transmit packet and just inserts the VLAN Type and VLAN Tag fields.

VLAN insertion, replacement, or deletion feature can be enabled for all Tx packets by programming the VLC and VLP fields of MAC_VLAN_Incl register.

VLAN insertion, replacement, or deletion for selective packets can be enabled by programming the VTIR field of TDES2 Normal Descriptor.

In addition, the VLP (VLAN Priority control) bit must be reset in MAC_VLAN_Incl register(for outer VLAN) and MAC_Inner_VLAN_Incl register (in inner VLAN) for the MAC to take the control inputs from the host, depending on the configuration.

1.6.1.6 Queue/Channel Based VLAN Tag Insertion on Tx

IP supports channel/queue based VLAN tag insertion on all transmitted packets. Queue/Channel specific VLAN tag registers are accessed using indirect addressing via the MAC_VLAN_Incl register. VLAN type and tag value can be independently programmed for each queue/channel. If CBTi field is set in the MAC_VLAN_Incl register, the VLAN tag is inserted on every packet that is transmitted from a queue/channel, with the programmed tag value taken from queue/channel specific VLAN tag register.

NOTE

See "Programming Sequence for Queue/Channel Based VLAN Inclusion Register"

1.7 Packet Filtering

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There are various packets filtering scheme available in MAC receiver. Filtering sequence in their precedence order of received packet is shown in figure.

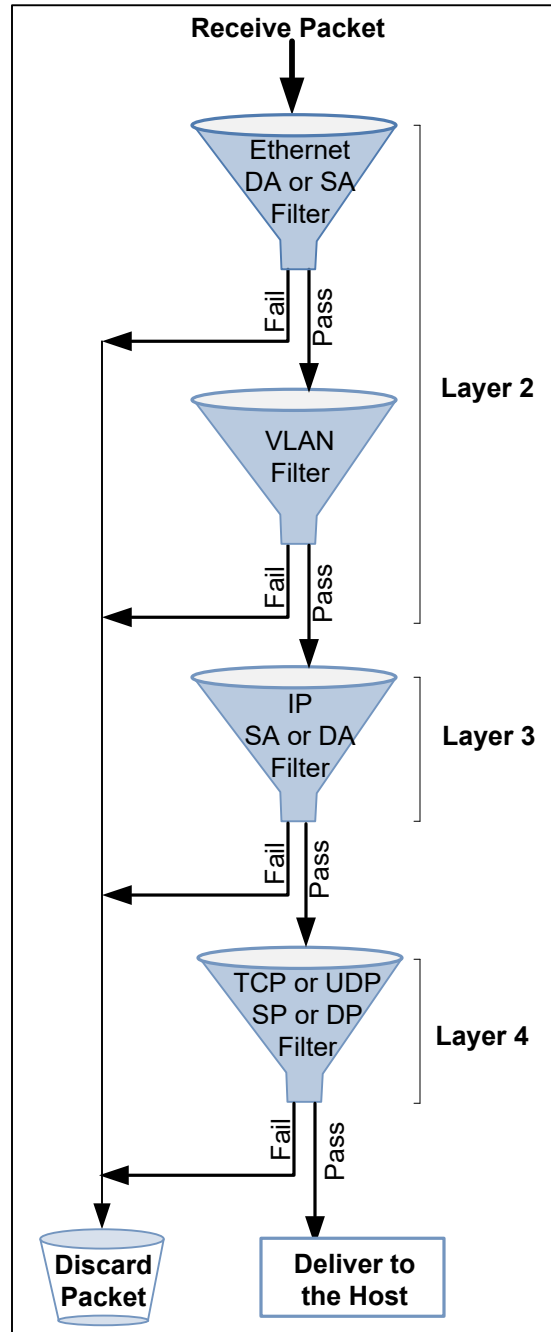


Figure 2. Packet Filtering Sequence

If any of the Layer filters are not enabled, that filter is bypassed and the subsequent filter is applied. A packet that fails any of the filters, is discarded. However, the discarded packet can be forwarded to the host based on the register control.

1.7.1 Source Address or Destination Address Filtering

The Address Filtering Module of the MAC checks the source address (SA) and destination address (DA) fields of each incoming packet. Programming of different type of address filtering is given below in this section.

1.7.1.1 Unicast Destination Address Filtering

MAC supports up to 2 MAC addresses for unicast perfect filtering. If perfect filtering is selected (HUC bit of MAC_Packet_Filter register is reset), the MAC compares all 48 bits of received unicast address with the programmed MAC address for any match. Default MacAddr0 is always enabled. The MacAddr1 address is selected with an individual enable bit. Each byte of MacAddr1 can be masked during comparison with corresponding received DA byte by setting the corresponding Mask Byte Control bit in the register. This enables group address filtering for the DA.

In hash filtering mode (when HUC bit is set), the MAC performs imperfect filtering for unicast addresses using a 64-bit Hash table. For hash filtering, the MAC uses the upper 6 bits CRC of the received destination address to index the content of the Hash table. A value of 00000 selects Bit 0 of selected register, and a value of 11111 selects Bit 63 of Hash Table register. If the corresponding bit (indicated by the 6-bit CRC) is set to 1, the unicast packet is considered to have passed the Hash filter; otherwise, the packet is considered to have failed the Hash filter.

1.7.1.2 Multicast Destination Address Filtering

PM bit of MAC_Packet_Filter register can be enabled to pass all multicast packets or disabled to do multicast addresses filtering based on HMC bit of MAC_Packet_Filter register. The multicast address is compared with the programmed MAC Destination Address registers (1–2). Also, Group address filtering is supported.

In Hash filtering mode, the MAC performs imperfect filtering using a 64-bit Hash table. The MAC uses the upper 6-bits CRC of received multicast address to index the content of the Hash table. A value of 000000 selects Bit 0 of selected register and a value of 111111 selects Bit 63 of the Hash Table register. If the corresponding bit is set to 1, the multicast packet is considered to have passed the Hash filter. Otherwise, the packet is considered to have failed the Hash filter.

1.7.1.3 Hash or Perfect Address Filtering

HPF bit of MAC_Packet_Filter register must be configured for enabling or disabling the destination address filtering by either Hash filter or Perfect filter. HUC and HMC bit must be configured to select the Hash filtering or Perfect filtering for Unicast/Multicast packet.

1.7.1.4 Broadcast Address Filtering

The MAC does not filter any broadcast packets by default. DBF bit of MAC_Packet_filter register must be set to reject the broadcast packets.

1.7.1.5 Unicast Source Address Filtering

By default, MAC compares the source address field with the value programmed in the source address registers. Bit 30 of MAC address register [1-2] can be configure to use it for source address instead of destination address comparison. The MAC also supports group filtering with source address(SA) . You can filter a group of addresses by masking one or more bytes of the address. The MAC drops the packets that fail the SA filter if the SAF bit is set in MAC_Packet_Filter register. Otherwise, the result of the SA filter is given as a status bit in the Receive Status word. When the SAF bit is set, the SA filter and DA filter result is AND'ed to decide whether the packet needs to be forwarded. This means that the packet is dropped if either filter fails. The packet is forwarded to the application only if the packet passes both filters in-order.

1.7.1.6 Inverse Filtering

For DA and SA filtering, you can invert the filter-match result at the final output by setting the DAIF and SAIF bits of MAC_Packet_Filter register. The DAIF bit is applicable for both Unicast and Multicast DA packets. The result of the unicast or multicast destination address filter is inverted in this mode. Similarly, when the SAIF bit is set, the result of unicast SA filter is reversed.

Following two tables summarize the DA and SA filtering based on the type of packets received.

Table 6. Destination Address Filtering

Packet Type	PR	HPF	HUC	DAIF	HMC	PM	DBF	DA Filter Operation
Broadcast	1	X	X	X	X	X	X	Pass
	0	X	X	X	X	X	0	Pass
	0	X	X	X	X	X	1	Fail
Unicast	1	X	X	X	X	X	X	Pass all packets
	0	X	0	0	X	X	X	Pass on Perfect/Group filter match
	0	X	0	1	X	X	X	Fail on Perfect/Group filter match
	0	0	1	0	X	X	X	Pass on Hash filter match
	0	0	1	1	X	X	X	Fail on Hash filter match
	0	1	1	0	X	X	X	Pass on Hash or Perfect/Group filter match
	0	1	1	1	X	X	X	Fail on Hash or Perfect/Group filter match
Multicast	1	X	X	X	X	X	X	Pass all packets
	X	X	X	X	X	1	X	Pass all packets
	0	X	X	0	0	0	X	Pass on Perfect/Group filter match and drop Pause packets if PCF = 0x
	0	0	X	0	1	0	X	Pass on Hash filter match and drop Pause packets if PCF = 0x
	0	1	X	0	1	0	X	Pass on Hash or Perfect/Group filter match and drop Pause packets if PCF = 0x
	0	X	X	1	0	0	X	Fail on Perfect/Group filter match and drop Pause packets if PCF = 0x
	0	0	X	1	1	0	X	Fail on Hash filter match and drop Pause packets if PCF = 0x
	0	1	X	1	1	0	X	Fail on Hash or Perfect/Group filter match and drop Pause packets if PCF = 0x

Table 7. Source Address Filtering

Packet Type	PR	SAIF	SAF	SA Filter Operation
Unicast	1	X	X	Pass all packets.

Table continues on the next page...

Table 7. Source Address Filtering (continued)

Packet Type	PR	SAIF	SAF	SA Filter Operation
	0	0	0	Pass status on Perfect or Group filter match but do not drop packets that fail
	0	1	0	Fail status on Perfect or Group filter match but do not drop packet
	0	0	1	Pass on Perfect or Group filter match and drop packets that fail
	0	1	1	Fail on Perfect or Group filter match and drop packets that fail

NOTE

When the RA bit of MAC_Packet_Filter register is set, all packets are forwarded to the system along with the correct result of the address filtering in the Rx Status.

1.7.2 VLAN Filtering

VLAN tag filter can be done by either perfect match or hash table. Depending upon the configuration, MAC can compare either lower 12 bits or all 16 bits of received VLAN tag for perfect match. MAC forwarded the VLAN tagged packet with match status while it drops all the packets which don't matches.

MAC uses 16-bit VLAN hash table for group address filtering based on the VLAN tag. If hash filtering is enabled then most significant 4-bit of CRC-32 of VLAN tag are used to index the content of the VLAN Hash Table Register. A value of 1 in the VLAN Hash Table register, corresponding to the index, indicates that the VLAN tag of the packet matched and the packet should be forwarded. A value of 0 indicates that VLAN-tagged packet should be dropped.

The MAC also supports the inverse matching for VLAN packets. In the inverse matching mode, when the VLAN tag of a packet matches the perfect or hash filter, the packet should be dropped. If the VLAN perfect and VLAN hash match are enabled, a packet is considered as matched if either the VLAN hash or the VLAN perfect filter matches. When inverse match is set, a packet is forwarded only when both perfect and hash filters indicate mismatch.

Table below shows the different possibilities for VLAN matching and the final VLAN match status. When the RA bit of MAC_Packet_Filter register is set, all packets are received and the VLAN match status is indicated in the VF bit of RDES2 Normal Descriptor (Write-Back Format). When the RA bit is not set and the VTFE bit is set in MAC_Packet_Filter register, the packet is dropped if the final VLAN match status is Fail.

When VLAN VID is programmed to 0 in the VL field of MAC_VLAN_Tag register, all VLAN-tagged packets are considered as perfect matched but the status of the VLAN hash match depends on the VTHM and VTIM bits in MAC_VLAN_Tag register

Table 8. VLAN Match Status

VID	VLAN Perfect Filter Match Result	VTHM Bit	VLAN Hash Filter Match Result	VTIM Bit	Final VLAN Match Status
VID = 0	Pass	0	X	X	Pass
	Pass	1	X	0	Pass
	Pass	1	Fail	1	Pass
	Pass	1	Pass	1	Fail
VID!= 0	Pass	X	X	0	Pass

Table continues on the next page...

Table 8. VLAN Match Status (continued)

VID	VLAN Perfect Filter Match Result	VTHM Bit	VLAN Hash Filter Match Result	VTIM Bit	Final VLAN Match Status
	Fail	0	X	0	Fail
	Fail	1	Fail	0	Fail
	Fail	1	Pass	0	Pass
	Fail	0	X	1	Pass
	Pass	X	X	1	Fail
	Fail	1	Pass	1	Fail
	Fail	1	Fail	1	Pass

In this table, X represents any value.

1.7.3 VLAN Filter Fail Packets Queue

When VLAN filtering is enabled, the VLAN Filter Fail Packets can be routed to a programmable Queue (VFFQ) when Bit RA = 1 or VTFE = 0 of MAC_Packet_Filter register and the enable bit (VFFQE of MAC_RXQ_CTRL4) for the queue is set.

The packets that passes the VLAN filtering, are routed to Rx queues based on the VLAN TAG priority field by programming the PSRQ in the corresponding MAC_RxQ_Ctrl2 and MAC_RxQ_Ctrl3 registers. The packets that fail the VLAN filter are discarded if RA=0 or VTFE=1. However, when RA=1 or VTFE=0, the VLAN filter fail packets are still forwarded to the application. In such a scenario, when the VLAN Filter Fail Queue Enable (VFFQE) bit is set, the VLAN filter fail packets are forwarded to the Rx queue number programmed in the VFFQ bit. If VFFQE=0, the Rx queue number is determined by the VLAN priority mapping as per the PSRQ fields.

Table below shows the Rx Queue routing table for Unicast tagged packets, with DA/SA Filter enabled.

Table 9. Rx Queue Routing table for Unicast Tagged Packets

RA	VFTE	SA/DA Filter Result	VLAN Filter Result	VFFQE	Queue Routing
X	X	PASS	PASS	X	PSRQ
0	0	PASS	FAIL	0	PSRQ
0	0	PASS	FAIL	1	VFFQ
0	X	FAIL	X	X	DROPPED
0	1	PASS	FAIL	X	DROPPED
1	X	FAIL	X	0	UFFQ*/PSRQ
1	X	FAIL	X	1	UFFQ*/VFFQ
1	X	PASS	FAIL	0	PSRQ
1	X	PASS	FAIL	1	VFFQ

X : Don't care condition * : When UFFQE is enabled else PSRQ

1.7.4 Extended Receive VLAN Filtering and Routing

When Extended Rx VLAN filtering and routing is enabled then both perfect filtering and Hash filtering can be enabled. The overall VLAN filter result is based on the Perfect filter result and Hash filter result (if enabled). Filter result is passed to application/software as part of the status bit. Extended Routing will happen only if VLAN filter has passed. Routing is only based on Perfect filter result. Each Perfect filter has a DMA channel enable and DMA channel number filed which should be programmed for routing.

For more details about routing, see "Extended VLAN Based DMA Selection".

1.7.4.1 Comparison mode

Software/Application has following comparison option for each VLAN tag filter

- It can program the MAC to compare the inner or outer VLAN tag with either 12 bits or 16 bits programmed VID.
- It can choose that VID comparison is for SVLAN or CVLAN type frames, if type check is enabled for a filter.

1.7.4.2 Filtering

Perfect Filtering is done based on the MAC_VLAN_Tag_Filter registers. For each VLAN Tag Filter, the MAC will compare the relevant VLAN Tag ID and gives a result. If any one of the VLAN Tag Filters gives a match then the frame is considered to have passed the VLAN Tag Filters. If the frame mismatches all the filters, then the frame is considered to have failed the VLAN filter. This behavior is applicable only when the Inverse Filtering is not enabled in MAC_VLAN_Tag_Ctrl Register.

If inverse Filtering is enabled and the frame has mis-matched all the relevant filters then it is considered to have passed the VLAN filter. If the frame matches any one of the relevant filters then it is considered as a fail. If none of the enabled filters can perform a comparison or if none of the filters are enabled, then the frame is bypassed to the application.

The overall filter result and the programming on the VTFE and RA bits of the MAC Filter Register determine if the frame will be dropped or forwarded to the application. If RA = 1 or VTFE = 0, it does not matter if the filter result is a pass or fail. The frame is always forwarded. If RA = 0 and VTFE = 1, only then, if the VLAN Tag Filter result is a pass does the MAC forward the frame. If the frame is forwarded to the application, then the relevant filter result is indicated through the Status bits.

1.7.4.3 Filter Status

The Extended Receive VLAN Filtering & Routing feature provides two status bits to indicate the comparison result of the VLAN tags.

By default, the MAC indicates the VLAN Filter Status through one bit in the status – VF field in RDES2. When Extended RX VLAN Filtering & Routing is enabled, two status bits are used to indicate the comparison result of VLAN tags. The Outer VLAN Tag Filter Pass and Inner VLAN Tag Filter Pass bits are defined in the following positions. The status indicated through these bits is highly dependent on the programming as explained below.

In RDES2:

- Bit 15 – Outer VLAN Tag Filter Status
- Bit 14 – Inner VLAN Tag Filter Status
- In ARI status: MAC Filter Status:
 - Bit 15 – Outer VLAN Tag Filter Status
 - Bit 14 – Inner VLAN Tag Filter Status
- In MRI Status:
 - Bit 47 – Outer VLAN Tag Filter Status
 - Bit 46 – Inner VLAN Tag Filter Status

Outer VLAN Tag Filter Status (OTS)

- In perfect Filtering, without inverse filtering enabled, if this bit is set, it indicates that the frame's Outer VLAN Tag has matched one of the VLAN Tag Filters.
- If this bit is reset, it indicates that the frame's Outer VLAN Tag has either failed the relevant Outer VLAN Tag Filters or bypassed them.
- If none of the filters are enabled for Outer VLAN Tag Comparison, then this bit is reset.
- If Inverse Filtering is enabled and this bit is set, then the frame's VLAN Tag has passed all the relevant VLAN Tag Filters. If it is reset, then it has failed at least one of or bypassed all the Filters programmed for Outer VLAN Tag Comparison.
- This bit is valid for both Single and Double VLAN Tagged frames.

Inner VLAN Tag Filter Status (ITS)

- In perfect Filtering, without inverse filtering enabled, if this bit is set, it indicates that the frame's Inner VLAN Tag has matched one of the VLAN Tag Filters.
- If this bit is reset, it indicates that the frame's Inner VLAN Tag has either failed the relevant Inner VLAN Tag Filters or bypassed them.
- If none of the filters are enabled for Inner VLAN Tag Comparison, then this bit is reset.
- If Inverse Filtering is enabled and this bit is set, then the frame's VLAN Tag has passed all the relevant VLAN Tag Filters. If it is reset, then it has failed at least one of or bypassed all the Filters programmed for Inner VLAN Tag Comparison.
- This bit is valid for only Double VLAN Tagged frames, when Double VLAN Processing is enabled.

The application must look at the status bits and the programming to determine if the Frame has passed or failed the VLAN Filter.

Following two tables 'OTS and ITS Bit Values with At Least 1 Perfect Filter Enabled' and 'OTS and ITS Bit Values with Only VLAN Hash Filter Enabled' show the possible Filter combinations and the corresponding filter results. These tables explain the scenarios when Double VLAN Processing and Hash VLAN filter are enabled in the design.

Legend for the table 'OTS and ITS Bit Values with At Least 1 Perfect Filter Enabled'

- VTIM: VLAN Tag Inverse Match Enable – bit 17 in VLAN_Tag_Ctrl Register.
- HFO: Hash Filter enabled for Outer VLAN Tag Comparison . bit 25 and bit 27 in VLAN_Tag_Ctrl Register.
- HFI: Hash Filter enabled for Inner VLAN Tag Comparison – bit 25 and bit 27 in VLAN_Tag_Ctrl Register.
- PFO – Perfect Filter comparison enabled for Outer VLAN Tag - Any of the MAC_VLAN_Tag_Filter Registers is enabled (bit 16 is set) and programmed for Outer VLAN Tag comparison (bit 20 is set to 0).
- PFI – Perfect Filter comparison enabled for Inner VLAN Tag - Any of the MAC_VLAN_Tag_Filter Registers is enabled (bit 16 is set) and programmed for Inner VLAN Tag comparison (bit 20 is set to 1).
- OTS – Outer VLAN Tag Filter Status
- ITS – Inner VLAN Tag Filter Status

Following table shows the possible values of status bits (OTS and ITS) when at least one Perfect filter is enabled.

Table 10. OTS and ITS Bit Values with At Least 1 Perfect Filter Enabled

VTIM	HFO	HFI	PFO	PFI	OTS	ITS
0	0	0	0	1	0	1/0
0	0	0	1	0	1/0	0
0	0	0	1	1	1/0	1/0
0	1	0	1	1	1/0	1/0

Table continues on the next page...

Table 10. OTS and ITS Bit Values with At Least 1 Perfect Filter Enabled (continued)

VTIM	HFO	HFI	PFO	PFI	OTS	ITS
0	1	0	1	0	1/0	0
0	1	0	0	1	1/0	1/0
0	0	1	1	1	1/0	1/0
0	0	1	1	0	1/0	1/0
0	0	1	0	1	0	1/0
1	0	0	0	1	0	1/0
1	0	0	1	0	1/0	0
1	0	0	1	1	1/0	1/0
1	1	0	1	1	1/0	1/0
1	1	0	1	0	1/0	0
1	1	0	0	1	1/0	1/0
1	0	1	1	1	1/0	1/0
1	0	1	1	0	1/0	1/0
1	0	1	0	1	0	1/0

Following table shows the possible values of status bits (OTS and ITS) when none of the perfect filters are enabled and only the VLAN Hash Filter is enabled.

Table 11. OTS and ITS Bit Values with Only VLAN Hash Filter Enabled

VTIM	HFO	HFI	OTS	ITS
0	0	0	0	0
0	1	0	1/0	0
0	0	1	1/0	1/0
1	0	0	1/0	0
1	1	0	1/0	0
1	0	1	1/0	1/0

With no perfect filters enabled, any VLAN packet is considered to have bypassed the perfect filter. If Hash Filter is enabled for one of the Tags, then the respective Status bit depends on the Filter's result. The Status bits are set to 0 if VLAN Hash Filter is also not enabled.

If the value of ITS/OTS is shown as 1/0; then it indicates that the final result is dependent on the enabled relevant filter's result.

Example 1: The second row of table 'OTS and ITS Bit Values with Only VLAN Hash Filter Enabled' indicates that at least one Perfect Filter is enabled for Outer VLAN tag comparison and none of the filters are enabled for Inner VLAN tag comparison. Inverse VLAN Filtering is not enabled. The bit OTS is given as 1/0. If the received frame passes atleast one of the enabled Outer VLAN Tag filters then the bit is set to 1. If the frame doesn't pass any of the enabled Outer VLAN Tag filters, then the bit is set to 0.

Example 2: Last Row of table 'OTS and ITS Bit Values with Only VLAN Hash Filter Enabled' indicates that Inverse Filtering is enabled, Hash Filter and at least one perfect filter is enabled for Inner VLAN Tag comparison, then if the received frame's Inner VLAN tag mismatches with both the Hash Filter and all the enabled Perfect filters, then the frame will have the ITS bit set to 1 else it is set to 0. OTS will be set to 0 as no comparison is performed.

1.7.4.4 Stripping

Each of the VLAN Tags has individual control over stripping. The programming options of always strip, never strip, strip on pass and strip on fail are available. Inner or Outer VLAN Tag Stripping is based on the pass or fail results of the individual tag. If a tag is bypassed by all the relevant filters, stripping is not applicable for the tag.

- If strip on Pass is enabled for the outer VLAN Tag, then the stripping occurs only if the Outer VLAN tag has passed the relevant Filters. The Outer VLAN Tag Filter Result bit will be set.
- If strip on Fail is enabled for the outer VLAN Tag, then stripping occurs only if the Outer VLAN Tag has failed relevant filters. The Outer VLAN Tag Filter Result Bit will be reset.
- If the Outer VLAN tag of the received frame is bypassed by the entire filter (no comparison has been made), then the tag is not stripped, though the Status Bit is still 0.
- As multiple filters are enabled, it is possible that the received VLAN frame could have matched any one or more of the filters. The VLAN Tag's value is not always deterministic from the filter status bits.

If the software/application strips the VLAN Tag based on the filter result, it might lose the VID. So the suggested use is, if Stripping is enabled for any of the tags, the tag can be put in the status. For this the software/application will have to enable the respective "VLAN Tag in Status" bit - 24 or 31 in the MAC VLAN Tag Control Register.

See section "Programming Guidelines for Extended VLAN Filtering and Routing on Receive"

1.7.5 Layer 3 and Layer 4 Filtering

IP supports the Layer 3 based packet filtering which is IP Source or Destination Address filtering in the IPv4 or IPv6 packets, and Layer 4 based packet filtering which is Source or Destination Port number filtering in TCP or UDP. When layer 3 and layer 4 packet filtering is enabled then packets are filtered in the following manner

- Matched Packets: The MAC forwards the packets that match all enabled fields to the application along with the status. The MAC gives the matched field status only if the IPC bit of MAC_Configuration register is set and one of the following conditions is true:
 - All enabled Layer 3 and Layer 4 fields match
 - At least one of the enabled field matches and other fields are bypassed or disabled

When multiple Layer 3 and Layer 4 filters are enabled, any filter match is considered as a match. If more than one filter matches, the MAC provides the status of the lowest filter where Filter 0 is the lowest filter and Filter 3 is the highest filter.

- Unmatched Packets: The MAC drops the packets that do not match any of the enabled fields. You can use the inverse match feature to block or drop a packet with specific TCP or UDP over IP fields and forward all other packets.
- Non-TCP or UDP IP Packets: By default, all non-TCP or UDP IP packets are bypassed from the Layer 3 and Layer 4 filters. MAC can optionally be programmed to drop all non-TCP or UDP over IP packets.
- IP has a control register "MAC_L3_L4_Control" to control the Layer L3 & L4 packet filtering along with address registers to program the Layer L3 & L4 fields to be matched.

1.7.6 Flexible Receive Parser

When this function is enabled, all incoming packets are parsed as per the programmable instruction stored in memory. It does the following function

- Packet Filter (Accept or Reject)
- DMA Channel Selection

Flexible receive parser block operates over the first 256 bytes data of receive packet based on 256 bytes of instruction on each packet. Following figure shows the functional block diagram of Flexible Receive Parser.

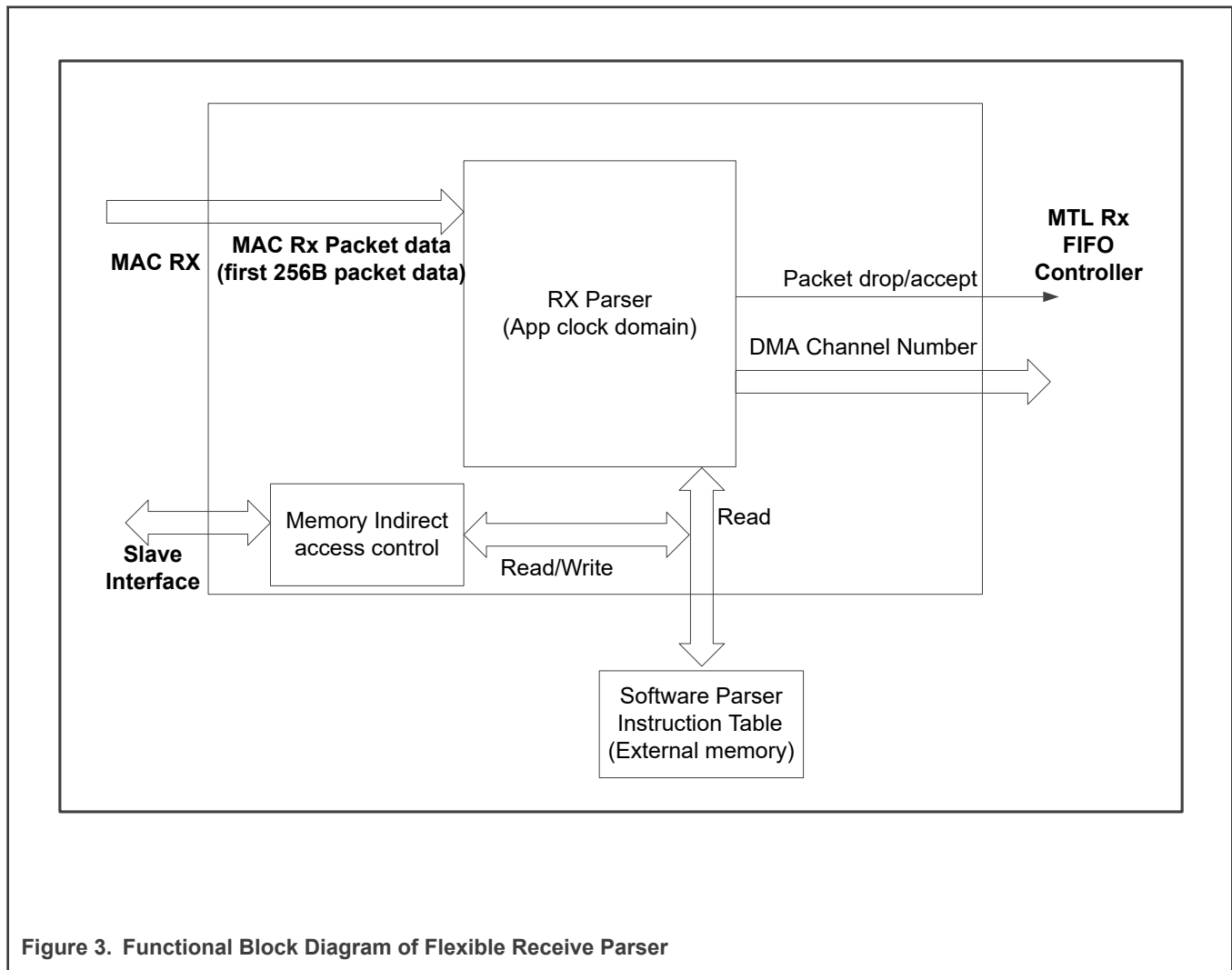


Figure 3. Functional Block Diagram of Flexible Receive Parser

1.7.6.1 Instruction Table Format of the Flexible Receive Parser

Software must build the 96-bit wide instruction before enabling the receive parser feature. Each instruction is programmed as an entry in the instruction memory.

The format of each entry in the Instruction table is shown in the following table.

Table 12. Rx Parser Instruction Entry Format

Field	Field Name	Description
31:0	MATCH_DATA	The 4-byte data. This data is used for comparing with incoming packet data starts at the frame offset as defined in [79:72] of the entry. The comparison is done only on those bits whose corresponding mask bits are set to 1 (MATCH_EN bits of [63:32]). See the Notes (below this table) on the byte order relative to Ethernet arrival data and endianness supported by QoS IP on the slave interface. The programming of this instruction table should adhere to the same.
62:32	MATCH_EN	When the MATCH_EN is set to 1, the corresponding Packet data bit is used for comparing. Otherwise, corresponding data bit is don't care.
64	AF	AF = Accept frame, RF = Reject Frame, IM = Inverse match, NIC= Next Instruction Controlmatch = MATCH_DATA & MATCH_EN Bits == (Packet data[frame offset*8 +: 8*4]) & MATCH_EN bitsIf IM=0 ENTRY_MATCH = matchIf IM=1 ENTRY_MATCH = !matchIf ENTRY_MATCH ==1 If AF = 1, then put frame into DMA Chanel (as indicated in DMA CH Number, the [95:88] of this table) else if RF = 1, then reject/drop frame else if !(AF RF) if(NIC==0) continue parse from OK Index else // (NIC ==1) continue sequentially (parse the next entry in the instruction table) If ENTRY_MATCH ==0 if(NIC==1) continue parse from OK Index else //(NIC ==0) continue sequentially (parse the next entry in the instruction table)Note: AF and RF should not be programmed together. However, if both AF and RF are programmed together, AF gets the priority in case of ENTRY_MATCH
65	RF	
66	IM	
67	NC	
71:68	Reserved	
79:72	Frame Offset	[77:72] indicates the frame offset in terms of 4 bytes. (Here [79:78] always 0)Frame offset in the packet data to be compared for Match.This is in terms of 4 bytes. Value Actual Frame offset in bytes0 01 42 8... ..63 252 The max value programmed in this Instruction Table should adhere to value 256. The 'Actual Frame Offset' should not cross this limit.
87:80	OK Index	If(NIC==0) Memory Index to be used next when ENTRY_MATCH==1 and neither AF or RF is set to 1If(NIC==1) Memory Index to be used next when ENTRY_MATCH==0
95:88	DMA CH NO	Indicate the DMA Channel Number (1-bit for each). This will be used when ENTRY_MATCH ==1 and AF = 1 (Accept frame)The encoding is given by the following.bit[0]- DMA Channel Number 0bit[1]- DMA Channel Number 1bit[2]- DMA Channel Number 2... bit[7]- DMA Channel Number 7Note: The DMA Channel number is encoded using bit wise as QoS IP support multicasting/broadcast across DMA Channels. Refer section 3.3.12 for more details and proper usage.
128:96	Reserved	0 (This field is only for software view and reserved for future enhancements. The memory width remains as 96-bits for now)

The parser begins parsing from the 0th entry, for each received packet. The subsequent parsing entry location (next entry or OK_INDEX[]) depends on the parser result of the current entry. Following figure shows the detailed flow.

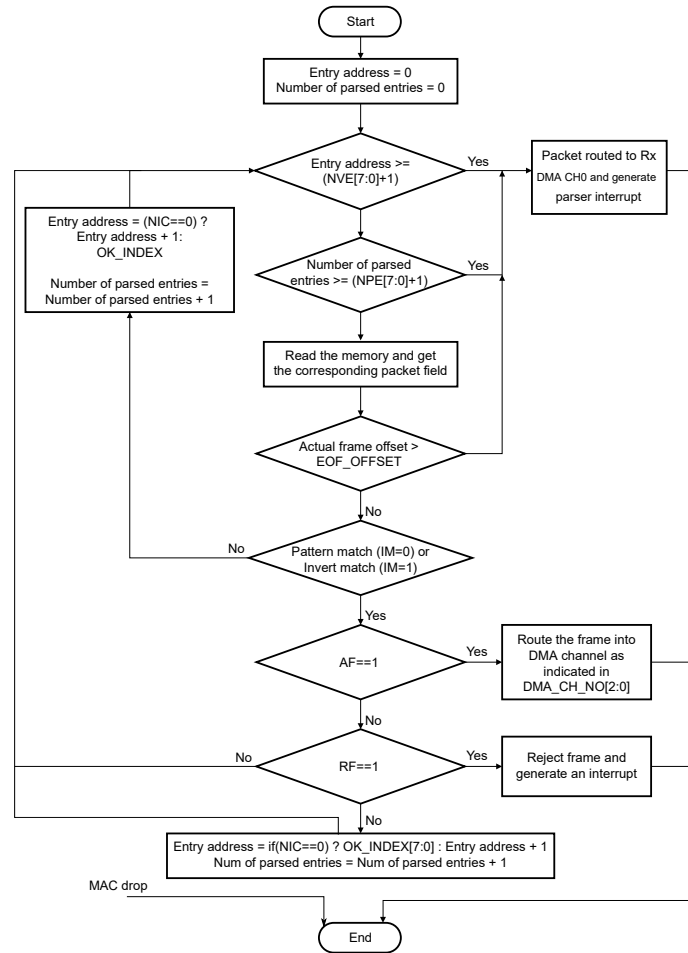


Figure 4. Flexible Receive Parser Flow Chart

The EOF_OFFSET shown in the figure is equal to the least among the following:

- Actual Packet size
- 128 Byte when frame pre-emption feature is not enabled/configured
- 128 Byte when frame pre-emption feature is enabled and packet is pre-emptible.
- 64 Byte when frame pre-emption feature is enabled and packet is Express Packet

1.7.6.2 Number of Valid Entries (NVE)

NVE indicates the number of valid entries in the instruction table. The default value is 64. Software can change this value by writing into MTL_RXP_Control_Status registers field NVE[5:0].

While parsing, if the Entry address is found to be greater than NVE[5:0]+1, an error is flagged in the MTL_RXP_Interrupt_Control_Status register bit[0], NVEOVIS (Number of Valid Entries overflow status). IP generates an interrupt status (MTL_Interrupt_Status, bit[23], MTL_PIS (MTL Parser Interrupt Status)).

1.7.6.3 Number of Parsable Entries (NPE)

NPE indicates the number of entries that can be parsed on an incoming packet. The default value is 64. Software can change this value by writing into MTL_RXP_Control_Status registers field NPE[5:0] (Number of Parsable Entries) bits [21:16].

While parsing, if the number of parsed entries is more than $NPE[5:0]+1$, an error is flagged in the MTL_RXP_Interrupt_Control_Status register bit[1], NPEOVIS (Number of Parsable Entries overflow status). IP generates an interrupt status (MTL_Interrupt_Status, bit[23], MTLPIIS (MTL Parser Interrupt Status).

While NVE[] indicates is the number of valid entries in the Rx parser memory built by software, NPE[] indicates the worstcase parsable entries considering the various possible paths that the parser can take. ($NVE[] \geq NPE[]$).

1.7.6.4 Frame Offset

It indicates the frame offset, in terms of 4 bytes which is used while comparing against the current table entry.

In an entry, an error is flagged in the TL_RXP_Interrupt_Control_Status register bit[2], FOOVIS (Frame Offset Overflow Interrupt Status) in any of the following cases:

When Frame Offset is more than

- Packet size
- 128B if pre-emption is disabled
- 128B if pre-emption is enabled and packet is pre-emptible
- 64B if pre-emption is enabled and packet is an Express packet

IP generates an interrupt status (MTL_Interrupt_Status bit[23]) and MTLPIIS (MTL Parser Interrupt Status).

NOTE

If the frame ends after the $(frame_offset * 4)$ bytes but before $((frame_offset * 4) + 4)$ bytes, the Receive parser declares Frame Offset Error if comparison is enabled (via MASK bits) for the non-received bytes; from $(frame_offset * 4)$ to $((frame_offset * 4) + 4)$.

1.7.6.5 Frame Reject

In cases where an Entry matches (considering the Inverse match, IM bit in the instruction table) and Frame Accept bit is not set, and Frame Reject bit (RF) is set, the frame is dropped.

When a frame is dropped due to $RF = 1$, IP generates an interrupt status (MTL_Interrupt_Status, bit[23], MTLPIIS (MTL Parser Interrupt Status).

1.7.6.6 Out of Order Processing

The Rx parser flow can be out of order. The decision tree need not be based on the order in which packet arrives. This implies that the next entry and next frame offset ($OK_INDEX[]/FRAME_OFFSET[]$) can be less than the current entry address and current Frame Offset.

1.7.6.7 DMA Channel Selection

When the Rx parser is enabled (FRPE bit[16] is set to 1 in MTL_Operation_Mode register) the Rx parser overrides the DMA selection criteria of the MAC.

When the Rx parser is disabled (FRPE bit[16] is set to 0 in MTL_Operation_Mode register) the MAC/MTL determines the DMA channel number.

1.7.6.8 Receive Queue Selection

The Rx Queue is determined by the existing MAC functionality, irrespective of the enabling/disabling of the Rx Parser. This is because the Rx parser is mainly designed for selecting the DMA channel and Rx queuing is decided by the MAC to enable the proper functioning of the Pause/PFC feature.

NOTE

When there are multiple queues and multiple DMA Channels are enabled, a particular DMA channel might receive out of order packets. This is because queue selection is based on MAC criteria (VLAN Priority and other criteria) and Rx DMA channel selection is based on Rx parser.

1.7.6.9 MAC Packet Filtering/Drop/Error Handling

The MAC does the packet filtering based on the received packet fields. Disable the Packet Filtering features inside the MAC when the Rx parser is enabled. To disable the Packet Filtering features inside the MAC,

1. Set Promiscuous Mode (PR, bit[0] of MAC_Packet_Filter register)
2. Set all other bits in the MAC_Packet_Filter register to its default values

When MAC decides to drop the packet (due to Rx MAC dependent Error such as GMII Error, Receive Watch Dog Error, CRC Error and Giant Frame Error) the packet is dropped irrespective of the Rx parser decision.

1.7.6.10 PAD Strip/CRC Strip Handling

The pad strip and CRC strip are controlled by the software and applicable to all the received packets. When these features are enabled, software must build the Rx parser instructions accordingly.

NOTE

Enable PAD and CRC stripping in real use case.

1.7.6.11 VLAN Strip Handling

The MAC supports stripping the VLANs, outer as well as inner VLAN. When MAC is programmed to strip the VLAN, the Rx parser considers those VLAN tags to be part of the incoming packet.

1.7.6.12 Multicast and Broadcast Support

IP supports multicast and broadcast packet and the packet is routed to the highest queue number.

When Rx Parser is enabled, the packet routing is decided by the DMA Channel Numbers (DMA CH NO, Bits [95:88]) in the Instruction Table.

When Rx Parser is disabled, the packet routing is determined by DCS (DMA Channel Select) field of the MAC_AddressX_High register. For more details, see "Broadcast/Multicast Packet Duplication".

NOTE

DCS/DMA CH NO field is per DMA Channel control and therefore, multiple DMA channels can be selected for routing the packet.

1.7.6.13 Pre-emption Support

IP supports frame pre-emption feature when Flexible Rx parser feature is enabled. Only limitation is that

- Number of bytes would be 64 that Rx parser can operate on the Express traffic
- Number of bytes would be 128 that Rx parser can operate on pre-emptible traffic.

1.7.6.14 Software Access to the Flexible Receive Parser Memory

Software can read/write into the Rx parser memory, using the following registers via indirect addressing.

- MTL_RXP_Indirect_Acc_Control_Status
- MTL_RXP_Indirect_Acc_data

For more details, see "Register Descriptions".

1.7.6.15 Statistical Counters

IP supports the following statistics counters for flexible receive parser.

1. MTL_RXP_Drop_Cnt
2. DMA_RXP_Error_Cnt
3. DMA_CH(#i)_Rxp_Accept_cnt (i=0; i<2)

For more details, see "Register Descriptions".

1.7.6.16 Changing the Instruction Table by Software

Software can update the instruction table in the memory in the following ways

- Disable the MAC Receiver and Receive Parser by setting the RE field of the MAC_Configuration register to 0 and FRPE bit of MTL_Operation_mode register to 0. Wait for Rx Parser to become inactive (RXPIA, MTL_RXP_Control_Status register bit[31])
- (Optional) Program in the Entry address 0 to unconditionally (all MATCH_EN bit 0) skip to OK_INDEX[] (to certain location in the memory) where it is programmed to reject or accept all packets.

1.7.6.17 Receive Cut-Through Functionality

In the case of cut-through mode, the Rx controller attempts to transfer the received packets to DMA just after the cut-through threshold amount of packet data is received (RTC field in the MTL_RxQx_Operation_Mode register).

However due to Rx parser's result worst case arrival time can be more than the programmed cut-through threshold. So, the Rx controller delays the packet transfer to DMA accordingly. So, the cut-through functionality is re-defined as follows

The Rx controller attempts to transfer packet to DMA after at least RTC number of bytes have been received and Rx parser results are available.

1.7.6.18 Receive Packet Drop Indication

In most cases, the packet drops internally in the Rx Queue. However, back-to-back packets can arrive for the same queue and first packet's Rx parser result is not available when the second packet arrives. In such cases, the packet cannot be flushed internally and forwarded to DMA or the Application Interface and status is indicated accordingly. For more details, see RDES2, bit 16(RXPD) in "Receive Normal Descriptor (Write-Back Format)".

1.7.6.19 Runt Packet Handling

The Rx parser can perform the operation with 64B minimum size packet. If the runt packet is received (should be indicated by MAC), the Rx parser assumes that it gets dropped in the MAC. This is because, back-to-back runt packets (< 64B) cannot be handled in the Rx parser.

1.7.6.20 Uncorrectable ECC Error Handling

When the parser detects an uncorrectable ECC error while parsing the parser memory, the packet is not dropped. Instead, the packet is sent to the application with an Error Indication. The RXPI (RX Parser Incomplete) is set in the packet status along with Error Summary (ES) bit.

For more details, see "Receive Normal Descriptor (Write-Back Format)".

1.8 IEEE 1588 Timestamp Support

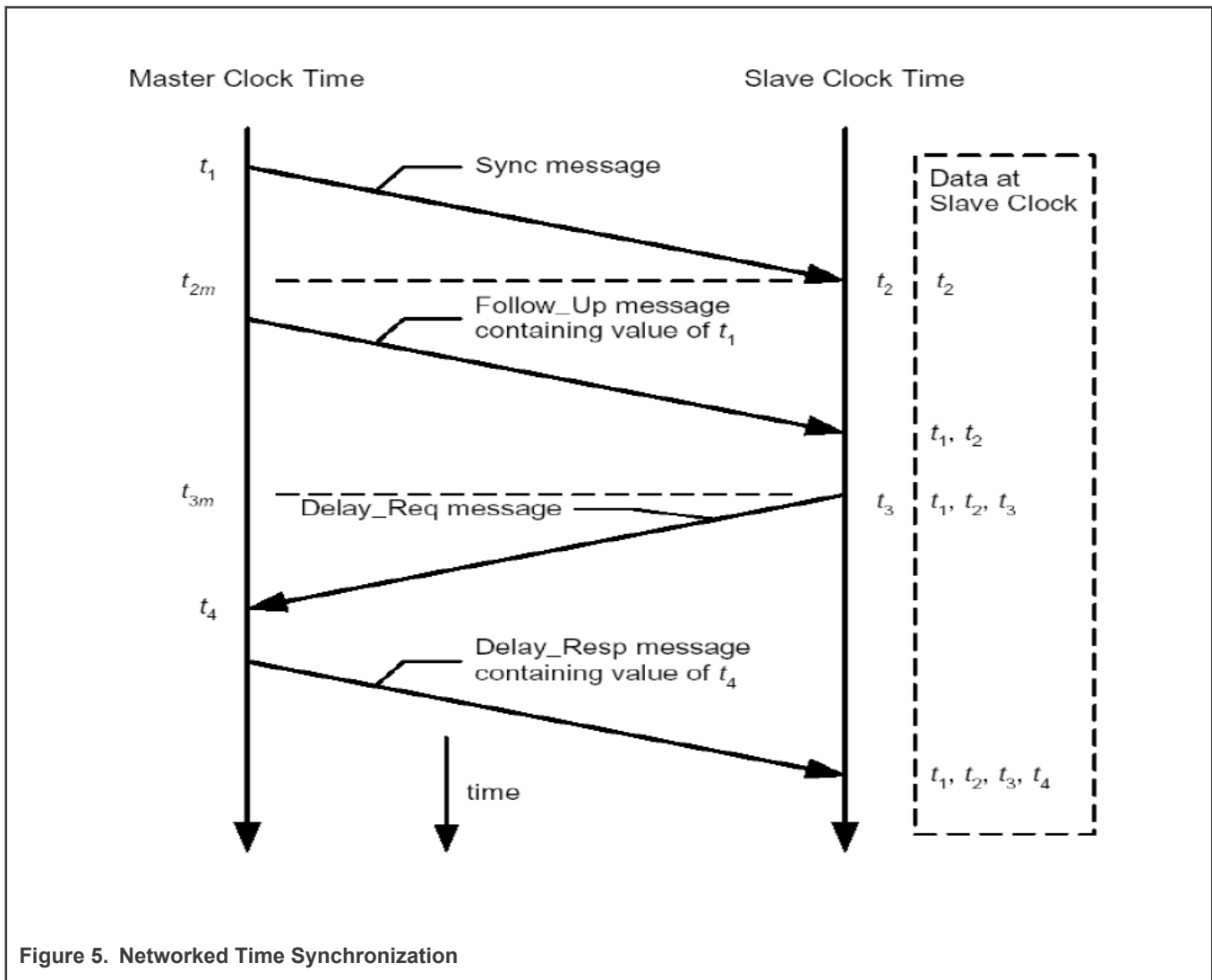
This section and all its sub-sections are Synopsys Proprietary. Used with permission.

IP supports the IEEE 1588 Precision Time Protocol (PTP). IP is able to do precise time stamping and capturing of incoming and outgoing frame because of PTP protocol implementation in the IP. IP supports the all clock types defined in IEEE 1588-2008. IP supports the frequency range from 5Mhz to 200Mhz. The IP supports the following features:

- Provides an option to take snapshot of all packets or only PTP type packets
- Provides an option to take snapshot of only event messages
- Identifies the PTP message type, version, and PTP payload in packets sent directly over Ethernet and sends the status
- Provides an option to measure sub-second time in digital or binary format

1.8.1 Delay Request-Response Mechanism

The system or network is classified into the master and slave nodes for distributing the timing and clock information. Following figure shows the process that PTP uses for synchronizing a slave node to a master node by exchanging PTP messages.



As shown in the above figure, PTP uses the following process:

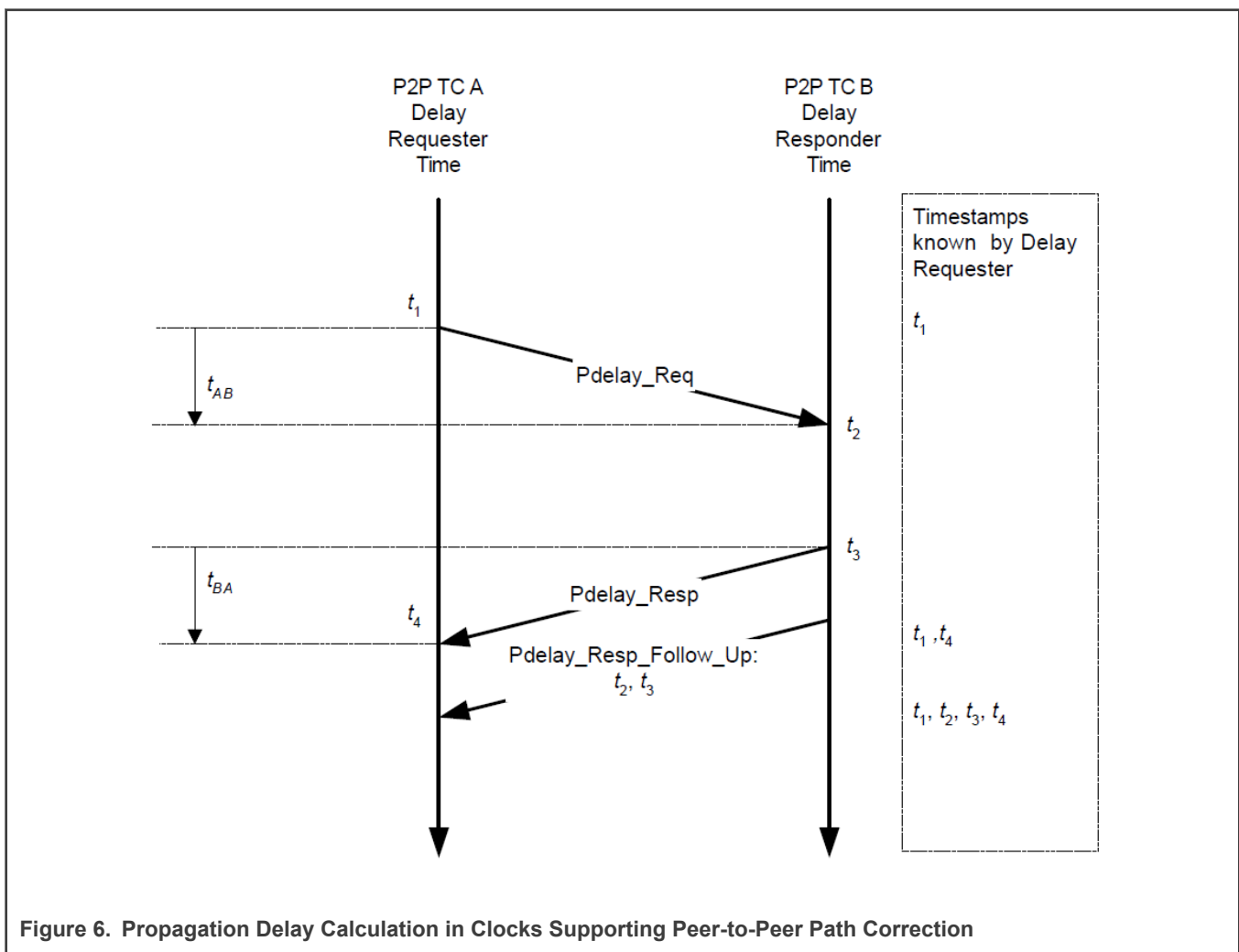
1. The master broadcasts the PTP Sync messages to all its nodes. The Sync message contains the reference time information of the master. This message leaves the system of the master at t_1 . This time must be captured for Ethernet ports at GMII or MII.
2. The slave receives the Sync message and also captures the exact time, t_2 , using its timing reference.
3. The master sends a Follow_up message to the slave, which contains t_1 information for later use.
4. The slave sends a Delay_Req message to the master and notes the exact time, t_3 , at which this packet leaves the GMII or MII interface.

5. The master receives the message, capturing the exact time t_4 , at which the message enters its system.
6. The master sends the t_4 information to the slave in the Delay_Resp message.
7. The slave uses the four values of t_1 , t_2 , t_3 , and t_4 to synchronize its local timing reference to the timing reference of the master.

Most of the PTP implementation is done in the software above the Ethernet layer. However, the hardware support is required to capture the exact time when specific PTP packets enter or leave the Ethernet port at the MII interface. This timing information must be captured and returned to the software for proper implementation of PTP with high accuracy.

1.8.2 Peer-to-Peer PTP Transparent Clock (P2P TC) Message Support

The IEEE 1588-2008 supports peer-to-peer PTP (Pdelay) message in addition to the SYNC, Delay Request, Follow-up, and Delay Response messages. Following figure shows the method to calculate the propagation delay in clocks supporting peer-to-peer path correction.



As shown in the above figure, the propagation delay is calculated in the following way:

1. Port 1 issues a Pdelay_Req message and generates a timestamp (t_1) for the Pdelay_Req message.
2. Port 2 receives the Pdelay_Req message and generates a timestamp (t_2) for this message.
3. Port 2 returns a Pdelay_Resp message and generates a timestamp (t_3) for this message. To minimize errors because of any frequency offset between the two ports, Port 2 returns the Pdelay_Resp message as quickly as possible after the receipt of the Pdelay_Req message. Port 2 returns any one of the following:

- Difference between the timestamps t2 and t3 in the Pdelay_Resp message
 - Difference between the timestamps t2 and t3 in the Pdelay_Resp_Follow_Up message
 - Timestamps t2 and t3 in the Pdelay_Resp and Pdelay_Resp_Follow_Up messages, respectively
4. Port 1 generates a timestamp (t4) on receiving the Pdelay_Resp message.
 5. Port 1 uses all four timestamps to compute the mean link delay.

1.8.3 Timestamp Correction

According to the IEEE 1588 specification, a timestamp must be captured when the PTP message timestamp point (leading edge of the first bit of the octet immediately following the Start Frame Delimiter octet) crosses the boundary between the node and the network. As the MAC takes the timestamp at an internal point far from the actual boundary of the node and network, this captured timestamp is corrected/updated for the ingress/egress path latency (including the delay in the PHY layers). Further correction is done for the inaccuracies/errors introduced due to the clock (MII Tx, Rx clock) being different at the capture point as compared to the PTP clock (clk_ptp_ref_i) that is used to generate the time. The resultant CDC (Clock Domain Crossing) circuits add error depending on the clock period of the MII and PTP clocks.

NOTE

The PTP clock must always be available for accurate Tx and Rx timestamping.

1.8.3.1 Ingress Correction

In the Receive side the timestamp captured at the internal snapshot point is delayed (later in time) as compared to the time at which that packet's SFD bit is received at the port's boundary. Therefore, the captured timestamp must be reduced by the ingress latency and the errors in CDC sampling. This correction value must be determined/calculated by the software and written into the MAC_Timestamp_Ingress_Corr_* registers.

The correction value consists of the following 3 components:

1. External latency in the PHY layer between boundary point and the input of the core. If the PHY is compliant to the IEEE 802.3 Clause 45 MMD registers, it has a register indicating the maximum and minimum ingress latency. The software can read these registers and determine the average ingress latency in the PHY. Alternatively (if the PHY does not support these registers), the ingress latency must be determined from its datasheet or timing characteristics.
2. Internal latency from the input of the core to the internal capture point. The internal ingress latency can be read from the MAC_Ingress_Timestamp_Latency register. This is a read-only register and gives the latency in scaled Nanoseconds format defined in IEEE 1588 Clause 5.3.2. The latency differs based on the active PHY interface (MII, RMII, so on) and the operating speed. Therefore, the software must read this register after any speed change in the MAC to determine the current internal latency.
3. CDC Synchronization: The CDC synchronization error is almost equal to 2 times the clock-period of the PTP clock (clk_ptp_ref_i). The values determined from these 3 components should be added by the software and must be written into the TSIC & TSICSNS fields of the MAC_Timestamp_Ingress_Corr_* registers.

NOTE

The value written to the register must be negative (two's complement as explained below), because it has to be subtracted from the captured timestamp. The MAC Receiver adds the value in this register to the captured timestamp and then gives the resultant value as the timestamp of the received packet.

When TSCTRLSSR bit in MAC_Timestamp_Control register is set, the nanoseconds field of the captured timestamp is in decimal format with a granularity of 1ns. So the Bit31 of TSIC must be set to 1 (for negative value) and bits[30:0] must be written with " 10^9 - total ingress_correction_value[nanosecond part]" represented in binary. For example, if the required correction value is -5 ns, then the value is 0xBB9A_C9FB.

When TSCTRLSSR bit in MAC_Timestamp_Control register is reset, the nanoseconds field of the captured timestamp is in binary format with a granularity of ~0.466ns. Therefore, bits[30:0] must be written with " 2^{31} - total ingress_correction_value" represented in binary with bit[31] = 1.

1.8.3.2 Egress Correction

In the Transmit side the timestamp captured at the internal snapshot point is earlier (advanced in time) as compared to the time at which that packet's SFD bit is output at the port's boundary. Therefore, the captured timestamp must be compensated by the egress latency and the errors in CDC sampling. This correction value must be determined/calculated by the software and written into the MAC_Timestamp_Egress_Corr_* registers.

The correction value consists of the following 3 components:

1. External latency in the PHY layer between the output of the core and the boundary of the port and the network. If the PHY is compliant to the IEEE 802.3 Clause 45 MMD registers, it has a register indicating the maximum and minimum egress latency. The software can read these registers and determine the average egress latency in the PHY. Alternatively (if the PHY does not support these registers), the egress latency must be determined from its datasheet or timing characteristics.
2. Internal latency from the internal capture point and the output of the core. This internal egress latency can be read from the MAC_Egress_Timestamp_Latency register. This is a read-only register and gives the latency in scaledNanoseconds format defined in IEEE 1588 Clause 5.3.2. The latency will differ based on the active PHY interface (MII, RMII, so on) and the operating speed. Hence the software must read this register after any speed change in the MAC to determine the current internal latency.
3. CDC synchronization error: The CDC synchronization error value for one-step timestamping would be = $(1 * \text{period of clk_ptp_ref_i} + 4 * \text{period of clk_tx_i})$. Otherwise (Two-step timestamping mode), the value = $-(2 * \text{period of clk_ptp_ref_i})$.

1.8.3.3 Frequency Range of Reference Timing Clock

The timestamp information is transferred across asynchronous clock domains, from the MAC clock domain to the application/system clock domain. Therefore, a minimum delay is required between two consecutive timestamp captures. This delay is 4 clock cycles of MII and 3 clock cycles of PTP clocks. If the delay between two timestamp captures is less than this delay, the MAC does not take a timestamp snapshot for the second packet.

1.8.3.4 PTP Processing and Control

Following table shows the common message header for the PTP messages. This format is taken from the IEEE 1588-2008.

Table 13. Message Format Defined in IEEE 1588-2008

Bits								Octet	Offset
7	6	5	4	3	2	1	0		
transportSpecific				messageType				1	0
Reserved				versionPTP				1	1
messageLength								2	2
domainNumber								1	4
Reserved								1	5
flagField								2	6
correctionField								8	8
Reserved								4	16

Table continues on the next page...

Table 13. Message Format Defined in IEEE 1588-2008 (continued)

Bits								Octet	Offset
7	6	5	4	3	2	1	0		
sourcePortIdentity								10	20
sequenceId								2	30
controlField (*)								1	32
logMessageInterval								1	33

(*) – control Field is used in version 1. In version 2, message Type field is used for detecting different message types.

There are some fields in the Ethernet payload that are being use to detect the PTP packet type and control the snapshot to be taken. These fields are given below in table for PTP packets over Ethernet.

Following table provides information about the fields that are matched to control the snapshots for the PTP packets sent over Ethernet for IEEE 1588 version 1 and 2. The octet positions for the tagged packets are offset by 4. This is based on the IEEE 1588-2008, Annex D and the message format.

Table 14. IPv4-UDP PTP Packet Fields Required for Control and Status

Field Matched	Octet Position	Matched Value	Description
MAC Packet Type	12, 13	0x0800	IPv4 datagram
IP version and Header Length	14	0x45	IP version is IPv4
Layer 4 Protocol	23	0x11	UDP
IP Multicast Address (IEEE 1588 version 1)	30, 31, 32, 33	0xE0, 0x00, 0x01, 0x81 (or 0x82 or 0x83 or 0x84)	Multicast IPv4 addresses allowed: • 224.0.1.129 • 224.0.1.130 • 224.0.1.131 • 224.0.1.132
IP Multicast Address (IEEE 1588 version 2)	30, 31, 32, 33	0xE0, 0x00, 0x01, 0x81 (Hex) 0xE0, 0x00, 0x00, 0x6B (Hex)	• PTP Primary multicast address: 224.0.1.129 • PTP Pdelay multicast address: 224.0.0.107
UDP Destination Port	36, 37	0x013F, 0x0140	• 0x013F: PTP event messages • 0x0140: PTP general messages
PTP Control Field (IEEE 1588 version 1)	74	0x00, 0x01, 0x02, 0x03, 0x04	• 0x00: SYNC • 0x01: Delay_Req • 0x02: Follow_Up

Table continues on the next page...

Table 14. IPv4-UDP PTP Packet Fields Required for Control and Status (continued)

Field Matched	Octet Position	Matched Value	Description
			• 0x03: Delay_Resp • 0x04: Management
PTP Message Type Field (IEEE 1588 version 2)	42 (nibble)	0x0, 0x1, 0x2, 0x3, 0x8, 0x9, 0xB, 0xC, 0xD	• 0x0: SYNC • 0x1: Delay_Req • 0x2: Pdelay_Req • 0x3: Pdelay_Resp • 0x8: Follow_Up • 0x9: Delay_Resp • 0xA: Pdelay_Resp_Follow_Up • 0xB: Announce • 0xC: Signaling • 0xD: Management
PTP Version	43 (nibble)	0x1 or 0x2	• 0x1: Supports PTP version 1 • 0x2: Supports PTP version 2

1.8.4 Transmit Path Functions

The MAC captures a timestamp when the Start Packet Delimiter (SFD) of a packet is sent on the MII interface. The packets, for which the timestamps has to be captured can be controlled on per-packet basis. Each Transmit packet can be marked to indicate whether a timestamp should be captured for it. The MAC does not process the transmitted packets to identify the PTP packets. You can specify the packets by using the control bits in the Transmit descriptor. The MAC returns the timestamp to the software inside the corresponding Transmit descriptor, thus connecting the time-stamp automatically to the specific PTP packet. The 64-bit timestamp information is written to the TDES0 and TDES1 fields. The TDES0 field holds the 32 least significant bits of the timestamp.

1.8.5 Receive Path Functions

The MAC can be programmed to capture the timestamp of all packets received on the MII interface or to process packets to identify the valid PTP messages. Use the following options of the MAC_Timestamp_Control register to control the snapshot of the time to be sent to the application:

- Enable snapshot for all packets
- Enable snapshot for IEEE 1588 version 1 or version 2 timestamp
- Enable snapshot for PTP packets transmitted directly over Ethernet
- Enable timestamp snapshot for the received packet for IPv4 or IPv6
- Enable timestamp snapshot only for EVENT messages (SYNC, DELAY_REQ, PDELAY_REQ, or PDELAY_RESP)
- Enable the node to be a master or slave and select the snapshot type

This feature controls the type of messages for which snapshots are taken.

Table 15. Timestamp Snapshot Dependency on Register Bits

SNAPTYPSEL	TSMSTRENA	TSEVNTENA	PTP Messages
00	x	0	SYNC, Follow_Up, Delay_Req, Delay_Resp
00	0	1	SYNC
00	1	1	Delay_Req
01	x	0	SYNC, Follow_Up, Delay_Req, Delay_Resp, Pdelay_Req, Pdelay_Resp, Pdelay_Resp_Follow_Up
01	0	1	SYNC, Pdelay_Req, Pdelay_Resp
01	1	1	Delay_Req, Pdelay_Req, Pdelay_Resp
10	x	x	SYNC, Delay_Req
11	x	x	Pdelay_Req, Pdelay_Resp

The DMA returns the timestamp to the software inside the corresponding Receive Descriptor. The extended status, containing the timestamp message status and the IPC status, is written in normal descriptor RDES1 and the snapshot of the timestamp is written in RDES0 and RDES1 fields of context descriptor. The RDES0 field holds the 32 least significant bits of the timestamp.

NOTE

Programming Guidelines for IEEE 1588 Timestamping (System Time Correction) See "System Time Correction" .

1.8.6 IEEE 1588 System Time Source

IP supports both external and internal time source for reference timestamping. The implementation varies per chip as external mode requires the timestamp input bus to have a valid source such as another Ethernet block on the chip. If no such option is highlighted users should program internal mode.

- External Time source uses the 64-bit external time reference and clock as input in IP. The clock input is used to synchronize the external timing reference into MAC clock domain. Upper 32-bit indicates the time in seconds while lower 32-bit indicates the time in nanoseconds.
- Internal time source uses the only clock input to generate timing reference internally for snapshot and capture timestamps. Internal time reference has two field.

— UInteger48 seconds Field: The seconds field is the integer portion of the timestamp in units of seconds. It is 48-bits wide.

For example, 2.000000001 seconds are represented as seconds Field = 0x0000_0000_0002.

- UInteger32 nano secondsField The nanoseconds field is the fractional portion of the timestamp in units of nanoseconds.

For example, 2.000000001 seconds are represented as nanoSeconds = 0x0000_0001. The nanoseconds field supports the following two modes:

Digital rollover mode: In this mode, the maximum value in the nanoseconds field is 0x3B9A_C9FF, that is, (10e9-1) nanoseconds.

Binary rollover mode: In this mode, the nanoseconds field rolls over and increments the seconds field after value 0x7FFF_FFFF. Accuracy is ~0.466 ns per bit.

There is a system time register module which is being used when internal time reference is being used. The 80-bit time is maintained in this module and updated using the input reference clock (clk_ptp_ref_i). This time is the source for taking snapshots (timestamps) of Ethernet packets being transmitted or received at the MII interface.

The system time counter can be initialized or corrected using the coarse correction method. In this method, the initial value or the offset value is written to the Timestamp Update register. For initialization, the system time counter is written with the value in the Timestamp Update register. For system time correction, the offset value is added to or subtracted from the system time.

In the fine correction method, the frequency offset and/or frequency drift of a slave clock (clk_ptp_ref_i) with respect to the master clock (as defined in IEEE 1588-2002) is corrected over a period of time instead of in one clock, as in coarse correction. This helps maintain linear time and does not introduce drastic changes (or a large jitter) in the reference time between PTP Sync message intervals.

NOTE

Programming Guidelines for IEEE 1588 Timestamping (For Internal Time-stamp Source Configuration) See "Initialization Guidelines for System Time Generation"

1.8.7 IEEE 1588 Higher Word Register

If system time source is internal, then higher word register can be invoked. MAC timestamp is 64-bit wide. The values of the upper 16-bits of the seconds field can be read from the CSR register.

1.8.8 IEEE 1588 Auxillary Snapshot

The auxiliary snapshot feature allows you to store a snapshot of the system time based on an external event. The event is considered to be the rising edge of the ptp_aux_ts_trig_i sideband signal. This feature is independent of whether the system time is generated internally or given as input (on ptp_timestamp_i[63:0] bus). IP supports the four auxiliary snapshot inputs signal.

The snapshots taken for any input are stored in a common FIFO. The application can read the MAC_Time-stamp_Status register to know the timestamp of which input is available for reading at the top of this FIFO. The MAC stores these snapshots in a FIFO. Only 64-bits of the timestamp are stored in the FIFO. You can read the upper 16-bits of seconds from the MAC_System_Time_Higher_Word_Seconds register when it is present. When a snapshot is stored, the MAC indicates this to the application with an interrupt. The value of the snapshot is read through a FIFO register access. If the FIFO becomes full and an external trigger to take the snapshot is asserted, a snapshot trigger-missed status (ATSSTM) is set in the MAC_Timestamp_Status register. This indicates that the latest auxiliary snapshot of the timestamp is not stored in the FIFO. The latest snapshot is not written to the FIFO when it is full. When an application reads the 64-bit timestamp from the FIFO, the space becomes available to store the next snapshot. You can clear a FIFO by setting the ATSFC bit in MAC_Auxiliary_Control register. When multiple snapshots are present in the FIFO, the count is indicated in Bits[27:25] of MAC_Timestamp_Status register.

1.8.9 Flexible Pulse-Per-Second Output

IP supports the flexibility of programming the start or stop time, width and interval of pulse generated on pulse per second output. IP supports 4 such output signal. By Default, IP is in fixed Pulse per second output mode with interval of 1 second.

Initially start time should be programmed in target time registers "MAC_PPSx_Target_Time_Seconds and MAC_PPSx_Target_Time_Nanoseconds" for all desired/enabled pps output. If start or stop time need to be re-programmed then it must be done after the synchronization of earlier programmed value. Bit 31 of MAC_PPS#_Target_Time_Nanoseconds register indicates that the synchronization is complete. If the application programs a start or stop time that has already elapsed, the MAC sets an error status bit indicating the programming error. If enabled, the MAC also sets the Target Time Reached interrupt event. The application can cancel the start or stop request only if the corresponding start or stop time has not elapsed. If the time has elapsed, the cancel command has no effect.

The PPS width and interval are programmed in terms of granularity of system time, that is, number of the units of sub-second increment value.

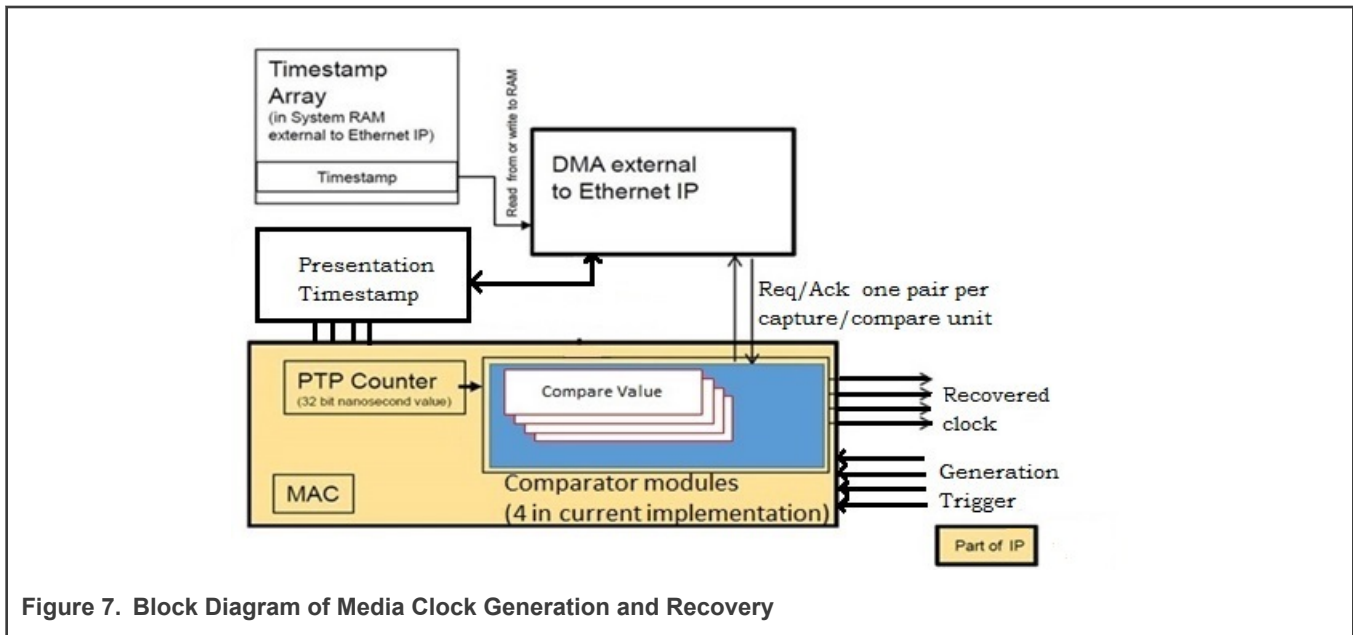
NOTE

See "Programming Guidelines for Flexible Pulse-Per-Second Output"

1.8.10 Media Clock Generation and Recovery

Media clock recovery and generation needs a dedicated hardware and software to complete the operation. IP generates a reference clock with the help of dedicated hardware which is then multiplied to get the desired media clock. Software extracts the presentation time from the incoming PTP 1722 frames and programs these value into CSR registers so IP can read them for media clock recovery.

As shown in the following figure, the timestamp array and the DMA module are external to IP. IP supports a 32-bit presentation time counter in nanoseconds that wraps at the full 32-bit value to match up with 1722 presentation time format and an handshake mechanism with external application (such as DMA) to program the presentation time into CSR register.



1.8.10.1 Presentation Time Counter

Presentation time counter is another view of the PTP System time (1588 timer). For a given PTP System time (PTP[63:32] represents time in seconds and PTP[31:0] represents time in nanoseconds), there exists a corresponding Presentation Time (32-bit value in nanosecond). The Presentation time computed is referred to as Current Presentation Time (CPT).

The Current Presentation Time (CPT) can be viewed as derived from 64bit PTP System Time. $PTPNS[63:0] = (PTP[63:32] * 32'd1,000,000,000) + PTP[31:0]$ where, PTPNS is the PTP system time converted into a 64bit nanosecond value.

Current Presentation Time[31:0] = PTPNS[31:0]

Media Clock Recovery is possible when the System Time is internally generated in IP and CPT is computed in the same way as the Internal PTP System Time generation. Because CPT and System Time values are in nanoseconds, and must be synchronous, the increment cycle of both the timers are same. The timer is updated at the same instance, but the updated value could be different. The software must compute a separate 32-bit update value in nanoseconds for the CPT update. CPT sampled at different edges of a triggering input results in the generation of Media Clock timestamps which are inserted in 1722 based AVBTP packets, to recover the clock at the destination.

1.8.10.2 Comparator Modules

IP supports four comparator modules to handle multiple media clocks that might be required. See the IOMUX sheet for details on the actual number of external signals available. The Comparator modules hand shake with the application to:

- Program the timestamps into the MAC_PPS(#)_Target_Time_Seconds register when PTGE field of MAC_Timestamp_Control register is set.
- Set the Presentation Time Control bits of the particular instance for recovery mode.

The timestamps received from the application are referred as Target Presentation Time (TPT).

When MCGREN#i field of MAC_PPS_Control register is set to 1, the IP operates in MCGR mode. The comparator module sends up to two requests to the application for TPT write, when PTGE field of MAC_Timestamp_Control register is set and the Presentation time control bits of the particular instance is set for MCGR mode. Subsequent requests are generated each time a Presentation Time match occurs. (CPT transitions from a value less than TPT, to a value greater than or equal to CPT). The first request is asserted when a specific comparator instance is set to MCGR mode with a non-zero Presentation Time Control and an additional request is made when the first data is received into the Comparator. This allows the application to write the next TPT value while IP processes the previous TPT value for a match.

The TPT read from the MAC_PPS(#i)_Target_Time_Second is considered as a future time. A toggle/pulse is generated in the next cycle when a match is detected. When a match is detected, the mcgr_dma_req_o#i request for that comparator is asserted (to obtain next TPT) until the corresponding application acknowledgment is set.

The presentation control bits (PPSCMD#i field of MAC_PPS_Control register) determine the shape of the generated waveform. Also, these bits can be programmed to either toggle or generate a high/low pulse for one PTP clock cycle, upon match.

Media Clock Recovery: A match is when the free-running CPT value matches the received TPT value. A output signal ptp_pps_o is asserted (toggle, low pulse, or high pulse) based on the presentation control value programmed.

Media Clock Generation: Based on the presentation control value programmed in MAC_PPS_Control register, the particular comparator captures the presentation time and programs it into MAC_PPS(#i)_Target_Time_Second register. A request is raised to the application to read the captured timestamp. Until the acknowledgment is received from the application, acknowledging that the read operation is complete, no new timestamps are captured.

1.8.10.3 Media Clock Generation and Recovery Flow

Following figure shows the media clock generation and recovery flow. For both media clock generation and recovery, set the PTGE field of MAC_Timestamp register for CPT generation and set the MCGREN#i field of MAC_PPS_Control Register for the corresponding instance to be enabled in the MCGR mode.

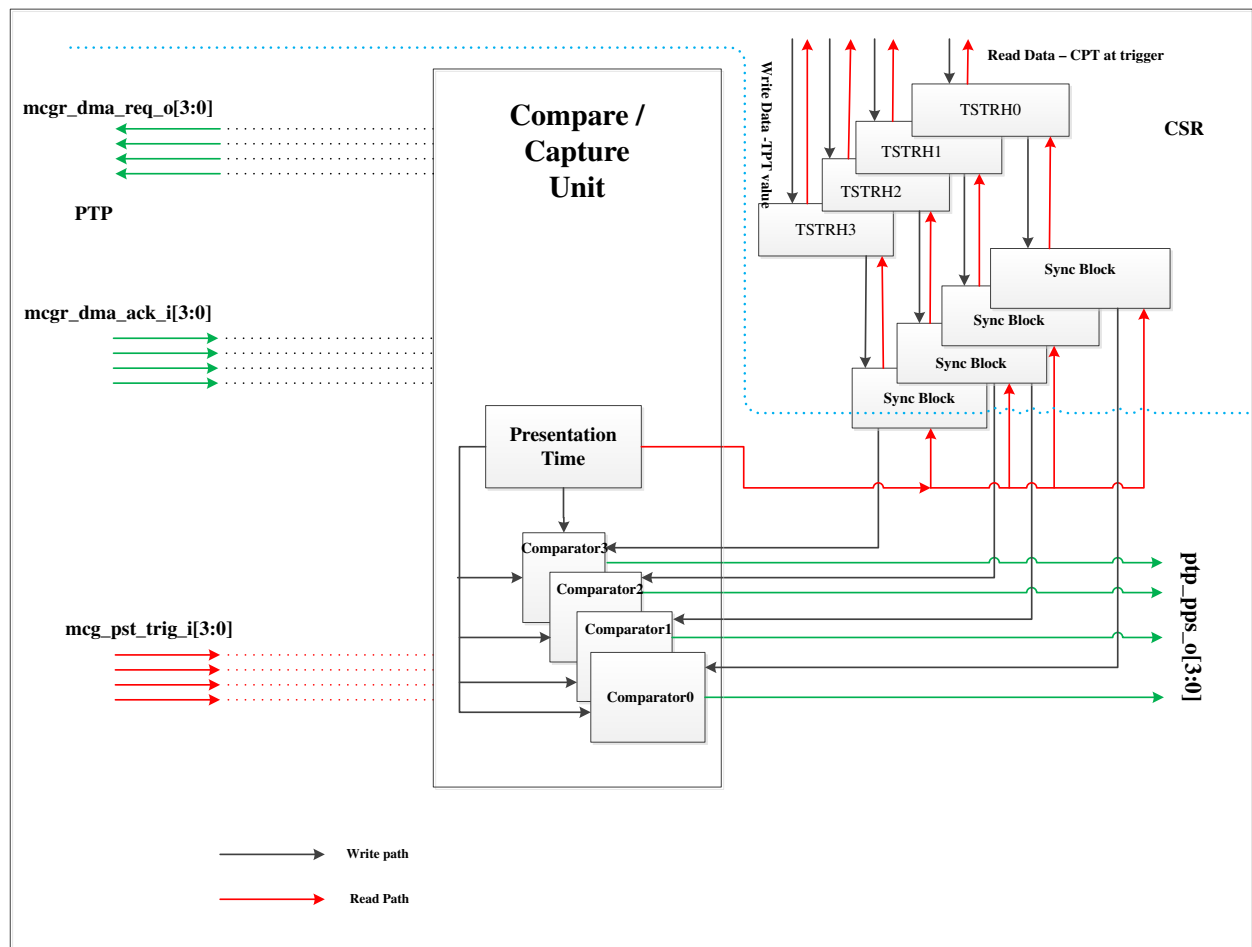


Figure 8. Media Clock Generation and Recovery Flow

NOTE

1. The consecutive triggers to sample the presentation time should always be asserted after a few PTP clock cycles to allow for synchronization delays. (This is not an issue when the input trigger maximum frequency is 8KHz and the PTP clock is running at least at 1MHz)
2. DMA acknowledgment can be both posted as well as non-posted, in the Media Clock Recovery Mode.
3. See "Programming Guidelines for Media Clock Generation and Recovery".

1.8.10.4 PTP Timestamp Offload Function

PTP Timestamp Offload Function is an optional feature that enables the automatic generation of specific PTP packets when the MAC is working as a specific node in the PTP network. These packets may be generated periodically or triggered by the host software. In other modes, this feature can parse the incoming PTP packets on the receiver, and automatically generate and respond to the required PTP packets. This helps in offloading certain functions of a PTP node with better accuracy and lower latency of responses.

1.8.10.5 Description of PTP Offload Function

Based on the programmed mode, the MAC generates PTP Ethernet messages periodically or from the application, or based on reception of a particular PTP message. Following table indicates the PTP message generation criteria.

Table 16. PTP Message Generation Criteria

Programming			Mode	Criteria for Generation of PTP Messages	PTP Message Type Generated
SNAPTYPSEL	TSMSTRENA	TSEVNTENA			
2'b00	0	1	Ordinary or Boundary Slave	SYNC message reception	Delay_Req
2'b00	1	1	Ordinary or Boundary Master	Periodic or on trigger from application	SYNC
				Delay_Req message reception	Delay_Resp
2'b01	0	1	Transparent Slave	Periodic or on trigger from application	Pdelay_Req
				Pdelay_Req message reception	Pdelay_Resp
				SYNC message reception	Delay_Req
2'b01	1	1	Transparent Master	Periodic or on trigger from application	Pdelay_Req
				Pdelay_Req message reception	Pdelay_Resp
				Periodic or on trigger from application	SYNC
				Delay_Req message reception	Delay_Resp
2'b11	X	X	Peer-to-Peer Transparent	Periodic or on trigger from application	Pdelay_Req
				Pdelay_Req message reception	Pdelay_Resp
All other programming combinations are invalid for PTP Offload feature.					

For example, when the MAC is set as an Ordinary or Boundary Slave clock in the PTPT network, it can respond to the reception of SYNC messages with an automatic generation and transmission of the corresponding Delay_Req message. Similarly, various other modes of operation are explained in the table.

The MAC supports the multicast communication model for the generation of SYNC and Pdelay_Req PTP messages. For instance, the Destination Address field of the generated PTP over Ethernet packet is the defined special multicast addresses (0x011B19000000 for all except peer delay mechanism messages and 0x0180C200000E for peer delay mechanism messages).

When the MAC responds to received SYNC, Delay_Req and Pdelay_Req PTP messages with special multicast destination address, it also uses the corresponding special multicast address in the DA field of the automatically generated Delay_Req, Delay_Resp, and Pdelay_Resp PTP messages, respectively.

When the MAC responds to received SYNC, Delay_Req and Pdelay_Req PTP messages with unicast destination address, it takes the SA field of the received packets and makes them as the DA field of the automatically generated Delay_Req, Delay_Resp, and Pdelay_Resp PTP messages, respectively.

At the same time, all the received PTP messages are forwarded to the application along with Rx status, indicating whether the response was generated by the MAC, if it satisfies the packet filtering logic of the MAC receiver.

When the MAC automatically generates a PdelayReq or responds with a Delay_Req, the egress timestamp of these two PTP messages are provided in the Tx TS status (Tx Timestamp Status Register and interrupt generated).

In addition to messageType and versionPTP fields match for basic PTP over Ethernet message detection, the following additional fields are matched to qualify the received PTP message type:

1. The domainNumber field is checked for a match against the value programmed in the CSR.
2. The twoStepFlag in flagField field is checked for one-step indication (1'b0).
3. The transportSpecific field is checked for Default PTP over Ethernet (4'h0) or 802.1AS mode (4'h1) when enabled.

1.8.10.6 PTP Packet Generation

This section explains the format and content of the automatically generated PTP packets by the MAC when this mode is enabled. The template of the Common PTP Message Header is provided, as well as the detailed description of the fields of specific PTP packets generated.

Table 17. Common PTP Message Header Fields

Bits								Octets	Offset
7	6	5	4	3	2	1	0		
transportSpecific				messageType				1	0
Reserved				versionPTP				1	1
messageLength								2	2
domainNumber								1	4
Reserved								1	5
flagField								2	6
correctionField								8	8
Reserved								4	16
sourcePortIdentity								10	20
sequenceId								2	30
sequenceId								2	30
controlField								1	32
logMessageInterval								1	33

messageType

The following encoded values are used for PTP message types:

- SYNC - 4'h0
- Delay_Req - 4'h1
- Pdelay_Req - 4'h2
- Pdelay_Resp - 4'h3
- Delay_Resp - 4'h9

transportSpecific

The following transport protocol encoding is used:

- Default PTP over Ethernet - 4'h0
- 802.1AS mode - 4'h1

versionPTP

It is always set to 2 because PTP version 2 is supported.

domainNumber

This contains the value from the MAC_PTO_Control Register.

flagField

The following values are used:

- alternateMasterFlag (Octet 0 bit 0) - 1'b0 for SYNC and Delay_Resp
- twoStepFlag (Octet 0 bit 1) - 1'b0 for SYNC and Pdelay_Resp
- unicastFlag (Octet 0 bit 2) - 1'b0 for Multicast Address, 1'b1 for Unicast Address **correctionField**
- For more information, see Table 7-12.

sourcePortIdentity

This field takes the value programmed in the MAC_Source_Port_Identityx registers. **sequenceId**

Pdelay_Resp and Delay_Resp use the same sequenceId field from received Pdelay_Req and Delay_Req PTP messages. For SYNC/Delay_Req, Pdelay_Req, a separate sequenceId counter is maintained. These sequenceId counters get incremented by 1 every time the corresponding message is generated and transmitted.

controlField

The following encoded values are used for controlField:

- SYNC - 8'h00
- Delay_Req - 8'h01
- Pdelay_Req - 8'h02
- Pdelay_Resp - 8'h05
- Delay_Resp - 8'h03

logMessageInterval

- SYNC - This contains logSyncInterval from the corresponding MAC_Log_Message_Interval register.
- Delay_Resp - This contains the sum of DRSYNCR and logSyncInterval value taken from the MAC_Log_Message_Interval register for a multicast PTP message and 8'h7F for unicast PTP message.
- Delay_Req, Pdelay_Req and Pdelay_Resp - 8'h7F

where logSyncInterval = log2 (Mean Value of Interval in seconds)

The MAC supports values of -15 to 15 for logSyncInterval fields, which translates to a range from 32.768 μ s (2-15) to 215 sec. For a given value of log sync interval (N), the time interval between two SYNC packets is given by the following:

- $2(30+N)$ nanoSec, when N is negative (-1 to -15)
- $2N$ Sec, when N is positive (0 to 15)
- For example:
- When logSyncInterval is programmed to 1, the interval is 21; therefore, the SYNC message is sent once every 2 seconds.
- When logSyncInterval is programmed to -1, the interval is $2^{-1} = 0.536$ seconds; therefore, the SYNC message is sent once every 536 milliseconds. The value is 0.536 seconds, because $2^{-30} = 1$ ns.
- When logSyncInterval is programmed to -5, the interval is $2^{-5} = 33.55$ ms; therefore, the SYNC message is sent once every 33.55 milliseconds.

1.8.10.7 PTP Message-Specific Fields

messageLength

There is no suffix supported, so this field contains the length of the PTP message that includes 34 byte PTP common header and the body specific to the message type. For SYNC and Delay_Req packets, this field contains 44, whereas for Delay_Resp, Pdelay_Req and Pdelay_Resp, it contains 54.

originTimestamp

This field is the captured egress timestamp for SYNC, Delay_Req, and Pdelay_Req PTP messages.

receiveTimestamp

For Delay_Resp PTP message, this is the ingress timestamp of the corresponding received Delay_Req PTP message.

requestingPortIdentity

For Delay_Resp and Pdelay_Resp PTP messages, this is the sourcePortIdentity field taken from the corresponding received Delay_Req and Pdelay_Req PTP messages. **requestReceiptTimestamp**

For the Pdelay_Resp PTP message, this field is set to 0.

1.8.11 One step timestamp

IP supports the one step time stamp feature. It can be enabled for a packet by setting the Bit 20 (OSTC) in the control word.

NOTE

See Programming Guidelines for IEEE 1588 Timestamping.

1.8.12 One-Step Time Stamping for PTP Over UDP

IP supports one-step timestamp operation for PTP messages sent over UDP/IPv4 and UDP/IPv6. Because origin timestamp and/or correction field (based on the PTP message type and the programmed PTP mode) in the PTP message are updated as part of the one-step timestamp operation, it also requires update to the checksum field of the PTP message sent over UDP/IP.

1.8.13 Checksum Update for One-Step Timestamping for PTP Over UDP

In IP, checksum is updated for a packet, in the MTL before the packet is sent to the MAC where the PTP message is updated for one step timestamp operation. PTP requires the timestamp to be captured while the packet (SFD of the packet) is on the line and one step timestamp requires the origin timestamp and/or correction field to be updated in the PTP header while the packet is being transmitted on the line. Because the PTP message is present in the UDP payload, it is not possible to update the checksum field in the UDP header for modifications made to the PTP message, because it is already transmitted by then. Instead, the following options are supported that are in line with specification mentioned in Annex D & E of IEEE 1588-2008:

- When PTP message is sent over UDP/IPv4, the UDP checksum may be set to 0. The application must set the UDP checksum to 0 and set CIC to 0 or 1 while providing the packet to the MAC and the MAC does not update checksum for changes in origin timestamp or correction field.
- When PTP message is sent over UDP/IPv6, a transmitting node extends the UDP payload of all PTP messages by 2 octets beyond the end of the PTP message. The contents of the UDP checksum field or the final 2 octets of the UDP payload might be modified to ensure that the UDP checksum remains intact after modification of PTP fields. The application must add the two extra bytes at the end of the PTP message while providing the packet to the MAC and the MAC updates these two bytes to keep the checksum correct.

A programmable option (CSC field in MAC_Timestamp_Control register) supports UDP with non-zero checksum for IPv4. When set, the two bytes at the end of the PTP message are updated to keep the UDP checksum correct, similar to PTP over UDP/IPv6. So, the application must add two bytes at the end of the PTP message sent to the IP and IP updates these two bytes to keep the checksum correct.

IP uses the transmit checksum engine to identify the start of PTP message in the UDP/IP packet. Checksum engine has the capability to detect some kind of errors in the header (L3) and payload (L4). Also is not computed for some type of packets, which is indicated by "No" in Table 9-1. For one step, operation the packet is free of those errors and have "Yes/Not applicable" for the corresponding packet type in Table 9-1. If an error is reported by the checksum engine or there is a "No" in Table 9-1, for PTP messages sent over UDP/IP, no update is performed on the PTP message.

See "Transmit Checksum Offload Engine" for more details about the errors detected by transmit checksum engine.

1.8.14 IEEE 1588 Sub Nanoseconds Timestamp

IP supports the ingress and egress correction accuracy in sub nanoseconds. It is programmed in the TSICSNS and TSECSNS fields of the MAC Timestamp Ingress Correction Subnanosecond and MAC Timestamp Egress Correction Subnanosecond registers, respectively. In this case, the correction has an unit of nanosecond, multiplied by 2^8 . The least significant 8 bits of the value should be programmed in the sub-nanoseconds register.

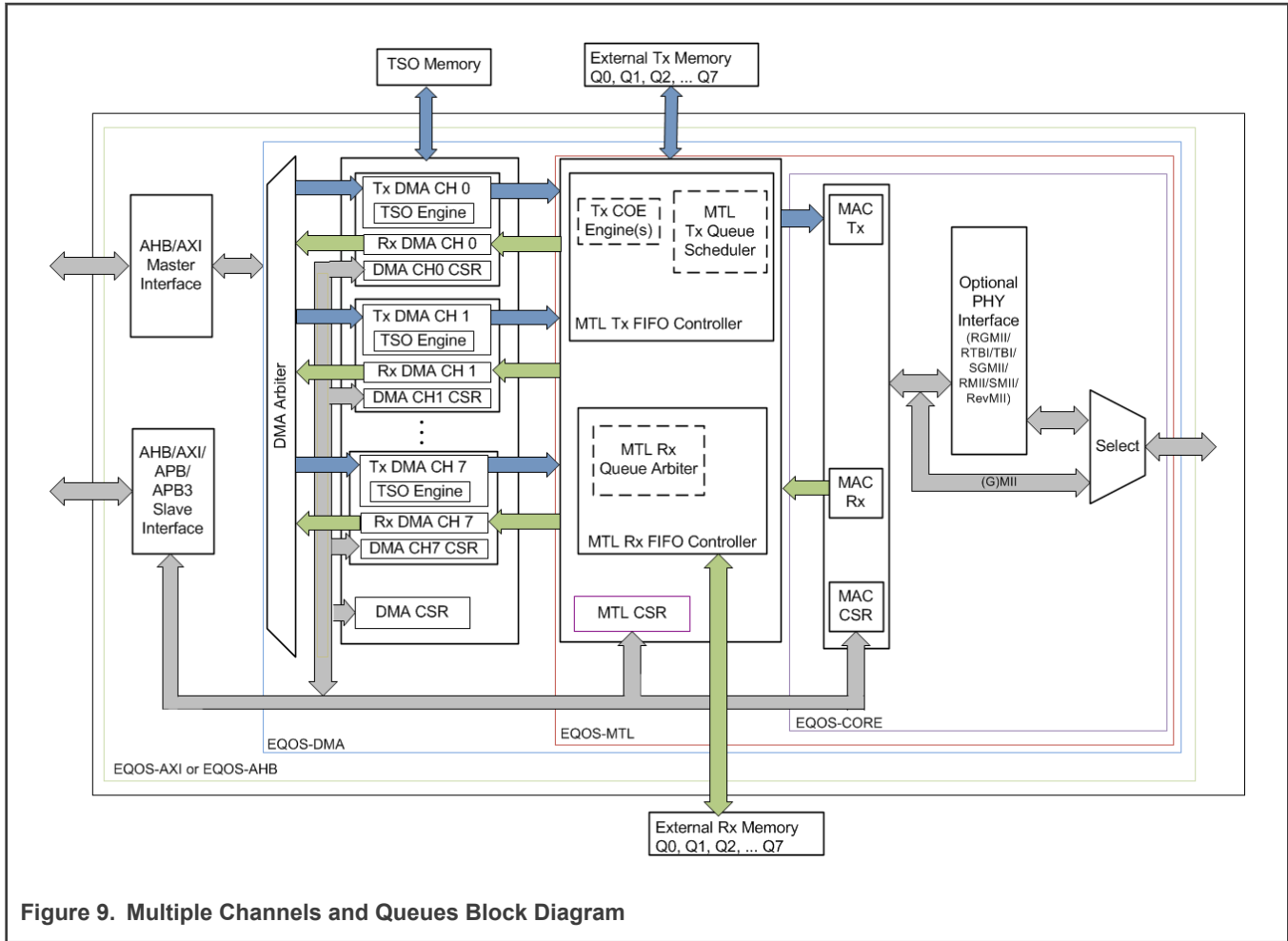
For example, if the required egress correction is 3.6 nS and TSCTRLSSR bit is set, then TSEC must be 0x3 and TSECSNS must be 0x99 ($0.6 * 256$). Similarly, if the required ingress correction is -3.6 nS and TSCTRLSSR bit is set, then TSIC must be 0xBB9A_C9FC and TSICSNS must be 0x66 (fractional part of $(10^9 - 3.6) * 256$).

1.9 Multiple Channels and Queues Support

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IP supports five queues and channels on Tx and Rx paths. Five queues with each queue of 4 K size leads to a total memory available of 20 K. Hence maximum value of TQS/RQS must be set such that $((TQS \text{ or } RQS) + 1) * 256 \leq 20 \text{ K}$.

Following figure shows the architecture of IP with queues and channel.

**NOTE**

In the above block diagram, only five queues are supported

DMA arbiter of IP supports two types of arbitration scheme, fixed priority and weighted round-robin. The DMA arbiter performs the arbitration between the Tx and Rx paths of DMA channels for accessing descriptors and data buffers. The DA bit of the DMA_Mode register specifies the arbitration scheme (fixed or weighted round-robin) between the Tx and Rx DMA of a channel.

Priority between the corresponding Tx DMA and Rx DMA can be set by using TXPR field of the DMA_Mode register. PR field of the DMA_Mode register can be set to specify the weighted priority between the Tx DMA and Rx DMA in round robin arbitration scheme.

Following table provides information about the priority scheme between Tx DMA and Rx DMA.

Table 18. Priority Scheme for Tx DMA and Rx DMA

Bit 14	Bit 13	Bit 12	Bit 11	Bit 10	Priority Schemes
x	x	x	0	1	Rx always has priority over Tx
0	0	0	0	0	Tx and Rx have equal priority. Rx gets the access first on simultaneous requests
0	0	1	0	0	Rx has priority over Tx in ratio 2:1

Table continues on the next page...

Table 18. Priority Scheme for Tx DMA and Rx DMA (continued)

Bit 14	Bit 13	Bit 12	Bit 11	Bit 10	Priority Schemes
0	1	0	0	0	Rx has priority over Tx in ratio 3:1
0	1	1	0	0	Rx has priority over Tx in ratio 4:1
1	0	0	0	0	Rx has priority over Tx in ratio 5:1
1	0	1	0	0	Rx has priority over Tx in ratio 6:1
1	1	0	0	0	Rx has priority over Tx in ratio 7:1
1	1	1	0	0	Rx has priority over Tx in ratio 8:1
x	x	x	1	1	Tx always has priority over Rx
0	0	0	1	0	Tx and Rx have equal priority. Tx gets the access first on simultaneous requests
0	0	1	1	0	Tx has priority over Rx in ratio 2:1
0	1	0	1	0	Tx has priority over Rx in ratio 3:1
0	1	1	1	0	Tx has priority over Rx in ratio 4:1
1	0	0	1	0	Tx has priority over Rx in ratio 5:1
1	0	1	1	0	Tx has priority over Rx in ratio 6:1
1	1	0	1	0	Tx has priority over Rx in ratio 7:1
1	1	1	1	0	Tx has priority over Rx in ratio 8:1

1.9.1 Multiple Queues and Channels Support in the Transmit Path

IP supports the five transmit queues and five channels.

Fixed priority scheme is the default priority scheme for the DMA channels and channel with highest priority always wins the arbitration when it requests for the bus. Channel 0 is always of lowest priority and Channel 1 is always of highest priority.

In Weighted Strict Priority (WSP), the weight corresponds to the number of burst transfers given to a channel in one arbitration cycle. The unused burst transfers of one or more channels are reallocated based on the priority. The channel with highest priority gets the unused burst transfer time before it is allocated to a channel with next highest priority. When a channel uses the allocated burst transfers, the channel with next lower priority is processed. After processing the allocated bandwidth of all channels that had packets to transmit, any unused burst transfer time is allocated to the channel of the highest priority (if required), and then next highest priority (if required), and so on.

In Weighted Round Robin (WRR) priority scheme, weight can be programmed through TCW field of the DMA_CH#_Tx_Control register. IN WRR scheme, all channels are serviced in round-robin order according to the weights settings. The TCW field of the DMA_CH#_Tx_Control register provides the weight for each Transmit channel as shown in the following table.

Table 19. Weight for DMA Channels

TCW Field	Transmit Channel Weight
000	1
001	2
010	3
011	4
100	5
101	6
110	7
111	8

The configured weights correspond to the number of burst transfers given to a channel in one arbitration cycle. The unused or excess burst transfers are distributed equally to all channels.

NOTE

See the "Programming Guidelines for Multi-Channel Multi-Queuing for transmit path"

1.9.2 Multiple Queues and Channels Support in the Transmit Path

IP supports five receive channels and five queues.

In the Receive direction, the MTL Rx Controller selects the Rx DMA for which it is transferring or reading the data from the Rx FIFO memory. This scheduling is based on the programming done in the respective MTL_RxQ[x]_Control register.

Each Rx DMA indicates when it is ready to transfer data and the size of the burst-length (number of beats) that it has to transfer. The scheduler checks whether sufficient data (of requested burst length) is available to be transferred to these DMAs and then selects the Rx DMA that gets serviced using the programmed priorities.

NOTE

See the "Programming Guidelines for Multi-Channel Multi-Queuing for receive path"

1.9.3 Rx Queue to DMA Mapping

The packets in the MTL Rx Queues can be routed to any one of the multiple DMA channels by programming the register MTL_RxQ_DMA_Map0 Register (for queues 0 and 1)

The following types of Rx Queue to DMA mapping is possible through programming:

1.9.3.1 Static Mapping

In this mode, all of the packets of an Rx Queue are connected to a specific DMA channel. For example, all the packets from Rx Queue 0 can be routed to a DMA channel by programming Q0MDMACH (bit[3:0]) and Q0DDMACH (bit[7] = 0) of the MTL_RxQ_DMA_Map0 Register.

Similarly, packets from other Rx Queues can be routed to any DMA channel by programming register fields corresponding to each Queue.

1.9.3.2 Dynamic (Per Packet) Mapping

In this mode, the destination DMA channel of a packet being read from a Rx Queue is not constant but decided independently for each packet. For example, if you set the Q1DDMACH bit of the MTL_RxQ_DMA_Map0 register, the static mapping is disabled for Rx Queue 1 and the value in Q1MDMACH is ignored. The destination DMA channel is decided by the MAC receiver for each packet, depending on the following in decreasing order of priority:

1. L3-L4 filter Based DMA Selection: The TCP/UDP and IP header fields of the received packet are matched against the corresponding values programmed and enabled for comparison in the MAC_L3_L4_Address_Control register. If the match is successful, the DMA channel number programmed in the DMCHN field of the MAC_L3_L4_Address_Control register is selected as the destination DMA channel number provided DMCHEN bit of the same register is set. If none of the L3-L4 Registers give a comparison match, then GMAC proceeds to the next step below.
2. Extended VLAN Based DMA Selection: Extended routing is applicable only if the VLAN Filter has passed. Routing is only based on Perfect Filter Result. Each perfect filter has a DMA Channel Enable and a DMA Channel Number field which can be programmed. Routing is applicable for a filter, only if the DMA Channel Enable bit is set. The frame is routed to the smallest Matching Filter's DMA Channel provided it is enabled. If that filter's DMA Channel number is not enabled, the frame gets routed to Channel 0. For example, if a frame's VLAN tag matches filters 7, 3, and 1. Then the MAC checks if Filter 1's DMA Channel Number is enabled through programming. If yes, the frame gets routed to the programmed value; otherwise it gets routed to DMA. When the inverse filter is enabled; is routed to the least mismatched filter's DMA channel number provided it is enabled. If the DMA Channel enable bit is not set, then the frame is routed based on DA based addressing or to Channel 0.

If Hash Filter is also enabled, it is used to determine the Filter result only. Routing will still depend on the Perfect Filters enabled. If none of the perfect filters are enabled or if all of them are bypassed, then the VLAN Filter based routing will not take place. The frame will be routed via DA Based addressing or to Channel 0. If all the perfect filters give a fail result and the Hash Filter has passed, then the VLAN Filter result is a pass but routing will be based on DA Based Addressing or to Channel 0. Similar behavior is seen when inverse Filtering is enabled as well.

3. Ethernet DA-Based DMA Selection: The DA address of the received packet is compared against the programmed DA values in MAC Address Registers. If the address matches any of the programmed values, the corresponding DCS field (when enabled) determines the destination DMA channel number.

If none of the previous operations is able to make a successful match/decision, then the packet is routed to DMA Channel 0 by default.

1.9.3.2.1 Broadcast/Multicast Packet Duplication

Broadcast/Multicast Packet Duplication feature provides a mechanism to send the received Broadcast/Multicast packets to multiple DMA channels.

The DCS field of the of MAC_Address0_High and other optional MAC_Address(#i)_High (i=1 to 2) registers determines the DMA channel number to which the received packet (that matches the MAC Address present in that register) must be routed.

Packet duplication is not supported in other DMA channel routing mechanisms such as Extended VLAN, or L3-L4 based routing.

The Packet Duplication feature is supported only on the highest MTL Queue configured. Therefore, packet duplication for Broadcast/Multicast packets requires that the MCBCQ field of MAC_RxQ_Ctrl1 register is programmed to the highest RxQ present in the configuration.

1.9.4 Selection of Tag Priorities Assigned to Rx Queues

The VLAN Tag priorities can be assigned to Rx Queues by programming the PSRQ field in the corresponding MAC_RxQ_Ctrl2 registers. The bit corresponding to the VLAN Tag priority can be set in the PSRQ field for assigning that priority to the Rx Queue. For example, if you want to assign VLAN Tag priority of 3 to Rx Queue 0, set bit [3] in PSRQ field of MAC_RxQ_Ctrl2 register. The VLAN Tag priority assigned to particular Rx Queue must be unique, that is, more than one Rx Queue cannot be assigned the same VLAN Tag priority. However, more than one VLAN Tag priorities can be assigned to same Rx Queue.

The settings in the PSRQ field is used for VLAN Tagged Rx packet routing to Rx Queues as well as for PFC based Tx flow control. The received VLAN Tagged Rx packet is routed to Rx Queue that has the VLAN Tag priority match. In PFC based Tx flow control,

PSRQ field corresponding to a particular Rx Queue is used for enabling VLAN Tag priorities in the PFC packet transmitted when corresponding Rx Queue threshold levels are reached.

1.9.5 Rx Side Routing from MAC to Queues

IP supports Rx Side Routing from the MAC to Queues. The MAC routes the Rx packets to the Rx Queues based on following packet types in that order

- Unicast/Multicast destination address packets that fail the destination address filter
- Multicast/Broadcast destination address packets that pass the destination address filter
- VLAN Tag Priority field in VLAN Tagged AV data packets
- AV Control packets
- VLAN Tagged IEEE 1588 PTP over Ethernet packets
- Untagged IEEE1588 PTP over Ethernet packets
- VLAN Tag Priority field in VLAN Tagged packets
- Untagged packets

The outer C-VLAN Tagged AV data Rx packets can be routed based on the priorities assigned to Rx Queues through PSRQ field in corresponding MAC_RxQ_Ctrl2 and MAC_RxQ_Ctrl3 registers and the corresponding Rx Queue is enabled for AV through RXQ#EN field in MAC_RxQ_Ctrl0 register. These packets may be single VLAN Tagged with C-VLAN type or Double VLAN Tagged with outer VLAN Tag of C-VLAN type when Double VLAN feature is enabled (EDVLP bit in MAC_VLAN_Tag register is set) with inner C-VLAN Tagged or inner S-VLAN Tagged when SVLAN processing is enabled (ESVL bit in MAC_VLAN_Tag register is set). This type of Rx packet routing is available when AV feature and Multiple Rx Queues are selected in the configuration.

The AV Control Rx packets can be routed based on the Rx Queue number specified in the AVCPQ field in MAC_RxQ_Ctrl1 register and corresponding Rx Queue is enabled for AV through RXQ#EN field in MAC_RxQ_Ctrl0 register. These packets may be single VLAN Tagged with C-VLAN type or Double VLAN Tagged with outer VLAN Tag of C-VLAN type when Double VLAN feature is enabled (EDVLP bit in MAC_VLAN_Tag register is set) with inner C-VLAN Tagged or inner S-VLAN Tagged when SVLAN processing is enabled (ESVL bit in MAC_VLAN_Tag register is set). This type of Rx packet routing is available when AV feature and Multiple Rx Queues are selected in the configuration.

The VLAN Tagged Rx packets can be routed based on the priorities assigned to Rx Queues through PSRQ field in corresponding MAC_RxQ_Ctrl2 and MAC_RxQ_Ctrl3 registers and the corresponding Rx Queue is enabled for DCB/generic through RXQ#EN field in MAC_RxQ_Ctrl0 register. This type of Rx packet routing is available when Multiple Rx Queues are selected in the configuration.

The untagged IEEE 1588 PTP over Ethernet Rx packets can be routed based on the Rx Queue number specified in the PTPQ field in MAC_RxQ_Ctrl1 register and the corresponding Rx Queue is enabled through RXQ#EN field in MAC_RxQ_Ctrl0 register. The VLAN tagged IEEE 1588 PTP over Ethernet Rx packets can be routed based on the priorities assigned to Rx Queues through PSRQ field in corresponding MAC_RxQ_Ctrl2 and MAC_RxQ_Ctrl3 registers or the Rx Queue number specified in the PTPQ field in MAC_RxQ_Ctrl1 register and the corresponding Rx Queue is enabled through RXQ#EN field in MAC_RxQ_Ctrl0 register. This is determined by programming in TPQC field of MAC_RxQ_Ctrl1 register. This type of Rx packet routing is available when IEEE 1588 Timestamp feature support and Multiple Rx Queues are selected in the configuration. The VLAN Tagged IEEE 1588 PTP over Ethernet Rx packets are detected only when 802.1AS mode is disabled (AV8021ASMEN bit in MAC_Timestamp_Control register is set to 0), otherwise this type of Rx packets are routed as generic VLAN Tagged Rx packets.

The multicast/broadcast destination address Rx packets that pass the destination address filter can be routed based on the Rx Queue number specified in the MCBCQ field of MAC_RxQ_Ctrl1 register when enabled through MCBCQEN bit of MAC_RxQ_Ctrl1 register and the corresponding Rx Queue is enabled through RXQ#EN field in MAC_RxQ_Ctrl0 register. This type of Rx packet routing is available when Multiple Rx Queues are selected in the configuration.

The untagged Rx packets can be routed based on the Rx Queue number specified in the UPQ field of MAC_RxQ_Ctrl1 register and the corresponding Rx Queue is enabled through RXQ#EN field in MAC_RxQ_Ctrl0 register. This type of Rx packet routing is available when Multiple Rx Queues are selected in the configuration.

The unicast destination address Rx packets that fail the destination address filter can be routed based on the Rx Queue number specified in the UFFQ field of MAC_RxQ_Routing_Ctrl register when enabled through UFFQE bit of MAC_RxQ_Routing_Ctrl register, RA bit of MAC_Packet_Filter register is set to 1 and the corresponding Rx Queue is enabled through RXQ#EN field in MAC_RxQ_Ctrl0 register. This type of Rx packet routing is available when Multiple Rx Queues are selected in the configuration.

The multicast destination address Rx packets that fail the destination address filter can be routed based on the Rx Queue number specified in the MFFQ field of MAC_RxQ_Routing_Ctrl register when enabled through MFFQE bit of MAC_RxQ_Routing_Ctrl register, RA bit of MAC_Packet_Filter register is set to 1 and the corresponding Rx Queue is enabled through RXQ#EN field in MAC_RxQ_Ctrl0 register. This type of Rx packet routing is available when Multiple Rx Queues are selected in the configuration.

If the Rx packet cannot be classified in any of the defined packet type for routing, it will be routed through Rx Queue 0.

1.9.6 Rx Side Arbitration Between DMA and MTL

IP controller supports Rx side arbitration between the DMA and the MTL.

After completing the current packet processing, the DMA Channel Controller fetches the next descriptor. After the descriptor fetching is complete, the DMA Channel Controller evaluates the amount of data to be transferred to the Rx Buffer based on the programmed PBL and Rx Buffer Length. Accordingly, it requests the MTL to transfer the data.

After servicing the current request, the MTL Rx Queue arbitration scheme selects Rx Queue based on the arbitration scheme (RAA field of the MTL_Operation_Mode register) and the weights programmed in the Queue <n> Receive Control Register. The arbitration is done among Queues for which DMA is ready to service. After the Rx Queue is selected, PBL (Programmable Burst Length) amount of data is read out from that Queue and is routed to the Rx DMA Channel based on the Rx Channel selection criteria.

The arbitration takes place when every PBL data transfer is completed and descriptors are ready for the processing from at least one DMA Channel.

1.9.7 Tx Side Arbitration between DMA and MTL

IP support Tx side arbitration between the DMA and the MTL. Since the number of Tx DMA channels is always equal to the number of Tx Queues in the MTL, they are always mapped directly. For instance, each Tx DMA pushes data into its respective Tx Queue assigned to it.

The data inside each Tx Queue is stored in packets. Hence, if two DMAs are allowed to transfer data into the same queue, when a Tx DMA starts a packet transfer, the other DMA cannot transfer data unless the previous packet is completely pushed-in. This means that the second DMA remains idle until the first packet is transferred. Hence, each DMA is always connected directly to its corresponding Tx Queue.

1.9.8 Data Centre Bridging

The Data Center Bridging feature supports three types of algorithms:

- **Weighted Round Robin:** In Weighted Round Robin (WRR) algorithm, each queue is assigned a weight based on the percentage of configured bandwidth. All queues are serviced in the round-robin order according to the weight settings. This algorithm is less complex and requires less hardware as compared to other algorithms. Large packets get more bandwidth in this algorithm.
- **Deficit Weighted Round Robin:** In Deficit Weighted Round Robin (DWRR) algorithm, each queue is assigned a weight based on the percentage of configured bandwidth. A deficit counter holds the credit for transmission. The quantum value, proportional to the bandwidth, is added to the deficit counter every time a queue is scanned. The packet in the queue is selected for transmission only if the credit in the deficit counter is less than or equal to the size of the packet available at the head of the queue. A selected queue is serviced until the outstanding credit is lesser than the packet size at the head of the queue during a particular queue scan cycle. After this, the next lower priority queue is serviced. When the packets in a queue are over, the credit becomes zero for that queue scan cycle. This algorithm is less complex, and it provides more fairness in scheduling as compared to other algorithms.
- **Weighted Fair Queuing:** In Weighted Fair Queuing (WFQ) algorithm, each queue is assigned a weight based on the percentage of configured bandwidth. The algorithm computes the finish time of packets at the head of the queues based

on the packet size and weights for the queue. The packets with earliest finish time is scheduled for transmission first. The algorithm is more complex and less scalable as compared to other algorithms.

1.9.9 Audio Video Bridging

IP supports the Audio Video bridging in 100 and 1000Mbps mode only which allows the transmission of time sensitive traffic over network. IP implements the IEEE 802.1Qbv-2015 (Enhancements to Scheduling Traffic), IEEE 802.3br (Interspersing Traffic) and IEEE 802.1Qbu (Frame Preemption) protocol for supporting the AVB feature.

The Transmit path of Queue 0 supports the strict priority algorithm, and it is used for best-effort traffic while transmit paths of additional queues support traffic management by using the credit-based shaper algorithm. For a queue, the credit-based shaper algorithm determines that a queue is available for transmission if the following conditions are true:

- The queue contains one or more packets.
- The credit for the queue is positive as per the algorithm
- AV traffic can be received on all queues. IP can also receive untagged PTP packets in addition to AV traffic on any queue.

1.9.10 Queue Modes

Both Tx and Rx queues are capable in handling the generic, AV, or all types of traffic.

Queueing is based on WRR (Weighted Round Robin) or WSP (Weighted Strict Priority) algorithm for generic traffic while it is based on CBS (Credit Base Shaper) and SP (Strict Priority) algorithm for AV traffic.

An Rx Queue is configured for generic or AV based routing depends on the RXQ#EN field of corresponding queue. Queueing is done based on VLAN tag priority. The VLAN Tag priority should match the PSRQ field of the MAC_RxQ_Ctrl2 registers. The Rx packets can also be routed to a particular DMA channel based on the DCS field of perfectly-matched MAC Address register.

By default, untagged packets in generic traffic are routed to Receive queue specified in UPQ field of MAC_RxQ_Ctrl1 register. Queue 0 is the default value of UPQ field. You can override the default value with any other value, for the UPQ field.

The AV control packets (tagged or untagged) are routed based on the AVCPQ field of the MAC_RxQ_Ctrl1 register. The PTP over Ethernet packets are routed based on the AVPTPQ field of MAC_RxQ_Ctrl1 register.

1.9.11 Queue Priorities

Priority of an Rx queue can be programmed in corresponding field of MAC_RxQ_Ctrl2 registers. AV queue (high priority) should be programmed as higher priority than the Best-effort queue (low priority).

1.9.12 Enhancements to Scheduled Traffic (EST)

The IEEE 802.1Qbv-2015 defines the schedule for each of the queues on every egress port which makes the implementation aware of traffic arrival schedule. This information can be used to block the lower priority traffic from transmission in this time window/slot. This ensures that scheduled traffic is forwarded from sender to receiver through all the network nodes with a deterministic delay.

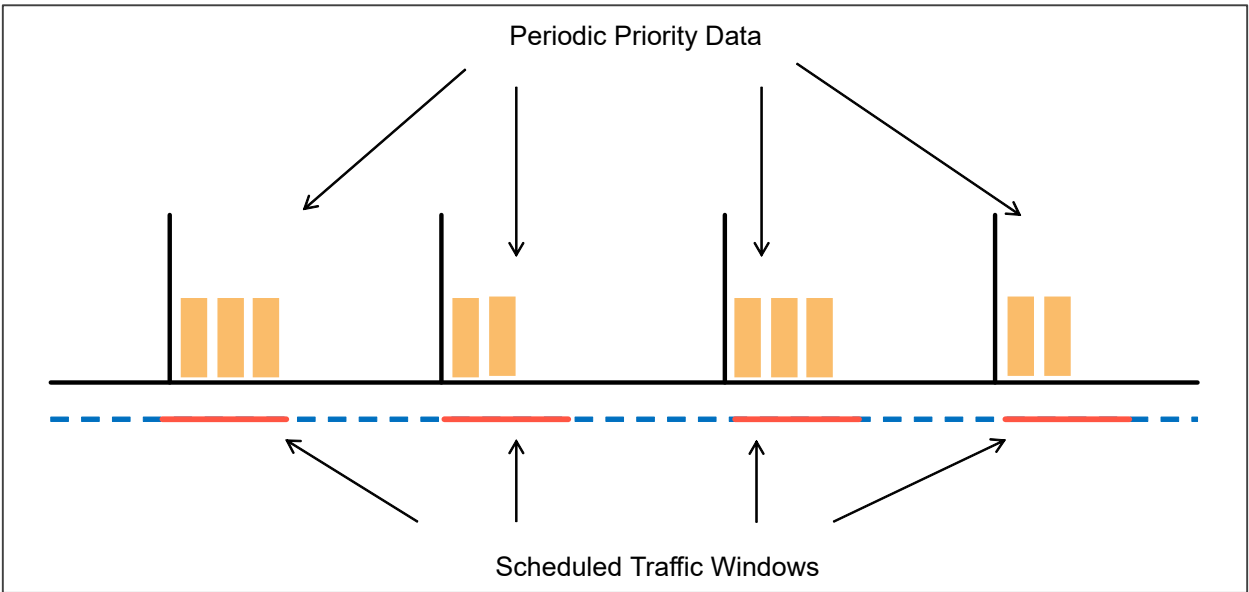


Figure 10. Time Aware Shaper Implementation for Traffic Scheduling

One of the important requirements to achieve a low latency is to ensure there are no interfering frames during the scheduled windows that are reserved for high priority traffic. The use of scheduled traffic imposes limitations while starting a transmission. As shown in the following figure, if an interfering frame begins transmission just before the start of a reserved time period, it can extend transmission into the reserved window, and potentially interfere with higher priority traffic. Therefore, a guard band whose size is equal to the largest possible interfering frame is required before the window starts.

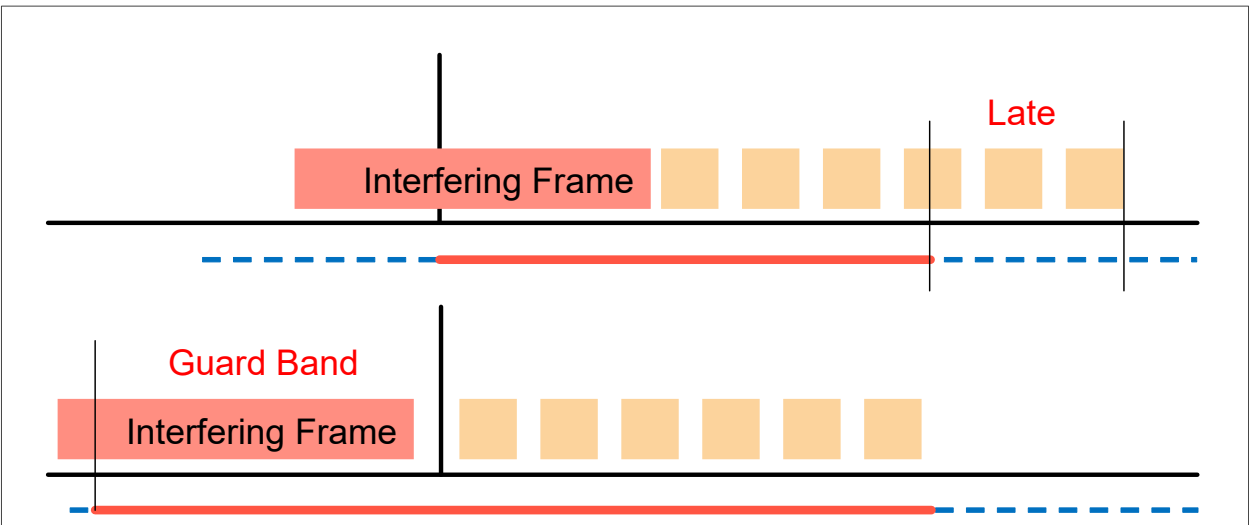


Figure 11. Implementing a Guard Band to Avoid Delays Due to Interfering Frames

A larger guard band equates to a less efficient use of network bandwidth. However, with the implementation of IEEE802.1Qbu (frame preemption), this issue is addressed. Frame preemption breaks the interfering frame into smaller fragments. Therefore,

the guard band needs to be only as large as the largest possible interfering fragment instead of the largest possible interfering frame.

During the guard band, only the frames that can complete the transmission of the entire frame before the next gate close event are permitted. This ensures that the high priority traffic can always start at the beginning of the window reserved for it.

1.9.12.1 Frequently Used Terms in EST

Following are some of the frequently used terms in the EST support and their definitions.

- Gate Control List: The list in the hardware memory that holds the gate controls and the associated time intervals.
- Gate Controls: For a given schedule (row in gate control list) there is a Gate Open (O) and Gate Close (C) state associated with each TC. The set of O or C values, whose width is same as configured TCs is called the Gate Controls. Example: CCOCOCOC means TC7=C, TC6=C, TC5=O and so on.
- Time Interval: Time interval (in nano seconds) is a 16, 20, or 24-bit configurable field in the Gate Control list that indicates the time for which the associated gate controls are valid.
- Base Time Register: Each Gate Control List is associated with a 64-bit Base Time Register that holds the start time (in PTP format) for the list.

1.9.12.2 Updates to the Transmit Scheduling to Support EST

To support EST, following updates are required to the transmit scheduling:

- Implementation of Gate Control List (GCL)
- Enforcing gate controls in the scheduler
- Accounting for Gate Open (O) duty cycle in the computation of idleSlope (CBS)

1.9.12.2.1 Implementation of Gate Control List (GCL)

Figure 8-7 shows block diagram from the IEEE 802.1Qbv spec that illustrates how the gate control list governs the gate close(C) and open(O) events based on the schedule provided for each event. GCL has the following two parts:

- Time Interval: Defines the time in nano seconds for which the gate controls are valid and should be applied before reading the next gate controls from the list. Configurable width of 16, 20 or 24 to represent a maximum of 64us, 1ms, or 16ms schedule interval respectively. Supports left shift of the time interval upto 7 bits to be able to apply a multiplication factor from 1 to 128ns (in steps of 2^n). The maximum value (post shifting) of this field must be 999,999,999 ns.
- Defines the Open (O) represented by logic 1 or Close (C) represented by logic 0 state for the gate of each TC.

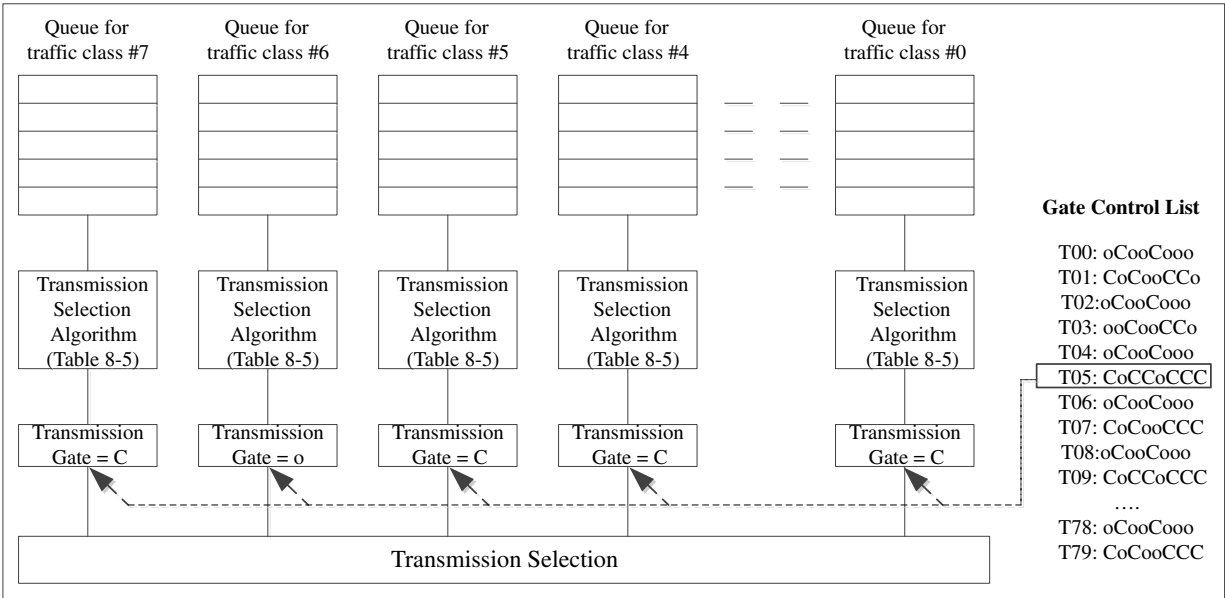


Figure 12. Block Diagram from IEEE 802.1 Qbv Specification - GCL Governing Gate Close and Open Events

The implementation of GCL consists of the following two Gate Control Lists:

- HWOL - Hardware Owned List which is a list for hardware access.
- SWOL - Software Owned List which is a list for software access.

The access to these lists is mutually exclusive. The ownership to the list is set by hardware in the SWOL field of MTL_EST_Status register. Following table provides the implementation details of GCL.

Table 20. External Memory Used for Holding the Two Gate-Control Lists

Gate Control (up to 8 bits)	Time Interval (ns) 16, 20, or 24 bits	
OCCCCCCC	T0	HWOL or SWOL
OOOCCCCC	T1	
CCCCOCCC	T2	
CCCCCCOO	T3	
I	I	
I	I	
OCOCOCOO		
OOOCCCCC	T last	

Table continues on the next page...

Table 20. External Memory Used for Holding the Two Gate-Control Lists (continued)

Gate Control (up to 8 bits)	Time Interval (ns) 16, 20, or 24 bits	
OOOOCCCC	T0	SWOL or HWOL
OOOOCCCC	T1	
OCCCCCCC	T2	
OOOOCCCC	T3	
I	I	
I	I	
OCOCCCCC		
OCCCCCCC	T last	

1.9.12.2.2 Registers Related to Gate Control List

Following are the set of 4 registers (one for each GCL) related to GCL. These registers are implemented through Indirect Addressing using the GCL_Control and GCL_Data registers.

1. 64-bit Base Time Register (BTR)
2. 40-bit Cycle Time (CTR)
3. m-bit Time Extension (TER)
4. n-bit [Gate Control] List Length (LLR) (n depends of the configured GCL depth)

1.9.12.2.2.1 Base Time Register (BTR)

Base Time Registers is a 64-bit register that holds the start time to execute the GCL. The format of the BTR is same as the PTP format (upper 32-bits holds time in seconds and lower 32 bits hold time in nano seconds). Once the execution of a given list begins, the implementation can update the value in BTR to indicate the next list execution begin time.

1.9.12.2.2.2 Cycle Time Register (CTR)

Cycle Time Register is a 40-bit register that holds the time at which the execution of the GCL should be repeated. The Cycle Time Register consists of an 8-bit value in seconds, and a 32-bit value in nano seconds (similar to the PTP time format with truncated seconds register). For a given Gate Control List the start time is "Base Time" + N * "Cycle Time" where N is an integer value representing the iteration number starting with 0 for first iteration. If the Gate Control List execution takes longer than the Cycle time, then the list is truncated at the Cycle Time and the subsequent loop begins at Cycle Time.

1.9.12.2.2.3 Time Extension Register (TER)

Time Extension Register (TER) is a m-bit (where m = Configured Time Interval width + 7) register that holds the amount of time (in nano seconds) the current Gate Control List can be extended before switching to the new Gate Control List. This will be useful in avoiding small fragments of the current list being executed before switching to a new list.

1.9.12.2.4 List Length Register (LLR)

List Length Register (LLR) is an n-bit register (when n is 7, 8, 9, 10, or 11 for a GCL configured depth of 64, 128, 256, 512, or 1024 respectively) that holds the integer value of the length of the GCL (that is the number of valid rows in each GCL). The processing of the GCL stops after the number of rows read equals to the LLR value.

1.9.12.2.3 Transmission Gating Implementation

A Bridge or an end station can be enhanced to allow transmission from each TC that is yet to be scheduled relative to a known timescale. To achieve this, a transmission gate is associated with each TC; the state of the transmission gate determines if the queued frames can be selected for transmission.

For a TC, the transmission gate can be in one of the following two states:

1. **OPEN** Queued frames are selected for transmission, in accordance with the definition of the transmission selection algorithm associated with the TC.

2. **CLOSED** Queued frames are not selected for transmission.

A frame on a traffic class queue is not available for transmission if the transmission gate is in the closed state or if there is insufficient time available to transmit the entirety of that frame before the next gate-close event associated with that queue.

The implementation has visibility to the current schedule of gate controls and the immediate next schedule of the gate controls. So the maximum Gate Open period does not exceed the sum of the two Time Intervals. This is because, a frame is selected for transmission only if the gate is currently Open and the duration of gate open interval is greater than the time taken to transfer the entire frame.

The implementation must know the frame size before the transmission, so that the transmission overruns can be avoided and only the frames that can complete are scheduled at all times.

The implementation adequately compensates for the implementation delays in the data transfer from the buffer to the line by offsetting the current time with all the relevant delays (provided by the CTOV field of MTL_EST_Control register). This ensures that the schedule provided is always accurately implemented at the line.

Software must ensure that the GCL slot interval is always greater than the expected packet size and overhead (scheduler delay, interframe gap (IFG), and preamble, all combined).

1.9.12.3 Idle Slope Computation Updates

When EST is enabled, credit is accumulated only when the gate is open; therefore, the effective data rate of the idleSlope must be increased to reflect the duty cycle for the transmission gate associated with the queue.

The idleSlope is computed based on the GateOpenTime and OperCycleTime values. Program the idleSlope registers (implemented one per CBS enabled TC) based on the following equation. The existing MTL register has sufficient bit width to accommodate the new values for idleSlope.

$$\text{idleSlope} = (\text{operIdleSlope}(N) * \text{OperCycle} / \text{GateOpenTime})$$

1.9.12.3.1 Operational Details of GCL

Set the switch to software List, the SSWL field of the MTL_EST_Control Register, so that hardware can access the programmed gate control list. The first set of gate controls are applied when the current time is equal to the value in the Base Time Register (BTR) and is held until the programmed "Time Interval" value.

To avoid transmission overruns, one additional gate control event is always read ahead from the list. This enables the GCL to determine the next gate close events (if any) for the TCs that are open.

The scheduling based on the Gate Open state and Time Interval of only the current and subsequent schedule. An internal accumulator is used to add the time intervals when gate controls are applied. BTR + Accumulator holds the time at which the next set of gate controls are to be applied.

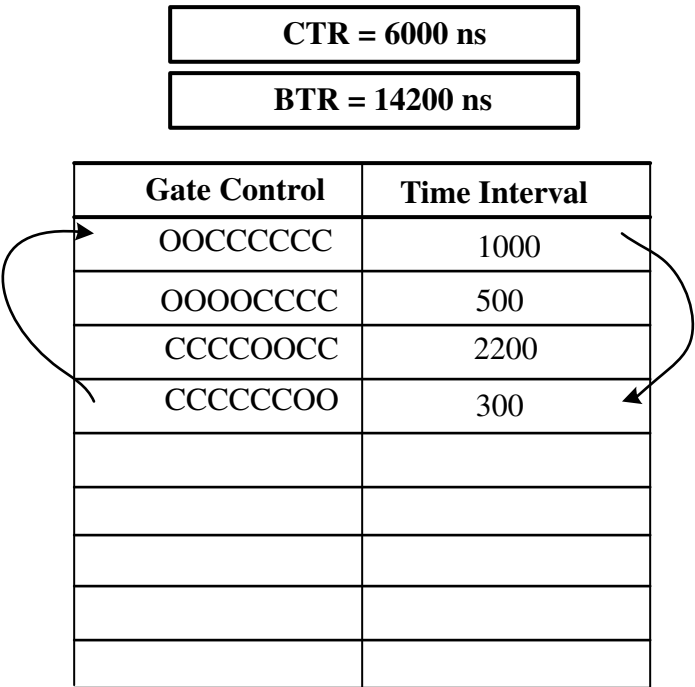


Figure 13. GCL and Associated Registers - BaseTime and CycleTime

The GCL is read progressively from the first row adhering to the schedule. The read operations continue until the List Length (from LLR Register) is reached and the execution of the list restarts at BTR + CTR time. At this point the BTR is incremented by value in CTR to mark the beginning of a new cycle. In the absence of any gate controls, all the gates are deemed to be in Open state, during the execution of the list.

In cases where the execution time of the list is greater than the CycleTime, the list is truncated and the next iteration starts when the current time equals BTR + CTR.

Table 21. GCL and Associated Registers - BTR and CTR

Current Time	Gate Control Applied	Accumulator Value	BTR (with updates)
14200	OCCCCCCC	1000	14200
15200	OOOOCCCC	1500	14200
15700	CCCCOOCC	3700	14200
17900	CCCCCCOO	4000	14200
18200	OOOOOOOO	0	20200
20200	OCCCCCCC	1000	20200
21200	OOOOCCCC	1500	20200

In the example in the table, the execution starts at 14200 and the first set of Gate Controls "OCCCCCCC" are applied immediately. As the time interval is 1000ns the next set of gate controls are applied at 14200 (BTR) + 1000 (Accumulator) = 15200 ns as

shown in Table 8-9. As there are no gate controls available after the execution of the last gate control and before the next iteration of the loop, the gates are deemed to be in open state during that time period as depicted at time 18200 in the following table.

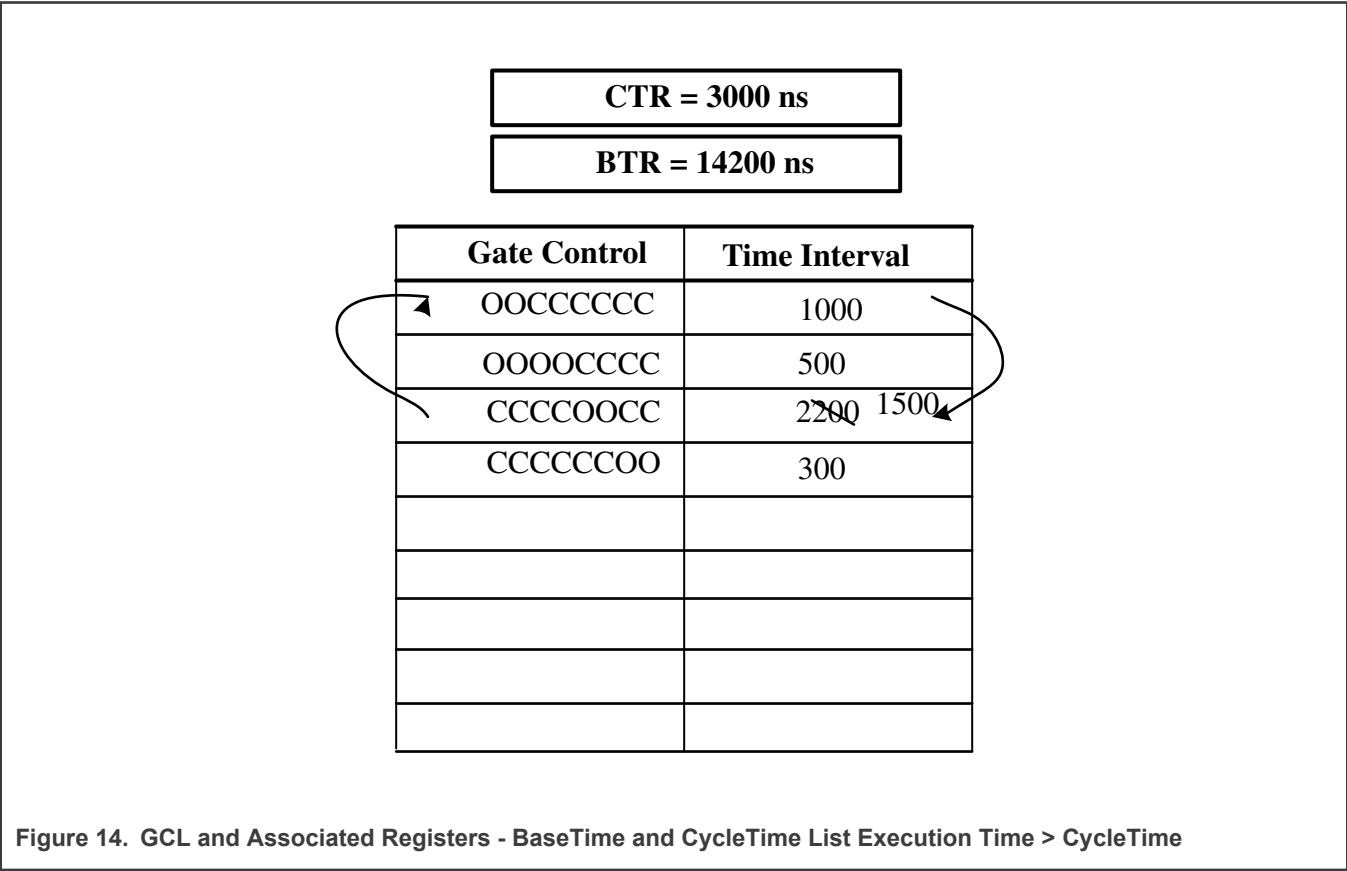


Figure 14. GCL and Associated Registers - BaseTime and CycleTime List Execution Time > CycleTime

As the list execution takes longer than the allocated CycleTime, the list is truncated and the list is started from the BTR+CTR as shown in the following table.

Table 22. GCL and Associated Registers - BTR and CTR, Execution Time > Cycle Time

Current Time	Gate Control Applied	Accumulator Value	BTR (with updates)
14200	O O C C C C C C	1000	14200
15200	O O O O C C C C	1500	14200
15700	C C C C O O C C	3700	14200
17200	O O C C C C C C	1000	17200
18200	O O O O C C C C	1500	17200
18700	C C C C O O C C	3700	17200

While applying the third set of gate controls, the BTR + CycleTime (17200) < BTR + Accumulator (17900), so the list is truncated and execution switches to a new iteration at 17200.

1.9.12.3.2 Installing a New GCL

When a new software programmed GCL is available and must be executed at the new BTR value, the switch to the new GCL can happen in one of the following ways.

- New Base Time aligned with current schedule
- New Base Time unaligned with current schedule

1.9.12.3.2.1 New Base Time Aligned with Current Schedule

If the choice of cycle time for the new gating cycle is unchanged from the cycle time for the current gating cycle, and if the BTR chosen for the new gating cycle (new BTR) is an integer multiple of the current cycle time (+ current BTR), then the new gating cycle exactly lines up with the old gating cycle, that is, the cycle start times for the new gating cycle is same as they would have been for the old configuration. This could be considered to be the ideal case and allows the new gating cycle to be installed and executed with no timing issues. The implementation completes the execution of an iteration of the current list and switches to the new list at the beginning of the BaseTime listed in the new list.

If (New BaseTime >= Current Time)

ConfigChangeTime = New BaseTime

Else If (New BaseTime < Current Time)

1. Set the BTRError
2. ConfigChangeTime = (New BaseTime + N* New CycleTime)

where N is the smallest integer for which the relation ConfigChangeTime >= CurrentTime and (N <= 8) is TRUE.

When N > 8 the hardware cannot auto recover and the loop count value in BTRError reporting is set to 1111 requiring the software to reprogram the New Base Time.

Following figure illustrates the installation of the new GCL along with the timelines.

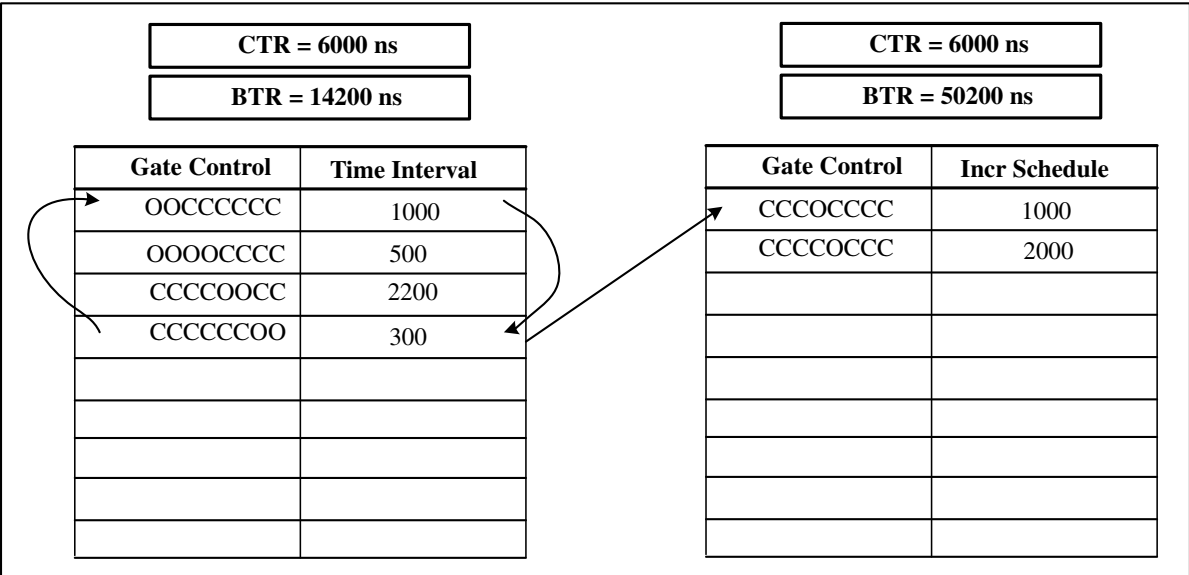


Figure 15. Switching to a New Configuration that is Aligned with the Existing Configuration

In the above example, after the sixth iteration of the first GCL, the BTR values of the old and new GCL are equal. At that point the new GCL is processed as a natural extension to the existing GCL.

Table 23. GCL and Associated Registers - BTR and CTR

Current Time	Gate Control Applied	Accumulator Value	BTR (with updates)
44200	OCCCCCCC	1000	44200
45200	OOOCCCCC	1500	44200
45700	CCCCOCCC	3700	44200
47900	CCCCCCOO	4000	44200
48200	OOOOOOOO	0	50200
50200	CCCOCCCC	1000	50200
51200	CCCOCCCC	3000	50200
53200	OOOOOOOO	0	56200

1.9.12.3.2.2 New Base Time Unaligned with Current Schedule

If new CycleTime differs from current CycleTime or New BaseTime is in the future and is not an integer multiple of current CycleTime, then the old and new cycles do not line up; when new BaseTime is reached (when the new configuration is installed and starts to execute), the last old cycle is normally truncated to start the first new cycle. This could be undesirable if it results in a very short last old cycle; arguably it would be better to simply extend the penultimate old cycle by that small amount, rather than starting a very short cycle. The Cycle Time Extension Register (related to the current config list) allows this extension of the last old cycle to be done in a defined way; if the last complete old cycle ends normally in less than current Cycle Time Extension (TER) ns before the new base time, then the last complete cycle before new BaseTime is reached is extended so that it ends at new BaseTime.

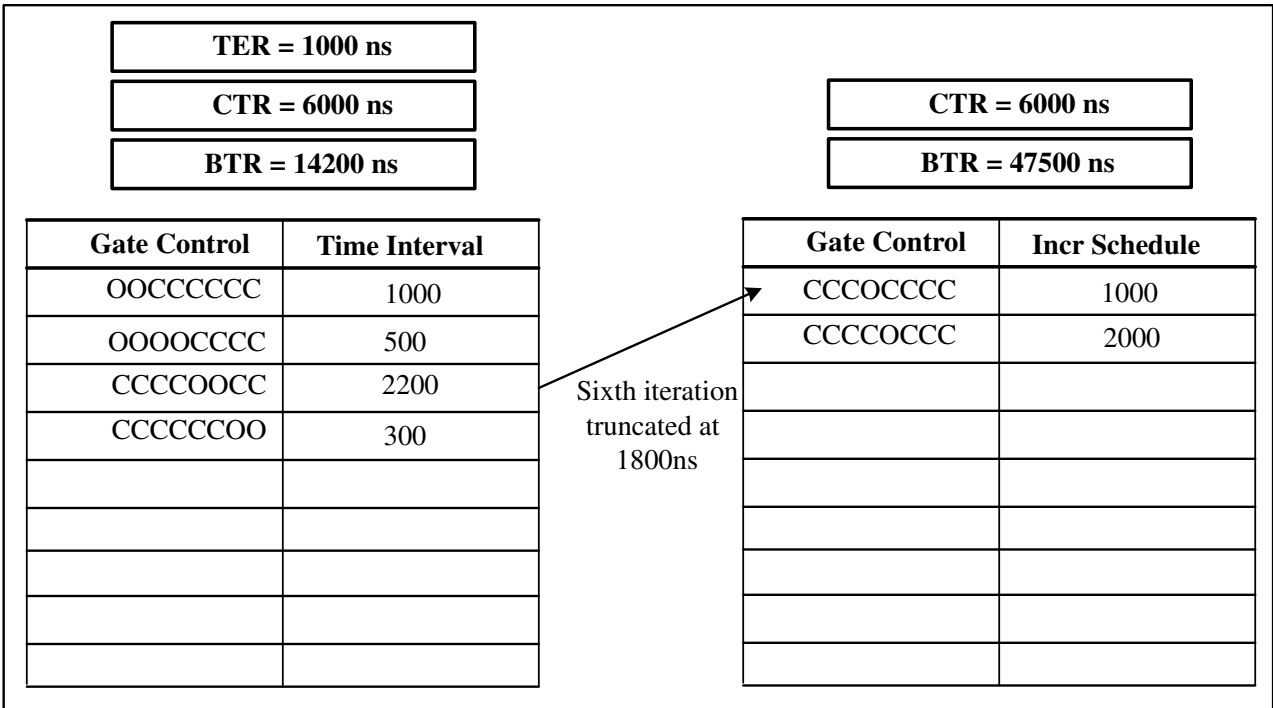


Figure 16. Switching to New List Truncating the Current List

At the end of the fifth iteration the Current time + Cycle time extension (TER) < New BTR so the sixth iteration of current configuration is started. During the sixth iteration of the current list when the new BTR value is smaller than the next schedule in the current list, it switches to the new list.

Table 24. Extending to New List By Truncating the Current List

Current Time	Gate Control Applied	Accumulator Value	BTR (with updates)
44200	OCCCCCCC	1000	44200
45200	OOOCCCCC	1500	44200
45700	CCCCOCCC	3700	44200
47500	CCCOCCCC	4000	44200
48500	CCCCOCCC	0	50200
50500	OOOOOOOO	1000	50200

Below is an example where the current config list is extended instead of starting a new iteration as the extension time of 800ns is less than the allowed cycle extension time (TER) of 1000ns.

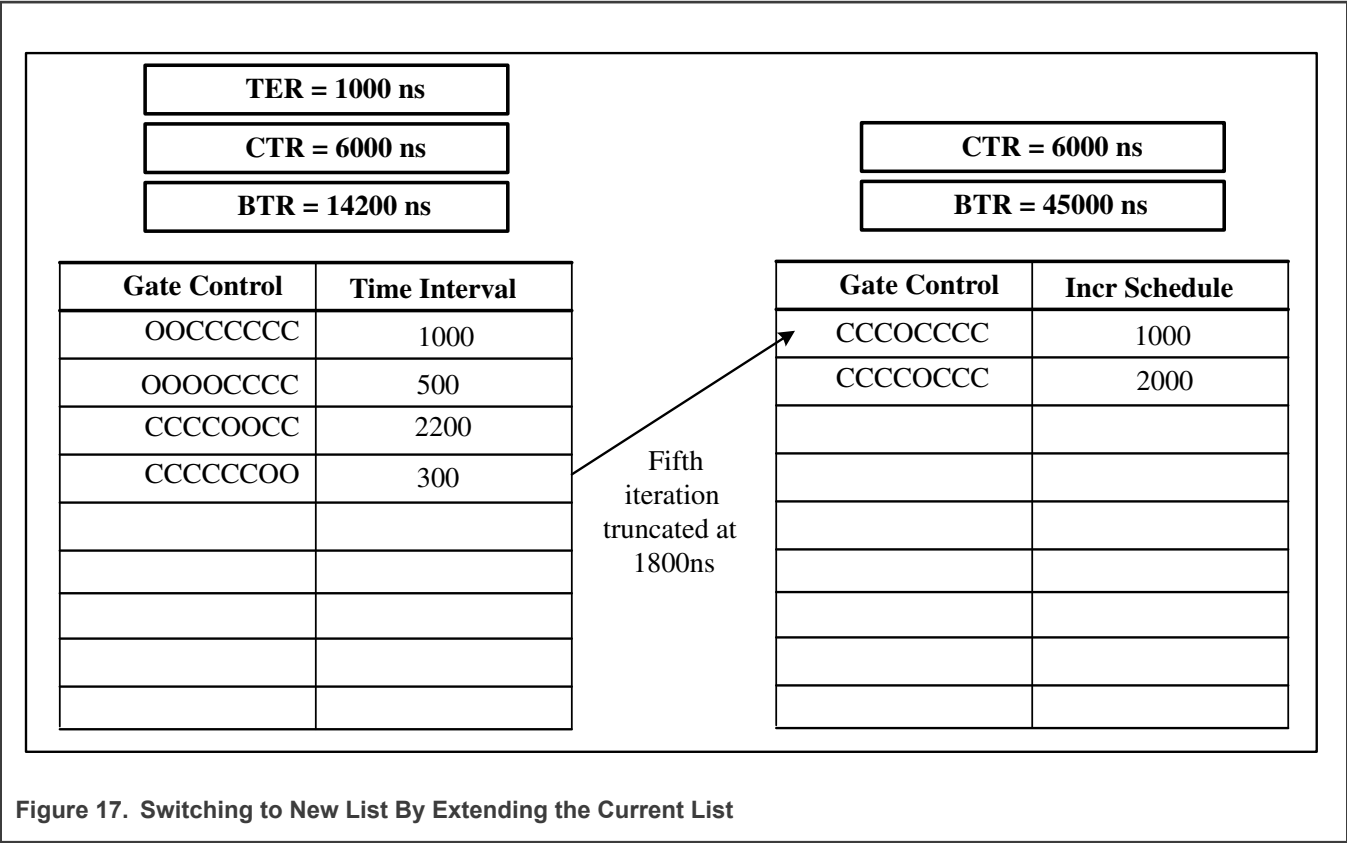


Table 25. Switching to New List By Extending the Current List

Current Time	Gate Control Applied	Accumulator Value	BTR (with updates)
38200	O O C C C C C C	1000	38200
39200	O O O O C C C C	1500	38200
39700	C C C C O O C C	3700	38200
41900	C C C C C C O O	4000	38200
42200	O O O O O O O O	0	44200
45000	C C C O C C C C	1000	45000
46000	C C C C O C C C	3000	45000
48000	O O O O O O O O	0	51000

1.9.12.4 Impact of Transmit Scheduling Algorithms on EST

When EST is used in isolation, the Gate Control List manages the final open/close state of the Queues along with the checks carried out by the Transmission Selection Algorithm in MTL. As the Gate Controls operate on a predefined repetitive schedule, it is recommended to use Strict Priority or Credit Based Shaper (CBS) scheduling algorithms.

Other algorithms such as the Weighted Round Robin (WRR), Deficit Weighted Round Robin (DWRR) and Weighted Fair Queuing (WFQ) implement masking of the queues based on the current winning queue. The algorithm is applied only among the group

of queues that open simultaneously. To ensure Queues whose gates are "Open" get priority, these algorithms are modified to treat "gate open" queues and "gate closed" queues as separate groups giving priority to gate open queues.

For example, consider 4 queues (Q3, Q2, Q1, Q0) with weights 4:3:2:1; Q3 & Q2 are in Open state in slot one slot, while Q1 & Q0 are in Open state in another slot. In this case, the scheduler works as follows:

1. In the first slot, the Q3 & Q2 are serviced in the ratio of 4:3 for the duration the slot is open.
2. In the second slot, the queues Q1 & Q0 are serviced in the ratio of 2:1.
3. Fresh arbitration is started every time a slot is opened.

In other words, the traffic does not get distributed in the intended ratio of 4:3:2:1; but as two groups with different ratios and only for the duration of the slot when the gates are open continuously

NOTE

See "Programming Guidelines for EST"

1.9.13 Frame Preemption (FPE)

Frame preemption breaks the interfering frame into smaller fragments. Therefore, the guard band needs to be only as large as the largest possible interfering fragment instead of the largest possible interfering frame. During the guard band, only the frames that can complete the transmission of the entire frame before the next gate close event are permitted. This ensures that the high priority traffic can always start at the beginning of the window reserved for it.

Preemption allows one or more higher priority (express) frames to interrupt the transmission of a lower priority (preemptable) frame; the preemptable frame transmission is resumed and completed after the express frame transmission is complete. To support frame preemption, the following two abstractions of the MAC are used:

- A preemptable MAC, called pMAC, which carries the preemptable traffic.
- An express MAC, called eMAC, which carries the express traffic.

In the implementation, only parts of the MAC that holds the state during preemption is replicated and represented as pMAC and eMAC. The MAC uses the following two ways to puts on hold, the transmission of the preemptable traffic, in the presence of express traffic:

- The MTL scheduler interrupts the preemptable traffic that is currently being transmitted.

When the preemption capability is active, the MAC interrupts the transmission and reception of preemptable frames. A preempted fragment can be followed by zero or more express frames, before the continuation fragments. The preemptable frame can be fragmented any number of times, however, the minimum final and non-final fragment length criterion must be met.

However, interleaving of more than 1 preemptable packet is not permitted. This implies that if a preemptable packet is fragmented by an express packet, another preemptable packet cannot be transferred until all the remaining fragments of the first preempted packet are transferred.

- The MTL scheduler prevents starting the transmission of preemptable traffic.

When the preemption capability is inactive, the pMAC entity is disabled and only express traffic is transmitted or received.

Frame preemption feature can be enabled by setting EFPE field of the MAC_FPE_CTRL_STS register.

If you are unable to schedule fragmented packets after a certain number of attempts because of some reason, there is a possibility that the remaining fragmented packets are dropped. For this, software can program [MTL_EST_Control\[LCSE\]](#) to increase number of iterations or [MTL_EST_Control\[DFBS\]](#) to disable dropping of packet.

NOTE

Frame scheduling (EST) and frame preemption (FPE) cannot be used at the same time.

1.9.13.1 GCL Modification to Support FPE

In the EST-only configuration, the GCL entry has up to 24 bits of Time Interval and up to 8 high order bits representing the Gate Open/Close state requirements as shown in the following figure.

Gate Control (up to 8 bits)	Time Interval(ns) 16, 20 , or 24 bits
OCCCCCCC	T0
OOOCCCCC	T1
CCCCOOCC	T2
CCCCCCOO	T3
OCOCOCOO	T126
OOOCCCCC	T127

Figure 18. Gate Control List When FPE is Disabled

EST only supports SetGate operation, which implies that the gates are set to either open or close at a given time interval. However, when both Frame Preemption (FPE) and EST are enabled, the GCL also supports Set-and-Hold-MAC and Set-and-Release-MAC operations, in addition to the SetGate operation. To enable the support of hold and release operations, the format of the GCL is slightly changed while configuring and enabling the FPE. The first Queue (Q0) is always programmed to carry preemption traffic and therefore it is always Open. The bit corresponding to Q0 is renamed as Release/Hold MAC control. The TX Queues whose packets are preemptable are indicated by setting the PEC field of the MTL_FPE_CTRL_STS register. The GCL bit of the corresponding Queue are ignored and considered as always "Open". So, even if the software writes a "0" ("C"), it is ignored for such queues.

Gate Control (up to 7 bits)	Release/Hold Indication	Time Interval(ns) 16, 20, or 24 bits
CCCCOOO	0	T0
CCCCOOO	0	T1
OCCCCOO	1	T2
COCCCCO	1	T3
CCOCCCC	1	T4
CCCCOOO	0	T5
CCCCOOO	0	T6

Figure 19. Gate Control List When FPE is Enabled

When the Release/Hold bit transitions from a '0' to '1', a Set-and-Hold-MAC operation is performed. This marks the cease of the preemptable traffic. This is achieved by sending a Hold request to MTL "ha" ns in advance (where ha is the time interval mentioned in the Hold Advance (HADV) field of the MTL_FPE_Advance register). When the Release/Hold bit transitions from a '1' to '0' a Set-and-Release-MAC operation is performed. This marks the resuming of the preemptable traffic. This is achieved by sending a Release request to MTL "ra" ns in advance (where ra is the time interval mentioned in the Release Advance (RADV) field of the MTL_FPE_Advance register). The preemptable traffic is blocked for the time duration the Release/Hold bit is set to a '1' in the Gate Control List.

1.9.13.2 Impact of Preemption on CBS

In Credit Base Shaper(CBS), the definition of "Transmit" is as follows:

- TRUE for the duration of frame transmission from the queue.
- FALSE when frame transmission from the queue is complete.

When CBS algorithm is used along with frame preemption, the value of "Transmit" is TRUE only while the frame is being transmitted by the MAC. If the frame transmission is delayed or interrupted (for instance, a preemptable frame transmission is interrupted to allow the transmission of an express frame from a different queue, or the start of express frame is delayed because a preemptable frame is being transmitted) the value of "Transmit" is FALSE until transmission of the frame commences or is resumed.

Also, the value of "Transmit" is FALSE during the transmission of any overhead that is a consequence of frame preemption. For example, additional frame overhead (mCRC, Fragment Count) that is added to the preemptable frame.

At any given time, if there are no frames in the queue, and the value of Transmit is FALSE, and credit is positive value, the credit value is set to zero if there is no preemptable frame from the queue for which transmission is in progress but has been interrupted.

1.9.13.3 mPacket Format

When the preemption capability is active, MAC sends mPackets to the PHY. An mPacket can be one of the following:

1. A express packet
2. A preemptable packet
3. An initial fragment of a preemptable packet

4. A continuation fragment of a preemptable packet

Figure 8-15 shows the format of the mPacket. It contains an express packet, a complete preemptable packet or the initial fragment of a preemptable packet. Figure 3.3(b) shows the format of an mPacket containing a continuation fragment of a packet.

Figure 8-15 mPacket Formats

Preamble

The preamble in the mPacket shown in Figure 8-15 (a) contains seven octets. The preamble in the mPacket shown in Figure 8-15 (b) contains six octets. Each octet contains the value of 0x55 (transmitted in order from left to right 10101010).

Start mPacket Delimiter (SMD)

The value of the SMD indicates whether the mPacket contains an express packet, the start of a preemptable packet (initial fragment or complete packet), or any of continuation fragments of a preemptable packet. Table 8-15 shows the valid SMD values.

Table 26. Possible SMD Values of mPacket

mPacket Type	Notation	Frame Count	Value
verify packet	SMD-V	-	0x07
respond packet	SMD-R	-	0x19
express packet	SMD-E	-	0xD5
preemptable packet start	SMD-S0	0	0xE6
	SMD-S1	1	0x4C
	SMD-S2	2	0x7F
	SMD-S3	3	0xB3
continuation fragment	SMD-C0	0	0x61
	SMD-C1	1	0x52
	SMD-C2	2	0x9E
	SMD-C3	3	0x2A

frag_count

A frag_count is a modulo-4 counter that increments for each continuation fragment of the preemptable packet. The frag_count protects against mPacket reassembly errors by enabling detection of the loss of up to 3 packet fragments.

The frag_count field is present only in mPackets with SMD-C notation (continuation fragment). The frag_count is zero in the first continuation fragment of each preemptable packet.

Table 27. Possible frag_count Values

frag_count	Value
0	0xE6
1	0x4C

Table continues on the next page...

Table 27. Possible frag_count Values (continued)

frag_count	Value
2	0x7F
3	0xB3

mData

The contents of the packet from the MAC, starting with the first byte after the SFD to the last byte before the FCS are sent in the mData fields of one or more mPackets for that frame. The minimum size of the mData field is 60 bytes.

CRC The CRC field contains a cyclic redundancy check (CRC) and has an indication of the final mPacket of a frame. In the final mPacket of a frame, the CRC field contains the last 4 octets of the MAC frame (the FCS field).

For other mPackets, the CRC field contains an mCRC value. The mCRC is calculated on the octets of the packet from the first octet of the frame (the octet following the SFD of preemption frames) to the last octet of the packet transmitted in that mPacket by performing an XOR of the calculated 32 bit CRC value of the fragment and a value of 0x0000FFFF.

Summary of Packet Formats

- Express Packet: 7bytes of PREAMBLE, SMD-E, Data, and CRC
- Complete Preemptable Packet: 7 bytes of PREAMBLE, <Current Preemptable packet SMD>, Data, CRC
- Initial Fragment (non-final) of Preemptable Packet: 7 bytes of PREAMBLE, <Current Preemptable packet SMD>, Data, mCRC
- Continuation fragments (non-final) of Preemptable packet: 6 bytes of PREAMBLE, <Current Preemptable continuation fragment SMD>, <Current Preemptable continuation fragment FC>, Data, mCRC
- Final fragment of Preemptable packet: 6 bytes of PREAMBLE, <Current Preemptable continuation fragment SMD>, <Current Preemptable continuation fragment FC>, Data, CRC

Table 28. Current and Previous SMD Values

Previous Preemptable packet SMD	Current Preemptable packet SMD
SMD-S0	SMD-S1
SMD-S1	SMD-S2
SMD-S2	SMD-S3
SMD-S3	SMD-S0

Table 29. Current and Previous SMD Values

Previous Preemptable fragment SMD	Previous Preemptable fragment FC	Current Preemptable continuation fragment SMD	Current Preemptable continuation fragment FC
SMD-S0	NA	SMD-C0	FC0
SMD-S1	NA	SMD-C1	FC0
SMD-S2	NA	SMD-C2	FC0

Table continues on the next page...

Table 29. Current and Previous SMD Values (continued)

Previous Preemptable fragment SMD	Previous Preemptable fragment FC	Current Preemptable continuation fragment SMD	Current Preemptable continuation fragment FC
SMD-S3	NA	SMD-C3	FC0
SMD-C0	FC0	SMD-C0	FC1
SMD-C0	FC1	SMD-C0	FC2
SMD-C0	FC2	SMD-C0	FC3
SMD-C0	FC3	SMD-C0	FC0
SMD-C1	FC0	SMD-C1	FC1
SMD-C1	FC1	SMD-C1	FC2
SMD-C1	FC2	SMD-C1	FC3
SMD-C1	FC3	SMD-C1	FC0
SMD-C2	FC0	SMD-C2	FC1
SMD-C2	FC1	SMD-C2	FC2
SMD-C2	FC2	SMD-C2	FC3
SMD-C2	FC3	SMD-C2	FC0
SMD-C3	FC0	SMD-C3	FC1
SMD-C3	FC1	SMD-C3	FC2
SMD-C3	FC2	SMD-C3	FC3
SMD-C3	FC3	SMD-C3	FC0

1.9.13.4 Transmit Preemption

When FPE is enabled (setting EFPE=1 in MAC_FPE_CTRL_STS register), the MTL preempts a preemptable frame, when a "hold" request is asserted (EST/Qbv configured and enabled) express frames are available for transmission, that is, frame is present in MTL FIFO and is qualified for arbitration, after ensuring that the minimum mPacket mData field size is met. Therefore, preemption occurs only if at least 60 bytes of the preemptable frame have been transmitted and at least 64 bytes (including the frame CRC) remain to be transmitted.

The earliest starting position of preemption is controlled by the AFSZ field of the MTL_FPE_CTRL_STS Register. Preemption does not occur until at least $64 * (1 + \text{AFSZ}) - 4$ bytes of the preemptable frame have been sent.

When preemption occurs, all the preemptable queues are blocked and only the express queues are allowed to arbitrate (if more than one express queue has traffic) and transmit.

Continuation fragment of the preempted frame is the first frame to be transmitted after "release" request is asserted (EST/Qbv configured and enabled) and all the express traffic transmission completes.

Note: All the PTP packets should be transmitted as express packets

MTL communicates the following frame-type information to the MAC using a 2-bit preemption control signal on the MTI interface qualified by SoF and EoF.

- Express Frame
- Preemption Frame (Full or Fragment)
- Continuation Fragment (non-Final or Final)

Table 30. Preemption Control Values on MTI Interface for Various Frame Types

Qualifier	Preemption Control Value	Frame Type
SoF	00	Start Express
SoF	01	Start Preemption
SoF	10	Continuation Fragment
SoF	11	Reserved
EoF	00	End of Frame
EoF	01	End of Fragment

MTL should wait for the previous fragment status to be received before resuming the continuation fragments of a preempted frame. Fragment status is described in detail in the Tx Fragment Status section.

1.9.13.4.1 MAC Tx Preemption

MAC supports preemption by implementing the functionality needed to generate the mPackets as described in "mPacket Format".

Based on the Preemption Control value received on the MTI interface (qualified with SoF and EoF), MAC determines the frame type (shown in the above table 'Preemption Control Values on MTI Interface for Various Frame Types') and generates mPackets accordingly.

When the preemption capability is active, MAC replaces the SFD of a preemption packet with an SMD-S value. A 2-bit rolling frame count is encoded in the SMD-S value.

The SMD-E value is the same as the SFD value so the SFD of an express packet does not need to be replaced.

1.9.13.4.1.1 Tx Fragment Status

MAC sends Tx fragment status to indicate successful transmission of fragmented mPackets.

In case of a transmission error (underflow, jabber, and so on) the frame status is sent (and not a fragment status) with an error indication along with all other relevant status fields. In case of receiving an error status for a transmitted fragment, the MTL drops the remaining fragments and does not send any more continuation fragments.

1.9.13.5 Receive Preemption

1.9.13.5.1 MAC Receive Preemption

When FPE is enabled, the MAC Receiver passes the incoming packets and differentiates between Express packets and preemptable packets. An SMD containing an SMD-E indicates express packet, and SMD containing an SMD-S indicates the first mPacket of a preemptable packet.

If an mPacket containing an SMD-S is received when MAC has not completed receiving the previous preempted packet, MAC sets a CRC Error status for the previously received partial packet.

When an SMD-S is detected, MAC records the frame count indicated by the SMD and then begins sending data on the MRI interface.

The MAC checks the last four bytes of the mPacket. If the last four bytes of the mPacket do not match CRC, that indicates the end of the packet with or without a CRC error as per the CRC check result. If the last four octets of the mPacket match, that indicates that the packet was preempted.

An SMD containing an SMD-C indicates an mPacket that continues the data for a preempted packet. Upon receiving an SMD value of SMD-C, MAC checks the following:

1. A preempted packet is in progress
2. The frame count indicated by the SMD matches the frame count of the packet in progress
3. The frag_count value indicates the next fragment count.

If any of the above check fails, the mPacket is discarded and MAC sets a CRC Error status for the partially received packet.

If all the checks pass, the next fragment count is incremented modulo 4.

When a packet is preempted, the MAC saves the state of the partially received packet (filter check status, timestamp, length fields etc.) and will be able to process any received Express packets before the continuation fragment is received.

The MAC Receiver sends a "dummy status" (all zeros in mri_rxstatus_o) for all the mPacket fragments successfully received and sends the Rx status with the final fragment. If an error is detected during any of the fragments the Rx status is sent and the fragment is marked as final fragment. All subsequent continuation fragments received for this packet are dropped in the MAC.

The MAC communicates the following frame type information to the MTL using a 2-bit preemption control signal.

1. Express Frame
2. Preemption Frame (Full or Fragment)
3. Continuation Fragment (non-Final or Final)

Table 31. Preemption Control Values on MRI Interface for Various Frame Types

Qualifier	Preemption Control Value	Frame Type
SoF	00	Start Express
SoF	01	Start Preemption
SoF	10	Continuation Fragment
SoF	11	Reserved
EoF	00	End of Frame
EoF	01	End of Fragment

If the mri_sof_o and mri_eof_o are sent in the same cycle on the MRI interface, the mri_preemp_ctrl (Preemption Control Value) for EoF is taken as '0' indicating End of Frame.

1.9.13.5.1.1 Data Alignment

When a received frame cannot be fragmented on any byte boundary, MAC retains the unaligned bytes of data in the previous fragment and resends them with the next fragment as shown in Figure 8-16.

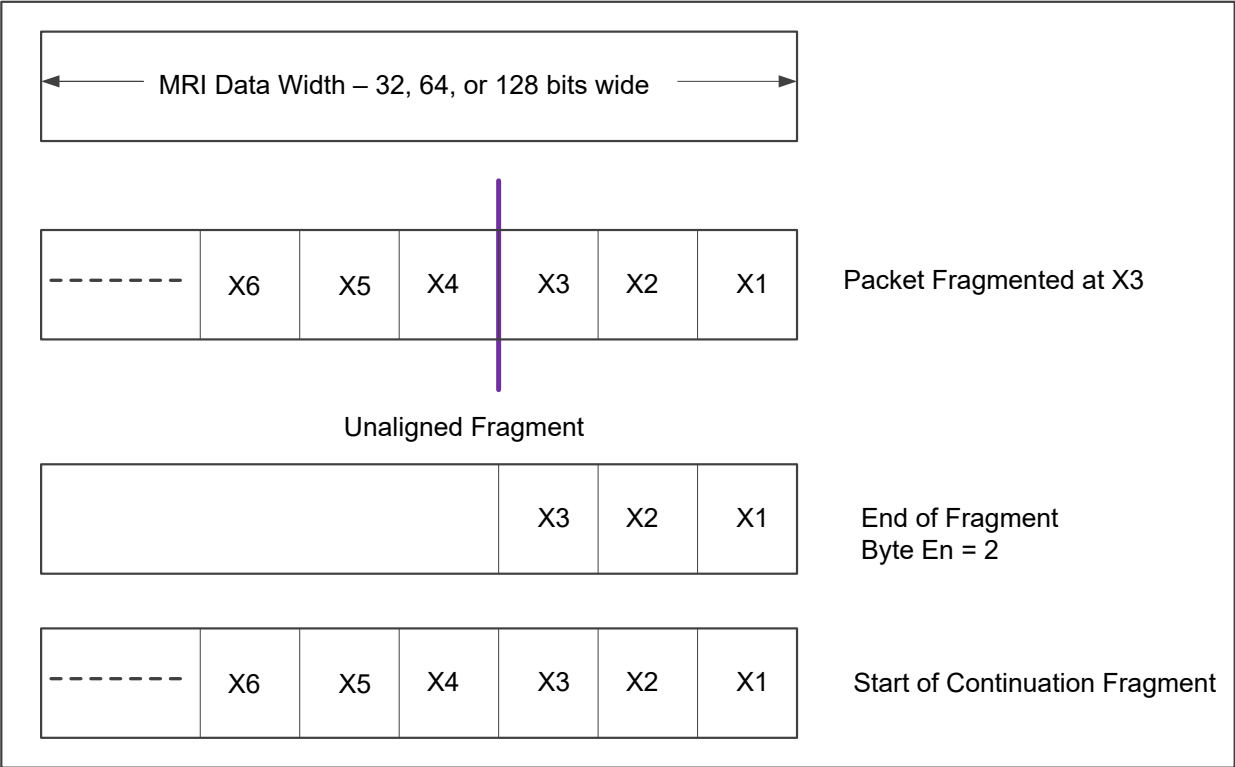


Figure 20. Data Alignment Feature of MAC

1.9.13.5.1.2 MTL Receive Preemption

The MTL Rx must have at least 2 Rx queues to support FPE function as the preemptable packets and the express packets must be routed to separate Rx queues. The destination Rx queue of a received packet is controlled by MAC_RxQ_Ctrl[n] registers. Program these registers such that Preemptable traffic and express traffic are not routed to the same Rx queue. As the queue mapping for tagged packets is based on VLAN user-priority field, this implies that priority of preemptable and express packets are mutually exclusive. In other words, packets of a certain priority (traffic class) are either express or preemptable but cannot be both.

For the preemptable frames, in addition to the PSRQ (Priority Selected in Rx queue) based queue routing, a programmable "Frame Preemption Residue Queue" (FPRQ) is supported to route all other Preemption Packets received (Untagged, SA/ DA or VLAN Filter Fails forwarded due to RA being set or VTFE being reset).

Table 32. Preemption Control Values on MRI Interface for Various Frame Types

Preemption Packet Type	Queue Routing
AV Tagged Packets passing the SA/DA and VLAN filters	PSRQ*
DCB/Generic Tagged Packets passing the SA/DA and VLAN filters	PSRQ**
Tagged Packets failing the SA/DA and VLAN filters without setting the Receive All (RA = 0) or setting VTFE = 1	Dropped
All other packets OR when RA = 1	FPRQ

1.9.13.5.1.3 MTL Receive Arbitration

On the ARI Interface, data is fetched from the MTL Rx FIFO based on the arbitration selected in the MTL_Operation_Mode register. Frame based arbitration can be used only when all the MTL Preemption Queues are operating in Store and Forward mode, else there will be loss of bandwidth on the ARI interface because all the fragments of a preemptable packet is not available. Therefore, any express packets received between the fragments are blocked until all the fragments are received and transferred; this defeats the purpose of express traffic.

When operating in either Threshold (cut through) or Store and forward modes of operation, PBL based arbitration is recommended over Frame based arbitration. In case of PBL based arbitration, the watermark check is always performed and the arbitration/ transfer of data in terms of chunks of "PBL" size of data which is almost similar to the concepts of "fragments". Therefore the express queue packets get blocked for less time as well as the ARI interface transfers the data without loss of efficiency.

1.9.13.6 Verify and Respond mPackets

When FPE function is present, the MAC can receive and detect the Verify and Respond mPackets, even when FPE is not enabled by software. When MAC detects valid Verify/Respond mPackets, it notifies the software by setting the RVER and RRSP fields of MAC_FPE_CTRL_STS register respectively. Optionally an interrupt can be generated. As such packets have empty (all-zero) data payload, they are dropped inside the MAC and not forwarded to the MRI.

Software can set the SVER and SRSP fields of MAC_FPE_CTRL_STS register to request MAC to transmit Verify and Respond mPackets respectively. Upon successful transmission of these frames, MAC clears the SVER/SRSP bits and sets the TVER & TRSP fields of MAC_FPE_CTRL_STS register. Optionally an interrupt can be generated when these events occur.

Software must ensure that frame preemption verify and response packet are not triggered at the same time. Trigger either of the packets and then wait for its completion before triggering another packet.

1.9.13.7 Frame Preemption and MMC Counter and Interrupt Registers

The following MMC counters and associated Interrupt registers are instantiated/present in the MAC when Frame Preemption feature is enabled.

Table 33. MMC Counters and Associated Interrupt Registers

Frame Assembly Error Counter	Description	Associated MMC Counter
Frame Assembly Error Counter	A 32-bit counter that provides the number of MAC frames with reassembly errors on the Receiver, due to mismatch in the Fragment Count value.	MMC_Rx_Packet_Assembly_Err_Cntr
Frame SMD Error Counter	A 32-bit counter that provides the number of received MAC frames rejected due to arriving with an SMD-C when there was no preceding preempted frame	MMC_Rx_Packet_SMD_Err_Cntr
Frame Assembly OK Counter	A 32-bit counter that provides the number of MAC frames that were successfully reassembled	MMC_Rx_Packet_Assembly_Ok_Cntr
MAC Rx Fragment Counter	A 32-bit counter that provides the number of additional mPackets received due to preemption	MMC_Rx_FPE_Fragment_Cntr
MAC Tx Fragment Counter	A 32-bit counter that provides the number of additional mPackets transmitted due to preemption	MMC_Tx_FPE_Fragment_Cntr
Hold Request Counter	A 32-bit counter that maintains the count of number of times a hold request is given to MAC	MMC_Tx_Hold_Req_Cntr

1.9.13.7.1 Additional Registers Associated With MMC Interrupts

Following are the additional registers associated with MMC interrupts for the MMC error counters:

- MMC_FPE_Tx_Interrupt
- MMC_FPE_Tx_Interrupt_Mask
- MMC_Rx_Interrupt
- MMC_FPE_Rx_Interrupt_Mask

1.9.14 Time-Based Scheduling

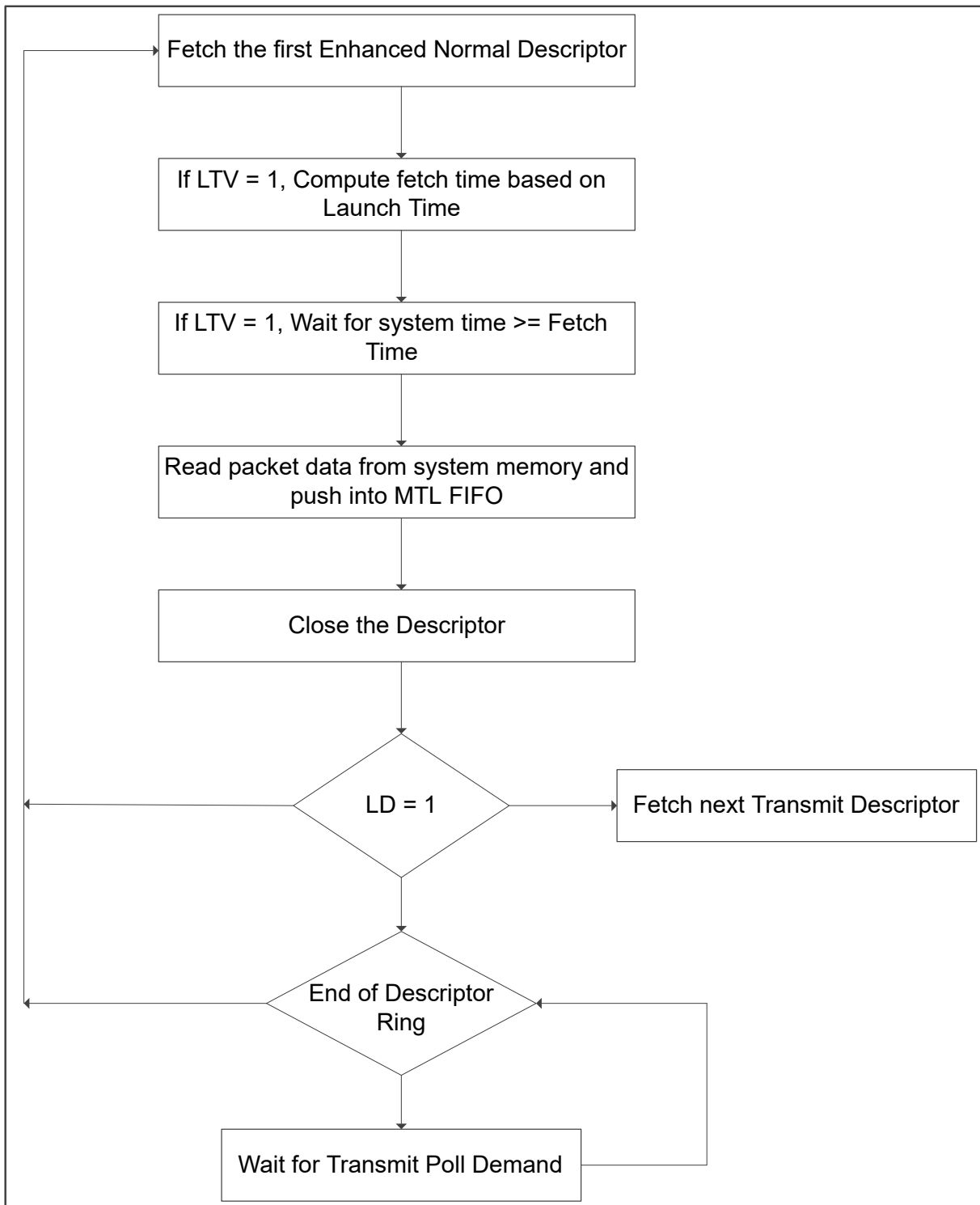
Time-based scheduling feature is suitable for traffic whose periodicity and rate are predictable. To improve the quality-of-service of such traffic,

- The transmit DMA fetches the packet from the host memory for transmission at designated time. This helps the software to setup the Transmit descriptors in advance even before packet is ready/available. It reduces the overhead on the software and avoids constant monitoring of the time and preparing descriptors just in time when the packet is targeted to be transmitted.
- The MAC transmits the packet only at the designated/pre-determined time even if the packets are fetched in advance. This helps in maintaining a constant transmission rate that can be consumed by the receiver station; therefore avoiding congestion and excessive buffering in the network.

The time-based scheduling feature supports fetching and launching an Ethernet packet at (or after) a pre-determined time. The time-based scheduling is supported only in the following modes/configurations:

- Full duplex mode
- Link speed is 100Mbps or higher

The following figure shows the time-based scheduling flow.



Following are the definitions specific to the time-based scheduling feature.

1.9.14.1 Launch Time

The time beyond which MTL can schedule the packet for transmission. The launch time is reset after the transmission of one frame. Following are the two formats of the launch time.

Normal/Absolute Format:

In this format, the Launch time is an absolute time value at which the packet is launched for transmission if Time-Based Scheduling feature is enabled. When both Time-Based scheduling and Enhancements to Scheduled Traffic, EST are enabled, then the launch time is interpreted in the Normal/Absolute format if the ESTM bit of the MTL_TBS_CTRL register is set to 0.

The launch time is a 32-bit value, where most-significant 8-bits represent the time in seconds and the rest 24 bits represent the time in 256 ns. The launch time is compared against the IEEE 1588 based System/PTP Time (bits[39:8]) and rolls over after 256 seconds.

The maximum value of the lower 32-bits of System Time is 999,999,999 decimal (0x3B9AC9FF) and it wraps to 0 when reaching this value (representing a full second). Therefore, the maximum value of the lower 24 bits of the Launch Time (after multiplying by 256) must be 0x3B9AC9.

As the maximum value of Launch time is 256 seconds,

- Launch Time is greater than current System Time when its value is between System Time[39:8] and System Time[39:8] + 128 sec
- Launch Time is less than current System Time when its value is between "System Time[39:8] + 128sec" and "System Time[39:8] + 256 sec", because this is a modulo 256 computation.

EST/Offset Format

In this format, the Launch time is a offset value relative to the time indicated by the Base Time Register (BTR) of the GCL list provided in the GCL Slot Number (GSN). The value in the BTR is always updated to the start time of the current loop of the GCL.

This format is enabled only when both Time-Based scheduling and Enhancements to Scheduled Traffic, EST features are enabled. For each packet, the GSN value and the Launch Time value are specified in the Enhanced Transmit Descriptor in the DMA configurations, or as Control Word in the MTL configurations.

The launch time offset is a 32-bit value; the upper 8-bits represent the time in seconds and the rest 24 bits represent the time in 256 ns, which is added to the BTR value corresponding to GCL Slot Number. The value of the Launch Time Offset must be smaller than the value of the Cycle Time (specified in the CTR register that is implemented when EST is enabled). If the CTR is greater than or equal to 1s, the maximum value of the lower-24 bits of Launch Time offset should be 999,999,999 decimal (0x3B9AC9FF).

In this format, the launch time is a 64-bit value, which is interpreted as, Launch Time = Launch GSN BTR[63:0] + (Launch Time Offset[31:0] << 8), which is compared with the System Time [63:0].

GSN BTR is the Base Time value at which the Launch GSN loop started execution.

GCL Slot Number (GSN) When Time-Based scheduling and Enhancements to Scheduled Traffic, EST features are enabled, a modulo 16 count of the GCL loop count is implemented and known as GSN. The count is incremented for every new GCL loop; installation of new GCL list does not impact the count. The current GCL slot number can be obtained by reading the CGSN field of the MTL_EST_Status register.

The maximum value of GSN is 15. Therefore GSN values between Current GSN (CGSN field of MTL_EST_Status register) and CGSN+8 represent current or future slots; all other GSN values are interpreted as elapsed slots or past slots. So, for the correct interpretation of time, GSN value must be between CGSN and CGSN+8, for correct interpretation of time.

1.9.14.2 Launch Expiry Time

Normal/Absolute Mode

In the normal mode, when the LEOV (Launch Expiry Offset Valid bit of MTL_TBS_CTRL register) is set, the Launch Expiry Offset (LEOS field of MTL_TBS_CTRL register) determines the maximum amount of time a frame is eligible for launch, starting from the time the frame becomes eligible for launch.

The Launch Expiry Offset is a 24-bit value defined in 256ns units, with a maximum possible value of 999,999,999 ns (0x3B9AC9FF).

Launch Expiry Time = (Launch Time[39:8] + LEOS[32:8]) * 256ns

The packet with a specific Launch time is considered eligible for transmission when the Launch time is less than the System Time and (if LEOV is set) the System Time is less than the Launch Expiry time.

When the System time is greater than the Launch Expiry time the frame is categorized as expired and will be dropped from the MTL FIFO.

Note: 1. For correct interpretation and meaningful operation, the Fetch, Launch, and Launch Expiry time should never be set to a value larger than Current System Time + 128 sec; such a value is interpreted as time that has already elapsed.

2. In full duplex mode, the frames dropped from the MTL FIFO have Error Summary (Bit 15) and Excessive Deferral (bit 3) of TxStatus set.

EST/Offset Mode

In the EST mode of operation, when the LEOV field of the MTL_TBS_CTRL register is set, the launch expiry GSN offset (LEGOS field of the MTL_TBS_CTRL register) and Launch Expiry Offset (LEOS field of MTL_TBS_CTRL register) determine the maximum amount of time a frame remains eligible for launch, starting from the time the frame becomes eligible for launch.

Launch Expiry Offset = (LEGOS: LEOS)

LEGOS holds the GSN offset (multiples of CTR time) and LEOS holds maximum value of CTR (sub CTR values) in ns.

Launch Expiry offset is a 24-bit value defined in the units of 256 ns, with a maximum possible value of the smaller of 999,999,999ns or CTR-1 ns.

The Launch Expiry GSN is computed as follows:

Launch Expiry GSN = (Launch GSN + LEGOS + CCMA) Where, CCMA is the CTR Carry due to Modulo addition. This value is 1 if ((Launch Time Offset + LEOS) << 8) is equal to or greater than CTR.

When CCMA = 1, Launch Expiry Offset = (Launch Time Offset+ LEOS) - CTR

When CCMA = 0,

Launch Expiry Offset = (Launch Time Offset + LEOS).

Launch Expiry Time = Launch Expiry GSN BTR[63:0] + Launch Expiry Time Offset.

When LEOV is not set, Launch Expiry Time is not checked.

When LEOV is set, and

- the system time is greater than the Launch Expiry time, the frame is dropped from MTL FIFO. The frame is considered as expired.
- the launch time is smaller than the System Time and Launch Expiry Time is greater than system time, the frame is considered eligible for launch.

NOTE

- Max value of LEGOS is 7. This implies that when LEOV is set, the frame has a maximum life time of <8 GCL loop iterations after it becomes eligible for Launch.
- The slot number of the First GCL List executed each time after EST is enabled, is zero.

1.9.14.3 Fetch Time

The fetch time accounts for all possible delays in the DMA fetch operation and ensures that the frame is present in the MTL FIFO before the launch time.

If FTOV field of DMA_TBS_CTRL register is set, then fetch time can be indicated for each packet. If the FTOV field is not set, the Fetch Time Offset is not valid and the DMA fetches packets without any time constraints.

Normal/Absolute Mode

In the Normal mode fetch time derived/calculated by reducing the time specified in the Fetch Time Offset (FTOS) field of DMA_TBS_CTRL register from the given Launch time.

Normal mode of operation, the Fetch Launch Time is computed as:

$$\text{Fetch Time}[39:8] = (\text{Launch Time}[39:8] - \text{FTOS}[31:8])$$

The Fetch time is 32-bits and is compared against System Time[39:8] to determine the Eligible for fetching the frame:

- The Fetch time is defined as "greater" than System Time if the Fetch Time is in the range of "System Time[39:8]" and "System Time[39:8] + 128 sec". The frame is considered as not-eligible for fetch.
- The Fetch time is defined as "smaller" than the System Time if the Fetch time is in the range of "System Time[39:8] + 128 sec" and "System Time[39:8] + 256 sec". The frame is considered as eligible for fetch.

This is a modulo 256 computation.

EST/Offset Mode

In the EST/Offset mode, the Fetch GSN Offset (FGOS field of MTL_TBS_CTRL register) provides the Slot Number offset to be deducted from the Launch GSN. In this case the FTOS value should be the smaller of 999,999,999ns or CTR-1 ns.

If (Launch Time Offset >= FTOS):

$$\text{Fetch Time Offset} = ((\text{Launch Time Offset} - \text{FTOS}) * 256\text{ns})$$

$$\text{CBFS(CTR Borrow for Fetch Subtraction)} = 0$$

$$\text{If (Launch Time Offset} < \text{FTOS) Fetch Time Offset} = \text{CTR} + ((\text{Launch Time Offset} - \text{FTOS}) * 256\text{ns})$$

$$\text{CBFS (CTR Borrow for Fetch Subtraction)} = 1$$

The Fetch GSN is computed as follows:

$$\text{Fetch GSN} = \text{Launch GSN} - \text{FGOS} - \text{CBFS}$$

$$\text{Fetch Time} = \text{Fetch GSN BTR}[63:0] + \text{Fetch Time Offset}$$

The frame is marked eligible for DMA fetch when the fetch time is smaller than the System Time.

DMA Operations Sequence

Following is the sequence of operation when FTOV = 1.

1. Fetch the first Enhanced Normal Descriptor. (FD is set).
2. If LTV is set and Fetch enabled, compute the Fetch Time based on the Launch Time. Wait for the System Time to be greater than Fetch Time
3. Read the Frame (Data) from the host memory and transfer to MTL FIFO
4. Close the Normal Descriptor
5. Fetch the next Normal Descriptor (if the previous Descriptor was not the Last)
6. Repeat steps 4 to 6, until the last descriptor of the frame (LD is set)

After the Last Descriptor of a frame, the next enhanced normal descriptor should be programmed with a new Launch Time and with LTV bit set. Otherwise, the subsequent frames will be processed without any time restrictions.

NOTE

See the "Programming the Launch Time in Time-Based Scheduling"

1.10 TCP/IP Offloading Features

This section and all its sub-sections are Synopsys Proprietary. Used with permission.

1.10.1 Transmit Checksum Offload Engine

The MAC has a Checksum Offload Engine (COE) to support checksum calculation and insertion in the Transmit path, using which, the software can offload the task of checksum insertion to the hardware. In the transmit path MAC calculates the checksum and inserts it in the Tx packet. This feature helps in reducing the load on the software and can improve the overall throughput of the system.

The checksum offload engine module supports two types of checksum calculation and insertion. The checksum engine can be controlled for each packet by setting the CIC bits (TDES3 Bits[17:16]).

NOTE

The checksum for TCP, UDP, or ICMP is calculated over a complete packet, and then inserted into its corresponding header field. Because of this requirement, when this function is enabled, the Tx FIFO automatically operates in the store-and-forward mode even if IP is configured for Threshold (cut-through) mode.

1.10.1.1 IP Header Checksum Engine

In IPv4 datagrams, the integrity of the header fields is indicated by the 16-bit Header Checksum field (the eleventh and twelfth bytes of the IPv4 datagram). The COE detects an IPv4 datagram when the Type field of Ethernet packet has the value 0x0800 and the Version field of IP datagram has the value 0x4. The checksum field of the input packet is ignored during calculation and replaced with the calculated value.

NOTE

IPv6 headers do not have a checksum field. Therefore, the COE does not modify the IPv6 header fields.

The result of this IP header checksum calculation is indicated by the IP Header Error status bit in the Transmit status (Bit 0 in Table 19-12 on page 1330). This status bit is set whenever the values of the Ethernet Type field and the Version field of IP header are not consistent, or when the Ethernet packet does not have enough data, as indicated by the IP header Length field. In other words, this bit is set when an IP header error is asserted under the following circumstances:

For IPv4 datagrams

- The received Ethernet type is 0x0800, but the Version field of IP header is not equal to 0x4.
- The IPv4 Header Length field indicates a value less than 0x5 (20 bytes).
- The total packet length is less than the value given in the IPv4 Header Length field.

For IPv6 datagrams

- The Ethernet type is 0x86dd but the IP header Version field is not equal to 0x6.
- The packet ends before the IPv6 header (40 bytes) or extension header (as given in the corresponding Header Length field in an extension header) is completely received.

1.10.1.2 TCP/UDP/ICMP Checksum Engine

The TCP/UDP/ICMP Checksum Engine processes the IPv4 or IPv6 header (including extension headers) and determines whether the encapsulated payload is TCP, UDP, or ICMP. The checksum is calculated for the TCP, UDP, or ICMP payload and inserted into its corresponding field in the header. The Tx COE can work in the following two modes:

- The TCP, UDP, or ICMPv6 pseudo-header is not included in the checksum calculation and is assumed to be present in the Checksum field of the input packet. This engine includes the Checksum field in the checksum calculation, and then replaces the Checksum field with the final calculated checksum.
- The engine ignores the Checksum field, includes the TCP, UDP, or ICMPv6 pseudo-header data into the checksum calculation, and overwrites the checksum field with the final calculated value.

NOTE

For ICMP-over-IPv4 packets, the Checksum field in the ICMP packet must always be 16'h0000 in both modes, because pseudo-headers are not defined for such packets. If it does not equal 16'h0000, an incorrect checksum may be inserted into the packet.

The result of this operation is indicated by the Payload Checksum Error status bit in the Transmit Status vector (Bit 12 in Table 19-3 on page 1324). This engine sets the Payload Checksum Error status bit when it detects that the packet has been forwarded to the MAC Transmitter engine in the store-and-forward mode without the end of packet (EOP) being written to the FIFO, or when the packet ends before the number of bytes indicated by the Payload Length field in the IP Header is received. When the packet is longer than the indicated payload length, the COE ignores them as stuff bytes, and no error is reported. When this engine detects the first type of error, it does not modify the TCP, UDP, or ICMP header. For the second error type, it still inserts the calculated checksum into the corresponding header field. Following table describes the functions supported by Transmit Checksum Offload engine based on the packet type. When the MAC does not insert the checksum, it is indicated as "No" in the table.

NOTE

You should not enable checksum insertion for IPv4 or IPv6 packets that are greater than the frame size constraint specified in Description of Transmit Checksum Offload Engine because it may result in incorrect checksum insertion or unexpected behavior.

Table 34. Transmit Checksum Offload Engine Functions for Different Packet Types

Packet Type	Hardware IP headerchecksum insertion	Hardware TCP/UDPchecksum insertion
Non-IPv4 or IPv6 packet	No	No
IPv4 packet is greater than 1,522 bytes (2,000 bytes when IEEE 802.3ad Support for 2K Packets is enabled in MAC) but less than or equal to the frame size constraint specified in Transmit Checksum Offload Engine .	Yes	Yes
IPv6 packet is greater than 1,522 bytes (2,000 bytes when IEEE 802.3ad Support for 2K Packets is enabled in MAC) but less than or equal to the frame size constraint specified in Transmit Checksum Offload Engine .	Not Applicable	Yes
IPv4 with TCP, UDP, or ICMP	Yes	Yes
IPv4 packet has IP options (IP header is longer than 20 bytes)	Yes	Yes
Packet is an IPv4 fragment	Yes	No
IPv6 packet with the following next header fields in main or extension headers: • Hop-by-hop options (in IPv6 main header) • Hop-by-hop options (in IPv6 extension header) • Destinations options • Routing (with segment left 0) • Routing (with segment left > 0) • TCP, UDP, or ICMP • Authentication • Any other next header field in main or extension headers	• Not Applicable • Not Applicable • Not Applicable • Not Applicable • Not Applicable • Not Applicable • Not Applicable • Not Applicable	• Yes • No • Yes • No • No • Yes • No • No
IPv4 packet has TCP header with Options fields	Yes	Yes

Table continues on the next page...

Table 34. Transmit Checksum Offload Engine Functions for Different Packet Types (continued)

Packet Type	Hardware IP headerchecksum insertion	Hardware TCP/UDPchecksum insertion
IPv4 Tunnels: - IPv4 packet in an IPv4 tunnel - IPv6 packet in an IPv4 tunnel	- Yes (IPv4 tunnel header) - Yes (IPv4 tunnel header)	- No - No
IPv6 Tunnels: - IPv4 packet in an IPv6 tunnel - IPv6 packet in an IPv6 tunnel	- Not applicable - Not applicable	- No - No
IPv4 packet has 802.3ac tag (with C-VLAN Tag or S-VLAN Tag when enabled).	Yes	Yes
IPv6 packet has 802.3ac tag (with C-VLAN Tag or S-VLAN Tag when enabled).	Not applicable	Yes
IPv4 frames with security features (such as encapsulated security payload)	Yes	No
IPv6 frames with security features (such as encapsulated security payload)	Not applicable	No

1.10.2 Receive Checksum Offload Engine

IP provides the Checksum Offload Engine that is used to detect any error in an IPv4 or IPv6 packet in the receive path. The MAC verifies the checksum field of the received packet with the internally calculated checksum and provides the status.

The Receive Checksum Offload engine is used to detect errors in IP packets by calculating the header checksum and further matching it with the received header checksum. This engine also identifies a TCP, UDP, or ICMP payload in received IP packets and calculates the checksum of such payloads appropriately.

Here, both IPv4 and IPv6 packet in the received Ethernet packets are detected and processed for data integrity. The MAC receiver identifies IPv4 or IPv6 packets by checking for value 0x0800 or 0x86DD, respectively, in the Type field of the received Ethernet packet. This identification is applicable to single VLAN-tagged packets. It is also applicable to double VLAN-tagged packets when the Enable Double VLAN Processing option is selected and the EDVLP bit of the MAC_VLAN_Tag register is set.

The Rx COE calculates the IPv4 header checksums and checks that they match the received IPv4 header checksums. The result of this operation (pass or fail) is given to the RFC module for insertion into the receive status word. The IP Header Error bit is set for any mismatch between the indicated payload type (Ethernet Type field) and the IP header version, or when the received packet does not have enough bytes, as indicated by the Length field of the IPv4 header (or when fewer than 20 bytes are available in an IPv4 or IPv6 header).

This engine also identifies a TCP, UDP, or ICMP payload in the received IP datagrams (IPv4 or IPv6) and calculates the checksum of such payloads properly, as defined in the TCP, UDP, or ICMP specifications. This engine includes the TCP, UDP, or ICMPv6 pseudo-header bytes for checksum calculation and checks whether the received checksum field matches the calculated value. The result of this operation is given as a Payload Checksum Error bit in the receive status word. This status bit is also set if the length of the TCP, UDP, or ICMP payload does not match the expected payload length given in the IP header.

Following table describes the functions supported by the Rx COE based on the packet type. When the payload of an IP packet is not processed (indicated as "No" in the table), the information (whether the checksum engine is bypassed or not) is given in the receive status.

NOTE

The MAC does not append any payload checksum bytes to the received Ethernet packets.

Table 35. Receive Checksum Offload Engine Functions for Different Packet Types

Packet type	Hardware IP header checksum checking	Hardware TCP/UDP/ICMP checksum checking
Non-IPv4 or IPv6	No	No
IPv4 packet is greater than 1,522 bytes (2,000 bytes when IEEE 802.3ad Support for 2K Packets is enabled in the MAC)	Yes	Yes
IPv6 packet is greater than 1,522 bytes (2,000 bytes when IEEE 802.3ad Support for 2K Packets is enabled in the MAC)	Not Applicable	Yes
IPv4 with TCP, UDP, or ICMP	Yes	Yes
IPv4 header's protocol field contains a protocol other than TCP, UDP, or ICMP	Yes	No
IPv4 packet has IP options (IP header is longer than 20 bytes)	Yes	Yes
Packet is an IPv4 fragment	Yes	No
IPv6 packet with the following next header fields in main or extension headers: • Hop-by-hop options (in IPv6 main header) • Hop-by-hop options (in IPv6 extension header) • Destinations options • Routing (with segment left 0) • Routing (with segment left > 0) • TCP, UDP, or ICMP • Any other next header field in main or extension headers	• Not Applicable • Not Applicable • Not Applicable • Not Applicable • Not Applicable • Not Applicable • Not Applicable	• Yes • No • Yes • Yes • No • Yes • No
IPv4 packet has TCP header with Options fields	Yes	Yes
IPv4 Tunnels: • IPv4 packet in an IPv4 tunnel	• Yes (IPv4 tunnel header) • Yes (IPv4 tunnel header)	• No • No

Table continues on the next page...

Table 35. Receive Checksum Offload Engine Functions for Different Packet Types (continued)

Packet type	Hardware IP header checksum checking	Hardware TCP/UDP/ICMP checksum checking
IPv6 packet in an IPv4 tunnel		
IPv4 Tunnels: - IPv4 packet in an IPv6 tunnel - IPv6 packet in an IPv6 tunnel	- Not Applicable - Not Applicable	- No - No
IPv4 packet has 802.3ac tag (with C-VLAN Tag or S-VLAN Tag when enabled)	Yes	Yes
IPv6 packet has 802.3ac tag (with C-VLAN Tag or S-VLAN Tag when enabled)	Not Applicable	Yes
IPv4 frames with security features (such as encapsulated security payload)	Yes	No
IPv6 frames with security features (such as encapsulated security payload)	Not Applicable	No

Set the IPC bit of the MAC_Configuration register for enabling Receive Checksum offload.

1.11 MAC Management Counters

This section and all its sub-sections are Synopsys Proprietary. Used with permission.

IP supports storing the statistics about the received and transmitted packets in registers that are accessible through the application/software. The MMC module maintains a set of registers for gathering statistics on the received and transmitted packets. The MMC counters are free running.

For MMC register details, see registers "MMC_Control" to "MMC_Rx_FPE_Fragment_Cntr" in "Register Descriptions" .

1.12 Flow Control

This section and all its sub-sections are Synopsys Proprietary. Used with permission.

1.12.1 Transmit Flow Control

The Transmit Flow Control involves transmitting Pause packets in full-duplex mode and backpressure in half-duplex mode to control the flow of packets from the remote end.

1.12.1.1 Flow control in Full duplex mode

IP supports the IEEE802.3x Pause Packets.

Pause packet structure is shown below in the following table.

Table 36. Pause Packet Fields

Field	Description
DA	Contains the special multicast address
SA	Contains the MAC address 0
Type	Contains 8808
MAC Control opcode	Contains 0001 for IEEE 802.3x Pause Control packets; 0101 for PFC packets
PT	Contains Pause time specified in the PT field of the MAC_Q#_Tx_Flow_Ctrl register

1.12.1.2 Pause Packet Control

When the FCB bit is set, the MAC generates and transmits a single Pause packet. If the FCB bit is set again after the Pause packet transmission is complete, the MAC sends another Pause packet irrespective of whether the pause time is complete or not. To extend the pause or terminate the pause prior to the time specified in the previously-transmitted Pause packet, the application should program the Pause Time register with appropriate value and then again set the FCB bit.

1.12.1.3 Flow Control in Half-Duplex Mode

In half-duplex mode, the MAC uses the deferral mechanism for the flow control (backpressure). When the application requests to stop receiving packets, the MAC sends a JAM pattern of 32 bytes when it senses a packet reception, provided the transmit flow control is enabled. This results in a collision and the remote station backs off. If the application requests a packet to be transmitted, it is scheduled and transmitted even when the backpressure is activated. If the backpressure is kept activated for a long time (and more than 16 consecutive collision events occur), the remote stations abort the transmission because of excessive collisions.

Following table describes the flow control in the Tx path for Queue 0 based on the setting of the following bits:

- EHFC bit of MTL_RxQ0_Operation_Mode register
- TFE bit of MAC_Q0_Tx_Flow_Ctrl register
- DM bit of MAC_Configuration register

Flow control is similar for all queues.

Table 37. Tx MAC Flow Control

EHF C	TFE	DM	Description
x	0	x	The MAC transmitter does not perform the flow control or backpressure operation.
0	1	0	The MAC transmitter performs back-pressure when Bit 0 of MAC_Q0_Tx_Flow_Ctrl register is set or the sideband signal sbd_flowctrl_i is 1.
1	1	0	The MAC transmitter performs back-pressure when Bit 0 of MAC_Q0_Tx_Flow_Ctrl register is set or the sideband signal sbd_flowctrl_i is 1. In addition, the MAC Tx performs back-pressure when Rx Queue level crosses the threshold set by Bits[10:8] of MTL_RxQ0_Operation_Mode register.
0	1	1	The MAC transmitter sends the Pause packet when Bit 0 of MAC_Q0_Tx_Flow_Ctrl register is set or the sideband signal sbd_flowctrl_i is 1.

Table continues on the next page...

Table 37. Tx MAC Flow Control (continued)

EHF C	TFE	DM	Description
1	1	1	The MAC transmitter sends the Pause packet when Bit 0 of MAC_Q0_Tx_Flow_Ctrl register is set or the sideband signal sbd_flowctrl_i is 1. In addition, the MAC Tx sends a Pause packet when Rx Queue level crosses the threshold set by Bits[10:8] of MTL_RxQ0_Operation_Mode register.

1.12.1.4 Enabling Transmit Flow Control

To independently enable Transmit Flow Control for each Tx queue, set the TFE bit in the MAC_Q#_Tx_Flow_Ctrl register.

1.12.2 Receive Flow Control

In the Receive path, the Flow Control is functional only in the full-duplex mode. If any Pause packet is received in the half-duplex mode, the packet is considered as a normal control packet.

NOTE

Receive pause packets should have a frame size of 64 bytes.

The Receive Flow Control is implemented by the MAC based on the bit value of the respective register, and the destination address and different fields of the received packet.

Following table describes the flow control in the Rx path based on the setting of the following bits:

- RFE bit of MAC_Rx_Flow_Ctrl register
- DM bit of MAC_Configuration register

Table 38. Rx MAC Flow Control

RFE	DM	Description
0	x	The MAC receiver does not detect the received Pause packets.
1	0	The MAC receiver does not detect the received Pause packets but recognizes such packets as Control packets.
1	1	The MAC receiver detects or processes the Pause packets and responds to such packets by stopping the MAC transmitter.

1.12.2.1 Enabling Receive Flow Control

To enable Pause Flow Control, set the RFE bit in the MAC_Rx_Flow_Ctrl register.

1.13 Loopback Mode

This section and all its sub-sections are Synopsys Proprietary. Used with permission.

The MAC supports Loopback of transmitted packets to its receiver.

The following are some guidelines for using the loopback mode:

- Enable loopback only with the full-duplex mode. In half-duplex mode, the carrier sense signal (crs) or collision (col) signal inputs get sampled which may result into issues such as packet dropping.
- If the loopback mode is enabled without connecting a PHY chip (for example, in FPGA setup), you should externally generate the Tx and Rx clocks and provide these clocks to the MAC.

- Do not loop back big packets. Big packets (>1522 bytes) may get corrupted in the loopback FIFO.

To enable this feature, program the LM bit of the MAC_Configuration register.

You can enable loopback for all PHY interfaces. The data is always looped back through internal asynchronous FIFO on to the internal Receive MII or GMII interface, irrespective of which PHY interface is selected. The loopback data is also passed through the corresponding transmit PHY interface block, on to the Ethernet line.

1.14 Automotive Safety Features

This section and all its sub-sections are Synopsys Proprietary. Used with permission.

IP supports the Automotive safety feature.

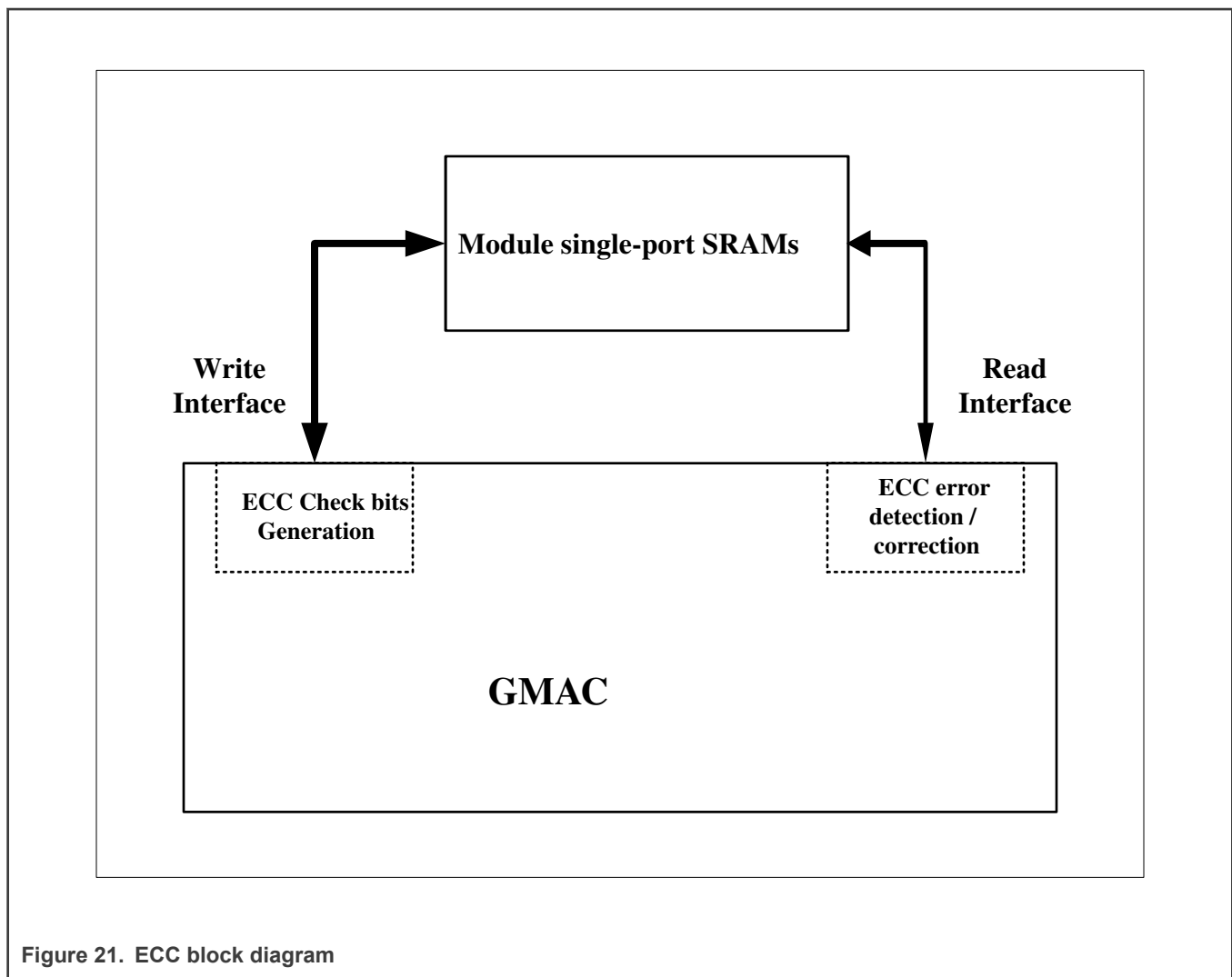
1.14.1 Error Correction Code (ECC) Protection for Memories

The Error Correction Code (ECC) block can correct single-bit error and detect double-bit error.

At the write interface, ECC checkbits are generated by computing ECC on the contents of the data bus and respective address is appended with the data that is written to the memory.

At the read interface, ECC checkbits are recomputed on the content of the read data and the respective address is compared with the received checkbits in the memory.

Following figure shows the block diagram of ECC protection for memories



1.14.1.1 Handling the Address Mismatch

The ECC is calculated over the sum of memory data and memory location address. However only data is written in the memory along with the ECC. (address is excluded).

On the read interface, the ECC is checked over the sum of memory DATA, memory ECC, and Internally generated address. When an ECC correctable error is detected over the range of the address bit position, it is treated as an uncorrectable error. This is because data might have been read from a different location.

1.14.1.2 Diagnostic Support for the Error Management

Following statistics are provided for monitoring the error behavior on each of the memory blocks.

- Error status provided to management
 - A separate status for correctable, uncorrectable, and address mismatch is specified in the MTL_EC-C_Interrupt_Status register.
 - Memory locations at which correctable and uncorrectable errors are detected is specified in the MTL_ECC_Err_Addr_Status register. In addition, a control bit (MEEAO field of MTL_ECC_Control register) decides, whether the first errored memory address is reported or the latest errored memory address is reported.
 - MTL_ECC_Err_Cntr_Status register has separate counters to count the number of correctable and uncorrectable errors
- Interrupts provided to management
 - Separate interrupts are generated for correctable and uncorrectable errors.
 - sbd_sfty_ue_intr_o interrupt is generated when uncorrectable errors are detected and these errors cannot be masked.
 - sbd_sfty_ce_intr_o interrupt is generated when correctable errors are detected. GMAC sets the respective interrupt enable bits (in the DMA/MTL_ECC_Interrupt_enable register).
 - To find the root cause of the error, read the DMA/MTL_Safety_Interrupt_Status and DMA/MTL_ECC_Interrupt_Status registers.
 - To clear the interrupt, write 1 in the respective interrupt status bit in DMA/MTL_ECC_Interrupt_Status registers.

NOTE

- The status/counters/interrupts are generated per memory.
- MTL Tx and Rx memory is logically divided into multiple queues. But, ECC diagnostics (status/counter/interrupts) are common to all queues.

1.14.1.3 ECC Error Injection Capabilities

IP supports error injection capabilities for each memory as a static configuration. The position where the errors are injected in the data word is random. For each memory, 3 control bits are provided to inject errors. The 3 bits are:

- A single bit to enable error injection
- Two bits to indicate the type of error to be injected
 - 00: 1 bit error
 - 01: 2 bit error
 - 10: 3 bit error
 - 11: 1 bit error in address field
- The control bit descriptions for
 - MTL Tx/Rx and DMA TSO memory are specified in the MTL_DBG_CTL register, for and for R.

- MTL EST memory are specified in the MTL_EST_GCL_Control register
- Rx parser memory are specified in the MTL_RXP_Indirect_Acc_Control_Status register

NOTE

While using debug mode for ECC error injection,

- There should be no traffic in GMAC
- When multiple CSR writes are required for writing single data word into the memory, the application should ensure that all the CSR writes corresponding to one memory write maintains the same value for the error injection control word.

NOTE

See "Programming Guidelines for ECC Protection for Memories"

1.14.2 FSM Parity and Timeout Protection

The FSM protection feature supports FSM parity and time out protection.

1.14.2.1 FSM State Parity Protection

Odd parity is implemented on all the FSM state register bits. Parity is monitored for every clock, after the reset is de-asserted. When a bit flips due to transient errors or permanent faults, the erroneous FSM is set to its default state and an uncorrectable error is indicated.

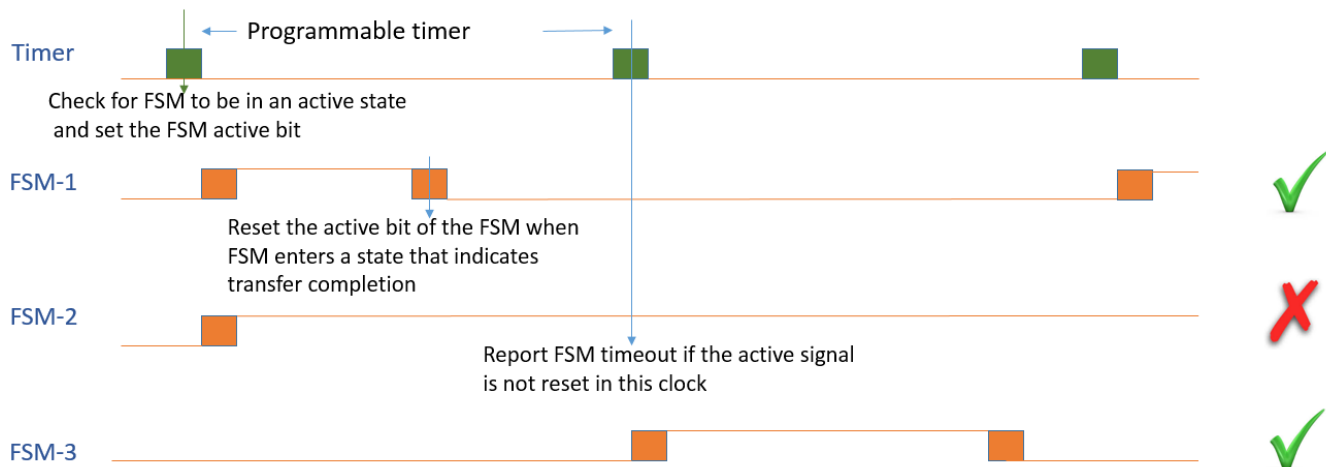
When the FSM state parity protection is enabled, by setting PRTYEN field of MAC_FSM_Control Register, the FSM state error is indicated by setting the FSMPEs field of the MAC_DPP_FSM_Interrupt_Status register. Also, the Safety Interrupt (sbd_sfty_ue_intr_o) is asserted when an FSM Error is detected.

Error Injection mode is also supported for FSM parity error check. Program the Error Injection enable for the respective clock domain denoted by [23:16] bits of MAC_FSM_Control Register.

1.14.2.2 FSM Timeout Protection

The FSM Timeout feature provides a mechanism to ensure that all FSMs in GMAC can complete a transaction and return to a known completion state (IDLE or any other state that indicates completion of a transaction/transfer) within the programmed timeout value.

Program the TMR field of MAC_FSM_ACT_Timer register with a value that indicates the number of CSR cycles required to generate a 1 microsecond tic. This microsecond tic is internally used to generate the programmed timeout duration. Two timeout tics (normal mode timeout and large mode timeouts) are generated to provide flexibility to choose one per clock domain, by using the bits [31:24] of MAC_FSM_Control Register. Supported values for Timeout duration are 1us, 1.024ms, 16.384ms, 65.5536ms, 262.144ms, 1048.576ms (~1sec), 4194.304ms (~4s), 8.388s, 16.777s, and 33.554s which is based on the programmable bits NTMRMD, LTMRMD fields of MAC_FSM_ACT_Timer register. Interface timeouts are based on the normal mode tic generation; only the FSM timeouts depend on either normal or large tick selection.



In the above figure, the timer ticks are generated based on the programmable timeout value. The timer ticks are synchronized and made available in all the clock domains that have the FSMs.

At every timer tick, all the FSMs are monitored for being in active state (non-IDLE state, which indicates the FSM is actively processing transactions or hand-shakes) and an FSM active flag bit is set. The active flag bit is implemented for every FSM that is monitored for timeout. When the FSM reaches an IDLE or transaction completion state, the active flag bit of that FSM is reset. At the subsequent timer tick, all the FSMs' active flag bits are monitored and any flag that is set indicates a timeout for that FSM. This process of setting the flag and checking at subsequent timer tick is repeated at every tick.

Timeout Error Injection per clock domain is enabled by programming [15:8] fields of MAC_FSM_Control register. When Timeout error injection enable is set for a clock domain, the FSMs in that clock domain automatically timeout and generate an interrupt even without traffic. As error injection is a debug mode, TMR value need not indicate 1us, but can be programmed to a smaller value at which debug mode ticks are needed. FSM Timeout and Parity Error Injection modes can be used for testing at key-on/key-off.

The FSM timeout status is specified in the [15:8] field of MAC_DPP_FSM_Interrupt_Status register as per the respective clock domains. One FSM timeout status bit is made available per clock domain. The safety interrupt (sbd_sfty_ue_intr_o) is asserted when the timeout error is set for any of the clock domains. IP does not attempt to recover from a FSM timeout condition and relies on the application to take the corrective action (resetting IP).

1.14.2.3 Enabling FSM Parity and Timeout Protection

FSM timeout feature is enabled by setting the TMOUTEN field of MAC_FSM_Control register.

NOTE

See "Programming Guidelines for FSM Parity and Timeout"

1.14.3 Application/CSR Interface Timeout Protection

This feature provides timeout protection to the application/CSR Interface. All the interfaces which has the handshake mechanism monitored for potential hangs due to the external agent (Master/Slave/Interconnect/Application client) not responding to the requests/transfers initiated by the QoS. After the request is initiated the response arrival interval is monitored. If the response does not arrive within a programmed time (TMR field of MAC_FSM_ACT_Timer) the timeout is triggered, using similar trigger generation as explained in the Section FSM protection.

The timeout is triggered when IP initiates SEQ, NON-SEQ and BUSY transfers on HTRANS[1:0]; and HREADY is not asserted by Slave within the programmed time.

The Timeout status for the AHB master interface is set in the MSTTES field of the MAC_DPP_FSM_Interrupt_Status register. The Safety interrupt (sbd_sfty_ue_intr_o) is asserted when application timeout status is set.

The hardware does not attempt to recover from Application/CSR Interface hangs and relies on software to take appropriate corrective action.

NOTE

1. Protection is not needed for AHB Slave interfaces because potential hangs can be only when IP does not respond to the requests. In such cases the IP internal protection logic (FSM timeout) detects such defects.
2. Based on the time at which the transfer is initiated relative to the timer ticks, the timeout period could be any value between the programmed timeout and 2x times the programmed timeout.
3. When an uncorrectable safety interrupt is issued and the read of the Interrupt status register returns all zeros, it implies that the CSR read is not functional. As soft reset of IP is not possible without the CSR access, and therefore, a hard reset of IP is recommended.

NOTE

See "Programming Guidelines for FSM Parity and Timeout" for details.

1.15 Descriptors

1.15.1 Overview

The DMA in the Ethernet subsystem transfers data based on a linked list of descriptors. The application creates the descriptors in the system memory. The module supports the following two types of descriptors:

- Normal Descriptor: Normal descriptors are used for packet data and to provide control information applicable to the packets to be transmitted.
- Context Descriptor: Context descriptors are used to provide control information applicable to the packet to be transmitted.

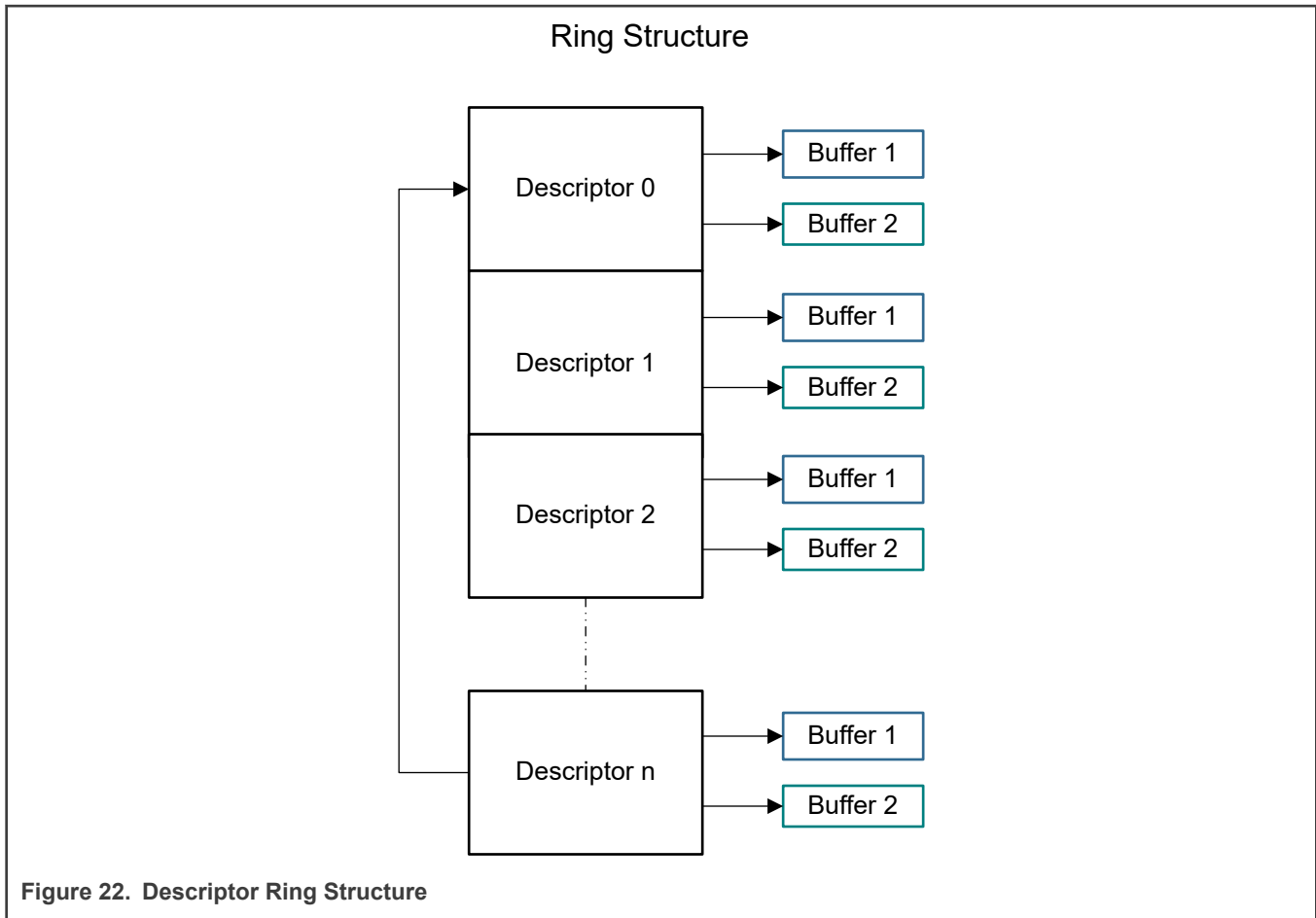
Each normal descriptor contains two buffers and two address pointers. These buffers enable the adapter port to be compatible with various types of memory management schemes.

NOTE

There is no limit for the number of descriptors that can be used for a single packet.

1.15.2 Descriptor Structure

The module supports the ring structure for DMA descriptor as shown in the following figure.



In Ring structure, descriptors are separated by the Word, DWord, or LWord number programmed in the DSL field of the DMA_CH#_Control register. The application needs to program the total ring length, that is, the total number of descriptors in ring span in the following registers of a DMA channel:

- Transmit Descriptor Ring Length Register (DMA_CH#_TxDesc_Ring_Length)
- Receive Descriptor Ring Length Register (DMA_CH#_RxDesc_Ring_Length)

The Descriptor Tail Pointer Register contains the pointer to the descriptor address (N). The base address and the current descriptor pointer decide the address of the current descriptor that the DMA can process. The descriptors up to one location less than the one indicated by the descriptor tail pointer ($N - 1$) are owned by the DMA. The DMA continues to process the descriptors until the following condition occurs:

Current Descriptor Pointer == Descriptor Tail Pointer;

The DMA goes into the Suspend mode when this condition occurs. The application must perform a write to the Descriptor Tail pointer register and update the tail pointer so that the following condition is true:

Current Descriptor Pointer < Descriptor Tail Pointer;

The DMA automatically wraps around the base address when the end of ring is reached, as shown in the following figure.

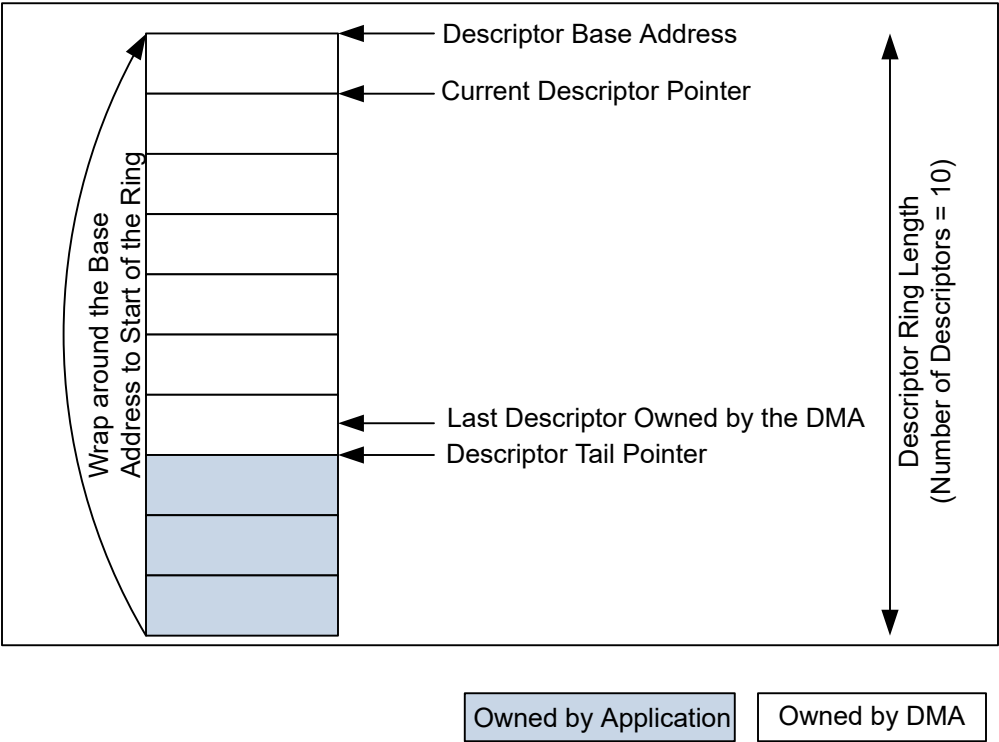


Figure 23. DMA Descriptor Ring

For descriptors owned by the application, the OWN bit of DES3 is reset to 0. For descriptors owned by the DMA, the OWN bit is set to 1. If the application has only one descriptor in the beginning, the application sets the last descriptor address (tail pointer) to Descriptor Base Address + 1. The DMA processes the first descriptor and then waits for the application to advance the tail pointer.

1.15.3 Transmit Descriptor

The DMA in the module requires at least one descriptor for a transmit packet. In addition to two buffers, two byte-count buffers, and two address pointers, the transmit descriptor has control fields which can be used to control the MAC operation on per-transmit packet basis. The Transmit Normal descriptor has two formats: Read format and Write-Back format

NOTE

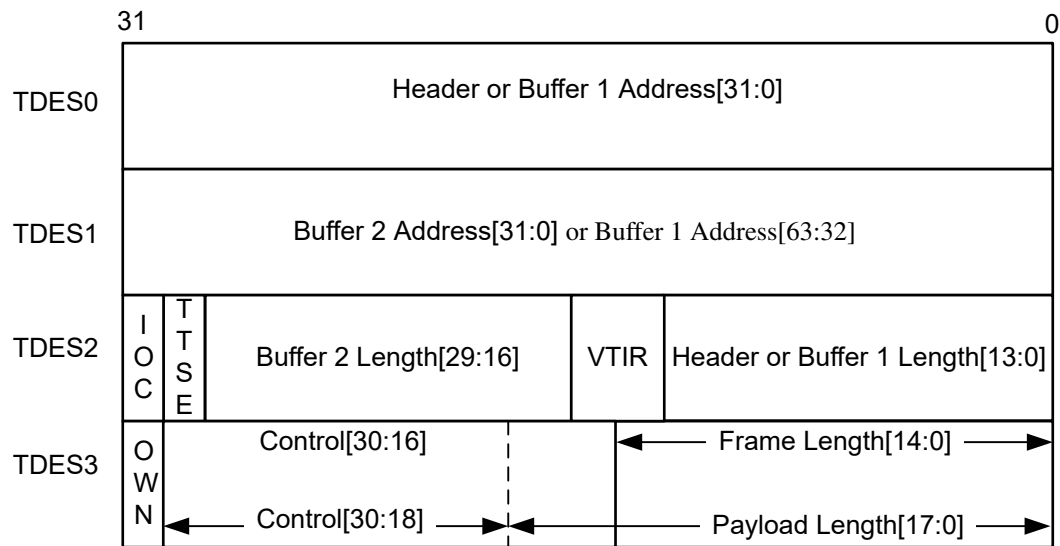
TSO Split header is not supported.

Software must not write same tail pointer addresses when DMA is in Suspended state before setting new descriptor(s) for further processing.

1.15.3.1 Transmit Normal Descriptor (Read Format)

Following figure shows the Read Format for a Transmit normal descriptor. Table: 'TDES0 Normal Descriptor (Read Format)' through Table: 'TDES3 Normal Descriptor (Read Format)' describe the read format for the Transmit Normal Descriptors: TDES0, TDES1, TDES2, and TDES3.

Figure 24. Transmit Descriptor Read Format



1.15.3.1.1 TDES0 Normal Descriptor (Read Format)

Table 39. TDES0 Normal Descriptor (Read Format)

Bit	Name	Description
31:0	BUF1AP	Buffer 1 Address Pointer or TSO Header Address Pointer These bits indicate the physical address of Buffer 1. These bits indicate the TSO Header Address pointer when the following bits are set: • TSE bit of TDES3 • FD bit of TDES3

1.15.3.1.2 TDES1 Normal Descriptor (Read Format)

Table 40. TDES1 Normal Descriptor (Read Format)

Bit	Name	Description
31:0	BUF2AP	Buffer 2 or Buffer 1 Address Pointer This bit indicates the physical address of Buffer 2 when a descriptor ring structure is used. There is no limitation for the buffer address alignment. In 40- or 48-bit addressing mode, these bits indicate the most-significant 8- or 16- bits of the Buffer 1 Address Pointer.

1.15.3.1.3 TDES2 Normal Descriptor (Read Format)

Table 41. TDES2 Normal Descriptor (Read Format)

31	30	29-16	15:14	13:0
IOC	TTSE/ TMWD	B2L	VTIR	HL or B1L

Table 42. TDES2 Normal Descriptor (Read Format)

Bits	Name	Description
31	IOC	Interrupt on Completion This bit controls the setting of TI and ETI status bits in the DMA_CH#_Status register. When ETIC = 1 and TDES2[LD] = 0, this bit sets the ETI bit. When TDES3[LD] = 1, this bit sets the TI status bit.
30	TTSE/ TMWD	Transmit Timestamp Enable or External TSO Memory Write Enable This bit enables the IEEE1588 time stamping for Transmit packet referenced by the descriptor, if TSE bit is not set. If TSE bit is set and external TSO memory is enabled, setting this bit disables external TSO memory writing for this packet.
29:16	B2L	Buffer 2 Length The driver sets this field. When set, this field indicates Buffer 2 length.
15:14	VTIR	VLAN Tag Insertion or Replacement These bits request the MAC to perform VLAN tagging or untagging before transmitting the packets. The application must set the CRC Pad Control bits appropriately when VLAN Tag Insertion, Replacement, or Deletion is enabled for the packet. The following list describes the values of these bits: • 2'b00: Do not add a VLAN tag. • 2'b01: Remove the VLAN tag from the packets before transmission. This option should be used only with the VLAN packets. • 2'b10: Insert a VLAN tag with the tag value programmed in the MAC_VLAN_Incl register or context descriptor. • 2'b11: Replace the VLAN tag in packets with the tag value programmed in the MAC_VLAN_Incl register or context descriptor. This option should be used only with the VLAN packets. These bits are valid when the Enable SA and VLAN Insertion on Tx option is selected while configuring the core.
13:0	HL or B1L	Header Length or Buffer 1 Length For Header length only bits [9:0] are taken. The size 13:0 is applicable only when interpreting buffer 1 length. If the TCP Segmentation Offload feature is enabled through the TSE bit of TDES3, this field is equal to the header length. When the TSE bit is set in TDES3, the header length includes the length in bytes from Ethernet Source address till the end of the TCP header. The maximum header length supported for TSO feature is 1023 bytes. The maximum header length supported for TSO feature is 1023 bytes. If the TCP Segmentation Offload feature is not enabled, this field is equal to Buffer 1 length.

1.15.3.1.4 TDES3 Normal Descriptor (Read Format)

Table 43. TDES2 Normal Descriptor (Read Format)

31	30	29	28	27:26	25:23	22:19	18	17:16	15	14:0
OWN	CTXT	FD	LD	CPC	SAIC	SLOTNUM or THL	TSE	CIC/TPL	TPL	FL/TPL

Table 44. TDES3 Normal Descriptor (Read Format)

Bits	Name	Description
31	OWN	Own Bit When this bit is set, it indicates that the DMA owns the descriptor. When this bit is reset, it indicates that the application owns the descriptor. The DMA clears this bit after it completes the transfer of data given in the associated buffer(s).
30	CTXT	Context Type This bit should be set to 1'b0 for normal descriptor.
29	FD	First Descriptor When this bit is set, it indicates that the buffer contains the first segment of a packet.
28	LD	Last Descriptor When this bit is set, it indicates that the buffer contains the last segment of the packet. When this bit is set, the B1L or B2L field should have a non-zero value.
27:26	CPC	CRC Pad Control This field controls the CRC and Pad Insertion for Tx packet. This field is valid only when the first descriptor bit (TDES3[29]) is set. The following list describes the values of Bits[27:26]: • 2'b00: CRC and Pad Insertion The MAC appends the cyclic redundancy check (CRC) at the end of the transmitted packet of length greater than or equal to 60 bytes. The MAC automatically appends padding and CRC to a packet with length less than 60 bytes. • 2'b01: CRC Insertion (Disable Pad Insertion) The MAC appends the CRC at the end of the transmitted packet but it does not append padding. The application should ensure that the padding bytes are present in the packet being transferred from the Transmit Buffer, that is, the packet being transferred from the Transmit Buffer is of length greater than or equal to 60 bytes. • 2'b10: Disable CRC Insertion The MAC does not append the CRC at the end of the transmitted packet. The application should ensure that the padding and CRC bytes are present in the packet being transferred from the Transmit Buffer. • 2'b11: CRC Replacement The MAC replaces the last four bytes of the transmitted packet with recalculated CRC bytes. The application should ensure that the padding and CRC bytes are present in the packet being transferred from the Transmit Buffer. This field is valid only for the first descriptor.

Table continues on the next page...

Table 44. TDES3 Normal Descriptor (Read Format) (continued)

Bits	Name	Description
		Note: When the TSE bit is set, the MAC ignores this field because the CRC and pad insertion is always done for segmentation.
25:23	SAIC	SA Insertion Control These bits request the MAC to add or replace the Source Address field in the Ethernet packet with the value given in the MAC Address 0 register. The application must set the CRC Pad Control bits appropriately when SA Insertion Control is enabled for the packet. Bit 25 specifies the MAC Address Register (1 or 0) value that is used for Source Address insertion or replacement. The following list describes the values of Bits[24:23]: • 2'b00: Do not include the source address • 2'b01: Include or insert the source address. For reliable transmission, the application must provide frames without source addresses. • 2'b10: Replace the source address. For reliable transmission, the application must provide frames with source addresses. • 2'b11: Reserved These bits are valid in the EQOS-DMA, EQOS-AXI, and EQOS-AHB configurations when the Enable SA and VLAN Insertion on Tx option is selected while configuring the core and when the First Segment control bit (TDES3 [29]) is set. This field is valid only for the first descriptor.
22:19	SLOTNUM or THL	SLOTNUM: Slot Number Control Bits in AV Mode These bits indicate the slot interval in which the data should be fetched from the corresponding buffers addressed by TDES0 or TDES1. When the Transmit descriptor is fetched, the DMA compares the slot number value in this field with the slot interval maintained in the RSN field DMA_CH#_Slot_Function_Control_Status. It fetches the data from the buffers only if a value matches. These bits are valid only for the AV channels. THL: TCP/UDP Header Length If the TSE bit is set, this field contains the length of the TCP/UDP header. The minimum value of this field must be 5 for TCP header. The value must be equal to 2 for UDP header. This field is valid only for the first descriptor.
18	TSE	TSO Split header is not supported. The value of this bit is always zero.
17:16	CIC/TPL	Checksum Insertion Control or TCP Payload Length These bits control the checksum calculation and insertion. The following list describes the bit encoding: • 2'b00: Checksum Insertion Disabled. • 2'b01: Only IP header checksum calculation and insertion are enabled. • 2'b10: IP header checksum and payload checksum calculation and insertion are enabled, but pseudo-header checksum is not calculated in hardware.

Table continues on the next page...

Table 44. TDES3 Normal Descriptor (Read Format) (continued)

Bits	Name	Description
		2'b11: IP Header checksum and payload checksum calculation and insertion are enabled, and pseudo-header checksum is calculated in hardware. This field is valid when the Enable Transmit TCP/IP Checksum Offload option is selected and the TSE bit is reset. When the TSE bit is set, this field contains the upper bits [17:16] of the TCP Payload (or IP Payload for UDP fragmentation). This allows the TCP/UDP packet length field to be spanned across TDES3[17:0] to provide 256 KB packet length support. This field is valid only for the first descriptor.
15	TPL	Reserved or TCP Payload Length When the TSE bit is reset, this bit is reserved. When the TSE bit is set, this is Bit 15 of the TCP payload length [17:0]. This field is valid only when the Enable TCP Segmentation Offloading for TCP/IP Packets option is selected while configuring the core.
14:0	FL/TPL	Frame Length or TCP Payload Length This field is equal to the length of the packet to be transmitted in bytes. When the TSE bit is not set, this field is equal to the total length of the packet to be transmitted: Ethernet Header Length + TCP /IP Header Length – Preamble Length – SFD Length + Ethernet Payload Length When the TSE bit is set, this field is equal to the lower 15 bits of the TCP payload length in case of segmentation and IP payload in case of UDP fragmentation. In case of segmentation, this length does not include Ethernet header or TCP/UDP/IP header length. In case of fragmentation, this length does not include Ethernet header and IP header. When DWRR/WFQ algorithm is NOT enabled, value written into this field is not used when TSE = 0.

1.15.3.1.5 Transmit Normal Descriptor (Write-Back Format)

The write-back format of the Transmit Descriptor includes timestamp low, timestamp high, OWN, and Status bits.

The write-back format is applicable only for the last descriptor of the corresponding packet. The LD bit (TDES3[28]) is set in the descriptor where the DMA writes back the status and timestamp information for the corresponding Transmit packet.

Following figure illustrates the write-back format of the Transmit Descriptor.

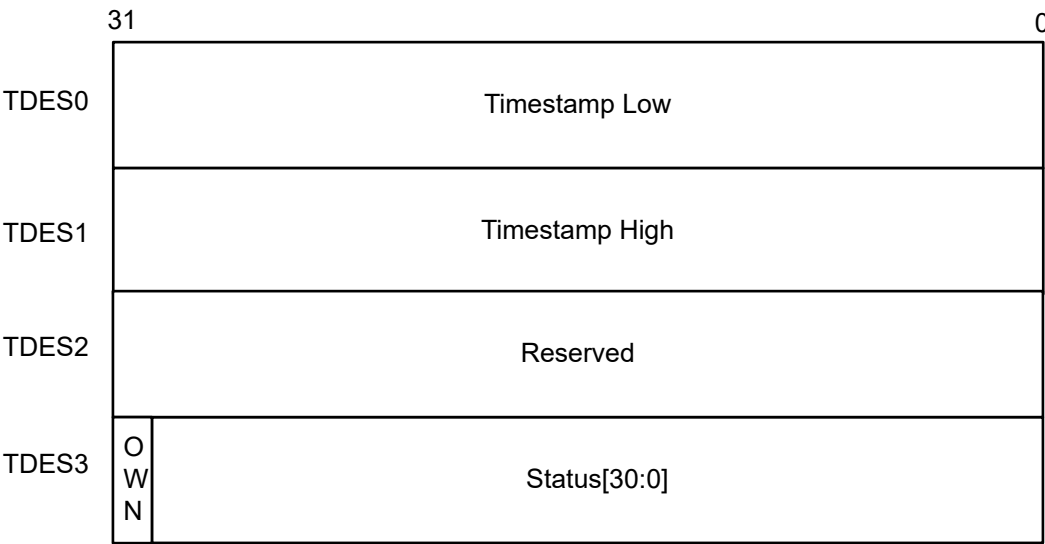


Figure 25. Transmit Descriptor Write-Back Format

TDES0 Normal Descriptor (Write-Back Format)

As described in following table, this format is only applicable to the last descriptor of a packet.

Table 45. TDES0 Normal Descriptor (Write-Back Format)

Bit	Name	Description
31:0	TTSL	Transmit Packet Timestamp Low The DMA updates this field with least significant 32 bits of the timestamp captured for the corresponding Transmit packet. The DMA writes the timestamp only if TTSE bit of TDES2 is set in the first descriptor of the packet. This field has the timestamp only if the Last Segment bit (LS) in the descriptor is set and the Timestamp status (TTSS) bit is set.

TDES1 Normal Descriptor (Write-Back Format)

As mentioned in the following table, this format is only applicable to the last descriptor of a packet.

Table 46. TDES1 Normal Descriptor (Write-Back Format)

Bit	Name	Description
31:0	TTSH	Transmit Packet Timestamp High The DMA updates this field with the most significant 32 bits of the timestamp captured for corresponding transmit packet. The DMA writes the timestamp only if the TTSE bit of TDES2 is set in the first descriptor of the packet. This field has the timestamp only if the Last Segment bit (LS) in the descriptor is set and Timestamp status (TTSS) bit is set.

TDES2 Normal Descriptor (Write-Back Format)

This format is applicable only to the last descriptor of a packet.

Table 47. TDES2 Normal Descriptor (Write-Back Format)

Bit	Description
31:0	Reserved

TDES3 Normal Descriptor (Write-Back Format)

This format is applicable only to the last descriptor of a packet.

Table 48. TDES3 Normal Descriptor (Write-Back Format)

31	30	29	28	27:18	17	16	15	14	13	12	11	10	9	8	7:4	3	2	1	0
OWN	CTXT	FD	LD	Rsvd	TTSS	EUE	ES	JT	FF	PCE	LoC	NC	LC	EC	CC	ED	UF	DB	IHE

Table 49. TDES3 Normal Descriptor (Write-Back Format)

Bit	Name	Description
31	OWN	Own Bit When this bit is set, it indicates that the the module DMA owns the descriptor. The DMA clears this bit when it completes the packet transmission. After the write-back is complete, this bit is set to 'b0.
30	CTXT	Context Type This bit should be set to 'b0 for Normal descriptor.
29	FD	First Descriptor This bit indicates that the buffer contains the first segment of a packet.
28	LD	Last Descriptor This bit is set 'b1 for last descriptor of a packet. The DMA writes the status fields only in the last descriptor of the packet.
27:24	Rsvd	Reserved
23	DE	Descriptor Error When this bit is set, it indicates that the descriptor content is incorrect. The DMA sets this bit during write-back while closing the descriptor. Descriptor Errors can be: • Incorrect sequence from the context descriptor. For example, a location after the first descriptor for a packet. • All 1s • CTXT, LD, and FD bits set to 1. Note 1: When Descriptor Error occurs due to All 1s or CTXT, LD, and FD bits set to 1, the Transmit DMA closes the transmit descriptor with DE and LD bits set to 1. When IOC bit in TDES2 of corresponding first descriptor is set to 1, Transmit DMA will set the TI bit in the DMA_CH#_Status register Note 2: Based on CTXT, LD, and FD bits of the transmit descriptor, the subsequent descriptor might be considered as the First Descriptor (even if FD bit is not set) and partial packet is sent.

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Table 49. TDES3 Normal Descriptor (Write-Back Format) (continued)

Bit	Name	Description
22:18	Rsvd	Reserved
17	TTSS	Tx Timestamp Status This status bit indicates that a timestamp has been captured for the corresponding transmit packet. When this bit is set, TDES0 and TDES1 have timestamp values that were captured for the Transmit packet. This field is valid only when the Last Segment control bit (TDES3 [28]) in a descriptor is set. This bit is valid only when IEEE1588 timestamping feature is enabled; otherwise, it is reserved.
16	EUE	ECC Uncorrectable Error Status Indicates the ECC uncorrectable error in the TSO memory. Note: Uncorrectable error in Transmit FIFO memory is reported with (Bit 13) FF = 1. This is because, all such packets are flushed by the module.
15	ES	Error Summary This bit indicates the logical OR of the following bits: • TDES3[0]: IP Header Error • TDES3[14]: Jabber Timeout • TDES3[13]: Packet Flush • TDES3[12]: Payload Checksum Error • TDES3[11]: Loss of Carrier • TDES3[10]: No Carrier • TDES3[9]: Late Collision • TDES3[8]: Excessive Collision • TDES3[3]: Excessive Deferral • TDES3[2]: Underflow Error This bit is also set when EUE (bit 16) is set.
14	JT	Jabber Timeout This bit indicates that the MAC transmitter has experienced a jabber time-out. This bit is set only when the JD bit of the MAC_Configuration register is not set.
13	FF	Packet Flushed This bit indicates that the DMA or MTL flushed the packet because of a software flush command given by the CPU.
12	PCE	Payload Checksum Error This bit indicates that the Checksum Offload engine had a failure and did not insert any checksum into the encapsulated TCP, UDP, or ICMP payload. This failure can be either because of insufficient bytes, as indicated by the Payload Length field of the IP Header or the MTL starting to forward the packet to the MAC transmitter in Store-and-Forward mode without the checksum having been calculated yet. This second error condition only occurs when the Transmit FIFO depth is less than the length of the Ethernet

Table continues on the next page...

Table 49. TDES3 Normal Descriptor (Write-Back Format) (continued)

Bit	Name	Description
		packet being transmitted to avoid deadlock, the MTL starts forwarding the packet when the FIFO is full, even in the store-and-forward mode. This error can also occur when Bus Error is detected during packet transfer. When the Full Checksum Offload engine is not enabled, this bit is reserved.
11	LoC	Loss of Carrier This bit indicates that Loss of Carrier occurred during packet transmission (that is, the gmii_crs_i signal was inactive for one or more transmit clock periods during packet transmission). This is valid only for the packets transmitted without collision and when the MAC operates in the half-duplex mode.
10	NC	No Carrier This bit indicates that the carrier sense signal from the PHY was not asserted during transmission.
9	LC	Late Collision This bit indicates that packet transmission was aborted because a collision occurred after the collision window (64 byte times including Preamble in MII mode and 512 byte times including Preamble and Carrier Extension in GMII mode). This bit is not valid if Underflow Error is set.
8	EC	Excessive Collision This bit indicates that the transmission was aborted after 16 successive collisions while attempting to transmit the current packet. If the DR bit is set in the MAC_Configuration register, this bit is set after first collision and the transmission of the packet is aborted.
7:4	CC	Collision Count This 4-bit counter value indicates the number of collisions occurred before the packet was transmitted. The count is not valid when the EC bit is set.
3	ED	Excessive Deferral This bit indicates that the transmission ended because of excessive deferral of over 24,288 bit times (155,680 bits times in 1000 Mbps mode or Jumbo Packet enabled mode) if DC bit is set in the MAC_Configuration register. When TBS is enabled in full duplex mode and this bit is set, it indicates that the frame has been dropped after the expiry time has reached.
2	UF	Underflow Error This bit indicates that the MAC aborted the packet because the data arrived late from the system memory. The underflow error can occur because of either of the following conditions: • The DMA encountered an empty Transmit Buffer while transmitting the packet • The application filled the MTL Tx FIFO slower than the MAC transmit rate The transmission process enters the suspended state and sets the underflow bit corresponding to a queue in the MTL_Interrupt_Status register.
1	DB	Deferred Bit

Table continues on the next page...

Table 49. TDES3 Normal Descriptor (Write-Back Format) (continued)

Bit	Name	Description
		This bit indicates that the MAC deferred before transmitting because of presence of carrier. This bit is valid only in the half-duplex mode.
0	IHE	IP Header Error When IP Header Error is set, this bit indicates that the Checksum Offload engine detected an IP header error. This bit is valid only when Tx Checksum Offload is enabled. Otherwise, it is reserved. If COE detects an IP header error, it still inserts an IPv4 header checksum if the Ethernet Type field indicates an IPv4 payload. In full duplex mode, when EST/Qbv is enabled and this bit is set, it indicates the frame drop status due to Frame Size error or Schedule Error.

1.15.3.2 Transmit Context Descriptor

The context descriptor is used to provide the timestamps for one-step timestamp correction, VLAN Tag ID for VLAN insertion feature.

The Transmit Context descriptor can be provided any time before a packet descriptor. The context is valid for the current packet and subsequent packets. The context descriptor is used to provide the timestamps for one-step timestamp correction and VLAN Tag ID for VLAN insertion feature. Write back is done on a context descriptor only to reset the OWN bit.

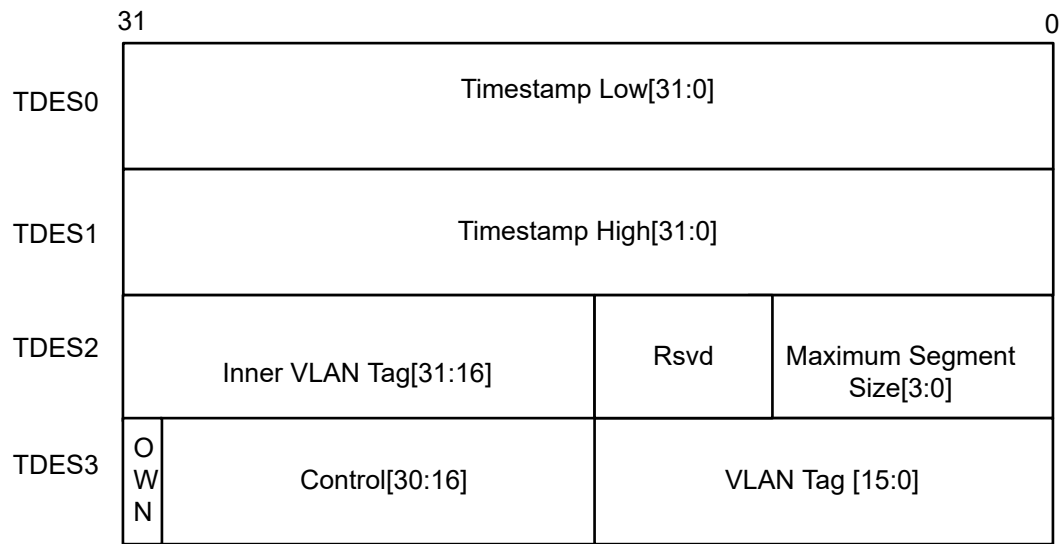
NOTE

The VLAN Tag IDs and MSS values, provided by the application in a context descriptor with their corresponding Valid bits set, are stored internally by the DMA. When the outer or inner VLAN tag is provided with the Valid bit set, the DMA always passes the last valid VLAN tag to the MTL. The application cannot invalidate the valid VLAN tag stored by the DMA. The VLAN tag is inserted or replaced based on the control inputs provided for the packet.

The Inner VLAN Tag Control input is used only for the next packet that immediately follows the context descriptor. The application must provide a context descriptor before the normal descriptor of each packet for which the DMA should use the inner VLAN Tag control input.

Following figure shows the format of the Transmit Context descriptor.

Figure 26. Transmit Context Descriptor Format



1.15.3.2.1 TDES0 Context Descriptor

Following table describes the format of the TDES0 Context descriptor.

Table 50. TDES0 Context Descriptor

Bit	Name	Description
31:0	TTSL	Transmit Packet Timestamp Low For one-step correction, the driver can provide the lower 32 bits of timestamp in this descriptor word. The DMA uses this value as the low word for doing one-step timestamp correction. This field is valid only if the OSTC and TCMSSV bits of TDES3 context descriptor are set.

1.15.3.2.2 TDES1 Context Descriptor

Following table describes the format of the TDES1 Context descriptor.

Table 51. TDES1 Context Descriptor

Bit	Name	Description
31:0	TTSH	Transmit Packet Timestamp High For one-step correction, the driver can provide the upper 32 bits of timestamp in this descriptor. The DMA uses this value as the high word for doing one-step timestamp correction. This field is valid only if the OSTC and TCMSSV bits of TDES3 context descriptor are set.

1.15.3.2.3 TDES2 Context Descriptor

Following table explains the format of the TDES2 Context descriptor.

Table 52. TDES2 Context Descriptor

Bit	Name	Description
31:16	IVT	Inner VLAN Tag When the IVLTV bit of TDES3 context descriptor is set and the TCMSSV and OSTC bits of TDES3 context descriptor are reset, TDES2[31:16] contains the inner VLAN Tag to be inserted in the subsequent Transmit packets.
15:14	Rsvd	Reserved
13:0	MSS	TSO Split header is not supported. The value of this bit is always zero.

1.15.3.2.4 TDES3 Context Descriptor**Table 53. TDES3 Context Descriptor**

31	30	29:28	27	26	25:24	23	22:18	17	16	15:0
OWN	CTXT	Rsvd	OSTC	TCMSSV	Rsvd	CDE	Rsvd	IVLTV	VLTV	VT

Following table details the format of the TDES3 Context descriptor.

Table 54. TDES3 Context Descriptor

Bit	Name	Description
31	OWN	Own Bit When this bit is set, it indicates that the module DMA owns the descriptor. When this bit is reset, it indicates that the application owns the descriptor. The DMA clears this bit immediately after the read.
30	CTXT	Context Type This bit should be set to 1'b1 for Context descriptor.
29:28	Rsvd	Reserved
27	OSTC	One-Step Timestamp Correction Enable When this bit is set, the DMA performs a one-step timestamp correction with reference to the timestamp values provided in TDES0 and TDES1.
26	TCMSSV	One-Step Timestamp Correction Input or MSS Valid When this bit and the OSTC bit are set, it indicates that the Timestamp Correction input provided in TDES0 and TDES1 is valid. When the OSTC bit is reset and this bit and the TSE bit of TDES3 are set in subsequent normal descriptor, it indicates that the MSS input in TDES2 is valid.
25:24	Rsvd	Reserved
23	DE	Descriptor Error

Table continues on the next page...

Table 54. TDES3 Context Descriptor (continued)

Bit	Name	Description
		When this bit is set, it indicates that the descriptor content is incorrect. The DMA sets this bit during write-back while closing the context descriptor. Descriptor Errors can be: • Incorrect sequence from the context descriptor. For example, a location before the first descriptor for a packet. • All 1s. • CD, LD, and FD bits set to 1. Note 1: When Descriptor Error occurs due to All 1s or CTXT, LD, and FD bits set to 1, the Transmit DMA closes the transmit descriptor with DE and LD bits set to 1. When IOC bit in TDES2 of corresponding first descriptor is set to 1, Transmit DMA will set the TI bit in the DMA_CH#_Status register Note 2: Based on CTXT, LD, and FD bits of the transmit descriptor, the subsequent descriptor might be considered as the First Descriptor (even if FD bit is not set) and partial packet is sent.
22:20	Rsvd	Reserved
19:18	IVTIR	Inner VLAN Tag Insert or Replace When this bit is set, these bits request the MAC to perform Inner VLAN tagging or un-tagging before transmitting the packets. If the packet is modified for VLAN tags, the MAC automatically recalculates and replaces the CRC bytes. The following list describes the values of these bits: • 2'b00: Do not add the inner VLAN tag. • 2'b01: Remove the inner VLAN tag from the packets before transmission. This option should be used only with the VLAN frames. • 2'b10: Insert an inner VLAN tag with the tag value programmed in the MAC_In-ner_VLAN_Incl register or context descriptor. • 2'b11: Replace the inner VLAN tag in packets with the tag value programmed in the MAC_Inner_VLAN_Incl register or context descriptor. This option should be used only with the VLAN frames. These bits are valid when the Enable SA and VLAN Insertion on Tx and Enable Double VLAN Processing options are selected.
17	IVLTV	Inner VLAN Tag Valid When this bit is set, it indicates that the IVT field of TDES2 is valid.
16	VLTV	VLAN Tag Valid When this bit is set, it indicates that the VT field of TDES3 is valid.
15:0	VT	VLAN Tag This field contains the VLAN Tag to be inserted or replaced in the packet. This field is used as VLAN Tag only when the VLTI bit of the MAC_VLAN_Incl register is reset.

1.15.4 Receive Descriptor

The DMA in the module attempts to read a descriptor only if the Tail Pointer is different from the Base Pointer or current pointer. It is recommended to have a descriptor ring with a length that can accommodate at least two complete packets received by the

MAC. Otherwise, the performance of the DMA is impacted greatly because of the unavailability of the descriptors. In such situations, the RxFIFO in MTL becomes full and starts dropping packets.

The following Receive Descriptors are present:

- Normal descriptors
- Context descriptors

All RX descriptors are prepared by the software and given to the DMA as "Normal" Descriptors with the content as shown in Receive Normal Descriptor (Read Format). The DMA reads this descriptor and after transferring a received packet (or part of) to the buffers indicated by the descriptor, the Rx DMA will close the descriptor with the corresponding packet status. The format of this status is given in the "Receive Normal Descriptor (Write-Back Format)".

For some packets, the normal descriptor bits are not enough to write the complete status. For such packets, the RX DMA writes the extended status to the next descriptor (without processing or using the Buffers/ Pointers embedded in that descriptor). The format and content of the descriptor write back is described in "Receive Context Descriptor".

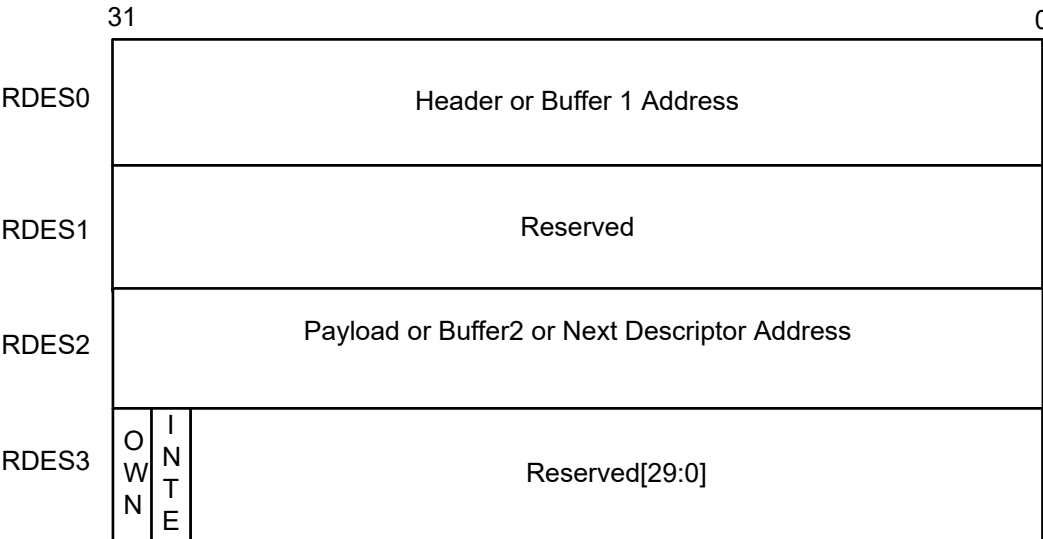
Software must allocate header buffers with start address aligned to data width or program the maximum header data size larger than the largest expected header size in receive packet by a number of bytes equal or more than one data width.

1.15.4.1 Receive Normal Descriptor (Read Format)

The read format for a Receive Normal descriptor is made up of a header or Buffer 1 address, reserved field, payload or Buffer 2 or Next Descriptor address, a 30-bit reserved filed, OWN bit, and an interrupt bit.

Following figure shows the Read Format for a Receive Normal Descriptor.

Figure 27. Receive Normal Descriptor Read Format



NOTE

In the Receive Descriptor (Read Format), if the Buffer Address field is all 0s, the module does not transfer data to that buffer and skips to the next buffer or next descriptor.

1.15.4.1.1 RDES0 Normal Descriptor (Read Format)

Following table explains the read format of the RDES0 Normal Descriptor.

Table 55. RDES0 Normal Descriptor (Read Format)

Bit	Name	Description
31:0	BUF1AP	Header or Buffer 1 Address Pointer When the SPH bit of Control register of a channel is reset, these bits indicate the physical address of Buffer 1. When the SPH bit is set, these bits indicate the physical address of Header Buffer where the Rx DMA writes the L2/L3/L4 header bytes of the received packet. The application can program a byte-aligned address for this buffer which means that the LS bits of this field can be non-zero. However, while transferring the start of packet, the DMA performs a Write operation with RDES0[1:0] (or RDES0[2:0]/[3:0] in case of 64-/128-bit configuration) as zero. However, the packet data is shifted as per actual offset as given by buffer address pointer. If the address pointer points to a buffer where the middle or last part of the packet is stored, the DMA ignores the offset address and writes to the full location as indicated by the data-width.

1.15.4.1.2 RDES1 Normal Descriptor (Read Format)

Following table describes the read format of the RDES1 Normal Descriptor.

Table 56. RDES1 Normal Descriptor (Read Format)

Bit	Name	Description
31:0	Reserved or BUF1AP	In 64-bit addressing mode, this field contains the most-significant 32 bits of the Buffer 1 Address Pointer. Otherwise, this field is reserved.

1.15.4.1.3 RDES2 Normal Descriptor (Read Format)

Following table provides the read format of the RDES2 Normal Descriptor.

Table 57. RDES2 Normal Descriptor (Read Format)

Bit	Name	Description
31:0	BUF2AP	Buffer 2 Address Pointer These bits indicate the physical address of Buffer 2. When the SPH bit of the DMA_CH#_Control register is set, the buffer address pointer must be bus width-aligned, that is, RDES2[3:0, 2:0, or 1:0] = 0 corresponding to 128, 64, or 32 bus width. LSBs are ignored internally. When the SPH bit of the DMA_CH#_Control register is reset, there is no limitations on the RDES2 value. However, the RxDMA uses the LS Bits of the pointer address only while transferring the start bytes of a packet. If the BUF2AP is giving the address of a buffer in which the middle or last part of a packet is stored, the DMA ignores BUF2AP[3:0 or 2:0 or 1:0] (corresponding to 128- or 64- or 32-bit data-bus) and writes to the complete location.

1.15.4.1.4 RDES3 Normal Descriptor (Read Format)

Following table describes the read format of the RDES3 Normal Descriptor.

31	30	19-26	25	24	23:0
OWN	IOC	Rsvd	BUF2V	BUF1V	Rsvd

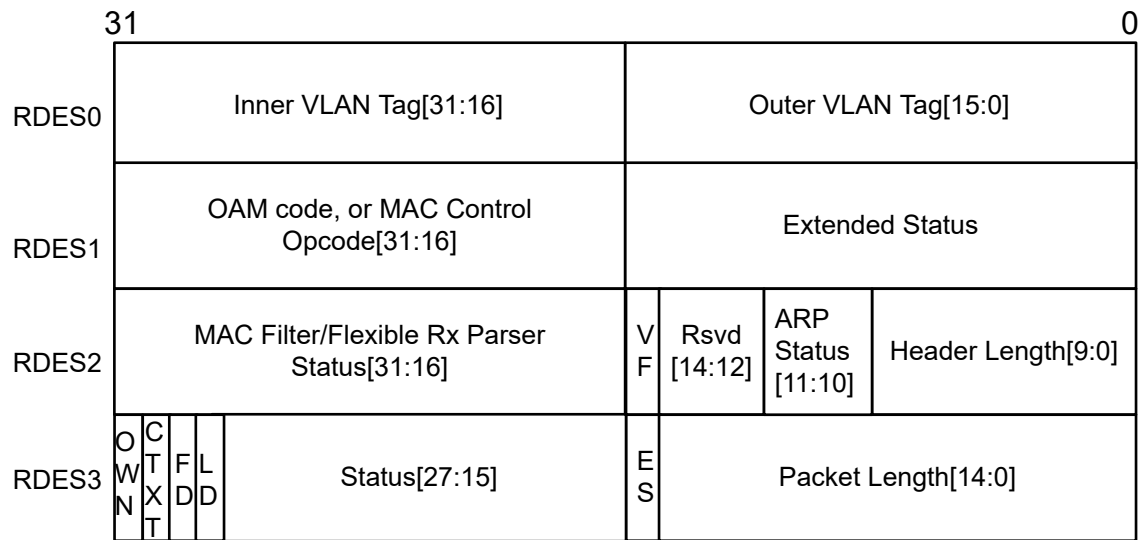
Table 58. RDES3 Normal Descriptor (Read Format)

Bit	Name	Description
31	OWN	Own Bit When this bit is set, it indicates that the module DMA owns the descriptor. When this bit is reset, it indicates that the application owns the descriptor. The DMA clears this bit when either of the following conditions is true: • The DMA completes the packet reception • The buffers associated with the descriptor are full
30	IOC	Interrupt Enabled on Completion When this bit is set, an interrupt is issued to the application when the DMA closes this descriptor.
29:26	Rsvd	Reserved
25	BUF2V	Buffer 2 Address Valid When this bit is set, it indicates to the DMA that the buffer 2 address specified in RDES2 is valid. The application must set this bit so that the DMA can use the address, to which the Buffer 2 address in RDES2 is pointing, to write received packet data.
24	BUF1V	Buffer 1 Address Valid When set, this indicates to the DMA that the buffer 1 address specified in RDES1 is valid. The application must set this value if the address pointed to by Buffer 1 address in RDES1 can be used by the DMA to write received packet data.
23:0	Rsvd	Reserved

1.15.4.1.5 Receive Normal Descriptor (Write-Back Format)

Following figure illustrates the write-back format for a Receive Normal descriptor.

Figure 28. Receive Normal Descriptor (Write-Back Format)



NOTE

When Flexible RX Parser is enabled, RDES2[31:16] indicates Parser status, and not the MAC filter status. MAC filter status is not available when Flexible RX parser is enabled.

1.15.4.1.5.1 RDES0 Normal Descriptor (Write-Back Format)

Following table describes the write-back format for the RDES0 Normal Descriptor.

Table 59. RDES0 Normal Descriptor (Write-Back Format)

Bit	Name	Description
31:16	IVT	Inner VLAN Tag This field contains the Inner VLAN tag of the received packet if the RS0V bit of RDES3 is set. This is valid only when Double VLAN tag processing and VLAN tag stripping are enabled.
15:0	OVT	Outer VLAN Tag This field contains the Outer VLAN tag of the received packet if the RS0V bit of RDES3 is set.

1.15.4.1.5.2 RDES1 Normal Descriptor (Write-Back Format)

The Status fields in write-back format are valid only for the last descriptor (RDES3[28] is set). Following table provides the details of the write-back format for RDES1 Normal Descriptor.

31:16	15	14	13	12	11:8	7	6	5	4	3	2:0
OPC	TD	TSA	PV	PFT	PMT	IPCE	IPCB	IPV6	IPV4	IPHE	PT

Table 60. RDES1 Normal Descriptor (Write-Back Format)

Bit	Name	Description
31:16	OPC	This field can have one of two meanings: • OAM Sub-Type Code: If bits 18:16 of RDES3 as described in RDES3 Normal Descriptor (Read Format) are 111b, this field contains the OAM sub-type and code fields. • MAC Control Packet opcode: Bits 15:8 of RDES3 as described in RDES3 Normal Descriptor (Read Format) contain the subtype and bits 7:0 contain the code.
15	TD	Timestamp Dropped This bit indicates that the timestamp was captured for this packet but it got dropped in the MTL Rx FIFO because of overflow. This bit is available only when you select the Timestamp feature. Otherwise, this bit is reserved.
14	TSA	Timestamp Available When Timestamp is present, this bit indicates that the timestamp value is available in a context descriptor word 2 (RDES2) and word 1(RDES1). This is valid only when the Last Descriptor bit (RDES3 [28]) is set. The context descriptor is written in the next descriptor just after the last normal descriptor for a packet.
13	PV	PTP Version This bit indicates that the received PTP message has the IEEE 1588 version 2 format. When this bit is reset, it indicates the IEEE 1588 version 1 format. This bit is available only when you select the Timestamp feature. Otherwise, this bit is reserved.
12	PFT	PTP Packet Type This bit indicates that the PTP message is sent directly over Ethernet. This bit is available only when you select the Timestamp feature. Otherwise, this bit is reserved.
11:8	PMT	PTP Message Type These bits are encoded to give the type of the message received: • 0000: No PTP message received • 0001: SYNC (all clock types) • 0010: Follow_Up (all clock types) • 0011: Delay_Req (all clock types) • 0100: Delay_Resp (all clock types) • 0101: Pdelay_Req (in peer-to-peer transparent clock) • 0110: Pdelay_Resp (in peer-to-peer transparent clock) • 0111: Pdelay_Resp_Follow_Up (in peer-to-peer transparent clock) • 1000: Announce • 1001: Management • 1010: Signaling • 1011–1110: Reserved

Table continues on the next page...

Table 60. RDES1 Normal Descriptor (Write-Back Format) (continued)

Bit	Name	Description
		• 1111: PTP packet with Reserved message type These bits are available only when you select the Timestamp feature.
7	IPCE	IP Payload Error When this bit is set, it indicates either of the following: • The 16-bit IP payload checksum (that is, the TCP, UDP, or ICMP checksum) calculated by the MAC does not match the corresponding checksum field in the received segment. • The TCP, UDP, or ICMP segment length does not match the payload length value in the IP Header field. • The TCP, UDP, or ICMP segment length is less than minimum allowed segment length for TCP, UDP, or ICMP. Bit 15 (ES) of RDES3 is not set when this bit is set.
6	IPCB	IP Checksum Bypassed This bit indicates that the checksum offload engine is bypassed. This bit is available when you select the Enable Receive TCP/IP Checksum Check feature.
5	IPV6	IPv6 header Present This bit indicates that an IPV6 header is detected. When the Enable Split Header Feature option is selected and the SPH bit of Control Register of a channel is set, the IPV6 header is available in the header buffer area to which RDES0 is pointing.
4	IPV4	IPv4 Header Present This bit indicates that an IPV4 header is detected. When the SPH bit of RDES3 is set, the IPV4 header is available in the header buffer area to which RDES0 is pointing.
3	IPHE	IP Header Error When this bit is set, it indicates either of the following: • The 16-bit IPv4 header checksum calculated by the MAC does not match the received checksum bytes. • The IP datagram version is not consistent with the Ethernet Type value. • Ethernet packet does not have the expected number of IP header bytes. This bit is valid when either Bit 5 or Bit 4 is set. This bit is available when you select the Enable Receive TCP/IP Checksum Check feature.
2:0	PT	Payload Type These bits indicate the type of payload encapsulated in the IP datagram processed by the Receive Checksum Offload Engine (COE): • 3'b000: Unknown type or IP/AV payload not processed • 3'b001: UDP • 3'b010: TCP • 3'b011: ICMP

Table continues on the next page...

Table 60. RDES1 Normal Descriptor (Write-Back Format) (continued)

Bit	Name	Description
		• 3'b110: AV Tagged Data Packet • 3'b111: AV Tagged Control Packet • 3'b101: AV Untagged Control Packet • 3'b100: IGMP if IPV4 Header Present bit is set else DCB (LLDP) Control Packet If the COE does not process the payload of an IP datagram because there is an IP header error or fragmented IP, it sets these bits to 3'b000.

1.15.4.1.5.3 RDES2 Normal Descriptor (Write-Back Format)

31:29	28	27	26:19	18	17	16	15	14	13:11	10	9:0
L3L4FM	L4FM	L3FM	MADRM	HF	DAF	SAF	OTS	ITS	Rsvd	ARPNR	HL

Table 61. RDES2 Normal Descriptor (Write-Back Format)

Bit	Name	Description
31:29	L3L4FM	Layer 3 and Layer 4 Filter Number Matched These bits indicate the number of the Layer 3 and Layer 4 Filter that matched the received packet: • 000: Filter 0 • 001: Filter 1 • 010: Filter 2 • 011: Filter 3 • 100: Filter 4 • 101: Filter 5 • 110: Filter 6 • 111: Filter 7 This field is valid only when Bit 28 or Bit 27 is set high. When more than one filter matches, these bits give the number of lowest filter. Note: This status is not available when Flexible RX Parser is enabled.
28	L4FM	Layer 4 Filter Match When this bit is set, it indicates that the received packet matches one of the enabled Layer 4 Port Number fields. This status is given only when one of the following conditions is true: • Layer 3 fields are not enabled and all enabled Layer 4 fields match • All enabled Layer 3 and Layer 4 filter fields match When more than one filter matches, this bit gives the layer 4 filter status of filter indicated by Bits[31:29]. Note: This status is not available when Flexible RX Parser is enabled.

Table continues on the next page...

Table 61. RDES2 Normal Descriptor (Write-Back Format) (continued)

Bit	Name	Description
27	L3FM	Layer 3 Filter Match When this bit is set, it indicates that the received packet matches one of the enabled Layer 3 IP Address fields. This status is given only when one of the following conditions is true: • All enabled Layer 3 fields match and all enabled Layer 4 fields are bypassed • All enabled filter fields match When more than one filter matches, this bit gives the layer 3 filter status of filter indicated by Bits[31:29]. Note: This status is not available when Flexible RX Parser is enabled.
26:19	MADRM	MAC Address Match or Hash Value When the HF bit is reset, this field contains the MAC address register number that matched the Destination address of the received packet. This field is valid only if the DAF bit is reset. When the HF bit is set, this field contains the hash value computed by the MAC. A packet passes the hash filter when the bit corresponding to the hash value is set in the hash filter register. Note: This status is not available when Flexible RX Parser is enabled.
18	HF	Hash Filter Status When this bit is set, it indicates that the packet passed the MAC address hash filter. Bits[26:19] indicate the hash value. Note: This status is not available when Flexible RX Parser is enabled.
17	DAF/RXPI	Destination Address Filter Fail When Flexible RX Parser is disabled, and this bit is set, it indicates that the packet failed the DA Filter in the MAC. When Flexible RX Parser is enabled, this bit is set to indicate that the packet parsing is incomplete (RXPI) due to ECC error. Note: When this bit is set, ES bit of RDES3 is also set.
16	SAF/RXPD	SA Address Filter Fail When Flexible RX Parser is disabled, and this bit is set, it indicates that the packet failed the SA Filter in the MAC. When Flexible RX Parser is enabled, this bit is set to indicate that the packet is dropped (RXPD) by the parser. Note: When this bit is set, ES bit of RDES3 is also set.
15	OTS	VLAN Filter Status When set, this bit indicates that the VLAN Tag of the received packet passed the VLAN filter. This bit is redefined as Outer VLAN Tag Filter Status (OTS). This bit is valid for both Single and Double VLAN Tagged frames
14	ITS	Inner VLAN Tag Filter Status (ITS) This bit is valid only for Double VLAN Tagged frames, when Double VLAN Processing is enabled.

Table continues on the next page...

Table 61. RDES2 Normal Descriptor (Write-Back Format) (continued)

Bit	Name	Description
		For more information, see Filter Status .
13:11	Rsvd	Reserved
10	ARPNR	ARP Reply Not Generated When this bit is set, it indicates that the MAC did not generate the ARP Reply for received ARP Request packet. This bit is set when the MAC is busy transmitting ARP reply to earlier ARP request (only one ARP request is processed at a time). This bit is reserved when the Enable IPv4 ARP Offload option is not selected.
9:0	HL	L3/L4 Header Length This field contains the length of the header of the packet split by the MAC at L3 or L4 header boundary as identified by the MAC receiver. This field is valid only when the first descriptor bit is set (FD = 1). The header data is written to the Buffer 1 address of corresponding descriptor. If header length is zero, this field is not valid. It implies that the MAC did not identify and split the header. This field is valid when the Enable Split Header Feature option is selected.

1.15.4.1.5.4 RDES3 Normal Descriptor (Write-Back Format)

Following table describes the write-back format for the RDES3 Normal Descriptor.

31	30	29	28	27	26	25	24	23	22	21	20	19	18:16	15	14:0
OWN	CTXT	FD	LD	RS2V	RS1V	RS0V	CE	GP	RWT	OE	RE	DE	LT	ES	PL

Table 62. RDES3 Normal Descriptor (Write-Back Format)

Bit	Name	Description
31	OWN	Own Bit When this bit is set, it indicates that the module DMA owns the descriptor. When this bit is reset, it indicates that the application owns the descriptor. The DMA clears this bit when either of the following conditions is true: • The DMA completes the packet reception • The buffers associated with the descriptor are full
30	CTXT	Receive Context Descriptor When this bit is set, it indicates that the current descriptor is a context type descriptor. The DMA writes 1'b0 to this bit for normal receive descriptor. When CTXT and FD bits are used together, {CTXT, FD} • 2'b00: Intermediate Descriptor • 2'b01: First Descriptor

Table continues on the next page...

Table 62. RDES3 Normal Descriptor (Write-Back Format) (continued)

Bit	Name	Description
		2'b10: Reserved 2'b11: Descriptor Error (due to all 1s) Note: When Descriptor Error occurs, the Receive DMA closes the receive descriptor indicating Descriptor Error. This receive descriptor is skipped and the buffer addresses are not used to write the packet data. Receive DMA will set the CDE bit in DMA_CH#_Status register but not the RI bit even when IOC is set, as this is not marked as last receive descriptor for the packet. The subsequent valid receive descriptor is used to write the packet data.
29	FD	First Descriptor When this bit is set, it indicates that this descriptor contains the first buffer of the packet. If the size of the first buffer is 0, the second buffer contains the beginning of the packet. If the size of the second buffer is also 0, the next descriptor contains the beginning of the packet. See the CTXT bit description for details of using the CTXT bit and FD bit together.
28	LD	Last Descriptor When this bit is set, it indicates that the buffers to which this descriptor is pointing are the last buffers of the packet.
27	RS2V	Receive Status RDES2 Valid When this bit is set, it indicates that the status in RDES2 is valid and it is written by the DMA. This bit is valid only when the LD bit of RDES3 is set.
26	RS1V	Receive Status RDES1 Valid When this bit is set, it indicates that the status in RDES1 is valid and it is written by the DMA. This bit is valid only when the LD bit of RDES3 is set.
25	RS0V	Receive Status RDES0 Valid When this bit is set, it indicates that the status in RDES0 is valid and it is written by the DMA. This bit is valid only when the LD bit of RDES3 is set.
24	CE	CRC Error When this bit is set, it indicates that a Cyclic Redundancy Check (CRC) Error occurred on the received packet. This field is valid only when the LD bit of RDES3 is set.
23	GP	Giant Packet When this bit is set, it indicates that the packet length exceeds the specified maximum Ethernet size of 1518, 1522, or 2000 bytes (9018 or 9022 bytes if jumbo packet enable is set). Note: Giant packet indicates only the packet length. It does not cause any packet truncation.
22	RWT	Receive Watchdog Timeout When this bit is set, it indicates that the Receive Watchdog Timer has expired while receiving the current packet. The current packet is truncated after watchdog timeout.
21	OE	Overflow Error

Table continues on the next page...

Table 62. RDES3 Normal Descriptor (Write-Back Format) (continued)

Bit	Name	Description
		When this bit is set, it indicates that the received packet is damaged because of buffer overflow in Rx FIFO. Note: This bit is set only when the DMA transfers a partial packet to the application. This happens only when the Rx FIFO is operating in the threshold mode. In the store-and-forward mode, all partial packets are dropped completely in Rx FIFO.
20	RE	Receive Error When this bit is set, it indicates that the gmii_rxr_i signal is asserted while the gmii_rxdv_i signal is asserted during packet reception. This error also includes carrier extension error in the GMII and half-duplex mode. Error can be of less or no extension, or error (rxd!= 0f) during extension.
19	DE	Dribble Bit Error When this bit is set, it indicates that the received packet has a non-integer multiple of bytes (odd nibbles). This bit is valid only in the MII Mode.
18:16	LT	Length/Type Field This field indicates if the packet received is a length packet or a type packet. The encoding of the 3 bits is as follows: • 3'b000: The packet is a length packet • 3'b001: The packet is a type packet. • 3'b011: The packet is a ARP Request packet type • 3'b100: The packet is a type packet with VLAN Tag • 3'b101: The packet is a type packet with Double VLAN Tag • 3'b110: The packet is a MAC Control packet type • 3'b111: The packet is a OAM packet type • 3'b010: Reserved
15	ES	Error Summary When this bit is set, it indicates the logical OR of the following bits: • RDES3[24]: CRC Error • RDES3[19]: Dribble Error • RDES3[20]: Receive Error • RDES3[22]: Watchdog Timeout • RDES3[21]: Overflow Error • RDES3[23]: Giant Packet • RDES2[17]: Destination Address Filter Fail, when Flexible RX Parser is enabled • RDES2[16]: SA Address Filter Fail, when Flexible RX Parser is enabled This field is valid only when the LD bit of RDES3 is set.
14:0	PL	Packet Length

Table continues on the next page...

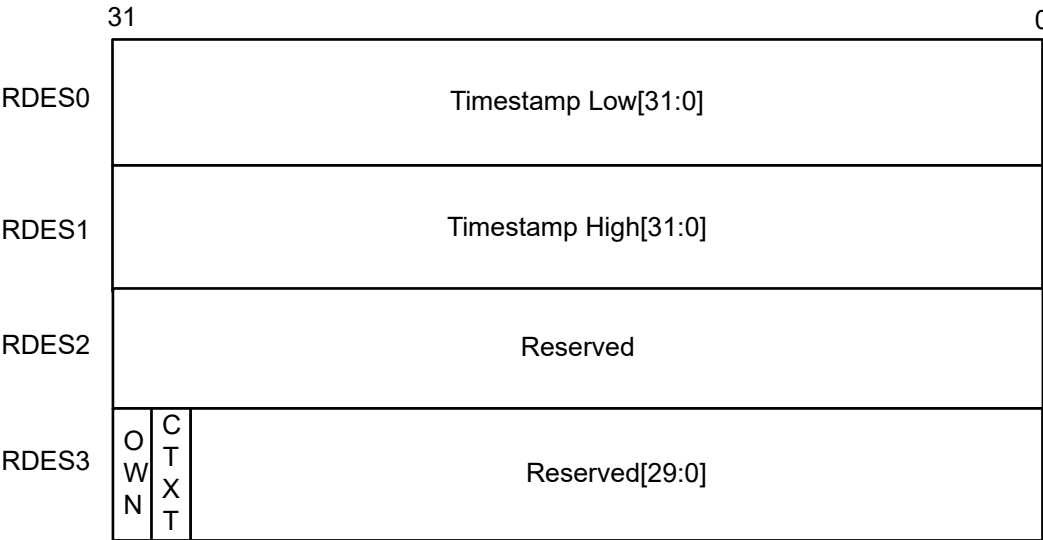
Table 62. RDES3 Normal Descriptor (Write-Back Format) (continued)

Bit	Name	Description
		These bits indicate the byte length of the received packet that was transferred to system memory (including CRC).
		This field is valid when the LD bit of RDES3 is set and Overflow Error bits are reset. The packet length also includes the two bytes appended to the Ethernet packet when IP checksum calculation is enabled and the received packet is not a MAC control packet.
		This field is valid when the LD bit of RDES3 is set. When the Last Descriptor and Error Summary bits are not set, this field indicates the accumulated number of bytes that have been transferred for the current packet.

1.15.4.1.6 Receive Context Descriptor

This descriptor is read-only for the application. Only the DMA can write to this descriptor. The context descriptor provides information about the extended status related to the last received packet. The Bit 30 of RDES3 indicates the context type descriptor.

Figure 29. Receive Context Descriptor Format



1.15.4.1.6.1 RDES0 Context Descriptor

Table 63. RDES0 Context Descriptor

Bit	Name	Description
31:0	RTSL	Receive Packet Timestamp Low The DMA updates this field with least significant 32 bits of the timestamp captured for corresponding Receive packet. When this field and the RTSH field of RDES1 show all-ones value, the timestamp must be considered as corrupt.

1.15.4.1.6.2 RDES1 Context Descriptor

Table 64. RDES1 Context Descriptor

Bit	Field	Description
31:0	RTSH	Receive Packet Timestamp High The DMA updates this field with most significant 32 bits of the timestamp captured for corresponding receive packet. When this field and the RTSL field of RDES0 show all-ones value, the timestamp must be considered as corrupt.

1.15.4.1.6.3 RDES2 Context Descriptor

Table 65. RDES2 Context Descriptor

Bit	Description
31:0	Reserved

1.15.4.1.6.4 RDES3 Context Descriptor

31	30	29	28:0
OWN	CTXT	DE	Rsvd

Table 66. RDES3 Context Descriptor

Bit	Name	Description
31	OWN	Own Bit When this bit is set, it indicates that the DMA owns the descriptor. When this bit is reset, it indicates that the application owns the descriptor. The DMA clears this bit when either of the following conditions is true: • The DMA completes the packet reception • The buffers associated with the descriptor are full
30	CTXT	Receive Context Descriptor When this bit is set, it indicates that the current descriptor is a context descriptor. The DMA writes 1'b1 to this bit for context descriptor. DMA writes 2'b11 to indicate a descriptor error due to all 1s. When CTXT and DE bits are used together, {CTXT, DE} • 2'b00: Reserved • 2'b01: Reserved • 2'b10: Context Descriptor • 2'b11: Descriptor Error Note: When Descriptor Error occurs, the Receive DMA closes the receive descriptor indicating Descriptor Error. This receive descriptor is skipped and the buffer addresses are not used to write the packet data

Table continues on the next page...

Table 66. RDES3 Context Descriptor (continued)

Bit	Name	Description
		Receive DMA will set the CDE bit in DMA_CH#_Status register but not the RI bit even when IOC is set, as this is not marked as last receive descriptor for the packet. The subsequent valid receive descriptor is used to write the packet data.
29	DE	Descriptor Error See the CTXT bit description for details of using the DE bit along with CTXT bit.
28:0	Rsvd	Reserved

1.15.5 Enhanced Descriptor for Time-Based Scheduling

The time-based scheduling feature needs 32 bytes Enhanced Descriptors to be enabled on all the DMA channels that intend to use the feature (by setting the EDSE bit of DMA_CH(#i)_TX_Control register).

The structure of 32-byte Descriptor for the Context and the Normal Descriptor in Read and Write formats are described in the following sections.

1.15.5.1 Enhanced Normal Descriptor - Read (32-Bit Mode)

The fields present in the first 16 bytes of the Enhanced Descriptor format of normal descriptor are as follows:

- LTV

Launch Time Valid indicates the Launch Time (LT) and GSN fields present in the descriptor are valid. The LTV must be set only if the FD bit of the descriptor is not.

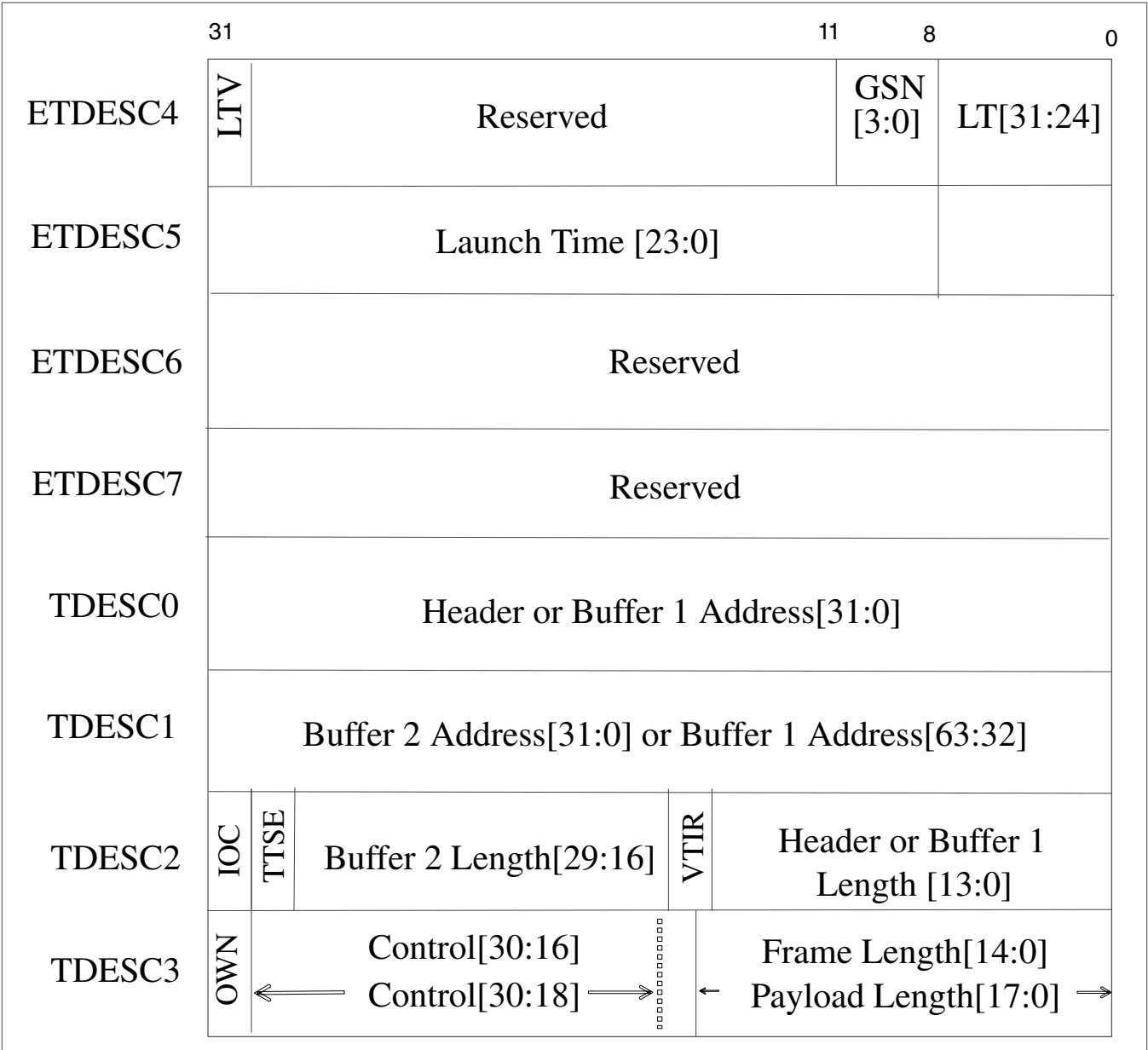
- GSN

This field indicates the GCL slot number associated with the packet.

- Launch Time (LT)

This field indicates the Launch Time associated with the packet.

Figure 30. Enhanced Normal Descriptor - Read (32-Bit Mode)



1.15.5.2 Enhanced Normal Descriptor (Write, 32-Bit Mode)

Figure 31. Enhanced Normal Descriptor Write (32-Bit Mode)

	31		11		8		0
ETDESC4	LTV	Reserved			GSN [3:0]	LT[31:24]	
ETDESC5	Launch Time [23:0]						
ETDESC6	Reserved						
ETDESC7	Reserved						
TDESC0	Timestamp Low						
TDESC1	Timestamp High						
TDESC2	Reserved						
TDESC3	OWN	Status[30:0]					

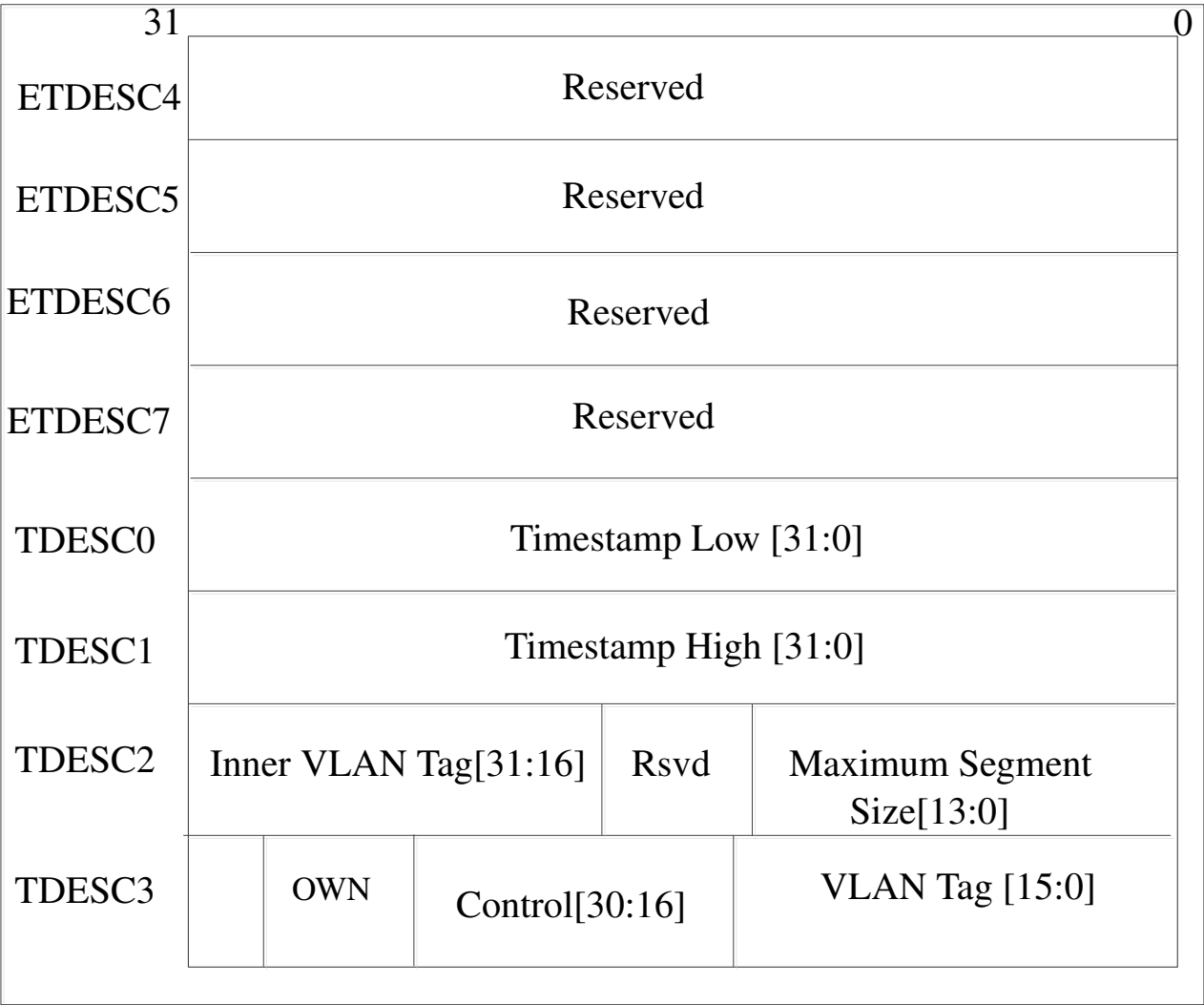
NOTE

- All the enhanced descriptor examples, assumes a 32-bit data width configurations. However, the same definitions are valid in 64 and 128 bit configurations.
- In both the write back formats, no modifications can be done to the extended 16 bytes. The rest of the 16 bytes (TDESC0 to TDESC3) are written back as per the previous 16 bytes Descriptor format.
- Fetch time when enabled overrides the AV slot function
- When an unaligned new GCL list (with a different CTR) is installed it is recommended not to have traffic during the installation of the new list. Any traffic during the switching of the lists might have unpredictable behavior regarding Fetch, Launch, and Launch expiry as the CTR and BTR values get updated while the frame is being processed

1.15.5.3 Enhanced Context Descriptor (Read, 32-bit Mode)

The first 16 bytes of the Enhanced context descriptor (ETDESC4 to ETDESC7) are reserved and must be zeros. The fields present in last bytes of the Descriptor (TDESC0 to TDESC3) are same as the context descriptor (TDESC0 to TDESC3) in 16 byte format.

Figure 32. Enhanced Context Descriptor - Read



1.15.5.4 Enhanced Context Descriptor (Write, 32 Bit Mode)

Figure 33. Enhanced Context Descriptor - Write

		31				0
ETDESC4	Reserved					
ETDESC5	Reserved					
ETDESC6	Reserved					
ETDESC7	Reserved					
TDESC0	Reserved					
TDESC1	Reserved					
TDESC2	Reserved					
TDESC3	OWN	CTXT	Reserved	CDE	Reserved	

1.16 Programming

This section and all its sub-sections are Synopsys Proprietary. Used with permission.

This section provides the instructions for initializing the DMA or MAC registers in the proper sequence.

1.16.1 Initializing DMA

Complete the following steps to initialize the DMA:

1. Provide a software reset. This resets all of the MAC internal registers and logic (bit-0 of DMA_Mode).
2. Wait for the completion of the reset process (poll bit 0 of the DMA_Mode, which is only cleared after the reset operation is completed).
3. Program the following fields to initialize the DMA_SysBus_Mode register:
 - a. AAL
 - b. Fixed burst or undefined burst
 - c. Burst mode values in case of AHB bus interface, OSR_LMT in case of AXI bus interface.
 - d. If fixed length value is enabled, select the maximum burst length possible on the AXI Bus (bits [7:1])

4. Create a descriptor list for transmit and receive. In addition, ensure that the descriptors are owned by DMA (set bit 31 of descriptor TDES3/RDES3). For more information about descriptors, see "Descriptors".
5. Program the Transmit and Receive Ring length registers (DMA_CH(#i)_TxDesc_Ring_Length (for i = 0; i <= 1) and DMA_CH(#i)_RxDesc_Ring_Length (for i = 0; i <= 1)). The ring length programmed must be at least 4.

NOTE

The descriptor address from the start to the end of the ring must not cross the 4GB boundary.

6. Initialize receive and transmit descriptor list address with the base address of the transmit and receive descriptor (DMA_CH(#i)_TxDesc_List_Address (for i = 0; i <= 4), DMA_CH(#i)_RxDesc_List_Address (for i = 0; i <= 4)). Also, program transmit and receive tail pointer registers indicating to the DMA about the available descriptors (DMA_CH(#i)_TxDesc_Tail_Pointer (for i = 0; i <= 4) and DMA_CH(#i)_RxDesc_Tail_Pointer (for i = 0; i <= 4))

NOTE

For 40-bit or 48-bit addressing mode, program the higher address List registers (DMA_CH[n]_TxDesc_List_HAddress, DMA_CH[n]_RxDesc_List_HAddress).

The tailpointer registers must be advanced to the location immediately after the descriptors that are set, for the DMA to know that additional descriptors are available.

7. Program the settings of the following registers for the parameters like maximum burst-length (PBL) initiated by DMA, descriptor skip lengths, OSP in case of TxDMA, RBSZ in case of RxDMA, and so on:
 - DMA_CH(#i)_Control (for i = 0; i <= 4)
 - DMA_CH(#i)_TX_Control (for i = 0; i <= 4)
 - DMA_CH(#i)_RX_Control (for i = 0; i <= 4)
8. Enable the interrupts by programming the DMA_CH(#i)_Interrupt_Enable (for i = 0; i <= 4) register.
9. Start the Receive and Transmit DMAs by setting SR (bit 0) of the DMA_CH(#i)_RX_Control (for i = 0; i <= 4) and ST (bit 0) of the DMA_CH(#i)_TX_Control (for i = 0; i <= 4) register.
10. Repeat steps 4 to 9 for all the Tx DMA and Rx DMA channels selected in the hardware.

1.16.2 Initializing MTL Registers

The Transaction Layer (MTL) registers must be initialized to establish the transmit and receive operating modes and commands.

Complete the following steps to initialize the MTL registers:

1. Program the Tx Scheduling (SCHALG) and Receive Arbitration Algorithm (RAA) fields in MTL_Operation_Mode to initialize the MTL operation in case of multiple Tx and Rx queues.
2. Program the Receive Queue to DMA mapping in MTL_RxQ_DMA_Map0 and MTL_RxQ_DMA_Map1 registers.
3. Program the following fields to initialize the mode of operation in the MTL_TxQ0_Operation_Mode register.
 - a. Transmit Store And Forward (TSF) or Transmit Threshold Control (TTC) in case of threshold mode
 - b. Transmit Queue Enable (TXQEN) to value 2'b10 to enable Transmit Queue0
 - c. Transmit Queue Size (TQS)
4. Program the following fields to initialize the mode of operation in the MTL_RxQ0_Operation_Mode register:
 - a. Receive Store and Forward (RSF) or RTC in case of Threshold mode
 - b. Flow Control Activation and De-activation thresholds for MTL Receive FIFO (RFA and RFD)
 - c. Error Packet and undersized good Packet forwarding enable (FEP and FUP)
 - d. Receive Queue Size (RQS)
5. Repeat previous two steps for all MTL Tx and Rx queues selected in the configuration.

1.16.3 Initializing MAC

The MAC configuration registers establish the operating mode of the MAC. These registers must be initialized before initializing the DMA.

The following MAC Initialization operations can be performed after DMA initialization. If the MAC initialization is completed before the DMA is configured, enable the MAC receiver (last step in the following sequence) only after the DMA is active. Otherwise, received frames fill the Rx FIFO and overflow.

1. Provide the MAC address registers: MAC_Address0_High and MAC_Address0_Low. If more than one MAC address is enabled in your configuration (during configuration in the coreConsultant), program the MAC addresses appropriately.
2. Program the following fields to set the appropriate filters for the incoming frames in the MAC_Packet_Filter register:
 - a. Receive All
 - b. Promiscuous mode
 - c. Hash or Perfect Filter
 - d. Unicast, multicast, broadcast, and control frames filter settings
3. Program the following fields for proper flow control in the MAC_Q0_Tx_Flow_Ctrl register:
 - a. Pause time and other Pause frame control bits
 - b. Transmit Flow control bits
 - c. Flow Control Busy
4. Program the MAC_Interrupt_Enable register, as required, and if applicable, for your configuration.
5. Program the appropriate fields in the MAC_Configuration register. For ex: Inter-packet gap while transmission and jabber disable.
6. Set bit 0 and 1 in MAC_Configuration registers to start the MAC transmitter and receiver.

1.16.4 Performing Normal Receive and Transmit Operation

During normal operation of the module, normal and transmit interrupts are read, descriptors polled, the DMA is suspended (if it does not own descriptors), and values of current host transmitter or receiver descriptor pointers are read for debugging.

For normal operation, complete the following steps:

1. For normal transmit and receive interrupts, read the interrupt status. Then, poll the descriptors, reading the status of the descriptor owned by the Host (either transmit or receive).
2. Set appropriate values for the descriptors, ensuring that transmit and receive descriptors are owned by the DMA to resume the transmission and reception of data.
3. If the descriptors are not owned by the DMA (or no descriptor is available), the DMA goes into SUSPEND state. The transmission or reception can be resumed by freeing the descriptors and writing the descriptor tail pointer to Tx/Rx tail pointer register (DMA_CH[n]_TxDesc_Tail_Pointer and DMA_CH[n]_RxDesc_Tail_Pointer).
4. The values of the current host transmitter or receiver descriptor address pointer can be read for the debug process (DMA_CH[n]_Current_App_TxDesc and DMA_CH[n]_Current_App_RxDesc).
5. The values of the current host transmit buffer address pointer and receive buffer address pointer can be read for the debug process (Register DMA_CH[n]_Current_App_TxBuffer and DMA_CH[n]_Current_App_RxBuffer).

1.16.5 Stopping and Starting Transmission

Complete the following steps to pause the transmission for some time. The steps are provided for Channel 0.

NOTE

The steps are provided for Channel 0.

1. Disable the Transmit DMA (if applicable) by clearing Bit 0 (ST) of DMA_CH(#)_TX_Control (for i = 0; i <= 4) Register.
2. Wait for any previous frame transmissions to complete. You can check this by reading the appropriate bits of MTL_TxQ0_Debug Register (TRCSTS is not 01 and TXQSTS=0).
3. Disable the MAC transmitter and MAC receiver by clearing Bit (RE) and Bit 1(TE) of the MAC_Configuration Register.
4. Disable the Receive DMA (if applicable), after making sure that the data in the Rx FIFO is transferred to the system memory (by reading the appropriate bits of MTL_TxQ0_Debug Register, PRXQ=0 and RXQSTS=00).
5. Make sure that both Tx Queue and Rx Queue are empty (TXQSTS is 0 in MTL_TxQ0_Debug Register and RXQSTS is 0 in MTL_RxQ0_Debug Register).
6. To restart the operation, first start the DMAs, and then enable the MAC Transmitter and Receiver.

NOTE

- Do not change the configuration (such as duplex mode, speed, port, or loop back) when the MAC is actively transmitting or receiving. These parameters are changed by software only when the MAC transmitter and receiver are not active.
- Similarly, do not change the DMA-related configuration when Transmit and Receive DMA are active.

1.16.6 Programming Guidelines for Switching to New Descriptor List in RxDMA

Switching to a new descriptor list is different in the Rx DMA compared to the Tx DMA. Switching to a new descriptor list is permitted when the RxDMA is in SUSPEND state, as clarified by the following points:

- Generally, RxDMA prepares the descriptors in advance.
- If the RxDMA goes to SUSPEND due to descriptors not being available, a major failure occurs (software is not able to free the filled-up descriptors/buffers). If this issue is not rectified immediately, frames will be lost because of an Rx FIFO Overflow. Therefore, the software is allowed to create a new descriptor list and program the RxDMA to start using it immediately, without going into STOP state.

1.16.7 Programming Guidelines for Multi-Channel Multi-Queuing

1.16.7.1 Transmit

1. Program the Transmit queue size in the TQS field of MTL_TxQ[n]_Operation_Mode register. Based on the value programmed in the TQS field, the size of the queue is determined.

In the Transmit operation, the number of channels is equal to the number of the queues. Due to this reason, the Channel-to-Queue mapping is fixed.

2. For a queue to be used, the queue needs to be enabled in TXQEN in the corresponding MTL_TxQ[n]_Operation_Mode Register. In DMA configurations, the ST bit of DMA_CH[n]_Tx_Control Register and corresponding TXQEN in MTL_TxQ[n]_Operation Mode Register needs to be enabled.
3. The scheduling method needs to be programmed in SCHALG of MTL_Operation_Mode register.
4. Program the MTL_TxQ[n]_Quantum_Weight for DCB queue as per the selected algorithm. In case of CBS algorithm in AVB queues, the MTL_TxQ[n]_ETS_Control, MTL_TxQ[n]_SendSlopeCredit, MTL_TxQ[n]_HiCredit and MTL_TxQ[n]_LoCredit registers also need to be programmed as required.
5. If DCB is enabled and PFC function is required, program MAC_TxQ_PrtY_Map0 Register to assign a fixed priority to the queue. This priority assigned is used for determining if the corresponding queue should stop transmitting packet based on the received PFC packet.

1.16.7.2 Receive

1. Program the Receive queue size in the RQS field of MTL_RxQ[n]_Operation_Mode Register. Based on the value programmed in RQS field, the size of the queue is determined.

2. Enable the Receive Queues 0 to 7 in the fields RXQ0EN to RXQ7EN in MAC_RxQ_Ctrl0 Register for AV or DCB. In DMA configurations, SR bit of statically or dynamically mapped DMA_CH[n]_Rx_Control Register and corresponding RXQ[n]_EN in MAC_RxQ_Ctrl0 Register needs to be enabled.
3. The MAC routes the Rx packets to the Rx Queues based on following packet types:
 - a. AV PTP Packets: Based on the programming of AVPTPQ in MAC_RxQ_Ctrl1 Register.
 - b. AV Untagged Control packets: Based on the programming of AVCPQ in MAC_RxQ_Ctrl1 Register.
 - c. Data Center Bridging (DCB) related Link Layer Discovery Protocol (LLDP) packets. Program DCBCPQ in MAC_RxQ_Ctrl1 Register to indicate to MAC which queue should get the DCB packets.
 - d. VLAN Tag Priority field in VLAN Tagged packets: Program PSRQ7-0 of the MAC_RxQ_Ctrl2 and MAC_RxQ_Ctrl3 Register for the routing of tagged packets based on the USP (user Priority) field of the received packets to the Rx Queues 0 to 7.
 - e. The AV tagged control and data packets are also routed based on PSRQ field of the MAC_RxQ_Ctrl2 and MAC_RxQ_Ctrl3 registers.

NOTE

The priorities set in PSRQ7-0 should be unique.

4. If multiple RX DMA channels are enabled, the following programming should be done for proper arbitration and mapping:
 - a. Program the RAA field of MTL_Operation_Mode register to select the arbitration algorithm to decide which RxQ is read out from the Rx FIFO memory.
 - b. Program the MTL_RxQ[n]_Control to decide the weights and the packet arbitration for each RxQ.
 - c. If static mapping is programmed in MTL_RxQ_DMA_Map[n] register (RXQ[n]DADMACH is reset to 0), bits RXQx2DMA and others need to be programmed to select the channel for which each queue is mapped.
 - d. Set RXQ[n]DADMACH bit in MTL_RxQ_DMA_Map0 Register to select dynamic mapping of packets in each RxQueue.
 - e. In dynamic channel mapping, the routing of a packet to a specific RxDMA channel is decided by the value of DCS field in the lowest MAC Address Register.

1.16.7.2.1 Programming Guidelines for Recovering from DMA Channel Failure

When the DMA channel issues a bus error, follow these steps to recover from the failure.

Recovering from the Receive DMA Channel Failure

Follow these steps if you get bus error in the Receive DMA channel:

1. Set the RPF bit to 1. This flushes all the packets one after the other.

This step is optional. However, setting this bit prevents HOL (head-of-line) blocking in the Rx queues when packets sent to the RXDMA are stopped due to bus error.

2. Re-program the specific Registers of the DMA channel.
3. Start the DMA channel.

Recovering from the Transmit DMA Channel Failure

1. Stop the specific DMA channel, even if it is in active state.
2. Flush the corresponding MTL queue.
3. Re-program the specific Registers of the DMA channel.
4. Start the DMA channel.

NOTE

Due to the known limitations in the design, reprogramming the DMA channel registers might not always be successful in recovering from a Bus Error. If the module is not fully functional after reprogramming the DMA, as a workaround, issue a Soft Reset to recover from the Bus Error.

1.16.8 Programming Guidelines for GMII Link State Transitions

1.16.8.1 Transmit and Receive Clocks Running when Link Down

Complete the following steps when the link is down but the Transmit and Receive clocks are running:

NOTE

The steps are provided for Channel 0.

1. Disable the Transmit DMA (if applicable) by clearing Bit 0 (ST) of DMA_CH(#)_TX_Control (for i = 0; i <= 4) Register.
2. Disable the MAC receiver by clearing Bit 2 (RE) of MAC_Configuration Register.
3. Wait for any previous frame transmissions to complete. You can check this by reading the appropriate bits of MTL_TxQ0_Debug Register (TRCSTS is not 01). Or, Flush the Tx FIFO for faster empty operation.
4. Disable the MAC transmitter by clearing Bit 1 (TE) of the MAC_Configuration Register.
5. Make sure that both Tx Queue and Rx Queue are empty (TXQSTS is 0 in MTL_TxQ0_Debug Register and RXQSTS is 0 in MTL_RxQ0_Debug Register).
6. After the link is up, read the PHY registers to know the latest configuration and accordingly program the MAC registers.
7. Restart the operation by starting the Tx DMA, and then enabling the MAC Transmitter and Receiver.

You do not need to disable the Rx DMA. As the Receiver is disabled, the FIFO does not get any data in the Rx FIFO.

1.16.8.2 Transmit and Receive Clocks Stopped when Link Down

Complete the following steps when the link is down and the Transmit and Receive clocks are stopped:

NOTE

The steps are provided for Channel 0.

1. Disable the MAC Transmitter and Receiver by clearing bits RE and TE of MAC_Configuration Register. This does not take effect immediately as the clocks are absent.
2. Wait until the link is up and the clocks are restored.
3. Wait for the completion of the transfer of any partial frame, if any, at time of stopping of Transmit/Receive clock. This can be checked by reading the MAC_Debug Register (should be all-zero). Some old packets may still remain in the TXFIFO as the MAC Transmitter is stopped.
4. Read the PHY registers to know the latest operating mode and accordingly program the MAC registers.
5. Restart the MAC Transmitter and Receiver by setting RE and TE bits.

1.16.9 Programming Guidelines for IEEE 1588 Timestamping

1.16.9.1 Initialization Guidelines for System Time Generation

You can enable the timestamp feature by setting Bit 0 of the MAC_Timestamp_Control Register. However, it is essential that the timestamp counter be initialized after this bit is set. Complete the following steps during the module initialization:

1. Mask the Timestamp Trigger interrupt by clearing the bit 16 of MAC_Interrupt_Enable Register.
2. Set Bit 0 of MAC_Timestamp_Control Register to enable timestamping.
3. Program MAC_Sub_Second_Increment Register based on the PTP clock frequency.

4. If you are using the Fine Correction approach, program MAC_Timestamp_Addend and set Bit 5 of MAC_Timestamp_Control Register.
5. Poll the MAC_Timestamp_Control Register until Bit 5 is cleared.
6. Program Bit 1 of MAC_Timestamp_Control Register to select the Fine Update method (if required).
7. Program MAC_System_Time_Seconds_Update Register and MAC_System_Time_Nanoseconds_Update Register with the appropriate time value.
8. Set Bit 2 in MAC_Timestamp_Control Register.
9. The timestamp counter starts operation as soon as it is initialized with the value written in the Timestamp Update registers. If one-step timestamping is enabled
 - a. To enable one-step timestamping, program Bit 27 of the TDES3 Context Descriptor.
 - b. Program registers MAC_Timestamp_Ingress_Asym_Corr and MAC_Timestamp_Egress_Asym_Corr to update the correction field in PDelay_Req PTP messages.
10. Enable the MAC receiver and transmitter for proper timestamping.

NOTE

If timestamp operation is disabled by clearing Bit 0 of MAC_Timestamp_Control Register, repeat all these steps to restart the timestamp operation.

1.16.9.2 System Time Correction

1.16.9.2.1 Coarse Correction Method

To synchronize or update the system time in one process (coarse correction method), complete the following steps:

1. Set the offset (positive or negative) in the Timestamp Update registers(MAC_System_Time_Seconds_Update and MAC_System_Time_Nanoseconds_Update).
2. Set Bit 3 (TSUPDT) of MAC_Timestamp_Control Register.

The value in the Timestamp Update registers is added to or subtracted from the system time when the TSUPDT bit is cleared.

1.16.9.2.2 Fine Correction Method

To synchronize or update the system time to reduce system-time jitter (fine correction method), complete the following steps:

1. With the help of the algorithm, calculate the rate by which you want to make the system time increments slower or faster.
2. Update the MAC_Timestamp_Addend with the new value and set Bit 5 of the MAC_Timestamp_Control Register.
3. Wait for the time for which you want the new value of the Addend register to be active. You can do this by enabling the Timestamp Trigger interrupt after the system time reaches the target value.
4. Program the required target time in MAC_PPS[n]_Target_Time_Seconds Register and MAC_PPS[n]_Target_Time_Nanoseconds Register.
5. Enable the Timestamp interrupt in bit 12 of MAC_Interrupt_Enable register.
6. Set bit 4 in Register MAC_Timestamp_Control.
7. When this trigger causes an interrupt, read MAC_Interrupt_Status Register.
8. Reprogram MAC_Timestamp_Addend Register with the old value and set bit 5 again

1.16.10 Programming Guidelines for AV Feature

After you enable the AV feature in the module controller, there are several programming tasks to be followed:

- Initializing the DMA for QOS-AHB configurations only

- Enabling slot number checking
- Enabling average bits per slot reporting
- Disabling flow control for AV-enabled queues (transmit and receive flow control)

1.16.10.1 Initializing the DMA in Audio Video Feature

The first step in programming the AV feature in a QOS-AHB configuration is to initialize the DMA.

Use this initialization sequence for QOS-AHB/QOS-AXI/QOS-AXI4 configurations with AV feature.

1. Provide a software reset to reset all QOS internal registers and logic (bit 0 in DMA_Mode register).
2. Wait for the completion of the reset process. Poll bit 0 of the DMA_Mode register, which is cleared only after the reset operation is completed.
3. Program the fields to initialize the DMA register by setting the values in DMA_Mode register.
4. Create a proper descriptor list for transmit and receive. In addition, ensure that the DMA owns the Transmit and Receive descriptors. When OSF mode is used, at least two TX descriptors are required.

For more information about descriptors, see "Descriptors".

5. Make sure that your software creates three or more different transmit or receive descriptors in the list before reusing any of the descriptors.
6. Program the Transmit and Receive Ring length registers (DMA_CH(#i)_TxDesc_Ring_Length (for i = 0; i <= 4) and DMA_CH(#i)_RxDesc_Ring_Length (for i = 0; i <= 4)). The ring length programmed must be at least 4.
7. Initialize receive and transmit descriptor list address with the base address of the transmit and receive descriptor (DMA_CH(#i)_TxDesc_List_Address (for i = 0; i <= 4), DMA_CH(#i)_RxDesc_List_Address (for i = 0; i <= 4)). In addition, you must program the Transmit and Receive tail pointer registers indicating to the DMA about the available descriptors (DMA_CH(#i)_TxDesc_Tail_Pointer (for i = 0; i <= 4) and DMA_CH(#i)_RxDesc_Tail_Pointer (for i = 0; i <= 4)).
8. Program the following fields to initialize the mode of operation in the MTL_TxQ0_Operation_Mode register:
 - a. Transmit Store And Forward (TSF)
 - b. Transmit Threshold Control (TTC)
 - c. Transmit Queue Enable (TXQEN) to value 2'b10 to enable Transmit Queue0
 - d. Transmit Queue Size (TQS)
9. Enable the interrupts by programming the DMA_CH(#i)_Interrupt_Enable (for i = 0; i <= 4) register.
10. Repeat steps 4 through 9 for all additional channels of AV feature.
11. Program the CBS control register, idleSlope, sendSlope, hiCredit, and loCredit registers of the AV Queues.
12. Start the Receive and Transmit DMA by setting bit 0 of the DMA_CH(#i)_TX_Control (for i = 0; i <= 4) register and bit 0 of the DMA_CH(#i)_RX_Control (for i = 0; i <= 4) register.

1.16.10.2 Enabling Slot Number Checking

You can enable slot number checking feature if you want to specify the intervals at which the DMA channels mapped to AV Queues fetch the frames from the AHB/AXI system bus.

These steps should be completed after step 11 and before step 12 of "Initializing DMA".

You can use the slot number check feature to specify the intervals at which the DMA Channels mapped to AV Queues fetch the frames from the AHB/AXI system bus. This feature is useful for a uniform and periodic transfer of the AV traffic from the host memory. The feature is available only when you enable timestamping and program the MAC_Sub_Second_Increment register. Complete the following steps to enable the slot number checking:

1. Enable timestamping by following the steps described in "Initialization Guideline for System Time Generation".

2. Make sure that the SLOTNUM field (bits 22:19) of TDES3 Normal Descriptor (Read Format) contains a valid slot number. You can read the current reference slot number from the DMA_CH(#)_Slot_Function_Control_Status (for i = 0; i <= DWC_EQOS_NUM_DMA_TX_CH-1).
3. Set Bit 0 (ESC) of the Slot Function Control and Status register of a channel to enable the slot number checking.

1.16.10.3 Enabling Average Bits Per Slot Reporting

You can enable the ability to report the average bits that are transmitted in a slot.

The CBS Status register of the additional AV channels provides information about the average bits that are transmitted in a slot. The software can asynchronously read this register to retrieve information about the average bits transmitted per slot. Complete the following steps to enable average bits per slot reporting:

1. Enable timestamping by following the steps described in the [Initialization Guidelines for System Time Generation](#).
2. Program Bits [6:4], SLC, of the MTL_TxQ[n]_ETS_Control register of a channel with the number of slots over which the average transmitted bits per slot need to be computed.
3. Enable Bit 9 (ABPSSIE) of the MTL_Q[n]_Interrupt_Control_Status register of a channel to generate the average bits per slot interrupt.

NOTE

- The frequency of this interrupt depends on the value programmed in step 2. For example, when you program value 0 in the SLC field, the interrupt is generated at every 125 microsecond.
- When not required, you can disable this interrupt to stop the interrupt flooding.

4. Read Bits [16:0], ABS, from the MTL_TxQ[n]_ETS_Status register of a channel on each interrupt.

NOTE

The software can read the ABS bits in polling mode even if the ABPSIE bit is not enabled. When high, bit 1 (ABPSIS) of the MTL_TxQ[n]_ETS_Status register indicates that a new value is updated in the ABS field.

1.16.10.4 Disabling Flow Control for AV Enabled Queues

1.16.10.4.1 Transmit Flow

Program the EHFC (Enable Hardware Flow Control) bit of corresponding Rx Queue's MTL_RxQ[n]_Operation_Mode register to 0.

1.16.10.4.2 Receive Flow Control

Program the PSTQ[n] field corresponding to AV enabled Tx Queue in MAC_TxQ_Prt_Map0/1 register to 0.

1.16.10.5 Programming Guidelines for Flexible Pulse-Per-Second Output

After you enable the Flexible Pulse-Per-Second Output feature in the module controller, you can perform the following tasks:

- Generating Single Pulse on PPS
- Generating Next Pulse on PPS
- Generating a Pulse Train on PPS
- Generating an Interrupt without Affecting the PPS

1.16.10.5.1 Generating Single Pulse on PPS

To generate single pulse on PPS:

1. Program 11 or 10 (for interrupt) in Bits [6:5], TRGTMODSEL, of MAC_PPS_Control Register. This instructs the MAC to use the Target Time registers (MAC_PPS0_Target_Time_Seconds and MAC_PPS0_Target_Time_Nanoseconds) for start time of PPS signal output.
2. Program the start time value in the Target Time registers (MAC_PPS0_Target_Time_Seconds and MAC_PPS0_Target_Time_Nanoseconds).
3. Program the width of the PPS signal output in MAC_PPS(#i)_Width (for i = 0; i <= 3) Register.
4. Program Bits [3:0], PPSCMD, of MAC_PPS_Control to 0001. This instructs the MAC to generate single pulse on the PPS signal output at the time programmed in the Target Time registers.

1.16.10.5.2 Generating Next Pulse on PPS

When the PPSCMD is executed (PPSCMD bits = 0), you can cancel the pulse generation by giving the Cancel Start Command (PPSCMD=0011) before the programmed start time elapses. You can also program the behavior of the next pulse in advance.

To program the next pulse, follow this procedure:

1. Program the start time for the next pulse in the Target Time registers. This time should be more than the time at which the falling edge occurs for the previous pulse.
2. Program the width of the next PPS signal output in MAC_PPS(#i)_Width (for i = 0; i <= 3).
3. Program Bits [3:0], PPSCMD, of MAC_PPS_Control to generate a single pulse after the time at which the previous pulse is de-asserted. This instructs the MAC to generate single pulse on the PPS signal output, at the time programmed in Target Time registers.

If you give this command before the previous pulse becomes low, then the new command overwrites the previous command and the QOS may generate only 1 extended pulse.

1.16.10.5.3 Generating a Pulse Train on PPS

Complete the following steps to generate a pulse train on PPS:

1. Program 11 or 10 (for interrupt) in Bits [6:5], TRGTMODSEL, of MAC_PPS_Control Register. This instructs the MAC to use the Target Time registers for start time of the PPS signal output.
2. Program the start time value in the Target Time registers.
3. Program the interval value between the train of pulses on the PPS signal output in MAC_PPS(#i)_Interval (for i = 0; i <= 3) Register.
4. Program the width of the PPS signal output in MAC_PPS(#i)_Width (for i = 0; i <= 3) Register.
5. Program Bits[3:0], PPSCMD, of MAC_PPS_Control Register to 0010. This instructs the MAC to generate train of pulses on the PPS signal output with start time programmed in Target Time registers.

By default, the PPS pulse train is free-running unless stopped by 'STOP Pulse train at time' or 'STOP Pulse Train immediately' commands.

6. Program the stop value in the Target Time registers. Ensure that Bit 31 (TSTRBUSY) of MAC_PPS(#i)_Target_Time_Nanoseconds (for i = 0; i <= 3) Register is reset before programming the Target Time registers again.
7. Program the PPSCMD field (bit 3:0) of MAC_PPS_Control to 0100. This stops the train of pulses on PPS signal output after the programmed stop time specified in Step 6 elapses.

You can stop the pulse train at any time by programming 0101 in the PPSCMD field. Similarly, you can cancel the Stop Pulse train command (given in Step 7) by programming 0110 in the PPSCMD field before the time (programmed in Step 6) elapses. You can cancel the pulse train generation by programming 0011 in the PPSCMD field before the programmed start time (in Step 2) elapses.

1.16.10.5.4 Generating an Interrupt without Affecting the PPS

The Bits [6:5], TRGTMODSEL, of the MAC_PPS_Control Register enable you to program the Target Time registers to do any one of the following:

- Generate only interrupts.
- Generate interrupts and the PPS start and stop time.
- Generate only PPS start and stop time.

Complete the following steps to program the Target Time registers to generate only interrupt event:

1. Program 00 (for interrupt) in Bits [6:5], TRGTMODSEL, of MAC_PPS_Control Register. This instructs the MAC to use the Target Time registers for target time interrupt.
2. Program a target time value in the Target Time registers. This instructs the MAC to generate an interrupt when the target time elapses.

If Bits [6:5], TRGTMODSEL, are changed (for example, to control the PPS), then the interrupt generation is over-written with the new mode and new programmed Target Time register value.

1.16.10.6 Programming Guidelines for VLAN Filtering on Receive

Complete the following steps to program VLAN filtering on receive:

1. Program MAC_VLAN_Tag register for the following bit to select the filtering method:
 - a. ETV: Enable 12-Bit VLAN Tag Comparison or 16-bit VLAN Tag comparison.
 - b. VTHM: VLAN Tag Hash Table Match Enable.
 - c. ERIVLT: Enable inner VLAN Tag or outer VLAN Tag (to enable the inner or outer VLAN Tag filtering, Double VLAN Processing should be enabled by setting EDVLP)
 - d. ERSVLM: Enable Receive S-VLAN Match or C-VLAN match (for S-VLAN processing to be enabled, set ESVL)
 - e. DOVLTC: Ignores VLAN Type for Tag Match
 - f. VTIM: to enable VLAN Tag Inverse Match instead of the normal VLAN Tag matching
2. Program VL of MAC_VLAN_Tag Register for the 12-bit or 16-bit VLAN tag.
3. If Hash filtering of VLAN tag is enabled, program MAC_VLAN_Hash_Table Register. When ETV bit is reset, upper 4 bits of the calculated CRC-32 of VLAN Tag are inverted and used to index the content of MAC_VLAN_Hash_Table register. When ETV bit is set, upper 4 bits of calculated CRC-32 of VLAN Tag are used to index the content of MAC_VLAN_Hash_Table register. For example, when ETV bit is set, a hash value of 4b'1000 selects Bit 8 of the VLAN Hash table. When ETV bit is reset a hash value of 4b'1000 selects Bit 7 of the VLAN Hash table.

1.16.10.7 Programming Guidelines for Extended VLAN Filtering and Routing on Receive

For the indirect access of the per VLAN Tag registers, follow these steps:

- Write
 - Write the required data into the MAC VLAN Tag Data register.
 - Program the VLAN Tag Control Register's OFS field with the required Filter Register's offset and command type to the CT field. For a write command, set this bit to 0.
 - Write 1 to the OB field and wait till the OB bit is reset to do the next write. This will guarantee that the appropriate VLAN Tag Filter Register has been programmed.
- Read
 - Program the VLAN Tag Control Register's OFS Field with the required register's offset and command type to the CT field. For a read command, set this bit to 1.

- Write 1 to the OB field and wait till the OB bit is reset. The appropriate VLAN Tag Filter register's value is available in the MAC VLAN Tag Data Register.

1.16.10.8 Programming Sequence for Queue/Channel Based VLAN Inclusion Register

To program the queue/channel-based VLAN inclusion register, complete the following steps:

NOTE

Indirect VLAN include registers are not accessed while setting the CBTI bit.

1. Set the CBTI bit of MAC_VLAN_Incl register to 1, to enable queue/channel based VLAN Tag insertion on all transmitted packets. This bit must be set before any indirect access to the queue/channel specific MAC_VLAN_Incl(#i) register.
2. Program the VLAN Tag and VLAN Type to be inserted in packets from a particular queue/channel in VLT and CSVL fields of MAC_VLAN_Incl register; corresponding offset address in ADDR field (0 for queue/channel 0, 1 for queue/channel 1, and so on) must be set. Set the RDWR bit to 0 to indicate write access. The write to byte 0 (byte 3 in Big Endian mode) of MAC_VLAN_Incl register initiates access to indirect access MAC_VLAN_Incl(#i) register.
3. The BUSY bit of MAC_VLAN_Incl register is set by the module to indicate the progress of access to indirect access MAC_VLAN_Incl(#i) register. On completion of the access, the BUSY bit is cleared. The application must not attempt subsequent access to MAC_VLAN_Incl(#i) register when the BUSY bit is 1.
4. Repeat step 2 and step 3 to program VLAN Tag and VLAN Type to be inserted in packets from the remaining queues/channels. The application must ensure that the required VLAN Tag and VLAN Type for all the queues/channels are programmed; otherwise unintended VLAN Tag and VLAN Type might be inserted.

1.16.11 Programming Guidelines for EST

Program the gate control values and time intervals in the Software Owned Gate Control List (SWOL) along with the other EST related registers that are described in [GMAC register descriptions](#) to appropriate values. The following sub-sections provide step by step details for programming the GCL and the other EST related registers.

1.16.11.1 Programming the GCL and GCL Linked Registers

Follow the below steps to program the Gate Control List (GCL) and the four other registers implemented per GCL:

1. The GCL and the four other GCL-linked registers should be accessed via indirect addressing using the MTL_EST_GCL_Control and MTL_EST_GCL_Data registers. The SWOL bit of the MTL_EST_Status register indicates if GCL0 or GCL1 is owned by software.
2. To program the GCL, write the 32-bit write data to MTL_EST_GCL_Data register. Then program TL_EST_GCL_Control register to write the write address and other control information.
3. In the MTL_EST_GCL_Data register, Write Data consists of up to 8 bits (configurable) of Gate Controls and up to 24 bits (configurable) of Time Interval. Gate Close is indicated by programming a '0' and Gate Open is indicated by programming a '1'. For a 4-TC and 20-bit Time Interval Configuration, the data width is 24-bits and the remaining 8-bits are reserved/read-only. The data should be written in the following format.
{8'h0, TC3, TC2, TC1, TC0, 20-bit Time Interval} where TCx = 0 or 1.
4. MTL_EST_GCL_Control register: Program the SRWO bit to 1 (to start a Write Op) and program the address and R/W fields appropriately.
5. Poll and check for the SRWO bit to be cleared by hardware to indicate the completion of the previous operation before initiating a new R/W operation via the same indirect addressing mode.
6. Repeat steps 3, 4, 5 until the programming of the GCL is completed.
7. Using the same indirect addressing method described above, program the BTR, CTR, TER and LLR registers. Set the GCRR bit in the MTL_EST_GCL_Control register appropriately. The GCRR bit interprets the address field as belonging to these registers (instead of the GCL).

After programming of the GCL and the related registers, program the MTL_EST_Control register to allow hardware to own and process the GCL. When the List Length (as indicated in LLR) is 1, the associated Time Interval should be smaller than the value of the Cycle Time Register. Otherwise, an error is reported (as detailed in the Error handling section) as a single set of gate controls add no value in the TSN context.

NOTE

The time unit in all the GCL related registers is Seconds and Nano-seconds. In cases where internally generated PTP System Time is used, the nano-seconds field must be programmed to use the Digital Rollover mode. (TSCTRLSSR field of MAC_Timestamp_Control must be set to value '1')

1.16.11.2 Programming the EST Registers

After completing the steps mentioned in "[Programming the GCL and GCL Linked Registers](#)", program the MTL_EST_Control register.

1. Set the Current Time Offset Value and Time Internal Left Shift fields of the MTL_EST_Control register appropriately. Also, set the Enable EST (EEST) and Switch to SWOL (SWOL) bits.

NOTE

The CTOV recommendations for GMII for SPRAM configurations are:

- 96 * Tx clock period, for 32 and 64 bit data width configurations.
- 128 * Tx clock period for 128 bit data width configurations.

The CTOV recommendations for GMII for non-SPRAM configurations are:

- 30 * Tx clock period, when SA/VLAN insertion is enabled.
- 22 * Tx clock period, when SA/VLAN insertion is not enabled.

2. This will enable the hardware to own and process the new GCL and make a switch to the new GCL at the BTR value. The hardware provides an interrupt (if enabled) when the switch to the new list happens.
3. Appropriate action has to be taken by software to address any other interrupts (explained in the Inter-rupts section) received during the hardware execution of the GCL.

Set the SSWL (Switch to Software Owned List) bit in the MTL_EST_Control register to handoff to hardware. Software is not allowed to write to the GCL and GCL linked registers when the SSWL bit is set, because the new GCL might be in use by the hardware.

The SSWL is reset/cleared by the hardware when it successfully switches to the new list. The hardware also flips the SWOL bit to indicate the new GCL that the software owns.

To install a new GCL, program the GCL it owns (indicated by SWOL bit) as described in "[Programming the GCL and GCL Linked Registers](#)" and then program the MTL_EST_Control register as described above. Ensure that the new BTR is set to an appropriate value to avoid BTR error that might need software intervention in some cases.

To avoid transmission overruns, the packet length (frame size) information should be available at all times. Therefore, in the DMA configurations, program the packet length in the first descriptor of every Tx frame. Similarly, in the MTL configuration, provide the packet length in the control word.

NOTE

To avoid transmission overruns, the Packet Length provided in the Transmit Descriptor should account for the SA and VLAN insertion, if applicable, that is, Packet Length should represent the actual packet length on the Ethernet line.

1.16.12 Programming the Launch Time in Time-Based Scheduling

The Launch Time is programmed in the Enhanced Normal Transmit Descriptors in DMA configurations and is driven as a control word in MTL configurations as follows:

The OSTC and Launch Time features are mutually exclusive and should not be used together; in case of a conflict (if OSTC = LTV = 1 in MTL configuration), priority is given to OSTC and the Launch Time is ignored.

In DMA configuration, if a context descriptor is received with a valid OSTC values immediately before receiving a first normal descriptor with LTV bit set, then the LTV is ignored.

1.16.13 Programming Guidelines for Media Clock Generation and Recovery

1.16.13.1 Programming Guidelines for Media Clock Generation

1. To set the PPS instance in Media Clock generation mode, program the appropriate Presentation Time Control (supported generation modes "0001-0011") to PPSCTRL_PPSCMD (for 0th instance) and PPSCMDi (for 1, 2, 3 instances) of MAC_PPS_Control register. Program all the MCGRENi of the MAC_PPS_Control register to enable MCGR mode
2. Based on the selected PPS instance, application must drive the appropriate trigger signal to the corresponding mcg_pst_trig_i[#i].
3. Based on the programmed mode, DUT captures the timestamp and programs it in the MAC_PPS(#i)_Target_Time_Seconds register. Then, mcgr_dma_req_o[#i] is set.
4. For every request generated by the DUT, the module reads the corresponding MAC_PPS(#i)_Target_Time_Seconds register and acknowledge the read request by asserting the corresponding mcgr_dma_ack_i[#i].

1.16.13.2 Programming Guidelines for Media Clock Recovery

- Enable the Current Presentation Time (CPT) Counter by setting the PTGE field of MAC_Timestamp_Control register. In addition to programming the initialization values for the System Time, update the MAC_Presn_Time_Updt register with the equivalent Presentation Time init value. Only then the TSINIT field of MAC_Timestamp_Control register is set.
- The Increment values used for the System time is also used for the Current Presentation time. This is because, the increment value is in Sub-seconds and Sub-nanoseconds.
- When an update is needed, along with programming the update values for the System Time, the MAC_Presn_Time_Updt register should be updated with the equivalent Presentation Time update value (32-bit ns). Finally, TSUPDT field of MAC_Timestamp_Control register must be set for the update to happen.
- To set the PPS instance in Media Clock recovery mode, program the appropriate Presentation Time Control (supported recovery modes "1001-1011") to PPSCTRL_PPSCMD (for 0th instance) and PPSCMD(#i) (for 1, 2, 3 instances) of MAC_PPS_Control register. Program all the MCGREN(#i) of the MAC_PPS_Control register to enable MCGR mode.
- After PPS instance is set in recovery mode, corresponding mcgr_dma_req_o[#i] is set by the DUT. For every request generated by the DUT, program the target presentation time into the MAC_PPS(#i)_Target_Time_Seconds register and acknowledge the request by asserting the corresponding mcgr_dma_ack_i[#i]. Acknowledgment can be given before/after programming the Target presentation time.
- Observe the recovered media clock on the corresponding ptp_pps_o[#i].

1.16.14 Programming Guidelines for ECC Protection for Memories

Programming guidelines for the ECC protection for memories:

- Enable the ECC feature for the respective memory by setting the appropriate bit in MTL_ECC_Control register.
- A correctable interrupt (sbd_sfty_ce_intr_o) or an uncorrectable interrupt (sbd_sfty_ue_intr_o) are generated to indicate to the application if any correctable/uncorrectable/address errors are detected. Appropriate status is indicated in the DMA/MTL_ECC_Interrupt_Status register.
- Debug mode is provided for each memory to specify errors.

NOTE

Enable the ECC feature before the traffic is online (or after the reset). Otherwise false interrupts might be triggered.

1.16.14.1 Programming Guidelines for ECC Hardware Error Injection (Debug Mode)

Follow the below steps for ECC Hardware Error Injection:

- Enable the ECC error injection feature for MTL TX/RX and DMA TSO, by setting the appropriate bit in the MTL_DBG_CTL register.
- Enable the ECC error injection feature for the EST memory by setting the appropriate bit in the EST_G-CL_Control register.
- Enable the ECC error injection for RX Parser memory by setting the appropriate bit in the MTL_RX-P_Indirect_Acc_Control_Status register. To access Rx Parser memory in debug mode, disable the Rx Parser feature by setting the FRPE bit of MTL_Operation_Mode register to 0.
- Where multiple CSR writes are required, for writing single data word into the memory, the application should ensure that all the CSR writes corresponding to one memory write should have the same value for the error injection control word.

1.16.15 Programming Guidelines for On-Chip Datapath Parity Protection

Follow the below steps for on-chip datapath parity protection:

- Set the EDPP bit of MTL_DPP_Control register to enable the generation of parity and detection of parity Mismatch on datapath.
- An uncorrectable interrupt (sbd_sfty_ue_intr_o) is generated to indicate to the application if any parity Mismatches are detected and the appropriate status will be indicated in DMA/MTL_Safety_Interrupt_Status and MAC_DPP_FSM_Interrupt_Status register.
- Debug mode is supported for each parity generator, to insert error.

NOTE

To avoid false safety interrupts, enable the datapath parity protection before the start of Rx or Tx traffic.

1.16.16 Programming Guidelines for FSM Parity and Timeout

Follow the below steps to program the FSM parity and timeout:

- Set the PRTYEN and TMOUTEN fields of MAC_FSM_Control register to enable the FSM parity and timeout feature respectively.
- For parity error detection, force the FSMs to enter into a state with even number of 1s and wait for the safety uncorrectable interrupt to be set with FSM_PES of MAC_DPP_FSM_Interrupt_Status.
- For Error Injection mode in FSM parity, select the appropriate clock domain for which parity error injection should be set using the [23:16] bits of MAC_FSM_Control register. Wait for the safety uncorrectable interrupt to be set with FSM_PES of MAC_DPP_FSM_Interrupt_Status.
- For FSM Timeouts, program the large and normal mode values indicated by LTMRMD and NTMRMD of MAC_FSM_ACT_Timer register.
- Select large or normal mode tic generation per clock domain by using the [31:24] bits of MAC_FSM_Control Register.
- Set the TMR field of the MAC_ACT_Timer field with the appropriate number of CSR clock cycles used to generate 1us tic.
- Force the FSM in a particular clock domain to be in an active state for Timeout duration (for example, T-2T). Check for the safety interrupt and the status bit for the corresponding clock domain getting set in [15:8] bits of MAC_DPP_FSM_Interrupt_Status register.
- For Error Injection mode in FSM timeout, select the appropriate clock domain for which timeout error injection must be set using the [15:8] bits of MAC_FSM_Control register. Wait for the safety uncorrectable interrupt along with the corresponding error status set in MAC_DPP_FSM_Interrupt_Status.
- For application /CSR interface timeout, only normal mode tic generation is used. Based on the time-outs of configured interface, appropriate fields (MSTTES, SLVTES) of MAC_DPP_FSM_Interrupt_Status is set along with the safety interrupt.

1.17 Revision history

Table 67. Revision history

Location	Technical changes
Unicast Destination Address Filtering	• Updated MAC addresses count

How To Reach Us

Home Page:

nxp.com

Web Support:

nxp.com/support

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